GASTROENTEROLOGY NURSING
A Core Curriculum

Gastroenterology Nursing

A Core Curriculum

Coordinated by the

Society of Gastroenterology Nurses and Associates, Inc.

illustrated

SGNA™
Society of Gastroenterology Nurses and Associates, Inc.

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This core curriculum should not be relied upon as a standard of practice in the field. Standards should be used as adjunctive study materials.

The Society of Gastroenterology Nurses and Associates hopes this material will enhance the knowledge and practice of the individual user. However, no guarantee can be made that mastery of this content will ensure success in future studies or qualifications for advanced levels of practice.

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CONTENTS

GASTROENTEROLOGY NURSING
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SECTION I

Gastroenterology Nursing Practice

Section I provides essential information associated with gastroenterology nursing as a specialty. It presents an overview of issues related to the development of gastroenterology nursing, as well as key expectations and responsibilities of gastroenterology nursing professionals. Additionally, notable topics such as Infection Prevention and Environmental Safety are thoroughly discussed.

Chapter 1

THE GASTROENTEROLOGY NURSE AND ASSOCIATE

This chapter reviews background information needed for health care workers to develop the skills required for the area of gastroenterology. It discusses the scope of practice and competencies required in this setting in order to provide quality care for patients who are having endoscopic diagnostic and therapeutic treatments. In addition, the chapter addresses educational preparation, job qualifications, practice settings, and the types of positions needed with an emphasis on the required responsibilities and functions.

Learning objectives

After reviewing the content of this chapter, the gastroenterology nurse should be able to:

1. Trace the evolution of gastroenterology nursing as an area of specialization.
2. Discuss the scope of practice in the gastroenterology setting and the educational requirements needed to practice quality gastroenterology nursing.
3. Describe the general responsibilities and functions of the gastroenterology nurse and associate.
4. Delineate the roles of the gastroenterology nurse and associate as specified by the Society of Gastroenterology Nurses and Associates, Inc. (SGNA).
5. Provide examples of the job responsibilities of nurse managers, staff nurses, and associates in a typical gastroenterology unit.

BACKGROUND AND HISTORICAL PERSPECTIVE OF NURSING

The concept of nursing has been around since the beginning of time and has transformed throughout history into what it is today. In the third century B.C., Greek physician Hippocrates referred to caregivers as assistants to the physician (Roshdahl & Kowalski, 2016). Nursing care was first differentiated from medicine in the fourth century by Hindu physician B.C. Charaka, who referred to the "aggregate of four" composed of the physician, the drug, the nurse, and the patient (Donahue, 2010). There are accounts from the pre-Christian era of nurses involved in caring for the sick. Archaeological discoveries describe early nursing procedures such as dressing wounds and feeding patients.

The Egyptians had a highly developed medical community, wherein the members of the medical profession were organized so they could protect the secrets of their practice. The Egyptians were as progressive as today's physicians in that they had specialists dedicated to one disease, such as treatment of the eyes, head, or stomach. Accounts of the Egyptians, Greeks, and Romans all refer to the existence of midwives, whose art was the care of child-bearing women. When their civilizations declined, medical care of women deteriorated and was not brought back to its former stage of development until the 17th century (Theofanidis & Sapountzi-Krepia, 2015).

During the Middle Ages, nursing was done primarily by women and was the function of many religious orders. Between 500 and 1300 A.D., nursing care consisted of only the most menial tasks, such as bathing, feeding, and bed-making. The nuns, who were chiefly responsible for this care, were assisted by women who were being punished for thievery or prostitution. It was not until the 19th century that education and dignity were brought to the nursing profession. Florence Nightingale emancipated upper-class women from idleness and encouraged them to serve humanity. As women became educated, the care of the sick began to improve.

Throughout the 19th century, prominent women became involved in promoting the cause of nursing.

School nursing, industrial nursing, and other nursing specializations began to emerge. As technology increased and physicians needed more time to learn and implement new techniques and practices, the responsibilities of nurses also increased.

In the late 19th and early 20th centuries, nurses worked to address the health care needs of men, women, and children by delivering direct care, promoting health, and preventing disease in remote and isolated areas. The need for nurses increased during the World Wars, with nurses being required to provide care without adequate training. During the 1940s, Frances Reiter first described the nurse with advanced education and clinical competence as a "nurse clinician" (Reiter, 1966). This was the precursor of nursing specialties with further education in certain fields that we recognize today.

Development of gastroenterology

As early as the time of Hippocrates, specific mention was made of the gastrointestinal (GI) tract in medicine. Hippocrates recorded use of a candle to inspect the rectum. In the 18th century, Bozzini documented the use of a rigid sigmoidoscope (Cyriac & Sahney, 2016). Almost 100 years later, Kussmaul made the first attempt to visualize the stomach with a rigid tube. Rigid esophagoscopy developed slowly over the next 50 years, with progress being dependent primarily on the quality of the light source. The rigid instrument was able to survey only a small portion of the stomach, and it carried a significant hazard because of the possibility of perforation.

In 1932, a semiflexible instrument was designed by Rudolph Schindler (Schäfer & Sauerbruch, 2004). The Schindler gastroscope remained the model for development of gastroscopes for the next 30 years. The semiflexible gastroscope was based on the principle that a series of convex lenses could transmit light undistorted through a flexible tube if the distal tube was not bent beyond a certain angle. Risk of perforation was reduced by the placement of a rubber obturator at the tip.

In 1958, Hirschowitz, Curtiss, Peters, and Pollard published their report of a new gastroscope, the fiberscope, which revolutionized gastroenterology. The development of fiberoptic scopes was made possible by the earlier work of **Professor Harold Hopkins** of University of Reading in the United Kingdom. Hopkins worked in parallel with John Logie Baird, the inventor of television, to design fiberoptic bundles that would transmit an image (Goddard, 2018). The optical principles are dependent on the total internal reflection of light in each fiber. The fiber bundles are of two types: noncoherent bundles that conduct light but not images and coherent bundles that produce high-quality images.

Modern flexible fiberoptic instruments have the same basic features as those developed in the early 1950s. The simplicity, ease of use, and safety compared to the earlier instruments caused the rapid adoption of this new technology. As new technology continues to evolve, conceptual, diagnostic, and therapeutic advances in gastroenterology have rapidly advanced. A clinical research study done at the end of the last century describes the ten most significant advances in gastroenterology during the 20th century:

1. *Helicobacter pylori* identification and treatment.
2. Use of fiberoptic endoscopy.
3. Gastrointestinal imaging by radiograph and computed tomographic scan.
4. Australia antigen, including vaccines for hepatitis A and B.
5. The molecular basis of colon cancer.
6. Liver transplantation.
7. Laparoscopic-assisted surgery.
8. Therapy for peptic ulcer disease, including H_2-receptor antagonists and proton pump inhibitors.
9. The discovery of gastrointestinal hormones, beginning with secretin.
10. The discovery of the role for gluten in celiac disease (Janowitz, Abittan, & Fiedler, 1999).

Development of gastroenterology nursing

Gastroenterology nursing was first recognized as an area of specialization in 1941, when a group of physicians called the American Gastroscopic Club met at Dr. Rudolph Schindler's home in Chicago. This group, later named the Gastroscopic Society, was the forerunner of the American Society for Gastrointestinal Endoscopy. At that time, Dr. Schindler was the recognized master gastroscopist, and his wife, Gabriele, was the first gastroenterology assistant. **Gabriele Schindler** was always at her husband's side, soothing the patient, helping with positioning, and assisting during the procedure (Schäfer & Sauerbruch, 2004). The memory of Gabriele personifies the spirit of professionalism and caring that has become the mark of excellence for today's gastroenterology nurses and associates. Since 1985, SGNA has annually presented the Gabriele Schindler Award to one of its members in recognition of high standards and outstanding achievement in gastroenterology nursing.

The next several decades brought about many changes in gastroenterology nursing practice and education. Fiberoptic instrumentation developed rapidly, as did the demand for skilled personnel to operate this instrumentation, resulting in the need for dedicated professionals to work with patients who were undergoing endoscopic procedures. Physicians demanded a specialized unit within the hospital to perform GI procedures. A nurse or assistant was required to attend the patient and assist the physician with the procedure. Initially, the role of the gastroenterology nurse was to support the patient while the physician inserted the scope. It soon became evident that a successful unit called for someone who would not only care for the patient, but also set standards and develop some order within the unit. There was a need for procuring and maintaining instruments according to industry standards, assisting physicians, monitoring patients, validating patient consent forms, dispensing and documenting medications, noting patient responses to treatment, and recording information for hospital reports.

As the complexity of procedures increased, government and regulatory agencies added another aspect to the knowledge base required. The Centers for Disease Control and Prevention (CDC) and other organizations put forth guidelines that required gastroenterology personnel to be informed of these new regulations and guidelines and to integrate them into practice settings. Persons functioning in gastroenterology units felt the need for a supporting network of individuals to unite those associated with the practice of gastroenterology.

Society of Gastroenterology Nurses and Associates, Inc.

The new guidelines and the subsequent need for support groups led to the formation of the Society of Gastrointestinal Assistants (SGA) in 1974. The first members of this group sought identification, respect, and national recognition of their activities. Their goals were to collect information, establish guidelines for future professionals, and expand specialized educational opportunities. SGA decided to hold annual meetings in May, concurrent with the annual educational meeting of gastroenterologists.

As the Society grew, the need to communicate information to the membership on an ongoing basis became evident. Therefore, in 1977, the first issue of the *SGA Journal* was published. Since 1989, the title of the journal has been *Gastroenterology Nursing*. Also in 1977, nine regional societies were established to give members the opportunity to meet throughout the year in geographically accessible areas. The number of regional societies has increased gradually to more than 60.

The American Nurses Association (ANA) and other independent specialty groups believed there was merit in establishing a mechanism to measure the competency of nurses in the practice setting. Certification for gastroenterology nursing had its beginning in the early 1980s, when SGA established a standing committee on certification to develop a certification program for gastroenterology clinicians. In 1985, the Certification Committee became the independent Certifying Council for Gastroenterology Clinicians, Inc. In 1986, a total of 666 candidates took the first certifying examination. This voluntary certification program is aimed at both assuring the patient that the caregiver has acquired a proficient level of knowledge and publicly recognizing the individual nurse or associate. The Society's membership continues to grow as it consistently demonstrates its commitment to further education and practice for the member.

In 1989, SGA changed its name to the **Society of Gastroenterology Nurses and Associates, Inc. (SGNA)** to better reflect the composition of its membership. SGNA members include 80 percent registered nurses (RNs); 8 percent licensed practical nurses/licensed vocational nurse (LPNs/LVNs); and 3 percent licensed medical technicians or medical technologists (LMTs or MTs), radiology technologists (RTs), and equipment technicians and associates. The remainder is comprised of affiliate members, such as vendors, managers, and consultants.

In 1990, to be consistent with the professional society, the Certifying Council for Gastroenterology Clinicians, Inc. changed its name to the Certifying Board for Gastroenterology Nurses and Associates (CBGNA). Further refinement of the certifying process began that same year. Until October 2006, different certification examinations were available for the RN and the LPN/LVN. After 2006, the certification exam was offered to the RN only. In addition, the organization's name was changed to the American Board of Certification for Gastroenterology Nurses (ABCGN).

Certification, as defined by the American Board for Specialty Nursing Certification (ABSNC), is the formal recognition of the specialized knowledge, skills, and experience demonstrated by the achievement of standards identified by a nursing specialty to promote optimal health outcomes (American Board of Nursing Specialties, 2016). ABSNC believes that the increasingly complex patient/client needs within the current health care delivery system are best met when RNs, certified in specialty practice, provide nursing care. The ABCGN has received accreditation from the ABSNC in recognition of the value of our certification program. For information about the examination contents and requirements for certification, contact ABCGN directly at ABCGN.org. Unlicensed Assistive Personnel and LPN/LVNs are encouraged to complete the SGNA Associate's Programs to become a GI technical specialist (GTS) and an advanced GI technical specialist (AGTS).

SCOPE OF PRACTICE

Scope of practice defines the services that an educated and trained health care professional is considered competent to perform. Delineating a **scope of practice** is the responsibility of a professional society. The practice of nursing is further defined by state regulatory agencies and institutional policies and procedures. The ANA supports the belief that changes in scope of practice and overlapping responsibilities are inevitable, so patients' interests are best served by a health care system that has the ability to increase access to specialty care (American Nursing Association, 2015).

As the preeminent professional society for nursing in the United States, ANA has assumed the responsibility of providing a current definition of nursing that reflects the dynamic, evolving role of professional nurses. In

1978, the Executive Committee of ANA's Division on Medical-Surgical Nursing Practice appointed an ad hoc committee to identify and describe the parameters of medical-surgical nursing. A statement on the scope of medical-surgical nursing practice was published in 1980. This statement defines the practice, delineates its dimensions, and outlines the functions of a professional nurse. This comprehensive definition serves as a guide to emerging specialty organizations and societies.

SGNA has since developed a scope of practice that defines gastroenterology nursing as the care of patients with known or suspected GI problems who are undergoing diagnostic or therapeutic endoscopic procedures. Nursing care includes the care and treatment necessary to provide comfort, assist individuals in the promotion and maintenance of health, meet the physical and emotional needs of the patient, and provide safe and proficient care during endoscopy and other specialized procedures. It encompasses patient assessment, nursing diagnosis, outcome identification, planning, implementation, and evaluation of patient care. The biological, psychological, and social components of the patient's response or adjustment to illness or disability are also taken into account. Gastroenterology nurses and associates use and adapt theories in microbiology, communication, ethics, and the behavioral sciences to form the basis of practice. That practice is continually influenced by the patient's physiological alteration, the patient's and family's needs for support and assistance, the collaboration between medicine and nursing, the continual attainment of quality care, and the level of professional autonomy in the practice setting.

EDUCATION AND TRAINING

SGNA continues to broaden its scope of practice and update standards for practice and the certification process. It is also essential to develop requirements for education, competence, and training of personnel who will be working in this practice setting because those factors have direct effect on the quality of patient care provided.

Educational requirements for RNs include graduation from an accredited school of nursing and a license to practice nursing. Additionally, RNs specializing in the field of gastroenterology would greatly benefit from at least one year of general medical-surgical experience. This enables the **gastroenterology nurse** to practice acquired skills and be exposed to "first-line" management. In general, individuals should possess the following backgrounds:

1. Solid education and training in the biological sciences.
2. Pertinent experiences in the health care field.
3. Familiarity with hospital and other practice settings.

Specific content areas that might be incorporated into a specialty training program include the following:

1. Anatomy and physiology of the GI tract and relationships to pathophysiology.
2. Relevant diagnostic and therapeutic procedures as they relate to routine and emergent GI situations.
3. Techniques of management of a gastroenterology unit.
4. Skills in patient care, teaching, and staff development.
5. Care and maintenance of clinical instruments.
6. Pharmacology and intravenous (IV) therapy.
7. Moderate sedation.
8. Emergency management.
9. Adult and/or pediatric advanced life support.
10. Research methods and application of peer-reviewed, evidence-based published research to the practice setting.
11. Ethical, professional, and legal standards inherent in professional conduct.
12. Cultural diversity.
13. Quality initiatives.
14. Patient- and family-centered care.

SGNA has developed position statements for the role delineation of the advanced practice registered nurse (APRN), registered nurse (RN), licensed practical/vocational nurse (LPN/LVN), and Unlicensed Assistive Personnel (UAP). A **gastroenterology nurse** includes the APRN, RN or LPN/LVN in gastroenterology (GI), hepatology, or endoscopy. A **gastroenterology RN** refers to the APRN and RN. A **gastroenterology associate** is an individual trained to function in an assistive role in the GI setting who has direct patient care responsibilities. The UAP may include GI technicians and GI assistants. The UAP is legally accountable to and supervised by an RN (ANA, 2017). The standards for the RNs are in accordance with the ANA Scope and Standards of Practice (2015). The LPN/LVN standards are in accordance with the National Federation of Licensed Practical Nurses, Inc. (2003).

The position descriptions, job responsibilities, and functions for all of these individuals vary according to the practice setting and its organizational structure. The duties also vary depending on the size of the facility and the number of procedures performed. In a hospital setting, the location of the gastroenterology department in the organizational structure also has an effect on the types of responsibilities assigned to different personnel. Another variable is the method used throughout the hospital to formulate position descriptions and the subsequent evaluation mechanisms.

Gastroenterology nurses also perform some gastroenterologic procedures independently, including esophageal and anal manometry, gastric and bowel analysis, and flexible sigmoidoscopy.

Patient care

Gastroenterology team members are responsible for the care of patients undergoing diagnostic and therapeutic gastroenterologic procedures. Preparation before a procedure includes instruction on physical preparation of the patient, such as food, medication restrictions, bowel preparations, and requirements related to moderate sedation or anesthesia. It is also important to meet the patient's psychological needs by supporting the patient, explaining the procedure, and ensuring that the patient is well-informed. A directed history is taken to evaluate and assess any pertinent medical, surgical, historical, or social conditions. Discharge needs, such as discharge instructions and post-procedure transportation, must be addressed during the pre-procedure assessment.

Pre-procedure phase (day of procedure)

When the patient arrives for his or her procedure, the patient's identity is validated by using two patient identifiers. An age-specific assessment is performed and documented by an RN. The components of pre-procedure evaluation should include but are not limited to: physical assessment; medical, surgical, and anesthesia history; psychosocial needs; current medications; validation of procedure; sedation type and consent; development of an individualized plan of care; and reinforcement of discharge instructions. The communication of patient assessment findings with health care team members promotes optimal outcomes.

Intra-procedure phase

Every patient undergoing a diagnostic, therapeutic, or invasive procedure requires monitoring by an RN or other qualified personnel. Each facility must comply with applicable regulations and guidelines, including state regulations, Centers for Medicare and Medicaid Services (CMS) requirements, and the facility's standards for monitoring of patients. Intra-procedure phase components include but are not limited to performing a time-out prior to the start of the procedure, administration of medications, and ongoing patient assessment. The communication of the procedure and any findings with health care team members promotes optimal outcomes.

During the procedure, care is focused on the patient's physical safety and psychological well-being. If an RN is giving conscious sedation under the direction of the doctor performing the procedure, his or her primary function is monitoring and assessing the patient and alerting the doctor to any untoward changes in the patient's physical condition. Additional functions of the RN include documentation and specimen labeling. The nurse and UAP must have a thorough understanding of the purpose of the procedure and how the procedure will be performed. He or she assists the physician with various technical aspects of the procedure, such as abdominal pressure, biopsy, polypectomy, coagulation, or other therapuetic manuvers. The nurse and UAP continue to offer reassurance to the patient throughout the procedure. The RN continually assesses the patient's response to sedation and to the procedure, intervening when appropriate. If a certified registered nurse anesthetist (CRNA) or anesthesiologist is giving sedation or anesthesia, the primary function of the nurse or UAP is assisting the doctor with the technical aspects of the procedure.

Post-procedure phase

An ongoing patient assessment is required until the patient is back to pre-procedure baseline. The nurse continues to monitor the patient's response to the procedure and to any medication administered. The frequency of the assessment is determined by institutional and departmental policy, the physician, and/or the nurse. During the initial post-procedure phase, attention should be focused on monitoring oxygenation, ventilations, circulation, level of conciousness, and temperature (American Society of Perianesthesia Nurses [ASPAN], 2019). Additional post-procedure components include, but are not limited to, patient monitoring for procedure-specific complications; age-specific, individualized discharge instructions; and validation of the patient's disposition.

In addition, post-procedure assessment is directed toward identification of potential and/or actual complications related to the procedure. ASPAN (2019) recommends that a patient is observed for at least 30 minutes following the last dose of medication unless he or she is being transferred to the same level of care. In addition, if a patient requires a reversal agent, it is recommended that observation be extended to a minimum of two hours. The nurse must ascertain that the patient has met the institution's discharge criteria before the patient is released. It is important to ensure that both the patient and the caregiver understand discharge instructions and that all questions have been answered. At discharge, the written explanation addressing follow-up care is reviewed with the patient, which should address the possible side effects of the procedure (including sedation), any changes in medication, diet and or activity restrictions, and additional resources on how to contact health personnel if problems arise. At this time, and with the patient's permission, the responsible adult providing the patient a ride home receives instruction on any post-procedure care, a review of procedure findings, and any discharge education to ensure a clear understanding of post-procedure instructions and to promote patient safety.

Documentation

Documentation of patient care and maintenance of records and department reports are other responsibilities of the gastroenterology team member. Documentation development is guided by the use of the nursing process (i.e., assessment, diagnosis, planning, intervention, and evaluation) to establish an individualized plan of care for the patient while in the endoscopy unit. Depending on institutional policy, the requirement for documentation may be less for non-sedated patients than for those receiving sedation. Documentation should clearly and uniformly record details that closely describe situations or events occurring to patients undergoing endoscopy or related procedures. Various members of the health care team may be responsible for documenting specific items in the patient record.

SGNA has compiled a documentation guideline that details pertinent documentation requirements and lists types of information that are recommended to be included during the pre-procedure, intra-procedure, and post-procedure phases and when providing discharge instructions to outpatients. The SGNA Guidelines for Nursing Documentation in the Gastrointestinal Endoscopy Setting can be found on the SGNA website.

Management

The scope of nursing management and the responsibilities of the manager of the gastroenterology department vary depending on the types of procedures performed and the services provided. Management functions include: developing and overseeing the organizational structure of the department (e.g., lines of communication and authority); establishing and enforcing standards for practice, policies, and procedures; creating position descriptions; and performing performance appraisals. To ensure optimal patient outcomes, a manager must coordinate services, develop policies, educate staff, and project for future success. Managers are fiscally accountable for planning for new and replacement equipment needs, developing and maintaining a budget, and determining staffing patterns. In addition, managers must make themselves aware of the ever-changing health care environment and keep up-to-date with the latest national standards and guidelines of the industry.

Care of instruments and equipment

Gastroenterology nurses and associates must demonstrate technical competency in instrument and equipment care in accordance with infection control recommendations, institutional and professional policies, and instructions for use (IFU) documents. Knowledge and understanding of the decontamination process, disinfection, and sterilization is essential to ensure safe patient care. Nurses and associates must also know how to maintain the instruments and equipment and arrange for repair when required.

Teaching and research

Individuals working in the gastroenterology department have a responsibility to be accountable for continual learning and willing to share knowledge. This may be done in staff development and orientation programs or through continuing education. Participation in professional organizations also offers an excellent opportunity to exchange information. The manager or educator of the department is responsible for assessing the level of expertise of the staff and encouraging employees to plan for their own personal and professional growth. It remains the responsibility of the individual nurse or UAP to identify their own learning needs, acquire necessary knowledge, and stay abreast of current technologies.

Participating in and communicating clinical research are also vital parts of the gastroenterology nurse's or associate's role. Research may entail participation in drug trial studies, testing new instruments or procedures, or evaluating the effectiveness of treatment. Whatever the research, the nurse or associate must have some background in data collection, documentation, analysis, and interpretation. Publication of findings is essential for communicating results of research to the health care community. In addition, presentation of a paper or poster at a professional meeting is an excellent way for the nurse or associate to meet the responsibility of expanding current knowledge. Knowledge of research techniques also allows for critical evaluation of published materials.

ROLE DELINEATIONS

Role delineation statements have been issued by SGNA to outline the scope of practice of nurses and associates.The complete and most current versions of the role delineations can be found on the SGNA website within the Standards of Clinical Nursing Practice and Role Delineations in the Gastroenterology Setting. An overview of the statements is to follow.

Role delineation of the APRN in gastroenterology

The changing health care environment has led to the development of expanded roles for nurses with graduate degrees in clinical specialties. Recognizing that the advanced practice nursing role in gastroenterology, hepatology, and endoscopy is still evolving, the following is a statement to broadly describe the responsibilities of the APRN specializing in gastroenterology and endoscopy nursing.

The APRN may be either a clinical nurse specialist or nurse practitioner who has completed an advanced degree in nursing (master's or doctorate) and who, through study and supervised clinical practice, has become an expert in a clinical area of nursing, such as gastroenterology, hepatology, or endoscopy. Additional specific training, credentials, and/or advanced practice certification may be required by the state in which the nurse practices. The scope of practice of the APRN is distinguished by the level of complexity, responsibility, and autonomy of practice. APRNs practice in a variety of settings, such as hospitals, private offices, ambulatory care centers, and clinics. The APRN functions within the scope of practice as defined by the state nurse practice act, job description of the employing facility, SGNA Standards of Clinical Nursing Practice and Role Delineations, the ANA Code of Ethics for Nurses (2015), the American Academy of Nurse Practitioners Standards of Practice and Core Competencies (2015), and the National Association of Clinical Nurse Specialists Statement on Practice and Education (2004).

The APRN provides service through core competencies of direct care, consultations, research, expert guidance, leadership, ethical decision-making, and collaboration (Hamric, Hanson, Tracy, & O'Grady, 2014). The specific patient populations to whom direct care is provided include adults, adolescents, or children with gastrointestinal or hepatic disorders and diseases. The care provided may include, but is not limited to, advanced assessment, diagnosis, treatment and care planning, implementation, evaluation, patient education, and endoscopy procedures.

APRNs build upon the role of the RN by exhibiting a greater depth and breadth of knowledge, an increased complexity of skills and interventions, and an advanced synthesis of data (ANA, 2015).

Role delineation of the RN in a staff position in gastroenterology

The role of the RN has expanded with the changes in advancing technology and newly defined patient needs. Recognizing that the role of the staff nurse in gastroenterology, hepatology, and endoscopy is still evolving, the role delineation broadly describes the responsibilities and functions of the RN in a staff role specializing in gastroenterology nursing. The roles the nurse assumes depend upon his or her basic nursing preparation, specialized formal or informal education, and clinical experiences. Certification as a certified gastroenterology registered nurse (CGRN) validates the acquisition of such skills and knowledge. The RN practices in a variety of settings, such as hospitals, private offices, ambulatory care centers, and clinics. The RN functions within the scope of practice as defined by state nurse practice acts, the job description of the employing facility, SGNA Standards of Clinical Nursing Practice and Role Delineations (2018), and the American Nurses Code of Ethics for Nurses with Interpretive Statements (2015).

The RN is accountable for the quality of nursing care rendered to patients. The RN assumes responsibility for assessing, planning, implementing, directing, supervising, evaluating direct and indirect nursing care, and identifying outcomes for patients in the gastroenterology setting. The RN is responsible for determining the education and competency level of assistive personnel to whom he or she is delegating work (ANA, 2015). The specific patient populations who receive direct care include adults, adolescents, and children with gastrointestinal disorders or diseases.

Role delineation of the LPN/LVN in the gastroenterology setting

The role of the licensed practical nurse/licensed vocational nurse (LPN/LVN) in gastroenterology, hepatology, and endoscopy is still evolving with changes in technology and patient needs. The following is a statement intended to broadly describe the responsibilities and functions of the LPN/LVN in this specialty. The roles which the LPN/LVN assumes depend on his or her basic nursing preparation, specialized formal or informal education, and clinical experiences. LPNs/LVNs practice in a variety of settings, such as hospitals, private offices, ambulatory surgery centers, and clinics. The LPN/LVN functions within the scope of practice as defined by state nurse practice acts, the job description of the employing facility, SGNA Standards of Clinical Nursing Practice and Role Delineations (2018), and the National Association of Licensed Practical Nurses (2015).

Under the supervision of an RN or physician, the LPN/LVN is accountable for the quality of nursing care he or she provides to patients, utilizing the nursing process and assuming responsibility for planning, implementing, directing, and evaluating nursing care for assigned patients in the gastroenterology and endoscopy setting. The specific patient populations who receive direct care include adults, adolescents, or children with gastrointestinal disorders and diseases.

Role delineation of UAP in gastroenterology

The changing health care environment has compelled the reassessment of the roles and tasks involved in providing patient care in gastroenterology, hepatology, and endoscopy practice settings. The following is a statement on this role.

UAP refers to individuals who are trained to function in an assistive role in the gastroenterology setting. These personnel who have specialized training or education in a specific area, such as gastroenterology (GI), may be further classified as technicians. Technicians may include GI assistants, GI technicians, or GI technical specialists (GTS) who have direct patient care responsibility and are supervised by an RN (ANA, 2015).

UAP will only accept or perform responsibilities that the individual knows he or she is competent to perform. After demonstrating required competencies, these personnel contribute to optimal patient outcomes by providing delegated patient care activities within specified limits (National Council of State Boards of Nursing [NCSBN], 2005).

POSITION DESCRIPTIONS

The position descriptions for all of these individuals vary according to the practice setting and its organizational structure. The duties also vary depending on the size of the facility and the number of procedures performed. In a hospital setting, the location of the gastroenterology department in the organizational structure also impacts the types of responsibilities assigned to different personnel. Another variable is the method used throughout the hospital to formulate position descriptions and the subsequent evaluation mechanisms.

Position descriptions usually include a number of common elements, including: the title of the position; the department in which the position belongs; the person to whom the individual is responsible; a job summary; job qualifications, such as level of education, experience required, and personal habits and characteristics; and specific duties or functions.

Nurse manager

Examples of the duties and responsibilities for the nurse manager of the gastroenterology department include, but are not limited to, the following:

Planning and organization

1. Plan, coordinate, and direct the flow of patients through the gastroenterology department.
2. Allocate space and physical resources according to patient and physician need.
3. Develop and review department performance improvement standards.
4. Develop and update clinical competencies.
5. Evaluate performance of staff as well as mentor and support staff to practice to their highest level of education.

Management

1. Conduct performance reviews for assigned staff.
2. Monitor clinical performance for compliance with established standards and policies.

Staff allocation

1. Adjust staffing to meet workload volume.

Fiscal responsibilities

1. Formulate and implement annual budget.

Education orientation

1. Support staff orientation and cross-training through education, case selection, assignment of preceptors, and ongoing evaluation.
2. Assist staff in ongoing educational process.

Professional commitment

1. Actively participate in meetings or committees as assigned.
2. Actively pursue certification in the specialty.

Developing healthy working relationships, building teams, enhancing communication, and practicing delegation are other areas that can be included in a manager's position description.

Staff nurse (RN)

Examples of some of the duties and responsibilities that may be assigned to an RN staff nurse include the following:

Nursing process

1. Provide assessment.
   a. Make initial observations of the patient on arrival at the gastroenterology unit.
   b. Formulate nursing diagnosis based on data collected.
   c. Communicate any significant patient assessment findings with the care team.
   d. Provide assessment documentation that reflects the full range of patient needs, including physical, psychosocial, spiritual, and safety.
   e. Continue assessment of the patient throughout entire stay in the gastroenterology unit.
2. Determine nursing diagnosis.
3. Identify outcomes.
4. Create plan.
   a. Implement the plan of care.
   b. Individualize the plan of care based on patient assessment.
   c. Prepare rooms, equipment, and supplies to accommodate the patient load.
5. Implement plan.
   a. Implement the plan of care.
   b. Administer medications and assist the physician with the procedure.
   c. Monitor the patient before, during, and after the procedure.
   d. Provide patient education.

Patient teaching

1. Assess the patient's needs to include family or significant other when appropriate.
2. Provide the patient with relevant information regarding diagnosis, medications, diet, and other therapy.

The above is only a sampling of the responsibilities that might appear. Other categories may include research and professional development.

Unlicensed Assistive Personnel

Examples of duties and responsibilities that might be expected of associates include the following:

1. Assist the physician and/or RN with procedures.
2. Implement the plan of care under direction of an RN.
3. Receive and decontaminate instruments, equipment, and reusable supplies.
4. Perform sterilization and reprocessing.
   a. Prepare items to be sterilized and maintain sterility through proper handling.
   b. Operate equipment used for cleaning and disinfecting endoscopic equipment.
5. Receive and place supplies in assigned storage area.

Examples of position descriptions, orientation programs, and competency tools are available in the *GI/Endoscopy Employee Development Resources,* published by SGNA (2015).

REVIEW TERMS

gastroenterology nursing assistive personnel, gastroenterology nurse, Professor Harold Hopkins, position descriptions, role delineation, Gabriele Schindler, scope of practice, Society of Gastroenterology Nurses and Associates, Inc. (SGNA)

REVIEW QUESTIONS

1. The first recorded gastrointestinal assistant was:
   a. Florence Nightingale.
   b. Frances Reiter.
   c. Gabriele Schindler.
   d. B.C. Charaka.
2. During what period was medicine so far advanced that they had what is equivalent to today's subspecialties?
   a. Egyptians.
   b. Pre-Christian era.
   c. Middle Ages.
   d. Greeks and Romans.
3. The individual who first developed the fiberoptic telescope used for GI procedures was:
   a. Hippocrates.
   b. Hopkins.
   c. Schindler.
   d. Baird.
4. The rationale for the rapid adoption of fiberoptic instruments includes:
   a. Simplicity, ease of use, and patient safety.
   b. Complex lens system and noncoherent bundles.
   c. Perfect imaging and ease of use.
   d. Combination of noncoherent bundles and coherent bundles to make a perfect image.
5. The first Society of Gastrointestinal Assistants (SGA) was formed in what year?
   a. 1968.
   b. 1976.
   c. 1972.
   d. 1974.
6. The practice of gastroenterology nursing requires application of the nursing process. The components include:
   a. Communication, planning, evaluation, follow-up.
   b. Assessment, diagnosis, planning, implementation, evaluation.
   c. Interviewing, observing, communicating, listening.
   d. Assessment, planning, evaluation, follow-up.
7. A **gastroenterology associate** is defined as:
   a. A nurse who is engaged in the field of gastroenterology.
   b. An individual trained to function in an assistive role in the GI setting having direct patient care responsibilities.
   c. An individual with advanced education who is engaged in gastroenterology.
   d. An individual who is responsible for transporting patients and caring for instruments and equipment.
8. Initial assessment of patients on arrival to the gastroenterology unit is usually the responsibility of the:
   a. Physician.
   b. Nurse manager.
   c. Staff nurse (RN).
   d. Associate.
9. A general requirement for a recently graduated nurse who is interested in practicing gastroenterology nursing is:
   a. To work for one year on a medical-surgical unit to gain experience.
   b. To work for a private GI physician to gain knowledge of diagnosis and procedures.
   c. To work in a special procedures unit for one year.
   d. To work in an ambulatory care unit to practice skills for one year.

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Chapter 2

NURSING ADMINISTRATION: LEADERSHIP AND MANAGEMENT OF THE GASTROENTEROLOGY DEPARTMENT

Nursing Administration requires an understanding of health care delivery systems as well as an ongoing commitment to nursing practice. The roles of the nurse leader and the nurse manager fall into this category. Their varied responsibilities complement one another and overlap on many fronts.

The American Nurses Association (ANA) (2015) states that the "registered nurse leads within the professional practice setting and the profession" p.5. The competencies related to this standard include but are not limited to developing an environment of trust and dignity, encouraging innovation, managing conflict, and mentoring colleagues.

Leadership can be defined as a multifaceted process of identifying a goal or target, motivating other people to act, and providing support and motivation to achieve mutually negotiated goals (Porter-O'Grady, 2003). To be a leader means developing the skills and behaviors necessary to occupy a leadership position. Managers are more involved with planning, organizing, directing, controlling, and staffing. Being a nursing leader requires a deep understanding of the interrelationship between leadership roles, as well as the principles and functions inherent in managing the human and material resources in a gastroenterology unit.

This chapter presents an overview of the **leadership attributes** of a nurse leader, leadership theory, and the components of nursing leadership. It will also review the fundamental concepts of management. These include the planning, organizing, directing, controlling, and staffing of an endoscopy unit. Nurses interested in future administrative leadership or managerial roles should continue formal education toward baccalaureate or graduate degrees, which are required for advancement in today's health care arena.

In addition to formal education, obtaining and maintaining certification in gastroenterology nursing such as the Certified Gastroenterology Registered Nurse credential (CGRN) demonstrates the nurse leader's professional competence and ability as an endoscopy clinician. Although certification may not be required for endoscopy nurse leaders or managers in health care organizations, it does validate professional goals and dedication to improving the quality of care for the gastroenterology and endoscopy patient.

Since it was established in 1990, there has been a continued trend for hospitals to earn Magnet recognition. This movement was the result of a study to investigate the problems associated with nursing shortages that improved a hospital's ability to attract and retain competent professional nurses (McClure & Hinshaw, 2002). Findings from this study described that leadership within Magnet organizations maintained a meaningful philosophy of patient care and a culture that supports the nursing staff. The participative management style was associated with the success of these organizations.

Learning objectives

After reviewing the content of this chapter, the gastroenterology nurse should be able to:

1. Describe leadership roles and the difference between the leadership and management functions in the administration of a gastroenterology department.
2. Describe the components of a group.
3. Discuss the five functions of management and provide examples of activities included in each function.
4. Describe the organizational structure of the gastroenterology unit.

5. Describe the function of budgeting and management of the gastroenterology unit.
6. Outline methods for accomplishing departmental goals through the efficient use of personnel.

THE FIRST NURSE LEADER AND MANAGER: FLORENCE NIGHTINGALE

The first and most discernible nursing leader was Florence Nightingale. In her book *Notes on Nursing* (1859), she discussed the basic functions of "taking charge" of a unit. Both men and women, she wrote, understand very little about the meaning of being in charge, from the most colossal calamities down to the most trifling accidents. She defined the individual in the charge or leadership role as one to not only carry out the proper measures by oneself, but also see that everyone else does so as well. The charge person must also verify that no one either willfully or ignorantly thwarts or prevents such measures. Being in charge also means ensuring that all involved individuals have carried out the tasks to which they were appointed. Finally, in caring for the sick, the charge person can make assignments for nurses based on an understanding of the acuity of the patients assigned to their care.

In regards to leadership, little has changed since Ms. Nightingale wrote these words. One difference, however, is that nurses today have the means to formally learn to be better-trained nurses and leaders and the capacity to practice what they learn in highly developed health care organizations. Nursing professionals also have the responsibility to seek out the resources they need in order to perform their role to the best of their ability within the interdisciplinary health care team.

LEADERSHIP

Nursing leaders today are challenged with increasingly complex health care environments. Decreasing supplies and staffing resources compounded by increasing technological and informational demands create a critical need for professional nurses to fully comprehend the leadership role and management functions.

The concepts of leadership and management may be better understood as relational instead of separate. An effective leader exhibits qualities and skills of both leadership and management. Leaders have a vision and create a long-term plan. They view situations in terms of the larger picture instead of the small and narrow focus. They are agents for change who will integrate new technologies and evidence-based research into organizational practice.

Leadership involves critical-thinking abilities, decision-making skills, and behaviors to bring about and accomplish change. Effective leaders accept challenges, take risks, and manage responsibility. Leadership skills and behaviors can be learned and continue to be developed over time.

To begin, leaders need followers. Leadership, therefore, requires an understanding of attributes that constitute a leader, as well as the attributes of the particular group of followers. A leader must be committed to a goal, plan, or idea; implement the leadership process; and be followed by at least one follower or group.

A group is generally defined as a number of individuals who work together toward a common goal. Groups may contain a formal and an informal leader. Formal leaders are appointed by organizational superiors, and the group has little or no input into the decision; informal leaders are individuals who are chosen by the group to lead. The group may come to a concensus consciously or subconsciously. The group process is generally referred to as the changes that occur within the group as members work together toward a mutual goal. The dynamics of the group refers to the interaction and communication patterns among group members. Both the group process and dynamics play an important role in forming the group climate, which is the level of cohesiveness or the loyalty, trust, respect, and understanding that exists among group members. A positive group climate supports higher productivity, while a negative situation presents inefficiency and increased challenges for the leader.

There are a number of interrelated qualities or leader attributes upon which the leader must possess the ability to self-reflect. These attributes include individual awareness, assertiveness, accountability, and advocacy.

- **Awareness** is a leader's inner capacity to honestly assess him- or herself and objectively evaluate the group members' abilities, maturity, endurance, and commitment to both organizational and departmental goals. In addition, a leader has an obligation to set limits and delegate responsibility in order to maximize the group's time and efficiency. Effective communication skills are essential for nurse leaders, as is an understanding of the use of power. Power may either be imposed on the group or shared with the group. An example of shared power is participative management.
- **Assertiveness** is the expression of feelings, needs, ideas, and rights in a professional manner.
- **Accountability** is defined as an obligation or willingness to accept responsibility for one's actions. Leaders are accountable to themselves, the group, the profession, and their superiors.
- **Advocacy** is a leader's ability to support and defend the group.

It is also important for a nurse leader to communicate a clear understanding of the organization and unit expectations. The nurse leader must be consistent in addressing critical issues such as availability, accountability, and conflict among the staff members in order to maintain credibility as the leader of the group.

LEADERSHIP: THEORY AND STYLES

It is important for nurses to understand the principles and complexities of nursing theory in order to become more effective leaders. These principles are scientifically acceptable and help to govern practice.

Early theories

Leadership theories have been based on behaviors or characteristics of leaders rather than on the leadership process itself. There has always been controversy among theorists about whether management is a function of leadership or leadership is a function of management. Nurses need to be knowledgeable of how leadership evolved to understand the current complexity of the leadership and management relationship (Marquis & Huston, 2017).

Great Man Theory
Great Man theory of Aristotle is one of the earliest leadership theories and is based upon the idea that leaders are born and not made. European monarchs are examples of this theory, which suggests that leadership ability is inherited rather than learned.
Trait Theory
Trait theory was developed through a retrospective review of individuals thought to be "great men" and included an examination of their traits and characteristics.
Contingency and Situational Theories
Contingency and Situational theories contend that work environments influence outcomes as much as a leader's style. This theory suggests that varying mixtures of leadership styles are more effective than a single style, and that the leadership model should be based upon the maturity levels inherent within the group dynamic (Hershey & Blanchard, 1977; Tannenbaum & Schmidt, 1958).
Empowerment Theory
Empowerment theory contends that formal and informal organizational structures influence a leader's empowerment (Kanter, 1989).

Contemporary theories

Motivational theories surfaced when advocates of the behavioral sciences became concerned about the lack of scientific validation behind the management theories. In the 1970s, theories focusing on an understanding of human needs and motivation became available.

Hierarchy-of-Needs Theory
Hierarchy-of-Needs Theory. Maslow (1943) was one of the first to initiate the human behavior school with his theory. He classified human needs into five categories: physiological, safety, love, esteem, and self-actualization.
Motivation-Hygiene Theory
Motivation-Hygiene Theory Herzberg's (1974) theory supports that job factors are classified as either satisfiers or dissatisfiers. Satisfiers, or motivators, include achievement, recognition, work, responsibility, advancement, and growth. For example, as an employee receives positive feedback, his or her level of performance improves. The job dissatisfiers, or hygiene factors, include supervision, company policy, working conditions, interpersonal relations, job security, and salary. These are not considered to be motivators because, for the most part, they do improve performance.

Interactive theories

Interactive theories became popular in the late 1970s and early 1980s. Contemporary managers were finding they were unsuccessful at motivating their employees to achieve maximum potential. A number of interactive theories were popularized as an alternative to motivational theory. McGregor's (1960) Theory X and Theory Y are examples.

Theory X
Theory X assumes that most people prefer to be directed, are not interested in assuming responsibility, and have to be constantly supervised.
Theory Y
Theory Y assumes that the employee is mature, independent, and self-motivated. Managers who believe that people are inherently lazy tend to use fear and threats to motivate personnel, delegate little responsibility, and do not include personnel in planning. Managers who philosophically believe that people are self-motivated and enjoy work use praise, offer recognition, and provide opportunities for growth.
Theory Z
Theory Z, also known as the Japanese or participative style of management, focuses on increasing employee loyalty to the company by providing a job for life with a strong focus on the well-being of the employee, both on and off the job (Ouchi, 1981).

The Transformational Leadership Theory

The Transformational Leadership theory emphasizes collective purpose and mutual growth.It is a "style of leadership in which the leader identifies the needed change and executes the change with the commitment of other (Sorensen, 2011).Transformational leadership style occurs when the leader inspires the team to succeed through empowerment (ANA, 2015).

A leadership style is related to how much control the leader allows the followers or the group. Each style exhibits a number of behaviors that characterize the leader. Regardless of which style is used, the leader needs an understanding of the role that behaviors play in managing the group to accomplish the goals of the unit.

Autocratic Style

Autocratic style demonstrates a behavior of control. The autocrat determines all the policies for the group, makes all the decisions, and details the implementation of goals. An example of this style is the military.

Democratic Leadership Style

Democratic leadership style involves individuals and groups in the decision-making process, and is particularly effective when cooperation and coordination among the group members is encouraged.

Laissez-Faire Style

Laissez-faire style empowers employees to do as they please. There are no policies and procedures in place, and no formal leadership structure.

THE ORGANIZATION

Most nurse leaders and managers practice in diverse health care environments that make up an organization. Regardless of the health care environment in which a nurse leader practices, it is imperative that the individual understand the nature, the structure, and the culture of the organization. The nurse leader must also be aware of the organizational hierarchy (or chain of command), be completely conversant with the organization's policies and procedures, and have the ability to access them as necessary.

The **organizational structure** determines the processes by which a specific group of people distributes responsibility, creates lines of communication, identifies relationships, and establishes authority.

Organizational structures can be formal or informal. An informal structure develops in all organizations. It is concerned primarily with the personal and social relationships that are not seen on the organizational chart. Formal organizational structures can be designed or rescinded, but informal structures are more difficult to dismantle because the manager has not instituted them. The manager needs to know that the informal structure exists, how it operates, and how to use it to advance the objectives of the unit. By knowing how the informal organization works, the manager can avoid activities that will unnecessarily threaten or disrupt it.

Formal organizational structures can be either bureaucratic or adaptive. Bureaucratic structures imply subdivision, specialization, technical qualifications, rules, and standards. The newer, adaptive organizational models are more free-form, open systems that are flexible and lend themselves to a more participative atmosphere.

Different types of formal organizational structures that may be appropriate for the gastroenterology unit include line authority, functional organization, staff authority, matrix organization, and project organization.

Line Authority

Line authority is the most traditional organizational structure, in which each position has authority over a lower one in the organization. This structure is better known as the chain of command or hierarchical structure.

Functional Organization

Functional organization authorizes a specialist from a given area to enforce recommendations within a clearly defined area. In an all-RN staff, one RN would act as the specialist or spokesperson for communication to the institution's administration.

Staff Authority Structure

Staff authority structure assigns the staff the primary responsibility of assisting personnel who are in a line authority or direct chain of command.

Matrix Organization

Matrix organization looks at individual subsystems within a complex structure. Depending on the complex structure, these subsystems can be totally dependent or totally autonomous. This is demonstrated in many decentralized systems in which subsystems are given the authority to work independently of other groups, toward a common goal.

Project Organization

Project organization is designed to accomplish a specific task. When the task is completed, the group is disbanded.

Organizational principles outline the efficiency of the organizational structure and help the manager and employees function more effectively within that structure. These principles include unity of command, requisite authority, and continuing responsibility:

Unity of Command

Unity of command implies clear lines of authority. The employee knows to whom they report and who has the final authority.

Requisite Authority

Requisite authority implies that when a subordinate has been delegated a task, he or she is given the final authority to accomplish the task.

Continuing Responsibility

Continuing responsibility implies that a manager who delegates responsibility for a function to a subordinate is still responsible for that function.

Recognition of these few basic principles assists the manager by enhancing the efficiency of the organizational structure.

Organizational cultures encompass how the organization thinks and behaves. Though not written, the culture of an organization includes values, beliefs, traditions, diversity, language, and customs, and is unique to every organization. Positive cultures focus on encouraging employees to proactively collaborate and interact in an effort to achieve common goals. Negative cultures are more reactive and confrontational in their approach to goals and act as a mechanism to protect individuals' status and security within the organization. Leaders must understand the culture of their organizations to implement effective strategies to bring about change (Marquis & Huston, 2017).

Organizational tools that are helpful to the gastroenterology manager include organizational charts and job descriptions. The organizational chart is a graphic representation of how the unit is organized. It depicts formal organizational relationships, areas of responsibility, and channels of communication. It is used for planning, administrative control, and policymaking.

Job descriptions are important because they assist the manager with organizing the administration of the various job duties. Descriptions specify the title of the position, the qualifications necessary to complete the duties listed, and the person to whom the employee is responsible.

Span of control, or span of management, is used to identify the scope of management responsibility. Although the manager remains the final authority, he or she delegates authority to subordinates who, in turn, supervise a group of employees.

Factors that influence the manager's capability to supervise include:

- The amount of experience and management training a manager has acquired.
- The amount and nature of the work a manager is required to perform.
- The willingness to accept responsibility.
- The characteristics of the employees themselves.
- The ability to design and define the complexity of procedures and activities that take place in the unit.

THE ROLE OF MANAGEMENT IN LEADERSHIP

Management can be defined as a process of getting tasks accomplished with the help of others. This means that individuals must be directed toward common organizational and departmental goals. Nurses in any level, particularly when in a position of responsibility, possess authority, which is the legal or rightful power to command and act. This authority reinforces order to alleviate disorganization and chaos. Regardless of how an individual applies that authority, it is important to have.

Management theory and function

In addition to leadership theory, there are a number of management theories that focus on interpersonal relationships.

- **Scientific management theory** focuses on the organization rather than the needs of the people within the organization.
- **Human relations theory** stresses a concern for the needs of the people versus the task. The human relations theory is based on the idea that the real power of the organization is the set of interpersonal relationships that develops within the working unit.
- **Managerial grid** includes both task and relationship concepts. This tends to be an attitudinal model because it measures the values and feelings of the manager. This theory is based on the manager's concern for tasks versus people within the organization.
- **Contingency model of leadership** suggests that leadership style can be effective or ineffective, depending on the situation. This model defines three aspects of a situation that structure the leader's role: leader-employee relations, task structure, and position power.

Five functions of management

The functions of management include planning, organizing, directing, controlling, and staffing. Nurse leaders and managers at all levels within the organization perform all five functions, although there will be variations in time spent on each function. For instance, the administrator probably spends more time planning and organizing, whereas the unit manager spends more time staffing, directing, and controlling.

Planning

Planning is the first and most important function performed by the gastroenterology unit nurse leader. Planning outlines a course of action that is realistic for the unit. It involves the development of a purpose or mission, philosophy, goals, and objectives for the organization, and the creation of "work maps" or work flows that depict how these goals and objectives are to be accomplished. Also important is the integration of the regulatory requirements and changes that are mandated by governmental and professional agencies in order to maintain practice.

Forecasting trends in patient treatment modalities and technologies are integral in the planning stage to ensure their implementation in the gastroenterology unit. Strategies to encourage good planning should include staff input. Staff and physicians can provide excellent feedback to help determine what equipment and supplies are needed.

Critical planning strategies must include projections for staffing based on the level of knowledge and skills needed to perform tasks in the gastroenterology unit. This includes physician staffing, expanded hours of operation, and the introduction of new technology for the unit.

Another planning task is to evaluate the unit's work flow. Determine if there are ways to become more efficient. Note any unnecessary activities that could be changed or eliminated. Finally, examine if there is a need to do any space planning. The physical space of the unit and the placement of equipment and furniture should be conducive to a smooth work flow.

In order to keep up with the speed of technology and its implications in health care, it is important to stay abreast of changes to help ensure efficiencies. Data resources suitable for forecasting are acquired by remaining abreast of current electronic and print information, attending professional meetings, and benchmarking evidence-based research and nursing informatics (Melnyk & Fineout-Overholt, 2014).

Managing the budget is one of the most important roles for the manager. The budget is prepared during the planning process and provides direction for the manager. The budget is simply a statement of estimated expenses and revenues for a predetermined time period (fiscal year). The two types of budgets are the operating budget and the capital budget.

The **operating budget** consists of consumable items and resources used to provide direct patient care. Operating expenses vary depending on the volume of patients and the procedures performed in the gastroenterology unit. Whatever budgetary goals are in place, the manager administering the budget should have a part in preparing it and be responsible for monitoring it throughout the fiscal year.

In most cases, the operating budget is further divided into direct or indirect expenses.

Direct expenses include items such as staffing salaries, medical and surgical supplies, repairs, and charges from other departments.

Indirect expenses include overhead cost including facility fees which contribute to the cost of running a business.

Table 2.1. Examples of Direct and Indirect Expenses

Direct expenses	Indirect expenses
• Salaries and wages • Per diem staff • Overtime • On-call personnel • Disposable items • Instruments • Medical and surgical supplies • Repairs (endoscopes, electrosurgical unit [ESU]) • Pharmacy	• Equipment depreciation • Building depreciation • Employee health benefits • Accounting • Purchasing • Utilities • Housekeeping • Clerical supplies

Calculating the staffing requirements is an important part of the operating budget. The goal is to secure enough budgeted positions to provide quality care in a cost-effective manner. There are various mathematical formulas and means to determine staffing requirements. Generally, the hospital administrator or departmental director will suggest a staffing formula. Examples of the most widely used methods for costing out nursing services are the **acuity level** and **direct care hours**.

Many health care providers use a patient classification system or **acuity level** to determine appropriate staffing levels. This is based on a system that assigns nursing care hours to an acuity level. This type of system needs further computation and assigns a relative value unit for each acuity level. It works best in intensive care units (ICUs) or inpatient units.

Total number of shifts per week required by staff category is an example of **direct care hours**. The staffing calculations are volume-driven and take into account the number and type of patients seen in the unit each day and the hours the unit will be staffed. Hours of operation include the time required for preparation of all needs of the unit. On-call staffing coverage for after-hours and emergency procedures are calculated separately and added to the unit's staffing budget. Meeting days, jury duty, and absenteeism must be added to the staffing budget, in addition to coverage for staff vacations. (Keep in mind that benefit hours increase with level of staff seniority.)

The supply part of the operating budget is usually based on the previous year's expenses and projection of costs for the next fiscal year. Other factors to consider include inflation rates, changes in workload by procedures or patients, the addition of new providers and/or specialty procedures, any new or different supplies and equipment technologies, and purchasing contracts.

The **capital budget** includes large purchases, such as equipment, furniture, or construction projects that amount to more than $500 (generally speaking) and usually depreciate over time. Most facilities expect the department manager to make a formal request for a capital budget in a proposal format that incorporates the following: data specific to the item requested; alternative solutions such as purchase, rent, or lease; priorities and recommendations; cost and benefit analysis; and an implementation plan. These costs are generally projected three to five years out and can include dollars for technology not yet available.

Organizing

Organizing, the second function of management, is the means by which a manager develops order and fosters productivity. Organizing involves the integration of resources and the assignment of activities within that unit for its maximum efficiency. This entails establishing lines of authority within activities and between departments, examining the contributions that individual employees make to the organization, and determining to whom they will report. The organizing process stems from the requirement among personnel for cooperation. The ultimate goals are to build, develop, and maintain a structure of working relationships that achieves the objectives of the unit in a positive manner.

Directing: Motivating and mentoring

After organizing, the function of management involves directing personnel and activities in such a way that the goals of the unit are met. Managers must demonstrate the ability to lead, motivate, and mentor employees by issuing directives, instructions, assignments, and orders. The main objectives are to complete work efficiently, to teach and serve as a resource to employees, and to be a role model.

Regardless of the level of management, directing is always an important function. It requires that a manager possess leadership skills and the ability to integrate a participative management style into the daily coordination of the gastroenterology unit. Every manager should become familiar with his or her own leadership style, managerial philosophy, and sources of power and authority.

Controlling: Quality assurance and performance improvement

The fourth management function of controlling includes setting standards, measuring performance, reporting results, and taking corrective actions. In controlling, the manager is concerned with making certain that the solution to a problem is properly implemented. This involves feedback of results and follow-up to determine the extent to which the predetermined goals have been accomplished.

Controlling entails three basic steps: communication of standards and scope of practice, evaluation of current process and strategies for performance improvement, and corrective action and strategies for implementation.

Communication of standards and scope of practice represent the overall objectives of the hospital. Standards of practice are broken down into relative objectives for each department. Standards and scope of practices must be realistic, achievable, and reflective of what the unit does and how it operates. These objectives include parameters such as quality, hours of operation, quotas, schedules, budgets, patient procedures, and professional standards of practice guidelines (Society of Gastroenterology Nurses and Assistants, Inc. [SGNA] practice statements). In addition, the scope of practice determines the staffing plan during optimum daily operation, as well as for potential emergencies. Because setting standards may seem overwhelming when considering the many types of activities in the department, the manager might break down the task to a more manageable size by focusing on selective strategic standards and specific critical performance indicators.

Evaluation of current processes and strategies for performance improvement is the next step. Today, the influence of continuous performance improvement permeates the hospital structure. This type of management philosophy results in many benefits, from reduced waste and improved services to better patient outcomes, increased efficiencies, and decreased cost of health care services.

Once the standards have been set, it is vital to take corrective action and strategies for implementation. It is the manager's responsibility to compare actual performance with standards and critical indicators within established timeframes. This is done by observing the work, analyzing organization satisfaction surveys, personally checking on employees, and reviewing benchmarking data and organization variance reports. Additionally, leaders should be current on research regarding current practice modifications. As the health care environment is moving toward evidence-based practice (EBP), the gap between research publications and the research's implementation into practice is of major concern for nursing leaders and federal agencies.

In every health care system, large or small, there is always the need to improve. Realistically, there will be discrepancies or variations that will need attention. When deviations do occur, the manager must analyze these variances in terms of the "total picture" that examines possible causes, outside influences, and new mandates by regulatory agencies. Corrective action and strategies for implementation are necessary to address the deviations or discrepancies.

Staffing

Finally, the staffing responsibility is ongoing and represents one of the greatest expenditures of time and effort for the manager. Staffing involves employee selection, placement, training, and compensation determinations. The manager's function is to select the right candidate for a position, place the candidate into the right role, develop an orientation pathway or plan, and provide training for the employee with the assistance of a preceptor. In addition, the manager evaluates the performance of employees and either rewards their efforts and abilities or disciplines (and at times terminates) employees who are not performing at acceptable levels.

The job description is the tool that outlines the skill requirements, educational competencies, physical demands (such as bending, lifting, or standing for any number of hours), and potential hazardous exposure (fluoroscopy, chemical, or chemotherapeutic) inherent in a position. The job description is designed to validate the tasks required by the position and screen potential applicants before an interview with the nurse leader, managers, and staff is arranged.

The manager needs to be familiar with organizational policies on probationary periods and the timeframe for orientation. Usually these overlap and follow the new employee's hire date. Orientation may be any determined amount of time but usually not longer than the probationary period. The probationary period is an organizationally determined "get-to-know-you" period until such time when the new employee becomes a permanent member of the organization.

Should the newly hired employee fall short of meeting probationary performance standards, the manager should discuss guidelines and recommendations with the organization's Human Resources department (HR) prior to terminating the employee. This is important for both union and nonunion hospitals and in dealing with the disciplinary processes within the organization.

The selection of competent, qualified employees represents one of the most challenging aspects of nursing management. HR may initially interview and screen job applicants; however, qualified individuals will ultimately be presented to the manager before being hired. The manager should obtain a copy of the applicant's curriculum vitae or resume to review the applicant's experience and qualifications and should conduct an interview for additional information about the applicant's job knowledge and personality.

The interview process should focus on the individual's competencies and qualifications to perform the required duties in the gastroenterology department. The interview questions must follow guidelines established by the Equal Employment Opportunity Commission (EEOC). The manager considers prospective employees' strengths and weaknesses to choose the one who best fits the needs of the department.

Staff interviews with prospective employees can be beneficial in a number of ways. The prospective employee receives a tour of the unit and has some interaction with the current staff. In a staff interview, the candidate has an opportunity to meet with prospective staff members to help decide whether or not the position is a good fit. Another option to accomplish this is to suggest or require a "shadowing" experience in the gastroenterology unit. Reference checks and diligent record-keeping of all of prospective employees are standard practice.

Orientation and staff development are important after the desired employee has been hired. These components of the staffing process coordinate the orientation of the new employee to the hospital and the gastroenterology unit. Most often, this is accomplished by utilizing a preceptor and competency statements. Orientation should include introduction to policies, procedures, rules, and regulations specific to the hospital and the departmental unit, as well as a thorough review of the tasks, duties, and responsibilities outlined in the employee's job description.

The purpose of professional development is to provide opportunities for an individual employee's professional growth and development. As such, it goes beyond orientation and should be available to all employees on an ongoing basis. Staff development opportunities might include hospital or departmental in-services, courses, seminars, and independent study or projects.

Staffing schedules provide staff with information as to their shift and the time to report for duty. Staffing assignments are also required by the organization as a means of providing for patient care on a daily basis. A variety of staffing patterns can be used. The delivery system may be point-of-service care, managed care, functional nursing, team nursing, or primary nursing. Any of these delivery systems will accommodate the various theoretical models of nursing.

Today, many health care organizations use internet-based software applications that have the capability of accommodating centralized and decentralized staffing schedules. This requires computer skills on the part of the employee.

Centralized scheduling is cost-effective and reduces the hours allocated to developing and implementing staffing needs. Decentralized scheduling can be a subschedule of a centralized system. It also provides up-to-the-minute details of organizational staffing required for disaster planning within the organization.

NURSING INFORMATICS

In today's administrative and clinical environments, information technology is affecting every facet of the health care delivery system. Health care is dominated by data-driven decisions and managed care. Nursing informatics (NI) represents the use of computer technology to support nursing. It helps to integrate nursing science with multiple information management and analytical sciences (ANA, 2015).

The NI model describes informatics as the epicenter of cognitive science, information science, computer science, and nursing science. Organizations of today may utilize organization-wide systems or one large system that connects the entire organization. While an organization-wide system would be beneficial, individual software applications designed for specific purposes do not interact well with larger systems; therefore, interfaces have to be developed by vendors so that multiple software applications can communicate effectively and data can be shared. This represents a major monetary and time expenditure.

The three overarching standards of NI practice are the incorporation of theories, concepts, and principles from appropriate sciences into informatics practice; the integration of ergonomics and human-computer interaction (HCI) into the informatics care plan; and the systematic determination of the social, legal, and ethical impact of an informatics solution within nursing and health care (ANA, 2015). Informatics clearly illustrates the shift from the old paper-driven recordkeeping to the current electronic medical records (EMR) and the explosion of clinical software applications designed to manage data and information.

EVIDENCE-BASED PRACTICE AND RESEARCH

The goal of nursing leaders is to ensure that patients are provided the best nursing care. Evidence-based practice (EBP) is based on empirical information resulting from nursing research. There is evidence that supports a positive relationship between successful EBP and the improvement of patient outcomes (Black et al., 2015).

EBP is defined as a problem-solving approach that integrates research, evidence-based theories, and clinical expertise with evidence from patient assessment, family input, and population preferences and values (ANA, 2015). The force behind EBP is the concern that without current best practices, the practice performed in health care today would be outdated and present serious concerns for the quality of nursing care.

The evidence-based movement was founded in the early 1970s by British epidemiologist Dr. Archie Cochrane. He published a book criticizing the medical profession for not providing more rigorous reviews of the research evidence so the policy-makers and health care organizations could make better decisions about health care (Melnyk & Fineout-Overholt, 2014).

Because EBP takes into consideration the expertise of the nurse or practitioner, as well as the patient's preferences and values, it is quite different in scope from research utilization. EBP is a critical and systematic appraisal of the most relevant evidence about a specific clinical question.

Once evidence has been established in the literature, it is the responsibility of the organization to ethically and objectively consider implementing the evidence into current practice. Evaluating EBP effectiveness is also vital in determining whether or not the change (implementation of EBP) resulted in the expected outcomes.

SUMMARY

Nursing leaders are continually challenged to foster new and innovative ways to achieve higher standards of quality, efficiency, and effectiveness. The advances in clinical technology, explosion of data and information technology, increasing clinical skill-sets required for

Table 2.2. Applications of Nursing Informatics into All Levels of Nursing Practice

Nursing practice	Nursing administration	Nursing education	Nursing research
• Work lists for nursing interventions • Computer-generated client documentation • Data sheets/ flow sheets to document vital signs and other measurements • Computer-generated nursing care plans and pathways • Automated billing for supplies and documentation • Reminders and prompts during documentation and charting • Accessibility to computer-archived patient data	• Automated staffing and scheduling • E-mail • Online variance reports • Quality assurance and outcome analysis • Online performance appraisals • Remote access to work computers • Presentation software slides and handouts	• Computerized record-keeping • Computer assisted instruction • Distance learning • Internet resources and Web-based education • Presentation software	• Computerized literature searching • Adoption of standardized language related to nursing terms • Explore trends in combined and aggregate data • Internet-based research and data collection tools

(Hebda, Czar, & Mascara, 2004)

patient care, and rising patient acuity have increased the demands on nursing at a time when nursing resources are decreasing. Nurses in all leadership capacities are challenged to develop new strategies that will improve productivity and increase the quality of nursing services. More important is the ongoing process of providing evidence-based nursing care in an effort to maintain positive patient outcomes.

Nurse leaders do not always possess skilled management abilities, and, conversely, managers do not always exhibit effective leadership qualities. With an understanding of the interrelationship between leadership roles and management functions, however, nurses in administrative capacities can perform the nursing process efficiently and effectively as they work toward attaining personal, departmental, and organizational goals.

CASE SITUATIONS

Scenario 1:

Vesna Hammer, RN, is the nurse leader of a gastrointestinal (GI) endoscopy unit in a metropolitan hospital. This unit is looking to purchase new technology for an ultrasound program that will be implemented at a date to be determined in the next fiscal year.

- What information will this nurse leader need to create a project plan for this program?
- What additional resources does this new program require: staffing the unit, procedure space, supplies, and equipment?
- What kind of a marketing plan should Vesna develop in order to implement the new program?

Scenario 2:

Vesna is a new manager to the GI endoscopy unit, but she has five years of experience in leadership and management from another hospital.

- What types of leadership attributes will Vesna exhibit?
- What types of activities demonstrate her management ability?
- What are some examples of behaviors that reinforce this management style?

Scenario 3:

Recently, Vesna had an employee who called in every day at 5:00 a.m. to request the day off.

- What type of plan might Vesna use to alter this employee's behavior?

Suggested responses

Scenario 1:

To prepare for the introduction of a new program, Vesna should meet with the group of stakeholders from the hospital who will support and assist in the implementation of this new program. Topics to discuss are:

- Perform a review of area hospitals using the same technology. Also, determine the population of patients targeted for this technology, and forecast the potential number of patients expected to seek this technology. Project the number of procedures expected to be performed with the new technology, and identify the potential for growth.
- Determine the capital equipment needed for this new procedure. Include requirements of additional supplies and equipment to support new technology and maintain equipment.
- Review current space limitations and project needs for additional space to accommodate equipment and supply storage, waste management, additional pre- and post-procedure areas, procedure rooms, family waiting rooms, and computer workspaces for dictation and documentation.
- Calculate staffing, orientation, and training program for new employees, as well as training needs for existing employees. Calculations of staffing projections should consider benefit time for paid time off (PTO).
- Develop a projected budget for the new program.
- Develop a written plan using dedicated software for project management or spreadsheets to document timelines for program implementation.

Scenario 2:

Leadership is a learned behavior that is represented innan individual's self awareness, assertiveness, accountability, and advocacy. Vesna should have an understanding of the organizational structure in her hospital, as well as knowledge of the management functions. Vesna believes that a democratic management style works best for her GI endoscopy unit. These skills will assist Vesna in effecting change.

Management entails the planning, organizing, directing, controlling, and staffing of the unit. Some specific activities are:

- Creating work schedules and making changes in staffing schedules based on requests of personnel.
- Reviewing monthly financial statements and justifying variances.
- Conducting staff meetings that include open discussion of unit issues.
- Providing education for staff supporting new devices, supplies, and technology.
- Conducting annual performance appraisals of staff.

A democratic management style involving individuals and groups in the decision-making process is particularly effective when cooperation and coordination among groups is encouraged. Some examples of this style of management behavior include:

- Staff involvement in the decision of weekend call coverage
- Staff meeting regarding general low morale throughout the institution.
- Staff meeting to obtain input regarding the replacement of obsolete equipment.

- Staff involvement in developing a policy for monitoring sedated pediatric patients.

Scenario 3:

The management of the human resource is one of the greatest challenges for the nurse leader. To improve this employee's behavior, Vesna might:

- Set up a meeting with the employee and attempt to gather objective data about the reasons for his or her behavior.
- Explain to the employee that the schedule is already complete and only needed personnel are assigned.
- Explain that the employee's request should be made in advance in accordance with written leave policies.
- Provide positive feedback on behaviors that should be recognized.
- Set mutual goals relative to requesting time off.
- Schedule a follow-up meeting to review progress.
- Document the counseling session, including anecdotal notes.

REVIEW TERMS

capital budget, direct costs, indirect costs, leadership attributes, operating budget, organizational culture, organizational structure, project management, span of control

REVIEW QUESTIONS

1. Department management is based upon a theoretical framework that involves:
   a. Coordinating the five functions of management.
   b. Delegating tasks to ensure that work is completed.
   c. Micromanaging all tasks in the unit.
   d. Ensuring that the employee knows to whom to report.
2. This management task outlines a course of action that is realistic for the unit and involves the development of a purpose, philosophy, goals, and objectives for the unit.
   a. Planning.
   b. Organizing.
   c. Directing.
   d. Staffing.
3. Managerial components of planning include:
   a. Forecasting and managing the budget.
   b. Maintaining a structure of working relationships.
   c. Assigning work activities.
   d. Hiring new personnel.
4. The gastroenterology nurse leader directs through:
   a. Counseling employees.
   b. Leading and guiding employees.
   c. Developing position descriptions.
   d. Conducting performance appraisals.
5. The following are all examples of leadership styles except:
   a. Democratic.
   b. Laissez-faire.
   c. Transformational.
   d. Autocratic.
6. To determine the staffing budget, it is necessary to calculate:
   a. Full-time equivalents and flexible staff.
   b. Volume of procedures and required staff.
   c. Qualifications of the required staff.
   d. Volume and cost of procedures.
7. A problem-solving approach that integrates research, evidence-based theories, and clinical expertise with evidence from patient assessment, family input, and population preferences and values is called:
   a. The tasks unassigned to staff.
   b. The tasks delegated to staff.
   c. Evidence-based practice.
   d. The tasks of other managers.
8. The process of getting tasks accomplished with the help of others and directed toward common organizational and departmental goals is defined as:
   a. Planning.
   b. Management.
   c. Staffing.
   d. Directing.
9. In an operating budget, salaries, overtime, benefits, and medical-surgical supplies are usually considered:
   a. Indirect expenses.
   b. Fixed assets.
   c. Direct expenses.
   d. Capital budget items.
10. This term represents the use of computer technology to support nursing and includes clinical practice, administration, education, and research.
   a. Nursing management.
   b. Nursing administrators.
   c. Performing a financial forecast.
   d. Nursing informatics.

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Chapter 3

INFECTION PREVENTION

This chapter describes the precautions that must be taken in the gastroenterology and endoscopy setting to control the risk of infection for patients and health care personnel. Standards adopted by the Society of Gastroenterology Nurses and Associates, Inc. (SGNA), the American Society for Gastrointestinal Endoscopy (ASGE), and the Association for Professionals in Infection Control and Epidemiology (APIC) are summarized. General infection prevention principles, including the Spaulding classification system, are reviewed. Guidelines for decontamination and high-level disinfection and/or sterilization of endoscopes and accessories are provided. In addition, automated reprocessors (AERs) and high-level disinfectants are discussed, as are recommendations for reprocessing reusable and single-use devices and accessory equipment. Patient populations and procedures that should be considered for antibiotic prophylaxis are presented.

Learning objectives

After reviewing the content of this chapter, the gastroenterology nurse should be able to:

1. Identify elements of an educational and continuous quality improvement program as they relate to infection prevention.
2. Discuss the Spaulding classification system as it applies to gastrointestinal endoscopes and accessories.
3. Explain the appropriate steps for decontamination and high-level disinfection of endoscopes.
4. Identify the special requirements for reprocessing endoscopic retrograde cholangiopancreatography (ERCP) elevator channels, endoscopic ultrasound (EUS) elevators, double-channel scopes, and probes for esophageal and rectal manometry.
5. Discuss specific requirements for reprocessing accessory devices, such as water pump tubing, water bottles, lithotripter devices, CO_2 gas hoses/tubes, and specialty Argon hoses.
6. Discuss the issues involved in the practice of reprocessing single-use endoscopy devices and accessories.

INFECTION PREVENTION PRINCIPLES

All health care personnel must be educated regarding appropriate infection prevention measures. Infection prevention education is a critical part of orientation and continuing education for all staff, including physicians, nurses, and assistive personnel, regardless of the setting in which endoscopy is performed (hospital, clinic, or office). An infection results when **microorganisms** enter the human body, multiply, and produce a reaction. For an infection to develop, all three of the following interlinking components in the chain of infection must be present: an infectious organism, a means of transmission, and a susceptible host. The infectious organism may be in the form of bacteria, viruses, fungi, protozoa, or helminthes (worm-like parasites). Bacteria are the most common infectious agents in the clinical setting. Both endogenous and exogenous microbes can cause infections related to endoscopic procedures.

1. **Endogenous infections** occur when the microflora colonizing the mucosal surfaces of the gastrointestinal or respiratory tract gain access to the bloodstream or normally sterile body sites as a consequence of a procedure. Examples of endogenous infections include cholangitis following the manipulation of an obstructed biliary duct or endocarditis in patients with mitral valve regurgitation who have sustained transient bacteremia during esophageal dilation. The highest rates of bacteremia reported have been associated with esophageal dilation, variceal sclerotherapy, and instrumentation of obstructed bile ducts (ASGE, 2015). The microbes most frequently isolated have been *Escherichia coli* (*E. coli*), *Klebsiella* species (spp.), *Enterobacter* spp., and enterococci (Kovaeva, Peters, van der Mei, & Degener, 2013).

2. **Exogenous infections** are the result of microorganisms introduced from a source outside the body. The most frequently isolated organisms have been *Pseudomonas aeruginosa*, *Salmonella* spp., and mycobacteria (Kovaeva, et. al, 2013). Bacteria less frequently isolated include *Helicobacter pylori* (*H. pylori*), *Klebsiella*, *Enterobacter* spp., and *Serratia* spp., as well as hepatitis C virus (HCV) and hepatitis B virus (HBV) (ASGE, 2008).

Pathogen transmission involving endoscopes has been associated with a breach in currently accepted cleaning and disinfection guidelines, use of an unacceptable liquid chemical germicide for disinfection, improper drying, or defective equipment (ASGE, 2018). In recent years, there have been reported outbreaks of infections caused by such multidrug-resistant organisms as carbapenem-resistant Enterobacteriaceae (CRE) in both Europe and the United States. These outbreaks have been related to contaminated duodenoscopes used during ERCPs. Despite reported adherence to reprocessing protocols, the transmission was attributed to persistent contamination at the elevator region and/or the elevator cable of the instrument. Complex instrument design, while increasing efficiency and effectiveness of endoscopic procedures, creates enormous challenges for effective processing (Association of periOperative Registered Nurses [AORN], 2017).

Direct and indirect contacts are the most common methods of transmission in the clinical setting. Other methods include airborne transmission; vehicle transmission, such as by endoscopes; and vector transmission via an animal or insect. A susceptible host can be anyone who has a lowered resistance. Factors that contribute to host susceptibility are:

a. Age, both the very young and the elderly.
b. Immune status or underlying disease.
c. Endovascular surface integrity.
d. Indwelling foreign material, e.g., prosthetic heart valves, joint replacement.
e. The presence of intrinsic infective foci.

Because of their incompetent immune systems, patients infected with the human immunodeficiency virus (HIV), transplant patients, and patients receiving immunosuppressant therapy or biologics that may reduce immunity are especially susceptible to infection.

MICROBIAL RESERVOIRS AND MECHANISMS OF TRANSMISSION

The ability of bacteria to form biofilms is an important factor in the pathogenesis of endoscopy-related infections, particularly as biofilms interfere with disinfection. Biofilms consist of colonies of organisms forming structures to maximize growth potential. Development of a biofilm begins when free-swimming bacteria attach to a surface. Cell-to-cell communication then signals formation of a biofilm with pillar-and-mushroom-like structures around which water can circulate. This allows both maximum exposure of the bacteria to circulating nutrients and a decrease in the accumulation of waste products. Biofilms have been found to adhere to the internal channels of endoscopes. This finding emphasizes the importance of strategies aimed at decreasing biofilm formation through meticulous mechanical cleaning of endoscopes.

To break the chain of infection, gastroenterology nurses and technicians must recognize the need for appropriate decontamination, disinfection, and in some cases, sterilization of instruments and patient care items. Each endoscopy setting must develop policies to identify whether disinfection or sterilization is indicated based on the item's intended use. A classification system for levels of disinfection and categorization of equipment and patient care items was devised in the 1960s by Earl H. Spaulding. Spaulding classified equipment and patient care items as critical, semi-critical, and non-critical based on the degree of risk for infection involved in their use.

1. **Critical items** are those items that present a high risk of infection if they are contaminated with any microorganism. These are objects or instruments that break the mucosal barrier and enter sterile tissue or the vascular system. Examples of critical items in the gastroenterology unit are sclerotherapy or injection needles, biopsy forceps, metal clips, and intravenous catheters. Sterilization is recommended for reprocessing critical items.

2. **Semi-critical items** are those items that come in contact with non-intact skin or mucous membranes. Endoscopes are considered semi-critical items. At a minimum, semi-critical items require high-level disinfection using chemical disinfectants such as glutaraldehyde, hydrogen peroxide, peracetic acid with hydrogen peroxide, or ortho-phthalaldehyde (OPA) (Rutala & Webber, 2013).

3. **Noncritical items** are those items that come in contact with intact skin, but not mucous membranes. There is relatively little risk of transmitting infection with these items. Some examples of non-critical items are stethoscopes, blood pressure cuffs, cardiac leads, and pulse oximeters, as well as bedside tables, bedrails, linens, patient furniture, and floors. These items require disinfection with a hospital-grade intermediate-level or low-level germicide (Rutala & Webber, 2013).

The Centers for Disease Control and Prevention (CDC) classifies levels of disinfection as sterilization, high-level disinfection, intermediate-level disinfection, and low-level disinfection.

Sterilization is the use of a physical or chemical procedure to destroy all forms of microbial life, including bacterial spores. Sterilization should be regarded as a process rather than a single event. Critical items should be purchased as sterile or be sterilized. Methods of sterilization include:

- Steam.
- Ethylene oxide.
- Hydrogen peroxide gas plasma.
- Ozone.
- Vaporized hydrogen peroxide.
- Liquid chemical sterilants.

Endoscope sterilization is required for such extraluminal applications as natural orifice transluminal endoscopic surgery (NOTES) or intraoperative endoscopy through open or laparoscopic access (ASGE, 2016). Supplemental measures for duodenoscope reprocessing following high-level disinfection may include SYSTEM 1E Liquid Chemical Sterilant Processing System (Steris Healthcare, Mentor, OH), ethylene oxide (ETO) sterilization, or another Food and Drug Administration (FDA)-cleared low-temperature sterilization (Centers for Disease Control and Prevention, 2017).

Disinfection is the process of cleaning that removes or kills organisms likely to cause disease, but effectiveness depends on the level of disinfection.

1. **Sterilization** refers to a validated process resulting in the complete elimination or destruction of all forms of microbial life.

2. **High-level disinfection** is a process that removes or kills organisms likely to cause disease. High-level disinfection destroys all mycobacteria, all viruses, all fungi, and all vegetative bacteria but not necessarily all bacterial spores. The FDA clears products with a high-level disinfectant claim to kill 100 percent of the test organism *Mycobacterium tuberculosis* (*M. tuberculosis*). High-level disinfectants may be referred to as disinfectants or sterilants because some are capable of sterilization with extended soak times.

3. **Intermediate-level disinfection** is a process that destroys *M. tuberculosis*, vegetative bacteria, most viruses, and most fungi. It does not kill bacterial spores.

4. **Low-level disinfection** is a process that destroys vegetative bacteria, some viruses, and fungi but is not effective against *M. tuberculosis* or bacterial spores.

STANDARDS FOR INFECTION PREVENTION IN THE ENDOSCOPY SETTING

The SGNA standards and practice guidelines, “Standard of Infection Prevention in the Gastroenterology Setting” (SGNA, 2019), “Standards of Infection Prevention in Reprocessing Flexible Gastrointestinal Endoscopes” (SGNA, 2018a), and “Guideline for Use of High-Level Disinfectants and Sterilants in the Gastroenterology Setting” (SGNA, 2017), address areas of concern in all endoscopy settings. Education on infection prevention is a vital component in the orientation and continuing education of all personnel, including physicians, nurses, and assistive personnel.

Concepts in infection prevention education include:

- Standard precautions (hand-washing and personal protective equipment [PPE]).
- Bloodborne pathogens (HBV, HCV, and HIV).
- Low-level disinfection of communication devices used in the GI setting such as phones and computers.
- Staff attire.
- Environmental conditions that support gastroenterology practice within gastroenterology, as well as infection prevention.
- Special considerations guidance in the handling of patients with *Clostridium difficile* (*C. difficile*), *M. tuberculosis* that causes tuberculosis (TB), vancomycin-resistant Enterococci (VRE), CRE, or various parasite or vermin infestations such as lice and scabies.
- Reprocessing room requirements.
- Quality assurance measures.
- Identification of infection prevention champions.
- Response to failure in infection prevention.
- Supporting a culture of safety.

The Occupational Safety and Health Administration (OSHA) mandates standards, which require exposure control plans, adequate ventilation systems, and use of sharps with engineered sharps injury protection where feasible, in an effort to reduce transmission of blood-borne pathogens (OSHA, n.d.). Personal protective devices, including gloves, masks, eye protection, and moisture-resistant gowns, should be used to protect personnel from blood or body fluid exposure. Eye wash stations should be readily available in case blood, body fluids, or chemicals splash into the eyes of caregivers, patients, or others. Staff should receive education and training, as well as prove competency, at hire; annually; and whenever new techniques, accessories, equipment, policies, or manufacturer instructions for use (IFUs) updates are introduced (CDC, 2017).

Personnel qualifications must be defined, including qualifications for staff who reprocess endoscopy equipment. Staff who reprocess endoscopes and accessories must have detailed knowledge of the instrument design and be proficient in device-specific methods required to reprocess an instrument so that it is safe for use. These staff members should be carefully selected, trained by a knowledgeable preceptor, and meet annual competency standards. The staff must understand the importance of reporting any breaches in reprocessing according to facility policies and protocols. Temporary personnel should never be given the responsibility for reprocessing instruments in either a manual or automated system until competency has been established.

An ongoing continuous quality improvement program should place special emphasis on strict adherence to published protocols for reprocessing flexible gastrointestinal endoscopes. A log or record of all endoscopy procedures performed must be maintained to assist in an outbreak investigation and should include the following information regarding each piece of equipment reprocessed:

1. Patient name.
2. Medical record number.
3. Date and time of the procedure.
4. Procedure(s) performed.
5. Clinician who performed the procedure.
6. Serial number (or other identifier) and model of endoscope.
7. Automated endoscope reprocessor (AER) model and serial number or other identifier (if applicable).
8. Minimum effective concentration (MEC) test results of high-level disinfectant.
9. Name of scope reprocessing staff member.

This record of all endoscopy reprocessing performed will be invaluable in the event that a reprocessing breach is detected and will facilitate a thorough investigation (Petersen et al., 2017). Supervisory personnel should be familiar with these standards and designate an individual in the setting to monitor compliance.

Periodic audits of the facility's reprocessing protocols, as well as compliance in completeness of documentation, should be performed (CDC, 2017). A process to trace the endoscopes and reusable accessories through the entirety of the cleaning and disinfection processes should be employed (SGNA, 2019).

Preventive maintenance of reprocessing equipment and liquid chemical germicide monitoring should be incorporated into the quality assurance plan. Procedures should be in place for reporting any identified transmission of infection to the proper authorities.

Procedure rooms should be separate from reprocessing rooms. The design of reprocessing areas should have a clear flow from dirty to clean areas. There should be a policy describing the correct storage, use, disposal, and containment of spills of liquid chemical germicides and other reagents. Equipment to contain and/or neutralize a spill should be readily available. A hand-washing sink or hand sanitizer is required in the reprocessing area.

Reprocessing Endoscopes and Accessory Equipment

Reprocessing of flexible gastroenterology endoscopes is accomplished in two distinct phases: decontamination and high-level disinfection.

DECONTAMINATION (CLEANING)

The first phase of endoscopic reprocessing is decontamination, which is a two-step process that includes pre-cleaning in the procedure room followed by manual cleaning in a reprocessing room. In any endoscopy setting, the primary objective of pre-cleaning is to remove gross debris and prevent secretions from adhering to the endoscope. Manual cleaning – the physical task, performed by hand, of removing secretions and contaminants from the endoscope with appropriate brushes, sponges, detergents, and water – allows for removal of a significant amount of organisms, organic material, and foreign debris from the endoscope. Manual cleaning is the most critical step in the disinfection process. It should be performed within the timeframe specified in the manufacturer IFUs (CDC, 2017).

According to Beilenhoff et al., (2008), in order for manual cleaning to be effective, it must be:

1. Performed by a person familiar with the structure and the particular characteristics of each model of endoscope to be reprocessed and trained in the proper cleaning procedure.
2. Undertaken immediately after the endoscope is used so that secretions and debris do not dry and harden.
3. Done in accordance with a protocol, which, using appropriate detergents and cleaning equipment, allows all surfaces of the endoscope – internal and external – to be cleaned.

4. Followed by thorough rinsing to ensure that all debris and detergents are removed prior to disinfection.

Several studies have shown manual cleaning alone to be effective in reducing microbial contaminants by as much as 4 log or 99.99 percent (Chu, McAlister, & Antonoplos, 1998; Vesley, Melson, & Stanley, 1998). Reprocessing standards endorsed by SGNA, ASGE, AORN, the American Gastroenterological Association (AGA), the American College of Gastroenterology (ACG), and the Association for Professionals in Infection Control and Epidemiology (APIC) require that manual cleaning of an endoscope, including all of its channels and removable parts, must precede either manual or automated disinfection (Petersen et al., 2017). If an endoscope is not clean, it cannot be high-level disinfected or sterilized. Decontamination (manual cleaning) is always necessary before disinfection or sterilization because any residual organic material may harbor microorganisms and allow the formation of biofilms. Formation of biofilms may prevent contact of the liquid chemical germicide with surfaces of the endoscope or accessory, resulting in ineffective disinfection and sterilization processes. Endoscopes and accessories must be completely submerged in liquid per the manufacturer instructions.

The SGNA "Standards of Infection Prevention in Reprocessing Flexible Gastrointestinal Endoscopes" (2018) outlines the nine steps of the endoscope reprocessing protocol. These steps are general, not manufacturer- or model-specific, so the health care team member performing the process must also refer to the specific manufacturer's instructions for each model.

1. **Pre-cleaning.**

   Immediately after removing the endoscope from the patient, wipe the insertion tube with the wet cloth or sponge soaked in a freshly prepared detergent solution. Dispose of, sterilize, or high-level disinfect the cloth or sponge between cases.

   Placing the distal end of the endoscope into the appropriate detergent solution, suction a large volume of solution through the endoscope until clear, then follow by suctioning air. According to the manufacturer instructions, flush and manipulate the forcep elevator of the duodenoscope or EUS scope, purge the air and water channels, and flush the auxiliary water channel. Remove the endoscope from the light source and suction pump and attach the protective video cap, if applicable. Note that non-video scopes and the newer generations of videoscopes do not have caps.

   After point-of-use pre-cleaning, transport the soiled endoscope to the reprocessing area for subsequent steps in high-level decontamination that must be completed before remaining soil dries. Reprocessing should occur in a room separate from the procedure room.

   During transportation, soiled endoscopes should be contained in a manner that prevents exposure of staff, patients, or the environment to any potentially infectious organisms. The container must be closed and labeled to indicate biohazardous contents. To ensure safe transfer to the decontamination area, the outside of the transport container should be clean and free from gross contamination. Containers, sinks, and basins in the decontamination room should be large enough that the endoscope will not be damaged by being coiled too tightly.

2. **Leak testing.**

   In the decontamination room, inspect the endoscope for damage, and perform a pressure/leak test according to the manufacturer instructions. Leak testing detects damage to the interior or exterior of the endoscope and is done before immersing the endoscope into reprocessing solutions in order to minimize damage to parts of the endoscope not designed for fluid exposure. Leak testing may be performed by dry methods, either manually or mechanically, or by wet methods, either mechanically or mechanically within an AER. Reference the manufacturer instructions regarding endoscope and leak tester compatibility.

   To perform a leak test, apply air pressure to the inside of the scope insertion tube and watch for air bubbles in water; a decrease in air pressure on the gauge; or a computerized report identifying a leak in the scope exterior, buttons, or knobs, or internally into one of the channels. Flex the distal bending portion of the scope in all directions to detect minute leaks.

   Leak testing is the single most important step to detect a problem with scope integrity that could lead to major fluid damage and infection risk for subsequent patients.

   If a leak has been identified, immediately remove the endoscope from service and follow the endoscope manufacturer instructions on how to proceed.

3. **Manual cleaning.**

   Detach all removable valves and parts, and discard those designated as disposable. Immerse the endoscope in a freshly prepared detergent solution that is low-foaming, diluted per manufacturer's instructions, and of neutral pH with or without enzymes. Wash all debris from the entire exterior of the endoscope by brushing and wiping the instrument with a lint-free cloth or sponge, keeping the endoscope submerged in the solution at all times to prevent aerosolization.

   Use a small, soft brush to clean all removable parts, as well as inside and under the valves and biopsy port covers and openings. Brush all accessible channels with compatible brushes, giving particular attention to crevices that are most likely to harbor contaminated organic material. Each channel should be brushed until all visible debris is removed. Gently wipe and/or brush the tip of the endoscope to remove any debris or tissue that may be lodged in or around the air and water nozzle.

When manually cleaning the duodenal (ERCP) scope and some of the EUS scopes (such as the linear ray), additional steps and brushes are required to ensure that no debris is lodged in the elevator movable part. The manufacturer's specific instructions must be referenced. Switchover levers (when present in some EUS scopes and double-channel scopes) also need careful attention during cleaning. Dial selectors and scope-stiffening control knobs must be set in a neutral position for reprocessing. Double-channel scopes require additional cleaning measures to address each channel.

Attach all channel-cleaning adaptors for suction, air, and water channels specific to the endoscope. All channels, including auxiliary water channels (even if not used), should be brushed and flushed with copious amounts of detergent to remove any dislodged debris. Ensure all air has been displaced. Leave the solution in contact with the instrument for the product-specified time. Automated pumps are available to accomplish the manual flush. Manufacturer instructions must be followed to accomplish this step for compatible endoscopes. The elevator channel of a duodenoscope may require manual flushing and manipulation by following the latest manufacturer instructions for specific steps.

Failure to complete the pre-cleaning and manual cleaning immediately following the endoscopy may necessitate initiation of additional interventions for delayed reprocessing. Manufacturers of endoscopes provide specific handling recommendations that include time-sensitive intervals of delay.

4. **Rinsing after manual cleaning.**

Purge the cleaning solution from all channels. Rinse outer surfaces, and flush all channels with clean water. It is essential that all detergent be removed prior to disinfection. Purge all channels using forced air, and dry the exterior of the scope with a soft, lint-free cloth. Discard enzymatic detergents after each use because these products are not microbicidal and will not retard microbial growth when stored (Petersen et al., 2017).

5. **Visual inspection.**

Visual inspection for cracks, corrosion, discoloration, or retained debris is recommended at this phase to ensure that the endoscope is visibly clean and free of conditions that could affect the disinfection process. Magnification and adequate lighting facilitate inspection. Discovery of an unclean endoscope prompts a repeat in the manual cleaning process. During this phase, a rapid indicator tool or rapid audit tool may be used as an additional measure to verify internal scope cleanliness. Recent studies have revealed that organic residues often remain after manual cleaning, resulting in endoscope contamination in institutions with documented adherence to reprocessing guidelines. The presence of residual material after cleaning reduces high-level disinfectant effectiveness, and researchers have recovered non-spore-forming microbes on 8%-64% of patient-ready endoscopes following high-level disinfection (Ofstead et al., 2017).

6. **High-level disinfection.**

After a thorough manual cleaning, the next step is high-level disinfection. Flexible gastrointestinal endoscopes are classified as semi-critical devices and require high-level disinfection using an FDA-cleared solution. High-level disinfection is the gold standard for the reprocessing of gastrointestinal endoscopes supported by professional associations and recognized as the appropriate standard of care by such agencies as the CDC and The Joint Commission (SGNA, 2018a). Sterilization of endoscopes is indicated on rare occasions, such as when used as critical medical devices where the potential for open surgical field contamination exists or per facility policy.

Reusable sterilants and/or disinfectants must be routinely tested according to the product label or facility policy to determine if the solution is above the minimum effective concentration (MEC). A brand-specific strip should be used to test the MEC of a disinfectant. A log or record of these tests must be kept in the endoscopy unit. Quality-control procedures recommended for certain chemical test strips should be done according to the manufacturer instructions. The sterilant and/or disinfectant should be changed when the MEC fails or when the solution use-life expires, whichever comes first. Never exceed the use-life claim of a high-level disinfectant. Occupational chemical exposure monitoring and the availability of current chemical-specific Safety Data Sheets (SDS) are two additional safety requirements.

High-level disinfection may be achieved either manually or automated via an AER.

The endoscope and all of its removable parts must be completely immersed in an adequately sized basin of FDA-cleared compatible disinfectant/sterilant. Using appropriate adaptors, all channels must be filled with the chemical solution, even if they were not used. Ensure no air pockets remain within the channels by forcing the solution out the opposite end of each channel until there are no further air bubbles visualized. Soak for the time and temperature required to achieve high-level disinfection. Monitor the process with a timer and thermometer to meet the manufacturer's parameters for both. Cover the soaking basins with a tight-fitting lid to contain the chemical solution and its vapors.

If an AER is used, ensure that the endoscope and endoscope components can be effectively reprocessed with the AER. Because the elevator wire channel of duodenoscopes may not be effectively disinfected by most AERs, this step should be performed manually. All channel adapters for the AER must be appropriately attached according to manufacturer instructions. Place the endoscope, valves, and other removable parts into the reprocessor's soaking basin. With the machine set for the appropriate time and temperature for the chemicals used, the scope should remain in the processor for all cycles or phases without interruption to ensure that high-level disinfection is completed.

Flush all channels with air before removing the endoscope from the disinfectant/sterilant. This may be accomplished manually or within the AER.

7. **Rinsing after high-level disinfection.**

   Rinse the endoscope exterior with clean water. Using appropriate adaptors, irrigate all channels and removable parts with clean water. This may require multiple separate rinses. Follow specific chemical instructions for use. Discard the rinse water after each cycle of use (ASGE, 2016).

   Adequate rinsing is essential to prevent toxic effects of residual chemicals after disinfection. Chemical colitis mimicking pseudomembranous colitis can result if the endoscope and all of its parts are not thoroughly rinsed.

8. **Drying.**

   Drying is a critical step in reprocessing, as moisture allows microorganisms to survive and multiply and also promotes biofilm development. The endoscope must be dried completely after every reprocessing cycle, both between patient procedures and before storage.

   Drying of the lumens is accomplished by flushing with 70%–90% isopropyl alcohol and then by using pressurized, filtered air. The concentration of alcohol is determined by the endoscope manufacturer and, if performed in the AER, by the manufacturer of the AER. When flushing manually, the alcohol should be seen exiting the opposite end of the channel and should be used even when sterile water is used for rinsing. The Alfa and Sitter study found that endoscopes that had received 10 minutes of forced air drying showed no microbial growth after 48 hours (British Society of Gastroenterology [BSG], 2017).

   When using pressurized, filtered air, refer to the manufacturer instructions regarding air pressure limits supported by the internal endoscope channels. If the final alcohol rinse cycle is not included in the AER, this step should be done manually, followed by forced air purges. The duodenoscope elevator and elevator channel must be manually flushed and dried per the manufacturer instructions.

   Dry the exterior of the endoscope, as well as all removable parts, with a soft, clean, lint-free towel. Do not attach the removable parts for storage. Dry the channel adapters used with the AER.

9. **Storage.**

   Store the endoscope without buttons, valves, or adaptors attached in a clean, well-ventilated, dust-free area, hanging freely so that the endoscope is not damaged by physical impact. It is the facility's responsibility to determine a method of documentation and traceability to the endoscope and reusable accessories. The high-level disinfected scopes should be clearly noted as having been reprocessed and when this occurred, so that users can readily identify that they are ready for use (Petersen et al., 2017). The selected storage cabinet should be made of a material that can be disinfected. Either conventional storage with endoscopes hanging vertically or drying cabinets that support either a horizontal or vertical storage may be selected. SGNA supports a 7-day storage interval for scopes that were reprocessed and stored according to professional guidelines and manufacturer instructions (SGNA, 2018a). Endoscope cases should never be used for storage.

The FDA provides a review and clearance process for sterilants and high-level disinfectants that make general claims for processing reusable medical and dental devices. A device manufacturer must comply with six FDA criteria regarding reprocessing instructions.

1. Labeling should reflect the intended use of the device.
2. Reprocessing instructions for reusable devices should advise users to thoroughly clean the device.
3. Reprocessing instructions should indicate the appropriate microbicidal process for the device.
4. Reprocessing instructions should be technically feasible and include only devices and accessories that are legally marketed.
5. Reprocessing instructions should be comprehensive.
6. Reprocessing instructions should be understandable.

The following are FDA-cleared germicides that have been deemed safe when used according to label directions for the reprocessing of such heat-sensitive medical devices as flexible endoscopes. Users should check with device manufacturers for information about germicide compatibility with their devices (Rutala, Weber, & Healthcare Infection Control Practices Advisory Committee, 2008).

DISINFECTANTS

Glutaraldehyde

Glutaraldehyde has been used by the health care industry for many decades to disinfect and sterilize medical devices, including those containing heat-sensitive materials. A large amount of information appears in the literature regarding its mechanism of action, the role of pH control, its compatibility with the various materials used to manufacture medical devices, the effect of temperature on the rate of microbial kill, and the procedures that should be used for the safe handling of glutaraldehyde solutions.

Disposal of glutaraldehyde must comply with local, state, and federal regulations. Some areas prohibit disposal into sewer systems, and others require neutralization (OSHA, 2006). Empty containers from freshly activated solutions should be thoroughly rinsed with water prior to disposal. Users should refer to the product SDS for specific disposal guidelines.

Advantages of glutaraldehyde

Advantages include:

- Proven effectiveness against microorganisms.
- Relatively inexpensive.
- Excellent material compatibility (Rutala & Weber, 2013).
- Not reported to cause adverse reproductive health effects or an increased incidence of cancer deaths (AAMI, 2013).

Disadvantages of glutaraldehyde

Disadvantages include:

- Recent evidence of resistance by some microorganisms to the antimicrobial effects of glutaraldehyde, such as with strains of nontuberculous mycobacteria (NTM) and Pseudomonas, as well as with some cyst and vegetative forms of protozoa (AAMI, 2013).
- Coagulates blood and fixes tissue to surfaces (Rutala & Weber, 2013).
- Irritant to the skin, eyes, and respiratory system. Although the symptoms are generally temporary, subsiding when leaving the exposure area, there is evidence that skin and respiratory irritant effects are exacerbated on repeated exposure to glutaraldehyde (AAMI, 2013).

  Skin. Skin contact irritation symptoms depend on contact time and can range from minor itching and slight local redness to moderate local redness and swelling.

  Eyes. Glutaraldehyde in a concentration of greater than 0.1% is considered to be irritating to the eyes and may result in moderate to severe pain, excessive blinking, and tear production. Contact with solutions containing concentrations of 2% or greater cause severe eye irritation and corneal damage that could result in permanent vision impairment (AAMI, 2013).

  Respiratory system. Inhalation of glutaraldehyde vapors may result in asthma-like symptoms, nose and throat irritation, and chest tightness. Additionally, there may be aggravation of pre-existing asthma and inflammatory or fibrotic pulmonary disease.

  Glutaraldehyde solutions should be used in use in a well-ventilated area or under a vented chemical fume hood. The lid should be kept on the soaking container, except when placing items into or removing them from the solution. The American Conference of Governmental Industrial Hygienists (ACGIH) recommends a maximum air concentration during any working exposure – a threshold limit value ceiling (TLV-C) – with glutaraldehyde of 0.05 parts per million volume (ppmv) (AAMI, 2013). Applying one of several mechanical techniques to monitor exposure will contribute to a safe work environment (Rutala et al., 2008).

Eye protection with goggles and face shield and skin protection with nitrile or butyl rubber gloves protect the user from injury. The equipment must receive the manufacturer's recommended number of plain water rinses to ensure that residual levels of glutaraldehyde are removed to protect the patient from mucous membrane irritation (AAMI, 2013).

Hydrogen peroxide

Hydrogen peroxide-based products are used for high-level disinfection of heat-sensitive, submersible medical devices, such as flexible endoscopes. Products may contain such concentrations as 2%, 3%, or 7.5% peroxide. The germicidal activity as a bactericidal, virucidal, sporicidal, and fungicidal works by producing destructive hydroxyl free radicals that can attack membrane lipids, DNA, and other essential cell components (Rutala et al., 2008).

Advantages of hydrogen peroxide

Advantages include:

- No activation is required.
- May enhance removal of organic matter and organisms. While it does not coagulate blood or fix tissues to surfaces, published studies indicate that it inactivates *Cryptosporidium* (Rutala & Weber, 2013).
- No disposal, odor, or irritation issues.

Disadvantages of hydrogen peroxide

Disadvantages include:

- Material compatibility concerns exist in the form of cosmetic (discoloration) and functional changes to brass, zinc, copper, nickel, and silver plating (Rutala & Weber, 2013).
- Mild eye and skin irritation may result at the 2% concentration, while severe and corrosive irritation with irreversible eye tissue damage and blindness may result at the 7.5% concentration.
- Vapors inhaled can be severely irritating to the nose, throat, and lungs (AAMI, 2013).
- The OSHA permissible exposure limit (PEL) for hydrogen peroxide vapor is 1 ppm as an 8-hour total weight average (TWA) (AAMI, 2013).

- Must be used in well-ventilated areas. Users should wear eye protection such as safety glasses, goggles or face shields, and skin protection such as rubber, neoprene, vinyl, or nitrile gloves (AAMI, 2013). Rinsing per manufacturer instructions should be followed to ensure that residual hydrogen peroxide is removed to protect the patient from mucous membrane irritation.
- Solutions must be diluted with a large volume of water, allowing the chemical to decompose prior to disposal into a suitable treatment system in accordance with federal, state, and local regulations.

Peracetic acid

Also known as peroxylacetic acid, peracetic acid is described as having rapid action against all microorganisms. It functions similarly to other oxidizing agents by denaturing proteins, disrupting the cell wall permeability, and oxidizing certain amino acid bonds in proteins, enzymes, and other metabolites. It will inactivate both gram-negative and gram-positive bacteria, fungi, yeasts, viruses, and all test strains of mycobacteria (Rutala et al., 2008).

The SYSTEM 1E Liquid Chemical Sterilant Processing System (Steris Healthcare, Mentor, OH) is the only system that is FDA-cleared specifically for liquid chemical sterilization of cleaned, immersible, reusable, heat-sensitive medical devices, including flexible endoscopes. The processor uses peracetic acid in an automated cycle to accomplish the sterilization of processed endoscopes (Steris Healthcare, n.d.).

Peracetic acid, once mixed in water to a 0.2% use solution, has been shown to be nontoxic and environmentally safe for disposal down the sewer (ASGE, 2016).

Advantages of peracetic acid

Advantages include (Rutala & Weber, 2013):

- Rapid sterilization cycle time in a fully automated format, using low-temperature liquid emersion.
- Production of environmentally friendly by-products (acetic acid, oxygen, water).
- Enhanced removal of organic material and endotoxins.
- Fewer adverse health effects to operators as compared to aldehydes.
- No coagulation of blood or fixing of tissues to surfaces.
- Rapidly sporicidal.
- Compatibility with many materials and instruments.
- Provision of procedure standardization (constant dilution, perfusion of channel, temperatures, and exposure).

Disadvantages of peracetic acid

Disadvantages include (Rutala & Weber, 2013):

- Potential material incompatibility, such as dulling of aluminum anodized coating.
- Allowing only one scope per cycle.
- More expensive (endoscope repairs, operating costs, purchase costs) as compared to aldehydes.
- Serious eye and skin damage with concentrated solution contact.
- Is a point of use system with no sterile storage.

Ortho-phthalaldehyde (OPA) 0.55%

Ortho-phthalaldehyde (OPA) received FDA clearance in 1999 as a high-level disinfectant, killing spores by blocking the spore germination process. Studies have shown that OPA has superior mycobactericidal activity to glutaraldehyde, with OPA's level of biocidal activity directly related to the temperature. OPA has been found to be effective against a wide range of microorganisms, including glutaraldehyde-resistant mycobacteria and *Bacillus atrophaeus* (*B. atrophaeus*) spores. The disinfection time for the solution at 20 degrees Celsius (68 degrees Fahrenheit) varies worldwide, but the standard in the United States is at least 12 minutes of chemical exposure (Rutala et al., 2008). As with other high-level disinfectants, meticulous manual cleaning of medical devices must precede exposure to this product for high-level disinfection (SGNA, 2018a).

OPA may not be compatible with all automated reprocessors. Check with manufacturers of automated reprocessors for specific compatibility statements. In an AER with an FDA-cleared capacity to maintain solution temperatures at 25 degrees Celsius (77 degrees Fahrenheit), the contact time for OPA is 5 minutes (Rutala et al., 2008).

Advantages of ortho-phthaladehyde

Advantages include:

- Reusable product with a maximum reuse life of 14 days as validated by testing to ensure that the solution remains at or above its MEC of 0.3% (Advanced Sterilization Products, 2015).
- Barely perceptible odor.
- Does not require exposure monitoring (Rutala et al., 2008).
- No required mixing or activation.
- Stable at a wider pH range of 3 to 9 (Beilenhoff et al., 2008).
- Lasting longer before reaching its MEC limit (about 82 cycles) compared with glutaraldehyde (after 40 cycles) in an automated reprocessor (Rutala et al., 2008).
- Excellent materials compatibility claims.
- Fast-acting high-level disinfectant.
- Does not coagulate blood or fix tissues to surfaces (Rutala & Weber, 2013).
- No reported reproductive or mutagenic effects in humans.

Disadvantages of ortho-phthaladehyde

Disadvantages include:

- Residues remaining on inadequately rinsed devices can result in staining of skin, mucous membranes, clothing, and environmental surfaces. Care should be taken to ensure that the recommended number of fresh-water rinses are performed as per the manufacturer IFUs (AAMI, 2013).
- More expensive than glutaraldehyde.
- Slow sporicidal activity (Rutala & Weber, 2013). Some studies indicate that some strains of mycobacteria, as well as some cyst and vegetative forms of protozoa, show resistance to OPA.
- Can cause redness and irritation with eye contact. PPE should be worn to prevent contact with OPA solutions, including gloves (polyvinylchloride and nitrile or butyl rubber), long sleeves, and splash-proof monogoggles.
- Can cause irritation with inhalation that could result in discharge, coughing, wheezing, tightness of chest and throat, difficulty breathing, stinging in the nose /throat/mouth, lip tingling, headache, loss of smell, and dry mouth. These symptoms are temporary and reversible. Pre-existing asthma may be aggravated by exposure (ASP, 2016). Solutions should be used in a well-ventilated area.

Small spills may be cleaned up with a damp sponge or absorbent pad. Larger spills should be deactivated with 25 grams of glycine (free-base) powder per one gallon of OPA solution spilled. The glycine should be thoroughly blended into the spill using a mop or other tool, allowing a 5-minute contact for neutralization. The neutralized product should be picked up and transferred to a properly labeled biohazard container. Neutralization should be allowed to continue for 1 hour, then the resulting material should be disposed of in accordance with all applicable federal, state, and local regulations (ASP, 2016). The SDS can be referenced for further instructions.

Peracetic acid-hydrogen peroxide solutions

The strong microbiocidal effects and broad-spectrum activity of peracetic acid at low concentrations have been known since the early 1900s. Peracetic acid alone is unstable and normally cannot exist without hydrogen peroxide and acetic acid. For this reason, all aqueous solutions of peracetic acid contain acetic acid and hydrogen peroxide. Peracetic acid-hydrogen peroxide formulations can be designed to include buffers, surfactants, and anticorrosives for specific applications.

There is one formulation in the United States designed for a single use in an automated system (AAMI, 2013). Medivators Rapicide PA Solutions A and B (Medivators, Inc., Minneapolis, MN) combine to produce a single-use liquid chemical germicide solution with peracetic acid in the concentration range of 1000–1300 ppm. It also contains corrosion inhibitors and water conditioning agents. The product is intended for use in the Medivators Advantage Plus or DSD Edge AERs (Medivators, 2011).

Advantages of peracetic acid-hydrogen peroxide solutions

Advantages include:

- No activation is required.
- Odor or irritation is not significant (Rutala & Weber, 2013).

Disadvantages of peracetic acid-hydrogen peroxide solutions

Disadvantages include:

- Material compatibility concerns with lead, brass, copper, and zinc, both cosmetic and functional.
- Limited clinical experience with its use.
- Potential for eye and skin damage with exposure (Rutala & Weber, 2013).
- Not compatible with some endoscopes (ASGE, 2016).

Selecting and using high-level disinfectants and sterilants

When considering chemicals and processes for high-level disinfection and sterilization of endoscopes and accessories, it is important to keep in mind that processes evolve over time. As such, FDA-cleared manufacturer label claims and IFUs change accordingly. It is essential that health care personnel obtain up-to-date information for the products they use, or are considering using, and refer to the current manufacturer label directions and written IFUs (AAMI, 2013).

Inadequate cleaning of instruments has been one factor cited in transmission of infection by flexible endoscopes (Edmiston & Spencer, 2014; Petersen et al., 2017; Rutala & Weber, 2015). In most cases, the cause has been attributed to a breach in the recommended reprocessing protocols. Because endoscopes may be used for patients with both recognized and unrecognized infections, such as HIV and the HBV, as well as bacteria that are resistant to the most powerful antibiotics, it is important that endoscopes be reprocessed according to SGNA's (2018) "Standards of Infection Prevention in Reprocessing Flexible Gastrointestinal Endoscopes" after each use. There is substantial evidence to show that the cleaning and disinfection processes described in these standards are highly successful in preventing the transmission of infectious disorders, including HIV, hepatitis, and tuberculosis.

Prions and other transmissible spongiform encephalopathies (TSE), including CJD and variant CJD, are resistant to conventional disinfectants and sterilants. For these patients, the European Society of Gastrointestinal Endoscopy (ESGE) recommends that an endoscopy be avoided. If the procedure must be done, special handling of the endoscope must occur, which may involve destruction of the used device (ASGE, 2008).

Automated endoscope reprocessors (AERs)

AERs have become commonplace in the endoscopy settings for standardizing parts of the reprocessing protocol and decreasing personnel exposure to sterilants and/or disinfectants. Some reprocessors have a validated cleaning cycle, but because the physical action of wiping and brushing channels is not achievable, the current accepted standard requires manual cleaning prior to using the AER. If the AER does not include a final alcohol and air rinse cycle, this step must be done manually. Because the elevator wire channel of duodenoscopes is not effectively disinfected by most AERs, this step should also be performed manually. Users should obtain and review model-specific reprocessing protocols for these scopes.

The FDA classifies AERs as medical devices that require Food, Drug and Cosmetic Act Section 510(k) clearance prior to introduction on the market or when modifications are made that impact safety and effectiveness. In October 2015, following ERCP-associated infectious outbreaks secondary to antibiotic resistant organisms, the FDA required the three manufacturers of duodenoscopes to revise and validate their reprocessing instructions, as well as to validate testing on their AER models (ASGE, 2016). Selection of an AER must be guided by compatibility with the equipment to be reprocessed, as well as by the chemicals selected for the washing and high-level disinfection cycles.

There are many types of AERs available commercially. An AER should have the following features (SGNA, 2018a):

- The machine should provide circulation of fluids through all channels at equal pressure without trapping air. Channel flow sensors provide an added measure of compliance.
- The detergent and disinfectant cycles should be followed by thorough rinse cycles and forced air pressure to remove all used solutions.
- The disinfectant should not be diluted with any fluids other than what is supplied through the AER.
- The machine should be self-disinfecting.
- No residual water should remain in hoses and reservoirs at completion of the cycle.
- Cycles for alcohol flushing and forced air drying are desirable.
- The machine should feature a self-contained or external water filtration system.
- A method to automatically store or print data verification of cycle completion is desirable.

Direct costs associated with use of the AERs include purchase price, installation, maintenance, chemicals, filters, test strips, and chemical disposal. Other costs may include installation of water softener equipment, installation of a special power supply, and personnel training and competency verification. However, these costs may be offset by decreased procedural delays, increased productivity, and decreased endoscope repairs due to the smaller potential for human error (ASGE, 2016).

ENDOSCOPIC ACCESSORY EQUIPMENT, INC.:

Reusable endoscopic accessories

Reusable endoscopic accessories, such as biopsy forceps and cutting instruments that break the mucosal barrier, are considered critical items. These require mechanical cleaning according to the manufacturer instructions, followed by sterilization, between each patient use (Peterson et al., 2017). They should be inspected for integrity and cleanliness before, during, and after use. Damaged items should be immediately removed from service, and soiled items should be reprocessed (SGNA, 2018a).

Water and irrigation accessories

Water and irrigation bottles are additional accessories requiring special attention. The water and irrigation bottles and connecting tubing should be manually cleaned and sterilized or high-level disinfected according to the manufacturer instructions at least daily. Water bottles used for endoscopy procedures should be filled with sterile water (SGNA, 2018b). Single-use water bottle caps and pump tubing using sterile water bottles are available.

Specialty lithotripters and Argon hoses

Specialty lithotripters are product-specific, and each manufacturer should be consulted regarding recommended cleaning, for not only a minimal of high-level disinfection but also for extending the life of the product. In addition, reprocessors may be used for high-level disinfection of Argon hoses according to each manufacturer's guidelines. Argon hoses have been manufactured as an integrated single-use system with probes and filters. Older Argon plasma generators can be adapted to accommodate this feature and to eliminate the need for reprocessing Argon hoses.

Manometry probes

Esophageal and anorectal manometry probes, as well as reusable CO_2 hoses, should be reprocessed and minimally high-level disinfected according to specific manufacturer recommendations.

REPROCESSING OF SINGLE-USE ACCESSORY EQUIPMENT

The gastroenterology nurse must be able to sort accessory devices according to the Spaulding classification system to determine how to reprocess each accessory. In some cases, the use of single-use devices (SUDs) may be desirable because they eliminate the need for reprocessing. In 2002, the FDA established statutory requirements for reprocessing specific SUDs by approved reprocessors. This approach remains controversial. In the absence of substantial scientific evidence that proves the safety and effectiveness of reprocessed SUDs of a critical classification within the endoscopy setting, SGNA maintains the position that critical medical devices originally manufactured and labeled for single-use should not be reused (SGNA, 2018b).

Disinfection

Safe and effective sanitation methods are an essential component of infection prevention practices within the gastroenterology unit. Such methods are necessary to remove any sources of contamination, thereby minimizing the hazards of cross-contamination and lessening the risk of nosocomial infections. Disinfection measures include the following:

1. Hands should be washed before and after each patient interaction, regardless of whether or not gloves are worn.
2. An Environmental Protection Agency (EPA)-registered, hospital-grade disinfectant should be used for general wipe-down of non-critical items, such as procedure carts, bedside and over-bed tables, and stretchers. Label instructions, including contact time, should be referenced for proper use.
3. Environmental surfaces such as walls and floors should also be wiped down on a routine basis. Facility policies and procedures must be developed and adhered to for routine housekeeping.
4. Visible spills of blood or other potentially infectious materials should be cleaned using products such as 10% bleach solution for blood, urine, or feces. Institution protocol should be followed.
5. Isolation precautions for patients with potential infections should be maintained when patients are transported to the endoscopy department for procedures (ASGE, 2008).
6. Specific methods to prevent infection transmission for patients with such infections as *C. difficile*, tuberculosis, VRE, and CRE should be followed. Examples include hand hygiene with soap and water for *C. difficile*, use of an EPA-registered hospital disinfectant with label claims to kill specific organisms, room incubation post exposure of patients with airborne transmitted illnesses such as tuberculosis, and enhanced environmental surface cleaning to eliminate such pathogens as CRE and VRE. Staff must give attention to the discovery, containment, and handling of patients with parasitic or vermin infections such as head lice, scabies, or bed bugs.
7. Endoscopy schedules should be planned so there is sufficient time for adequate processing of instruments and for cleaning/setting up the unit for the next patient.

Waste management

Waste management policies and procedures should be developed by each health care facility using local, state, and federal waste management regulations. Contaminated items should be discarded in compliance with these regulations. Using PPE and following facility processes are required for handling specimens, contaminated wastes, and linens (SGNA, 2019).

Facility policies and procedures should include proper methods of handling and disposing of waste and should clearly identify the designated parties responsible for compliance with the waste management regulations. Adherence to these specific procedures will benefit the health and safety of both staff and patients.

Special attention to the policies and procedures is required in the disposal of specific medications and devices (needles, sharps, and wires) frequently used in the endoscopy setting, as these may pose a risk to facility personnel or the environment.

Unresolved issues requiring further study

Unresolved issues relating to gastrointestinal endoscopy that require further study to develop guidelines include the following:

1. Replacement of water bottles and tubing. The optimal frequency for replacing clean water bottles, tubing for insufflation of air, lens wash water, waste vacuum canisters, and suction tubing has not yet been determined. Concerns involve the potential for backflow from a soiled endoscope against the direction of forced fluid and air passage into the clean air or water source and from a contaminated tubing and collection chamber against a vacuum into clean instruments used on subsequent patients (ASGE, 2016).
2. Surveillance and indicators. The use of surveillance culturing of endoscopes, as well as alternative indicators of adequate reprocessing (rapid indicators), are ongoing topics of investigation. At present, facilities are not specifically required to conduct these tests.
3. Duodenoscope reprocessing. The optimal methods for cleaning and disinfection or sterilization of duodenoscopes remain incompletely defined (ASGE, 2016).
4. Endoscope cabinet features. Endoscope cabinet features that are optimal for preventing contamination have not yet been determined. This includes cabinet ventilation parameters and the capacity to store accessories (CDC, 2017).
5. Replacement of endoscopes. While endoscopes should be serviced no less frequently than indicated in the FDA-cleared manufacturer IFUs, the optimal time interval for replacing the endoscope and its accessories has not yet been determined (CDC, 2017).

CASE SITUATION

Alex Holmes is a 28-year-old female who presents to the endoscopy unit for a colonoscopy. She has several tattoos and multiple body piercings. She has had diarrhea off and on for the past month. Recently, she has experienced bloody bowel movements.

Points to consider

1. Keeping in mind that Alex has increased risk factors for being hepatitis B- and hepatitis C-positive, what protection should be considered for personnel during the colonoscopy?
2. Upon completion of the procedure, how should the accessories be cleaned?
3. How should the environment be cleaned at the conclusion of the procedure?
4. After the procedure, what steps must the gastroenterology nurse take to properly decontaminate and disinfect the flexible colonoscope before use on subsequent patients?

Suggested responses

1. Regardless of the patient's suspected infection status, personnel should apply consistent infection prevention practices, including:
   a. Providing a water-absorbent pad for procedure equipment set-up.
   b. Wearing gloves when touching blood or blood-contaminated fluids or instruments.
   c. Wearing masks and eye protection.
   d. Wearing fluid-resistant gowns or aprons when assisting with the procedure, cleaning, and reprocessing of the colonoscope, and when soiling of clothing with blood or body fluids may be anticipated.
   e. Washing hands thoroughly and changing gloves between endoscopy procedures and patient contacts.
   f. Changing PPE between activities.
   g. Ensuring no cross-contamination by contact with surfaces.
2. Disposable accessories are preferred. If nondisposable accessories are used, they must be meticulously cleaned and disinfected or sterilized according to the manufacturer instructions. If biopsy forceps are being reprocessed, they should receive both ultrasonic cleaning and steam sterilization, following manufacturer's instructions for use.
3. At the conclusion of the colonoscopy, the environment should be cleaned by:
   a. Confining and removing organic material.
   b. Using moisture-proof disposable drapes as protective covering for procedure carts and equipment.
   c. Using an EPA-registered tuberculocidal hospital disinfectant for general wipe-down of all non-critical equipment, following the disinfectant's instructions for use for the appropriate contact time.
   d. Using a 10% bleach solution for visible blood or body fluids.
   e. Discarding moist, grossly contaminated waste in a disposable, leak-proof container.
   f. Transporting grossly soiled linens with minimal handling in leak-proof bags labeled as biohazard.
4. To properly decontaminate and disinfect the flexible colonoscope, the nurse or assistant should:
   a. Consider the flexible colonoscope as a semi-critical item that has been in contact with mucous membranes.
   b. Wash hands; don gloves, mask, and eye protection; and protect skin and clothing with a fluid-resistant gown.
   c. Thoroughly inspect the colonoscope for damage and perform a leakage test.
   d. Manually pre-clean and then high-level disinfect the colonoscope and accessories using appropriate sterilant and/or disinfectants, and dry and store the endoscope in accordance with SGNA infection prevention guidelines and manufacturer instructions.

Additional considerations

Health care workers in all gastroenterology and endoscopy settings should be immunized with hepatitis B vaccine. Personnel should also be screened annually for tuberculosis.

General guidelines have been established by ASGE for the prevention of endoscopically induced systemic or distant infections for some patient populations. Patient populations that should receive antibiotic prophylaxis prior to endoscopic procedures include:

- Patients with high-risk cardiac conditions, such as prosthetic cardiac valve or a history of infective endocarditis.
- Patients with congenital heart disease that is unrepaired, within 6 months post repair surgery, or repaired with residual defects, who also have gastrointestinal tract infections in which enterococci may be a part of the infecting bacteria.
- Patients who have had liver transplantation or who have known or suspected biliary obstruction where there is possible incomplete biliary drainage; antibiotics should be given prior to having an ERCP.
- Patients with endoscopic ultrasound-guided fine needle aspiration (EUS-FNA) of mediastinal cysts, as well as pancreatic or peripancreatic cysts.
- Patients undergoing percutaneous endoscopic gastrostomy (PEG) or jejunostomy (PEJ) tube placement for feeding.
- Patients with cirrhosis admitted with gastrointestinal bleeding; antibiotic therapy should be given on admission.

• Patient receiving continuous ambulatory peritoneal dialysis; antibiotics should be given prior to endoscopy of the lower GI tract.

ASGE recommendations are based on the 2007 American Heart Association (AHA) guidelines (Wilson et al., 2008; ASGE, 2015).

Endoscopy-related infections should be reported to all of the following:

1. Persons responsible for infection prevention at the facility.
2. The appropriate public health agency (state or local health department, as required by state law or regulation).
3. The FDA (www.fda.gov/medwatch).
4. The manufacturer(s) of the endoscope, disinfectant and/or sterilant, and AER (if used).

REVIEW TERMS

critical, decontamination, endogenous, exogenous, glutaraldehyde, high-level disinfection, intermediate-level disinfection, low-level disinfectants, minimum effective concentration, noncritical, peracetic acid, semicritical, sterilization

REVIEW QUESTIONS

1. The most common infectious agents in the clinical setting are:
   a. Viruses.
   b. Fungi.
   c. Helminthes.
   d. Bacteria.
   e. Protozoa.
2. Examples of endogenous infections include:
   a. Cholangitis following manipulation of an obstructed bile duct.
   b. Endocarditis in patients with mitral valve regurgitation who have sustained transient bacteremia during esophageal dilatation.
   c. Both a & b.
3. The most frequent exogenous infection associated with endoscopic transmission is:
   a. Hepatitic C virus.
   b. Mycobacteria.
   c. Hepatitis B virus.
   d. Human immunodeficiency virus (HIV).
4. According to the Spaulding classification system, examples of critical items are:
   a. Injection needles.
   b. Endoscopes.
   c. Biopsy forceps.
   d. a & c.
   e. b & c.
5. Circumstances for endoscope sterilization may include:
   a. Reprocessing option for duodenoscopes.
   b. Use of endoscopes in sterile operative field.
   c. Use of endoscopes for natural orifice transluminal endoscopic surgery.
   d. For patients with known HIV, HCV, or HBV.
   e. a, b, & c.
6. The steps for decontamination of endoscopes are:
   a. Pre-clean, leak test, manual clean, inspect, high-level disinfect (or AER), rinse, and dry.
   b. Pre-clean, manual clean, high-level disinfect (or AER), inspect, rinse, dry, and leak test.
   c. Pre-clean, leak test, inspect, high-level disinfect, manual clean, rinse, and dry.
7. The standard of care for reprocessing of gastrointestinal endoscopes is:
   a. Sterilization.
   b. High-level disinfection.
8. Agents approved for high-level disinfection include:
   a. Hydrogen peroxide, peracetic acid, green soap, ortho-phthalaldehyde (OPA).
   b. Isopropyl alcohol, hydrogen peroxide, peracetic acid, ortho-phthalaldehyde (OPA).
   c. Glutaraldehyde, hydrogen peroxide, peracetic acid, ortho-phthalaldehyde (OPA).
   d. Glutaraldehyde, hydrogen peroxide, peracetic acid, isopropyl alcohol.
9. Reprocessing of a duodenoscope requires:
   a. additional steps in all phases of reprocessing
   b. manual flushing of alcohol and drying post high-level disinfection per the manufacturer's instructions
   c. a & b.
10. The statement that best describes the SGNA position on reprocessing of single-use accessory equipment is:
    a. Reprocessing is accepted by the FDA as long as third-party processors are registered.
    b. Critical medical devices originally manufactured and labeled for single-use should not be reused.

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Chapter 4

ENVIRONMENTAL SAFETY

This chapter identifies environmental hazards in gastroenterology settings, including those associated with the use of electrical equipment, chemical agents, ionizing radiation, electrosurgery, and laser technology; latex sensitivity-allergens; accidental injury; and workplace violence.

The chapter also discusses the precautions that nurses and associates can take to protect both patients and health care personnel. These practices are intended as recommendations representing what is believed to be an optimal level of nursing practice. Policies and procedures of institutions, individual states' nurse practice acts, and national regulatory agencies will reflect variations in practice settings and/or clinical situations that determine how the recommended practices can be implemented. The importance of disaster planning will also be covered, including how the gastroenterology team will respond to such an event.

Learning objectives:

After reviewing the content of this chapter, the gastroenterology nurse should be able to:

1. Identify potential hazards in the Endoscopy Unit that may cause harm to patients or personnel, including electrical equipment, chemical agents, radiation, latex sensitivity, accidental injury, electrosurgery, laser, workplace violence, and disaster planning.
2. Describe steps that may be taken to minimize the dangers associated with these hazards.
3. Explain the responsibilities of gastroenterology personnel for ensuring a safe environment.

Basic safety regulations that govern institutional and health care facilities are found in the standards, guidelines, and codes of local, state, and federal agencies such as the Food and Drug Administration (FDA) and the Occupational Safety and Health Administration (OSHA), as well as voluntary standard-setting groups such as The Joint Commission and the National Fire Protection Association (NFPA).

It is the legal, ethical, and moral responsibility of all gastroenterology personnel to be aware of these safety standards and how they affect patient care in the gastroenterology suite.

ELECTRICAL SAFETY

Electrical equipment and potential hazards associated with the use of electrical equipment should be identified and safe practices established. Light sources, electrosurgical and cautery devices, laser technology, and video imaging equipment are all powered by electrical energy, as are lights, radiographic equipment, and operating tables in surgery. The most common hazards when working with electricity are fire, burns, and shock (electrocution).

Staff members working in the endoscopy suite should demonstrate competence in using electrical equipment. All electrical equipment should be inspected prior to each use. Inspection should include, but not be limited to, the following activities:

1. Check outlets and switch plates for damage.
2. Check power cords and plugs for fraying or other damage.
3. Check that all new equipment has been inspected by a biomedical technician or electrical safety officer before it is introduced into the practice setting and at established intervals for preventative maintenance.

Any direct contact with 110- or 220-volt wiring has the potential for **electrocution**. If the voltage is high enough, it can damage the brain, impair respiration, and cause cardiac dysrhythmia (ventricular fibrillation), resulting in cardiopulmonary arrest. Complications of electrocution may include vascular injury, tissue destruction from the inside out, loss of consciousness, damage to the respiratory center, infection, or eye damage (CDC, 2014).

FIRES

To prevent fires and reduce the risks of burns and electrocution associated with electrically powered equipment, gastroenterology personnel can take the following measures:

1. Use appropriate precautions when using flammable gases or liquids.
2. Before use, check the integrity of all electrical equipment, including cords, switches, plugs, and wall receptacles. When possible, use an emergency power outlet.
3. Ensure that all electrical equipment is properly grounded.
4. Do not use multiple-outlet adapters or two-wire extension cords, and do not remove ground pins from three-pin plugs.
5. Avoid routing power cords through heavy traffic areas, and avoid rolling equipment over electrical cords. Store cords loosely coiled, without kinks.
6. Follow manufacturers' recommendations and standards for all electrical equipment. Attach user manuals to equipment. When that is not possible, keep information within easy reach of staff in an area close to where the equipment is used.
7. Follow institutional policies and procedures regarding testing electrical equipment. Never use an electrical device that that does not have appropriate documentation of preventative maintenance inspections from the biomedical department (this may include a biomedical label with date inspected).
8. Remove any electrical equipment for which physicians or staff members have noted possible operational problems. Do not use if damaged or in need of repair. Report the item as defective, and tag it with the facility-approved process for repair.
9. Never place containers of liquid on or near electrical equipment.
10. Keep smaller electrical equipment secured on carts that do not tip easily.

Even when staff members take all reasonable precautions, it is still possible for a fire to occur in the endoscopy suite. To handle a fire quickly and effectively, gastroenterology personnel must participate in periodic fire drills in which they may practice using fire-fighting equipment. In addition, there should always be written policies and procedures for handling a fire emergency. Fire policies and procedures should be reviewed regularly and whenever a new type of equipment is installed.

If a fire occurs, it is important to proceed with the following steps:

1. Remember the "RACE" acronym:
   a. **RESCUE** patients and staff from the immediate danger zone, if feasible.
   b. Sound the **ALARM** and proceed according to institutional fire policy. In the procedure area, the alarm is to call out there is a fire. The procedure team response includes recognition and acceptance of the event in order to react. If the fire is not immediately put out, the fire alarm is pulled to notify the team in the general area of the need for help.
   c. Close all doors to **CONTAIN** smoke and flames.
   d. Smother or **EXTINGUISH** flames, if possible, using a **Halon (ABC) extinguisher** for an electrical fire. A Halon extinguisher contains a liquefied, compressed gas that stops the spread of fire by chemically disrupting combustion, leaving no residue behind. Halon is rated for class A, B, and C. It is safe for human exposure. If unable to extinguish, **EVACUATION** will be necessary. This should be done swiftly with good communication. Evacuation is done horizontally first, then vertically.

The RACE process should be posted and reviewed during all fire drills.

2. Know different types of fire extinguishers classified according to source of fire. These include A, B, C, or multipurpose fire extinguisher, B-C or A-B-C. Additionally, D is for metals and K is for oils/fats.
   a. A= common combustibles like paper.
   b. B= flammable liquids.
   c. C= electrical.
3. Never use water to extinguish an electrical fire.
4. Activate circuit breakers and close gas supply valves as directed by institutional policy.
5. Treat materials that are melting, dripping, or smoking as if they are actually burning to eliminate the possibility of their igniting or reigniting.
6. Direct firefighters to the location of the fire.

When fire involves a patient directly, personnel should remove burning articles or smother the flame with a towel or blanket. Smoldering, charred debris should be extinguished and removed from the patient area immediately. Personnel should never use fire extinguishers directly on patients because of the danger of cryogenic tissue damage or disposition of dangerous residue.

All burned material and equipment should be isolated and saved for further investigation after the fire has been extinguished (OSHA, n.d. a).

ENDOSCOPIC IMAGING SYSTEMS

Like any other electrical device, endoscopes and their imaging systems must be checked for proper functioning before each use. Gastroenterology personnel should adhere to the following procedure (ASGE, 2014):

1. Check the integrity of all electrical cables before use. Frayed, cracked, or kinked cables are dam-

aged. Do not use them.
2. Do not handle equipment when hands, feet, body, or floor is wet.
3. Never use the light source as a supply table.
4. Ensure that all components of the imaging system are correctly connected.
5. Check the video image on the monitor(s) for clarity, and ensure additional equipment (e.g., image manager, printer, slide maker, computer) set-up is correct before use.
6. Turn on all system components and suction to make sure they are functioning properly.
7. Turn the light source or processor off when not in use. If doing hourly scheduled procedures, do not turn light source or processor off between each case.

ELECTROSURGICAL AND CAUTERY DEVICES

The **electrosurgical unit (ESU)** and cautery unit are among the most hazardous electrical devices in the endoscopy suite. Proper care and handling of electrosurgical equipment is essential to patient and personnel safety. These devices use high radiofrequency electrical current to cut and coagulate body tissue. These recommended practices do not endorse any specific product. Biomedical services personnel in practice settings should develop detailed routine safety and preventive maintenance inspections and maintain records.

Burns can result from:

- Poorly applied patient **dispersive electrodes** (formerly called grounding pads).
- Electrical current seeking an alternate pathway out of the body, such as through ECG electrodes or metal objects like intravenous (IV) poles and x-ray tables.

It is critical for safety to follow the manufacturer's instructions for testing all electrosurgical equipment before and during the procedure.

With **bipolar** units, the circuit is completed locally; therefore, no dispersive electrode is required in contrast to monopolar electrocautery (ASGE, 2013). Bipolar is safer for patients with pacemakers and automatic internal cardiac defibrillators. Because there is no dispersive electrode, current is not traveling through the body.

If using the **monopolar** mode, it is important to place the dispersive electrode (return electrode) properly on the patient. The dispersive electrode should not be placed near or on a bony prominence, on skin over an implanted metal prosthesis, on scar tissue, or on places where circulation is likely to be impaired. If a dispersive electrode must be repositioned or becomes loose, it must be replaced with a new one.

To ensure patient safety, use smaller pediatric dispersive electrodes on the neonate or pediatric patient. The child's upper thigh may not provide enough skin surface for proper application of a grounding plate. In these cases, the lower back area may be used safely and effectively when electrocautery is anticipated. Follow the manufacturer's instructions for use for adults and pediatric patients.

The dispersive electrode should never direct current flow close to the site of a cardiac pacemaker. Considerations required for patients with an implantable cardiac defibrillator (ICD) may include suspending therapy by the device representative before the procedure or before applying and securing a magnet in the gastroenterology lab where there is a likelihood of an ESU or cautery device being used (ECRI Institute, 2018). Keep in mind that these considerations may vary based on facility requirements.

Older pacemaker models may interpret electrocautery as cardiac activity and inhibit the pacemaker from initiating a heartbeat. When caring for patients with pacemakers and ICDs, personnel should take additional precautions that include, but are not limited to:

1. Know the manufacturer of the device (e.g., make a copy patient's wallet device ID card).
2. Validate battery life for the ICD/pacemaker with the patient's identified device clinic/practitioner. Battery life and device checks are scheduled every 3-6 months after implantation.
3. Coordinate ICD therapy suspension with a magnet or by a device representative before using any electrical energy tools that may cause interference. Provide continuous cardiac monitoring and position the defibrillator / resuscitation equipment for emergency care.
4. Ensure that the distance between the active and dispersive electrodes is as short as possible.
5. Place both electrodes as far from the pacemaker site as possible. Follow the manufacturer's instructions for use for dispersive electrode placement.
6. Keep all ESU cords and cables away from the pacemaker/ICD and its leads.
7. Use the lowest possible power setting on the ESU.

ESUs and cautery devices should be used on the lowest effective power setting for the endoscopic procedure. These settings should be confirmed orally to the endoscopist before activation to ensure they are correct. An audio tone should be heard when the device is activated so that everyone knows when it is in use. The ESU or cautery unit should be turned off or placed in standby mode after each use.

If power settings above those expected are required to cut or coagulate, check the electrical pathway from the patient to the machine to rule out faulty or improper connections. Most generators will sound an alarm if the electrical circuit is incomplete. If the problem is not obvious and easily correctable, the machine should be removed from service. Incomplete circuitry may lead to patient burns or to shock to those touching the unit (ECRI Institute, 2018; Baeg et al., 2016; ASGE, 2013).

Argon enhanced coagulation (AEC), also called **argon plasma coagulation (APC)**, technology should be used according to the manufacturer's instructions for use (IFU). As the AEC unit uses electrical current, the considerations and safety issues are the same as using any monopolar ESU. This will require personnel to take precautions including, but not limited to, the following:

1. Observing all safety measures used for the ESU according to IFU.
2. Avoiding placing the electrode in direct contact with tissue. This can cause gas embolism to occur.
3. Activating the argon gas flow and the ESU simultaneously.
4. Limiting the argon gas flow to the lowest level possible.
5. Purging the argon gas line of air before each procedure and flushing air out of the argon gas line by pushing the "purge" button twice, allowing gas to travel through line between each purge.
6. Activating the system after moderate delays between uses.
7. Testing the system to assure it is working properly before handing the physician the probe (ASGE, 2013).

EQUIPMENT MAINTENANCE

Every piece of electrical equipment must have an operation manual and receive routine preventive maintenance checks performed by the biomedical engineering department. In-services should be held on a regular basis to update staff members on new equipment or changes in old equipment.

Health care facility policies and procedures for the use and maintenance of equipment used on patients must be in compliance with the Safe Medical Devices Act of 1990 (SMDA). When patient or personnel injuries occur or when equipment failures occur, the ESU machine and the active and dispersive electrodes should be isolated in accordance with institutional policies based on the SDMA.

Patient and personnel injury requires immediate attention. Follow procedure for thorough examination of the patient and emergency care, if needed. Injured personnel will need to be relieved and appropriate care provided via employee health or emergency services.

Regarding equipment, follow the institution's policy and procedure. Some probable steps include notifying department leadership who will contact Risk Management and/or the Legal Department immediately. Provide device identification, preventative maintenance, service information, and adverse event information in the report. Remove, replace, and sequester all equipment involved in the event, including the ESU, the active and dispersive electrodes, and all packaging. Facilities Engineering and Risk Management will review and submit the report to the FDA (US FDA, 2018).

LASER SAFETY

The most dangerous aspects of **laser** use are misdirection, scattering of the laser beam, smoke inhalation, and fire hazards. Whenever lasers are in use, there is a potential for fire, damage to the skin or eyes, and respiratory tract irritation of patients and personnel. To prevent such injuries, personnel should follow these guidelines:

1. Place a warning sign on the door leading into the room where the laser is in use. The sign should include "LASER IN USE" in bold letters, the specific laser type, the international laser symbol, wavelength, and maximum voltage.
2. Provide protective eyewear for all patients and personnel. Extra eyewear should be left hanging on the procedure room door so additional or unexpected personnel may put them on before entering the room. The optical density of the eyewear should be equivalent to the wavelength of the laser used.
3. Use instruments that are anodized or covered with non-reflective coating to prevent inadvertent beam reflection.
4. Place the laser in the standby mode when it is not in use, and provide the physician access to only one foot pedal when it is in use.
5. Be knowledgeable about the laser key control, how it functions, and the emergency stop or "shutdown" button.
6. Use lasers with a tamperproof audio tone that sounds when the beam is being activated.
7. Use smoke evacuators and high-filtration masks to minimize the inhalation of the smoke produced in laser procedures.
8. Do not use flammable or combustible materials near the laser beam. Follow all guidelines for electrical safety.
9. Complete a laser safety "check-list" for each case so quality safety measures can be tracked.

Policies and procedures should establish authority, responsibility, and accountability, and should serve as operational guidelines. To build knowledge, skills, and attitudes that enhance patient care, personnel orientation and ongoing education should include an introduction and review of laser safety policies and procedures.

All health care personnel working in laser treatment areas should attend a laser safety educational program that includes periodic updates to reinforce safe use of lasers. This program should provide participants a thorough understanding of all laser procedures and technology required for establishing and maintaining a safe environment during laser procedures (OSHA, n.d. c).

RADIATION SAFETY

Certain waves of electromagnetic energy can penetrate matter by disrupting bonds between atoms and creating electrically charged ions. Such waves are termed **ionizing radiation**. Radiation is a hazard because of its ability to modify molecules within body cells, thus causing cell dysfunction, alteration or stoppage in cell replication, or cell destruction. The effects of radiation may be somatic or genetic.

The primary source of ionizing radiation in the endoscopy unit is radiography equipment. Fluoroscopy allows the physician and staff to view and monitor the placement of catheters, stents, dilators, or endoscopes. In addition, portable x-ray machines may be used for diagnostic purposes. The incident beam is the primary source of radiation exposure for the patient. The "primary" radiation beam produced is focused and directed through the area to be examined. The personnel in the room are exposed to "secondary" radiation or scatter radiation. This is the major source of radiation to the endoscopist and staff. In addition, a third source of radiation is leakage radiation from the radiographic machine itself.

Standards provide recommendations related to the annual maximum permissible dose (MPD) of radiation for a whole body (trunk and head), active blood-forming organs, and gonads of 5 rem. The lens of the eye has an MPD of 15 rem per year, whereas the extremities and skin have an MPD of 50 rem per year. A declared pregnant woman should be monitored to limit the dose of radiation to the embryo or fetus to 0.5 rem. If the dose equivalent to the embryo/fetus is found to have exceeded 0.5 rem (5 mSv) or is within 0.05 rem (0.5 mSv) of this dose by the time the woman declares the pregnancy to the licensee, the licensee shall be deemed to be in compliance with paragraph (a) of this section if the additional dose equivalent to the embryo/fetus does not exceed 0.05 rem (0.5 mSv) during the remainder of the pregnancy (NRC, 2007). Employers and the radiation safety officers are required to monitor the film badges and report exposure higher than compliance rates

Any level of ionizing radiation is potentially harmful. Unnecessary exposure can be minimized by:

1. Decreasing the time of exposure.
2. Increasing the distance from the source.
3. Placing a shield between the radiation source and the body.
4. Questioning all female patients of childbearing age about the possibility of pregnancy before radiographic procedures. (Maternity leaded aprons can provide double thickness in the area where protection is needed most.)

The following protective guidelines may help minimize the exposure of gastroenterology patients and personnel to ionizing radiation (US FDA, 2018; US NRC, 2018):

1. Wear lead-lined, flexible wraparound aprons and gloves, at a minimum, in accordance with institutional guidelines. Consider thyroid collars and protective eyewear (leaded, with side shields).
2. Use patient gonadal shielding as appropriate. Aprons should fully cover the breast area for females, and aprons must be long enough to cover long bones of the operator's body.
3. Wear film badges during procedures in which radiation exposure is encountered. These badges should be worn in the same place each time. A monitor at the neckline measures the amount of exposure to the head, neck and lens of the eye. Endoscopists who perform biliary procedures should wear a finger monitor in the same location for each procedure.
4. Analyze exposure levels monthly. Film badges should be stored at the hospital and not worn or taken away from the workplace, as they may collect ionizing radiation from unrelated sources like the sun, soil, or airport scanners.
5. Alternate personnel for procedures that require radiation, and limit exposure time for personnel holding patients and/or films.
6. Ensure personnel do not turn their unshielded backs to the radiographic equipment and maintain as great a distance from the x-ray beam as possible when the fluoroscope is in use.
7. Allow only trained personnel to operate and maintain all radiographic equipment.
8. Be certain that the contrast medium selected is effective for the type of study being done.
9. Make sure the patient is properly prepared and positioned. Expose only the targeted area of study to the film or fluoroscopy to help reduce unnecessary repeat procedures.
10. Consult with a radiation safety office regarding equipment inspection, analysis of radiation exposure badges, and educational in-services.
11. Check newly purchased leaded protective devices for cracks or holes that may compromise their integrity to protect personnel. The leaded devices should also be checked at least annually, and according to facility protocol, to determine the need for replacement. Follow the institutional policies, and participate in yearly in-service.
12. Inspections are documented and will be available for regulatory agency reviews.
13. Store leaded aprons flat or hung vertically. Do not fold leaded aprons.
14. Clean leaded protective devices with an EPA-registered hospital disinfectant after every use.

In addition, personnel should be aware of their documentation for fluoroscopic procedures. Documentation must include the type of protection provided to the patient and the area protected in all post-procedural documentation. Quality care requires a team member to be responsible for reporting results that are higher than the established threshold. Leadership will establish steps to be taken including staff notification and sign-off procedures. Specific scenarios may require further apparent cause analysis or root cause analysis for safety and compliance. Use facility policy and procedure for safety reporting. When handled properly and efficiently, ionizing radiation can be safe and effective during diagnostic and therapeutic procedures in the Endoscopy Unit (US FDA, 2018; US NRC, 2018).

LATEX SENSITIVITY OR ALLERGY

Latex allergy continues to be a health risk. Identifying patient allergies, including **latex allergy**, is important before any endoscopic procedure is performed. A latex-free environment can make procedures safer for the latex-allergic patient. Gastroenterology nurses, associates, and personnel have the responsibility to assess and educate patients about latex allergy to promote positive patient outcomes in the endoscopy setting.

The following individuals are at a higher risk for developing latex allergy:

1. Workers with ongoing latex exposure, such as health care personnel who frequently change latex gloves.
2. Individuals with a tendency toward multiple allergic conditions.
3. Persons with allergies to certain foods, especially avocados, potatoes, bananas, tomatoes, chestnuts, kiwi fruits, and papaya.
4. Persons with spina bifida, asthma, or a history of multiple surgical procedures.

Patient care should always be individualized. Some of the following recommendations may be unnecessary if low-allergen and powder-free gloves are utilized throughout the health care facility. Some guidance for the endoscopy staff practicing in facilities where powdered gloves are still in use include:

1. Notify the endoscopy unit as soon as the known or suspected latex allergy is discovered. Place a "latex alert" on the schedule for that patient.
2. Schedule the procedure as the first case of the day, if possible.
3. Remove all latex items from the patient care area, including the procedure room, admission, and post-procedure recovery area.
   - Remove all known latex supplies and equipment. Check blood pressure cuffs for latex.
   - Check supplies and tubing, ampules, etc. Remove the rubber stoppers before withdrawing the medications instead of withdrawing medication with a needle through the rubber stopper.
4. Remove boxes of latex gloves from all areas and replace them with latex-free gloves, both sterile and nonsterile.
5. Educate the patient, family, or significant other(s) regarding the latex-safe plan.
6. Document the "LATEX ALLERGY" in the electronic medical record and place an allergy band on the patient.
7. Monitor for severe allergic response.
8. Discuss treatment options available during the Time Out related to IV fluids, emergency drugs (i.e. antihistamine, oral steroid tablets, and epinephrine), and resuscitation equipment.

The above recommendations are designed to promote a safe environment for latex-sensitive or latex-allergic patients and health care workers. Follow the facility latex-safe policies, procedures, and protocols (Premier Safety Institute, n.d.; OSHA, n.d. b).

CHEMICAL SAFETY

Gastroenterology personnel come into daily contact with a large number of chemicals, some of which carry high exposure risks. OSHA regulations require secondary containers to include the name of the chemical, hazard warnings regarding target organs and applicable personal protective equipment (PPE), plus the name and address of the chemical manufacturer. There are a number of common-sense rules for handling chemicals (Safety Management Group [SMG], 2018):

1. Always read and follow label directions.
2. Ensure that all care providers have easy access to and an understanding of the information provided on the safety data sheet (SDS) that accompanies all hazardous chemicals.
3. Familiarize all employees with a spill-containment plan specific for the liquid chemical germicide used. The information from the specific SDS should be incorporated into the plan.
4. Mix chemicals only in accordance with directions.
5. Follow the rules of standard precautions as if the chemicals were body fluids. If it will come in contact with hands, wear gloves. If it might splash, wear a face shield and protective clothing. Always wash hands after exposure to any chemical.
6. Follow institutional policies and procedures for chemical spills, which may include but are not limited to the following:
   a. Drip or splash – Wipe up with gauze and dispose in biohazard bag.
   b. Larger than a drip or a splash – Use appropriate spill kit. Dispose in spill kit disposal bag after material solidifies. Use spill kit brush and pan. Call Environmental Services personnel to cleanse floor and transport segregated spill kit bag to Chemical Disposal Room.
   c. Larger than the available spill kits – If fumes are overbearing, evacuate. Call Protective Services to secure area. Call Facilities Engineering to evaluate airflow, and make changes to prevent fumes from entering patient care areas.

      Call Professional Disposal Team to clean up spilled chemical with special equipment.
7. After the chemical is removed, rinse instruments and surfaces thoroughly to remove any harmful residue.

One of the most frequently encountered hazardous chemicals in the endoscopy unit is **glutaraldehyde**. Because glutaraldehyde is water-soluble and vaporizes readily, its main effects are on the mucous membranes and the eyes. Short-term exposure can cause nose and throat irritation, burning of the eyes, and headaches.

When handling glutaraldehyde, gastroenterology personnel should wear rubber gloves (such as nitrile or butyl gloves), goggles, a mask or a face shield, and an impervious gown. Glutaraldehyde should only be used in areas that have adequate ventilation. It should be stored in a tightly covered container in a cool, dry place. Sinks, showers, and eyewash stations should be available. If glutaraldehyde comes in contact with the eyes, flush with water for a full 15 minutes. Skin should be washed thoroughly. Clothing should be removed and washed before reuse. In case of excessive inhalation, the individual should be moved to fresh air and treated medically as symptoms warrant (CDC, 2014).

Another hazardous chemical that gastroenterology personnel are exposed to is **formaldehyde**, which is primarily a tissue fixative. In the endoscopy lab, this could be in the form of formalin small specimen containers. When using small volumes of this chemical, open the container for short intervals when the specimen is ready to be placed into the fixative and label appropriately.

Formaldehyde is water-soluble and, therefore, mainly affects the mucous membranes and the eyes. To minimize exposure, workers should wear gloves, goggles, aprons, and masks. Formaldehyde should be stored in an airtight container. Personnel should be informed about possible adverse reactions. First-aid measures are the same as for glutaraldehyde (OSHA, 2011).

Peracetic acid is another variety of liquid chemical germicide. In its reconstituted form, it is extremely irritating to the eyes, mucous membranes, skin, and respiratory tract. It should be stored in a dry area. Staff should be familiar with written procedures for disposing of partially or fully reconstituted peracetic acid. The supply of peracetic acid should be rotated and expiration dates routinely checked (CDC, 2015).

Isopropyl (rubbing) alcohol is also found in endoscopy suites. Gloves should be worn when using this chemical, and care should be taken to avoid splashing because it may cause corneal burns and eye damage. Isopropyl alcohol is highly flammable and, therefore, should be stored in tightly closed containers away from heat, flames, or sparks. In case of excessive inhalation, the individual should be moved to fresh air and given medical support. In case of ingestion, the individual must receive immediate medical treatment (CDC, 2016).

Cytology fixatives can cause eye and skin irritation, as well as respiratory irritation with repeated or prolonged exposure. Full PPE, including gowns, gloves, and masks, must be worn when handling such chemicals (Hologic, 2011).

Mercury-filled dilators must be inspected on a regular basis to ensure they are intact to avoid leakage. Always follow the manufacturer's recommendations and guidelines regarding replacement and expiration dates. It is illegal to put mercury-containing devices into the garbage or biomedical waste ("red bag" trash). Expired devices or items no longer in use must be disposed through a mercury reclamation facility or a hazardous waste management company. Mercury-filled dilators are becoming rare. Many are now filled with tungsten, which is a safer alternative to mercury and may be disposed of in general waste (CDC, 2017).

ACCIDENTAL INJURY

A few simple preventive measures help gastroenterology personnel reduce the risk of accidental injury to themselves and their patients.

Human errors related to the operation of medical devices in the endoscopy unit can be minimized by utilizing built-in safety features such as the audible activation indicator on an ESU, providing adequate operator training and in-services, and performing regular preventative maintenance of equipment. Mental or physical exhaustion is a contributor to the risk of accidents. If problems develop, the FDA has the authority to recall devices.

Accidental falls are another potential source of injury. To avoid accidental falls in the endoscopy unit, the following steps should be taken:

1. Hallways and doorways should be free of equipment, supplies (with special attention to incoming supplies delivered by non-department personnel), and carts.
2. Rooms should be kept neat and orderly.
3. Electrical cords, cables, and accessories should be placed so that personnel do not trip or fall.
4. Beds should be locked using floor brakes, and side rails should be up when stepping away from a patient's bedside.
5. Post-sedation patients should be assisted in dressing and going to the restroom.

Proper lifting techniques must be practiced to prevent staff physical injury. Devices such as transfer boards should be used for lifting, and requesting additional help to assist should be encouraged. Comfortable temperatures, noise control, ventilation, and adequate lighting are essential for a safe environment.

OSHA is the federal agency responsible for regulating workplace environments to ensure safety. It is important that the endoscopy unit comply with all applicable OSHA standards (Calderwood et al., 2014; OSHA, 2016).

EMERGENCY PATIENT CARE

The endoscopy unit must be prepared for a patient emergency by having:

1. Oxygen, via different methods of delivery (nonrebreather mask, facemask, nasal cannula).
2. Crash cart with appropriate equipment and medications.

3. Immediate availability of personnel trained in cardiopulmonary resuscitation (CPR) and Advanced Cardiac Life Support (ACLS). All endoscopy staff should be proficient in CPR, with annual validatons.
4. Immediate availability of personnel knowledgeable in the use of medications, recognizing side effects, and understanding peak effect and reversal medication.
5. Immediate availability of personnel trained in Pediatric Advanced Life Support (PALS) and Neonatal Advanced Life Support (NALS), if applicable. The pediatric crash cart should be stocked with size-appropriate equipment for the age groups receiving care in the facility. All medical facilities involved in the care of children must be prepared for the possibility of a life-threatening emergency (Calderwood et al., 2014).

DISASTER PLANNING

Health care personnel must be prepared to respond quickly and effectively to internal or external disasters. Although rare, the occurrence of such events is a real possibility and must be anticipated. Examples of internal disasters could be unit-based fire, utility failures, large chemical spills, or an active shooter. External disaster examples include severe weather and mass casualty.

An internal disaster plan that includes institutional policies and procedures for moving patients out of the unit or evacuating them from the building should be reviewed and posted or accessible. Simulated disaster drills help personnel evaluate how they would function in a crisis.

WORKPLACE VIOLENCE

Workplace violence is increasing across all clinical areas. The National Institute for Occupational Safety and Health (2015) describes workplace violence as “any physical assault, threatening behavior or verbal abuse occurring in the workplace setting.” The gastroenterology personnel are as vulnerable in the endoscopy unit as any other workplace. Increasing staff’s sensitivity and awareness of potential violent incidents may help to identify a potentially violent situation before it escalates.

A form of violence that has been given more attention in recent years is that of lateral violence or harassment between co-workers. Examples of lateral violence are nonverbal negative innuendos (e.g., raising eyebrows, face-making), covert or overt verbal affront (snide remarks, withholding information, abrupt responses), undermining clinical activities (not available to help, turning away when asked for help), sabotage, bickering among peers, using a scapegoat (always assigning blame to one person when things go wrong), backstabbing, failure to respect the privacy of others (gossip or talking about others without their permission), and broken commitments and/or broken confidences (repeating something that was meant to be confidential). In the past, these behaviors have often been ignored, but increased awareness and efforts to educate staff have taken precedence in efforts to prevent lateral violence from distracting staff from delivering safe patient care.

Harassment between staff and physicians has also become a concern of accrediting bodies such as The Joint Commission and DNV GL. Accreditation standards have been developed to guide hospitals and ambulatory facilities in dealing with these issues. Education and prevention is the best protection to empower staff toward a safe work environment through organization policies for reporting and follow-up that staff and physicians agree upon and are held accountable to.

Policies and procedures to prevent or address workplace violence should be established for all health care facilities. On the institutional level, OSHA makes the following recommendations for abatement of workplace violence:

1. Develop a crisis management plan.
2. Require mandatory training for all workers in prevention techniques, restraint and seclusion procedures, and multicultural diversity techniques.
3. Increase staff-to-patient ratios, when applicable.
4. Identify patients with increased potential for violence.

To keep the gastroenterology personnel healthy, physically and mentally, workplace violence should be reported, and periodic training should be mandatory. Personnel should learn to spot violence and defuse it before an event takes place. Recognizing that workplace violence is not “part of the job” is the first step to change (OSHA, n.d. e).

CASE SITUATION

Christopher Little, age 68, was admitted to the endoscopy unit for a gastroscopy after radiographic x-rays were suggestive of a possible gastric mass. Christopher has an extensive medical history, which includes multiple surgeries, drug allergies, and food allergies.

Points to think about

1. The initial assessment should include all surgeries, drug allergies, and food allergies.
   a. What particular foods are potentially significant for latex sensitization?
   b. Is there significance if the patient has had multiple surgeries? If so, why?
2. Did the primary care physician relay any information regarding a potential latex alert?
3. Had the endoscopist seen the patient prior to his arrival for his appointment in endoscopy?
4. What immediate nursing responsibilities are implemented if a “latex allergy/ sensitivity” is documented?

Suggested responses

1. If the initial interview determined that the patient has a latex allergy or sensitivity, the following must be completed:
   (a) Immediately remove all latex items from the patient preparation area.
   (b) Prepare the procedure room with removal of all latex-containing items, such as gloves and IV accessories, and replace with non-latex items.
   (c) Communicate allergy to the health care team members, including the physician.
   (d) Be prepared to treat the patient with specific medications before the procedure is needed.
2. During the interview and implementation of care of the patient with latex allergy, be sure to include the patient, family, and/or significant other.

CASE SITUATION

The Endoscopy Team is at lunch and the team is discussing a situation that occurred in the procedure earlier that day. The case involved an endoscopist who is known to be verbally abusive to members of the team, especially the team member who is assisting the endoscopist with the procedure.

Points to think about

3. Individuals listen to each other related to scenarios.
   a. Knowing workplace violence is a "hot topic," how can the team support each other? What is different about this compared to years ago?
      1) Management needs to be involved.
      2) Team involved in the case needs to speak up.
      3) Safety Officer notification - when workplace violence is evident; teams will be less likely to speak up in an emergency or ask clarifying questions.
      4) Patient safety and staff safety can be compromised.
      5) Due to hierarchy with roles and titles, workplace violence can occur between physician and staff member; staff member and staff member; patient/family and staff member.

Suggested responses

1. Staff should support one another as the situation is discussed and identified.
   a. First the team needs to recognize the situation as inappropriate.
   b. Who should they involve in the discussion in order to make a change in the workplace environment?
   c. The Manager suggests a team meeting to discuss what is not acceptable in today's health care environment related to workplace violence.
   d. Employee assistance may be available relatedto the stress of these events.
   e. A safety report is filed related to professional conduct, which will involve peer review and a plan to eliminate this behavior in the department.

REVIEW TERMS

electrosurgical unit (ESU), equipment failure, evacuation plan, fire safety, glutaraldehyde safety, ionizing radiation safety, laser safety, latex allergy

REVIEW QUESTIONS

1. An electrical fire that does not involve the patient directly should be handled by:
   a. Using a Halon extinguisher.
   b. Throwing water on it.
   c. Suffocation with a fire blanket.
   d. Running from the room.
2. When not being used during the procedure, an ESU device should:
   a. Be set down on a drape.
   b. Be placed in the safety holster.
   c. Remain in the endoscopist's hand.
   d. Be handed to an assistant.
3. For laser procedures, the optical density of the protective eyewear used is determined by:
   a. The physician's preference.
   b. The procedure that will be performed.
   c. Institutional policies and procedures.
   d. The wavelength of the laser being used.
4. When ionizing radiation is in use, pregnant health care workers should:
   a. Wear a lead-lined apron, thyroid collar, and protective eyewear in accordance with institutional guidelines.
   b. Wear a film badge.
   c. Limit the dose to the embryo or fetus to 0.5 rem and monitored during the pregnancy.
   d. All of the above.
5. If a piece of electrical equipment malfunctions during a procedure, it should be:
   a. Fixed by gastroenterology personnel.
   b. Removed, replaced, and sequestered.
   c. Reported per institutional policy.
   d. Both b and c.
6. If glutaraldehyde accidentally comes in contact with a health care worker's skin, he or she should:
   a. Wash the area thoroughly for the time recommended on the Safety Data Sheet.
   b. Rinse the area with water only.
   c. Apply burn ointment.
   d. Cover the area with a bandage.
7. Plans for evacuating patients in the event of an external or internal disaster should be:
   a. Posted in an obvious location.

b. Recorded in institutional policies and procedures manuals.
c. Practiced in simulated disaster drills.
d. All of the above.

8. If a procedure is scheduled for a latex-allergic patient, the following should be done:
a. Schedule the procedure the first of the day.
b. Report to FDA immediately.
c. Refer the patient to the operating room.
d. Both a and b.

9. To avoid accidental injuries to patients and personnel when a laser is in use, it is important to:
a. Use protective eyewear.
b. Post "Laser in Use" signs.
c. Use nonreflective instruments.
d. All of the above.

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Chapter 5

STANDARDS OF PRACTICE

Nursing is a profession with a distinct body of knowledge and a specialized practice. Nurses are accountable for the judgments they make and the actions they take during the course of their practice. The nursing profession is based on a model of critical thinking utilized to deliver evidence-based practice that supports positive patient outcomes. Encompassing the diagnosis and treatment of human responses to both actual and potential health problems, nursing is based on a model of caring and respect for human dignity.

Standards of practice guide, define, and direct the practice of professional nursing in all practice settings. They reflect a contract between nursing practitioners and the public to deliver a safe level of care. They delineate the professional responsibilities of professional nurses in all practice settings. They are dependent on the nurse's education, experience, and role, as well as the population being served.

Nursing standards reflect the philosophical and moral values of the profession and describe the responsibilities for which practitioners are accountable. They describe the minimal levels required for competence of practice in the professional role. Nursing standards guide the practitioner and ensure that he or she does not practice beyond the limitations of educational training or expertise or the scope of delineated practice. Additionally, they recognize that nursing is a profession where patient care is individualized for each patient and his or her unique situation. Standards recognize nursing as a partnership between the practitioner, the multidisciplinary health care team, and patients and their significant others.

Standards are developed by professional nursing organizations and range from general to specialty-specific. They are authoritative statements that delineate the responsibilities of the practitioners within the profession. The standards are defined and promoted by the profession and present a medium by which the quality of practice, service, or education can be evaluated (ANA, 2015b). Professional nursing organizations have a responsibility to both their members and the general public to develop and promote universal and uniform standards of practice.

The nationally recognized professional organization of registered nursing, the **American Nurses Association (ANA)**, has developed general standards that apply to the practice of all professional nurses (ANA, 2015b). Specialty nursing organizations further tailor and define specialty-specific scope of practice statements, standards, and associated measurement criteria.

Competencies in individual specialty areas of practice may be defined by their disparate specialty scope and standards of practice documents, which are authored by specific specialty nursing associations. Many specialty nursing organizations recognize individual expertise through national certification in their specific specialty area.

The specialty organization of gastroenterology nursing is the Society of Gastroenterology Nurses and Associates, Inc. (SGNA). Gastroenterology-specific standards of practice will be discussed in this chapter. Also explored will be the regulatory agencies that oversee and set guidelines for safe professional practice.

Learning objectives

After reviewing the content of this chapter, the gastroenterology nurse will be able to:

1. Outline the expected behavior of the gastroenterology nurse as set forth in the SGNA "Standards of Clinical Nursing Practice and Role Delineations in the Gastroenterology Setting."

2. Discuss accreditation organizations and their role in health care.
3. Explain how regulations and guidelines developed by specific government agencies guide practice in the endoscopy unit.

DEFINITIONS

Standards

Standards are an acknowledged measure of comparison for quantitative or qualitative value, such as a model, criterion, or rule of professional behavior. They can be used as guidelines either in the workplace or in legal and ethical arenas. Standards are indicators of competent practice. They can be revised to reflect changing practice requirements and needs based on scientific and technological advances.

Standards of Practice

Standards of practice are authoritative statements of the duties that all registered nurses, regardless of role, population, or specialty, are expected to perform competently. They describe a competent level of nursing care as demonstrated by the critical thinking model known as the nursing process. They are not intended to specify how an outcome is to be achieved, but rather what is to be achieved. Standards of care can be defined as measurable statements that demonstrate the means by which practice outcomes are accomplished. They define criteria against which the nurse's level of performance can be measured. Standards of practice should be continually updated and validated in the practice setting.

Standards of Professional Nursing Practice

Standards of professional nursing practice consist of standards of practice and standards of professional performance.

Standards of Professional Performance

Standards of professional performance describe a competent level of behavior in the professional role. They include activities related to ethics, culturally congruent practice, communication, collaboration, leadership, education, evidence-based practice and research, quality of practice, professional practice evaluation, resource utilization, and environmental health. All registered nurses are expected to engage in professional role activities, including leadership appropriate to their education and position. Nurses are accountable for their professional actions to themselves, their health care consumers, their peers, and ultimately to society at large (ANA, 2015b). The profession of nursing is both a science and an art, requiring both judgment and skill.

Standards and guidelines in gastroenterology

Standards delineate a focused framework for practice. Gastroenterology standards are specialty-specific and adapted from guidelines issued by relevant professional associations. They must be congruent with co-existing standards and guidelines relevant to current practice.

Guidelines describe a process of patient care management. They are statements based on available scientific evidence or nationally accepted practices and address specific patient populations or phenomena.

SGNA "STANDARDS OF CLINICAL NURSING PRACTICE AND ROLE DELINEATIONS IN THE GASTROENTEROLOGY SETTING"

The SGNA gastroenterology-specific standards were first published in 1991 and have been subsequently updated by the SGNA Practice Committee and approved by the SGNA Board of Directors several times to account for changes in best practices. The standards of practice are qualitative, conceptual outcome statements that specify the expected result of the care provided. They describe what is to be accomplished, not how those outcomes are to be achieved. The criteria necessary to meet these standards should be individualized for each work setting, based on the policies, procedures, and practices of the organization, as well as the educational backgrounds and professional licensures of the setting's staff.

The SGNA "Standards of Clinical Nursing Practice and Role Delineations in the Gastroenterology Setting" focus on the collaborative efforts of a multidisciplinary team of health care professionals dedicated to promoting optimal patient outcomes during the performance of diagnostic and therapeutic gastrointestinal (GI) procedures. Interventions are directed toward health maintenance, restoration, and palliation. The standards are broad enough to include a varied array of gastroenterology staff members: advanced practice nurses, registered nurses (RN), licensed practical nurses/licensed vocational nurses (LVN/LPN), and unlicensed assistive personnel/associates.

The standards have been designed to be applicable across a broad continuum that encompasses independent, collaborative, and dependent aspects of practice. Gastroenterology nurses and associates perform some tasks independently, while others are dependent on the performance or judgment of other personnel. These tasks will differ significantly and are dependent on each state's Nurse Practice Act, professional or vocational licensure, and educational background, as well as the unique and specific policies and constraints of individual organizations.

The SGNA "Standards of Clinical Nursing Practice and Role Delineations in the Gastroenterology Setting" contains:

- "Standards of Practice for the Registered Nurse in the Gastroenterology Setting"
- "Standards of Professional Performance for the Registered Nurse in the Gastroenterology Setting"
- "Standards of Practice for the Licensed Practical/Vocational Nurse in the Gastroenterology Setting"

The standards are reviewed and updated by SGNA at regular intervals and describe a competent level of behavior in the professional role. All registered nurses are accountable for their professional actions to themselves, their patients, their peers, and ultimately, to society.

Based on ANA's "Nursing: Scope and Standards of Practice" (2015a), each SGNA standard is described below. The competencies that accompany each standard may be evidence of compliance with the corresponding standard.

"Standards of Practice for the Registered Nurse in the Gastroenterology Setting"

Standard 1. Assessment

The Registered Nurse is accountable for collecting comprehensive data pertinent to the patient's health or situation.

Competencies:

1. Prioritizes data collection based on the patient's immediate condition or anticipated needs and the relationship to the proposed intervention.
2. Collects comprehensive, pertinent data using appropriate evidence-based assessment techniques, instruments, and tools.
   a. Obtains data by interview, examination, observation, and review of health records.
   b. Obtains data related to age-appropriate assessment and culturally sensitive needs.
      Data includes, but is not limited to:
      1) Preferences, values, expressed needs, and knowledge of the health care situation.
      2) Function and significance to the gastroenterology/endoscopy patient, such as: airway patency, body image/need for privacy, current level of comfort or pain, physical limitations, communication barriers, elimination patterns, nutrition and hydration status, safety measures, self-care deficits, skin integrity/color/turgor, venous access, and the ability to swallow.
      3) Knowledge of health maintenance and practice of health promotion and disease prevention activities.
      4) Educational needs/developmental level.
      5) Previous access to and utilization of the health care system.
      6) Current diagnosis(es), medication(s), and treatment(s).
      7) Environmental, occupational, recreational, psychosocial, cultural, and spiritual information.
      8) Past medical history.
      9) Review of body systems.
3. Synthesizes available data, information, and knowledge relevant to the situation to identify patterns and variations.
4. Assesses the impact of family dynamics on the patient's health and wellness.
5. Considers the interaction between the patient, family, and health care providers, as well as the environment, for holistic data collection.
6. Documents relevant data accurately and in a manner accessible to the interprofessional team.
7. Applies ethical, legal, and privacy guidelines and policies to the collection, maintenance, use, and dissemination of data and information.

Standard 2. Diagnosis

The Registered Nurse analyzes the assessment data to determine the actual or potential nursing diagnoses or issues.

Competencies:

1. Develops the nursing diagnosis from the assessment data. Examples may include:
   a. Potential risk for bleeding related to alterations in blood clotting mechanism.
   b. Knowledge deficit related to newly diagnosed gastrointestinal disorder (e.g., Crohn's disease, cancerous polyps).
2. Validates nursing diagnoses and identifies actual or potential risks with the patient, family, and health care providers, when possible and appropriate.
3. Prioritizes diagnoses, problems, and issues based on mutually established goals to meet the needs of the patient across the health continuum.
4. Documents diagnoses, problems, and issues in a manner that facilitates the determination of the expected outcomes and plan.

Standard 3. Outcomes Identification

The Registered Nurse identifies expected outcomes for a plan individualized for the patient or situation.

Competencies:

1. Engages the patient, family, and interprofessional team in formulating expected outcomes.
2. Formulates culturally sensitive expected outcomes through collaboration with the patient by

integrating his or her culture, values, and ethical considerations with assessment findings and nursing diagnoses.
3. Applies clinical expertise and current evidence-based practice to identify health risks that can impact expected outcomes.
4. Documents expected outcomes as measurable goals with a time frame for attainment. Examples may include:
   a. *The patient will meet discharge criteria within a specified time frame.*
   b. *The patient will identify the name, dose and frequency, purpose, and potential side-effects of the medications prescribed.*
5. Develops and modifies expected outcomes according to the status of the patient and evaluation of the situation for the actual outcomes that will facilitate the continuity of care.

Standard 4. Planning

The Registered Nurse develops a plan that prescribes strategies to attain expected outcomes.

Competencies:

1. Develops an individualized, evidence-based plan of care by partnering with the patient and interprofessional team by considering examples, such as:
   a. The ability of the patient to tolerate preparation for diagnostic or therapeutic interventions.
   b. The patient's/family's readiness to learn.
2. Establishes the plan priorities with the patient, family, and interprofessional team.
3. Ensures that the plan reflects current evidence-based nursing strategies to address identified diagnoses, problems, and expected outcomes. Examples may include, but are not limited to, the following:
   a. Consideration of the patient's rights and expectations.
   b. Priorities for nursing actions and interventions.
4. Modifies the plan according to the ongoing assessment of the patient's response and other outcome indicators.
5. Documents the plan using standardized language or recognized terminology.

Standard 5. Implementation

The Registered Nurse implements the identified plan by coordinating care and providing health teaching and health promotion.

Competencies:

1. Partners with the patient to implement the plan in a safe, effective, efficient, timely, patient-centered, and equitable manner.
2. Demonstrates and employs caring behaviors to foster therapeutic relationships among the patient and interprofessional team.
3. Utilizes evidence-based interventions and strategies to achieve the mutually identified goals and outcomes specific to the needs of the patient and the diagnosis or problem. Examples may include:
   a. Ensuring patient safety.
   b. Acting as a patient advocate.
   c. Ensuring the patient's privacy and confidentiality are maintained at all times.
   d. Providing dignified, culturally sensitive care that focuses on the patient.
4. Implements interventions that comply with existing professional practice standards, regulatory agencies, accrediting bodies, and organizational policies and procedures. Examples may include:
   a. Infection prevention measures.
   b. Sedation and analgesia safety.
   c. Radiation safety.
   d. Handling of hazardous materials.
   e. Health promotion, teaching, and safety.
5. Implements the plan in a timely manner in accordance with the patient's safety goals.
6. Documents implementation and any modifications, including changes or omissions, of the identified plan.
7. Collaborates with a diverse, interprofessional team to implement a coordinated plan focused on the mutually identified expected outcomes.

Standard 6. Evaluation

The Registered Nurse evaluates progress toward attainment of goals and outcomes.

Competencies:

1. Conducts a systematic, ongoing, criterion-based evaluation of the outcomes, including elements such as:
   a. Response to diagnostic or therapeutic interventions.
   b. Level of understanding of health promotion and maintenance.
2. Collaborates with the patient and others involved in the care or situation in the evaluation process.
3. Uses ongoing assessment data to revise diagnoses, outcomes, plan, and implementation strategies as new needs are identified for the patient.
4. Shares evaluation data and conclusions with the patient/significant others and interprofessional team.
5. Documents the results of the evaluation.

"Standards of Professional Performance for the Registered Nurse in the Gastroenterology Setting"

Standard 7. Ethics

The Registered Nurse practices ethically.

Competencies:

1. Utilizes the *Guide to the code of ethics for nurses with interpretive statements: Development, interpretation, and application* (2nd ed.) (Fowler, 2015) to guide practice.
2. Maintains patient confidentiality.
3. Serves as a patient advocate by assisting the patient with self-advocacy skills and informed decision-making.
4. Delivers care in a manner that preserves and protects patient autonomy, dignity, rights, values, and beliefs.
5. Takes appropriate action in instances of unethical or inappropriate behavior.
6. Advocates for equitable patient care.
7. Maintains responsibility for nursing practice through continued professional development.
8. Contributes to the advancement of gastroenterology nursing through scholarly inquiry, professional standards development, generation of policy, and collaboration with other health care professionals.

Standard 8. Culturally Congruent Practice

The Registered Nurse practices in a manner that is congruent with cultural diversity and inclusion principles.

Competencies:

1. Demonstrates respect, equity, and empathy in actions and interactions with all patients.
2. Participates in lifelong learning to understand cultural preferences, worldview, choices, and decision-making processes of diverse patient populations.
3. Applies knowledge of variations in health beliefs, practices, and communication patterns in all nursing practice activities.
4. Uses the skills and tools that are appropriately vetted for the culture, literacy, and language of the population served.
5. Communicates with appropriate language and behaviors, including the use of medical interpreters and translators, in accordance with patient preferences.
6. Respects the patient's decisions based on age, traditions, beliefs, family influence, and stage of acculturation.

Standard 9. Communication

The Registered Nurse communicates effectively in all areas of practice.

Competencies:

1. Assesses own communication skills and effectiveness.
2. Seeks continuous improvement of communication.
3. Assesses the patient's communication ability, health literacy, resources, and preferences.
4. Conveys accurate information.
5. Contributes the nursing perspective in discussions with the interprofessional team.
6. Maintains communication with the interprofessional team to minimize risks associated with transitions in care delivery.
7. Incorporates appropriate alternative strategies to communicate effectively with patients who have visual, speech, language, or communication challenges.

Standard 10. Collaboration

The Registered Nurse collaborates with the patient, family, and others in the conduct of nursing practice.

Competencies:

1. Identifies the areas of expertise and contributions of other professionals to optimize the attainment of desired outcomes.
2. Communicates and coordinates with the patient, family, and interprofessional team regarding patient care and the nurse's role in the provision of care.
3. Participates in building consensus or resolving conflicts in the context of patient care.
4. Adheres to the standards and applicable codes of conduct to create a work environment that promotes cooperation, respect, and trust.
5. Partners with the patient, family, and others in creating a comprehensive plan of care.

Standard 11. Leadership

The Registered Nurse demonstrates leadership in the gastroenterology setting and in the profession.

Competencies:

1. Contributes to the establishment of an environment that supports and maintains respect, trust, and dignity.
2. Manages change and addresses conflict.
3. Mentors others for the advancement of gastroenterology nursing practice and the profession.
4. Promotes advancement of the profession and gastroenterology nursing practice through participation within SGNA and certification through the

American Board of Certification for Gastroenterology Nurses (ABCGN).
5. Seeks ways to advance gastroenterology nursing autonomy and accountability.
6. Demonstrates a commitment to continuous, lifelong learning for self and others.

Standard 12. Education

The Registered Nurse attains knowledge and competence that reflects current nursing practice and promotes futuristic thinking.

Competencies:

1. Acquires knowledge and skills relative to gastroenterology nursing.
2. Demonstrates a commitment to lifelong learning through self-reflection and inquiry.
3. Demonstrates accountability for maintaining competency and participates in educational activities relevant to professional issues and changing needs in gastroenterology nursing practice.
4. Shares educational findings, experiences, and ideas with peers.
5. Facilitates a work environment of role-modeling, encouraging, and mentoring.

Standard 13. Evidence-Based Practice and Research

The Registered Nurse integrates the findings of peer-reviewed, published scientific evidence and research into practice.

Competencies:

1. Utilizes current evidence-based nursing knowledge, including valid research findings, to guide practice and coordination of patient care.
2. Incorporates evidence when initiating changes in nursing practice.
3. Promotes ethical principles of research in practice and the health care setting.
4. Participates in the formulation of evidence-based practice through research by:
   a. Identifying clinical problems suitable for nursing research.
   b. Participating in data collection.
   c. Participating in a unit, organization, or community research committee or program.
   d. Sharing research activities with peers and colleagues.
   e. Conducting research.
   f. Reading and critiquing research for application to practice.
   g. Using knowledge gained from research findings in the development and revision of policies and procedures.

Standard 14. Quality of Practice

The Registered Nurse contributes to quality nursing practice.

Competencies:

1. Participates in quality improvement initiatives, which may include:
   a. Identifying barriers and opportunities to improve health care safety, effectiveness, efficiency, equitability, timeliness, and patient-centeredness.
   b. Identifying indicators used to monitor quality, safety, and effectiveness of nursing care (e.g., use of reversal agents, return to baseline status).
   c. Collecting data to monitor quality and effectiveness of nursing practice.
   d. Analyzing quality data to identify opportunities to improve nursing practice.
   e. Formulating recommendations to improve nursing practice.
   f. Implementing activities to enhance the quality of nursing practice.
   g. Collaborating with interprofessional teams to implement quality improvement plans and interventions.
   h. Developing, implementing, and evaluating policies and procedures to improve quality of practice.
2. Utilizes the results of quality improvement initiatives to change nursing practice.
3. Achieves professional certification.

Standard 15. Professional Practice Evaluation

The Registered Nurse evaluates one's own nursing practice in relation to professional practice standards, guidelines, and relevant statutes and regulations.

Competencies:

1. Engages in self-evaluation of practice on a regular basis, identifying strengths and areas for professional growth.
2. Accepts feedback regarding own practice.
3. Takes action to achieve goals identified during the evaluation process.
4. Participates in peer review.
5. Provides the evidence for practice decisions and actions as part of the evaluation process.
6. Collaborates with peers and colleagues to enhance own professional nursing practice.

Standard 16. Resource Utilization

The Registered Nurse utilizes appropriate resources to plan and provide safe, effective, and fiscally responsible nursing services.

Competencies:

1. Identifies patient care needs, potential for harm, complexity of the task, and desired outcome when considering resource allocation.
2. Delegates elements of care to appropriate team members in accordance with any applicable legal or policy parameters or principles.
3. Advocates for resources, including technology that enhance nursing practice.
4. Assists the patient and family in identifying and securing appropriate services to address needs across the health care continuum.

Standard 17. Environmental Health

The Registered Nurse practices in an environmentally safe and healthy manner.

Competencies:

1. Promotes a safe and healthy workplace and professional practice environment.
2. Attains knowledge of environmental health concepts.
3. Assesses the environment to identify risk factors.
4. Maintains a practice environment that reduces environmental health risks.
5. Communicates environmental health risks and exposure reduction strategies to patients, families, colleagues, and communities.
6. Participates in developing strategies to promote healthy communities and practice environments.

SGNA's specific standards of practice

Specific standards of practice are developed, monitored, and updated by SGNA as appropriate. Standards of practice are applied to meet the daily practice requirements of gastroenterology personnel in a multitude of practice settings. Based on standards of evidence-based practice, SGNA's standards specifically address, but are not limited to, the following:

- "Standards of Clinical Nursing Practice and Role Delineations in the Gastroenterology Setting"
- "Statement on the Use of Sedation and Analgesia in the Gastrointestinal Endoscopy Setting"
- "Standards of Infection Prevention in Reprocessing of Flexible Gastrointestinal Endoscopes"
- "Standards of Infection Prevention in the Gastroenterology Setting"
- "Guideline for Use of High-Level Disinfectants & Sterilants in the Gastroenterology Setting"

ACCREDITATION

In order for a health care organization to participate in and receive federal payment from Medicare or Medicaid programs, the health care organization is required to meet the government requirements for program participation. These requirements include a certification of compliance with the health and safety requirements called Conditions of Participation (CoPs) and Conditions for Coverage (CfCs). CoPs and CfCs are set forth by federal regulatory agencies such as the Centers for Medicare & Medicaid Services (CMS). The certification is achieved based on either a survey conducted by a state agency such as CMS on behalf of the federal government or by a national accrediting organization that has been approved by CMS as having standards and a survey process that meets or exceeds Medicare's requirements. This approval process is called "deeming."

The **accreditation** process includes a self-assessment and external peer assessment of health care organizations to accurately assess their level of performance in relation to established standards. The organizations then utilize this information to implement continuous improvement initiatives. The process of review allows health care organizations to demonstrate their ability to meet regulatory requirements and standards established by a recognized accreditation organization.

Accreditation connotes something more than fulfilling minimum conditions. Earning this recognition also provides a competitive advantage in the health care industry and strengthens community confidence in the quality and safety of care, treatment, and services. Overall, accreditation helps to improve risk management, foster risk reduction, and organize and strengthen patient safety efforts to create a culture of safety. Accreditation also may enhance recruitment and staff education and development, as well as assess all aspects of management.

Accrediting organizations

The selection of an accrediting organization is based on several considerations. When choosing an accrediting body, organizations should consider the various factors related to their individual needs and goals, including cost. Accreditation costs vary by the size of the health care organization, the complexity of the services offered, and the facets of care being examined.

Examples of accrediting organizations include, but are not limited to, the following:

- The Joint Commission
- DNV GL Healthcare
- Center for Improvement in Healthcare Quality (CIHQ)
- Accreditation Association for Hospitals/Health Systems, Inc. (AAHHS)
- Accreditation Association for Ambulatory Health Care, Inc. (AAAHC)

The Joint Commission

Founded in 1951, The Joint Commission is the nation's oldest and largest standards-setting and accrediting body in health care. It is an independent, not-for-profit organization that accredits and certifies nearly 21,000 health care organizations and programs in the United States. Joint Commission accreditation and certification is recognized nationwide as a symbol of quality that reflects an organization's commitment to meeting certain performance standards.

DNV GL Healthcare

DNV GL received CMS deeming authority in 2008. Since that time, they have accredited nearly 500 hospitals of all sizes and in every region of the United States. It is the first and only accreditation program to integrate the CMS Conditions of Participation (CoPs) with the International Standards Organization (ISO) 9001 Quality Management Program.

DNV GL is a merger of a risk management company and a classification society, now dedicated to safeguarding life, property, and the environment.

Center for Improvement in Healthcare Quality (CIHQ)

CIHQ is the nation's newest accreditation provider approved by CMS to deem acute care hospitals as meeting Medicare CoPs. Its program strives to ensure that patients receive care in organizations that exceed minimum standards set forth by the federal government. To date, there are almost 50 hospitals accredited by CIHQ.

Accreditation Association for Hospitals/Health Systems, Inc. (AAHHS)

AAHHS seeks to help health care organizations better serve the community through accreditation, education, and research. AAHHS assumed management of HFAP (Health Facilities Accreditation Program) from the American Osteopathic Association (AOA) in 2015.

Accreditation Association for Ambulatory Health Care, Inc. (AAAHC)

AAAHC is a private, non-profit organization formed in 1979. It is the leader in developing standards to advance and promote patient safety, quality care, and value for ambulatory health care through peer-based accreditation processes, education, and research. AAAHC currently accredits more than 6,000 organizations in a wide variety of ambulatory health care settings. With a single focus on the ambulatory care community, AAAHC offers organizations a cost-effective, flexible, and collaborative approach to accreditation.

FEDERAL REGULATIONS AND GUIDELINES

Several governmental agencies and federal regulations impact to varying degrees the safe delivery of nursing care in the gastroenterology department.

Centers for Disease Control and Prevention (CDC)

The CDC is the nation's leading public health agency, dedicated to saving lives and protecting the health of Americans. One of the eleven major operating divisions of the U.S. Department of Health and Human Services (HHS), the focus of the CDC is the prevention, identification, and control of infectious and chronic diseases; workplace hazards and injuries; and environmental health threats. It is globally recognized for its action-oriented approach and brings a research-driven response to health emergencies. The CDC helps save lives by responding to emergencies, providing expertise, developing vaccines, and detecting disease outbreaks wherever they arise.

The CDC formulated general guidelines for the prevention and control of hospital-acquired infections. Most applicable of these for patient care in the endoscopy unit is the "Guideline for Hand Hygiene in HealthCare Settings," which covers handwashing; cleaning, disinfecting, and sterilizing patient care equipment; microbiological sampling; managing infective waste; housekeeping; and laundering (CDC, 2002).

CDC guidelines have no force of regulation or law and may be modified as appropriate in individual health care settings. However, they represent the best available compilation of practical, well-founded infection control practices in the nation. Their recommendations are based on well-documented epidemiological studies and theoretical rationales.

U.S. Environmental Protection Agency (EPA)

The mission of the EPA is to protect both human health and the environment. Composed of engineers, scientists, policy analysts, and others, this federal agency develops and enforces regulations that implement environmental laws enacted by the United States Congress. The EPA is responsible for maintaining environmental quality through research, education, and assessment.

The EPA is the federal agency that approves the products used for varying levels of disinfection. Registration is awarded to a product after all labeling information and supportive data are submitted for approval and the product is thoroughly investigated for efficacy, safety, and compliance with existing standards. The EPA classifies a spectrum of products that includes chemical germicides, sporicides, general disinfectants, hospital disinfectants, and sanitizers, among others. High-level disinfection agents and protocols in the endoscopy suite are under the purview of the EPA.

Another function of the EPA is to regulate the disposal of solid waste. This includes hazardous waste such as toxic, ignitable, corrosive, reactive, and biohazardous waste. In November 1988, this agency was commissioned to establish a demonstration program for tracking medical waste after federal regulations were enacted. Specific guidelines were developed that address the handling of infectious waste. These guidelines include the designation of waste; the handling, storage, packaging, and transportation of waste; the selection of appropriate treatment and disposal methods; the monitoring of treatment methods; and the oversight and enforcement of compliance with federal, state, and local requirements.

U.S. Food and Drug Administration (FDA)

The FDA is the federal regulatory agency responsible for protecting the public health by ensuring that foods are safe, wholesome, sanitary, and properly labeled and that human and veterinary drugs, vaccines, and other biological products and medical devices intended for human use are safe and effective. The FDA does not regulate meat from livestock, poultry, and some egg products. These are regulated by the U.S. Department of Agriculture (USDA).

The complexity of medical devices and the increasing number of new products resulted in Congress adding the comprehensive "Medical Device Amendments of 1976" to the "Federal Food, Drug, and Cosmetic Act." The primary reason for the implementation of these amendments was to ensure that any new device developed and put into widespread use was safe and effective prior to market release. Congress divided medical devices into subsets. These categories were based on the potential hazards of the device. Criteria for device evaluation were established based on six basic categories: preamendment, postamendment, substantially equivalent, implant, custom, and investigational.

Essentially, there are two ways in which a new device reaches the market.

- Premarket notification. If the manufacturer can establish substantial equivalence, premarket notification is all that is required.
- Premarket Approval Application (PMA). If the new device is not similar to a preamendment device, a PMA must be filed.

Most manufacturers file a "premarket notification," which is referred to as a 510(K), before introducing a new product. This is a request by the manufacturer to the FDA to market a device. If the FDA does not approve, the only alternative is the lengthy process of submitting a PMA.

In some situations, the manufacturer may need to conduct clinical studies to support a PMA or 510(K) submission. If this is the case, the device may need to be distributed and used for investigational purposes. An investigation device exemption (IDE) is obtained to ensure that the preclinical testing makes use of a predetermined protocol.

The FDA has also established a reporting system whereby health care practitioners must report problems with medical devices. The "Safe Medical Devices Act (SMDA)" of 1990 requires health care organizations to submit a report twice a year regarding device-related injuries and serious illnesses. Also, the malfunction of a device while in use during patient care must be reported to the FDA. Based on the user reports, the FDA has the authority to recall devices.

Occupational Safety and Health Administration (OSHA)

The OSHA is the federal regulatory agency responsible for enforcing safety and health regulations in the workplace. The primary function of OSHA is to protect workers by making sure that employers comply with the health and safety provisions of federal laws. Workplaces are subject to inspection at any time at the request of an employee.

OSHA's "Bloodborne Pathogens Standard" (29 CFR 1910.1030) and the 2000 "Needlestick Safety and Prevention Act" outline safeguards to protect workers against the health hazards caused by bloodborne pathogens. The standard addresses items such as exposure control plans, universal precautions, engineering and work practice controls, personal protective equipment, housekeeping, laboratories, hepatitis B vaccination, post-exposure follow-up, hazard communication and training, and recordkeeping. The standard places requirements on employers whose workers can be reasonably anticipated to come in contact with blood or other potentially infectious materials (OPIM), such as unfixed human tissues and certain body fluids.

Health Insurance Portability and Accountability Act (HIPAA)

With its "Privacy Rule" and "Security Rule" standards, this federal legislation from 1996 impacts every aspect of the health care industry. These regulations are considered the most significant legislative changes made to the health care industry since the advent of Medicare diagnosis-related groups (DRGs).

The HHS Office for Civil Rights provides oversight and enforcement of several aspects of this legislation, including the protection of the privacy of an individual's personal health information, both written and electronic. Focusing on patient confidentiality and privacy, the rule allows patients greater control over their health information and dissemination. Protected health information (PHI) is defined as individually identifiable health information that is oral or maintained in any form or medium. Consent must be obtained from the patient before anything can be done with his or her personal health information.

Only health care professionals with a "need to know" status should have access to a patient's personal health information. All communication must be safeguarded for privacy. Every person working in a health care facility is responsible for the protection of a patient's personal health information. Providers must be continually on the alert to eliminate areas where patient confidentiality might be breached and institute steps to prevent the inadvertent dissemination of private information. These measures include shutting doors, pulling curtains in semi-private areas, and taking patients to private areas where discussions can be held without the risk of other people hearing or obtaining private information.

Endoscopy units must review their operational guidelines and policies on a regular basis to ensure all patients' personal and health information is secured and protected. In the endoscopy unit, this includes all verbal communications with patients, verbal communications with others about patients, written and electronic patient records, and telephone and e-mail correspondence to or about a patient.

CASE SITUATION

Dawn Stephens, a 56-year-old woman, has just arrived on the endoscopy unit for her scheduled colonoscopy. As the admitting nurse directs her to the changing area, Dawn relates some of her concerns and anxieties. She has had rectal bleeding for two months and is concerned about what the doctor will find when he performs the test. Preferring to believe that "what you don't know won't hurt you," she has never had the screening colonoscopy her internist had been reminding her was long overdue. Having read about colon cancer on various social media sources, Dawn is now concerned that she might have been foolish to wait and is afraid that she might have cancer. She hasn't shared her fears with her husband of 30 years, Tom, as she doesn't want to worry him until a diagnosis is definitively confirmed.

Points to think about

1. What activities might an endoscopy nurse perform on behalf of Dawn to ensure her right to privacy and confidentiality in the endoscopy unit?
2. In preparing for Dawn's colonoscopy, what types of activities might be conducted to meet the standard of "preventing harm to the patient?"
3. What information related to pain should the nurse document that would demonstrate nursing actions and the patient's responses to those actions?

Suggested Responses

1. To ensure Dawn's right to privacy and confidentiality, the endoscopy nurse might:
   a. Establish mutually with the patient what information she would like discussed in front of her family.
   b. Make sure that any information discussed with other staff and the patient's physician is communicated discreetly and in a professional manner so it is not overheard by the patient's family or by other patients and their families.
   c. When preparing for the colonoscopy, take care to keep the patient covered with an appropriate drape.
2. Types of activities that might be performed to protect the patient from harm would include:
   a. Decontaminating and disinfecting endoscopic instruments.
   b. Ensuring that informed consent has been obtained before the procedure.
   c. Performing a safety check of all electrical equipment that has the potential to burn the patient or cause fires.
3. An example of a nursing action and patient response related to pain that the nurse would document is:
   a. Nursing action: Administration of pain medication.
   b. Response: The patient is relaxed and cooperative.

REVIEW TERMS

American Nurses Association (ANA), Centers for Disease Control and Prevention (CDC), Health Insurance Portability and Accountability Act (HIPAA), Occupational Safety and Health Administration (OSHA), standards of care, standards of practice, U.S. Environmental Protection Agency (EPA), U.S. Food and Drug Administration (FDA)

REVIEW QUESTIONS

1. Selecting an accrediting organization might be based on, but not limited to, the following considerations:
   a. Costs.
   b. Complexity of the services offered.
   c. Facets of care being examined.
   d. All of the above.
2. Measurement of comparison for quantitative or qualitative value is called a:
   a. Standard.
   b. Guideline.
   c. Standard of practice.
   d. Criteria.
3. Standards of professional nursing practice consist of which of the following:
   a. Standards of professional performance.
   b. Standards of practice.
   c. State nurse practice acts.
   d. Both a & b.

4. Which of the following is not listed as a standard of practice?
   a. Assessment.
   b. Planning.
   c. Communication.
   d. Evaluation.
5. Which federal agency established the “Safe Medical Devices Act?”
   a. OSHA.
   b. EPA.
   c. FDA.
   d. CDC.
6. The CDC guidelines for the prevention and control of hospital-acquired infections:
   a. Have the force of law.
   b. Are an accepted compilation of practical, well-founded recommendations.
   c. Are applicable only to hospital settings.
   d. Are all based on conclusive scientific research.
7. Standard ____________ speaks to the Registered Nurse developing an individualized, evidence-based plan of care by partnering with the patient and interprofessional team.
   a. Standard 4: Planning.
   b. Standard 5: Implementation.
   c. Standard 9: Communication.
   d. Standard 16. Resource Utilization.
8. The Standard of Professional Performance 8: Culturally Congruent Practice includes all of the following, except:
   a. Demonstrates professional accountability and responsibility for nursing practice.
   b. Demonstrates respect, equity, and empathy with all health care consumers.
   c. Creates inventory of one’s own values and beliefs.
   d. Considers the effects of discrimination and oppression on practice within and among vulnerable groups.
9. Name the federal regulatory agency responsible for enforcing safety and health regulations in the workplace:
   a. OSHA.
   b. EPA.
   c. CDC.
   d. DNV GL.
10. Which accrediting body has a single focus on the ambulatory care community?
   a. The Joint Commission.
   b. AAHHS.
   c. HFAP.
   d. AAAHC.

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Chapter 6

PROCESS IMPROVEMENT

Transforming the commitment to quality care from an idea into a reality is one of the challenges health care professionals face. The emphasis on quality is driven by customer demand for health care that is appropriate, timely, safe, efficient, and cost-effective. Customers also expect services to be provided in a caring and respectful manner. This chapter provides guidelines for integrating a process improvement plan into the gastroenterology unit to ensure that individual nurses and their facilities are meeting customer expectations and promoting positive health care outcomes.

Learning objectives

After reviewing the content of this chapter, the gastroenterology nurse should be able to:

1. Define the term *quality* as it pertains to patients undergoing gastroenterology procedures.
2. List the steps to develop a process improvement plan.
3. Identify the participants responsible for a continuous quality initiative.
4. Apply continuous quality improvement principles to assess a practice problem.
5. Identify indicators for measuring process improvement for gastroenterology patients.
6. Formulate a plan for improving quality within the gastroenterology unit.
7. Discuss the importance of communication in driving engagement and change.

RATIONALE FOR A QUALITY PROGRAM

The quality of health care provided to consumers has been a focus of health care organizations for many years. Much of this interest is in response to dramatic transformations toward health care systems and reimbursement. Hospitals and policies continue to focus on evidence-based medicine, while insurance providers emphasize cost-effective, value-based health care in relation to producing positive health outcomes. Both of these major initiatives have a strong focus on quality outcomes. Dramatic transformations within the health care industry, such as reimbursement on quality care outcomes, require organizations to reassess their focus and priorities (Mainz, 2003).

A case in point is that of the Premier Hospital Quality Incentive Demonstration (HQID) project of the Centers for Medicare & Medicaid Services, which links both evidence-based medicine and cost-effective health care (Centers for Medicare and Medicaid Services [CMS], 2013). This project raised overall quality initiatives by almost 16 percent over a three-year period by incorporating 30 standards and widely accepted care measures that were based on scientifically informed guidelines (Swensen et al., 2010). Health care professionals continue to seek new ways to improve patient outcomes that ultimately reduce mortality and morbidity, hospital-acquired infections, and costs, while concurrently improving functional status, patients' experiences, and access to health care (Swensen et al., 2010).

The health care industry has since moved from an environment of quality assurance to a model of continuous **process improvement**. The rationale for this change is that quality assurance is a retrospective process, while process improvement is a prospective or concurrent process. Process improvement is a methodology designed to allow for scientific analysis of current workflow. Although there are several models of process improvement utilized in health care organizations, they all have the same basic principles:

a. Most problems are process issues rather than people issues.
b. The people closest to the process are the experts who have an interest in improving the process.
c. Decisions should be made based on measurable data.
d. Changes should be evaluated for effectiveness.

Below are some examples of process improvement models.

- **Plan-Do-Study-Act (PDSA)** is a best-practice model of continuous improvement that focuses on promoting learning from experiences through repetitive and systematic reflection. The model's various steps include forming a team; setting aims; establishing measures; and selecting, testing on a small scale, implementing, and spreading changes.
- **Six Sigma** is a quality improvement method that uses a defined series of steps to improve quality by identifying and eliminating errors, while minimizing variability for complex problems within large organizations.
- **DMAIC** is an acronym for define, measure, analyze, improve, and control. Specifically, the model uses these steps to define the problem, measure baseline performance, analyze root causes of clinical dilemma, improve by implementing change, and control and ensure that gains from change are sustained. It is a prescribed improvement process for Six Sigma methodology that focuses on learning more about the root causes of a problem through structured analysis and prudent use of data, without prematurely applying solutions to the complex problem.
- **Lean management** is a continuous process improvement model focused on eliminating and reducing waste with the goal of transforming organizational behaviors and culture through incremental changes and, as a result, improving efficiency and quality. Its key principles are to identify value, map the value stream, create flow, establish pull, and seek perfection.
- **Positive deviance** is a tool for improving processes based on the premise that individuals involved in the process are able to find the best solution.
- **Standards for Quality Improvement Reporting Excellence (SQUIRE)** is a model that follows the principle that improvement is an applied science rather than an academic discipline. Its immediate purpose is to change human performance rather than generate new knowledge. It is driven mostly by experiential learning.
- **Chronic Care Model (CCM)** is a process that synthesizes evidence-based system changes that are known to improve outcomes with an emphasis on six areas: organization of health care, community linkages, self-management support, delivery system design, decision support, and information systems.

Quality was once considered unquantifiable and, therefore, not measurable. In *Service Quality Improvement: The Customer Satisfaction Strategy for Health Care,* Leeboy and Scotts (1994) defined quality as conforming to customer expectations on a consistent basis. These needs and expectations can be precisely measured with appropriate and relevant **indicators**. Quality improvement can also be defined as the combined and unceasing effort to make changes that will lead to better patient outcomes (health), system performance (care), and professional development.

Indicators are quantitative measures that assess, monitor, evaluate, and improve the quality of patient care, clinical support services, and organizational functions that affect patient outcomes or health care processes. Indicators provide a measurement tool for clinicians and administrators to achieve improvements in patient care. Clinical indicators are expressed as numbers, rates, or averages and measure the extent to which predetermined targets have been met (Mainz, 2003). Evidence-based clinical indicators are capable of predicting and measuring patient outcomes and are considered true measures of quality. When evidence-based research is lacking, indicators are chosen through professional consensus and aid in decision-making and process improvement planning. According to Mainz (2003), indicators make it possible to:

- Document the quality of care.
- Make comparisons through benchmarking.
- Make decisions based on data.
- Set priorities.
- Support accountability, regulation, and accreditation.
- Support quality improvement.

Process improvement involves more than care of the patient: It encompasses all of the work performed in and around the specific department or office setting. Each department needs to identify and respond to all of its customers. Customers can be physicians, patients, visitors, other hospital departments such as radiology and pathology, insurance providers, and staff within the department. To meet the changing expectations and needs of these customers, the health care industry must change and acclimate as technology improves and consumers become more educated. The result of this cyclic process is a change in health care processes.

The Joint Commission (2017c) has provided definitions of process (a collaborative and interdisciplinary series of interrelated and goal-directed actions) and process improvement (a systematic method of process design, performance measurement, and assessment/improvement across an entire institution).

In 1998, The Joint Commission's ORYX initiative was the first national program for the measurement of hospital quality. In its early development, it only required the reporting of non-standardized data on performance measures (The Joint Commission, 2017b). In 2002, hospitals accredited by The Joint Commission were required to collect and report data on performance for at least two of the four core measure sets (acute myocardial infarction, heart failure, pneumonia, and pregnancy), and these data were made public by The Joint Commission in 2004 (Chassin, Loeb, Schmaltz, & Wachter, 2010).

In 2006, The Joint Commission launched a Standards Improvement Initiative (SII) with the goals of:

- Clarifying standards language.
- Ensuring that the standards are program-specific.
- Deleting redundant or nonessential standards.
- Consolidating like standards.

In 2010, The Joint Commission categorized its process core measures into accountability and non-accountability measures, placing greater emphasis on accountability measures. Beginning in 2012, a minimum performance requirement was incorporated into the standards for the ORYX accountability measures, which called for Requirements for Improvement (RFI) when measures did not meet the requirements.

INTEGRATING QUALITY INTO THE GASTROENTEROLOGY DEPARTMENT

To implement any process improvement plan, the management team must be educated and committed in the principles of the process. Staff must then be trained in the principles of **continuous process improvement** and be empowered to make changes in how they do their work. Staff members need to be engaged in the process and challenge the status quo. Over time, some processes may break down, and customer satisfaction may decrease. Even effective processes should be examined for further improvement.

Before a process improvement plan is developed, the gastroenterology unit's mission (objectives) and vision (goals) must be in place. It is from the mission and vision statements that quality initiatives are construed. If the unit is hospital-based, it is important that the mission and vision statements reflect the global vision and mission of the organization. The statement must be reviewed regularly and updated as needed to account for changes in facility management and patient expectations.

A mission statement for a gastroenterology unit might be:

The gastroenterology unit will provide quality diagnostic and interventional therapeutic care to all individuals utilizing the unit.

The vision statement for a gastroenterology unit might be:

To be the most cost-effective, efficient, state-of-the-art environment for all persons utilizing the unit.

MEASURING QUALITY

The gastroenterology department's mission and vision statements guide the unit in planning a process change. The challenge is to translate these broad statements into measurable indicators. Providing a solid foundation for a comprehensive process improvement plan requires a working knowledge of the process and the tools necessary to achieve the goal. These include:

a. Flowcharting the process through process mapping or value stream mapping (Environmental Protection Agency [EPA], 2017).
b. Establishing work teams with defined roles.
c. Collecting and interpreting data.

When developing the plan for a process improvement initiative, there are several criteria to consider. First, the plan should be kept simple. For example, if lunch relief for staff is an area of desired improvement for the unit, convene a representative group from the staff and establish the group leader. Select a staff member to take the meeting minutes and be the timekeeper to keep the meeting on track. The group then makes a list of reasons why it is difficult to get lunch. After the list is created, similar reasons are grouped together to identify the underlying issue. The reasons are then analyzed and discussed. The end objective is the synthesis of ideas toward the formulation of a solution. When staff members address the issue and are involved in the development of the improvement plan, the likelihood of compliance with the solution is much greater. The next step is to decide when and how the solution will be evaluated to measure its efficacy.

Quality indicators and thresholds

A clinical indicator is a measurable variable used to assess the degree to which outcomes or expectations are met. The identification of indicators for the gastroenterology unit is a complex process that requires a high level of clinical expertise. Those responsible must develop suitable indicators so that important aspects of care are measured and evaluated effectively.

According to Mainz (2003), the following are criteria for developing effective quality indicators:

- Objectivity.
- Specificity.
- Measurability.
- Comprehensiveness.
- Validity.
- Reliability.
- Relevancy.
- Efficiency.

Indicators are used to measure standards, of which there are three types: structure, process, and outcome.

- **Structure indicators** denote the characteristics of the health care setting where patient care occurs and describe the health care system's ability to meet the health needs of its patients. Material, human, and financial resources, as well as organizational structure, are examples of structure indicators.
- **Process indicators** assess what the provider did for the patient and how well it was done. They denote what transpired in the process of giving and receiv-

ing care. In other words, process indicators measure the activities and tasks in episodes of care.

- **Outcome indicators** describe the effects of care on the health status of the patient(s) or patient populations. Outcome measures sometimes need to be risk-adjusted for factors operating outside the health system in order for fair comparisons to be made. Examples of factors included in risk adjustments are patient demographic variables, psychosocial characteristics, severity of illness, health status, and comorbid conditions (Mainz, 2003).

The following are some examples of indicators that may be appropriate for the gastroenterology unit:

- Proper cleaning and disinfection of equipment (structure).
- Emergency equipment available and functional for all procedures (structure).
- Qualifications of gastroenterology nurses in the procedure suites (structure).
- Number of incomplete procedures because of poor bowel prep (process).
- Number of patients who do not have necessary lab work pre-procedure (process).
- Patient or family who rate satisfaction with nursing care as 80 percent or above (outcome).
- Patients having sclerotherapy who have blood pressure and pulse within the range of plus or minus 5 percent of their baseline value, if available, before discharge (outcome).

Indicators must be stated in a concise manner and be easy to locate to facilitate the data collection. For example, for patients having sclerotherapy, it would be simple and convenient to check the patient's chart for pre-procedure and post-procedure vital signs to see if the last outcome standard noted above was met. Indicators must be easy to use, or they will probably not be used at all.

Indicators ideally should be:

- Based on agreed upon definition.
- Specific and sensitive (i.e., detects few false positives or false negatives).
- Valid and reliable.
- Able to permit useful comparisons.
- Supported by evidence-based literature (Mainz, 2003).

Indicators can be designated as either rate-based indicators or sentinel-event indicators. Once indicators are developed, thresholds or acceptable rates of activity need to be established that determine when to evaluate care.

Rate-based indicators

Rate-based indicators use data about events expected to occur with some frequency and are expressed as proportions or rates where, for example, the numerator represents the population at risk for an event and the denominator is the period of time over which the event takes place (Mainz, 2003). Rate-based indicators are established on the premise that there are usually exceptions to standards of care. Acceptable thresholds are established for clinical events based on the literature, expert opinion, and/or previous hospital experience.

Sentinel-event indicators

Sentinel-event indicators identify individual events that are intrinsically undesirable, always trigger further analysis and investigation, and represent the extreme of poor performance (Mainz, 2003). Sentinel-event indicators represent standards of care that, if not followed, may result in serious outcomes. An example might be a negative reaction to transfusion or perforation of the bowel during a procedure. Typically, established sentinel-event indicators are managed by a quality committee.

Monitoring quality

In The Joint Commission E-dition (2017b), the chapter "Performance Improvement" requires that a hospital has a planned, systematic, hospital-wide approach to process design and performance measurement, assessment, and improvement. These activities need to be collaborative and interdisciplinary. The standards state that the facility is required to collect and analyze data to monitor and take action to improve performance.

Regarding the collection of data to monitor performance, organizational leaders should consider examples of The Joint Commission requirements for performance improvement.

- Set priorities for data collection.
- Identify the frequency for data collection.
- Collect data on performance improvement priorities identified by leaders (e.g., adverse events related to using moderate sedation; significant medication errors; patient perception of safety; and quality of care, treatment, or services).

Regarding the compilation and analysis of data, organizational leaders must:

- Use statistical tools and techniques.
- Compare internal data over time to identify levels of performance, trends, and variations.
- Use results of data analysis to identify improvement opportunities.

Regarding improving performance within the organization, organizational leaders must:

- Take action on improvement priorities.
- Take action when planned improvements are not achieved or sustained.

Organizations should have a written plan for the process improvement program that describes the program's objectives, organization, scope, and mechanisms for overseeing the effectiveness of monitoring, evaluating, and problem-solving activities. However, simply having the goals, objectives, standards, and indicators in written form is not enough. There must also be a formal process for monitoring the process improvement, and the entire staff should be involved in this process because it gives insights into measuring quality and empowers staff.

In *Achieving Impressive Customer Service: 7 Strategies for the Healthcare Manager*, Leebov, Scott, and Olson (2012) emphasize the benefit customer satisfaction surveys have for an organization. In one example of a process improvement plan, nurses were assigned to ask satisfaction-related questions to three patients or family members each week. Each nurse was able to receive immediate feedback on his or her unit's work. These results were then added together with all the other nurses' reports to provide an overall picture of the unit's processes. This knowledge is empowering for the gastroenterology nurse.

Hospital-based gastroenterology units usually have a quality improvement program in place with a coordinator to facilitate the process. Free-standing gastroenterology units should delegate the responsibility of monitoring and evaluating the quality of care delivered to a staff member. The Joint Commission's *2018 Comprehensive Accreditation Manual for Ambulatory Care (CAMAC)* (The Joint Commission, 2017a) provides a guideline for this function.

Monitoring the quality of care delivered requires data collection, tabulation, evaluation, and reporting. Data collection specifications must detail the indicators to be used, how much data to collect, and the frequency of the data collection task. It not always necessary to create new forms or tools if existing forms, such as patient records, incident reports, patient satisfaction questionnaires, and infection rate reports, are current and appropriate for data collection. Some data only needs to be reported internally within the department, organization, or organization system. Other data are mandated to be reported to the public. In general, the more transparent the reporting mechanism, the greater the value of the process. "Transparency is fundamental to the integrity and spread of high-value care" (Swensen et al., 2010, p. 3).

The frequency with which information is collected should be based on the volume of patients and recurrence of a problem. Data should be gathered, reviewed, and reported regularly. Processes may be reviewed monthly, quarterly, semi-annually, or annually, depending on the degree of importance to the unit. Any indicators exceeding the established threshold require investigation of the quality and appropriateness of care.

Once information has been collected, it must be tabulated so that it can be evaluated and compared to acceptable variance levels, which should have been determined in advance. An example in the gastroenterology unit is that all patients' allergies must be verified prior to the delivery of any medication. The variance in this case would be zero; the expectation is that this standard is met 100 percent of the time.

Evaluation is a two-part process that involves determining (1.) the degree to which the key quality elements were met and (2.) whether the care being delivered is still considered evidence-based by current literature.

The final step is to report the findings. This may require developing a form or tool to serve as the tracking mechanism or adapting an existing form. Using control charts that depict trends or patterns of performance can be very helpful.

In summary, to measure the quality of care provided, it is necessary to establish the following components:

- Written indicators and thresholds.
- Tools for data collection and tabulation.
- Predetermined acceptable levels of variance for evaluating each indicator.
- A means for reporting the findings.

Following these steps provides a means of measuring and documenting the process improvement plan as recommended by The Joint Commission.

IMPROVING QUALITY

Once data have been collected and evaluated, the next step is to identify the area(s) that need improvement. Following is a brief list of strategies for implementing change and improvement. Effective methods for improving quality always start with a commitment to making changes and providing staff with frequent updates on current process improvement activities.

STRATEGIES FOR IMPLEMENTING CHANGE AND IMPROVEMENT
Education and training
Education is essential when implementing change. Staff cannot be expected to understand the need for change, or the effect of a change on their role, without adequate knowledge. Training in process improvement should be incorporated into job orientation and also be a part of the annual review process.
Team approach
The entire unit needs to be involved. A basic tenet of process improvement is that those people closest to the issue are the ones who understand it best and must be involved in the development of the solution. The Joint Commission advises that, when a process involves more than one department, the group improving the process must reflect that cross-departmental activity.
Easily attainable
Start with an easily attainable, timely process improvement.

Once strategies for the process improvement have been established, they must be written, communicated to all parties, and then monitored. The effectiveness of the process improvement relies on a clearly written plan of action, which The Joint Commission has made a requirement. The written plan for hospital-based gastroenterology units should also include the overall hospital-wide process improvement plan. Once the written plan is developed, it must be evaluated on a regular basis to ensure that the changing needs, expectations, and outcomes of customers are being met in a timely, effective manner.

FUTURE ENDEAVORS

In keeping with the theme of continuous process improvement, there is always room for improvement. Current measures should be under constant review so that poorly performing measures can be adapted or eliminated and replaced with better quality measures. Chassin, Loeb, Schmaltz, and Wachter (2010) cite the following criteria to use in this effort:

- Employing measures that are based on a strong foundation of research and show the process addressed by the measure.
- Capturing accurately whether the evidence-based care has been delivered (e.g., the measurement strategy is not a check mark in a box on a form).
- Addressing a process that is as close to the desired outcome as possible with very few intervening processes.
- Having minimal or no unintended adverse consequences for the measure.

Chassin et al., (2010) believe measures that meet all of these criteria will be more likely to result in improved patient outcomes. They propose that only measures meeting all four of these criteria should be used for accreditation, public reporting, or pay-for-performance. These authors have found that only 22 of the 28 core measures of The Joint Commission 2010 standards aligned with CMS met all four criteria of *accountability measures*. The criteria is rigorous and does not minimize the importance of other accountability measures used internally. The goal for all types of continuous process improvement tools is to produce the necessary results to provide efficient, cost-effective, quality processes for patients, physicians, and other customers.

CASE SITUATION

Kelsey Smith presents in the emergency room at 8:00 p.m. with food lodged in her esophagus. The gastroenterologist is called at midnight after a barium swallow has been attempted, leaving the esophagus full of barium. The endoscopy call team is contacted shortly after midnight by the night supervisor to come to the hospital for removal of a foreign body. The team finishes the procedure at 2:30 a.m.

Points to think about

1. How often does this situation happen?
2. Is there a hospital policy, protocol, or process to determine when a gastroenterology consult is appropriate?
3. Why was a barium swallow done before calling the gastroenterologist?
4. Does the barium in the patient's esophagus put the patient at increased risks during the endoscopic procedure?
5. What does the hospital policy or evidence-based literature state is the gold standard of care for a patient with a food impaction?

Suggested responses

To address the above questions, apply process improvement principles, such as the following:

1. Keep a logbook with all procedures involving the gastroenterology unit call team to track events for patients presenting in the emergency room.
2. Document the number and types of gastroenterology procedures that are being performed after normal gastroenterology unit working hours or on weekends.
3. Flowchart the process from the time the patient presents in the emergency room until the gastroenterologist is called. Possibilities include:
   a. Actions taken by the emergency room physicians before calling the gastroenterologist.
   b. Length of time the patient spends in the emergency room before seeing a gastroenterologist.
4. Form a team to look at the data. Include physician and nurse representatives from the emergency room and the gastroenterology unit.
5. Develop a mutually agreed upon protocol for patients presenting with food lodged in the esophagus.
6. Develop objectives or goals.
7. Monitor results for six months.
8. Evaluate findings.
9. Initiate improvements, if needed, and continue to monitor and evaluate the process until objectives are consistently met.

REVIEW TERMS

DMAIC, indicators, lean management, outcome indicators, Plan-Do-Study-Act (PDSA), process indicators, quality improvement, sentinel events, Six Sigma, structure indicators

REVIEW QUESTIONS

1. Leaders in health care have defined quality as:
   a. Indefinable and not measurable.
   b. A continuous conformance to meet the needs and expectations of the customers.
   c. The degree to which actions maximize the probability of having satisfied customers.
   d. A distinguishable attribute.

2. Models of process improvement must be based on which of the following principles?
   a. Most problems are related to process rather than people.
   b. The people closest to the process should be involved in improving it.
   c. Decisions should be made on measurable data.
   d. All of the above
3. For indicators to be effective, they must be:
   a. Subjective.
   b. Vague.
   c. Complex.
   d. Measureable.
4. Improving quality in the patient care setting should be based on:
   a. Correcting the immediate problem.
   b. Finding the person who caused the problem.
   c. Regulatory rules alone.
   d. Improving the process.
5. A quality improvement program should include:
   a. Written, measurable indicators and thresholds.
   b. Tools for data collection and tabulation.
   c. Means of evaluating the plan.
   d. All of the above.
6. True or False: Once an improvement plan is in place, it does not need to be reevaluated unless a problem occurs.
7. One effective strategy for improving the quality of care is to:
   a. Provide education and training.
   b. Assign the manager complete responsibility for making changes.
   c. Discipline those staff members who are responsible for failure to meet objectives.
   d. Change the goals and objectives.
8. The mission statement of the unit should be:
   a. Independent of the organization's statement.
   b. Formulated before a process improvement plan can be developed.
   c. Reflective of the overall philosophy of the patients.
   d. Written as the final step in developing a process improvement plan.
9. A best-practice model of continuous improvement that focuses on promoting learning from experiences through repetitive and systematic reflection is:
   a. Six Sigma.
   b. Positive deviance.
   c. Plan-Do-Study-Act.
   d. Chronic Care Model.
10. Quantitative measures that assess, monitor, evaluate, and improve the quality of patient care, clinical support services, and organizational functions affecting patient outcomes or health care processes are:
    a. Indicators.
    b. Transparency.
    c. Brainstorming.
    d. Validity and reliability.
11. An example of a structure indicator in a gastroenterology unit is:
    a. Ensure the electrocautery machine is present for the insertion of a PEG tube.
    b. Assess patient satisfaction regarding services rendered.
    c. Tabulate the monthly case report.
    d. Determine the number of patients who had a delayed discharge related to blood pressure problems.

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Chapter 7

THE RESEARCH PROCESS

THE RESEARCH PROCESS AND INTEGRATING THE EVIDENCE INTO NURSING PRACTICE: AN INTRODUCTION

This is an exciting time for the nursing profession. As nurses embrace the movement toward integrating findings gleaned from nursing research into practice, nurses can make better clinical decisions that ultimately improve the quality of patient care.

The impetus for conducting nursing research is to generate new knowledge. It is accomplished through the **Research Process** and involves an organized method of identifying a research question, hypothesis, or phenomenon or topic of interest; exploring the current literature to determine what is known about the topic; designing the study; measuring the variables; analyzing the data; and presenting the results (Polit & Beck, 2017). This chapter provides an overview of the steps involved in the nursing research process for both **quantitative research** and **qualitative research** studies. Measurements and basic statistics strategies used in nursing research will be discussed. In addition, this chapter ends with a case study in which nursing research is conducted in a clinical setting and integrated into clinical practice.

The process of conducting nursing research is of great significance for the nursing discipline and the focus of this chapter. However, it is also important to understand the differences between the research process and evidence-based practice (EBP). The *evidence-based practice* approach involves the methodical integration of relevant information that embodies current **best clinical practices found in the literature**, as well as other key points that include patient values, health care costs, care delivery, and clinical expertise (Grey, Grove, & Sutherland, 2017). The EBP process starts with the identification of a question or practice problem, using a PICOT format: Patient (group or population), Intervention, Comparison, Outcomes, and Timeframe. PICOT helps to identify specific search terms, keywords, or phrases that will be used to conduct the literature search. The literature used in EBP is evaluated by levels of evidence (Melnyk & Finehout-Overholdt, 2015), and further analyzed and synthesized for relevance to the original question of practice problem.

Conducting nursing research and disseminating findings from published research provides the clinical evidence that is pulled together, or *synthesized*, though the EBP approach methods. Grey et al., (2017) cite an example demonstrating the benefit of this approach in the Evidence Based Practice Guidelines. These guidelines represent the gold standard of excellence for patient care outcomes as an integration and "consensus of the best information for making clinical practice decisions" (p. 33).

The utilization, or *translation*, of research findings into EBP changes interventions supports clinical decision-making, improves cost-effectiveness of care, and most importantly, improves the outcomes of patient care (Polit & Beck, 2017).

The gastroenterology nurse should refer to the bibliography at the end of this chapter to find sources for more in-depth details on research methods and statistical testing.

LEARNING OBJECTIVES

After reviewing the content of this chapter, the gastroenterology nurse will be able to:

1. Identify basic principles and methods for conducting nursing research.

2. Identify types of research methods and their importance for nursing practice.
3. Describe how to utilize the evidence to improve nursing practice.
4. Define basic terminology used to conduct nursing research.
5. Identify the steps involved in conducting a research study.
6. Define basic statistics used in analyzing nursing research.
7. Outline methods for collecting data.

HISTORICAL PERSPECTIVE OF NURSING RESEARCH

Florence Nightingale was nursing's first researcher and statistician. In *Notes on Nursing: What it is, and what it is not* (Nightingale, 1859), she documented her experiences of caring for the sick and dying; her observations of illness, disease processes, health care facilities, environmental sanitation; and her vision of nursing care attributes. Since that time, nursing has embarked on a consistent journey to establish research within the profession. In 1900, the first nursing journal was published, followed by the founding of the first doctoral program for nurses in 1923. Additional strides include the creation of the National Center for Nursing Research (NCNR) within the National Institutes of Health (NIH) in 1985, which provided needed financial support for nursing research within the United States. In 1993, the NCNR was promoted to full institute status within the NIH and renamed the National Institute of Nursing Research (NINR), elevating nursing research to the same level as that of other health disciplines (Polit & Beck, 2017).

THE RESEARCH PROCESS

The scientific research process is reiterative, meaning it is marked by repetition of a process. Investigation at one stage of the study often casts light on an earlier stage, inducing the researcher to return to the earlier step to refine a research problem or to revise study methods. Rarely does discovery proceed in a linear, sequential manner. Seldom does a researcher start at the first step in the process and proceed in an orderly fashion through the remaining sequence of steps.

All research, whether quantitative or qualitative in design, follows specific steps.

1. Identify the problem.
   a. State the purpose of the study.
   b. State the research question or hypothesis.
2. The literature review: Find the relevant evidence.
   a. Determine what is currently known about the topic.
   b. Find the relevant literature and identifying the sources of evidence.
   c. Appraise the evidence: What to look for in research reports.
3. Identify the implications for nursing and nursing practice.
   a. The importance for the nursing profession, clinical decision-making, and nursing practice.
4. Determine the methodology: How the study will be executed.
   a. Define basic research terminology.
   b. Describe a theoretical framework.
   c. Determine the study design for either qualitative or quantitative research.
   d. Identify the sample or population and study setting.
   e. Determine the number of participants needed for the sample.
   f. Identify ethical considerations: Protection of Human Subjects and Institutional Review Board.
5. Data collection plan: Implement the study and collect the data.
   a. Plan the method of implementing the study.
   b. Plan how to protect the data.
6. Analyze the data and interpret the findings.
   a. Plan how data will be analyzed.
   b. Cover the basic statistical terminology.
7. Report the findings in the study.
   a. Illustrate how the statistical findings relate to the research questions.
   b. Incorporate basic statistics when reporting study findings.
   c. Describe the conclusions found in the study.
   d. Explain any limitations found in the study.
   e. Discuss the findings, and include finding or implications that the findings could have for nursing and/or nursing practice.
8. Disseminate the results.
   a. Present as a poster presentation.
   b. Present as a podium presentation.
   c. Submit a written manuscript.

Step 1. Identify the problem, state the purpose, and determine the research question or hypothesis in the study

Research often begins with a problem that is identified in daily practice. The act of stating a research question or identifying the problem initially gives the project direction.

In the beginning, problem statements are often too broad and vague. The problem statements must be refined and narrowed to make the project manageable.

The purpose of the study

The purpose of the study identifies the specific focus of the study. Further explained in this section are the aims or objectives, as well as research questions and/or hypotheses. These are general statements that delineate what the researcher hopes to accomplish or learn from a research study.

Research questions and **hypotheses** are more specific than the aim or purpose. They address the relationship between variables to be tested during the study. A study may have several research questions or several research hypotheses, or the same study could have both hypotheses and research questions.

Research questions are written in the present tense and ask general questions about the relationship between variables.

An example of a research question is *"What is the relationship between the amount of dietary fiber ingested and the frequency of bowel movements?"*

Hypotheses are more specific, are written as declarative sentences, and state the relationship between two or more variables. Relationships are also directional. A positive relationship means that, as one variable increases, the second variable also increases. A negative relationship means that, as one variable decreases, the second variable also decreases. An inverse relationship means that, as one variable increases, the second variable decreases and vice versa.

An example of a hypothesis is *"Adult subjects who eat a high-fiber diet will have more stools than those who do not."*

The *null hypothesis*, or statistical hypothesis, states that there is no difference or relationship between the variables. The null hypothesis is abbreviated as *HO* (Plichta, & Kelvin, 2013).

An example of a null hypothesis is *"Adult subjects who have a high-fiber diet will have the same number of stools as those who do not."*

In other words, the independent variable (dietary fiber) has no effect on the dependent variable (frequency of stools). If the statistical test demonstrates no relationship between the variables, the null hypothesis is accepted. If a relationship is supported by statistical testing, the null hypothesis is rejected (Plichta & Kelvin, 2013; Polit & Beck, 2017).

Searching existing literature will help to clarify and refine the focus of the research, enabling the researcher to define the scope of the study so it becomes more feasible or practical. To determine whether a study is feasible, a *pilot study* may be conducted. When a study appears feasible, it is important to consider many factors before pursuing a research idea. These factors may influence the feasibility of a project and can include:

- **Interest level.** Is there interest among the professional community in the findings to come out of the proposed research?
- **Motivation.** Is the researcher sufficiently interested in the topic to sustain motivation to study the problem until the work is completed?
- **Time.** Is the scope of the problem such that it can be studied with substantial outcome data within the time allotted for the study?
- **Availability of subjects.** Can the researcher obtain a sufficient number of subjects to investigate the problem?
- **Cooperation of others.** Researchers rarely conduct investigations independently. Are the necessary resources available to enable the research to complete the project? Will it be a problem to get the necessary permissions required from subjects, guardians, institutional administrators, and so on?
- **Facilities and equipment.** What facilities and equipment will be needed, and will they be available to complete the study?
- **Money.** Will funding be necessary? If so, how much? Does the anticipated cost outweigh the value of the expected findings?
- **Experience of the researcher.** Is the problem from a field in which the researcher has some experience? If not, is there another more experienced investigator with the substantive knowledge of existing concepts, findings, and theories who can guide the researcher in developing methods of study and help avoid research problems that would require sophisticated measuring instruments and/or complex statistical analyses?

Step 2. The literature review: Finding the relevant evidence

A critical step in beginning a research study is to review existing literature related to the topic. The researcher should define the potential population or patient sample as the search terms for the study when reviewing the literature to determine what is currently known about a specific group and/or problem. Search terms are used to facilitate searching databases for appropriate studies relevant to the topic.

There are several objectives when conducting a literature review.

- Identify whether the proposed problem has already been solved.
- Determine whether the problem has been studied, but not sufficiently to rule out further investigation. This is termed a "research gap" in the literature. In this case, it may be feasible to replicate the study according to methods outlined in the literature.
- Determine whether certain research designs are appropriate for the proposed topic. There are many resources to use when designing a study. Some may not be appropriate. Other researchers have made mistakes that must be avoided.
- Identify the sources of the evidence and critically appraise the strength of the evidence (Melnyk et al., 2017).

Finding the relevant literature and identifying sources of evidence

The researcher should critically consider all sources of information.

A *primary source* of information is when the research report is written by the person or persons who performed the study. Primary information sources should be used as much as possible.

When reviewing a study, the researcher should examine the relevance, the validity, and the reliability of the instruments used (as discussed in the section on study design). The review should also consider the control of extraneous variables, the sample size, and the statistical and clinical significance of the results.

A *secondary source* describes research performed by other investigators. Reviews of the literature, for example, are secondary sources.

Tools that allow for more efficient retrieval of relevant data are excellent resources for a literature search.

- The librarian is an excellent resource who is skillful in searching databases for current literature that is relevant to the project.
- If the topic requires information from disciplines other than nursing (e.g., chemistry, genetics, or psychology), a librarian can identify relevant indexes on these subjects.
- Commonly used databases for nursing include CINAHL Complete, Embase, Ovid, PubMed, Medline, and PsycINFO. Other resources such as Web of Science and Scopus also may be helpful. A librarian can assist with a search strategy, if needed.

These tools may be a fruitful means of retrieving pertinent literature on the topic of interest. Some tools include:

- **Abstracts.** Abstracts contain summaries of journal articles. Although not detailed, they are useful in determining whether an article will be relevant to the topic.
- **Indexes.** Printed indexes, such as *Index Medicus*, can unlock vast stores of literature on every subject imaginable.
- **Computer databases.** Databases are associated with specific types of literature and are readily accessible, usually for free, in large medical institutions.
- **References.** References provide compilations of books, periodicals, and reports on topics. References can also identify other investigators with experience in the area of interest.
- **Personal contacts.** Contacting other investigators can help clarify information they published and help the researcher refine his or her own purpose and methods.

Appraising the evidence: What to look for in research reports.

When reviewing previous research, various types of information may be uncovered.

- **Facts, statistics, or findings.** This is the most important category of information because it represents the results of other research efforts. It includes progress other investigators have made when examining the topic.
- **Theory or interpretation.** This type of information is significant because it concerns the relationship between the idea or topic of interest and an existing body of knowledge surrounding that topic.
- **Methods and procedures.** Literature focused on research instruments and procedures is useful because it saves the researcher from redeveloping and revalidating measurement tools that already exist. Determine whether the research instruments are appropriate to measure the intended variables.
- **Opinions, speculation, anecdotes, clinical impressions, or narrations of incidents and situations.** These categories of information are value-laden but subjective. They serve to broaden the researcher's understanding of the problem and help deal with snags that may occur. Because it is important to remain as objective as possible, the researcher should avoid the temptation to rely heavily on such sources.
- **Negative findings.** Negative findings in research studies should be considered if relevant to your study population. It is not uncommon to encounter conflicting reports, which may be an indication that more research is necessary.

Step 3. Identify implications for nursing and nursing practice

The implication and importance for nursing answers the "so what" of your research. In this section, the research highlights the meaning or advantage this new knowledge provides the discipline of nursing, as well as how it will improve nursing practice and patient care.

Research is important for nursing and practice because research generates new knowledge about a phenomenon or topic of interest. Research involves systemic processes and uses measurements to answer research questions or to solve specific problems. Nurses are at the front line of nursing care and, through their observation of practice, can identify many possible topics to study. Nursing research studies generate trustworthy evidence and acquire new knowledge that is important for the nursing profession across such areas as clinical practice, education, nursing informatics, and administration (Polit & Beck, 2017; Gray et al., 2017).

Step 4. Determine methodology: How the study will be executed

Methodology is the technique researchers use to design a study and determine how to collect the information and data related to the research questions (Brown, 2014; Polit & Beck, 2017; Gray et al., 2017). There are basic research terms that should be understood.

Basic research terminology

Before reviewing the research process, the following terminology should be understood.

- **Researcher.** The individual conducting the study is called the *principle investigator (PI)* or *researcher*.

The term co-principle investigator (Co-PI) occurs when there are two or more PIs, and responsibility of the study is shared among all PIs.

- **Sample.** People or populations included in the study sample may be referred to as *participants* or *subjects* in quantitative research and *informants* in qualitative research. Whether the study is quantitative or qualitative, the total group of participants is known as the *sample population* or simply the *sample*. The way to determine sample size is calculated through *power analysis*.
- **Setting.** A research *setting* or *site* refers to the location where the nursing study takes place. To increase the sample size, some studies include more than one location of the same setting and so are called *multi-site* studies.
- **Statistical significance.** The *p-value* is a data-based probability determining that the results of the study were attributed to chance variation. This value is the level of significance and indicates that the relationships or differences among the variables were statistically significant (Brown, 2014).
- **Measurement.** An *instrument*, when used in research, is a tool to measure the variables in a study.
- **Process.** The *study design* refers to the overall plan for how a study will be implemented, how variables will be controlled, and how interventions will be carried out. *Methodology* is the process or technique used to gather and analyze information that is relative to the research question. Methods also include who will be collecting the data, how often data will be collected, and the timeframes for when the data is collected (Melnyk et al., 2017).

Nursing researchers often use a **theoretical framework** to guide the study. A theory is an organized set of assumptions, concepts, definitions, and propositions (or connections among the theoretical concepts) that describe how the concepts exist in the physical, psychological, and social world (Brown, 2014). The purpose of a theoretical framework is to facilitate the selection of the variables for a study and how they relate to each other. They also provide guidance in developing the intervention for an experimental study (Melnyk et al., 2017).

Quantitative research

Quantitative research is a disciplined process used to acquire information. The researcher follows a logical series of steps in accordance with a specified plan of action (Polit & Beck, 2017). Quantitative researchers gather data that are embedded in objective reality obtained through any of the five senses: taste, smell, hearing, sight, and touch. The collection of information for quantitative research is generally numeric and, therefore, can be analyzed statistically.

Examples of research designs used in quantitative studies are *experimental*, *quasi-experimental*, *correlation*, and *descriptive*.

In quantitative studies, the *variable* is something that varies, such as weight or blood pressure, or may be something created, such as the effectiveness of patient-controlled analgesia. The main variables in quantitative research are the independent variable and the dependent variable.

An *independent variable* is referred to as the presumed "cause," which has a presumed "effect" on the dependent variable (Polit & Beck, 2017).

The *dependent variable*, also called the outcome variable, is the variable affected by the independent variable (Brown, 2014).

An example of a quantitative study is *"Investigating the extent to which lung cancer (dependent variable) results from smoking (independent variable) in a population of Army veterans."*

An example of a qualitative study is *"What are the lived experiences of Army veterans after discharge from active duty?"*

In **experimental studies**, the researcher is interested in the relationship between an *independent variable* and a *dependent variable*. The classic experimental design is a *randomized control trial (RCT)*. The importance of experimental studies includes (Brown, 2014):

- Ensuring the creation of a control and comparison group.
- Random assignment for control and intervention groups.
- Strict control of how the intervention is provided to the intervention group.

In **non-experimental studies** such as *descriptive* or *observational* studies, the researcher observes the sample population in the natural setting and does not attempt to manipulate the variables in the study. The variables in descriptive study can be characteristics such as blood pressure, age, gender, and weight (Brown, 2014).

Correlation studies employ a research design that describes a relationship or connection between variables in terms of direction that is either positive or negative, and strength that is strong, moderate, or weak. These relationships are determined using statistical testing (Brown, 2014).

Quasi-experimental studies are similar in design to experimental studies, but when the researcher cannot control any one of the components of experimental design, the lack of control could potentially include uncontrolled, or *extraneous*, variables (Reeves, Wells, & Waddington, 2017).

A quasi-experimental research design is used to identify the impact of treatment or preventative measures on disease. This research design establishes a causal connection between the intervention and the outcome (Polit & Beck, 2017).

Also, in quasi-experimental research design, the *intervention group* receives the experimental treatment, and the comparison group does not receive the experimental intervention. This type of research is also referred to as an *interventional study*. In this case, the intervention is the independent variable inthe study.

Qualitative research

Qualitative research focuses on understanding the human experience. A qualitative study is research that focuses on an understanding of the world or phenomena viewed in the natural setting and interpreting their significance and meaning of being (Creswell & Poth, 2018). In qualitative studies, there is a method for subject sampling, and data is collected through interviews, observations, and focus groups (Brown, 2014). Measurement of data is not numerical but categorized, or *coded*, into *themes*. Computer software is available that is specifically designed to assist with categorizing data—coding and organizing themes—so that they may be analyzed.

The sample or population of the study

Sampling refers to the process of selecting a study group to represent the entire **population**. The major importance in assessing a sample is to ensure that the sample selected for the study is representative of the general population to be studied. For example, if the study sample targets endoscopy nurses working in hospital settings, a sample would be nurses who perform endoscopy procedures in a hospital. Nurses who work on inpatient units would not be included. The location where the study will take place is called the research setting.

The sample or population is important for quantitative research because the emphasis is on what factors influence a phenomenon in the specific population sample, rather than one person in the sample. Understanding the extent to which the phenomenon occurs in the sample is critical in determining whether the finding can be generalized to individuals other than those who participated in the study (Polit & Beck, 2017).

Determining the appropriate number of participants for the study

Power analysis in research is a calculation for determining the appropriate number of participants, known as the sample size, for a specific study. This is specifically important for whether the study hypothesis is appropriately rejected or accepted. Calculating the sample size is based on factors that include power (.80), effect size (small, medium, or large), and significance level (alpha = 0.05) (Gray et al., 2017; Plichta & Kelvin, 2013; Polit & Beck, 2017).

Power is defined as the ability to detect true relationships among variables. Generally, a power of 80 percent (or 0.80) is sufficient to detect relationships between the independent and dependent variables (Polit & Beck, 2017).

Effect size refers to how much of an impact the independent variable will have on the dependent variable. Effects are noted as small, medium, or large and must be estimated to determine the sample size (Plichta & Kelvin, 2013; Polit & Beck, 2017).

Significance level is called *alpha (p-value)* and is generally set at *alpha (p=0.05)*. It is the most important reported value that judges the extent that the HO is supported. The p-value ranges from 0.00 to 1.0 (Plichta & Kelvin, 2013).

Two of the most common types of sampling plans are:

- *Probability sampling*, or *random sampling*. This is the simplest of the probability sampling designs. It involves selection from the population (or a subpopulation) at large and is performed in such a way that each member of the population has equal probability of being included in the sample (Brown, 2014). Probability sampling is the more respected of the two types because greater confidence can be placed in how representative the sample is of the general population.
- *Non-probability sampling*, or *accidental sampling*. This is a sampling technique that involves using the most readily available subjects or objects as participants in the study (Brown, 2014). Researchers must be careful about generalizing results to the population at large when sampling by this non-probability technique, as it may not be representative.

Study subject groups

In **experimental study** designs, there is an experimental group, and either a control group or a comparison group.

The *experimental group* is made up of subjects who receive an experimental treatment or intervention.

The *control group* is made up of subjects who do not receive an experimental treatment or intervention. Their performance provides a baseline against which the effects of the treatment can be measured.

A *comparison group* is a selection of subjects whose scores on a dependent variable are used as the basis for evaluating the scores of a group of primary interest. The phrase "comparison group," rather than the phrase "control group," is used when the investigation does not use a true experimental design (Melnyk et al., 2017; Thiese, 2014).

In **quasi-experimental study** designs, similar to the experimental study designs, there is manipulation of the independent variable. However, the subjects are not randomly assigned to the treatment group or comparison group (Polit & Beck, 2017).

The terms *double-blind* and *placebo effect* refer to design features of research studies. A double-blind experiment is one in which neither the subjects nor the investigators are aware of which subjects are in the experimental group and which ones are in the control group.

A *placebo* is an inert material whose appearance is identical to the drug under investigation. In an investigation of the effects of a drug, for example, neither the drug administrator nor the subjects are aware of whether the drug administered is the experimental drug or the placebo. Using a placebo allows researchers to separate psychological effects induced by drug administration (the *placebo effect*) from the true effects of the drug.

Ethical considerations: Protection of human subjects and Institutional Review Board

Ethical considerations. Can the researcher ensure that the study involving human beings does not impose unfair or unethical demands on the participants? These rights of human subjects include the principles of beneficence (protection from harm or discomfort), justice (equally fair treatment for all participants), and privacy (Melnyk et al., 2017).

Historically, there were scientific experiments on human subjects that were considered unethical. Ethical considerations in research are based on federal regulations that protect the rights of human subjects, so they will never be mistreated during the process of conducting research (Polit & Beck, 2017). The principles for the ethical rights of human subjects should guide nurse researchers (Brown, 2014; Melnyk et al., 2017; Polit & Beck, 2017) when planning studies that include human subjects in the population sample.

Before a study involving human subjects can begin, it must be reviewed by an approving body, known as an Institutional Review Board (IRB) (Melnyk et al., 2017; Polit & Beck, 2017). The IRB maintains specific guidelines for research processes and includes the rights of human subjects. They are the governing bodies that enforce the ethical conduct of research. The ethical principles and rights of all participants in a research study must be clearly outlined for the Board's review. The ethical rights of human subjects are:

- **Self-determination:** All participants make their own decisions to participate in the study after the researchers have disclosed the purpose of the project; its general value; any procedures, questionnaires, or treatments involved; and any potential risks and benefits. These points are included in an **informed consent** that must be signed by the participant to take part in the study. All ethical principles and rights of human subjects are included and described in an informed consent.
  - Moral and ethical considerations must be strenuously applied in protecting the rights of **vulnerable participants** who may be incapable of giving informed consent or are at high risk for unintended side effects. Specific restrictions apply to these individuals and include:
    - Children.
    - Subjects who are emotionally, mentally, or physically disabled.
    - Institutionalized individuals.
    - Incarcerated individuals.
    - Pregnant women.

  Participation in research must always be **voluntary**. Although participants sign consent for inclusion into the study, they may voluntarily and freely withdraw from the study at any time.
- **Privacy:** Participants have the right to ask how and with whom their personal and health information will be shared.
- **Confidentiality:** Data collected is only shared with others identified in the consent signed by the participant.
- **Anonymity:** Data collected should not identify the participants or be linked to the participants' identity.
- **Justice:** Participants must by treated fairly and without bias.
- **Beneficence:** Participants are protected from harm, pain, or discomfort. The phrase "harm and discomfort" is not limited to physical injury and should also address emotional concerns. The risks, benefits, and alternatives must also be disclosed.

Step 5. Data collection plan: Implement the study and collect the data

Data collection is the methodological and systematic way that the study will be carried out. Data is collected following the methods designed for the study and should be approved by the organization's Internal Review Board. Adherence to how the data will be collected is the most important in terms of the outcome and reliability of the results.

- A plan for how the data will be collected must be clearly stated, and time frames for completion of the study should be identified. Because this phase of research is labor-intensive, resources needed to execute the study should also be included in the planning.
- The researcher must include careful observation and ongoing assessment during the data collection process to ensure that data is collected and documented as planned. In addition, the collected data needs to be protected. Preparation should include a secure location for data storage. Locked cabinets may be used but should be limited. Data is best stored in a password-protected and secure databases specifically for this purpose.

Step 6. Analyze the data and interpret the findings

When planning a research study, some beginning researchers give inadequate attention to plans for data analysis. A clear plan for the types of statistical tests to be performed should be identified in the design of the study. Although computer packages make statistical analysis quick and easy, researchers must ensure that data are accurate, appropriate statistics were chosen, and the research design met the assumptions of the statistics chosen (Brown, 2014; Polit & Beck, 2017).

- Assistance from a biostatistician is helpful to ensure reliability of results. Failure to consider how the data will be analyzed could result in collecting useless data and inappropriate findings for the study.
- Once the data analysis is complete, the data are examined to understand and interpret their implications, determine whether the findings are significant, and identify whether research questions and hypotheses are supported (Brown, 2014; Polit & Beck, 2017).

An outline of basic statistical terminology follows in the next section.

Statistical Terminology—the Basics

There are many resources available to assist the individual researcher in understanding basic statistics, and he or she should be proactive is seeking assistance from supportive resources. Every researcher must understand the basic statistical terminology to:

- Systematically critique the literature.
- Determine whether the findings are statistically significant.
- Determine whether the hypotheses or research questions are supported.

Statistics is a mathematical methodology that involves collecting, organizing, and interpreting data. Using statistics enables the researcher to quantify and mathematically describe and summarize information. Statistical significance, or the p-value, is the point of divergence between two groups. Statistics are used to make predictions or generalizations based on observations and to identify associations, relationships, or differences between the study variables (Plichta & Kelvin, 2013; Gray et al., 2017).

Inferential statistics are statistical techniques that provide predictions about a specific population's characteristics based on the sample from the specific population (Plichta & Kelvin, 2013).

Descriptive statistics are used to describe or characterize data and are found in tables, charts, frequencies, and measures of central tendencies.

The most basic statistical inferences, involving **central tendency**, are the mode, the median, and the mean.

The *mode* is the numerical value that occurs most frequently in a set of values. It is not necessary to perform calculations in order to derive the mode. The mode is determined by simply inspecting the set of values.

The *median* is an index of average position in a set of values. It does not consider the quantitative values of individual scores. Specifically, the median is the point in a set of values above which and below which 50 percent of the values lie (Plichta & Kelvin, 2013).

The *mean* is the measure of central tendency that is usually referred to as the average. The mean is simply the sum of the values in a set, divided by the number of elements in a set.

Of the three indexes of central tendency, the mean is the most stable. If repeated samples are drawn from a population and some characteristic is measured, the means for these samples would fluctuate less than the modes or the medians. When researchers work with interval or ratio-level measurement, the mean, rather than the mode or median, is almost always the statistic reported.

Variability

The concept of **variability** is concerned with the degree to which subjects in a sample vary from one another with respect to some critical attribute.

Sampling strategies can have a dramatic effect on variability. If, for instance, the aim of a study is to determine the variability of systolic pressure in any given sample of the population, and the sample studied is a convenience sample of patients seen in one emergency room, the variability of systolic pressures is likely to be greater than if the sample were selected from the junior class of a local high school. Because emergency room patients are more likely to present with extremes of pressure, the range of pressures may be greater than those among a young, healthy population. Consequently, it is necessary to describe variability in ways that account for situations like this one.

Range is simply the highest minus the lowest score (or value) in a given set of values (Polit & Beck, 2017).

Standard deviation is a more complicated measure of variability. The standard deviation measures the spread among values or simply how far away the numbers in a list are from their average (Polit & Beck, 2017). The standard deviation is considered more reliable than the range as a measure of variability. Like the mean, it considers every value in the set of values.

What does the standard deviation indicate about variability?

In general, the greater the standard deviation, the more variable the data will be. Conversely, the smaller the standard deviation, the less variable the data will be. If the mean blood pressure of two populations is equal, yet the standard deviation in one sample is 17, while the standard deviation in the second sample is 3, clearly the blood pressures of the first sample are more variable than the blood pressures of the second sample.

The standard deviation provides a picture of the distribution of the data for a particular variable or score. In a normal distribution, 67 percent of the sample will be within one standard deviation of the mean, 95 percent within two standard deviations of the mean, and 99 percent within three standard deviations of the mean.

The standard deviation can also illustrate the relative value of using the mean to describe typicality of an entire data set. The larger the standard deviation, the less reliable the mean is as an indicator of typicality. Conversely, the smaller the standard deviation, the more reliable the mean is as an indicator of typicality.

Statistical testing techniques

Statistical testing techniques include the following:

Non-parametric testing is used to answer research questions that are targeted at whether a relationship exists between two variables.

Non-parametric statistics are used to test the relationship between variables measured at the nominal or ordinal level. Examples of non-parametric statistics include the Chi-square test (for nominal level data) and the Mann-Whitney U test (for ordinal level data) (Plichta & Kelvin, 2013).

Parametric testing is used to determine whether groups differ on outcomes measures, also called interventions.

Parametric statistics are used when evaluating an outcome measure from groups within one population sample.

The t-test and analysis of variance (ANOVA) are used to test hypotheses whose variables are measured at the interval or ratio level (Plichta & Kelvin, 2013).

Measurements

Measurement is the assignment of numbers to objects according to specific rules (Brown, 2014; Polit & Beck, 2017). These rules allow researchers to assign values consistently from one subject or event to another. Nurses taking measurements for research studies follow the same kinds of rules they follow when performing measurements in nursing practice. For example, patients are weighed at the same time each morning, in the same way, with the same amount of clothing each time. These rules of measurement are used in research to ensure the accuracy of the results. There are four levels of measurement: nominal, ordinal, interval, and ratio (Brown, 2014).

Nominal data are the weakest form of measurement and are used to organize data into categories such as gender, race, religion, and marital status.

Ordinal data can be arranged according to rank, such as pain intensity, anxiety level, and coping ability. When measuring pain, the researcher might ask the subject to rate his or her pain on a scale from 0 to 10. A rating of 10 would be greater than a rating of 7, which would be greater than a rating of 3. However, the measurement method is too crude to consider these ratings as exact distances apart. Another example of a rank-order scale is: strongly disagree, disagree, neutral, agree, and strongly agree.

Interval level is characterized by rank-order data that occur at specific intervals. An example is temperature. Because the intervals on the thermometer have equal gradations, you can say that a measurement of 101 degrees is 4 degrees higher than a measurement of 97 degrees. However, because interval-level data do not have a genuine zero point, you cannot say that 40 degrees is twice as warm as 20 degrees, or that 0 degrees means that there is no temperature.

Ratio is the highest level of measurement and has all the characteristics of interval-level data, plus a genuine zero point. Weight and height are ratio levels of measurement. Therefore, we can say that 40 pounds is twice as much as 20 pounds.

Although there is a difference between interval and ratio-level measurements, they are treated the same way statistically. The level of measurement is the major concern when selecting the appropriate statistic to test a hypothesis.

Validity and reliability: How credible are the measurements?

Before initiating the research study, researchers must ensure the **validity** and **reliability** of all instruments used to measure the variables. This means that the nursing study results or findings reflect accurately what is being studied. Assessing the quality of the study is known as *scientific merit*.

Validity is a concept that broadly defines whether the findings are unbiased and well-grounded. Validity is also a criterion for measurement tools and is concerned with whether it is measuring the concepts that it is supposed to measure (Polit & Beck, 2017).

Reliability assures that the study findings are accurate and consistent. In relating reliability to the measurement tools used to obtain information, it refers to the reproducibility or consistency of the measurement (Polit & Beck, 2017). For example, the researcher would expect a person to weigh the same if he or she stepped on and off the scale several times in succession. The researcher would expect two nurses to obtain the same heart rate if they were auscultating the same person's heart at the same time.

Questionnaires and surveys are valuable measurement instruments and may also be called "tools." However, establishing their validity and reliability—known as *psychometric testing*—is often complicated. When evaluating study instruments, the research must determine whether the instrument is appropriately measuring the intended variable. New researchers should try to find data collection instruments that have already undergone extensive validity and reliability testing.

Step 7. Report findings of the study

Once the data have been analyzed, the findings relate to the research questions or the hypothesis in the study. Does the data statistically answer the research questions? Does the data statistically support or reject your hypothesis?

Include basic statistics in the report of study findings. An example of statistical data may include descriptive statistics, which address the population sample in the study and are generally reported in a table. Relationships between variables can also be reported in an outcome table that presents the data simply and is easier to read and understand.

The study **conclusions** are based on the outcomes of the data analysis an identify the meaning of the study findings. Based on the findings in the study, the conclusions may include any implications for nursing and/or nursing practice as well as recommendations for future research.

The **limitations** of the study identify any characteristics of the study that could decrease the generalizability of the study findings. How valid are the study findings for all populations? For example, if the study population was small, the sample was a limited representation of a larger population and cannot be generalized to the larger population.

- Every study, no matter how sophisticated, has some limitations or flaws. Investigators are inevitably faced with compromises and constraints that must be acknowledged when conclusions are drawn. Consumers of research need to understand these limitations to evaluate the usefulness of the information provided.

The **discussion of study** describes the researchers' interpretation of the study findings. The researcher may compare their study findings to other relevant studies or discuss how the findings were different and explain possible rational for why the differences occurred.

Step 8. Disseminate study findings: Spread the word

Disseminating results of a study is essential, whether the findings were those that were hypothesized or not. Both expected and unexpected findings add to our knowledge and our understanding of the world of nursing and health care.

The results of a research study may be reported in the following ways:

- Writing and publishing an article that describes the study and the results obtained.
- Giving an oral presentation of the study findings at a local or national meeting.
- Presenting the findings with a poster at a local or national meeting.
- Presenting the findings within the local area at informal group meetings, in-services, or work conferences.

CASE SITUATION

Denise Heckel is a baccalaureate-prepared gastroenterology nurse in a large metropolitan hospital. Over the next few months, the staff will cooperate with Denise by helping her prospectively study the effects of preprocedural teaching by an endoscopy nurse on the anxiety level of endoscopy patients. The subjects Denise selects for the study must be adults (18 years or older) who are scheduled for upper and lower endoscopies and who are English-speaking, oriented to their surroundings, and able to hear and give accurate information. Denise is also measuring the amount of midazolam used for patient sedation before and during the endoscopic procedure.

Points to think about

- What steps should Denise follow to study her target population?
- What purpose(s) (e.g., description, exploration, explanation, prediction,and control) might Denise fulfill in performing her study?
- What factors should Denise explore to determine whether her study will be feasible?
- What hypothesis might Denise test in a study of this nature?
- What variables might influence her results? Which variables has she controlled? Which variables may she yet need to control?

Suggested responses

1. Denise should take steps when conducting her study to:
   - Identify the research problem.
   - Determine the importance for nursing and nursing practice.
   - Perform a literature review.
   - State the purpose of her study.
     - Formulate a hypothesis or research question.
   - Determine the study design: Methods or methodology.
   - Implement the study and collect the data.
   - Analyze the data and interpret the findings.
   - Disseminate study findings.
2. To determine whether her study will be feasible, Denise should explore the following factors:
   - Timeframe allotted to complete the study.
   - Calculation of the sample size and identification of the availability of a sufficient number of subjects.
   - Cooperation of others and availability of the necessary permissions.
   - Availability of facilities and equipment.
   - Amount of funding needed, if any, to perform the study, and whether the anticipated cost outweighs the value of the expected findings.
   - Availability of an experienced investigator who can provide guidance in:
     - Developing methods of study and helping avoid research problems requiring sophisticated measuring instruments and/or complex statistical analyses. She may consider the inclusion of a biostatistician as a helpful resource.

- ◦ Ensuring the rights of human subjects in her research.
- ◦ Determining whether the study could impose unfair or unethical demands on the participants.
- ◦ Evaluating whether she can obtain informed consent and, if not, whether she can justify not obtaining informed consent in terms of the benefits to patients versus the harm done.

3. Some hypotheses that Denise might test in a study of this nature include:
   - That sensory information combined with procedural information reduce patient anxiety, rather than procedural information alone.
   - That patients who receive information the evening before their procedure experience less anxiety than patients who receive information immediately before endoscopy.
   - That negative prior experience influences preprocedural anxiety: Pre-procedure information is more effective in reducing both anxiety and the need for sedation during endoscopy when patients have had positive experiences or no experience with outpatient surgery.
   - That the midazolam premedication did not provide a benefit of reducing anxiety, compared to not providing any premedication, when pre-procedure visits have been performed.
4. Variables that might influence Denise's results include patient age and sex, timing of pre-procedure visits, content of information provided, predisposition toward anxiety, length of visit, format of visit, past experiences, present emotional circumstances, reason for the procedure, and many others.
5. Variables that Denise has controlled are age, type of procedure, and the ability to communicate with an English-speaking investigator. She has yet to control all other variables that potentially may confound or confuse findings in the study.

REVIEW TERMS

Research designs, statistics, data types, research processes, sampling, variables

REVIEW QUESTIONS

Questions 1–8 are based on the following study:

A research study is examining the effects of preprocedural teaching by an endoscopy nurse on the anxiety level of patients undergoing upper and lower endoscopies. The subjects are adults (18 years or older) who are scheduled for upper and lower endoscopies, and who are English-speaking, oriented to their surroundings, and able to hear and give accurate information.

The patients participating in the study are assigned to one of two groups using a table of random numbers. Group 1 receives the current pre-procedure care, which includes an instructional pamphlet. Group 2 receives the current pre-procedure care, which includes an instructional pamphlet, as well as preprocedural teaching from an endoscopy nurse.

During a study, the researcher measures the patients' level of anxiety by asking them to rate it on a scale from 0 to 10.

1. What is the *dependent* variable?
   a. Anxiety level.
   b. Pre-procedure visits by an endoscopy nurse.
   c. The amount of sedation required.
   d. The type of endoscopic procedure scheduled.
2. What is the *independent* variable?
   a. Amount of sedation used for the endoscopy procedure.
   b. Preprocedural teaching by endoscopy nurses.
   c. The anxiety level of the endoscopy patient.
   d. Preprocedural teaching by the unit clerk.
3. What does the following phrase taken from the study above describe: "English-speaking adults (18 years or older) who are scheduled for upper and lower endoscopies and who are able to hear and give accurate information."
   a. All patients seen in the endoscopy clinic.
   b. The population selected for the study.
   c. A random sample from the population at large.
   d. All of the variables in the study.
4. Which statement best describes Group 2?
   a. Group 2 is a random sample.
   b. Group 2 is the experimental group.
   c. Group 2 is the control group.
   d. Group 2 is not necessary.
5. Which statement best describes Group 1?
   a. Group 1 is a random sample.
   b. Group 1 is the experimental group.
   c. Group 1 is the comparison group.
   d. Group 1 is the control group.
6. Asking subjects to rate their anxiety level on a scale from 0 to 10 is what level of measurement?
   a. Interval.
   b. Nominal.
   c. Ordinal.
   d. Ratio.
7. To calculate how long it will take to complete the study, the researcher needs to compute the average number of patients treated each day in the clinic. How might the researcher find this average?
   a. Read it in a review of the literature.
   b. Scan clinic records and see that, on most days, there are 20 patients treated.
   c. Add the total number of patients treated last year, and divide it by the number of days the clinic operated.
   d. Compute the difference between the highest number of patients treated and the lowest number of patients treated per day over the last year.

8. If the most sedation required by a member of the study group was 1.7 mg/kg and the least amount required was 0.8 mg/kg, the range of sedation required for this group is:
   a. The least and the greatest amount required to produce sedation.
   b. The average of the greatest and least amounts required to produce sedation.
   c. 1.7 mg/kg minus 0.8 mg/kg.
   d. 2.5 mg/kg.

Questions 9–22 are related to research in general:

9. Which statement best describes a null hypothesis?
   a. A conclusion statement saying that the study results are inconclusive.
   b. A hypothesis that assumes no relationship between the variables under study.
   c. A hypothesis to be ignored.
   d. A hypothesis that was never pursued because the idea was too preposterous.
10. When conducting research, a placebo is a:
    a. Treatment.
    b. Measure of central tendency.
    c. Theory.
    d. Sample.
11. The measure that focuses on the spread among values (calculates how far the numbers in a list are from the average) is called:
    a. The mean of a sample.
    b. The mode of a sample.
    c. The standard deviation.
    d. The deviation scores.
12. When written by the person who performed the study, a research source is considered a ___________.
    a. Primary source of information.
    b. Secondary source of information.
    c. Clinical impression.
    d. Theory or interpretation.
13. The simplest of the probability sampling designs is ___________.
    a. The control group.
    b. The comparison group.
    c. The random sample.
    d. The experimental group.
14. The levels of measurement are:
    a. Nominal, ordinal, standard deviation, and mean.
    b. Nominal, ordinal, interval, and ratio.
    c. Nominal, range, variability, and mean.
    d. Nominal, validity, variability, and range.
15. The mode is ___________.
    a. An index of the average position in a set of values.
    b. A measure of variability.
    c. The empirical value that occurs most frequently in a set of values.
    d. Theoretical.
16. Standard deviation is an indicator of ___________.
    a. The average.
    b. The sample size.
    c. The mean.
    d. Variability.
17. To ensure that human subjects' rights are maintained, the research study needs approval from ___________.
    a. The hospital administrator.
    b. The nursing leadership.
    c. The Institutional Review Board.
    d. The Executive Management Committee.
18. The evidence-based practice approach involves the methodical integration of relevant information that embodies current best clinical practices found in the literature.

    This statement about evidence-based practice is:
    a. True
    b. False
19. The acronym PICOT stands for:
    a. Preamble, Indication, Climate, Opportunity, Timeframe.
    b. Presentation, Improvement, Concentration, Optimism, Timeframe.
    c. Population, Interventions, Comparison, Outcome, Timeframe.
    d. Population, Intervention, Concentration, Optimum, Timeframe.
20. In descriptive research design, there is a strict random sampling of the subjects into two distinct groups.
    a. True.
    b. False.
21. Correlation studies determine relationships between variables. The relationship is positive or negative.
    a. True.
    b. False.
22. In any research study, it is imperative that moral and ethical rights are ensured. Human subject protection involves protecting the rights of all study subjects, as well as vulnerable subjects who may be incapable of providing informed consent.
    a. True.
    b. False.

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SECTION II

Nursing Process

Section II presents a series of chapters that explore the nursing process. Although nurses' experiences and education may vary considerably, the nursing process provides a shared vision of practice that brings continuity to the nursing care of a patient. The nursing process is composed of five interrelated steps: assessment, diagnosis, planning, implementation, and evaluation.

Chapter 8

ASSESSMENT

In today's health care environment, patients have become the drivers behind their care. Consumer-driven care means that the concept of health will depend on the individual defining it; nurses must understand that two patients with the same diagnosis or condition might not agree on their level of wellnesss and acknowledge that the determinations of both patients are legitimate (Carpenito, 2016). Nursing diagnoses define the art and science of nursing, and this chapter serves to enhance nurses' ability to assess the patient's concept of "health" amid the challenges of ongoing rapid change in the nursing profession.

Learning objectives

After reviewing the content of this chapter, the gastroenterology nurse should be able to:

1. Delineate the essential elements that constitute the basis of a nursing assessment.
2. Highlight the nursing assessment as a part of collaborative practice.
3. Identify the steps of a nursing assessment, specifically as it applies to the assessment of the gastroenterology patient.
4. Classify the types of data collection and the types of data specific to the gastroenterology patient.
5. Discuss the nursing assessment of the patient receiving sedation and analgesia in the gastrointestinal (GI) endoscopy setting.

ASSESSMENT PROCESS

The first step in the **nursing process** is **assessment**. It involves systematically gathering **data** about a patient and analyzing it to form a nursing diagnosis. Assessment is a continuous activity the nurse performs throughout contact with a patient, with constant reassessment as needed and documentation of findings (Alfaro-Lefevre, 2013). The initial assessment is only the beginning of a process that will repeatedly evaluate the effectiveness of therapy and determine the course of clinical care.

A nursing assessment differs in scope from a medical assessment. The aim of a medical assessment is to define the existence of medical problems and identify underlying pathology. The purpose of a nursing assessment is to identify human responses to biological, psychological, and social stressors, as well as responses to health problems and their treatments. In addition to the identification of adverse patient responses, the classification system for nursing diagnoses has expanded to include assessment of patient strengths and a nursing focus on patient health maintenance and health promotion behaviors.

The American Nurses Association (ANA) has designated the North American Nursing Diagnosis Association-International (NANDA-I) as the official organization for the development of evidence-based nursing diagnoses. NANDA-I performs continuous reviews of current nursing diagnoses and considers new diagnoses for inclusion. This constantly evolving classification system relies on input from nurses, who should consider it part of their patient care duties to submit missing diagnoses whenever noted (Carpenito, 2016).

NANDA-I currently identifies three types of nursing diagnosis (more fully covered in Chapter 9).

1. **Problem-focused nursing diagnosis** is a clinical judgment concerning an undesirable human response to a health condition/life process that exists in an individual, family, group, or community.
2. **Risk nursing diagnosis** is a clinical judgment concerning the vulnerability of an individual, family, group, or community for developing an undesirable human response to a health condition/life process.

3. **Health promotion nursing diagnosis** is a clinical judgment concerning motivation and desire to increase well-being and to actualize human health potential. These responses are expressed by a readiness to enhance specific health behaviors and can be used in any heath state. Health promotion responses may exist in an individual, family, group, or community.

Collaborative problems are patient conditions best managed by a multidisciplinary approach. Some agencies require documentation of multidisciplinary team efforts in planning health interventions, delivering care, and organizing the patient for discharge. The concept of collaborative care can be expanded to include the overall care of a patient that may involve several hospitals and multiple providers to treat an existing condition. The nurse makes independent decisions for both collaborative problems and nursing diagnoses. The difference is that, for nursing diagnosis, nursing prescribes the definitive treatment to achieve the desired outcome. In contrast, for collaborative problems, prescription for definitive treatment comes from both nursing and medicine (Alfaro-LeFevre, 2013; Carpenito, 2016).

Over time, patients may receive a range of care in multiple settings from several different providers. The Joint Commission continues to look for ways to improve safety and quality as patients experience these "transitions of care." The mandate from The Joint Commission's (2013) initiative is that collaboration among all care providers is crucial in keeping patients safe and healthy during such transitions.

TYPES OF NURSING ASSESSMENTS

The appropriate type of assessment is determined by the clinical situation, client status, time available, and purpose of the data collection. Assessments can be divided into four general categories: initial assessment, focused assessment, time-lapse reassessment, and emergency assessment (Craven, Hirnle, & Henshaw, 2017).

1. The **initial assessment** is performed on initial contact with the patient to identify normal functional status and collect data concerning actual or potential dysfunction. The initial assessment is the baseline for reference and future comparison. It can be described as a comprehensive assessment of patient health. The initial assessment also may be referred to as the Database Assessment or Start-of-Care Assessment (Alfaro-LeFevre, 2013).
2. The goal of the **focused assessment** is to determine the status of a specific problem identified during a previous or initial assessment.
3. In a **time-lapsed reassessment**, the nurse compares the patient's current status to the baseline obtained previously and detects changes in all functional health patterns after an extended period of time has passed.
4. The **emergency assessment** is the identification of a life-threatening situation that can occur any time a patient experiences a physical or psychological crisis.

STEPS OF A NURSING ASSESSMENT

A patient's entire plan of care is based on the assessments made by a variety of health care providers. The nursing assessment is used to determine a need for nursing service and to assist other professionals (e.g., physicians, pharmacists, nutritionists, case managers) in determining their plan of care. Alfaro-LeFevre (2013) has identified six key phases of assessment:

Step 1	Collecting data
Step 2	Identifying cues (abnormal data) and making inferences
Step 3	Validating (verifying) data
Step 4	Clustering related data
Step 5	Identifying patterns and testing first impressions
Step 6	Reporting and recording data

Step 1. Collecting data

Systematic guidelines for collecting data ensure that comprehensive, holistic health information will be collected. If data collection is well planned, nursing diagnoses follow easily. Problems in data collection arise when the data are inappropriately organized, pertinent data are omitted, irrelevant or duplicate data are collected, erroneous or misinterpreted data are collected, too little information is acquired, interpretation of data (rather than observed behavior) is recorded, or the **database** is not updated.

The nurse uses the interview, observation, physical examination, review of records and diagnostic reports, and collaboration with colleagues to establish the database. The nurse should always consider direct observation and assessment of the patient as the primary source of information, as opposed to relying solely on lab values and readings from a monitor to indicate the patient's status (Alfaro-LeFevre, 2013).

Interview

The intent of the **nursing interview** is to record a patient's unique qualities. When this goal is met, the planning of individualized care becomes possible. The nursing interview will be a reflection of the type of assessment needed at that time. A patient who comes to the primary care setting will most likely require a more complete and comprehensive assessment than the patient with an acute problem who is being seen in an emergency setting. Data collection focuses on identification of a patient's present and past health status. This includes factors that positively or negatively influence health, how health status impacts daily life, changes that the patient has made to adapt to the current health status, response to previous nursing/medical therapy, risk for potential problems, and desire for higher level of wellness. Family members or caregivers may provide insight into the patient's state of illness.

All nurse-patient interactions are based on communication. Listening and observing are central to building a therapeutic relationship and performing an assessment. The nurse who establishes rapport with the patient will help ease feelings of vulnerability and instill confidence in the patient that his or her health concerns will be addressed (Alfaro-Lefevre, 2013).

Therapeutic communication describes techniques that encourage the patient and/or family to share views and feelings openly. A skilled interviewer uses interpersonal skills to convey a sense of warmth and concern that will make the patient more at ease and willing to share necessary personal information. The health care setting often makes people anxious, and the interview provides the nurse an opportunity to set a reassuring tone for the health care encounter. The interviewer should first introduce him- or herself to the patient and family/caregiver, then explain what will happen during the health care encounter.

Nonverbal communication is just as important as verbal communication. The skilled interviewer takes note of patients' nonverbal cues – their facial expressions, posture, pauses between answering questions, gestures, and body language. The interviewer is also conscious of his/her own nonverbal communication with patients, aiming to convey attentiveness, interest, and concern. If possible, the interviewer should position herself on the same level as patients, rather than hover over them.

The nursing interview gives the patient an opportunity to provide a personal account of his or her present health status. **Subjective data** collection compiles information from the patient's perspective as it relates to his or her clinical presentation. Subjective data usually include feelings of anxiety, discomfort, or mental stress. The presence of pain is an important subjective finding. Only the patient, family, and or caregiver can provide information about the frequency, location, duration, or intensity of pain, as well as any attempts to relieve pain and how the pain affects daily life. Tools such as pain scales are available to help measure and document pain. Nurses should always acknowledge the patient's verbalizations of discomfort or pain, even if they cannot do anything to provide immediate relief.

Interviewing techniques should be responsive to the individual qualities of the patient that may affect the ability of the nurse to collect pertinent data. The nursing assessment should consider religious and cultural practices, emotional barriers, desire and motivation to learn, physical and cognitive limitations, language barriers, and the financial implications of care choices. Educational level, level of consciousness, and language barriers are examples of possible limiting factors for an effective interview. The nurse should be aware of any cultural or religious factors that would impact the interview or assessment. Using available medical records, the interviewer should try to identify potential barriers prior to the encounter so she can bring the necessary tools to facilitate the interview (e.g., a translator for a patient with a language barrier or a writing board for a patient who is unable to communicate verbally).

Assessment of the spiritual domain is another important aspect of competent nursing care. **Spirituality** involves a person's inner resources and values that guide and give meaning to life. It can contribute profoundly to a person's sense of strength, hope, reassurance, and security. **Spiritual needs** are those items stemming from an individual's belief system that provide comfort by connecting the person with others and with whatever supreme power the individual ascribes to (Craven et al., 2017). All areas of nursing assessment provide important data for developing nursing diagnoses that address the patient's spiritual needs. Assessment findings that indicate anger, doubt, loss of hope or purpose in life, questions about moral and ethical implications of a therapeutic regimen, or self-blame are manifestations of spiritual distress in the patient. Accurate assessment of the patient's spiritual well-being contributes to the development of a holistic plan of care for the patient.

Observation

Craven et al., (2017) note that **Observation** begins from the first moment the nurse and patient meet. The nurse uses all sensory modalities to continually observe the patient's state of health, combining the observed facets with her in-depth knowledge of nursing care throughout the duration of the interaction.

Visual clues can give insight to a patient's level of comfort or discomfort. Visually observed data can include signs of distress such as pallor or shortness of breath. The patient's body size, hydration, and nutritional status can be seen. A patient's manner and interactions with others are also important components to an assessment.

A keen sense of smell is also used when assessing a patient. Body or breath odors may indicate an underlying physical condition. For example, foul-smelling breath may signify an oral or pulmonary infection, poor dentition, or even esophageal or gastric cancer. A fruity breath odor may indicate a metabolic disorder such as ketoacidosis in diabetes mellitus. Alcohol on a patient's breath may be one explanation for physical findings. These findings may also alert the nurse to possible self-care deficits.

Hearing is important in order to listen to any physical abnormalities that may need further investigation, such as in an abdominal assessment. Listening to what the patient says is not only important in terms of subjective data collection but can also give insights into the patient's level of mental capability, mood, consciousness, and education. These are aspects to consider, especially when planning teaching strategies for the patient and family.

Touch is used not only as a part of physical assessment, but also to provide compassion and reassurance. Touch usually begins with an introductory handshake between the patient and the nurse. Even this brief encounter can provide the nurse with valuable information such as the patient's skin temperature and texture and the strength of their grip. Be aware of any cultural beliefs of the patient before initiating touch.

Physical examination

The **nursing physical examination** verifies information uncovered in the **nursing history** interview and contributes new **objective data**. A physician's physical assessment focuses on pathology and etiology (cause of disease). The nursing exam is used to verify and expand on the data that has been gathered during the nursing interview.

The nursing physical exam and observation of the patient contribute much of the objective data to the nursing evaluation of the patient. In comparison to the subjective data surrounding a patient's perception of his or her situation, the objective data compiled provides information that can be measured and validated by others. Some nurses in expanded roles perform comprehensive physical examinations. All nurses conduct selected aspects of physical assessment for nursing purposes.

Physical measurements of any patient must include vital signs. Height and weight may also be included. In addition, the patient's heart, lungs, and abdomen should be assessed. Assessment of other systems may be appropriate, depending on the patient's condition and health problems.

Physical assessment of a patient is performed in four phases: inspection, auscultation, palpation, and percussion. The steps of a general assessment will be covered briefly at this time, with a more detailed explanation later in this chapter as it relates to the gastrointestinal patient.

Table 8.2. Steps to the Physical Assessment
1. **Inspection** is concentrated watching. It is close, careful scrutiny, first of the individual as a whole and then each body system. During inspection, the patient's anatomic structures are considered and any abnormalities are noted.
2. **Auscultation** is listening to sounds produced by different structures of the body such as the heart and blood vessels, the lungs, and the abdomen. Certain sounds, such as congested breathing or bowel sounds, can be heard without a special device, but most body sounds are very soft and must be channeled through a stethoscope.
3. **Palpation** follows and often confirms points noted during inspection. Palpation applies the sense of touch – via fingertips or the palm of the hand – to assess texture, temperature, moisture, consistency, organ location and size, fluid, and any swelling or masses. Light palpation always precedes deep palpation. Palpation always follows auscultation in the assessment of the the abdomen, as to not disrupt bowel sounds.
4. **Percussion** employs the examiner's sense of hearing to gather data. The examiner taps the patient's skin with short, sharp strokes to assess the underlying structures. The strokes yield a palpable vibration and a characteristic sound that depicts the location, size, and density of the underlying organ.

Review of records and diagnostic reports

Lab values and diagnostic reports are important measures that guage a patient's wellness (Alfaro-LeFevre, 2013). These data points provide vital information for the formulation of a medical diagnosis. They are also very useful in developing the nursing diagnosis and formulating a plan of care.

A nurse's role often involves more than the review and interpretation of data. A nurse generally supervises patient teaching and appropriate preparation for diagnostic tests. Diagnostic work is an interdisciplinary function that involves coordination and communication among nurses, physicians, and staff from the laboratory, radiology department, and diagnostic specialty units. The nurse's role is often pivotal in the organization of services and the acquisition of patient diagnostic data.

Collaboration with colleagues to establish the database

Nurses often consult and collaborate with many specialty team members such as physicians, pharmacists, physical therapists, case managers, counselors, and spiritual advisors, as well as specialist members of the primary care team like technicians and patient care assistants. A collaborative care approach can provide continuity of care and improve patient outcomes. Communication with other team members is imperative as the patient is evaluated and re-evaluated. It is important for team members to constantly review data collected by their colleagues. Information gathered from other team members can help validate one's own data about the patient.

Step 2. Identifying cues and making inferences

The subjective and objective data the nurse identifies act as cues. Cues give the nurse insights into the patient's health or illness. The interpretation, perception, or conclusion drawn about the cue is called an **inference**.

Step 3. Validating data

To keep data free from error, bias, and misinterpretation, it must be confirmed or verified periodically. The act of verifying or confirming data is also known as validation. To verify data, return to the five steps of data collection (1. interview, 2. physical examination, 3. observation, 4. review of records and diagnostic reports, and 5. collaboration with colleagues) and review the data from the various sources used to establish the database. Does information strengthen the initial impression or point out inconsistencies? Review records and test results to compare information with acquired data. Do the sources of information appear reliable? Any discrepancies will require further investigation and re-evaluation.

Assumptions occur when information has been taken for granted without being verified. The process of validating data helps the nurse expose assumptions and make corrections (Alfaro-LeFevre, 2013). Validation becomes particularly necessary when data discrepancies exist. For example, a nurse who receives verbal communication of no known allergies, yet notes a recorded allergy to penicillin, should validate the patient's drug allergies. Validation is also critical when the source of the assessment data may not be reliable or when serious harm to the patient could occur as a result of inaccurate data.

Step 4. Clustering data

The data are then clustered into groups of information that help to identify patterns of health or illness. A variety of frameworks exists for the orderly collection and recording of assessment data. To approach a nursing diagnosis, a nurse clusters data about a patient's strengths, talents, and functional health patterns, as well as information derived from the physical assessment of the patient. Physical examination is extremely valuable to the assessment of the patient and is separated from functional assessment for organizational purposes only.

Step 5. Identifying patterns and testing first impressions

After clustering data into groups of related information, initial impressions of patterns of human functioning will become apparent. These impressions should be tested to determine if the patterns are as they appear. Testing first impressions involves deciding what is relevant, making tentative decisions about what the data may suggest, and focusing the assessment to gain more information to fully understand the situations at hand. Combining subjective and objective information from health care assessments and the patient and family can help confirm or invalidate initial data. Deciding what is relevant or irrelevant will continue the process to relay appropriate information to other members of the team verbally and through written documentation.

Step 6. Reporting and recording data

Data collected in an assessment should be documented in the medical record and verbally reported to other members of the health care team. Data are of little benefit unless effectively communicated in a timely manner. Acceptable timeframes for communication will depend on the patient's condition. Critical information must be recognized and communicated immediately to ensure timely intervention.

Documentation of patient data fulfills the standard of care that requires the communication of a patient's health status. Data should be recorded legibly, concise grammar should be used, and only standard medical abbreviations should be included. To speed data retrieval, the data should be formatted under headings and organized categorically whenever possible. Some information is more appropriately recorded in narrative form, whereas other information should be charted on a flow sheet.

Documentation of patient data must comply with certain legal requirements. Written records are the most readily acceptable as evidence in a trial. Specifically, the nurse must chart everything observed, carried out, changed, taught, evaluated, or initiated. The database should contain descriptive, subjective, and objective information as supported by documented facts.

Electronic medical records have replaced much of the hand-written documentation required by nurses, and many institutions have adopted electronic decision-support systems. Decision-support systems are applications that analyze patient data and provide recommendations/algorithms to guide clinicians in treatment/care planning. Computerized systems have been associated with improved communication within the health care system and increased safety during the delivery of clinical care. However, electronic systems do not replace nursing expertise and critical thinking (Alfaro-LeFevre, 2013). It is important to remember that electronic records may be audited for purposes of quality assurance and reimbursement. To avoid ambiguity when documenting data derived from a conversation or interview, the patient should be quoted verbatim or the entry should be noted as a paraphrase. Similarly, clarifying errors avoids miscommunication of data. A nurse can correct a written error by striking the notation with a single line, enclosing it in parentheses, writing “error” above it, and initialing it. For more detailed directions regarding documentation, specifically as it applies to the GI nurse, see the SGNA’s *Guidelines for Nursing Documentation in Gastrointestinal Endoscopy* (2013).

The gastroenterology nursing assessment

Because of the vague nature of many symptoms of gastrointestinal (GI) disorders, digestive diseases are among the most difficult to assess. Patients with illnesses unrelated to the GI tract may nonetheless present with presumed GI-associated symptoms such as nausea, vomiting, abdominal pain, altered appetite, and/or bowel changes. On the other hand, certain GI disorders may produce symptoms that are extraintestinal. For example, one of the most common extraintestinal manifestations of inflammatory bowel disease is arthritis. To complicate matters further, some digestive symptoms may not be disease-related at all but may be a side effect of medication. For this reason, all complaints are potentially meaningful and therefore merit attention.

The nursing assessment of the gastroenterology patient follows the same guidelines as the general assessment, with an added focus on GI data collection. The steps for a focused GI assessment include the interview, physical exam, observation, and review of records and diagnostic reports.

Interview

The first objective is to arrive at a clear statement of the patient’s health problem or chief complaint. The nurse must be alert to manifestations of pain, as evidenced by both verbal and nonverbal cues (e.g., grimacing). The nurse will ask the patient to describe the location, intensity, and any radiation of pain, along with any precipitating or aggravating factors. The nurse should also determine whether the patient has symptoms that are known to be associated with certain gastrointestinal conditions. For example, testicular atrophy, gynecomastia, and alopecia may be associated with hepatic cirrhosis.

Commonly experienced gastrointestinal symptoms should be evaluated. These include nausea, vomiting, diarrhea, constipation, appetite, significant weight loss or gain, dysphasia (difficulty swallowing), early satiety, odynophagia (painful swallowing), chest or abdominal pain, jaundice, and the passage of blood in vomitus or stool. The timing of symptoms in relation to the ingestion of food is also important. Ingestion of food constitutes physiologic stress to the gut, much the same as exercise can apply stress to the heart and elicit symptoms in an ailing heart.

Equally important is the pattern with which symptoms occur. Relevant history of GI symptoms, including the duration of the diagnosed GI problem (if it has been diagnosed), and any evidence of complications stemming from GI disease should be documented. The nurse should establish the patient’s normal GI status, clarify how the patient’s status has changed, and note any strategies the patient may have used to manage the problem.The presence of risk factors for GI disease should be explored and documented. Possible familial tendency toward a GI disease, as manifested in relatives, should lead the nurse to consider a genetic testing referral (Goodman & Chung, 2016). A genetic predisposition to a disease is a collaborative problem that would require input of other members of the multidisciplinary team.

Nutritional assessment is an important aspect of a comprehensive GI assessment. The provision of adequate nutrition and its ability to be metabolized is vital to the maintenance and repair of the body. Alteration in nutritional status may be manifested by recent significant weight loss or gain; overweight or obese appearance; underweight or malnourished appearance; decreased energy levels; altered bowel patterns; and altered appearance of the skin, hair, teeth, and mucus membranes. Documentation should include patterns of consumption and elimination, relationships between diet and symptoms (e.g., the development of epigastric pain upon ingestion of spicy food), diet and medications, diet and lifestyle, and diet and emotional states.

Nutritional problems can be the result of the body’s inability to use ingested nutrients. Diseases such as inflammatory bowel disease can cause poor nutrient absorption. The inability to tolerate certain foods can cause malabsorption syndromes. Celiac disease, which is characterized by intolerance to the gluten found in wheat, rye, oats, and barley, can cause small bowel mucosal villi to atrophy, decreasing their absorptive abilities. Lactose intolerance occurs when there is a deficiency of lactase, the digestive enzyme that breaks down the sugar found in milk. Any obstruction of the intestinal tract caused by scar tissue, benign or cancerous growths, or structural abnormalities will alter nutritional status. Surgical removal of absorptive sections of the intestine will interfere with the ability to maintain a balance of the associated nutrients. Decreased bile salts and the absence of normal digestive secretions from disease (e.g., secondary to pancreatic insufficiency) or from medication regimes may also result in decreased utilization of nutrients. Increased excretion or protein loss characterizes many GI disorders, including abscesses, fistulas, Crohn’s disease, or ulcerative colitis.

Care should be taken to evaluate and address the nutritional needs of each patient. Periods of stress can alter nutritional needs and leave the patient at risk for nutritional deficiencies. Increased metabolic demands can be a response to the emotional stresses that accompany hospitalization or physical stresses such as infection or wound healing. In addition, increased metabolic demands frequently accompany conditions marked by increased tissue destruction, including cancer, ulceration, and necrosis. Psychological states can affect a patient's desire to eat. Anorexia nervosa is an eating disorder in which the patient refuses to eat due to fear of becoming overweight. Bulimia nervosa is an eating disorder in which the patient ingests an abnormally large amount of food and attempts to avoid gaining weight by purging (vomiting, over-exercising, and/or using laxatives or diuretics).

Nutritional problems can also include conditions of weight gain. Obesity can be a physiologic stress to the body that imposes physical limitations on a patient, inhibiting mobility and putting the patient at increased risk for certain conditions such as heart disease.

Medications are frequently the cause of GI distress and can disrupt the normal uptake of certain medications. A medication profile should be obtained, including all medication and/or food allergies. The patient's past and present use of medications for GI problems should be explored. If the patient has experienced or is presently experiencing side effects from these medications, this fact should also be recorded. Likewise, it is important to evaluate whether the patient is experiencing or has experienced the desired effects of the medication. Finally, current use of other prescription and nonprescription medications should be described. Many patients do not think to tell their doctor about dietary supplements/herbal medicines that they may be taking. Ascertaining this information is crucial, however, as many supplements have been found to cause drug-to-drug interactions, perioperative complications, and other harmful effects (SGNA, 2013).

Individuals identified as having nutritional risks during the initial interview should undergo a comprehensive nutritional assessment. This includes a dietary history and intake information, physical examination for clinical signs, anthropometric measures (measurements of the human body), and laboratory tests. Nutritional assessment skills are specialized, but identification of a patient at risk can begin a process of communication with other health team members who will ensure an appropriate evaluation.

Routine laboratory tests are of particular value in nutritional assessment because they are objective, can detect preclinical nutritional deficiencies, and can be used to confirm subjective findings. Laboratory indicators of nutritional status are hemoglobin, hematocrit, cholesterol, triglycerides, total lymphocyte count, and serum albumin. Glucose, low- and high-density lipoproteins, prealbumin, transferrin, and total protein levels also provide meaningful information (Jarvis, 2016).

Physical examination and observation

This section describes techniques used in inspection, auscultation, palpation, and percussion of the abdomen. For the purposes of a comprehensive GI assessment, inspection and auscultation should always precede percussion and palpation. This order is different from the suggested sequence for the general physical assessment of a patient, but it is particularly important when assessing the abdomen. Stimulation of the abdomen caused by percussion and palpation affects normal bowel activity.

Observation: Observation of the gastroenterology patient in the clinical setting can provide information that can be combined with subjective and objective data to create an individualized plan of care. Follow guidelines suggested for the observation of the general patient and apply them to GI concerns.

Inspection: The examiner should note the patient's breathing pattern (chest versus abdominal breathing) and be alert for scars, distended veins, and signs of abdominal trauma that may be relevant to the present illness. In addition, striae, which may evidence significant weight loss, should be noted. Upon inspection of the contour, the examiner should observe symmetry of the abdomen (above and below the umbilicus). The abdomen is normally convex (slightly rounded) but may be flat in an athletic person. Distention will present as a rounded contour. The umbilicus is useful as an aid when distinguishing causes of distention. It may be everted in ascites, whereas it remains unchanged in gaseous distention. Peristaltic waves can sometimes be observed, particularly if the patient is thin. This observation becomes significant when an obstruction is present.

Auscultation: Auscultation of the abdomen must be performed before palpation or percussion for a GI comprehensive assessment, as it may stimulate intestinal activity and change the quality or frequency of bowel sounds (Craven et al., 2017). Bowel sounds originate from the movement of air and fluid through the small intestine. The nurse should observe the frequency and rhythm of intestinal peristaltic sounds by listening with the diaphragm of a stethoscope. Bowel sounds vary considerably in healthy individuals. Normal bowel sounds do not rule out bleeding or other pathology. Normal bowel sounds are soft, high-pitched sounds. Because bowel sounds are normally intermittent, it is important to listen in each quadrant for a total of four to five minutes before declaring that no bowel sounds are present. Bowel sounds should be documented as normal, hypo/ hyperactive, or absent. The nurse should assess for an abdominal bruit (possible abdominal aortic aneurysm [AAA]) before attempting deep palpation.

Table 8.3. Significant Auscultation Findings
1. **Bruits**, which may indicate cardiovascular abnormality rather than GI pathology.
2. High-pitched **tinkles** and peristaltic **rushes**, which are audible when intestinal obstruction occurs.
3. Decreased or absent bowel sounds, which can suggest paralytic ileus, gangrene, peritonitis, or inflammation.

Percussion: Percussion is based on much the same concept as ultrasound. Sound waves produced by striking one object against another allow an examiner to note the presence of air, fluid, and solid matter in the abdomen. To perform percussion, the examiner should place the distal phalanx of the middle finger of the nondominant hand flat against the area to be percussed. This finger is the only part of the examiner's hand that should be in contact with the abdomen. With the tip of the flexed middle finger of the dominant hand, the examiner should sharply strike the distal joint of the phalanx on the abdomen. Sounds generated by this action reflect the size, density, and characteristics of the underlying abdominal structures and can be described as flat, dull, resonant, hyperresonant, or tympanic. Tympany generally denotes large amounts of gas in the stomach and intestines, whereas dullness is normally noted over a full bladder, a mass, or organs such as the liver or spleen. The presence of fluid is not always easy to differentiate from other conditions but is suspected when there is abdominal distention with bulging flanks, a fluid wave, and shifting dullness when the patient is turned on one side.

Palpation: Palpation is the last physical assessment performed during the examination of the abdomen (Craven et al., 2017) for a comprehensive GI assessment. Touch is the most useful sense in identifying tenderness, temperature changes, and masses. Both shallow (light) and deep palpation should be performed during assessment. Light palpation provides the nurse with information regarding firmness of abdominal muscles and degree of abdominal distention and is performed using one hand. Deep palpation may be used to assess deep abdominal masses and organs such as the liver, kidney, and spleen and is performed using two hands. Findings suggestive of digestive diseases include fluid waves, abnormal organ size, rebound tenderness, masses, and hernias. Examination of the size, shape, position, consistency, mobility, and tension of the major organs concludes physical assessment of the abdomen. Abdominal tenderness should be assessed and characterized.

Review of records and diagnostic reports

The gastroenterology nurse may be working in a variety of settings to assist with data collection and/or interpretation of lab values and diagnostic tests. Nursing functions include not only the understanding of test results, but also the ability to use diagnostic information to guide the nursing care of a patient. In the endoscopy setting, lab values and prior test results may change the focus of the GI procedure or the approach of the endoscopic exam.

The following table lists some of the tests and studies that are often used in the diagnostic work-up of the gastroenterology patient.

Table 8.4. Potential Diagnostic Data in Gastroenterology
Laboratory Tests
• CBC: Hb, HCT, differential leukocyte count • PT, PTT, INR • Electrolytes • Ketones and protein • Amylase • Lipase • Creatinine • Serum cholesterol • LDH, AST, ALT, ALP • Glucose • Bilirubin • Serum albumin • CPK • HBV, HCV • FOBT • Stool culture • FE, TIBC, Ferritin • Vit B_{12} • Vit D • HP Breath/stool tests • Celiac panel • IBD panel • CEA, CA-19-9
Imaging Studies
Type 1: Simple radiographic pictures in which structure are enhanced either by natural differences in radio density or by differences enhanced through instillation of a contrast agent. • Chest radiograph • Flat plate of abdomen • Upper GI series, may include esophagogram or SBFT • Lower GI series/barium enema • Contrast radiographs • Percutaneous transhepatic cholangiogram (C) PTC • Defecography
Type 2: A cross-sectional look at internal organs by sound waves, x-ray beams, or measurement of magnetic resonance signals. • Computed Tomography – CT scan • Ultrasonography • Magnetic Resonance Imaging – MRI/MRCP

Type 3: Pictures of internal organs detected by radioactive tracers and gamma cameras after the intravenous injection of a radioisotope.
- Nuclear imaging scan
- HIDA scan
- Gastric emptying scan

Endoscopic Exams
- Esophagogastroduodenoscopy
- Colonoscopy
- Proctosigmoidoscopy
- Anoscopy
- Endoscopic Retrograde Cholangiopancreatography
- Endoscopic Ultrasound (EUS)
- Single/double balloon enteroscopy

Tests of Gastrointestinal Motility
- Manometry

Microscopic Examination of Tissue
- Liver biopsy
- Cytology
- Mucosal biopsy

Test of Digestive Function
- Gastric analysis
- Fecal analysis

Misc. Tests
- pH studies
- Bernstein test
- Virtual colonoscopy
- Video capsule
- Positron emission tomography (PET scan)
- Hydrogen breath test

Nursing care of the patient in the endoscopy setting

Today, the care of the patient undergoing an endoscopic procedure has become more technologically complex and more comprehensive. As the field of gastroenterology advances, the role of the GI nurse expands. Nurses who work in GI perform many roles, including preoperative assessment, care of the patient during the procedure (e.g., providing gastroenterologists with technical support), and monitoring the patient as he or she recovers post-procedure. In certain institutions, the GI nurse is also responsible for administering anesthesia.

Assessment in the gastrointestinal endoscopy setting

Pre-procedure evaluation is required of all patients prior to endoscopic procedures, regardless of setting. The gastroenterologist should have performed a recent history and physical. The pre-operative nurse will perform a focused assessment, which will include measuring vital signs, performing any necessary labwork, and reviewing current medications, allergies, and NPO status. If the patient is to receive anesthesia during the procedure, he or she will be seen by an anesthesiologist, who will assess the patient's airway, review his or her medical status, identify potential risks, and assign the patient an ASA class. It is important to note that, even when moderate sedation is planned, deep sedation may occur. Thus, all providers must have the training and tools required to rescue patients who become sedated deeper than was intented (e.g., airway support, intubation equipment, reversal agents, cardiopulmonary resuscitation).

SGNA has developed specific guidelines for the documentation of care provided for the patient in the gastrointestinal endoscopy setting. The *Statement on the Use of Sedation and Analgesia in the Gastrointestinal Endoscopy Setting* (SGNA, 2017) offers guidelines that delineate the essential components of the nursing documentation and provides an excellent example of a focused nursing assessment. Nursing assessment occurs in the Pre-Procedure, Intra-Procedure, and Post-Procedure stages of the endoscopic procedure. Data to be gathered during each of the stages are listed below:

- **Pre-Procedure:** baseline vital signs, oxygenation status, pain assessment, falls risk assessment, pregnancy status, results of bowel prep, and nothing by mouth (NPO) status. Discuss the use of anticoagulants, as well as any implantable devices such as an AICD/pacemaker or stimulator.
- **Intra-Procedure:** vital signs, oxygenation status through the use of pulse oximetry, attention to the I.V. (site, type, and amount of fluids administered), assessment of the patient's overall comfort, and clinical status.
- **Post-Procedure:** vital signs, level of consciousness/mental status, airway/respiratory status, oxygen saturation, gag reflex if applicable, feel of the abdomen (e.g., a distended or tense abdomen may be due to accumulation of gas from insufflation).

All three phases of assessment should be tailored to the patient's age, the patient's individual needs, and the procedure performed. Throughout the endoscopy procedure, it is the nurse's duty to provide the patient with care that maintains or improves wellness and comfort.

CASE SITUATION

Eleanor Russell, age 73, arrives in the gastroenterology department holding area. She originally went to the emergency room (ER) complaining of nausea and vomiting, severe right upper quadrant abdominal pain, and sternal pain with pressure radiating to her back. The ER nurse communicates that Eleanor described a family history of gallbladder disease and was diagnosed with cholecystitis several months ago. In addition, he reports that Eleanor said her symptoms became worse after eating a peanut butter sandwich and a milkshake yesterday for lunch. Her electrocardiogram (ECG) is normal, and she is currently on Coumadin for thrombophlebitis in her right leg. The ER nurse reports that

Eleanor's stool is negative for occult blood, but that her prothrombin time (PT) and partial thromboplastin time (PTT) are abnormal. He also adds that the patient is febrile.

Eleanor has been diagnosed with cholecystitis and cholelithiasis. The physician awaits the results of further laboratory studies and an abdominal ultrasound examination to confirm or rule out gallstone obstruction of the common bile duct. With this, the physician asks the nurse to prepare for endoscopic retrograde cholangiopancreatography (ERCP).

The nurse notices that Eleanor's face is rigid and pale. She is scanning her environment nervously, and her eyes are open wide. Her hands are cool, moist, and trembling. A large bruise near her left wrist is noted. Her respirations are rapid and shallow as she mentions, "I've never had anything like this before. My daughter was supposed to have talked to my doctor about my diet."

Points to think about

1. What steps should a nurse carry out in assessing Eleanor before diagnostic ERCP?
2. What subjective data are evident thus far? What objective data?
3. Which data might need validation?
4. Which data require documentation in Eleanor's records?
5. What additional pre-procedure assessment should the nurse undertake before preparing for the ERCP procedure?

Suggested responses

1. The nursing assessment prior to Eleanor's ERCP should include the following steps:
   a. Collect data.
   b. Validate data.
   c. Organize data.
   d. Identify patterns and test first impressions.
   e. Report and record data.
2. The following subjective and objective data are evident.
   a. Subjective data: Nausea and sternal pain with pressure radiating to her back; symptoms became worse after eating a peanut butter sandwich and a milkshake.
   b. Objective data: Eleanor's ECG is normal; her face is rigid and pale; she is scanning her environment nervously with eyes open wide; her hands are cool, moist, and trembling; she has a large bruise near her left wrist; respirations are rapid and shallow; she has a family history of gallbladder disease and a personal history of cholecystitis; a nursing assessment reported fever, occult-negative stool, and an abnormal PT and PTT.
3. The ER nurse communicates that Eleanor was diagnosed with cholecystitis several months ago, her symptoms became worse after lunch yesterday, her stool is negative for occult blood, her PT and PTT are abnormal, and she presents with fever. This information requires further investigation and validation.
4. The nurse must document everything observed, carried out, changed, taught, evaluated, or initiated.
5. During a pre-procedure phase, the nurse should perform and document a nursing assessment. The assessment factors should include physical factors; psychosocial factors; current medications and treatment; and previous medical, anesthetic, and drug history. Review of Eleanor's symptoms and history will supply any pertinent information to be documented.

REVIEW TERMS

assessment, collaborative problems, therapeutic communication, data, database, nursing physical examination, nursing history, nursing interview, nursing process, objective data, observation, subjective data, validation, collaborative practice, spirituality

REVIEW QUESTIONS

1. A nursing assessment:
   a. Is a systematic approach to nursing care.
   b. Is always comprehensive.
   c. Is a process of identifying a patient problem.
   d. Should precede a nursing history.
2. Which statement is true of a general physical assessment of the abdomen?
   a. The order in which inspection, palpation, percussion, and auscultation are performed is only important if the patient is in pain.
   b. Percussion and palpation should be performed before auscultation and inspection.
   c. Bowel sounds are normal if none is heard in four abdominal quadrants over a period of four to five minutes.
   d. Bowel sounds characterized by high-pitched tinkles and peristaltic rushes are abnormal.
3. A pre-procedural assessment includes:
   a. Determination of baseline vital signs.
   b. Verification of current medications in use.
   c. Verification of nothing by mouth (NPO) status.
   d. All of the above.
4. Validation is the act of:
   a. Clarification.
   b. Verification.
   c. Repeating a patient's responses twice.
   d. Checking to be sure a nursing history was taken.

5. Which of the following might the gastroenterology nurse record as objective nursing assessment data concerning a patient who presents in the ER with apparent biliary colic?
   a. A medical diagnosis of choledocholithiasis.
   b. "Patient is anxious."
   c. "Patient ate a peanut butter sandwich and some Swiss cheese yesterday at lunch."
   d. "Patient is scanning her surroundings with wide open eyes."
6. By which method(s) could a patient's medication history be validated?
   a. By asking the patient what medications he or she takes.
   b. By reading the prescription labels on the bottles of medicines the patient provides.
   c. Both a and b.
   d. Neither a nor b.
7. A nurse's sense of vision, hearing, smell, and touch are important features of:
   a. The observation of a patient.
   b. Collection of subjective data.
   c. A patient diagnostic review.
   d. Patient interview and data analysis.
8. The purpose of nursing assessment is to:
   a. Identify underlying pathology.
   b. Identify teaching needs.
   c. Identify human responses to medical conditions, treatments, and changes in activities of daily life.
   d. Analyze data to identify actual or potential health problems.
9. A multidisciplinary team approach to patient care is characterized by:
   a. A coordination of care among practitioners.
   b. A continuous flow of patient care service.
   c. A collaborative approach to manage health problems.
   d. All of the above.
10. A patient returns to the clinic after two weeks of taking a new prescription for reflux. The most appropriate assessment for a follow-up visit is:
   a. A comprehensive assessment.
   b. A focused assessment.
   c. A time-lapsed reassessment.
   d. A limited assessment.

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Chapter 9

NURSING DIAGNOSIS

This chapter discusses the nurse's role in diagnosing actual or potential health problems specific to patients with gastrointestinal (GI) disorders. Nursing diagnoses are defined and differentiated from medical diagnoses. Examples of functional health patterns and the subsequent nursing diagnoses for gastroenterology patients are presented.

Learning objectives

After reviewing the content of this chapter, the gastroenterology nurse should be able to:

1. Define nursing diagnosis.
2. Differentiate between nursing diagnosis, medical diagnosis, and collaborative diagnosis.
3. Formulate a nursing diagnosis using an approved classification system.
4. Discuss actual, potential, and possible nursing diagnoses applicable to gastroenterology patients.

The term **nursing diagnosis** first appeared in the literature during the 1950s. Because of the implications associated with the word "diagnosis," and the idea that only physicians made diagnoses, nurses were reluctant to use the term. They were more comfortable using the word "problems" to refer to a patient's need requiring nursing intervention. Periodically through the 1960s and 1970s, the literature introduced descriptions of nursing diagnoses. Finally, an article by Lester King, MD, in the *The Journal of the American Medical Association (JAMA)* refuted the idea that only physicians could diagnose, thereby clearing the way for more exploration of the concept of nursing diagnosis (McFarland & McFarlane, 1996).

King outlined three criteria that must be present to make a diagnosis:

1. A pre-existing series of categories or classes that provide a reference for the diagnosis.
2. An entity to be diagnosed.
3. A judgment that the assessed response or phenomenon belongs to a particular class or category.

At that point, nurses began to develop a classification system for nursing diagnoses. This work was started through conferences on classification of nursing diagnoses and is still in progress today through NANDA International (NANDA-I) – previously called the North American Nursing Diagnosis Association (NANDA). Other individuals and groups have played a role in identifying and testing human response patterns that result in a diagnostic label.

NURSING DIAGNOSIS DEFINED

Nursing diagnosis is a clinical judgment about actual or potential individual, family, or community experiences and responses to health problems and life processes. A nursing diagnosis provides the basis for selection of nursing interventions to achieve outcomes for which the nurse has accountability (Herdman & Kamitsuru, 2014). Gallagher-Lepak (2014) made a clarifying distinction that physicians use the International Classification of Disease (ICD-10) taxonomy and that mental health professionals use the Diagnostic and Statistical Manual of Mental Disorders (DSM-V). Nurses, meanwhile, manage human responses to health problems/life processes and use the NANDA International classification system.

Nursing diagnosis provides a useful and practical method for organizing nursing knowledge so that the expertise of nursing can be defined, clarified, and developed.

Nursing diagnoses are based on data obtained from the patient during the course of a nursing assessment. Assessment is the initial component of the nursing process. It should always be done when the patient is admitted to the pre-procedure area. The plan of care is initiated when the nurse interprets assessment findings and determines pertinent nursing diagnoses.

A nursing diagnosis provides a concise statement of the interpretation of data collected. This concise statement describes the nature, source, and manifestations of health changes the nurse is licensed to identify and treat through independent and interdependent nursing interventions. Components of a nursing diagnosis are name of diagnosis; description of diagnosis; observable manifestations; environmental, physiological, psychological, genetic, or chemical risk factors; and factors that may contribute to related nursing diagnoses.

Nursing diagnosis has become a critical link in the nursing process. The process of determining a nursing diagnosis requires clinical reasoning and judgment, which result in labeling the patient's health problem. This labeling is the product of collecting data, interpreting the information, grouping and clustering related facts, and assigning a name to the groupings. The same problem-solving skills are required to analyze the data obtained during an assessment and to make decisions about the problems that prevent patients and families from responding in a normal, healthy fashion.

Carpenito (2014) has described nursing diagnosis as both process and structure. Nursing diagnosis as a process refers to the second step of the nursing process, when the nurse analyzes the data collected during the assessment step and evaluates the health status. Often, the diagnostic statement that results from the process of nursing diagnosis is also referred to as a nursing diagnosis. Nursing diagnosis as a structure refers to specific labels or titles assigned to diagnostic categories that describe the health states nurses can legally diagnose and treat.

From either perspective – process or structure – the value of the nursing diagnosis is that it enhances communication among nurses and the health care team by standardizing nursing language (Oreofe & Oyenike, 2018). It provides a scientific basis for nursing practice so that desired patient outcomes and planned interventions are consistent with the patient's health problems.

NURSING DIAGNOSIS VERSUS MEDICAL DIAGNOSIS

The process used to identify a diagnosis or a patient problem is the same for nursing and medicine. Both nursing and medical diagnoses are concerned with gathering, sorting, interpreting, and analyzing data. **Medical diagnosis** focuses on identification of a disease based on pathology and etiology, whereas nursing diagnosis focuses on the patient's present **health problems**, strengths and limitations, and methods of adapting to health problems. Table 9.1 provides examples of how nursing diagnoses and medical diagnoses differ.

NANDA-I identifies three types of nursing diagnosis (Herdman & Kamitsuru, 2014).

1. **Problem-focused nursing diagnosis** is a clinical judgment concerning an undesirable human response to a health condition/life process that exists in an individual, family, group, or community.
2. **Risk nursing diagnosis** is a clinical judgment concerning the vulnerability of an individual, family, group, or community for developing an undesirable human response to a health condition/life process.
3. **Health promotion nursing diagnosis** is a clinical judgment concerning motivation and desire to increase well-being and to actualize human health potential. These responses are expressed by a readiness to enhance specific health behaviors and can be used in any heath state. Health promotion responses may exist in an individual, family, group, or community.

Syndrome is a clinical judgment describing a specific cluster of nursing diagnoses that occur together and are best addressed together through similar interventions (Herdman & Kamitsuru, 2014).

Table 9.1. Nursing Diagnosis Versus Medical Diagnosis

Nursing diagnosis	Medical diagnosis
Focuses on care.	Focuses on etiology.
Records the human response to actual or potential health problems.	Diagnoses a disease or medical condition.
Identifies patient problems and situations the nurse is licensed to treat.	Indicates a course of treatment the physician is licensed to treat.
Represents the intention that nurses will perform interventions to alleviate, diminish, modify, or prevent a state of unwellness and maintain an optimal health state.	Constitutes the intention of prescribing specific treatments to cure the disease or reduce injury.
May change from day to day as the patient's response to therapy, health, and illness change.	Remains the same as long as the disease persists.

COLLABORATIVE PROBLEMS

Carpenito (2014) introduced the Bifocal Clinical Practice Model to describe the two foci of clinical nursing: nursing diagnoses and collaborative problems. **Collaborative problems** are physiologic conditions that nurses monitor to identify onset or change of status. These are problems for which the nurse identifies a need to work with other members of the health care team toward resolution. Collaborative diagnoses require both nursing and medical intervention to diagnose, prevent, or treat.

Medical diagnoses, medical pathologies, diagnostic tests, treatments, and equipment names are not nursing diagnoses. Although not nursing diagnoses, the following data are considered when identifying health problems: therapeutic patient needs, therapeutic patient goals, a single sign or symptom, and invalidated nursing inferences.

FORMULATING A NURSING DIAGNOSIS

Chapter 8 discusses collection of data. It is essential that nurses understand what data are significant and how to obtain that important information. Knowledge of nursing science and other related biopsychosocial sciences is important to interpret the data and to accurately formulate the nursing diagnosis. Interpretation means sorting information and making a hypothesis for the occurrence of related events, conditions, or behavior. The nurse uses clinical inferences with inductive reasoning when looking at functional patterns and observing patient responses. The nurse then asks, “What does this mean?” or “Why is the patient saying this?”

Cues are explored, data are collected, and the cues are then clustered. Clustering involves making a judgment about whether cues are consistent or inconsistent. The conclusion may be that the evidence does not support the existence of an actual problem, but that the patient is at risk for the problem to occur (a potential problem). Inconsistencies may be caused by conflicting reports from other members of the health care team, the patient, or the family. Unrealistic expectations on the part of the nurse may be the result of inexperience or lack of knowledge.

The diagnosis statement gives the health problem as related to the etiology and as manifested by the signs and symptoms. An example would be constipation (problem) related to a diet high in dairy products (etiology) as evidenced by complaints of fullness, abdominal pain, and irregular bowel movement (signs and symptoms).

There are seven steps to a nursing diagnosis:

1. Establish database by collecting information from the patient and/or family, reviewing the medical record, and assessing the patient.
2. Analyze the data obtained by asking:
   a. Are there changes in the patient’s health status? (i.e., Is the patient healthy or unhealthy?)
   b. Does the patient’s medical record show values that deviate from the norm?
   c. Is the patient physiologically healthy?
   d. Is the patient psychologically healthy?
   e. Is the patient sociologically healthy?
   f. Is the patient spiritually healthy?
3. Organize the data by summarizing the pattern of the problems discovered:
   a. There is no problem, and intervention is unnecessary.
   b. There is an actual or potential problem.
   c. There is a clinical problem other than a nursing problem (a collaborative problem) that requires the nurse to consult with and sometimes cooperate with appropriate health care professionals.
4. Choose a diagnostic label specific to the patient. Check your facility’s policy in the use of diagnostic labels. Most facilities use NANDA-I’s list.
5. Identify the factors that appear to influence that patient, specific to the diagnostic label identified during the nursing assessment.
6. List defining characteristics, which include signs, symptoms, and other assessment findings.
7. Confirm the sufficiency and accuracy of the database by evaluating the appropriateness of diagnoses to nursing interventions and ultimately determining that other qualified practitioners would arrive at same nursing diagnosis.

Identifying specific nursing problems and prioritizing them helps to focus gastroenterology nursing practice and improve the quality of nursing care. Table 9.2 contains a partial list of approved nursing diagnoses classified by functional health problems. Included are some new and revised diagnoses released by NANDA-I. The classification system reflects areas in which assessment takes place.

Table 9.2. Potential Nursing Diagnoses for Gastroenterology Patients
Nutritional metabolic pattern
Body temperature, altered
Dentition, altered
Fluid volume deficit
Growth and development delayed
Infection, risk for
Nutrition, altered: less than body requirements
Nutrition, altered: potential for more than body requirements
Obesity*
Oral mucous membranes, altered
Overweight*
Skin integrity, impaired
Swallowing, impaired
Elimination pattern
Bowel incontinence
Constipation, chronic functional*
Diarrhea

Table 9.2. Potential Nursing Diagnoses for Gastroenterology Patients (continued)
Activity-exercise pattern
Activity intolerance
Mobility, impaired physical
Tissue perfusion, altered
Cognitive-perceptual pattern
Knowledge deficit
Sensory-perceptual alteration
Sleep-rest pattern
Sleep pattern disturbance
Self-perception/self-concept pattern
Anxiety
Body image disturbance
Fatigue
Fear
Hopelessness
Powerlessness
Self-concept disturbance
Role relationship pattern
Communication, impaired
Family process, altered
Grieving
Loneliness, risk of
Role performance, altered
Social isolation
Social interaction, impaired
Sexuality pattern
Sexual dysfunction
Sexuality patterns, ineffective
Coping-stress tolerance pattern
Adjustment, impaired
Coping, ineffective
Personal identity disturbed
Powerlessness
Self-concept, disturbed
Self-esteem, disturbed

**New in the 2015-2017 NANDA*

CASE SITUATION

Chapter 8 introduced Eleanor, who was medically diagnosed with cholecystitis and cholelithiasis. She is scheduled for a diagnostic endoscopic retrograde cholangiopancreatography (ERCP). During the pre-procedural assessment, certain subjective data were identified, including:

- Nausea.
- Sternal pain with pressure radiating to her back.
- Symptoms becoming worse after eating peanut butter, Swiss cheese, and a milkshake.

Objective data were also gathered, including:

- Normal electrocardiogram (ECG).
- Febrile.
- Face rigid and pale.
- Scanning environment nervously with eyes open wide.
- Hands cool, moist, and trembling.
- Large bruise near left wrist.
- Respirations rapid and shallow.

Points to think about

1. Based on observations thus far, what are at least three functional health problems for Eleanor?
2. According to Table 9.2, what are specific nursing diagnoses for Eleanor's health problems?
3. What type(s) of nursing diagnoses were formulated?
4. Is there a collaborative diagnosis for Eleanor? If interdependent nursing and medical interventions are necessary, what would the difference be in the nursing focus versus medical focus?

Suggested responses

1. Functional health problems the gastroenterology nurse might identify are:
   a. Health perception-health management pattern.
   b. Coping-stress tolerance pattern.
   c. Nutritional metabolic pattern.
   d. Cognitive-perceptual pattern.
2. Specific nursing diagnoses that correspond to these health problems might be:
   a. Functional health problem.
   b. Health perception
   c. Management pattern.
   d. Noncompliance with prescribed regimen.
   e. Self-perception and self-concept pattern.
   f. Anxiety.
   g. Nutritional metabolic pattern.
   h. Injury, potential for bruising, hemorrhage.
   i. Cognitive-perceptual pattern.
   j. Knowledge deficit related to disease and therapy.
3. Nursing diagnoses might include the following types:
   a. Actual problems: noncompliance with prescribed regimen, anxiety, knowledge deficit.
   b. Potential problem: injury, potential for bruising, hemorrhage.

4. In Eleanor's case the diagnosis of cholecystitis with cholelithiasis is a medical diagnosis. The medical diagnosis focuses on the medical history and physical examination, with the desired outcome being treatment of the disease. The nursing diagnosis focuses on collecting data related to functional needs, with the outcome being care of the patient.

REVIEW TERMS

collaborative diagnoses, health problems, medical diagnosis, nursing diagnosis

REVIEW QUESTIONS

1. Which of the following was not one of King's criteria for making a diagnosis?
   a. A preexisting series of categories to provide a reference.
   b. An entity to be diagnosed.
   c. The existence of a medical pathology.
   d. A judgment that the assessed phenomenon belongs to a particular category.
2. The organization responsible for classification of nursing diagnoses is the:
   a. American Nurses Association.
   b. NANDA-I.
   c. Society of Gastroenterology Nurses and Associates.
   d. American Medical Association.
3. "Cholecystitis with cholelithiasis" is an example of a:
   a. Collaborative diagnosis.
   b. Nursing diagnosis.
   c. Medical diagnosis.
   d. Medical history.
4. A syndrome is a:
   a. Problem attended to by all health care professionals.
   b. Clinical judgment describing a specific cluster of nursing diagnoses that occur together
   c. Behavior.
   d. Condition related to health.
5. A collaborative problem:
   a. Requires both nursing and medical intervention to diagnose, prevent, and treat.
   b. Requires cooperation between nurses responsible for pre-, intra- and post-procedure patient care.
   c. Involves the nurse in identification but not treatment.
   d. Is identified by more than one member of the health care team.
6. A nursing diagnosis states:
   a. Important assessment data from patient's health problem.
   b. The health problem as related to the etiology and as manifested by the signs and symptoms.
   c. Interdependent nursing interventions for signs and symptoms.
   d. Outcome criteria for evaluation.
7. Nursing diagnosis focuses on the patient's:
   a. Pathology and etiology.
   b. Pathophysiology.
   c. Present health problems.
   d. Health perceptions.
8. An example of a potential nursing diagnosis for a gastroenterology patient would be:
   a. Potential burn resulting from electrosurgery.
   b. Swallowing impaired.
   c. Fluid volume deficit, possible, caused by nausea and pain.
   d. Skin integrity, potential for impairment.
9. A nursing diagnosis for a patient admitted to the endoscopy unit for a diagnostic esophagogastroduodenoscopy (EGD) who is not responding to treatment for a gastric ulcer might be:
   a. Knowledge deficit.
   b. Sensory perception, altered.
   c. Ineffective therapeutic regimen.
   d. Airway clearance, ineffective.

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Chapter 10

PLANNING

This chapter will acquaint learners with the planning component of the nursing process. **Planning** is the development of goals to prevent, reduce, or eliminate problems and to identify nursing interventions that will assist patients in meeting outcomes or expected outcomes. Trends in care planning are discussed, followed by an elaboration of the purposes of planning; the responsibility for planning; and the activities involved in initial, ongoing, and discharge planning. The identification of treatment options, documentation of **nursing orders**, and outlining of care plans are also covered. Finally, considerations in planning care for gastroenterology patients are summarized.

Learning objectives

After reviewing the content of this chapter, the gastroenterology nurse should be able to:

1. List the advantages of planning patient care based on solid nursing diagnoses.
2. Compare nursing and medical plans of patient care.
3. Describe three basic types of planning, including who performs each type.
4. Describe four means by which nurses can expand their existing repertoire of treatment options.

TRENDS IN PATIENT CARE PLANNING

Whenever nurses respond to an actual or potential health problem by determining expected patient outcomes and by identifying nursing activities for preventing, reducing, or resolving health problems, they are planning patient care. A plan of care may be a formally documented plan, or it may be implied by a set of nursing interventions. The Joint Commission no longer requires that a formal nursing care plan be documented in a patient's hospital records. However, planning and documentation of care based on accepted nursing diagnoses, etiology, and scientifically sound intervention remain the hallmark of professional nursing practice. Furthermore, the Society of Gastroenterology Nurses and Associates, Inc. (SGNA) has delineated its own position statement, *Role Delineation of the Registered Nurse in a Staff Position in Gastroenterology* (2018), within the Standards of Clinical Nursing Practice and Role Delineations in the Gastroenterology Setting, to accomplish the following:

1. Establish nursing diagnoses.
2. Establish priorities and/or expected outcomes.
3. Document patient data to ensure continuity in care.
4. Manage follow-up care.
5. Collaborate with other health care professionals.

Rationale for care planning

Although formal written care plans are no longer required, the planning of patient care is essential. Once assessment data are obtained, a nursing diagnosis is made, and desired outcomes are identified, a plan of action must be developed. Planning is the deliberate, systematic process of identifying nursing actions directed toward achieving desirable outcomes and preventing, reducing, or resolving patient health problems. This process involves **setting priorities** for care, determining patient goals and expected outcomes, selecting nursing actions and strategies, and documenting the plan of care. This is done so that the problem causing the greatest danger, discomfort, or pain to the patient will be addressed first. For example, if a patient is anxious about an impending procedure, the gastroenterology nurse assesses the patient's fear and considers the types of interventions that will alleviate the patient's anxiety. One intervention to help decrease anxiety is for the nurse to explain the procedure in detail.

There are a number of important reasons for developing a plan of care:

1. A plan of care can identify patient problems that may be prevented, reduced, or resolved by nursing activities. It must be recognized that the nurse cannot solve all patient problems. Sometimes it is necessary for the nurse to seek assistance from other members of the health care team or the patient (or patient's significant others).
2. A plan of action helps the gastroenterology nurse assign priorities for care and select nursing interventions or actions that meet the unique needs of the individual patient.
3. A documented plan provides a means of communicating information that will help other members of the health care team provide continuity of care.
4. The planning of care based on nursing diagnoses uses a universal language with which nurses can communicate about health problems, thus helping to build on the existing knowledge base of nursing science.
5. The planning of care based on nursing diagnoses gives a professional quality and character, rather than merely vocational assistance, to the act of nursing. It serves to separate nursing and medicine, bringing the two professions into a collegial relationship in which nurses can demonstrate their unique knowledge base and contribute to their patients' health.

The planning of patient care also has economic importance. Medicare's system of prospective payment is based on diagnosis-related groups, and providers are under to include in ng pressure to render quality care while accepting decreasing compensation for increasing costs. More than ever, it is important for the nursing profession to delineate its product and to demonstrate that this product is cost-effective.

Nursing and medical plans of care

Nursing and medical diagnoses share certain characteristics. Both are abstractions derived from assessment and inferences based on scientific knowledge that summarize a cluster of signs and symptoms. Both nursing and medical plans of care, therefore, prescribe monitoring of signs and symptoms, both are instituted and refined following initial and ongoing assessments, and both prescribe measures based on bodies of scientific knowledge.

Because nursing and medical diagnoses differ (see Chapter 9), there are also differences between nursing and medical plans of care. Medical diagnosis identifies pathologic basis for illness, focuses on the physical condition of the patient, addresses actual problems, and uses a plan of care that has standard goals or treatments. Nursing plans of care address psychological, environmental, and socioeconomic factors contributing to disturbances in health, as well as biological factors. Moreover, because the act of diagnosis carries legal implications regarding the right to initiate treatment, nursing care plans based on nursing diagnoses focus on patient responses to medical treatment or on health problems nurses can independently treat. Thus, interventions prescribed in nursing care plans include only actions that nurses can lawfully perform.

Clinical pathways

Within literature, synonymous terms for clinical pathways include care pathways, general critical element pathways, standardized care pathways, multidisciplinary pathways of care, care maps, and integrated care pathways.

The evolution of health care delivery during the 1980s and 1990s demanded quality patient care with controlled cost and, thus, clinical pathways evolved. Hospitals needed to control costs to remain in compliance with the Centers for Medicare and Medicaid (CMS) payment system and reduced payments from insurance companies. Clinical pathways helped to decrease costs by standardizing patient care, which in turn facilitated improving care and patient safety, decreasing hospital length of stay, and decreasing mortality rates.

A clinical pathway is not a care plan, but in many ways they are similar. Based on a clinical asessment tool, specific patient outcomes are listed. Unlike nursing care plans that are usually developed by nurses, a clinical pathway is a multidisciplinary managed plan that is standardized, evidence-based, identifies an appropriate sequence of clinical interventions along a specific timeline, and plans out expected outcomes for a specified patient group. The goals of clinical pathways are to implement evidenced-based practice, improve processes, reduce duplication through standardization, and reduce variation in health care delivered.

A care coordinator or case manager, frequently a nurse, may be assigned to oversee the patient's care. Documentation of variances and continuous multidisplinary feedback provides a basis for pathway revision, patient care evaluation, and continuous process improvement.

There are multiple benefits to implementing clinical pathways. In addition to supporting the use of evidence-based practice, they encourage patient and family involvement; improve health care team comunication and use of their resources; support standards of care/practice; ensure quality of care which leads to quality improvement; provide a baseline for care; improve clinical outcomes; and reduce costs.

There are also some disadvantages to clinical pathways. These include the appearance of depersonalized care; possible increase in litigation risks; adaptability if patient condition deteriorates; possible lag time in the inclusion of new technology; and insufficient time and resources to dedicate to the process. Finally, the success of the clinical pathway is contingent upon the support of the entire health care team.

COMPREHENSIVE PLANNING

There are three basic types of planning that take place in the course of comprehensive care planning: initial planning, ongoing problem-oriented planning, and discharge planning.

Initial planning

Initial planning is usually performed by the nurse who admits the patient and completes a nursing history and physical examination. Standardized care plans, including computerized care plans, agency-developed plans, care and clinical maps, and textbook plans, should be tailored to the patient's needs and may be initiated at this time.

To illustrate initial planning, the nurse's role in endoscopic variceal ligation (EVL) is reviewed. When the nurse is notified that a patient is going to undergo EVL, he or she completes initial planning, which includes assessing the supplies and equipment needed, checking the equipment for proper functioning, and verifying the competency of staff performing the procedure. Supply/equipment items to consider might include the following checklist:

1. Flexible gastroscope.
2. Overtube with bite block.
3. Suction apparatus, including suction tip.
4. Positioning devices.
5. Ligation kit.
6. Topical anesthetic.
7. Sedative, depending on patient's hemodynamic stability.
8. Devices for intra-procedure monitoring of vital signs.

The policies and procedures of the institution guide the initial planning phase. Ensuring the accessibility of an emergency cart, typing and crossmatching blood for patients with potential bleeding problems, and providing the staff with personal protective equipment are all integral components of comprehensive planning.

The gastroenterology nurse uses assessment data obtained upon admission to outline the plan of care. This includes any current medications, allergies that might cause complications, history of bleeding problems, medical history, transmittable diseases, history of malignant hyperthermia, and any procedures the patient has already undergone that are related to the reason for the planned procedure. In addition, the nurse reviews laboratory reports, EKG readings, radiology reports, and recent vital signs to identify any values of concern that should be communicated to other members of the health care team. The information obtained from the assessment, as well as sufficient knowledge of anatomy and physiology to plan for complications and unexpected outcomes, will help to produce the best patient outcomes.

To ensure that the patient will understand the purpose and expected outcomes of an endoscopic procedure, planned nursing interventions might include:

1. Allowing the patient to ask questions and verbalize any concerns.
2. Supporting the patient by listening, showing concern, and encouraging questions.
3. Being prompt when performing procedures to avoid delays or postponement.
4. Involving family and significant others in discussions and questions about the procedure and care needed.

Ongoing planning

Ongoing problem-oriented planning is carried out by any nurse who has contact with the patient. This type of planning involves updating and individualizing nursing interventions. Typically, ongoing planning encompasses the following activities:

1. Clarifying nursing diagnoses.
2. Revising expected outcomes to make them more realistic and patient-centered.
3. Developing new outcome statements and/or new diagnoses as indicated by analyses of new data.
4. Identifying nursing actions that promote achievement of identified outcomes.
5. Documenting patient responses to nursing interventions.

Actual or potential changes in the patient's health status that might occur before, during, or after the planned procedure should be documented. During the procedure, monitoring of physiological and hemodynamic parameters should continue. Changes may affect the nursing diagnosis, identified outcomes, and/or original plan of care. When changes occur, immediate plan modification is essential to better accomplish the desired end results of care. Again, documentation is needed to ensure continuity of care and to facilitate measurement of outcomes.

Discharge planning

Discharge planning should be multidisciplinary and involve nurses, physicians, dieticians, other health care providers, and the patient's significant others. It may also involve social workers and/or case managers. Discharge planning begins on admission.

A patient admitted for an endoscopic procedure usually signs an informed consent form agreeing to the prescribed care; however, informed consent does not affect the patient's right to participate in or refuse treatment at a later date. Patient participation in the plan of care should begin when the patient is first seen by the gastroenterology nurse and should continue to discharge. Patients and their significant others may or may not take an active role in therapeutically managing the illness. During the nursing history, the nurse should assess the role the patient wants to assume.

The establishment of a nurse-patient relationship with a participative patient will result in the patient's active role as a member of the health care team. This means that before, during, and after endoscopic diagnosis and treatment, the participative patient will want to stay informed about his or her health status, methods available to alter it, and ways to assist health professionals. If desired, the patient can be involved in all decisions, beginning at admission and ending at discharge. The patient should also be involved in establishing the outcomes expected from prescribed treatment.

The nurse must also anticipate and plan for the patient's needs after discharge. In discharge planning, the nurse working with the patient assists the patient and significant others in identifying appropriate community resources or options such as home health care, respiratory therapy, or other appropriate agencies. In some institutions, a discharge planner or care manager may be available to assist the gastroenterology nurse in planning for care after discharge from the hospital, or for care following procedures or treatments performed in a physician's office or outpatient setting. With today's shorter hospital stays, discharge planning is crucial.

The gastroenterology nurse must be aware of certain factors that influence a patient's ability to participate in planning his or her own care. First are the patient's basic beliefs about health. A patient who does not believe in his or her susceptibility to illness may not want to take preventive measures. A typical example is the patient who consumes large amounts of alcohol and has cirrhosis of the liver. In many cases, family superstitions or folklore affect behavior toward health care. Other barriers to patient participation include distrust, lack of knowledge, pride, modesty, fear, impatience, religious beliefs, and cultural diversity. A patient's level of involvement in his or her care can vary depending on the ability to overcome many of these barriers. Nurses can influence their patients by providing them with support and encouraging them to be self-directed consumers of health care.

DETERMINING NURSING ACTIVITIES

Care planning involves determining age-specific nursing activities that will help the patient achieve predetermined health outcomes. Although patient outcomes are related to nursing diagnoses, nursing interventions stem from the etiology of the health problem. Consequently, whenever they are known, measures that address the etiology of the problem must be prescribed. For example, if the problem identified is "ineffective individual coping related to fear of hospitalization," a nurse might tactfully discuss coping strategies with the patient. Similarly, the nurse might write the orders, "apply warming blanket to procedure table" and "increase room temperature to 75° F" in response to the diagnosis "body temperature, altered; potential" for an infant who is scheduled to undergo an endoscopic procedure.

Identifying treatment options

There are no substitutes for knowledge, experience, and resourcefulness when it comes to prescribing care. The array of treatment options documented in research and in the empirical literature ranges from skilled nursing procedures to teaching and counseling to simple acts of humanity such as silence, humor, and touch. Strategies that may help nurses broaden their repertoire of nursing actions include:

1. Consulting with successful colleagues and studying their successful practices.
2. Researching the nursing literature for suggestions to improve care.
3. Talking with patients and significant others about measures they have found most helpful in addressing their problems.
4. Reviewing standards of care, including those of the American Nurses Association (ANA) and SGNA, professional guidelines, institutional standards, and accrediting agencies such as The Joint Commission or CMS.

Selected treatment strategies should be tailored to each individual patient's total plan of care. They must be consistent with the patient's values, beliefs, culture, and psychosocial background. They must be realistic in terms of the abilities, time, and resources available.

Writing nursing orders

The following requirements pertain when writing nursing orders:

1. Use directive verbs to describe clearly and concisely the action to be taken (i.e., what, where, who, when, and how).
2. Date and sign the order, making a note when the plan is reviewed.
3. Use only accepted abbreviations.
4. Refer to policy manuals or other agency guidelines for routine steps for lengthy procedures.

Below are examples of adequate nursing orders written in regard to endoscopy care:

1. "Assess patient's breath sounds immediately after procedure and with each set of post-procedural vital signs."
2. "Instruct parent not to lift the child up by the arms or sides for a period of 24 hours after the procedure."
3. "Explain to caregiver signs and symptoms of bleeding or hematoma and what to do if bleeding occurs."

Documenting the plan of care

The final phase of care planning is to document the plan. The plan documents nursing diagnoses, desired outcomes, and nursing orders. A well-documented plan directs nursing efforts, includes evidence-based practice, and advances the four goals of nursing: to promote wellness, to prevent disease or illness, to promote coping, and to prevent injury. It is tailored to the individual patient, is based on scientific principles, and incorporates findings of nursing research. An effective plan of care also addresses dependent and interdependent nursing functions and thus is compatible with the medical plan of care and other interdisciplinary efforts. It includes nursing responsibilities for fulfilling the medical plan of care, while guiding nursing assessment priorities, teaching and counseling activities, and advocacy behaviors. It is designed to meet the patient's psychological, environmental, sociological, and physiological needs.

PLANNING CARE IN GASTROENTEROLOGY

To aid in gastroenterology care planning, Table 10.1 contains nursing care plans based on nursing diagnoses relevant to patients who undergo endoscopic procedures. Diagnoses in the first column represent health problems that gastroenterology nurses frequently treat. The patient outcomes listed are realistic and achievable within the short duration of the patient's stay in the endoscopy unit. Notice that all patient outcomes are behaviors or states that nurses can readily and objectively observe. The right column lists the actions that

Table 10.1. Sample Endoscopy Care Plan

Nursing diagnosis	Patient outcome	Nursing activity
Anxiety & fear	Patient demonstrates effective coping mechanisms. Patient exhibits reduced physiological manifestations of anxiety & fear.	Instruct in relaxation techniques (e.g., deep breathing and imagery), pre-procedure. Refer patient concerns to physician, when appropriate, all phases of endoscopy experience. Allow time for questions, pre-procedure. Involve significant others for support, pre-procedure and post-procedure. Minimize noxious stimuli; keep environment calm and unhurried, all phases. Use therapeutic communication and touch, all phases. Monitor patient for subjective and objective signs of anxiety (e.g., shakiness, perspiration, increased heart rate, extraneous movement, hesitation, poor eye contact), all phases.
Alteration in comfort, actual or potential.	Patient tolerates procedure, asking for medication as needed.	Teach patient the system of ranking pain, pre-procedure. Support patient through discomfort by coaching patient in relaxation techniques (e.g., deep breathing, visualization), intra-procedure. Assess positioning for maximum comfort, pre-procedure and intra-procedure.
Knowledge deficit related to unfamiliar environment and procedure.	Patient describes expected physiological and psychological responses to endoscopy.	Provide information about impending procedure using visual aids or appropriate literature, pre-procedure. Provide sensory information about procedure and sedation, pre-procedure.
Injury related to change in vital signs related to sedation, potential.	Patient remains stable while sedated.	Obtain baseline vital signs, on admission. Monitor pulse, blood pressure, ventalitory status, cardiac electrical activity and level of sedation before procedure, after administration of medication, at least every 5 minutes during procedure, during recovery, and just before discharge (ASGE, 2018). Have supplemental oxygen available.
Injury related to instrumentation trauma (perforation and/or hemorrhage), potential.	Patient experiences minimal trauma from endoscope.	Check lab work (CBC, PT & PTT, platelets), pre-procedure. Insert bite block to protect teeth and pharynx, intra-procedure. Verify that emergency equipment is in working order, pre-procedure. Monitor intake and output, intra-procedure.
Aspiration, potential.	Patient's airway remains clear and unobstructed.	Position patient on left side, pre-procedure. Suction secretions PRN, pre-procedure, intra-procedure, and post procedure.

nurses caring for these patients would take to treat each problem. These actions are based on scientific principles and incorporate the most recent findings of nursing research. Finally, notice that each intervention has a time frame. Time frames become important in evaluating patient care. When caring for patients in the endoscopy unit, these time frames may be short or long, which suggests that follow-up is required beyond the expected length of stay.

The therapy modes that physicians and nurses may undertake are independent of one another and are defined by medical and nurse practice acts. Because the nursing profession has defined its role as having independent, interdependent, and dependent aspects, nursing diagnoses focus on the independent aspects of nursing practice. After collating and interpreting patient data, identifying nursing diagnosis, and establishing desired outcomes, the nurse caring for the patient begins the plan of care. In endoscopy, planning requires use of nursing knowledge and information about the patient and the intended procedure to prepare for the procedure. Clinical judgment and decision-making are the hallmarks of professional nursing.

CASE SITUATION

Two-year-old Nathan Mitchell has experienced persistent diarrhea for more than 10 days before admission and is severely dehydrated. The gastroenterology unit nurse reviews Nathan's nursing history and visits Nathan, who is combative and uncooperative despite his parents's best efforts to comfort him. The nurse who admitted Nathan initiated a standard care plan used in the department for pediatric patients. He diagnosed Nathan by selecting the following from a list of health problems:

1. Impaired verbal communication.
2. Potential for injury, secondary to development (and combativeness).
3. Defensive coping.
4. Noncompliance, secondary to development.
5. Powerlessness.
6. Anxiety, secondary to hospitalization.

He also added the following health problems:

7. Actual fluid volume deficit.
8. Impaired skin integrity (rectal excoriation).
9. Altered health maintenance.
10. Ineffective family coping, disabling.

On examination, the gastroenterology nurse notices that the IV in Nathan's forearm has infiltrated, leaving it quite swollen and sore. The nurse discontinues Nathan's IV and elevates his arm on his teddy bear while applying a warm compress.

Nathan's parents accuse the doctors of giving Nathan medicine that he does not need, saying that rice cereal and milk have always been good for diarrhea. They believe the medicine is responsible for their child's poor appetite and vomiting. They also are grumbling about the day care center where they take Nathan, reporting that Nathan is the eighth child from the center to come down with severe diarrhea. While listening to their account of the day care center, another nurse takes the gastroenterology nurse aside to show the result of Nathan's stool culture. Nathan is infected with *Giardia lamblia*.

Points to think about

1. Which basic types of planning have occurred in this scenario?
2. In choosing three priority health problems, what factors would be taken into consideration?
3. The nursing diagnosis "altered health maintenance," which was selected by the admitting nurse, is unfamiliar to the gastroenterology nurse. How would its meaning be determined?
4. In what ways will the nursing plan of care differ from the medical plan of care?
5. The gastroenterology nurse in this case usually cares for adults and, therefore, feels somewhat limited in his existing repertoire of pediatric treatment options. How might he expand his skillset?

Suggested responses

1. The basic types of planning that have occurred in this scenario include:
   a. Initial planning, which was instituted by the admitting nurse.
   b. Ongoing problem-oriented planning, which occurred when the gastroenterology nurse responded to a health problem (i.e., impaired skin integrity, secondary to IV infiltration) by setting a patient-centered outcome (to reduce pain and swelling caused by fluid extravasation) and identifying means of reducing or resolving the problem (i.e., discontinuing the IV, applying the warm compress, elevating the arm).
   c. Discharge planning with the parents (this planning may include contacting the public health department to investigate the day care center for the presence of *Giardia lamblia*).
2. Factors the nurse may take into consideration in choosing three priority health problems include:
   a. The actual or potential threat of each problem to Nathan's well-being.
   b. The parents' preferences and values, which will determine their participation in Nathan's therapy.
   c. The potential problems posed by the risks unique to Nathan's age, health status, and medical management.
3. To determine the meaning of the nursing diagnosis "altered health maintenance," the gastroenterology nurse should refer to NANDA International's (NANDA-I) axonomy of nursing diagnoses (Herdman & Kamitsuru, 2017). This reference not only

defines health problems associated with each nursing diagnosis, but also provides defining characteristics of health problems that differentiate them from other problems.

4. The nursing plan of care differs from the medical plan of care in that the nursing plan of care:
   a. Addresses psychological, environmental, and sociological factors contributing to disturbances in health, as well as biological causes.
   b. Focuses on patient responses to medical treatment or health problems that nurses can independently treat. Interventions prescribed in it include only those actions that nurses can lawfully perform.
5. The gastroenterology nurse might expand his repertoire of treatment options by:
   a. Consulting with and observing successful colleagues.
   b. Researching the nursing literature for suggestions to improve care.
   c. Talking with patients and family about measures they have found most helpful in addressing their problem.
   d. Consulting standards of care, including ANA and SGNA standards, institutional standards, and standards of accrediting agencies.

REVIEW TERMS

nursing orders, planning, setting priorities

REVIEW QUESTIONS

1. Which of the following is not a valid reason for planning patient care based on nursing diagnoses?
   a. It is required by managed care companies.
   b. It helps nurse to assign priorities for care.
   c. It improves the clarity of communication among nurses.
   d. It emphasizes the collegial relationship between nursing and medicine.
2. The nursing care plan and the medical care plan are both:
   a. Derived from assessment and inferences based on scientific knowledge.
   b. Are instituted and refined following initial and ongoing assessments.
   c. Prescribe measures based on bodies of scientific knowledge.
   d. All of the above.
3. A nursing plan of care:
   a. Addresses only the biological factors of the patient's health problems.
   b. Identifies pathologic basis for illness.
   c. Focuses on the physical condition of the patient.
   d. Includes only actions that nurses can lawfully perform.
4. Assessment of the supplies and equipment needed to perform an endoscopic procedure is an example of:
   a. Initial planning.
   b. Admission planning.
   c. Ongoing planning.
   d. Discharge planning.
5. Intra-procedural changes in the patient's health status may affect:
   a. The nursing diagnosis.
   b. The expected outcomes.
   c. The plan of care.
   d. All of the above.
6. Discharge planning begins on admission and:
   a. Is the responsibility of the physician.
   b. Is the responsibility of the nurse.
   c. Is a multidisciplinary task.
   d. Should not include patient participation.
7. To help broaden the array of treatment options available to them, nurses should consult successful colleagues, the literature, standards of care, and:
   a. The taxonomy of nursing diagnoses.
   b. The plan of care.
   c. Patients and their families.
   d. Physicians.
8. In response to a patient's fluid deficit secondary to persistent diarrhea, which of the following nursing orders would be appropriate for the gastroenterology nurse to write?
   a. "Increase the IV rate following bouts of diarrhea."
   b. "Prepare patient for electrosurgery."
   c. "Monitor vital signs every 2 hours until diarrhea stops. Observe for signs of hypotension with widening pulse pressure."
   d. "Collect stools for culture in am and pm."
9. The final product of the planning phase of the nursing process is:
   a. A well-documented plan of care.
   b. The clinical pathway.
   c. A series of outcome statements.
   d. Nursing intervention.
10. The benefits of implementing clinical pathways includes all of the following except:
    a. Supporting the use of evidenced based practice.
    b. Appearing to provide depersonalized care.
    c. Improving health care team comunication.
    d. Providing a baseline for care.

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Chapter 11

IMPLEMENTATION

This chapter will familiarize gastroenterology nurses with the implementation phase of the nursing process. General information about a variety of issues that may arise during implementation of planned care is provided, followed by a discussion of variables that potentially influence the way care is implemented and specific guidelines for nursing intervention. The role of the gastroenterology nurse in communicating the care provided in the gastroenterology unit is explored. The last part of the chapter examines nursing activities that surround teaching, counseling, and advocacy of patient rights and provision of care in gastroenterology nursing.

Learning objectives

After reviewing the content of this chapter, the gastroenterology nurse should be able to:

1. Describe the three broad areas of nursing activity that occur in the implementation phase of the nursing process.
2. Distinguish between independent, interdependent, collaborative, and dependent nursing functions when implementing a plan of care.
3. Discuss six variables that influence the way a plan of care is implemented.
4. Define general guidelines for implementing the care of the gastroenterology patient.
5. Describe four means of communicating the nursing actions performed during the implementation phase of the nursing process.
6. List documentation expectations when implementing care of the gastroenterology patient undergoing endoscopy procedures.
7. Discuss the nurse's role as teacher and counselor when implementing care of the gastroenterology patient.
8. Discuss the nurse's advocacy role as it applies to informed consent and ethical decision-making.

MOVING TO IMPLEMENTATION

Prior to the implementation phase, a patient's health problems are identified. Risks specific to the individual patient – including age, current health status, and medical management – are considered to anticipate and prevent potential problems. During the planning phase, these health problems are prioritized in order of actual or potential threat to the patient's well-being. Nursing interventions should progress the patient toward achievement of these prioritized outcomes.

Once identified, the care plan outlines the set of nursing actions in a logical sequence that is designed to promote wellness, prevent disease and illness, stimulate recovery, and facilitate coping with altered functioning. It specifies what will be done, as well as how, when, where, and by whom (e.g., nurse, patient, or significant other). Each plan of care is tailored to the patient's individual needs and incorporates criteria for evaluating the effectiveness of nursing actions in assisting the patient's achievement of desired outcomes.

In the implementation phase, the plan becomes a blueprint that guides nursing care. Each nursing action is based on scientific principles and reflects the medical needs as well as the rights and desires of the patient and significant others. All actions are carried out safely, skillfully, and efficiently, and reports should be given at each and every transition point to ensure accurate continuity of the plan during each phase of health care.

During implementation, the following three broad areas of nursing activity occur:

1. The plan is put in motion.

2. The data regarding patient care is updated as data collection continues.
3. The nursing care and patient progress are documented and communicated.

This chapter examines these three activities and the role of the gastroenterology nurse as teacher, counselor, and patient advocate during these activities.

PUTTING THE PLAN IN MOTION

During implementation, all of the steps outlined in the plan of care should be carried out efficiently and effectively. However, it is important to note that the amount of time an institution and its nursing professionals allocate for teaching, counseling, and advocacy ultimately depends on the value and benefits they associate with each activity.

Carrying out the plan of care requires cognitive ability, interpersonal skill, and technical skill.

Cognitive ability is necessary to not only think critically about a patient's health problem but also apply nursing theories to solve problems and find appropriate solutions.

Interpersonal skill is a component of overall professional skills. The ability to communicate clearly, competently, and compassionately helps elicit patient trust and cooperation. A nurse's well-developed intellectual and interpersonal skills not only maximize the chance that healthy outcomes will be achieved but also ensure efficient implementation of care.

Technical skill is necessary when a procedural intervention is required. Technical ability ranges from minimal to extensive, depending on the nature of equipment used to execute procedures and the complexity of the procedure. Examples of implementation measures that require technical skill are starting IVs, holding abdominal pressure, maintaining a patent airway, collecting specimens, using clips/snares/other instruments, measuring arterial hemoglobin oxygen saturation with a pulse oximeter, inserting a duodenal or nasogastric tube, performing esophageal manometry studies, and monitoring cardiac status using an ECG monitor.

Independent, interdependent, and dependent nursing functions are addressed in the plan of care and carried out during the implementation phase.

Independent intervention is action initiated without direction or supervision from other health care professionals and is instituted as the result of a nursing assessment. Actions taken in the course of independent intervention are those for which a nurse is legally accountable. Teaching, counseling, and advocating for patient rights are examples of activities nurses can initiate freely.

Interdependent intervention or collaborative intervention is action performed in concert with the efforts of other health care professionals. For example, the "time out" or surgical safety pause initiated at the beginning of a procedure reflects the importance of collaborative or interdependent team function. Each practitioner has a specific role to play in providing safe patient care.

Dependent intervention is action performed under the supervision or direction of a physician and is most often directly related to the patient's disease. Dependent interventions might include assisting with obtaining a good biopsy specimen or with a procedure.

Nursing practice standards and legislation in all states clearly indicate that nurses are to fulfill orders given by physicians. Physicians are ultimately responsible for actions performed at their direction or under their supervision, and nurses are responsible for carrying out these tasks safely and appropriately. It is important for nurses to understand their responsibility in dependent intervention and that they are therefore required to advocate on the patient's behalf by clarifying any doubts concerning the activities prescribed. The nurse who fails to do so risks liability.

Safety is the focus of many nursing activities carried out during implementation. Examples of typical nursing activities performed to safeguard patient safety include but are not limited to:

- Adhering to disinfection guidelines during reprocessing of contaminated equipment.
- Monitoring the environment for safety.
- Positioning the patient according to physiological principles during or after operative procedures to avoid neurovascular damage and skin breakdown.
- Verifying patient identification and informed consent.
- Routinely inspecting hospital equipment to ensure proper function.

Managing time constraints

There are some ways to free available time for implementing comprehensive nursing care which may include;

1. Delegating technical and non-nursing tasks.
2. Revising policies to delete obsolete and/or unnecessary practices.

Delegating tasks

Making high-quality nursing care possible is a matter of understanding nursing priorities and supporting them. This frequently requires delegation of non-nursing tasks. Many practice settings have instituted professional practice models or collaborative team nursing models to allow nurses to use their time more efficiently and effectively. A professional practice model is recognized as an overarching conceptual framework for nurses, nursing care, and interprofessional patient care. As such, it provides a description of a system that illustrates how nurses practice, collaborate, communicate, and develop professionally to provide high-quality care (American Nurses Credentialing Center [ANCC], 2014).

Revising practice policies

To make high-quality nursing care possible through increased efficiency, nurses must cast off unnecessary rituals and rethink outdated processes to accommodate best practices. Nurse leaders review, observe, and update procedural practice in collaboration with evidence-based research to support current practice.

Preventing complications

In addition to using evidence-based guidelines to define best practice and procedural process, nurses contribute to research by providing data that is integral in determining best practice. Adapting procedures to incorporate new research findings can also free up time by minimizing the risk of complications. For example, current best practices suggest using shorter, smaller IV catheters to minimize irritation to local veins while still allowing blood transfusion. Taking this simple measure minimizes the risk and severity of a complication of IV therapy, potentially increasing time available for important nursing functions.

Variables that affect the way care is implemented

Variables that influence how a plan of care is implemented fall into the following six categories:

1. Patient variables.
2. Nurse variables.
3. Standards of care.
4. Research findings.
5. Resources.
6. Ethical and legal guides to practice.

Patient variables

Every plan of care must be tailored to individual needs, a requirement that frequently calls for creative solutions to health problems. For example, a patient's previous response to nursing measures may affect his or her receptiveness to further intervention. It is likely that the patient whose past experiences and/or preconceived notions about health care were unpleasant or unrewarding is less likely to be receptive. Diminished ability to participate in self-care also influences the way a plan of care is implemented. A geriatric patient with arthritis may require assistance with ostomy care, whereas the otherwise healthy young ostomate will not. Sometimes a patient's willingness to participate in care becomes a factor when the nurse implements planned care. Psychological perceptions, religious beliefs, or socioeconomic factors may prompt a patient to reject all or some nursing interventions. For instance, an otherwise receptive patient whose diet does not include meat may not comply with a high-protein diet regimen for weight loss that includes meat. In this case, interim measures to attain the outcome of the patient getting aedquate protein may include teaching the patient how to identify complete proteins in combinations of vegetable proteins (e.g., grains, legumes) and dairy products.

Developmental differences can substantially influence planned care, especially among pediatric patients. Not only do actual and potential health problems (and therefore desired outcomes) differ within developmental age groups (Table 11.1), but means of achieving outcomes also vary according to the child's unique developmental task for his or her age group. Thus, when taking action (e.g., to reduce the fear and anxiety children and adolescents may experience during endoscopy), it is necessary to identify the developmental task for the patient's age group and select age-specific and age-appropriate treatment options. Similarly, it is necessary to consider whether outcomes and treatment options are realistic, given the patient's developmental circumstances.

Nursing variables

A nurse's cultivated level of expertise affects the type of treatment options implemented. Nurses vary in the degree to which they seek out patient contact. The nurse who enjoys patient contact will more likely have a greater motivation to provide comprehensive care during implementation. Additionally, effective time management increases a nurse's efficiency and ability to carry out all measures planned. However, many nurses continually struggle to find time to implement total patient care and remain frustrated by unpredictable and uncontrollable events.

Standards of care

Current standards of care determine nursing responsibilities and suggest liability limits. **Liability** refers to the legal responsibility for nursing acts or failure to act, including financial restitution in the event of demonstrable damages resulting from negligent acts. The Society for Gastroenterology Nurses and Associates, Inc. (SGNA) has a number of standards, guidelines, and position statements that address these and other issues related to the standard of care. Among them are the "Standards of Clinical Nursing Practice and Role Delineations in the Gastroenterology Setting" that are discussed in greater detail in Chapter 5.

Resources

Even the best-laid plans are doomed to fail without adequate resources to enable their implementation. Inadequate staffing, limited equipment, and insufficient procedural time booked may reduce the quality or alter the type of care rendered.

Table 11.1. Health Problems Specific to Four Pediatric Age Groups

Age	Developmental stage	Actual or potential health problems
Birth to 1 year	Trust versus mistrust	Sleep pattern disturbance Swallowing, impaired Body temperature, altered Fluid volume, altered: deficit or increase Skin integrity, impaired Aspiration, potential: related to existing or coexisting conditions Infection, potential: related to existing or coexisting conditions Impaired gas exchange Ineffective airway clearance Self-concept disturbance: maturational, related to deprivation
1 to 3 years	Autonomy versus shame and doubt	Impaired verbal communication Sleep pattern disturbance Skin integrity, impaired Infection, potential: related to existing or coexisting conditions Self-concept disturbance Health maintenance, altered: maturational, related to inadequate health practice Injury, potential for: related to development (e.g., poisoning, falls) Social interaction, impaired Powerlessness, maturational or situational Coping, ineffective: situational or maturational Anxiety, situational Fear, situational Self-care deficit, inability to feed or toilet self
3 to 6 years	Initiative versus guilt	Coping, ineffective: situational or maturational Powerlessness, maturational or situational Anxiety, situational Fear, situational Sleep pattern disturbance, related to nightmares or fears Injury, potential for: related to development (e.g., poisoning, falls) Infection, potential: related to existing or coexisting conditions Self-care deficit, inability to dress self Self-concept disturbance: maturational, related to deprivation
7 to 11 years	Industry versus inferiority	Self-concept disturbance: maturational, related to peer pressure Noncompliance, situational Anxiety, situational Fear, situational Self-care deficit, inability to bathe self completely Impaired social interaction Violence, potential for: situational, related to inability to control behavior Health maintenance, altered: maturational, related to inadequate health practice

Modified from Herdman, T.H. & Kamitsuru, S. , 2014; NANDA International Nursing diagnoses: Definitions & classification 2015–2017.

Ethical and legal guides to practice

Within the past decade, laws governing delivery and ethical dimensions of health care have become more complex. In many settings, hospital risk managers and ethical committees counsel hospital decision-makers whose policies, in turn, influence nursing practice.

Guidelines for nursing intervention

Suggested guidelines for implementation of nursing care include:

1. Before instituting any measure to treat a health problem, the patient must be assessed to be sure action is still necessary.
2. The nurse should be fully prepared to perform the planned action when he or she approaches the patient. All equipment should be ready, and the nurse should either know how to perform the nursing action or come with a nurse associate who does. The nurse should tell the patient why the action is being taken and what the potential adverse responses to the procedure are.
3. The nurse should approach the patient with a caring attitude, using language the patient understands. By communicating genuine concern for what the patient is experiencing, the nurse conveys regard for the patient's well-being.
4. The nurse should develop a large repertoire of skilled nursing interventions. The larger an array of options he or she has to choose from when treating health problems, the greater the likelihood of success.
5. The nursing actions chosen must comply with standards of care and be within the range of ethical and legal institutional guidelines to practice.
6. The nurse should think critically about the plan of care, always questioning whether routines are really the best method of treatment. He or she should consult immediate colleagues, colleagues in related nursing fields, and relevant literature to discover more effective ways of managing health problems. The effectiveness of each action should be evaluated in terms of its positive and negative effects on the outcome.
7. The nurse should modify the prescribed interventions to accommodate the patient's developmental and psychosocial circumstances, ability, and willingness to participate in achieving desired outcomes; the patient's previous responses to nursing measures; and any progress the patient has already made toward expected outcomes.

DATA COLLECTION UPDATING

Data collection continues throughout the implementation phase, and is frequently updated. These activities serve three important functions during implementation by:

1. Comparing new data against the baseline data to enable the nurse to identify patterns and trends in the patient's health.
2. Collecting data following nursing intervention to allow the nurse to evaluate the effectiveness of nursing actions according to evaluation criteria listed in the care plan.
3. Revising the plan of care as the data is updated.

Just as during the assessment phase, nurses collect both subjective and objective data. During implementation, however, subjective data are derived from validation of responses to therapy and perceived progress as noted by the patient or significant others. Objective data are obtained by monitoring both the patient's response to nursing activity and the medical plan of care.

COMMUNICATING ABOUT CARE

Communication takes many forms, including documentation, discussion and verbal reports, conferences or consultation, and referrals. Communicating about care with the patient and family, colleagues, and other health professionals is vitally important for several reasons. It not only sparks the involvement of patients and significant others, but it also ensures continuity of care as the responsibility for the patient's care shifts from one provider to the next. Communication is essential to the coordination and continuity of care, and it also promotes efficiency among members of the health care team. Informing colleagues of action enables nurses to supplement and complement each other's efforts, avoiding duplication and omission of effort. Communication about care should focus not only on nursing care provided and patient progress toward specified outcomes, but also on the total plan of care rendered by physicians and all involved health professionals.

Documentation

Documentation of care in written records is the most visible and permanent medium of communication with regard to implementation of nursing care. Documentation can refer to an action or to a written record. Keep in mind that written documentation encompasses both handwritten notations and electronic documentation systems. As a process, documentation includes the acts of collecting, abstracting, and coding patient data and therapeutic processes for the purposes of communicating about patient care, supplying a supporting reference concerning the status or progress of a patient, and archiving evidence of care rendered.

Documentation, the most formal of all methods of communication, constitutes written, legal evidence of all pertinent intervention involving a patient. Documentation of care that has been rendered serves purposes beyond communication between colleagues, including:

Planning
Because patient records document not only baseline and ongoing data but also how a patient is responding to therapies, they are useful as resources for planning and modifying planned care.
Process improvement audit
Records provide a medium for studying the quality of the care provided and the competence of the nurses rendering the care. Charts selected at random are audited against standards of care to determine if standards are being met. If discrepancies between practice and standards are found, action may be initiated to remedy substandard practices, involving in-services, policy changes, counseling, and so on.
Research
The nursing record serves as a valuable source of data in nursing research. Each record represents a unique case study from which researchers hope to learn how to improve recognition and treatment of patient health problems.
Education
By reading patient charts, nurses can learn a great deal about clinical manifestations of disease, effective or less-effective treatment modalities, and factors that affect patients' abilities to achieve health outcomes.
Legal evidence
The nursing record serves an important function in detailing timelines and events during a procedure. Typically it is timely, fact-based, unbiased documentation that may be entered into court proceedings as evidence.
Historical document
Because records are dated and retained for many years, they are useful as an accurate indicator of a patient's past health.

Documentation requirements in the endoscopy unit

The following are examples of the types of documentation required for endoscopic procedures:

a. Procedure performed.
b. Date and time of the procedure.
c. Equipment used.
d. Staff present.
e. Anesthesia and/or sedation administered and patient's response.
f. Medications and response.
g. Vital signs and monitoring methods.
h. Oxygen therapy used.
i. Use of cautery, electrocoagulation, or laser.
j. Dilators used.
k. Solution injected.
l. Specimens taken.
m. Photographs, videos, and/or x-rays taken.
n. Post-procedural diagnosis.
o. Nursing notes and procedure nurse signature.

In addition to these procedural notes, SGNA's "Guidelines for Documentation in the Gastrointestinal Endoscopy Setting" also lists the information required for pre-procedural, post-procedural, and discharge documentation.

Discussion and verbal reporting

Verbal communication is common in health care and occurs in a variety of situations.

Errors can occur if there is miscommunication between nursing staff during a patient handoff. Organizations accredited by The Joint Commission have been required to standardize the handoff process to reduce communication errors since 2006 (Streeter, Harrington, & Lane, 2015). Verbal communication is faster and can convey elements that do not come across in written communication such as tone, inflection, and body language. In addition, verbal communication, whether face-to-face or via telephone, allows for real-time discussion and for questions to be addressed.

Because clarity is so important in health care communication, it is important for common communication events (such as shift-to-shift handoff, procedure time-out, or reporting abnormal lab values) to follow a specific structure or format. Additionally, it is important to establish policy and processes where critical information (labs, orders, test results, etc) are repeated back for confirmation to avoid communication errors.

Conferences or consultation

Two types of conferences, care conferences and nursing rounds, are held in many practice settings for the purpose of communicating about care:

Care conferences
Care conferences are group meetings during which a specific case situation is reviewed. Impressions are shared, treatment options are explored, and care is planned. Organizations may use this method to review a sentinel or critical incident.
Nursing rounds
Nursing rounds are on-site discussions during which patient and family input is sometimes elicited and from which direct observation can be made. At the same time, nursing care and patient progress can be evaluated. Nurses frequently use these multi-disciplinary rounds to review best possible treatment options for patients.

Consultations with other health care professionals to seek advice, instruct or inform, or exchange ideas concerning patient care may be considered another form of conference communication. For example, gastroenterology nurses consult with clinical nurse specialists in their area concerning mutual patients.

Referrals

Nursing is increasingly recognized for its role in screening and referring patients who require or would benefit from adjunct services or existing community resources. The process of sending a patient to another source for aid, or to another professional for appropriate action, is known as **referral**. Both internal referrals to departments elsewhere within a facility and external referrals to other hospital facilities or outside agencies are useful in providing comprehensive holistic care.

Most agencies have policies governing referrals that require submitting special forms. In addition, these policies specify who can make the referral, how it is to be done, and so on. When a referral is appropriate, the nurse must give information to the receiving agency or department that will ensure continuity of patient care. Answering the question, "What would I need to know if I were to continue care of this patient?" will help to accomplish this end.

INDEPENDENT INTERVENTION: TEACHING, COUNSELING, AND ADVOCACY

During implementation, nurses actively determine their patients' needs for assistance and ability to perform activities of daily living (ADLs). The promotion of self-care through teaching, counseling, and advocacy frequently becomes important when assisting patients to meet desired outcomes.

Patient teaching

Whether teaching patients about an upcoming endoscopic examination or about self-care during or after their hospital stay, a nurse's role as educator remains the same. Nurses implement patient education through the following four broad areas of activity:

1. Diagnosing and assessing a patient's learning needs, learning style, learning capacity, knowledge deficit, and readiness to learn.
2. Planning a learning activity.
3. Providing learning opportunities.
4. Evaluating learning.

Diagnosing knowledge deficits

When identifying a knowledge deficit as a health problem, nurses base this diagnosis on an assessment of the patient's learning needs. This assessment must include not only evidence that the patient is unaware of information that might improve health behaviors but also that the patient is ready – and fully able – to learn and agrees that knowledge is a priority. Before nurses can effectively implement a teaching plan, they must also assess their own knowledge on the subject and, if necessary, take steps to become sufficiently informed. It is often necessary to contact appropriate resource persons or obtain additional information to provide accurate, up-to-date facts to substantiate the lessons.

Planning the learning activity

Setting goals for the learning activity is an action very similar to that of identifying expected outcomes (see Chapter 10). Learning goals must be realistic, and they may be short-term or long-term. The identified goals are developed into the nursing care plan and discussed with the patient. The nursing care plan is a product of the nursing process. In this way, nurses and patients identify measurable behaviors to help each other monitor progress toward the learning goals. The following are examples of measurable goals:

a. "Nursing staff will provide written home bowel preparation instructions by 5/12."
b. "Mr. Franklin will review these instructions and verbally describe home bowel preparation procedure at next office visit on 5/15."

Providing the learning opportunity

In providing an environment conducive to learning, finding a room or space where distractions can be kept to a minimum also augments the patient learning opportunity. Adequate preparation for learning activities ensures that the learning content can be delivered in a logical and comprehensive manner. Involving the patient and significant others also facilitates learning, encourages reinforcement, and makes proper follow-up more likely. Finally, it is important for gastroenterology nurses to convey an attitude of support. The nurse-teacher who exercises accomplished communication skills ensures that information is imparted in a clear, concise, comprehensive manner and that comprehension occurs. The learning opportunity may be facilitated by continually reminding the patient about the care plan.

Evaluating the learning

Methods for evaluating whether the patient is learning correspond to the type of skill learned. Cognitive skills are best evaluated by oral questioning or written questionnaire. Affective skills are better evaluated through observation of behavioral response. Psychomotor skills may be evaluated by return demonstration. Short-term goals (i.e., those that may be attained within 1 week) may often be evaluated during the period of hospitalization, but long-term goal evaluation must be referred to home health nurses, office nurses, or long-term care nurses.

Counseling

Counseling may be one of the most important gastroenterology nursing activities. Several factors influence the patient's vulnerability to crisis, including personality, cultural factors, and availability of support systems. Moreover, anxiety about hospitalization, the gastroenterology procedure, an outcome of surgery, or the results of laboratory studies can manifest itself in subtle ways that impede learning and healing. Fear, when suppressed, may precipitate a crisis marked by a total breakdown in a patient's coping mechanisms. To reduce fear and anxiety, gastroenterology nurses should counsel patients throughout the course of therapy.

Counseling is the act of rendering guidance to a patient and/or their significant other, an act that sometimes entails assisting the patient in problem-solving. Counseling may be short-term, long-term, or motivational.

Short-term counseling

Short-term counseling is given when ineffective coping patterns surface that require immediate attention. The severity of the underlying emotional issue may cause disturbances ranging from a minor dilemma to a major crisis. In the gastroenterology unit, short-term counseling to enable patients to overcome anxiety and fear is common.

Long-term counseling

Long-term counseling is rendered consistently over a period of weeks or months and involves repeated contact with a nurse via telephone or personal visit.

Motivational counseling

Motivational counseling is performed either to stimulate a patient's inner drive to get well or to enhance motivation to cooperate in performing or learning to perform self-care. Motivational counseling generally requires an exploration of attitudes, values, and feelings underlying the disinterest in or indifference to recovery. To implement motivational counseling, nurses typically design incentives to stimulate drive and ambition.

Advocacy

The act of supporting patient rights is receiving greater emphasis by both consumers and nursing professionals. Consumer expectations and demands have evolved with changes in health care delivery. At the same time, nursing professionals have come to value the promotion of individual well-being and to respect the individual's right to self-determination.

Advocacy behaviors in nursing exhibit two components: informing patients and supporting them in their decisions. That information enables patients to make educated decisions. Gastroenterology nurses can help patients make educated decisions by providing information about their rights and their options.

The following elaborates on two areas in which the gastroenterology nurse can act to safeguard patient rights in gastroenterology: informed consent and assisting patients to make informed decisions when confronted by ethical dilemmas.

Informed consent

Endoscopic procedures are becoming more varied and technical, and the role of the gastroenterology nurse as an educator and advocate is becoming more important. When gastroenterologists and nurses work together to educate patients, there is no reason why any physically and mentally competent patient should not be totally informed about impending GI procedures. Yet nurses in the endoscopy unit frequently encounter circumstances in which they question a patient's comprehension of a proposed procedure or the risks involved. They also witness situations in which a patient's ability to freely consent to treatment is altered. The effects of anxiety, pain, medication, depression, or temporary or permanent disorientation are among the factors that can influence an individual's ability to make judgements about his or her health care.

Although it is common to think of informed consent as a thing (i.e., a legal document), it is actually a process. **Informed consent** is an interaction between physician and patient in which a meaningful exchange of information concerning an impending health care administration occurs. Only the patient or a legally authorized guardian may give consent, and a legal guardian may give consent only in the event that the patient is incompetent by reason of age (i.e., the patient is a minor and neither married nor self-supporting), physical inability, or legal incompetence (see below).

Informed consent is required on three occasions: on admission, before any diagnostic procedure or medical or surgical treatment is performed, and/or before any human experimentation is enacted. Furthermore, the conditions under which consent can be waived are very specific, as listed in the following explanations:

1. Consent is not needed in an emergency if there is immediate threat to life or health; if experts agree that an emergency exists; and/or if the patient is unable to consent and a legally authorized person cannot be reached.
2. Consent is not required for an action necessary to treat an unanticipated complication incurred during surgery when a legally authorized person cannot be reached.

A patient's refusal of treatment constitutes a slightly different circumstance. Refusal to consent must be documented and, when appropriate, accompanied by an explanation of the consequences that the patient may incur by refusing. A release form should be signed and witnessed, to relieve the nurses, doctors, and facility of all liability for any negative outcome of the treatment refusal.

Failure to obtain consent may result in charges of battery against the nurse, doctor, and hospital caring for the patient. Even when given, consent is legally valid only when all of the four following conditions are met:

1. The physician and hospital make full disclosure concerning the proposed treatment or experiment.
2. The patient possesses competent judgment and decision-making ability.
3. The patient claims to comprehend the procedure, attending risks, and after-effects.
4. The patient gives consent voluntarily of his or her own free will.

Responsibility for obtaining consent rests with the person(s) performing the therapeutic or diagnostic procedure or conducting the research study. Gastroenterology nurses may play a role in evaluating comprehension, assessing impediments to comprehension (including deficits in reasoning or judgment processes), or detecting deleterious influences on patient decision-making (e.g., force, coercion, and/or manipulation).

Advocacy in ethical dilemmas

Respect for the inherent dignity, worth, unique attributes, and human rights of all individuals is a fundamental principle. Through building relationships of trust and providing care according to the needs of the patients and their families, nurses set aside any bias or prejudice (American Nurses Association [ANA], 2015).

Ethical conflicts in nursing are on the rise with an aging society, changes in the financial health care arena, technological advances, decreased resources, greater cultural and religious diversity, and changing public expectations of the health care system. Nurses need to build a strong knowledge base that supports the recognition of ethical issues. In addition, nurses need to hone their skills to take moral action when required in any clinical situation (Pavlish, Brown-Saltzman, Jakel, & Fine, 2014).

There are three circumstances in which nurses may experience moral conflict when implementing a plan of care:

1. Moral uncertainty, which is an inability to recognize the nature of an ethical problem.
2. Moral dilemmas, as when a conflict arises between two or more ethical principles with no obvious solution.
3. Moral distress, which is a conflict between an individual's knowledge of ethically appropriate action and institutional constraints preventing action.

One dilemma frequently encountered in the gastroenterology unit concerns whether or not to maintain nutrition of severely debilitated or vegetative patients through percutaneous endoscopic gastrostomy tubes.

ANA's position statement, "The Nurse's Role in Ethics and Human Rights: Protecting and Promoting Individual Worth, Dignity, and Human Rights in Practice Settings Recommendations" (2016) includes but is not limited to the following recommendations for the nurse's role in ethics:

- All nurses advocate for human rights of patients, colleagues, and communities.
- Nurses advocate for the ethical and just practice of nursing by creating and sustaining environments that support accepted standards of professional practice, as the practice environment and rights of nurses influence the practice and moral context of nursing.
- Nurses strengthen practice environments by refusing to act in ways that would create a negative impact on the quality of care.
- Through their professional organization, nurses must reaffirm and strengthen nursing values and ideals with a united voice that interprets and explains the role of nursing in society.
- Health care agencies pay close attention to the potential for human rights violations as they relate to patients, nurses, health care workers, and others within their institutions.
- Nurses must examine the conflicts arising between their own personal and professional values and the values and interests of others who are also responsible for patient care and health care decisions, and they must address these conflicts in ways that ensure patient safety and promote the best interests of the patient.

CASE SITUATION

Janet Estelle is a 48-year-old executive with rectal polyps who was undergoing a colonoscopic polypectomy. She sought medical attention when she began to notice occasional blood in her stool. She reported having confidence in her physician, Dr. MacElroy, but also heard "a few horror stories" about hospital care. Janet was somewhat apprehensive but cooperated in her care. In fact, Janet responded favorably to pre-procedural teaching, listening carefully to the information given and asking many questions. She claimed to feel "more in control of the situation."

While Janet was under moderate sedation for the colonoscopic polypectomy, Dr. MacElroy reported difficulty maintaining light in the operative field just after the procedure began. The gastroenterology nurse attempted to troubleshoot the equipment difficulty when Dr. MacElroy noticed that the patient began to hemorrhage.

Points to think about

1. In removing the rectal polyps, Dr. MacElroy encountered one of the possible complications (hemorrhage) of the procedure and must now

treat it. What intervention must the gastroenterology nurse take to help Dr. MacElroy correct the immediate problem?
2. Of the actions identified, which might the gastroenterology nurse be able to delegate?
3. Which of the actions identified are activities the nurse may initiate independently? Which must occur under Dr. MacElroy's direction and supervision?
4. What variables may influence how the nurse implements the actions identified?

Suggested responses

1. The gastroenterology nurse might identify the following actions to help Dr. MacElroy correct the problem:
   a. Call for additional help.
   b. Increase the IV flow rate.
   c. Place the patient in the Trendelenburg position.
   d. Insert a large-bore IV line.
   e. Set up cautery equipment.
   f. Continue to troubleshoot equipment failure.
   g. Type and cross and send for blood (stat).
   h. Increase the frequency of vital sign monitoring.
   i. Administer oxygen through nasal cannula.
   j. Plan to take further action (for example, a reversal agent).
   k. Comfort Janet as appropriate.
   l. Document all actions, observations, and patient responses.
2. The following list includes actions the gastroenterology nurse might delegate.
   a. Sending for blood.
   b. Changing the suction canister.
   c. Setting up the cautery.
   d. Continuing to troubleshoot equipment failure.
3. Actions the gastroenterology nurse might take independently include the following:
   a. Increasing the frequency of vital sign monitoring.
   b. Planning further actions.
   c. Counseling Janet when appropriate.
   d. Documenting actions, observations, and patient responses.
4. Actions the nurse might implement, depending on Dr. MacElroy's directions and under his supervision, include the following:
   a. Increasing the IV flow rate.
   b. Placing the patient in the Trendelenburg position.
   c. Applying and increasing oxygen.

In this circumstance, the nurse may need to prompt Dr. MacElroy to elicit direction because he may be preoccupied with the hemorrhaging vessel.

5. Variables that may affect the way the nurse treats this situation may include the following:
   a. Janet's level of awareness of the problem.
   b. The Janet's coping style.
   c. The rapidity of change in Janet's metabolic condition.
   d. Whether or not the nurse has available support personnel.
   e. Whether or not the nurse has backup equipment.
   f. The nurse's experience in similar situations.

REVIEW TERMS

advocacy, care conferences, consultations, counseling, dependent intervention, documentation, independent intervention, informed consent, interdependent intervention, liability, nursing rounds, referral

REVIEW QUESTIONS

1. The implementation phase of the nursing process is characterized by all of the following except that:
   a. Evaluation criteria are developed against which the effectiveness of nursing intervention can be measured.
   b. The nursing plan of care is put in motion.
   c. The database is updated as data collection continues.
   d. Nursing care and patient progress are documented and communicated.
2. The act of rendering guidance or assisting a patient with problem-solving is:
   a. Referring.
   b. Consulting.
   c. Counseling.
   d. Teaching.
3. Carrying out the plan of care requires all of the following except:
   a. Cognitive ability.
   b. Technical skill.
   c. Interpersonal skill.
   d. Delegation.
4. A gastroenterology nurse might vary the way he or she comforts an anxious 10-year-old boy based on:
   a. The developmental task of children in the 7- to 11-year-old age group.
   b. His willingness to participate in counseling.
   c. Recent findings concerning the impact of certain words in calming or provoking anxiety.
   d. All of the above.
5. Assisting with a procedure is:
   a. An independent nursing activity.
   b. An interdependent task.
   c. A dependent nursing activity.
   d. A non-nursing chore.
6. Nurses accomplish patient teaching in four phases, including planning the learning activity, providing learning opportunity, evaluating learning, and:
   a. Correcting mistakes.
   b. Diagnosing a patient's knowledge deficit.

c. Explaining the patient's privacy needs to significant others.
d. Helping patients make informed decisions.

7. Data collection activities serve three important functions during implementation by each of the following except:
   a. Comparing new data against the baseline to enable the nurse to identify patterns and trends.
   b. Collecting data following nursing intervention to allow the nurse to evaluate the effectiveness of nursing actions according to evaluation criteria listed in the care plan.
   c. Revising the plan of care as the database is updated.
   d. Implementing motivational counseling to stimulate drive and ambition.
8. A research nurse wants a gastroenterology nurse's patient to participate in a research study. Whose responsibility is it to obtain the patient's consent?
   a. The research nurse.
   b. The physician.
   c. The legal department.
   d. The gastroenterology nurse.
9. Suggested guidelines for implementation of nursing care include all of the following except:
   a. Think critically about the plan of care, always questioning whether routines are really the best method of treatment.
   b. Consult immediate colleagues, colleagues in related nursing fields, and relevant literature to discover more effective ways of managing health problems.
   c. Assess the patient to be sure that the action is still necessary before instituting any measure to treat a health problem.
   d. Document his or her action.
10. Ethical conflicts in nursing are on the rise due to several factors. These include all of the following except:
   a. An aging society.
   b. Increased resources.
   c. Changes in the financial health care arena.
   d. Changing public expectations of the health care system.

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Chapter 12

EVALUATION

The final component of the nursing process is evaluation. It is a dynamic, ever-changing indicator of the effectiveness of nursing interventions. It is a reflection of continuous reassessment of the plan of care, as priorities are modified to meet the patient's changing needs and health care goals.

Learning objectives

After reviewing the content of this chapter, the gastroenterology nurse should be able to:

1. Explain the tasks involved in the evaluation process.
2. Review the purpose of the evaluation process.
3. Describe the function of evaluation as it coordinates with the progressive steps of the nursing process.
4. Explain the continuous, cyclic evolution of the nursing process.
5. Identify therapeutic nursing interventions derived from each level of the nursing process that are appropriate to the plan of care developed to meet each individual patient's physical and behavioral needs.

THE NURSING PROCESS

The nursing process, an accepted framework for nursing practice, has evolved to become more expansive and inclusive over the years. Responsive to evidence-based changes in practice, it has grown into a process that is currently composed of five distinct areas of patient-focused care. These core stages of modern nursing delivery are assessment, nursing diagnosis, outcome identification/planning, implementation, and evaluation.

The **assessment phase** of the nursing process consists of the collection of data, both real and potential. Common words associated with this phase of the nursing process include: *observe, gather, collect, differentiate, assess, recognize, detect, distinguish, identify, display, indicate, and describe.*

The most recent addition to the nursing process, the **nursing diagnosis phase**, represents an actual or potential clinical determination made by the nurse in response to a patient's perceived placement on the health continuum. The basis for the nursing care plan, the diagnosis segment recognizes the patient as a holistic entity.

Continuing down the continuum of care, the expected outcomes of an individualized patient care plan are identified during the **outcomes identification/planning phase**. The goals set during this step of the nursing process should be both measurable and achievable. The goal of outcomes identification/planning is to design a plan of care that results in the prevention, reduction, and ultimate resolution of the health issues for which the patient is being treated. This step consists of utilizing the data collected to formulate patient goals and outcomes. It must be measurable and prioritized, and include the patient, significant others, and all members of the health care team. Common words associated with this phase of the nursing process include: *rearrange, reconstruct, determine, outcomes, formulate, include, expect, designate, plan, generate, short- and/or long-term goal, and develop.*

The **implementation phase** of the nursing process is the action phase. During this phase of the process, the plan of care is implemented. It involves direct care of the patient and includes all treatments, procedures, and activities involving the patient. This phase of the process involves documentation of patient response to interventions. Common words associated with this phase of the nursing process include: *document, explain, give, inform, administer, implement, encourage, advise, provide, and perform.*

The final phase of the nursing process, and the portion we will deal with in greater detail in this chapter, is evaluation. The **evaluation phase** of the nursing process is an analytical portion that determines whether the interventions delivered were effective and if the goals previously determined require reassessment. It is a time for reassessment of the plan of care, with necessary modifications initiated based on the patient's objective reponse to the plan of care and the extent to which goals were realized. Some common words associated with this phase of the nursing process include: *monitor, expand, evaluate, synthesize, determine, consider, question, repeat, outcomes, demonstrate, and re-establish.*

Just as the nursing process is the manner in which we approach nursing practice, the standards of practice are the delineation of the values that comprise them. The scientific basis upon which the practice of professional nursing is based has been set forth in guidelines by the American Nurses Association (ANA). The evaluation portion of the nursing process is addressed in ANA's *Nursing: Scope and Standards of Practice, 3rd edition (2015)*, Standard 6: Standards of Practice.

The process of evaluation is the analysis of the patient's progress toward the attainment of preset outcomes (goals). More specifically, ANA (2015) states that:

1. Evaluation is systematic, ongoing, and criterion-based.
2. The patient, significant others, and other health care providers are involved in the evaluation process as appropriate.
3. Ongoing assessment data are used to revise diagnoses and outcomes, modifying the plan of care as needed.
4. Revision in diagnoses, outcomes, and the plan of care are documented.
5. The effectiveness of the interventions is evaluated in relation to outcomes. Documentation of the patient's progress toward achievable outcomes is retrievable.
6. The patient's responses to interventions are documented.
7. The nurse acts as the patient's advocate.

The nursing roles implicit within the framework of the nursing process are multidimensional and include the provision and the coordination of care. It is inherent in professional accountability and includes the appropriateness of nursing actions, the need to rearrange priorities, and an analysis of all aspects of the delivery of care.

Evaluation itself is a multidimensional process of multiple facets. Patient care is evaluated through direct observation, patient interviews, and review of the patient's documented records. Interventions are recorded as "met," "not met," or "ongoing." They are consistent with the established plan of care and reflect the rights and desires of the patient, the patient's significant others, and other members of the health care team who are involved with the patient's care. As a health care provider and coordinator of patient care, the nurse focuses on the patient and direct delivery of care. The nursing role is largely predetermined by the profession's scope of practice. As members of a professional discipline, nurses accept accountability for their actions and thereby become advocates for patients and the profession.

The foundation of nursing practice is a scientific framework that involves **critical thinking**, **communication** with patients and peers, and adherence to a **standard of care** that can be measured. **Criteria** are measurable qualities, attributes, or characteristics that specify skills, knowledge, or health states. When analyzed within the framework of standards of care, these three elements delineate nursing behaviors. Every standard of nursing care is accompanied by one or more measurable criteria that must be satisfied in order to meet the pre-established standard of care.

Direct and indirect nursing care interventions are derived through critical thinking and are documented as real and potential patient outcomes. They are set into motion by the implementation of predetermined tasks that address the patient's physical and emotional needs. The effectiveness of these interventions is the framework for the evaluation phase of the nursing process.

Evaluation is a collaborative process that involves a continuous and ongoing interaction among the patient, the nurse, and the environment on a verbal and nonverbal level. It is continually reassessed as patient priorities change and preset goals are satisfactorily met or not met. Often, the unmet goals – those that were preset but never achieved – become more important as they can indicate where the biggest changes are required.

Important points to remember when reassessing a patient for satisfactory goal resolution include:

1. Does the patient have the ability to achieve the goals set? Are the goals realistic?
2. Can the patient realistically manage the totality of the activities of daily living (ADL) upon discharge, or does his or her physical or emotional state make this an unattainable goal?
3. Was the patient involved in establishing the goals?

More than the patient's health care status must be taken into account in the predetermination of patient goals and outcomes. These include, among many other factors, the patient's:

1. Past and present coping mechanisms.
2. Spiritual beliefs.
3. Support systems available at the current time and in the future.
4. Cultural influences.
5. Current level of activity and future activity expectations.
6. Disease process and treatment impact on self-image, comfort level, and ability to function independently.

STANDARDS OF CARE

Standards of care serve as a blueprint for quality nursing care. They ensure that data collection is systematic and well-documented, reflecting the patient's actual or potential health problem. They are individualized for each patient, based on current knowledge, and address both the patient's and the caregiver's priorities in the pursuit of health restoration and maintenance. They are assessed jointly by the nurse, the patient, and significant others, reassessed and revised as tasks are completed and goals are met or unmet.

Standards of care also provide a foundation for the development of performance evaluation tools, which help establish sound quality assurance programs. They allow for peer review, provide a measurable body of knowledge that can be utilized for research purposes, and promote interdisciplinary collaboration and cooperation. This ensures that the **quality of care** remains consistent with existing standards.

Gastroenterology nurses apply standards of care in their professional practice on an ongoing basis. The gastroenterology nurse continually evaluates the quality of care provided to patients undergoing endoscopic procedures to determine the effectiveness of the intervention, as well as the quality and safety of the patient care. Professional practice is evaluated within the scope of professional practice standards and relevant regulatory agencies.

The gastroenterology nurse follows a specified plan of care to achieve desired patient outcomes in a safe, efficient, and cost-effective manner. The criteria to be assessed include patient comfort and stability, and maintenance of equipment at the optimum level. Examples of the application of these standards and criteria in professional practice include those below.

Standard: The gastroenterology nurse will maintain all equipment appropriately to ensure safe, quality care of the patient undergoing an endoscopic procedure.

Criteria: All endoscopes will be reprocessed following manufacturer's written guidelines and institutional policy.

Standard: The gastroenterology nurse will correctly and thoroughly complete the documentation required for patients undergoing endoscopic procedures.

Criteria:

1. Vital signs are recorded pre- and post-procedure.
2. Procedure start and completion times are accurately and appropriately documented.
3. The patient's tolerance to sedation is recorded.
4. The patient's post-procedure discharge criteria are satisfactorily achieved.

In the plan of care, expected patient outcomes represent the heart of the evaluation process because they reflect the desired outcomes as an actual task completed and goal achieved. Potential patient outcomes are assessed relevant to the patient's health status at the time of assessment and the environmental factors impacting the patient's ability to complete the tasks. Examples of actual and potential outcome influences are airway clearance and knowledge deficit.

Outcome: Ineffective airway clearance; potential.

Criteria: The patient is free from respiratory injury relative to sedation.

Outcome: Knowledge deficit relative to endoscopic procedure; actual.

Criteria: The patient will verbalize an understanding of the endoscopic procedure and preparation requirements.

Evaluation of patient care is performed jointly by the nurse, the patient, and the rest of the health care team involved with the patient's care. It assesses the extent to which the desired outcome was achieved, evaluates the effectiveness of nursing interventions, and determines the success of the overall plan of care.

The evaluation of an implemented plan of care is performed continually and systematically. It encompasses a series of tasks to quantify the quality of nursing planning and interventions. ANA standards guide the evaluation process with principles for patient-specific and institution-specific plans of care, a framework for quality assurance programs, and a scientific basis for professional and patient education.

The ANA's statement on *Nursing: Scope and Standards of Practice* consists of 17 standards. These Standards of Practice define criteria against which the nurse's level of performance can be measured, while the Standards of Professional Performance describe a competent level of behavior in the professional role (American Nurses Association [ANA], 2015).

ANA Standards of Practice address the following:	
Standard 1	**Assessment**
	The registered nurse collects pertinent data and information relative to the health care consumer's health or the care situation.
Standard 2	**Diagnosis**
	The registered nurse analyzes assessment data to determine actual or potential diagnoses, problems, and issues.
Standard 3	**Outcomes Identification**
	The registered nurse identifies expected outcomes for a plan individualized to the health care consumer or the care situation.
Standard 4	**Planning**
	The registered nurse develops a plan that prescribes strategies to attain expected, measurable outcomes.

Standard 5	**Implementation**
	The registered nurse implements the identified plan.
Standard 6	**Evaluation**
	The registered nurse evaluates progress toward attainment of goals and outcomes.
ANA Standards of Professional Performance address the following:	
Standard 7	**Ethics**
	The registered nurse advocates for the patient, demonstrates professional accountability, and maintains the principles of medical ethics during all patient interactions and situations.
Standard 8	**Culturally Congruent Practice**
	The registered nurse develops a values and belief system based upon consumer diversity and cultural variations, and delivers care with the respect, knowledge, and compassion required when caring for a widely diverse patient population.
Standard 9	**Communication**
	The registered nurse demonstrates effective communication skills, conveying accurate information in a caring and intelligible manner.
Standard 10	**Collaboration**
	The registered nurse aptly demonstrates the role of the profession, while concurrently recognizing the contributions of all stakeholders in the implementation of a comprehensive plan.
Standard 11	**Leadership**
	The registered nurse acts as a model for enhanced practice and, as an active mentor and innovator, contributes to the progression and evolution of nursing care.
Standard 12	**Education**
	The registered nurse demonstrates a commitment to learning through lifelong faithfulness to the enhancement of knowledge, the support of ongoing education, and the mentoring of others.
Standard 13	**Evidence-Based Practice and Research**
	The registered nurse recognizes the importance of research, including its impact upon nursing practice and the evolution of the profession.
Standard 14	**Quality of Practice**
	The registered nurse ensures safe and effective nursing care delivery through an ongoing process of quality improvement initiatives that include recommending innovations for the improvement of nursing care, engaging in formal and informal peer review programs, and collaborating and exchanging ideas with all other members of the health care team. Professional performance is improved by identifying observed deficiencies in care and responding to them with increased educational strategies designed to meet currently recognized deficits.
Standard 15	**Professional Practice Evaluation**
	The registered nurse evaluates her own practice, as well as the practice of others, for the purpose of ensuring that nursing care delivery not only meets all regulatory standards and codes but also moves nursing practice along a continuum to an augmented level of care.
Standard 16	**Resource Utilization**
	The registered nurse recognizes and uses the appropriate resources to plan and provide safe, effective, evidence-based care.
Standard 17	**Environmental Health**
	The registered nurse supports a safe and professional workplace environment that is conducive to the generation of new ideas and advocates for a healthy workplace environment for both the patient and the staff.

The plan of care includes interventions based upon those required for patient safety, effectiveness of care, cost containment of care, and efficient delivery of care. The analysis of measurable outcomes is reviewed as concisely and systematically as they were assessed, planned, and implemented. The body of knowledge known as the evaluation process comprises three steps:

collecting data, analyzing outcomes, and reassessing the plan.

Step 1. Collecting evaluative data

The goal of the first analytical tool inherent in the evaluation of patient task accomplishment and goal achievement answers the question, “To what degree were predetermined patient outcomes achieved?” The answer can be found by employing several methods of analysis. The evaluative technique utilized will be determined by whether the outcome evaluated is cognitive, psychomotor, or affective (Michlich, 2011).

1. **Cognitive outcomes** define increases in knowledge. Data are collected by asking the patient to repeat or apply information to familiar situations.
2. **Psychomotor outcomes** address the achievement of new, learned skills. Asking the patient to demonstrate the new skill is an effective way of measuring goal achievement.
3. **Affective outcomes** are more difficult to evaluate, as they define less measurable behavioral differences. These outcomes relate to changes in the patient’s values, beliefs, and attitudes. Behavior and social interactions are observed to evaluate the achievement of these outcomes.

When all evaluative data have been collected, they are compared with the standards of nursing care, expected outcomes, and professional regulations and policies.

Step 2. Analyzing outcome achievements and effectiveness of nursing interventions

Once the care team has determined whether identified patient outcomes have been achieved, an analysis of the variables that contributed to their success or failure becomes a valuable evaluative tool. The value of this step is apparent when the actual influences promoting outcome achievement, or the lack of it, are identified and reassessed.

Step 3. Reassessment of the plan of care

The final portion of the evaluation phase of the nursing process is reassessment. The relevant data have been collected, actual outcomes and potential outcomes compared, and reassessment made accordingly. It is during the reassessment step of the evaluation phase that the nursing outcomes, goals, and plans of care are reviewed and revised as necessary.

QUALITY IMPROVEMENT AS A FUNCTION OF THE NURSING PROCESS

Quality improvement, as an extension of the nursing process, is a bridge between direct patient care and the balance of the members of the health care team. The desired goal is the achievement of a standard of excellence in an objective and straightforward manner. The ultimate goal is the improvement of patient care.

Monitoring the quality of care is the responsibility of all health care workers and is an important function of the evaluation phase of the nursing process. Documentation of every patient intervention ensures a more accurate evaluation of **outcome criteria** and provides the methodology by which standards of care can be monitored and analyzed. This, in turn, provides important information that promotes the development of quality improvement programs.

Effective quality improvement programs are consistent with both scope of practice and regulatory and institutional policies. The evaluation of patient care outcomes should be systematic, be facility-wide, and provide data that can be utilized to improve delivery of care. Although quality of care cannot be ensured, it can be monitored continuously, thereby improving patient care substantially. Historically, quality improvement is designed to evaluate which nursing interventions are the most optimal. These may include patient satisfaction with the services provided, quality of the technical aspects of the services provided (and the risk of injury or illness that are associated with them), and the evaluation of the efficient utilization of available resources (including cost containment of physical resources, and time and type of utilization of medical personnel). It is a continuous process that compares actual practice against preset standards of practice for the purpose of ensuring acceptable and current levels of patient care delivery.

EXAMPLES OF EVALUATION OF CARE IN THE ENDOSCOPY SUITE

Example 1

Standard

Patients receiving moderate sedation are to be monitored continually for the development of adverse reactions.

Criteria

1. Personnel monitoring the sedated patient are fully conversant with adverse reactions and the therapeutic interventions for each. These include appropriate dosage indications and medication contraindications, interactions with other medications, and the onset and duration of action of each medication that has been used.
2. All patients receiving moderate sedation have adequate and continual intravenous (IV) access.
3. Emergency equipment, including appropriate reversal agents and equipment (i.e., oxygen administration equipment and defibrillators), should be readily available whenever moderate sedation is administered.

Evaluation

1. Continuous IV access is adequately maintained throughout the duration of the patient's procedure and recovery.
2. Discharge criteria reflect that the patient has achieved a baseline level of functioning. These include, but are not limited to, the patient's return to a physiological level inclusive of adequate respiratory levels, a pre-procedural level of consciousness, healthy skin color and intact condition, fully functional protective reflexes, and stability of vital signs.

Example 2

Goal

The gastroenterology nurse will provide patient education that is commensurate with the patient's procedure, level of understanding, and personal beliefs (cultural, religious, etc).

Criteria

1. Throughout the evaluation process, the nurse will assess the patient's ability and readiness to learn by evaluating learning needs and deficits.
2. In collaboration with the patient and/or significant others, the nurse will deliver patient teaching based on identified need and deficits.
3. The nurse will teach the patient and/or significant others in an environment that is conducive to their assessed learning needs and sensitive to their learning deficits.

Evaluation

1. The nurse will evaluate the patient's ability to meet the learning criteria through verbalization and discussion.
2. Teaching interventions will be modified according to the patient's assessed ability to meet the goal criteria.

CASE SITUATION

Leadville Community Hospital is continually reviewing and revising its standards of care as its staff ceaselessly strive to deliver the safest and most cost-efficient patient care in Colorado. Delivery of care is continually reviewed and upgraded to meet the challenges of the technologically and medically complex hospital environment.

The facility is currently re-evaluating the patient discharge criteria used for their post-endoscopy outpatient population. Currently the endoscopy physician must evaluate the patient before he or she can be discharged from the hospital setting. The hospital is planning to implement a discharge program based on a nursing evaluation of pertinent criteria. Hospital personnel suggest that this will help the endoscopy unit flow more efficiently. It will allow the physician to continue with procedures uninterrupted, while at the same time the post-procedure nursing staff will ensure that patients adequately meet all criteria for a safe discharge from the hospital.

The preset criteria for the nursing staff to evaluate a patient's ability to be safely discharged from the hospital include:

1. Return of vital signs to pre-procedural levels.
2. Return of level of consciousness (LOC) to pre-procedural levels.
3. Oxygen saturation within predetermined parameters.
4. Evaluation of ability to safely ambulate.
5. Evaluation of skin color, warmth, dryness, and turgor.
6. Evaluation of patient's perception of pain on a scale of one to 10.

Gastroenterology nurse Sally Caulfield is monitoring patient Winston MacBride as he awaits discharge from the endoscopy unit at Leadville Community Hospital following a colonoscopy with polypectomy of a large cecal polyp. Winston had been medicated with propofol (Diprivan) during the course of the procedure. His physician, Dr. Elkins, has written orders to discharge Winston when he is stable and to have Winston follow up with him in his office in one week.

Points to think about

1. How might the gastroenterology nurse evaluate her patient's readiness for discharge within the discharge criteria currently set forth by Leadville Community Hospital?
2. Sally noted that pre-procedural teaching was completed and documented by the nurse who admitted Winston to the endoscopy unit. This teaching covered potential and actual post-procedure changes in bowel movements; dietary restrictions, if necessary; medication restrictions, if necessary; and possible reasons he may need to contact his physician before his scheduled post-procedure appointment. These reasons might include rectal bleeding in large amounts, an elevated temperature over 100 degrees Fahrenheit, severe abdominal pain, and a rigid and distended abdominal area. When should the evaluation of teaching take place, and how might an evaluation be performed?
3. Assuming that Winston achieved all expected outcome criteria (met all necessary criteria set forth for patient discharge from the endoscopy unit), what evaluative statements would Sally document on the patient's chart? These evaluative statements should be pertinent to goals met and specific outcome criteria achieved, thereby adequately meeting the standards of the hospital for discharge.
4. When Sally evaluates Winston's progress one hour post-procedure, his vital signs are stable (within

20 percent of his pre-procedural vital signs). He is alert and oriented to person, place, and time. He is denying abdominal pain, and his abdomen is soft and non-distended. He is, however, complaining of nausea and is declining oral intake of fluids. What actions might Sally implement as she reassesses Winston's requirements based on these new actual outcomes?

5. Sally is a member of her unit's Quality Improvement Committee. The committee is redesigning their post-procedure discharge policy to meet the needs of a growing and evolving patient population. What suggestions could she offer that would accumulate measurable data for evaluation of the efficacy of the endoscopy unit's nurse-driven, post-procedure discharge policy?

Suggested responses

1. Evaluation of a patient's progress toward a preset goal is a measurable process with preset outcomes evaluated against preset criteria. The preset criteria, in this instance, are the parameters for discharge that the patient must achieve before he can be safely discharged from the hospital following a polypectomy. The actual outcomes are based on the patient's ability to achieve these preset criteria. As the nurse proceeds through each individual criterion for discharge, she will evaluate the patient's actual outcomes against these preset criteria. By carefully evaluating Winston's progress in achieving these predetermined outcome criteria (stable vital signs, LOC return to pre-procedural levels; oxygen saturation level; ambulation ability; skin color, warmth, dryness, and turgor; and pain perception), Sally can make a determination of his readiness for discharge.
2. Patient teaching is the responsibility of every health care worker who comes into contact with the patient. Multiple staff members care for the patient as the plan of care progresses. It is the function of every person treating the patient to assess the patient's level of understanding and address all of the patient's questions or concerns as he proceeds with the plan of care. Patient perceptions of both physical and emotional experiences are altered on an ongoing basis as new information and experiences are processed. This is why the evaluation process is an ongoing process that must be continually reassessed. Methods to evaluate patient teaching will help in achieving the desired patient outcomes. For example, a positive cognitive outcome such as a patient's understanding of potential changes in bowel function immediately post-colonoscopy might be best assessed by asking the patient to repeat the information and verbalize understanding. At the very minimum, all patients should be evaluated for their understanding of the plan of care at implementation, during the pre-assessment telephone interaction with the nurse, and through the pre-discharge instructions. Encouraging patients to verbalize their learning needs both before and after the procedure, and then evaluating the extent to which these individual needs were met, will evaluate their effectiveness. A new post-procedure plan can then be designed and implemented, based upon actual educational deficits assessed during this evaluative process.
3. At the time of Winston's discharge from the hospital, after the successful completion of his polypectomy, Sally's evaluation of her patient's readiness for discharge is based on his having achieved the predetermined outcome criteria as set forth in the hospital's discharge policy. It is also dependent upon her evaluation and documentation of his understanding of any possible ADL restrictions, complications to notice in the immediate post-procedure period, and his responsibility to follow up with his doctor within a specified time frame. At the time of discharge, Sally documents Winston's discharge status, along with the degree to which each outcome criteria was met, partially met, or not met. All predetermined discharge criteria are addressed, along with any discharge teaching/instructions and Winston's level of understanding of each. Sally's discharge note might read as follows:

   Patient is alert and oriented to person, place, and time, and describes minimal discomfort. Vital signs are stable and within 20 percent of their pre-procedural levels. Oxygen saturation is 100 percent. Ambulating without assistance and with a steady gait. Skin is pink, warm, and dry to the touch, with normal turgor. Oral intake tolerated well. Patient describes post-discharge care measures and self-evaluation criteria, and verbalizes understanding of all teaching. Discharged to home in stable condition in the care of his brother, Mike. Patient will follow up at his physician's office in one week, per physician instructions.
4. The evaluation process is dynamic and continually altered by the patient's response to interventions. After each patient interaction, the nurse reassesses the degree to which the expected outcomes have been achieved. It is at this point that the plan of care can either be terminated (the patient is discharged to home), modified (new interventions are determined as necessary for the patient to achieve the outcome objectives desired), and/or continued. Planned care may be terminated if each outcome has been achieved, i.e., the patient may be discharged to home when all discharge criteria have been successfully met. If the patient is continually having difficulty achieving the desired outcome, the plan of care may require reassessment and

modification. In the instance described here, Winston is refusing oral intake. Sally must now reassess Winston to determine how best to help him meet the outcome criteria. She must decide whether to give him ice chips and begin oral intake slowly, or to first administer an anti-emetic and then offer oral intake again. If the patient just requires more time to achieve the desired outcome, the plan of care is continued unaltered. In this instance, Sally has decided to continue monitoring her patient until the cause of his nausea is determined or the nausea subsides without intervention.

5. A quality improvement program implemented to assess the efficacy of a criteria-based endoscopy discharge plan, as assessed by the nurse recovering the patient, would include predetermined discharge criteria and documentation of the patient's achievement of each. Assessing the efficacy of previously determined discharge parameters enables the nurse to determine which interventions were the most successful in achieving desired patient outcomes. Multiple factors must be considered in an evaluation of such a program. Is the patient satisfied with the services provided? Was the quality of the patient's technical care adequate to meet his individual needs? Was there an efficient use of personnel and resources? What are the possible risk factors that may be generated by a criteria-based discharge program? This quality improvement program involves a review of the nursing care and patient outcomes performed while the patient receives this care. It is performed through direct observation of nursing care, patient interviews, and chart reviews (the **concurrent audit**). This program offers the advantage of immediate feedback and, therefore, immediate corrective actions as deemed necessary. A quality improvement program is an important tool for the evaluation of patient care and enables the caregiver to change and improve interventions and quality. The result is an immediate improvement in the quality of care and future patient interventions.

Sally's proposal will include a checklist of patient needs met, not met, and/or partially met for all of the unit's predetermined outcome criteria for discharge. From this checklist, she can determine compliance with the plan of care and note where improvement is required. She can then document the patient's understanding of the discharge instructions delivered and identify areas of learning deficits to address more succinctly at this time.

REVIEW TERMS

affective outcome, cognitive outcome, concurrent audit, outcome criteria, psychomotor outcome, quality of care, standards of care

REVIEW QUESTIONS

1. The reasons why nursing professionals evaluate the quality of care include all of the following except:
   a. Nurses recognize that a classification system for nursing diagnoses is important.
   b. Nurses aim to promote excellence in nursing care.
   c. Nurses must be accountable to society for the quality of care they provide.
   d. Nurses want to improve professional performance by identifying deficiencies in care provided and, therefore, educational needs, and to analyze and explain the differences in patterns of practice and results of care.
2. Criteria are:
   a. Nationally recognized standards.
   b. Facts.
   c. Interventions.
   d. Measurable.
3. Evaluation of nursing care consists of:
   a. Examining the cost-effectiveness of nursing actions, the efficiency of admission and discharge, and the rate of patient readmission.
   b. Determining whether a patient has achieved outcomes or is making progress toward goals developed in the outcome identification stage of care, whether nursing interventions chosen to treat identified health problems are effective in reducing or resolving identified health problems, and whether health care provided has been effective overall.
   c. Verifying whether policies and procedures are eliminating problems, whether the doctors are practicing according to the dictums of their specialty, and whether adequate staffing is available in nursing units.
   d. Examining nursing values; identifying structure, process, and outcome criteria; and executing performance appraisals to correct individual resistance to quality practices.
4. Evaluative statements:
   a. Must include the patient's response to care provided.
   b. Must describe actual outcomes of care.
   c. May describe ongoing assessment data used to revise diagnoses and outcomes.
   d. All of the above.
5. Changes in values, beliefs, and attitudes are difficult to evaluate because:
   a. They define less measurable behavioral differences.
   b. Survey questionnaires used to evaluate them are rarely valid.
   c. There is little value placed on casual conversation with patients.
   d. All of the above.

6. A quality improvement (QI) review is:
   a. An acceptable, expected level of performance established by authority, custom, or consent.
   b. An ongoing process of review in which data collected over a period of time are used to compare actual practice against standards of practice to determine if the care actually rendered meets a level of quality deemed acceptable within a practice setting.
   c. A description to help determine whether specific objectives were met during the period of time outlined.
   d. A statement defining an actual outcome (e.g., skills developed, knowledge obtained, change in health status).
7. The evaluation of patient care outcomes should be:
   a. An independent role of the nurse leader.
   b. Only performed in the event of a bad outcome.
   c. Performed jointly by the nurse, the patient, and the rest of the health care team involved with the patient's care.
   d. Performed by the physician post-procedure.
8. Cognitive outcomes:
   a. Define increases in knowledge.
   b. Relate to changes in the patient's values, beliefs, and attitudes.
   c. Address the achievement of new, learned skills.
   d. None of the above.
9. Throughout the evaluation phase of the nursing process:
   a. Expected patient outcomes are identified.
   b. Actual patient interventions are performed.
   c. Outcome achievement is determined.
   d. A nursing diagnosis is synthesized.

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SECTION III

Anatomy, Physiology, and Pathophysiology

Section III presents the anatomy, physiology, and pathophysiology specific to gastroenterology. Focusing on a specific organ or system, an in-depth discussion on key areas such as clinical features, symptoms, diagnoses, pathology, and treatment options is provided.

Chapter 13

ESOPHAGUS

This chapter will acquaint the gastroenterology nurse with the normal anatomy and physiology of the esophagus and the clinical features, diagnosis, and treatment of selected esophageal disorders.

Learning objectives

After reviewing the content of this chapter, the gastroenterology nurse will be able to:

1. Describe the anatomy and physiology of the esophagus, including the upper and lower esophageal sphincters and the esophageal body.
2. Explain the normal motility of the esophagus.
3. Discuss pathological conditions of the esophagus and the corresponding pathophysiology, diagnosis, and treatment options.

ANATOMY AND PHYSIOLOGY

The **esophagus** is a hollow, muscular tube, approximately 23-25 centimeters (cm) (10 inches) in length and 2-3 cm (approximately 1 inch) in diameter in adults. The esophagus serves as a channel for food going from the mouth to the stomach (Figure 13.1).

The esophagus is considered the third organ of digestion, after the mouth and the pharynx. The esophagus is located posterior to the trachea and larynx, passing through the diaphragm and into the abdomen at an opening called the **diaphragmatic hiatus**. Almost immediately after the esophagus passes through the hiatus, it enters the stomach (Figure 13.2).

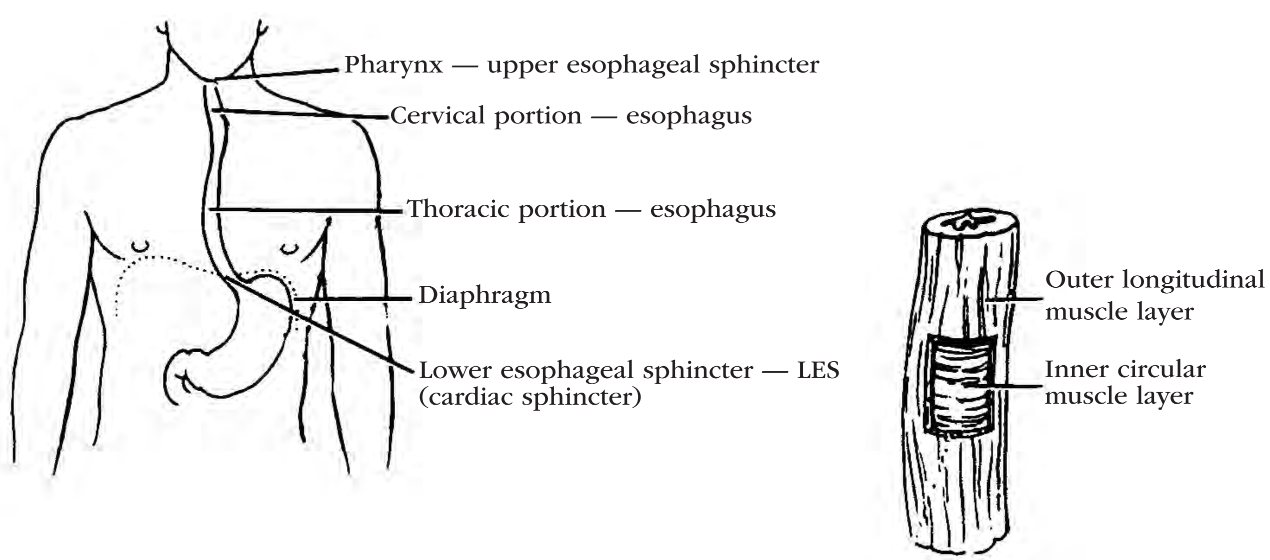

Figure 13.1. Esophageal Anatomy. Insert shows muscular layers of the esophagus.

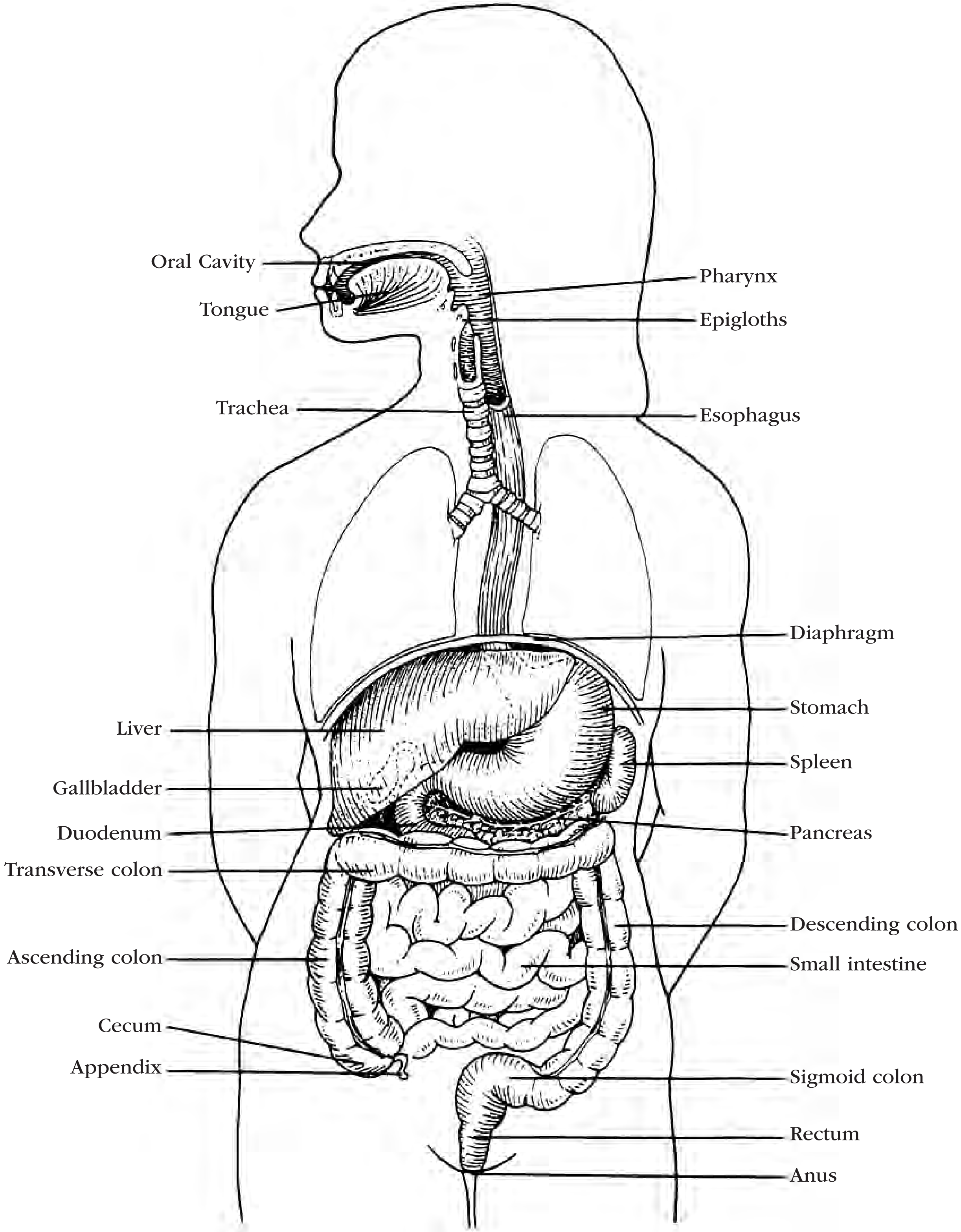

Figure 13.2. Anatomy of the GI System

Layers of the esophagus

The wall of the esophagus is made up of three layers: the mucosa, the submucosa, and the muscularis. Unlike most of the gastrointestinal (GI) tract, the esophagus is not surrounded by serosa.

- **Mucosa.** The inner mucosal layer is covered with a layer of stratified squamous epithelium. Beneath the epithelium is the lamina propria, which consists of loose connective tissue. Extensions of the lamina propria, called the dermal pegs, protrude into the epithelium. Particularly at the upper and lower ends of the esophagus, the mucosa contains well-organized, mucus-producing glands. A thin band of smooth muscle, the muscularis mucosa, separates the lamina propria from the underlying submucosa.

- **Submucosa.** The submucosa is the middle layer of the esophagus. It contains loose connective tissue with both fibrous and elastic elements, as well as blood vessels and nerve fibers.
- **Muscularis.** The muscularis is the outermost layer and consists of an inner layer of circular muscle and an outer layer of longitudinal muscle fibers. Between the two muscular layers is the intramuscular Auerbach's nerve plexus. Approximately the first 5% of the esophagus is made up strictly of striated muscle, and 50% to 60% of the distal esophagus is entirely smooth muscle. A transition zone of both striated and smooth muscles covers as much as 35% to 40% of the length of the esophagus.

Sphincters located at each end of the esophagus include:

- **Upper Esophageal Sphincter (UES)**. At the upper end of the esophagus is the hypopharyngeal sphincter or UES. Posteriorly and laterally, the UES is made up of the cricopharyngeal muscle. Its anterior border is the cricoid cartilage.
- **Lower Esophageal Sphincter (LES).** At the lower end of the esophagus, directly above the angle of His, which marks the junction between the tubular esophagus and the stomach, is the LES. The LES is also known as the gastroesophageal (GE) sphincter or cardiac sphincter. It controls the passage of ingested food into the stomach. The LES is a physiological rather than an anatomical sphincter and is 2-4 cm in length.

Vascular supply

Arterial

The esophagus receives arterial blood via the esophageal arteries of the aorta, the inferior thyroid artery, and the left gastric artery. The gastroesophageal junction (GEJ) receives arterial blood from branches of the left gastric artery and the inferior phrenic arteries.

Venous

Blood is returned by way of the azygos, thyroid, and left gastric veins.

Innervation

The esophagus receives both sympathetic and parasympathetic innervation. The swallowing center in the medulla initiates peristaltic contractions via the vagus nerve, which innervates striated muscle in the upper esophagus. Sympathetic innervation, which controls secondary peristalsis that arises from within the esophagus, is derived from cervical and thoracic ganglia, and from preganglionic fibers of the greater and lesser splanchnic nerves. The LES is believed to receive both sympathetic and parasympathetic innervation.

Gastroesophageal reflux (non-cardiac chest pain) can sometimes be misdiagnosed as heart-related due to the same sympathetic and parasympathetic innervation involved in myocardial infarctions.

Motility

At rest, both the UES and the LES are closed. When a bolus of food is pushed through the pharynx, the UES opens, allowing the bolus to pass into the esophagus. At this point, gravity and peristalsis combine to advance the bolus down through the esophagus.

Peristalsis is defined as a distally progressive band of circular muscle contraction. When initiated by swallowing, it is known as primary peristalsis. Secondary peristalsis originates below the hypopharynx with no antecedent swallowing movement and is elicited by esophageal distention. In the esophagus, peristalsis begins in the pharynx and moves distally at a rate of 3-5 cm per second, effectively transporting material toward the stomach.

When the bolus reaches the LES, the sphincter opens to allow the bolus to enter the stomach and then closes to prevent a reflux of food and acid.

DISORDERS OF THE ESOPHAGUS

Significant disorders of the esophagus include **gastroesophageal reflux disease (GERD)**, esophageal varices, tumors, **diverticula**, and strictures. Other noteworthy pathological conditions include esophageal rings and webs, foreign-body obstruction, inflammatory diseases, infectious diseases, Mallory-Weiss tears, motility disorders, caustic injuries, Barrett's esophagus, congenital defects, and fistulas.

Gastroesophageal reflux disease

Gastroesophageal reflux develops when gastric or duodenal contents flow back into the esophagus. All adults and children normally have some amount of reflux, particularly after eating. Esophageal reflux is considered a pathological condition—**gastroesophageal reflux disease (GERD)**—only when it causes undesirable symptoms, such as pain or respiratory distress.

Risk factors

Patients with excessive or symptomatic esophageal reflux have an incompetent LES. An incompetent LES does not have sufficient intraluminal pressure to prevent reflux. This is often seen in the case of hiatal hernia, a condition where the uppermost portion of the stomach migrates above the diaphragm, thus compromising the effectiveness of the LES. Esophageal reflux may also occur as a result of pyloric stenosis, intestinal malrotation, or a motility disorder.

Symptoms

The most common symptoms of GERD are **dyspepsia** (epigastric discomfort or pain), **heartburn** (pyrosis), and **regurgitation**. Other symptoms may include **dysphagia** (a sensation of difficulty in swallowing), **odynophagia** (painful swallowing), bleeding from erosions or **esophagitis**, asthma, and aspiration pneumonia.

Diagnosis

Diagnostic testing associated with mild or infrequent GERD includes:

- **Intraesophageal pH study.** An extended esophageal pH study is used to determine the frequency of reflux episodes and the relationship of these episodes to symptoms as they occur (e.g., respiratory symptoms, chest pain, bradycardia). Intraesophageal pH is recorded for 24 or 48 hours. Symptoms are recorded manually on a diary or electronically using the pH monitoring equipment. Twenty-four-hour pH monitoring is generally considered to be the most accurate test for GERD.
- **Barium swallow with or without upper GI series.** A barium swallow will demonstrate reflux but does not allow for evaluation of the frequency of reflux or its association with other symptoms (e.g., respiratory distress, pain). Barium swallow may also show inflammation associated with esophagitis but is not always a reliable test for esophagitis, especially in children. Barium swallow also allows for evaluation of swallowing, motility, and anatomy. The upper GI series is a valuable tool for determining if an anatomical abnormality (e.g., malrotation, stricture, obstruction) is responsible for the reflux symptoms.
- **Esophageal manometry.** This procedure may be indicated if a motility disorder is suspected as the cause of symptoms. Manometry is sometimes a useful adjunct to evaluation for GERD, but the diagnosis cannot be confirmed by manometry alone.
- **Esophagoscopy with biopsy.** This procedure is used to determine the presence of esophagitis, Barrett's esophagus, and anatomical abnormalities. Esophagoscopy with biopsy allows direct visualization to aid in the diagnosis of medical or surgical therapy.
- **Bernstein test.** While not typically used, this test may be ordered to differentiate between cardiac and non-cardiac chest pain for the diagnosis of GERD. A feeding tube is placed transnasally so that the tip is positioned in the esophagus. Hydrochloric acid is instilled, and the resultant symptoms are compared with the patient's presenting symptoms. Intraesophageal pH study has widely replaced the Bernstein test, though it may be valuable for chest pain related to nonacid reflux.
- **Gastric emptying study.** This study is used to evaluate the speed at which the stomach empties. It is done by having the patient eat a meal where the solid, the liquid, or both components of the meal are mixed with radioactive material. A scanner over the patient's stomach monitors the amount of radioactive material in the stomach for a period of hours.
- **Nuclear scintiscan.** This is a nuclear medicine test that evaluates reflux a patient may be having and checks for aspiration. Radioisotope is swallowed by the patient and then assessed for its reappearance in the esophagus, its appearance in the lungs indicating aspiration, and the rate it disappears out of the stomach.

Complications

Complications of GERD may include esophageal stricture, esophageal ulcer, Barrett's esophagus, pulmonary aspiration, or upper GI bleeding. Other complications can include severe esophagitis and aspiration pneumonia.

Treatment

There are three objectives in the treatment of GERD: relief of symptoms, healing of damaged mucosa, and prevention of complications. These can be accomplished by lifestyle modifications, drug therapy, and antireflux surgery.

Lifestyle modifications

The first step in treatment of GERD is lifestyle modification by incorporating these recommendations:

- Dietary adjustment with avoidance of foods and beverages that lower LES pressure, including alcohol, tomatoes, peppermint, and licorice, as well as caffeine-containing foods and beverages such as coffee, tea, chocolate, and colas.
- Weight loss and avoidance of tight-fitting garments.
- Elevation of the head of the bed on blocks. Using pillows to elevate the patient's head causes bending at the waist, increasing intra-abdominal pressure, and increasing reflux.
- Smoking cessation.
- Avoidance of food or drink 2 hours before bedtime.

Drug therapy

The next step in the treatment of GERD is drug therapy using one or more of the following:

- Antacids.
- Histamine2 (H_2) blockers such as famotidine (Pepcid, Pepcid AC).
- Sucralfate (Carafate).
- Proton-pump inhibitors (PPIs) such as omeprazole (Prilosec, Prilosec OTC), esomeprazole magnesium (Nexium), rabeprazole (Aciphex), lansoprazole (Prevacid, Prevacid 24hr), or dexlansoprazole (Dexilant).

If these drugs are not helpful, the addition of a medication to increase motility and strengthen the LES such as a dopamine antagonist or a cholinergic agent such as bethanechol (Urecholine) may be useful. Metoclopramide (Reglan) can be used for motility but must be given cautiously due to the risk of tardive dyskinesia and should not be used long term.

There is some data to suggest that PPIs can increase the risk of dementia. Recent observational studies have suggested that regular use of PPIs had a 44% increased risk of dementia compared with patients not using the drugs (Nehra, Alexander, Loftus, & Nehra, 2018).

Appropriate use of OTC medications for gastroesophageal reflux without a medical evaluation offers some benefits to the patient. If typical symptoms of GERD do not improve, however, further evaluation may demonstrate the existence of GERD and evaluate for an alternate diagnosis (Badillo & Francis, 2014).

Antireflux surgery

In patients debilitated by disabling symptoms, antireflux surgery may be needed to provide an artificial closing mechanism for the LES. Surgical procedures include the Hill gastroplexy, laparoscopic Nissen fundoplication, and magnetic sphincter augmentation. The Nissen fundoplication is the gold standard and is aimed at reducing a hiatal hernia. Antireflux surgery is typically performed by a thoracic surgeon.

- Hill gastroplexy. Complicated cases may require other procedures such as the Hill gastroplexy. The GEJ is securely fixed with sutures to its normal intra-abdominal location.
- Nissen fundoplication involves wrapping the distal esophagus circumferentially, or partially with a portion of the stomach, to prevent reflux. The laparoscopic procedure has been shown to decrease morbidity, shorten the recovery period and hospital stay, and minimize discomfort. It is considered a potential cure of GERD. It may be augmented by a Hill gastroplexy. Reportedly, belching and vomiting are difficult with a fundoplication.
- Magnetic sphincter augmentation requires the laparoscopic placement of a circular string of titanium beads at the GEJ. The magnetic bond between the beads augments the sphincter function, expanding during swallowing, belching, or vomiting. A long-term retrospective study demonstrates efficacy and safety of the magnetic sphincter augmentation (Schwameis et al., 2018).

GERD in pregnancy

Gastroesophageal reflux in pregnancy is a frequent occurrence. Symptoms of heartburn begin during early pregnancy and resolve with delivery. Increased reflux symptoms coincide with increased intra-abdominal pressure from the growing fetus and a decrease in LES pressure in early pregnancy, both of which return to normal after delivery.

Lifestyle modifications may be enough to manage heartburn. Careful consideration must be given to the use of systemic medications because of the potential risk to the fetus.

Esophageal varices

Esophageal varices are enlarged veins. Although varices may occur in other parts of the GI tract, they develop most commonly in the submucosal veins of the distal esophagus, the stomach, and the hemorrhoidal plexus.

Risk factors

Esophageal varices are related to portal hypertension, which is often associated with alcoholic cirrhosis but may also be seen in patients with cirrhosis resulting from other causes such as chronic hepatitis, and portal vein thrombosis, as well as congenital disorders such as biliary atresia. Fibrotic liver changes and hepatic vein obstruction cause increased pressure within the portal venous system. This pressure is transmitted to preexisting collateral circulation, which results in dilation of submucosal esophageal veins.

Symptoms

Although varices may be asymptomatic and variceal bleeding is painless, there is a high risk of rupture of the distal esophageal veins, which causes life-threatening bleeding. In an acute episode, bright red blood gushes from the patient's mouth, and the patient shows signs and symptoms of hypovolemia.

Diagnosis

An upper endoscopy is useful to diagnose esophageal varices and to evaluate the risk of bleeding. Varices are classified on a scale of grade I to IV, depending on their size. Larger varices (grade III or IV) carry a higher risk of bleeding.

Complications

Overall prognosis for patients with acute variceal bleeding is poor. While there has been a very significant improvement in the prognosis of patients with acute variceal bleed over the last two decades, mainly due to the implementation of new and more effective protocols, the mortality rate remains high at around 20% (de Souza et al., 2015).

Treatment

The treatment for variceal bleeding is esophageal variceal ligation (EVL), which involves the endoscopic placement of rubber bands or O-rings on the target vessel(s).

The drug of choice for the control of variceal bleeding is IV infusion of octreotide (Sandostatin). Side effects include bradycardia and seizures. Although vasopressin appears to stop variceal bleeding, it is not often used for this purpose due to its association with serious systemic side effects, including hyponatremia, hypertension, myocardial ischemia or infarction, cardiac arrhythmias, and decreased cardiac output.

A last resort for bleeding esophageal varices may be balloon tamponade, a procedure that makes use of either a four-lumen Minnesota tube or a triple-lumen Sengstaken-Blakemore tube.

The nonsurgical procedure called transjugular intrahepatic portosystemic shunt (TIPS) is used to control and prevent esophageal variceal bleeding. For long-term prevention of recurrent bleeding, a portal-systemic shunt may be used to divert esophageal blood from the portal circulation to the systemic circulation. Using imaging guidance, a stent is placed between the portal and hepatic veins, sending blood from the bowel back to the heart, bypassing the liver. Because operative morbidity and mortality are high in patients with end-stage liver disease and active variceal bleeding, this procedure is most useful as a long-term measure and may not be appropriate in emergency situations.

Supportive measures are required if shock occurs. The physician may order replacement of blood volume with packed red blood cells, albumin, and intravenous hydration, along with correction of concurrent coagulopathy with fresh frozen plasma.

Tumors

The esophagus may develop benign or cancerous tumors. Globally, squamous cell carcinoma remains the most common histological type (Abbas & Krasna, 2017). The incidence of adenocarcinoma of the esophagus occurs in a small percentage of patients with Barrett's esophagus.

Risk factors

Chronic irritation of the esophageal mucosa caused by caustic ingestion, chronic or persistent reflux, Barrett's esophagus, obesity, or excessive smoking or drinking may predispose a patient to esophageal cancer. Smoking, obesity, and GERD are significant risk factors for GEJ adenocarcinoma.

Helicobacter pylori (*H. pylori*) and high dietary fiber intake have been linked to reduced incidence and lower disease risk.

Symptoms

The most common indications of an esophageal tumor are dysphagia and odynophagia. A steady pain may be located substernally or occasionally in the back. Other symptoms may include anorexia and weight loss, anemia, hoarseness, and cough. Blood loss is usually slow and steady, rather than by acute hemorrhage.

Diagnosis

Diagnostic testing includes barium swallow, computerized tomography (CT) scan, magnetic resonance imaging (MRI), positron emission tomography (PET) scan, and esophageal ultrasound. The primary diagnostic method is direct visualization by esophagogastroduodenoscopy (EGD). Tissue biopsy and/or cytological examination confirms the diagnosis.

Complications

By the time the diagnosis is made, esophageal tumors are usually large. The overall survival rate for patients with esophageal tumors is 19.9% at 5 years after diagnosis (American Cancer Society [ACS], 2019b).

Treatment

Because a cure is uncommon, surgery is primarily palliative and intended to restore normal swallowing. Surgical procedures include esophageal reconstruction such as transhiatal esophagectomy excision and reconstruction or bypass.

A multimodal approach may be the best option for cure. This would include preoperative chemotherapy and radiation therapy for 6-8 weeks, then surgery, with or without additional chemotherapy following surgery.

Radiation therapy or endoscopic laser therapy, such as photodynamic therapy (PDT), may be recommended for advanced, untreatable tumors. A patient with an obstructing tumor may need dilation or stent placement to allow limited oral nutrition and to relieve symptoms. If surgery is impossible and palliative treatment is unsuccessful, a gastrostomy tube may be placed to bypass the esophagus and allow the patient to be fed directly into the stomach.

Psychosocial support is warranted for patients undergoing care for esophageal cancer. Psychological care is also important for patients who undergo a gastrostomy or who have inoperable cancer or an esophageal obstruction. Such patients may experience feelings of grief or altered body image and should be encouraged to express them.

Diverticula

A **diverticulum** is an outpouching of one or more layers of the esophageal wall.

Risk factors

Diverticula (plural for diverticulum) probably result from esophageal pressure and motor abnormalities or from extrinsic inflammation. They are relatively uncommon and usually nonproblematic. Esophageal diverticula are classified according to location. An esophageal diverticulum may occur immediately above the UES (Zenker's diverticulum), near the esophageal midpoint (traction diverticulum), immediately above the LES (epiphrenic diverticulum), or intramurally along the body of the esophagus (intramural diverticulum).

Symptoms

More often than not, patients with esophageal diverticulum are asymptomatic, however, some patients may experience:

- Difficulty swallowing.
- Feeling like food is caught in the throat.

- Regurgitating food when bending over, lying down, or standing up.
- Pain when swallowing.
- Chronic cough.
- Bad breath.
- Chest or neck pain.
- Weight loss.
- Vocal changes or Boyce's sign (a gurgling sound when air passes through the diverticulum).

Diagnosis

Esophageal diverticula may be diagnosed by barium swallow, EGD, or manometry.

Zenker's diverticulum

A Zenker's diverticulum may be associated with a dysfunctioning UES and is typically found in middle-aged and elderly patients. This esophageal diverticulum occurs at the junction of the hypopharynx and the esophagus, an area known as Killian's triangle.

Symptoms

Patients may present with cervical dysphagia, halitosis, or aspiration pneumonia.

Complications

Complications of a Zenker's diverticulum may include malnourishment caused by poor oral intake, aspiration pneumonia, or perforation of the diverticulum, leading to severe inflammation of the mediastinum with possibly fatal sequelae. The most common life-threatening complication is aspiration. Other complications include massive bleeding from the mucosa or from fistulization into a major vessel, esophageal obstruction, and fistulization into the trachea.

Treatment

A Zenker's diverticulum is typically left untreated, except when complications occur, because it is primarily a benign condition. Patients may be encouraged to sleep with the head of the bed elevated on blocks and should not eat or drink within 3 to 4 hours of bedtime.

If surgical intervention is necessary, a small diverticulum may be treated simply by myotomy (cutting the cricopharyngeal muscle under local anesthesia), by diverticulectomy, or, more commonly, in a two-stage operation involving both myotomy and diverticulectomy.

Traction diverticula

Traction diverticula are commonly small and nonretentive. These esophageal diverticula occur at the esophageal midpoint.

Risk factors

Evidence suggests that they may be caused by esophageal motility abnormalities, including spasm, achalasia, and LES hypertension.

Symptoms

A traction diverticulum may cause no signs or symptoms.

Treatment

Traction diverticula usually do not require therapy. Occasionally an underlying abnormality requires a long myotomy.

Epiphrenic diverticula

Epiphrenic diverticula are rare, occurring at a rate of about 1 in 500,000 annually. These distal esophageal outpouchings occur most often on the anterior wall of the esophagus, immediately above the LES (Abdollahimohammad, Masinaeinezhad, & Firouzkouhi, 2014).

Risk factors

A lack of coordination between esophageal contraction and LES relaxation may be the cause of epiphrenic diverticulum. Another causal theory is a combination of esophageal obstruction, functional or mechanical dysfunction, and a point of weakness of the muscularis propria.

Symptoms

A patient with an epiphrenic diverticulum may regurgitate massive amounts of fluid, usually at night when lying down.

Diagnosis

Chest radiograph, barium esophagogram, endoscopy, and manometry can be used to aid in the diagnosis of an epiphrenic diverticulum.

Complications

Large diverticula can result in severe dysphagia, regurgitation, food retention, and the risk of aspiration pneumonia.

Treatment

If surgical intervention is indicated, an epiphrenic diverticulum is best treated with removal and a long myotomy.

Intramural diverticula

An intramural diverticulum is an obscure condition characterized by numerous small intramural outpouchings that probably represent dilated ducts of submucosal glands. These diverticula occur intramurally along the body of the esophagus.

Risk factors

They are associated with strictures, esophageal dysmotility, or infection.

Symptoms

Intramural esophageal diverticula are thought to be associated with symptoms of gastroesophageal reflux.

Treatment

An intramural diverticulum is generally treated by treating the coexisting conditions, such as stricture or infection.

Strictures

An esophageal **stricture** is an abnormal formation of white fibrous tissue that is usually at the lower end of the esophagus and may or may not be circumferential.

Risk factors

Esophageal strictures are common complications of caustic injuries or may be the result of candidiasis or prolonged and severe reflux.

Symptoms

Progressive dysphagia is the most common clinical feature.

Diagnosis

To rule out malignancy as the cause of the stricture, endoscopic examination with multiple biopsies and a brush exfoliative cytological examination are mandatory.

Complications

Perforation is the primary complication of dilation used to treat strictures. The most common symptom of esophageal perforation is persistent pain after dilation—even mild pain is abnormal, though minor bleeding is common. Mediastinitis is a common complication following perforation, and death can occur in less than 24 hours. Bacteremia, another complication, is a serious problem in immunocompromised patients.

Treatment

In treating a stricture, the physician may use weighted tungsten bougies, also known as Maloney or Hurst dilators; hydrostatic balloon dilators; or graduated plastic Savary-Gilliard or American dilators. Note that mercury-weighted bougies may still be used in some practice settings. Many patients require follow-up with further dilation at variable intervals.

A long, narrow stricture is seen more often in children and may be treated surgically by colon interposition.

Rings and webs

Esophageal rings and webs are thin, circumferential, mucosal shelves within the esophagus.

Esophageal webs

Webs consist of mucosa and submucosa and generally appear in the upper esophagus.

Risk factors

The association of a cervical esophageal web and iron deficiency anemia in middle-aged women is known as Paterson-Kelly or Plummer-Vinson syndrome. In these patients, the webs seem to regress spontaneously with treatment of the iron-deficiency anemia.

Symptoms

Esophageal webs often present as intermittent dysphagia.

Diagnosis

Although endoscopic examination can provide the diagnosis, it usually ruptures the web.

Complications

Paterson-Kelly or Plummer-Vinson syndrome is also associated with an increased incidence of postcricoid carcinoma.

Treatment

If the obstruction remains after an endoscopy, bougienage with Maloney dilators may be used. In rare cases, a patient may require dilation with a pneumatic balloon. If symptoms recur, further dilation may be necessary.

Esophageal rings

Rings, which are usually thicker than webs, consist of mucosa and muscle, and usually arise in the lower esophagus at the GEJ. Lower esophageal rings, also known as **Schatzki's ring** or B-rings, are thin, concentric membranes located at the esophagogastric junction.

Risk factors

The cause of esophageal rings is not entirely known but is thought to be associated with GERD.

Symptoms

Esophageal rings are more likely to cause symptoms than webs. A characteristic symptom is episodic dysphagia, or food impaction, which is often related to eating rapidly.

Diagnosis

Diagnosis is by barium swallow. Endoscopic investigation may also be needed to rule out the presence of a peptic stricture.

Complications

The most common complication of an esophageal ring is a food impaction that requires endoscopic removal. Dilation follows at a later time due to the inflammation caused by the food impaction.

Treatment

To relieve symptoms, the esophagus may be dilated using balloon dilation, bougie, or Savary-Gilliard or Maloney dilators.

Foreign-body obstruction

The esophagus is the most common site of acute foreign-body obstruction.

Risk factors

The majority of foreign-body ingestions occur in children, with coins being the most frequently ingested object. In adults, the most frequently observed foreign body is meat. Most ingested foreign bodies pass spontaneously from the esophagus into the stomach and through the GI tract. Some individuals with mental illness may be at risk at ingesting other foreign objects, removal of which may require endoscopy or surgical intervention.

If a pill is swallowed with little or no fluid intake when the patient is in a supine position, or just before going to bed, it can remain in the esophagus, eroding the esophageal tissue and causing an ulcer. Substances known to damage the esophagus include doxycycline, tetracycline, clindamycin, potassium chloride, ferrous sulfate, quinidine, aspirin, nonsteroidal anti-inflammatory drugs (NSAIDs), and vitamin A.

If small alkaline batteries (such as those found in electronic equipment) are swallowed, they must be retrieved as soon as possible, and any sign of perforation should be treated surgically because local corrosive effects may be fatal.

Symptoms

Patients with an acute esophageal obstruction usually experience local pain, dysphagia, or odynophagia.

For pediatric patients, a chronic foreign body (where a foreign object has been in the esophagus for 1-12 months) presents itself as a recurrent cough or abdominal pain.

Diagnosis

If the object is radiopaque, plain X-ray films may be ordered to provide information on the location and size of the obstruction. If barium X-ray films are used to visualize an obstruction, it is important to be aware of the danger that the patient may aspirate the barium.

Complications

Potential complications of foreign-body esophageal obstructions include esophageal ulceration, perforation, or penetration of the aorta or its branches followed by hemorrhage. The risk of perforation can be minimized by the use of overtubes and hoods that cover objects as they are being withdrawn.

Treatment

Larger foreign bodies tend to lodge at the GEJ. Once objects enter the stomach, most will pass uneventfully.

The treatment method used to remove the obstruction depends on the type of foreign body and the etiology of the obstruction. These methods may include pharmacologic agents, endoscopic removal, or surgery.

Pharmacologic agents such as sublingual nitroglycerin or IV glucagon may be used to relax the LES and allow passage of the item in adults. In the past, meat boluses have been dissolved enzymatically using papain. However, the use of papain is not recommended because it has been associated with esophageal perforation and pulmonary aspiration.

Treatment for damage to the esophagus from medications involves stopping the medication, relieving odynophagia, providing adequate nutrition, and watching for complications. In 3-6 weeks, the patient's symptoms should be relieved.

Chapter 30 provides in depth information on the endoscopic and surgical removal of foreign bodies.

INFLAMMATORY DISEASES

Inflammatory diseases that can affect the esophagus include Crohn's disease and eosinophilic esophagitis.

Crohn's esophagitis

Crohn's disease can produce inflammation anywhere in the GI tract. Inflammation isolated to the esophagus can occur but is usually associated with disease elsewhere in the GI tract. (For information on Crohn's disease, see Chapter 15.)

Eosinophilic esophagitis

This is a chronic, allergic, inflammatory disease characterized by excessive histamine production and mast cell degradation. It is usually found in the upper GI tract.

Risk factors

Eosinophilic esophagitis occurs in people who are highly allergic and have multiple food allergies, allergic rhinitis, asthma, and eczema. It is associated with a heightened immune response, causing inflammation and luminal narrowing in the esophagus. It can also occur secondary to eosinophilic gastritis.

Symptoms

Patients experience dyspepsia symptoms, nausea, vomiting, and a feeling of food getting stuck in the esophagus. Aphthous ulcers in the mouth are often associated with this disease.

Diagnosis

Patients often undergo an upper endoscopy for associated symptoms. The esophageal appearance shows circular rings, linear furrows, whitish papules, and plaques. Diagnosis is made by identifying eosinophil counts >10 per high-power field on microscopic examination of biopsies.

Complications

Failure to thrive, nausea, and vomiting in children and food impaction in adults are common complications in eosinophilic esophagitis.

Treatment

Treatment for eosinophilic esophagitis is aimed at stopping the allergic response and identifying and avoiding the allergy triggers.

Treatment also includes proton pump inhibitors (PPIs), topical (inhaled) steroids, and possibly H_2 blockers. Mast cell stabilizers such as montelukast (Singulair) or cromolyn (Gastrocrom) may also be required.

INFECTIOUS DISEASES

Viruses, bacteria, fungi, and mycobacteria can all cause esophageal infection. The most common causes are *Candida*, herpes simplex virus (HSV), and cytomegalovirus (CMV). Infection of the esophagus is more commonly found in patients who have an underlying disease that compromises the immune system such as human immunodeficiency virus (HIV) infection.

Esophageal candidiasis

Candidiasis is caused by the yeast *Candida*. It is a fungal infection of the esophagus.

Risk factors

Esophageal candidiasis rarely develops in patients who do not have an underlying disease such as diabetes, immune deficiency, or malignancy.

Symptoms

The main symptoms of esophageal candidiasis are dysphagia and odynophagia.

Diagnosis

Esophageal candidiasis may be diagnosed on barium swallow, but a specific diagnosis of candidiasis is not possible using radiographic studies. An endoscopy will reveal whitish plaques with a normal mucosal pattern between the plaques. Definitive diagnosis is by demonstration of mycelial forms in tissue samples obtained from a biopsy or cytology brushing.

Complications

Severe *Candida* infection can destroy esophageal innervation, causing abnormal motility. Serious complications include hemorrhage, yeast dissemination, and, rarely, perforation.

Treatment

Esophageal candidiasis is treated with antifungal medications such as clotrimazole, miconazole, nystatin, or Diflucan.

Herpetic esophagitis

Herpetic esophagitis is caused by the herpes simplex virus (HSV) in immunocompromised patients.

Risk factors

Predisposing factors to herpetic esophagitis include HIV infection, organ transplant, lymphoma, leukemia, or any immunocompromised state.

Symptoms

Patients with this form of esophagitis may have dysphagia, nausea, chest pain, heartburn, fever, or sores in the back of the throat or mouth.

Diagnosis

Separate ulcers or shallow plaques are visible on endoscopy. Diagnosis is by virus tissue culture or by characteristic cellular changes on cytology or biopsy.

Complications

Spontaneous esophageal perforation has been reported in cases of esophageal HSV.

Treatment

Oral or intravenous antiviral medication such as acyclovir may be used to treat herpetic esophagitis, depending on the severity of the infectious process.

CMV esophagitis

Cytomegalovirus (CMV) is the cause of CMV esophagitis.

Risk factors

Patients who have had organ transplantation, are undergoing long-term renal dialysis, or have HIV infection or acquired immunodeficiency syndrome (AIDS) have been reported to develop CMV esophagitis.

Symptoms

Patients with CMV esophagitis may experience dysphagia, odynophagia, retrosternal pain, nausea and vomiting, abdominal pain, fever, diarrhea, weight loss, chest pain, or hemoptysis.

Diagnosis

Diagnosis may be made by double contrast barium swallow or by endoscopy. CMV esophagitis may appear on endoscopic exam as a spectrum of lesions in the esophagus, ranging from superficial mucosal inflammation to giant ulcerations. Esophageal biopsy should be performed and sent for viral culture. Cytology brushings are sent for stain to detect infected cells.

Treatment

Antiviral medications are used to treat CMV esophagitis, including intravenous (IV) ganciclovir, valganciclovir (Valcyte), cidofovir (Vistide), or foscarnet (Foscavir). In some cases, combination therapy of two or more drugs may be necessary.

Symptomatic relief of CMV esophagitis can be achieved by using viscous lidocaine, H_2 blockers, sucralfate (Carafate), or antacids.

HIV esophagitis

HIV esophagitis is caused by the HIV virus.

Risk factors

HIV esophagitis may be present in patients who are HIV-positive. Usually it is secondary to one of the other infectious esophagitis types, such as infections with *Candida*, HSV, or CMV.

Symptoms

Patients who are HIV-positive may present with complaints of odynophagia.

Diagnosis

Endoscopy with tissue cultures can differentiate between HIV and CMV ulcers.

Treatment

HIV esophagitis responds dramatically to treatment with steroids and antiviral medications.

Mallory-Weiss tears

A **Mallory-Weiss tear** is a mucosal tear at the GEJ.

Risk factors

These tears are associated with prolonged, forceful vomiting; trauma; childbirth; or complications of EGD. Patients who have a Mallory-Weiss tear often have a history of alcohol misuse.

Symptoms

Typically, prolonged emesis or dry heaves is followed by vomiting of bright red blood.

Diagnosis

A Mallory-Weiss tear can be diagnosed and treated during an upper endoscopy.

Complications

In severe cases, Mallory-Weiss tears can cause severe bleeding.

Treatment

The amount of blood lost is usually small, and the patient with a Mallory-Weiss tear is generally treated conservatively because bleeding stops spontaneously. Profuse bleeding may be controlled endoscopically with a coagulating contact probe or by placing several clips to close the tear.

Motility disorders

Esophageal motility disorders are primary if the etiology of the abnormality is unknown but the esophagus is the site of major involvement, or secondary if the esophageal abnormalities are features of a more generalized disease process. Primary esophageal motility disorders include achalasia, diffuse esophageal spasm, and jackhammer (previously called nutcracker) esophagus.

Achalasia

Achalasia is a combined defect of peristalsis in the esophageal body and elevated LES pressure.

Risk factors

The etiology of the abnormality is unknown.

Symptoms

Patients with achalasia present with dysphagia to solids and liquids, regurgitation, and weight loss.

Diagnosis

Diagnostic evaluation procedures may include a radiographic examination, manometric study, and EGD. On X-ray films, the esophagus appears dilated with a narrow “bird beak” at the distal end. Microscopically, a loss of non-argyrophilic ganglion cells in the myenteric plexus is evident. On manometry study, peristalsis may be absent, with very high pressures in the LES. On endoscopic exam, it may be difficult to pass the endoscope through the LES.

Complications

Complications of achalasia are related to retention of food and stasis in the esophagus. Esophagitis may be evident endoscopically. Aspiration of esophageal contents is a danger and may be followed by bronchopneumonia. The prevalence of esophageal carcinoma is higher than normal in patients with achalasia, possibly because it may be precipitated by stasis and mucosal irritation. Regular follow-up care is recommended.

Treatment

Symptoms may be relieved by eating slowly, chewing well, drinking fluids with meals, and sitting up while eating. Many patients benefit from pneumatic balloon dilation. A cardiomyotomy (Heller’s operation) may be performed to reduce LES pressure in patients who experience unsuccessful treatment with pneumatic dilation or who respond poorly to dilation.

A newer treatment is peroral endoscopic myotomy (POEM). A submucosal tunnel is created by the endoscopist, followed by a myotomy of the circular muscle of the lower esophagus.

Rarely, in carefully selected patients, a transhiatal esophagectomy may be recommended for patients with achalasia.

In patients who are not considered suitable for dilation or surgery, long-acting nitrates such as sublingual isosorbide dinitrate (Isordil) or calcium channel blockers such as nifedipine (Adalat CC) may be used to reduce LES pressure.

The use of botulinum toxin has been shown to provide an alternative therapy for treatment of achalasia. Botulinum toxin is injected into the four quadrants of the LES using a sclerotherapy needle. The toxin acts as a paralytic agent, reducing the LES pressure.

Diffuse esophageal spasm

Diffuse esophageal spasm (DES) is one of the most common motility disorders. It is defined manometrically as repetitive or prolonged simultaneous contractions (nonperistaltic esophageal contractions that occur all at once up and down the length of the esophagus, independent of pharyngeal contractions) with intermittent normal peristalsis.

Risk factors

Diffuse esophageal spasm has an unknown cause and pathophysiology.

Symptoms

One symptom may be non-cardiac substernal chest pain, which may be severe and may closely mimic angina pectoris. Dysphagia may be present with both solids and liquids and is most severe when the patient ingests extremely hot or cold foods. In most cases, the distal two-thirds of the esophagus shows muscular thickening.

Diagnosis

A barium swallow may show isolated, uncoordinated movements of the lower two-thirds of the esophagus. The entire two-thirds of the esophagus may contract as a unit, propelling barium both retrograde and into the stomach. Manometric examinations may reveal a simultaneous high-amplitude and abnormally long contraction in the lower two-thirds of the esophagus.

Complications

Severe pain is associated with DES.

Treatment

DES may be relieved by pharmacologic therapy with anticholinergics, nitrates, smooth muscle relaxants such as hydralazine, or calcium channel blockers such as nifedipine (Adalat CC) or verapamil (Calan). When the condition is severe and unresponsive, dilation may be used to relieve symptoms. Some patients benefit from a long esophagomyotomy, which involves cutting the esophageal muscles to prevent peristalsis.

Jackhammer (Nutcracker) esophagus

Jackhammer esophagus, also known as nutcracker esophagus and symptomatic esophageal peristalsis, is a manometric disorder.

Risk factors

The etiology of the abnormality is unknown.

Symptoms

Patients present with non-cardiac chest pain.

Diagnosis

Jackhammer esophagus is defined manometrically as peristalsis with contractile amplitude that is two to three times the normal value.

Treatment

If treatment is necessary, pharmacological therapy may be tried as described for DES.

Caustic injury

Accidental or deliberate ingestion of highly alkaline or acid compounds may result in injury to the esophagus.

Risk factors

Alkaline compounds such as lye, drain cleaners, and bleaches typically cause more damage than acidic compounds such as toilet bowl cleaners, soldering fluxes, and battery fluids.

Symptoms

The most common symptom is odynophagia, but patients may also complain of dysphagia and chest pain, or they may drool profusely. Prognosis depends on the depth and extent of injury. Symptoms may take as long as 14 days to completely manifest due to the necrotizing process of the esophageal tissue.

Diagnosis

Diagnostic evaluation of caustic injuries involves determining the nature of the ingested agent; radiographic examinations of the neck, chest, and abdomen to exclude perforation or pneumonia; and endoscopic examination to document the extent of injury, unless there is evidence of a perforation or extensive necrosis. Because tissue damage is usually not immediately evident, endoscopy is usually delayed for 12-24 hours after the ingestion. This allows the endoscopist to fully assess the extent of injury. Once an area of severe damage is visualized, the endoscopist may choose to discontinue the procedure to reduce the risk of perforation.

Esophageal burns may be clinically staged:

- **First-degree:** hyperemia and edema of the mucosa; damage limited to the mucosa.
- **Second-degree:** exudate, erosions, and shallow ulcers destructive of the mucosa and submucosa with penetration of the injury into the muscle layers.
- **Third-degree:** deep ulceration, circumferential necrosis, often the presence of a black coagulum, and full-thickness injury with extension into the pleura and mediastinum.

Complications

Complications of caustic ingestion may include formation of strictures or eventually squamous cell carcinoma. Post-corrosive complications can be fatal due to perforation or tracheal necrosis. Fistula formation, pneumonia, and anemia can prolong hospital stay and are another cause of mortality.

Treatment

Treatment following caustic injury depends on the extent of the injury, but patients should always be kept nothing by mouth (NPO). In the case of acid ingestion, the physician may order large volumes of milk or water to dilute the acid.

Hemodynamic stabilization and adequacy of the patient's airway should be supported as needed, and intake and output should be carefully documented. Since strictures can occur as a result, nasogastric tube insertion to ensure patency of the esophageal lumen should be done with caution. If the patient shows evidence of laryngeal involvement or respiratory difficulties, endotracheal intubation or tracheostomy may be required. Dilation may be required for second-degree injury, and surgery is indicated for patients who deteriorate despite intensive medical management.

The physician may order PPIs and H_2 blockers to minimize esophageal injury by suppressing acid reflux. Corticosteroids and broad-spectrum antibiotics are primarily used with airway involvement. Surgical management, ranging from repair of perforations to esophagectomy, may be required in third-degree injury.

Barrett's esophagus

Barrett's esophagus is defined as epithelial metaplasia in which normal squamous epithelium is replaced by one or more of the following types of columnar epithelium: a distinctive, specialized columnar epithelium; a junctional type of epithelium; and/or a gastric fundus type of epithelium.

Risk factors

Barrett's esophagus occurs in a small percentage of patients with chronic GERD. Despite this risk, the American Gastroenterological Association (AGA) (2011) and American College of Gastroenterology (ACG) guidelines do not recommend screening the general population of patients with GERD for adenocarcinoma (Shaheen, Falk, Iyer, & Gerson, 2016). However, screening is recommended for patients with multiple risk factors for adenocarcinoma, including chronic GERD, hiatal hernia, age ≥50 years, male gender, white race, central obesity, cigarette smoking, and a confirmed history of Barrett's esophagus or esophageal adenocarcinoma in a first-degree relative.

Symptoms

Barrett's esophagus typically has no symptoms. Symptoms may be associated with GERD and obstructing tumor growth.

Diagnosis

Diagnosis of Barrett's esophagus is by endoscopic visualization of the mucosa supported by examination of a tissue biopsy.

Complications

The prevalence of adenocarcinoma in patients with Barrett's esophagus is greater than in the general population, although the actual frequency of progression from dysplasia to cancer is unknown. The rate of progression to adenocarcinoma varies based on published estimates. Barrett's esophagus is the only known precursor to adenocarcinoma (di Pietro, Alzoubaidi, & Fitzgerald, 2014).

Treatment

Treatment is directed at the underlying gastroesophageal reflux esophagitis. Stricture dilation may be required for scar tissue formation.

Patients diagnosed with Barrett's esophagus with high-grade dysplasia (HGD) may require surgery. Most patients referred for evaluation of HGD or early cancer will have at least one visible abnormality detected during endoscopic inspection.

Endoscopic ablative therapies include endoscopic radiofrequency ablation, cryoablation, and argon plasma ablative therapies, and endoscopic mucosal resection (EMR) may be done with curative intent and are increasingly advocated as minimally invasive management alternatives for Barrett's esophagus with early neoplasia. Published reports describe the use of EMR to treat focal areas of high-grade dysplasia or intramucosal carcinoma occurring with Barrett's esophagus (Singh, Sanaka, & Thota, 2018).

Currently, there are limited research studies showing that combining PPIs and aspirin prevents the neoplastic progression of Barrett's esophagus (Jankowski et al., 2018). Long-term aspirin use has showed promise in improving survival in patients with esophageal cancer.

Fistulas

A fistula is a communication channel between the bronchi and the esophagus (bronchoesophageal) or the trachea and the esophagus (trachoesophageal). An aortoesophageal fistula extends through the wall of the esophagus and into the aortic arch.

Risk factors

Bronchoesophageal and tracheoesophageal fistulas in adults are usually caused by cancer but may also be caused by a benign inflammatory process such as infectious esophagitis or trauma. Aortoesophageal fistulas may develop if an ingested foreign object lodges in the region above the aortic arch or if a tumor extends through the wall of the esophagus.

Symptoms

Symptoms of bronchoesophageal and tracheoesophageal fistulas include chronic cough, fever, and recurrent pulmonary infections, with or without dysphagia. Fistulas related to cancer cause acute episodes of coughing after eating. Aortoesophageal fistulas usually result in minor bleeding as erosion connects the aorta and esophagus.

Diagnosis

Diagnosis is by contrast tests such as barium swallow, CT scan, or MRI.

Complications

With an aortoesophageal fistula, massive bleeding may begin at any time and almost always causes death.

Treatment

The treatment for esophageal fistulas is surgery to remove necrotic and irreversibly damaged tissue and to close the fistula. When surgical procedures are not possible, endoscopic repair using stents is an option.

Esophageal fistulas can be fatal, apart from the carcinoma itself. Many of these patients cannot undergo surgery due to their poor overall condition. In these cases, stents offer an alternative to major surgery and can often help to alleviate symptoms. In general, two main types of stents are used. Covered self-expanding stents can firmly expand and cover the damaged area, and have a low migration rate. Silicone stents are empty cylinders with a fully closed outer shape with external studs to prevent dislodgement. While this new type of stent is flexible, migration and adhesion are still major issues affecting airway tissues.

Other nonsurgical treatments include transplantation of mesenchymal stem cells derived from bone marrow, which can successfully treat 3 mm bronchopleural fistulas; one-way umbrella-shaped endobronchial valves; and fibrin glue injection.

CASE SITUATION

Doris Johnson is a 51-year-old woman who lives with her husband and four grown, unmarried sons on a large dairy farm in Wisconsin. For the past 4 years, Doris has been experiencing increasing retrosternal discomfort or a burning sensation shortly after eating or on sudden bending. Recently, the discomfort, described as heartburn, has progressed to a painful ache that radiates to her neck, shoulders, and upper arms. Because both gallbladder and coronary artery disease are in her family history, Doris has become concerned enough to seek medical assistance. Doris is 5 feet 7 inches tall and weighs 186 pounds. Her vital signs are: blood pressure 170/90; pulse 90; respiration 22; temperature 99.2 degrees Fahrenheit. She denies use of all habit-forming substances, including tobacco, although she admits she may have a small amount of alcohol and a cigarette on festive occasions. She takes no regularly prescribed drugs, but she has used an antacid to help the burning sensation and the occasional NSAID for pain. Her primary care physician has ruled out cardiac chest pain and has made a tentative diagnosis of GERD and prescribed a PPI. After 2 weeks, her symptoms continue, and she is referred to a gastroenterologist for a diagnostic workup. She is scheduled for an EGD.

Points to think about

1. Doris comes to the endoscopy unit for her EGD. Besides the information on the usual assessment form, what other information would be helpful to the gastroenterology nurse?
2. What finding during EGD would help make Doris's diagnosis?
3. What nursing diagnosis would fit this patient?
4. How might a nurse assist Doris in managing the stress associated with EGD?
5. What can the nurse teach Doris?

Suggested responses

1. Additional data about Doris that might be helpful would include:
   a. All previous studies in her recent workup, including the cardiac workup and its results.
   b. The name, dosage, frequency, and length of use of all medications Doris is currently taking, including over-the-counter (OTC) medications such as NSAIDs that may be contributing to her gastric reflux; any allergies to medications; and the effectiveness of the antacid she is using.
   c. The presence of symptoms specific to gastric reflux such as regurgitation, dysphagia, or odynophagia, and the duration, time, and other factors associated with discomfort or pain episodes (e.g., eating, bending over, exercise, tight clothing, sleeping, the possibility of pregnancy).
2. A finding of gross esophagitis would give a definitive diagnosis of GERD; however, some patients may not have this finding with endoscopic examination.
3. As Doris' BMI is above normal, an appropriate nursing diagnosis might be "altered nutrition: more than body requirements, related to caloric intake exceeding metabolic need." Excessive weight gain can increase intra-abdominal pressure and exacerbate reflux. In women, obesity can be a precursor to Barrett's esophagus. Eating certain foods can add calories, decrease LES tone, exacerbate pain, and increase gastric acid secretion. If a full nutritional assessment seems warranted, an appropriate referral to a dietitian should be provided.
4. Measures the nurse may use to help Doris manage stress during the EGD might include:
   a. Reiterate the procedural process, focusing on Doris's questions and concerns.
   b. Repeat the explanation of the method and type of anesthetic to be used, rationale for its use, its duration, and related postprocedural nursing precautions and measures.
   c. Provide periodic reassurance before the procedure that she will be able to breathe normally despite the bite-block mouthpiece and tube.
   d. Assign a nurse who can support Doris throughout the procedure, both verbally and through touch.

5. Once a diagnosis of GERD has been made, to be sure that Doris has adequate knowledge about her condition and proposed treatment, the gastroenterology nurse might:
   a. Use anatomical drawings to describe the physical process of reflux, including a discussion of the LES area and, if identified on endoscopy, hiatal hernia, and the location of inflammation or strictures.
   b. Explain how to use gravity to decrease gastric reflux by not lying down after meals and elevating the head of the bed on 6-inch blocks to decrease acid reflux at night.
   c. Explain the medications the physician has prescribed for this problem, how they work, when to take them, and any side effects.
   d. Review any OTC or other medications Doris is already taking to help avoid those that may be harmful to this condition, including the use of NSAIDs that can contribute to gastric irritation. Other such medications may include anticholinergics, sedatives, tranquilizers, theophylline, and calcium channel blockers.
   e. Explain dietary modifications that may be helpful, including avoiding foods that lower LES pressure such as alcohol, tomatoes, peppermint, licorice, and caffeine-containing foods and beverages.
   f. Suggest other lifestyle modifications that may be helpful, including cessation of smoking (cigarette smoking reduces LES tone), avoiding clothing that is tight around the abdomen, initiating a supervised diet, and using proper body mechanics to avoid bending.
   g. Advise Doris that chronic irritation of the esophagus can lead to complications such as ulceration, stricture formation, Barrett's esophagus, and pulmonary problems.
   h. Encourage her to follow the prescribed treatment regimen carefully and to seek follow-up care.

REVIEW TERMS

achalasia, atresia, Barrett's esophagus, candidiasis, diaphragmatic hiatus, diffuse esophageal spasm (DES), diverticula, dyspepsia, dysphagia, esophageal webs and rings, esophagitis, esophagus, fistula, gastroesophageal reflux disease (GERD), heartburn, jackhammer (nutcracker) esophagus, lower esophageal sphincter (LES), Mallory-Weiss tear, odynophagia, peristalsis, regurgitation, Schatzki's rings, stricture, upper esophageal sphincter (UES), varices

REVIEW QUESTIONS

1. In adults, the approximate length of the esophagus is:
   a. 15 cm.
   b. 25 cm.
   c. 35 cm.
   d. 45 cm.
2. The outermost layer of the esophagus is made up of:
   a. Mucosa.
   b. Submucosa.
   c. Muscularis.
   d. Serosa.
3. The progressive circular muscle contraction initiated by esophageal distention is known as:
   a. Achalasia.
   b. Diffuse esophageal spasm.
   c. Primary peristalsis.
   d. Secondary peristalsis.
4. The first step in the treatment of GERD is:
   a. Lifestyle modification.
   b. Drug therapy.
   c. Dilation.
   d. Antireflux surgery.
5. A form of metaplasia in the esophagus in which normal squamous epithelium is replaced by one or more types of columnar epithelium results in:
   a. Esophageal reflux.
   b. Esophageal candidiasis.
   c. Barrett's esophagus.
   d. Esophageal cancer.
6. Life-threatening bleeding is a frequent complication of:
   a. Esophageal varices.
   b. Esophageal reflux.
   c. Esophageal tumors.
   d. Zenker's diverticulum.
7. Outpouchings of the esophageal wall located immediately above the lower esophageal sphincter are known as:
   a. Zenker's diverticula.
   b. Traction diverticula.
   c. Epiphrenic diverticula.
   d. Intramural diverticulosis.
8. The most frequently observed foreign-body obstruction in the esophagus of adults is a:
   a. Coin.
   b. Piece of bone.
   c. Hard candy.
   d. Piece of meat.
9. The most common causes of infectious disease in the esophagus include all of the following except:
   a. Cytomegalovirus.
   b. Herpes simplex virus.
   c. Candidiasis.
   d. Giardiasis.

10. Esophageal fistulas in adults are most often caused by:
    a. Trauma.
    b. Foreign-body obstruction.
    c. Cancer.
    d. Congenital defects.

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Chapter 14

STOMACH

This chapter will acquaint the gastroenterology nurse with the normal anatomy and physiology of the stomach. The symptoms, diagnosis, and treatment of selected gastric disorders will be reviewed.

Learning objectives

After reviewing the content of this chapter, the gastroenterology nurse will be able to:

1. Describe the anatomy of the stomach, including both macroanatomy and microanatomy.
2. Identify the normal motor and secretory functions of the stomach.
3. Discuss the pathophysiology, symptoms, diagnosis, and treatment of certain disorders that affect the stomach.

ANATOMY AND PHYSIOLOGY

Through a complex combination of exocrine secretory mechanisms and smooth muscle contraction and relaxation, the stomach has several important physiological purposes:

- Maintains relatively low levels of microbes in the upper digestive tract.
- Serves as a reservoir for swallowed food, drink, and digestive secretions.
- Originates signals for hunger and satiety.
- Digests food and prepares nutrients for absorption.
- Mixes and delivers chyme to the small intestine for further digestion and absorption.

The stomach is a muscular, J-shaped, distensible organ located just to the left of and below the diaphragm, between the esophagus and the duodenum (Figure 14.1). The stomach measures approximately 12 inches (30.5 cm) in length and 6 inches (15.2 cm) at its widest point, but stomach size varies. The position and shape of the stomach can also vary based on these factors:

- Volume or amount of food in the stomach.
- Position of the body.
- Pressure exerted on the stomach from the intestines.
- Stage of digestion.

The stomach includes the:

- **Cardia**, the portion of the stomach that immediately adjoins the esophagus.
- Gastric **fundus**, the dome-shaped part of the stomach that extends to the left above the cardia.
- Gastric **body**, which is below the fundus and extends to the incisura angularis, a notch-like indentation located on the upper, lateral border of the stomach.
- **Antrum**, which is below the incisura angularis and extends to the narrow, tubular pylorus.
- **Lesser curvature**, the upper lateral border of the stomach.
- **Greater curvature**, the lower lateral border of the stomach (Figure 14.1).

Movement of food into and out of the stomach is controlled by two sphincters.

- **Lower esophageal sphincter (LES)**, also called the cardiac sphincter, is a group of thickened circular muscles at the distal end of the esophagus. The LES regulates the opening and closing of the esophageal lumen, controlling the entry of food into the stomach.
- **Pyloric sphincter** is the thick, muscular wall of the **pylorus**, which is located at the distal end of the stomach. The pyloric sphincter controls the movement of stomach contents into the duodenum.

Layers of the stomach

The stomach wall consists of four layers.

- **Mucosa** lines the interior of the stomach. When it is not filled, the stomach interior has wrinkled ridges of muscle tissue called **rugae**. The rugae allow the stomach to distend to hold a large quantity of food without any substantial increase in pressure, helping to control the rate at which food enters the duodenum. The gastric mucosa contains several types of epithelial cells.
- **Submucosa** lies below the muscular layers of the stomach. The submucosa is composed mainly of areolar connective tissue. It contains the gastric blood vessels, **lymphatics**, and nerves.
- **Muscularis propria** is a three-tiered muscular layer, also known as muscularis externa. The muscularis propria consists of an outer layer of longitudinal muscle fibers, a middle layer of circular smooth muscle fibers, and an inner layer of oblique fibers. These three muscle layers contract to produce the peristaltic motion of the stomach while it churns and compresses food during digestion.
- **Serosa** is the outer layer and is a continuation of the visceral peritoneum that covers the exterior of the stomach. The serosa includes both the **greater omentum**, which hangs in a double layer from the greater curvature over the anterior side of the abdominal viscera, and the **lesser omentum**, which connects the lesser curvature to the underside of the liver. The subserosa is support tissue for the serosa.

Gastric vascular supply

Arterial

The stomach receives arterial blood from branches of the celiac artery (also known as the celiac axis or trunk): the common hepatic, the left gastric, and the splenic arteries. These arteries send branches to the lesser curvature and the greater curvature. Along the lesser curvature, the left gastric artery flows down from the cardia. At this point, it joins with the right gastric artery, which is a branch of the hepatic artery. The greater curvature receives blood from the gastroepiploic artery, which runs from the fundus to the pylorus. In addition, short gastric arteries derived from the splenic artery supply blood to the fundus of the stomach.

Venous

The venous drainage of the stomach accompanies the arterial supply, emptying into the portal vein. The greater curvature is drained by the right and left gastroepiploic veins, and the lesser curvature is drained by both the right gastric vein and the coronary vein. There is no gastroduodenal vein.

Gastric innervation

Parasympathetic

Parasympathetic innervation of the stomach is via the vagus nerves: the left vagus (anterior gastric nerve) and the right vagus (posterior gastric nerve). Over the lower esophagus, the right and left vagus nerves split. The left vagus divides into anterior gastric and hepatic branches. The right vagus also divides into two branches: one that supplies the posterior wall of the fundus and the body of the stomach, and another that supplies the posterior wall of the antrum. Vagal innervation to the body and fundus primarily stimulates acid secretion, whereas vagal innervation to the antrum mainly stimulates motility.

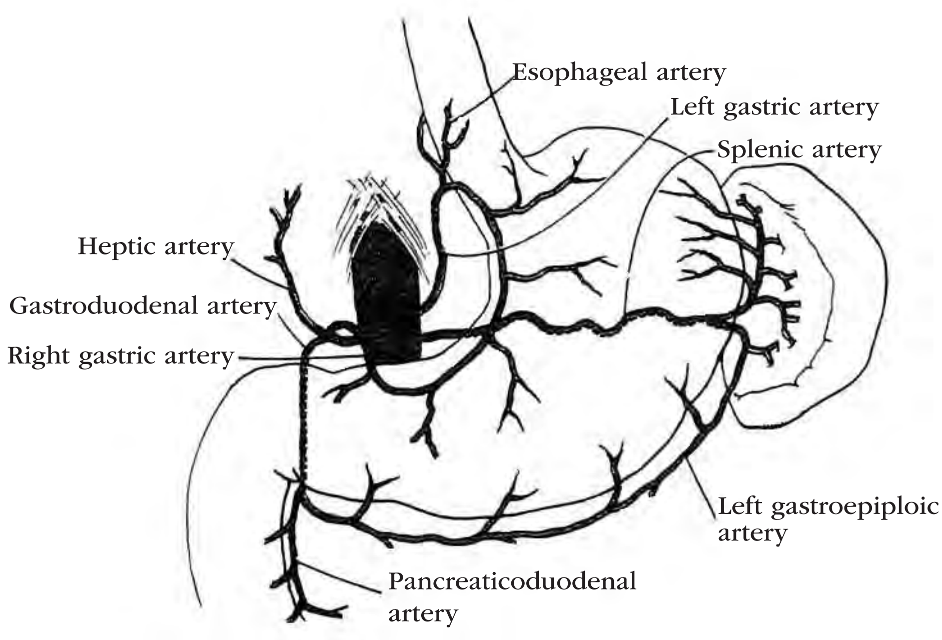

Figure 14.1. Blood Supply of the Stomach and Duodenum (From: Price, S. A., & Wilson, L. M. (2002). *Pathophysiology: Clinical Concepts of Disease Processes* (5th ed.). St. Louis, MO: Mosby.)

Sympathetic

Sympathetic innervation of the stomach is derived from the greater splanchnic nerves and the celiac ganglia. The afferent fibers conduct visceral gastric pain impulses, and the efferent fibers inhibit gastric secretion and motility. Intrinsic innervation of the stomach involves Auerbach's and Meissner's plexuses. Auerbach's plexus influences gastric motility, and Meissner's plexus is believed to be involved in gastrin release.

Gastric secretion

The primary function of the stomach is to initiate digestion by using both chemical secretions and mechanical movements. Gastric secretions also promote the absorption of vitamin B_{12}. In addition, the stomach serves as a reservoir for ingested food and liquid and regulates their entry into the duodenum.

The mucosa contains three types of glands, which differ from one region of the stomach to another.

- **Cardiac glands** are located just distal to the gastroesophageal junction. These glands secrete mucus and pepsinogens, which are converted into pepsin by hydrochloric acid.
- **Oxyntic glands**, also called the fundus glands, are located in the proximal two-thirds of the stomach. At least four types of cells have been identified in these glands:
  - Zymogen cells, also called **chief cells**, which secrete pepsinogens.
  - Oxyntic cells, also called **parietal cells**, which secrete hydrochloric acid and intrinsic factor, a glycoprotein necessary for the absorption of vitamin B_{12}.
  - Mucous neck cells, which primarily secrete mucus.
  - Endocrine or endocrine-like cells.
- **Pyloric glands** are located in the antrum and pylorus. These glands contain mucous cells, which secrete mucus and pepsinogens, and **G cells**, which secrete gastrin.

The pyloric and oxyntic glands also have **enterochromaffin cells**, which secrete serotonin. The cardiac, oxyntic, and pyloric glandular mucosae contain at least nine different types of endocrine cells that secrete hormonal products such as somatostatin and glucagon. In addition, the parietal and pyloric glands both secrete bicarbonate.

The stomach normally secretes about 2-3 quarts of digestive juices each day.

Gastric motility

The movements of the stomach serve the following two functions:

- Mixing and grinding of food, which takes place in the distal areas.
- Controlled emptying of the gastric contents into the duodenum.

The peristaltic movement of the stomach combines each bolus of food with a mixture of digestive juices to form a relatively homogeneous, semiliquid mass called **chyme**. Normally, peristalsis moves the chyme slowly toward the pylorus, regulated by a small region in the middle of the gastric body known as the gastric pacemaker. Before the chyme is ready to enter the duodenum, it must be the proper consistency and acidity, and the duodenum must be receptive. When these conditions are present, antral peristaltic contractions force the chyme through the pyloric sphincter and into the duodenum.

Only a portion of the chyme that is advanced toward the pylorus actually moves through the pyloric sphincter and into the duodenum. The rest is propelled backward, colliding with and further breaking down larger food particles in the antrum and body. The duodenum dilates as the chyme passes through the first segment of the small intestine.

The gastric emptying rate is controlled by neural impulses, the composition of the chyme, and hormones secreted by the small intestine.

DISORDERS OF THE STOMACH

Pathological conditions that are frequently seen in the stomach include peptic ulcer disease, gastric cancer, gastritis, stress ulcers, Dieulafoy's lesion, gastric varices, and gastric polyps. Other important gastric disorders include hiatal hernia, gastric outlet obstruction, congenital anomalies, gastric motor abnormalities, caustic injury, and bezoars. Infectious disorders include infectious gastritis and *Helicobacter pylori (H. pylori)* infection.

Peptic ulcer disease (PUD)

For **gastric ulcers** to develop, a variety of irritants initially disrupt the gastric mucosa. When the normal balance between factors that promote mucosal injury (e.g., gastric acid, pepsin, bile acids, and ingested substances) and those that protect the mucosa (e.g., an intact epithelium, mucus, and bicarbonate secretion) is upset, inflammation or ulcers may arise. The unique causal event may be mechanical, chemical, infectious, or ischemic.

In some patients, decreased pyloric sphincter pressure permits an increased reflux of duodenal material into the stomach, which disrupts the gastric mucosal barrier and leads to an increased back-diffusion of acid onto the mucosa. Subsequent histamine release causes mucosal damage and edema. Continued erosion can damage mucosal capillaries and submucosal blood vessels, leading to hemorrhage and shock. If erosion continues through the serosa, perforation and peritonitis can result.

Risk factors

The incidence of PUD ranges from an estimated 0.1% to 0.3% each year, and the incidence of PUD increases with age (Vakil, 2017). The two major risk factors associated with PUD are ***Helicobacter pylori (H. pylori)*** infection and the chronic use of nonsteroidal anti-inflammatory drugs (NSAIDs).

Risk factors for peptic (gastric or duodenal) ulcer disease include:

- *H. pylori* infection. Despite eradication, *H. pylori* is still present in half of the world's population. The prevalence of *H. pylori* has declined in developed countries, and the ulcer incidence in individuals infected with *H. pylori* is close to 1% annually (Vakil, 2017).
- Chronic use of salicylates or NSAIDs such as indomethacin or ibuprofen. PUD is increasingly noted in the elderly secondary to the increased use of NSAIDs in this age group (Wongrakpanich et al., 2018).
- Family history of gastric ulcers.
- Cigarette smoking.
- Excessive acid production from gastrinomas (gastrin-secreting tumors) seen in Zollinger-Ellison syndrome.
- High-dose and prolonged use of corticosteroids, bisphosphonates, potassium chloride, and some chemotherapeutic agents.

The role of emotional and physical stress and personality type in the development of gastric ulcers remains controversial.

Symptoms

Most patients with PUD have abdominal pain. However, the sensitivity and specificity of abdominal pain as markers for gastric ulcer are low. In general, gastric ulcer symptoms include:

- Abdominal pain that is localized to the epigastrium and does not radiate. It is most often described as having a burning quality.
- Pain after meals or on an empty stomach, with relief of pain after a meal or after taking an antacid or an antisecretory medication.
- Nonspecific complaints, such as nausea, vomiting, decrease in appetite, indigestion, and heartburn.
- Presence of pain that awakens the person from sleep, which is the most discriminating (although inexact) symptom.
- Complaints of black or bloody stools.
- Vomit that resembles coffee grounds or contains blood.
- Sharp, severe, and continuous upper abdominal pain (sometimes radiating to the back). These are alarming symptoms and require immediate medical attention.

Symptoms that mimic a gastric ulcer in patients who have no objective evidence of an ulcer are called non-ulcer dyspepsia. Dyspepsia, also known as indigestion, is a vague term that refers to epigastric discomfort, burning or pain, nausea, belching, and bloating. Medications, health conditions, digestive tract diseases, and lifestyle can cause dyspepsia. It is a common condition that may be occasional, chronic, or functional, but is not considered a disease. Dyspepsia affects about 1 in 4 individuals in the United States annually (National Institute of Diabetes and Digestive and Kidney Diseases, 2016).

Diagnosis

To aid in the diagnosis of PUD, the physician may order:

- Studies that may include an upper gastrointestinal (GI) series with barium or a water-soluble contrast such as Gastrografin; a computerized tomography (CT) scan for the evaluation of abdominal pain; or an esophagogastroduodenoscopy (EGD). Patients with unknown PUD who present to the emergency department with acute abdominal pain will commonly undergo a CT scan as part of the diagnostic workup. EGD is the procedure of choice for a patient with active bleeding and includes rapid urease testing, histology, culture, and/or polymerase chain reaction on gastric biopsies.
- Laboratory tests may include a urea breath test and a stool sample to detect active *H. pylori* infection.
- Blood samples to detect *H. pylori* antibodies and a complete blood count to test for anemia.
- Gastric acid secretion tests or serum gastrin levels may be ordered to determine the cause of existing peptic ulcers but are not helpful in the diagnosis of an ulcer.

Prepyloric ulcers are typically benign and best diagnosed endoscopically. Because gastric ulcers are potentially malignant, multiple tissue biopsies are necessary to exclude the presence of cancer.

After medical therapy and when there is no clear etiology, a repeat EGD should be performed in 12 weeks to evaluate and document gastric ulcer healing.

Complications

Complications of peptic ulcer disease include hemorrhage, perforation, penetration, and gastric outlet obstruction.

- **Hemorrhage**, the most common complication, causes hematemesis and/or melanotic stool. Peptic ulcer bleeding is more common in older patients. In some cases, life-threatening hemorrhage is the first indication of an ulcer.
- **Perforation** is less common in peptic ulcer patients. Gastric ulcers tend to perforate along the anterior wall of the lesser curvature of the stomach. Patients who have a perforation experience upper abdominal pain, guarding, rebound tenderness, and absent bowel sounds. Initial therapy includes nasogastric (NG) tube insertion, intravenous (IV) fluids for volume replacement, IV proton pump inhibitors (PPIs), and antibiotics. Perforation has the highest mortality rate, and surgery may be required.
- **Penetration** is pathologically similar to perforation, except that the ulcer crater extends into an adjacent

organ, most commonly the left lobe of the liver. Rarely, gastric ulcers can penetrate into the colon, resulting in a gastrocolic fistula. Symptoms may include back pain, night distress, changing epigastric pain, or pain that is refractory to agents that previously provided relief. Surgical intervention is often necessary.

- **Gastric outlet obstruction** is the least frequent complication of PUD in developed countries. It can result from impaired antral motility caused by inflammation and edema of the ulcer or from obstruction caused by scarring at the gastroduodenal junction. Symptoms of obstruction include abdominal distention, tympany, succussion splash, vomiting, and early satiety. Vague abdominal pain and early satiety may be symptoms of a partial obstruction. Diagnosis is made by EGD or an upper GI series with barium or a water-soluble contrast such as Gastrografin. Obstruction must be treated by decompressing the stomach using nasogastric aspiration, restoring fluid and electrolyte balance, and administering IV PPIs or histamine2 (H_2) blockers. Surgery may be indicated if these therapies fail.

Treatment

In general, treatment for nonmalignant gastric ulcers is pharmacological, and the first line of therapy is usually an acid suppression medication, also known as a proton pump inhibitor (PPI), and the discontinued use of NSAIDs.

The effectiveness of diet manipulation in ulcer healing is unknown. Foods known to increase acid secretion include milk and milk products, alcoholic and caffeinated beverages, and decaffeinated coffee. Other than encouraging patients to avoid foods that cause symptoms, dietary restrictions are not generally applied.

If all forms of medical management have failed, a definitive acid-reducing surgical procedure, such as parietal cell vagotomy or vagotomy with drainage or partial gastrectomy, may be considered in emergent situations. Definitive acid-reducing surgery should not be performed in patients with acid hypersecretion due to an increased risk of postoperative ulcer recurrence.

Drug therapy for ulcer patients without *H. pylori* may consist of the following agents:

- **PPIs** such as lansoprazole (Prevacid), omeprazole (Prilosec, Zegerid), esomeprazole (Nexium), esomeprazole strontium, pantoprazole (Protonix), dexlansoprazole (Dexilant), and rabeprazole (Aciphex) are effective in decreasing gastric acid by inhibiting the enzyme hydrogen-potassium adenosine triphosphatase (HK-APTase) responsible for completing the final step of acid secretion.
- **H_2 blockers** such as cimetidine (Tagamet HB), famotidine (Pepcid), nizatidine (Axid AR), or ranitidine (Zantac) reduce the amount of acid produced by the stomach by binding to the histamine receptor on parietal cells and blocking histamine-stimulated acid production.
- **Sucralfate (Carafate)**, a complex of aluminum hydroxide and sulfated sucrose, forms a protective gel at the site of disrupted mucosa, providing a protective covering for the ulcer.
- **Antacids** act to neutralize gastric acid, strengthen the gastric mucosal barrier, and heighten the tone of the LES. Optimally, antacids should be taken 1 and 3 hours after meals and before sleep. The use of antacids during the treatment for a gastric ulcer disease should be discussed with a physician.
- **Prostaglandins** such as misoprostol (Cytotec) have antisecretory and cytoprotective effects. Women who are or may be pregnant should not take this medication.

Investigational drugs used to treat PUD include tricyclic antidepressants such as trimipramine (Surmontil) and doxepin (Silenor), which block acid production by interfering with acetylcholine-stimulated acid secretion; pirenzepine; and colloidal bismuth compounds, which act by coating the ulcer crater.

Surgery for benign gastric or duodenal PUD is rare since the advent of H_2 blocker medications, but it may be indicated in emergencies.

- **Gastric PUD surgery.** Emergency gastric surgery is required for uncontrolled hemorrhage from a gastric ulcer. Surgical options include vagotomy with pyloroplasty, Billroth I (gastroduodenostomy and hemigastrectomy), Billroth II (gastrojejunostomy), total gastrectomy (esophagojejunostomy), or gastric resection (antrectomy). The complications of these surgeries include dumping syndrome, hypoglycemic symptoms, nutrient deficiency states, weight loss, diarrhea, and recurrent ulceration. In addition, the risk of gastric carcinoma may be increased after certain types of surgery for peptic ulcer disease.
- **Duodenal PUD surgery.** If emergency surgery is required to treat complications from a duodenal ulcer, options may consist of a truncal vagotomy and drainage (pyloroplasty or gastrojejunostomy), a selective vagotomy and drainage, or a highly selective vagotomy.

Helicobacter pylori infection

Researchers have identified a strong link between the bacteria *H. pylori* and peptic ulcer disease. The association was first noted in 1982 when Marshall and Warren (1983) cultured a gram-negative spiral rod-shaped bacterium. The bacterium was originally thought to be *Campylobacter*-like but later was named *Helicobacter pylori*. The bacterium has many flagella on one end, so is highly mobile. It can adapt to and avoid the acidic nature of the stomach by burrowing beneath the mucosa of the stomach and duodenum.

Risk factors

The transmission of *H. pylori* is thought to be from person to person via fecal-oral or oral-oral pathways.

Research studies postulate that the spread of the bacterium is through contaminated food or water sources, as well as through contact with the feces and/or vomit of an infected person. *H. pylori* is found in all parts of the world but is more prevalent in less-developed countries with overcrowding and poor sanitation.

Diagnosis

Multiple methods of diagnosis are available to detect the presence of *H. pylori*. Noninvasive tests include serology for antibody or antigen detection, a fecal antigen test, and carbon^{13} or carbon^{14} urea breath testing. Invasive tests include biopsy during an EGD for either rapid urease testing or histology. Culture is considered the gold standard but is usually prohibited by cost and time.

Complications

Consequences of *H. pylori* infection include chronic gastritis and peptic ulcer disease. It is also noted that patients with *H. pylori* have a higher rate of gastric cancer. *H. pylori* infection is associated with risk of gastric adenocarcinoma and mucosa-associated lymphoid tissue lymphoma, although researchers continue to investigate this association.

Treatment

Treatment for *H. pylori* continues to be debated and is ever-changing. Presently, treatment may be suggested in patients who are *H. pylori*-positive with non-ulcer dyspepsia. The goal of treatment is eradication of the bacteria. After initial treatment with one of the regimens listed in Table 14.1, eradication of *H. pylori* infection may be confirmed by a urea breath test, a stool antigen test, or upper GI endoscopy-based testing. Therapy involves a combination of drugs ranging from high-dose, dual coverage to quadruple therapy (Table 14.1). Choice of therapy depends on compliance, effectiveness, and cost considerations.

Gastric cancer

Gastric cancer is one of the most common types of cancers in the world. Its incidence varies with geographic regions. Over the past few decades, the incidence of gastric cancer has rapidly declined, and the cause has been linked to the discovery of *H. pylori*, use of refrigeration for food storage, frequent use of antibiotics to treat infections, and reduction of other dietary and environmental risk factors.

Risk factors

The incidence of gastric cancer increases with age and is more common in males. Although the etiology of gastric cancer is not known, risk factors include:

- Cigarette smoking.
- Family history of gastric cancer and/or adenomatous polyposis.

Table 14.1. Examples of Drug Therapy for *H. pylori*

Treatment Regimens			
Regimen	**Agents Used**	**Dose**	**Duration**
P A C	Proton Pump Inhibitor (PPI)* Amoxicillin Clarithromycin	Bid (or QD) 1 gram bid 500 mg bid	10-14 days**
P M C	PPI* Metronidazole Clarithromycin	Bid (or QD) 500 mg bid 500 mg bid	10-14 days**
P B M T	PPI* Pepto-Bismol Metronidazole Tetracycline	Bid (or QD) (525 mg) qid 250 mg qid 500 mg qid	10-14 days**
P B M T	PPI Pepto-Bismol Metronidazole Tetracycline	Bid (or QD) 420 mg qid 375 mg qid 375 mg qid	10-14 days**

*PPIs include lansoprazole 30 mg (BID), omeprazole 20 mg (BID), pantoprazole 40 mg (BID), rabeprazole 20 mg (BID), or esomeprazole 40mg (once daily). (Xin et al., 2016)
** (Chet et al., 2017)
Note: Reduced success rates have been related to metronidazole resistance (Safavi, Sabourian, & Foroumadi, 2016)

- Diet high in starch, food preservatives, nitrates, pickled vegetables, and salted fish and meat, and low in fruits and vegetables.
- Having certain conditions such as:
  - **adenomatous polyps.**
  - Ménétrier's disease (hypoproteinemic hypertrophic gastropathy).
  - Common variable immune deficiency (CVID).
  - Peutz-Jeghers syndrome (PJS) or Li-Fraumeni syndrome (LFS).
  - Gastric ulcers.
  - Previous gastric surgery.
  - Achlorhydria.
  - Pernicious anemia.
  - Chronic atrophic gastritis.
  - Intestinal metaplasia.
  - *H. pylori* infection.

Almost all (90% to 95%) of all gastric cancers are adenocarcinoma (American Cancer Society [ACS], 2017c). The remaining types of gastric cancers are lymphoma, leiomyosarcoma or leiomyoma, gastrointestinal stromal tumor (GIST), carcinoid tumor, adenoacanthoma, small-cell carcinoma, and squamous cell carcinoma.

The Lauren classification consists of two main pathological variants of gastric adenocarcinoma: intestinal (well-differentiated) and diffuse (poorly differentiated). Each type has a distinct morphologic appearance, epidemiology, pathogenesis, and genetic profile. The Lauren classification is based on the cells' behavior and appearance under the microscope.

Historically, most gastric carcinomas developed in the antrum or along the lesser curvature, although cancer stemming from gastric atrophy tends to affect the upper portion of the stomach. However, over the past several decades, gastric cancer located in the proximal stomach and gastroesophageal junction has increased. Benign lesions on the greater curvature are rare. All lesions in this area should be considered malignant until proven otherwise.

Symptoms

Signs and symptoms of gastric cancer include:

- Epigastric discomfort.
- Vomiting.
- Occult blood.

Depending on the location of the lesion, patients may also present with:

- Unexplained weight loss, early satiety, or anorexia.
- Anemia.
- Abdominal or epigastric mass.
- Gastric outlet obstruction, including nausea and vomiting.
- Ascites.
- Difficulty swallowing.
- Enlarged lymph nodes in the supraclavicular areas.
- Low hematocrit or **hypoalbuminemia**.

Gross hematemesis is rare.

Gastric cancer can metastasize via these major routes:

- Direct extension to the greater and lesser omentums, liver, pancreas, diaphragm, spleen, transverse colon, mesocolon, biliary tract, and the duodenum.
- Through the lymphatics to local perigastric nodes; regional to celiac, common hepatic, left gastric, or splenic nodes; distant to supraclavicular, left axillary, or umbilical nodes.
- Through hematogenous to the liver, pulmonary system, bone, or central nervous system (brain).
- Within the perineum to the pelvis (ovary or rectum) or general dissemination.

Diagnosis

Gastric cancer may be revealed on a stomach or chest X-ray, double-contrast upper GI series, magnetic resonance imaging (MRI) scan, positron emission tomography (PET) scan, CAT scan, endoscopic ultrasound (EUS), or EGD.

During an EGD, biopsies may be taken from suspicious or abnormal-looking areas in the stomach to identify gastric cancer. An upper endoscopy with multiple tissue biopsies and cytological examination is needed to make a definitive diagnosis. This procedure is generally preferred for patients who have symptoms suggestive of gastric cancer.

EUS is currently the gold-standard test to evaluate the gastric cancer extension locally. An abdominal CT scan is used to identify distant metastasis and may be ordered to determine the spread of the tumor.

Complications

Routine screening for gastric (stomach) cancer is not commonly done in the U.S. Testing for gastric cancer is not usually done until patients have symptoms suggestive of the disease. Earlier diagnosis of stomach cancer increases the chances of surviving 5 years after diagnosis. Unfortunately, in the U.S., gastric cancer is frequently diagnosed at an advanced stage, so prognosis is poor. Most patients survive less than 5 years.

According to the American Cancer Society (2017d), the 5-year survival rate for patients depends on the spread and stage:

- Localized gastric cancer is 68%.
- Regional or the spread of cancer outside the stomach to lymph nodes and nearby structures is 31%.
- Distant or the spread of cancer to distant body parts is 5%.

The worst prognosis is for patients diagnosed with linitis plastica, also known as Brinton's disease or leather bottle stomach, which is a submucosal tumor that spreads diffusely and is associated with the formation and development of fibrous tissue. In this condition, the stomach becomes narrowed and nondistensible.

Complications of gastric cancer metastasis, including perforation and obstruction, should be managed as they develop.

Treatment

Surgery is the treatment of choice for curable gastric cancer lesions.

In the U.S., most surgeons perform a total gastrectomy for negative margins (if needed), an esophagogastrectomy for tumors located in the cardia and gastroesophageal junction, and a subtotal gastrectomy for tumors located in the distal stomach. Treatment for advanced gastric cancer is largely palliative. Palliative treatment involves tumor resection to relieve obstruction or dysphagia or control chronic bleeding or gastrojejunostomy to provide temporary relief or prevent obstruction.

Following surgery, combination chemotherapy may be used alone or with local radiation therapy. Chemoradiotherapy, adjuvant (postoperative) therapy, neoadjuvant (preoperative) therapy, and combined modality or multimodality therapy are used to treat gastric cancer. These treatment modalities continue to evolve.

Nursing responsibilities include, but are not limited to, meeting the patient's emotional needs, providing adequate nutrition, and assisting with pain management.

Gastric polyps

Gastric **polyps** are defined as any circumscribed, discrete stomach tumor and are relatively uncommon. Single polyps appear more often than multiple polyps and frequently arise in the antrum and along the lesser curvature. Polyps most often appear in mid-to-late adult life. They are seldom found in young people. The different types of gastric polyps include the following:

Hyperplastic (regenerative) polyps are commonly associated with chronic inflammatory disorders, such as chronic gastritis, *H. pylori* gastritis, pernicious anemia, and chemical gastritis. They are usually located in the antrum, body, fundus, and cardia. These polyps consist of normal gastric epithelium and are the most common type of gastric polyp where *H. pylori* infections are common. They are usually benign, but the risk of malignancy increases if the polyp is greater than 1 cm and pedunculated (having an elongated stalk).

Fundic gland polyps are mostly benign polyps located in the gastric corpus. Long-term use of proton-pump inhibitors (PPIs) is associated with increased risk for these polyps. Fundic gland polyps are sporadic and are also seen in 20% to 100% of patients diagnosed with familial adenomatous polyposis (FAP) and in 11% of patients diagnosed with MUTYH-associated polyposis (MAP) (Mahachai, Graham, & Odze, 2019).

Adenomatous polyps, also called raised intraepithelial neoplasia, are precursors to gastric cancer. Histologically, they are similar to colon adenomas and are commonly associated with chronic atrophic gastritis and intestinal metaplasia. Polyps that are greater than 2 cm pose a risk for neoplasia. They are also linked to FAP. These polyps are typically isolated and found in the antrum but can be located anywhere in the stomach.

Gastrointestinal stromal tumors (GISTs) are rare connective tissue tumors (a subset of GI mesenchymal tumors) that are histologically classified as leiomyoma, leiomyosarcoma, leiomyoblastoma, or schwannoma. They usually develop in the muscularis propria layer of the intestinal wall.

Hamartomatous polyps are typically mucosal-based and are derived from three embryonic layers. They are rare and commonly associated with certain syndromes: hereditary mixed polyposis syndrome (HMPS), juvenile polyposis syndrome (JPS), Cowden syndrome, and PJS. These polyps are usually benign with abnormal mixtures of tissue indigenous to the organ. Some have an increased risk of cancer.

Inflammatory fibroid polyps are commonly referred to as Vanek's tumors and are very uncommon. Typically, patients do not have any symptoms. These polyps have been associated with gastric outlet obstruction or bleeding. They arise in the submucosa and are usually non-neoplastic.

Neuroendocrine tumors are divided into four types and derived from enterochromaffin-like cells. Type 1 is the most common form. Types 1 and 2 are associated with hypergastrinemia and are usually located in the fundus and body of the stomach. Type 3 is usually solitary and found throughout the stomach. Type 4 is sporadic and may arise anywhere in the stomach. Type 4 is the most aggressive form with the worst prognosis.

Risk factors

Gastric polyps are most prevalent where *H. pylori* infections are common. Risk factors for gastric polyps include, but are not limited to: **achlorhydria syndrome**, atrophic gastritis, pernicious anemia, gastric cancer, and gastric resection.

Symptoms

Unless a polyp bleeds, it usually causes no symptoms.

Diagnosis

Gastric polyps are usually detected by an upper GI series, MRI, ultrasonography, angiography, CT scan, or endoscopy with tissue biopsy and cytology.

Treatment

Endoscopic polypectomy or partial gastrectomy may be used to remove polyps.

Gastritis

Gastritis is an inflammation of the gastric mucosa (stomach lining).

Several classification systems for gastritis (and gastropathy) have emerged, such as the Sydney system and the Operative Link on Gastritis Assessment (OLGA) staging system. However, there is no universally accepted classification for the condition. The Sydney System basically combines the morphological, topographical, and etiological information for the classification of gastritis based on the:

- Location of gastritis: antrum, fundus - body, cardia, focal, or diffuse.
- Type of gastritis: acute, chronic, lymphocytic, granulomatous, eosinophilic, etc.
- Grading the presence or activity of the following:
  - *H. pylori* infection (the density);
  - glandular atrophy;
  - intestinal metaplasia of the epithelium;
  - neutrophilic infiltrates of the lamina propria, pits, or surface epithelium; and the
  - type and pattern of inflammation (chronic, active or both).

There are also specific forms of gastritis, including Ménétrier disease, eosinophilic gastritis, sarcoidosis, and certain infections, but they are less common.

Risk factors

Gastritis is most commonly caused by *H. pylori* infection, damage to the stomach lining, or an autoimmune response. Other causes include irritants, such as gastric acid, bile reflux, alcohol, cocaine, radiation exposure, radiation treatments, medications (NSAIDS and aspirin), or toxins. This exposure is often combined with an impairment of natural protective mechanisms. Some noninfectious causes of gastritis include a high degree of stress and hypersensitivity reactions.

Acute gastritis may be associated with serious or life-threatening illness, alcoholism, localized gastric trauma, and gastrectomy.

Chronic gastritis is common in adults and may be associated with noninfectious and infectious causes.

- Noninfectious causes are normal aging, chemical gastropathy (linked to irritants), uremic gastropathy, chronic noninfectious granulomatous gastritis (linked to Crohn's disease and sarcoidosis), lymphocytic gastritis, eosinophilic gastritis, radiation injury to the stomach, graft-versus-host disease, and ischemic gastritis. Chronic gastritis may also occur due to drug therapy, autoimmunity, and allergic reactions.
- Infectious gastritis may be caused by *H. pylori* infection; *Helicobacter heilmannii* infection; viruses (cytomegalovirus [CMV], herpes virus); and parasites (*Strongyloides*, *Schistosoma*, and *Diphyllobothrium latum*).

Symptoms

If symptoms are present, a patient will complain of pain or discomfort in the upper part of the abdomen. In chronic gastritis, patients may have gradual blood loss that goes unnoticed for years. Patients may test positive for occult blood in stool, have flecks of blood in vomitus, or exhibit chronic anemia.

Diagnosis

In patients with acute GI bleeding, the most effective tool for diagnosing gastritis is an EGD. In rare instances such as severe gastritis and recurrent gastric ulcers, serum gastrin and a gastric analysis may be ordered to determine the cause.

Treatment

Treatment for gastritis depends on the underlying cause—illness or injury—and the patient's signs and symptoms.

Pharmacological treatment may include treating *H. pylori* infection with antibiotics and may involve the use of antacids, sucralfate (Carafate), PPIs, or H_2 blockers. Some patients who bleed from diffuse gastritis cease bleeding spontaneously after treatment.

Contributing factors must be assessed, including medications, stress from severe sepsis disorders, alcohol, *H. pylori*, smoking, and other illnesses. Gastritis or gastric erosions may heal very quickly when offending mucosal irritants are discontinued, often within 48 hours, so early endoscopy is important. Research has not found an association between the cause and prevention of gastritis to eating, diet, and nutrition.

Stress ulcers

Stress ulcers are gastric mucosal stress erosions that are usually associated with serious illness. They are commonly found in the fundus and the body of the stomach, but can develop in the antrum, duodenum, or distal esophagus. Cushing's ulcers may be located in the esophagus, stomach, or duodenum.

Risk factors

Stress ulcers seem to arise in patients who:

- Sustain severe trauma; have ongoing sepsis; have been admitted to an intensive care unit; or are being treated for serious illness such as environmental electrical injuries, stroke complications, or liver failure.
- Have significant burn injuries (**Curling's ulcer**).
- Sustain intracranial trauma such as craniotomy or traumatic head injuries (**Cushing's ulcer**).
- Chronically ingest substances that have adverse effects on the gastric mucosa such as NSAIDs or alcohol.

Symptoms

A symptom of a stress ulcer is massive upper GI bleeding. Stress ulcers rarely cause classic ulcer signs and symptoms before bleeding begins.

Complications

Deeper and more distal stress ulcerations can deteriorate the submucosa and cause severe hemorrhaging. Perforation is a rare complication.

Cushing's ulcers tend to be deep, full-thickness ulcers and therefore are more prone to perforation than gastric mucosal ulcerations induced by trauma or sepsis.

Treatment

Treatment is focused on controlling bleeding, correcting shock, and treating the underlying disorder.

Dieulafoy's lesion

Dieulafoy's lesion is a dilated, aberrant, submucosal artery that erodes the mucosa in the absence of an ulcer. It is an important etiology of acute upper GI bleeding. About 70% of the lesions occur in the stomach, 6 cm from the gastroesophageal junction along the lesser curvature. Other common locations include the duodenum, the distal stomach, and the esophagus. These lesions are responsible for 1.5% of acute upper GI bleeding (Nojkov & Cappell, 2015).

Symptoms

Patients typically present with acute, profuse bleeding, which results in hematemesis, melena, or hematochezia.

Diagnosis

EGD is usually preferred for diagnosis due to the presentation. Diagnostic accuracy increases when EGD is performed within 24 hours of the bleed.

Endoscopic biopsies are contraindicated because of the risk of severe bleeding from the exposed artery. If the endoscopy is nondiagnostic, interventional angiography maybe required to establish the diagnosis and treat the vessel in the setting of acute bleeding.

Complications

Bleeding can be massive, life-threatening, and recurrent.

Treatment

Standard treatments include:

- Endoscopic hemostasis using hemoclips, band ligation, injection therapies, epinephrine injection, ablation therapy, heater probe, electrocoagulation, or Argon plasma coagulation (APC).
- PPIs IV.
- Volume resuscitations.

Gastric varices

Varices are abnormally dilated and tortuous veins. Esophageal varices are by far the most clinically significant type. However, some patients with esophageal varices also have **gastric varices**. Less common sites of varices due to gastrointestinal bleed are the duodenum, ileum, colon, and rectum.

Risk factors

Portal hypertension, which is most often the result of cirrhosis, may lead to the development of collateral circulation with formation of varices that carry blood away from the portal circulation.

Diagnosis

An upper endoscopy is needed to diagnose variceal bleeding.

Complications

Upper GI bleeding from gastric varices is potentially life-threatening. Patients who bleed from gastric varices appear to have a higher mortality. The chance of recurrent variceal bleeding is common.

Treatment

Reducing variceal pressure by pharmacological or other means may decrease the risk of bleeding in early cirrhosis. Therapeutic treatments may consist of endoscopic variceal ligation (EVL) and endoscopic sclerotherapy (ES). The preferred initial treatment is EVL due to complications related to ES. If EVL fails after several attempts to treat the varices, ES may be performed. However, self-expanding metal stents have been used to treat acute refractory esophageal varices.

The objectives for managing esophagogastric varices can include:

- Hemodynamic stabilization.
- Stopping acute variceal bleeding using two methods: endoscopic schlerotherapy using a sclerosant and/or an endoscopic variceal ligation using a rubber-band ligation.
- Reducing portal pressure by surgical insertion of a portosystemic shunt or by using therapeutic radiologic procedures such as the percutaneous transhepatic embolization of the gastroesophageal varices or placement of a transjugular intrahepatic portosystemic shunt (TIPS).

Balloon tamponade with a Sengstaken-Blakemore tube, Linton-Nachlas tube, or Minnesota tube may be used as temporary management in some patients.

Hiatal hernia

In most people, the muscle structure at the diaphragmatic hiatus anchors the stomach under the diaphragm. A **hiatal hernia** occurs when part of the stomach protrudes through the diaphragm and into the thoracic cavity.

Most are sliding hiatal hernias in which a portion of the stomach slides up through the opening, so that the gastroesophageal junction lies above the level of the diaphragm. A sliding hiatal hernia is often associated with a weakening of the LES.

In the less common rolling hiatal hernia, the gastroesophageal junction is located below the level of the diaphragm, but a part of the greater curvature of the stomach herniates through the diaphragm and into the thoracic cavity.

Risk factors

Hiatal hernias are common in people over 50 and people who are overweight. They are more common in females than in males.

Symptoms

A sliding hiatal hernia is often associated with esophageal reflux.

With a rolling hernia, patients may have a feeling of fullness and discomfort after meals, but reflux is not common.

Diagnosis

Diagnostic tests for hiatal hernia include a chest X-ray, barium swallow, and endoscopic examination. High-resolution manometry with esophageal pressure topography may be helpful in providing information about esophageal motility and the esophagogastric junction, but it is not diagnostic for a hiatal hernia.

Complications

Strangulation and infarction are potential complications of a rolling hiatal hernia. Ulceration may occur with either type of hiatal hernia. Other complications of hiatal hernias include reflux esophagitis, heartburn, acid regurgitation, water brash, and dysphagia.

Treatment

A hiatal hernia is treated in much the same way as gastroesophageal reflux disease (GERD).

Congenital abnormalities

The second most common disorder requiring surgery in the first few months of life, after inguinal hernia, is **infantile hypertrophic pyloric stenosis** (IHPS or pyloric stenosis), which affects approximately 1-4 per 1,000 live births (Al-Mayoof & Doghan, 2018). The muscular valve between the stomach and the small intestine—the pylorus—thickens, compromising food passage out of the stomach.

Risk factors

Infant males are more affected than infant females, and the condition is seen more in preterm infants than term infants. Environmental conditions and genetic susceptibility may influence the likelihood of developing the IHPS.

Symptoms

The first symptoms of pyloric stenosis usually occur at 3-6 weeks of age and rarely occur after 12 weeks of age. Patients have a history of progressive nonbilious vomiting, which becomes projectile. In infants diagnosed with IHPS, the hypertrophied pylorus is palpable, and they may look dehydrated, very thin, and weak.

Diagnosis

An upper GI series demonstrates an unchanging elongation and narrowing of the antrum and pylorus. IHPS may also be diagnosed by abdominal ultrasound, barium studies (UGI with contrast), and upper endoscopy.

Complications

Infants with the condition may become dehydrated, have electrolyte disturbances, and commonly are in metabolic alkalosis with hypokalemia.

Treatment

The treatment of choice is the Ramstedt pyloromyotomy, even though laparocopic pyloromyotomy has been used. Nonsurgical treatments for this condition are being researched.

Other Congenital Abnormalities of the Stomach

These include:

- **Gastric, antral, and pyloric atresias,** in which the stomach ends blindly or is totally occluded by two apparent membranes connected by a strand of each mucosa and submucosa.
- **Pyloric or antral membranes,** which may produce no obstructive symptoms until late in life.
- **Microgastria,** also called hypoplasia, a rare condition with limited life expectancy, in which the stomach never becomes differentiated from the primitive foregut into a true fundus, body, and pylorus.
- **Gastric duplication,** a rare condition in which a distinct mass lesion that contains all layers of the gastric wall develops in the stomach.
- **Neonatal perforation of the gastric wall,** which is a rare condition associated with prematurity, peptic ulceration, and distal small intestinal obstruction.
- **Gastric diverticula,** most of which are congenital, and are located high on the posterior wall of the stomach, below the gastroesophageal junction. Prepyloric diverticula, which are relatively rare, are usually associated with previous peptic ulceration. Occasionally, gastric diverticula may cause dyspepsia or symptoms of postprandial vomiting and nausea.

Gastric motor abnormalities

A number of gastric motor abnormalities have been identified, ranging from conditions that are of little or no clinical importance to those that result in severe or chronic disability. Both excessively slow and excessively rapid gastric emptying can produce disabling symptoms.

Risk factors

Symptomatic gastric motor abnormalities may be seen after vagotomy and pyloroplasty, or in patients with mechanical obstructions, acute metabolic disorders, inflammatory diseases, or long-standing diabetes mellitus, or as a side effect of certain medications.

Serious gastric motor dysfunction may also be idiopathic, as in antral tachygastria, in which an aberrant pacemaker in the antrum cycles 3–4 times faster than the usual pacemaker area in the gastric body.

Gastric surgery that includes vagotomy (e.g., gastroenterostomy, gastrojejunostomy, or Billroth II) may lead to rapid gastric emptying and a group of disabling symptoms known as **dumping syndrome**. The cause of dumping syndrome is complex and may be related to rapid fluid shifts from plasma to the intestinal lumen as a result of the rapid introduction of hyperosmolar solutions into the jejunum and the release of numerous hormones and vasoactive intestinal polypeptides into the bloodstream.

Symptoms

Early symptoms of dumping syndrome appear within 30 minutes after the start of a meal and include anxiety, weakness, dizziness, tachycardia with a pounding pulse, diaphoresis, flushing, abdominal cramps, and diarrhea. Within 90-120 minutes after a meal, symptoms may result from reactive hypoglycemia, weakness, diaphoresis, and tachycardia. Sometimes a decreased level of consciousness may be experienced.

Treatment

Management of acute gastric retention is directed at the underlying cause, if known. Gastric muscle stimulants such as metoclopramide and bethanechol or short-term use of erythromycin may be of benefit.

Treatment of dumping syndrome centers on slowing the intestinal delivery of nutrients and minimizing the release of endogenous vasoactive peptides. Standard recommendations include small meals high in fat and protein and low in carbohydrates, as well as minimal fluids with meals. Fluids should be taken before or after ingestion of solid food. Clinical research continues to be conducted on the use of medications to inhibit gastric emptying.

Infectious gastritis

Gastritis can be caused by infectious agents, including, but not limited to: *H. pylori* infection, mycobacterial, syphilitic, viral, parasitic, and fungal infections. These are also etiology-based classifications of gastritis.

Risk factors

Immunocompromised patients are particularly at risk, including those with acquired immune deficiency syndrome (AIDS) or those who have undergone organ transplantation or chemotherapy.

The agents implicated in infectious gastritis may be bacterial, fungal, viral, or parasitic.

- Bacterial inflammations other than *H. pylori* are relatively rare but may include phlegmonous and emphysematous gastritis, tuberculosis, and syphilis.
- Fungal infections such as *Candida albicans* or *Candida glabrata* may be found in gastric ulcer or erosion beds in immunocompromised hosts.
- Gastric cytomegalovirus (CMV) has been observed in patients with a transplant, CMV mononucleosis, or AIDS, and in children.
- Parasitic infestations found in the stomach are infrequent but include cryptosporidiosis, anisakiasis, strongyloidiasis, and giardiasis.

Gastric outlet obstruction

Obstruction of the pyloric sphincter at the outlet of the stomach blocks the flow of gastric contents into the duodenum.

Symptoms

Patients may vomit partially digested gastric contents and may complain of gastric pain or a feeling of fullness that is relieved only by vomiting. The pain is aggravated by eating, and a patient may have weight loss. The patient may complain of abdominal tenderness or distention and succussion splash may be noted.

Diagnosis

Abdominal X-ray, an upper GI series, barium studies, nuclear scanning, endoscopic examination, or a saline load test may be ordered to confirm gastric retention and to rule out atony as the cause.

Complications

Patients with gastric outlet obstruction may be anemic. Prolonged vomiting may lead to an electrolyte imbalance, such as hypokalemia or metabolic alkalosis.

Treatment

Treatment begins with restoration of fluid and electrolyte balance, decompression of the stomach, and correction of nutritional deficiencies. To eliminate the obstruction, the pylorus may be dilated using an endoscope with a balloon. If surgery is indicated, the procedure used depends on the cause of the obstruction.

Caustic injury

The ingestion of acid or alkaline chemicals may cause tissue injury on contact with the oropharynx, esophagus, stomach, or duodenum. The severity of the tissue injury after caustic ingestion depends on the nature, concentration, and quantity of the caustic agent ingested and the duration of tissue contact.

Risk factors

Caustic ingestions are most commonly seen in children ages 1-3 years of age from accidental ingestion.

- **Alkaline agents.** The oropharynx, hypopharynx, and esophagus are most frequently injured by ingestion of alkaline agents. Contact with strong alkaline agents causes liquefactive necrosis, which is the complete destruction of entire cells and their membranes.
- **Acid agents.** In the case of acid ingestion, the caustic agent tends to pass rapidly through the esophagus, thus producing shallow burns. Contact with strong acids promotes coagulation necrosis and the formation of a firm, protective eschar, which limits the depth, penetration, and injury produced by the acid. In the stomach, acids usually collect in the antrum, where the most severe damage occurs.

Symptoms

The severity of tissue injury ranges from diffuse gastritis to hemorrhagic ulceration and necrosis, leading to perforation of the stomach. Gastric perforation, in turn, may lead to mediastinitis, peritonitis, and shock. Patients with severe gastric injury may present with epigastric pain and retching or emesis of tissue, blood, or material resembling coffee grounds.

Diagnosis

X-ray examinations can assess for signs of aspiration pneumonia or perforation. Cautiously performing endoscopy can establish the extent and severity of tissue damage.

Complications

Stricture formation is a common late complication of caustic ingestion and is usually apparent by the eighth week following injury. Strictures in the stomach may cause gastric outlet obstruction, which leads to early satiety, postprandial vomiting, and weight loss.

There is no definitive evidence of an increased risk of gastric carcinoma in patients with a history of caustic ingestion.

Treatment

Management of caustic injuries includes:

- Determining the type of agent ingested and keeping the patient NPO (nothing by mouth).
- Emergency surgery, which may be necessary in the event of perforation, peritonitis, mediastinitis, or severe hemorrhage.
- Dilation, if strictures develop.
- Partial or total gastrectomy in the case of severe gastric burns, or antral and pyloric stenosis.

Bezoars

Bezoars are concretions of foreign material found in the stomach. They are usually composed of either vegetable and plant material (phytobezoar), ingested medications (pharmacobezoar), milk curd (lactobezoar), or hair that has been chewed (trichobezoar).

Risk factors

Phytobezoars, which are more common, are composed of partially digested fibers, leaves, roots, and skins of almost any plant matter. Excessive ingestion of rinds from melons and oranges is a frequent cause of phytobezoars. People with diabetes with gastroparesis are at risk of developing bezoars.

Pharmacobezoars are concretions of medications that occur due to changes in the GI tract anatomy or problems with motility. They can be composed of drugs that are difficult to dissolve such as enteric-coated or extended-release drugs, sucralfate, or sodium alginate.

Lactobezoars, also known as milk curd bezoars, have been reported in infants as a result of ingesting a powdered formula diluted with an inadequate amount of water. Continuous-drip feeding of preterm infants seems to be the most important predisposing factor.

Trichobezoars are composed of large quantities of hair or hair-like fibers and decaying food particles. These bezoars can resemble a solid mass when it becomes matted together, and they can also assume the shape of the stomach. This is the most common type of bezoar found in the pediatric population. They are commonly seen in females under the age of 30 and commonly linked to individual diagnosed with a psychiatric condition.

Symptoms

Patients with bezoars may present with vague complaints of fullness in the upper quadrants, anorexia, halitosis, weight loss, epigastric pain, and periodic attacks of nausea and vomiting. A palpable mass can be evident in phytobezoars and trichobezoars. Signs and symptoms depend largely on the size, location, and degree of disturbance in gastric physiology.

Diagnosis

EGD is the best technique for diagnosing and classifying bezoars, although plain X-ray films of the abdomen or an upper GI series may show evidence of an abdominal mass. Pharmacobezoars are rare and may be difficult to diagnose.

Complications

Bezoars can result in anorexia, vomiting, ulceration, bleeding, perforation, constipation, obstipation, or small bowel obstruction.

Treatment

In general, treatment for bezoars includes:

- During endoscopy, fragmentation of the bezoar with forceps, wire snare, jet spray, or laser may be used, allowing them to be passed or extracted.
- Metoclopramide may be given for several days after fragmentation to increase peristalsis and aid passage of the material.
- Enzymatic agents such as papain, acetylcysteine, or cellulase may be used to dissolve a bezoar or assist with fragmentation. Use of enzymatic preparations can result in gastric perforation, especially if the bezoar has caused ulceration of gastric tissue.

If these methods fail, a gastrotomy may be necessary. Surgical mortality is minimal; however, the mortality of untreated bezoars may be significant.

Management of pharmacobezoars depends on the medications involved and their toxicity. Decomtamination may be necessary.

Phytobezoars may be disrupted endoscopically or enzymatically. To prevent postgastrectomy phytobezoars, it may be helpful to improve dentition and counsel the patient to avoid pulpy, fibrous fruits (especially oranges) and vegetables.

Symptoms of lactobezoars may be resolved by withholding feedings for 48 hours, implementing gastric lavage with saline solution, and ensuring proper hydration.

Trichobezoars must be treated surgically because there is no way to dissolve the matted hair in vivo.

CASE SITUATION

Carter Williams, age 74, passed out during his daily walk and was brought to the emergency room (ER). Two months earlier, Mr. Williams had passed his annual physical with flying colors. His only complaint at that time was right arm and shoulder pain, attributed to the exertion of building a fence, and his physician prescribed an NSAID. Carter stopped smoking 10 years ago and drinks alcohol only at occasional social functions. He also drinks 2-3 cups of coffee daily.

In the ER, Carter's chest X-ray films and ECG are normal. His CBC shows a hemoglobin of 9g, but he denies any abdominal pain. Carter reports having occasional black-looking bowel movements for the last 6 weeks, but he attributes them to something in his diet. A rectal examination in the ER confirms that there is blood in his stool.

Points to think about

1. Because Carter denies abdominal pain, is there any easy way to identify upper GI bleeding?
2. Carter is admitted. Before going to his room, he is taken to the GI endoscopy unit for an upper GI endoscopy. What items in this history are important?
3. Carter is found at endoscopy to have hemorrhagic ulceration and necrosis secondary to peptic ulcer disease (PUD). What nursing diagnosis might apply?
4. What are specific expected outcomes and associated nursing interventions for this nursing diagnosis?
5. What else can the gastroenterology nurse tell Carter about treatment and follow-up care?

Suggested responses

1. In the ER, aspiration of gastric contents through an NG tube reveals coffee-ground-like material that tests positive for occult blood. Blood in the nasogastric aspirate is good evidence of a source of upper GI bleeding, but a negative test does not rule it out. Care must be taken to avoid trauma to the nares or to the esophageal or gastric mucosa because it might cause bleeding and obscure the diagnosis.
2. Some things the gastroenterology nurse has learned that will help with the evaluation are:
   a. Gastritis is an inflammation of the gastric mucosa caused by an irritant material, such as gastric acid, bile reflux, medications, or toxins, and is often combined with an impairment of natural protective mechanisms.
   b. Carter was taking an NSAID, which is known to be a gastric mucosal irritant and may cause gastritis or a gastric ulcer. Older patients are more likely to bleed from NSAID use and are more likely to suffer upper GI tract perforations.
   c. Gastritis or gastric erosions may heal very quickly when offending mucosal irritants are discontinued, often within 48 hours, so early endoscopy is important.
   d. Carter had noticed occasional black stools, which may indicate blood loss leading to anemia, which is even more indicative of gastritis.
3. The applicable nursing diagnosis for Carter is, "fluid volume deficit related to blood loss."
4. Defining characteristics of this nursing diagnosis include:
   a. Fluid volume deficit related to blood loss.
   b. Decreased urine output.
   c. Output greater than intake.
   d. Decreased venous filling.
   e. Thirst.
   f. Increased pulse rate.
   g. Decreased pulse amplitude.
   h. Increased body temperature.
   i. Dry mucous membranes.
   j. Concentrated urine.
   k. Hypotension.
   l. Narrowed pulse pressure.
   m. Decreased skin turgor.
   n. Change in mental status.
   o. Weakness.

Some, but not necessarily all, expected outcomes and nursing interventions for this nursing diagnosis might include the following:

Table 14.2. Expected Outcomes and Nursing Interventions

Expected outcomes	Nursing interventions
There will be an adequate fluid volume	• Monitor and record vital signs and central venous pressure • Weigh daily • Monitor intake and output • Assess for signs and symptoms of hypovolemic shock • Assess for signs and symptoms of dehydration • Administer drugs and IV and/or oral fluids as ordered
Laboratory values will be within normal limits	• Monitor laboratory values pertinent to fluid volume deficit

5. Because Carter's gastric mucosa is sensitive to injury, he should discontinue use of NSAIDs and aspirin-containing products. Caffeine and alcohol are also recognized as gastric irritants and

should be discontinued, at least during treatment. Depending on what medication is prescribed, the gastroenterology nurse should inform Carter of the following:

a. H_2 blockers inhibit the action of histamine at receptor sites in gastric parietal cells, which inhibits the production of gastric acid, thus decreasing further mucosal injury by acid contact. Tablets should be taken with meals, or at bedtime if the patient is on once-daily therapy. Some H_2 blockers interact with other medications, so it is important to let the physician know all medications the patient is taking so the physician can choose an H_2 blocker accordingly.
b. Antacids are usually taken between meals and at bedtime to neutralize gastric acid, but they are rarely prescribed since the introduction of H_2 blockers. Liquid antacids should be shaken well and taken with a little water to ensure passage into the stomach. Some antacids can cause diarrhea or constipation and may need to be alternated. Carter should discuss with his physician whether low-sodium antacids are recommended.
c. Sucralfate (Carafate) is an oral medication that adheres to the gastric mucosa to form a protective barrier against further damage, particularly where cell protein is exposed because irritants have eroded the protective mucosa. The medication should be taken on an empty stomach for best results, one hour before meals and at bedtime. Since sucralfate is not readily absorbed systemically, there are few side effects. Occasionally, patients experience constipation.
d. Misoprostol (Cytotec) is a synthetic prostaglandin E, which replaces prostaglandins that are depleted by NSAIDs, and may be given concomitantly with NSAIDs to help prevent gastric mucosal damage. This drug is strictly contraindicated in pregnant women because it can cause miscarriage.
e. The gastroenterology nurse should teach Carter how to use paper stool test cards at home and how to evaluate the color changes on the card for the presence of occult blood in his stool. He should be told to report any further signs of bleeding and/or epigastric pain to his physician.
f. The gastroenterology nurse should provide a list of foods high in iron to help raise hemoglobin levels to normal.
g. If Carter feels he will not be able to comply with the recommended treatment regimen, he should let the physician know so that alternative treatment can be discussed.

REVIEW TERMS

achlorhydria syndrome, adenomatous polyps, antrum, bezoars, body, cardia, cardiac glands, chief cells, chyme, Curling's ulcer, Cushing's ulcer, dumping syndrome, enterochromaffin cells, fundus, G cells, gastric ulcers, gastric varices, gastritis, greater curvature, greater omentum, *Helicobacter pylori* (*H. pylori*), hiatal hernia, hypoalbuminemia, infantile hypertrophic pyloric stenosis, lesser curvature, lesser omentum, oxyntic gland, parietal cells, pernicious anemia, polyps, pyloric glands, pyloric sphincter, pylorus, rugae, stress ulcers, ulcers

REVIEW QUESTIONS

1. Entry of food into the stomach is controlled by the:
   a. Lower esophageal sphincter.
   b. Fundus.
   c. Pyloric sphincter.
   d. Antrum.
2. The layers of the stomach are the mucosa, the submucosa, the muscularis, and the:
   a. Rugae.
   b. Cardia.
   c. Serosa.
   d. Connective tissue.
3. The parietal cells secrete:
   a. Mucus.
   b. Hydrochloric acid and intrinsic factor.
   c. Pepsinogens.
   d. Gastrin.
4. Intrinsic factor is necessary for the:
   a. Conversion of pepsinogens to pepsin.
   b. Secretion of mucus.
   c. Absorption of vitamin B_{12}.
   d. Secretion of hormones.
5. The gastric emptying rate is controlled by neural impulses, hormones secreted by the small intestine, and:
   a. The amount of food ingested.
   b. The composition of the chyme.
   c. The amount of gastric secretions.
   d. Vitamin B_{12} absorption.
6. *H. pylori* infection:
   a. Has been associated with gastric cancer.
   b. Is a virus of the stomach.
   c. Should be treated by surgical resection.
   d. Is not detectable by biopsy.
7. Most gastric cancers are of which of the following types?
   a. Adenocarcinomas.
   b. Leiomyosarcomas.
   c. Sarcomas.
   d. Lymphomas.

8. Cushing's ulcers are a form of:
   a. Specific gastritis.
   b. Nonerosive, nonspecific gastritis.
   c. Peptic ulcers.
   d. Stress ulcers.
9. Gastric surgery may lead to rapid gastric emptying and a group of disabling symptoms that can cause reactive hypoglycemia. This syndrome is called:
   a. Curling's syndrome.
   b. Paterson-Kelly syndrome.
   c. Dumping syndrome.
   d. Cushing's syndrome.
10. The best technique for diagnosing a bezoar is:
   a. Plain X-rays.
   b. Palpation.
   c. Gastroscopy.
   d. Upper GI series.

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Chapter 15

SMALL INTESTINE

This chapter will acquaint the gastroenterology nurse with the normal anatomy and physiology of the small intestine, also sometimes called the small bowel. In addition, a number of disorders of the small intestine are described in terms of their pathophysiology, diagnosis, and treatment.

Learning objectives

After reviewing the content of this chapter, the gastroenterology nurse will be able to:

1. Describe the normal anatomy of the small intestine.
2. Discuss the physiology of intestinal absorption, secretion, and motility.
3. Describe the pathophysiology, diagnosis, and treatment of a number of important disorders of the small intestine.

ANATOMY AND PHYSIOLOGY

The **small intestine** is a tubular structure that begins distal to the pyloric sphincter of the stomach and extends to the cecum of the large intestine. The small intestine has digestive, absorptive, secretory, barrier, and immunologic functions. It is approximately 275 cm (approximately 9 ft) in the neonate, growing to 5-6 m (approximately 16-19 ft) in the adult. The small intestine is divided into three sections.

- **Duodenum** is the first 30 cm (1 ft) section of the small intestine. It is the c-shaped, muscular section that begins at the pyloric sphincter and ends at the **ligament of Treitz**. The common bile duct empties into the duodenum at the **ampulla of Vater** (Figure 15.1).
- **Jejunum** is the section located after the duodenum and is the proximal two-fifths of the small intestine.
- **Ileum** is the distal three-fifths of the small intestine and extends from the jejunum to the **ileocecal valve** This valve controls the flow of chyme into the large intestine and prevents reflux from the large intestine into the ileum.

Layers of the small intestine

The small intestinal wall consists of the following concentric layers:

- **Mucous layer (mucosa)** is the inner layer containing simple epithelial cells that overlie the **lamina propria** and rest on the smooth muscle layer known as the muscularis mucosae.
- **Submucous layer (submucosa)** consists of dense connective tissue containing numerous arteries, as well as venous and lymphatic plexuses. This layer also contains Meissner's plexus, a collection of autonomic nerves that communicate with Auerbach's plexus.
- **Muscular layer (muscularis propria)** contains outer longitudinal and inner circumferential muscles, separated by intrinsic ganglion cells and nerve fibers called the myenteric or Auerbach's nerve plexus.
- **Serous layer (serosa)** is the outer layer composed of mesothelial cells that overlie loose connective tissue.

The mucosa and submucosa are arranged in circular folds called the **plicae circulares**, which provide a greater surface area for secretion and absorption during the 3-6 hours the chyme remains in the small intestine. In addition, the mucosa forms an estimated 4-5 million small, finger-like villi that project into the intestinal lumen (Figure 15.2). Each **villus** is lined with simple columnar epithelial cells.

Below the epithelium, the lamina propria contains nerve fibers, smooth muscle, lymphatic vessels called lacteals, blood vessels, and connective tissue. A brush border consisting of multiple **microvilli** covers the surface of each columnar cell.

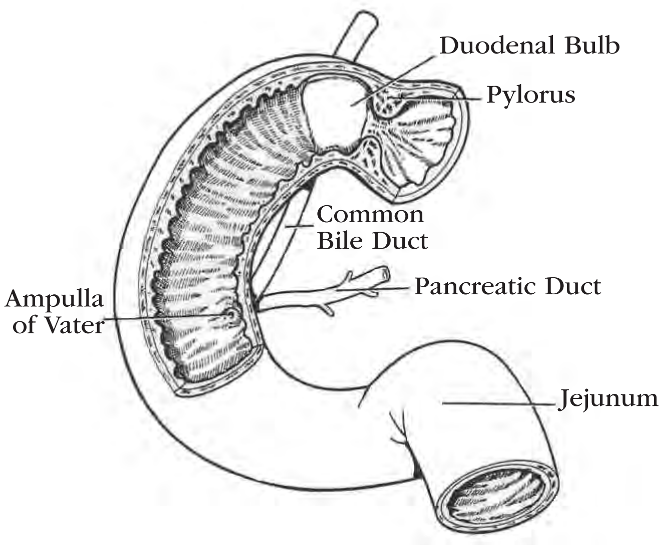

Figure 15.1. Duodenal Anatomy

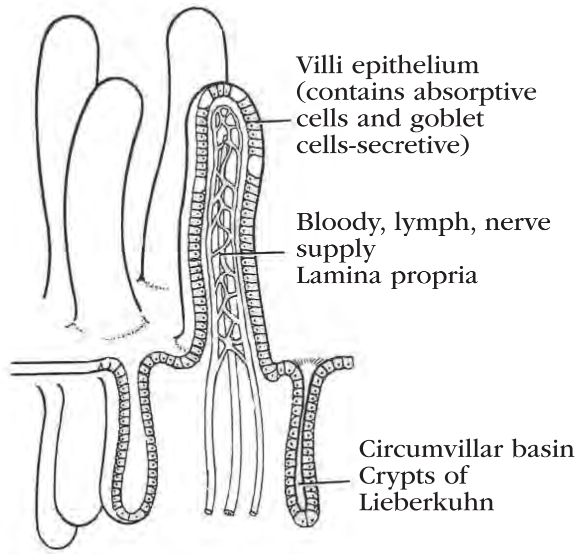

Figure 15.2. Small Intestine Villi Anatomy

The presence of the mucosal folds, the villi, and the microvilli increases the total surface area, increasing the absorptive capacity of the small intestine approximately 600-fold.

Between the villi are small, tubular glands known as the **crypts of Lieberkühn**. Extremely mitotic undifferentiated cells that lie deep in the crypts serve to replace the epithelium. The entire intestinal epithelial surface is replaced every 32 hours. **Paneth's cells** are also located at the base of the crypts. These cells secrete antimicrobial and growth protein substances. A third cell type that resides in the crypts is **enterochromaffin cells**, which serve a neuroendocrine function.

In the proximal duodenum, the normal villous pattern is interrupted at intervals by **Brunner's glands**, which are elaborately branched acinar glands that contain both mucus cells and serous secretory cells. Brunner's glands empty into the crypts of Lieberkühn.

Lymphoid tissue makes up approximately 25% of the intestinal mucosa. The gastrointestinal (GI)-associated lymphoid tissue is made up of three distinct types.

1. **Peyer's patches** are circular, aggregated lymph nodes that lie in the mucosa and submucosa of the ileum. They participate in antibody synthesis and in the body's immune response.
2. **Lymphocytes and plasma cells** are located in the lamina propria. Approximately 70% to 80% of these cells produce immunoglobin A (IgA), which is the major immunoglobulin in intestinal secretions. This secretory IgA plays an important role in the resistance of the mucous membranes to pathological microorganisms and dietary antigens.
3. **Intraepithelial lymphocytes**, which are another type of lymphocyte, lie between the intestinal epithelial cells. A large proportion of these intraepithelial lymphocytes are T cells.

Vascular supply

Arterial

The duodenum receives arterial blood from the hepatic artery. The rest of the small intestine derives its blood supply from the superior mesenteric artery.

Venous

The entire small intestine blood supply drains through the superior mesenteric vein.

Innervation

The wall of the digestive tract is innervated by the **enteric plexus**, which is an autonomic nerve plexus made up of the submucosal plexus (**Meissner's plexus**), myenteric plexus (**Auerbach's plexus**), and the subserosal plexus.

- **Parasympathetic stimulation,** by way of the vagus nerve, increases the tone of the small intestine and the frequency, strength, and velocity of smooth muscle contractions. Vagal stimulation also enhances motor and secretory activities.
- **Sympathetic stimulation,** via spinal nerves from thoracic vertebrae levels T6 to T12, reduces peristalsis and inhibits GI activity. The intrinsic nerve supply, which initiates motor function, passes through Auerbach's and Meissner's plexuses.

Secretion

In addition to their absorptive function, cells located in the small intestine secrete digestive juices, mucus, and a variety of hormones. The small intestine also receives secretions from the liver and pancreas.

- Microvilli contain the peptidases and disaccharides that are required for digestion of proteins and carbohydrates.

- Brunner's glands located in the proximal duodenum secrete a clear, alkaline (pH 8.2 to 9.3), viscous fluid that protects the duodenal mucosa from gastric acid secretions.
- **Goblet cells**, located on and between the villi on the mucosa, secrete a protective mucus.
- Crypts of Lieberkühn secrete 2-3 liters of succus entericus per day. This watery fluid supplies a carrier substance for absorption of nutrients when the villi come in contact with the chyme. There are ten different types of endocrine cells located in the crypts that produce a number of peptides and hormones, including secretin, cholecystokinin, gastrin, somatostatin, enteroglucagon, motilin, neurotensin, gastrin inhibitory peptide, vasoactive intestinal peptide, and serotonin.

Motility

Within the small intestine, three types of movement contribute to the mixing of the chyme:

- Concentric, segmenting contractions, which normally take place in the jejunum, give the small intestine the look of a chain of sausage links. The concentric movement helps to mix secretions of the small intestine with the chyme particles.
- Short, propulsive contractions called peristaltic waves slowly push the chyme in the direction of the colon. Peristaltic contractions are found predominantly in the first portions of the duodenum and jejunum.
- The continuous shortening and lengthening of the villi constantly stir the intestinal contents.

As the chyme nears the large intestine, contractions in the ileum increase. After a meal, the ileocecal sphincter relaxes, allowing the chyme to move from the ileum into the cecum.

Absorption

The primary function of the small intestine is to absorb nutrients from the chyme. The small intestine receives up to 8 liters of fluid per day and passes only 500-1,000 ml into the large intestine. The remaining fluid is absorbed by the columnar cells of the villous epithelium.

Absorption of different nutrients takes place at different locations in the small intestine. The duodenum is the primary site of iron and calcium absorption, and the jejunum is the site of absorption of fats, proteins, and carbohydrates. The ileum absorbs vitamin B_{12} and bile acids.

There are five basic mechanisms for absorption in the small intestine: hydrolysis, nonionic movement, passive diffusion, facilitated diffusion, and active transport. Water, for instance, diffuses passively across the wall of the small intestine, with only a small portion left unabsorbed.

Sodium is absorbed in the jejunum through active transport. Magnesium, phosphate, and potassium are absorbed throughout the small intestine. Chloride diffuses with sodium in the jejunum. In the ileum, chloride is actively transported. Dietary iron is absorbed predominantly by the duodenum and proximal jejunum.

Calcium absorption occurs by two mechanisms: 1) a transcellular route that takes place largely in the duodenum and proximal jejunum; and 2) a concentration-dependent process that occurs along the length of the small intestine.

Absorption of most water-soluble vitamins takes place by diffusion. The exception is vitamin B_{12}, which combines with intrinsic factor for active transport and is absorbed in the ileum.

Carbohydrates are hydrolyzed by intestinal enzymes into simple sugars (glucose, fructose, and galactose), which are then absorbed into the bloodstream via the intestinal mucosa, using either active transport or facilitated diffusion. Proteins are hydrolyzed into amino acids, which are absorbed by active transport.

Fats are emulsified and then broken down primarily by the enzyme pancreatic lipase into glycerides, fatty acids, and glycerol. Fatty acids and monoglycerides are made water-soluble by the formation of micelles with bile acids. Fat-soluble vitamins also combine briefly with bile salts to form water-soluble micelles.

DISORDERS OF THE SMALL INTESTINE

Disorders of the small intestine can affect absorption of nutrients in the affected area. To best understand the impact of a particular condition on nutrient absorption, consider the location of disease. Duodenal disease may lead to deficiencies in iron and calcium, whereas jejunal disease may lead to malabsorption of fats, proteins, and carbohydrates. Ileal disease may result in vitamin B_{12} deficiency and bile acid malabsorption.

Common disorders of the small intestine include duodenal ulcers, bacterial and viral infections, parasitic disease, Crohn's disease, Meckel's diverticulum, vitamin B_{12} deficiency, motility disorders, small intestine tumors, and lymphangiectasia. A variety of malabsorption syndromes also affect the small intestine, such as celiac disease, tropical sprue, Whipple disease, short bowel syndrome, lactase deficiency, and albetalipoproteinemia.

Common small intestinal disorders affecting children include Meckel's diverticulum, celiac disease, acute infectious diarrhea, and short bowel syndrome. Other common large or small intestinal disorders affecting children include intussusception, inflammatory bowel disease, Hirschsprung's disease, appendicitis, and constipation.

Duodenal ulcers

Peptic ulcers can develop in the lower esophagus, stomach, pylorus, duodenum, or jejunum. The yearly incidence of peptic ulcers is greater than 5 cases per 1,000 people. Most peptic ulcers are duodenal ulcers. They occur when the protective mucosa of the duodenum cannot resist corrosion by above-normal hydrochloric acid levels.

Risk factors

Duodenal ulcers are most common in men between the ages of 20 and 50 years of age, and in people with type O blood.

Helicobacter pylori (*H. pylori*), a spiral bacteria most likely transmitted by the gastro-oral route, is present in almost all adults with duodenal ulcers, most patients with gastric ulcer, and virtually all patients with histological evidence of gastritis. *H. pylori* is associated with peptic disease much less frequently in children than in adults. Children with peptic inflammation of the small intestine are also much less likely to develop duodenal ulcers than adults with the same condition.

H. pylori penetrates the mucous layer of the duodenum. The duodenal ulcer has a round, punched-out appearance and is on average about 0.5 cm across.

Symptoms

Symptoms of duodenal ulcers include gnawing or burning epigastric pain occurring after meals, heartburn, and intermittent nighttime pain or discomfort that is localized in the epigastrium. Pain is exacerbated by fatty foods, but may be relieved by other foods. Attacks usually occur about 1-3 hours after meals when the stomach is empty or after consumption of orange juice, coffee, aspirin, or alcohol. Many patients report symptoms that are unlike those expected in this classic presentation. Almost 50% of patients with ulcers may present with bleeding (ASGE, 2012).

Diagnosis

Diagnosis may be made by upper GI radiographic examination and/or upper GI endoscopy. The sensitivity of individual radiologists in detecting ulcers varies, with a double-contrast technique being somewhat more sensitive than a single-contrast technique. Endoscopy may be more sensitive than radiography, with experienced endoscopists detecting most gastroduodenal lesions.

Guidelines from the American Society for Gastrointestinal Endoscopy (ASGE) (2010) advise an esophagogastroduodenoscopy (EGD) in patients over age 50 years with new-onset dyspepsia, in patients at any age with alarm features that suggest significant structural disease or malignancy, or for any patient with persistent dyspeptic symptoms. Endoscopy is not recommended to evaluate benign-appearing uncomplicated duodenal ulcers identified on radiologic imaging.

During endoscopy with biopsy, testing for the presence of *H. pylori* may be performed by placing the biopsy sample into a medium designed for a rapid urease test. Although several manufacturers make different types of this test, most commercially available tests show a color change in the presence of the enzyme urease, which is produced by *H. pylori*.

Non-endoscopic methods of testing for this bacteria include the performance of a C-urea (carbon 13 or 14) breath test or measurement of *H. pylori* antibodies in the blood. An additional noninvasive method for diagnosis is a stool antigen test, which can also serve as a monitor for effectiveness of therapy.

Complications

In peptic ulcer disease, erosion of the affected mucosa can lead to hemorrhage, perforation, and peritonitis.

Treatment

Treatment of *H. pylori* infection is directed toward eradication of the infection through the use of a combination of antibiotics, acid suppression, and bismuth, along with discontinuation of NSAIDS. Greater than 90% of duodenal ulcers heal with 4 weeks of proton pump inhibitor (PPI) therapy (ASGE, 2010).

Indications for surgery include intractable bleeding, perforation, and difficulty using medications. Surgical procedures include:

- Direct suture ligation of the bleeding vessel: a duodenotomy or duodenopyloromyotomy must be performed to expose the ulcer.
- Anti-secretory procedures:
- Truncal vagotomy with drainage: destruction of anterior and posterior vagi.
- Highly selected vagotomy: dissection of vagal distribution along the stomach.
- Truncal vagotomy and antrectomy.
- Complementary gastrojejunostomy or antrectomy: increases drainage through the pylorus.

Patients with perforations may undergo a simple closure or a closure combined with one of the definitive operations described above.

Bacterial and viral infections

A number of bacteria and viruses can infect the small intestine.

Risk factors

The small intestine may become infected by any of the following types of agents:

- Enterotoxigenic bacteria produce enterotoxins that stimulate the active secretion of electrolytes into the lumen of the proximal small intestine. Examples are toxigenic *Escherichia coli* (*E. coli*), *Vibrio cholerae* (*V. cholerae*), *Bacillus cereus* (*B. cereus*), *Clostridium perfringens* and *Staphylococcus aureus* (*S. aureus*).

- Invasive bacteria invade and damage the intestinal mucosa of the distal small intestine and colon. Examples are Salmonella species, Shigella species, *E. coli*, *Vibrio parahaemolyticus* (*V. parahaemolyticus*), *V. cholerae*, *Clostridium difficile* (*C. difficile*) and *S. aureus*.
- Penetrating bacteria invade the mucosa, usually in the distal small intestine, but do not produce extensive mucosal ulcerations. *Yersinia enterocolitica* (*Y. enterocolitica*) and *Salmonella* Typhi are examples of penetrating bacteria.
- Viruses such as coronavirus, rotavirus, Norwalk virus, and adenovirus may also invade the small intestine mucosa.

Symptoms

Symptoms may be based on the type of infectious agent.

- Enterotoxigenic bacteria produce watery, voluminous diarrhea that is not usually accompanied by fever, and fecal leukocytes are seldom present.
- Invasive bacteria produce scant, bloody, mucoid stools; fever, and fecal polymorphonuclear leukocytes.
- Penetrating bacteria often cause extraintestinal disease (e.g., sepsis), fever, and fecal leukocytes.
- Viruses result in malabsorption and diarrhea.

Diagnosis

Identification of the etiological agent in GI infections requires taking a careful history to obtain information on the symptom pattern, any recent exposure to infected individuals or contaminated food or water, or any recent history of foreign travel. Patients should be evaluated for fluid status and abdominal tenderness, and stools should be examined for the presence of fecal leukocytes.

Complications

GI tract infections are a major cause of morbidity and mortality throughout the world.

Treatment

In most cases, infectious diarrhea can be treated with rest and fluid replacement. Intravenous (IV) fluids may be needed for patients with severe dehydration.

Generally, for both adults and children with acute infectious diarrhea, consumption of an unrestricted diet should continue, with additional fluids or oral rehydration solutions provided to replace increased stool water losses.

Antidiarrheal medications are contraindicated as they may prolong mucosal contact with the infectious organism and exacerbate intestinal stasis, gas, and discomfort. If the etiological agent is identified, specific antibiotics may be indicated. Table 15.1 lists some of the causes of infectious diarrhea in which antibiotic treatment is indicated.

Table 15.1. Treatment for Infectious Diarrhea

Infectious Agent	Treatment
Staphylococcus aureus	None; self-limited
Toxigenic *Escherichia coli* (also called enterotoxigenic *E. coli* [ETEC])	trimethoprim-sulfamethoxazole (Cetraxal)
Vibrio cholerae	Doxycycline (Oracea, Monodox, Doryx); ciprofloxacin (Cipro); trimethoprim-sulfamethoxazole
Clostridium botulinum	polyvalent antitoxin (Botulism Antitoxin Heptavalent [A,B,C,D,E,F,G] [Equine]), penicillin, and colonic lavage
Clostridium perfringens	None
Vibrio parahaemolyticus	Tetracycline (Sumycin, Actisite, Panmycin [oral])
Shigella	For severe infection, ampicillin or trimethoprim-sulfamethoxazole
Salmonella	For debilitated patients, ampicillin, chloramphenicol (Chloracol), amoxicillin (Amoxicot, Amoxil, DisperMox, Moxatag, Trimox) or trimethoprim-sulfamethoxazole; cefotaxime (Claforan) or ceftriaxone; or fluoroquinolones—either ciprofloxacin or levofloxacin (Levaquin)
Clostidium difficile	metronidazole (Flagyl, Flagyl ER) or vancomycin (Vancocin)

PARASITIC DISEASES

There are a number of parasitic diseases that affect the small intestine, including giardiasis, coccidiosis, cryptosporidiosis, strongyloidiasis, ascariasis, and diphyllobothriasis.

Giardiasis

Giardiasis is caused by the protozoan *Giardia*, also known as *Giardia lamblia* and *Giardia intestinalis*. It may be present in as many as 20% to 30% of people in the developing world and in as many as 2% to 5% of people in the industrialized world (Floch et al., 2010). Endemic areas in the United States include upstate New York and the Rocky Mountain region, especially Colorado. It is the most frequently identified intestinal parasite in the United States.

Risk factors

Giardiasis is most often associated with ingestion of contaminated food or water. The two forms of the parasite are cysts and trophozoites. The cysts are transmitted readily in water or through contamination from a number of hosts, including domestic and wild animals. A heavily infected host may pass thousands of cysts into the environment, with as few as 10 cysts possibly resulting in infection. Hosts may be without symptoms and become carriers (Floch et al., 2010).

After the cyst is ingested orally or nasally, it matures and, once in the stomach, releases trophozoites. These trophozoites adhere to the wall of the proximal small intestine, causing an inflammatory response. The incubation period lasts between 7 and 14 days, although cysts do not appear in the stool until 1 week after the development of symptoms.

Symptoms

Most adults with giardiasis are asymptomatic, although nonbloody diarrhea occurs in 25% to 30% of patients, and malabsorption with weight loss may occur in 50% (Floch et al., 2010). Clinical symptoms exhibited by children range from none to acute illness characterized by abdominal discomfort and distention, nonbloody diarrhea, headaches, nausea, and vomiting. Infants, children, the elderly, and immunocompromised patients are at high risk for infection.

Diagnosis

Diagnosis is by examination of multiple fresh stool specimens for cysts, stool specimen for *Giardia* antigen (most reliable), or duodenal aspiration with small intestinal biopsy examination. Additionally, serology testing is available.

Treatment

If treatment is necessary, the drug of choice is metronidazole (Flagyl). The dose is 250 mg for adults and 5 mg/kg for children, given three times a day for 5 to 7 days. This treatment is more than 90% effective (Floch et al., 2010). Other drugs that are effective include tinidazole (Tindamax), nitazoxanide (Alinia – one year and older, liquid; 12 years and older, tablet), paromomycin, furazolidone (oral), and quinacrine (Atabrine). An alternate option for pediatric patients is furazolidone.

For patients who are resistant to metronidazole alone, a combined regimen of quinacrine and metronidazole for 2 weeks may be effective.

Stools and/or duodenal aspirate may be rechecked 2 weeks after treatment in symptomatic patients or in those who are carriers. A second course of treatment is instituted if necessary.

Coccidiosis

Coccidiosis is caused by the intracytoplasmic protozoa *Cystoisospora belli* (formerly Isospora belli), *Isospora hominis* (human coccidiosis) and *Isospora natalensis* (human parasite). It is endemic throughout the world but is rarely seen in the United States. The parasite multiplies in the small intestine, causing mucosal damage. It has occasionally been identified in travelers' diarrhea (Floch et al., 2010).

Symptoms

Clinical features include acute fulminant diarrhea, continuous or intermittent chronic diarrhea, and **steatorrhea** (fatty stools).

Diagnosis

Diagnosis is by stool examination, small intestinal biopsy examination, and blood tests for eosinophilia.

Treatment

Most patients have a benign, self-limited course that lasts from a few days to 6 months. Adult patients with a mild case may be treated with fluid and electrolyte replacement. In children, severe diarrhea with malabsorption may necessitate temporary parenteral nutrition. For severe cases, the drug of choice is furazolidone (Fumedil Dis).

Cryptosporidiosis

Cryptosporidiosis is caused by the protozoan parasite *Cryptosporidium* (Crypto). The colon and small intestine are the most common sites of infection, but *Cryptosporidium* has been reported in all parts of the GI and respiratory tracts. After the oocysts are ingested, they embed in the surface of the bowel where they complete their life cycle. After about 12 days, oocysts are passed in the stools.

Risk factors

Transmission is fecal-oral. It may be transmitted between animals and humans, but it is rarely transmitted from human to human. Water-borne transmission is common.

Before 1981, only eight cases had been reported in humans. However, because immunocompromised patients are particularly susceptible to *Cryptosporidium*, the beginning of the AIDS epidemic marked a noteworthy increase in its prevalence in humans. Although low and reported less frequently, the incidence is significant and should be considered. It is more prominent in the elderly and immunocompromised patient populations. Children in developing countries are also susceptible candidates and have a high morbidity from this infection.

Symptoms

The severity of the illness ranges from self-limited diarrheal episodes lasting from 3 days to 4 weeks to more severe symptoms in immunocompetent hosts, sometimes leading to death in immunocompromised individuals. Onset is acute, with malaise and fever, followed by abdominal pain, diarrhea, vomiting, steatorrhea, and occasionally mild rectal bleeding and nonspecific proctitis.

Diagnosis

The diagnosis may be made by serial stool examinations for identification of acid-fast oocysts or intestinal biopsy examination. The most accurate staining method for identification of the parasite is a three-step procedure that involves concentration evaluation using a sugar flotation technique, iodine staining, and a modified acid-fast stain. Immunoassays or PCR-based assays may increase detection sensitivity.

Treatment

Therapy focuses on correcting the severe fluid and electrolyte imbalance. For immunocompromised patients such as HIV-infected individuals, the approach is to maintain immune system function by using highly active antiretroviral therapy (HAART). Antibiotics to consider for immunocompromised patients include paromomycin, azithromycin (Zithromax, Zmax), and nitazoxanide (NTZ) (Alinia). *Cryptosporidium* has some resistance to antimicrobial drugs.

Strongyloidiasis

Strongyloidiasis occurs in warm, moist climates and is caused by the nematode *Strongyloides stercoralis* (*S. stercoralis*) (American threadworm), a soil-transmitted helminth. Filariform larvae are the infectious form, while rhabditiform larvae are noninfectious.

Entry is through skin penetration from the soil, followed by migration through the bloodstream with entry into the lungs. The larvae are then swallowed and enter the duodenum, where the adults attach to or penetrate the wall. Multiple cycles of the parasite may occur in a single host, thereby allowing some patients to harbor *S. stercoralis* for 30-50 years.

Symptoms

Most patients are asymptomatic, with larvae discovered in the stool identifying the infestation. Persistence of the presence of these nematodes, however, can cause symptoms for many organs (Floch et al., 2010).

Early symptoms of strongyloidiasis include fever; migratory, erythematous, macular eruptions over the distal extremities (cutaneous lesion at the entry site in the perianal area or on the feet); and productive or nonproductive cough. With progression, GI symptoms predominate. Neurologic disease is rare.

Patients with hyperinfective syndrome may have heavy worm burdens, which are manifested by esophagitis, pneumonitis, gastritis, enterocolitis, hepatitis, myocarditis, intestinal obstruction or perforation, shock, or meningitis. There may be diarrhea, malabsorption, and abdominal pain.

Diagnosis

Diagnosis is by blood tests, stool examination, duodenal aspirate, chest X-ray, or upper GI series with small intestine follow-through. Eosinophilia is present, but may be absent in immunocompromised patients. Serologic tests vary in sensitivity. The agar plate method is the most sensitive test to identify the larvae.

Complications

In some patients, repeated auto-infection may produce an overwhelming fatal hyperinfective syndrome.

Treatment

For both symptomatic and asymptomatic patients, the well-tolerated drug of choice is ivermectin (Stromectol) as 200 mcg/kg orally for 2 days. Safety of this drug in children less than 15 kg or in pregnant women has not yet been determined. An alternative, yet less effective drug therapy is albendazole (Albenza) as 400 mg BID for 7 days. Patients with hyperinfective syndrome also require fluid and electrolyte maintenance and appropriate bacterial cultures.

Stools and/or duodenal aspirate should be rechecked 1-2 weeks after completion of therapy, with a second course of treatment initiated if necessary. Worms may be difficult to find. Checking three stool samples can increase the identification rate to 50%, and seven stool samples improve the identification rate to 100% (Floch et al., 2010).

With treatment, most patients with mild to moderate cases have an excellent prognosis. In patients with hyperinfective syndrome, however, the mortality rate approaches 50%.

Ascariasis

Ascariasis is caused by the roundworm *Ascaris lumbricoides*, which is the largest intestinal nematode to affect humans and one of the most prevalent soil-transmitted helminths. Endemic areas in the United States include the southeastern Appalachian range and the Southern states.

The adult worms live in the small intestine for 10-18 months, then penetrate intestinal blood vessels and lymphatics. At this point, they are carried through the portal circulation and pass through the liver into the lungs. The adult male measures 15-25 cm and is smaller than the female, which may be as large as 35 cm. The mature female may produce up to 200,000 eggs daily.

Risk factors

Transmission is by ingestion of eggs that are passed in the stool.

Symptoms

Reactions to the larvae include high fever; frequent, spasmodic coughing; and hemoptysis. Reactions to the adult worms include colicky mid-epigastric pain, abdominal pain and distention, loss of appetite, nausea, vomiting, diarrhea, and constipation. A rare reaction is pancreatitis with symptoms of biliary or pancreatic disease. Extraintestinal manifestations have been reported in as many as 50% of patients (Floch et al., 2010).

Diagnosis

Definitive diagnosis requires that the egg or the adult worm be identified in the stool. The worms and eggs are easy to identify. The larvae reach the lungs via the pulmonary artery into the alveoli and bronchi. The larvae develop into adult worms in the small bowel. Lung involvement precedes intestinal phases by 8-10 weeks. Eosinophils may be present during the pulmonary phase with Charcot Leyden crystals, as well as larvae visualized in the sputum. Live worms have been reported in sputum and vomitus.

Complications

The most frequent complication is partial or complete intestinal obstruction. Obstruction is more common in children. Obstruction may occur in the appendix. A hypersensitivity reaction in the lung may occur, causing the self-limiting disease known as Löffler syndrome. Extraintestinal conditions, such as pancreatitis, may be serious problems.

Treatment

Uncomplicated ascariasis may be treated with albendazole (Albenza) as 400 mg orally in a single dose. Other medications that may be used are mebendazole (Emverm, Vermox) as 100 mg orally BID for 3 days or 500 mg as a single dose, or ivermectin (Stromectol) as 150–200 mcg/kg orally as a single dose. Safety in children and pregnant women is not yet established for ivermectin.

For patients with complete intestinal obstruction, surgical intervention may be necessary.

Prognosis is usually good, but it is dependent on the worm burden and the presence of complications.

Diphyllobothriasis

Diphyllobothriasis is caused by the fish tapeworm *Diphyllobothrium latum*. It occurs mainly in northern temperate regions where a major part of the diet consists of freshwater fish. These areas include Baltic countries, northern Europe, northern Siberia, Asia, Japan, and northern Manchuria. Endemic areas in the United States include northern Wisconsin, Michigan, and Minnesota. The tapeworm lives with its head attached to the small intestinal mucosa and grows into adulthood within 3 weeks, reaching up to 10 mm in length. A single parasite may live within the host for years.

Risk factors

Infection results from ingestion of infected, raw fish.

Symptoms

Although the effects on the intestinal mucosa are minimal, rarely, profound vitamin B_{12} deficiency may develop. The majority of patients with this infection are asymptomatic. When symptoms do develop, those most often seen are abdominal pain, ataxia, paresthesia, fatigue, nausea, vomiting, diarrhea, and weight loss.

Diagnosis

Definitive diagnosis requires finding the operculated egg (an egg with a cap on one end) in the stool. Mild eosinophilia may occur.

Treatment

Diphyllobothriasis is most often treated with niclosamide (Niclocide) as a 2 gm single dose. An alternative drug is praziquantel (Biltricide) as 5-10 mg/kg in a single dose.

Follow-up stool exams should be done in 6-8 weeks. Patients may pass a very long worm following treatment. With proper treatment, prognosis is excellent.

All fish should be thoroughly cooked or frozen for 24 hours to prevent the infection.

Crohn's disease

The first documented case of Crohn's disease was noted in 1761 by Morgagni. Nine cases of inflammatory bowel disease were described in 1913 by Dalziel, a Scottish surgeon. However, the disease received the name of "Crohn's disease" when later described in a detailed paper by Crohn, Ginzburg, and Oppenheimer (Geller & de Campos, 2014).

Crohn's disease, also known as **regional enteritis**, granulomatous colitis, or transmural colitis, is a transmural, predominantly submucosal inflammation that may affect any part of the GI tract from the mouth to the anus. Crohn's disease most commonly occurs in the terminal ileum and the colon. Thirty percent of cases involve only the small intestine, 40% involve both the small intestine and the large intestine, and 25% involve only the colon (Floch et al., 2010).

Both Crohn's disease and ulcerative colitis are chronic intestinal disorders. The two are often grouped together under the term **inflammatory bowel disease (IBD)**. The asymmetric and patchy distribution of the lesions in Crohn's disease helps to differentiate the condition from ulcerative colitis. (For information on ulcerative colitis, see Chapter 16.)

Risk factors

Crohn's is a complex genetic disorder with a combination of genetic predisposition and possible environmental triggers such as bacteria, a western diet, and smoking.

The incidence of Crohn's disease is higher in persons of Jewish descent in Europe and North America. There is a higher incidence in Caucasians than nonwhites and in urban than rural settings. There is a lower incidence in Japan, South America, and Africa. In western countries, the incidence is from 6-10 per 100,000 people and the prevalence is 130 per 100,000 people, with a rising trend. In the United States, there is an incidence of 6-15 per 100,000 and a prevalence of 50-200 per 100,000. Crohn's disease most commonly affects people during the second and third decades of life. In some surveys, males and females are equally affected. In other surveys, females predominate by as much as 1.6 to 1 (Chen, 2014).

Potential infectious agents include *Mycobacterium avium paratuberculosis* (MAP) and the measles virus.

Genetic factors appear to be the single strongest risk factor for developing the disease. There is a strong familial association with a positive history in 25% of people with the disease (Floch et al., 2010). Chromosomes 16, 3, 7, and 12 have been linked to Crohn's disease. *NOD2* gene, also called *CARD15* gene, of chromosome 16, found in monocytes involved in immune response to pathogens, is responsible for the increased susceptibility. Immunologic factors suggest autoimmune phenomenon as a cause. Stress may play a role in exacerbation of symptoms but is not an identified cause of illness.

Symptoms

Inflammation spreads slowly and progressively, with periods of remission often alternating with exacerbations. The segmental inflammation and rectal sparing of Crohn's disease are features that distinguish it from ulcerative colitis.

The disease process begins with lacteal blockage and lymphedema in the submucosa. Peyer's patches appear in the intestinal mucous membrane. Lymphatic obstruction causes edema with inflammation, mucosal ulceration, stenosis, and development of fissures, abscesses, and possibly granulomas. Typically, deep longitudinal "rake ulcers" appear in the bowel. If deep ulcers appear between islands of edematous inflamed mucosa, the bowel wall takes on a cobblestone appearance. As the disease progresses, fibrosis occurs and serositis develops, causing diseased bowel loops to adhere to other normal or diseased loops. Perianal lesions may be present.

During periods of acute inflammation, patients with Crohn's disease may report intermittent and colicky abdominal pain, particularly in the lower abdomen, with diarrhea the next most frequent symptom. Patients with involvement at the thoracic vertebra level T1 may present with right lower quadrant pain. Systemic symptoms may include low-grade fever, malaise, loss of strength, and weight loss. Bloody diarrhea and malabsorption may be severe, leading to **malnutrition** and weight loss. Growth failure or a decline in the rate of linear growth may be the presenting symptom in children and adolescents.

Chronic symptoms are more persistent but less severe. They include diarrhea, abdominal pain, steatorrhea, anorexia, weight loss, and nutritional deficiencies.

Extraintestinal manifestations occur in 43% of patients and may include hematologic symptoms such as hypercoagulability; dermatologic symptoms such as erythema nodosum and pyoderma gangrenosum; articular symptoms such as peripheral arthritis and spondylarthritis; ophthalmic symptoms such as sclera-conjunctivitis, uveitis, and episcleritis; and hepatic symptoms such as primary sclerosing cholangitis (Chen, 2014).

Diagnosis

The diagnosis of Crohn's disease is based on clinical history and physical examination.

Physical exam. Exam may reveal no findings or a right lower quadrant mass. The patient may be anemic, pale, and febrile.

Labs. Lab may reveal anemia, increased erythrocyte sedimentation rate and C-reactive protein, leukocytosis, thrombocytosis, and hypoalbuminemia. Serologic markers in 60% to 70% of cases demonstrate an increase in the anti-*Saccharomyces cerevisiae* antibodies (ASCA (Floch et al., 2010). If the perinuclear antineutrophil cytoplasmic antibody (pANCA) is negative, the diagnosis is positive.

Endoscopic exams. Endoscopic exams, including double-balloon enteroscopy and barium contrast studies, identify stricture, inflamed bowel loop, and enteroenteric fistulas, and histologic exams of blood and tissue are commonly used in diagnosing Crohn's disease. Colonoscopic exam may reveal aphthous ulcerations of the bowel with skipped areas. The terminal ileum should be biopsied. Capsule endoscopy may reveal ulcerations and strictures.

Ultrasonography. Ultrasound may be useful in assessing undiagnosed right lower quadrant pain.

Computerized tomography (CT) scan. CT scan shows increased fat pad deposition and smudgy areas outside the bowel, as well as areas of bowel wall thickness.

Complications

The most common complication of Crohn's disease is the development of fistulas. These occur in 20% to 40% of patients and are usually enteroenteric or enterocutaneous (Floch et al., 2010). Fistula occur between sites of perforation and adjacent organs.

Other complications include perforation and obstruction. Perforation can lead to general peritonitis in rare situations. Localized abscesses occur near perforation sites in 15% to 20% of patients with Crohn's disease (Floch et al., 2010).

Malabsorption of bile acids and vitamin B_{12} is common in disease of the ileum. Toxic megacolon with a presentation of colonic dilatation, fever, and leukocytosis is another possible complication.

Long-standing Crohn's disease may predispose patients to certain cancers. These include adenocarcinoma of the colon and small intestine, most commonly in the ileum; squamous cell carcinoma of the vulva and anal canal; and Hodgkin's and non-Hodgkin's lymphomas.

Long-term data demonstrate that nearly all patients will develop structuring or penetrating complications over a 20-year course. Approximately half of patients will require surgical resection within 10 years of diagnosis, with a similar number of patients experiencing postoperative recurrence after 10 years (Dudley-Brown, 2017).

Treatment

There is no cure for Crohn's disease. Medical and surgical therapies are palliative and aimed at relieving acute exacerbations or complications. Dietary therapies aim to replenish depleted nutrient stores, allow for intestinal protein synthesis and healing, and offer preoperative preparation. Therapeutic measures may include the following.

- **Diet.** During exacerbation of symptoms, recommendations include:
  - Diet low in fiber and high in protein, calories, vitamins, and minerals to replace nutrient losses associated with the inflammatory process and provide sufficient nutrients to promote energy and nitrogen balance.
  - Small, frequent meals.
  - Lactose avoidance by patients who are lactose-intolerant.
  - Chemically defined elemental and polymeric diets, especially in patients whose disease is limited to the small intestine (shown to decrease disease activity in some studies).
  - Liquid polymeric diets (may be better tolerated by patients).
  - Total parenteral nutrition (TPN) for patients with severe disease or short bowel syndrome refractory to oral therapy or specialized enteral tube feeding.
- **Medications.** FDA-approved pharmacologic treatment options for Crohn's disease include:
- **Induction.** Locally-acting oral steroid capsules: controlled-release budesonide (Entocort EC, Uceris).
- **Induction and maintenance.** Anti-TNF agents: adalimumab (Humira, Amjevita, Cytezo) subcutaneous injection (subQ); certolizumab pegol (Cimzia) subQ; infliximab (Remicade, Inflectra, Ixifi, Renflexis) intravenous (IV) infusion. While using these medications, regularly monitoring complete metabolic panels, signs and symptoms of infusion reactions for IV medications, and drug and antibody levels (for infliximab and adalimumab) is recommended. Screening for latent tuberculosis (TB) and hepatitis B infection should be done before initiating therapy with these agents.
- **Induction and maintenance.** Anti-integrin IV infusion. Monitoring for signs and symptom of infusion reactions and screening for infections such as TB are recommended.
- **Surgery.** Surgical indications include intestinal obstruction, intestinal perforation with fistula formation or abscess, free perforation, GI bleeding, urologic complications, cancer, and perianal disease. Children who acquire such systemic symptoms as retardation of growth may benefit from resection. Surgery should be limited toward the specific complication with only the involved bowel segments being resected. A laparoscopic approach may be used in select patients. Procedures include segmental resection, subtotal colectomy with ileoproctostomy, ileorectal anal anastomosis, and total colectomy with ileostomy. Operative procedures provide symptomatic relief but are not curative. During the first year, the endoscopic recurrence rates are between 35% and 85%, with 10% to 38% of the patients showing symptoms. Three years post-operatively, the recurrence rates are 85% to 100% (Gklavas, Dellaportas, & Papaconstantinou, 2017). Recurrence within 5 years will most likely be close to the area of resection (Floch et al., 2010).
- **Psychosocial support.** Patients with Crohn's disease should be encouraged to rest, reduce tension in their lives, and communicate their feelings about this chronic and often disabling disorder. Education about the disease and its process and treatments should be provided. Contact information for community agencies and support groups such as the Crohn's and Colitis Foundation can be provided, and participation should be encouraged. Depression is a common emotion experienced by patients with Crohn's disease; therefore, the provision of emotional support is an important part of the treatment process.

Studies have shown that positive disease outcomes, including steroid-free remissions, decreased risk of major abdominal surgery, and overall improvement in quality of life, are all possible with mucosal healing. Treatment models have changed from a "step-up" approach, where the treatment progresses in intensity as the disease worsens, to a more personalized format, in which there is earlier, more aggressive treatment for those patients who are found to be at risk for more complicated forms of Crohn's disease (Dudley-Brown, 2017).

Meckel's diverticulum

Meckel's diverticulum is a congenital anomaly of the GI tract, resulting from incomplete separation of the fetal gut from the yolk sac. This is a true diverticula of the intestine, with all layers of the intestinal wall affected. The diverticulum range in size from 1-10 cm in length and 1-3 cm in width and are located 30-90 cm proximal to the ileoceal junction.

Risk factors

Meckel's diverticulum is the most frequent congenital anomaly of the GI tract and occurs in 1% to 3% of the population (Floch et al., 2010).

Diagnosis

Meckel's diverticulum cannot be diagnosed by endoscopic or radiological contrast studies. A Meckel scan (technetium-99m pertechnetate radionuclide study) allows diagnosis by administration of technetium, which is excreted by gastric mucosa, followed by a nuclear medicine scan to identify areas of excretion. Administration of cimetidine (Tagamet HB) or ranitidine (Zantac) orally for several days before the scan or given IV before the scan increases the accuracy of this diagnostic tool by decreasing anion secretion from the gastric mucosa. Crohn's disease or other inflammatory disorders can lead to false-positive scans.

Additional diagnostic studies may include small intestine follow-through, enteroclysis examination, or angiography. CT scans can provide a differential diagnosis between appendicitis, cholecystitis, diverticulitis, and salpingitis.

Complications

This outpouching of the ileum is lined by typical ileal mucous membrane but may contain ectopic gastric and/or pancreatic tissue in its wall. Secretion of acid or pepsin from this ectopic tissue may cause ulceration of adjacent ileal tissue, resulting in hemorrhage. Hemorrhage usually occurs in children under 2 years of age and typically presents as painless passage of bright red blood from the rectum.

Intestinal obstruction may result from Meckel's diverticulum, either from **volvulus** (twisting of the bowel) or **intussusception** (prolapse of the diverticulum into the lumen of the bowel). Symptoms of obstruction include abdominal pain, bilious or feculent vomiting, and/or passage of bright red or "currant jelly" stools.

Unusual complications in adults include development of carcinomas, such as carcinoids, sarcomas, and adenocarcinomas. Other complications include development of enteroliths that may lodge in the diverticulum and result in vomiting, obstruction, and bleeding.

Treatment

Treatment of Meckel's diverticulum requires diverticulectomy (resection of the diverticulum) or, if necessary, resection of the adjacent ileum. Because there is a low risk of complications of Meckel's diverticulum in adults, the incidental finding of such a lesion may be left intact. When Meckel's diverticulum causes bleeding, surgical correction is required.

In the rare Littre hernia, Meckel's diverticulum is a component of the hernia sac. These are difficult to diagnose, but the clinical picture includes localized pain over a pre-existing hernia that is accompanied by intestinal obstruction, bleeding, perforation, fistula, and malignancy.

Vitamin B_{12} deficiency

In the stomach, protein-bound vitamin B_{12} is freed by the action of gastric pepsin. The free B_{12} binds to two molecules of intrinsic factor, which is a glycoprotein secreted by gastric parietal cells. This complex protects vitamin B_{12} from use by bacteria and from the formation of unabsorbable aggregates while it is in the small intestine. In the ileum, the complex of B_{12} and intrinsic factor binds to receptor sites on the brush border, thus facilitating B_{12} entry into the enterocytes. After passage through the enterocytes, B_{12} is transported in the blood bound to transcobalamins, which deliver the vitamin to the tissue. Intrinsic factor remains bound to the receptor site, where it may promote the uptake of more B_{12}. When the amount of ingested B_{12} is very large, absorption also occurs by diffusion, probably at all levels of the intestine.

Risk factors

Vitamin B_{12} deficiency may occur if any one of the steps of production, transport, or absorption is impaired. Disorders or conditions that may result in vitamin B_{12} deficiency include gastrectomy, pancreatic insufficiency, Zollinger-Ellison syndrome, Crohn's disease, bacterial overgrowth, ileal disease or resection, familial cobalamin malabsorption, or transcobalamin deficiency. Additionally, infestation with the tapeworm *diphyllobothrium latum* may be a cause when a competitive situation for vitamin B_{12} occurs.

Symptoms

Deficiency of vitamin B_{12} may be subtle, with the patient appearing with neurologic findings such as degeneration of the spinal cord and progressive neuropathy, or there may be a full-scale megaloblastic or pernicious anemia.

Diagnosis

The source of vitamin B_{12} malabsorption may be pinpointed using the three-part Schilling test, which traces the excretion of radiolabeled B_{12} in the urine with or without the addition of intrinsic factor or a broad-spectrum antibiotic.

Treatment

Treatment is directed at the underlying disease state. Parenteral vitamin B_{12} intramuscularly may be required in cases of severe disease that includes glossitis and peripheral neuropathy or in extensive ileal resection.

MALABSORPTION SYNDROMES

Malabsorption syndromes include abnormalities of mucosal transport and/or intraluminal digestion of one or more dietary constituents. Because all nutrients are absorbed across the mucosa of the small intestine, disorders that affect the mucosa may affect the absorption of fat, protein, carbohydrates, vitamins, and/or minerals. The clinical significance of malabsorption depends on the site and extent of involvement. The primary feature of malabsorption syndromes is chronic diarrhea. Other symptoms may include weight loss and anorexia despite normal or high caloric intake, physical weakness, abdominal distention and cramping, and malaise.

A wide variety of disorders of the digestive organs may cause malabsorption or **maldigestion**. These include small intestine disease, such as celiac disease, tropical sprue, Whipple disease, lactase deficiency, abetalipoproteinemia, or eosinophilic gastroenteritis; and short-bowel syndrome (SBS), usually following significant small intestine resection.

Other disorders can also result in malabsorption syndromes, including pancreatic exocrine deficiency, bile acid insufficiency, lymphatic disorders, gastric hypersecretory states, and GI stromal tumors (GISTs, previously referred to as leiomyosarcomas of the GI tract). Small intestinal bacterial overgrowth (SIBO) results in weight loss and malnutrition from malabsorption. Additional conditions where malabsorption may be manifested in varying degrees include infection with *Giardia lamblia* and *Strongyloidiasis*.

To pinpoint the cause of malabsorption, the physician may order blood tests and specific absorption tests. Chemical analysis of stool such as fat analysis via a 72-hour stool collection can assist in the evaluation of small intestine malabsorption. Small intestine X-ray may be used to detect mucosal disease. A small intestine biopsy may be ordered to diagnose celiac disease, Whipple disease, abetalipoproteinemia, or agammaglobulinemia.

Treatment of most malabsorption conditions is directed at the underlying disease.

Celiac disease

Celiac disease, also known as gluten-sensitive enteropathy and celiac sprue, is characterized by poor food absorption and intolerance of gluten, which is a protein found in wheat, oats, rye, barley, and their products. Malabsorption in the proximal small intestine results from atrophy of the villi and a decrease in the activity and amount of enzymes in the surface epithelium. Samuel Gee first described the disease in 1888 as "celiac affliction." Dicke and colleagues found in 1950 that removing wheat from the diet caused the signs and symptoms to disappear (Floch et al., 2010). The term celiac disease should be used only when gluten is demonstrated to be the cause.

Non-celiac gluten sensitivity (NCGS), also known as gluten sensitivity, is a condition that involves an intolerance of gluten without small intestine damage. Gluten-related disorders has replaced the former term gluten intolerance.

Risk factors

Celiac disease is an inherited inflammatory autoimmune disease that affects the lining of the small intestine. It most likely results from a combination of environmental factors and genetic predisposition.

Risk factors for celiac disease include being female (occurrence is at a female-to-male ratio of 2:1), family history, and Northwestern European ancestry. It affects an estimated 1% of the world's population. One in every 133 people in the United States has celiac disease. Ireland, Italy, and the Netherlands have the highest incidental rates of celiac disease in Northwestern Europe (Armone, 2012).

This autoimmune disease can affect individuals at any time across the developmental lifespan. However, the typical age of onset is between 8 and 24 months, after a child has been receiving gluten in his or her diet. In some cases, clinical signs disappear during adolescence and reappear in adulthood. The second peak age of presentation is in the 20s in adults who have been previously asymptomatic. The ratio of occurrence between children and adolescents is 1:104 (Armone, 2012).

Symptoms

Symptoms of celiac disease may include:

- Recurrent attacks of diarrhea, constipation, vomiting, steatorrhea, abdominal distention, flatulence, stomach cramps, weakness, fatigue, and lack of energy.
- Anorexia or increased appetite without weight gain.
- Unexplained weight loss or gain.
- Irritability, uncooperativeness and/or apathy, or depression.
- Growth failure, muscle wasting, delayed development, iron deficiency anemia, bone or joint pain, and osteoporosis.
- Arthritis.
- Dental enamel defects with discoloration, canker sores, and dermatitis herpetiformis.
- Vitamin K deficiency.
- Unexplained anemia.
- Missed menstrual periods, infertility, or spontaneous miscarriages.

Diagnosis

Diagnosis is made based on clinical history, physical examination, laboratory tests, and upper GI endoscopy. Early diagnosis is possible via serum testing that includes IgA endomysial antibodies and anti-tissue transglutaminase antibodies, which reach close to 98% sensitivity and specificity in detecting celiac disease. Serum IgA and IgG antigliadin antibodies are less sensitive and less specific. There are fecal tests for celiac disease-associated antibodies, but these have not been well validated in adults and have not been studied in children (Floch et al., 2010).

Definitive diagnosis requires an initial small biopsy examination that shows typical characteristics of celiac disease such as villous atrophy with an increase in intraepithelial lymphocytes (IELs). Four to six biopsies should be taken from the second or third portion of the duodenum. Capsule endoscopy is an additional diagnostic tool.

The diagnostic evaluation is complete if symptoms resolve within weeks following a gluten-free diet.

Complications

In adults, potential complications include an increased risk of malignant small intestine lymphoma and carcinoma of the esophagus and stomach. In individuals who have had celiac disease for many years, osteoporosis, anemia, and infertility may occur.

Treatment

Treatment of celiac disease involves lifelong elimination of gluten from the diet, which means avoidance of all wheat, barley, rye, and oats. It is important that the patient also avoid meats containing wheat fillers and certain commercially prepared foods that may use wheat as an extender, like ice cream and candy bars. Rice, corn, soy, and the flours of these grains are acceptable.

In cases of severe intestinal damage, a lactose-free diet may also be recommended until intestinal mucosa heals. Parents and children require continuing education and support to maintain compliance with a gluten-free diet, even when overt symptoms disappear. Consideration should also be given to ensure medications prescribed to patients are gluten-free.

In addition to diet restrictions, therapy may include enteral nutrition support to correct nutritional deficiencies, including macro- and micronutrients.

For patients with refractory disease, glucocorticoids may be given.

Possible adjunctive therapies under investigation include bacterial prolyl endopeptidases, development of a site-specific inhibitor of intestinal tissue tranglutaminase (TTG), T-cell silencing, cytokine therapy, genetically modified wheat, and monoclonal antibodies.

Additional therapies in various trial stages include enzyme-based treatment with the intent to inhibit toxic potential of gluten, as well as antibody targets to both lessen the disease and treat the symptoms. Hookworm larvae have a potential role in treating gluten sensitivity through the anti-inflammatory proteins that the hookworms secrete.

The clinical response to treatment is often dramatic, beginning with an improved disposition and general appearance.

Tropical sprue

Tropical sprue, a malabsorption syndrome acquired in endemic tropical areas, is characterized by progressively more severe alterations of the jejunum and ileum. Histological changes associated with tropical sprue consist of a lengthening of the crypt area, a broadening and shortening of the villi, epithelial cell changes, and infiltration by chronic inflammatory cells.

Risk factors

The disease is associated with an abnormal small intestinal bacterial flora, with the jejunum becoming contaminated with colonic bacteria. This results in mucosal injury. Bacterial contamination of the jejunum may occur during acute GI infection.

Symptoms

Symptoms include chronic diarrhea, malaise, fatigue, and weight loss associated with malabsorption of fat, carbohydrates (glucose and d-xylose), folate, and vitamin B_{12}.

Treatment

Antibiotic treatment improves vitamin B_{12} malabsorption and serves to reduce intestinal bacteria. However, vitamin B_{12} may not return to normal levels. Treatment with folate and vitamin B_{12} may improve the epithelium in 3-6 days, with an increase in mitotic activity, redevelopment of villi, and reversion of nuclear alterations. D-xylose malabsorption may persist in greater than 50% of the patients, despite treatment with folate, vitamin B_{12}, or antibiotics (Antunes & Singhal, 2018). Steatorrhea is treated with a low-fat diet.

Whipple disease

Whipple disease is a rare bacterial infection primarily affecting the small intestine. It is characterized by chronic diarrhea and progressive wasting. It is also known as intestinal lipodystrophy. Only about 700 cases have been reported, primarily in the United States (many from the state of Connecticut), England, continental Europe, and South America. George Whipple first described the syndrome in 1907 (Antunes & Singhal, 2018).

Risk factors

The bacterium that causes the disease was isolated and identified as *Tropheryma whipplei* in 1992 (Antunes & Singhal, 2018). Transmission is believed to be the fecal-oral route, often from consuming bacteria-contaminated food or water, coupled with a genetic immune system response defect that increases infectious tendency with exposure. Whipple disease most often affects Caucasian men between the ages of 20 and 67.

Symptoms

The most common signs and symptoms are weight loss, diarrhea, arthralgias, and vague abdominal pain. Other signs and symptoms include bloating; steatorrhea; impaired intestinal absorption; slight fever; hyperpigmentation; peripheral, mesenteric, periaortic, and celiac lymphadenopathy; and occasionally splenomegaly. There may be central nervous system (CNS) involvement, including signs of aging, dementia, ophthalmoplegia (paralysis of muscles around the eye), myoclonus, and cardiac involvement with fibrinous pericarditis and endocarditis.

Diagnosis

Diagnosis is by biopsy examination of the small intestine, which shows lipid-filled macrophages and large cytoplasmic granules that stain a brilliant magenta with the periodic acid-Schiff (PAS) stain. The bacterium can be demonstrated on electron microscopy.

Complications

Some neurologic signs (dementia) may not reverse. PAS positive macrophages may persist between 1 and 8 years. Untreated, the disease can be fatal.

Treatment

The treatment algorithm for classic Whipple disease includes doxycycline (Vibramycin, Oracea) 200 mg/day and hydroxychloroquine (Plaquenil) 600 mg/day for 12 months, followed by lifetime treatment of doxycycline 200 mg/day. Chronic localized infections should be treated with Doxycycline 200 mg/day and hydroxychloroquine 60 mg/day for 12-18 months.

During the acute phase, corticosteroids may be administered. Patients with iron deficiency anemia need iron supplements.

Most extraintestinal manifestations and positive signs disappear within a year.

Short bowel syndrome

Short bowel syndrome (SBS) is a malabsorptive disorder that occurs as a result of decreased functional mucosal surface area.

Before the days of adequate nutritional intervention, infants with <15 cm of small intestine with the ileocecal valve, or <40 cm of small intestine without the ileocecal valve, rarely survived. SBS is considered among the most lethal disorders in young children. However, more recently, the prognosis for children with even less residual small intestine without an ileocecal valve has improved. A few children with as little as 12 cm of residual small intestine have survived and been successfully weaned from parenteral nutrition support.

Risk factors

SBS usually occurs due to massive resection of the small intestine, when <200 cm of small intestine remain after surgery.

The most common causes of SBS in infants include congenital anomalies (jejunal and ileal atresia, and gastroschisis), necrotizing enterocolitis, and volvulus with malrotation. Other causes of SBS in infants and children include small intestine resection due to long-segment Hirschsprung's disease, omphalocele, Crohn's disease, and trauma. Although rare, congenital short bowel may also be a cause of SBS in a small number of infants.

Radiation enteritis may lead to functional SBS if injury to small intestine mucosa is extensive.

In adults, surgery-related causes of SBS are vascular compromise; Crohn's disease; volvulus; radiation enteritis; trauma to the bowel; neoplasm; and postgastrectomy disorders, especially following a Billroth II operation.

Symptoms

The malabsorption of macro- and micronutrients, fluid, and electrolytes may lead to severe undernutrition and disturbances in fluid, acid/base balance, and electrolytes. Malabsorption may be exacerbated by complications frequently associated with SBS, such as dysmobility and bacterial overgrowth and possible D-lactic acidosis with a need for antibiotic therapy.

These problems lead to a host of symptoms including diarrhea, abdominal pain, distension, and vomiting. The severity of these disturbances varies with age (if occurring in infancy or childhood), the length and location of the bowel that is removed, and the condition of the remaining bowel.

Location of the resection impacts symptoms.

- **Duodenum.** Although isolated duodenal resection is uncommon, the removal or bypassing of the duodenum may result in malabsorption of micronutrients and lead to clinically significant deficiencies. Malabsorption of iron, folate, and calcium may lead to anemia and osteopenia in children who undergo duodenal resection. Malabsorption of fat and lipid-soluble vitamins may occur in the absence of adequate digestion and absorption that is aided by duodenal pancreatic and biliary secretions.
- **Jejunum.** Extensive jejunal resection often leads to significant carbohydrate malabsorption due to the loss of the primary intestinal site of disaccharidase activity. Undigested and malabsorbed carbohydrate contributes to osmotic diarrhea. The duodenum and jejunum are also the primary site of absorption of proteins, fats, copper, and lipid-soluble vitamins. Although resection of the ileum and preservation of the jejunum may initially cause fewer complications associated with fluid and nutrient absorption, the long-term capacity for adaptation is greater in the ileum after a proximal

small intestine resection than in the jejunum after a distal small intestine resection.
- **Ileum.** Extensive resection of the ileum may lead to severe, early consequences. There may be tremendous fluid and electrolyte malabsorption due to the inability of the jejunum to compensate. Bile salt malabsorption following ileal resection may also contribute to watery diarrhea and fat malabsorption. Steatorrhea and malabsorption of fats and lipid-soluble vitamins are the consequences.
- **Ileocecal valve.** Preservation of the ileocecal valve affects prognosis because the ileocecal valve performs functions that ultimately influence enteral feeding tolerance. The ileocecal valve regulates the flow of fluid and nutrients from the small intestine into the colon and the rate of small intestine transit. The ileocecal valve also prevents migration of colonic bacterial flora into the small intestine. Bacterial counts in the small intestine increase dramatically when the ileocecal valve is resected, as a result of the migration of colonic flora.

Diagnosis

Diagnosis can be confirmed based on surgical records, upper GI and small intestine contrast imaging, endoscopy esophagogastroduodenoscopy (EGD), and colonoscopy. The serum level of citrulline is a quantitative biomarker to assess remaining functional absorptive capacity. Evaluation for anemia and iron, vitamin B_{12}, and lactose absorption and tolerance provides a clear therapeutic guideline.

Complications

Intestinal adaptation and long-term outcomes depend, in part, on the length and location of the residual small intestine and whether or not the ileocecal valve was preserved. SBS is considered among the most lethal disorders in young children.

Complications of SBS can include cholestasis, a common complication in children and adults with SBS who require long-term parenteral nutrition. Although the mechanisms contributing to this problem are not fully understood, there is evidence that both prolonged parenteral nutrition and a lack of enteral feeding play independent and significant roles.

Treatment

The capacity for adaptation and eventual ability to absorb all nutrients necessary to sustain growth by the enteral route may not be known for weeks or months following the initial resection as attempts to feed progress. Following the initial surgical interventions, management of nutritional therapy should become the focus of care for infants and children with SBS.

The prognosis for children and adults has dramatically improved in the past 25 years, primarily due to advances in parenteral nutrition and the understanding of the importance of enteral nutrition. The prognosis is favorable if the duodenum and at least 60 cm of jejunum or ileum remain intact. If the colon is present, it can become a digestive organ. Small intestine transplantation has been effective in select cases.

Following extensive small intestine resection, a marked adaptation response occurs in the residual bowel. These compensatory changes begin within days of bowel resection and last for many months. The most striking feature of adaptation is hyperplasia of the villus cells. The villus height increases with an increase in the number of epithelial cells and elongation of the crypts. The degree of hyperplasia appears to be directly proportional to the amount of small intestine resected.

Several factors appear to be trophic for small intestine adaptation, and the most important stimulus for adaptation is enteral feeding. Nutrients in the small intestine lumen stimulate mucosal adaptation through several mechanisms, including:
- Direct contact and absorption of nutrients by enterocytes.
- Stimulation of pancreatic and biliary secretions.
- Stimulation of trophic hormones and growth factors in the bowel and elsewhere.

For all patients with SBS, treatment involves maintaining optimal nutrition while addressing complications, including:
- **Acidosis.** The combined problems of small intestine bacterial overgrowth and carbohydrate malabsorption can lead to metabolic acidosis. Acidosis occurs when bacterial fermentation of unabsorbed carbohydrates produces short-chain fatty acids, which are then absorbed into the circulation. Treatment should be aimed at decreasing the bacterial population in the small intestine with antibiotics, administering low-carbohydrate formulas, and adding citrate or bicarbonate to the enteral or parenteral nutrition solutions. Small intestine bacterial overgrowth is determined by a quantitative culture of aspirated small intestine fluid. Choices for treating bacterial overgrowth include enteral metronidazole (Flagyl, Metro I.V.), gentamicin, neomycin, rifaximin (Xifaxan), and trimethoprim and sulfamethoxazole (Bactrim DS). Probiotics may be used to enhance effectiveness of microflora in the colon.
- **Motility disturbances.** Motility disturbances following small intestine resection occur frequently and may compound problems associated with SBS, including malabsorption and bacterial overgrowth. Antimotility agents may be helpful, such as diphenoxylate/atropine (Lomotil), loperamide (Immodium A-D), codeine, or tincture of opium. Cholestyramine (Questran, Prevalite) can help to bind bile salts. The condition of the remaining bowel may have a more significant impact on motility and absorption than the total length and location of bowel resected. Stasis

due to fibrosis, surgical narrowing, or poor perfusion can be as detrimental to enteral feeding success as fast transit.

- **Gastric acid hypersecretion.** Gastric acid hypersecretion is a common problem associated with SBS. Treatment with histamine (H_2) receptor antagonist drugs such as ranitidine (Zantac) or famotidine (Pepcid) or more potent proton pump inhibitors (PPIs) such as omeprazole (Prilosec) may be indicated if upper GI symptoms or biopsy-proven endoscopic findings of peptic disease are present.

Maintaining optimal nutrition

Maintaining optimal nutritional status and growth with a combination of parenteral and enteral nutrition support should be a priority for all patients with SBS.

Successful management of complex nutritional support has provided the opportunity for long-term survival and a desirable quality of life for many people with SBS. Early and aggressive enteral feeding is the primary stimulus for intestinal adaptation, which is the key to ultimate independence from parenteral nutrition and prevention of associated complications.

Parenteral nutrition therapy

Many patients with SBS require parenteral nutrition therapy (total parenteral nutrition [TPN]), but many are managed without this therapy. The goals of parenteral nutrition therapy include maintenance of fluid and electrolyte balance and support of rapid growth during infancy and childhood while bowel adaptation occurs. Adjustments in total caloric delivery via parenteral nutrition should be adjusted based on the child's growth or the adult's energy needs.

The first major goal for children and adults with SBS is achieved when parenteral nutrition support is no longer required, thus the goal of parenteral nutrition weaning during advancement of enteral feeding should remain a high priority.

Enteral nutrition

Enteral feeding should begin as soon as possible following small intestine resection, once fluid and electrolyte balance is achieved and GI motility has returned. During the advancement of enteral feeding, parenteral nutrition should be continued for the balance of caloric needs.

Malabsorption of protein, carbohydrate, and fat all occur with SBS. Initial enteral feedings should be provided as continuous infusion via nasogastric or gastrostomy tube. A formula containing hydrolyzed protein is optimal in most cases. Protein hydrolysates have the advantage of being more readily absorbed than whole protein.

The role of carbohydrate in dietary management of SBS requires much attention. Severe malabsorption of carbohydrate contributes to osmotic diarrhea and metabolic acidosis. In this situation, carbohydrate often needs to be restricted. Complex starches should be avoided initially because they require considerable digestion before absorption can occur.

Researchers and clinicians have shown that high-fat diets resulted in improved weight gain without increased fat malabsorption or stool volume in patients with cystic fibrosis with protracted diarrhea in infancy. If a fraction of the increased fat is absorbed with high-fat enteral feedings, even when total stool fat increases, high-fat feedings can prove beneficial. Medium-chain triglycerides are more readily absorbed than long-chain triglycerides in the absence of sufficient bile acids, as medium-chain triglycerides may pass directly into the portal circulation.

Supplemental enteral multivitamin and mineral products should be administered. Severe deficiencies may be corrected with additional individual micronutrients administered enterally or parenterally.

Advancements in enteral feeding should be made at regular intervals, as tolerated, with increased volume or concentration of formula. Once complete enteral nutrition is achieved, compression of continuous feedings can be attempted, with increased hourly infusion rates with fewer hours of feedings during the day. Oral feedings (low fructose and lactose) can be encouraged and advanced as tolerated.

Lactase deficiency

The most common disorder of carbohydrate absorption is acquired **lactase deficiency**, in which a deficiency in the brush-border enzyme lactase causes malabsorption of the disaccharide lactose. Holzel first described defective lactose absorption in infants in 1959 (Jarrett & Holman, 1966).

Risk factors

Lactase deficiency is the result of a genetically determined reduction of lactase activity, with genetic control at the transcriptional level. This disorder is often called lactose intolerance.

Congenital lactase deficiency is a rare disorder in which mucosal lactase levels are low at birth.

Secondary lactase deficiency can result from diseases that damage the small intestine mucosa, such as celiac disease, tropical sprue, Crohn's disease, bacterial or viral enteritis, and radiation enteritis.

Symptoms

Patients with lactase deficiency vary in symptomatology, but most typically experience distention, flatulence, abdominal cramps, and watery diarrhea after ingesting cow's milk or milk products.

Diagnosis

The diagnosis is suggested by relief of symptoms on a lactose-free diet and reappearance of symptoms on reintroduction of lactose. Definitive diagnosis may be made by a lactose breath hydrogen test or by biochemically assaying the lactase content of the small intestine mucosa obtained by perioral or endoscopic biopsy examination.

Treatment

Treatment consists of restricting milk and milk products or the use of lactose enzyme (lactase) supplements.

Abetalipoproteinemia (ABL)

Abetalipoproteinemia is a rare, autosomal recessive disorder with a frequency <1 in 100,000 people that impairs the normal absorption of fat and certain vitamins from the diet (Hussain, 2018).

Risk factors

The fundamental biochemical defect is a complete absence of plasma apolipoprotein (apo) beta-lipoproteins, specifically chylomicrons, very-low density lipoprotein (VLDL), and low-density lipoprotein (LDL). A deficiency of microsomal triglyceride transfer protein (MTP) activity has been suggested to be the central cause of abetaliproteinemia (ABL).

Symptoms

Abetalipoproteinemia is characterized by fat malabsorption, acanthocytosis, and hypocholesterolemia in infancy. Later in life, deficiency of fat-soluble vitamins is associated with development of atypical retinitis pigmentosa, coagulopathy, posterior column neuropathy, and myopathy.

Additional manifestations of abetalipoproteinemia may include:

- GI affectations, including diarrhea and fat-soluble vitamin deficiency.
- Hepatic symptoms of elevated serum transaminases with hepatomegaly and cirrhosis.
- Hematologic symptoms include acanthocytosis.
- Neurological changes of the central and peripheral nervous system and demyelination.
- Muscle involvement of the striated and smooth muscle and myositis.
- Ophthalmic changes of pigmentary retinal degeneration, loss of night vision, loss of color vision, and complete loss of vision.

Treatment

Standard treatment is dietary modification and replacement of fat-soluble vitamins. A low-fat diet improves the steatorrhea and allows absorption of other nutrients essential for growth and development. Additionally, high dose oral fat-soluble vitamin supplementation (high oral doses of combined vitamin A and E) introduced early in therapy may help clinical outcomes.

Chronic intestinal pseudo-obstruction

Chronic intestinal pseudo-obstruction (CIPO) is a rare disorder of GI motility. This clinical syndrome mimics the signs and symptoms of intestinal obstruction, but no mechanical obstruction is present on evaluation. CIPO is thought to be a form of dysmotility where the muscles and nerves that control peristalsis are affected, resulting in altered and inefficient movement of material through the GI tract.

Risk factors

CIPO may be related to a functional disturbance in the interstitial cells of Cajal. The condition may be congenital or may develop with age.

Symptoms

CIPO can manifest at any age. Symptoms may include nausea, vomiting, distention, abdominal pain, inability to have a bowel movement, and failure to thrive.

Diagnosis

Diagnostic evaluation includes fluoroscopic studies (X-ray, barium contrast studies); nuclear medicine studies; CT scan; gastric emptying study; antroduodenal, esophageal, and intestinal motility studies; and laparoscopic biopsy. A thyroid evaluation, as well as tests for collagen and vascular diseases, may be done. Additional testing may involve a deep biopsy of intestinal muscle.

Complications

Malnutrition, with impaired growth and development in children, may occur in the absence of prompt intervention. Patients may die within a few months to years.

Treatment

Treatment is generally supportive in nature. The mainstays include nutritional therapy and treatment for bacterial overgrowth with antibiotics.

- **Nutrition.** Enteral feedings may be given via nasogastric, nasojejunal, gastrostomy, or jejunostomy tube. However, many children with pseudo-obstruction tolerate enteral feeding only on an intermittent basis or not at all. In these children, parenteral nutrition is required to meet nutritional needs. Diet should be low in fat and fiber and administered in frequent small feedings.
- **Pharmacotherapy.** Use of motility agents may be effective in alleviating symptoms in some patients. Pharmacological agents may include metoclopramide (Reglan, Metozolv ODT) 10–20 mg TID–QID daily; domperidone (Motilium) 10–20 mg TID–QID daily; and octreotide (Sandostatin), a prokinetic. Teduglutide (Gattex), a recombinant analog of human glucagon-like peptide 2, may be used for adults dependent on parenteral support (10-15 mg. qid).

- **Decompression.** Gastric and/or intestinal decompression may be necessary to maintain comfort.
- **Surgery.** Palliative surgery in the form of decompression of the bowel and colectomy may be done.

Future progress with small intestine transplantation may provide an alternative treatment option.

Small intestine tumors

Benign or malignant neoplasms of the small intestine make up less than 5% of GI tumors. Benign tumors have a 0.16% incidence, and malignant tumors 0.1% (American Cancer Society, 2019). Small intestine tumors usually only occur in patients who are over the age of 50.

Risk factors

Small intestine neoplasms may be hamartomas, lymphomas, alpha heavy chain disease, primary or secondary carcinoma, or carcinoid tumors.

- **Hamartomas** occur in **Peutz-Jeghers syndrome** (PJS). PJS is characterized by multiple GI hamartomatous polyps, as well as an increased risk of various neoplasms and melanocytic macules of the lips, buccal mucosa, and digits. A consistent feature is jejunal polyp identification with intussusception and bleeding as usual symptoms. PJS is an inherited mendelian-dominant syndrome.
- **Lymphomas** of the GI tract may be primary, with or without involvement of the adjacent mesenteric nodes, or secondary, having spread from elsewhere. In primary small intestine lymphomas, patients present with abdominal pain or obstruction. In secondary lymphomas, non-alimentary complaints usually precede involvement of the gut.
- **Alpha heavy chain disease** is a proliferation of lymphoid tissue that involves the IgA secretory system, leading to the production of incomplete immunoglobulin molecules made up of IgA heavy chains. Clinical features of alpha heavy chain disease include severe malabsorption syndrome and frequently finger clubbing. Histopathological features include plasma cell proliferation in the small intestinal mucosa and in the nodes, followed by the development of overt malignant lymphoma in most cases.
- **Primary carcinomas** of the small intestine are most likely to occur in the duodenum or the jejunum and make up 30% to 50% of the small intestine tumors—virtually all are adenocarcinomas (Neugut et al., 2001). Predisposing factors are alcohol abuse, Crohn's disease, celiac disease, neurofibromatosis, and pre-existing adenoma. Symptoms include pain, vomiting, anorexia, malaise, and nausea. Diagnosis is by barium X-ray. Small intestine carcinomas are usually flat, stenosing, ulcerative, or polypoid.
- **Secondary carcinomas** of the small intestine are relatively common as a site of metastasis. They most often arise from the breast or bronchus or are metastatic malignant melanomas.

Carcinoid tumors are often found by chance during appendectomy and occur most frequently in the ileum and appendix. Carcinoid tumors that metastasize to the liver may be associated with carcinoid syndrome.

Symptoms

General presenting symptoms of small intestine tumors may include small intestine obstruction; abdominal pain and/or vomiting; GI bleeding; weight loss; distended, tympanic abdomen; and high-pitched bowel sounds. Stool may contain occult blood.

Diagnosis

Diagnostic tests may include blood laboratory studies, an upright plain X-ray of the abdomen, barium contrast X-rays, endoscopy and biopsy examination for lesions in the proximal duodenum, ultrasonography, and CT scan. Selective arteriography of the celiac axis and superior mesenteric artery may be done.

Video capsule endoscopy is a diagnostic tool that may reveal such tumor suggestive findings as a bleeding, protruding mass; disruption of mucosa; surface irregularity; white villi; and area of discoloration.

Treatment

Most small intestine tumors are treated surgically. Even secondary metastatic tumors may require palliative surgery to relieve obstruction or bleeding. Protocols for treatment of lymphomas may include surgical resection, in addition to radiotherapy and chemotherapy. Radiation and chemotherapy of other malignant small intestine tumors have been largely ineffective. In Peutz-Jeghers syndrome, multiple enterotomies are usually required over the lifetime of the patient to remove large polyps that may be responsible for severe colicky pain.

Lymphangiectasia

Lymphangiectasia of the small intestine in adults is a disorder with dilated intestinal lacteals, causing loss of lymph into the lumen of the small intestine. It results in hypoproteinemia, hypogammaglobulinemia, hypoalbuminemia, and a reduced number of circulating lymphocytes of lymphopenia. There is a resultant impairment in the absorption of chylomicrons and such fat-soluble vitamins as vitamin D.

Risk factors

Lymphangiectasia of the small intestine may occur as a primary congenital disorder or secondary to abdominal or retroperitoneal carcinoma, lymphoma, chronic pancreatitis, Crohn's disease, Whipple disease, celiac disease, or congestive heart failure.

Symptoms

Patients present with malabsorption, as well as lymphocytopenia with hypogammaglobulinemia and evidence of protein-losing enteropathy. There may be bilateral lower limb edema, sometimes with lymphedema, pleural effusion, and chylous ascites, and low serum proteins, particularly serum albumin. "Yellow nail syndrome" has been described in primary intestinal lymphangiectasia with dystrophic ridging and loss of nail lunula. Osteomalacia (marked softening of bone) associated with vitamin D deficiency has been noted, as well as tetany secondary to hypocalcemia.

GI symptoms may include diarrhea with steatorrhea, abdominal pain, distention, nausea and vomiting, and GI bleeding. There may be increased fecal loss of protein, manifested in increased fecal alpha-1-antitrypsin. Iron deficiency anemia may be at times associated with occult small intestinal bleeding.

Diagnosis

Diagnosis is based on jejunal biopsy that indicates dilated lymphatic lacteals. Endoscopy may reveal white opaque spots in the duodenum. Double-balloon enteroscopy has been useful in evaluating malabsorption, and a finding in some patients may include yellow and white submucosal plaques, most likely lymphangiectasis. Video capsule endoscopy and enteroscopy may reveal more extensive changes along the small intestine.

Measurement of GI clearance of alpha-1 antitrypsin and stool fat quantitation may be done. Prothrombin time, serum calcium, and carotene levels may be determined to assess the degree of malabsorption and nutritional deficiency. Magnetic resonance imaging (MRI) can evaluate intestinal, mesenteric, and thoracic duct changes. CT scan may be performed. Lymphangiography may be used in congenital cases.

Treatment

Treatment is directed toward the underlying disease that resulted in lymphangiectasia.

Therapy includes a low-fat diet with medium-chain triglyceride supplementation that leads to portal venous rather than lacteal uptake. Management of lower limb edema, usually with elastic bandages or stockings, is important.

Localized resection of involved areas of the small intestine may be done.

Other therapies include tranexamic acid (Cyklokapron, Lysteda), an antiplasmin, in an effort to normalize immunoglobulin and lymphocyte numbers. Octreotide (Sandostatin) has been useful, but the mechanism of action has yet to be determined.

CASE SITUATION

Kelsey O'Malley, age 43, was referred to a gastroenterologist after a 2-year history of diarrhea, abdominal distention with increased flatulence, stomach cramps, and fatigue. She has experienced two to four daily stools that are usually soft and bulky and a 15-pound weight loss. She describes herself as "tense and nervous" because of family problems. Her referring physician has treated her for irritable bowel syndrome (IBS) with antispasmodics and sedatives without improvement in her symptoms. She had upper and lower GI barium studies 18 months ago, which were normal. Her blood work shows she is anemic and hypoalbuminemic, but her stool is negative for occult blood.

Points to think about

1. Given Kelsey's symptoms, what might be possible medical diagnoses?
2. What factors point to a diagnosis of celiac disease?
3. Kelsey had normal upper and lower GI barium studies. Would further barium studies be helpful?
4. It would be necessary to obtain stool specimens for ova and parasites to rule out infectious or parasitic causes for Kelsey's symptoms. What should the gastroenterology nurse tell her about collecting stool specimens?
5. The gastroenterology nurse knows that small intestine biopsy examination of the distal duodenum and proximal jejunum is the most definitive test for celiac disease. It will show that the mucosal surface is flat with shortened villi and elongated crypts, with changes in surface epithelial cells from tall columnar to cuboidal. What can the gastroenterology nurse tell Kelsey to expect when she goes for her biopsy examination?
6. What is one nursing diagnosis the gastroenterology nurse might consider for a patient with celiac disease?
7. What might the gastroenterology nurse teach Kelsey about her follow-up care?

Suggested responses

1. Possible medical diagnoses in Kelsey's case might include the following:
   a. IBS might explain her symptoms, but weight loss, anemia, and hypoalbuminemia are not usually seen in IBS. Also, her primary physician has tried treating her for IBS without success.
   b. An inflammatory bowel disease could cause these symptoms, particularly Crohn's disease.
   c. Lymphoma of the small intestine can cause obstruction of lymphatic drainage necessary for fat absorption.

   d. Chronic pancreatitis, causing pancreatic insufficiency and inadequate amounts of lipase for fat digestion, might account for steatorrhea.
   e. Tropical sprue or some parasitic disorders can cause these symptoms, but Kelsey denies any travel history.
   f. Celiac disease with malabsorption in the proximal small intestine and intolerance of gluten could cause these symptoms, but her age may extend beyond the typical range.
2. Factors that point to a diagnosis of celiac disease include:
   a. Female sex.
   b. Presentation of features of malabsorption syndrome, including anemia, hypoalbuminemia, and steatorrhea.
   c. Age factor. Although celiac disease is often diagnosed by 2 years of age, it can occur at any time in the lifespan.
3. Because celiac disease affects the small intestine, an upper GI series with small intestine follow-through should be done. This may show excessive secretions and dilatation of mucosal folds in the proximal small intestine.
4. Regarding collection of stool specimens, Kelsey should receive the following instructions:
   a. Obtain stool specimens before a barium X-ray because barium interferes with microscopic examination.
   b. Use a clean, dry container.
   c. Do not use stool that has been in contact with toilet bowl water or urine.
   d. Take the specimen to the laboratory immediately to ensure accurate results.
   e. It is often necessary to test several specimens, for example, the first morning stool each day for 3 days.
5. Regarding her upcoming small intestine biopsy examination, Kelsey should be told that:
   a. The physician may try to get a large enough biopsy specimen by doing a standard esophagogastroduodenoscopy (EGD). Kelsey may have this done in the hospital endoscopy unit or in the office endoscopy setting. The gastroenterology nurse should explain this procedure to her, with reassurance that a gastroenterology nurse will be there to assist her through all phases of the procedure.
   b. Normally, a submucosal biopsy examination is done as an outpatient procedure. Therefore, Kelsey will not be admitted to the hospital. The gastroenterology nurse should instruct her to immediately report any severe abdominal pain or distention, nausea, vomiting, or bleeding after the procedure.
6. The gastroenterology nurse might consider the nursing diagnosis, "knowledge deficit related to necessary dietary modifications for celiac disease." In many instances, the gastroenterology nurse can begin intervention predicated on such a diagnosis. However, it is important to recognize that the act of acquiring and recalling new information does not necessarily mean that the learner will change his or her behavior as a consequence.
   a. The "knowledge deficit" diagnosis is broad and involves three learning domains: receiving and retaining new information, responding emotionally by personalizing the information, and changing behavior by alteration in neuromuscular pathways. There may be a deficit or deficits in any one or more of these learning dimensions, although most deficits occur in the cognitive or psychomotor areas.
   b. Other factors that might influence the patient's response to nursing interventions based on this diagnosis include cognitive function, lack of exposure to information, visual or auditory sensory deficits, pathophysiological state or conditions, state of motivation or readiness to learn, cultural or language barriers, or coping strategies (e.g., denial, anxiety).
7. Regarding her ongoing care, the gastroenterology nurse should explain the following to Kelsey:
   a. It will be necessary for her to make permanent, lifelong adjustments in her diet based on written gluten-free diet instructions, which the gastroenterology nurse will provide.
   b. Fecal fat studies and laboratory blood studies will be required at specified intervals as part of ongoing follow-up.
   c. Foods containing lactose may not be tolerated well because they may cause diarrhea and flatulence and often must be eliminated from the diet.
   d. Response to dietary changes can be dramatic. Within a few days or weeks, general appearance will improve, fatigue will lessen, stools will be more formed and less frequent, and abdominal distention will decrease.
   e. Lost weight is usually recovered within several months.
   f. The villous architecture in the small intestine should return to normal. A follow-up biopsy examination may be done to confirm improvement.
   g. There are several support groups for families with celiac disease.
   h. Many companies produce gluten-free products, and many restaurants offer gluten-free options.
   i. Check labeling on packaged food products for a gluten-free label. Wheat-free does not necessarily mean gluten-free. Check the allergen labeling also as well as ingredient listings (wheat, barley, rye, malt, Brewer's yeast, oats).

REVIEW TERMS

abetalipoproteinemia, ampulla of Vater, argentaffin cell, Auerbach's plexus, Brunner's glands, celiac disease, Crohn's disease, crypts of Lieberkühn, duodenum, enteric plexus, giardiasis, goblet cells, ileocecal valve, ileum, inflammatory bowel disease, intussusception, jejunum, lactase deficiency, lamina propria, ligament of Treitz, malabsorption, maldigestion, malnutrition, Meckel's diverticulum, Meissner's plexus, microvillus, Paneth's cells, peptic ulcers, Peutz-Jeghers syndrome, Peyer's patches, regional enteritis, short bowel syndrome, small intestine, steatorrhea, tropical sprue, villus, volvulus, Whipple disease

REVIEW QUESTIONS

1. The proximal two-fifths of the small intestine is known as the:
   a. Duodenum.
   b. Ileum.
   c. Jejunum.
   d. Cecum.
2. The mucous layer (mucosa) of the small intestine forms villi and is lined with:
   a. Simple epithelial cells.
   b. Columnar epithelium.
   c. Connective tissue.
   d. Peritoneum.
3. The principal function of Brunner's glands is to:
   a. Synthesize antibodies.
   b. Absorb nutrients.
   c. Secrete a viscous, alkaline fluid.
   d. Secrete hormones.
4. The primary site of absorption for vitamin B_{12} and bile acids is the:
   a. Duodenum.
   b. Jejunum.
   c. Ileum.
   d. Stomach.
5. The intestinal contents are constantly being stirred by:
   a. Active transport.
   b. Segmenting contractions.
   c. Peristalsis.
   d. Shortening and lengthening of the villi.
6. Segmental submucosal inflammation and a cobblestone appearance of the bowel wall are associated with:
   a. Ulcerative colitis.
   b. Crohn's disease.
   c. Duodenal ulcers.
   d. Celiac sprue.
7. The most effective treatment for symptomatic Meckel's diverticulum is:
   a. Diverticulectomy.
   b. Antibiotic therapy.
   c. Bowel resection.
   d. Dietary modification.
8. Most peptic ulcers occur in the:
   a. Stomach.
   b. Duodenum.
   c. Ileum.
   d. Jejunum.
9. A malabsorption syndrome characterized by gluten intolerance is:
   a. Cystic fibrosis.
   b. Whipple disease.
   c. Tropical sprue.
   d. Celiac disease.
10. Primary carcinomas of the proximal small intestine are virtually all:
    a. Lymphomas.
    b. Melanomas.
    c. Adenocarcinomas.
    d. Sarcomas.

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Chapter 16

LARGE INTESTINE

This chapter will acquaint the gastroenterology nurse with the normal anatomy and physiology of the large intestine. Selected diseases and disorders of the large intestine in terms of pathophysiology, diagnosis, and treatment are discussed.

Learning objectives

After reviewing the content of this chapter, the gastroenterology nurse will be able to:

1. Explain the normal macroanatomy and microanatomy of the large intestine.
2. Outline the physiology of absorption, secretion, and motility in the large intestine.
3. Discuss the pathophysiology, diagnosis, and treatment of the presented diseases and disorders of the large intestine.

ANATOMY AND PHYSIOLOGY

The large intestine is 120-150 cm (4-5 ft) long and approximately 4-6 cm (2 in) in diameter. It extends from the ileocecal valve to the anus. The large intestine is sometimes referred to as the large bowel or the colon. It is divided into the following eight sections: (Figure 16.1 and Figure 16.2)

- Cecum.
- Ascending colon.
- Hepatic flexure.
- Transverse colon.
- Splenic flexure.
- Descending colon.
- Sigmoid colon.
- Rectum.

The ascending and descending colon are retroperitoneal, while the transverse and sigmoid colon are intraperitoneal.

At the distal end of the small intestine, a flap valve known as the ileocecal valve acts as a sphincter both to control the passage of intestinal contents from the terminal ileum into the **colon** and prevent the reflux of bacteria from the colon back into the small bowel, preserving the relative sterility of the small bowel. In the adult, the resting pressure of the ileocecal valve is about 20 mmHg above the colonic pressure, and the length of the high-pressure zone is approximately 4 cm.

Sections of the large intestine include:

- **Cecum** is the first section of the large intestine to receive material from the small bowel. The cecum is a short blind sac that is wider in diameter than long, and to which the **vermiform appendix** is attached. The appendix is a thin, tubular structure about 8 mm in diameter and 2–20 cm in length, but is somewhat longer in children. As a vestigial structure, it has no known physiologic role.
- **Ascending colon** (right colon) is immediately above the cecum. The ascending colon passes up the right side of the abdomen to the lower border of the liver, where it bends to the left at the right colic flexure, also known as the **hepatic flexure**.
- **Transverse colon** is the portion of the bowel that crosses the abdomen from right to left, from the hepatic flexure to the left colic flexure, also known as the **splenic flexure**.
- **Descending colon** (left colon) runs down the left side of the abdomen from the spleen to the iliac crest, where it makes an s-curve at midline to form the **sigmoid colon**.The large intestine terminates at the rectum and anal canal, extending from the rectosigmoid junction at the third sacral vertebra (S3) level and 8-13 cm (4–6 inches) down to the anorectal line.

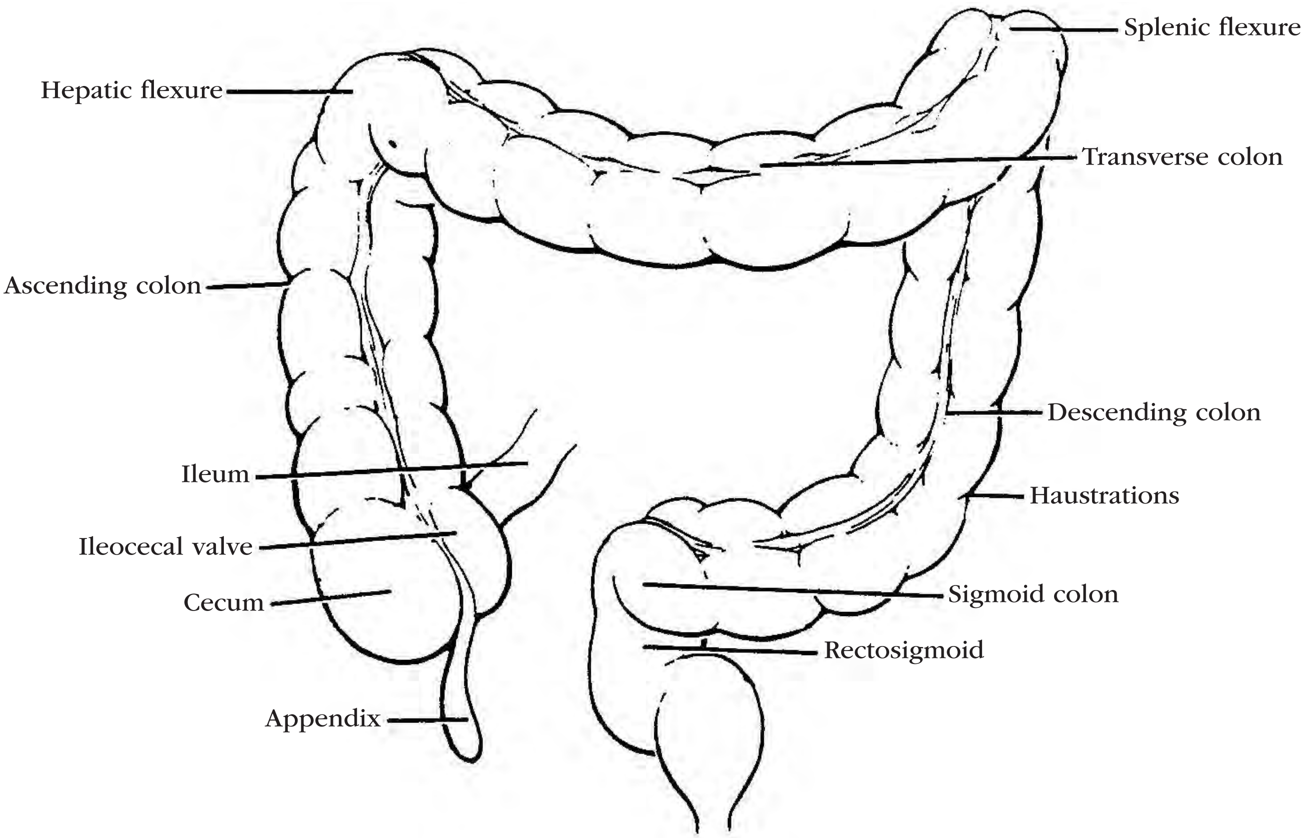

Figure 16.1. Large Bowel Anatomy

The last major portion of the large intestine is the **rectum**, which follows the curvature of the lower sacrum and coccyx. The rectum length is about 9 cm in infants, increasing to 15 cm in adults. The distal portion forms the anal canal. Movement of feces through the anal canal is controlled by an inch-wide internal sphincter composed of involuntary smooth muscle and an external sphincter made up of voluntary striated muscle. Levator ani muscles surrounding the rectum also help keep defecation under voluntary control. The external opening is called the **anus**.

The main functions of the large intestine are storage and movement of intestinal contents; absorption of water, some electrolytes, and bile acids; and, to a lesser extent, the excretion of mucus, potassium, and bicarbonate.

Layers of the large intestine

The wall of the large intestine has four layers:

- **Serosa.** The outer serous layer is formed by the visceral peritoneum found on the ascending, transverse, and sigmoid colon. The rectum does not have a serous layer.
- **Muscularis propria.** The muscularis propria includes an inner circular layer and an outer longitudinal layer that is composed of three heavy longitudinal bands called **taenia coli**. The taenia coli are shorter than the intestine, thus causing it to pucker and form small sacs called **haustra**. The size and shape of the haustra vary with the state of contraction of the circular and longitudinal muscle layers. Auerbach's nerve plexus is located between the circular and longitudinal muscle layers. A prominent extracellular matrix contains increased amounts of collagen and elastin than is present in the small bowel. The amount of collagen and elastin increases with advancing age.
- **Submucosa.** The submucosa is loose connective tissue that contains arteries, veins, lymphatic vessels, and the nerve fibers and ganglion cells that form Meissner's plexus. The submucosa is separated from the mucosa by the muscularis mucosae, a layer of smooth muscle cells.
- **Mucosa.** The mucosa is smooth-surfaced and arranged in folds called the **plicae semilunares coli**. As in the small intestine, the tubular crypts of Lieberkühn open into the lumen of the large intestine. Intestinal villi, however, are absent in the large bowel. The absorptive surface is flat and lined by columnar epithelial cells and a number of goblet cells. There are fewer microvilli on the colonic columnar cells than in the small intestine. The epithelium in the lower half of the crypts is composed of proliferative undifferentiated columnar cells, mucus-secreting goblet cells, and a few endocrine cells.

The mucous membrane that lines the rectum is arranged in longitudinal rows called rectal or anal columns. Each rectal column contains an artery and a vein.

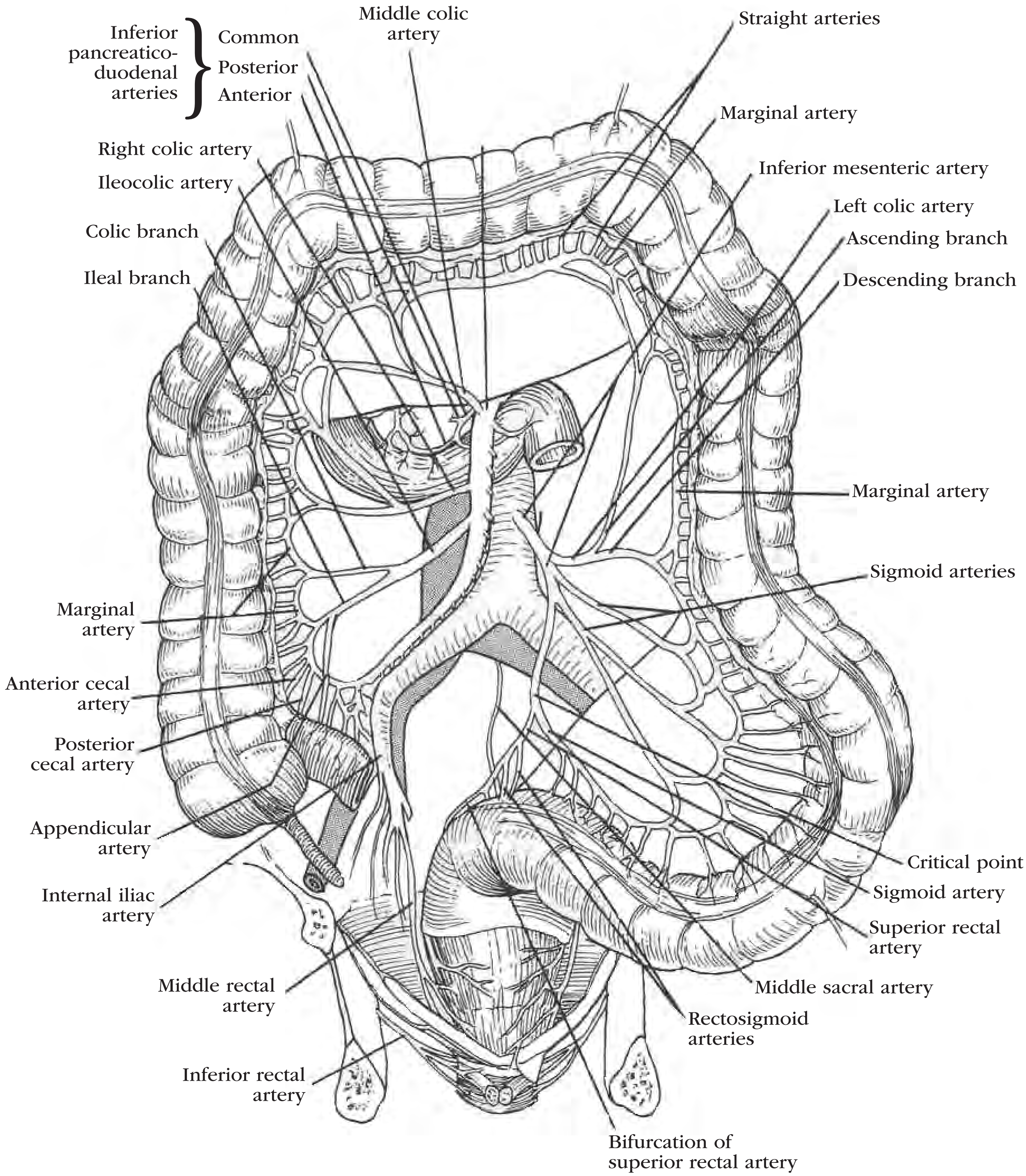

Figure 16.2. Arterial Blood Supply to the Large Intestine

From Principles of Ostomy Care by D. C. Broadwell and B. S. Jackson, 1981, St. Louis, MO: Mosby, Inc. Adapted with permission.

These rectal columns end about 2 cm from the anal orifice, where they join transverse tissue folds. At the anorectal line, also known as the mucocutaneous border or the dentate line, the colonic mucosa meets the external anal canal. The anorectal line also marks a change in nerve supply and venous drainage.

Vascular supply

Arterial

The right half of the large intestine receives arterial blood from the branches of the superior mesenteric artery, and the left or lower portion receives arterial blood from the branches of the inferior mesenteric artery.

The rectum and anal canal receive arterial blood from the hemorrhoidal artery, which is a branch of the inferior mesenteric artery. The rectum is also supplied by the branches of the hypogastric artery.

Venous

Venous blood from the colon is drained mainly through the inferior and superior mesenteric veins (Figure 16.2).

Above the anorectal line, venous blood drains upward to the superior hemorrhoidal veins and from there to the portal veins. Below the anorectal line, venous blood drains downward to the inferior hemorrhoidal veins.

Innervation

Colonic nerve supply lies close to the arterial vessels.

Parasympathetic innervation of the right portion of the colon is derived from the vagus nerve. The remaining portion of the colon receives parasympathetic innervation through branches of the sacral nerves. In general, stimulation of the parasympathetic nerves increases intestinal contraction and mucus secretion and inhibits the rectal sphincter.

Sympathetic innervation is through the spinal nerves. Stimulation of sympathetic fibers inhibits colonic motility and secretions and stimulates the rectal sphincter.

Secretion

Colonic secretion is scanty and consists primarily of water, mucus, potassium, and bicarbonate. The alkaline mucus secreted by goblet cells in the crypts lubricates the intestinal walls, protects the mucosa from acidic bacterial action, and helps lubricate the passage of stool. Chemical and mechanical irritation increases the secretion of the fluid.

Motility

The colon has three basic patterns of movement.

1. Periodic, uncoordinated tonic contractions or segmentations of both the longitudinal and circular muscles bunch up the folds of the mucosa, forming the haustra.
2. Phasic, random, nonpropulsive contractions last 30 seconds to 2 minutes and include peristalsis and retrograde peristalsis, which mix the stool material and help absorb its liquid contents without advancing the material toward the anus.
3. Spontaneous mass movements occur 3 or 4 times a day when the colon becomes filled and distended. This rapid caudal movement of material is much more evident in the colon than are the phasic or mixing movements. When mass movements of the sigmoid colon move feces into the rectum, the urge to defecate is stimulated, causing peristaltic waves in the rectum and relaxation of the internal and then the external anal sphincter.

If relaxation of the internal and external sphincters is not enough to provide for defecation, a Valsalva **maneuver** assists in the process. Through the Valsalva maneuver, the involuntary movement of the bowel wall and the relaxation of the external sphincter are assisted by contraction of the diaphragm and the thoracic and abdominal muscles. This strain and downward push lasts approximately 8 seconds, thus increasing the intra-abdominal pressure and frequently emptying the colon from as high as the splenic flexure.

Under normal conditions, gas moves down the colon at a rate of about 1-4 inches per second. Liquids are moved by gravity, peristalsis, and by mass movements, but solid materials move primarily by mass movements. Overall motility of the large intestine is slow compared to the small intestine, and it can take as long as 7 days for material to clear the colon.

Absorption and elimination

Of the 1,000-2,000 ml of liquid chyme that enters the colon daily, only 150-250 ml of fluid is evacuated in the stool. The colon removes 80% to 90% of the water, making it much more efficient than the small intestine. The mechanisms for water and ion transport are the same as in the small intestine (see Chapter 15). The epithelial lining of the colon is considered tight compared to the small intestine, allowing less back diffusion of ions and, therefore, more water absorption. The colon absorbs sodium, chloride, and water, with most absorption being accomplished in the ascending colon.

The colon also serves an excretory function as an elimination route for such metals as lead, mercury, bismuth, silver, and calcium.

Feces are normally brown in color because of the metabolism of bile pigments to stercobilin. Odor is the result of indole, skatole, mercaptan, hydrogen sulfide, and the breakdown products of cystine.

Indigenous bacteria

The colon is sterile at birth, but rapidly becomes colonized by swallowed bacteria. The bacterial composition is established primarily from maternal contamination within 1 week of birth and is maintained throughout the lifetime.

Normal feces contain intestinal bacteria that help break down and putrefy body wastes; trace fluids, such as water; mucus; undigested cellulose and connective tissue; and toxins from meat proteins, undigested fats, and bile pigments. An estimated 500 species of 100 trillion organisms live in the healthy colon.

The normal adult colon contains approximately 109-1,011 organisms per gm of stool, which consist primarily of both gram-negative anerobic cocci and aerobic bacterium.

- Gram-negative anaerobic cocci, such as *Bacteroides*, *Lactobacillus*, *Eubacterium*, *Bifidobacterium*, *Veillonella*, and *Fusobacterium*.
- Aerobic bacterium, such as *Escherichia*, *Enterococcus*, *Streptococcus*, *Bacillus*, *Citrobacter*, and *Klebsiella*.

The positive effects of these resident bacteria include breaking down and fermenting cellulose and other waste material, deconjugating and transferring bile acids to extrahepatic circulation, synthesizing vitamin K, controlling the overgrowth of harmful bacteria, thickening the mucosal lining of the bowel, and destroying pancreatic enzymes that are harmful to the skin and other structures. Products of carbohydrate fermentation are absorbed and make up 5% to 10% of absorbed energy.

Indigenous bacteria also have a number of potentially harmful effects, including the manufacture of gas; the production of the toxins or chemicals that produce colitis or diarrhea; the invasion of the bowel wall, producing infectious colitis; and the production of ammonia, which can cause hepatic encephalopathy in patients with hepatic cirrhosis.

DISORDERS OF THE LARGE INTESTINE

Among the diseases and disorders that affect the large intestine are polyps; angiodysplasia; colitis, also known as inflammatory bowel disease (IBD), including ulcerative colitis (UC) and Crohn's colitis; irritable bowel syndrome (IBS); chronic and recurrent abdominal pain syndromes; parasitic infestations; diverticular disease; colorectal cancer; intestinal obstructions; and anorectal disorders.

Polyps

Colonic **polyps** are discrete tissue masses found in the lumen of the bowel. Some polyps may be flat and not easily visualized. Polyps usually develop in the rectum, sigmoid, and descending colon with a greater incidence in the ascending colon or right side. Polyps may be pedunculated or sessile:

- Pedunculated polyps are attached to the intestinal wall by a stem.
- Sessile polyps are broad-based and attach directly to the intestinal wall.

Risk factors

Familial polyposis coli, Gardner syndrome, and Peutz-Jeghers syndrome are the most common polyposis syndromes. Polyposis syndromes include:

- **Familial polyposis coli**. Also called familial adenomatous polyposis (FAP), familial polyposis coli is inherited as an autosomal dominant trait. FAP occurs from germline mutations of the adenomatous polyposis coli (*APC*) gene. Incidence varies from 1 in 7,000 to 1 in 22,000 (NIH, 2019). There is no geographic or ethnic variation. The average age for FAP to appear in children is during their mid-teens. The disorder results in hundreds of adenomatous polyps that develop throughout the colon. FAP is diagnosed by digital rectal examination, colonoscopy or proctosigmoidoscopy, and barium enema. Patients may also present with nonspecific symptoms, such as **hematochezia** (passage of bloody stools), **diarrhea**, abdominal pain, and intestinal obstruction. Because these polyps may become cancerous, surgery is strongly recommended—most likely ileal pouch anal anastomosis because of the lower incidence of pouchitis or total colectomy with ileorectal anastomosis.
- **Gardner syndrome**. Like familial polyposis coli, Gardner syndrome is hereditary. In Gardner syndrome, multiple adenomatous polyps appear along the colon or other parts of the gastrointestinal (GI) tract, and osteomas appear on the mandible, skull, and long bones. Symptoms of both diseases are similar and thought to be variable expressions of a single disease. Polyps associated with Gardner syndrome tend to become cancerous in 10-15 years, so surgical treatment is recommended.
- **Peutz-Jeghers syndrome (PJS)**. PJS is inherited as an autosomal dominant trait caused by mutations in the *STK11/LKB1* gene. Peutz-Jeghers syndrome results in adenomatous or hamartomatous polyps of the GI tract, but polypoid lesions have also been noted outside the GI tract. Abnormal brown pigmentation of the lips, oral mucosa, and skin may also occur. The freckle-like pigmentation of the lips appears in childhood, during the first decade of life, and tends to fade with age. Pigmentation of the buccal mucosa is a more reliable indicator in older individuals and may appear earlier in infancy than cutaneous lesions.

Rectal bleeding and abdominal pain secondary to polyps appear during the second decade of life. Abdominal pain caused by mechanical blockage or intussusception is the most common symptom. The average age of diagnosis age 23 for males and age 26 for females. The extent of involvement should be documented via upper and lower endoscopy and small bowel capsule endoscopy every 2 years throughout life.

Peutz-Jeghers syndrome has been associated with precocious puberty in girls, feminizing gonadal tumors in boys (PJS should be suspected in males with prominent gynecomastia), hepatic cysts, and thyroid papillary adenocarcinoma. Most individuals do well over many decades, although there is a definite increased risk of developing GI, pancreatic, breast, ovarian, and cervical cancer.

Juvenile polyps. Nearly all polyps found in children are benign. These are referred to as juvenile polyps, also called retention or inflammatory polyps. They are not associated with a genetic disorder, but are related to an inflammatory process. These types of polyps are relatively common, and most occur before age 10, peaking between ages 2 and 5. They are rarely seen in older children. The child presents with painless bleeding and with less than five polyps revealed on colonoscopy. Polyps may be larger than 2 cm.

Symptoms

Most polyps are asymptomatic, but some patients experience bleeding.

Diagnosis

Diagnosis is by colonoscopy or proctosigmoidoscopy and air-contrast barium enema.

Complications

Complications including bleeding, obstruction, diarrhea, and colon cancer.

Treatment

Research shows that removal of polyps dramatically reduces the incidence of subsequent colon cancer.

Angiodysplasia

Angiodysplasias (arteriovenous malformations) are vascular dilatations in the submucosa that consist of arterial, venous, and capillary elements. They are often multiple and occur most frequently in the cecum and right colon but may be found throughout the GI tract.

Risk factors

Most angiodysplasias are associated with normal aging, but their etiology remains uncertain. In the hereditary autosomal-dominant disorder Osler-Weber-Rendu disease, also known as hereditary hemorrhagic telangiectasia, angiodysplasias occur throughout the GI tract, on the skin, in the nail beds, and in the mucosa of the mouth and nasopharynx.

Individuals over the age of 60, with no history of GI bleeding, may have colonic vascular **ectasia**, the type of angiodysplasia found in the colon.

Symptoms

Some people with angiodyslasias are asymptomatic. In symptomatic individuals, the usual presentation is acute lower GI bleeding, which may be attributed to another coexisting disorder. It is not uncommon for a person to experience several bleeding episodes before the proper diagnosis is made.

Diagnosis

Diagnosis of angiodysplasia is by colonoscopy or angiography. Selective mesenteric angiography during active bleeding is the most accurate differentiating diagnostic technique when determining origination of bleeding, such as diverticula versus ectasia.

Complications

Bleeding from vascular ectasia may range from occult to torrential.

Treatment

Asymptomatic individuals require no treatment. The treatment of choice for symptomatic angiodysplasias is endoscopic treatment with bipolar electrocoagulation. Argon plasma coagulation (APC) is an alternate option, depending on the area to be treated.

If electrocoagulation is not successful, resection of the bowel segment containing the lesion may be performed. Bleeding can recur in patients who are treated surgically.

COLITIS

Colitis is broadly defined as an inflammation of the colon. Several different forms of this disorder are recognized, including ischemic colitis, necrotizing **enterocolitis**, ulcerative colitis, pseudomembranous colitis, Crohn's colitis, radiation enteritis, and food protein-induced enterocolitis.

Ischemic colitis

Ischemic colitis is an acute vascular insufficiency of the colon. It may occur in any portion of the small or large bowel, but usually affects the descending colon (most often in the region of the splenic flexure) or the sigmoid colon—portions of the colon that are supplied by the inferior mesenteric artery.

Risk factors

Ischemic colitis may be either occlusive or nonocclusive.

- Occlusive ischemic disease represents a mechanical obstruction in the blood vessels that supply the bowel. It can often be traced to emboli from an artificial mitral valve; a fibrillating atrium; or surgical bypass, especially vascular surgery involving the distal aorta.
- Nonocclusive ischemia is usually related to a severe derangement in the central circulation consequent to such conditions as congestive heart failure, myocardial infarction, major hemorrhage, sepsis, or cardiac arrhythmia. Hypotensive agents may also cause intestinal ischemia, as may drugs causing arterial or venous thrombosis or intense arterial constriction.

Symptoms

Symptoms of ischemic colitis include abrupt onset of pain at the left iliac fossa, bloody diarrhea, low-grade fever, abdominal distention, and minor abdominal tenderness.

Diagnosis

The classic radiological sign is thumbprinting in the affected area of the bowel, resulting from localized elevation of the mucosa by submucosal hemorrhage or edema. Cautious use of sigmoidoscopy may reveal nonspecific colitis, ulcerations, and/or bluish soft nodules. Histology may reveal characteristic apoptosis (programmed cell death).

Complications

Bowel infarction and peritonitis require emergent attention. Ischemic strictures may occur during healing.

Treatment

Most patients recover spontaneously, although a minority develop peritoneal signs and absent bowel sounds, signifying bowel infarction and peritonitis. Such patients require surgical intervention consisting of segmental resection and either primary anastomosis or temporary colostomy.

Follow-up barium enema X-rays or colonoscopy should be done 6-8 weeks after recovery to rule out ischemic strictures.

Necrotizing enterocolitis

Necrotizing enterocolitis (NEC) is a disease of focal or diffuse ulceration of the GI tract that usually occurs in the distal small bowel and/or colon.

Risk factors

It occurs almost exclusively in the neonatal period. No single etiological factor has been found to be responsible for NEC. Three factors acting either individually or in concert have been associated with occurrence. These are:

- Formula feeding as opposed to breast-feeding.
- Intestinal ischemia.
- Mucosal or systemic invasion of enteric bacteria.

Immaturity due to prematurity or low birth weight, including an inefficient intestinal barrier, has also been implicated as a risk factor. An undeveloped mucosal barrier allows bacterial invasion into the intestinal wall. Sepsis and hypoxia may also contribute to the condition.

Symptoms

Presentation varies from insidious to acute and fulminant. The classic signs of NEC include abdominal distention, bilious vomiting, and bloody stools. Other systemic signs and symptoms of illness may include hypothermia, apnea, and tachycardia or bradycardia.

Diagnosis

The diagnosis of NEC is confirmed radiologically by the presence of pneumatosis intestinalis (air within the intestinal wall or air in the hepatic portal venous system) and dilated bowel loops. Colonoscopy is contraindicated due to the risk of progression to perforation as a result of insufflating more air into the colon.

Complications

Infection may result from perforation.

Treatment

The management of a child with NEC is largely supportive. Oral feedings are withheld, and a nasogastric tube placed to suction is used. Intravenous (IV) nutrition and broad-spectrum parenteral antibiotic therapy are provided. Surgery involving conservative bowel resection of gangrenous bowel may be necessary in cases of perforation.

Ulcerative colitis

Ulcerative colitis is chronic, recurrent inflammation that affects the mucosa and submucosa of the large intestine. Inflammation often originates in the lower colon and then spreads proximally. Ulcerative colitis and Crohn's disease are often described together as inflammatory bowel disease. (For information on Crohn's disease, see Chapter 15.)

Signs and symptoms of ulcerative colitis and Crohn's disease are similar. Ulcerative colitis is usually continuous from the rectum and is limited to the large intestine. Crohn's disease, however, typically affects all layers of the bowel wall, not just the mucosa. Crohn's disease is often segmental and most often involves the small intestine, particularly the terminal ileum, but may affect any area of the GI tract from the mouth to anus.

Table 16.1. Differentiation of Ulcerative Colitis and Crohn's Disease

Ulcerative colitis	Crohn's disease
Continuous inflammation	"Skip areas" or segmental areas of ulceration with normal tissue between ulcerations
Mucosal ulceration only	Ulcerations typically affect all layers of the bowel wall
Most often seen in the left colon and rectosigmoid areas, but at times may involve the entire colon	Usually seen in the right colon, involving the terminal ileum, but may involve any area of the GI tract from the mouth to the anus
Characterized by exacerbations and remissions	Often slow and progressive
May cause a shortening effect on the bowel	Can narrow the lumen with stricture formation
Inconsistent mucosal cobblestoning effect with no patches of healthy tissue visualized in the diseased section	Consistent mucosal cobblestoning effect with intermittent pattern of healthy and diseased tissue visualized
Diarrhea is bloody and mucopurulent	Diarrhea is watery and sometimes associated with steatorrhea
Pseudopolyps are common	Pseudopolyps are rare
Inflammatory masses are rare	Inflammatory masses are common

Risk factors

The etiology of ulcerative colitis is unknown. The incidence and prevalence have drastically and rapidly grown over the last 20 years. Environmental and genetic factors create altered immune response that result in inflammation of the bowel mucosa. Alterations in the T-cell and cytokine responses occur.

Symptoms

The colon mucosa develops diffuse ulceration and hemorrhage with congestion, edema, and exudative inflammation in the lamina propria and submucosa. Crypt abscesses form and become necrotic, leading to bloody, mucoid stools. Periods of remission often alternate with exacerbations. As crypt abscesses heal, granulation tissue forms, causing the colon to narrow, shorten, and lose its haustra.

Symptoms of ulcerative colitis may begin abruptly. They commonly include:

- Diarrhea, the cardinal symptom, which occurs most of the time. It is usually, but not always, associated with blood in the stool. The blood may be seen on the surface of the stool only or mixed in with the stool, depending on extent of inflammation. Onset may be sudden or may manifest as a gradual change in bowel habits.
- Urgency, which may accompany bloody stools.
- Pain in either the lower quadrant or **tenesmus** (painful rectal cramping), if ulcerations are in to the rectum.
- Fever, malaise, and weight loss with malnutrition, if all or most of the colon is involved.
- Anemia.
- Constipation due to rectal spasm that interferes with passage of stool, particularly in elderly patients.

Atypically, the skin, joints, eyes, and liver may also be affected. For example, the patient may experience a skin rash or inflamed joints.

Pediatric patients present with moderate signs of systemic illness such as abdominal pain, diarrhea, hematochezia, cholangitis, and growth failure.

Diagnosis

Diagnosis is made by:

- **History and physical.** The most important part of the clinical evaluation is the history and physical examination. A good initial guide to ascertaining disease severity is the intensity and frequency of diarrhea. Severe disease is generally identified by more than six daily bowel movements.
- **Diagnostic exams.**
  - Colonoscopy is usually performed to evaluate the extent of the disease. The disease may be active from the rectum to the cecum. Findings consistent with UC include granularity and increased friability of the mucosa, ulceration, and pseudopolyps. Biopsies are done for histological verification of the disease process.
  - Computerized tomography (CT) scan reveals thickened bowel.
  - Barium enema may be ordered, but is contraindicated in patients with very active or fulminant colitis. Findings are spicules of mucosa, strictures, and loss of haustral markings.
- **Laboratory tests** serve to confirm clinical findings of severe disease. Other blood studies may be ordered in addition to:
  - CBC, with the results of most aid being hemoglobin, leukocyte count, and erythrocyte sedimentation rate.
  - Serologic markers are helpful, revealing an elevation of perinuclear antineutrophil cytoplasmic antibody (pANCA) and anti-Saccharomyces cerevisiae antibody (ASCA). A combination of positive pANCA and negative ASCA are indicative of a positive predictive value.
  - An additional sign of extensive disease is hypoalbuminemia. Severe diarrhea may cause such electrolyte imbalances as hypokalemia.
  - Stool analysis, stool cultures, and stool for ova and parasites (O&P) may be ordered.

Complications

Following are common complications of UC.

- **Potential minor local complications** include pseudopolyps, hemorrhoids, anal fissures, perianal or ischiorectal abscess, and rectal prolapse.
- **More serious local complications** may include massive colonic hemorrhage, colonic strictures, colonic perforation, and perianal disease.
- **Pouchitis or inflammation of an ileal pouch** after surgery for UC may occur, with symptoms similar to those seen with UC. Treatment with antibiotics and anti-inflammatory agents may be required.
- **Arthritis**, a frequent extraintestinal manifestation in patients with UC, correlates with disease activity.
- **Sacroiliitis and ankylosing spondylitis**, shown to be higher in patients with ulcerative colitis (Chan, et al, 2018), are defined as inflammation of one or more vertebrae and the sacroiliac joint. These conditions are rare and occur in patients with histocompatibility marker HLA-B27. These conditions follow a course independent of intestinal disease.
- **Liver complications** of minor elevations of serum transaminase or alkaline phosphatase levels are not uncommon in UC and may be secondary to hepatic steatosis. The major chronic liver disease associated with UC is sclerosing cholangitis, which is generally progressive and independent of intestinal disease. This may occur before UC develops and is a part of an autoimmune process. It may progress to liver failure.
- **Ophthalmological complications** such as uveitis (inflammation of the vascular middle layer of the eye), episcleritis (reddened eye from inflammation between the conjunctiva and the sclera), and conjunctivitis may occur in children with UC.

- **Hematological abnormalities** may occur, including thromboembolic disease.
- **Dermatological manifestations** such as erythema nodosum and pyoderma gangrenosum (a classic manifestation of extraintestinal activity) may occur.
- **Renal disease** may also occur.
- **Osteoporosis** has been reported at increased incidence in patients with UC.
- **Toxic megacolon** is an acute dilatation of the diseased colon (with colonic diameter of greater than 6-7 cm) that is associated with systemic toxicity. It is the most severe, life-threatening complication of UC. Mortality rates are higher in patients with UC perforation but may also be associated with any severe inflammatory conditions of the bowel, including Crohn's disease, bacterial colitis, and amebiasis. In some patients with UC, toxic megacolon represents the first attack, but it is relatively rare in young patients.

Prominent symptoms of toxic megacolon are fever, elevated heart rate, abdominal pain, vomiting, dehydration, mental changes, electrolyte disturbance, hypotension, and fatigue. Patients may experience a progressive decrease in the frequency of stools, as well as markedly hypoactive or absent bowel sounds resulting from loss of colonic propulsive activity. Leukocytosis, anemia, hypomagnesemia, and hypoalbuminemia are common laboratory findings. Clinical symptoms such as fever and abdominal tenderness may be masked by high-dose corticosteroid therapy or immunosuppressive agents. Plain X-ray films of the abdomen will reveal dilatation of the entire colon or only the transverse colon. Catastrophic complications of toxic megacolon include colonic perforation, gram-negative sepsis, and massive hemorrhage.

The patients who are most susceptible to toxic megacolon are those with pancolitis. They are more apt to develop the condition early in the course of the disease, most often with the first episodes of colitis. Early recognition of the condition with prompt intervention is key to a positive outcome.

- **Adenocarcinoma of the colon** is an increased risk for patients with UC. The risk of cancer is directly related to the extent of colon involvement and the duration of disease. The risk of developing cancer becomes greater 8-10 years after diagnosis and increases with time. The cancers that develop are largely submucosal. For surveillance purposes, it is recommended that colonoscopy be performed yearly after 10 years of disease identification. If no dysplasia or adenoma is found on the initial colonoscopy, monitoring should be done every 2 years. Specimens for mucosal biopsy are taken every 5-10 cm. Ten areas should be sampled.

Treatment

Dietary and drug therapy may be ordered to reduce inflammation, maintain fluid and electrolyte balance, and relieve symptoms. The ultimate goal is to enable patients to return to a normal, active lifestyle. Treatment methods may include:

- In adults, avoidance of foods that appear to aggravate symptoms.
- In children, nutritional therapy to correct nutritional deficiencies and provide supplemental calories, optimal healing, and growth requirements.
- In those with strictures, fiber restriction.
- In those who are lactose intolerant, lactose avoidance.
- Drug therapy includes:
  - Anti-inflammatories: sulfasalazine (Azulfidine), mesalamine (Asacol HD, Pentasa, Rowasa, Apriso, Delzicol, Lialda).
  - Corticosteroids: Budesonide (Entocort EC, Uceris). Other steroids such as prednisone are used for acute exacerbation or flares.
  - Immunosuppressive agents, also called tumor necrosis factor (TNF) inhibitors: 6-mercaptopurine, azathioprine (Imuran), methotrexate (Trexall), cyclophosphamide (Cytoxan).
  - Biologicals: infliximab (Remicaide), adalimumab (Amjevita, Cytezo, Humira), vedolizumab (Entyvio), golimumab (Simponi).
- Bedrest, possibly in the hospital, for an acute exacerbation.
- Surgery is required in 20% of patients during the course of the disease (Kühn & Klar, 2015). Surgery includes treatment for conditions such as exsanguinating hemorrhage, perforation, documented or strongly suspected carcinoma, severe colitis with or without **toxic megacolon**, unresponsiveness to conventional maximal medication therapy, intractable symptoms, or intolerable steroid side effects. If surgery is required, the surgeon may perform a total proctocolectomy with a permanent ileostomy (Brooke ileostomy), a one-stage procedure.

Growth failure in children is considered a form of intractability of the disease. In such cases, therapy should be initiated prior to epiphyseal closing that occurs during puberty. Colectomy is a curative procedure. Emergency colectomy may be required in toxic megacolon or in a severe fulminating attack of toxic megacolon.

Treatment of toxic megacolon includes nothing by mouth (NPO) status; nasogastric suction; and IV fluids, electrolytes, steroids, and broad-spectrum antibiotics. A Miller-Abbott tube or Cantor tube may provide more therapeutic results. If toxic megacolon does not improve within 24-48 hours, surgical intervention is indicated. Signs and symptoms of deterioration include rebound tenderness, increasing abdominal girth, and cardiovascular collapse. The most common procedure is colon resection with ileostomy placement. There may be a two-stage approach, but a proctocolectomy is the usual practice.

Pseudomembranous colitis

Pseudomembranous colitis is defined as an acute inflammation of the bowel mucosa with the formation of pseudomembranous plaques overlying an area of superficial ulceration. It is also called **pseudomembranous enterocolitis**, enteritis, antibiotic-associated colitis or diarrhea, or *Clostridium difficile (C. difficile)*-associated colitis or diarrhea.

Community-acquired *C. difficile*-associated diarrhea is reported at an estimated 450,000 cases reported; 12.1% is reported among hospitalized patients (CDC, 2018). Any disturbance in the microflora ecosystem within the colon can account for the increased incidence among hospitalized patients.

Risk factors

The great majority of cases of pseudomembranous colitis involve a toxin or toxins produced by *C. difficile*, which is a spore-forming, gram-negative, anaerobic rod capable of producing many toxic factors. These include toxin A (an enterotoxin) and toxin B (a cytotoxin).

Pseudomembranous colitis caused by other pathogens is rare but has been attributed to organisms such as *Salmonella*, *Clostridium perfringens*, *Plesiomonas shigelloides*, and other clostridia. *Staphylococcus aureus* (Staph) has also been identified as a possible etiologic agent.

Pseudomembranous colitis is commonly associated with oral or IV antibiotic therapy and less often with chemotherapeutic agents. It should be suspected in any patient who develops diarrhea during or within 10 weeks of antibiotic therapy.

Symptoms

Symptoms of pseudomembranous colitis may range from mild diarrhea to fulminant colitis and include nonbloody, watery diarrhea; crampy abdominal pain; fever; and leukocytosis.

Diagnosis

The preferred method of diagnosis of pseudomembranous colitis is endoscopy to detect typical red, edematous, fragile mucosa with yellow-white raised plaques that may extend from the rectosigmoid area to the proximal colon.

Sigmoidoscopy is usually adequate, but up to one-third of patients have lesions restricted to the right colon, thus necessitating colonoscopy. Biopsies can identify pseudomembranes too small to visualize endoscopically.

The most useful diagnostic test is the demonstration of toxins A and B in the stool, using either enzyme-linked immunosorbent assay (ELISA) or counterimmunoelectrophoresis (CIE).

Complications

Complications may include toxic megacolon, pneumoperitoneum, severe electrolyte imbalance, hypoalbuminemia, and acute migratory polyarthritis.

Treatment

Pseudomembranous colitis is usually self-limited, although diarrhea can last for 4-6 weeks. One-third of patients require hospitalization.

Pseudomenbranous colitis is treated by discontinuing antibiotics and administering:

- An agent that binds the *C. difficile* toxin such as cholestyramine or colestipol (Colestid); or
- An agent that eradicates the organism altogether such as metronidazole (Flagyl) for 10 days, oral vancomycin (Vancocin) for 10 days (if intolerant to Flagyl), bacitracin (Baciim), fidaxomicin (Dificid), or bezlotoxumab (Zinplava).

Cholestyramine binds with vancomycin, so the two are not given concurrently.

C. difficile may recur in approximately 20% of patients following 7-14 days of therapy (Eyre et al, 2012). Probiotics with *Saccharomyces boulardii* or *Lactobacillus rhamnosus* (GG strain) may be used to prevent reinfection.

The current treatment for intractable *C. difficile*-associated diarrhea is fecal transplant therapy, also called fecal microbiota therapy or fecal microbiota transplantation, in which healthy donor stool is used to restore the flora of the colon that is resistant to *C. difficile*.

Crohn's colitis

Crohn's disease confined to the colon may be referred to as **Crohn's colitis**. However, the disease can extend anywhere from the mouth to the anus. (For information on Crohn's disease, see Chapter 15.)

Risk factors

Risk factors are discussed in Chapter 15: Small Intestine.

Symptoms

Clinically, Crohn's colitis may resemble UC, but the ulceration is often longitudinal and involves the full thickness of the bowel. The disease is often segmental.

Diagnosis

Crohn's colitis is most often diagnosed by colonoscopy.

Complications

Stricture formation is common, and fistulas are a frequent complication, especially in the perineum.

Treatment

Treatment options are discussed in Chapter 15: Small Intestine.

Radiation enteritis

Radiation enteritis is inflammation or damage of the intestinal lining usually caused by radiation therapy. Many patients experience mild, acute symptoms during treatment, but some experience severe, chronic damage.

Risk factors

When radiation injury occurs to the GI tract, it is usually a result of radiotherapy for pelvic, intra-abdominal, or retroperitoneal malignancies such as carcinoma of the cervix, prostate, bladder, testicle, or ovary.

Symptoms

The acute phase of radiation enteritis occurs during treatment and is related to alterations in epithelial cell function. It is characterized by nausea, cramping, and altered bowel habits. If the colon is involved, there may be acute proctocolitis with diarrhea, tenesmus, and rectal bleeding.

The chronic phase of radiation enteritis results from abnormalities in vascular and connective tissues. Symptoms may appear 3 months to 30 years after radiation treatment. When the colon and rectum are involved, proctocolitis, bleeding, strictures, perforation, and fistula formation may result.

Diagnosis

Valuable diagnostic studies include upper GI contrast study with small intestine follow-through, lactose breath hydrogen test, fecal fat collection, and small intestine biopsy.

Complications

Chronic enteritis is often an intractable and progressive disease associated with significant morbidity and mortality.

Treatment

In the acute phase of radiation enteritis, the extent of injury can be controlled by altering the size and timing of radiation dosage and by using antiemetics, bulk-forming agents, and antidiarrheal agents.

Medical management for radiation enteritis consists of administration of steroid enemas, sulfasalazine (Azulfidine), and transfusions for chronic bleeding.

Strictures can be dilated manually or by using a colonoscope.

Radiation proctitis is treated with argon plasma coagulation. Most times it takes several return treatments for effective control of bleeding.

Surgery may be used selectively in patients with complete obstruction, uncontrolled bleeding, perforation, or fistulas. Complications and disease progression following surgery are common. Resection is recommended for localized disease, but bypass is preferred for more extensive disease.

Food protein-induced enterocolitis

Food protein-induced enterocolitis, sometimes referred to as cow's milk protein-induced enterocolitis, may present soon after birth or up to 6 months of age.

Risk factors

The exact mechanism of the hypersensitivity reaction to cow's milk and other food proteins is not known, and there is a wide spectrum of clinical presentation. This syndrome has been found to occur even in breastfed infants who are sensitized to cow's milk protein through the mother's diet. Reactivity to soy proteins may occur in as many as 40% of those with cow's milk protein-induced enterocolitis (Nowak et al., 2017).

Symptoms

Classic manifestations include vomiting, diarrhea, poor weight gain, hematochezia, and irritability.

Diagnosis

Diagnosis is made by withdrawal and challenge with the offending protein. Supportive evidence can be gained by small intestine and/or rectosigmoid biopsies revealing increased plasma cell content of the lamina propria, along with eosinophilia.

Treatment

Treatment of food protein-induced enterocolitis may involve simply switching to a hypoallergenic formula or, if severe, total parenteral nutrition with bowel rest, followed by slow reintroduction of feedings via a nasogastric tube using a synthetic elemental formula.

Irritable bowel syndrome

Irritable bowel syndrome (IBS), characterized by motility disorders, is the most common GI disorder in the United States and the most frequent reason for outpatient visits to gastroenterologists at about 3 million office visits per year (International Foundation for Gastrointestinal Disorders, 2016). It ranks close to the common cold as a frequent cause of work absenteeism.

Risk factors

The anatomical cause of IBS is unknown. The cause is thought to be disturbance in the autonomic and enteric nervous systems of the gut and the gut-brain axis. The results are abnormal motility and visceral hypersensitivity, and it has been suggested that these abnormalities are secondary to psychological disturbances. Additionally, evidence regarding the role of immune activation in the etiology of IBS continues to increase (Saha, 2014).

IBS is more common in Jewish and white populations, but the female predominance that is found in adults does not exist in children. In 75% of children with IBS, at least one parent or sibling has a functional GI disorder. In western countries, 6% of middle school and 14% of high school children are found to experience this condition (Zhu et al., 2014).

Symptoms

Symptoms for IBS include altered bowel habits and abdominal pain that is directly related to a change in the pattern of bowel movements. Patients with IBS present with complaints of lower abdominal pain, distention, and constipation and/or diarrhea. With constipation, stools may be described as pebble-like.

Symptoms vary in pattern and intensity. Additional symptoms include increased straining, urgency, feeling of incomplete evacuation, appearance of mucus, and change in the caliber or appearance of stool.

Diagnosis

Diagnosis is by identification of the characteristic symptom complex of IBS and by careful exclusion of other organic GI disorders.

Complications

Although it is associated with severe discomfort, IBS is not related to other chronic or life-threatening conditions or with decreased longevity.

Treatment

The usual treatment for IBS involves the following regimen:

- Sympathetic diagnosis and reassurance that organic causes have been ruled out, including thyroid disease, lactose intolerance, and fructose intolerance.
- A high-fiber diet at 25-35 grams daily.
- Anticholinergic agents to decrease abdominal pain such as diphenoxylate/atropine sulfate (Lomotil) and dicyclomine (Bentyl).
- Antidiarrheals such as loperamide (Imodium A-D).
- Laxatives.
- Psychological support and drug therapy, which may include antidepressants aimed at decreasing anxiety that may be related to stress such as desipramine (Norpramin) or amitriptyline.
- Addition of probiotics containing *Bifidobacterium infantis* such as Bifantis, Align, Florastor, Phillips Colon Health, or VSL#3.
- Serotonin receptor agonists (5-HT4) to decrease colonic motility such as alosetron (Lotronex).
- Rifaximin (Xifaxan) for bacterial overgrowth.
- Peppermint oil as an antispasmodic to relax intestinal smooth muscle. It is available in an enteric coated preparation (Colpermin).

Chronic and recurrent abdominal pain syndromes

Chronic abdominal pain (CAP) and **recurrent abdominal pain (RAP)** are common disorders in childhood and adolescence. Apley first described CAP of either continuous abdominal pain or intermittent abdominal pain, also called recurrent abdominal pain, occurring in school-age children, persisting for more than 3 months, and affecting normal activity (Greenberger, 2018).

Risk factors

Chronic or recurrent abdominal pain is thought to be a manifestation of IBS in pediatric patients. This syndrome is reported to occur in 10% of children and about 2% of adults, mostly females (Greenberg, 2018).

The etiology and pathogenesis of the pain with this syndrome are unknown; however, the prevailing viewpoint revolves around a disorder of GI motility and/or visceral hypersensitivity.

Symptoms

Children with chronic or recurrent abdominal pain experience the pain with only occasional bloating, dyspepsia, and alteration of bowel patterns. Associated symptoms in children include headache, pallor, dizziness, and nausea.

Diagnosis

In most cases, only a limited evaluation is indicated to rule out organic causes.

Complications

Although IBS and chronic or recurrent abdominal pain may be associated with severe pain and discomfort, they do not predispose a patient to other chronic or life-threatening conditions, and there is no evidence that they interfere with longevity.

Treatment

Careful counseling and explanation regarding the condition is key. Emphasis on acknowledging the presence and severity of the pain is important for both the parent and child.

Use of a high-insoluble fiber diet is somewhat controversial but has been shown to be of some benefit. There is no data to support the use of drug therapy in children with chronic or recurrent abdominal pain, and it may lead to hypochondriasis. Consultation with a child psychologist may be beneficial if stress or maladaptive coping mechanisms are identified.

PARASITIC INFESTATIONS

Parasitic diseases that affect the large intestine include amebiasis, trypanosomiasis, and trichuriasis.

Amebiasis

Amebiasis is a form of colitis caused by the protozoan *Entamoeba histolytica* (*E. histoytica*). In the United States, *E. histolytica* represents the third most frequently identified protozoan from human specimens (Kantor et al., 2018).

Risk factors

Amebiasis is transmitted primarily by fecal contamination of food or water, but other modes of transmission include close interpersonal contact, venereal spread, and houseflies and cockroaches. Ingested cysts form trophozoites in the small bowel, which are then released into the colon. It may take only one cyst to cause infection in a susceptible host. Approximately 50,000 cysts are produced annually in a patient with invasive *E. histolytica*. The amebae penetrate host tissues, causing necrosis without inflammation. Lesions are located throughout the colon, most often in the cecum, ascending colon, and rectosigmoid colon areas.

Symptoms

Carriers of *E. histolytica* are often asymptomatic, but symptoms include:

- Amebic diarrhea, an afebrile illness that is usually intermittent and followed by constipation.
- Amebic dysentery, associated with headache, nausea and vomiting, fever and chills, **tenesmus**, abdominal cramps, and stools with a large amount of blood-tinged mucus.
- Amebic appendicitis, an obstruction of the appendix lumen.
- Ameboma, a large, mass-like lesion in the wall of the intestine, most often in the cecum.
- Visceral abscess, particularly a single abscess in the right lobe of the liver, which ruptures in a small number of cases.

Diagnosis

Diagnosis of amebiasis requires demonstration of *E. histolytica* trophozoites and cysts in the stool. Sigmoidoscopy and indirect hemagglutination titers may be useful diagnostic tools, and a barium enema may be helpful in diagnosing amebomas.

Additional serologic tests may be helpful. These include counter immunoelectrophoresis and agar gel diffusion. EIA is positive in 95% of patients with extraintestinal infection and 70% of patients with active infection (CDC, 2013).

Final diagnosis is based on identifying the pathology or trophozoites on examination of the aspirated material.

In cases of suspected liver abscess, liver-spleen scan, abdominal ultrasonography, or CT scan, radiography, magnetic resonance imaging (MRI), or percutaneous needle aspiration may be used. When there is liver involvement, patients have elevated alkaline phosphatase.

Complications

Complications include colon perforation and pleuropulmonary conditions. Amebic brain abscesses claim a high mortality rate.

Treatment

Amebiasis is treated with an oral antiprotozoal, most often diiodohydroxyquinoline, also called iodoquinol (Yodoquin, Diquinol, Yodoquinol, Yodoxin), with or without metronidazole (Flagyl).

With appropriate treatment, response to therapy is often dramatic and prognosis is excellent.

Trypanosomiasis

Trypanosomiasis is a chronic illness caused by the protozoan *Trypanosoma cruzi* (*T. cruzi*), which causes Chagas disease, also known as South American trypanosomiasis.

The parasite multiplies rapidly at the site of inoculation, producing a severe inflammatory reaction. It may be disseminated through the bloodstream to any organ, particularly the heart, esophagus, or colon. At the site of infection, the parasite causes degeneration of the intramuscular autonomic nerve plexuses, resulting in megaesophagus, **megacolon**, or dilatation of other tubular organs many years after the initial infection.

Symptomatic Chagas disease has been confined largely to rural areas of South and Central America, but it has been found in all Western Hemisphere countries except Guyana, Surinam, and Canada.

Risk factors

Trypanosomiasis is transmitted primarily by the bite of the reduviid bug, but may also be transmitted by damage to the placenta during birth, blood transfusions, or contaminated food.

Symptoms

Acute Chagas disease is most often seen in children under age 2. Symptoms include anorexia, nausea, vomiting, generalized lymphadenopathy, diarrhea, unilateral conjunctivitis, edema of the face and eyelids, swelling of lacrimal and submaxillary glands, hepatosplenomegaly, cardiac arrhythmias, congestive heart failure, and frequently death.

Diagnosis

Diagnosis requires identification of *T. cruzi* in blood or tissue. Trypanosomes may be found in cerebrospinal fluid (CSF) cultures.

Treatment

For active infections, prolonged oral administration of nifurtimox (Lampit) or benznidazole (Radanil, Ro7-1051, Rochagan) is effective. Ultimately, surgery may be required to correct the "mega syndromes," which include megaesophagus and megacolon. Treatment of the illness is unsatisfactory after a certain period of time.

Trichuriasis

Trichuriasis is caused by the nematode *Trichuris trichiura*. It is also known as whipworm because of the whip-like appearance of the parasite. Whipworm is most often found in tropical areas or in areas of poor sanitation. The life span of the adult worm can range from 1-8 years. Each female worm can produce as many as 3,000–20,000 eggs. 604-795 billion people are affected worldwide (CDC, 2013).

Risk factors

Transmission is fecal-oral. The ingested eggs hatch in the duodenum. When the larvae mature into adult worms, they migrate down the GI tract and accumulate in the cecum and ascending colon.

Symptoms

The majority of patients are asymptomatic. Heavy infestations may cause dehydration, weakness, abdominal distention and cramping, anorexia, weight loss, acute and chronic diarrhea, bloody mucoid stools, anemia, and occasionally rectal prolapse. There is usually low-grade eosinophilia.

Complications

In children, chronic heavy infestation may be an important contributor to impairment of growth and development.

Diagnosis

Diagnosis requires identification of the characteristic eggs in the stools or, rarely, detection of the worms by sigmoidoscopy or colonoscopy. The worms may be visualized hanging in the intestinal lumen or buried within the wall.

Treatment

Treatment is dependent on the type of the infection (T. b. gambiense or T. b. rhodesiense) and the stage of the disease. The recommended drug for first-stage T. b. gambiense infection is Pentamidine, which is widely available in the U.S.

DIVERTICULAR DISEASE

Diverticular disease is characterized by herniation at weak points on the intestinal mucosa and submucosa, most often on the left side in the descending and sigmoid colon. A diverticulum typically has a narrow neck that contains all four mucosal layers and a spherical sac that contains intestinal serosa and mucosa only.

Diverticular disease is found in approximately 35% of the population in the United States (NIH, n.d.). Asian countries have traditionally seen lower incident rates, but with westernization of diets, rates are starting to increase. In migrants from low-prevalence areas to Westernized countries, diverticular disease incidence increases within 10 years. Prevalence is strongly correlated with advancing age.

Diverticular disease may take the form of diverticulosis or diverticulitis.

- **Diverticulosis** is uncomplicated diverticular disease.
- **Diverticulitis** is an inflammation in the wall of the diverticulum at its apex and, rarely, at its neck. It occurs most often in the sigmoid colon and adjacent structures and may last for several weeks.

Risk factors

Contributing factors for diverticular disease include:

- Hypertrophy of segments of the colon's circular muscle.
- Increased intraluminal, intracolonic pressure from uncoordinated peristalsis.
- Age-related atrophy or weakness in the bowel muscles (rare in people younger than age 40).
- Chronic constipation and straining.
- Lack of dietary fiber.
- Obesity.
- Inadequate luminal contents.

Symptoms

Symptoms depend on whether inflammation of the diverticulae occurs.

- **Diverticulosis.** Patients with diverticulosis show no signs of infection. In fact, most are asymptomatic. Approximately 20% of individuals with diverticulosis present with symptoms (Barroso & Quigley, 2015). When symptoms are reported, they include bloating, excessive flatulence, irregular defecation, and intermittent abdominal pain.
- **Diverticulitis.** The most common symptoms of diverticulitis are low-grade fever and localized or diffuse abdominal pain. Other symptoms include nausea and vomiting, constipation, palpable sigmoid, mild leukocytosis (predominantly of polymorphonuclear leukocytes), and obstipation.

In children, the most common type of diverticulum of the GI tract to become symptomatic is Meckel's diverticulum.

Diagnosis

Diagnosis for:

- **Diverticulosis** is confirmed by barium enema or colonoscopy.
- **Diverticulitis** is by patient complaint of severe abdominal pain, X-ray films, proctosigmoidoscopy, and CT scan. CT scan is preferred, as it will reveal abscess formation, perforation, bowel thickening, and disease extent. White blood cell count (WBC) is an additional evaluative test.

Complications

Possible complications of diverticular disease include rupture of an inflamed diverticulum, with localized or generalized peritonitis; abscess formation around the diverticulum; edema or spasm related to inflammation; vesicocolonic fistula formation; erosion of an artery or vein with bleeding; and colonic fibrosis and narrowing, possibly leading to obstruction. Patients may develop bacteremia and septicemia. Repeated attacks of diverticulitis may cause abscesses.

Treatment

Treatment depends on the severity and complications of diverticular disease.

- Mildly symptomatic diverticular disease usually involves dietary management, most often in the form

of a high-fiber diet and use of hydrophilic colloids or bulk-forming laxatives to help maintain a regular, soft stool.
- Acute attacks may require hospital admission, bed rest, IV antibiotics, bowel rest, and analgesics. Slow progression of diet from clear liquids to low fiber then to high fiber will follow bowel rest.
- Colonic strictures, peritonitis, or septicemia may require surgical intervention in the form of a colon resection (one stage) or a temporary diverting colostomy (two-stage procedure when continuity of bowel function is restored). Surgical mortality rates are 3% (Haas et al., 2016).
- Abcesses may be treated using percutaneous drainage.
- Bleeding is treated through arteriography with injection of sclerosing agents to occlude the bleeding vessel, vasopressin (Vasostrict), or vessel embolization.

Mesalamine (5-aminosalicylic acid) may aid in decreasing diverticulitis attacks.

COLORECTAL CANCER

Colorectal cancer is the third most common cancer type in adults in the United States, after lung cancer in men and breast cancer in women. It occurs most frequently in people between ages 50 and 80. Incidence is 1 in 23 in the United States (American Cancer Society, 2019) and 16.1-32.2 per 100,000 people in Asian countries (Arnold et al, 2017).

Approximately 95% of intestinal cancers are adenocarcinomas. Types of adenocarcinoma include mucinous or colloid varieties. In adults, mucinous tumors comprise 15% of colorectal carcinomas and as many as 40% in children. African Americans have a higher incidence than other racial groups in the United States. Currently, the highest percentage of colorectal cancers in the United States in Caucasians are located in the cecum and ascending colon (22% in males and 27% in females) and in the sigmoid colon (25% in males and 23% in females). Cancers of the colon in children tend to be more evenly distributed throughout the large intestine.

Risk factors

Colon cancer is believed to be an acquired genetic disease caused by exposure to environmental carcinogens.

Risk factors for colorectal cancer include:
- Lifestyle factors, including:
  - Low-fiber diet with high intake of fat and high-fat red meat and/or processed meats.
  - A sedentary lifestyle.
  - Smoking.
  - Being overweight or obese.
  - Heavy alcohol use.
- Increasing age.
- Family history of colon or rectal cancer.
- Previous colon cancer.
- Personal history of adenomatous polyps.
- Familial adenomatous polyposis (FAP) or Gardner syndrome.
- UC for more than 7 years.
- Genital cancer or breast cancer (in women).
- Genetic alterations in *KRAS*, *APC*, *DCC* and *TP53* genes in hereditary nonpolyposis colorectal cancer (HNPCC) or Lynch syndrome.

Screening for colon cancer

Periodic serial stool screening for occult blood is important to detect colorectal lesions before metastasis occurs.

The American Cancer Society (ACS, 2018) recommends that both males and females at average risk for developing colorectal cancer should use one of the following screening tests to identify polyps and cancer:
- Flexible sigmoidoscopy every 5 years, colonoscopy every 10 years.
- Double-contrast barium enema every 5 years.
- CT colonography (virtual colonoscopy) every 5 years.

If there are positive findings from any of these studies, a colonoscopy should be done.

Acceptable tests that can be used to identify cancer include:
- Fecal occult blood test (FOBT) every year.
- Fecal immunochemical test (FIT) every year.
- Stool DNA test (sDNA) every 3 years.

Any positive finding from these tests also warrants a colonoscopy.

Patients with predisposing factors for colorectal cancer should receive more frequent and lifelong surveillance beginning at an earlier age, for example, earlier than 45 years of age or earlier than the age at which a close blood relative was first diagnosed with colon cancer.

The following options are acceptable choices for colorectal cancer screening in average-risk adults. Since each of the following tests has inherent characteristics related to accuracy, prevention potential, costs, and risks, individuals should have an opportunity to make an informed decision when choosing a screening test (ACS, 2018).

Table 16.2. Cancer Screening Options for Average-Risk Adults

Test	Interval (beginning at age 45)	Comment
Flexible sigmoidoscopy	Every 5 years	Colonoscopy should be done if test results are positive.
Fecal occult blood test (FOBT)	Yearly	The recommended take-home, multiple-sample method should be used. Colonoscopy should be done if test results are positive.

Table 16.2. Cancer Screening Options for Average-Risk Adults (continued)

Test	Interval (beginning at age 45)	Comment
Colonoscopy	Every 10 years	Colonoscopy provides an opportunity to visualize, sample, and/or remove significant lesions.
Double contrast barium enema (DCBE)	Every 5 years	Colonoscopy should be done if test results are positive.
CT colonography (virtual colonoscopy)	Every 5 years	Colonoscopy should be done if test results are positive.
Fecal immunochemical test (FIT)	Yearly	The take-home, multiple-sample method should be used. Colonoscopy should be done if test results are positive.
Stool DNA (sDNA) test	Every 3 years	Colonoscopy should be done if test results are positive.

ACS guidelines (2018) on screening and surveillance for the early detection of colorectal adenomas and cancer for people at increased risk or at high risk are included in Table 16.3. The indications that make risk higher than average are:

- Personal history of colorectal cancer or adenomatous polyps.
- Personal history of inflammatory bowel disease (ulcerative colitis or Crohn's disease).
- Strong family history of colorectal cancer or polyps.
- Known family history of a hereditary colorectal cancer syndrome, such as familial adenomatous polyposis (FAP), hereditary non-polyposis colon cancer (HNPCC), or Lynch syndrome.

Symptoms

Signs and symptoms of colorectal cancer can depend on the location.

- Right-sided tumor symptoms may include melenic stools, dull abdominal pain referred to the upper abdomen and back, anorexia and weight loss, malaise, lethargy, weakness, and indigestion.
- Left-sided tumor symptoms frequently include a distinct change in bowel habits, including difficulty passing stools, abdominal distention, ribbon- or pencil-shaped stools, constipation, cramping, flatulence, and a feeling of rectal pressure or incomplete stool evacuation.

Table 16.3. American Cancer Society Guidelines on Screening and Surveillance for the Early Detection of Colorectal Adenomas and Cancer for People at Increased Risk or at High Risk

Risk Category	Age to Begin	Recommended Test(s)	Comment
Patients with increased risk due to a history of polyps on prior colonoscopy			
Small rectal hyperplastic polyps	Same as those at average risk	Colonoscopy or other screening options at same intervals as for those at average risk	Those with hyperplastic polyposis syndrome are at increased risk for adenomatous polyps and cancer and should have more intensive follow-up.
1 or 2 small (less than 1 cm) tubular adenomas with low-grade dysplasia	5 years after the polyps are removed	Colonoscopy	Time between tests should be based on other factors, such as prior colonoscopy findings, family history, and patient and doctor preferences.
3 to 10 adenomas, a large (1 cm or greater) adenoma, or any adenomas with high-grade dysplasia or villous features	3 years after the polyps are removed	Colonoscopy	Adenomas must have been completely removed. If colonoscopy is normal or shows only 1 or 2 small tubular adenomas with low-grade dysplasia, future colonoscopies can be done every 5 years.
More than 10 adenomas on a single exam	Within 3 years after the polyps are removed	Colonoscopy	Doctor should consider the possibility of a genetic syndrome, such as FAP, HNPCC, or Lynch syndrome.
Sessile adenomas that are removed in pieces	2–6 months after adenoma removed	Colonoscopy	If the entire adenoma has been removed, further testing should be based on doctor's judgment.

Risk Category	Age to Begin	Recommended Test(s)	Comment
Patients at increased risk with colorectal cancer			
Diagnosed with colon or rectal cancer	Time of colorectal surgery, or can be 3–6 months later if person doesn't have cancer spread that can't be removed	Colonoscopy to view entire colon and remove all polyps	If the tumor presses on the colon or rectum and prevents colonoscopy, CT colonoscopy (with IV contrast) or double-contrast barium enema may be done to look at the rest of the colon.
Had colon or rectal cancer removed by surgery	Within 1 year after cancer resection or 1 year after colonoscopy to make sure the rest of the colon and rectum is clear	Colonoscopy	If normal, repeat exam in 3 years. If 3-year follow up exam is normal then, repeat exam every 5 years. Time between tests may be shorter if polyps are found or there is reason to suspect HNPCC or other hereditary polyp syndromes. After low anterior resection for rectal cancer, exams of the rectum may be done every 3–6 months for the first 2–3 years to look for signs of recurrence.
Patients at increased risk due to a family history			
Colorectal cancer or adenomatous polyps in any first-degree relative before age 50 or in 2 or more first-degree relatives at any age (if not a hereditary syndrome)	Age 40 or 10 years before the youngest case in the immediate family, whichever is earlier	Colonoscopy	Every 5 years
Colorectal cancer or adenomatous polyps in any first-degree relative age 50 or younger or in at least 2 second-degree relatives at any age	Age 40	Same options as those at average risk	Same intervals as for those at average risk
Patients at high risk			
Familial adenomatous polyposis (FAP) diagnosed by genetic testing or suspected FAP without genetic testing	Age 10–12 years	Yearly flexible sigmoidoscopy to look for signs of FAP; counseling to consider genetic testing if it hasn't been done	If genetic test is positive, removal of the colon (colectomy) should be considered.
Hereditary non-polyposis colon cancer (HNPCC) or Lynch syndrome or at increased risk based on family history without genetic testing	Age 20–25 years, or 10 years before the youngest case in the immediate family	Colonoscopy every 1–2 years; counseling to consider genetic testing if it hasn't been done	Genetic testing should be offered to first-degree relatives of people found to have HNPCC or Lynch syndrome mutations by genetic tests. It should also be offered if 1 of the first 3 of the revised Bethesda Guidelines criteria is met (ASGE, 2018).
Inflammatory bowel disease: • Chronic ulcerative colitis • Crohn's disease	Cancer risk begins to be significant 8 years after the onset of pancolitis (involvement of entire large intestine) or 12–15 years after the onset of left-sided colitis	Colonoscopy every 1–2 years with biopsies for dysplasia	Recommend referral to a center with experience in the surveillance and management of inflammatory bowel disease.

Diagnosis

To diagnose colorectal cancer, the physician may order a stool test for occult blood; sigmoidoscopy or colonoscopy with biopsy examination and brush cytology; barium enema, usually with air contrast; upper GI series if small intestine involvement is suspected; and possibly a serum carcinoembryonic antigen (CEA) test, but this is less sensitive in children.

Complications

Early detection of colon cancer is of primary importance because, for patients with carcinoma limited to the bowel wall, the five-year survival rate is 75%, compared to only 5% for patients with disseminated disease. According to the Centers for Disease Control and Prevention (CDC) statistics in 2016, 141,270 new cases of colorectal cancer were reported and 52,286 people died of colorectal cancer in the United States. Unfortunately, large intestine tumors often exhibit no signs or symptoms until they become rather large, thereby reducing the likelihood of early diagnosis and making surgical cure difficult. Metastasis usually involves the liver.

Based on delayed recognition and types of tumors found in children, the prognosis for children with carcinoma of the colon is poor.

Treatment

Treatment of colorectal cancer depends on the stage of the disease.

Staging

Two different methods are used to stage colon cancer: Duke's classification and the TNM Staging System (Tumor, Nodal involvement, and Metastasis) classification.

Duke's classification recognizes four stages of tumor involvement.

- **Class A:** limited to the mucosa and submucosa.
- **Class B:** penetration of the entire bowel wall and serosa or pericolic fat.
- **Class C:** Class A and B, plus invasion of the regional draining lymph node system.
- **Class D:** advanced and widespread regional metastasis.

A comparison of staging of adult versus young patients with colon cancer is displayed in Table 16.4.

Table 16.4. Comparison of Staging of Adult Vs. Young Patients with Colon Cancer

Stage	Adult	Child
Limited to mucosa	60%	2.2%
Through mucosa, but no nodes	60%	16.2%
Nodes involved	40%	47.6%
Distal metastasis	40%	34.0%

The TNM classification describes the anatomical extent of the primary tumor depending on:

- T: Size, invasion depth, and surface spread of the tumor.
- N: Extent of nodal involvement.
- M: Presence or absence of metastasis.

Surgery

Most patients with colon cancer require either curative or palliative surgery such as bowel resection and anastomosis or stoma formation (colostomy). Surgery involves resection of the tumor and associated blood vessels, lymphatic channels, and affected structures as a unit. In some cases, the surgeon will irrigate the peritoneum with an antineoplastic agent and do a biopsy examination of the liver and abdominal lymph nodes.

The need for a permanent or temporary colostomy depends on the part of the colon involved and the extent of the disease. Formation of a colostomy involves making an outside opening, or **stoma**, on the abdomen using a section of the colon. There are three types of stomas.

- **Single-barrel stoma** is when the functioning proximal end of the bowel is brought through the abdominal wall, then folded in on itself and sutured to form a cuff.
- **Double-barrel stoma** is when both the active and inactive bowel ends are temporarily brought through the abdominal wall, thereby creating two stomas.
- **Loop stoma** is a temporary procedure formed by bringing an intact bowel loop through the abdominal wall and making an incision in the top of the loop.

Conscientious ostomy care is required, with priority given to protecting the skin from corrosive enzymes, controlling odor and fecal drainage, and providing emotional support and encouragement.

Radiation and chemotherapy

Additional interventions for patients with colorectal cancer may include radiation therapy (for rectal cancer), chemotherapy, and/or immunotherapy. A fluorouracil (5-FU) and leucovorin combination for all patients with stage III is suggested. Diet for patients with stage III should be a high-fiber, non-Western diet.

In patients with stage C tumors, adjuvant therapy has improved recurrence rates. The standard of care for stage III colon adenocarcinoma after resection is chemotherapy and provides a 22% to 32% overall survival advantage and a 30% reduction in recurrence (Kannarkatt et al., 2017).

INTESTINAL OBSTRUCTION

Intestinal obstructions may occur in either the large or small intestine. The source of an obstruction may be mechanical, neurogenic, or vascular.

Mechanical obstruction

Mechanical obstructions may result from congenital defects or acquired causes.

Risk factors

Mechanical obstructions are related to:

- Congenital defects, including atresia, tenosis, hernia, **malrotation**, meconium ileus, or aganglionic megacolon (Hirschsprung's disease).
- Acquired causes, including intraluminal causes such as polyps, tumors, and feces; intramural causes such as stricture or pyloric stenosis; and intrinsic causes such as postoperative adhesions or hernias.

Hirschsprung's disease is the congenital absence of intramural ganglia of the anorectum and variable lengths of the distal colon, resulting in failure of relaxation of the contracted segment and internal sphincter, followed by obstructive symptoms and dilatation of the more proximal normal colon. Intestinal ischemia may occur secondary to distention of the bowel wall and contribute to the development of enterocolitis, the leading cause of death in children with Hirschsprung's disease.

Hirschsprung's disease occurs in 1 out of 5,000 live births, shows a male predominance of 3.8:1, and is familial (NIH, 2018). Approximately 2% to 15% of patients with this disorder have Down syndrome (National Down Syndrome Society, n.d.). If not diagnosed within the first 24 hours of life, other symptoms that may appear include constipation, diarrhea, vomiting, and perforation of the appendix or colon. Poor nutrient intake and growth may also be seen.

Diagnosis

Although 80% of cases are diagnosed by 1 year of age, as many as 8% may not be recognized until 3 to 12 years of age. Most of these children have short-segment Hirschsprung's disease. The vast majority of Hirschsprung's cases are confined to the colon. The aganglionic segment is usually limited to the rectum and sigmoid colon. In rare cases, the aganglionic segment extends throughout the large intestine. In extremely rare cases, the aganglionic segment may extend into the small intestine, resulting in serious nutritional, medical, and surgical challenges.

Diagnosis

Hirschsprung's disease may be associated with other abnormalities involving the neurological, cardiovascular, urological, and GI systems. Most of these associated abnormalities are also associated with early errors in embryonic development.

Hirschsprung's disease becomes apparent shortly after birth when the infant passes little meconium and develops a distended colon. It is diagnosed by barium enema and anorectal manometry and confirmed by a full-thickness biopsy examination of the rectal mucosa. The definitive findings are aganglionosis or a patchy area in the segment.

Treatment

Surgical treatment is required for Hirschsprung's disease, and a number of operations have been proposed to remove or to counterbalance the obstructing effect of the aganglionic segment. Obstruction in infants is relieved through creation of a stoma proximal to the aganglionic segment. In older children, however, a preliminary colostomy may not be required and a one-stage surgical procedure may be performed.

Prognosis is good after successful surgery for Hirschsprung's disease. Residual fecal soiling may occur in some cases.

Neurogenic obstruction (intestinal pseudo-obstruction)

In this rare condition, also called **intestinal pseudo-obstruction**, there is no mechanical blockage. Instead, this anomaly results from ineffective intestinal peristalsis. Paralysis is usually incomplete.

Risk factors

Intestinal pseudo-obstruction is a failure of the intestinal pump due to alteration of the enteric nervous system. It may be primary, also known as chronic idiopathic intestinal pseudo-obstruction (CIP), and associated with generalized visceral neuropathy, or it may be secondary and associated with identifiable smooth muscle, endocrine, neurological, or pharmacological causes.

Abdominal surgery may be a cause of neurological impairment of peristalsis (adynamic or paralytic ileus). In such cases, it is a reversible, self-limited form of pseudo-obstruction that usually disappears 2-3 days after surgery.

Symptoms

Characteristic physical findings of intestinal pseudo-obstruction include cachexia and variable degrees of abdominal distention. Pain results from abdominal distention and heightens with increasing distention.

Diagnosis

Radiographically, a paralytic ileus with dilated bowel loops and intermittently present air fluid levels are seen.

Because many patients have esophageal motor abnormalities, esophageal manometry shows poor peristalsis of the esophagus and incomplete relaxation of the lower esophageal sphincter.

Diagnosis of neurologic impairment of peristalsis secondary to abdominal surgery is made by observation of abdominal distention, gastric aspiration, lack of flatus, and **constipation**.

Complications

Inflammation within the myenteric ganglia may cause severe progressive neuropathic CIP in conjunction with autoimmune disease and circulating anti-enteric neuronal antibodies.

Paralytic ileus secondary to abdominal surgery may be prevented by maintaining fluid and electrolyte balance and by handling the bowel gently during surgery. It is treated by nasogastric suction and IV fluid administration to correct electrolyte imbalance. In some cases, rectal tube decompression or colonoscopic decompression may be performed.

Vascular obstruction

A vascular obstruction occurs when emboli or atherosclerotic narrowing interrupts the blood supply to the bowel. Another source of vascular obstruction is telangiectasia, in which capillaries and venules are dilated in the mucus membranes throughout the GI system.

Symptoms

Symptoms of intestinal vascular obstruction include abdominal pain, vomiting, **obstipation** (intractable constipation), failure to pass flatus, and abdominal distention.

Diagnosis

Diagnosis can be made by taking a careful history and conducting a thorough physical examination. Other tests may be ordered such as adominal radiographic examinations; a barium enema or swallow; and blood tests to detect dehydration, tissue necrosis, and electrolyte imbalance.

Complications

Vascular obstruction inhibits peristalsis and can lead to life-threatening intestinal ischemia in as little as 40 minutes.

Treatment

Intervention strategies for patients with intestinal vascular obstructions involve the following:

- Restoring bowel patency.
- Regulating fluids and electrolytes, often intravenously. Correction and prevention of specific deficiencies of vitamins and minerals are often necessary.
- Administering medications to control pain and nausea.
- Administering antibiotics to treat bacterial growth.
- Using a nasogastric or intestinal tube to relieve abdominal distention.
- Using a decompressive gastrostomy or colostomy during periods of exacerbation.

A feeding jejunostomy placed endoscopically may allow for slow refeeding and tolerance of enteral nutrition without the prolonged use of parenteral nutrition that may be required in some patients, especially those with CIP.

After abdominal distention and compression have been controlled, surgical intervention may be necessary. The type of surgery performed depends on the source of the obstruction but may include hernia reduction, adhesion division, intestinal bypass, lesion excision, or a diverting colostomy. After surgery, the patient should be monitored for signs of recurrent obstruction or paralytic ileus. Any abdominal drain should be monitored for patency and drainage amount.

Drug therapy using metoclopramide (Reglan, Metzolv) and erythromycin has been tried, with scattered reports of success. In general, however, it has been less than successful.

A diet lower in fat, lactose, and fiber may help to prevent or control a recurrence of symptoms.

ANORECTAL DISORDERS

Anorectal disorders typically cause rectal pain, rectal bleeding, and/or a change in bowel habits. The cause may be hemorrhoids, fecal impaction, **encopresis**, anorectal abscess, fistula, fissure, or rectal prolapse. Other anorectal disorders include rectal cancer and trauma.

Hemorrhoids

Hemorrhoids are vascular masses in the anal canal. They are varicose dilatations of the radicals of the superior or inferior plexus of the hemorrhoidal veins. Internal hemorrhoids bulge into the rectal lumen above the internal sphincter and the anorectal line. External hemorrhoids are dilatations of the inferior hemorrhoidal plexus. They lie below the anorectal line and protrude below the external sphincter.

Hemorrhoids affect 1 in 20 Americans, usually affecting people age 50 and older (NIH, 2016).

Risk factors

The following are predisposing factors for hemorrhoids:

- Genetic predisposition.
- Absence of valves in the portal venous system attributed to the erect human posture.
- Conditions that cause transient or constant increased pressure or stasis within the rectal venous plexuses, including:
  - Straining from constipation.
  - Frequent bowel movements (diarrhea).
  - Rectal tumors.
  - Strictures.
  - Pregnancy.
  - Pelvic tumor.

Primary internal hemorrhoids are rarely seen in pediatric patients but may be associated with portal vein obstruction.

Symptoms

Symptoms include bright red rectal bleeding, occasionally leading to iron-deficiency anemia; rectal pain, which may be severe; and in the case of external hemorrhoids, a sensation of a bulging mass on exertion.

Diagnosis

Diagnosis is by visual examination and possibly proctoscopy or anoscopy. Internal hemorrhoids are graded one to four, depending on their degree of prolapse.

Treatment

Medical interventions for patients with hemorrhoids may include:

- High-fiber diet and adequate fluid intake if chronic dehydration is suspected.
- Comfort measures, such as warm compresses; topical analgesic ointments; and sitz baths, with or without salts in warm water; or astringents (witch hazel).
- Stool softeners (psyllium seed).
- Hydrocortisone suppository or cream.
- Mesalamine (Canasa) suppository to decrease inflammation.
- Benzocaine 20% spray or dibucaine 1% topical ointment.
- Injection sclerotherapy.
- Surgical intervention, such as rubber band ligation, hemorrhoidectomy, cryosurgery, infrared photocoagulation, or electrocautery. The first-line nonmedical therapy is rubber band ligation, which is superior to sclerotherapy.

Fecal impaction

Fecal impaction is the formation of a large, firm, immovable mass of stool that occurs when normal movement of the feces is impaired. This allows the bowel to absorb more water than usual, thereby hardening the fecal matter and making it difficult to pass.

Risk factors

Fecal impaction may be caused by chronic severe dehydration, chronic constipation, inactivity, medications, anal disease that causes painful defecation, neurological problems, or barium retention after radiological studies. It is relatively common in children, resulting from voluntary withholding of stool subsequent to a painful bowel movement.

Symptoms

Symptoms of fecal impaction are abdominal discomfort; a sensation of rectal fullness; nausea and vomiting; headache; passing small amounts of watery, malformed stool; or a large, firm mass in the left lower quadrant.

Complications

Possible complications of an untreated impaction include intestinal or urinary tract obstruction or spontaneous perforation.

Treatment

Treatment options include digital breakup of low-lying impactions, administration of enemas and/or fluids, or medications to soften the stool and encourage its passage.

Encopresis

Encopresis is a condition of chronic constipation that results in involuntary leakage of feces after the developmental age of 4 years.

Risk factors

Encopresis is not directly related to organic illness. This is more common in boys than in girls.

Symptoms

Symptoms of encopresis are chronic constipation with involuntary leakage, causing soiling.

Diagnosis

Diagnosis is made based on exclusion of organic causes of constipation and psychological disorders that may be associated with incontinence.

Treatment

Treatment involves the use of non-habit-forming stool softeners and bowel habit retraining. The goal of stool softening therapy should be to promote the passage of one to two large, loose bowel movements daily. Maintenance of an empty rectal vault assists in the return of bowel tone and proper functioning. Biofeedback training in combination with stool softening may be beneficial to help some patients speed their recovery.

Dietary therapy in pediatrics is rarely sufficient to overcome encopresis; however, it is used as an adjunctive therapy when stool softeners are weaned.

Anorectal abscess

Anorectal abscess occurs when infection causes localized accumulation of pus in the tissue spaces around the anorectum.

Risk factors

An anorectal abscess usually results from the invasion of the normal rectal flora into the perirectal or perianal tissues.

In children, pruritis, perianal cellulitis, and blood-streaked stools may be caused by group A beta-hemolytic streptococci. Patients with Crohn's disease are particularly susceptible to abscess formation. Other predisposing conditions include malignancy, radiation proctitis, leukemia, lymphoma, tuberculosis, actinomycosis, and lymphogranuloma venereum.

Symptoms

One symptom may be throbbing pain in the anorectum, which worsens when the patient walks or sits. Other symptoms include fever, fatigue, and rectal discharge.

Diagnosis

Diagnosis is by endosonography and MRI.

Complications

Complications of untreated abscess may include abscess extension or anorectal fistula. Mortality may increase if not treated.

Treatment

Typically, the physician will drain the abscess surgically; prescribe follow-up treatment of antibiotics to treat both aerobic and anaerobic flora; and recommend analgesics, sitz baths, and stool softeners.

Anorectal fistula

An anorectal fistula is a hollow, fibrous tract leading from the anal canal or rectum to the perianal skin. The types of anorectal fistulas are complete internal, blind internal (sinus), external (intrasphincteric and transsphincteric), branching (complex), and horseshoe.

Usually, a primary drainage port drains into an anal crypt, while a secondary port drains into the perianal skin or rectal mucous membrane.

Risk factors

An anorectal fistula often results from an anorectal abscess in the chronic phase.

Symptoms

The predominant symptom of anorectal fistula is a purulent drainage of pus, blood, mucus, or, less commonly, stool. Females may notice passage of flatus or feces through the vagina. Other symptoms may include pruritus, pain, or odor and the presence of a palpable tract on rectal examination.

Diagnosis

Anoscopy or sigmoidoscopy may locate the source of the fistula or abscess.

Treatment

Anal fistulas are treated by:

- Fistulectomy to repair superficial, straight fistulas or fistulotomy to correct deep fistulas.
- Closure of fistulae with fibrin glue.
- Placement of setons (temporary drains).
- Surgical creation of a rectal flap.

Anal fissure

An anal fissure is a thin tear of the superficial anal mucosa. It occurs most commonly in the distal anal canal, usually in the posterior midline.

Risk factors

Although unknown, the cause is associated with increased resting anal pressure. The most common condition is trauma following passage of a large, firm stool. If the fissure's edges adhere immediately after defecation, thereby preventing pus drainage, the resulting edema and fibrosis of adjacent tissue may cause the development of a sentinel pile or tag at the lower end of the fissure.

Symptoms

Symptoms of anal fissures include acute, severe pain (tearing or burning sensations) after defecation, anal itching, and discharge of bright red blood on the toilet paper or in the bowl. The pain may persist for hours.

Diagnosis

Diagnosis is by inspection and anoscopy. Digital rectal examination is extremely painful and should be avoided. Sigmoidoscopy should be deferred until the symptoms abate.

Treatment

Most fissures can be treated with analgesic ointments or creams (lidocaine plus hydrocortisone cream), sitz baths, and bulk agents or stool softeners. Psyllium fiber supplement softens the stool. An emollient suppository may provide some relief. Surgical excision or lateral subcutaneous sphincterotomy (for internal sphincter) may be required for chronic fissures with 90% success rate (ACRS, n.d.). Additional treatment includes topical nitrates (glycerol trinitrate), calcium channel blockers, and Botox.

Rectal prolapse

Rectal prolapse occurs when rectal mucosa (one or more layers) bulges through the anal orifice. It can be partial or complete. Among adults, females are more susceptible than males. Rectal prolapse is uncommon in children and must be distinguished from a prolapsed rectal polyp.

Risk factors

Prolapse may result from increased intra-abdominal pressure when straining, relaxed anal sphincters and weak pelvic muscles, or tissue damage or trauma during childbirth. In the elderly or debilitated patients, the cause is loss of sphincteric tone. In pediatrics, both cystic fibrosis and previous imperforate anus repair have been associated with rectal prolapse.

Symptoms

Symptoms are a protruding rectal mass that is apparent on defecation or other exertion, mucus discharge, rectal bleeding, and fecal incontinence.

Treatment

Rectal prolapse may be prevented by avoiding constipation and prolonged straining.

Primary treatment involves reduction of the prolapse, either digitally or by surgery, depending on the severity of the prolapse. If constipation is a predisposing factor, it should be treated with non-habit-forming stool softeners titrated to a dose to maintain loose stool passage with a minimal amount of time spent sitting on the toilet. Injection treatment with a sclerosant may be used if simple measures are unsuccessful.

Surgery involves a combined procedure in which the rectum is fixed to the sacral hollow, and the redundant sigmoid colon is removed. The extra-abdominal approach through the perineum to accomplish a rectosigmoidectomy is preferred for the elderly. If the prolapse is considered small or the patient is not medically stable for surgery, supportive garments can help.

CASE SITUATION

David Rosenstein is a 24-year-old medical student who has been doing an internship rotation in pediatrics. In the last few months, he has experienced fatigue and aching joints. Feeling overworked and stressed, he took a week off to go camping with his wife. During the following month, he began to notice frequent episodes of bloody, mucopurulent diarrhea. He then began having abdominal cramps and more joint tenderness. He thought he must have gotten "a bug" and visited a gastroenterologist because his symptoms seemed to be getting worse. After a battery of tests and a colonoscopy with biopsy exam, David was diagnosed as having ulcerative colitis and was placed in the hospital for treatment because of his debilitated condition.

Points to think about

1. After learning of David's diagnosis and history, what physical findings might the gastroenterology nurse expect?
2. The gastroenterology nurse knows that colonic biopsy can show submucosal inflammatory reactions and granuloma. What other findings might be noted on colonoscopy?
3. Initial planning and nursing intervention for David would, of necessity, focus on restoration of his physiologic equilibrium. The gastroenterology nurse's data search and assessment could substantiate at least five possible nursing diagnoses in the physiologic realm. What are they?
4. A diagnosis of ulcerative colits requires a great deal of reorganization of David's life. Establishing a trusting relationship with David and his wife is critical if the gastroenterology nurse is to help them in these efforts. Once such a relationship is tentatively established, what is the most important psychological area to which assessment should be directed?
5. Because it is quite likely that David is anxious, substantiation of the nursing diagnosis "anxiety," possibly related to threats to bodily health, self-concept, altered role performance, and lack of control over events, is necessary. What are the defining characteristics of this diagnosis?
6. There are three general goals for the nursing care of a patient with the nursing diagnosis "anxiety." What are they?

Suggested responses

1. Physical findings that might be expected in a patient with ulcerative colitis include:
   a. Fever, tachycardia, and signs of dehydration and hypovolemia; imminent cardiovascular collapse in severe abrupt-onset cases.
   b. Pallor, secondary to anemia.
   c. Tenderness on abdominal palpation and increased bowel sounds on auscultation.
   d. Redness, swelling, and tenderness in large joints, most often unilateral.
   e. Abdominal distention; profuse bloody, mucopurulent diarrhea; vomiting; and debilitation in severe cases with toxic megacolon.
   f. Skin manifestations on the arms and legs, such as erythema nodosum or pyoderma gangrenosum.
   g. Ankylosing spondylitis.
   h. Conjunctivitis or uveitis in the early course of ulcerative colitis.
2. Colonoscopy may reveal the following findings in ulcerative colitis:
   a. Continuous inflammation.
   b. Mucosal ulceration only.
   c. Most often seen in the left colon and rectosigmoid areas, but at times may involve the entire colon.
   d. Characterized by exacerbations and remissions.
   e. May cause a shortening effect on the bowel.
   f. Inconsistent mucosal cobblestoning effect with no patches of healthy tissue visualized in the diseased section.
   g. Bloody and mucopurulent diarrhea.
   h. Pseudopolyps are common.
   i. Inflammatory masses are rare.
3. Possible nursing diagnoses in the physiologic realm include:
   a. Altered nutrition: less than body requirements.
   b. Change in bowel habits related to bowel irritability.
   c. Fluid volume deficit.
   d. Acute pain related to bowel irritability.
   e. Activity intolerance related to weakness, debilitated condition, and joint swelling.
4. The most important psychological priority toward which assessment should be directed is determining David's coping mechanisms and skills. Without knowing the mechanisms by which he commonly manages problems and stresses in his daily life, a gastroenterology nurse cannot initiate a plan to aid him in adapting his life.
5. The defining characteristics of the nursing diagnosis "anxiety" are:
   a. Reported apprehension, nervousness.
   b. Escape or avoidance behavior.
   c. Physiologic arousal.
   d. Narrowed perceptual field, difficulty concentrating, self-focusing.

   e. Behaviors that may interfere with coping such as excessive anger, fear, guilt, and regression.
   f. Denial.
   g. Withdrawal.
   h. Increased wariness.
6. Three general goals for the nursing care of a patient with the nursing diagnosis “anxiety” are:
   a. Prevention of severe anxiety or panic status.
   b. Elimination or reduction of incapacitating anxiety symptoms.
   c. Use of effective coping skills.

The gastroenterology nurse must realize that specific, concrete, personalized patient outcomes within these broad goals may be multiple and based on a variety of factors and possible patient responses.

REVIEW TERMS

amebiasis, anus, ascending colon, cecum, chronic or recurrent abdominal pain, colitis, colon, constipation, Crohn’s colitis, descending colon, diarrhea, diverticulitis, diverticulosis, ectasia, encopresis, enterocolitis, familial adenomatous polyposis, food protein-induced enterocolitis, Gardner syndrome, haustra, hematochezia, hepatic flexure, Hirschsprung’s disease, intestinal pseudo-obstruction, irritable bowel syndrome (IBS), ischemic colitis, malrotation, megacolon, necrotizing enterocolitis (NEC), obstipation, paralytic ileus, Peutz-Jeghers syndrome, polyps, pseudomembranous colitis, pseudomenbranous enterocolitis, radiation enteritis, rectum, sigmoid colon, splenic flexure, stoma, tenesmus, toxic megacolon, transverse colon, trypanosomiasis, ulcerative colitis (UC), Valsalva maneuver, vermiform appendix

REVIEW QUESTIONS

1. The first portion of the large intestine to receive material from the small bowel is the:
   a. Ileum.
   b. Cecum.
   c. Appendix.
   d. Ascending colon.
2. Small sacculations in the large intestinal wall that are formed by the taenia coli are the:
   a. Haustra.
   b. Diverticula.
   c. Crypts of Lieberkühn.
   d. Plicae semilunares.
3. The colonic mucosa is:
   a. Made up of thousands of finger-like projections called villi.
   b. Covered with a layer of squamous epithelium.
   c. Arranged in folds called the plicae circulares.
   d. Smooth-surfaced.
4. Colonic secretion consists primarily of:
   a. Sodium, chloride, and water.
   b. Water, mucus, potassium, and bicarbonate.
   c. Mucus and hormones.
   d. Bile pigments and toxins.
5. The appearance of multiple adenomatous polyps in the GI tract and osteomas of the mandible, skull, and long bones is symptomatic of:
   a. Osler-Weber-Rendu disease.
   b. Colorectal cancer.
   c. Familial adenomatous polyposis (FAP).
   d. Gardner syndrome.
6. Toxic megacolon is a potentially serious complication of:
   a. Ischemic colitis.
   b. Ulcerative colitis.
   c. Pseudomembranous colitis.
   d. Transmural colitis.
7. Mildly symptomatic diverticular disease is most often treated by:
   a. Diverticulectomy.
   b. Colon resection.
   c. Dietary management and hydrophilic colloids or bulk-forming laxatives.
   d. Colostomy.
8. Metastatic colorectal cancer most often involves the:
   a. Small bowel.
   b. Liver.
   c. Pancreas.
   d. Stomach.
9. A hollow, fibrous tract leading from the anal canal or rectum to the perianal skin is called a(n):
   a. Hemorrhoid.
   b. Anorectal fissure.
   c. Anorectal abscess.
   d. Anorectal fistula.
10. Encopresis commonly seen in school-age children, particularly boys, is defined as:
    a. Involuntary nocturnal passage of urine.
    b. Involuntary leakage of stool.
    c. Nocturnal awakening with abdominal pain.
    d. Infectious diarrhea with fecal incontinence.

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Chapter 17

BILIARY SYSTEM

This chapter will acquaint the gastroenterology nurse with the normal anatomy and physiology of the biliary system. Selected diseases and disorders of the gallbladder and its associated duct system in terms of pathophysiology, diagnosis, and treatment are discussed.

Learning objectives

After reviewing the content of this chapter, the gastroenterology nurse will be able to:

1. Describe the normal gross structure and histology of the biliary system, including the gallbladder and its associated duct system.
2. Explain the motility and secretory functions of the gallbladder.
3. Discuss the pathophysiology, diagnosis, and treatment of the diseases and disorders of the biliary system presented.

ANATOMY AND PHYSIOLOGY

The functions of the biliary system are to collect, concentrate, and store bile and to release it into the duodenum when needed for digestion.

The **gallbladder** is a pear-shaped, sac-like bile storage structure. It is attached to the undersurface of the liver by connective tissue, peritoneum, and blood vessels. The gallbladder is approximately 7-10 cm (3 inches) long and 2.5-3.5 cm (about 1 inch) wide and is capable of holding up to 50 ml of bile.

The four anatomical divisions of the gallbladder are:

- **Fundus**, also known as the distal blind sac.
- **Funnel-shaped body**, which connects the fundus and the infundibulum.
- **Infundibulum**, which is a transitional region between the body and neck. The infundibulum is notable because occasionally a shallow diverticulum, referred to as a Hartmann's pouch, may be present on the inferior surface. Gallstones may become impacted within the Hartmann's pouch, causing cystic obstruction and acute cholecystitis (Podolsky et al., 2016).
- **Neck**, which narrows into the cystic duct.

As it emerges from the gallbladder, the mucous membrane of the gallbladder neck forms spiral valves (Heister valves) that are involved in regulating the flow into and out of the gallbladder. The **cystic duct** combines with the **hepatic duct** to form the **common bile duct**, which then joins with the main pancreatic duct to form the orifice called the ampulla of Vater that empties into the duodenum.

During passage through the duodenal wall, the common bile duct, the pancreatic duct, and the ampulla of Vater are surrounded by a complex arrangement of smooth muscles called the **sphincter of Oddi**.

The functions of the sphincter of Oddi are to:

- Regulate the flow of bile and pancreatic juices into the intestine.
- Inhibit entry of bile into the pancreatic duct.
- Prevent reflux of intestinal contents into the ducts.

The gallbladder is not an essential organ. If it is removed, bile flow continues to be regulated by the sphincter of Oddi.

Layers

The wall of the gallbladder has three layers:

1. **Outer serosa** that is derived from the peritoneum.
2. **Fibromuscular layer** that contains longitudinal and spiral smooth muscle and fibrous tissue.
3. **Inner mucosa** that is made up of simple columnar epithelium and arranged in folds, or rugae, similar to those of the stomach.

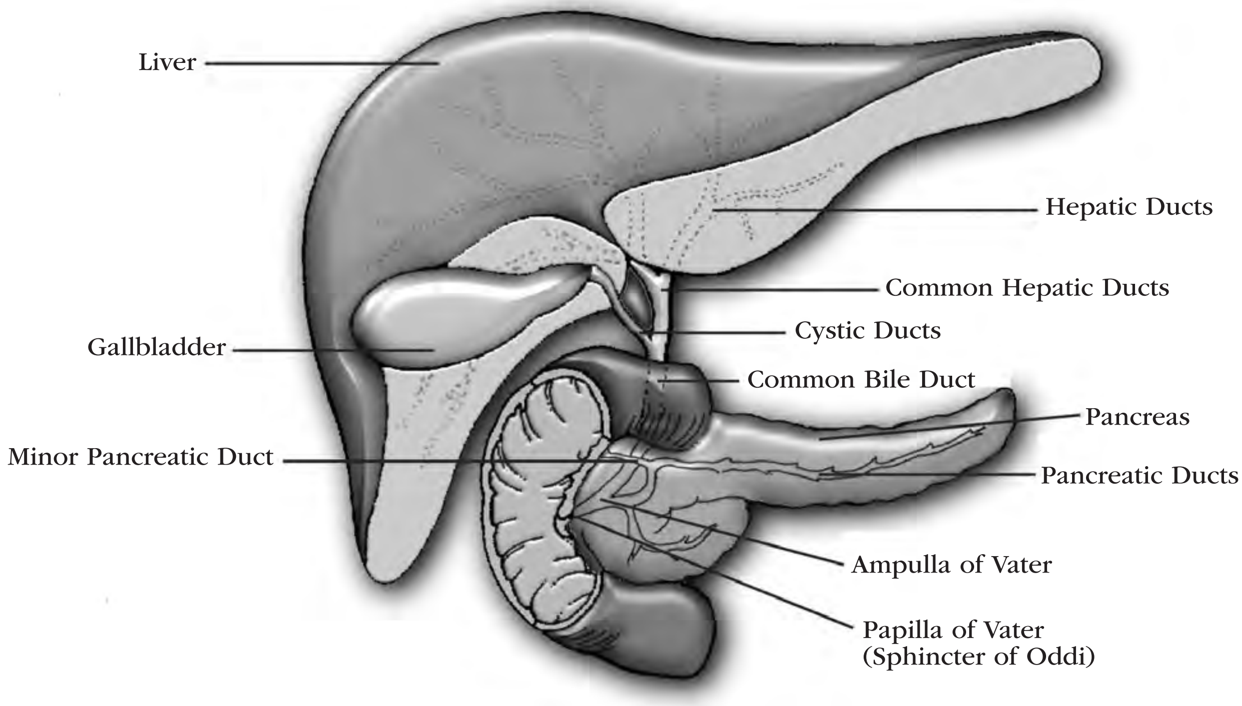

Figure 17.1. Anatomy of the Liver, Biliary, and Pancreatic Ductal Systems

Vascular supply

Arterial

Blood supply to the biliary system is delivered by the hepatic artery.

Venous

Blood is drained through the cystic vein.

Innervation

Innervation is both parasympathetic and sympathetic:

- **Parasympathetic innervation** is derived from the right branch of the vagus nerve. Mild parasympathetic stimulation causes the gallbladder to contract and relaxes the sphincter of Oddi at the duodenal junction.
- **Sympathetic innervation** is derived from the splanchnic nerve and the seventh through tenth thoracic segments. Sympathetic stimulation inhibits smooth muscle gallbladder contraction.

Secretion

Bile is an alkaline, greenish-yellow fluid that is secreted continuously by the liver. Its major components are water (which makes up 97% of hepatic bile), bile salts, fatty acids, lipids (mainly cholesterol and lecithin), inorganic electrolytes, conjugated bilirubin, and other organic substances. Bile flows from the liver through the cystic duct and into the gallbladder for storage. Within 4 hours of the bile's arrival in the gallbladder, up to 90% of the water in bile is removed, leaving a very concentrated mixture of sodium, bile salts, and other electrolytes.

Bile has a variety of functions, which include:

- Emulsification of undigested fats.
- Facilitation of the absorption of fat-soluble vitamins.
- Activation of intestinal and pancreatic enzymes.
- Provision of a route for the excretion of bilirubin; cholesterol; and certain sex, thyroid, and adrenal hormones.

Bile also affects the absorption of certain minerals and helps to neutralize gastric acid in the duodenum.

If the flow of bile through the common bile duct or hepatic ducts is obstructed by gallstones or other abnormalities, bile builds up in the blood, resulting in hyperbilirubinemia. Bile pigments, mainly bilirubin, are deposited in the skin, mucus membranes, and sclera, causing a yellow discoloration of these areas that is commonly known as **jaundice**. Jaundice also may result from destruction of red blood cells or from liver cell dysfunction.

Motility

Under normal conditions, the sphincter of Oddi remains slightly opened, allowing for a constant but miniscule amount of bile to enter the duodenum. During normal digestion, as food enters the duodenum, the hormone cholecystokinin-pancreozymin (CCK-PZ) is released, causing the gallbladder to contract, which in turn pushes increased amounts of bile into the duodenum. Other factors, such as certain drugs, sensory input, disease, and emotional states, also may affect the filling and contracting of the gallbladder.

DISORDERS OF THE BILIARY SYSTEM

Gallbladder and duct diseases may be short-term and easily correctable or long-term and debilitating. Generally, they present during middle age, and incidence increases with age. In people between ages 20 and 50, gallbladder and duct diseases are six times more common in women than in men; after age 50, the incidence in men and women equalizes (Scott & Rosh, 2017).

Biliary tract disorders include cholelithiasis, choledocholithiasis, cholecystitis, cholangitis, cholangiocarcinoma, congenital anomalies, and sphincter of Oddi dysfunction.

Cholelithiasis

Cholelithiasis is the presence of stones or calculi in the gallbladder (gallstones). As many as 1 million new cases of cholelithiasis are diagnosed each year, and approximately half of those patients undergo biliary surgery. Cholelithiasis is the fifth leading cause of hospitalization among adults and accounts for 90% of all diseases of the biliary system (Chemmanur & Smith, 2018).

There are two types of gallstones.

- **Cholesterol stones** include both pure cholesterol stones and mixed stones. These make up about 80% of all gallstones (Podolsky, 2016). Cholesterol stones are usually yellow-green in color and are made primarily of hardened cholesterol, although they also contain calcium salts, bile acids, fatty acids, protein, and phospholipids. Cholesterol stones are associated with either the hepatic production of bile that is supersaturated with cholesterol or reduced bile-salt secretion. When cholesterol is no longer soluble in this supersaturated system, it forms crystal nucleates that grow and cluster with each other and with other bile constituents to form recognizable stones in the gallbladder.
- **Pigment stones** are less common than cholesterol stones. They include black pigment stones made up of bilirubin polymers and inorganic calcium salts, and brown pigment stones composed principally of calcium bilirubinate and organic fatty-acid salts of calcium.

Gallstones may move around in the gallbladder as it empties and refills. If they remain in one place, they may be asymptomatic. However, stones that move to the neck, cystic duct, or common duct may obstruct these passages, resulting in mucosal irritation and subsequent bacterial invasion. Stones also may pass through the biliary ducts, causing pain, or they may obstruct the flow of bile if they become lodged there. It is possible for small stones to be located in any part of the biliary system without causing distress or to pass into the duodenum and be discharged in the stool. Gallstones can be as small as a grain of sand or as large as a golf ball.

Risk factors

Risk factors for the formation of cholesterol gallstones include:

- Being over 60 years of age.
- Being female.
- Pregnancy or use of oral contraceptives.
- Hormone therapy.
- Being of Mexican American, Native American, or European decent.
- Ileal disease (e.g., Crohn's disease), resection, or bypass.
- Certain forms of hyperlipidemia.
- Obesity.
- Rapid weight loss, weight-reduction diets, or bariatric surgery.
- Increased blood triglycerides.
- Gallbladder stasis.
- Spinal cord injury.
- Diabetes, hemolytic anemia, or sickle cell disease.
- Fasting.

Risk factors for the formation of pigment stones include:

- Increasing age.
- Chronic hemolysis, such as sickle cell disease or thalassemia.
- Alcoholism or alcoholic cirrhosis.
- Biliary infection, usually *Escherichia coli* (*E. coli*) or parasitic infestation, such as *Clonorchis sinensis* or *Ascaris lumbricoides*.
- Total parenteral nutrition (TPN).
- Vagotomy.
- Periampullary diverticula.
- Gallbladder stasis.

The prevalence of pigment stones is not influenced by gender.

Symptoms

Gallstones are asymptomatic in at least 50% of patients (Podolsky, 2016). In symptomatic patients, the severity, extent, and nature of the symptoms vary considerably, depending on the movement of the stones, the degree of obstruction, if any, and the presence or absence of inflammation.

Symptomatic patients usually experience steady pain, which often stems from gallbladder distention or spasms of the sphincter of Oddi or ductal muscles. This pain is referred to as **biliary colic**. Pain most often occurs 3-6 hours after the heaviest meal of the day, frequently during the early hours of the night. Pain often radiates to other parts of the abdomen or back and sometimes to the scapula, the middle of the back, or the tip of the right shoulder. Nausea and vomiting may occur, and pain may be relieved by vomiting. Patients may have vague symptoms of upper abdominal discomfort, increased eructation (belching), and dyspepsia. Some patients also have fever and chills, resulting from a common duct stone, acute cholecystitis, or associated pancreatitis.

In children, mild to moderate jaundice is a common symptom, but gallbladder enlargement is infrequent. Because the diagnosis of cholelithiasis is seldom considered in children, the delay between onset of symptoms and diagnosis in this age group may be from 1 to 5 years. Stones are generally found in the gallbladder and seldom occlude either the cystic or the common bile duct. Most children have a specific etiological factor, such as prolonged TPN, congenital hemolytic disease, or ileal disease.

Diagnosis

Ultrasonography is the most effective diagnostic technique for cholelithiasis. Ultrasonography can identify stones that are as small as 2 mm. Other tests that evaluate for gallstones include computerized tomography (CT) scan, magnetic resonance cholangiogram (MRC), magnetic resonance cholangiopancreatography (MRCP), cholescintigraphy (hepatobiliary [HIDA] scan), transabdominal or endoscopic ultrasonography, blood tests (liver and pancreas enzymes), oral cholecystogram (OCG), endoscopic retrograde cholangiopancreatography (ERCP), and examination of duodenal bile for cholesterol crystals or bilirubinate granules.

Complications

Potential complications of cholelithiasis include cholecystitis, cholangitis, abscess or fistula formation, perforation of the gallbladder, gangrene, pancreatitis, sepsis, jaundice, liver disease, and hepatic damage. Cholelithiasis also has been linked to gallbladder cancer.

Approximately 2% to 10% of cholecystectomy patients later exhibit postcholecystectomy syndrome, defined as abdominal pain or dyspepsia in patients who have had a cholecystectomy (Podolsky, 2016). Postcholecystectomy syndrome is most often seen in women who are between the ages of 40 and 49. Their distress cannot be attributed to the surgical procedure itself. Potential causes include residual biliary tract disease, nonbiliary digestive disease, nonspecific digestion dysfunction, and psychiatric disorders.

Treatment is determined by the specific diagnosis. Retained common bile duct stones, for example, are not uncommon following cholecystectomy. These stones may be removed by endoscopic introduction of a basket or balloon into the common bile duct, by a second surgical procedure, or through nonoperative manipulation of a basket inserted through a T-tube.

Treatment

Noninterventional "expectant" management is preferred for asymptomatic patients because neither medical dissolution therapy nor elective cholecystectomy is warranted in such cases.

Management of patients with minor symptoms may include pain relief and dietary control to reduce fat intake. Small, frequent meals can sometimes help to prevent future attacks. In children, cholecystectomy is probably indicated, regardless of the severity of symptoms.

Surgery is the treatment of choice for most symptomatic patients. Typically, a cholecystectomy is used to excise the gallbladder and ligate the cystic duct. The most common operation is laparoscopic cholecystectomy. Prognosis is generally good but may vary depending on the patient's age, sex, and the presence of complicating conditions.

In addition to cholecystectomy, a variety of techniques exist for the dissolution or removal of cholesterol stones, including:

- **Oral dissolution therapy.** This therapy consists of drugs made from bile acid that are used to dissolve cholesterol stones. Ursodeoxycholic acid (ursodiol) and chenodeoxycholic acid (chenodiol) work best for small cholesterol stones. Recurrence rate of stone formation is high, so oral dissolution therapy is usually reserved for patients who are unable to tolerate surgery.
- **Contact dissolution therapy.** This treatment consists of a continuous infusion of methyl tert-butyl ether into the gallbladder to dissolve stones.
- **Extracorporeal shock-wave lithotripsy (ESWL).** Biliary lithotripsy, which involves fragmentation of gallstones by use of extracorporeal acoustic shock waves, is often combined with dissolution therapy. This treatment method can be done on an outpatient basis. Patients have less pain, a shorter recovery period, and less chance of infection than those who have cholecystectomies.

Disadvantages of these medical therapies are the probability of recurrence of cholelithiasis after cessation of the therapeutic modality and concerns regarding the effects of long-term use of these treatments

No medical intervention options exist for pigment stones. Surgery is the only treatment for symptomatic patients with stones of this type.

Choledocholithiasis

The presence of gallstones in the common bile duct or the hepatic duct is called **choledocholithiasis**. This problem occurs when stones pass out of the gallbladder or biliary tree and lodge in the hepatic duct or the common bile duct and obstruct the flow of bile into the duodenum.

Risk factors

Stone formation in the common bile duct can still happen after the gallbladder is removed. A few people form gallstones in the common bile duct (primary stones). Primary stones are always pigment stones. Pigment stones can also be secondary stones, meaning that they were formed in the gallbladder and traveled to the common bile duct. Cholesterol stones found in the duct are always secondary stones. About 40% of all common bile duct stones are pigment stones (Feldman, 2016).

Symptoms

Patients with choledocholithiasis may have no symptoms or may present with any combination of the following:

- Biliary colic, with constant epigastric or right upper quadrant pain or tenderness.
- Obstructive jaundice and pruritus.
- Cholangitis, with fever, chills, and jaundice (known as Charcot's triad).
- Loss of appetite, nausea, and vomiting.
- Acute gallstone pancreatitis, manifested by severe abdominal pain that radiates into the back.

Diagnosis

Differential diagnosis may require laboratory blood tests (bilirubin, complete blood count, pancreatic enzymes, liver enzymes), ultrasonography, abdominal CT scan, radioisotope imaging, OCG, ERCP, MRCP, or percutaneous transhepatic cholangiography (PTC). Other diagnostic modalities include endoscopic ultrasound (EUS) and MRC, which, although highly specific to identifying choledocholithiasis, lack therapeutic options.

Complications

Complications of choledocholithiasis may include cholangitis, cirrhosis with hepatic failure, portal hypertension, hepatic abscess formation, or gallstone pancreatitis.

Treatment

Patients with choledocholithiasis should be kept with nothing by mouth (NPO). They should be stabilized using intravenous (IV) hydration and electrolyte replacement. Analgesia should be provided, nasogastric (NG) suction may be used, and antibiotics may be prescribed if there is evidence of sepsis or cholangitis.

Therapeutic ERCP is the preferred method of treatment. During ERCP, endoscopic sphincterotomy and stone removal can be done. This involves cutting the muscle fiber of the sphincter of Oddi and using a balloon catheter or basket to extract the stones. Large stones may be broken up using mechanical, electrohydraulic, or laser lithotripsy before removal. Alternatively, the common bile duct may be explored surgically through a choledochotomy.

Cholecystitis

Cholecystitis is an acute or chronic inflammation that causes painful distention of the gallbladder. Of patients who require surgery for gallbladder disease, 10% to 20% have cholecystitis (Feldman, 2016).

Most cholecystitits in the pediatric population is chronic and associated with gallstones. The presenting symptom is right upper quadrant abdominal pain, sometimes radiating to the back, with complaints of vomiting. In 25% to 30% of children, jaundice and fever are seen and are more common in young infants.

Cholecystitis can also occur without evidence or presence of gallstones. This is rare in adults but occurs with surprising frequency in children. The cause is unknown but has been associated with immediate postoperative states, trauma, or burns. This condition usually presents with fever, abdominal pain, and jaundice. A laparoscopic cholecystectomy is the surgical treatment of choice for cholecystitis.

Acute calculous cholecystitis

More than 90% of the time, cholecystitis is associated with a gallstone that is impacted in the cystic duct, a condition known as acute calculous cholecystitis (Podolsky, 2016). The obstructed gallbladder becomes distended, and the walls become edematous, ischemic, and inflamed. Secondary infections with enteric organisms may compound the inflammation, leading to cholangitis and sepsis.

Risk factors

Predisposing factors to acute calculous cholecystitis include older age, ethnicity (especially persons of Italian, Jewish, or Chinese descent), obesity, a sedentary lifestyle, pregnancy, hemolytic anemias, and insulin-dependent diabetes mellitus.

Symptoms

Acute calculous cholecystitis produces the following symptoms, which are similar to those of cholelithiasis:

- Acute abdominal pain, usually midepigastric or localized to the right upper quadrant, with radiation to the shoulders and back.
- Nausea, vomiting, and anorexia.
- Fever, headache, and leukocytosis.
- Tachycardia and tachypnea.
- Tenderness, guarding, and rebound tenderness in the right upper quadrant.
- Intolerance of fatty foods and heavy meals.

Diagnosis

To confirm the diagnosis, the physician may order blood tests, ultrasonography, and radioisotope imaging.

Complications

Potential complications of acute calculous cholecystitis include perforation with subsequent peritonitis; cholecystenteric fistula; and gallstone ileus, a form of intermittent or permanent intestinal obstruction caused by the impaction of a large gallstone that has entered the intestine through a cholecystenteric fistula. Such obstructions are most often found in the ileum because of its relatively small diameter compared to the rest of the intestine. Gallstone ileus requires an emergency laparotomy.

Treatment

Patients with acute calculous cholecystitis should be stabilized with NG suctioning, IV fluid and electrolyte replacement, and analgesia. Antibiotics should be given in cases of severe illness, sepsis, or complications. Early cholecystectomy is the preferred treatment approach.

If cholecystectomy is contraindicated by the patient's condition, an operative or percutaneous cholecystostomy may be performed. In this procedure, the gallbladder is evacuated of stones and infected bile, and a catheter drains to the outside of the body.

Temporary or short-term biliary decompression can also be accomplished using ERCP to place a nasobiliary catheter (NBC) or internal stent above the impacted stone. An NBC consists of a long polyethylene tube, one end of which is placed inside the biliary tree, with the other end exiting through the nostril and connecting to a bile drainage bag. This allows the NBC to function as a T-tube. An internal stent is a plastic tube placed into the biliary tree, which allows bile to flow into the duodenum without the patient having a tube in the nose.

Acalculous cholecystitis

Acalculous cholecystitis, which is cholecystitis in the absence of stones, is relatively rare.

Risk factors

Acalculous cholecystitis usually occurs in otherwise severely ill, hospitalized patients. Conditions may include:

- Intercurrent illnesses such as bacterial enteric infections, including typhoid fever, shigellosis, or *E. coli* infection; viral gastroenteritis; scarlet fever; respiratory infections; pneumonia; human immunodeficiency virus (HIV) infection; or hepatitis A infection.
- Parasitic infestations with *Giardia* or *Ascaris* found in the gallbladders of some patients.
- Hospitalization for burns, trauma, or major surgery.
- Mechanical ventilation with positive end-expiratory pressure (PEEP). Most patients requiring PEEP have been receiving TPN for longer than 3 months. The pathogenesis of cholecystitis may result from the absence of oral intake associated with gallbladder stasis, sludge formation, and increased biliary pressure caused by narcotic drugs that increase the tone at the sphincter of Oddi.

Symptoms

Symptoms of acute acalculous cholecystitis include an acute onset of abdominal pain on the right side, abdominal tenderness and guarding, vomiting, and nausea.

Diagnosis

Usually, the diagnosis is made by an abdominal CT scan or abdominal ultrasonography and laboratory blood tests, such as complete blood count and liver function tests. Blood cultures should also be performed.

Complications

The incidence of gangrene, necrosis, and perforation is high in patients with acalculous cholecystitis, and mortality may be as high as 50% (Feldman, 2016).

Treatment

Because of the high incidence of complications, the treatment of choice for acalculous cholecystitis is urgent cholecystectomy or cholecystostomy. Patients should also be treated with antibiotics to cover enteric organisms and enterococcus, and other supportive measures should be taken as needed. High-risk surgical patients may receive an endoscopic gallbladder stent placement as a palliative treatment until a cholecystectomy can be performed.

Emphysematous cholecystitis

In emphysematous cholecystitis, the gallbladder walls and bile ducts contain gas that has been produced by infective organisms, such as *E. coli*, *Clostridium*, and other anaerobes.

Risk factors

This condition is usually seen in males older than 60 years of age with a history of type II diabetes mellitus.

Symptoms

Common complaints of emphysematous cholecystitis are fever and right upper quadrant pain that often radiates to the back.

Complications

Complications for emphysematous cholecystitis include gallbladder gangrene and perforation (Feldman, 2016).

Treatment

The treatment for emphysematous cholecystitis is acute cholecystectomy.

Cholangitis

Cholangitis is an uncommon bacterial infection of the bile duct that is often associated with choledocholithiasis or with obstruction of the hepatic or common bile ducts by strictures, cysts, fistulas, neoplasms, or parasites. Cholangitis is equally seen in females and males who are age 50 and older (Scott & Rosh, 2017). Widespread inflammation may cause fibrosis and stenosis of the common bile duct.

Risk factors

Cholangitis is caused by an impacted stone in the biliary system, resulting in bile stasis. In Western countries, choledocholithiasis is the most common cause of acute cholangitis, followed by ERCP and tumors (Feldman, 2016).

Any condition that leads to stasis or obstruction of bile in the common bile duct can result in bacterial infection and cholangitis. Partial obstruction is associated with an increased rate of infection compared with that of complete obstruction.

Causes of cholangitis include:

- Obstructive tumors, including pancreatic cancer, cholangiocarcinoma, ampullary cancer, and porta hepatis tumors or metastasis.
- Parasitic infection, including *Ascaris lumbricoides* infections.
- Benign or malignant strictures or stenosis.
- Endoscopic manipulation of the common bile duct.
- Choledochocele.
- Extrinsic compression by the pancreas.
- Sclerosing cholangitis (from biliary sclerosis).
- Acquired immunodeficiency syndrome (AIDS) cholangiopathy.

Symptoms

Most patients experience a transient, self-limited illness that is characterized by a fever spike, chills, dark urine, and abdominal pain.

Complications

In some patients, the illness is devastating, consistent with profound toxic sepsis with shock and impaired mental function. Prognosis for these patients is poor.

Treatment

Cholangitis is a medical and surgical emergency. The patient should be stabilized with IV hydration and antibiotics, which should be followed by prompt surgical or endoscopic decompression of the common bile duct. Drainage of the biliary tree should produce immediately beneficial results.

If the common bile duct is explored surgically to remove retained stones, a T-tube is inserted to provide a means of decompressing the biliary ducts. The tube can be removed in the clinician's office 3 weeks after surgery. Before postoperative extraction of the T-tube, a cholangiogram should be performed to detect residual stones.

Retained common bile duct stones can be removed by inserting a basket through the T-tube or by endoscopically introducing a basket or balloon for extraction. Large stones that are difficult to remove may be fragmented by a mechanical lithotripter basket and the fragments pulled out by a balloon.

Primary sclerosing cholangitis

Primary sclerosing cholangitis (PSC) is an inflammatory process that results in multiple strictures of the bile ducts, causing chronic cholestatic liver disease.

Risk factors

The etiology of PSC is unknown. Some 75% to 90% of patients with PSC have inflammatory bowel disease (IBD)—Crohn's disease or ulcerative colitis (UC)—but only 5% of patients with IBD develop PSC. PSC has no correlation with UC exacerbations or remissions. Men are affected twice as often as women, and the average age of diagnosis is 40 (Feldman, 2016).

Symptoms

Clinical manifestations of PSC are progressive fatigue, jaundice, pruritis, abdominal pain, and elevated serum alkaline phosphatase.

Diagnosis

Diagnosis is by ultrasonography, ERCP or PTC, MCRP, and liver biopsy examination. ERCP may show strictured areas in the intrahepatic and extrahepatic ducts, with areas of dilatation between the strictures, giving a beaded appearance to the ducts.

Complications

In patients requiring liver transplantation for advanced stages of PCS, progression to cirrhosis and portal hypertension is expected, and death occurs from liver failure.

Treatment

Subclinical PSC requires no treatment. If pruritis occurs, it can be treated with bile-salt binding agents such as cholestyramine. To bypass tight strictures, surgery or endoscopic balloon dilatation and/or placement of biliary stents may be required.

Liver transplant may be the only viable treatment in the advanced stages of the disease and is the third most common reason for transplant in the United States. Patients with PCS may require more than one liver transplant (Feldman, 2016).

Cholangiocarcinoma

Cholangiocarcinoma is cancer of the biliary system. It may involve the gallbladder or, less often, the bile ducts.

Gallbladder cancer

Cancer of the gallbladder accounts for approximately 3% of all cancers. Approximately 80% of gallbladder cancers are adenocarcinomas. The remainder are squamous cell carcinomas, adenocanthomas, and others. Benign tumors are rare. Most patients have evidence of local or metastatic spread, often through the lymph nodes, at the time of diagnosis (Feldman, 2016).

Risk factors

Cancer of the gallbladder occurs more frequently in women than in men, and the incidence peaks at age 70 years. Approximately 80% of patients with gallbladder cancer also have gallstones (Feldman, 2016).

Symptoms

Patients with gallbladder cancer usually present with vague abdominal symptoms, including nausea, vomiting, substantial weight loss, anorexia, fat intolerance, jaundice, and right upper quadrant pain. If a palpable right upper quadrant mass can be felt, the lesion is almost always incurable.

Diagnosis

Diagnostic tests for gallbladder cancer include abdominal ultrasound, carcinoembryonic antigen (CEA) assay, CT scan, cholecystogram, cholangiogram, PTC, EUS, ERCP, and possibly magnetic resonance imaging (MRI). In practice, cancer of the gallbladder is rarely diagnosed preoperatively because signs and symptoms are similar to those for cancer of the liver, cancer of the pancreas, and obstructive cholelithiasis.

Complications

For most patients, resection, radiotherapy, and chemotherapy seem to do nothing to prolong survival.

Treatment

Medical treatment should be supportive and symptomatic, with the goals of comfort and short-term rehabilitation. If surgery is performed, the average postoperative survival time is only 8 months (Feldman, 2016). Cholecystectomy is indicated only for patients with small, localized tumors. If the lesion is inoperable, an internal bile drainage system may be inserted to permit the flow of bile directly from the liver into the intestine. The objectives of this procedure are to relieve symptoms and prolong the quality and length of life.

To minimize malnutrition and dehydration, the patient may be given antiemetics or small sips of carbonated beverages to control nausea; vitamin and mineral replacements; frequent small meals; small amounts of pain medication before meals; and, if necessary, a tube feeding. In the terminal stages of the disease, skin care, pain relief, and emotional support are of vital importance.

Bile duct cancers

Bile duct cancer is cancer of the extrahepatic biliary tree. Adenocarcinoma is the most common form of extrahepatic bile duct cancer, and benign neoplasms are relatively rare.

Risk factors

Bile duct cancer is associated with gallstones in only 30% of patients. It may also be associated with longstanding UC, Crohn's disease, PSC, and/or congenital dilatations of the bile ducts (e.g., choledochal cysts) (Feldman, 2016).

Symptoms

Often, patients with bile duct cancer present with painless obstructive jaundice, a serious symptom that represents a probable clinical presentation of bile duct carcinoma. Later, pruritus, nausea, vomiting, weight loss, and intermittent or steady right upper quadrant pain may develop. Serum alkaline phosphatase is always elevated.

Diagnosis

Bile duct cancer is most often diagnosed during a laparotomy for other biliary tract disease.

Ultrasonography (endoscopic or laparascopic), MRI, MRCP, magnetic resonance angiography (MRA), cholangiography, positron emission tomography (PET) scan, or CT scan may be used to visualize dilated intrahepatic bile ducts, and PTC or ERCP are used preoperatively to define the level and cause of the obstruction.

Complications

Most patients have localized extension or metastatic disease at the time of diagnosis.

Treatment

Common surgical treatments for bile duct cancer may consist of surgical removal of the duct, a partial hepatectomy, the Whipple procedure, or liver transplantation. However, treatment options are based on the type and stage of the cancer, possible side effects, the overall health of the patient, and the patient's preferences.

Other forms of treatment may include stent placement, often using a permanent wall stent, and surgical bypass; radiation therapy; chemotherapy; or palliative and supportive care.

Although rare, the most common pediatric neoplasm of the biliary tract is botryoid embryonal rhabdomyosarcoma. Symptoms include pruritus, jaundice, right upper quadrant and epigastric abdominal pain, and sometimes a palpable mass. The tumor is usually locally invasive. Even with a large surgical resection, radiation, and drug therapy, prognosis can be poor.

Congenital abnormalities

Congenital anomalies may occur in either the gallbladder or the bile ducts. Anomalies of either the gallbladder or the bile ducts may impair the normal flow of bile, resulting in **cholestasis**. This leads to sludging, the formation of cholesterol monohydrate crystals and bilirubin granules embedded in a matrix of mucous gel.

Congenital anomalies of the bile ducts often present in infancy, whereas congenital gallbladder anomalies are rarely of clinical significance until adulthood. Biliary anomalies do not follow recognizable patterns of genetic inheritance, nor are they associated with other developmental abnormalities. This suggests that they may be caused by pathogenic factors, such as viruses, drugs, or toxins transmitted by the mother.

Gallbladder anomalies

Congenital anomalies of the gallbladder include:

- Agenesis, or congenital absence of the gallbladder, which most likely occurs as a result of embryonic maldevelopment.
- Anomalies of location (ectopic gallbladder), which may occur anywhere in the abdomen and often require cholecystectomy.
- Anomalies of form, in which more than one cystic structure is found in the gallbladder fossa, suggesting either double gallbladder, bilobed gallbladder, folded

fundus, or Ladd's bands across a normal gallbladder. If two organs exist and one is found to be diseased at the time of surgery, both should be removed.
- Anomalies of fixation, which occur when the mesenteric supporting structures of the gallbladder are elongated, thus leaving a normally functioning gallbladder to "float" below the inferior surface of the liver and occasionally into the pelvis. If this floating gallbladder twists, vascular occlusion, ischemic necrosis, or perforation may occur, requiring cholecystectomy.

Bile duct anomalies

Congenital anomalies of the hepatic and common bile ducts include:
- Anomalies of extrahepatic duct configuration such as atresia, accessory ducts, abnormal lengths of ducts, and variations in the junction of the cystic and hepatic ducts. Liver transplantation has greatly improved the survival rate of children with biliary atresia.
- Cystic anomalies of the common bile duct, including cystic dilatation of the common bile duct (choledochal cyst), which makes up the majority of these anomalies; congenital choledochocele; and congenital diverticulum of the common bile duct. In patients with choledochal cyst, symptoms generally appear before the age of 10 years. Diagnosis is by ultrasonography, and the recommended treatment is Roux-en-Y choledochocystojejunostomy with cholecystotomy. Complete excision of the cyst is recommended (Feldman, 2016).
- Cystic dilatation of the intrahepatic ducts, called Caroli disease. This rare genetic disorder most often occurs in childhood or young adulthood. Symptoms include bile stasis, cholangitis, and intrahepatic stone or abscess formation.

CASE SITUATION

Dr. Ross Lazar, a pediatrician, was diagnosed with UC at age 24. In the following 5 years, he had two exacerbations of the disease that required medication, but they were not as serious as his first attack. For the last 13 years, the disease has been in remission, and Ross has required no medication.

Now, at age 42, Ross has experienced fatigue and pruritus for several months. Last week, he noticed yellowing of his skin and eyes and sought out the gastroenterologist again. He says he has been feeling fine for the last 13 years, except for the recent fatigue and itching. He denies any pain, gallbladder disease, hepatitis, IV drug use, or alcohol abuse. He has had no blood transfusions. His blood work shows elevated total bilirubin and alkaline phosphatase, with normal serum transaminases. His hepatitis panel is all negative. Ultrasonography shows that the biliary system and pancreas are normal.

Points to think about

1. Because of the negative hepatitis panel and normal ultrasonography, and given the past history of UC, what might be the diagnosis for Ross?
2. Ross will undergo an ERCP to visualize the biliary tree. The ERCP will most likely show strictured areas in the intrahepatic and extrahepatic ducts, with areas of dilatation between the strictures, which can produce a beaded appearance to the ducts. How can the gastroenterology nurse help Ross before, during, and after this procedure?
3. What data are available in relation to the nursing diagnosis "alteration in comfort?"
4. What outcome criteria and associated nursing interventions may be devised for this diagnosis?

Suggested responses

1. Based on Ross's negative hepatitis panel, normal ultrasound examination, and past history of UC, primary sclerosing cholangitis (PSC) might be the diagnosis. Up to 5% of patients with UC develop PSC, but PSC has no correlation with UC exacerbations or remissions. Males with UC who are younger than age 45 account for 60% to 70% of patients with PSC. It can be diagnosed before or after a diagnosis of UC and can occur even after a total colectomy (Feldman, 2016).
2. Educational endeavors by the gastroenterology nurse must take into account the patient's current level of understanding and his readiness to learn. In this case situation, there are two possible inferences about Ross's level of understanding:
   a. Ross has the usual medical school foundation with reference to UC and its accompanying problems and diagnostic procedures.
   b. Because he has lived with the diagnosis of UC for many years, Ross has extensive knowledge of UC and its accompanying problems and diagnostic tests. He may have already made a tentative diagnosis of PSC before he consulted the gastroenterologist.

   Although the second inference would seem the most likely, it should not be assumed without testing its truth. With regard to the ERCP procedure, the gastroenterology nurse can determine Ross's level of knowledge through the use of straightforward or open-ended questions such as:
   a. "Do you have any questions about the ERCP procedure?"
   b. "Are you aware of what happens during the ERCP procedure?"

   If responses do not elicit sufficient details, leading statements and other open-ended questions can focus on specific aspects of the procedure. The gastroenterology nurse must be aware that Ross may choose not to respond.

Depending on the patient's knowledge of the procedure, the gastroenterology nurse might consider the following pre-procedure activities:

a. Assess the patient's knowledge and offer any necessary explanations of what will happen during the procedure.
b. Encourage questions.
c. Explain the duration of the procedure.
d. Explain the X-ray room and that radiographic examinations will be taken during the procedure.
e. Explain the medications given and their expected effects.
f. Reinforce awareness that the gastroenterology nurse will be with the patient during the entire procedure, monitoring the patient's response to sedative drugs and to the procedure and taking vital signs regularly.
g. Review the patient's record before the procedure to refresh recall of laboratory data and be aware of incipient problems during the procedure.
h. Review probable post-procedure events with the patient.

3. Data collected from the physical exam and the patient's report of symptoms demonstrate that Ross is still experiencing pruritus that, although not severe, is bothersome and widespread to his torso, legs, and arms. Ross does not have open lesions that are associated with severe scratching on affected areas of his skin. The nursing diagnosis "alteration in comfort related to pruritus caused by PSC" seems applicable.
4. Outcome criteria and associated nursing interventions for the diagnosis "alteration in comfort related to pruritus caused by PSC" are:

Table 17.1. Nursing Diagnosis "Alteration in Comfort Related to Pruritus Caused by PSC"

Outcome criteria	Nursing interventions
Patient will report some relief from pruritus	• Cool skin with wet compresses or baths • Pat skin dry after bathing • Apply prescribed antipruritic drugs • Apply pressure to itchy areas instead of scratching
Patient will sleep for 2- to 3-hour intervals during the night	• Encourage therapeutic bath before retiring • Have patient keep a log of sleeping/waking periods • Place a clock and writing material at bedside

REVIEW TERMS

bile, biliary colic, cholangitis, cholecystitis, choledocholithiasis, cholelithiasis, cholestasis, common bile duct, cystic duct, gallbladder, hepatic duct, jaundice, primary sclerosing cholangitis (PSC), sphincter of Oddi

REVIEW QUESTIONS

1. The gallbladder wall is made up of:
   a. Serosa, muscularis, submucosa, and mucosa.
   b. Serosa, a fibromuscular layer, and mucosa.
   c. Muscularis, submucosa, and mucosa.
   d. Serosa, a fibromuscular layer, submucosa, and mucosa.
2. Blood is supplied to the gallbladder by the:
   a. Superior mesenteric artery.
   b. Hepatic artery.
   c. Celiac artery.
   d. Inferior phrenic artery.
3. What is the maximum amount of bile that can be stored in the gallbladder?
   a. 5 ml.
   b. 10 ml.
   c. 50 ml.
   d. 500 ml.
4. What is the major component of the bile that is produced by the liver?
   a. Cholesterol.
   b. Bilirubin.
   c. Bile salts.
   d. Water.
5. By far, the most common disease affecting the biliary system is:
   a. Choledocholithiasis.
   b. Cholecystitis.
   c. Cholelithiasis.
   d. Cholangitis.
6. Chronic hemolytic disease, total parenteral nutrition (TPN), and alcoholism are among the risk factors for the formation of:
   a. Cholesterol gallstones.
   b. Pigment gallstones.
   c. Cholangitis.
   d. Gallbladder cancer.
7. The primary disadvantage of most nonsurgical treatment alternatives to cholecystectomy is:
   a. Potential recurrence of cholelithiasis after cessation of treatment.
   b. Unpleasant side effects.
   c. The need for specialized equipment.
   d. The need for specially trained personnel.
8. For most patients, the treatment of choice for choledocholithiasis is:
   a. Sphincteroplasty.

   b. Cholecystectomy.
   c. Dissolution of stones using ursodiol or chenodiol.
   d. Endoscopic retrograde cholangiopancreatography (ERCP) with sphincterotomy and stone removal.
9. The most common cause of cholecystitis is:
   a. Cholecystenteric fistula.
   b. Bacterial infection.
   c. A gallstone impacted in the cystic duct.
   d. Gallstone ileus.
10. Treatment for cancer of the gallbladder most often involves:
   a. Supportive and symptomatic measures only.
   b. Cholecystectomy.
   c. Pancreaticoduodenectomy.
   d. Insertion of an internal drainage system.
11. The sphincter of Oddi functions to:
   a. Regulate bile and pancreatic juice flow.
   b. Inhibit entry of bile into the pancreatic duct.
   c. Prevent reflux of intestinal content into the biliary duct.
   d. All of the above.

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Chapter 18

PANCREAS

This chapter will acquaint the gastroenterology nurse with the normal anatomy and physiology of the pancreas and will discuss the pathophysiology, diagnosis, and treatment of selected pancreatic disorders.

Learning objectives

After reviewing the content of this chapter, the gastroenterology nurse will be able to:

1. Describe the normal anatomy of the pancreas.
2. Explain the normal physiological functions of the pancreas, both endocrine and exocrine.
3. Discuss pathophysiology, diagnosis, and treatment of disorders of the pancreas.

ANATOMY AND PHYSIOLOGY

The **pancreas** is a fish-shaped, lobulated gland that lies behind the stomach (Figure 13.2). It is approximately 15-20 cm (6-8 inches) long and 5 cm (2 inches) wide and has an average weight of less than 110 g (about 4 ounces). The pancreas may be referred to in five distinct components.

- The **head** is the widest part of the pancreas and is found in the right side of the abdomen. The head lies over the vena cava in the c-shaped curve of the duodenum.
- The **uncinate process** is the part of the head of the pancreas that hooks toward the back of the abdomen. The uncinate hooks around two very important blood vessels: the superior mesenteric artery and the superior mesenteric vein.
- The **neck** is the thin section of the gland between the head and the body of the pancreas.
- The **body** is the middle part of the pancreas, which lies behind the duodenum and extends across the abdomen, behind the stomach, and across the spine. The superior mesenteric artery and vein run behind this part of the pancreas.
- The **tail** is the thin tip of the pancreas in the left side of the abdomen in close proximity to the spleen.

Layers of the pancreas

The pancreas contains two basic cell types: exocrine cells and endocrine cells.

Exocrine cells

The pyramidal acinar cells are exocrine cells that make up the majority of the pancreatic tissue. Groups of acinar cells form an **acinus**, and groups of acini (plural of acinus) form grape-like lobules. The lobules, in turn, are joined together by connective tissue into lobes, which unite to form the entire gland.

The acini are arranged around a small, central lumen into which they drain their enzymes. The central lumina—of which the most proximal portion is lined by clear, cuboidal cells called centroacinar cells—is drained by means of ductules. The ductules drain into multiple intralobular ducts, which in turn drain into the **duct of Wirsung**, the main pancreatic duct. The duct of Wirsung runs the whole length of the pancreas from left to right and joins the common bile duct, emptying into the duodenum at the **papilla of Vater**. Most individuals have an accessory duct called the **duct of Santorini**, which leads from the head of the pancreas and drains into the duodenum at an accessory or minor papilla that lies just proximal and anterior to the papilla of Vater.

Endocrine cells

Endocrine cells make up the remaining 1% of the pancreatic cells. They are located in the **islets of Lang-**

erhans, which are embedded in the loose connective tissue between the lobules, mainly in the tail.

Vascular supply

Arterial

Branches of the splenic, superior mesenteric, and celiac arteries supply blood to the pancreas.

Venous

Blood is drained away from the pancreas into the portal or splenic circulation by the superior mesenteric and splenic veins.

Innervation

Sympathetic nerve fibers control pain sensation, blood flow, and enzyme secretion in the pancreas. Parasympathetic fibers control exocrine and endocrine function.

Secretion

The pancreas is both an exocrine and an endocrine organ and has two main functions.

- **Endocrine function.** The pancreas produces hormones that regulate blood sugar levels.
- **Exocrine function.** The pancreas produces juices (enzymes) that digest food.

Endocrine function

The three types of endocrine cells in the pancreas are:

- **Alpha cells**, which produce glucagon. When the blood sugar level falls below normal, the alpha cells are stimulated to secrete glucagon, which accelerates the conversion of glycogen to glucose in the liver.
- **Beta cells**, which produce insulin. When the blood sugar level is above normal, the beta cells secrete insulin, which promotes both the metabolism of glucose by the tissue cells and the conversion of glucose to glycogen, which is then stored in the liver and muscles.
- **Delta cells**, which produce the hormone somatostatin. Somatostatin secretion inhibits insulin and glucagon secretions, as needed.

These endocrine products are released directly into the circulation. Despite being only 1% of the pancreatic mass, **islet cells** receive about 10% to 15% of the pancreatic blood flow. This allows hormones secreted by the cells close access to the circulation. Their innervation by parasympathetic and sympathetic neurons modulates secretion of insulin and glucagon.

Exocrine function

The exocrine **acinar cells** secrete 500 ml to 1.5 L of pancreatic juice daily. This colorless fluid has a pH of 8.3 and consists of water, bicarbonate, enzymes, potassium, sodium, chloride, and calcium.

The three major types of enzymes secreted by the pancreas include:

- **Amylase**, predominantly a-amylase, which hydrolyzes carbohydrates into glucose and maltose.
- **Lipase**, including pancreatic lipase and phospholipase A, which are important in early stages of fat digestion.
- **Protease**, including trypsinogen, the precursor of trypsin, which break amino acid bonds of protein chains and convert other proenzymes to their active forms.

Pancreatic secretion

Pancreatic secretions are controlled by hormones:

- **Secretin.** When the chyme is mainly acidic, secretin stimulates the release of pancreatic juice that is rich in bicarbonate and water.
- **Cholecystokinin-pancreozymin (CCK-PZ).** When the chyme is made up predominantly of undigested proteins and fats, the duodenum releases CCK-PZ, which stimulates the release of enzyme-rich pancreatic juice.

Pancreatic juice enters the duodenum with the biliary secretions at the papilla of Vater.

Pancreatic secretion occurs in four phases:

- **At rest.** When at rest, the pancreas secretes bicarbonate at about 2% of the maximal rate and enzymes at about 15% of the maximal rate.
- **Cephalic phase.** This phase occurs when the sight and smell of food stimulates a modest output of enzyme-rich pancreatic juice.
- **Gastric phase.** Distention of the stomach stimulates the secretion of a moderate amount of pancreatic juice that is rich in enzymes but low in bicarbonate.
- **Intestinal phase.** The delivery of food into the proximal intestine evokes secretion of pancreatic enzymes at about 70% of the maximal rate. The volume of pancreatic juice and bicarbonate output increases as the pH of the meal decreases and the acid load increases.

In the absence of pancreatic enzymes, up to 40% of dietary fat and protein may be assimilated using other, less efficient pathways of digestion such as salivary amylase, pharyngeal lipase, and peptidases from the brush borders of the gut mucosa. It has been estimated that pancreatic enzyme secretion must fall below 10% of normal before maldigestion or malabsorption occurs (Singh et al., 2017).

DISORDERS OF THE PANCREAS

Noteworthy pathological conditions that affect the pancreas include pancreatitis, pancreatic fluid collection, pancreatic fistula, pancreatic carcinoma, pancreatic tumors, pancreatic exocrine insufficiency, cystic fibrosis, congenital anomalies, and pancreatic diseases in children.

Pancreatitis

Pancreatitis is an inflammatory condition of the pancreas that can result from obstruction of the pancreatic duct and from other mechanisms such as trauma, toxicity from alcohol, infections, and certain drugs. It may be acute or chronic and represents a wide spectrum of severity and clinical presentations. Many aspects of the pathogenesis of acute and chronic pancreatitis are poorly understood. Therapy is often supportive and may be directed at a specific etiology or at morphological changes in the pancreatic duct (e.g., stricture, obstruction, disruption, stones).

Acute pancreatitis

Acute pancreatitis is a clinical condition defined when a patient presents with two of the three following criteria:

- Abdominal pain (often in the epigastric region).
- Serum amylase or lipase level greater than three times the normal limit.
- Radiologic imaging that is consistent with the diagnosis, usually computerized tomography (CT) scan or magnetic resonance imaging (MRI).

The characteristic anatomical changes associated with acute pancreatitis are the result of enzymatic digestion of the pancreatic parenchyma and peripancreatic tissues by enzymes that are normally present in the pancreas in their inactive proenzyme form. This produces varying degrees of edema, necrosis, hemorrhage, and the release of harmful substances (cytokines), which can injure distant organs and lead to severe complications, including death.

Risk factors

Common causes of acute pancreatitis include alcohol misuse, gallstones, abdominal trauma, hyperparathyroidism, hyperlipidemia, infections, and certain drugs.

This condition also occurs after **endoscopic retrograde cholangiopancreatography (ERCP)**, and the incidence varies. Studies have indicated patient risk factors including age, therapeutic **endoscopic sphincterotomy**, sphincter of Oddi dysfunction, and prior history of post-ERCP pancreatitis. Technical factors such as difficult cannulation, the number of pancreatic duct injections, and the amount of contrast injected are also attributed to the development of acute pancreatitis (Thaker, Mosko, & Berzin, 2015; American Society for Gastrointestinal Endoscopy [ASGE], 2017).

Symptoms

Signs and symptoms of acute pancreatitis include:

- Persistent, severe pain in the midepigastrium, left chest, right upper quadrant, shoulder, and back.
- Nausea and vomiting that persists for several hours.
- Low-grade fever.
- Abdominal swelling and tenderness.
- Shock, hypovolemia, and hypotension.
- Anorexia and excessive weight loss.
- Hypocalcemia.
- Hypoxia.
- Very rarely, Grey Turner's sign (a bluish flank discoloration) or Cullen's sign (a bluish periumbilical discoloration) caused by blood tracking from the retroperitoneum to these sites.

Diagnosis

Diagnosis of acute pancreatitis is made by detecting an elevation of serum amylase and lipase more than three times the upper normal limits. Symptoms of post-ERCP pancreatitis typically develop within 2-4 hours of the procedure (Thaker et al., 2015).

An etiology for acute pancreatitis may be identified by a combination of laboratory tests, radiological evaluations (CT scan, MRI, magnetic resonance cholangiopancreatography [MRCP]), abdominal and/or endoscopic ultrasonography, and ERCP. Laboratory values may assist in making the diagnosis. For example, an elevated serum alanine aminotransferase (ALT) concentration, and possibly the aspartate aminotransferase (AST) concentration, may predict the cause as a gallstone (Tetangco et al., 2016).

Acute pancreatitis ranges in severity from mild and self-limiting to life-threatening necrotizing pancreatitis.

- **Interstitial pancreatitis.** This is a milder form of acute pancreatitis that is characterized by pancreatic interstitial edema with intact pancreatic acini and ducts without necrosis. Although mortality from this form is less than that of necrotizing pancreatitis, it requires the same degree of intensive care when severe. Pseudocyst formation can also occur after interstitial pancreatitis.
- **Necrotizing pancreatitis.** This is the most severe form of acute pancreatitis (Shyu et al., 2014). Necrotizing pancreatitis is diagnosed when there is >30% nonenhancement of the pancreas on a contrast-enhanced CT scan. MRI is also useful in diagnosing this grave condition (Feldman, Friedman, & Brandt, 2016). The morphological findings include destruction (necrosis) of pancreatic tissue, blood vessels, and fat, with extensive fluid accumulation in the retroperitoneum.
- **Infected necrosis.** Also called an inflammatory mass, this occurs when the collection of tissues resulting from necrotizing pancreatitis becomes infected. Infected necrosis usually occurs at 2-4 weeks during the course of necrotizing pancreatitis (van Brunshot et al., 2012). Stable patients are treated with antibiotics and maximum supportive care. Surgical debridement is only considered for patients who are persistently ill or who have organ failure.
- **Pseudocyst and organized necrosis.** Massive necrosis frequently leads to the formation of large pancreatic and peripancreatic collections of fluid, blood, and necrotic debris, which are initially diffuse. If the fluid collection persists beyond 4 weeks after the attack and becomes walled off by adjacent anatomical structures, it is called a pseudocyst. If this contains significant debris, it is termed organized necrosis.

Overall mortality in patients with necrotizing pancreatitis is up to 30%, irrespective of etiology, with infected necrosis carrying the highest mortality (Boumitri et al., 2017).

Complications

Complications of pancreatitis within the first 30 days include secondary infection of pancreatic necrosis, pulmonary infections, pleural effusion, adult respiratory distress syndrome, **multiple organ dysfunction syndrome (MODS)**, renal failure, cardiac dysfunction, third-spacing, coagulopathy, persistent hypotension, encephalopathy, liver failure, immunocompromise with opportunistic infections, ischemia of limbs and internal organs, and malnutrition.

New complications occurring later than 30 days are mostly in or around the pancreas and include pseudocyst, organized necrosis, abscess, pancreatic fistula, and duodenal and bile duct stricture.

Treatment

The majority of patients who have interstitial pancreatitis improve within several days with supportive care alone. Treatment of acute pancreatitis is aimed at hemodynamic stabilization, correction of metabolic abnormalities, and reducing the risk of organ failure.

Supportive medical treatment includes:

- Adequate pain relief.
- GI rest by withholding of foods and fluids by mouth (NPO).
- Nasogastric suctioning, if there is significant vomiting.
- Bed rest.
- Electrolyte and fluid replacement.
- Blood transfusions, as required.
- Insulin to control hyperglycemia or diabetes.
- Antibiotics, as determined by clinical trials.
- Full intensive care support, as appropriate, such as endotracheal intubation and ventilation, renal dialysis, coagulation support, sedation, pressor support, and CT-guided fine needle aspiration of the pancreas if infected necrosis is suspected.

When the cause of pancreatitis is gallstones, surgery (laparoscopic cholecystectomy) is recommended during the same admission. If there is evidence of biliary tract obstruction from stones or if the attack is severe, urgent ERCP is indicated, with endoscopic sphincterotomy where necessary (Lee et al., 2018).

Chronic pancreatitis

Typically, the persistent inflammation of chronic pancreatitis produces irreversible morphological changes by fibrosis and atrophy in the pancreas, resulting in exocrine and endocrine functional loss. At least 90% of the exocrine function must be lost before significant malabsorption is noticed and diabetes develops late in the disease (Singh et al., 2017).

Risk factors

Approximately 50% to 55% of **chronic pancreatitis** is associated with long-term, heavy alcohol use (Herreros et al., 2013). Worldwide, it may also be the result of protein calorie malnutrition, cystic fibrosis, obstruction of the pancreatic duct by tumor, tropical pancreatitis, and familial pancreatitis. Chronic pancreatitis is generally found in males who are between the ages of 40 and 50 years (Kocher & Kadaba, 2011).

Hereditary pancreatitis, although rare, presents at a median age of 5 to 19 years (Raphael & Willingham, 2016). Chronic obstructive pancreatitis can be caused by tumor, post-traumatic stricture, pancreatic calculi, tropical pancreatitis, and possibly by pancreas divisum.

Symptoms

Symptoms and signs of chronic pancreatitis may include:

- Epigastric, subcostal, or umbilical pain radiating to the back.
- Excessive weight loss and debilitation.
- Steatorrhea.
- Diabetes mellitus.
- Obstructive jaundice from bile duct stricture.
- Epigastric mass.

Diagnosis

Diagnosis is usually made by having a high index of suspicion, evaluating with appropriate imaging examinations (CT scan, MRI, MRCP, ERCP, ultrasound, endoscopic ultrasound [EUS]), laboratory tests during acute exacerbations, malabsorption tests, and tests of pancreatic function. The main value of ERCP is the identification of the extent of the disease, anatomy, and presence of potentially correctable lesions. EUS may be the most sensitive test for detecting early changes of chronic pancreatitis.

Treatment

Treatment for chronic pancreatitis is initially nonsurgical and includes abstinence from alcohol, acute and chronic pain relief, nutritional support, and oral replacement of pancreatic enzymes when steatorrhea is detected.

Endoscopic intervention may reduce or eliminate the need for surgery. Surgical intervention may be necessary in situations where symptoms such as pain and/or recurrent attacks of pancreatitis are uncontrolled.

Celiac plexus block to relieve chronic pain can be achieved percutaneously or by EUS. Many patients with chronic pancreatitis who are in permanent pain require the support of pain services and mental health professionals.

Pancreatic fluid collections

There are different types of pancreatic fluid collections, which include but are not limited to the following:

- **Pseudocyst.** The most common is a pancreatic pseudocyst. These collections are called pseudocysts because they lack an epithelial layer typically present in "true" cysts.
- **Walled-off pancreatic necrosis (WOPN).** Previously called a pancreatic abscess, this type of fluid collection is similar to a pseudocyst in that it results from injury to the pancreas. In contrast, however, these are filled with solid debris and usually do not resolve on their own without endoscopic or operative intervention.

Risk factors

Pancreatic fluid collections result from many causes, including injury to the pancreas and premalignant or malignant conditions.

Symptoms

Most pseudocysts are small and asymptomatic. However, fluid collections can be large and cause symptoms such as pain and fevers. Persistent intra-abdominal symptoms such as early satiety, nausea and vomiting, increase in cyst size, and cyst-related adverse events such as biliary obstruction and infection may indicate a need for treatment.

Treatment

While most fluid collections resolve without treatment, symptoms related to the cyst, increase of cyst size, and cyst-related biliary obstruction and infection may indicate the need for treatment. Occasionally, pseudocysts need to be drained if pain, fever, or infection develops.

EUS-guided pancreatic fluid collection (PFC) drainage is the process of performing a transgastric or transduodenal cystostomy and placement of either a plastic or metallic stent to facilitate the drainage of fluid. Mature, fully-encapsulated fluid collections may require necrosectomy, but this is not indicated in acute fluid collections.

Occasionally, cysts will need to be removed operatively when they are first diagnosed, but this is not necessary for most cysts. Cysts should be monitored carefully to make sure they do not transform into a malignancy. Other types of fluid collections such as inclusion cysts are benign and do not need to be removed.

Pancreatic fistulas

Pancreatic fistulas are disruptions in the pancreatic duct with communication through the skin (external) or into other body structures (internal) such as the retroperitoneum. The majority of clinically significant pancreatic fistulas are external.

Risk factors

Causes of pancreatic fistulas include:
- Trauma with pancreatic duct injury.
- External drainage of a pseudocyst.
- Pancreatic surgery involving a ductal anastomsosis.
- Spontaneous pancreatic duct disruption as a complication of pancreatitis.

Symptoms

Clinically, external pancreatic fistulas should be suspected when clear fluid drains from the cutaneous foreface of a high-risk patient.

Diagnosis

Diagnosis of an external pancreatic fistula is confirmed when amylase content of the drainage fluid is evaluated.

Complications

Internal fistulas can cause massive ascites (pancreaticoperitoneal fistula), pleural effusion (pancreaticopleural fistula), or pseudocysts.

Treatment

The majority of pancreatic fistulas will close spontaneously within a few months. High-output (greater than 200 ml per day) fistulas should always be treated.

Medical treatment includes gut rest and octreotide (octapeptide of somatostatin) for suppression of pancreatic secretion. If medical treatment fails, endoscopic therapy can be effective. Surgery is reserved for complex fistulas, those with additional indications for surgery, or those not responding to less invasive treatment.

Pancreatic carcinoma

Pancreatic cancer is the fourth-most common cancer-related cause of death in the United States (Saad, Turk, Al-Husseini, & Abdel-Rahman, 2018). The location of this small gland deep in the abdomen and its proximity to major organs and blood vessels contribute to the late diagnosis of this disease that is often without symptoms (silent) until the disease is late stage.

As with all cancers, prognosis is stage-dependent. Ninety percent of pancreatic tumors are solid malignant tumors and arise from the ductal epithelium, making exocrine tumors of the pancreas more common than those of endocrine origin (Zhang et al., 2016).

Risk factors

Risk factors for pancreatic cancer include cigarette smoking, alcohol intake, diabetes mellitus, chronic pancreatitis, and genetic predisposition. Although familial pancreatic cancer is rare, the genetic component may be present in up to 5% of patients diagnosed with the disease (Klein, 2012). A germline mutation of the p16 gene has been reported in families with pancreatic cancer and melanoma, as well as an excess of pancreatic cancer seen in families with the BRCA-2 (breast cancer susceptibility gene-2) mutations.

Symptoms

Pancreatic cancer often presents as right upper quadrant pain that radiates to the back, epigastric pain, back or spine pain, anorexia, and weight loss. Painless jaundice occurs when the tumor arises at the head of the pancreas, causing blockage of the biliary duct. Additional symptoms include dark urine, light colored stools, fatigue, diarrhea, mental depression, nausea, dyspepsia, unexplained blood clots, and sudden onset of type 2 diabetes in patients 50 years or older. These symptoms are the result of the inability of the gland to perform its normal enzyme function.

Diagnosis

Diagnostic work-up in patients presenting with the described symptoms includes abdominal ultrasound, MRI, ERCP, MRCP, and CT scans of the abdomen. Once the presence of a mass has been established with these studies, a tissue diagnosis is essential before a treatment plan can begin. This is accomplished with either EUS-guided, CT-guided, or ultrasound-guided fine-needle aspirate (FNA) biopsy.

Determination of surgical resectability requires staging of the tumor. EUS and/or radiology imaging (CT scans by pancreatic protocol) are used to evaluate tumor involvement of the blood vessels, such as the superior mesenteric artery. Involvement of major blood vessels is most times decisive in ruling out surgery as part of the treatment plan. In rare cases, laparoscopy is used to stage and diagnose prior to a planned resection if staging is not identified by other imaging or testing.

Laboratory studies, such as the CA 19-9 tumor-associated antigen, are markers for this tumor in many patients. However, the CA 19-9 tumor-associated antigen is nonspecific and can be elevated in benign conditions, such as pancreatitis. A decrease in this tumor marker post surgery and chemotherapy has been found to correlate with survival of pancreatic cancer patients if elevated prior to treatment. After surgery, it is used to help monitor for reoccurrence.

Imaging tests such as EUS and ERCP and genetic testing for changes in certain genes such as KRAS are used in high-risk individuals (those with a strong family history) to provide early detection of this deadly disease. Additionally, trials are underway to test the role of vaccines in the treatment of patients with existing pancreatic cancer. The vaccination causes an immune response that targets the disease (American Cancer Society [ACS], 2019).

Approximately one in four patients diagnosed with pancreatic carcinoma will survive one year. The five-year survival rate is 3%, with 34% of those patients treated with surgery (ACS, 2019).

Treatment

As with other cancers, most providers refer to treatment recommendations and guidelines formulated by the National Comprehensive Cancer Network (NCCN) (2019). The NCCN, a non-profit alliance of 20 of the world's leading cancer centers, is dedicated to improving the quality and effectiveness of care provided to cancer patients.

Treatment is stage-dependent, and patient comorbidities are factors in recommendations for care. Often, symptom management such as relief of biliary obstruction with **endoscopic stent placement** and pain relief with narcotics and/or endoscopic celiac plexus block is a beginning adjunct to the treatment plan.

The only curative therapy for pancreatic cancer is resection of the tumor and surrounding pancreatic tissue with negative margins. Clinical outcome after surgery correlates with the experience of the surgeon and the treatment center. As such, the NCCN recommends referral to high-volume centers.

Depending on the tumor size and location, the surgery may involve a partial pancreatectomy or a pancreaticoduodenectomy (Whipple procedure), which removes the head of the pancreas, a portion of the stomach, part of the duodenum, the gallbladder, and the bile duct. Surgery is also performed to diminish symptoms due to tumor bulk and associated complications. For a borderline resectable tumor, neoadjuvant therapy of chemotherapy and radiation are used to shrink the tumor away from a major blood vessel, making surgical resection feasible. For patients who are undergoing resection, adjuvant (postoperative) therapy is indicated to reduce the likelihood of recurrence and to prolong overall survival rate.

Patients with unresectable tumors may receive a palliative surgical bypass procedure for biliary obstruction caused by the tumor. For patients who have an unresectable tumor or have metastatic disease at the time of diagnosis, systemic chemotherapy or chemoradiation—as standard therapy or clinical trials—is recommended.

Nursing care focuses on providing a thorough patient assessment to facilitate pain management and nutrition support, in addition to networking for such needs as home health care and hospice.

PANCREATIC TUMORS

Pancreatic tumors include cystic tumors and endocrine tumors.

Cystic tumors

A different spectrum of tumors, called cystic neoplasms, involve any part of the pancreas, but most commonly the head, and contain fluid (Farrell & Fernandez-del Castillo, 2013). Depending on the type of neoplasm, they range from benign to malignant and from very small to massive.

Terminology is complex, but serous cystadenoma or cystadenocarcinoma and mucinous cystadenoma or cystadenocarcinoma are the dominant types. Intraductal papillary mucinous neoplasm (IPMN), a variant that causes the ducts to dilate and fill with mucin, carries a better prognosis than the more common pancreatic ductal adenocarcinoma.

Diagnosis

Cystic tumors may be indistinguishable from pseudocysts on CT scan, but the absence of a history of pancreatitis should raise suspicion. Diagnosis of the type of cystic tumor is obtained by FNA of the cysts to determine the pathology and cytology of the fluid content.

Treatment

Treatment recommendations often include the Whipple procedure for cure.

ENDOCRINE TUMORS

Islet cell tumors are classified on the basis of the predominant hormone they secrete, although most secrete more than one hormone. Types of islet cell tumors include gastrinomas, insulinomas, glucagonomas, somatostatinomas, and VIPomas (vasoactive intestinal polypeptide) producing tumor. They may be benign or malignant.

Gastrinoma islet cell tumor

Zollinger-Ellison syndrome (ZES) is a rare gastrinoma (or non-beta islet cell tumor of the pancreas) usually located in the pancreas and duodenum. These tumors release an excess of the hormone gastrin into the circulation (hypergastrinemia), which stimulates gastric acid hypersecretion. The hypergastrinemia is due to loss of the feedback inhibition of gastrin secretion, which then rises unchecked. In turn, this causes atrophic gastritis and pernicious anemia, resulting in severe ulcers in the upper GI tract, predominantly in the distal duodenum or jejunum, and often severe diarrhea.

ZES has two common variants:

- Sporadic type, which is usually malignant and is seen later in life.
- Genetic type, which is associated with multiple endocrine neoplasia type 1 (MEN1) syndrome and causes tumors or hyperplasia of the parathyroid, pancreatic islet, and pituitary glands.

Risk factors

ZES occurs most frequently between the ages of 35 and 65 and is more common in men than in women (Schroeder, Deftereos, Lupetin, & Hartman, 2015).

Symptoms

Patients with ZES have pain from peptic ulcer disease as the predominant symptom, and diarrhea and steatorrhea are common symptoms.

ZES is suspected in patients with:

- A history that suggests aggressive peptic ulcer disease (PUD) or gastroesophageal reflux disease (GERD) refractory to therapy or with recurrent complications.
- Peptic ulcer disease with associated diarrhea.
- Any feature that suggests the presence of MEN1 syndrome (e.g., hyperparathyroidism, pituitary adenoma, and pancreatic endocrine tumor).

Diagnosis

Hypergastrinemia refers to an increased fasting serum gastrin level and is one of the characteristics of ZES. Fasting serum gastrin level is elevated in a majority of patients with ZES. If elevated, the fasting gastric pH should be checked to exclude achlorhydria.

Confirmatory diagnostic tests may include secretin infusion to stimulate gastrin levels further, when the fasting level is equivocal. The best methods for determination of tumor site are CT scan, EUS, and octreotide nuclear scanning.

Complications

One-third of ZES patients have metastatic liver disease on presentation (Aamar, Madhani, Virk, & Butt, 2016). Progression of this disease is usually slow, with metastasis normally beginning in regional lymph nodes, then to the liver, and later to the bone.

The extent of morbidity and mortality in patients with ZES is related principally to ulcer complications, fistulas, hemorrhage, or perforation. Most patients experience complications of ulcer disease such as bleeding or perforation at some time during the course of their disease.

Treatment

Medical treatment of ZES is highly effective and involves long-term administration of proton pump inhibitors (PPIs)—often at doses much larger than normally prescribed—which suppress acid production and promote healing. For metastatic disease, chemotherapy is promising. When medical treatment fails, surgery may be recommended to remove the primary tumor if it can be located. In patients who do not have MEN1, some gastrinomas can be completely resected with resultant cure. If resection is possible, this is the optimal treatment. Total gastrectomy to control acid secretion is very rarely needed.

Other islet cell tumors

Other islet cell tumors include:

- **Insulinoma.** The most common type of islet cell tumor is an insulinoma, which arises from the beta cells. The tumor is typically round, firm, and encapsulated and is most frequently located in the body or tail of the pancreas. Ninety percent are solitary benign tumors (Okabayashi et al., 2013). Patients with insulinomas present with symptoms of hypoglycemia on fasting after exercise. These patients may also exhibit neuropsychiatric manifestations, ranging from subtle personality changes to confusion, coma, or seizure disorders. Insulin-to-glucose ratio is also elevated in these patients.

- **Glucagonoma.** Glucagon-secreting tumors of the alpha islet cells are called glucagonomas. They are far less common than insulinomas or gastrinomas. The most distinctive feature is a skin disorder, necrolytic migratory erythema, which appears as an erythematous area that develops a central blister and is followed by crusting, healing, and sometimes bronze hyperpigmentation. Most patients also have diabetes mellitus. Because of the catabolic activity of glucagon, patients with glucagonomas often exhibit profound weight loss and anemia.
- **Somatostatinoma.** Patient who have somatostatinomas may present with steatorrhea, mild diabetes mellitus, and/or cholelithiasis. These symptoms are consistent with somatostatin's inhibitory effect on the secretion of gastrin, secretin, insulin, glucagon, and cholecystokinin.
- **VIPoma.** Tumors that produce vasoactive intestinal peptide are termed VIPoma. Patients generally experience profuse watery diarrhea, hypokalemia, and hypochlorhydria or achlorhydria, also known as the pancreatic cholera syndrome or Verner-Morrison syndrome.

Complications

More than 50% of patients with a glucagonoma have metastases at the time of diagnosis (Al-Faouri, Ajarma, Alghazawi, Al-Rawabdeh, & Zayadeen, 2016).

Treatment

The treatment of choice for most islet cell tumors is surgical excision. Patients with metastatic disease may improve after surgical reduction of the tumor mass. Streptozocin (Zanosar) has been shown to decrease tumor size and prolong survival. A long-acting analog of somatostatin effectively controls symptoms in patients with several types of islet cell tumors.

Pancreatic exocrine insufficiency

Also referred to as exocrine pancreatic insufficiency (EPI), problems with digestion result from a deficiency of the exocrine pancreatic enzymes needed to digest food.

Risk factors

Pancreatic exocrine insufficiency generally results from chronic pancreatitis leading to acinar loss. Pancreatic duct obstruction should be excluded, since this may be a correctable cause.

Symptoms

Because pancreatic enzymes are necessary for the digestion of fat, protein, and carbohydrate, pancreatic insufficiency leads to malabsorption and steatorrhea. Weight loss is a common symptom.

Diagnosis

Plain X-ray films may show calcification of the pancreas, but CT scan is more sensitive. A 72-hour fecal fat analysis with stool fat output greater than 7% of intake is evidence of fat malabsorption.

Treatment

Pancreatic exocrine insufficiency is treated with oral enzyme preparations and concomitant acid suppression to improve delivery of sufficient enzyme into the small intestine. Some patients may benefit from supplemental calcium, vitamin D, and other fat-soluble vitamins.

Cystic fibrosis

Cystic fibrosis is the most common lethal genetic defect in Caucasian populations, occurring in one out of every 3,000 newborns (Rang & Wilson, 2019). This disease of the exocrine glands affects not only the pancreas but also the respiratory system, the sweat glands, and the reproductive system.

Risk factors

Mutations in the cystic fibrosis transmembrane conductance regulator (CFTR) gene are the cause of cystic fibrosis. The transcriptional regulation of CFTRs and underlying molecular mechanisms remain poorly understood.

It is possible that alteration in a common intracellular mediator or inhibitor protein distal to the site of generation of cyclic adenosine monophosphate is likely to be the underlying abnormality in affected tissues of patients with cystic fibrosis.

Cystic fibrosis is an autosomal-recessive disease and is most common among people of Northern European ancestry. About one in 29 people is a carrier (Castellani et al., 2015). The genetic mutations for this disease have been identified and can be measured.

Symptoms

The majority of children with cystic fibrosis have obvious clinical signs and symptoms resulting from both pulmonary and pancreatic involvement. Most patients with cystic fibrosis have pancreatic insufficiency at birth, although the extent of pancreatic involvement is highly variable.

Symptoms of cystic fibrosis include:

- Poor fat and protein digestion, leading to oily and foul-smelling stools.
- Weight loss despite an increased appetite.
- Repeated episodes of pneumonia and bronchitis.
- Diminished or absent pancreatic enzymes.
- Pulmonary changes. The production of thick, sticky, dry mucus leads to noisy respirations and intermittent wheezing and coughing, particularly when lying down.

At birth, approximately 13% to 17% of all patients with cystic fibrosis present with meconium ileus, which is a plug of meconium in the terminal ileum (van der Doef et al., 2011). Meconium ileus is acquired in utero and is perhaps the first overt manifestation of diminished pancreatic function.

Diagnosis

Cystic fibrosis is frequently diagnosed during the first year of life. Elevation of sodium and chloride concentrations in the sweat is the most characteristic finding of cystic fibrosis.

The definitive diagnostic test for cystic fibrosis is the sweat electrolyte test, where quantitative pilocarpine iontophoresis shows elevated sodium and chloride levels in affected patients. Sweat electrolyte testing may not be reliable in neonates and in individuals with edema.

DNA testing can be performed to identify patients with cystic fibrosis and carriers of the cystic fibrosis gene.

Complications

A hallmark of cystic fibrosis is bacterial infection, and most people with the condition die from respiratory failure (Bush, Bilton, & Hodson, 2016). The airways show an increased viscoelasticity, and the resulting mucus thickens over time, leads to bacterial overgrowth, and prevents mucus clearance. Cystic fibrosis is recognized as a polymicrobial disease.

Gastrointestinal (GI) complications include intestinal obstruction (meconium ileus equivalent), intussusception, constipation, and rectal prolapse. Liver disease, esophageal varices, hyperglycemia, and diabetes mellitus may also be present. Literature suggests that approximately 5% to 30% of patients with cystic fibrosis develop liver cirrhosis (Flass & Narkewicz, 2013).

Treatment

Cystic fibrosis is managed by controlling respiratory complications and by aiding digestion through dietary regulation; supplements of vitamins A, D, E, and K; and pancreatic enzyme replacement. The pancreatic exocrine dysfunction requires the patient to rely on oral pancreatic enzymes. Gene therapy is promising.

Much of the morbidity, and virtually all of the mortality, beyond the neonatal period in patients with cystic fibrosis is attributable to chronic obstructive pulmonary disease (COPD). In 1960, the mean survival age after diagnosis was less than 5 years. In 2010, survival is estimated to be 37 years for women and 40 years for men (MacKenzie et al., 2014). This prolongation of life can be attributed largely to the development of pancreatic replacement therapy, use of antibiotics, and vigorous pulmonary hygiene.

CONGENITAL ANOMALIES

Congenital anomalies of the pancreas may include pancreatic rest, pancreas divisum, and annular pancreas.

Pancreatic rest

Pancreatic rest is an uncommon condition that is defined as the presence of ectopic pancreatic tissue, usually in the gastric antrum, that forms a polyp-type lesion.

Symptoms

When symptomatic, the condition may produce ulceration, bleeding, or clinical pancreatitis.

Diagnosis

Pancreatic rest is usually an incidental finding during an upper GI endoscopy.

Treatment

Symptomatic lesions are treated with simple excision, which can be accomplished by endoscopic mucosectomy.

Pancreas divisum

Pancreas divisum is the most common congenital pancreatic anomaly, occurring in approximately 10% of individuals (Bulow et al., 2014). This anomaly occurs when the two embryonic precursors of the pancreas—the ventral and the dorsal buds—fail to fuse, resulting in separate dorsal and ventral pancreatic ducts. The dorsal gland is drained by the duct of Santorini through the minor or accessory papilla into the duodenum, whereas the ventral gland is drained by an abnormally short duct of Wirsung and enters the duodenum through the major papilla together with the bile duct.

Symptoms

Studies have shown that pancreas divisum is associated with pancreatitis. It may be that both pancreas divisum and a stenotic accessory papilla must be present for clinically evident pancreatitis to occur. Recurrent acute pancreatitis may be attributable to pancreas divisum when the pancreatic ducts appear normal, but chronic pancreatitis may occur consequently and often affects both ducts.

Diagnosis

Diagnosis is made by pancreatography. On ERCP, the major papilla is often difficult to cannulate, but on injection with contrast, the duct of Wirsung appears stunted and of small diameter. Additionally, there is rapid filling of small accessory ducts. If possible, the accessory duct should be cannulated, revealing a duct of Santorini running the entire length of the pancreas that is not in communication with the duct of Wirsung.

Pancreatic manometry and secretin ultrasound scanning is sometimes used if stenosis of the papilla is not demonstrated by MRCP.

Treatment

For patients with frequent episodes of recurrent acute pancreatitis, intervention may be considered. Treatment options include surgical sphincterotomy or sphincteroplasty of the accessory papilla, and endoscopic sphincterotomy of the minor papilla with a temporary protective stent, both of which give similar results.

Annular pancreas

Annular pancreas occurs when the embryonic dorsal and ventral glands fail to fuse, and part of the ventral pancreas encircles the duodenum, often resulting in duodenal obstruction.

Risk factors

Annular pancreas is the most common anomaly obstructing the duodenum in infancy and may also be associated with other congenital anomalies, such as atresia of the duodenum and Down syndrome.

Symptoms

Annular pancreas may not cause any symptoms until adulthood. At that time, presenting complaints include intermittent epigastric discomfort that is relieved by vomiting.

Diagnosis

Upper GI X-ray films show a narrowing of the second portion of the duodenum. Additional tests include abdominal ultrasound, abdominal X-ray, and CT scan.

Treatment

Symptomatic cases are best treated by surgical bypass using a duodenojejunostomy.

PANCREATIC DISEASES IN CHILDREN

Pancreatic diseases in children are rare and may prove to be a diagnostic challenge. Pancreatic disorders occur with less frequency in the pediatric age group compared to the adult population. A well-documented symptomatology is needed to reveal the occurrence of pancreatic diseases. Pancreatic insufficiency should be considered in children with significant weight loss, failure to thrive, malabsorption, or steatorrhea.

Shwachman-Diamond syndrome

Shwachman-Diamond syndrome, first described in 1964, represents the second most common cause of pancreatic insufficiency.

Risk factors

The suggested mode of inheritance is autosomal recessive.

Symptoms

The main features of Shwachman-Diamond syndrome are pancreatic cysts, cyclic neutropenia, metaphyseal dysostosis, and growth retardation. Associated manifestations include dental abnormalities, renal dysfunction, hepatomegaly, abnormal lung function, delayed puberty, and ichthyosis.

- **Growth retardation.** Stunted growth is seen as the most prevalent clinical feature in Shwachman-Diamond syndrome. This is related to metaphyseal dysphasia of the long bone. There is a noted delay in the growth spurt of preadolescence, which resolves during puberty. Patients are usually below the third percentile for height, but linear growth is maintained. As adults, they rarely reach above the 25th percentile for height (Feldman et al., 2016).
- **Malnutrition.** Malnutrition, along with growth failure, is the result of pancreatic insufficiency.
- **Malabsorption.** Malabsorption is manifested during infancy with steatorrhea. Stools are greasy, pale, and foul-smelling.
- **Hematological changes.** Significant hematological changes are noted, including neutropenia, thrombocytopenia, and anemia. These hematological changes make the patient susceptible to infection and bleeding tendencies.

Diagnosis

Diagnosis of Shwachman-Diamond syndrome is confirmed by performing a secretin pancreozymin stimulation test, which results in a very low or absent pancreatic zymogen enzyme or an elevation in stool fat content. Blood counts are performed twice weekly for 3 weeks as a confirmation of diagnosis.

A negative sweat test excludes cystic fibrosis.

Complications

Patients with Shwachman-Diamond syndrome are susceptible to bacterial infections and predisposed to orthopedic complications. They are at risk for aplastic anemia and leukemia, which contribute further to their morbidity and mortality. There is a prevalent association of pancreatic insufficiency with hematopoietic manifestations and leukemia.

Treatment

The treatment of Shwachman-Diamond syndrome is primarily symptomatic and supportive. Pancreatic enzyme replacement is often the treatment of choice. Fat-soluble vitamins may be needed as a replacement because of severe malabsorption. Because of increased susceptibility to infections, antibiotic therapy is considered. The patient's orthopedic and hematological needs are monitored and treated as necessary.

Acute pancreatitis in children

Acute pancreatitis is the same acute inflammatory process of the pancreas that can occur in adults with the same risks for complications. The difference, however, is the incidence of different etiologies. In adults, obstruction from gallstones accounts for 40% of acute pancreatitis and alcohol misuse is responsible for 30% of acute pancreatitis (Forsmark, Santhi, & Wilcox, 2016). This is usually not the case in the pediatric population.

Risk factors

The five leading causes of pancreatitis in children are biliary disorders, medications, an idiopathic cause, systemic disease, and trauma. Other common causes of this medical condition are infectious, metabolic, and hereditary diseases.

Symptoms

Often the symptoms of acute pancreatitis will mimic other emergency situations. A mild presentation might be confused with gastritis, whereas a severe case may present as a small bowel obstruction or perforation. The patient may have midepigastric tenderness and/or pain with a sudden onset, increasing gradually and changing in severity.

Diagnosis

Diagnosis can be challenging. A careful history of possible trauma, exposure to viral infection, or drug ingestion must be obtained. Any family history of metabolic or other conditions associated with pancreatitis should be investigated.

The most frequently used diagnostic tools are serum amylase and lipase and abdominal ultrasonography. Studies indicate that the sensitivities of abnormal pancreatic enzyme levels in testing may vary due to developmental differences in pancreatic enzyme expression in different pediatric age groups (Suzuki, Sai, & Shimizu, 2014). It is important to note that although serum amylase is the most frequently used diagnostic tool, normal serum amylase can be seen in patients with severe pancreatitis.

Ultrasound is a common diagnostic tool because it is widely available, can complement serology, and is useful in evaluating other causes for abdominal pain such as appendicitis and intussusception. Unless the diagnosis is unclear, CT scan of the abdomen is not initially recommended to evaluate children for pancreatitis due to exposure to ionizing radiation.

As in the adult population, children with recurring episodes of pancreatitis should undergo ERCP to assess for congenital anomalies, such as pancreas divisum, sphincter of Oddi dysfunction, and microlithiasis.

CASE SITUATION

Mrs. Florinda Jones, a 55-year-old female, went to her doctor because of severe epigastric pain. For 3 days she endured the pain, along with nausea and vomiting, thinking she had the stomach flu. Her doctor noted that she was dehydrated. He suspects she has acute pancreatitis and put her in the hospital for further workup.

Points to think about

1. In making a nursing assessment, what questions might the gastroenterology nurse ask the patient about her pain to support the physician's preliminary diagnosis of acute pancreatitis?
2. Florinda is scheduled for ultrasonography and a CT scan. What might be found on these tests?
3. Under what circumstances might the gastroenterology nurse expect the patient to have an ERCP?
4. What nursing interventions may help to avoid post-ERCP pain or fever or identify it early?
5. What can the gastroenterology nurse teach Florinda to help her avoid future attacks of acute pancreatitis?

Suggested responses

1. In performing a nursing assessment, the gastroenterology nurse might ask Florinda the following questions about her epigastric pain:
   a. "How long have you had the pain?" "Can you describe it?" Onset, duration, and character of pain are always important clues. The pain of pancreatitis can be gradual or sudden, and usually becomes severe, constant, and laboring.
   b. "Can you point to where the pain is?" Pancreatic pain usually localizes in the left epigastric area and often radiates to the back and left shoulder. The pancreas is located in the left epigastric area and lies posterior to the stomach, which accounts for the location and radiation of the pain.
   c. "Does pressing here, over the left side of your upper abdomen, cause pain?" The inflamed pancreas is often tender to palpation.
   d. "Does the pain come and go, or is it there all the time?" "Do you notice that it is worse at certain times?" Pain is usually constant and worsens several hours after eating a large meal or ingesting alcohol.
   e. "Is the pain better after you vomit?" Vomiting does not relieve the pain of pancreatitis as it might for patients with intestinal obstruction.
   f. "Is there anything you do that helps relieve the pain?" Patients with pancreatitis often assume a characteristic position of bending over at the torso with hands over the left epigastric area.
2. Most of the time, pancreatitis in women of this age group is a result of gallstones. The second leading cause of pancreatitis is alcohol misuse. The gastroenterology nurse might expect the following findings when Florinda undergoes ultrasonography and CT scan:
   a. Ultrasonography would be a first-line test for cholelithiasis. It may also identify pseudocysts, abscesses, common duct dilation, and calcifications.
   b. CT scan may further define and evaluate the above findings and may also evaluate the architecture of the liver, biliary tree, and pancreas.
3. At present, the only indication for ERCP in acute pancreatitis is relief of gallstone-related pancreatitis in selected patients with severe indicators or evidence of concomitant biliary obstruction. In such patients, a sphincterotomy may be done to drain the common bile duct and remove any stones that may be causing an obstruction. This allows drain-

age of the pancreatic duct to resume, and pancreatitis will usually subside uneventfully.

In patients who have had two or more attacks of acute pancreatitis, ERCP should be considered when looking for potentially correctable lesions such as stones (biliary and pancreatic), strictures, obstructions of the main duct (benign and malignant), or pseudocysts.

4. To ensure that Florinda will have no increase in pain or fever after ERCP, appropriate nursing interventions might include:
   a. Be aware of the increased risk of pancreatitis after ERCP.
   b. Watch the X-ray monitor while injecting contrast to confirm cannulation of the pancreatic duct.
   c. Slowly and carefully inject the minimal amount of dye, taking care not to produce acinar filling of the pancreas.
   d. Use sterile accessory equipment.
   e. Instruct the patient to report any chills, fever, or increase in pain.
5. To prevent future attacks of pancreatitis, instruct the patient to adhere to the following instructions:
   a. Avoid alcohol completely until an etiology is established.
   b. Follow a diet that is low in fat to aid in digestion during possible disruption of the pancreatic function due to inflammation.
   c. Avoid large meals.
   d. Avoid drugs that are known to cause pancreatitis, including estrogens, birth control pills, tetracycline, furosemide, thiazide diuretics, azathioprine, and sulfonamides.

Let Florinda know that if a cause of her pancreatitis can be found, future attacks may be eliminated by its treatment, and that she should contact her physician with any changes in her condition.

REVIEW TERMS

acinus, alpha cells, annular pancreas, beta cells, cholecystokinin-pancreozymin (CCK-PZ), chronic pancreatitis, cystic fibrosis, delta cells, duct of Santorini, duct of Wirsung, endoscopic retrograde cholangiopancreatography (ERCP), endoscopic sphincterotomy, endoscopic stents, glucagonoma, infected necrosis, insulinoma, interstitial pancreatitis, islets of Langerhans, multiple organ dysfunction syndrome (MODS), necrotizing pancreatitis, organized necrosis, pancreas, pancreas divisum, pancreatic exocrine insufficiency, pancreatic fistulas, pancreatic rest, pancreatitis, papilla of Vater, pseudocyst, Shwachman-Diamond syndrome, secretin, somatostatinoma, Zollinger-Ellison syndrome (ZES)

REVIEW QUESTIONS

1. The majority of the pancreatic tissue is made up of:
   a. Acinar cells.
   b. Alpha cells.
   c. Beta cells.
   d. Delta cells.
2. The endocrine cells of the pancreas are located in the:
   a. Crypts of Lieberkühn.
   b. Islets of Langerhans.
   c. Duct of Wirsung.
   d. Pancreatic lobules.
3. The beta cells secrete:
   a. Somatostatin.
   b. Glucagon.
   c. Vasoactive intestinal peptide.
   d. Insulin.
4. The cephalic phase of pancreatic secretion is stimulated by:
   a. Gastric distention.
   b. The sight and smell of food.
   c. The presence of acidic chyme in the small intestine.
   d. The presence of alkaline chyme in the small intestine.
5. The most severe form of pancreatitis is:
   a. Chronic pancreatitis.
   b. Necrotizing pancreatitis.
   c. Interstitial pancreatitis.
   d. Alcoholic pancreatitis.
6. A sac-like structure that is filled with pancreatic fluid 6 weeks after an attack of pancreatitis is called a:
   a. Pancreatic rest.
   b. Pancreas divisum.
   c. Annular pancreas.
   d. Pseudocyst.
7. The most specific diagnostic and staging test for pancreatic cancer is:
   a. ERCP.
   b. PTC.
   c. Ultrasound.
   d. EUS.
8. Zollinger-Ellison syndrome often results in what clinical manifestation?
   a. Peptic ulcer disease.
   b. Steatorrhea.
   c. Necrolytic migratory erythema.
   d. Pancreatic cholera syndrome.
9. The preferred treatment for most islet cell tumors is:
   a. Supportive and palliative measures only.
   b. Radiotherapy.
   c. Chemotherapy.
   d. Surgical excision.

10. The definitive diagnostic test for cystic fibrosis is the:
    a. Sweat electrolyte test.
    b. Schilling test.
    c. Bernstein test.
    d. Serum amylase and lipase level.
11. Pancreatic insufficiency in children is often seen in:
    a. Malabsorption.
    b. Steatorrhea.
    c. Maldigestion.
    d. Cystic fibrosis.
12. The most prevalent clinical feature of Shwachman-Diamond syndrome is:
    a. Stunted growth.
    b. Jaundice.
    c. Pigmented retinopathy.
    d. Edematous extremities.
13. The five leading causes of acute pancreatitis in children are:
    a. Malabsorption, steatorrhea, maldigestion, infection, and tumor.
    b. High-fat diet, high potassium level, high sugar consumption, high calcium level, and high amylase level.
    c. Biliary disorders, medications, an idiopathic cause, systemic disease, and trauma.
    d. Parasitic diseases, immunocompromised state, congenital defects, tumor, and steatorrhea.
14. Which organization's guidelines and recommendations are used in the care and treatment of the cancer patient?
    a. American Cancer Society (ACS).
    b. National Institutes of Health (NIH).
    c. National Comprehensive Cancer Network (NCCN).
    d. Public Health Department.

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Chapter 19

LIVER

This chapter will acquaint the gastroenterology nurse with the normal anatomy and physiology of the liver, along with certain pathological conditions that affect this organ.

Learning objectives

After reviewing the content of this chapter, the gastroenterology nurse will be able to:

1. Describe the normal anatomy of the liver.
2. Explain the normal physiological functions of the liver, including its role in bile formation and secretion, metabolism, coagulation, detoxification, and vitamin storage.
3. Discuss the pathophysiology, diagnosis, and treatment of pathological conditions that affect the liver.
4. Elaborate on the process of orthotopic liver transplantation.

ANATOMY AND PHYSIOLOGY

The liver is a large, highly vascular organ, weighing between 1,200 g and 1,500 g (approximately 3 pounds). It is located behind the ribs in the right upper quadrant of the abdomen beneath the diaphragm, with a small portion extending into the left upper quadrant. The liver accounts for approximately 2% to 3% of average body weight.

Protected by the rib cage, the liver is held in position by avascular, ligamentous attachments. Although not true ligaments, they are attached to the Glisson capsule, the connective tissue, or the visceral peritoneum that surrounds the liver.

Layers of the liver

The liver can be described in terms of morphological anatomy (external appearance of the liver) and functional anatomy (the internal features of the liver).

The liver consists of two main lobes, with the larger left lobe containing the falciform ligament. A French surgeon and anatomist, Claude Couinaud, developed a classification system in 1957 that further divides the liver into eight functionally independent segments. Each segment within the Couinaud classification of liver anatomy has its own vascular inflow, outflow, and biliary drainage. The surface of the liver is covered by the visceral peritoneum.

The structural-functional units of the liver are hepatic lobules that contain blood vessels and bile ducts within, as well as between, each lobule. Liver tissue consists of laminae (sheets) of hexagonal liver cells (hepatocytes), which account for approximately 80% of the liver volume and 60% of all liver cells.

Microscopic anatomy of the liver includes:

- **Lobules.** This uniform structure of clusters of cells carries out the vital functions of the liver. Each lobule consists of numerous rows of liver cells (hepatocytes).
- **Hepatocytes.** These rows of liver cells are one cell thick with finger-like projections (microvilli). Hepatocytes radiate from central veins called terminal hepatic venules toward a thin layer of connective tissue that separates each lobule.
- **Sinusoids.** These spaces (also called hepatic capillaries) separate rows of hepatocytes from one another on several surfaces. Sinusoids are lined by thin endothelial cells that have openings through which the microvilli of the hepatocytes extend. This allows hepatocytes direct access to the bloodstream in the sinusoids.
- **Canaliculi.** Hepatocytes are bound to one another by dense, tight junctions perforated by these small channels (canaliculi) that are the farthest end of the biliary system. Canaliculi receive bile from the hepatocytes.

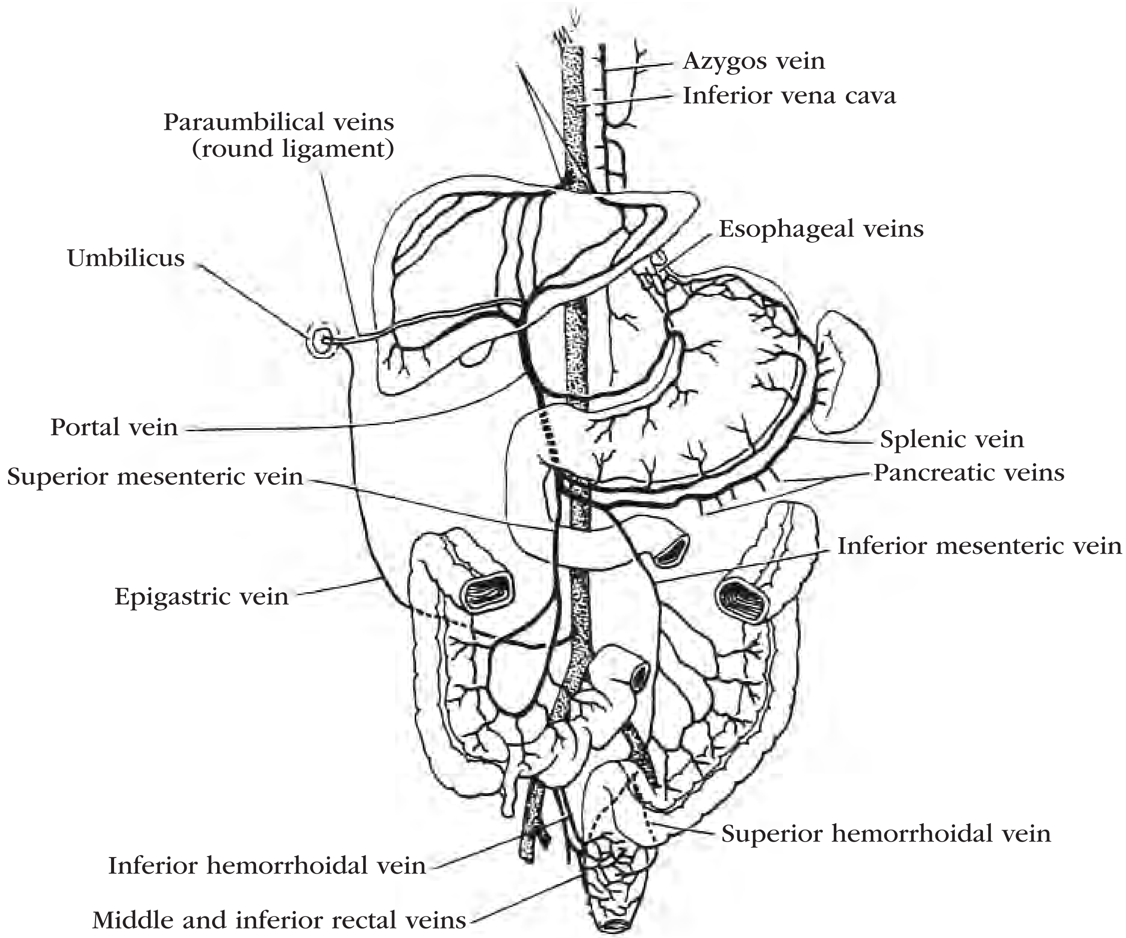

Figure 19.1 Hepatic Portal System

Blood is carried from the stomach, intestines, spleen and pancreas into the liver sinusoids. Hepatic veins convey it to the inferior vena cava. Clinically significant sites of anastomosis between the hepatic and systemic circulations are (1) the esophageal veins (portal tributary), which anastomose with the azygos veins; (2) the paraumbilical veins in the round ligament, which originate in the left branch of the portal vein and connect with the superficial veins of the anterior abdominal wall (systemic tributaries) in the area of the umbilicus; (3) the superior rectal or hemorrhoidal veins (portal tributary), which anastomose with the middle and inferior rectal veins; (4) the portal tributaries to the intestines, pancreas and liver, which anastomose with the phrenic, renal and lumbar veins (systemic tributaries not shown). In portal hypertension and chronic liver disease, blood may be backed up in these veins and shunted around the liver through the points of anastomosis (Price & Wilson, 1992).

The bile canaliculi branch and combine, eventually forming the right and left hepatic ducts, which merge to form the common hepatic duct. The common hepatic duct joins the cystic duct of the gallbladder to form the common bile duct. The common bile duct then joins the duct of Wirsung at the ampulla of Vater just before entering the duodenum.

- **Kupffer cells.** These other major cells of the liver adhere to the wall of the sinusoid and project into its lumen. These cells function as phagocytes—cells that engulf and destroy foreign material or other cells.

Liver cells have the ability to regenerate themselves within 3 weeks. Normal liver function can be restored within 4 months in conditions that do not lead to progressive damage to the liver. It is possible for the liver to function even with damage to 90% of its mass, but liver removal or total destruction leads to death within about 10 hours.

Vascular supply

As much as 1,500 ml of blood enter the liver each minute, making the liver one of the most vascular organs in the body (Figure 19.1).

Arterial

The portal vein supplies about 75% of the liver's blood, bringing in about 50% of its oxygen supply and nutrients that are absorbed from the gastrointestinal (GI) system. The hepatic artery supplies the other 25% of the blood flow and the remaining 50% of the oxygen to the liver.

Venous

Blood leaves the sinusoids by entering the central lobule vein. It then enters the hepatic veins and follows the normal venous circuit into the inferior vena cava.

The liver receives blood from the hepatic portal vein and hepatic arteries. The hepatic portal vein supplies blood, which comprises the majority of the liver's blood from the gastrointestinal circulation (i.e. esophageal veins; paraumbilical veins; superior rectal or hemorrhoidal veins; and the portal tributaries to the intestines, pancreas, and liver). Blood from this source provides half of the liver's oxygen demand. Carbohydrates and vitamins in this blood are brought to the liver for storage before being sent to the heart.

The remainder of the liver's blood, along with the other half of its needed oxygen, is supplied by the hepatic arteries. Blood from both the hepatic portal vein and hepatic arteries flows through the liver tissue and empties into the central vein of each lobule, ultimately merging into hepatic veins to direct blood back to the heart.

Innervation

Nerve fibers to the liver come from the vagus nerve and the thoracolumbar system. Sympathetic nervous stimulation may cause hepatic artery and portal vein vasoconstriction and a dull pain over the area of the liver. Bile ducts receive sensory nerves, as well as sympathetic and parasympathetic innervation.

Physiologic functions

The major physiological functions of the liver are:

- Bile formation and secretion.
- Metabolism of carbohydrates, proteins, fats, and steroids.
- Manufacture of substances necessary for coagulation and anticoagulation.
- Detoxification of foreign and toxic substances.
- Vitamin storage.

Bile formation and secretion

Bile is a combination of water, bile acids, bile pigments, cholesterol, bilirubin, phospholipids, potassium, sodium, and chloride. The liver synthesizes and transports the bile salts and bile pigments that are necessary for fat digestion.

- Primary bile acids are produced from cholesterol. When bile acids are conjugated in the liver, they become bile salts.
- Bile pigments are formed from the breakdown of erythrocytes by the cells of the reticuloendothelial system. In this process, heme is converted first into biliverdin and then into bilirubin, the main bile pigment.
- The water-insoluble, indirect (unconjugated) bilirubin is released into the blood, where it is bound to albumin. In the hepatocytes, indirect bilirubin is conjugated with glucuronic acid to form water-soluble, direct (conjugated) bilirubin, which is secreted into the bile canaliculi and excreted in bile.

Metabolism

Carbohydrate, protein, fat, and steroid metabolism are important functions of the liver.

Carbohydrate metabolism

Liver cells:

- Convert glucose, fructose, and galactose to glycogen (glycogenesis) for storage in the liver.
- Break down glycogen to glucose (glycogenolysis) to maintain blood glucose levels when there is a decrease in carbohydrate intake.
- Synthesize glucose from noncarbohydrate nutrients (gluconeogenesis) to maintain blood glucose levels (e.g., synthesizing glucose from proteins or fats).

Protein metabolism

Liver cells:

- Deaminate amino acids to produce ketoacids and ammonia, from which urea is formed and excreted in the urine.
- Synthesize about 50 g of new protein daily, including most of the plasma proteins such as albumin, fibrinogen, transferrin, ceruloplasmin, haptoglobin, and the lipoproteins.

Fat metabolism

Digested fat is converted in the intestine to triglycerides, cholesterol, phospholipids, and lipoproteins. These substances are then taken up by the liver and hydrolyzed to glycerol and fatty acids through a process known as ketogenesis.

Steroid metabolism

The liver metabolizes adrenocortical steroids, glucocorticoids, estrogens, testosterone, progesterone, and aldosterone.

Coagulation

Most of the clotting factors used in the body are produced in the liver, including prothrombin and fibrinogen. The liver also produces the anticoagulant heparin and releases vasopressor substances after hemorrhage.

Standard measures of hemostasis include platelet count, prothrombin time (PT), and partial thromboplastin time (PTT). PT serves as a key element of most of the grading systems designed to quantitate the severity of liver disease in order to determine organ allocation priorities.

The correction of coagulopathy should not be undertaken routinely as it may complicate the determination of liver diseases. All coagulation factors should continue to be monitored, especially prior to procedures or with symptoms of clinical bleeding.

Detoxification

Detoxification is another unique function of the liver. The hepatocytes attempt to make medications, foreign and toxic substances, and certain endogenous substances more water-soluble so they can be eliminated in urine or bile.

Mechanisms of detoxification include reduction; hydrolysis; conjugation; oxidation; excretion in bile; degradation; and storage of certain substances such as morphine, curare, and strychnine to be released at a later time into the circulation. Through these processes, the hepatocytes protect the body from drug-related hepatic injury.

Storage Functions

Lipid-soluble vitamins (A, D, E, K), cyanocobalamin (B_{12}), and minerals such as iron and copper are stored in the liver.

DISORDERS OF THE LIVER

Pathological conditions that affect the liver are cirrhosis, portal hypertension, ascites, hepatic encephalopathy, hepatorenal syndrome, different types of hepatitis, fulminant hepatic failure, Budd-Chiari syndrome, tumors, Wilson's disease, porphyria, hemochromatosis, alpha-1 antitrypsin deficiency, intrahepatic biliary dysplasia, biliary atresia, and Gilbert syndrome.

Cirrhosis

Cirrhosis is an outcome of progressive hepatic damage, either as a result of chronic inflammation or cholestasis. The anatomical hallmarks of this disorder are hepatic parenchymal inflammation and necrosis, nodular regeneration, loss of the centrilobular vein, and formation of new connective tissue (fibrosis). This makes it increasingly difficult for the liver to carry out essential functions.

Risk factors

In the United States, the most common causes of cirrhosis are hepatitis C, hepatitis B, alcoholic liver disease, and nonalcoholic fatty liver disease. Less common causes include autoimmune hepatitis, primary sclerosing cholangitis, medications, Wilson's disease, alpha-1 antitrypsin deficiency, celiac disease, idiopathic adulthood ductopenia, granulomatous liver disease, idiopathic portal fibrosis, polycystic liver disease, infection, congestive heart failure, hereditary hemorrhagic telangiectasia, and hepatic sinusoidal obstruction syndrome (previously termed hepatic veno-occlusive disease).

Symptoms

In patients with cirrhosis, symptoms include:

- Ascites, which develops as a consequence of increasing hydrostatic pressure. The change in pressures within the splanchnic bed forces proteinaceous fluid to leak into the abdominal cavity, eventually leading to hypoalbuminemia. The accumulation of fluid in the abdominal cavity reduces systemic volume and stimulates the production of aldosterone, a form of mineralocorticoid that further promotes fluid retention.
- Caput medusae, a cardinal feature of portal hypertension that appears as engorged veins radiating from the umbilicus. With the increased portal pressure, blood is shunted through the umbilical veins into the veins of the abdominal wall, resembling the serpent hair of Medusa from Greek mythology.
- Cruveilhier Baumgarten syndrome, a rare symptom of portal hypertension, which is a loud venous murmur heard over the epigastric area. This murmur is caused by blood flow from the portal vein to the umbilical vein branches. The murmur can be present even in the absence of caput medusae.
- Spider angiomata or spider telangiectasia, which are swollen blood vessels under the skin surface, mainly on the face, arms, and trunk, where small radiating vessels surround a central arteriole resembling a spider web. Spider angiomata, and also palmar erythema, may be caused by the liver's decreased ability to degrade estradiol.
- Gynecomastia and male-pattern hair loss can occur due to the enhanced conversion of androstendione to estrone and estradiol.
- Splenomegaly, which is believed to occur as a result of portal congestion, tissue hyperplasia, and liver fibrosis.
- Jaundice, the yellow discoloration of skin, oral mucus membranes, and sclera that is caused by an abnormal elevation of bilirubin attributed to the liver's impaired excretory function. The discoloration is usually detectable when serum bilirubin level is >2 mg/dL (34 mmol/L).

Diagnosis

In the clinical setting, liver biopsy collected either by percutaneous, transjugular, laparoscopic, or radiographically guided fine-needle aspiration (FNA) is useful in diagnosing cirrhosis.

The simplest approach to staging cirrhosis is to categorize it according to compensated and decompensated stages.

- **Compensated cirrhosis.** Patients typically have normal portal pressures and are generally asymptomatic or may report nonspecific symptoms, such as poor appetite, easy fatigability, weakness, and weight loss.
- **Decompensated cirrhosis.** Patients are at a higher risk of death due to severe associated complications such as bleeding varices, ascites, hepatic encephalopathy, spontaneous bacterial peritonitis, and hepatorenal syndrome. Clinical features include ascites, caput medusae, Cruveilhier Baumgarten syndrome, hormonal imbalance, splenomegaly, and jaundice.

Complications

Two major complications of cirrhosis include portal hypertension and hepatocellular carcinoma (HCC).

Portal hypertension

Portal hypertension leads to the development of collateral vessels that bypass the liver. Because of this bypass, toxins that are normally removed by the liver are instead introduced into the general circulation. The collateral vessels become enlarged, convoluted (varices) and can develop in the esophagus, stomach, abdominal wall, and rectum. These varices are fragile and can bleed with occasional fatal results. Portal hypertension also can lead to ascites and splenomegaly, a condition that further predisposes the patient to bleeding and infection. (Fig. 19.2). Esophageal varices are discussed in Chapter 13.

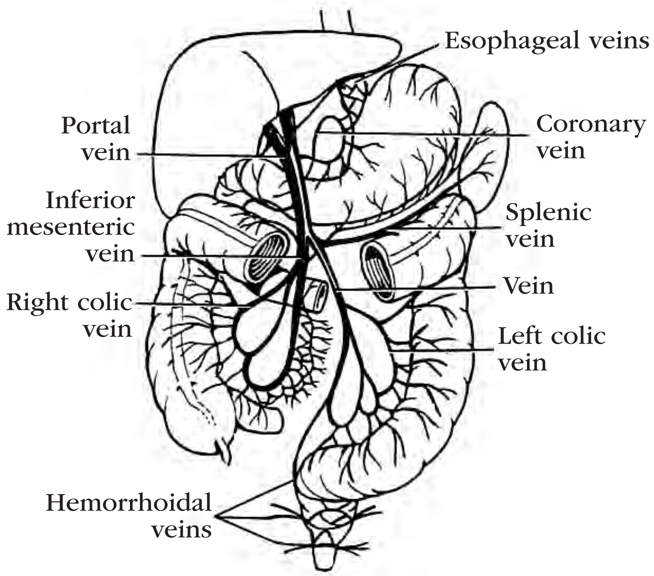

Figure 19.2. Portal Circulation Showing Collateral Circulation

Hepatocellular carcinoma (HCC)

The mechanism of development of HCC is thought to be caused by changes to the protective tips of chromosomes (telomere shortening) and decreased hepatocyte proliferation. With telomerase dysfunction, the regenerative capacity of the liver is reduced and furthers the progression to fibrosis formation. The activation of stellate cells also leads to the release of cytokines, growth factor, and products of oxidative stress, which decrease hepatocyte proliferation. These changes are believed to contribute to tumor formation.

The highest incidence of HCC is documented in sub-Saharan Africa and Eastern Asia. Areas such as North America, South America, Northern Europe, and Oceania have traditionally reported low HCC incidence, although recent reports found that HCC incidence in these areas, including the United States and Canada, is increasing (El-Serag, 2012).

HCC is discussed in more detail further in this chapter under Tumors.

Treatment

Treatment of the underlying cause in the early stage of cirrhosis may reverse the condition, although liver transplantation may be the only option for advanced disease that is considered to be irreversible.

Portal Hypertension

Portal hypertension is increased pressure within the portal vein that brings blood from digestive organs into the liver. Increased pressure is usually caused by resistance entering the liver such as from cirrhotic scarring.

Diagnosis

Assessing portal hypertension in liver cirrhosis is done by measuring the hepatic vein pressure gradient (HVPG), an estimate of the difference in pressure between the portal vein and the inferior vena cava. This technique is safer and less invasive than direct measurement of portal pressure.

HVPG >4 mmHg is considered portal hypertension, and higher scores are associated with more severe clinical manifestations and increased risk for complications.

Table 19.1. HVPG Thresholds in Relation to their Prognostic Significance

Compensated	
HVPG 10 mmHg	Formation of gastroesophageal varices, HCC, decompensation after HCC surgery
HVPG 12 mmHg	Variceal bleeding
HVPG 16 mmHg	First clinical decompensation in patients with varices, mortality
Decompensated	
HVPG 16 mmHg	Variceal bleeding, mortality
HVPG 20 mmHg	Failure to control active variceal bleeding, low one-year survival
HVPG 22 mmHg	Mortality in patient with alcoholic cirrhosis and acute alcoholic hepatitis
HVPG 30 mmHg	Spontaneous bacterial peritonitis

(Bleibel et al., 2017)

Complications

Portal hypertension progresses to the eventual development of such complications as ascites, hepatic encephalopathy, variceal hemorrhage, spontaneous bacterial peritonitis, hepatorenal syndrome, portal hypertensive gastropathy, hepatic hydrothorax, hepatopulmonary syndrome, portopulmonary hypertension, and cirrhotic cardiomyopathy.

Treatment

A transjugular intrahepatic portosystemic shunt (TIPS) is currently the most efficient treatment for portal hypertension. It creates a connection between a branch of hepatic vein and a branch of portal vein (usually the right branch) to improve blood flow and decrease portal hypertension.

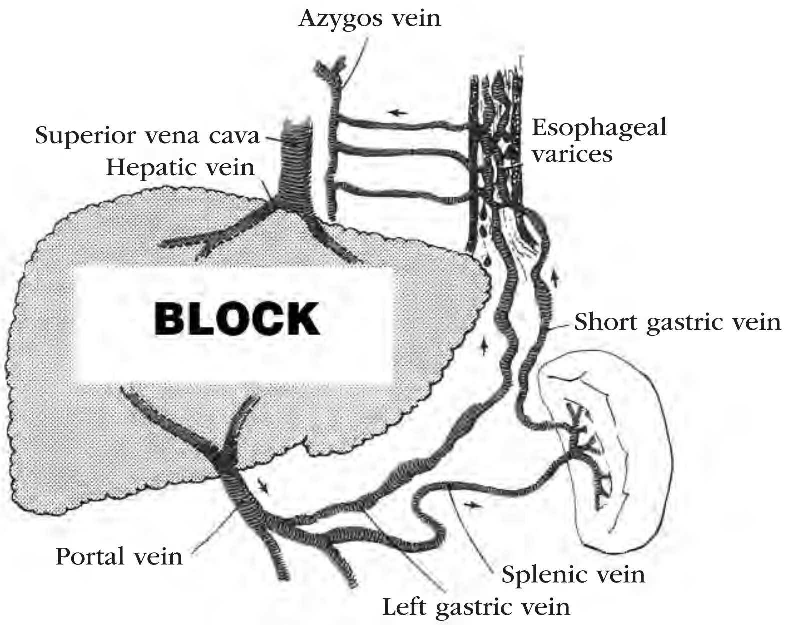

Figure 19.3. Hemodynamic Changes in Liver Cirrhosis Leading to the Development of Esophageal Varices

During the TIPS procedure, a needle catheter (Colapinto needle) is inserted via the transjugular vein and passed into the hepatic vein. The needle is advanced through the liver parenchyma to the portal vein. Finally, an intrahepatic metal stent is deployed to make the connection between the hepatic vein and the portal vein. This procedure is less invasive compared to previously used open surgical shunt procedures.

One of the most serious complications of TIPS is hepatic encephalopathy, which results from placement of a large intrahepatic shunt. Because of the intrahepatic stent, blood from the splanchnic circulation bypasses the liver, so circulating blood avoids detoxification before entering the systemic circulation and reaching the brain. Hepatic encephalopathy is manifested by neuropsychiatric symptoms, such as confusion, asterixis, and coma. Using a small diameter stent (<8 mm) can reduce the rate of encephalopathy.

Another complication of TIPS is stent occlusion, a rare occurrence seen more often in previously used bare stents. The introduction of polytetrafluoroethylene-covered stents has been highly effective in preventing occlusion.

Ascites

In patients with cirrhosis, ascites develops as a consequence of increasing hydrostatic pressure. The change in pressures within the splanchnic bed forces proteinaceous fluid to leak into the abdominal cavity, eventually leading to hypoalbuminemia. The accumulation of fluid in the abdominal cavity reduces systemic volume and stimulates the production of aldosterone, a form of mineralocorticoid that further promotes fluid retention.

Risk factors

Portal hypertension secondary to cirrhosis is the most common cause of ascites, accounting for most of ascites. Other causes are cancer (malignant-related ascites), cardiac ascites with a history of congestive heart failure, acute hemorrhagic pancreatitis or ruptured pancreatic duct in chronic pancreatitis, and trauma.

Symptoms

With peritoneal fluid accumulation, patients with ascites may have an increased waist size and weight. Severe ascites may cause abdominal distention and discomfort. The swollen abdomen may put pressure on the lungs leading to dyspnea and sometimes may lead to anorexia with increased pressure on the stomach.

Diagnosis

Ascites can be diagnosed with physical examination, ultrasonography, or computerized tomography (CT) scan. Paracentesis helps with fluid sampling for laboratory analysis to determine the cause.

Complications

Spontaneous bacterial peritonitis can occur as a complication of ascites. Manifestations of this complication include abdominal tenderness, fever, confusion, disorientation, and drowsiness. Appropriate use of antibiotics is key in preventing the deadly effects of this complication.

Treatment

The treatment strategy for patients with cirrhotic ascites is based on the grade of ascites.

Grade 1 ascites

Typically, this grade does not require treatment, as fluid accumulation is mild and only detectable by ultrasound.

Grade 2 ascites

This grade is manifested by moderate symmetrical abdominal distention. Treatment includes these recommendations:

- Dietary modification, which is directed toward reducing sodium intake.
- Administration of diuretics. Aldactone and CaroSpir, both belonging to the spironolactone family, are aldosterone receptor antagonists. These are diuretics that promote mobilization of fluid retention. Spironolactones are more effective than loop diuretics, and effects are seen in 3-5 days. However, a loop diuretic such as furosemide (Lasix) is sometimes added to counter the potassium-sparing effect of spironolactone and prevent hyperkalemia.

Grade 3 ascites

This grade involves severe abdominal distention. Treatment includes:

- Large volume paracentesis (LVP), which can safely be performed in the outpatient setting.
- Administration of IV human albumin 8 g/L for LVP >5 L. Albumin substitution is a rapid, safe, and effective way to reduce paracentesis-induced circulatory dysfunction (PICD). PICD is a complication of LVP that leads to hyponatremia, faster recurrence of ascites, kidney impairment, and poor survival.

Hepatic encephalopathy

One of the liver's functions is to convert nitrogen-containing compounds into urea to be excreted safely by the kidneys. In patients with liver cirrhosis, this conversion process can be severely impaired. The liver's inability to metabolize ammonia increases ammonia in the blood and the concentration in the brain, resulting in **hepatic encephalopathy (HE)**, also called portosystemic encephalopathy (PSE). Gamma-amino butyric acid (GABA), an inhibitory neurotransmitter, may play a role in HE development.

Risk factors

HE may arise as a complication of liver cirrhosis and advanced hepatic dysfunction.

Symptoms

Ammonia is toxic and causes impaired brain function that ranges from minor symptoms to neuropsychiatric symptoms, including coma.

Sleep pattern disturbance such as insomnia or hypersomnia are early symptoms. Neuropsychiatric signs and symptoms are seen in the advanced stages of the disorder and can include hand-flapping tremors (asterixis), hyperactive deep tendon reflexes, and transient decerebrate posturing.

Diagnosis

In addition to a thorough history, HE diagnosis is based on tests including CBC, liver function tests (LFTs), ammonia levels, and electroencephalogram. CT scan and MRI are used to exclude other brain conditions such as tumors.

Based on the 2014 joint guideline from the American Association for the Study of Liver Diseases (AASLD) and the European Association for the Study of Liver Diseases (EASL), treatment depends on the severity of a patient's HE based on grade as follows (Vilstrup et al., 2014):

- Minimal: Psychometric or neuropsychological alterations of tests exploring psychomotor speed/executive functions without evidence of mental change.
- Grade I: Changes in behavior, mild confusion, slurred speech, disordered sleep.
- Grade II: Lethargy, moderate confusion, asterixis, inappropriate behavior.
- Grade III: Marked confusion (stupor), incoherent speech, sleeping but arousable, bizarre behavior.
- Grade IV: Coma, unresponsive to pain.

Treatment

Treatment should be directed toward reversing any predisposing conditions, including infection and GI bleeding, and reducing ammonia level using lactulose. Treatment also includes determining the appropriate setting for care (e.g., critical care).

After precipitating causes are treated, the treatment of HE includes:

- Lowering blood ammonia concentrations with medications such as lactulose or lactitol. Both are available as an enema for patients who cannot tolerate the oral form. Drug therapy, the mainstay of treatment to lower blood ammonia concentration, works by preventing ammonia absorption in the GI tract.
- Administering antibiotics such as rifaximin (Xifaxan) or neomycin to reduce GI bacterial load known to be responsible for toxin production.
- Correcting hypokalemia, as hypokalemia increases renal ammonia production.
- Including dietary modifications that adjust intake to 35-40 kcal/kg/day. Protein restriction that avoids protein intake has been associated with mortality and is no longer recommended. However, consumption of protein should be limited to 1.2-1.5 g/kg/day. Fasting should be avoided because it leads to deamination, a metabolic process that promotes ammonia production after converting amino acid to glucose.

Patients who have minimal hepatic encephalopathy (MHE) and are at an increased risk for developing overt HE may benefit from drug therapy with lactulose or lactitol.

Hepatorenal syndrome

Hepatorenal syndrome (HRS) is a progressive renal failure that occurs in patients with significant liver disease, especially liver cirrhosis. Toxins build up due to kidney failure and contribute to the liver damage.

Risk factors

The exact cause of HRS is unknown but is attributed to the increase in nitric oxide release due to portal hypertension. Nitric oxide causes vasodilation of splanchnic circulation that subsequently reduces circulatory blood volume. The kidney's juxtaglomerular apparatus responds to the low volume by activating the renin-angiotensin-aldosterone mechanism, creating vasoconstriction and reduced blood flow to the kidneys. As a result, glomerular filtration rate, sodium excretion, and mean arterial pressure decline.

Symptoms

There are two forms of HRS: Type 1 and Type 2:

- Type 1 HRS has severe symptoms (e.g. decreased urination, edema, azotemia, hepatic encephalopathy) associated with rapid progression to kidney failure. Creatinine level rises to >2.5 mg/dL in less than 2 weeks. There is also a reduction in urine output to <500 ml/day.
- Type 2 HRS is less severe and is associated with slow progressive kidney failure.

Treatment

Management of HRS includes the identification, removal, and treatment of any factors known to precipitate renal failure.

The short-term treatment for HRS includes:

- Albumin, the most effective plasma expander used for HRS.
- Vasoconstrictors such as vasopressin (Vasostrict and Pitressin), norepinephrine (Levophed), midodrine (Orvaten), and octreotide (Sandostatin).

The combination of albumin, norepinephrine, midodrine, and octreotide is effective as a regimen. Dialysis may be helpful in patients with reversible forms of liver disease or while on the liver transplant waiting list. Liver transplantation will resolve hepatorenal syndrome in appropriate candidates.

HEPATITIS

Hepatitis is an inflammation of the liver that may be accompanied by parenchymal liver damage. It may be acute or chronic.

Viral forms of hepatitis include hepatitis A (HAV), hepatitis B (HBV), hepatitis C (HCV), hepatitis D (HDV), and hepatitis E (HEV).

Other types of hepatitis include alcoholic hepatitis, drug-induced hepatitis, autoimmune hepatitis, and nonalcoholic steatohepatitis (NASH).

Hepatitis A

Hepatitis A virus (HAV), a member of genus *Hepatovirus* in the family *Picornaviridae*, is the causative agent for hepatitis A infection. Once ingested, the virus replicates in the hepatocyte cytoplasm before being released into the circulatory system and feces. The incubation period is 15-50 days, during which patients are considered infectious. The host's immune response causes the liver injury.

Risk factors

HAV is transmitted from person to person by the fecal-oral route or by contaminated food and water.

Symptoms

Symptoms at the onset of HAV infection include nausea, vomiting, abdominal pain, fever, and malaise. After several days, infected patients may manifest dark colored urine, acholic stools, jaundice, scleral icterus, pruritus, fever, hepatomegaly, skin rashes, and arthralgia.

Complications

Fulminant hepatic failure associated with HAV infection is rare but can occur among individuals with underlying liver disease.

Treatment

HAV infection is self-limiting. Treatment is supportive care. Recovery is expected within 3-5 months, although relapse is possible after 6-12 months from onset. Patients infected by HAV become immune to the virus.

Measures to prevent HAV infection include handwashing, avoiding tap water and raw foods in areas with questionable sanitation, heating food to a minimum of 185°F (85°C), and using solutions that inactivate HAV (chlorine, iodine, and bleach) for cleaning.

In the United States, implementation of the hepatitis A vaccine has resulted in a decline in the incidence of HAV infection. The Advisory Committee on Immunization Practices (ACIP) of the U.S. Centers for Disease Control and Prevention (CDC) recommends vaccination for (CDC, n.d.a.):

- All children at 1 year. The vaccine is not given to children less than 12 months of age.
- Children between 2 and 18 years of age, if not given between 12 and 23 months.
- People at increased risk for complications related to hepatitis A infection.
- Anyone wishing to obtain immunity from HAV.

The ACIP also recommends vaccination against HAV for high-risk individuals, including:

- People traveling to or working in countries with intermediate or high rates of HAV infection.
- Men who have sex with men.
- People who use recreational drugs, both injectable and noninjectable.
- People who have risk of exposure based on their occupation such as working in a research laboratory. Hepatitis A vaccine is not routinely indicated for childcare, health care, sewage, and food service workers.

- People with chronic liver disease and clotting factor disorders.
- People in close contact, during the first 60 days following arrival in the United States, with an international adoptee from a country with intermediate to high HAV infection incidence.
- People who have direct contact with or recent exposure to someone with HAV.

In the United States, available hepatitis A vaccines include Havrix, Vaqta, and Twinrix (a combination hepatitis A and B vaccine). Although the hepatitis A vaccine is superior, the immune globulin GamaSTAN S/D can provide up to 2 months of protection from HAV. This injection is indicated for individuals who are:

1. Older than 40 years or younger than 12 months.
2. Immunocompromised and not responsive to the hepatitis A vaccine.
3. Allergic to the vaccines.
4. Suffering from a chronic medical condition such as chronic liver disease who are unable to wait for the second dose of the vaccine.

Hepatitis B

The causative agent of hepatitis B virus infection is hepatitis B virus (HBV), a member of *Hepadnavirus* family. The virus attaches itself to the hepatocyte membrane and penetrates the cells' nuclei for replication via reverse transcriptase. Some replicated viruses are then released, while some are recycled to repeat the replication process. The host's cytotoxic T lymphocyte response induces severe inflammatory liver disease.

Risk factors

Persons at risk for HBV infection include individuals who:

1. Use recreational injectable drugs.
2. Have multiple sexual partners or men who have sex with men.
3. Have household contacts with unvaccinated individuals with chronic HBV infection.
4. Are developmentally disabled and living in long-term facilities.
5. Are incarcerated.
6. Work with patients who are HBV positive.
7. Are on hemodialysis.
8. Have HCV infection.
9. Travel to countries where HBV is endemic.
10. Have chronic liver disease, HIV, or diabetes.

Symptoms

Early symptoms are similar to an allergic reaction or serum sickness-like syndrome and occur in the early stages of HBV infection. Following this prodromal period, signs and symptoms associated with acute infection include fever, fatigue (malaise), loss of appetite (anorexia), nausea and/or vomiting, jaundice, scleral icterus, dark urine, alcoholic stool, abdominal pain, myalgia, and arthralgia. Symptoms can last from 1 week to 1 month.

Diagnosis

The serological tests that are used to define different clinical states of HBV infection are hepatitis B surface antigen (HBsAg) and hepatitis B surface antibody (anti-HBs). These markers are used to screen asymptomatic individuals and diagnose HBV infection.

In acute infection, HBsAg becomes detectable around 1-10 weeks after exposure. The level subsequently drops to an undetectable level at 4-6 months, denoting recovery. In most patients, the disappearance of HBsAg is followed by the appearance of anti-HBs, a marker indicating long-term immunity. HBsAg that persists more than 6 months implies chronic infection. Other useful markers include hepatitis B core antigen (HBcAg), hepatitis B core antibody (anti-HBc), isolated anti-HBc, hepatitis B e antigen (HBeAg), and hepatitis B e antibody (anti-HBe).

Alanine and aspartate aminotransferases are also elevated.

Complications

Complications of chronic HBV infection are liver damage (cirrhosis), hepatocellular carcinoma, and fulminant hepatic failure and are associated with about 880,000 deaths per year (Ringehan, McKeating, & Protzer, 2017).

Chronic hepatitis, an inflammatory reaction of the liver that lasts more than 6 months, develops in less than 5% of adult patients with HBV infection (Kim & Kim, 2018; Trepo, Chan, & Lok, 2014). Chronic hepatitis may manifest itself in different ways.

- Mild hepatitis, which has a protracted course. Cirrhosis is rare in patients with mild hepatitis, but the incidence of hepatocellular carcinoma is increased after age 50. Symptoms are usually mild, if any. No therapy is needed and prognosis is generally good.
- Moderate to severe hepatitis, which is uncommon in patients with HBV infection. Compared to mild hepatitis, moderate to severe hepatitis is associated with higher levels of serum transaminases, along with mild hyperbilirubinemia and more severe portal and periportal inflammation with erosion of the limiting plate of peripheral hepatocytes (piecemeal necrosis). In more severe cases, necrosis may span the lobules (bridging necrosis), or multilobular collapse may be seen, starting a progressive process that leads to fibrosis, scar formation, and cirrhosis.

Treatment

The goals of treatment of HBV infection include decreasing serum HBV DNA levels in the blood; lessening infectivity; decreasing the level of hepatic inflammation; and preventing or slowing the development of cirrhosis, liver failure, and liver cancer. The ultimate goal in the treatment of HBV infection is the clearance of HBV from the liver and from sites of extrahepatic infection. This treatment process is completed when the patient clears HBsAg and develops detectable anti-HBs. The majority of adults who acquire acute HBV recover uneventfully. However, in patients with chronic HBV, prognosis depends on the severity of the disease and the patient's response to antiviral treatment.

There are no known therapies to eliminate HBV infection, nor is there documentation that any antiviral therapy against HBV leads to the prevention of cirrhosis, liver failure, or liver cancer. Compounding problems in the development of effective therapy for HBV infection include the presence of mutant forms of HBV, an altered immune system in immunosuppressed patients, consumption of alcohol, and coinfection with other viruses.

In-hospital management may be required for elderly patients and for patients who develop ascites or encephalopathy.

Treatment for HBV infection includes:

- One to 2 weeks of rest from strenuous physical activity.
- Nutritionally balanced diet.
- IV hydration and electrolyte management, if necessary.
- Subcutaneous vitamin K and fresh frozen plasma, in some cases.
- Medications used sparingly and alcohol avoidance.
- Liver transplantation as an option, although the risk of damage to the transplanted organ from HBV is high.
- Antivirals for chronic hepatitis B infection.

Antivirals:

Antivirals used for chronic HBV infection include nucleoside analogs or pegylated interferon (PegINF).

Nucleoside analogs, especially tenofovir (Vemlidy, Viread) and entecavir (Baraclude), are recommended as initial treatment because of their potent antiviral activity and that drug resistance is less likely to occur. Other nucleoside analogs currently available are lamivudine (Epivir-HBV, Epivir), and adefovir (Hepsera). Lamivudine is associated with a high drug resistance rate. Adefovir's effect in viral suppression is slow, and increased doses may lead to nephrotoxicity.

PegINF is also an initial treatment option for HBV infection, but it has been found to cause further liver damage as a result of necroinflammation and is associated with more side effects.

Vaccination:

Protection against HBV infection is available as a vaccination. The AICP recommends yeast-derived, single-antigen hepatitis B vaccines such as Recombivax HB, Engerix-B, and Heplisav B (CDC, n.d.b.). Twinrix (a combination of hepatitis A and B vaccines) and Pediarix (a combination of DTaP, hepatitis B, and inactivated polio vaccines) are also approved in the United States. The prevalence of HBV infection among children in the United States has significantly decreased – by 68% – since the initiation of universal hepatitis B vaccination (Nelson et al., 2016).

Hepatitis B vaccine is recommended for:

- All neonates.
- Individuals without HBV infection who live, travel to, or work in intermediate or high endemic areas.
- People who are at high risk, including those who are in sexual, household, and needle-sharing contact with HBV carriers, individuals in a setting with high HBV prevalence such as prisons and facilities for people who are developmentally disabled, and health care and public safety workers.
- People with health disorders such as end-stage renal disease, chronic liver disease, HIV infection, HBV infection, and diabetes.

Hepatitis C

Hepatitis C infection, formerly known as non-A, non-B hepatitis, is caused by the hepatitis C virus (HCV). It is estimated that 2.4 million people in the United States are living with HCV infection. The incidence of acute HCV infection has been increasing (CDC, n.d.c.).

Risk factors

The leading cause of HCV transmission is sharing needles of recreational injectable drugs. Other methods of transmission are needle-stick injury to health care workers and transmission by infected mother to baby. Sharing personal articles such as razors and toothbrushes, sexual contact with an infected person, and getting a tattoo in an unregulated setting are less-common methods of HCV spread.

In the past, HCV was commonly transmitted by blood transfusion and organ transplantation. In the United States, however, widespread screening of the blood supply has all but eliminated this source of infection.

Symptoms

The majority of patients with acute HCV infection are asymptomatic. For some, mild and flu-like symptoms may occur within 2 weeks to 6 months after HCV exposure. These symptoms may include fatigue, sore muscles, joint pain, fever, anorexia, nausea, abdominal pain, itching, dark urine, and jaundice.

Diagnosis

The HCV antibody test is a serologic test to diagnose HCV. A reactive result may indicate an active infection or that the patient had past exposure to HCV that has resolved. A positive HCV RNA test (a qualitative test that uses a process called polymerase chain reaction [PCR] or transcription-mediated amplification [TMA]) confirms chronic HCV infection.

Complications

Some 75% to 85% of acute cases of HCV infection progress to chronic HCV infection, and 10% to 20% of these patients develop cirrhosis (CDC, n.d.c.).

Hepatitis C infection is also associated with extrahepatic disease. The most important of these is cryoglobulinemia. This immune-related disease can cause kidney failure and a rash on the lower legs. Alcohol consumption increases the risk of developing cirrhosis and accelerates the time to cirrhosis. Patients who have HCV infection and cirrhosis are at a markedly increased risk of developing liver cancer.

Treatment

There are about 15% to 25% of infected patients who were able to clear HCV in their bodies without treatment (CDC, n.d.c.). However, reinfection is possible even in those whose infection was cleared by treatment.

Selection of treatment regimen is based on prior treatment history, which includes exposure and prior response to therapy. For example, there are differences in establishing the treatment plan for treatment-naive patients versus for treatment-experienced patients who previously did not benefit from treatment with peginterferon, ribavirin (Rebetol, Ribasphere, Virazole), and protease inhibitors.

In the treatment of HCV infection, knowing the genotype helps in finding the most effective treatment. There are six known genotypes, and all pose a risk for developing liver cirrhosis. There are two subtypes of genotype 1: 1a and 1b. Cirrhosis and liver cancer risk are higher among those infected with subtype 1b and genotype 3 (Ringehan, McKeating, & Protzer, 2017).

Direct-acting antivirals (DAAs):

In 2016, the Food and Drug Administration (FDA) approved oral direct-acting antivirals (DAAs) that are pangenotypic, meaning they are effective against all genotypes of HCV. DAAs target the virus itself by disrupting viral replication at specific steps within the lifecycle of HCV. They have stronger sustained virological response (SVR) when combined with one another. There are four classes of DAAs:

1. nonstructural proteins 3/4A (NS3/4A) protease inhibitors (PIs).
2. NS5B nucleoside polymerase inhibitors (NPIs).
3. NS5B non-nucleoside polymerase inhibitors (NNPIs).
4. NS5A inhibitors.

DAAs are significantly more tolerable in contrast to prior therapies. The duration of these newer regimens is for as little as 12 weeks, with few side effects and rare occurrence of serious adverse events. DAAs are an effective cure, even for patients who may be difficult to treat such as those with compensated cirrhosis or end-stage renal disease or who have not benefited from previous DAA treatment. Overall, DAAs provide a high rate of success and safety, and eventual elimination of HCV may be possible.

Interferon-free regimens:

Several interferon-free regimens are available, appear effective, and are relatively safe for patients with decompensated cirrhosis. The following medications are FDA approved:

- Epclusa (sofosbuvir; velpatasvir).
- Harvoni (ledipasvir; sofosbuvir).
- Sovaldi (sofosbuvir).
- Vosevi (sofosbuvir; velpatasvir; voxilaprevir).
- Daklinza (daclatasvir).

Patients with decompensated cirrhosis or a MELD score >10 should be evaluated for liver transplantation prior to initiation of HCV therapy.

Development of a hepatitis C vaccine is underway.

Hepatitis D

Hepatitis D infection is caused by the hepatitis D virus (HDV), also referred to as hepatitis delta virus or delta agent. For HDV to complete virion assembly and secretion, the patient must also be infected with hepatitis B virus, so patients with HDV infection are always dually infected with HBV and HDV. HDV infection occurs in about 5% of those with chronic HBV infection (Hung et al., 2014).

Risk factors

HDV is endemic in Italy and other Mediterranean countries and several regions of South America. In non-endemic areas such as North America and northern Europe, HDV is transmitted primarily by serum and is found mainly in children who have had multiple transfusions, men who have sex with men, and people who use recreational injectable drugs.

Symptoms

Patients with acute HDV infection may develop fatigue, anorexia, nausea, vomiting, right upper quadrant pain, acholic stool, dark urine, and jaundice. Patients with chronic HDV infection have few symptoms for several years before severe chronic liver complications develop, then are likely to experience such symptoms as weakness, fatigue, weight loss, abdominal swelling, edema, pruritus, and jaundice.

Diagnosis

The diagnosis of HDV infection is made with high titers of IgG and IgM anti-HDV. Detection of HDV RNA confirms the diagnosis. Additional tests may include elastography and liver biopsy.

Complications

Although rare, acute HDV infection can lead to acute liver failure. Chronic HDV infection may lead to cirrhosis, liver failure, and liver cancer.

Treatment

Specific treatment for acute HDV infection is not established. However, fulminant hepatitis related to HDV and HBV infection has been reversed by foscarnet sodium (Foscavir) in some patients. The success rate of interferon alfa (IFNa) is low, but it is currently the only drug approved for chronic HDV infection.

There is no vaccination available for HDV. Protection against HBV, which is required for HDV replication, is an important measure in preventing HDV infection.

Hepatitis E

The causative agent of hepatitis E infection is the hepatitis E virus (HEV) in the genus *Hepevirus* in the family *Hepeviridae*. There are five known genotypes of HEV, four of which are associated with human infection. HEV is found worldwide, but the disease is most common in East and South Asia. The incubation period for HEV infection is 15-60 days (deWit & Kumagai, 2014). In chronic HEV infection, the disease continues to be detectable in blood or stool for longer than 6 months.

Risk factors

Transmission of HEV can be fecal-oral, consumption of raw or undercooked meat, blood transfusion, and mother-to-child. Chronic HEV infection almost exclusively occurs among patients who are immunocompromised.

Symptoms

Acute HEV infection is self-limiting and primarily asymptomatic or mildly symptomatic. Symptoms may include jaundice, malaise, anorexia, nausea, vomiting, joint pain, pruritus, skin rashes, and elevated ASL/ALT.

Complications

Acute hepatic failure associated with HEV infection is a potential complication, although this is rare (i.e. 0.4%) in the United States (Fontana et al., 2012).

Treatment

Preventive measures for HEV infection include avoiding food from a setting where preparation is questionable and avoiding eating raw or undercooked meat, especially deer, wild boar, and pig liver sausage.

The treatment of acute HEV infection is mainly supportive, although a liver transplant may be necessary with fulminant hepatic failure.

The antiviral ribavirin (Rebetol, Ribasphere, Virazole) is recommended for chronic HEV infection, along with reduction of immunosuppressive therapy. Blood level monitoring is essential to prevent drug toxicity. Ribavirin is teratogenic and should not be used for women who are pregnant. For patients who do not respond to ribavirin therapy, peginterferon and sofosbuvir may be considered.

Hepatitis G virus

Hepatitis G virus (HGV), referred to as GB virus C (GBV-C), was identified in 1995. The virus is a single-strand RNA belonging to the *Flaviviridae* family.

Risk factors

Blood and sexual contact are the most likely mode of transmission.

Symptoms

Most individuals with GBV-C are asymptomatic, and the incubation period is unknown.

Diagnosis

A detectable HGV RNA by reverse transcriptase PCR indicates GBV-C infection. There is no serologic test available for GBV-C.

Complications

Most of GBV-C infections are cleared, although, in some cases, it can become chronic.

Patients with HCV infection are often co-infected with GBV-C, and a higher GBV-C infection incidence is associated with B cell non-Hodgkin's lymphoma.

Treatment

There is no current recommended treatment for GBV-C.

Alcoholic hepatitis

Alcoholic hepatitis is inflammation of the liver that is the result of alcohol use.

Risk factors

The major risk factor for alcoholic hepatitis is heavy consumption of alcohol (i.e. drinking >3.5 oz daily for 20 years or more). However, sex (i.e. being female), obesity, genetics, race and ethnicity (i.e. African-American and Hispanics), and binge drinking are also identified as other risk factors affecting those who consume less alcohol.

Symptoms

Symptoms of alcoholic hepatitis include right upper quadrant abdominal pain, fever, vomiting, anorexia, and dark urine. Unlike other forms of hepatitis, patients do not present with jaundice. Physical signs include an enlarged and tender liver, splenomegaly, signs of chronic alcohol misuse and cirrhosis, and mental status abnormalities.

Diagnosis

Biopsy of the liver in patients with a history of alcoholism may show scattered fatty deposition, hepatocyte degeneration and necrosis, pericellular fibrosis and inflammation, cholestasis, and cirrhosis.

Complications

Alcoholic hepatitis is considered a precirrhotic lesion. If not reversed, alcoholic hepatitis can be deadly with significant short-term mortality. Patients who are untreated are at higher risk of dying 1 month from the onset of signs and symptoms (Pang et al., 2015).

Treatment

Treatment for alcoholic hepatitis can include the following measures:

- Alcohol abstinence is critical to improve the outcome of treatment and prognosis.
- Corticosteroid is a first-line drug used to reduce liver inflammation associated with alcoholic hepatitis. This drug increases the survival rate for patients with severe alcoholic hepatitis, although it is linked to side effects such as increased fungal infection risk, hyperglycemia, weight gain, and electrolyte imbalance. Corticosteroids are not recommended for patients with coexisting sepsis, acute pancreatitis, and GI bleeding.
- Pentoxifylline (Pentoxil) is an alternative drug that also has a beneficial effect for patients with alcoholic hepatitis who do not respond to corticosteroids. Although pentoxifylline's exact action is not clearly understood, the drug may have a protective mechanism to prevent hepatorenal syndrome.
- Nutritional supplementation should also be considered, especially in patients with alcoholic hepatitis who are prone to protein-calorie malnutrition.
- Liver transplantation may be beneficial for patients with alcoholic hepatitis. Before being considered for liver transplantation, however, patients are required to satisfy the criterion of at least 6 months of alcohol abstinence. Other barriers include the shortage of donor organs, sociocultural factors, and public opinion regarding the self-inflicted nature of alcoholic hepatitis.

Toxic hepatitis

Toxic hepatitis is inflammation of the liver that is cause by a drug or toxic substance.

Risk factors

Drugs and toxic substances can cause a wide range of hepatic injuries, including:

- A predictable dose-related hepatotoxic reaction such as that produced by acetaminophen (Tylenol), carbon tetrachloride, and methotrexate.
- An unpredictable non-dose-related viral-like hepatitis, with or without cholestasis, which is produced by such drugs as isoniazid (Laniazid), flurazepam, and methyldopa.
- Cholestasis, related to the use of anabolic steroids, birth control pills, and haloperidol (Haldol).

Symptoms

The clinical presentation of drug-induced hepatitis ranges from asymptomatic with transaminase elevations to patients who develop fulminant hepatic failure.

Complications

Patients who ingest massive doses of acetaminophen are at risk for fulminant hepatic failure.

Treatment

Most patients recover within 1 to 2 weeks of discontinuing the offending agent.

For acetaminophen toxicity, appropriate therapy must be instituted within 24 hours of ingestion, including ipecac-induced emesis and administration of N-acetylcysteine in a loading dose, followed by maintenance doses until plasma acetaminophen falls below hepatotoxic levels.

Autoimmune hepatitis

Autoimmune hepatitis is believed to be a result of an environmental agent that causes an abnormal T-cell response that specifically targets the liver, causing inflammation, elevation of antibodies, and hypergammaglobulinemia.

Risk factors

Autoimmune hepatitis affects genetically susceptible individuals and affects more women than men (Ngo, Styen, & McCombe, 2014).

Complications

Autoimmune hepatitis may result in cirrhosis and fulminant hepatic failure.

Treatment

Prednisone is the drug of choice for initial treatment of autoimmune hepatitis. However, due to Predisone's side effects, azathioprine (Imuran, Azasan) is more favored as a long-term maintenance drug, especially during remission. In patients who are unable to tolerate azathioprine, mycophenolate mofetil (Cellcept) may be used as an alternative.

Nonalcoholic fatty liver disease and nonalcoholic steatohepatitis

Nonalcoholic fatty liver disease (NAFLD) is the accumulation of fat in the liver (hepatic steatosis) that is not associated with alcohol exposure and shows no evidence of liver inflammation.

Nonalcoholic steatohepatitis (NASH) is NAFLD with evidence of liver inflammation.

NAFLD includes a range of conditions from Type 1 to Type 4, with Types 3 and 4 being considered NASH.

Risk factors

Risk factors for developing NASH include obesity, type 2 diabetes, and hyperlipidemia. Fat accumulates in the liver when the liver cannot degrade or export the lipids. The diagnosis of NASH will likely increase as millions of American adults are overweight or obese and the incidence of obesity has increased throughout the world (Estes et al., 2017).

Symptoms

The most common signs and symptoms of NASH are fatigue, vague right upper quadrant pain, weight loss, and weakness. The advanced stage of NASH includes such symptoms as lack of appetite, nausea, varices, loss of interest in sex, ascites, itching, edema, and encephalopathy.

Diagnosis

NAFLD is best detected by ultrasonography and NASH is diagnosed by liver biopsy.

Complications

NASH can develop into cirrhosis.

Treatment

The goal of treatment is to remove the cause of the fatty infiltration of the liver and remove exposure to factors that may worsen the condition. Weight loss (even if slightly overweight), control of blood sugar, low-fat diet, and aerobic exercise continue to be the mainstay management for NASH. Vitamin E and pioglitazone (Actos) continue to show good results in reducing disease activity.

Bariatric surgery is an option for selected patients, usually young adults (<40–50 years of age) with a body mass index (BMI) of >35. This surgery is high risk with the potential for serious complications (see Chapter 32).

There are several agents being studied to treat NASH, including peroxisome proliferator-activator receptors (PPARs), farsenoid X receptor (i.e., obeticholic acid [OCA]), glucagon-like peptide (GLP-1), lipid-altering agents such as steatoryl-CoA desaturase (SCD), SCD-1 inhibitor (Aramchol), and statins (HMG-CoA reductase inhibitors). Further studies are needed to fully confirm their efficacy (Oseini & Sanyal, 2017).

Fulminant hepatic failure

Fulminant hepatic failure, also referred to as acute liver failure, is loss of liver function that has developed over a period of less than 26 weeks in a patient without cirrhosis or pre-existing liver disease.

Risk factors

Causes include acetaminophen toxicity, idiosyncratic drug reactions, viral hepatitis, alcoholic hepatitis, autoimmune hepatitis, Wilson's disease, ischemic hepatopathy, Budd-Chiari syndrome, SOS, acute fatty liver of pregnancy / HELLP (hemolysis, elevated liver enzymes, low platelets) syndrome, malignant infiltration of the liver, partial hepatectomy, toxin exposure (including mushroom poisoning), sepsis, heat stroke, and hemophagocytic lymphohistiocytosis (primarily in children).

Symptoms

Initial nonspecific symptoms may include jaundice, right upper quadrant pain, fatigue, malaise, anorexia, nausea, and vomiting.

Diagnosis

The defining characteristics of fulminant hepatic failure are encephalopathy and abnormal prothrombin time (INR ≥1.5).

Complications

In its progressive stages, patients may develop ascites, coagulopathy, GI bleeding infection, and cerebral edema resulting in seizures and mental status changes such as lethargy, confusion, and eventually coma.

Treatment

Patients should be managed in centers with an active liver program, as liver transplantation is the only treatment for fulminant hepatic failure.

Care should include frequent neurologic checks and a quiet, restful environment. Admission to an intensive care unit is indicated in the later stages of the disease. Routine blood, urine, sputum cultures, and chest radiographs are recommended as patients are at high risk for bacterial and fungal infections in the blood and urinary and respiratory tracts.

Care options include the following:

- Initial fluid resuscitation using normal saline is important for patients with hemodynamic derangements. Half-normal saline with 75 mEq/L of sodium bicarbonate can be used for patients with acidosis. Overhydration should be avoided to prevent worsening cerebral edema.
- A vasopressor and norepinephrine may be required for patients with poor response to fluid resuscitation.
- A proton pump inhibitor (PPI) or histamine2 (H_2) blocker is given prophylactically to prevent bleeding in the GI tract, a common site of bleeding in fulminant hepatic failure.
- Antibiotic prophylaxis remains controversial and is only recommended when there is evidence of active infection or clinical deterioration. Piperacillin/tazobactam (Zosyn) or fluoroquinolones are drugs of choice, while aminoglycosides are avoided due to their nephrotoxic effects.
- An intake of 60 g of protein per day is recommended to prevent catabolism of protein stores in the body. Severe protein restriction should be avoided.
- Mannitol (0.5 to 1.0 g/kg) is given to patients with increased intracranial pressure (ICP) to achieve an ICP <25 mmHg and a cerebral perfusion pressure >50 mmHg.

Routine use of lactulose for encephalopathy secondary to acute liver failure is no longer recommended, as the drug does not improve overall outcomes and causes bowel distention that results in technical difficulties during liver transplantation.

Budd-Chiari syndrome

Budd-Chiari syndrome (BCS) is a rare condition characterized by an obstruction of the hepatic venous outflow tract at the level of the small hepatic veins, large hepatic veins, and suprahepatic segment of the inferior vena cava.

Risk factors

Causes include malignancy, thrombogenic disorders, inflammatory disorders, and other risk factors such as hormone disorders.

Symptoms

In its classic form, BCS is a nearly complete obstruction of flow caused by the acute formation of a clot at the opening or an acute phlebitis of the hepatic veins into the inferior vena cava. This sudden event is followed by the onset of hepatomegaly, pain, ascites, and liver failure.

Diagnosis

The diagnosis of BCS is usually established by Doppler ultrasonography, CT scan, or magnetic resonance imaging (MRI).

Complications

The course of BCS is unpredictable. In some patients, complete spontaneous remission occurs with or without remitting aggravation of liver insufficiency or portal hypertension. In other patients, secondary BCS is related to invasion or compression of the veins by a lesion.

Treatment

Treatment is based on cause. Anticoagulation and thrombolytic therapy are possible treatments. Angioplasty with or without stenting is an option. If angioplasty fails or is impossible, transjugular intrahepatic portosystemic shunting (TIPS) may be performed. Liver transplant may increase survival in the most severe cases.

Tumors

Liver lesions that are considered benign include hemangiomas, focal nodular hyperplasia, hepatic adenoma, idiopathic noncirrhotic portal hypertension, and regenerative nodules. Malignant liver lesions include hepatocellular carcinoma (HCC), cholangiocarcinoma, and metastatic disease.

Risk factors

Primary cancerous liver tumors originate in either the hepatocytes, bile duct epithelium, or mesenchymal tissue. Predisposing factors include a history of cirrhosis, hepatitis B, hepatitis C, alcohol misuse, androgen therapy, and exposure to hepatotoxic chemicals. Metastasis is generally to the regional lymph nodes, lungs, and peritoneum. Primary liver tumors are less common than metastatic or secondary liver tumors, which often spread from the lungs, breasts, GI tract, thyroid, prostate, or skin.

Symptoms

The majority of solid liver lesions are asymptomatic, although abdominal pain can occur in ruptured and bleeding adenoma and HCC.

Diagnosis

Although the differential diagnosis of solid liver lesions is broad, many of them can be diagnosed noninvasively. Diagnosis is made by ultrasonography, CT scan, or arteriography.

Treatment

Treatment for very large tumors involves tumor excision to reduce the risk of tumor rupture and hemorrhage.

Hepatocellular carcinoma

Hepatocellular carcinoma (HCC) is the most common primary malignant tumor of the liver. HCC is the third most common cause of cancer-related death in the world and seventh in the United States (Yezaz, Mian, & Rowe, 2017). The incidence of HCC increases progressively with age in all populations and is rare in children (Khanna & Verma, 2018).

Risk factors

Risk factors for HCC are hepatitis B, hepatitis C, cirrhosis, repeated exposure to environmental toxins such as aflatoxin (from contaminated corn, soybeans, and peanuts), chewing betel nut, and microcystin (from contaminated pond-ditch water). Other important factors include tobacco and alcohol misuse, diabetes mellitus, obesity, NAFLD, hemochromatosis, alpha-1 antitrypsin deficiency, acute intermittent porphyria, cholelithiasis, and cholecystectomy.

Symptoms

Most patients with liver cancer are asymptomatic until the disease is well advanced. Symptoms are similar to all liver failure diagnoses, including abdominal pain, weight loss, weakness, nausea, and vomiting. Physical signs are hepatomegaly, hepatic bruit, ascites, splenomegaly, jaundice, wasting, and fever. Paraneoplastic manifestations are hypoglycemia, polycythemia, hypercalcemia, sex hormone changes, portal hypertension, watery diarrhea, porphyria, neuropathy, osteoarthritis, and carcinoid syndrome.

Diagnosis

Screening of individuals in high-risk groups for liver cancer using ultrasound and assessing serum levels of alpha fetoprotein (AFP), des-y-carboxy prothrombin, or a-l-fucosidase are essential steps to finding small tumors and applying curative procedures.

Hepatic tumor diagnosis is by CT scan, MRI, hepatic angiography, and needle biopsy examination. Large elevations in serum AFP are virtually diagnostic for primary hepatocellular carcinoma.

Complications

Prognosis is grim if the patient has a large tumor or has any evidence of metastatic disease.

Treatment

For patients with a localized mass without lymph node, bile duct, or blood vessel involvement or distal metastases, up to 80% of the liver may be resected. Curative procedures include lobe resection and liver transplantation.

Other treatment options are alcohol injection or radiofrequency ablation, chemoembolization, and chemotherapy. Pain may be relieved with chemotherapeutic liver perfusion via the hepatic artery. Ligation or occlusion of the hepatic artery may temporarily slow cell growth and activity. If the tumor obstructs the bile duct, endoscopic or transhepatic insertion of a biliary decompression catheter or stent may provide some relief.

Wilson's disease

Wilson's disease is a rare autosomal-recessive disorder that is characterized by defective excretion of copper into bile, which leads to excessive amounts of copper accumulating in the brain, liver, kidneys, and cornea. This accumulation of copper causes tissue necrosis, hepatic disease, and potential hepatic failure.

Risk factors

Wilson's disease is linked to a mutation in the ATP7B gene acquired genetically from both parents in an autosomal recessive pattern. This gene plays an important role in the transport of copper from the liver to other parts of the body.

Symptoms

Clinical features in patients with Wilson's disease include:

- Physical signs. Patients with Wilson's disease have a characteristic rusty brown ring of pigment around the periphery of the cornea called a Kayser-Fleischer ring and a bluish discoloration on the fingernails (lunulae ceruleae).
- Hepatic disorders. These include jaundice, ascites, asymptomatic hepatomegaly, isolated splenomegaly, persistently elevated aspartate transaminase (AST) and alanin transaminase (ALT), acute or chronic hepatitis, fatty liver, and cirrhosis.
- Neurologic symptoms. Symptoms include dysarthria, ataxia, tremors, Parkinsonism, drooling, seizures, urinary incontinence, autonomic dysfunction, and abnormal involuntary body movements such as dystonia, risus sardonicus, chorea, athetosis, hyperreflexia, and myoclonia.
- Psychiatric disorders. Disorders include depression, bipolar disorder, schizophrenia, and dementia.
- Other disorders. These include renal abnormalities, ocular sunflower cataracts, pancreatitis, hypoparathyroidism, menstrual irregularities, cardiomyopathy, and skeletal abnormalities such as premature osteoporosis and arthritis.

Diagnosis

Wilson's disease is diagnosed by a combination of diagnostic tests such as blood and urine tests, eye exam, liver biopsy, and genetic testing. Once the diagnosis is confirmed, all first-degree relatives should be screened for the disease.

Complications

Without treatment, patients with Wilson's disease invariably succumb to liver disease or neurological complications.

Treatment

Treatment of Wilson's disease involves life-long therapy with penicillamine (Cuprimine, Depen). Trientine (Syprine) is also an effective treatment and is indicated especially for patients who are intolerant of penicillamine. Both drugs are chelators and promote excretion of copper into the urine. Zinc, which interferes with the absorption of copper, is sometimes used as maintenance therapy and has minimal side effects. Zinc appears preferable for pre-symptomatic children under age 3.

Liver transplantation is the only option for patients with acute liver failure or those with decompensated liver disease unresponsive to medical therapy.

Patients should avoid the intake of high concentrations of copper in foods (shellfish, nuts, chocolate, mushrooms, and organ meats) and water (well water or water transported in copper pipes). Copper containers or cookware should not be used to cook or store food. A water-filtering system that filters out copper is advisable.

Porphyria

Porphyria is a hereditary or acquired enzyme defect in which the biosynthesis of heme in either the bone marrow or liver leads to an overproduction of porphyrins or their precursors.

Types of porphyria:

- Acute intermittent porphyria (AIP) is an autosomal-dominant inherited disease that usually occurs as episodes over the course of several hours or a few days. The most common symptom is severe abdominal pain, nausea, vomiting, constipation or diarrhea, and abdominal distention. Ileus and urinary retention, peripheral neuropathy, psychiatric symptoms (e.g., paranoia and hallucination), and coma can also occur. Porphyrins may be normal or slightly elevated in stool and plasma. The erythrocyte porphobilinogen deaminase (PBGD) in blood is approximately half the normal level.

- Hereditary coproporphyria (HCP) is an autosomal-dominant inherited disease characterized by large amounts of coproporphyrin III in the feces and the urine. Protoporphyrin is also mildly elevated in the feces. Patients develop photosensitivity.
- Variegate porphyria (VP) is an autosomal-dominant disease characterized by large amounts of protoporphyrin and copoporphyrin III in the feces. Plasma porphyrin and plasma fluorescence are also elevated. VP is most common in the South African white population. It can produce photosensitivity.
- Delta-aminolevulinic acid (ALA) dehydratase (ALAD) porphyria (ADP) is an autosomal-recessive inherited disease and a rare form of acute hepatic porphyria. Its clinical features are indistinguishable from other forms of acute porphyrias. Urine detection of delta-aminolevulinic acid and molecular genetic testing are used to confirm the diagnosis.

Symptoms

Porphyrias can be classified as either erythropoietic or hepatic. However, categorizing porphyrias based on their distinctive clinical features is more useful in establishing a diagnosis and selecting the appropriate treatment. The three main clinical categories of porphyrias are:

- Blistering cutaneous porphyrias. These cause chronic, blistering, cutaneous lesions on parts of the skin that are exposed to the sun. The most common porphyria under this category and the most common in adults is porphyria cutanea tarda (PCT).
- Acute nonblistering cutaneous porphyrias. These cause photosensitivity. The most common porphyria under this category is erythropoietic protoporphyria (EPP). It is the most common porphyria in children and the third most common in adults (Dawe, 2017).
- Acute hepatic porphyrias. These are characterized by neuropathic abdominal pain, autonomic changes such as tachycardia and hypertension, muscle weakness, sensory loss, and pain in the back, chest, and extremities. The four types of hepatic porphyrias include (ADP), variegate porphyria (VP), hereditary coproporphyria (HCP), and acute intermittent porphyria (AIP).

Diagnosis

Differentiating the type of porphyria is based on symptoms and types:

- In blistering cutaneous porphyrias, the initial screening test is total plasma or urine porphyrins. Additional tests include fractionation and fluorescence screen.
- In acute nonblistering cutaneous porphyrias, screening tests include erythrocyte total protoporphyrin and measurement of metal-free and zinc protoporphyrin.
- In acute hepatic porphyrias, the initial test is a spot urine porphobilinogen (PBG). Additional tests to determine the specific type of porphyria under this category include urine ALA, plasma and fecal porphyrins, erythrocyte ALAD activity, lead level, and urine organic acids.

Treatment

Treatment depends of the symptoms and type of porphyria:

- For blistering cutaneous porphyrias, treatment includes phlebotomy or low-dose hydroxychloroquine (Plaquenil). Generally, sun exposure should be avoided.
- For acute nonblistering cutaneous porphyrias, avoiding sun exposure is also important.
- For acute hepatic porphyrias, even in the absence of biochemical confirmation, hemin (Panhematin), a drug that inhibits delta-aminolevulinic acid synthetase and limits the rate of the porphyrin/heme biosynthetic pathway, should be considered early in the diagnosis for patients with known porphyria to reduce risk of prolonged complications (e.g., nerve damage) and death.

Hemochromatosis

Hemochromatosis consists of three classifications of iron overload syndromes:

- Hereditary hemochromatosis.
- Secondary iron overload.
- Miscellaneous disorders.

Risk factors

Hereditary hemochromatosis is a recessively-inherited gene disorder often affecting people of north European descent (National Library of Medicine, 2019b). It is caused by a mutation of the homeostatic iron regulator (HFE) gene that causes problems with iron metabolism and is characterized by excessive tissue iron deposition. Screening to detect early disease and prevent complications is recommended in first-degree relatives of patients with HFE gene-related hereditary hemochromatosis.

There are four main categories of pathophysiological mechanisms of hereditary hemochromatosis:

- Increased absorption of dietary iron in the upper intestines.
- Decreased expression of the iron regulatory hormone hepcidin.
- Altered function of HFE protein.
- Tissue injury and fibrogenesis induced by iron.

Secondary iron overload, also known as secondary hemochromatosis, is typically seen in patients with hemoglobinopathies (e.g., sickle cell disease, thalassemia, sideroblastic anemia), congenital hemolytic anemias, myelodysplasia, excessive oral iron intake, increased iron absorption, and repeated blood transfusions (significant tissue deposition of iron is seen with >40 units transfused). Risk factors for secondary iron overload include alcoholism, family history of diabetes, heart disease, liver disease, and excessive intake of dietary iron especially when taken with vitamin C supplements.

Miscellaneous disorders refer to other disorders that are also associated with abnormal iron deposition such as congenital atransferrinemia and aceruloplasminemia. Miscellaneous iron overload disorders are rare conditions.

Symptoms

Hemochromatosis can be asymptomatic, frequently with no physical findings, although most patients have hepatomegaly.

Symptomatic patients will have complaints of weakness, malaise, loss of libido, weight loss, changes in skin color, right upper quadrant abdominal pain, and arthralgias. Physical findings can present as hepatomegaly, splenomegaly, ascites, arthritis, swollen joints, chondrocalcinosis, cardiomyopathy, congestive heart failure, increased skin pigmentation, hypothyroidism, porphyria cutanea tarda, testicular atrophy, or any sign of liver failure. Hemochromatosis is sometimes called "bronze diabetes" because clinical findings usually include hyperglycemia and darkening of the skin.

Diagnosis

Early diagnosis of hereditary hemochromatosis is crucial in preventing complications. Clinical diagnosis is based on:

- Lab tests that show increased iron stores, demonstrated by an elevated serum ferritin and serum iron concentration and by transferrin saturation.
- Plain X-ray films of the hands that show chondrocalcinosis, sclerosis of subchondral bone, loss of articular cartilage, and subchondral cyst formation.
- CT scans of the liver that show marked increases in density.
- Liver biopsy, which is recommended for diagnosis, staging the degree of liver disease, and identifying the prognosis. Biopsy of the liver also permits estimation of tissue irons by histochemical staining and chemical analysis of hepatic iron concentration.
- Genetic testing that can detect gene mutations associated with the disorder.

Secondary iron overload is diagnosed by measuring serum ferritin, serum iron, and trasferrin saturation; fasting serum iron; and iron binding capacity. Additionally, genetic testing along with a thorough history may be done to rule out hereditary hemochromatosis.

The diagnoses of atansferrinemia and aceruloplasminemia are established by detailed history and genetic testing.

Complications

Iron accumulation in the organs may lead to liver cirrhosis and hepatocellular carcinoma. Although hemochromatosis is generally not common in infants, if present, it can lead to the need for a liver transplant.

Treatment

Phlebotomy is a treatment modality that can reduce iron overload. During the procedure, about 200-250 mg of iron is removed with every 500 ml of whole blood. Initially, patients may tolerate removal of 0.5-2 units per week, depending on age, sex, BMI, and presence of cardiopulmonary disease. A more accelerated phlebotomy schedule is usually permissible after a few weeks. As maintenance, most patients require a 500 ml phlebotomy every 2-4 months. Patients should be encouraged to keep hydrated and avoid strenuous exercise 24 hours after phlebotomy to prevent hypovolemia.

For patients who are poor candidates for phlebotomy—for example, patients with moderate to severe anemia, symptomatic hypovolemia, or aceruloplasminemia—chelating agents can help decrease iron blood levels. The following are chelating agents used to treat secondary iron overload:

- Deferoxamine mesylate (Desferal)
- Deferasirox (Jadenu, Exjade, Jadenu Sprinkle)
- Deferiprone (Ferriprox)

Erythrocytapheresis can remove larger quantities of iron than phlebotomy, but the procedure is associated with higher cost and technician time consumption (Sundic et al., 2014).

Moderate consumption of iron-containing foods is allowed for patients who undergo phlebotomy, although dietary and nutritional supplements with iron and vitamin C should be avoided. Fresh fruits and vegetables are encouraged, except citrus fruits and juices. Tannates in tea, phytates, oxalates, calcium, and phosphates can bind with iron and inhibit absorption, but these are unnecessary if the patient is already undergoing phlebotomy.

Alcohol consumption should be limited because ethanol can increase iron absorption, and some red wines have a high iron content. Patients with hereditary hemochromatosis who consume large quantities of alcohol are more likely to develop hepatocellular carcinoma and cirrhosis.

Bacteria commonly found in seafood, including *Listeria monocytogenes*, *Yersinia enterocolitica*, and *Vibrio vulnificus*, thrive in a high-iron environment. Patients with the disorder should avoid uncooked seafood to prevent bacterial infection.

Alpha-1 antitrypsin deficiency

Alpha-1 antitrypsin (AAT) deficiency is an autosomal-recessive inherited disorder that can result in lung and liver disease. In the United States, approximately 80,000-100,000 people have severe AAT deficiency (Meseeha & Attia, 2018). The disorder is thought to be as common as cystic fibrosis but is still under-recognized despite the availability of extensive educational and evidence-based publications.

Risk factors

AAT deficiency is the result of an inherited genetic mutation in the SERPINA 1 gene. The most common phenotype is PiMM, which is associated with normal circulating levels of AAT. The homozygous PiZZ phenotype, and most likely the heterozygous PiMZ phenotype, play an important role in the development of this disease.

Symptoms

AAT deficiency is associated with pulmonary symptoms such as early emphysema, hepatic disorders such as cirrhosis, and skin diseases such as panniculitis. In the liver, pathologic polymerization of AAT is thought to be the cause of AAT accumulation in the hepatocytes, which causes liver injury.

Diagnosis

Severe AAT deficiency is diagnosed by AAT serum level <11 mmol/L and phenotyping. Isoelectric focusing, a phenotype test, is the gold standard for identifying AAT variants.

Complications

Long-term outcomes vary. Affected patients may develop cirrhosis.

Treatment

The treatment of AAT deficiency includes medical therapies and supportive care for specific symptoms. For patients with lung problems, augmentation therapy helps in preventing lung disease progression. Augmentation therapy is an intravenous AAT replacement that increases AAT protein blood level. In the United States, there are five available alpha-1-proteinase inhibitors (human) for augmentation therapy: Prolastin-C, Aralast NP, Zemaira, and Glassia. Liver transplantation may be performed if liver failure develops.

Intrahepatic biliary dysplasia

Intrahepatic biliary dysplasia (IHBD), also known as Alagille syndrome, incorporates a combination of anomalies that occur in conjunction with chronic cholestasis.

IHBD is a unique, autosomal-dominant liver disease that occurs in approximately 1 in 70,000 live births (National Library of Medicine, 2019a).

Symptoms

Patients with IHBD display characteristic facial features; vertebral malformations; delayed physical, mental, and sexual development; and various cardiac anomalies, most often peripheral pulmonic stenosis. Liver changes are progressive, with a decrease in the number of portal zones, followed by bile plugging of small interlobular bile ducts, hepatocytes, and small canaliculi, then leading to absence or decrease in portal bile ducts and collapsed portal bile ducts.

Infants and children with IHBD frequently have cholestatic jaundice in the neonatal period, often accompanied by failure to thrive, prematurity, or small size for gestational age. The liver is enlarged and firm or hard on palpation. Most infants and children with IHBD are more affected by their cardiac abnormalities than by their liver disease.

Diagnosis

The diagnosis of IHBD is established by a detailed history, liver biopsy, liver function tests, eye examination, X-ray, and abdominal ultrasound. The diagnosis is confirmed by genetic testing.

Complications

The overall life expectancy for children with IHBD is unknown but depends on several factors, including the severity of scarring in the liver, whether heart or lung problems develop, and the patient's nutritional status. Many adults with IHBD lead normal lives.

Treatment

Treatment of IHBD involves symptomatic and supportive care, including good nutrition and fat-soluble vitamin supplementation. Cholestyramine and phenobarbital may be given to relieve pruritus as a result of cholestasis. Liver transplantation is considered when cirrhosis and liver failure develop.

Biliary atresia

Biliary atresia is a bile duct disorder that results in rapid and progressive scarring, leading to obstruction of the flow of bile through the extrahepatic bile duct system. The overall incidence is low, about 1 in 10,000 live births (Sanchez-Valle et al., 2017).

Symptoms

Biliary atresia is suspected when the infant's jaundice persists for >3 week after birth.

Diagnosis

Blood tests to measure bilirubin and liver function, ultrasound, hepatobiliary scan, and liver biopsy are diagnostic tests for biliary atresia.

Complications

Infants with biliary atresia have the inability to digest and absorb nutrition. This leads to poor growth and vitamin (A, D, E, and K) deficiencies.

Treatment

No procedure has yet been developed to correct biliary atresia, except for liver transplantation.

Upon diagnosis of biliary atresia, the Kasai procedure (hepatoportoenterostomy [HPE]) is performed. HPE creates a roux-en-Y loop using a length of the infant's small intestine, which is then anastomosed to the hilum of the liver, creating a duct into the distal bowel and restoring bile flow. The goal of the Kasai procedure is to allow excretion of bile from the liver into the intestine. However, some patients have poor response to the procedure because of obstructed intrahepatic and extrahepatic bile ducts. The Kasai procedure is often not effective in restoring bile flow for infants over two months of age, so early diagnosis of biliary atresia is critical.

Unfortunately, despite bile flow, the Kasai procedure is not a cure for biliary atresia. Liver damage often continues, leading to cirrhosis and the need for transplantation. Biliary atresia is the most common indication for liver transplantation in young children.

Neonatal hepatitis

Neonatal hepatitis is inflammation of the liver that occurs only in early infancy, usually between 1 and 2 months of age.

Risk factors

Some infants with neonatal hepatitis were infected by a virus that caused the inflammation before or shortly after birth while the remaining infants have no identifiable virus. These viruses include cytomegalovirus (CMV), rubella, and hepatitis A, B, or C.

Complications

Neonatal hepatitis caused by rubella or CMV may lead to cirrhosis and may be accompanied by mental retardation. Neonatal hepatitis caused by hepatitis A usually resolves within 6 months, but cases that are the result of the hepatitis B or C viruses will most likely result in chronic liver disease.

Treatment

There is no specific treatment for neonatal hepatitis. Attention to healthy nutrition and growth and development is essential to the long-term prognosis. Infants who develop cirrhosis may require liver transplantation.

Gilbert syndrome

Gilbert syndrome is a relatively common and benign congenital disorder. It is estimated that from 3% to 7% of Americans have Gilbert syndrome (National Library of Medicine, 2019d).

Risk factors

The disorder occurs more frequently in males and is probably hereditary (Harper & Hernandez, 2014).

Symptoms

Gilbert syndrome is characterized by a mild, fluctuating increase in serum bilirubin. There are rarely significant symptoms, but occasionally jaundice may occur. Except for elevated serum bilirubin, liver tests and cholangiography are normal.

Diagnosis

The diagnosis of Gilbert syndrome is established primarily by documenting the persistence of an increased serum bilirubin when other liver function tests are repeatedly normal. A liver biopsy may occasionally be necessary to rule out other liver diseases.

Treatment

There is no treatment necessary for Gilbert syndrome, and it will not interfere with a normal lifestyle.

ORTHOTOPIC LIVER TRANSPLANTATION

Since the 1984 Congressional enactment of the National Organ Transplant Act (NOTA), the Organ Procurement and Transplantation Network (OPTN) manages and directs the allocation and distribution of life-saving organs within the Unites States. The United Network for Organ Sharing (UNOS) is the nonprofit policy-making organization that administers the OPTN. UNOS is the only OPTN in the country. Membership includes transplant centers, organ procurement organizations, transplant recipients, and patients awaiting transplants.

In 2002, the Model for End-stage Liver Disease (MELD) score was implemented in the United States. The MELD model is continuously updated, and in 2016, the MELD-Na was implemented. This model is a reliable tool to predict survival of patients with cirrhosis who are 12 years old and above prior to liver transplant. Use of this model has resulted in a notable improvement in liver allocation and a reduction in waitlist registration and waitlist mortality over previously used methods of scoring.

The MELD scoring system mainly focuses on laboratory data. It incorporates objective data, including creatinine, bilirubin, and international normalized ratio (INR), which provides a better outcome in identifying mortality risks. The MELD-Na added the serum sodium level, as abnormal sodium levels have a correlation to low survival rates after liver transplant.

In addition to the laboratory data, standard MELD exceptions are designated to provide MELD point upgrades for eligible patients based on the presence of other conditions that may have bearing on the patient's survival. The exceptions include hepatocellular carcinoma (HCC), hepatorenal syndrome, hepatopulmonary hypertension, familial amyloid polyneuropathy, primary hyperoxaluria, cystic fibrosis, hilar cholangiocarcinoma, and hepatic artery thrombosis. The upgrades allow sicker patients to be prioritized on the list.

For liver transplant candidates younger than 12 years old, the Pediatric End-stage Liver Disease (PELD) score is useful as a predictor of waitlist mortality and post-transplant survival. The PELD score incorporates age, growth failure, bilirubin, albumin, and prothrombin time expressed as INR.

A high MELD-Na or PELD score is associated with a higher mortality rate. UNOS prioritizes patients with higher MELD/PELD scores.

To calculate the MELD-Na score, the following information is needed:
• Plasma bilirubin (mg/dL)
• INR
• Plasma creatinine (mg/dL)
• Serum sodium

To calculate the PELD score, the following information is needed:
• Age at listing
• Serum albumin (g/dl)
• Total bilirubin (mg/dL)
• INR
• Growth failure (based on gender, height, and weight)

Indications

Transplantation may be an option in the treatment of various pathological conditions affecting the liver that result in fulminant hepatic failure and chronic liver disease.

Procedure

After evaluation by the transplant physician, a patient is added to the national UNOS waiting list by the transplant center. Lists are specific to both geographical area and organ type. Each time a donor organ becomes available, a list of potential recipients is generated based on factors that include genetic similarity, organ size, and medical urgency. Although most liver transplants are performed using organs from cadaver donors, living related donor programs have been created at some centers, utilizing a portion of the liver from a living relative for transplant in small children. A portion of the liver may be used for adult-to-adult transplants from nonrelated donors, as well.

Considerations

Following transplantation, the organ recipient will require life-long medications that suppress the immune system (e.g., cyclosporin, tacrolimus) to prevent rejection of the transplanted organ. The post-transplant patient is monitored closely for organ rejection. Other potential complications include infection, hemorrhage, liver function problems, vascular graft obstruction, graft-versus-host disease, and complications from medications.

CASE SITUATION

Mr. Rolando Stern, age 34, was admitted to the emergency room for hematemesis. His blood work showed a prolonged prothrombin time and elevated transaminase values. A hepatitis screen and an emergency gastroscopy are ordered to determine the cause of his GI bleeding. He has a large-bore nasogastric tube in place, which is draining copious amounts of dark red blood. In the endoscopy room, the gastroenterology nurse sets up equipment for gastric lavage and proceeds to lavage Rolando's stomach until the return is clear enough to begin the endoscopy.

Rolando is found to have a large gastric ulcer with an oozing visible vessel, which the physician cauterizes with a bipolar probe. The bleeding stops and Rolando is moved to the intensive care unit. The next day, however, he shows signs of rebleeding. He is taken to surgery and a partial gastrectomy is performed.

When Rolando's hepatitis screen comes back, it is positive for HBsAg, anti-HBc, and hepatitis A IgG antibody (IgG-anti-HAV).

Points to think about

1. Before the gastroenterology nurse was able to put on gloves, she had significant contact with Rolando's blood when he vomited before the lavage. Because the gastroenterology nurse has severe dermatitis on her hands, would she have been exposed to HAV, HBV, or both?
2. Health care personnel in high-risk areas for HBV (i.e., those personnel who are in frequent contact with blood and body fluids) should be vaccinated against HBV. If the gastroenterology nurse in this case has put off this important vaccination, what prophylaxis is available?
3. If the gastroenterology nurse were never informed of Rolando's hepatitis and subsequent possible exposure, what symptoms would alert the nurse to HBV infection?
4. The law requires that certain communicable diseases be reported. The gastroenterology nurse should determine who in the institution is responsible for reporting a communicable disease, to whom it should be reported, and what information is required in the report.
5. Although precise data are not available, the prolonged prothrombin time, Rolando's tendency to bleed, and his diagnosis of viral hepatitis would lead the nurse to tentatively consider what nursing diagnosis?
6. What is an appropriate outcome criterion for this nursing diagnosis? What nursing interventions will lead to this outcome?
7. What special infection-control precautions should the gastroenterology nurse take in the endoscopy setting?

Suggested responses

1. HBV is transmitted via serum, semen, and saliva. The nurse has been exposed to HBV, which is a major health care-associated infection that can have serious sequelae. Finding HBsAg in the blood is diagnostic of HBV infection. HBsAg is first detectable in serum during the incubation period 30–60 days after exposure. This marker will usually be present in the blood until the end of the acute phase of the illness, or longer.

Patients are considered infectious as long as HBsAg is detectable in serum. Anti-HBc appears in the blood about the same time as serum transaminase levels rise.

HAV is transmitted by the fecal-oral route and not in the blood, although dirty needles can serve as a means of transmission. The presence of IgG-anti-HAV is indicative of prior infection with HAV and a naturally acquired immunity. Rolando had HAV in the past and is now immune, so this is not a contagious disease risk for the gastroenterology nurse.

2. If the gastroenterology nurse has not been vaccinated against HBV, the treatment of choice is HBIG, given within 1 week of exposure for passive immunity. It is recommended that hepatitis B vaccine be given in conjunction with the HBIG. The injections are given at different sites. Hepatitis B vaccine is safe in healthy adults and is given in three serial intramuscular doses of 1.0 ml within 1 week of exposure, at 1 month from exposure, and at 6 months from exposure.
3. If the gastroenterology nurse has been infected with HBV, the incubation period is 6 weeks to 6 months, with an average of 12 weeks. Symptoms may be relatively mild or quite severe. If symptoms are mild, they can mimic a flu-like syndrome. Symptoms may include fatigue, malaise, joint or muscle pain, anorexia, nausea, vomiting, fever and chills, jaundice, clay-colored stools, dark urine, and pruritus. There may be abdominal pain and tenderness in the right upper quadrant. Even though signs and symptoms may be absent during the incubation period, the person with hepatitis is contagious during that time.
4. Ordinarily, the patient's physician is responsible for reporting both HAV and HBV infections to the state health department. However, the gastroenterology nurse should know what diseases are reportable by law. Such cases should be reported to the infection-control nurse in the hospital (or a counterpart), and that person can make certain the state health department has been informed. The Centers for Disease Control and Prevention (CDC) will include the case in its national statistics.

   The information to be reported may vary from state to state. Usually the health department will want, at a minimum, the patient's name, age, address, any possible contacts, the diagnosis, and the name of the physician treating the patient. The health department may refer the case to its nursing division for post-discharge follow-up, or the gastroenterology nurse may refer the patient to the visiting nurse service of a local health department, if available.
5. This information would permit the nurse to consider the nursing diagnosis "potential for injury: bleeding, related to altered clotting mechanisms."
6. An appropriate outcome in relation to this nursing diagnosis would be "Rolando will experience a resolution of bleeding episodes." Related nursing interventions would include the following:
   a. Administer prescribed medications.
   b. Observe incision and nasogastric drainage every 2 hours.
   c. Take vital signs every hour.
   d. Check the intravenous (IV) site for bleeding, proper cannulation, and flow every 2 hours.
   e. Monitor neurological vital signs every 4 hours.
7. Special infection-control practices that should be used by gastroenterology nurses in the endoscopy setting include:
   a. Following standard precautions with all patients. When there is a potential for contact with the blood or body fluids of any patient, the nurse should wear gloves, protective gowns, masks, and eyewear.
   b. Strict adherence to the endoscopy reprocessing protocol is essential for safety. This protocol should be reviewed and updated according to institutional policy.
   c. Using information obtained in the nursing assessment to help identify patients at risk for hepatitis B virus infection. Persons at risk include individuals who use recreational injectable drugs; have multiple sexual partners or men who have sex with men; have household contacts with unvaccinated individuals with chronic HBV infection; are developmentally disabled and living in long-term facilities; are living in correctional facilities; work with patients who are HBV-positive; are on hemodialysis; have HCV infection; travel to countries where HBV is endemic; or have chronic liver disease, HIV, or diabetes.

REVIEW TERMS

alpha-1 antitrypsin (AAT) deficiency, ascites, biliary atresia, Budd-Chiari syndrome, cirrhosis, fulminant hepatic failure, Gilbert syndrome, hereditary hemochromatosis, hepatic encephalopathy, hepatitis, hepatocytes, hepatorenal syndrome, intrahepatic biliary dysplasia (IHBD), Kupffer cells, nonalcoholic steatohepatitis (NASH), porphyria, portal hypertension, sinusoids, splanchnic vasodilation, Wilson's disease

REVIEW QUESTIONS

1. In the liver, bile is secreted by:
   a. Sinusoids.
   b. Hepatocytes.
   c. Glisson's capsule.
   d. Kupffer cells.

2. Seventy-five percent of blood flow into the liver is delivered via the:
   a. Inferior vena cava.
   b. Hepatic artery.
   c. Portal vein.
   d. Splenic artery.
3. In the liver, carbohydrates are metabolized to:
   a. Glycogen.
   b. Amino acids.
   c. Ammonia.
   d. Glycerol.
4. The primary physiological function of Kupffer cells is:
   a. Metabolism.
   b. Production of prothrombin and fibrinogen.
   c. Vitamin storage.
   d. Phagocytosis of harmful substances.
5. A definitive diagnosis of cirrhosis is obtained through:
   a. Observation of a Kayser-Fleischer ring on the cornea.
   b. Ultrasonography or CT scan.
   c. Palpation of the liver.
   d. Biopsy of the liver.
6. Patients with hemochromatosis must avoid:
   a. Iron and vitamin C supplements.
   b. Vegetables.
   c. Red meat.
   d. Deep fried seafood.
7. HBV infection treatment includes all of the following except:
   a. Nutritionally balanced diet.
   b. Rest from strenuous physical activity.
   c. Octreotide.
   d. IV hydration and electrolyte management.
8. Accumulation of excessive amounts of copper in certain body tissues is characteristic of:
   a. Intrahepatic biliary dysplasia.
   b. Primary liver cancer.
   c. Wilson's disease.
   d. Porphyria.
9. Chronic, blistering, cutaneous lesions on parts of the skin that are exposed to the sun, mild liver disease, and a lack of neuropsychiatric manifestations are characteristics of:
   a. Acute intermittent porphyria.
   b. Hereditary porphyria.
   c. Variegate porphyria.
   d. Porphyria cutanea tarda.
10. The most common cause of ascites is:
    a. Cancer.
    b. Portal hypertension secondary to cirrhosis.
    c. Hepatitis C virus infection.
    d. Heart failure.
11. Treatment for alcoholic hepatitis can include the following except:
    a. Alcohol abstinence.
    b. Corticosteroids.
    c. Tylenol (acetaminophen).
    d. Pentoxil (pentoxifylline).
12. Assessing portal hypertension in cirrhosis of the liver is done by:
    a. Administer a splanchnic vasoconstrictor such as octreotide and monitor its effect.
    b. Liver biopsy under ultrasound.
    c. Measure the hepatic vein pressure gradient (HVPG).
    d. Perform an EGD to assess for esophageal varices.
13. The most common risk factors for NASH are:
    a. Poisoning and infectious diseases.
    b. Obstruction of the hepatic venous outflow tract.
    c. Obesity, type 2 diabetes, and hyperlipidemia.
    d. Cholestasis and drug-related hepatotoxic reaction.

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SECTION IV

Pharmacology, Intravenous Therapy, and Nutrition

The topics of pharmacology, intravenous therapy, and nutrition as they relate to gastroenterology nursing are reviewed in this section.

Chapter 20

PHARMACOLOGY

The primary goal of this chapter is to provide the gastroenterology nurse with the critical thinking tools necessary to effectively use pharmacotherapeutics in the delivery of medications commonly used to diagnose and treat gastrointestinal (GI) disorders. Gastroenterology nurses must know not only the names, actions, and uses of drugs, but also be able to employ effective clinical decision-making when monitoring drug therapy.

General information is provided on techniques and routes of drug administration, dose calculations and dosage, patient education, and documentation. The terms "dose" and "dosage" are often used interchangeably, but dose refers to the amount of medication (e.g., one tablet), while dosage refers to the means and frequency of administration (e.g., one tablet three times daily).

The classes of drugs used most often in gastroenterology practice are described in general terms, including indications, contraindications, and possible side effects. An extended table is provided at the end of this chapter with the generic and brand names of representative medications, including primary use, usual adult and pediatric doses, and common adverse effects.

Learning objectives

After reviewing the content of this chapter, the gastroenterology nurse will be able to:

1. Discuss general aspects of drug therapy that are relevant to nursing practice, including techniques and routes of administration, dose calculations and conversions, dosage, patient education, and documentation.
2. Describe the general classes of medications that are important in gastroenterology, including general indications, contraindications, and potential adverse effects.
3. Identify representative drugs used in gastroenterology practice, their respective generic and trade names, indications, usual adult and pediatric dosages, and possible adverse effects.

BASIC PRINCIPLES

The gastroenterology nurse should be able to recognize expected actions of drugs prescribed, be confident that ordered doses and dosages are within the safe range for a particular patient, and have knowledge of possible adverse effects or undesirable drug interactions. If, based on knowledge and experience, the gastroenterology nurse has any concerns or notes any serious adverse effects related to the medication, the physician should be consulted immediately.

The gastroenterology nurse should be cognizant of the differences in medication administration, dosing, and effects specific to differences in neonatal, pediatric, adult, and geriatric patients.

- **Neonates** have underdeveloped metabolic enzyme systems and inadequate renal function, so they need highly individualized doses and careful monitoring.
- **Pediatric patients** require frequent review of medication treatment and therapies because of rapid changes throughout childhood. A child who is ill can experience growth spurts with rapid weight gain or acute weight loss.
- **Geriatric patients** present special concerns because they are prone to drug-drug interactions caused by polymedication regimens for multiple system problems (e.g., heart disease, arthritis). Most drugs are metabolized in the liver. Geriatric patients can have decreased hepatic function, which may affect a drug's usual metabolism. Most drugs are excreted by the kidney and geriatric patients may have diminished renal

function. Drug doses may need to be modified based on BUN and creatinine values. Geriatric patients may have a gradual decline in vision, hearing, and memory loss, making medication compliance difficult.

Over-the-counter (OTC) medications, herbs, and vitamins may affect drug therapy in all age groups.

Patient education

Patient education is an important aspect of drug therapy. The patient and/or the responsible party should understand the purpose of any medication that is ordered. If there is a change in medication or dosage, the patient should be informed. He or she should be instructed, as appropriate, about possible side effects and should be encouraged to report any symptoms that may indicate adverse effects (adverse drug reaction). The Joint Commission (2018) has developed goals and standards regarding medication reconciliation. This helps ensure essential information is shared with both the health care team and the patient.

Patient education should also encompass:

- Recognizing drug-diet and drug-drug interaction.
- Knowing brand name and generic name.
- Understanding the instructions on taking and storing medications.
- Notifying the pharmacist and prescribing physician of other medications, such as OTC medications, herbs, and vitamins.
- Reading labels carefully.
- Not changing brands without consulting the prescriber.
- Filling all prescriptions with one pharmacy, if possible.
- Checking expiration dates on medications.
- Storing medications in the original container.
- Making sure to have an adequate supply of medication when on vacations and/or carrying an extra prescription for long trips.

Unless gastric irritation is a problem, it is usually best for patients to take medications on an empty stomach because food can interfere with drug absorption. In addition, most drugs are more efficiently absorbed if taken with 100-250 ml of fluid. Smaller amounts of fluid are used for children and should be based on the size of the child. It is important for the patient to be aware of these considerations to maximize the effectiveness of drug therapy.

The issue of drug-drug interactions is a significant and increasingly complex area of concern. For example, a stimulant **laxative** that increases GI motility may decrease the rate and extent of absorption of time-released drugs and drugs that require prolonged intestinal contact. Anticholinergics that decrease motility may alter the absorption of certain drugs. Medications that alter gastric pH such as antacids decrease the absorption of some weakly acidic drugs such as phenobarbital and aspirin. Patients must be cautioned to report to their physician all the prescription and OTC medications, recreational drugs, vitamins, supplements, and herbal therapies they are taking, and patients should be informed of any potentially problematic interactions. Patients should also be encouraged to report any known drug allergies to their physician and pharmacist.

Routes of administration

Medications may be administered orally, parenterally, or topically. The route of administration is determined by the required speed of onset of the drug's effect; patient comfort and safety considerations; pharmacokinetic properties, such as absorption and excretion; and the organ or system targeted by the drug. IV injections, for example, take effect quickly because they are immediately accessible in the bloodstream, whereas drugs administered by topical routes must be absorbed before they can be effective. However, injections are more uncomfortable for the patient than oral or topical drugs and carry a greater risk of side effects. When the GI system is the target of drug therapy, oral administration may be the most effective.

The **oral route** is the most common route of drug administration. Oral medications may take the form of tablets, capsules, syrups, elixirs, oils, liquids, suspensions, powders, or granules. Most oral medications are swallowed by mouth (po), but they may also be given through a nasogastric or gastrostomy tube or may be dissolved between the cheek and gum (buccal) or under the tongue (sublingual). Nausea, vomiting, inability to swallow, or unconsciousness may contraindicate oral administration of medications.

Parenteral medications are administered by injection using a subcutaneous (subcut), intramuscular (IM), intravenous (IV) or intradermal route.

- Subcut injections are administered into the adipose tissue (the fatty layer) beneath the skin. The subcut route is faster than oral administration but allows slower, more sustained drug action than an IM injection. Subcut injections also cause minimal tissue trauma and carry little risk of striking large blood vessels and nerves. The most common subcut injection sites are the outer aspect of the upper arm muscles.
- Muscles used for IM injections include the deltoids, dorsogluteal, and ventrogluteal areas, and vastus lateralis (lateral muscle of the quadriceps group).
- The addition of drugs to IV solutions and IV push administration are discussed in Chapter 21.
- Intradermal injections deposit medication into the dermis.

If the volume of medication to be injected is not excessive, the combination of two drugs in one syringe avoids the discomfort of two injections. Drugs can be combined from two multidose vials, from one multidose vial and one ampule, or from two ampules. Drugs may be combined only if their compatibility has been documented. Rarely are more than two drugs combined in one syringe.

Topical drugs, depending on their purpose, may be administered to gastroenterology patients in the form of ointments or sprays (e.g., antiseptics, anesthetics), transdermal patches, rectal suppositories, or rectally administered foams or enemas. Medicated shampoos, eye medications, ear drops, nasal sprays and drops, vaginal medications, and inhaled products are also considered topical medications. The site of topical drug application should be inspected frequently for side effects, such as skin irritation or an allergic reaction.

Techniques of administration

Before administering any medication, the gastroenterology nurse must compare the physician's order with the patient's medication record. There should be a written order for every medication given. If the medication order is verbal, organizational policies and state laws will determine the time frame in which verbal orders must be signed by the physician.

In the medication process, there are "eight rights" to medication administration. They are:

1. Right drug.
2. Right patient.
3. Right dose.
4. Right time.
5. Right route.
6. Right reason.
7. Right response.
8. Right documentation.

The eight rights include verifying medication, having two identifiers, checking the label for expiration dates, noting appropriateness of the drug ordered, noting the patient's response to the medication, and describing nursing intervention.

The process to prevent medication errors covers ordering and prescribing, transcribing and verifying, dispensing and delivering, administering, and monitoring and reporting of medication errors.

Privacy should be provided as needed for injections and for application of topical drugs. The procedure should be explained to the patient. Regardless of the route of administration, hand washing prior to delivery is essential.

Technology plays an important role in patient safety. Computerized physician order entry (CPOE), where the prescriber enters medication orders into the computerized record, eliminates errors due to illegible handwriting. CPOE also helps identify correct dosages, routes, drug interactions, and allergy checks, and it can be linked to drug information databases. Bar coding to decrease medication errors is also endorsed by the Health and Medicine Division (HMD), previously the Institute of Medicine (IOM) of the National Academies of Science, Engineering, and Medicine (the National Academies); The Joint Commission; the Agency for Healthcare Research and Quality (AHRQ); and the Institute for Safe Medication Practices (ISMP).

Calculations and conversions

To administer the correct dose of a medication, it may be necessary for the gastroenterology nurse to convert the physician's order to a different unit of measure or to dilute available doses. Usually, physicians order medications in grains, grams, milligrams, or other units of weight. Before the medication can be administered to the patient, the order may need to be converted to the corresponding number of tablets, capsules, minims, drams, ounces, milliliters, or other units of volume. Abbreviations for these units and for other terms used in drug administration are given in Table 20.1.

Table 20.1. Abbreviations Used in Medication Administration

Abbreviation	Definition
ac	before meals (ante cibum)
ad lib	as desired (ad libitum)
amp	ampule
bid	two times daily (bis in die)
buc	buccal
cap(s)	capsule(s)
DC	discontinue
DS	double strength
EC	enteric coated
Elix	elixir
ER	extended release
fld or fl	fluid
g, gm	gram
gr	grain
gtt(s)	drop(s)
h or hr	hour
hs or HS	at bedtime
IM	intramuscular
IV	intravenous
IVPB	intravenous piggyback
kg	kilogram
lb(s)	pound(s)
mcg	microgram
mEq	milliequivalent
mg	milligram

Table 20.1. Abbreviations Used in Medication Administration (continued)

Abbreviation	Definition
m	minim
ml	milliliter
neb	nebulizer
NPO	nothing by mouth (nil per os)
OTC	over-the-counter
pc	after meals (post cibum)
PCA	patient controlled analgesia
po	oral (per os)
prn	when necessary (pro re nata)
daily	every day
qh	every hour
q2h	every 2 hours
qid	four times daily (quater in die)
R	rectal
stat	immediate and once only
subcut	subcutaneous
supp	suppository
SL	sublingual
tab(s)	tablet(s), pill(s)
Tbsp, T, tbs	tablespoon
Tsp, t, tsp	teaspoon
tid	three times daily (ter in die)

Doses may be specified in any one of three different systems of measurement:

1. Metric units (e.g., milliliters, grams).
2. Apothecary units (e.g., ounces, drams, minims).
3. Household units (e.g., teaspoons, drops, cups).

Some of the more common equivalents are contained in Table 20.2. It is important that the gastroenterology nurse memorize the equivalents necessary to convert between these three systems of measurement. By applying these conversion factors to simple arithmetic calculations involving ratios and proportions, any dose problem can be easily solved.

In addition to the metric, apothecary and household systems of measurement, some drugs are measured in milliequivalents (mEq) or units:

- Milliequivalent refers to the number of ionic charges of an element or a compound. It is a measure of the chemical combining power of a substance.
- Units mean something different for every drug. One United States Pharmacopeia (USP) unit of insulin, for example, is defined as the quantity that promotes the metabolism of about 1.5 g of dextrose.

Labels on drugs that are measured in this way indicate the number of milliequivalents or units in a particular volume measure (e.g., 400,000 units/ml or 20 mEq/15 ml).

When administering medications in tablet or capsule form, remember that only scored or marked tablets can be divided accurately. Capsules cannot be divided. It is best to obtain tablets or capsules of the desired dose.

Most volume doses of oral medications can be measured out in medicine cups, which are frequently marked in drams, ounces, milliliters (or cubic centimeters), teaspoons, and tablespoons. Smaller volume amounts, such as minims and drops, must be measured using droppers, calibrated pipettes, or syringes. Drops are an approximate household unit of measure that can be safely measured using a medicine dropper. A minim is a more accurate apothecary unit of measure and must be measured using a minim pipette or syringe.

Table 20.2. Approximate Equivalents of Common Measurements

Weight Units			Fluid Units		
Metric	Apothecary	Household	Metric	Apothecary	Household
1 gram	15–16 grains		1 ml or 1 cc	15–16 minims	15–16 drops
30–32 grams	1 ounce	2 tablespoons	30 ml	1 fluid ounce	2 tablespoons
40 grams	8 drams	6 teaspoons	30 ml	8 fluid drams	6 teaspoons
1 kilogram	2.2 pounds*		240 ml	8 ounces	1 cup
			1,000 ml	1 quart	4 cups
			4,000 ml	1 gallon	16 cups

**using the avoirdupois system, a system of weights based on a 1 pound of 16 ounces or 7,000 grains*

Solutions for parenteral administration may be ordered in units of volume but are more frequently ordered in units of weight such as grams or milligrams. The gastroenterology nurse must then convert the weight ordered to a unit of volume such as drams, ounces, or milliliters. In many cases, labels on drugs in solution indicate a certain weight measure in a particular volume (e.g., 500 mg/5 ml). Labels may also state a percentage or ratio strength of the solution (e.g., 5% or 1:20 solution).

Occasionally, it may be necessary to prepare oral or topical solutions from crystals, powder, or stock solutions, or to prepare children's doses from adult dose tablets. Preparation of solutions is a simple calculation in proportions. For pediatric doses, tablets that are soluble in water can be dissolved in a specified amount of water, and the ordered dose measured using a syringe. Although such preparations are acceptable, commercially prepared liquid medications are more accurately measured and should be used whenever possible.

Most drug references include pediatric dose ranges for drugs that are approved for use in children. Doses are given in milligrams per kilogram of body weight. Pediatric doses may be calculated based on the child's weight in kilograms.

Consideration of body surface area can also be used. Use of surface area requires consultation of one of the various charts or nomograms used to determine body surface area from height and weight. If surface area can be determined, the following formulas are used to calculate the pediatric dose from the usual adult dose:

- (Surface area in m^2 × Adult dose) / 1.7 = Child dose
- Surface area in m^2 × 60 = Percentage of adult dose

If surface area is not known, the most commonly used procedures for determining approximate pediatric dose include:

- **Young's rule (children from 1 or 2 to 12 years)**
  (Adult dose × child age / child age + 12 = Child dose)
- **Fried's rule (infants and children up to 1 or 2 years)**
  (Age in months / 150) × Adult dose = Child dose)
- **Clark's rule (infants or children)**
  (Weight in pounds / 150) × Adult dose = Child's dose)

Again, if there is any question about the appropriateness of the ordered dose for any patient, the physician or pharmacist should be consulted.

Documentation

Documentation describes the care or service provided for a patient. In addition, documentation facilitates communication and is a valuable source of data for making decisions about funding and resource management, as well as facilitating nursing research, which has the potential to improve the quality of nursing practice and client care. Medical records must reflect care and services provided.

Requirements for documentation of medication administration are drawn from federal and state legislation, case law, professional standards of practice, regulatory agencies, and organizational policies and procedures. The key principles of documentation remain the same whether the gastroenterology nurse uses the traditional (i.e., paper) or electronically generated (i.e., computer) documentation methods. The gastroenterology nurse is accountable for safeguarding the confidentiality of patient information regardless of which method is used.

Documentation must include:

- Name of the drug administered.
- Person administering the drug.
- Dose and route of administration. Parenteral medication requires documentation of the injection site. In the case of topical administration, note the condition of the skin and the location at the time of application.
- Date and time of administration.
- Patient's response or adverse reaction.
- Any nursing intervention.

If the patient refuses a drug, the refusal should be documented. The nurse manager and the patient's physician should be notified, and this notification should be documented, as well. If a drug is omitted or withheld for other reasons such as radiology or laboratory tests, that omission and the reason should be documented according to organizational policy.

If the patient's intake and output are being monitored, the fluid volume of any medications administered orally, intravenously, or through a nasogastric or enteral tube should be noted on the intake and output sheet.

DRUGS COMMONLY USED FOR PATIENTS WITH GASTROINTESTINAL DISORDERS

The gastroenterology nurse must have a working knowledge of the various classes of drugs used in the practice of gastroenterology and GI endoscopy. Some of the medications prescribed for gastroenterology patients are intended to alter GI secretions and/or motility, including anticholinergics and cholinergics, some laxatives, and certain antiulcer agents. Others are prescribed for their antacid, anti-inflammatory, antiemetic, or antidiarrheal effects. Still other medications are used in diagnostic testing or to facilitate endoscopic or invasive procedures, which often require the use of topical anesthetics, sedatives, and/or narcotics. In addition, prophylactic antibiotics and therapeutic antiparasitic agents may be prescribed.

The following pages outline the general characteristics of various drug classifications that should be familiar to the gastroenterology nurse.

Antacids

Antacids still have a place in the treatment of acid peptic disorders, even with the widespread availability of proton pump inhibitors (PPIs), histamine2 (H_2)-receptor antagonists, and sucralfate. Antacids commonly prescribed have at least one of the following elements as their main ingredient: aluminum salts, calcium carbonate, and magnesium salts. The ingredient alginate is not an antacid but may provide barrier protection. Antacids are available without a prescription.

Action

Antacids have no direct effect on gastric acid secretion and do not coat or protect the mucous lining. Instead, antacids neutralize gastric acid, reducing the total acid load in the GI tract and leading to a transient rise of gastric pH. This decreases pepsin activity because pepsin is rendered inactive in alkaline conditions. It also increases lower esophageal sphincter (LES) tone, reducing reflux.

Aluminum ions inhibit smooth muscle contraction and gastric emptying. This action is counterproductive because in gastroesophageal reflux disease (GERD), prompt gastric emptying is beneficial.

Indications

Antacids are indicated in the treatment of:

- Hyperacidity—heartburn (pyrosis).
- Gastroesophageal reflux disease (GERD).
- Acid indigestion.
- Hyperacidity associated with peptic ulcer disease.

Research studies demonstrate that antacids are clearly superior to placebos in healing gastric ulcers.

Because antacids leave the stomach rapidly, they are best used for intermittent symptoms. Taken on an empty stomach, the effects of antacids last less than 1 hour. After meals, the buffering effect may last as long as 3 hours. For optimal effect, antacids should be given about 1 hour after meals or feedings and during periods of acid rebound.

Adverse effects

Prolonged use of magnesium- or calcium-containing antacids may cause systemic absorption of toxic quantities of these ions. Excessive use of aluminum-containing antacids may lead to hypophosphatemia. Aluminum and calcium preparations tend to be constipating, whereas magnesium preparations tend to produce a laxative effect.

In chronic patients with renal failure, administration of aluminum-containing antacids is best at mealtime. Antacids containing sodium can precipitate edema in patients with cirrhosis, hypertension, or renal or congestive heart failure.

Contraindications

The use of antacids is contraindicated in patients taking tetracycline, iron, or H_2-receptor antagonists. Calcium binds and prevents the absorption of tetracycline, whereas antacids decrease the absorption of iron and H_2 blockers.

Magnesium-containing antacids can precipitate hypermagnesemia in patients with chronic renal failure and should be avoided in these patients, as central nervous system depression, skin irritation, and, rarely, muscle paralysis with respiratory failure can occur.

Probiotics and prebiotics

Probiotics are good bacteria naturally found in the GI tract. **Prebiotics** are non-digestible plant fibers or carbohydrates such as psyllium and oligosaccharides, found in fruits, vegetables, and whole grains.

Action

Probiotics are widely used to help maintain and restore the gut microbiome. Probiotics are believed to have a positive effect on the GI system by modifying the cytokines, creating an anti-inflammatory response. Probiotics add to the epithelial barrier, reduce visceral hypersensitivity, and, most significantly, maintain appropriate GI flora. Probiotics include *Saccharomyces boulardii*, *Lactobacillus casei* (*L. casei*), *Lactobacillus plantarum* (*L. plantarum*), *Lactobacillus bulgaricus* (*L. bulgaricus*), *Bifidobacteria*, and naturally occurring *Lactobacillus acidophilus* (*L. acidophilus*) and *Escherichia coli* (*E. coli*). A variety of single strain and combination strain probiotics are available.

Prebiotics may promote positive bacterial growth in the GI tract by providing food for beneficial bacteria.

The combination of both prebiotics and probiotics act synergistically to promote the greatest impact on healthy intestinal flora.

Indications

Alterations in the gut microbiome are thought to be associated with a host of GI conditions, including bloating, diarrhea, infectious diarrhea (*Clostridium difficile* [*C. difficile*]), irritable bowel syndrome (IBS), allergic disorders, and inflammatory bowel disease (IBD), as well as non-GI related disorders attributed to an autoimmune response. Probiotics are widely used in patients with IBD.

Adverse effects

The safety of probiotics has been questioned when severe illness is present. Rare cases of bacteremia have been noted in severely immunosuppressed patients.

Contraindications

There is a remote chance that probiotics may cause a pathological infection in the severely immune-compromised individual. Likewise, the potential for allergic response is always present, but rare. Probiotic strains of *Lactobacillus* have also been reported to cause bacteremia in patients with short-bowel syndrome, possibly due to impaired gut integrity.

Antibiotics

Antibiotics may be prescribed for pediatric and adult gastroenterology patients in both ambulatory and hospital care. The gastroenterology nurse must have a basic knowledge of the mechanism of action, antimicrobial spectrum, typical uses, and toxicity of commonly used antibiotics. Cost and dosing schedules must be considered in the decision-making process when prescribing, in order to enhance patient compliance. Knowledge of patient drug allergies, recommended antibiotic dosages, and bacterial taxonomy is of critical importance for gastroenterology nurses in order to provide safe medical care and promote patient and family education.

Examples of antibiotics that may be prescribed include clindamycin, ampicillin, tetracycline, ciprofloxacin, levofloxacin, moxifloxacin hydrochloride, metronidazole, azithromycin, clarithromycin, trimethoprim-sulfamethoxazole (TMP-SMX), and vancomycin.

Action

Antibiotics are antimicrobial agents. The specific antibiotic drug selection depends on the pathogen, drug adverse-effect profile, patient drug allergies, age of the patient, and any comorbid illness.

Indications

Antibiotics may be prescribed for GI infections. Some examples of infectious agents found in gastroenterology include:

- Mouth infections: *Streptococcus mutans* and other *Streptococcus* species (spp.) and *Actinomyces* spp.
- Gastroenteritis: viral, bacterial, and parasitic infections.
- Traveler's diarrhea: *Escherichia coli* (*E. coli*), *Shigella*, *Salmonella*, *Campylobacter*, *C. difficile* and amebiasis caused by *Entamoeba histolytica* (*E. histolytica*), *Cyclospora*, *Cryptosporidium*, *Giardia*, and *Cystoisospora belli*.
- Severe diarrhea: *Shigella*, *Salmonella*, *Campylobacter jejuni* (*C. jejuni*), *E. coli 0157H7*, *C. difficile*, and *Entamoeba histolytica*.
- Diverticulitis: Enterobacteriaceae, *Pseudomonas aeruginosa* (*P. aeurginosa*), *Bacteroides splanchnicus*, (*Bacteroides*) spp., and enterococci.
- Gastritis and ulcers: *Helicobacter pylori* (*H. pylori*).

Prophylactic antibiotics may be prescribed for at-risk patients undergoing endoscopic procedures that may produce bacteremia. Criteria for the use of prophylactic antibiotics continues to be controversial.

The American Society for Gastrointestinal Endoscopy's (ASGE) (2015) guideline, "Antibiotic prophylaxis for GI endoscopy", states that antibiotic prophylaxis solely to prevent infective endocarditis (IE) is no longer recommended before endoscopic procedures, except for patients with these cardiac conditions listed by the American Heart Association (AHA) (2017):

1. Prosthetic cardiac valves, including transcatheter-implanted prostheses and homografts.
2. Prosthetic material used for cardiac valve repair such as annuloplasty rings and chords.
3. Previous IE.
4. Unrepaired cyanotic congenital heart disease or repaired congenital heart disease with residual shunts or valvular regurgitation at the site of or adjacent to the site of a prosthetic patch or prosthetic device.
5. Cardiac transplant with valve regurgitation due to a structurally abnormal valve.

For patients with these cardiac conditions who have established infections of the GI tract in which enterococci may be part of the infecting bacterial flora (such as cholangitis), and particularly for those who are about to undergo an endoscopic procedure that will increase the risk of bacteremia (such as an ERCP), the AHA suggests that it may be reasonable that the antibiotic regimen include an agent active against enterococci. Amoxicillin or ampicillin should be included in the antibiotic regimen for enterococcal coverage. Vancomycin may be substituted for patients allergic to or unable to tolerate amoxicillin or ampicillin. However, the AHA reiterates that no studies demonstrate that such therapy would prevent enterococcal IE (Nishamura, 2017; ASGE, 2015).

Adverse effects

Side effects may vary depending on agent given and can include nausea, vomiting, diarrhea, rash, urticaria, headache, and anaphylaxis. The administering provider should be familiar with the medication, its potential side effects, and their contraindications prior to drug administration.

Contraindications

Contraindications include a known sensitivity to the antibiotic agent.

Anticholinergics and cholinergics

Anticholinergic medications include atropine sulfate, hyoscyamine, and dicyclomine hydrochloride (Bentyl).

Cholinergic medications include bethanechol (Urecholine, Duvoid). Side effects such as abdominal cramping, blurred vision, and other cholinergic symptoms have limited the use of bethanechol.

Action

Anticholinergic medications inhibit gastric acid secretion at its source by blocking the acetylcholine receptor on gastric parietal cells. These medications decrease the output of pepsin and block vagal stimulation of smooth muscle, thus decreasing GI tone and motility. They also decrease gastric emptying time, presumably through their inhibition of vagal and cholinergic-mediated motility.

Cholinergic agents, in contrast to anticholinergic agents, increase GI tone and motility. They produce the same effects as stimulation of the parasympathetic nervous system, thereby stimulating GI secretion and motility.

Indications

Anticholinergics

Anticholinergics may be used in the treatment of diffuse esophageal spasm, peptic ulcer disease, IBS, diverticulitis, hypermotility disorders, and ulcerative colitis. They can help to relieve the gastric distress caused by gastric spasms, hyperperistalsis, and rapid emptying of the stomach.

Because of their side effects, anticholinergics are used primarily as an adjunctive therapy for peptic ulcer disease, in combination with antacids or H_2 blockers.

Anticholinergics are best given approximately 1 hour after meals, when food-stimulated acid is at its peak. Their effects persist for 4-5 hours.

Cholinergics

The cholinergic agent bethanechol may be used to increase LES pressure in patients suffering from gastroesophageal reflux. It is also used in children with gastroesophageal reflux who are unresponsive to metoclopramide (Reglan).

Domperidone is a peripheral dopamine antagonist that has a safety profile similar to metoclopramide used to enhance gastric emptying. There have been only a few large studies performed to determine its efficacy, but it has shown promise. This drug is not FDA-approved in the United States.

Cisapride was introduced into the American drug market in 1993 to relieve symptoms of reflux esophagitis by increasing contractions in the stomach to improve stomach emptying. The drug was removed from the market in late 2000, with much controversy for post-market reports of cardiac dysrhythmias and fatalities associated with the combination of cisapride and several agents that are metabolized by the cytochrome P-450 system (particularly antifungal agents and some antimicrobials). The drug is available under a strict controlled access plan for extreme medical necessity.

Adverse effects

Adverse effects of anticholinergics include dry mouth, nose, and throat; hoarseness; tachycardia; blurred vision; urinary hesitancy or retention; flushing of the skin; and constipation.

Cholinergics cause the same parasympathetic symptoms as anticholinergics, which may include diaphoresis and bladder contractions.

Contraindications

Anticholinergic medications are contraindicated in patients who experience bleeding or who have tachycardia, closed angle glaucoma, achalasia, obstruction, or suspected toxic megacolon.

Cholinergic medications are contraindicated in the presence of peptic ulcer or possible GI obstruction.

Antidiarrheals

Antidiarrheal agents are used for symptomatic relief of diarrhea.

Action

Antidiarrheals include drugs that decrease intestinal motility (opium alkaloids and synthetic opium alkaloids) and drugs that decrease the fluid content of the stool or inhibit intestinal secretions. Bismuth subsalicylate acts by inhibiting intestinal secretions. In addition, it may absorb toxins and provide a protective coating for the mucosa.

Indications

Indications for antidiarrheals include:

- Opium alkaloids including morphine, methylmorphine, camphorated tincture of opium, and synthetic opium alkaloids such as loperamide (Imodium A-D) may be used as antidiarrheal agents, in addition to anticholinergics such as Lomotil, which is a combination of diphenoxylate hydrochloride and atropine. These agents act primarily by inhibiting intestinal motility.
- Bismuth subsalicylate (e.g., Pepto-Bismol) is useful for mild diarrhea and upset stomach. Bismuth may also be used in combination with antibiotics for the treatment of (*H. pylori*) infection.

Antidiarrheal agents also may be used judiciously to decrease the frequency of bowel movements in patients with ulcerative colitis or Crohn's disease.

For patients with IBS with chronic diarrhea (IBS-D) who have failed to respond to conventional therapies of anticholenergics and antidiarrrheals, a drug such as eluxadoline (Virbezi) may be trialed. Eluxadoline is a mu-opioid receptor agonist. It is contraindicated in patients without a gallbladder or in patients with hepatic impairment, a history of pancreatitis, a history of excessive alcohol use (greater than three drinks daily), or sphincter of Oddi dysfunction.

Adverse effects

Continued use of antidiarrheal agents over an extended period is not recommended. If it is not possible to control diarrhea promptly, diagnostic tests should be ordered. In seriously ill patients, the etiological agent should be isolated and specific antimicrobial therapy should be started.

Because opium alkaloid drugs may be habit-forming, patients must be cautioned not to exceed the recommended dosage.

Contraindications

In general, antidiarrheal agents delay the intestinal clearance of pathogens. They are not recommended for patients with fever or bloody diarrhea or in patients younger than 2 years old.

Agents that inhibit intestinal motility may cause toxic megacolon in patients suffering from pseudomembranous enterocolitis, acute dysentery, and acute ulcerative colitis. They should be used in acute exacerbation of IBD only after an infectious cause for the symptoms has been ruled out.

Opium alkaloids are contraindicated in patients with toxic causes of diarrhea and should be used cautiously by patients with asthma, liver disease, prostatic hypertrophy, and narcotic dependence.

Antiemetics

Antiemetics produce symptomatic relief of nausea and vomiting. They include phenothiazines and antihistamines, as well as trimethobenzamide hydrochloride (Tigan) and ondansetron (Zofran).

Action

The action of antiemetic drugs depends on their type:

- **Phenothiazines** such as prochlorperazine and chlorpromazine are the most common type of antiemetic drugs. They appear to exert their effects on the cells of the chemoreceptor trigger zone (CTZ) located in the medulla of the brain stem, preventing the vomiting center from being activated.
- **Certain antihistamines** such as dimenhydrinate (Dramamine) act on the CTZ to suppress centrally mediated nausea and vomiting.
- **Trimethobenzamide hydrochloride** has a mechanism of action that is unknown, but it may act on the CTZ.
- **Ondansetron** is a selective 5-HT3 antagonist antiemetic, a subtype of serotonin receptor capable of controlling nausea and vomiting by acting on receptors within the vagas nerve and CTZ, without the sedation that accompanies phenothiazines.

Metoclopramide (Reglan) may also be considered an antiemetic drug, although its principal function is to increase LES pressure and the rate of gastric emptying. Metoclopramide appears to produce its antiemetic effects as a result of its antagonism of central and peripheral dopamine receptors.

Indications

Indications for antiemetics include:

- Phenothiazines: effective in relieving vomiting associated with gastroenteritis, radiation sickness, and drug therapy, but they do not relieve motion sickness.
- Antihistamines: suppress nausea and vomiting associated with motion sickness, drug or radiation therapy, or following surgery.
- Trimethobenzamide hydrochloride: effective for the treatment of postoperative nausea and vomiting and nausea associated with gastroenteritis.
- Ondansetron: a preferable choice for long-term use such as during chemotherapy.

Adverse effects

Adverse effects can include:

- Phenothiazines: sedation, hypotension, restlessness, dry mouth, blurred vision, constipation, and muscle twitching. They should be used only when nondrug antiemetic measures or other drugs fail.
- Antihistamines: drowsiness.
- Trimethobenzamide hydrochloride: dizziness, headache, rash, Parkinson-like symptoms, muscle cramps, and seizures.
- Ondansetron: headache, drowsiness, diarrhea, fatigue, dystonia, hypotension, cardiac arrhythmias, atrioventricular (AV) block, and Stevens-Johnson syndrome.

Metoclopramide has been associated with an increased risk of extrapyramidal movement disorders such as tardive dyskinesia.

Contraindications

Contraindications include patients with the following conditions:

- Phenothiazines: asthma, benzyl alcohol hypersensitivity, phenothiazine hypersensitivity, sulfite hypersensitivity, CNS depression, leucopenia, seizure disorder, tardive dyskinesia, a history of urinary retention, and previous coronary disease, liver disease, or pulmonary disease.
- Antihistamine: renal and liver disease, a history of urinary retention, coronary disease, hypertension, seizure disorder, narrow angle glaucoma, bowel obstruction, hyperthyroidism, asthma, or COPD.
- Trimethobenzamide hydrochloride: renal failure and impairment, hepatic disease, pregnancy, breastfeeding, encephalopathy, fever, Reye's syndrome, or in geriatric and pediatric populations.
- Ondansetron: hepatitis, liver disease, phenylketonuria, GI obstruction, coronary artery disease, electrolyte imbalance, alcoholism, bradycardia, or hypertension.

Metoclopramide is contraindicated in patients with congestive heart failure (CHF), seizure disorder, tardive dyskinesia, Parkinson's disease, GI bleeding, hypertension, pheochomocytoma, G6PD deficiency, or breast cancer (prolactin is increased by metoclopramide, which may enhance tumor growth).

Antiflatulents

Antiflatulent agents such as simethicone (Mylicon) are used to relieve painful symptoms of excess gas in the GI tract.

Action

These agents act by dispersing and preventing the formation of mucus-surrounded air or gas pockets in the GI tract. Simethicone is also added to some antacid preparations to help break up gas bubbles.

Indications

Antiflatulants relieve gas that may be caused by air swallowing, postoperative gaseous distention, peptic ulcers, spastic or irritable colon, diverticulosis, or infantile colic.

Adverse effects

Adverse effects include stool discoloration, diarrhea, and nausea.

Contraindications

Simethicone should not be taken with oral thyroid medications, as simethicone chelates with oral thyroid medications reducing thyroid absorption in GI tract. Patients with phenylketouria must check for the presence of aspartame in the chosen drug preparation.

Antiparasitic/antifungal agents

There are a number of agents available to combat the effects of intestinal parasites (antiparasitic agents) and fungi (antifungal agents).

Action

Antiparasitic agents. One important agent is metronidazole (Flagyl), which demonstrates antimicrobial activity with antibacterial, antiprotozoal, and anti-inflammatory effects against a wide variety of bacterial and protozoan infections. Nitazoxanide (Alinia) is an effective broad-spectrum drug used in the treatment of helminth, protozoa, and viral infections.

Antifungal agents. Antifungal treatments can be topical, oral, or injectable. Examples of commonly prescribed antifungals are:

- Nystatin, available as an oral tablet or suspension and topical cream, powder, or ointment.
- Clotrimazole, available OTC as topical cream or solution, oral lozenge, or vaginal cream.
- Ketoconazole, available as an oral tablet and topical cream, foam, gel, or shampoo.
- Itraconazole, available as an oral solution or tablet.
- Amphotericin–B, available as an injectable.
- Fluconazole, available as an oral tablet or suspension.

Indications

Metronidazole is effective in treating bacterial and protozoan infections, including *Bacteroides fragilis* (*B. fragilis*) infections and *C. difficile*-associated colitis, and as surgical prophylaxis. Indications for use of metronidazole in Crohn's disease include perineal disease, Crohn's colitis, and sulfasalazine intolerance or allergy. Metronidazole has shown no benefit in the therapy of ulcerative colitis. A common organism of the GI tract that can invade tissues when the host defenses are altered is *Candida albicans*.

Nitazoxanide (Alinia) is another drug effective in the treatment of cryptosporidiosis or giardiasis caused by *Cryptosporidium parvum* or *Giardia lamblia*, respectively. It has been used effectively for rotavirus, *C. difficile* and the eradication of *H. pylori*.

Antifungal treatments are effective against oropharynx, esophageal, or vaginal fungal infections to which patients may be predisposed due to disease and drugs.

Adverse effects

Adverse effects of nitazoxanide include headache, nausea, diarrhea, abdominal pain, and sinus tachycardia.

Adverse effects of metronidazole include headache, nausea, pruritus, candida, Stevens-Johnson syndrome, seizures, elevated liver function tests (LFTs), and cardiac arrhythmias.

Contraindications

Contraindications include:

- Metronidazole: use with caution in patients with liver and renal failure and a history of alcoholism or in patients on corticosteroid therapies. Patients with a history of CAD, arrhythmia, electrolyte imbalance, sodium restriction, and newly diagnosed primary malignancy should be monitored carefully.
- Nitazoxanide: use with caution in patients with renal and hepatic or biliary disease. Insufficient data is available regarding use in pregnancy, breastfeeding, and pediatric and geriatric populations.

Antiulcer agents

The classes of drugs used as antiulcer agents include antacids, anticholinergic agents, H_2 blockers, sucralfate (Carafate), prostaglandins, and proton pump inhibitors (PPIs). Antacids and anticholinergics have been discussed previously.

Action

The various drugs prescribed for peptic ulcer disease promote healing by reducing gastric acid secretion, buffering secreted gastric acid, and/or enhancing intrinsic mucosal defenses.

- **Histamine$_2$ (H_2) blockers** include cimetidine (Tagamet HB), famotidine (Pepcid, Pepcid AC), nizatidine (Axid AR), and ranitidine (Zantac). Depending on the drug, they are available OTC or at prescription doses. They reduce the secretion of gastric acid by blocking histamine's action on the H_2 receptors in the parietal cells.
- **Sucralfate (Carafate)** is a basic aluminum salt of sucrose octasulfate, which forms a viscous adhesive gel that adheres to the ulcer crater, preventing further digestive action by both acid and pepsin.
- Synthetic **prostaglandins** such as misoprostol (Cytotec) have both antisecretory and cytoprotective effects.
- **Proton pump inhibitors (PPIs)** provide more complete control of acid than H_2 blockers. PPIs are prodrugs and need to be administered 30 minutes prior to eating in order to maximally activate proton pump blockade. The agents available in the United States are omeprazole (Prilosec), lansoprazole (Prevacid), pantoprazole (Protonix), rabeprazole (Aciphex), esomeprazole (Nexium), and dexlansoprazole (Dexilant). Many of these PPIs are now available OTC.

Indications

Indications differ by type of antiulcer agent.

- H_2 blockers reduce the secretion of gastric acid and may also be used to reduce gastric acidity in patients with upper GI bleeding that stems from a peptic ulcer.

- Sucralfate has been approved for the treatment of duodenal ulcers but not for gastric ulcers.
- Prostaglandins may be used to prevent the gastric ulcers and mucosal injury that have been associated with the use of nonsteroidal anti-inflammatory drugs (NSAIDs).
- PPIs are the most effective agents in the therapy of GERD. Use of these agents is indicated for the treatment of patients with erosive esophagitis or active duodenal ulcers. Long-term treatment may be indicated in some hypersecretory conditions such as Zollinger-Ellison syndrome, systemic mastocytosis, chronic erosive esophagitis, and Barrett's esophagus. Omeprazole and lansoprazole may be used in combination therapy with clarithromycin and amoxicillin in patients with positive *H. pylori* infection.

Adverse effects

Side effects, adverse effects, and precautions differ by type of antiulcer agent.

- H_2 blockers: The most common side effects of H_2 blockers are diarrhea, headaches, dizziness, fatigue, muscle pain, rash, impotence, mild gynecomastia, leukopenia, and thrombocytopenia.
- Sucralfate: The main side effect of sucralfate is constipation, which occurs in approximately 2% of patients who take this drug. Other rare side effects include dizziness, vertigo, sleepiness, dry mouth, skin rashes, pruritus, back pain, diarrhea, nausea, gastric discomfort, and indigestion.
- Prostaglandins: The most common side effect of prostaglandins is diarrhea, followed by abdominal pain.
- PPIs: Patients taking anticoagulants, anticonvulsants, or diazepam along with a PPI will require special monitoring because of potential drug-drug interactions. Adverse reactions to PPIs may include abdominal pain, asthenia, constipation, diarrhea, nausea, vomiting, and headaches. Although rare, several severe adverse reactions have been attributed to long-term use and include increased risk of fractures, hypomagnesemia, thrombocytopenia, vitamin B_{12} deficiency (especially in patients with Ellison-Zollinger syndrome), interstitial nephritis, enteric infections (including *C. difficile*), and neoplasm. There is some concern that prolonged use may contribute to increased cardiovascular risk and dementia.

Contraindication

- Increases in H_2 blocker drug therapy may increase or produce a confused state in women who are breastfeeding, patients with renal and hepatic disease, and the geriatric population.
- Sucralfate is contraindicated in patients with renal failure and in patients with dysphagia where aspiration is a potential risk. It should be used with caution in patients with diabetes due to the glucose content of the drug.
- Because of its abortifacient properties, misoprostol is contraindicated in pregnant women and generally is not recommended for women of child-bearing age.
- PPIs should be used with caution in patients with known liver disease, osteoporosis, vitamin B_{12} deficiency, and hypomagnesemia.

Corticosteroids

Action

Corticosteroids are used in gastroenterology practice primarily for their anti-inflammatory properties.

Indications

Corticosteroids are indicated in GI patients for:

- Treatment of inflammatory conditions, such as IBD, autoimmune hepatitis, collagenous sprue, severe cases of celiac disease, collagenous colitis, and radiation injury. Corticosteroids include hydrocortisone, prednisone, and synthetic analogs such as prednisolone and methylprednisolone (Medrol, Solu-Medrol injectable, Depo-Medrol injectable, and Medrol oral tabs).
- Possible relief of dysphagia in patients with benign esophageal strictures of various causes following intralesional injection therapy. Injection of the synthetic corticosteroid triamcinolone acetonide (Kenalog, Trisence injectable) and triamcinolone hexacetonide (Aristospan) may be used for this purpose.
- Crohn's disease, including proctitis or distal colitis. Corticosteroids may be administered topically (by rectal suppository, foam, or retention enema). Oral administration is effective for patients with more extensive disease and more prominent symptoms. For patients with severe and fulminant forms of the disease, IV hydrocortisone, methylprednisolone, or prednisolone may delay the course of the disease and improve symptoms.

Adverse effects

Patients on prolonged corticosteroid therapy should be monitored for signs of Cushing's syndrome, which is characterized by rapidly developing adipose deposition of the face, neck, and trunk; kyphosis; hypertension; diabetes mellitus; amenorrhea; hypertrichosis; impotence; osteoporosis; bruising; mood alterations; insomnia; muscular wasting; and weakness.

Because of the immunosuppressive nature of corticosteroids, patients must be monitored for infection and cautioned to avoid exposure to infectious disease. The physician should be notified of such exposures (such as chicken pox). The physician should be consulted before the patient considers receiving immunizations prepared with live virus (e.g., measles, mumps, rubella, varicella, and oral polio vaccines and nasal flu vaccine).

Contraindications

Corticosteroids should not be used in patients with concurrent bacterial or viral infection, communicable disease (varicella, measles, tuberculosis), myasthenia gravis, or recent acute myocardial infarction, or while on anticholinesterase agents.

Immunomodulators

Immunomodulators are medications used to regulate or normalize the immune system in an effort to promote disease control or remission. They are immunosuppressants that weaken the immune response, which appears to be overactive in patients with IBD.

Action

Immunomodulators have a "steroid-sparing" effect, allowing the patient to eventually discontinue the use of corticosteroids. Commonly used immunomodulators include mercaptopurine, (6-MP, Purinethol), azathioprine (Imuran), cyclosporine, methotrexate, and tacrolimus (Prograf).

Indications

Immunomodulators are used in IBD when the patient is unable to wean or discontinue corticosteroids without exacerbation of symptoms.

Cyclosporine has a more rapid onset of action than mercaptopurine or azathioprine. Tacrolimus and cyclosporine have been effective in treating severe ulcerative colitis until either one of the slower immunomodulators begins to work or surgery is performed. Tacrolimus is often used in Crohn's disease when corticosteroids proved ineffective or fistula development is present.

Adverse effects

Because of the tendency for immunomodulator medications to suppress bone marrow activity, it is vital that patients be instructed to keep all follow-up appointments with their physician and to have laboratory studies drawn when requested.

Patients on an immunomodulator regimen should avoid exposure to communicable illness and should see a physician promptly when ill. Annual flu shots are a must for people who take these medications. Other immunizations should be discussed with the physician because immunizations prepared with live virus should be avoided. Parents who are taking immunomodulators should check with their child's pediatrician before their child receives oral polio vaccine because the virus is shed in stool and urine for a period of time following immunization.

Contraindications

Some immunomodulator agents have a teratogenic effect. Women considering pregnancy should consult their physician while taking these drugs.

Agents used in diagnostic tests

A number of different pharmacological agents may be used in the diagnosis of GI disorders.

Action

These agents are used in a variety of ways to improve data collection and visualization during diagnostic and therapeutic exams.

Indications

Various agents are used in diagnosing GI disorders:

- Pentagastrin stimulates gastric acid secretion.
- Secretin (Chirhostim-secretin) stimulates pancreatic secretions to aid in diagnosis of pancreatic exocrine dysfunction. It can be used to stimulate gastric secretions to aid in the diagnosis of gastrinomas.
- Edrophonium chloride (Enlon injectable), a cholinergic agent, is used for provocative testing in patients with noncardiac chest pain.
- Glucagon (Glucagen) is used in diagnostic and therapeutic procedures, primarily to reduce GI motility.
- Sincalide (Kinevac), the GI hormone cholecystokinin, is given by exogenous administration and may be used in diagnostic procedures in which gallbladder contraction is desired.

Adverse effects

Adverse effects vary by medication.

- Pentagastrin: often produces nausea, vomiting, increased gastric distress, and abdominal cramping.
- Secretin: nausea, vomiting, GI distress, and flushing.
- Edrophonium: watery eyes, muscle twitching, nausea, vomiting, and diarrhea.
- Glucagon: rash, urticaria, and respiratory distress.
- Sincalide: cramping, nausea, headache, shortness of breath, and flushing.

Contraindications

Contraindications to the use of these drugs include:

- Pentagastrin: concurrent medication use, especially antacids, anticholinergics, H_2-receptor antagonists, or PPIs, and acute bleeding peptic ulcer.
- Secretin: patients with a history of liver disease, as they may be hyperreactive to secretin. Patients with a history of IBD or vagotomy may be hyporeactive to secretin, giving a false positive to testing. Alternative diagnostic testing should be considered.
- Edrophonium: bowel or urinary obstruction, narrow angle glaucoma, and pyloric stenosis.
- Glucagon: insulinoma and pheochromocytoma.
- Sincalide: cholelithiasis, as this can lead to common bile duct (CBD) obstruction.

Gallstone therapeutic agents

Ursodeoxycholic acid or ursodiol (Actigall, Urso 250, Urso Forte) is an agent that may dissolve cholesterol gallstones. Urso 250 and Urso Forte oral tabs are available in different strengths.

Action

Ursodiol decreases the rate of secretion of cholesterol into bile, thus causing the bile to become desaturated with cholesterol. This unsaturated bile dissolves cholesterol molecules from gallstones and holds them in micellar or vesicular solution until gallbladder contraction discharges them into the duodenum. Therapy must be continued until the stones dissolve completely, typically in 6-24 months.

Indications

The efficacy of medical dissolution therapy differs according to the size of the gallstone and length of therapy. Studies show the disappearance of stones with a diameter of less than 5 mm in about 90% of cases, whereas the chance of dissolution is approximately 40%-50% for larger or multiple stones (Portincasa et al., 2012). Ursodeoxycholic acid studies have shown clinical significance in treatment of postcholestectomy microlithiasis, also referred to as sludge, biliary sand, biliary sediment, and reversible cholelithiasis. Clinical conditions associated with the formation of microlithiasis are prolonged fasting, total parenteral nutrition (TPN), rapid weight loss, pregnancy, and chronic illness.

Adverse effects

Adverse effects include abdominal pain, diarrhea, headache, and fatigue.

Contraindications

Ursodiol is contraindicated for patients without patent cystic ducts, women who are or may be pregnant, and patients with radiopaque (pigment) stones, cholangitis, biliary obstruction, biliary-gastrointestinal fistula, and stones that are larger than 15 mm in diameter.

Laxatives

Laxatives are classified into six primary categories: bulk, stool softener, osmotic, saline, enema, and stimulant.

Action

The mechanism of action and effects of all laxatives vary according to their respective categories.

Bulking agents hold on to water and soften the stool, making it easier to pass, which is particularly important in patients with anorectal disorders. These agents are also used to relieve chronic constipation. The best example of a bulking agent is the nonabsorbable complex carbohydrate called fiber. Other examples include bulking agents that contain processed husk from psyllium (e.g., Metamucil, Konsyl); agents based on methylcellulose, a synthetic fiber (e.g., Citrucel); and a complex nonabsorbable starch-calcium polycarbophil (e.g., FiberCon). The addition of natural fiber from fruits and vegetables to the diet has the same bulking effect.

Stool softener and emollients or lubricating agents prevent the stool from becoming compacted and dry by:

- Lowering the surface tension, thus allowing the fecal mass to be penetrated by intestinal fluid.
- Acting as surfactants that soften the fecal mass by facilitating the mixture of aqueous and fatty substances.
- Inhibiting fluid and electrolyte reabsorption by the intestine.
- Increasing the secretion of water, both in the small bowel and the colon.

These agents include mineral oil, docusate sodium (e.g., Colace), and a combination of fiber and mineral oil in emulsion form (e.g., Kondremul).

The absorption of the fat-soluble vitamins A, D, E, and K appear to be moderately reduced with prolonged use of mineral oil, but there appears to be no significant clinical significance when used in limited quantities as directed.

Osmotic laxatives are largely nonabsorbable sugars, thus making sugar a high concentration in the bowel. Water from the tissues moves into the bowel to equalize the osmotic pressure and increase the bulk. This results in increased colonic peristalsis. Types of osmotic laxatives include:

- Lactulose (e.g., Enulose, Kristalose)—most osmotic laxatives are disaccharides.
- Sorbitol—a nonabsorbable sugar.
- Sodium phosphate monobasic monohydrate (OsmoPrep)—often referred to as the pill prep. It lost favor because of documented cases of renal failure associated with its use.
- Glycerin—a naturally occurring trivalent alcohol suppository that acts via hyperosmotic action and a local irritation and lubrication.
- Polyethylene glycol (e.g., CoLyte, GoLytely, NuLytely)—an example of a polyethylene glycol 3350 (PEG-3550) and electrolytes colon lavage preparation that may be administered orally or by nasogastric tube infusion. MiraLax, Plenvu, and MoviPrep also contain PEG-3550.
- Sodium sulfate, potassium sulfate, and magnesium sulfate (SuPrep)—an osmotic laxative intended for cleansing the colon prior to colonoscopic exam.
- Clenpiq (sodium picosulfate, magnesium oxide, and anhydrous citric acid) oral solution—a stimulant and osmotic laxative intended for colon preparation prior to procedure.

Saline laxatives increase the intraluminal pressure by attracting and retaining water in the small intestine and colon and inducing contractions. The most common saline laxative is magnesium salts—magnesium hydroxide (e.g., Milk of Magnesia) and magnesium citrate. Magnesium hydroxide also stimulates the release of cholecystokinin, which increases intestinal secretion and stimulates peristalsis and transit.

Enemas primarily work by inducing evacuation as a response to colonic distention and by lavage. They soften the stool by preventing colonic reabsorption of fecal fluid resulting from creation of a barrier between the feces and the colon. They also have a lubricant effect that eases the passage of feces through the intestine. Mineral oil retention enemas also work by lubricating the rectum and colon. Soap sud enemas provide an irritant action that stimulates peristalsis.

Stimulant laxatives work by directly signaling the muscles and nerve plexus of the intestines to contract and expel their content. These laxatives include:

- A derivative of the senna leaf (e.g., Senokot).
- Alkaloid chemicals such as bisacodyl (e.g., Ducolax).
- A combination of sodium picosulfate, magnesium oxide, and anhydrous citric acid (Prepopik), an oral stimulant laxative intended for the preparation of the colon prior to a diagnostic exam.

Indications

Laxatives vary in indication. They may be used to:

- Cleanse the bowel before radiographic examination, colonoscopy, flexible or rigid sigmoidoscopy, or surgery.
- Eliminate a substance from the GI tract.
- Prevent straining.
- Obtain a stool specimen.
- Treat constipation.

For patients who experience irritable bowel syndrome with constipation (IBS-C) and for whom laxatives and bulking agents are ineffective, medications such as linaclotide (Linzess) may be trialed. Linaclotide is a guanylate cyclase-C agonist.

Adverse effects

None of the bulking or lubricating agents or osmotic laxatives carries long-term side effects. Results vary from 5 minutes to several days.

Side effects of osmotic preparations include headache, nausea, vomiting, bloating, electrolyte imbalance, and dehydration.

Stimulant laxatives may be habit-forming, necessitating increasingly higher doses to produce the same stimulating effect.

Side effects of Prepopik include headache, nausea, vomiting, bloating, electrolyte imbalance, and dehydration.

The most significant side effect of linaclotide is diarrhea.

Contraindications

Stimulant laxatives are contraindicated in the presence of obstruction or peritonitis or immediately after bowel surgery.

Prepopik should be used with caution in patients with known renal insufficiency.

Linaclotide is contraindicated in children under 6 years of age and for those with known or suspected bowel obstruction.

Narcotics or opioids and antagonists

Narcotics, including meperidine (Demerol) and fentanyl (Sublimaze), are used in gastroenterology. Other narcotics include tramadol and morphine.

Action

Overall, narcotic painkillers work by reducing nerve excitability that leads to the sensation of pain. Narcotics bind to special receptors in the brain (central nervous system) and in other areas of the body (peripheral nervous system, including the gastrointestinal tract) called opioid receptors. The receptors block the release of neurotransmitters involved with pain sensation.

Indications

Narcotic analgesics, particularly meperidine and fentanyl, may be used for premedication of patients undergoing endoscopic procedures. They may also be used for postoperative pain relief.

Adverse effects

Narcotics should be used sparingly because they tend to mask symptoms and complications and may cause physical and psychological dependence. Abrupt discontinuance of the drug may precipitate withdrawal symptoms, including convulsions and a decrease in bowel motility.

Fentanyl appears to spur less emetic activity than either morphine or meperidine. In addition, clinically significant histamine release rarely occurs with fentanyl.

The most dangerous side effect of narcotic medications is respiratory depression. Respiratory depression, sedation, and hypotension can be reversed by administration of naloxone, which is thought to antagonize the opioid effects by competing for the same receptor sites (see section on reversal agents).

Contraindications

Morphine should not be given to patients with biliary or pancreatic problems, either preoperatively or postoperatively, because it may increase smooth muscle spasm. Meperidine is usually the drug of choice for these patients. Morphine must also be restricted in patients with severe liver disease.

Sclerosing agents

Sclerotherapy is rarely used anymore. Endoscopic variceal banding is the recommended treatment to stop bleeding from varices. For recurrent gastric varices, beta blockers, transjugular intrahepatic portosystemic shunt (TIPS), and tissue adhesives are used to create hemostasis.

Action

Sclerosing agents such as sodium tetradecyl sulfate (Sotradecol), ethanolamine oleate (Ethamolin), and 100% ethanol are injected intravariceally or paravariceally to promote intima inflammation and thrombus formation in patients with esophageal varices. The subsequent formation of fibrous tissue results in partial or complete vein obliteration. Vasoactive medications are recommended in the treatment of acute variceal bleeding and have shown to reduce mortality and re-bleeding rates. The two classes of vasoactive drugs—vasopressin, including terilpressin, and somatostatin—and their respective analogues octreotide and vapreotide can be highly effective for prolonged hemostatsis control.

Indications

Sclerotherapy is used to stop bleeding from esophageal varices or to obliterate varices that previously bled.

Adverse effects

Potential post-procedure complications for sclerotherapy may be various degrees of bleeding, esophageal ulcers, esophageal perforation, esophagopleural fistula, and sepsis. Sclerotherapy also may precipitate or increase hepatic encephalopathy.

Contraindications

Sclerosing agents are pregnancy category C, meaning that there is unknown risk to pregnant women and unborn children. There is a risk of fatal anaphylaxis for larger than normal dose administration.

Sedatives and antagonists

The use of **sedatives** to achieve moderate sedation requires the gastroenterology nurse to complete specialized training and competency. Patients require uninterrupted monitoring throughout the procedure once moderate sedation is initiated, once the desired sedation level is achieved, and until sedation is resolved. Resuscitative equipment must be readily available. When using sedatives, the individual administering the drug must have no duty other than patient monitoring and must be capable of full cardiopulmonary rescue.

Action

Propofol (Diprivan) works by increasing gamma-aminobutyric acid (GABA)-mediated inhibitory tone in the central nervous system.

Indications

Sedatives and antianxiety agents such as diazepam (Valium) and midazolam are used in the practice of gastroenterology primarily to medicate patients before and during endoscopic or invasive procedures. They may be used to relieve pre-procedural anxiety and tension and to decrease recall of the procedure. Sedatives are most commonly used to achieve moderate sedation for patients during an endoscopic examination.

Propofol is given for deep sedation or monitored anesthesia care (MAC). In most states, only licensed anesthesia providers can administer propofol. Propofol is administered via slow IV push and is titrated according to need.

Adverse effects

The most serious side effect of moderate sedation with midazolam or propofol is a centrally mediated respiratory depression. Apnea is associated with rapid IV injection of either drug. The extent of respiratory depression depends on dosage, rate of administration, and individual susceptibility. Sedative effects are accentuated by the concomitant administration of other central nervous system depressants such as opiates, barbiturates, or alcohol. Midazolam is more potent, is faster-acting, and has a greater amnesic effect than propofol. Propofol is associated with more injection site complications such as thrombophlebitis.

During MAC with propofol, hypotension, oxyhemoglobin desaturation, obstruction, and apnea may occur.

Contraindications

Allergy to eggs and soy are contraindications to propofol use.

Contraindications to moderate sedation include sensitivity to the anesthetic agent; previous complications with sedation, including airway difficulty; morbid obesity; history of sleep apnea; and a compromised cardiovascular state. The inability to effectively communicate with the physician or nurse may be taken into consideration.

Reversal agents

Reversal agents include medications to reverse the effects of benzodiazepines and opioids.

Action

The purpose of reversal agents is to shorten or reverse the action of the narcotic or benzodiazepine administered.

Indications

Benzodiazepine reversal. Flumazenil may reverse the sedative effects of benzodiazepines (i.e., midazolam, propofol) but not the resulting respiratory depression. Flumazenil is given intravenously in individual doses of 0.2 mg/min, not to exceed 1 mg at a time and not more than 3 mg/hr. The drug acts in minutes, but its duration is considerably shorter than that of the benzodiazepines. Careful monitoring for resedation must be carried out.

Opioid reversal. Naloxone reverses both sedation and respiratory depression associated with administration of opioids. As with flumazenil, multiple doses may be necessary because the drug's duration is shorter than that of the opioid. Careful monitoring for resedation must be carried out.

Adverse effects

When given a reversal agent, outpatients in the endoscopy unit should not be discharged home until the gastroenterology nurse is reasonably certain that the effect of the reversal agent has worn off (usually a minimum of one 1 hour) and the institutional guidelines for discharge have been met.

Contraindications

Contraindications include hypersensitivity to the drug. Reversal agents should be used with caution in patients with known cardiovascular disease or in patients with known addiction, as rapid reversal can precipitate an acute withdrawal syndrome.

Smooth muscle relaxants

Action

These agents have a direct relaxant effect on the smooth muscle fibers of the LES.

Indications

Smooth muscle relaxants can alleviate esophageal symptoms of achalasia and diffuse esophageal spasm in some patients. Sublingual isosorbide dinitrate (Dilatrate-SR oral tab; Isordil oral tab) can improve symptoms and radionuclide transit time. Calcium channel blockers such as nifedipine (Procardia) also have recognized relaxant effects on LES muscle.

Adverse effects

Adverse effects of smooth muscle relaxants include hypotension, flushing, tachycardia, syncope, nausea, vomiting, and rash.

Contraindications

Smooth muscle relaxants are contraindicated in patients with hypotension, hypovolemia, cardiomyopathy, closed angle glaucoma, and hepatic disease. Safety during pregnancy and breastfeeding and use in children is not established.

Topical anesthetics

Action

Topical sprays and gels work by reversibly blocking nerve conduction responsible for pain.

Indications

Topical **anesthetics** such as benzocaine (Cetacaine) and lidocaine (Xylocaine) are used in gastroenterology to suppress the gag reflex and control pain for upper GI endoscopic procedures.

Adverse effects

Common adverse effects of topical anesthetics include contact dermatitis, localized swelling, and edema. On rare occasions, methemoglobinemia has been reported in connection with the use of benzocaine-containing products.

Contraindications

Benzocaine should never be used near the eyes, in any large denuded area, or in patients with a cholinesterase deficiency. Lidocaine should be not used in patients with G6PD deficiency, an amide allergy, or significantly impaired lung function.

Other agents

Other drugs used in the treatment of GI disorders include medications containing 5-aminosalicylic acid (5-ASA), tumor necrosis factor blockers, pancrelipase, cholestyramine, penicillamine, neomycin, lactulose, and medications for hepatitis C.

5-aminosalicylic acid (5-ASA)

The clinical response of 5-ASA is thought to be due to a localized effect. Preparations are designed to deliver drug to the large intestine where its therapeutic action is driven by intestinal lumen absorption. 5-ASA absorption is variable and depends on disease activity and colonic pH.

Medications containing 5-ASA include mesalamine, olsalazine, and sulfasalazine.

5-ASA may be administered orally as mesalamine (Asacol, Pentasa, Apriso, Delzicol) and olsalazine sodium (Dipentum). Lialda and Apriso are FDA-approved mesalamines that are a once-daily prescription tablet. These drugs have proven as effective as sulfasalazine in preventing relapse of ulcerative colitis but without the side effects associated with the sulfapyridine moiety.

Mesalamine

Mesalamine (Rowasa) may be administered in the form of a suspension retention enema for patients with distal ulcerative colitis, proctosigmoiditis or proctitis, and radiation enteritis and colitis. Research has demonstrated that some forms of mesalamine (Asacol, Pentasa) are effective when disease is proximal to the colon (Ye et al., 2015).

Adverse effects

The mesalamine suspension enema also contains potassium metabisulfite, which may cause life-threatening allergic reactions in patients with sulfite sensitivity. Epinephrine is the preferred treatment for serious allergic or emergency situations. Other side effects include nausea, headache, diarrhea, worsening or ulcerative colitis symptoms, and renal impairment.

Contraindications

5-ASAs are contraindicated in patients with a salicylate hypersensitivity and those with renal impairment. The Rowasa brand of rectal suspension may produce an allergic response in patients with a sulfite allergy because it contains potassium bisulfite. The Apriso brand tablets contain aspartame, which is contraindicated in patients with phenylketonuria.

Sulfasalazine

Sulfasalazine (Azulfidine) is a complex of sulfapyridine and 5-ASA. It is effective in the maintenance of clinical remission and in the treatment of mildly to moderately severe attacks of ulcerative colitis. It is also effective in active Crohn's colitis and ileocolitis, although it does not appear to be as effective in ileitis alone.

Adverse effects

Side effects are common and include nausea, vomiting, headache, rash, and skin sensitivity to sun exposure. Temporary male infertility may also result.

Contraindications

Sulfasalazine should be used with caution in patients with a hypersensitivity to salicylate 5-ASA, carbonic anhydrase inhibitor, sulfonamide, sulfonylurea, or thiazide diuretic, and in patients with G6PD deficiency. Safety is not established in pregnancy, breastfeeding, and in children. It should be used with caution in patients with chronic or recurrent infections, as immunosuppression can lead to sepsis and death.

Thiopurines

Action

Azathioprine and 6-mercaptopurine are immunosuppressive immunomodulators (acting on isolated genes). They deactivate the T lymphocytes responsible for inflammation. They often take months to be effective.

Indications

Thiopurines are often employed when 5-ASA therapy along with two course of steroids have failed.

Adverse effects

Fatigue, malaise, flu like symptoms, nausea, lymphopenia, neutropenia, liver toxicity, and pancreatitis are common side effects.

Contraindications

Thiopurines should not be used in patients with preexisting liver or pancreatic disease.

Tumor necrosis factor (TFN) blockers

Action

Adalimumab (Humira), infliximab (Remicade), golimumab (Simponi), vedolizumab (Entyvio), and certolizumab pegol (Cimzia) are TNF blockers. Vedolizumab is a monoclonal antibody that specifically targets $\alpha 4\beta 7$.

Indications

TNF blockers are used to reduce the signs and symptoms of moderate to severe Crohn's disease and/or ulcerative colitis in adults. Some TNF blockers also may be used to treat other autoimmune disorders such as rheumatoid arthritis.

Golimumab is used to treat ulcerative colitis in patients who have not responded to 5-ASA, corticosteroids, and immunomodulators such as 6-MP.

Vedolizumab is used to treat both Crohn's disease and ulcerative colitis.

Adverse effects

Patients taking TNF blockers are at increased risk for developing serious opportunistic infections, including tuberculosis, fungal infections, histoplasmosis, coccidioidomycosis, legionellosis, and others. Live vaccines are not recommended in this population. Patient education regarding preventative health care, annual screenings, open communication with providers, and participation in care is strongly encouraged during therapy.

Prior to starting therapy with TNF blockers, hepatitis B and varicella serologies, PPD testing, and a chest X-ray are required. These should then be repeated annually while on therapy. Adalimumab, infliximab, certolizumab pegol, and vedolizumab require monitoring of liver function and hematology tests, as these drugs can affect immune and liver function. Often c-reactive protein (CRP) is monitored to determine the effect on inflammation.

Dosage for infliximab is weight-based, so weight must be taken on each medical visit.

Contraindications

Active disease, including viral infections, upper respiratory infections or other illness is a contraindication to TNF therapy. Patients with mild to moderate heart failure are not suitable candidates for treatment. Patients with preexisting demyelinating disease are at greater risk for developing muscular sclerosis or optic neuritis if treated with TNF blockers.

Any previous acute reaction to an infusion is a contraindication to proceed with further treatment.

Biosimilars

A biosimilar drug is one that is highly similar to its reference drug in terms of structure, purity, safety and efficacy, biological activity, receptor binding, and potency. They are FDA-approved drugs that perform identically to their reference drug. These drugs differ in their inactive compounds and the number of Phase III trials required by the FDA.

Biosimilars are medications contained under the Biologics Price Competition and Innovation Act of 2009. The goal of this law is to promote access to medication that is otherwise too costly.

Renflexis (infliximab-abda) is one such drug. It carries all the same prescribing, dosing, and side effect information as its originator drug.

Pancrelipase

Action

Pancrelipase, which includes Creon and Zenpep, is a combination of digestive enzymes, including lipase, protease, and amylase.

Indications

Pancrelipase is used to decrease the number of bowel movements and improve stool consistency in patients with pancreatic exocrine insufficiency, cystic fibrosis, steatorrhea, and other disorders of fat metabolism.

Adverse effects

In patients with cystic fibrosis or malabsorption syndrome, high doses of pancreatic enzymes containing 20,000 USP units of lipase per capsule may increase the risk of strictures in the ileocecal region and/or ascending colon. Other side effects may include constipation, diarrhea, and abdominal pain.

Contraindications

Pancrelipase is contraindicated in patients with acute pancreatitis and hypersensitivity to pork protein. It should be used with caution in patients with asthma and gout.

Cholestyramine

Action

Cholestyramine is an anion exchange resin. It absorbs and combines with bile acids in the intestine to form an insoluble complex that is excreted in the feces. This increased fecal loss of bile acids leads to an increased oxidation of cholesterol to bile acids, a decrease in plasma low-density lipoproteins (LDLs), and a decrease in serum cholesterol.

Indications

Cholestyramine is used to lower plasma cholesterol levels. In patients with partial biliary obstruction, the reduction of serum bile acid levels reduces excess bile acids deposited in the skin, resulting in a decrease in pruritus.

Adverse effects

The most common adverse reactions are constipation, abdominal discomfort, and nausea.

Contraindications

Cholestyramine is contraindicated in patients with complete biliary obstruction.

Penicillamine

Action

Penicillamine (Cuprimine) chelates to copper, which is then excreted in the urine. Trientine may also be used as a chelating agent.

Indications

Penicillamine is the drug of choice for patients with Wilson's disease, which is a hereditary disorder of copper metabolism. Patients with Wilson's disease must be committed to lifelong administration of a daily dose of penicillamine or trientine. Complete reversal or improvement of hepatic, neurological, and psychiatric abnormalities can be expected in most patients.

Other medications used to treat Wilson's disease include GABA antagonists, anticholinergics, baclofen, CNS depressants, and levodopa to treat dystonia and accompanying tremors; antiepileptics to manage seizures; other neuroleptics to manage psychiatric symptoms; and lactulose to treat hepatic encephalopathy.

Adverse effects

A small portion of patients develop serious toxic reactions to penicillamine and may require administration of an alternative chelating agent.

Contraindications

Use of penicillamine is contraindicated during pregnancy, and women should not breastfeed while on therapy. Penacillamine also is contraindicated in patients with a history of penicillamine-related aplastic anemia or agranulocytosis.

Neomycin and lactulose

Action

Neomycin acts by reducing the production of the nitrogenous breakdown products that cause encephalopathy. Rifaximin (Xifaxan), a non-aminoglycoside and non-systemic antibiotic, is a derivative of rifampin and is often better tolerated than neomycin for hepatic encephalopathy.

Bacterial breakdown of lactulose acidifies the colonic contents, resulting in the retention of ammonia in the colon and a concomitant reduction in blood ammonia levels.

Indications

Neomycin, rifaximin, and lactulose are used in the treatment of hepatic encephalopathy.

Rifaximin is also used at lesser doses to treat IBS-D and traveler's diarrhea.

Adverse effects

Most common side effects include nausea, dizziness, abdominal pain, peripheral edema, and myalgias.

Contraindications

Rifaximin is contraindicated in patients with a hypersensitivity to rifamycin. Safety and effectiveness have not been established in pediatric and geriatric populations. Use during pregnancy or lactation is not recommended. Teratogenic effects on animals were established, but no human studies have been completed.

Medications for hepatitis C (HCV)

Recent research has shown that non-A/non-B hepatitis is the most common infectious cause of chronic liver disease. In 2015, the Centers for Disease Control and Prevention (CDC) found an 11% increase in HCV among nonurban young Caucasians (CDC, 2017). Several kinds of interferon products have been developed to fight hepatitis C virus (HCV). The most commonly used ones are shown on Table 20.3.

Action

Interferon, such as interferon alfa-2B (Intron A), works against HCV by:

- Preventing the virus from entering into cells.
- Interfering with the virus's ability to make the proteins it needs to thrive.
- Stimulating the production of other immune system defenders.
- Increasing the immune system's ability to recognize viral cells so that it can destroy them.

Ribavirin is an oral antiviral agent that is used in combination with interferon. Ribavirin is a nucleoside analogue that does not work when used alone against HCV.

The length of treatment course for HCV depends on viral genotype. Both interferon and ribavirin can cause side effects. Some of the more common side effects include nausea, vomiting, diarrhea, headache, fatigue, mood changes, thrombocytopenia, and anemia

Research into antiviral drug therapy for HCV has resulted in a growth in therapies. There are a variety of therapies based on genotype and comorbidity.

CASE SITUATION #1

Six-month-old Davy Simms is admitted to the endoscopy unit with a persistent reflux problem manifested by vomiting or spitting up. Regurgitation occurs within 1.5-2 hours following ingestion of food or fluids. Davy was diagnosed with a patent ductus arteriosus at the age of 4 weeks. He is a poor feeder, highly irritable and fussy, and on the low borderline developmental scale for his age. He weighs 13.5 pounds (6.1 kg), is 24.5 inches in length, and has a body surface area of 0.33 meters (based on a precalculated approximation). Because he lives in a high streptococcus incidence area of the country and has two siblings, ages 5 and 7, he is on a daily dose of prophylactic ampicillin 20 mg. He is scheduled to have general anesthesia for an endoscopic procedure. His pediatrician orders 65 mg of ampicillin and 12 mg of gentamicin to be given by IV drip over 1 hour. Sedation will be by diazepam (Valium) given orally 1 hour before the procedure.

Points to think about

1. What might be the rationale for using general anesthesia rather than moderate sedation with midazolam to sedate this patient?
2. What are the benefits of using diazepam as an adjunct to an anesthetic agent?
3. Why would diazepam be given orally in this patient, rather than intravenously or intramuscularly?

Suggested responses

1. Moderate sedation in a 6-month-old is risky because of the inherent danger associated with esophagoscopy in an infant who cannot be adequately sedated or cannot understand the need to

Table 20.3. Antiviral Drug Therapy for HCV Based on Genotype

Genotype	Drugs	Length of treatment
1,3	Daklinza (daclatasvir) with Savoldi (sofosbuvir); sometimes with ribavirin	12 weeks
1,2,3,4,5,6	Epclusa (sofosbuvir/velpatasvir); only given with ribavirin in cases of decompensated (advanced) cirrhosis	12 weeks
1,4,5,6	Harvoni (ledipasvir/sofosbuvir); sometimes given with ribavirin	8, 12, or 24 weeks, depending on type
1	Olysio (simeprevir) plus Sovaldi (sofosbuvir); sometimes given with ribavirin	24 weeks
1	Viekira Pak (dasabuvir/ombitasvir/paritaprevir/ritonavir) combo pack; given with ribavirin if genotype 1a	12-24 weeks
1	Viekira XR (dasabuvir/ombitasvir/paritaprevir/ritonavir) extended release formula; given with ribavirin if genotype 1a	24 weeks
1,4	Zepatier (elbasvir/grazoprevir); sometimes given with ribavirin	12-24 weeks
2,3,4	Savoldi (sofosbuvir); often given with ribavirin	12-24 weeks
4	Technivie (ombitasvir/paritaprevir/ritonavir) plus ribavirin	12 weeks
1,2,3,4,5,6	Vosevi (sofosbuvir/velpatasvir/voxilaprevir) not taken with ribavirin	12 weeks

remain still during the procedure. Although general anesthesia has its own associated difficulties, it precludes movement during the procedure.

2. The benefits of diazepam as a premedication include:
   a. Reduction of fear and anxiety resulting from the unknown and, in this case, parental separation.
   b. Sedation and amnesia.
   c. Probable better acceptance of a face mask.
   d. Few nightmares.
   e. Not likely to cause hypotension, tachycardia, dizziness, excitement, and/or postoperative nausea and vomiting.
3. Diazepam probably would be given orally to this patient for the following reasons:
   a. It absorbs rapidly from the GI tract.
   b. The rate of absorption from the IM route is erratic, and absorption depends on the muscle into which the drug is injected (deltoid and upper thigh preferred, rather than buttock). These muscles are relatively small in an infant.
   c. The drug cannot be mixed with other drugs for IV use.
   d. The drug may cause persistent pain at the injection site if given intramuscularly and may cause superficial, painful venous thrombosis when given intravenously in small veins.

CASE SITUATION #2

Larry West is a 23-year-old male with a history of Crohn's disease since age 11. He is 73 inches tall and weighs 130 pounds. He has been maintained on mesalamine 4.2 grams daily and has been well controlled until recently. His last colonoscopy, 3 years ago, showed active ileitis and granulomas of the large bowel. He was asymptomatic at that time, and there was no change in therapy. Larry now presents at the physician's office with complaints of urgency, pain, frequency of stools (up to 7 times a day), and weight loss. A colonoscopy is repeated, showing increased inflammation of the terminal ileum with ulceration and stenosis. The physician decides to initiate infliximab (Remicade) therapy.

Points to think about

1. What are the preliminary lab and diagnostic tests that are required prior to initiating Remicade therapy?
2. How is the Remicade dosage determined?
3. What are the known risks when taking Remicade?

Suggested responses

1. Before starting Remicade, the doctor will likely order the following labs and diagnostic tests:
   a. Chest X-ray.
   b. PPD testing.
   c. Hepatitis B and varicella serologies.
2. Remicade dosage is determined by weight.
3. The known risks when taking Remicade include:
   a. Tuberculosis.
   b. Coccidioidomycosis.
   c. Immunosupression.
   d. LFT elevations.

CASE SITUATION #3

Susan Jones is a 35-year-old, 165-pound (77 kg) female who underwent an upper endoscopy. She was highly agitated throughout the endoscopic procedure. She received 7.5 mg of midazolam IV during the procedure in titrated doses for moderate sedation. Susan's first postoperative vital signs are blood pressure 120/82, pulse 80, respirations 14, oxygen saturation level 98%.

Points to think about

1. What should the post-procedure gastroenterology nurse anticipate in the course of recovery for Susan?
2. Why was a reversal agent not used immediately after the procedure?
3. What are the responsibilities of the gastroenterology nurse when moderate sedation is used on an adult patient?

Suggested responses

1. The usual dose of midazolam for moderate sedation does not normally exceed 5 mg titrated throughout the procedure. Susan was highly agitated throughout the procedure, requiring continuous titration of midazolam for a total of 7.5 mg. After the procedure, the gastroenterology nurse should expect Susan to be highly sedated once the stimulus of the endoscopy is concluded.
2. A reversal agent for midazolam is not necessarily warranted in a patient who is hemodynamically stable but sedated. Some physicians will order the reversal agent flumazenil immediately post-procedure to block any further uptake of the benzodiazepine midazolam. This will generally reverse the effects of the midazolam in 1-3 minutes and cause the patient to awaken from sedation. Most physicians will not use a reversal agent if vital signs are stable with oxygen saturations above 90% to 92% and will allow the patient to awaken slowly.
3. The responsibilities of the registered nurse in the gastroenterology unit when moderate sedation is used for an adult are as follows.

   Pre-procedure phase:
   a. Perform a nursing assessment.
   b. Verify presence of signed informed consent.
   c. Establish venous access.
   d. Explain the procedure, the medications, and what the patient can expect before, during, and after the procedure, and review post-endoscopy instructions.

Intra-procedure phase:
a. Monitor the patient's physical parameters.
b. Document all events occurring during the procedure and the patient's status on completion.
c. Reassure the patient verbally and by touch.

Post-procedure phase:
a. Monitor the patient's condition and level of consciousness.
b. Document all findings and events.
c. Provide verbal and written instructions regarding diet, medications, activities, and signs and symptoms of complications, with action to take if complications develop.
d. Ensure that the patient meets institutional discharge criteria.

REVIEW TERMS

anesthetic, antacid, antibiotic, anticholinergic, antidiarrheal, antiemetic, antiflatulent, cholinergic, corticosteroid, histamine2 (H_2) blocker, laxative, narcotics, oral, parenteral, proton pump inhibitors (PPI), prostaglandin, sedative, topical

REVIEW QUESTIONS

1. When used in a medication order, the abbreviation "ac" indicates that the drug is to be administered:
   a. At mealtime.
   b. At bedtime.
   c. Before meals.
   d. As needed.
2. One kilogram is equivalent to approximately:
   a. 2.2 pounds.
   b. 22 ounces.
   c. 20 pounds.
   d. 1,000 ounces.
3. Pediatric dosages are usually calculated based on the child's:
   a. Weight in kilograms.
   b. Height and weight.
   c. Age.
   d. Surface area.
4. For optimal effect, antacids should be given:
   a. Before meals.
   b. Immediately after meals.
   c. One hour after meals.
   d. At bedtime.
5. Antacids work by:
   a. Decreasing acid secretion.
   b. Coating or protecting mucous lining.
   c. Reducing total acid load.
   d. Decreasing gastric pH.
6. Anticholinergics are primarily used as adjunct therapy for peptic ulcer disease in combination with:
   a. Cholinergics.
   b. Sedatives.
   c. Antacids or H_2 blockers.
   d. Antiflatulents or antiemetics.
7. Corticosteroids are used most often in gastroenterology in patients who have:
   a. Inflammatory bowel disease.
   b. Peptic ulcers.
   c. Pancreatic exocrine insufficiency.
   d. Cholelithiasis.
8. The purpose of glucagon in GI diagnostic testing is to:
   a. Reduce GI motility.
   b. Stimulate gastric acid secretion.
   c. Stimulate production of bile.
   d. Increase GI motility.
9. The use of sedatives to achieve moderate sedation for endoscopic procedures requires:
   a. Ability to continuously monitor the patient.
   b. Sedation competency by the practitioners.
   c. Resuscitative equipment readily available.
   d. All of the above.
10. Omeprazole (Prilosec) is a proton pump inhibitor (PPI) that:
    a. Inhibits production of stomach acid.
    b. Can cause bleeding.
    c. Increases the production of stomach acid.
    d. Is used to treat ulcerative colitis.

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Chapter 21

INTRAVENOUS THERAPY

This chapter will acquaint the gastroenterology nurse with the principles of intravenous (IV) therapy. The primary goal of IV therapy is to maintain and restore circulation and electrolyte replacement. IV lines are used in gastroenterology practice to administer **crystalloids** for the maintenance of fluid and electrolyte balance; to administer certain drugs; and to transfuse **colloids** such as whole blood, packed red blood cells (RBCs), fresh frozen plasma (FFP), platelets, and albumin. IV lines are also used for parenteral hyperalimentation, which is discussed in Chapter 22.

Learning objectives

After reviewing the content of this chapter, the gastroenterology nurse will be able to:

1. Discuss the gastroenterology nurse's role in IV therapy care.
2. Discuss the basic principles of fluid and electrolytes balance with early detection and reporting of imbalances.
3. Explain the correct method for administration of IV solutions.
4. Describe IV administration risks both to the gastroenterology nurse and the patient.
5. Explain IV equipment, types of vascular access devices (VADs), and administration of IV drug therapy for gastroenterology patients.
6. Describe techniques for transfusing blood and blood products in gastroenterology patients, along with signs, symptoms, and treatment of adverse reactions.

GASTROENTEROLOGY NURSE'S ROLE IN IV THERAPY

The overall role for gastroenterology nurses during IV therapy is to enhance the patient's experience and provide compassionate, holistic care through the nursing process. (For more information on the nursing process, see Chapters 8, 9, 10, 11, & 12.)

Gastroenterology nurses must have a working knowledge of standards of practice, principles, and application in IV therapy to ensure a safe working environment for patients, themselves, and colleagues. Issues on minimizing the risks and complications to the patient will be discussed, including medication errors, infection, phlebitis, speed shock, extravasation, and infiltration.

The complexity of IV therapy practice demands a high level of expertise and training. Many organizations utilize designated IV teams to maximize continuity and quality of care. Even in those organizations, the gastroenterology nurse must maintain a skill level related to IV therapy administration. The gastroenterology nurse is responsible for assessment, observation, and monitoring to determine the patient's response to treatment, whether optimal or adverse. In addition, the gastroenterology nurse is often responsible for peripheral IV access and routine IV care.

Components of IV therapy include the following:

- **Assessment** is the first step and includes assessing the patient's knowledge and experiences with IV therapy; the patient's diagnosis, co-morbid conditions, activity level, and mental state; and the duration and type of therapy.
- **Planning** includes identification of the appropriate person to place the vascular access device (VAD), the type of VAD, the dressing necessary for the VAD, and assessment for the best site to place the VAD.
- **Implementation** includes preparing the patient, the environment, and the equipment, and placing the VAD.

- **Evaluation and assessment** includes determining the patient's response or outcome to the IV therapy administered and revising the plan of care to achieve desirable outcomes.

Competencies in IV therapy for the gastroenterology nurse should include:

- Knowledge of the types of VADs used.
- Skill in performing the procedure competently and safely.
- Skill in inspection and assessment of the insertion site.
- Ability to problem-solve as necessary.
- Ability to monitor the patient's condition and report changes.
- Skills for proper VAD documentation and record-keeping.
- Knowledge of medication and fluids infused via the VAD.

Table 21.1. Gastroenterology Nurse's Responsibilities for IV Therapy

1.	Identify the right patient.
2.	Verify the prescription and the right medication. Check the infusion fluid, drug, and container for any obvious faults or contamination.
3.	Confirm the right dosage.
4.	Ensure administration of the prescribed drug or fluid via the right route. Establish that the VAD is patent and comfortable for the patient.
5.	Verify the right times and frequency for the medication.
6.	Maintain appropriate documentation. Monitor the condition of the patient and report any changes.
7.	Verify the rationale for the medication delivered.
8.	Monitor the condition of the patient to determine if the medication has led to the desired effect. Report any changes.

(Bonsall, 2011)

FLUIDS AND ELECTROLYTES

Maintaining the patient's fluid and electrolyte balance is an important part of the gastroenterology nurse's role.

Basic principles

Water makes up about 60% of the average adult's total body weight. Of this amount, approximately two-thirds of the total body water is held within the intracellular space—the fluid within the cell membrane. The remaining one-third is found in the extracellular space—the fluid between the cells (interstitial space) and the **plasma** (intravascular space). The interstitial space normally contains 80% of the extracellular fluid, with the plasma containing the other 20%.

The body's fluid space is a **solution** composed of **solutes** (electrolytes and nonelectrolytes) and **solvents** (water) (see Table 21.2).

Osmosis is the movement of water (the solvent) across a semipermeable membrane from an area of higher water concentration to a lower water concentration. Osmolality, a measure of the concentration of molecules per weight of water, controls the distribution and movement of water between body compartments and plays a key role in fluid balance. Osmotic pressure is defined as the drive for water to move in one direction or the other and is dependent on the number of ions present in the water solution. Osmotic pressure can be exerted by the intracellular space or the extracellular space.

Water moves freely between the intracellular and extracellular compartments, maintaining an osmotic equilibrium. For example, if the extracellular electrolyte concentration increases, water moves from the intracellular compartment (area of lower concentration) to the extracellular compartment (area of higher concentration), diluting the extracellular compartment and maintaining the osmotic equilibrium. When intracellular electrolyte concentration increases, water moves in the reverse. The ability of extracellular water to move in and out of a cell by osmosis is referred to as cellular tonicity.

Body fluids (solutions) are composed of two types of solutes, electrolytes and nonelectrolytes.

- **Electrolytes** are electrically charged and dissociate into ions—cations and anions—when placed in solution. Electrolytes account for 95% of the solute molecules in body water. Cations are ions with a positive charge and include sodium (the predominant extracellular cation), potassium (the predominant intracellular cation), calcium, and magnesium. Anions are ions with a negative charge and include chloride, phosphate, and sulfate. These electrolytes help to maintain osmotic pressure within cells. The concentration of particles is expressed as milliequivalents per milliliter (mEq/ml). One milliequivalent of any cation is able to react with one milliequivalent of any anion. Ordinarily, electrolyte solutes are balanced so there is an equal distribution of cations and anions.
- **Nonelectrolytes**, such as glucose, urea, bile salts, creatinine, and cholesterol, do not dissociate in water into ions, but they do affect the acid-base balance and osmotic pressure gradients.

Table 21.2. Relationship Between Solution, Solute, and Solvent

Solution	A liquid that contains dissolved substances
Solvent	The liquid portion of the solution
Solute	The substance dissolved in the solution

(McCance & Huether, 2018)

Fluid and electrolyte imbalances

The body receives approximately 2 liters of water daily through the ingestion of fluids and solid foods. Through metabolic reactions in the body, approximately 300-400 ml of additional fluid are produced. Under normal conditions, the body loses water daily in the urine, feces, through perspiration, and via insensible (immeasurable) losses through the skin and the lungs. In the healthy patient, these daily outputs are balanced by fluid input. A fluid or electrolyte imbalance may develop as a result of vomiting, diarrhea, suction, draining wounds, intestinal obstructions, draining fistulas, hemorrhage, infections, fever, or prolonged use of enemas and laxatives. Infants are especially vulnerable to fluid loss because of their higher proportion of body fluid (75% at birth, 65% at one year), immature kidneys, increased heat production, and rapid growth.

Excessive loss of body water can result in dehydration. The goal for dehydrated patients is to restore the circulating volume of fluid without causing an overload. Careful nursing observation, recording, and reporting of the patient's signs and symptoms and fluid intake and output are essential. In addition to fluid imbalances, patients should be monitored for possible electrolyte disturbances. The most common electrolyte imbalances occurring in gastroenterology patients are excesses or deficits of chloride, magnesium, sodium, potassium, bicarbonate, calcium, or hydrogen ions.

Systematic observations to detect fluid and electrolyte imbalances include changes in temperature, pulse rate, respirations, and blood pressure. Changes in skin and mucous membranes, and/or changes in speech, behavior, facial appearance, skeletal muscle, sensations, fatigue, and body weight are also significant. Desire for food, anorexia, and thirst are important signs in detecting imbalances. Electrolyte imbalance may contribute to cardiovascular changes and arrhythmias. Monitoring urine output for the specific gravity, pH, volume, and character is important. Significant observations should be relayed to the physician to facilitate early diagnosis and treatment before serious imbalances occur.

ADMINISTRATION OF FLUIDS AND ELECTROLYTES

One way of correcting fluid and electrolyte disturbances is by IV administration of solutions containing the necessary electrolytes and nutrients. All IV solutions are considered medications, and their infusion requires a physician's order. Physician's assistants and advanced practice nurses who are authorized by the organization to prescribe treatments and medications may also write orders for IV solutions. The order should include the name of the solution and/or medication, volume or dose, rate, frequency, and route of administration. Verbal consent of the patient or a legally authorized representative and the patient's identity must be confirmed before initiation of IV therapy.

Considerations when selecting IV fluids and electrolytes

The **tonicity** of body fluids refers to the effective osmolality (concentration of solute; the total number of solute particles per liter) of a solution. While tonicity and osmolality are related, they are not interchangeable as tonicity also takes into consideration the ability of the solute to cross the cell membrane. When the human body is in balance, the intracellular and extracellular fluids maintain the same osmolality. IV fluid administration is based on the hydration status of the patient's intracellular and extracellular space.

Several types of IV fluids are available for administration:

- **Hypotonic.** Hypotonic solutions have a lower concentration of solutes and are more dilute than body fluids, causing water to pull into the cells. An example of a hypotonic solution is 0.45% NaCl.
- **Isotonic.** Isotonic solutions have the same concentration of solutes (285 mOsm/kg) as the intracellular or extracellular space. Examples of isotonic solutions include **normal saline** solution (0.9% NaCl), dextrose 5% in water (D5W) solution, and **Lactated Ringer's (LR) solution**. Multiple formulations of a balanced crystalloid solution that closely resemble blood plasma (PlasmaLyte) are also isotonic and may be used in some preoperative units.
- **Hypertonic.** Hypertonic solutions contain a higher concentration of solutes (greater than 285 mOsm/kg) than the intracellular space and cause water to be pulled from the cells. Examples of hypertonic solutions include 3% NaCl or 10% dextrose.

IV fluids may be administered for maintenance or replacement purposes.

- **Maintenance therapy** meets the patient's standard needs by providing approximately 3,000 ml of fluid per 24 hours, along with added electrolytes, nutrients, and vitamins. For maintenance therapy, a balanced hypotonic electrolyte solution with the addition of 5% dextrose for calories is ideal. This balanced electrolyte solution typically contains sodium, potassium, magnesium, chloride, and acetate. These solutions are available in concentrations appropriate for pediatric and adult patients.
- **Replacement therapy** restores lost fluids on a volume-to-volume basis, often in excess of 3,000 ml of fluid per 24 hours. The electrolyte concentration of replacement therapy is generally equal to the concentration of electrolytes in the extracellular fluid. Ideally, replacement therapy solutions should contain ions in the same composition as that of the plasma and interstitial fluid. Depending on the caloric and fluid needs of the patient, 5% dextrose may or may not be included. In patients with preserved renal function who have a nothing by mouth (NPO) status, replacement potassium should be provided in IV fluids to prevent hypokalemia, as the kidneys routinely excrete potassium.

IV therapy administration risks for the gastroenterology nurse

Primary risks to the gastroenterology nurse are injury and disease transmission due to needlestick injury. Spills and splashes of contaminated solutions or blood and body fluids also pose a threat. Because universal precautions require the use of gloves when dealing with blood and body fluids, development of latex allergies pose concern for health care workers and patients.

Needlesticks

In November 2000, the United States introduced groundbreaking legislation that resulted in the Needlestick Safety and Prevention Act, also known as Public Law 106-430 (Occupational Safety and Health Administration [OSHA], 2000). Effective April 2001, the Act requires health care facilities to select and implement safer medical devices such as sharps with engineered sharps injury protections and needleless systems to protect health care workers from needlestick injuries.

Diseases of greatest concern for transmission to the gastroenterology nurse through exposure to the blood of an infected patient include viral hepatitis B and C and human immunodeficiency virus (HIV). The risk of all infections increases if the needlestick injury involves deep penetration with a blood-filled hollow bore needle such as those used to establish venous access.

Adherence to universal precautions has had major impact on decreasing the risks of needlestick injuries over the past 30 years. Although these injuries still occur, the incidence is at a much lower rate. The risk of acquiring infection from a single needlestick is 6% to 30% for hepatitis B, 1.8% for hepatitis C, and 0.3% for HIV (Centers for Disease Control and Prevention [CDC], 2008).

There are a number of sharps used in IV therapy such as intravenous, injection, and suture needles; **cannulas** and introducers; scalpels; glass tubes; and glass medication vials, ampules, or bottles. As a result of the Act, a variety of needleless or protected needle devices have been developed to limit sharps exposure in the workplace.

Latex allergy

Latex allergies remain an issue for health care workers. Reports suggest that the average prevalence of latex allergy worldwide remains 9.7% among health care workers and 7.2% among susceptible patients. The development of alternative materials for latex and increased efforts to identify and label latex-derived products have proven to be effective approaches to control the health risks associated with latex.

Symptoms of latex sensitivity can range from irritant contact dermatitis to anaphylaxis. The most commonly seen reaction is irritant contact dermatitis, which is exhibited by the development of dry, itchy, irritated areas on the skin. More severe reactions include hives, itching, rhinorrhea, itchy eyes, scratchy throat, and asthma symptoms. The risk of allergic reaction from gloves increases as a result of the modified starch powder used in some gloves. When using latex gloves, a powder-free product should be used to decrease exposure to allergy-causing antigens (SGNA, 2014).

Vascular access devices

IV therapy can be initiated by a variety of devices. A new, sterile VAD should be used for each catheterization attempt (Infusion Nurses Society, 2016). Peripheral IV therapy can be delivered via a peripheral short IV catheter or **needle** via a peripheral long (midline) **catheter**. Central venous access is obtained for individuals receiving long-term IV therapy or in those who have poor peripheral venous access. Central venous access devices (CVADs) include non-tunneled central venous catheters, tunneled central venous catheters, and implanted devices or ports.

Types of VADs include the following:

Steel, Winged Infusion Sets

Steel, winged infusion sets (butterfly-winged devices or butterfly needles) are used for peripheral short IV access and should be limited to short-term or single-dose administration. These metal needle devices pose a higher risk of needlestick injury for the health care worker, as well as a greater incidence of dislodgement and infiltration. Use of a steel, winged infusion set necessitates the use of an armboard or device to immobilize the involved extremity. Steel, winged infusion sets are used for short-term therapy in cooperative adult patients, for therapy in infants and children, or in elderly patients with veins that are fragile or sclerotic (hardened and thickened).

Peripheral Short IV Catheters

Peripheral short IV catheters are made of radiopaque Teflon, Silastic, or polyvinyl chloride. These radiopaque devices appear light or white on X-ray films. Over-the-needle catheters are routinely used for IV infusion or IV medication administration and are commonly called angiocatheters (angiocath). They are composed of a metal stylet that is used to pierce the skin, with a catheter cover that is threaded into the vein. The catheter remains for the instillation of fluid or medications. The flexible catheters are associated with lower infection rates and should be used for routine peripheral IV therapy. In adults, a 16- or 18-gauge angiocath is indicated for major surgery; an 18- or 20-gauge for rapid infusion of IV fluids, blood components, or viscous medications; a 20-gauge for most IV applications and for blood products; and a 22- or 24-gauge with a three-quarter-inch catheter for elderly patients.

Peripheral Long or Midline IV Catheters

Peripheral long or midline IV catheters are typically placed in the basilic, cephalic, or median cubital veins of the upper arm or antecubital fossa, with the tip resting in the cephalic, basilic, or brachial vein, distal to the shoulder but not entering the central vein. The basilic vein has a larger vein diameter and is the preferred placement site. The midline catheter is indicated for peripheral IV therapy prescribed for a duration of 1-4 weeks (Infusion Nurses Society [INS], 2016). The advantages of the midline catheter over the peripheral short IV catheter are increased drug hemodilution and improved catheter dwell time. The risk of **phlebitis** and **infiltration** are reduced with the midline catheter due to the increased blood flow dilution of an administered drug's pH and osmolarity. Midline catheters must be placed only by health care professionals who have completed a training program, finished the supervised practicum, and have demonstrated competency. Midline catheters may be open-ended, which require heparinization when not in use, or closed-ended, which require only saline flushing to maintain patency when not in use (INS, 2016).

Peripherally Inserted Central Catheters (PICC)

Peripherally inserted central catheters (PICC) are typically placed with the distal tip dwelling in the lower one-third of the **superior vena cava** via access in the basilic or cephalic vein in the antecubital fossa. A PICC line is typically placed in patients when IV therapy needs are expected to last up to a year. These catheters must be placed only by health care professionals who have completed both a training program and the supervised practicum and have demonstrated competency. Line placement must be confirmed by chest X-ray prior to infusing fluids. Flushing to maintain patency when PICC lines are "not in use" or "locked off" is heparin flushing for open-ended catheters and saline flushing for closed-ended catheters (INS, 2016).

Non-tunneled Central Venous Catheters (CVC)

Non-tunneled central venous catheters (CVC) are placed in the jugular, subclavian (preferred site), or femoral central venous sites under aseptic technique, most commonly by a physician or advanced practice nurse. The tip of the CVC dwells in the lower one-third of the superior vena cava. X-ray confirmation of placement must be obtained prior to using the device for IV therapy. These devices may have a single, double, triple, or quadruple lumen for IV access. These devices can be open-ended, which require heparinization when not in use, or closed-ended which requires saline flushing only.

Tunneled Central Venous Catheters

Tunneled central venous catheters are most commonly referred to as Groshong catheters. The tip resides in the lower one-third of the superior vena cava. These devices have a closed end and do not require heparinization when not in use. However, Groshong catheters do require sterile normal saline flushes every 7 days when not in use. These devices are placed in an operating room or vascular procedural setting and can be used for several years.

Totally Implanted Devices or Ports

Totally implanted devices or ports are placed in the operating room or vascular procedural setting and can be used for several years. The access port of this central line may be located in the upper chest wall or in the brachial area and may be a single or a double lumen port. The portal septum is palpated through the skin. The port is accessed using a special non-coring needle inserted into the center of the portal septum. These devices are available with open ends or closed-valve ends.

Equipment

The method of IV administration and choice of equipment depends on the patient's condition and desired effects of therapy. The three main methods for delivering fluids and/or drugs in the gastroenterology setting are continuous infusion, intermittent infusion, and intermittent injection.

The equipment needed for IV therapy generally includes the venous access device, IV fluid containers, tubing, filters, tourniquets, tape, antimicrobial agents, gloves, IV poles/stands, clamps, and electronic pumps or controllers.

- **IV fluid containers** should be checked before use for cracks or tears, foreign matter, cloudiness, precipitation (settling out of solid particles), any other signs of contamination. The expiration date should be verified for currency.
- **IV tubing** may be a regular (macrodrip) solution administration set that delivers 10-20 drops per ml, or a microdrip set that delivers 60 drops per ml. Tubing with a secondary injection port permits separate or simultaneous infusion of two solutions. This tubing, with a piggyback port and a back-check valve, permits intermittent infusion of a secondary solution. Vented tubing is used for solutions contained in a non-vented bottle. Non-vented tubing is used for solutions in vented bottles or containers. A reduced-volume chamber (e.g., Buretrol) may be used in pediatric patients to prevent accidental fluid overload and to provide greater accuracy in the rate of delivery of fluids.

- **Intermittent infusion sets** consist of a flexible catheter with tubing ending in a resealable rubber injection port (peripheral lock). The device is flushed with saline to prevent blood clot formation. Peripheral locks are used both to provide immediate access in case IV therapy is needed during a procedure and to maintain venous access in patients who are receiving IV medication regularly or intermittently but do not require continuous infusion of fluids.
- **Filters** are available in several different types for use in IV therapy. Bacteria- and particulate-retentive, air-eliminating filters of 0.2 micron are used for central venous access lines to decrease the complications of infection and air embolism. Particulate matter filters of 1 or 5 micron are used to remove particulate matter in situations where bacteria- and particulate-retentive filters are contraindicated. Blood filters range from microaggregate filters that are 20-40 micron to standard 170 micron. The use of filters is directed by manufacturer's instructions for use and is based on organizational policy, as well as filtration requirements of the therapy.
- **Tourniquets** are used for peripheral access and are applied above the intended insertion site to distend a vein with a larger-than-average amount of blood. Venous distention may also be enhanced by application of warm, moist heat; light tapping or massage of the skin over the veins; or having the patient open and close a fist several times. The tourniquet should impede venous, but not arterial, flow. This can be assessed by palpating for a radial pulse after applying the tourniquet. Tourniquets should be a single-use device routinely discarded after every procedure. To avoid circulatory impairment, they should be applied only for the short time needed to perform the venipuncture.
- **Transparent semipermeable membrane dressings** are frequently used as a VAD dressing. Consider the patient's age, skin turgor and integrity, and any previous adhesive injury to the skin when deciding on the most suitable method for VAD stabilization. A gauze dressing is preferable if the patient is diaphoretic or if there is oozing from the access site. Use of a catheter stabilization device is preferred over tape for device stabilization, when feasible, as there is a demonstrated reduction in overall site complications and improved dwell time. Hypoallergenic tape is used for patients who have tape allergies. Tape or medical wrap is used to secure armboard devices or loops of tubing. Roller bandages obstruct visualization of the IV site and may impair circulatory flow. They are not recommended on an extremity where an IV cannula is placed (INS, 2016).
- **Antiseptic agents** are used based on organizational policy to cleanse the IV site before venipuncture. Antiseptic formulations containing 2% chlorhexidine gluconate are recommended for pre- and post-insertion site catheter care (CDC, 2017). In patients with an allergy to chlorhexidine gluconate or who are under 2 months of age, alcohol followed by povidone iodine is preferred. Alcohol should not be applied after the application of povidone iodine solution. The antiseptic preparation should be allowed to air dry completely before site puncture (INS, 2016).
- **IV stands/poles** may be portable or may be attached to the bed or wall. To achieve the maximum flow rate, the infusion tubing drip chamber should be suspended approximately 3 feet above the access site.
- **Infusion controllers** can include mechanical and electronic devices. Mechanical flow-control clamps, including roller, screw, and slide clamps, are suitable for the regulation of most infusions. Electronic infusion pumps or controllers should, however, be used when warranted by the patient's age and condition, the setting in which the therapy is delivered, and the prescribed therapy. Controllers regulate gravity flow either by counting drops or by measuring volume. Infusion pumps generate flow under positive pressure and operate independently of gravity flow. An infusion pump is frequently used to administer medications via the continuous infusion route in order to maintain a prescribed drug delivery. Infusion pumps may be used in pediatric patients to prevent accidental fluid overload and to provide greater accuracy in the rate of delivery of fluids.

Patient-controlled analgesia (PCA) devices may be used to deliver IV analgesic agents. When these devices are used, patients must be educated as to the purpose of the PCA therapy, operating instructions, expected outcomes, precautions, and potential side effects. PCA may be delivered as a combination of continuous and patient demand doses or as patient demand only. Medications administered through PCA devices are controlled substances and must be obtained, delivered, administered, documented, and discarded in accordance with state and federal regulations.

Insertion sites

Selection of a peripheral insertion site for IV therapy depends on the type of solution; the type, frequency, and duration of therapy; the patient's age, size, diagnosis, and condition; and the patency, size, and location of available veins. The vein chosen must accommodate the size of the catheter. When a saline or peripheral lock is used to administer conscious sedation during endoscopic procedures, it is usually placed in a large vein in the arm. A large vein is more appropriate because the medications used to produce sedation can irritate the vessel wall.

When choosing an insertion site, it is important to distinguish between veins and arteries. If a vessel pulsates, it's an artery. Puncture and the inadvertent injection of certain medications into arteries can lead to complications such as inflammation, necrosis, sloughing, and even gangrene and loss of function of the affected part.

To avoid injury to the radial nerve, avoid the area approximately 3 inches above the crease of the wrist, above the thumb, and on the inner aspect of the wrist.

Peripheral IV infusions are typically inserted in a superficial vein on the nondominant arm. Preferred sites for peripheral venous access include upper extremity veins in adults, including the superficial dorsal and metacarpal veins on the dorsum of the hand and cephalic, basilic, and median veins on the lower and upper arm. Avoid extremities that have impaired circulation or injury such as lymphedema, postoperative swelling, recent trauma, a dialysis shunt, a hematoma, axillary lymph node dissection, local infection or cellulitis, phlebitis, or open wounds. Avoid sites distal to previous venipuncture sites, sclerosed or hard veins, phlebotic vessels, areas of previous IV infiltration, and areas of venous valves or bifurcation. It is best to begin with distal veins and then proceed proximally towards the body.

Cannulation of the lower extremities should be avoided in adults because it increases the risk of thrombophlebitis and embolism (INS, 2016). If an IV cannula must be placed in a lower extremity, a physician's order should be obtained, and the IV site should be changed as soon as an alternative site can be established.

In infants and children, lower extremities may be used safely and effectively. These sites generally allow for access to the IV site even when the child has been "bundled" to provide restraint. Scalp veins are acceptable IV sites for infants.

Insertion technique

Meticulous hand hygiene must be performed before initiating IV therapy. No more than two attempts at IV insertion should be made per nurse (INS, 2016).

Prior to insertion, a visibly soiled site should be cleansed first with soap and water. Hair should be removed using scissors or an electric clipper, if necessary (INS, 2016).

Steps for insertion include:

1. Prepping the selected site with the organizationally approved antiseptic agent and permit it to air dry.
2. Using topical anesthetics to numb the insertion site, especially in children, according to organizational policies and procedures (INS, 2016).
3. Applying a tourniquet above the intended insertion site.
4. Using the smallest gauge and length of angiocatheter capable of accommodating the prescribed treatment (INS, 2016).
5. For peripheral insertion, holding the skin taut and anchoring the vein with the thumb below the injection site. Insert the needle slowly into the vein until blood return is noted, then advance the needle or catheter, and release the tourniquet.
6. Connecting the primed tubing and running the fluid at a fast rate to flush blood from the tubing. Once blood is cleared, regulate the IV drip at the prescribed rate.
7. Stabilizing the cannula so it does not interfere with assessment and monitoring of the IV site or impede delivery of the prescribed therapy.
8. Applying a sterile gauze or transparent, semipermeable membrane dressing. All edges of the dressing should be securely taped. An armboard or handboard may be used if the cannula is placed close to an area of flexion.
9. Looping a 3- to 6-inch portion of tubing and securing it near the insertion site. The loop allows some slack to prevent dislodgement of the catheter caused by tension on the line.
10. Documenting all pertinent information.

Monitoring and assessment

Frequent monitoring and assessment of patients receiving IV therapy minimizes the risk of potential complications. Monitoring and assessment should include observation of the insertion site, flow rate, clinical data, and patient response to the prescribed therapy.

IV therapy should be discontinued on the order of a physician; on completion of therapy; for needle or catheter changes; when infection, phlebitis, or infiltration is suspected; or in accordance with organizational guidelines. The patient and/or a legally authorized representative also has the right to request discontinuation of treatment.

To remove a peripheral line:

- Clamp the tubing.
- Remove the securement device from the skin.
- Slowly and smoothly withdraw the needle or catheter.
- Maintain pressure with a gauze pad at the insertion site after withdrawal of the VAD until bleeding stops.
- Apply a dry, sterile bandage.
- Inspect the removed VAD for appropriate length and any defects.

Complications of intravenous therapy

IV therapy is an invasive procedure that is not without risks. Because the IV system provides direct access into the vascular system, stringent infection-control measures must be applied. Hand hygiene must be performed before and after all clinical procedures (INS, 2016). If an IV-related infection is suspected, the catheter, any purulent drainage, and the infusate should be cultured to identify the responsible microorganism. If an IV site or line infection is suspected, the IV set and cannula is sent for culture and laboratory examination. Blood cultures may be considered to determine the extent of the infection.

Other potential local, systemic, and mechanical complications associated with IV therapy include:

Infiltration or extravasation—the inadvertent administration of IV solution or medication into surrounding tissue, with escape of fluid from its physiologic space.

Hematoma—a swelling of tissue with blood caused by a break in a blood vessel.

Phlebitis—inflammation of a vein, marked by redness and irritation of the vein, with or without thrombus formation.

Pyrogenic reactions—conditions causing fever, including septicemia (blood poisoning) and bacteremia (presence of bacteria in the blood).

Air embolism—obstruction of a blood vessel caused by inadvertent air introduction into the IV line and vascular system.

Catheter embolism—catheter fragments in the vascular system.

Pulmonary edema—a potentially life-threatening accumulation of fluid in the lungs resulting from fluid overload.

Speed shock or overload—a systemic reaction that occurs when a foreign substance is introduced too rapidly into the circulation.

Nerve injury—injury to a nerve during venipuncture, usually to the distal portion of the radial nerve, just above the thumb.

Complications of IV therapy with respective causes and signs and symptoms, along with appropriate interventions are listed in Table 21.3. Other potential risks for patients are drug incompatibilities and medication errors.

Dosage calculations

The rate of flow of IV fluids can be affected by many factors such as height of the fluid container; bent or kinked tubing; use of filters; medication additions; needle or catheter size, position, or occlusion; or infiltration of IV fluid into the surrounding tissue. Patient movement or manipulation of the clamp may also be a factor.

It is important to assess the flow rate, volume infused, and patient status every 1-2 hours on adults and at least hourly on infants and children. These assessments should be done according to organization and medication guidelines, dependent on the patient and the therapy that is being administered.

Reduced-volume chamber devices are useful to limit the amount of fluid that can be infused between nursing assessments and are commonly used on infants and small children. IV infusion devices can be used to administer set volumes of medication or fluid over a prescribed time frame and are commonly used on infants, children, and elderly patients.

Calculated infusion flow rates are only guidelines. Maintaining the calculated rate of flow does not relieve gastroenterology nurses of the responsibility for observing each individual patient for signs that the infusion is too rapid or too slow. Observation of patients for indications of infusion rate intolerance is imperative.

Table 21.3. Potential Complications of Intravenous Therapy

Complication	Possible Causes	Signs and Symptoms	Intervention
Infiltration	• Puncture of the vein wall • Needle or cannula slipping out of the vein	• Coolness, blanching, edema, pain, burning • No blood return • Leakage at the venipuncture site	• Discontinue infusion • Remove cannula • Restart IV at a different, more proximal location • Elevate the affected part • Apply heat or cold compresses based on infiltrated solution
Extravasation	• Administration of drugs into tissue	• Redness along with blistering • Tissue necrosis and ulceration • Burning, stinging pain, swelling, leakage	• Requires immediate physician notification • Do not pull IV catheter—treatment should be determined prior to catheter removal, as some vesicants require drugs to be infused directly into the extravasation site via the IV

Complication	Possible Causes	Signs and Symptoms	Intervention
Hematoma	• Unsuccessful attempt at venipuncture • Infiltration of a blood transfusion	• Painful, raised area, with blue or purplish coloring	• Remove needle or cannula • Apply pressure and cold compresses
Phlebitis, with or without clot formation	• Vein irritation and inflammation • May occur post-infusion	• Redness with or without edema along the affected vein • Sore, hard, cordlike, warm vein	• Discontinue infusion • Remove cannula • Apply warm, moist compresses • Notify the physician
Pyrogenic reactions	• Contaminated equipment or solutions	• Fever, chills, nausea and vomiting, backache, malaise	• Discontinue infusion • Record lot number of solution • Culture cannula and solution • Notify the physician
Air embolism	• Air in tubing • Loose connections allowing air to enter tubing	• Hypotension • Cyanosis • Heart murmur • Tachycardia • Syncope • Vascular collapse • Loss of consciousness	• Place the patient in the left lateral decubitus position • Notify the physician • Check the system for leaks and clamp line proximal to any leaks or breaks noted
Catheter embolism	• Portion of the plastic cannula breaks off and flows into the vascular system • Attempting to rethread the catheter with a needle • Unsecured catheter	• Vein discomfort • Cyanosis • Decreased blood pressure • Weak, rapid pulse • Loss of consciousness	• Discontinue infusion • Apply tourniquet above insertion site • Notify the physician • Per physician order, X-ray film is taken to locate fragment
Pulmonary edema	• Circulatory overload • Excessive infusion flow rate	• Headache • Venous dilation • Hypertension • Coughing • Dyspnea • Tachycardia • Coarse crackles auscultated in lung fields	• Stop infusion until patient can be assessed • Elevate the head of the bed and lower the patient's legs to dependent position • Notify the physician
Speed shock/ overload	• Too-rapid administration of solutions and medications	• Shock • Syncope • Cardiac arrest	• Slow infusion to keep-open rate • Resuscitate • Notify the physician
Nerve injury	• Puncture of the nerve during venipuncture	• Immediate "electric shock" sensation radiating to the thumb and index finger when the catheter is inserted • Can result in complex regional pain syndrome	• Immediately remove angiocath • Place dry, sterile dressing.

(Camp-Sorrell, 2010; INS, 2016; Perry & Potter, 2018)

Patient education prior to IV therapy

The establishment of a holistic rapport and an effective system of compassionate communication between the gastroenterology nurse and the patient is essential. It is important to explain the procedure to ensure cooperation and reduce anxiety, which can cause a vasomotor response resulting in venous constriction. IV therapy can only be initiated safely after the patient has exhibited signs of understanding and acceptance.

Before starting a continuous infusion, patient education should cover:

- The purpose of the therapy.
- An estimate of the duration of therapy.
- Patient mobility.
- Avoiding pressure to the IV site.
- Keeping the site dry.
- Maintaining the fluid container at an appropriate height.
- Not adjusting the flow rate (for optimal patient safety).
- Instructions to notify the gastroenterology nurse in the case of pain, burning, redness, or swelling at the IV site or other complications or concerns.

Patient education and understanding is documented in the medical record in accordance with the organization's standards and policies and procedures.

Insertion documentation

Documentation of IV therapy should be legible and accessible to all health care professionals involved in the patient's care. Documentation of any venipuncture should include:

- Date and time of initiation.
- Amount and type of solution used.
- Type and length of needle or catheter, and its gauge.
- Number of venipuncture attempts.
- Location of the venipuncture site.
- Type of dressing and any stabilization device used.
- Infusion devices, if used.
- Rate of flow recorded in milliliters per hour (ml/hr).
- Any complications, anxiety, or untoward reactions on the part of the patient, along with nursing interventions taken (INS, 2016).
- Patency during therapy.
- Site condition during infusion therapy and following removal of the VAD.

INTRAVENOUS MEDICATION ADMINISTRATION

IV administration of medications has a number of advantages. Some drugs, including antibiotics and opioids, are frequently given by the IV route to provoke a quick, continuous, therapeutic response. Other drugs may be given by the IV route because they are ineffective or dangerous by other routes. Some drugs are contraindicated for IV administration, including certain nonaqueous or suspension medications, because they obstruct blood flow.

Techniques and routes of administration

Hand hygiene must be performed before and after all clinical procedures involving IV therapy (INS, 2016). Aseptic technique should be used for admixing. The injection port site must be cleansed with an organizationally approved antiseptic agent prior to accessing (INS, 2016).

Prior to administration

Before medication administration, the gastroenterology nurse must verify and assess:

- The physician's or other provider's written order for the IV medication.
- The dose, route, and rate of the medication ordered.
- The label on the medication against the order on the patient's medication record and against the physician's order.
- The appropriateness of the prescribed therapy.
- The patient's age and condition.
- Any patient medication allergies.
- The medication's indications, actions, side effects, and the appropriate nursing interventions in the event of adverse reactions.
- The drug compatibility with the solution before adding any drug to an IV solution. The physical and chemical compatibility of the delivery systems used must be confirmed. Incompatibilities between two or more drugs cause a chemical reaction and can result in clouding or crystallization of IV fluid. Mixing of drugs in a continuous infusion should be done only after consultation with incompatibility lists and with the pharmacist.
- The expiration date of solutions and medications.
- The patient's identity.

The medication should not be administered if any discrepancies or concerns are found. Any discrepancies or concerns need to be brought to the physician's attention for clarification.

The addition of a drug to a blood transfusion complicates identification of the source of any adverse reactions and should never be performed.

Drug administration methods

Various methods can be used for IV drug administration, including the addition of drugs to the IV solution (continuous infusion), infusion through a secondary line (piggyback or add-a-line method), use of an intermittent infusion injection device (saline lock), and IV push or IV bolus injection.

- **Continuous infusions** are diluted in a large quantity of fluid, from 250 to 1000 ml, and delivered over a 2-hour to 24-hour period. The continuous infusion is typically a premixed solution, or a medication is added to the infusion container using a needle and syringe. After adding medications to IV containers, it is important to gently rotate the bottle or bag to mix the solution thoroughly to allow equal distribution of the drug and maintain the integrity of the proteins. The solution should be observed after the medication is added for any precipitation, discoloration, or cloudiness. Once the medication has been added to the IV fluid container, a label documenting the drug name, drug amount, date, time, and the gastroenterology nurse's initials should be placed on the IV container. Most IV admixtures are routinely performed in the pharmacy department.

- **Piggyback method** uses a moderate quantity of fluid (usually 50-100 ml) administered over a 5-minute to 2-hour period. This method uses a separate small-volume IV fluid container that is attached either through the top of the drip chamber for the primary container or through a Y-connector on the tubing of the main line using a secondary infusion tubing. This secondary container must be elevated above the level of the main line IV container. If the drug to be added through the piggyback set is incompatible with the primary IV solution, the line must be flushed with 0.9% NaCl before starting the drug infusion.
- **Intermittent infusion sets** (peripheral locks) may be used for patients who are receiving IV medications regularly, either by the push method or intermittently, but who do not require continuous infusion of fluids. After insertion, saline (and/or heparin, depending on type of VAD) is injected every 6-12 hours to maintain the patency of the infusion set. For VADs that require heparinization, the VAD is flushed with saline prior to drug administration, then the medication is injected in the time recommended, followed by an injection of saline to flush the medication, then a heparin lock flush per organization policies and procedures.
- **IV push or IV bolus route** introduces a concentrated dose of a drug directly into the systemic circulation. IV push medications may or may not be diluted in 0.25-50 ml of fluid. They are administered over a period based on the manufacturer recommendations or drug reference materials by direct venipuncture or by using an in-progress infusion setup or a peripheral lock. Medications should be timed as they are administered, as exceedingly rapid administration can produce speed shock. If administered through an in-progress setup, the IV tubing above the injection site should be clamped during injection so the medication does not flow up to the bottle or bag. The IV fluid should be permitted to flow rapidly for 30-60 seconds after injection to move the medication through the tubing. The IV push method is often used in endoscopic procedures to administer conscious sedation medications. It also may be used to provide an immediate drug effect in an emergency, to achieve peak drug levels in the bloodstream, or to deliver drugs that cannot be diluted or administered intramuscularly.

Indications for intravenous medication in gastroenterology

IV medications may be administered to gastroenterology patients for the following purposes:

- Moderate sedation/analgesia or monitored anesthesia care (MAC), for example diazepam (Valium), meperidine (Demerol), midazolam (Versed), fentanyl (Sublimaze), or propofol (Diprivan).
- Treatment of opioid-induced respiratory depression with naloxone (Narcan).
- Treatment of benzodiazepine-induced respiratory depression with flumazenil (Romazicon).
- Treatment of cardiac dysrhythmias, for example atropine for bradycardia or adenosine for supraventricular tachycardia.
- Reducing peristalsis or intestinal spasms, for example glucagon.
- Treatment of nausea, for example ondansetron (Zofran) or proclorperazine (Compazine).
- Use of adjunct medicines for synergistic effect, for example diphenhydramine (Benadryl) given in combination with narcotics and benzodiazepines to enhance sedation.
- Biologic therapy such as infliximab (Remicade), vedolizumab (Entyvio), and ustekinumab (Stelara) in the treatment of ulcerative colitis and Crohn's disease.

Clinically significant infections (bacteremia) are rare after endoscopic procedures. All cirrhotic patients with GI bleeding should have IV antibiotic therapy initiated on admission (ASGE, 2015). Esophageal dilation, sclerotherapy of varices, and instrumentation of obstructed bile ducts have the highest reported rates of bacteremia. Endoscopic retrograde cholangiopancreatography (ERCP) in patients with a non-obstructed bile duct has a 6.4% rate of bacteremia, which rises to 18% in the setting of biliary tree obstruction (ASGE, 2015). Peri-procedural antibiotic prophylaxis is recommended in the following cases:

- ERCP in those with bile duct obstruction.
- Endoscopic ultrasound fine-needle aspiration (EUS-FNA) of a cystic lesion along the GI tract.
- All patients undergoing percutaneous endoscopic feeding tube placement.

Routine antibiotic prophylaxis solely to prevent infectious endocarditis, infection of prosthetic joints, or infection of synthetic vascular grafts or other nonvascular cardiovascular devices is no longer recommended prior to endoscopic procedures (ASGE, 2015).

Adverse reactions

After administering any drug, it is important to observe the patient for signs of drug sensitivity or intolerance. If adverse effects occur, the medication is discontinued, and the physician is notified.

Infections and vein irritation

Appropriate steps to reduce the risk of infection include meticulous hand hygiene. To reduce vein irritation, the following steps may be taken:

1. Use the smallest gauge needle possible for IV push medications.
2. Consider diluting drugs in 10-20 ml of saline to decrease irritation.
3. When possible, use normal saline as the IV fluid or diluent.
4. Use at least 50-100 ml of diluent with all antibiotics.

Complications specific to the method of administration

Certain complications of IV drug administration are specific to the method of administration:

- Continuous infusion complications can result from excessively high drug concentration in the IV solution. These complications can include sclerosis, thrombosis (clot formation), **hemolysis**, or phlebitis.
- Piggyback infusion side effects and reactions to the infused drug can occur. Repeated punctures of the secondary injection port can cause an imperfect seal, with possible leakage or contamination.
- The most common complications with intermittent infusion devices are infiltration and a specific reaction to the infused drug.
- IV push injection effects are often immediate, and signs of an acute allergic reaction or anaphylaxis can develop rapidly. If signs of anaphylaxis occur (i.e., dyspnea, cyanosis, convulsions, or increasing respiratory distress), the physician should be notified immediately, and emergency measures should be instituted as necessary.

IV infiltration and extravasation

Infiltration is the leakage of infused solution from a vein into surrounding tissue as a result of a needle puncturing a vascular wall or leakage around the venipuncture site. It causes local pain and itching, edema, blanching, and decreased skin temperature in the affected extremity. Extravasation is the infiltration of some drugs that can severely damage tissue through vesicant, sclerotic, corrosive, or vasoconstrictive actions.

Treatment of infiltrations of IV solutions and nonirritating drugs involves such routine comfort measures as application of warm compress. The heat from the compress will alleviate discomfort and increase circulation to aid in the absorption of the isotonic solution. Extravasation of corrosive drugs requires emergency treatment to prevent tissue necrosis. If signs of infiltration occur such as swelling, the absence of blood backflow, and/or a sluggish flow rate, the injection should be stopped, the amount of infiltration estimated, and the physician notified. The infiltrated IV line should be left in place until post-infiltration treatment has been ordered. Antidotes to specific infiltrated drugs may be administered through the existing IV site.

Administration documentation

Documentation requirements for IV medication infusions are similar to those that apply when administering fluids and electrolytes, including the following:

- For patients receiving IV admixture additives, document the medication added, the infusion rate, and who added the medication.
- In patients receiving IV bolus medications, document the drug and dose administered, the time, and the patient's response.
- For patients who arrive in an endoscopy unit with a line in place, document the condition of the IV.

All unusual occurrences should be reported to the physician immediately. Written reports of adverse effects include all signs, symptoms, treatments, and outcomes. Indicate the location of pain, infiltration, phlebitis, or other complications.

In the case of infiltration or extravasation, document the site of the infiltration, the patient's symptoms, the estimated amount of infiltrated solution, the nursing treatment, the time, and the name of the physician notified. Continue to document the appearance of the infiltrated site and any associated symptoms until resolution.

BLOOD AND BLOOD COMPONENTS

The United States blood supply is safer than ever before, although the transmission of some bacteria, viruses, prions, and parasites can still occur via blood transfusions (CDC, 2019). The improvement in processing methods for blood products has reduced the number of infections resulting from these products. The estimated risk of a transfusion-transmitted disease is low, with the risk of HIV transmission being 1 in 1,467,000; hepatitis B 1 in 765,000 - 1,006,000; and hepatitis C 1 in 1,149,000 (American Red Cross, 2017).

The more common complications associated with blood transfusion include allergic and febrile reactions. Proper blood administration guidelines and crossmatch compatibility protocols can greatly reduce the risk of a hemolytic reaction caused by blood type of Rh incompatibility. A 17% fatality risk exists for patients transfused with ≥50 mL ABO-incompatible blood (Janatpour et al., 2008).

Federal, state, and organization regulations require registered nurses to receive education and demonstrate competency before administering blood components. Blood therapy administration requires:

- Obtaining blood samples for type and crossmatching.
- Ordering blood components.
- Checking the blood product with patient identification before administration.
- Initiating blood component administration.
- Regulating blood component infusion rate.

- Observing and assessing the patient for adverse reactions during and following administration.
- Discontinuing blood components.
- Documenting the therapy provided.
- Documenting any adverse reactions or complications during administration of blood component.

Types of blood component administration

Blood therapy is generally administered as fractionated blood components, as opposed to whole blood, to avoid circulatory overload. Whole blood can be fractionated into red blood cells (**erythrocytes**), platelet suspensions and concentrates, white blood cells (**leukocytes**), fresh frozen plasma (FFP), cryoprecipitates or factor VIII/IX concentrates, fibrinogen, and albumin.

Whole Blood

Whole blood contains red and white blood cells, **serum**, platelets, proteins (albumin, globulin, fibrinogen), and other intravascular nutrients and substances. While rarely used for transfusion and not readily available, whole blood is useful in actively hemorrhaging patients.

Packed Red Blood Cells (RBCs)

Packed red blood cells (RBCs) are indicated for the treatment of patients who require an increase in oxygen carrying capacity and/or red cell mass. RBCs raise the hemoglobin and hematocrit faster than whole blood with less risk of circulatory overload. Packed RBCs are used in patients with severe anemia or in patients whose bleeding has ceased and whose vascular volume has been replenished with saline or Lactated Ringer's solution. RBCs are used cautiously in patients with underlying cardiac disease or renal failure. RBCs may be frozen and then thawed for autologous transfusions.

Leukocyte-poor Blood

Leukocyte-poor blood is produced by removing leukocytes and platelets from whole blood. Leukocyte reduction may lower the risk of febrile nonhemolytic reactions, cytomegalovirus (CMV) transmission, and HLA alloimmunization, which may lead to patients becoming refractory to platelet transfusions. Leukocyte filtration may be performed in the blood bank or via filtration at the time of transfusion.

Platelets

Platelets are administered to patients with thrombocytopenia (a decreased number of platelets), usually with a platelet count of 10,000 microliters or less. Platelets are ordinarily pooled into batches of four to eight units for transfusion to adults.

Fresh Frozen Plasma (FFP)

Fresh frozen plasma (FFP) is the commonly used form of plasma. FFP is used in patients with multiple blood factor deficiencies (factors II, V, X, XI), liver disease, dilutional coagulopathy after massive blood loss and replacement with crystalloids or colloids, and for warfarin reversal or active bleeding in patients receiving anticoagulation therapy.

Cryoprecipitates

Cryoprecipitates are fibrinogen products administered to correct deficiencies of factor VIII, von Willebrand factor, factor XIII, and fibrinogen. Cryoprecipitate has also been used for intravenous supplementation of fibrinogen in disseminated intravascular coagulation (DIC) and to ameliorate the platelet dysfunction seen in patients with uremia.

Factor VIII

Factor VIII is used to treat moderate to severe factor VIII deficiency (hemophilia A).

Factor IX

Factor IX is used to treat hemophilia B (Christmas disease), with the treatment regimen based on severity and type of bleeding.

Human Albumin

Human albumin provides volume expansion, colloid replacement, and plasma protein fraction without the risk of transfusion-transmitted disease.

Autologous Transfusion

Autologous transfusion can be performed using the patient's own blood. The blood for autologous transfusion can be obtained by preoperative donation or by postoperative salvage via the operative site. Autologous transfusion eliminates the risk of incompatibility reactions (except those due to clerical error) and the risk of blood-borne virus exposure. GI endoscopy procedures do not allow for salvage of blood for autologous transfusion due to the contaminants in the GI tract.

Equipment

The equipment required for transfusing blood and blood products includes:

- Ordered blood component.
- Standard blood infusion set with an inline filter.
- Microaggregate filter for RBC administration.
- 0.9% NaCl flush IV bag.
- Proper gauge needle or catheter access.
- Infusion equipment.
- Portable infusion stand.

Blood components should only be administered with 0.9% NaCl (INS, 2016). Blood component tubing and filters should be changed every 4 hours or should coincide with blood administration changes (INS, 2016). A leukocyte reduction filter may be required based on the patient's transfusion history. Antihistamine premedication can be considered in patients with a history of allergic transfusion reactions.

To preserve blood and prevent contamination, blood is stored at 40 degrees Fahrenheit. Blood product transfusions are typically initiated within 30 minutes of product release from the blood bank or transfusion service. Many blood banks will not accept back into inventory blood products that have been at room temperature 30 minutes or longer.

Nursing responsibilities

The gastrointestinal nurse's responsibilities in the administration of blood and blood components include, but are not limited to, the following:

- Confirm the informed patient consent, verbal and/or written, as required by state law.
- Provide patient education.
- Inspect the blood product.
- Verify the product expiration date.
- Verify the blood component administration order.
- Confirm crossmatch and compatibility between the patient recipient and the donor unit.
- Adhere to aseptic technique.
- Begin the transfusion.
- Monitor the patient and vital signs at 15 minutes after administration is started and at completion of transfusion, or per organizational policy.
- Identify any immediate and delayed reactions.
- Intervene in the case of adverse reactions.
- Discontinue the completed transfusion.
- Complete written documentation.
- Communicate with other health care providers.

Ensuring accurate recipient patient identification and identification of the transfusion component is one of the most important safety steps in the transfusion process (INS, 2016). Organizational protocols should be followed to ensure that the blood product to be transfused is identified and double-checked with the patient's name, assigned hospital number, the number on the unit of blood, and the date of collection and crossmatching. It is vital that this identifying information be absolutely accurate. **Transfusing the wrong blood product to a patient can be fatal.** In many organizations, two individuals must check the blood component before transfusion to ensure the correct product is given to the correct patient (INS, 2016). Any discrepancies should be reported to the blood bank before administration.

After verifying the identifying information, blood administration steps include the following:

- Connect the Y-tubing to an IV infusion container of normal saline, then flush the tubing, drip chamber, and filter(s) with 30-60 ml of sterile normal saline. IV filters should be fully wetted and drip chambers no more than half full to ensure optimal flow rates. Then connect the tubing to the VAD.
- Gently rotate the blood bag to distribute the blood cells.
- Connect the blood component to the filter and the other branch of the Y-tubing.
- Obtain pre-transfusion vital signs.
- Begin the blood product administration slowly at approximately 2 ml/minute, as catastrophic reactions become apparent after a small volume of blood product enters the patient's circulation.
- Closely observe the patient during the first 15 minutes of transfusion for any evidence of transfusion reaction or per organizational policy.
- After that time, if the patient is tolerating the transfusion, increase the rate to meet the physician's order or patient's needs. Transfusions of single units of blood products should be administered within 4 hours (INS, 2016).
- Continue to observe the patient periodically during the transfusion and up to one hour after completion for tolerance of transfusion and evidence of adverse reactions.
- If the patient develops any evidence of a transfusion reaction, stop the transfusion immediately.

When the transfusion is complete, the gastroenterology nurse should obtain vital signs and assess the patient's condition and response to the transfusion. The blood bag, used tubing, copy of the completed transfusion form, and tag may be returned to the blood bank, based on organization policy. If an infusion is to be administered following the transfusion, the tubing should be changed, the filter removed, and the needle or cannula flushed with 10-20 ml normal saline before reconnecting the infusion.

In the gastroenterology arena, blood-warming devices are recommended for massive transfusions, trauma situations, and rapid administration rates of blood products. Blood product warming should be performed using a U.S. Food and Drug Administration (FDA)-cleared warming device to avoid hemolysis of the blood product (INS, 2016).

Adverse reactions

Any gastroenterology nurse who administers blood must be able to recognize the signs and symptoms of adverse reactions. There are a number of potential adverse reactions to blood and blood component therapy.

Circulatory overload

Circulatory overload occurs when the amount of fluid administered exceeds the pumping capacity of the heart. At highest risk for overload are patients with underlying cardiac, renal, liver, or pulmonary disease. Patient findings include dyspnea, hypertension, jugular venous distension, bounding pulse, tachycardia, tachypnea, cough, confusion, restlessness, and pulmonary congestion. The gastroenterology nurse should slow the transfusion, elevate the head of the patient's bed, notify the physician, and administer oxygen and diuretics as ordered.

Bacterial sepsis

Bacterial sepsis is caused by bacterial contamination of a blood product. It is marked by rapid onset of chills, high fever, abdominal cramping, vomiting, diarrhea, and profound hypotension. The gastroenterology nurse should stop the transfusion, maintain IV access, and contact the physician for further orders.

Allergic reactions

Allergic reactions are caused by a blood product recipient being allergic to a protein or antigen in the donor's blood. Symptoms range from local erythema, hives, and urticaria to cough, respiratory distress, hypotension, loss of consciousness, and possible cardiac arrest. The gastroenterology nurse should immediately stop the transfusion, maintain IV access, and contact the physician. Treatment includes antihistamines and possibly corticosteroids or epinephrine, depending on the severity of the reaction.

Hemolytic reactions

Hemolytic reactions occur as a result of the infusion of ABO-incompatible RBCs, with symptoms evident within the first 5-15 minutes of the transfusion. The patient may complain of or display fever, increased heart rate, chills, low back pain, headache, nausea, chest pain, dyspnea, and hypotension. From hemolysis of the blood cells, the patient may develop acute renal failure, DIC, and possibly cardiac arrest. The gastroenterology nurse must immediately stop the transfusion, maintain IV access, and contact the physician. Vital signs should be monitored at least every 15 minutes.

Hepatitis B, hepatitis C, and HIV

Hepatitis B, hepatitis C, and HIV are viruses that can be transmitted during blood transfusion and pose a risk of chronic infection in the recipient of the infected component. Most individuals who contract these viruses do not have obvious symptoms or physical evidence of disease at the time of transfusion.

Other transfusion-related infectious agents

Other transfusion-related infectious agents (viral, bacterial, parasitic) have been reported.

Citrate toxicity

Citrate toxicity may develop when large volumes of fresh frozen plasma, whole blood, or platelets are transfused rapidly in patients with liver disease or those with circulatory collapse. Citrate is an anticoagulant added to blood products to prohibit clotting of the product and is rapidly metabolized by a healthy liver. Symptoms develop from the hypocalcemia that occurs. Ventricular dysrhythmias can develop, and the patient may also experience paresthesia, muscle cramps, fasciculations, spasms, nausea, and hyperventilation.

If an adverse reaction occurs, the gastroenterology nurse should immediately stop the transfusion, save the substance being transfused, and keep the vein open with normal saline. The patient's vital signs should be assessed and the physician notified immediately. A urine specimen should be collected if a hemolytic reaction is suspected. The blood bank should be notified of a possible transfusion reaction. The labels of all blood containers should be compared to corresponding patient identification forms. Blood samples, all transfusion containers, and the administration set should be sent immediately to the blood bank.

The patient should be treated symptomatically and supportively. If ordered, oxygen, epinephrine, or other drugs are administered. A hypothermia blanket may be ordered to reduce fever in patients not responding to acetaminophen. The patient is made as comfortable as possible and reassured as necessary. In the case of an anaphylactic reaction, emergency resuscitative measures must be instituted immediately.

Patient education

Blood transfusions always require an informed consent from the patient, unless an emergency situation exists. The physician should inform the patient of the risks, benefits, and alternatives to transfusion administration. Organization practice regarding signed consent forms for blood product transfusion should adhere to state requirements. Nursing staff should educate the patient on how the transfusion is given, how long it will take, what symptoms to report, and the need to take vital signs during the transfusion process. Patients who experience allergic reactions should be educated on the cause and preventative treatment for future transfusions.

Transfusion documentation

Complete nursing documentation for recording transfusion information should include the following:
- Verification of the correct unit for the correct patient.
- Type of blood product administered, unit number, and blood type.
- Date and start time of the transfusion, and the completion time.
- Pre-transfusion vital signs, vital signs at 15 minutes, and vital signs at the completion of transfusion per organizational policy.
- Total fluid volume infused, including saline flush and blood component volume.
- Any reaction noted during the transfusion.

In the case of a transfusion reaction, the gastroenterology nurse should record:
- Time and date of the reaction.
- Type and amount of transfused blood or blood products.
- Clinical signs of the transfusion reaction in order of occurrence.
- The patient's vital signs.
- Any specimens that were sent to the laboratory.
- Any treatment, and the patient's response to treatment.
- Completion of a transfusion reaction form, if required by the organization.

For patients receiving an autologous postoperative salvage transfusion, the gastroenterology nurse should record:
- Duration of collection.
- Collection device type.
- Anticoagulant type and amount.
- Duration of the transfusion.
- Use of a blood filter or washed cells.
- Amount and characteristics of drainage.
- Any complications.

CASE SITUATION

Mary James is an 87-year-old female scheduled for a percutaneous gastrostomy feeding device placement under moderate sedation in the gastroenterology suite. She has a history of breast cancer of the left breast, diabetes, and a recent cerebrovascular accident that has left her with left-sided paralysis of her arm and leg. Mary is unable to communicate verbally and becomes easily agitated. She has no known allergies to any drugs.

Points to think about

1. What points should be considered prior to initiating IV access for Mary's procedure?
2. What type of peripheral short IV catheter would be recommended for Mary?
3. Should Mary receive prophylactic antibiotic therapy?

Suggested responses

1. Mary has a history of breast cancer. The gastroenterology nurse should explore whether Mary has a central venous access device in place for chemotherapy administration. Placement of a peripheral IV is contraindicated if Mary has had a left axillary node dissection or has experienced lymph edema of the left arm.
2. Based on the recommendations given above, accessing an existing central venous access device or inserting a 22-gauge peripheral short angiocatheter in the arm would be appropriate. With Mary's history of agitation, use of a steel, winged infusion set would not be appropriate due to the risk of needle displacement.
3. Mary is undergoing a procedure classified as a high-risk GI endoscopy procedure, which carries an increased risk of bacteremia. She should receive pre-procedural IV antibiotics.

REVIEW TERMS

cannulas, catheters, colloids, cryoprecipitates, crystalloids, electrolytes, erythrocytes, hemolysis, hypertonic, hypotonic, infiltration, isotonic, Lactated Ringer's solution, leukocytes, needles, normal saline, osmolality, osmosis, phlebitis, plasma, platelets, serum, solutes, solutions, solvents, superior vena cava, tonicity, whole blood

REVIEW QUESTIONS

1. Salts that dissociate in solution into positive and negative ions are called:
   a. Anions.
   b. Cations.
   c. Electrolytes.
   d. Colloids.
2. The preferred instrument for delivering short-term peripheral IV therapy is a(n):
   a. Stainless-steel needle.
   b. Over-the-needle catheter.
   c. In-the-needle catheter.
   d. Cannula.
3. A peripheral IV cannula would most likely be inserted in the:
   a. Cephalic vein.
   b. Femoral vein.
   c. Superior vena cava.
   d. Radial artery.

4. The recommended antiseptic for venipuncture site preparation in adults is:
   a. Alcohol.
   b. Povidone iodine.
   c. Chlorhexidine gluconate.
   d. Neosporin.
5. Drugs should never be added to blood transfusions because:
   a. They are incompatible.
   b. It complicates determination of the source of any adverse reaction.
   c. Drugs can cause clotting.
   d. The rate of infusion is too slow.
6. Central venous catheters are used for long-term IV therapy or for repeated venous access, and:
   a. Are inserted into large veins.
   b. Are threaded into the lower third of the superior vena cava.
   c. Can be used for medication administration.
   d. All of the above.
7. The following are examples of isotonic solutions except:
   a. 0.9% NaCl.
   b. 0.45% NaCl.
   c. D5W.
   d. Lactated Ringers.
8. What is the best substance to transfuse in patients with a decreased oxygen carrying capacity?
   a. Packed RBCs.
   b. Platelets.
   c. Plasma.
   d. Whole blood.
9. If the patient experiences an adverse reaction to a blood transfusion, the first thing the gastroenterology nurse should do is:
   a. Assess the patient's vital signs.
   b. Stop the transfusion.
   c. Compare labels on blood containers with the patient's identification forms.
   d. Return the blood to the blood bank.
10. Hemolytic reactions to blood transfusions usually occur:
    a. Immediately.
    b. Within the first 5-15 minutes of the transfusion.
    c. Within 24 hours.
    d. As long as 6 months after the transfusion.

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Chapter 22

NUTRITIONAL THERAPY

This chapter will acquaint the gastroenterology nurse with the basic principles of nutrition and nutritional therapy. These principles are followed by detailed discussions of special diets in the practice of gastroenterology and the care of patients who are receiving enteral or parenteral nutrition.

Learning objectives

After reviewing the content of this chapter, the gastroenterology nurse should be able to:

1. Outline the basic principles of good nutrition, nutritional assessment, and nutritional therapy.
2. Describe a number of special diets that may be prescribed for patients with gastrointestinal (GI) disorders.
3. Discuss the techniques used in the administration of enteral and parenteral nutrition, as well as factors to be considered in the care of patients receiving this type of therapy.

BASIC PRINCIPLES

A patient's nutritional status depends on the balance between nutrient intake and energy expenditure. This balance may be affected by internal factors such as age and physical condition or by external factors such as the quantity and quality of food available.

In patients with GI disorders, nutritional status may also be affected by the following:

1. Factors that interfere with food consumption such as impaired appetite, disease, or a special diet.
2. Factors that increase tissue destruction such as cancer, ulceration, or necrosis.
3. Factors that interfere with the patient's ability to absorb nutrients, including absence of normal digestive secretions, altered intestinal motility, or decreased absorptive surface.
4. Factors that interfere with nutrient utilization or storage such as impaired liver function, neoplasms, or pancreatitis.
5. Factors that increase nutrient excretion or loss, including hemorrhage, abscess or fistula formation, nausea and vomiting, surgery, or diarrhea.
6. Factors that increase nutritional requirements, including fever, chronic infection, and malignancy.

For any patient, **nutrition** is an integral part of the total plan of care. A poor nutritional state can affect recovery, contribute to complications, and cause increased morbidity and mortality.

There are six general classes of essential nutrients: carbohydrates, lipids, proteins, vitamins, minerals, and water.

Carbohydrates

Carbohydrates, a class of energy-yielding nutrients that contain carbon, hydrogen, and oxygen, are classified as simple (mono- and disaccharides) or complex (polysaccharides-starch and fiber) carbohydrates. Food sources of carbohydrates include grains, vegetables, fruits, milk and dairy products, syrups, and sugars. Carbohydrates are the primary fuel for energy in the body and must be converted (catabolized) into the simplest component or sugar (glucose) to be utilized for energy metabolism. Glucose is circulated in blood and stored in liver and muscle tissue as **glycogen**. It is recommended that carbohydrates account for 45 percent to 60 percent of total daily calories each day (U.S. Department of Health and Human Services [HHS] and U.S. Department of Agriculture [USDA], 2015).

Ingested carbohydrates come in the form of polysaccharides, mainly starch; **disaccharides** such as sucrose and lactose; and **monosaccharides**, including glucose, fructose, and galactose. In order for the body to utilize carbohydrates, they must be **hydrolyzed** or broken down to monosaccharides by a chemical process that involves the splitting of a bond by the addition of water molecules. Some plant carbohydrates cannot be hydrolyzed by intestinal or pancreatic enzymes and are not absorbed. These are called dietary fibers and can be partially hydrolyzed by enzymes of colonic bacteria (Gropper, Smith, & Carr, 2018).

Digestion of carbohydrates begins in the mouth where the enzyme alpha-amylase, secreted by the salivary glands, hydrolyzes the ingested starches into disaccharides in the mouth and stomach. In the duodenum, pancreatic amylase hydrolyzes the remaining ingested starch. In the jejunum, these newly formed disaccharides are split into monosaccharides by brush border enzymes for absorption by the enterocyte. Any condition that compromises the duodenum or jejunum inhibits carbohydrate absorption (Gropper et al., 2018).

The end product of carbohydrate **catabolism** is **glucose**, a monosaccharide that is the chief source of energy for all living organisms. Glucose is taken up by all of the cells of the body and used for immediate energy. Excess glucose can be converted to glycogen, a long-chain polymer that is stored mainly in the liver and muscles. When necessary, liver glycogen can be converted to glucose for systemic distribution, but muscle glycogen is used primarily by the muscle itself. Excess dietary carbohydrates are also converted to triglycerides for storage in adipose tissue (Gropper et al., 2018).

Lipids

Lipids consist of triglycerides, phospholipids, sterols, sphingolipids, and terpenes. These compounds serve as sources of energy, thermal insulation, precursors for steroid hormones, structural components of cell membranes, and carriers of essential nutrients. Dietary lipids are composed mainly of triglycerides, which contain saturated and unsaturated long-chain fatty acids. The liver can produce most fatty acids; however, it cannot synthesize omega-3 (linolenic) or omega-6 (linoleic) fatty acids. These are considered essential and must be consumed through diet. The primary sources of lipids are oils, fats, dairy products, egg yolks, meats, fish oils, and nuts, and seeds. It is recommended that total fat intake should be between 20% and 35% of calories (HHS & USDA, 2015).

Ingested **triglycerides** are first broken down by oral (lingual) and gastric lipase into diglycerides and fatty acids. Short-chain triglycerides are broken down by gastric lipase and pancreatic lipase. Long-chain triglycerides are emulsified by bile, which allows pancreatic lipase to cleave them into **monoglycerides**, free **fatty acids** and **glycerol**. In the jejunum, these products form micelles, or emulsified fat particles. In the enterocyte, fatty acids are reformed into triglycerides, packaged into chylomicrons, and enter into circulation by the thoracic duct (Gropper et al., 2018).

Fat malabsorption can be a result of diseases such as Crohn's disease, chronic pancreatitis, or cystic fibrosis, or due to fat enzyme deficiencies or small bowel resections. Medium chain triglycerides (MCTs) are a modular source of lipids that may be tolerated by those with impaired ability to digest or absorb long-chain fatty acids. They are more rapidly hydrolyzed by lipases due to their water solubility. MCTs do not require bile salts and pass directly into the portal blood for faster metabolism by the liver (Gropper et al., 2018).

Proteins

Proteins are essential components of all living cells. They are composed of building blocks called **amino acids**. Nine of the 22 recognized amino acids are considered essential because our bodies cannot synthesize them and they must be consumed through diet (Gropper et al., 2018). Proteins are needed for body structure, hormone production, enzymes, fluid and electrolyte balance, transport molecules, and energy. The chief dietary sources of protein are meats, fish, poultry, dairy products, nuts, seeds, and legumes.

Ingested proteins may either be used to synthesize the proteins needed to build new tissue or be catabolized for energy. It is recommended that protein intake is 10% to 35% of total daily calories. For nonstressed adults, 0.8 grams per kilogram (g/kg) of body weight should maintain protein balance. Protein needs increase for physically active adults, individuals in postsurgical stress, and those in other stress-related situations and can range up to 2.5 grams per kilogram in burn patients (Mcclave et al., 2016).

Nitrogen balance (N-balance) is a method of measuring protein turnover in the body. In the clinical setting, it is determined by comparing nitrogen intake with nitrogen excretion over a 24-hour period. First, a 24-hour urine sample is collected and sent to a lab for analysis of nitrogen content. N-balance is then calculated by dividing the total grams of protein consumed by 6.25 grams (protein is 16% nitrogen by weight) and subtracting the urinary urea nitrogen (UUN) and adding a factor for insensible losses.

N-balance = grams of protein/6.25 – (UUN + 4)

Depending on the patient's condition, N-balance may be neutral, positive, or negative.

1. In a healthy adult, protein synthesis is equal to protein degradation, and the individual is said to be in nitrogen balance.
2. Positive nitrogen balance occurs when protein synthesis exceeds protein degradation. This state is normal and to be expected in children, pregnant women, and people recovering from an injury who are building new tissue. In adults, it may signify a rebuilding of wasted tissue.

3. Negative nitrogen balance occurs when protein losses are greater than protein intake. It indicates that protein breakdown exceeds protein synthesis.

Metabolism of ingested protein begins in the stomach, where pepsin, in the presence of hydrochloric acid, hydrolyzes proteins into amino acids and polypeptides of varying lengths. Then, in the duodenum and jejunum, pancreatic enzymes reduce these products to their basic peptides and amino acids. In the lower jejunum and ileum, the remaining peptides are hydrolyzed into amino acids and are absorbed. Protein absorption may be reduced by disorders that reduce gastric hydrochloric acid secretion or impair production of proteolytic enzymes, by duodenal or jejunal inflammation or infection, or by gastric or intestinal resection (Gropper et al., 2018).

Vitamins

Vitamins are a group of organic compounds that work together with enzymes, co-factors (substances that assist enzymes), and other substances that are essential for normal metabolism, growth, and development, and regulation of cell function. Required in small amounts, vitamins are classified as either fat-soluble or water-soluble.

Fat-soluble: Vitamins A, D, E, and K are soluble in fat. They are absorbed with dietary fats and tend to be stored in the body in moderate amounts.

Water-soluble: Vitamin C, all of the B vitamins [thiamin (B_1), riboflavin (B_2), niacin (B_3) pantothenic acid (B_5) pyroxidine (B_6), cobalamin (B_{12})], folic acid, and biotin are soluble in water. They are excreted in urine and are not stored in the body in appreciable amounts.

Minerals

Minerals are inorganic nutrients that serve to regulate enzyme metabolism, maintain nerve and muscle integrity, and facilitate membrane transfer of essential compounds. They include major minerals and trace minerals, also called trace elements. Major minerals include calcium, chloride, sodium, magnesium, potassium, sulfur, and phosphorus. These essential mineral nutrients contribute more than 0.005 percent of body weight and are needed by the body in quantities of 100 mg or more per day. Trace minerals include chromium, cobalt, copper, fluorine, iodine, iron, manganese, molybdenum, selenium, and zinc. These essential mineral nutrients are found in the human body in amounts less than 0.005 percent of body weight, and the body needs them only in microgram (mcg) amounts.

Water

Water acts as an intracellular and extracellular solvent and provides a medium for transportation of nutrients and metabolic waste products. It also lubricates the tissues and helps maintain body temperature. Recommended consumption is 1 gram of water per 1 calorie (Gropper et al., 2018).

FOOD-DRUG INTERACTIONS

Some medications may interfere with the body's use of vitamins and nutrient absorption and processing.

Anticoagulants

Patients who take warfarin may experience a decrease in the anticoagulant effect if they consume large amounts of vitamin K. In the past, it was recommended for patients taking the medication to completely avoid foods containing vitamin K. Recent studies have indicated that a consistent consumption of vitamin K in small doses is a preferred method over abstinence (Mahtani, Heneghan, Nunan, & Roberts, 2014). The nurse or physician can establish proper drug dose according to the patient's consumption of vitamin K. Foods that are rich in vitamin K include dark green leafy vegetables, kale, broccoli, Brussels sprouts, green onions, parsley, spinach, and chard.

In addition to maintaining moderate yet consistent consumption of vitamin K-containing foods, those who take warfarin should also avoid consumption of cranberry juice and cranberry products (Violi, Lip, Pignatelli, & Pastori, 2016). Flavonoids in cranberries may inhibit cytochrome P450 system activity and predispose the patient to hemorrhage.

Anti-inflammatories

Patients who take steroids, including methylprednisone, prednisolone, or prednisone, may experience fluid retention, hyperglycemia, or ulcers. These conditions subside when patients discontinue the use of steroids.

Patients who develop hyperglycemia will need to limit consumption of carbohydrates and consult with a registered dietitian to establish a glucose-controlled nutrition plan. They should avoid foods high in simple carbohydrates such as sugared breakfast cereals, canned fruit in syrup, sugared flavorings, sugar, syrup, jams, jellies, candy, honey, and regular soft drinks.

Patients who develop hypertension are advised to reduce sodium intake by limiting the consumption of foods high in sodium such as croissants, bagels, biscuits, baked beans, canned or creamed corn, scalloped potatoes, canned mushrooms, sauerkraut, marinara sauce, canned vegetables, vegetable juice, cheese (especially feta), miso, pickles, soy and soy sauce, barbecue sauce, table salt, ketchup, and teriyaki sauce.

Potassium-sparing diuretics

Patients who take spironolactone or triamterene for high blood pressure, edema, cirrhosis, or cystic acne may experience increased urinary output. Because these drugs limit the potassium urinary excretion, elevated serum levels may be noted. Potassium is present in abundance in fruits, vegetables, dairy, meats, and legumes. Because potassium is present in all major food groups, evaluation from a registered dietitian is essential to avoid overconsumption. Some foods may have to be limited. If increased potassium consumption is unavoidable, patients may be switched to a non-potassium sparing diuretic such as furosemide (Lasix).

NUTRITIONAL ASSESSMENT

A thorough assessment of a patient's nutritional status includes completion of the following:

1. A health and dietary history, including a report of any recent weight gain or loss; recurrent nausea, vomiting, or diarrhea; any chronic illness; all medications, over-the-counter (OTC) drugs, and supplements; possible dysphagia; poor dentition; food allergies or intolerances; psychosocial issues; and changes in appetite associated with current nutrition status.
2. A physical examination, including height, weight, body mass index (BMI), mid-arm circumference, and skin fold thickness. Signs of nutritional deficiency (edema, loss of subcutaneous fat, and muscle wasting, for example) and compromised skin integrity (such as decubiti) should be noted.

   In the case of a pediatric patient, the child's height, weight, and head circumference should be correctly plotted on a standardized growth chart. The pediatric care team should complete a record of growth patterns and velocity of growth, which are very important in pediatric assessment because prior growth parameters help to define growth patterns and velocity.
3. Diagnostic studies, including laboratory tests for serum albumin, transferrin, prealbumin, retinol-binding protein, and nitrogen balance calculations.

For patients with recognized nutritional deficits, the physician may order oral, enteral, or parenteral nutritional therapy. Oral intake is the preferred means of administering a special diet, but it requires the patient to have a functioning GI tract and be able to ingest foods. Enteral tube feedings are used when patients are unable or unwilling to use the oral route, such as a patient suffering from dysphagia, altered central nervous system function, or severe **anorexia** secondary to the primary underlying illness. Parenteral feeding (intravenous nutrition) is provided through a central venous or peripheral venous access for patients without a functional GI tract or for those patients requiring prolonged bowel rest.

Determining a patient's calorie needs can be estimated by using a range of 25 - 35 kcals/kg (kilocalories per kilogram), depending on the patient's current nutrition status and illness or injury. A widely used method of calculating energy requirements is the Harris-Benedict equation for basal energy expenditures (BEE), where W is the patient's actual or usual weight in kilograms, H is the height in centimeters, and A is the age in years.

WOMEN: BEE = 655 + (9.6 ´ W) (1.8 ´ H) - (4.7 ´ A)
MEN: BEE = 66 + (13.7 ´ W) (5 ´ H) - (6.8 ´ A)

A stress factor or activity factor can be multiplied by the BEE to determine total energy needs for maintenance or weight restoration. Factors may range from 1.2 to 1.5 for severely stressed patients. Protein needs also vary depending on the present state of protein balance and illness. For post-surgery patients, protein needs can range from 1.1 - 1.2 g/kg for patients under mild stress, and 1.5 - 2.0 g/kg for severely stressed patients (including those afflicted with burns or severe wounds) (Mcclave et al., 2016).

SPECIAL ORAL DIETS

Many diets used in the treatment of GI illnesses involve restricting a particular dietary component such as fats or protein. Others alter the consistency of the diet or the amount of dietary fiber. When prescribing a special diet, it is important to remember that dietary habits or customs are difficult to change. As much as possible, when designing a special diet, consider the patient's cultural, socioeconomic, and religious background.

High-fiber diets

Dietary fiber is composed of polysaccharides and lignins that are not digested in the bowel. The polysaccharides include cellulose, hemicelluloses, gums, mucilages, and pectins. Soluble fiber is a type of fiber that can dissolve in water and increases transit time in the intestines. Sources include oat bran, barley, nuts, and flax seed. Insoluble fiber does not dissolve in liquids and moves bulk through bowel, decreasing transit time. Food sources of insoluble fiber include whole-grain breads and cereals and fresh fruits and vegetables. The recommended intake for fiber is 25 - 38 grams (25g in females and 38g in males for ages 18 - 50; 21g in females and 30g in males for ages 50+) of fiber per 14 grams/1000 Kcal (Mahan & Raymond, 2017). The high-fiber diet is recommended as an aid in the prevention of constipation and diverticulosis and has been associated with a decreased risk of colon and breast cancer.

Low-fiber diets

A low-fiber diet is recommended for those with active inflammatory bowel disease, diverticulitis, and acute ulcerative colitis or Crohn's disease. It may also be indicated as a transitional diet after intestinal surgeries, including colostomy and ileostomy procedures. Typically, a low-fiber diet used for chronic problems involves a reduction in the intake of whole grains, fruits, and vegetables. The nutritional implications of a low-fiber diet are that it may provide an inadequate amount of many vitamins and minerals, and patients adopting this may require supplementation.

Gluten-free diets

Gluten is a protein found in wheat, rye, and barley. A lifelong gluten-free diet is used in the treatment of celiac disease (gluten enteropathy), an autoimmune disease characterized by malabsorption of nutrients and the absence of normal intestinal villi on the small bowel mucosa. Symptoms of this condition include diarrhea, weight loss, bloating, and weakness. It may also present with non-GI-related complaints such as skin disorders and neurologic or metabolic disturbances, including a genetic factor that has been determined for celiac disease. Symptoms can present at any age in life, usually related to stress or a stressful event.

Biopsies and serological tests are most accurate when the duodenal mucosa is abnormal or/damaged. Patients should be tested before removing gluten from their diet. Diagnosis of celiac disease usually involves an endoscopic intestinal biopsy and serological tests at the time of symptoms, including IgG anti-gliadin antibody. Only the gliadin fraction of gluten protein is implicated in mucosal damage. To remove all grains containing gliadin (specifically wheat, barley, rye, spelt, kamut, and, in some cases, oats), it is necessary for patients with celiac disease to avoid the following:

- Obvious sources of gluten, including baked goods, cereals, pastas, and soups containing wheat, barley, rye, and oats (that are not labeled as gluten free).
- Less obvious sources of gluten, including wheat and its derivatives that are used as an extender in processed foods and beverages such as ice cream, salad dressings, soy sauce, canned foods, catsup, mustard, candy bars, and instant coffee.
- Foods containing modified food starch, hydrolyzed vegetable protein, and malt.

Many sources of gliadin are not readily apparent to the consumer. Even the adhesive in postage stamps and envelopes may contain gliadin. It is very important that patients on gluten-free diets read all labels and carefully study the ingredients of all processed foods. Flours made from rice, soybean, corn, buckwheat, potato, nut (almond, macadamia, pecan, walnut, etc.), coconut, and tapioca are appropriate substitutes or alternatives for gluten-allergic and gluten-sensitive patients.

Health care providers must remember that lifelong adherence to a gluten-free diet is a big commitment and may represent a significant social liability, especially for children and teenagers. Patient education regarding the importance of compliance for continued good health is essential. Young adult patients with celiac disease should be informed that failure to comply with a gluten-free diet can impact weight and physical development. Closely monitored follow-up and continued counseling are recommended. There are many resources that offer excellent educational materials, cookbooks, and support for the celiac patient. Resources are available at the Academy of Nutrition and Dietetics at www.eatright.org, the Celiac Disease Foundation at www.celiac.org, and at the Celiac Sprue Association at www.csaceliacs.info.

Within a week of removing gluten from the diet, patients may experience a decrease in diarrhea, improved appetite, and an overall improvement in attitude and behavior. In other patients, clinical improvement may take several months. The most common cause of failure to respond to a gluten-free diet is incomplete removal of gluten.

Some patients with celiac disease may develop other nutritional deficiencies secondary to their gluten intolerance. If a secondary lactase deficiency develops, milk and milk products should be limited initially. In young children, initial dietary management also may require adequate calorie and protein to sustain catch-up growth. Multivitamin, iron, and folic acid supplementation are also commonly needed for children with anemia. Most patients develop an increased tolerance to lactose and decrease their need for supplementation after gluten withdrawal, as their intestinal structure and function begin to return to normal.

Low-lactose diets

A low-lactose diet is indicated for patients with symptoms of lactose intolerance, which is manifested by a history of abdominal pain, diarrhea, bloating, and flatulence following ingestion of lactose. Tests to measure lactase deficiency include the hydrogen breath test, urine and fecal tests, and measurement of lactase within intestinal biopsies.

Lactose intolerance may be the result of a lactase deficiency secondary to a disease process such as celiac disease or Crohn's disease. Lactase deficiency may also occur in the absence of disease. Lactose intolerance may also result from decreased time of exposure to the intestinal mucosa, as in short bowel syndrome or dumping syndrome. A low-lactose diet may be used in the treatment of patients with irritable bowel syndrome (IBS) or during the acute phase of a diarrheal illness in which intestinal transit is rapid or lactase deficiency is transient, as in acute gastroenteritis, ulcerative colitis, or Crohn's disease. A low-lactose diet may also be helpful in the initial phase of therapy for celiac disease.

Patients with lactose intolerance are quite variable in their ability to tolerate smaller amounts of lactose. Some patients with severe intolerance may need to avoid milk and milk products, including ice cream, cheeses, and butter. Aged cheeses, lactose-free milk, butter, and even milk in limited amounts can be tolerated by some patients with less severe symptoms. Patients should read food labels to become familiar with processed foods containing milk solids, lactose, milk sugar, galactose, or skim milk powder. Yogurt containing active cultures may be well-tolerated because fermentation continues in the intestinal lumen. Additionally, use of lactase tablets prior to the consumption of lactose-containing foods can be effective for many.

Most patients with lactose intolerance can tolerate up to 3g of lactose per day. However, more severe restrictions may be necessary in certain patients such as those with galactosemia. Dairy products typically provide a significant percentage of dietary calcium. Thus calcium supplements may be necessary for patients on low-lactose diets, particularly for postmenopausal women. Calcium can also be found in dark green leafy vegetables, sardines, and salmon, as well as some fruit juices, soy milk, and rice milk fortified with calcium. The treatment for lactose intolerance is relief of symptoms through a low-lactose diet. After a period of time, small amounts of lactose can be carefully reintroduced into the diet, as long as the patient remains asymptomatic.

Low-protein diets

Restriction of dietary protein is indicated in patients with end-stage liver disease who are at risk for portosystemic encephalopathy. For patients with acute encephalopathy, protein restriction of 0.6 - 0.8 g/kg/d (grams per kilogram per day) was recommended in the past; however, the more recent standard of care recommends normal protein intake to meet the patient's needs and avoid worsening his/her nutrition status (Cabral & Burns, 2011). Dietary restriction of protein requires limiting intake of milk, cheese, meat, fish, poultry, and eggs; focusing on high-fiber foods like vegetables; and eating small, frequent, carbohydrate-rich meals. Low-protein diets are also indicated for patients with chronic renal failure (pre-dialysis) to delay the progression of renal failure and the need for dialysis. Emphasis on high biological value proteins (animal products) is important to maximize protein intake for those on restricted diets. Vitamin and mineral supplementation may be required for patients with end-stage renal disease.

Low-fat diets

Low-fat diets are used to control symptoms of GI disease, particularly steatorrhea and diarrhea. They are indicated in all acute and chronic diseases that impair the functions of lipolysis, micellar solubilization, mucosal absorption, transport out of the absorptive cell, or transport in the lymphatic system. A short-term, low-fat diet may also be helpful in patients with acute gastroenteritis or pancreatitis. Medium-chain triglyceride (MCT) oil can be added to the diet to supplement extra calories if long-term adherence to a very low-fat diet is needed.

The major sources of dietary fats in the American diet are processed foods such as baked goods, fried foods, and whole-fat dairy products. Because small amounts of cooking fats can add a significant amount of dietary fat, all foods prepared on a low-fat diet must be broiled, boiled, or baked. Consumption of breads, cereals, vegetables, and fruits should be emphasized. Dairy products made with whole or 2 percent milk should be avoided, as well as most desserts, cheeses, nuts, olives, high-fat meats, mayonnaise, salad dressings, and cream sauces or gravies. A registered dietitian can instruct patients on careful adherence to a low-fat diet. If a diet is severely restricted in fat, supplements containing preparations of fat-soluble vitamins A, D, E, and K may be necessary.

2-gram sodium diets

The restriction of dietary sodium is indicated for gastroenterology patients with ascites and edema caused by severe liver disease. A combination of sodium restriction and diuretic administration is the mainstay of treatment for ascites. Sodium restriction prevents further expansion of the extracellular fluid volume and may be used alone for outpatients with minimal ascites. Once diuretic therapy has significantly reduced the fluid retention, patients may be maintained on a low-sodium diet without diuretics.

Patients on 2-gram sodium diets should not add salt to their food, nor should they use salt in food preparation. Foods that are particularly high in sodium should be restricted, including most frozen meals, smoked meats, cheese, commercial mixes, commercially prepared desserts, sodium-containing seasonings such as monosodium glutamate (MSG), and commercially prepared soups.

Low-FODMAP diets

Research provides new dietary strategies for IBS management with the low FODMAP diet (Tuck et al., 2014). The FODMAP acronym stands for fermentable oligosaccharides, disaccharides, monosaccharides, and polyols. These short-chain carbohydrates are poorly absorbed in the small intestine. Symptoms of IBS and other functional GI disorders have been found to improve with the restriction of FODMAPs.

GI RESTORATION PROGRAM: MAKING THIS COMPLEX SIMPLE

A framework with which to target therapies aimed at improving GI function has been developed. The *4R GI restoration program* is a conceptual process to focus the evaluation of GI function status, resulting in targeted nutritional, digestive enzyme, and probiotic support that promotes the normalization of important GI functions.

The program simplifies the complex interactions and systems by asking four main questions:

1. What does this patient need to have **removed** (e.g., pathogenic growth in the intestinal tract, allergenic foods in the diet) for healthy GI function?
2. What does this patient need to have **replaced** (e.g., stomach acid, digestive enzymes) to support improved GI function?
3. What does this patient need to support and/or **reinoculate** a healthy balance of microflora? That is, does he or she require probiotic reinoculation and/or prebiotic support?

4. What does this patient need to support the **regeneration** of the mucosal layer? That is, does he or she require targeted nutritional support for GI barrier regeneration?

1. Remove

REMOVE focuses on eliminating pathogenic bacteria, viruses, fungi, parasites, and other environmentally derived toxic substances from the GI tract. Several studies have shown that avoiding the suspected foods leads to substantial improvement in clinical symptoms. REMOVE typically introduces oligoantigenic diets that feature foods with the least possible risk of allergic reaction.

Pathogens are also important to consider because they are a source of antigenic stimuli and promote other symptoms. For example, *Helicobacter pylori (H. pylori)* colonizes the upper GI and has been associated with many health conditions, including gastritis and peptic ulcer disease. Pathogens can take up colonization anywhere in the GI mucosa, and cell wall components of bacteria, yeasts, and parasites can be absorbed, which may cause systemic symptoms that are difficult to pinpoint to the pathogen directly. Laboratory tests evaluating microbiology, parasites, and presence of serum anti-bodies to pathogens are often useful in verifying unwanted microbial organisms in the GI tract and monitoring response to treatment.

2. Replace

REPLACE refers to the replenishment of enzymes and other digestive factors lacking in an individual's GI environment. GI enzymes that may need to be replaced include proteases, lipases, and saccharidases normally secreted by cells of the GI tract or by the pancreas. Other digestive elements that may require replenishment include hydrochloric acid and intrinsic factor, normally produced by parietal cells in the stomach wall.

3. Reinoculate

REINOCULATE aims to reintroduce desirable bacteria or "probiotics" into the intestine to reestablish microflora balance. Probiotics synthesize various vitamins, produce short-chain fatty acids (SCFAs) necessary for colonic cell growth and function, degrade toxins, and prevent colonization by pathogens. Fecal microbial transplantation (FMT) therapy has been used to treat *Clostridium difficile (C. difficile)* by implanting healthy stool to reestablish healthy microflora. Current literature reinforces the use FMT in the treatment of *C. difficile* to be a safe, well-tolerated, effective treatment with few adverse events (Kelly et al., 2015).

4. Regenerate

REGENERATE provides support for the healing and regeneration of the GI mucosa. Part of the support for healing comes from removing insults that continually re-injure or irritate the mucosa and promoting healthy microflora. Key to any successful regeneration strategy is providing the nutrients necessary for the mucosal epithelial cells to proliferate and differentiate, thereby producing the tight connections that constitute the intestinal barrier. Pre- and post-testing with inert sugar combinations, including lactulose or mannitol, can be performed to evaluate the intestinal barrier function.

REGENERATE benefits from all of the clinical approaches discussed so far. Still, some factors are particularly helpful for healing and regeneration of the GI tract, and these are discussed below.

L-glutamine is a preferred fuel for the rapidly replicating cells of the GI mucosa. It is the most abundant and versatile amino acid in the body. Glutamine may promote production of glutathione in the GI cells as well, which provides additional protection from oxidative insults during inflammation and supports healing. The uptake of glutamine by the mucosal cells from both the intestinal lumen and arteriolar circulation is increased in catabolic states and during glucocorticoid therapy. Traditionally, glutamine was thought to play a metabolic role of supporting the GI mucosa respiration, regulating acid-base balance through the production of urinary ammonia, and acting as a precursor for nucleic acids and proteins. More recent research data suggest that glutamine plays a more vital role in the GI system, particularly that it can induce certain genes involved in growth and differentiation of the GI mucosa cells (Kim & Kim, 2017).

Clinically, dietary glutamine is known to be important for the maintenance of GI mucosa. Dietary deficiency of glutamine is associated with atrophy and degenerative changes in the small intestine following intestinal injury, infection, stress, surgery, and radiation. It is, therefore, considered a "conditionally essential" amino acid.

Overall nutrition must be considered in an individual who has compromised gut function. Micronutrients particularly important for support of GI regeneration include vitamins C and E, carotenoids for their antioxidant effects, and zinc and vitamin B_5 (pantothenic acid) for their roles in supporting the healing mechanisms. A diet that is high in nutritional value and has a low antigenicity profile (that only minimally stimulates an immune response) provides a basis for regeneration of the GI tract.

INTESTINAL MICROFLORA: THE USE OF PROBIOTICS TO PROMOTE INTESTINAL BALANCE

Probiotics are viable organisms and supportive substances that improve intestinal microbial balance such as *Lactobacillus acidophilus (L. acidophilus)* and bioactive proteins. There is evidence that links the use of fermented dairy products such as yogurt with the promotion of intestinal health (Gonzalez et al., 2019; Kok & Hutkins, 2018).

The ability of the probiotic *L. acidophilus* to help prevent pathogenic bacteria from proliferating and healthy bacteria from becoming toxic is well documented. When the proper strain is chosen, it may help to maintain a population equilibrium or balance between the different forms of microorganisms, curtailing their potential overgrowth and pathogenicity.

Bifidobacteria is another probiotic naturally occurring in the human intestine, with *Bifidobacterium infantis* being the first flora to colonize the intestines of newborns. Research studies (Cronin et al., 2011) have documented several beneficial effects of bifidobacteria when given to infants.

L. acidophilus and bifidobacteria help maintain the proper balance between the different forms of microorganisms in the intestine by producing organic acids that reduce intestinal pH and thereby inhibit the growth of acid-sensitive bacteria, including many pathogenic species. Lactobacilli, which are frequently more acid-tolerant than other organisms, produce lactic acid, hydrogen peroxide, and possibly acetic and benzoic acids. Acids produced by bifidobacteria include SCFAs such as acetic, propionic, and butyric acids, as well as lactic and formic acids. At optimal pH values, they exert several inhibitory influences on bacterial cell growth. The most plentiful SCFA produced by bifidobacteria is acetic acid, which exerts a wide range of antimicrobial activity against yeasts, molds, and bacteria.

In addition to lactic and other acids, lactobacilli have the capacity to secrete numerous metabolites, or endotoxins, that kill pathogenic bacteria. A variety of antibacterial and anti-yeast substances have been isolated, including lactocidin, lactobicillin, lactobreven, and acidolin. Because these substances are difficult to isolate and stabilize, their value can best be obtained through the administration of those strains known to secrete these agents as a part of their life cycle.

Prebiotics

Prebiotics are defined as nondigestible food ingredients that may beneficially affect the host by selectively stimulating the growth and/or the activity of a limited number of bacteria in the colon. Thus, to be effective, prebiotics must escape digestion in the upper gastrointestinal tract and be used by a limited number of the microorganisms comprising the colonic microflora. Prebiotics are principally oligosaccharides (carbohydrate molecules composed of a relatively small number of monosaccharide units). Because they mainly stimulate the growth of bifidobacteria, they are referred to as bifidogenic factors.

The following describes the various oligosaccharides, which are classified as prebiotics.

- Fructo-oligosaccharides (FOS) typically refer to short-chain oligosaccharides containing from three to five monosaccharide units. FOS, also called neosugar and short-chain FOS, are produced on a commercial scale from sucrose using a fungal fructosyltransferase enzyme. FOS are resistant to digestion in the upper gastrointestinal tract. They act to stimulate the growth of bifidobacterium species in the large intestine. FOS are marketed in the United States in combination with probiotic bacteria and in some functional food products.
- Other oligosaccharides classified as prebiotics include inulins, isomalto-oligosaccharides, lactilol, lactosucrose, lactulose, pyrodextrins, soy oligosaccharides, transgalacto-oligosaccharides, and xylo-oligosaccharides.

ENTERAL NUTRITION

Enteral nutrition is the administration of a prescribed diet by means of a flexible tube inserted into the stomach or small bowel either transnasally, surgically, radiographically, or endoscopically.

Indications and contraindications

Enteral nutrition is indicated for maintenance of nutritional status in patients who have a functioning GI tract but cannot ingest sufficient food and nutrients to meet energy requirements. Some examples of conditions for which enteral nutrition may be indicated are dysphagia (including achalasia), anorexia, malabsorption syndromes, chronic malnutrition, failure to thrive in infants, major burns, severe trauma, hepatic or renal failure, cerebrovascular accident, or esophageal tumor.

For patients in whom the GI tract is functional, but oral feeding is not possible, enteral nutritional support is preferred over total parenteral nutrition (TPN). Compared to TPN, enteral nutrition is safer, less expensive, reduces risks of catheter-related infections, and helps to maintain gut integrity.

Enteral nutrition is contraindicated in patients with hemodynamic instability, intestinal fistulas, bowel obstruction, or severe motility dysfunction. It is also not appropriate in end of life care where use of nutrition support will not positively affect outcome of care.

Administration of enteral nutrition

Enteral nutrition is administered through a tube that may be inserted transnasally, surgically, radiographically, or endoscopically. Transnasal administration is the most common delivery type.

Transnasal insertion of a feeding tube involves the use of a nasogastric, nasoduodenal, or nasojejunal tube. Nasogastric tubes are indicated when pharyngeal reflexes are intact and there is little risk of aspiration. Nasojejunal or nasoduodenal intubation is indicated when the potential for aspiration is high, although this is not recommended because it is more difficult to confirm duodenal placement of a tube and may become easily dislodged. In addition, duodenal and jejunal tubes tend to be smaller in diameter and easily become obstructed with medication granules or thick formulas. Discomfort and esophagitis as a result of lumen size or reflux may occur with any transnasal tube.

Tubes made of silicone or polyurethane plastics are thinner and more flexible than older tubes made of polyvinyl plastic. In addition, they do not stiffen or become brittle in the GI tract. Nasogastric tubes come in a variety of lengths and diameters. Tubes for gastric feeding measure approximately 36 inches, whereas the longer tubes used for intestinal feeding are up to 60 inches. Some tubes are weighted at the distal end with tungsten to facilitate transpyloric passage of the tube. Guidewires or stylets may also be used to place a soft, flexible feeding tube into the duodenum with fluoroscopic guidance. Nasoduodenal tube feedings are currently preferred for enteral nutrition because they offer less potential for aspiration than nasogastric tubes.

For patients who require long-term enteral nutrition, the feeding tube may be inserted by gastrostomy or jejunostomy. Intermittent bolus feeding through a gastrostomy tube (G-tube) is most suitable for non-critical patients; ambulatory patients, as in rehabilitation; or patients requiring home tube-feedings. Both nasogastric tubes and G-tubes have a risk of aspiration of gastric contents, but gastrostomy tubes also carry the risk of infection, drainage, and cellulitis at the site.

Jejunostomy tubes (J-tubes) are most useful for patients who will undergo surgery for esophageal, gastric, pancreatic, or biliary disease. Enteral feedings via jejunostomy are usually administered through continuous drip infusion. With continuous drip infusion, a defined amount is given continuously with the use of an infusion pump. This method minimizes the risk of aspiration, abdominal distention, and diarrhea. Intermittent "bolus" infusions are not recommended for feeding through the jejunum due to risk of dumping syndrome, nausea, vomiting, and diarrhea.

Initiation of feedings

It is important to check for proper tube placement before each intermittent feeding and at least once a shift during continuous feedings. Radiographic verification of tube placement is the best method of ensuring proper placement of nasoenteric tubes prior to beginning any tube feeding and after any event that may predispose a patient to tube dislocation. Other methods to check placement include auscultation of the abdomen while air is introduced into the tube or withdrawing fluid to check for volume, appearance, and pH.

The patient's head and thorax should be elevated between 30 degrees and 45 degrees to prevent aspiration of formula. To promote better tolerance, tube feedings may be started at a rate of 10 - 40 ml/hr. The rate is dependent on the physician's orders, based on input from the nutrition support team and calculated to each specific patient's nutritional needs. It can then be advanced every 4 to 8 hours as tolerated, until the goal rate is reached. Periodic flushing with water throughout the day helps to maintain tube patency. Flushing the tube with 15 - 30 ml of water before and after giving medications, flushing with 20 - 60 ml of water every 4 hours during continuous infusions, and flushing with 20 - 60 ml of water after intermittent feeding or when feedings are interrupted is recommended. If a tube becomes clogged, a 60 ml syringe with 30 - 60 ml flush of warm water is recommended. To maintain optimum patency, the tube should be considered for replacement every 4 weeks (Mcclave et al., 2016).

For pediatric patients, enteral feedings are best tolerated when initiated slowly at an initial rate of 1 - 2 ml/kg/hr (milliliter per kilogram per hour) of formula via continuous infusion (Hay, 2018).

Enteral formulas

The type of formula selected for a patient depends on the status of the patient's GI tract and on his or her caloric and nutrient requirements. Most commonly available tube feeding formulas in the United States are lactose-free, gluten-free, and kosher.

1. Polymeric formulas contain intact mixtures of proteins, fats, and carbohydrates. Polymeric formulas are used when GI function is normal or near normal. They are available in fiber-containing and low-residue formulations; in a variety of protein levels; in formulations specialized for renal disease, pulmonary disease, and diabetes mellitus; and in formulations for those under high metabolic stress (e.g, trauma).
2. Monomeric formulas, called elemental or semi-elemental, are predigested formulas composed of hydrolyzed starch, peptides or amino acids, and LCTs and MCTs. These are nutritionally complete, low-residue, and relatively non-stimulating to pancreatic, biliary, and GI secretions. Elemental diets are used for patients with definite evidence of impaired digestion and malabsorption.
3. Disease-specific formulas are designed for special nutritional needs of patients with conditions such as renal disease, respiratory disease, diabetes, and hepatic failure.
4. Modular formulas contain only one nutritional component (fats, proteins, or carbohydrates). Modular preparations may be added to formulas to increase calories or protein to meet specific nutritional or metabolic needs of patients.

Improper handling of formulas can lead to bacterial contamination, increasing the risk of food-borne illness. Open systems where the formula is poured from the original container into a feeding container can increase a patient's risk of infection. Closed systems where the formula container directly attaches to the tubing for administration are recommended to reduce the risk of infections.

Potential complications of enteral feedings

Gastrointestinal Complications:

1. Nausea, vomiting, or cramping may be caused by delayed gastric emptying due to medications.
2. Constipation may result from dehydration or inadequate fiber intake.
3. Abdominal distention may be the result of GI ileus, obstruction, or ascites.
4. Diarrhea is commonly caused by medications such as antibiotics, sorbitol-containing elixirs, or infections such as *Clostridium difficile*.
5. Dumping syndrome may be due to rapid infusion of a high volume of formula.
6. Malabsorption and maldigestion may result from Crohn's disease, diverticular disease, radiation enteritis, fistulas, and pancreatic insufficiency.

Mechanical Complications:

1. Tube placement can cause tracheal perforation, respiratory complications, or GI tract perforation leading to abscess formation and peritonitis.
2. Presence of feeding tubes can cause pharyngeal discomfort; nasal or pharyngeal irritation; or necrosis, fistulas, and cellulites.
3. Tube obstruction and clogging, common in concentrated and fiber-containing formulas, can be prevented by frequent flushing of tube.
4. Aspiration is the most common complication and may be minimized by placing nasojejunal or jejunostomy tubes well beyond the ligament of Treitz and by elevating the patient's head and shoulders 30 - 45 degrees.

Metabolic Complications:

1. Fluid and electrolyte imbalances – hypo- or hypernatremia, hypo- or hyperkalemia, overhydration, hypertonic dehydration, and hyperglycemia.
2. Trace elements, vitamin, and mineral deficiencies.
3. Excessive carbon dioxide production.
4. Refeeding syndrome.

Nursing care

- Frequent monitoring is necessary to reduce complications associated with feeding that result in compromised nutritional status. Monitoring may include the following:
- Gastric residuals every 4 hours. All aspiration secretions should be returned to the stomach because they contain nutrients, electrolytes, and digestive enzymes. In general, if gastric residuals are greater than 150 ml for percutaneous endoscopic gastrostomy (PEG) feedings or greater than 200 ml from a nasogastric tube (NGT), feedings are held for about 2 hours and residuals are checked prior to resuming feeding. The American Society of Parenteral and Enteral Nutrition (ASPEN) guidelines state that gastric residual volumes should not be used as part of routine care to monitor ICU patients receiving enteral nutrition (Mcclave et al., 2016).
- Daily weight to detect fluid shifts, and daily intake and output (I&O).
- Consistency, volume, and frequency of bowel movements.
- Signs and symptoms of intolerance such as abdominal distention, vomiting, nausea, diarrhea, and constipation.
- Daily electrolyte levels, BUN, and creatinine until the goal rate is achieved and, thereafter, two to three times per week. Minerals and a weekly blood count may be ordered.
- Inspection of tube site for placement, patency, or infection.

Additional support

The effectiveness of enteral support should be assessed periodically by testing for metabolic imbalances and repeating nutritional assessments. A registered dietitian can also provide a comprehensive assessment with recommendations to adjust nutrition regimens as needed. The nurse should note any changes in feeding tolerance, significant fluctuations in weight, or other signs of nutritional compromise, all of which warrant a consultation with a dietitian.

Patients who are receiving enteral feedings also need psychological support. Ambulation should be encouraged, if possible, and both the patient and significant others should be reassured that these feedings are necessary. As appropriate, the patient may be able to resume eating regular foods at some point. Patients who are receiving enteral nutrition at home must be instructed in techniques of administration, I&O recordkeeping, and what physical symptoms to report to the physician.

PARENTERAL NUTRITION

Parenteral nutrition (PN) is the intravenous (IV) infusion of nutrients, including protein (amino acids), carbohydrates (dextrose), fat (lipid), and additives (vitamins, minerals, electrolytes, trace elements). It can be delivered through central or peripheral veins.

Indications and contraindications

Parenteral nutrition is indicated for patients who are moderately to severely malnourished, or for those who are in negative nitrogen balance and are not expected to meet their nutritional requirements orally within a short period of time. Central parenteral nutrition, also called **total parenteral nutrition (TPN)**, should only be utilized when it is determined that the patient is likely to need PN for more than 4 to 5 days and cannot be fed enterally. Otherwise, the risks of complications may outweigh the potential benefit of therapy.

Parenteral nutrition may be ordered in the following situations:

- For patients with GI disorders such as radiation enteritis, mesenteric ischemia, small bowel obstruction, acute pancreatitis, and short bowel syndrome.

- In preoperative preparation of malnourished patients who cannot tolerate enteral feedings.
- For patients with postoperative surgical complications, particularly fistulas or paralytic ileus.
- In postoperative care of neonates who cannot tolerate enteral feeding.
- For infants with intractable diarrhea.

Note: Parenteral nutrition is usually not used if oral or enteral nutrition is possible.

Administration of parenteral nutrition

To be practiced safely and successfully, parenteral nutrition should be administered by a trained TPN team, consisting of a physician, dietitian, pharmacist, and nurse. Most often, parenteral feedings are administered through an indwelling subclavian vein catheter via the superior vena cava. In some cases, the catheter is inserted via the femoral vein or antecubital vein to the subclavian vein (peripherally inserted central catheter [PICC]).

For some patients, **peripheral parenteral nutrition (PPN)** may be administered through a peripheral vein such as a small vein of the hand or forearm rather than a central vein. PPN may be ordered for a patient who cannot be fed enterally for 5 to 7 days, for patients in whom a central line is precluded, and for patients who are eating but not getting sufficient calories. The most common complication associated with peripheral access is venous thrombophlebitis, which causes inflammation of the cannulated vein, pain, erythema, and tenderness of the site.

PN solutions can be 2-in-1 or 3-in-1 formulations in which lipids are either included in the formulation or excluded, as in 2-in-1 solutions with only **dextrose** and amino acids. Concentration of dextrose in solutions can range from 2.5% to 70%; however, it should not be greater than 10% for PPN due to the risk of thrombophlebitis. Lipids may be given with peripheral alimentation to further dilute the dextrose concentration. Because of the limitations on dextrose content, the number of calories that can be given with PPN may be inadequate for long-term maintenance nutrition requirements.

The initial infusion should be administered gradually, beginning with a rate of approximately 1.0 liter (L) in the first 24 hours. The concentration of the solution is increased slowly until the desired amount of nutrients is delivered.

Parenteral formulas

Parenteral formulas are concentrated solutions composed of protein, carbohydrates, lipids, electrolytes, vitamins, trace minerals, and water. In most hospitals, TPN solutions are ordered by the physician on a daily solution order form. Formulation of a parenteral solution must be carefully calculated so that it meets the complete needs of the patient in terms of calories, nitrogen, fatty acids, vitamins, trace elements, and water. Dextrose solutions and lipid emulsions are the main sources of energy in patients who are fed by central catheter.

Energy needs, as previously discussed, can be calculated by estimating 25 - 35 kcal/kg, or using the Harris-Benedict equation with stress and activity factors. Protein requirements are usually supplied by crystalline amino acids. The amount of protein needed can be estimated based on the specific needs or illness of the patient. A range of 0.8 - 1.0 g/kg is considered a maintenance requirement, while the range of 1.2 - 1.4 g/kg may be needed for patients experiencing wound healing, visceral protein repletion, or hemodialysis.

The recommended daily fat allowance is 1.0 g/kg, approximately 30% of total calories (Hajishafiee, Azadbakht, & Adibi, 2015). Usually, a lipid solution is added to the carbohydrate and protein solutions, and the combined mixture is administered to the patient. Some patients receive lipid solution only one or two times per week to prevent fatty acid depletion. Vitamins and trace minerals are provided in parenteral alimentation solutions in amounts designed to meet daily requirements of patients with disease. Medications and insulin can also be added to the mixture as indicated.

Daily fluid requirements

Daily fluid requirements are calculated according to the following formula:
1000 ml for the first 10 kg body weight (100 ml/kg)
500 ml for the next 10 kg (50 ml/kg)
Ages < 50: 20 ml for each kg of body weight thereafter
Ages > 50: add 15 ml/kg for each kg
Fluid requirements can be estimated:
Most adults 18 to 55 yr: 35 ml/kg of body weight
Older adults > 55 yr: 30 ml/kg of body weight

Potential complications of total parenteral nutrition

Infection and Sepsis Complications:

1. Catheter contamination during insertion.
2. Long-term indwelling catheter.
3. Catheter seeding from bloodborne or distant infection.
4. Contaminated solution.

Metabolic Complications:

1. Dehydration, hypovolemia.
2. Bone demineralization.
3. Hyperglycemia, rebound hypoglycemia.
4. Hyperosmolar hyperglycemic nonketotic syndrome.
5. Azotemia.

6. Electrolyte disturbances, including hypocalcemia, hypophosphatemia, hyperphosphatemia, hypokalemia, and hypomagnesemia.
7. Deficiencies of essential fatty acids, trace elements, vitamins, and minerals.
8. Altered acid-base balance.
9. Elevated liver enzymes, hepatomegaly, fatty liver, and cholestasis.
10. Fluid overload.

Mechanical Complications Related to Catheterization:

1. Catheter misplacement.
2. Hemothorax (blood in the chest).
3. Pneumothorax (air or gas in the chest).
4. Hydrothorax (fluid in the chest).
5. Hemomediastinum (blood in the mediastinal spaces).
6. Subcutaneous emphysema.
7. Hematoma.
8. Arterial puncture.
9. Myocardial perforation.
10. Catheter embolism.
11. Cardiac dysrhythmia.
12. Air embolism.
13. Endocarditis.
14. Nerve damage at the insertion site.
15. Laceration of lymphatic duct, chylothorax, and lymphatic fistula.
16. Thrombosis.

Serious complications occur in 5% to 10% of patients receiving TPN. The most common complication is catheter-related sepsis, which occurs in less than 5% of TPN patients. TPN-related infection may result from contamination of the catheter by skin flora, contamination of the TPN solution or tubing, or bacteremia originating from another source in the body. To avoid catheter infections, specific aseptic techniques must be followed in the care of the catheter and dressing. Fever in a patient with a central catheter should receive immediate attention, including blood culture and an attempt to determine the source of infection. If the patient has a positive blood culture, the catheter may need to be removed and replaced after treatment with antibiotics.

Potential mechanical complications of parenteral nutrition include thoracic injury during catheter insertion such as subclavian artery puncture or myocardial perforation, air embolism from air entering the catheter during a line change, or venous thrombosis and thrombophlebitis.

Nursing care for patients receiving parenteral nutrition

ASPEN guidelines require daily weights, accurate I&O, and careful blood glucose monitoring. Additionally, daily labs are needed to allow for appropriate adjustments in electrolytes. Nursing goals in the administration of parenteral nutrition include prevention of infection, maintenance of the prescribed rate of flow, ongoing patient assessment, and the provision of education and support for the patient and significant others.

All aseptic precautions must be observed when preparing parenteral solutions and when caring for patients who are receiving parenteral nutrition. Careful aseptic insertion of the central venous catheter is the first step. Over the long term, care of the catheter dressing, the tubing, and the catheter must be meticulous. Hands should be washed with an antimicrobial solution before any contact with the TPN system or dressing. Dressings are usually changed when they become soiled, wet, or nonocclusive. Institutional policies and procedures for insertion site care should be consulted because they may vary.

Parenteral feeding solutions should be kept refrigerated until needed, then allowed to reach room temperature before infusion to prevent hypothermia. The solution should be administered at a steady rate. To prevent air embolism when changing the tubing, the patient should be placed in the Trendelenburg position, flat on the back with the feet higher than the head by 15 - 30 degrees.

The hospitalized patient on TPN should be monitored daily for fluid I&O and weight gain or loss. Vital signs and blood glucose should be monitored regularly by the care team throughout the hospital stay. Regular laboratory tests should include creatinine and urea nitrogen, serum chemistries, electrolytes, magnesium, phosphorus, transferrin, triglycerides, and complete blood count with differential and platelet count. When parenteral nutrition is to be stopped, the glucose concentration should be decreased gradually.

For patients who continue to need parenteral nutrition but no longer require skilled nursing care, home parenteral nutrition can be safe and cost-effective. Home TPN with infusion done overnight permits the patient to return to a reasonably normal daytime routine and aids in social adjustment. Both the patient and significant others need to be trained before discharge regarding dressing changes, addition of fluids, I&O recordkeeping, what symptoms to note, and how to contact a health care provider.

CASE SITUATION

Mrs. Ellen Marshall is a 60-year-old woman who is 5 feet 6 inches and weighs 179 pounds. She has been diagnosed by endoscopy with reflux esophagitis. The gastroenterology nurse's initial assessment notes that Ellen is overweight. Excessive weight gain can increase intra-abdominal pressure and exacerbate reflux. Furthermore, eating the wrong foods can add calories, decrease lower esophageal sphincter tone, exacerbate pain, and increase the gastric acid secretion rate. One appropriate nursing diagnosis in Ellen's case is "Altered nutrition: More than body requirements, related to caloric intake exceeding metabolic need."

Points to think about

1. How might the gastroenterology nurse differentiate between the meanings of overweight and obese?
2. What are the critical indicators in making a nursing diagnosis of “Altered nutrition: More than body requirements, related to caloric intake exceeding metabolic need”?
3. How might the gastroenterology nurse describe the approach to obtaining these indicator data to confirm or refute such a nursing diagnosis?
4. What additional data are needed to complete a basic dietary assessment on Ellen?
5. What nutritional advice might the nurse give Ellen that might alleviate symptoms of esophageal reflux?

Suggested responses

1. The generally accepted definitions of overweight and obese are as follows:
   a. Overweight is any weight in excess of the ideal or desirable body weight for age, height, and body size.
   b. Obese is an excess of body fat that is 20% or more above the ideal or desirable body weight.
   c. A person may be overweight but not obese because of muscle mass.
   d. Overweight: BMI of 25 - 29.9; Obese: BMI of 30 or more; Morbid obesity: BMI of 40 or more (CDC, 2017).
2. The critical indicators for the nursing diagnosis “Altered nutrition: More than body requirements, related to caloric intake exceeding metabolic need” are as follows:
   a. Body weight 10% to 30% over ideal weight for height and frame.
   b. In women, triceps skin fold greater than 25 mm (a measurement of body fat).
3. To confirm or refute such a nursing diagnosis, it would be appropriate to do the following:
   a. Obtain accurate weight, height, triceps skin fold, and wrist measurements.
   b. Calculate body frame size as small, medium, or large (height in cm divided by wrist circumference in cm). For a woman, small is <11.0; medium is 10.1 to 11.0; large is >10.1.
   c. Using derived frame size, use weight and height norms for age to arrive at ideal weight.
   d. Calculate actual weight percentage over ideal weight. Note that there are other methods for calculating the amount of body fat.
4. In addition to weight, height, and body frame, data needed to complete a dietary assessment on Ellen might include the following:
   a. Usual food and fluid intake, including frequency of meals, snacks, method of food preparation, portion size, and types of foods normally consumed.
   b. Elimination patterns.
   c. Activity and energy levels.
   d. Ethnic or cultural background.
   e. Factors affecting food consumption.
5. Nutritional advice that the nurse could give Ellen to alleviate symptoms of esophageal reflux might include the following:
   a. Avoid large meals and refrain from snacking before going to bed to keep gastric volume at a minimum.
   b. Keep dietary fat to a minimum because fat slows gastric emptying and decreases lower esophageal sphincter pressure.
   c. Minimize dietary intake of chocolate, alcohol, and coffee, which also reduce basal lower esophageal sphincter pressure.
   d. Avoid carminative substances (used to relieve flatulence) such as spearmint and peppermint, and avoid citrus juices, tomato products, and coffee because they all impair lower esophageal sphincter function.
   e. Encourage other dietary modifications to promote weight loss such as meeting with a registered dietitian or joining a weight loss program.

REVIEW TERMS

amino acids, anorexia, carbohydrates, catabolism, dextrose, disaccharides, enteral nutrition, fatty acids, glucose, glycerol, glycogen, hydrolyzed, lipids, minerals, monoglycerides, monosaccharides, nitrogen balance, nutrition, parenteral nutrition (PN), peripheral parenteral nutrition (PPN), proteins, total parenteral nutrition (TPN), triglycerides, vitamins, water

REVIEW QUESTIONS

1. Excess glucose is stored in the liver and in the muscles in the form of:
   a. Adipose tissue.
   b. Glycogen.
   c. Disaccharides.
   d. Triglycerides.
2. If a patient is in negative nitrogen balance, that means he or she is:
   a. Building new tissue.
   b. Ingesting the proper amount of protein.
   c. Eating too much protein.
   d. Losing more protein than he or she is taking in.
3. Patients with celiac disease must avoid eating:
   a. Gluten.
   b. Fiber.
   c. Sodium.
   d. Protein.

4. Low-fat diets are usually used to control:
   a. Steatorrhea and diarrhea.
   b. Intestinal gas and bloating.
   c. Ascites.
   d. Constipation.
5. The primary disadvantage of using a nasogastric feeding tube is:
   a. Risk of aspiration.
   b. Patient discomfort.
   c. Risk of infection.
   d. High cost.
6. Peripheral parenteral nutrition (PPN) is most appropriate for:
   a. Patients who need nutritional supplementation for less than 1 week.
   b. Patients with inadequate fat stores.
   c. Patients who require concentrated feeding solutions.
   d. Patients who are eating and getting sufficient calories.
7. The most common complication in patients receiving total parenteral nutrition (TPN) is:
   a. Thoracic injury.
   b. Air embolism.
   c. Metabolic imbalance.
   d. Catheter-related sepsis.
8. Utilizing the 4R GI restoration program, "Remove" focuses on eliminating all of the following except for:
   a. Parasites.
   b. Viruses.
   c. GI enzymes.
   d. Fungi.
9. "Regenerate" refers to:
   a. Introduction of probiotics.
   b. Providing support for prebiotics.
   c. Re-introduction of desirable bacteria.
   d. Providing support for healing and regeneration of GI mucosa.
10. Re-introduction of probiotics into the intestine to reestablish microflora balance would be part of which program:
    a. "Regenerate."
    b. "Modify."
    c. "Reinoculate."
    d. "Support."
11. The enzymes proteases, lipases, and saccharides are part of which concept of GI restoration:
    a. "Regenerate."
    b. "Modify."
    c. "Remove."
    d. "Replace."
12. The probiotic responsible for being the first flora to colonize the intestines of newborns is:
    a. *Lactobacillus acidophilus*.
    b. *Lactobacillin*.
    c. *Lactobacillus bulgaricus*.
    d. *Bifidobacterium infantis*.
13. Prebiotics mainly stimulate the growth of:
    a. Fructo-oligosaccharides.
    b. Probiotics.
    c. a & b.
    d. Bifidobacterium.
14. Oligosaccharides are classified as:
    a. Probiotics.
    b. Prebiotics.
    c. Disaccharides.
    d. Short chain fatty acids.
15. Fructo-oligosaccharides, lactilol, and inulins belong to a group known as:
    a. Prebiotics.
    b. Short chain fatty acids.
    c. a & b.
    d. Oligosaccharides.

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SECTION V

Diagnostic Procedures and Tests

Section V offers a discussion and update of important procedures such as endoscopy and tests such as manometry, biopsy, and cytology. The section ends with an overview of other procedures and tests that may be performed in a gastroenterology setting.

Chapter 23

ENDOSCOPY

This chapter will acquaint the gastroenterology nurse with endoscopic techniques that are used to diagnose disorders of both the upper and lower gastrointestinal (GI) tracts. Specific diagnostic and therapeutic endoscopy procedures are discussed in subsequent chapters.

Learning objectives

After reviewing the content of this chapter, the gastroenterology nurse will be able to:

1. Describe the different types of endoscopes used in GI procedures and their main components.
2. Discuss the responsibilities of the gastroenterology nurse with respect to intravenous (IV) sedation and analgesia of gastroenterology patients.
3. Explain the indications, contraindications, techniques, and potential complications of endoscopic investigations of the upper and lower GI tract, including esophagogastroduodenoscopy (EGD), endoscopic retrograde cholangiopancreatography (ERCP), small bowel enteroscopy (SBE), anoscopy, proctosigmoidoscopy, flexible sigmoidoscopy, and colonoscopy.
4. Discuss nursing considerations involved in each of these procedures.
5. Describe additional endoscopic diagnostic procedures, including capsule endoscopy, endoscopic ultrasonography, and endoscopy though an ostomy.

TYPES OF ENDOSCOPES

Gastrointestinal **endoscopy** is the direct visual examination of the lumen of the GI tract. It is a safe, effective way of evaluating the appearance and integrity of the GI mucosa, detecting lesions, and providing access for therapeutic procedures.

The first endoscopes were rigid, metal instruments. Most have now been replaced by flexible video scopes, which increase visibility and promote patient safety and comfort. Depending on the section of the GI tract to be explored, the endoscopist may use one of the following:

- A flexible forward-viewing or side-viewing (oblique) endoscope, used to visualize the esophagus, stomach, and proximal duodenum in esophagogastroduodenoscopy (EGD), to cannulate the biliary tract in endoscopic retrograde cholangiopancreatography (ERCP), or to cannulate beyond the ligament of Treitz in small bowel enteroscopy (SBE).
- An anoscope, which is a rigid plastic or metal speculum, used to inspect the anal canal.
- A proctosigmoidoscope, or rectosigmoidoscope, a rigid endoscope that is used to examine the rectum and sigmoid colon.
- A flexible sigmoidoscope, used to examine the rectum, sigmoid, and descending colon.
- A colonoscope, used to visualize the entire lower GI tract from the rectum to the ileocecal valve and the terminal ileum.

Although each are designed for specific uses, most endoscopes have certain common parts, which include:

- A flexible insertion tube that is usually 8-12 mm in diameter. The insertion tube contains air, water, and biopsy channels; fiberoptic bundles; and cables. The tube extends from the distal end of the scope to the control head.
- A universal cord (also called an umbilical cord or light guide tube) that extends from the control head and inserts into the light source.
- An optic system, which consists of **fiberoptic bundles** that conduct light through the shaft and trans-

mit the image to the eye using a lens system that focuses the image at the eyepiece. In a video endoscope, the optic system consists of a one-piece, solid-state video camera (including the camera head, coupler, and focusable optics), which transmits the image to a video monitor without the need for fiber optics.
- A control head that houses the lenses, which includes controls for maneuvering the tip up and down and left and right; valves that regulate irrigation, air, or carbon dioxide insufflation; and suction.
- Cables that extend the length of the insertion tube and serve to control the movement of the flexible tip.
- Channels for air and water flow.
- A suction and biopsy channel. The channel allows the passage of accessories, such as biopsy forceps, cytology brushes, polypectomy snares, laser fibers, electrocautery devices, banding equipment, aspiration and injection needles, prostheses (stents), and minimally invasive surgical equipment. The channel also allows suctioning of fluid that obstructs the endoscopist's vision.
- Optional cameras that can be attached to the endoscope to allow the taking of still 35-mm or instant photographs or video recordings.

Video endoscopes

Video endoscopes have no coherent fiberoptic image bundle. Instead, a distal sensing device in the tip of the instrument electronically transmits an image to a video processor for display in color or black and white on a video monitor. Endoscopy is performed by referencing the monitor. An advantage of this technology is that all members of the endoscopy team are able to view the procedure, thus enhancing collaboration among team members. High-definition capabilities and narrow-band imaging provide high-quality views that more clearly define mucosa and fine capillaries inside the GI tract.

The controls on the video endoscope for tip deflection are similar to those of conventional fiberoptic scopes, with two coaxial knobs for left and right and up and down deflections. Valves are provided for air and water insufflation and for suction. An accessory and suction channel is also provided.

New computer programs are continually being developed to enhance the capabilities of **video endoscopy**. Patient data can be accessed for statistical purposes, and reports of procedures can be generated by computer.

Digital capture devices

Digital capture devices (DCDs) may be used to store endoscopic images and video. Comprehensive software systems provide options for still image capture, generation of endoscopy reports, and videos. External DCDs plug into the endoscope processor or are transmitted through an integration system.

SEDATION AND ANALGESIA

Many diagnostic and therapeutic procedures in the endoscopy unit are performed using moderate sedation and analgesia. The American Society of Anesthesiologists (ASA) "Practice Guidelines for Moderate Sedation and Analgesia" (2018) serve as a guideline in the endoscopy setting.

The ASA guidelines define four levels of continuum of sedation depth:

1. **Minimal sedation (anxiolysis)** is a drug-induced state during which cognitive function and coordination may be impaired, but patients are able to maintain ventilatory and cardiovascular functions and respond normally to verbal commands.
2. **Moderate sedation/analgesia (conscious sedation)** is a drug-induced state of depressed consciousness in which patients can still respond to verbal commands, perhaps accompanied by light tactile stimulation, and are able to maintain a patent airway and spontaneous ventilation and cardiovascular function.
3. **Deep sedation/analgesia** is a drug-induced depression of consciousness during which patients cannot be easily aroused but respond with repeated or painful stimulation. Patients may require assistance to maintain a patent airway and ventilatory function, but cardiovascular function is usually maintained.
4. **General anesthesia** is a drug-induced loss of consciousness during which patients are not arousable, even by painful stimulation, require assistance maintaining a patent airway and ventilation, and may have impaired cardiovascular function.

Because sedation is a continuum, predicting how an individual patient will respond is not always possible. Practitioners who provide sedation should be able to rescue patients experiencing a deeper-than-intended level of sedation resulting in adverse physiologic consequences, such as hypoventilation, hypoxia, and hypotension (ASA, 2018).

Indications

For endoscopy procedures, the ASA has set forth practice guidelines for sedation and analgesia provided by nonanesthesiology practitioners that have been approved by the American Society for Gastrointestinal Endoscopy (ASGE, 2018). These guidelines are intended to ensure the performance of safe, effective diagnostic and therapeutic procedures. The goals of moderate sedation and analgesia include the following:
- Maintain intact protective reflexes.
- Allow relaxation to allay anxiety and fear.
- Minimize changes in vital signs.
- Ensure cooperation.
- Provide decreased pain perception.
- Ensure easy arousal from sleep.
- Maintain patient ability to respond to commands.
- Provide some degree of retrograde amnesia.

Procedure

In endoscopy settings, moderate sedation and analgesia is most often induced by the IV administration of the benzodiazepines diazepam or midazolam. These agents may be administered in combination with atropine and narcotic analgesics, such as meperidine (Demerol), morphine, and fentanyl.

Titration of IV sedation and analgesia medications to achieve the desired effect improves clinical efficacy and reduces adverse outcomes (Amornyotin, 2013), In the event of complications or adverse reaction to sedation, the gastroenterology registered nurse, under the order of the physician, must be prepared to administer the antagonist reversal agents of flumazenil for benzodiazepines and naloxone for narcotics.

Medications

Diazepam and midazolam

- Both diazepam and midazolam must be slowly titrated in small incremental doses until the desired endpoint is reached, usually exhibited by onset of slurred speech and decreased responsiveness.
- Respiratory depression may be potentiated by the additional use of an analgesic, emphasizing the need to reduce the amount of incremental doses and closely monitor the level of consciousness and respiratory function.
- Individual response varies with age, physical status, and current medications. Particular care must be taken with pediatric, elderly, and debilitated patients.

Propofol

The ASA has specific guidelines for the administration of drug agents that can provide the sudden onset of deep sedation such as propofol (Diprivan) due to the rapid and profound changes in sedation and anesthetic depth and the lack of antagonist medications.

Monitoring during anesthesia

The involvement of an anesthesiologist in the care of every patient undergoing anesthesia is optimal. When that is not possible, nonanesthesia personnel should be qualified to rescue patients whose level of sedation becomes deeper than initially intended and who enter, if briefly, a state of general anesthesia. Rescue of a patient from a deeper level of sedation than intended is done by a practitioner proficient in airway management and advanced life support (ASA, 2018).

During endoscopy with sedation:

- A registered nurse (RN) trained in moderate sedation administration is responsible for monitoring and assessing the patient receiving moderate sedation and analgesia throughout diagnostic and therapeutic endoscopic procedures (Society of Gastroenterology Nurses and Associates, Inc. [SGNA], 2017).
- The moderate-sedation-trained RN administers the sedation and analgesia during endoscopic procedures in the presence of and by the order of a physician.
- In procedures that are complicated by the severity of the patient's illness, age, and/or complex technical requirements of the procedure, a second gastroenterology nurse may be needed to assist the physician while the first gastroenterology nurse assesses and monitors the patient.
- The anesthesia department may need to assess and monitor pediatric or high-risk patients, as determined by the physician. This may be standard practice in some institutions.

Patients who need special consideration for increased risk during moderate sedation and analgesia include those:

- Over age 60.
- Unable or unwilling to cooperate.
- With a history of severe cardiac, pulmonary, hepatic, renal, or central nervous system (CNS) disease.
- Who are morbidly obese or pregnant.
- With sleep apnea, a recent history of drug or alcohol abuse, metabolic imbalance, or airway difficulties.
- Who have not been properly prepared for the procedure such as with an emergency procedure.

Before sedation administration

Before administration of sedation and analgesia, it is important that the gastroenterology RN assess the patient for:

- Communication barriers or developmental stage factors.
- Cultural or religious needs.
- Age, height and weight, blood pressure, pulse, respiratory rate, oxygen saturation level, electrocardiogram (ECG), and circulation and perfusion.
- Allergies, including drug and latex allergies.
- Current medications.
- Alcohol and recreational or illegal drug use history.
- Relevant medical-surgical history.
- Level of comfort.
- Level of consciousness and cognitive ability.
- Mobility and associated safety measures, including a falls risk assessment.
- Type of bowel preparation and results, if indicated.
- Laboratory results, including pregnancy testing, if indicated.

Other pre-procedure responsibilities include:

- Verify signed informed consent.
- Verify current history and physical (complete and on the chart).
- Verify post-procedure transportation with a responsible adult (for outpatients).
- Establish venous access.
- Provide patient education, including what to expect during all phases of the procedure, as well as discharge instructions.
- Verify NPO status to reduce the risk of aspiration. Institutional policies for length of fasting should be followed.

- Administer pre-procedure medication as ordered.

The ASA guideline (2018) for fasting before moderate sedation and analgesia recommends:

- Children under 6 months of age should not be allowed solids or full liquids (including infant formula) for 6 hours and breast milk for 4 hours before sedation, but may be allowed clear liquids up until 2-3 hours before sedation.
- Children 6-36 months of age should not be allowed solids or full liquids for 6 hours before sedation, but may be allowed clear liquids up until 2-3 hours before sedation.
- Children over 36 months of age and all adult patients should not be allowed solids or full liquids for 6 hours before sedation, but may be allowed clear liquids up until 2-3 hours before sedation.
- Patients who have delayed gastric emptying may require longer fasting times.

Re-evaluation of the patient is performed by the practitioner immediately prior to the administration of sedation.

The Joint Commission's, "National Patient Safety Goals" (2019) have been adopted by various accrediting bodies. As part of the pre-procedure verification process, Joint Commission recommends a "time-out" be conducted immediately before starting any invasive procedure. The time-out has the following characteristics:

- It is standardized as defined by the institution.
- It is initiated by a designated member of the team. The provider performing the procedure assumes responsibly for the time-out and engages the entire procedural team.
- During the time-out, the team members agree (at a minimum) on the correct patient identity, the correct site, and the correct procedure to be done.

During the procedure

During any procedure performed under moderate sedation and analgesia, it is the responsibility of the gastroenterology RN to monitor the patient's vital signs and response to both sedation and the procedure. Procedural monitoring for patients of all ages includes:

Observation of oxygen saturation, blood pressure, pulse, respiratory rate and effort, ECG, level of consciousness (i.e., response to verbal and/or tactile stimuli and purposeful communication), warmth and dryness of skin, and pain tolerance. Using capnography and a measure of ventilation by end-tidal carbon dioxide (CO_2) monitoring during procedural sedation is recommended by the ASA (2018).

Documentation of physiological data from continuous monitoring at least every 5 minutes during medication titration or more often if needed, and at least every 15 minutes thereafter until the patient meets pre-established discharge criteria. These intervals should be shortened as changes in the patient's status, need for additional medication, and the increased complexity of the procedure occur.

Although automatic monitoring devices may enhance the gastroenterology nurse's ability to assess the patient accurately, they are no substitute for watchful, educated assessment.

Documentation includes:

- Diagnostic or therapeutic technique(s) used.
- Any unusual events, including intervention and subsequent patient response.
- Status of the patient after completion of the procedure.
- All drugs, fluids, and blood products administered, including dose or amount, route, time given, and patient response.

After the procedure

After the procedure is complete, it is the gastroenterology nurse's responsibility to:

- Monitor the patient's vital signs and level of consciousness. The patient is observed for signs of malignant hyperthermia, which is a rare but life-threatening reaction to certain triggering anesthetics.
- Continue to observe and document any further unusual events or post-procedural complications and their nature. Patients who received reversal agents (flumazenil or naloxone) require longer periods of observation because the half-life of the offending agent may exceed that of the reversal medication and lead to resedation.
- Review with the patient and responsible caregiver the written post-procedure instructions that address diet, medications, safety-focused activity restrictions according to age or level of mobility, follow-up care, and course of action if a post-discharge complication develops.
- Ensure that institutional discharge criteria are met. These usually include return to pre-sedation and analgesia oxygen saturation, vital signs, and level of consciousness; no nausea, vomiting, or abdominal pain; and steady ambulation.

Considerations

The safe administration and maintenance of sedation and analgesia by order of the physician is one of the most important responsibilities of the gastroenterology RN working in the GI endoscopy setting. The gastroenterology RN must be trained and educated in endoscopy; must possess the knowledge of medications used and the skills to assess, diagnose, and intervene in the event of complications; and may be given the responsibility for the administration of reversal agents prescribed by the physician.

Whether or not the gastroenterology RN administers the medications, the gastroenterology RN is responsible for monitoring and assessing the patient receiving sedation and analgesia throughout diagnostic and therapeutic GI procedures. It is imperative that the gastroenterology RN receives specialized training to provide rescue with advanced life support and follows his or her state's nurse practice act and the institutional policies and procedures regarding sedation and discharge.

Because of increased potential adverse reactions, it is important that pediatric patients undergoing sedation and analgesia be monitored by individuals who are familiar with normal age-specific pediatric vital signs and are trained in pediatric basic life support and the use of respiratory emergency equipment.

It is recommended that an individual trained in adult or pediatric advanced life support is immediately available in each setting. The emergency cart must be stocked with equipment and medications appropriate for the age and size of the individual patient receiving sedation and analgesia.

Potential adverse reactions to sedation and analgesia include anaphylaxis, respiratory distress, arrhythmias, and seizures. Rarely, malignant hyperthermia may occur.

Some patients may have a paradoxical reaction to sedation and analgesia, becoming incoherent, uncontrollable, and inconsolable. Caregivers should be educated regarding this possible response before the procedure.

Unsedated endoscopy

Selected patients may be able or may choose to undergo endoscopic procedures without sedation. Small diameter endoscopes (less than 6 mm) can improve the tolerability of upper endoscopy when sedation is not used. Topical anesthesia may be used. Successful flexible sigmoidoscopy and colonoscopy may be performed in selected patients who receive no sedation or sedation only if needed.

For procedures performed without medications, the types and level of monitoring should be individualized. Patient preparation should be the same as described for sedation in the event that sedation is administered (ASGE, 2018).

ESOPHAGOGASTRODUODENOSCOPY

In **esophagogastroduodenoscopy (EGD)**, a flexible endoscope less than 10 mm in diameter is passed into the upper GI tract. The esophagus, stomach, and the proximal duodenum are visualized. Smaller pediatric endoscopes may be used for younger patients or for patients with strictures. Larger-diameter gastroscopes with a larger suction channel or two suction channels may be used for therapeutic procedures.

EGD allows the physician to diagnose and document GI abnormalities through the use of direct vision and still video photography.

Indications

EGD may be indicated for diagnosis, or in conjunction with diagnostic or therapeutic procedures, including:

- Dysphagia or odynophagia.
- Dyspepsia.
- Anemia.
- Esophageal reflux that persists despite appropriate therapy.
- Persistent, unexplained vomiting.
- Upper GI X-ray showing lesions that require biopsy.
- Acute or chronic upper GI bleeding (hematemesis or melena).
- Suspected esophageal or gastric varices.
- Suspected esophageal stenosis, esophagitis, hiatal hernia, gastritis, obstructive lesions, and gastric or peptic ulcers.
- Epigastric or chest pain.
- Chronic abdominal pain.
- Suspected polyps or cancer.
- Follow-up of patients with Barrett's esophagus; large, indeterminate ulcers; or previous gastric or duodenal surgery.
- Removal of ingested foreign bodies.
- Caustic ingestion.
- Oral aversion.
- Dilation of the upper GI tract.
- Placement or removal of a feeding tube.
- Pre-surgical screening.

EGD may be contraindicated for patients with:

- Suspected perforated viscus.
- Shock.
- Seizures.
- Recent myocardial infarction.
- Severe cardiac decompensation.
- Thoracic aortic aneurysm.
- Respiratory compromise.
- Severe cervical arthritis.
- Acute oral or oropharyngeal inflammation.
- Acute abdomen.
- Known Zenker's diverticulum.
- Unwillingness or inability to cooperate.
- Noncompliance with NPO guidelines.

Procedure

Prior to the procedure

- The patient must be NPO before the procedure to decrease the risk of aspiration. Guidelines for fasting before sedation should be followed.
- A thorough and current medical and drug history and physical examination are important, with special attention given to any history of drug reactions, bleeding disorders, or associated cardiac, pulmonary, renal, hepatic, or central nervous system (CNS) disease.
- The mouth should be inspected for loose teeth and orthodontic appliances that could become dislodged.
- If ordered, a topical anesthetic may be applied to the oropharynx to suppress the gag reflex.

- A bite-block should be placed into the patient's mouth to protect the patient's teeth and to prevent damage to the endoscope.

During the procedure

- The patient is placed in the left lateral position. The chin should be tilted toward the chest, keeping the head in the midline. Reassure the patient and hold his or her head and shoulders to help maintain the proper position. Keeping the chin tilted toward the table allows secretions to drain.
- Before insertion, the endoscope is lubricated with a water-soluble lubricant.
- The endoscope is passed in stages, examining each structure as the scope advances.
- To obtain the best possible view, mucus or other secretions are suctioned and air is instilled to distend structures. CO_2 is an alternative to room air for insufflation. CO_2 insufflation has been associated with decreased post-procedure pain, flatus, and bowel distention.
- As the endoscope passes through the pylorus, the patient may experience some abdominal discomfort or may retch. At this time, it may help to have the patient breathe deeply and slowly to help relax the abdominal muscles. The patient may also experience a feeling of fullness or an urge to defecate as air passes into the stomach and duodenum.

Occasionally, duodenal spasm makes visualization of this area difficult. Administration of a smooth-muscle relaxant such as glucagon decreases contractions so the mucosa and contour of the duodenum can be examined thoroughly.

After the procedure

After the procedure, the patient should remain NPO until the gag reflex returns, which may take up to 1-2 hours. After the gag reflex returns, any residual sore throat or hoarseness may be relieved by drinking liquids, gargling with normal saline, or taking anesthetic throat lozenges. Because of the risk of aspiration, throat lozenges cannot be given to patients who are still mildly sedated, nor should they be given to small children.

The gastroenterology nurse should monitor the patient's oxygen saturation and vital signs regularly to observe the patient for signs of bleeding, vomiting, change in vital signs, pain, and abdominal distention. The patient should be instructed to report any symptoms of complications such as hematemesis, pain, or difficulty breathing.

Considerations

During EGD, it is important to maintain the patient's oral airway, suctioning secretions and regurgitated material from the pharynx when necessary. Nasal oxygen should be used to treat hypoxia, and resuscitation equipment should be immediately available in the event of adverse cardiopulmonary reactions.

Because of the narrow and collapsible airway in infants and small children, introduction of the endoscope may result in respiratory distress such as stridor, apnea, or oxygen desaturation. A simple jaw-lift maneuver applied by the gastroenterology nurse is likely to reduce compression of the trachea and result in resolution of these symptoms. To apply a jaw lift, the gastroenterology nurse's fingers are gently placed along the lower jawbone and used to guide the jaw forward.

An additional threat to the child's respiratory status is the increased potential for over-inflation of the stomach (especially in infants), which decreases chest expansion. Symptoms may include abdominal distention, growing tachypnea, and oxygen desaturation. Symptoms are usually relieved by removal of the air.

EGD is generally regarded as safe. The rate of complications increases when therapeutic maneuvers are performed. Patients at the highest risk are the elderly and those with advanced cardiac, pulmonary, hepatic, or CNS disease. Adverse events may include:

- Respiratory depression or arrest.
- Perforation of the esophagus, stomach, or duodenum.
- Hemorrhage related to trauma or perforation.
- Pulmonary aspiration of blood, secretions, or regurgitated gastric contents.
- Infection.
- Cardiac arrhythmia or arrest.
- Hypotension.
- Vasovagal response.
- Allergic reaction to the topical anesthetic or IV medications.

ENDOSCOPIC RETROGRADE CHOLANGIOPANCREATOGRAPHY

Endoscopic retrograde cholangiopancreatography (ERCP) uses a combination of endoscopic and radiologic techniques to visualize the biliary and pancreatic ducts.

Indications

ERCP is indicated for:

- Evaluation of signs or symptoms suggesting pancreatic malignancy when results of ultrasonography and/or a computerized tomography (CT) scan is normal or equivocal.
- Evaluation of acute, recurrent, or chronic pancreatitis of unknown etiology.
- Before therapeutic endoscopy procedures of the biliary tree such as removal of retained common bile duct stones, endoscopic sphincterotomy, balloon dilatation of strictures, or placement of a stent or biliary drain.
- Unexplained chronic abdominal pain of suspected biliary or pancreatic origin.

- Evaluation of jaundiced patients suspected of having treatable biliary obstruction.
- Evaluation of patients without jaundice whose clinical presentation suggests bile duct disease.
- Preoperative or postoperative evaluation to detect common duct stones in patients who undergo laparoscopic cholecystectomy.
- Manometric evaluation of the ampulla of Vater and common bile duct.

The use of ERCP in pediatric patients is limited to rare cases of obstructive jaundice without dilated biliary ducts such as in sclerosing cholangitis or congenital stricture of the common hepatic duct, relapsing pancreatitis of unknown cause, and preparation of patients for pancreatic surgery when knowledge of the ductal anatomy is important.

ERCP is contraindicated in patients who are unable or unwilling to cooperate and in patients who are physically unable to tolerate the procedure. It is also contraindicated in patients with recent myocardial infarction, severe pulmonary disease, coagulopathy, or pregnancy.

ERCP may also be contraindicated in patients with acute pancreatitis, unless the clinical situation necessitates the procedure.

Procedure

Prior to the procedure

Barium studies should not be conducted within 72 hours before ERCP because the residual barium can obstruct the view of the contrast medium in the ducts. Because ERCP usually takes longer than routine EGD, IV moderate sedation and analgesia is likely to be heavier.

Before the procedure, the patient should fast for 6 hours. All equipment should be set up and tested in a radiographic examination room. The physician should be notified if the patient has a known allergy to contrast media, and all necessary precautions should be taken.

In the event that therapeutic intervention is required, all ERCP therapeutic accessory equipment must be readily available.

During the procedure

- The patient is placed in either the prone or left lateral position.
- A bite block is placed into the patient's mouth to protect the patient's teeth and to prevent damage to the endoscope.
- A side-viewing duodenoscope is passed into the second part of the duodenum.
- Glucagon may be injected intravenously to suppress duodenal peristalsis and enhance visualization.
- When the endoscope is in the proper position to view the ampulla of Vater, the patient is moved to the prone position.
- The endoscopist passes a plastic cannula through the endoscope and maneuvers it into the orifice of the ampulla of Vater. Further adjustment of the cannula using the endoscope's elevator control allows it to enter the pancreatic duct or the common bile duct.
- To be certain that the contrast medium is free of air bubbles, the cannula must be primed with contrast before being inserted into the endoscope. Radiocontrast material is injected through the cannula. When the contrast medium is injected, the amount injected should be stated verbally. Contrast should be injected slowly to avoid overfilling the duct.
- X-ray films are taken to identify the configuration of the appropriate ductal system.
- The patient is observed for any allergic reactions to the radiocontrast dye.
- Before the scope is withdrawn, a biopsy examination or cytological brushing may also be done.

After the procedure

After the procedure is completed, delayed X-ray films may be ordered to determine the time and amount of drainage of the ducts.

The gastroenterology nurse is responsible to:

- Assess, monitor, and document oxygen saturation and vital signs.
- Observe the patient for abdominal distention and signs of pancreatitis such as chills, low-grade fever, pain, vomiting, and tachycardia.
- Maintain NPO status until the patient's gag reflex returns or further orders are written.
- Administer antibiotics as ordered.
- Check the patient's temperature every 4 hours for 48 hours for possible signs of perforation or infection.
- Offer a light meal 2-4 hours after the procedure. The day after the procedure, a full diet may be resumed.

Considerations

The most significant complications associated with ERCP are pancreatitis and sepsis. Injury to the pancreas can be the result of mechanical, chemical, enzymatic, microbiological, thermal, or hydrostatic factors. If pancreatitis does result, it usually occurs within 2-4 hours after the procedure.

Another frequent complication of ERCP is biliary sepsis, especially in patients with partial obstruction of either the pancreatic or common bile ducts. The introduction of infection into a stagnant duct system can result in cholangitis, pancreatitis, and septicemia. If a pseudocyst is present, ERCP should be used only as an immediate preoperative procedure under antibiotic prophylaxis. Parenteral antibiotics, usually gentamicin sulfate and/or ampicillin, should be given immediately to patients diagnosed by ERCP as having biliary or pancreatic stasis.

Additional potential complications include aspiration, bleeding, perforation, respiratory depression or arrest, and cardiac arrhythmia or arrest. Patients may experience a rise in temperature, chills, nausea, vomiting, abdominal pain, or ascending cholangitis.

In many patients, an asymptomatic rise in serum amylase is noted following ERCP. Unless accompanied by abdominal pain, this rise in amylase is usually insignificant and subsides shortly.

SMALL BOWEL ENTEROSCOPY

Small bowel enteroscopy (SBE) permits visualization of the small bowel beyond the ligament of Treitz. Small bowel enteroscopes are 230 cm (centimeters) in length and are designed to examine the small bowel to the full extent of their length.

Indications

SBE is indicated for patients with GI bleeding of suspected small bowel origin, with continued or intermittent blood loss, in whom a GI bleeding site has not been found despite exhaustive testing.

SBE is useful in small bowel tissue sampling for the diagnosis of celiac disease and is indicated for patients with suspected small bowel abnormality out of the reach of the standard endoscope.

Because of the length of the enteroscope, it is important to check the length and compatibility of accessories before beginning a procedure. Fluoroscopy may be used to assist in the examination but is not required.

Contraindications for this examination are the same as those for EGD.

Procedure

Several SBE techniques are used, including the following:

- **Small bowel enteroscopy.** This technique uses a small bowel enteroscope. Depending on the type of enteroscope (i.e., single or double balloon), enteroscope channels range from 2.2 cm to 3.8 cm with diameters from 8.5 cm to 11.6 cm (ASGE, 2015). The enteroscope is similar to a standard GI endoscope, which accepts forceps and hemostasis devices in its biopsy channel. The patient is positioned in the left lateral position, and a bite-block should be placed into the patient's mouth to protect the patient's teeth and prevent damage to the enteroscope. With the use of a protective supporting overtube, the enteroscope is inserted through the mouth, passing through the esophagus and stomach and into the small intestine, where it can advance to its full length to examine this area of the GI tract.
- **Balloon-assisted enteroscopy.** This technique allows the advancement of a long endoscope into the small intestine for both diagnostic and therapeutic purposes. Balloon-assisted enteroscopy is either single-balloon enteroscopy or double-balloon enteroscopy. A spiral overtube device assists the scope to pass deep into the small intestine. The endoscope is advanced by alternately inflating and deflating the balloon(s) and pleating the small bowel over the insertion tube. Accessories such as biopsy forceps, dilating devices, and cautery probes can be passed through the channels of the scope in order to biopsy or treat abnormal findings in the small bowel. The procedure may take several hours. Most procedures are performed through the mouth, although a rectal approach may allow better access to lesions in the lower part of the small bowel.
- **Sonde enteroscope or peristalsis method.** This technique uses a pediatric colonoscope as a push enteroscope to advance a long, thin, flexible sonde enteroscope into the small bowel. A balloon on the tip of the small bowel enteroscope is inflated to hold it in place, and the pediatric colonoscope is withdrawn. IV metoclopramide is administered, and the patient is sent to a recovery area for several hours while peristalsis advances the enteroscope through the small bowel. Once the enteroscope is positioned in the distal ileum, the small bowel is examined thoroughly while the enteroscope is slowly withdrawn. The complete procedure lasts 8-10 hours and enables visualization of all 20 feet of small bowel. A major disadvantage of this method is that, because this scope does not accommodate accessories, no therapeutic interventions can be done.

Considerations

Complications with SBE are perforation, pancreatitis, and gastric mucosal stripping. Patients must be observed post-procedure for significant abdominal distention as a result of the prolonged procedure and air insufflation.

ANOSCOPY

Anoscopy is done using an anoscope, which is a clear plastic or metal speculum designed for examining the anus and lower rectum. One type is cylindrical, with one side that is incomplete or slotted. The mucosa to be evaluated protrudes through this slot. By withdrawing and reinserting the anoscope in multiple orientations, excellent visualization of the entire circumference of the anal canal is possible.

Indications

Anoscopy is indicated for:

- Evaluation of bright red rectal bleeding in adults and children. The most common causes are hemorrhoids and fissures, although hemorrhoids are not common in children.
- Suspected protein allergy, the most common cause of rectal bleeding in neonates.
- Before sigmoidoscopy or colonoscopy.

Procedure

For anoscopy, the patient is positioned in the Sims left lateral position or the knee-chest position, which allows the sigmoid colon to straighten, promoting better visualization. Special proctological tilt tables (breakaway tables) are available that permit the patient to be placed in an inverted position with the head down against a headrest. To minimize embarrassment, the patient is draped, exposing only the lower buttocks area.

First, a visual and digital rectal examination is performed using a gloved, well-lubricated index finger. The little finger is used for infants under 2 years of age. At this time, the examiner looks for anal anomalies, sphincter tone, polyps or adenomas, rectal prolapse, extracolonic masses, and hemorrhoids. Any stool remaining on the examining finger may be checked for blood.

A warm, well-lubricated anoscope is inserted slowly, with the beveled surface of the tip facing laterally. The general appearance of the mucosa is noted, and the area is examined for fissures, abscesses, or signs of anal papillitis or cryptitis. Biopsies may be necessary to determine if abnormal mucosa is present.

Considerations

Severe spasm induced by digital examination usually signifies low-lying inflammatory bowel disease, such as proctitis or ulcerative colitis (UC).

PROCTOSIGMOIDOSCOPY

Proctosigmoidoscopy, also known as **rectosigmoidoscopy**, is an examination of the rectum and the sigmoid colon using a rigid proctosigmoidoscope. The proctosigmoidoscope is a small, hollow, stainless steel or plastic disposable tube that is 25-30 cm long and approximately 1.5 cm in diameter. A smaller **proctoscope** may be used with newborns or infants or for patients with strictures. A light source is attached to the end of the scope.

Indications

Indications for proctosigmoidoscopy include:

- Melena or bleeding from the anorectal area.
- Persistent diarrhea.
- Change in bowel habits.
- Passage of pus and mucus.
- Suspected chronic inflammatory bowel disease.
- Bacteriology and histological studies.
- Surveillance of known rectal disease.
- Rectal pain.
- Screening for suspected polyps or tumors.
- Foreign body removal.
- As an adjunct to a barium enema.
- Surveillance following rectal surgery.

Contraindications include severe necrotizing enterocolitis, toxic megacolon, painful anal lesions, severe cardiac arrhythmia, and patients who are unwilling or unable to cooperate.

Procedure

Prior to the procedure

Most patients are given a hypertonic phosphate or saline enema the morning of the examination. Patients who have UC or acute diarrhea can be examined without the use of cleansing enemas or laxatives. Medication is rarely needed for adult or adolescent patients. Young children may require sedation and analgesia. Infants under 3 months of age usually tolerate the procedure well without sedation.

During the procedure

With the patient in either a knee-chest or Sims left lateral position, a visual and digital rectal examination is conducted. Then the proctosigmoidoscopy is performed:

- While the patient bears down, the instrument is easily passed into the rectum, the obturator is removed, and the instrument is passed slowly into the colon. If spasm or difficulty is encountered, the scope should be withdrawn until the full lumen is seen, then a second attempt is made.
- During insertion, the patient should breathe deeply with the mouth open to keep the abdominal muscles relaxed. Cramping or a desire to defecate can be relieved by having the patient relax and take deep breaths or pant.
- If air is insufflated into the lumen to enhance visualization, the patient will typically experience some flatulence as the air moves down the bowel and escapes.
- Once the desired depth is reached, the proctosigmoidoscope is gradually withdrawn, thus allowing examination of all sides of the bowel.
- A large cotton swab or suction may be used to remove any fecal matter, blood, or mucus that is obscuring vision.
- The examiner notes the color and friability of the mucosa, bleeding sites, petechiae, and ulcers. Mucosal friability is established if pinpoint bleeding is noted immediately after application of mechanical pressure with a cotton swab or cytology brush.
- If a biopsy examination is indicated, the site of the lesion or specimen and its distance from the anus should be recorded.

Throughout the procedure, the gastroenterology nurse should monitor the patient's vital signs and assess abdominal distention, pain tolerance, and the warmth, color, and dryness of the skin. Sudden changes in position during the procedure should be prevented. Infants and children may need to be swaddled or held.

After the procedure

After the procedure is finished, the table is returned to a low position and the patient should be brought slowly to a seated position before standing. The patient should be observed for signs of bleeding or perforation. Diet, fluids, and activity can return to normal following the procedure unless complications occur.

Considerations

Potential complications of rigid proctosigmoidoscopy include perforation, minimal bleeding from lacerations, transient abdominal discomfort, and cardiac arrhythmias.

If perforation is suspected and confirmed by clinical symptoms and radiographic examination, prompt surgical intervention is indicated.

If there is excessive bleeding from a biopsy site, the patient's bleeding status (i.e., PT, PTT) should be checked and the biopsy site reexamined. Either silver nitrate or electrocautery of the bleeding area will usually stop the bleeding.

FLEXIBLE SIGMOIDOSCOPY

Flexible sigmoidoscopy is the examination of the rectum, the sigmoid, and the descending colon using a flexible sigmoidoscope or colonoscope. The flexible fiberoptic or video sigmoidoscope has largely replaced the standard 25-30 cm rigid sigmoidoscope for routine examination. The flexible fiberoptic or video sigmoidoscopes, which measure up to 65 cm in length, have a much greater range than the older, rigid scopes. Flexible instruments are capable of reaching the descending colon in a great majority of patients, and it is possible to reach the splenic flexure. In addition, patients seem to tolerate flexible sigmoidoscopy better than rigid proctosigmoidoscopy.

Considerations

Flexible sigmoidoscopy is indicated for:

- Routine screening of adults over age 45 (American Cancer Society [ACS], 2018).
- Inflammatory bowel disease.
- Chronic diarrhea.
- Pseudomembranous colitis.
- Radiation colitis.
- Sigmoid volvulus.
- Foreign body removal.
- Lower GI bleeding.

Sigmoidoscopy is contraindicated in patients who have fulminant colitis; toxic megacolon; severe, acute diverticulitis; peritonitis; or the inability or unwillingness to cooperate.

Procedure

Prior to the procedure

Except in patients with watery diarrhea or suspected colitis, preparation for flexible sigmoidoscopy includes two warm, tap water or sodium biphosphate (Fleet) enemas 1-2 hours before the examination. Warm tap water enemas are not given to pediatric patients due to the risk of electrolyte imbalances. A more thorough bowel preparation may be indicated by the patient's disease status.

Compliance with the bowel preparation ordered by the physician is verified. A medical history is obtained, including medications, allergies, and any information pertaining to the current complaint. Patients usually do not require sedation and analgesia.

During the procedure

- After a visual and digital rectal examination, the patient is instructed to breathe slowly and deeply to relax the anal sphincters while the well-lubricated endoscope is inserted.
- While the endoscope is being advanced to the rectosigmoid junction, air is insufflated to facilitate passage into the sigmoid and descending colon.
- While the instrument is slowly withdrawn, the physician examines the sigmoid colon, and then the rectal and anal mucosa. Suction may be used to remove residual matter, blood, or mucus that obscures vision.

During the procedure, the gastroenterology nurse monitors the patient's vital signs; color, warmth, and dryness of the skin; abdominal distention; level of consciousness; pain tolerance; and vagal response. The physician should be notified if the abdomen is becoming excessively distended secondary to air insufflation. To offer support and to help the patient cooperate, it may be helpful to offer back rubbing or instructions in breathing techniques.

After the procedure

After the procedure is completed, the patient may resume normal activity if no sedation has been administered. The patient should be observed for post-biopsy bleeding and persistent abdominal pain or distention.

Considerations

Potential complications of flexible sigmoidoscopy include bleeding and perforation.

COLONOSCOPY

Colonoscopy involves direct visualization of the lower GI tract using a long, flexible endoscope. Fiberoptic and video colonoscopes are similar in design to upper GI endoscopes but are longer, ranging in length from 120 cm to 180 cm. Suction removes liquid secretions that obscure vision, thereby permitting safe advancement of the instrument. Air and water controls help keep the lumen distended and the optics at the tip of the instrument clean.

An experienced endoscopist may view the entire colon from the rectum to the ileocecal valve, reaching the cecum in most patients and often the distal ileum.

Indications

Colonoscopy is indicated for diagnosis and diagnostic and therapeutic procedures, including:

- Active or occult lower GI bleeding, such as hematochezia, melena with a negative upper GI investigation, unexplained fecal occult blood, and unexplained iron-deficiency anemia.
- Abnormalities found on radiographic examination.
- Suspected cecal or ascending colonic disease.
- Surveillance for colon neoplasia in patients who have had previous colon cancer or previous colon polyps.
- Screening in patients 45 years of age or older. Patients with an increased or high risk of colorectal cancer may need to start colorectal cancer screening before age 45 and be screened more often (ACS, 2018).
- Surveillance in patients with chronic UC of several years' duration.
- Diagnosis or management of chronic inflammatory bowel disease.
- Chronic, unexplained abdominal pain.
- Suspected polyps, rectal or colonic strictures, or cancer.

Colonoscopy is contraindicated in patients with fulminant ulcerative colitis; acute ischemic colitis; acute radiation colitis; suspected toxic megacolon; suspected perforation; acute, severe diverticulitis; the presence of barium; imperforate anus; massive colonic bleeding; shock; acute surgical abdomen or a fresh surgical anastomosis; or patients who are physically unable to tolerate the procedure.

Relative contraindications include massive hematochezia, infectious bowel disease, pregnancy, coagulation abnormalities, unstable cardiovascular status, and patients who are unable or unwilling to cooperate.

Procedure

Prior to the procedure

The goal of a colonoscopy preparation is to rapidly purge the colon of all fecal material without causing alteration of the colonic mucosa. Dietary modifications are often used along with oral cathartics and include a low residue diet for several days before the procedure. This has been shown to be at least as effective as a clear liquid diet.

A variety of colonoscopy preparations are available. Use of a split-dose bowel cleansing regimen is strongly recommended for elective colonoscopy (ASGE, 2015). Ideally, the second dose of split preparations should begin 4-6 hours before the colonoscopy, with completion at least 2 hours before the procedure time.

Adequate hydration is an important adjunct to any bowel preparation before colonoscopy. The patient should follow the instructions provided with the bowel preparation regarding the amount and type of fluids to be consumed.

If rectal bleeding or severe abdominal pain occurs during bowel preparation, the physician should be notified immediately.

A variety of colonoscopy preparations are used for children. The child's age, size, and reason for the procedure must be considered before deciding what type of preparation will be used. Osmolar electrolyte solutions are often not well-tolerated in the volume necessary for adequate bowel preparation and may be administered via nasogastric infusion. Children may tolerate and have good results with a regimen of a clear liquid diet for 3 days before the procedure, in addition to enemas and cathartics.

During the procedure

- The patient is given an opportunity to urinate before the procedure.
- Because colonoscopy takes longer and is more uncomfortable than flexible sigmoidoscopy, it is typical to administer IV moderate sedation and analgesia to promote relaxation and diminish discomfort. General anesthesia is used routinely in pediatric patients.
- Colonoscopy is usually begun with the patient in the left lateral decubitus position.
- After a visual and digital examination, the lubricated colonoscope is inserted into the rectum while the patient breathes slowly and deeply.
- The physician insufflates a small amount of air to help dilate the bowel lumen.
- During insertion of the colonoscope, the objective is to reach the cecum as quickly and as safely as possible.
- When appropriate, the gastroenterology nurse assists the physician in repositioning the patient smoothly to facilitate passage through the splenic flexure, the transverse colon, the hepatic flexure, the ascending colon, and the cecum. In addition, the gastroenterology nurse may apply pressure to areas of the abdomen as requested by the physician to assist with passage of the instrument.
- During withdrawal, the objective is meticulous inspection. The mucosa is scanned by changing the position of the flexible tip while the instrument is slowly withdrawn. The bowel wall is examined for abnormalities such as ulcerations, bleeding sites, polyps, inflammation, or tumors.
- Direct visualization allows for therapeutic procedures such as polypectomy, dilation, decompression, and fulguration of bleeding sites.

Throughout the procedure, the gastroenterology nurse monitors the patient for changes in oxygen saturation, vital signs, and ECG; color, warmth, and dryness of the skin; abdominal distention; level of consciousness; vagal response; and pain tolerance. The physician is notified if the abdomen becomes excessively distended. To help the patient cooperate more readily, instructions in breathing technique may be offered.

The physician may order the administration of atropine or glucagon in an attempt to decrease bowel spasms and/or motility. If either drug is used, the gastroenterology nurse should closely monitor the patient for hypotension and irregular or rapid pulse.

After the procedure

Following colonoscopy, the gastroenterology nurse monitors and documents vital signs frequently per institutional policy until the patient is stable and observes for signs of complications.

Considerations

Complications include bleeding, vomiting, a change in vital signs, severe or persistent abdominal pain and/or distention, and abdominal rigidity.

Major complications occur in less than 1% of patients undergoing colonoscopy (ASGE, 2011). The two major complications of perforation and hemorrhage are most likely to occur during or after polypectomy, which is discussed in Chapter 30.

Other potential complications of colonoscopy include medication reactions such as cardiac arrhythmias or arrest, respiratory depression or arrest, vasovagal reactions, and cardiac failure or hypotension related to overhydration or underhydration of a susceptible patient during bowel preparation. Excessive bleeding from biopsy sites is rare, unless the patient has a coagulation disorder or has been on products that contain aspirin. Unsuspected serosal tears and retroperitoneal emphysema have occasionally been found during laparoscopy.

Colonoscopic complications may be minimized by advancing the colonoscope with care and with a clear view and by avoiding overdistention of the colon with air. Fluid and electrolyte status of elderly patients and those with renal and cardiac disease should be considered during bowel preparation.

ADDITIONAL ENDOSCOPIC DIAGNOSTIC PROCEDURES

The technology available for endoscopic visualization of the GI tract continues to advance. Additional diagnostic procedures available to the endoscopist include esophageal and small bowel enteroscopy by capsule (capsule endoscopy), endoscopy through an ostomy, and endoscopic ultrasonography.

Capsule Endoscopy

Capsule endoscopy is useful to help diagnose difficult and challenging esophagus and small bowel conditions where conventional methods have failed.

- **Capsule enteroscopy** is a diagnostic tool approved for imaging of the small intestine. Capsule enteroscopy is indicated for diagnosis of diseases of the small bowel, including obscure bleeding, irritable bowel syndrome, Crohn's disease, celiac disease, chronic diarrhea, malabsorption, and small bowel cancer.
- **Capsule esophagoscopy** is a diagnostic tool approved for imaging of the esophagus. Capsule esophagoscopy is indicated for diagnosis of diseases of the esophagus, including gastric esophageal reflux disease (GERD).

See Chapter 26 for complete information regarding capsule endoscopy.

Endoscopy through an ostomy

Occasionally it may be necessary to directly visualize a segment of the small or large intestine using a flexible endoscope that has been inserted through a stoma.

Indications

Using the ostomy as an entry for endoscopy is indicated for evaluation of an anastomotic site, identification of recurrent disease (e.g., Crohn's disease or cancer), or visualization and/or treatment of GI bleeding.

This technique is contraindicated in patients with a recent ostomy or bowel surgery, poor bowel preparation, suspected bowel perforation, presence of a large peristomal hernia, or massive lower GI bleeding.

Procedure

Before the procedure, the ostomy site should be examined to determine the size of the stoma, the type of effluent that can be expected during the procedure, and what ostomy appliance will be needed following the procedure. Compliance with bowel preparation orders should be confirmed and their effectiveness ascertained. The old ostomy appliance is gently removed, working from the top downward, taking care to push the skin away from the appliance.

During the procedure, the patient is placed in the supine position, with fluid-impervious towels draping the stoma. A large supply of sponges should be provided, especially for an ileostomy. The endoscope is held at a right angle to the abdominal wall to facilitate entry through the stoma. The gastroenterology nurse maintains a tight seal around the endoscope as it enters the stoma so that the endoscopist is able to accomplish adequate insufflation. Throughout the procedure, the gastroenterology nurse also monitors the patient's oxygen saturation and vital signs; color, warmth, and dryness of the skin; abdominal distention; level of consciousness; and pain tolerance and provides emotional support.

After the procedure is completed, the gastroenterology nurse monitors and documents oxygen saturation and vital signs. The peristomal area should be cleaned carefully with a mild soap and water, and the skin should be thoroughly dried. An appropriate skin barrier and collecting pouch should be applied immediately.

Considerations

The patient is observed for stomal bleeding, vomiting, change in vital signs, severe or persistent abdominal pain, and abdominal rigidity.

Endoscopic ultrasonography

Endoscopic ultrasonography (EUS) technology combines the endoscope with ultrasonography to enhance visualization of the GI tract, which is obscured in conventional ultrasonographic examinations by intra-abdominal gas and bony structures. With an oblique-viewing endoscope that has an ultrasonic transducer built into the tip, high-frequency ultrasonic beams can be targeted in close proximity to existing lesions.

Indications

EUS offers better-quality resolution, which enhances evaluation of the histological structure of targeted lesions. In addition, esophageal wall thickness may be evaluated and assessed. The walls of the esophagus, stomach, duodenum, and colon may be visualized, as well as the structure of several contiguous organs.

The act of scanning through the gastric wall, combined with changes in the patient's position, enables the study of the gastric wall, the gallbladder, the pancreas, the kidneys, the left lobe of the liver, the spleen, the aorta, the inferior vena cava, and various tributaries of the extrahepatic portal vein system.

Procedure

An inflatable balloon covers the ultrasonic transducer and can be filled with water to compress the esophageal wall, eliminate the air space, and improve the quality of resolution, which enhances the evaluation of the targeted lesion.

Considerations

EUS, as compared with radiographic and endoscopic examinations, has many advantages for detecting and staging of lesions in the wall of the GI tract. With the addition of needle aspirate and biopsy potential, EUS has become a valuable tool in the identification of GI cancers and treatment decisions.

CASE SITUATION

The gastroenterology nurse manager of a busy hospital endoscopy unit has hired a new gastroenterology nurse. Hannah Day has finished her hospital orientation and started in the unit. She is vaguely familiar with gastroenterology procedures but has no gastroenterology experience. The gastroenterology nurse manager needs to set up a training program for Hannah.

Points to think about

1. The gastroenterology nurse manager is showing Hannah how to assist with a colonoscopy. What types of things must she be taught?
2. What might the supervisor consider regarding procedural information when providing an educational experience for Hannah?

Suggested responses

1. Before Hannah assists with a colonoscopy, it is important that she know:
   a. Basic colon anatomy and physiology.
   b. Indications for colonoscopy.
   c. Techniques of preparing the colon for colonoscopy.
   d. How to relate possible diagnosis with the equipment setup.
   e. How the gastroenterology nurse can facilitate the passage of the scope by using gentle external hand pressure to prevent the scope from bowing in the sigmoid, and how different patient positions can facilitate passage of the scope in difficult cases.
   f. How to comfort the patient by talking gently and providing encouragement and praise, and that gently stroking the patient's back or forehead can be comforting and reduce the need for medication.
   g. Pertinent information that should be provided to the patient before discharge.
2. Patients are usually receptive and supportive of a teaching situation if they feel they will not be left alone in the care of a novice. The gastroenterology nurse manager should let the patient know that he or she will be teaching a new gastroenterology nurse but will be present throughout the procedure. The manager should leave technical discussions of any pathological conditions encountered and the treatment that will be used for them until after the procedure to avoid causing the patient any additional stress.

When teaching, the gastroenterology nurse manager should always remember to support statements with a rationale.

REVIEW TERMS

anoscopy, capsule endoscopy, colonoscopy, endoscopic retrograde cholangiopancreatography (ERCP), endoscopy, esophagogastroduodenoscopy (EGD), fiberoptic bundles, flexible sigmoidoscopy, proctoscope, proctosigmoidoscopy, rectosigmoidoscopy, small bowel enteroscopy (SBE), video endoscopy

REVIEW QUESTIONS

1. The endoscopes used in EGD can visualize the upper GI tract as far as the:
   a. Pylorus.
   b. Ampulla of Vater.
   c. Proximal duodenum.
   d. Ileocecal valve.

2. According to ASA guidelines, before moderate sedation or analgesia is provided for an EGD, the adult patient should be NPO from solids or full liquids for at least:
   a. 2 hours.
   b. 6 hours.
   c. 12 hours.
   d. 24 hours.
3. The most significant complication(s) associated with ERCP is (are):
   a. Perforation.
   b. Adverse effects of medication.
   c. Hemorrhage.
   d. Pancreatitis and sepsis.
4. The most common cause(s) of bright red rectal bleeding in adults and children is (are):
   a. Inflammatory bowel disease.
   b. Perforation.
   c. Hemorrhoids and fissures.
   d. Bleeding ulcers and varices.
5. One contraindication for rigid proctosigmoidoscopy is:
   a. Severe cardiac arrhythmias.
   b. Previous rectal surgery.
   c. Rectal bleeding.
   d. Rectal pain.
6. The usual bowel preparation for flexible sigmoidoscopy in adults is:
   a. A single warm tap-water enema.
   b. Two warm tap-water or sodium biphosphate enemas.
   c. A two-day liquid diet, followed by a strong laxative.
   d. Electrolyte lavage.
7. For an enteroscopy procedure, the patient should be positioned in the __________.
   a. Prone position.
   b. Supine position.
   c. Right lateral position.
   d. Left lateral position.
8. Patients who will undergo colonoscopy usually receive:
   a. Local anesthesia only.
   b. IV moderate sedation and analgesia.
   c. General anesthesia.
   d. No sedation and analgesia or anesthesia.
9. Distention of the abdomen during flexible sigmoidoscopy is most likely caused by:
   a. Excessive insufflation of air.
   b. Excessive amounts of water used for irrigation.
   c. Perforation.
   d. Colonic obstruction.
10. Small bowel enteroscopy (SBE) is indicated for patients with:
   a. Peptic ulcers.
   b. Inflammatory bowel disease.
   c. Persistent blood loss with no identifiable source.
   d. Intestinal polyps.

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Chapter 24

MANOMETRY

This chapter will provide the gastroenterology nurse with information on manometry procedures used in the diagnosis of motility abnormalities. These manometry procedures include esophageal manometry, anorectal manometry, antroduodenal manometry, and sphincter of Oddi manometry.

Learning objectives

After reviewing the content of this chapter, the gastroenterology nurse will be able to:

1. Describe the equipment and techniques typically used in manometry studies.
2. List the indications and contraindications for manometry procedures involving the esophagus, stomach, small bowel, sphincter of Oddi and anorectum.
3. Discuss the types of manometry tracings that are observed in patients with normal test results and in patients with common abnormal findings.

BASIC PRINCIPLES OF GI MANOMETRY

Gastrointestinal (GI) **manometry** is a diagnostic test that measures changes in intraluminal pressure and the coordination of activity in the muscles of the GI tract. The most common manometry studies include esophageal and anorectal manometry. Sphincter of Oddi and antroduodenal manometry studies are performed less frequently.

Obtaining pressure measurements involves using catheters with external transducers or direct intraluminal transducers. The electrical signals from the transducers are transmitted to a computer that produces a pressure display. The pressures are plotted on a graph or color plot in mmHg.

Types of manometry catheters

Catheters are available with several different technologies. Each type has distinct advantages and disadvantages. They include solid-state catheters, disposable air-filled catheters, water-perfused pressure sensors, and high-resolution catheters.

Solid-state catheters contain small, direct intraluminal transducers. Solid-state catheters are available in several configurations, ranging from three to 38 sensors. The sensors positioned in the upper esophageal sphincter (UES) and lower esophageal sphincters (LES) are circumferential sensors that provide an average sphincter pressure. Solid-state catheters are very accurate for UES and pharyngeal studies, as they are not position-dependent. The patient can sit up for a more accurate study, and the response time is faster to record the rapid pharyngeal upslope.

Disposable air-filled catheters are connected to external transducers. These catheters are configured with four circumferential sensors. The advantages of air-filled catheters are that they have a smaller diameter, are single-patient use, and require minimal equipment maintenance.

Water-perfused pressure sensors are connected to external transducers on a water-perfusion pump. Compressed air pushes water through capillaries—the **pressure transducers**—and the catheter at a steady rate. The number of sensors and the spacing vary with catheter models. There is added maintenance and nursing responsibilities with the water-perfusion system. This includes changing the water reservoir daily, ensuring that no air bubbles are in the transducers or catheter lumens, checking that the system capillaries and catheter are patent, and confirming that the pump maintains a constant pounds per square inch (psi).

High-resolution anorectal manometry (HRAM) catheters have approximately 23 sensors spanning the entire sphincter length. This allows visualization of the entire sphincter length and rectum, requiring the catheter to be positioned only once. This configuration is often more comfortable for the patient, as they don't feel the movement of the catheter with the station pull-through. HRAM catheters also allow visualization of the entire sphincter and rectal dynamics. Colors are assigned to the pressures to allow clear visualization of the multiple pressure channels on a color plot.

ESOPHAGEAL MANOMETRY

Esophageal manometry measures pressure throughout the regions of the esophagus, which include the upper esophageal sphincter (UES), the esophagus body, and the lower esophageal sphincter (LES).

Indications

Indications for performing esophageal manometry include:

- Evaluation of dysphagia after a mechanical obstructing lesion has been excluded by barium esophagram or endoscopy.
- Evaluation of chest pain after previous cardiac testing has ruled out a cardiac origin.
- Preoperative evaluation for antireflux, hiatal hernia repair, and bariatric surgeries to assess peristaltic pressure in the esophageal body.
- Evaluation of gastroesophageal reflux disease (GERD) to help define the mechanism for the reflux (e.g., low LES pressures, weak or absent peristalsis in the distal esophagus) and the degree of reflux-related changes in the esophagus body.
- Determination of the location of the LES for placement of reflux detecting probes, which require placement of the pH sensor 5 cm above the proximal LES border for pH impedance and 6 cm above the proximal LES border for Bravo probe placement.
- Exclusion of esophageal involvement in generalized GI tract diseases, such as scleroderma and pseudo-obstruction.

Esophageal manometry is generally a safe procedure with minimal risks and contraindications. It is, however, contraindicated in patients who:

- Are unwilling or unable to cooperate.
- Have cardiac instability, severe coagulopathy, suspected complete or near complete obstruction, or recent gastric surgery.

The recent administration of sedatives or narcotics for endoscopic placement may not be a contraindication in some cases.

Considerations

The complications for esophageal manometry are rare but can include aspiration and vagal stimulation, resulting in slowing of the heart rate. This could cause faintness, dizziness, and syncope. The treatment for bradycardia is administration of atropine. Perforation has never been described.

Types of esophageal manometry testing

High-resolution manometry

High-resolution manometry (HRM) uses multiple pressure sensors (32-38) spaced 1 cm apart to enable visualization of the entire esophagus and sphincters. HRM has the advantage of displaying the dynamics of the entire esophagus. The catheter stays at the same position throughout the study, simplifying the procedure, decreasing the procedure time, and increasing patient tolerance. Colors are assigned to the pressures to allow clear visualization of the multiple pressure channels. This color system graphic is called the "Clouse plot" in honor of the innovator Dr. Ray Clouse.

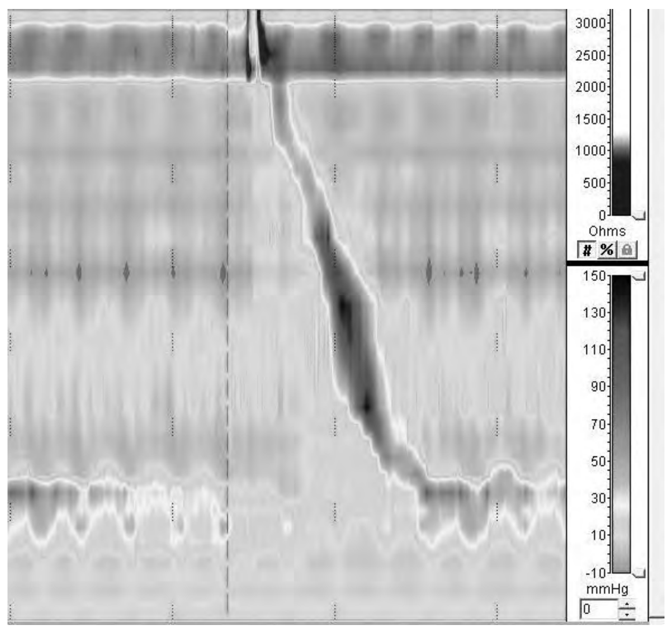

Figure 24.1. High-Resolution Esophageal Manometry with Clouse Plot Display

Impedance manometry

Impedance manometry is the combination of impedance testing and manometry. It provides simultaneous information on esophageal pressure changes and bolus movement with test swallows. This technology measures changes in resistance to alternating current between two metal electrodes. Intraluminal impedance (resistance) rapidly decreases as the high-ionic content (saline swallow) of the bolus passes between the electrodes and rapidly increases as the bolus clears the electrodes. Multiple impedance channels span the esophagus to determine direction and bolus transit. Normal bolus transit is <12 seconds. The esophagus is also challenged with viscous swallows to assess bolus transit.

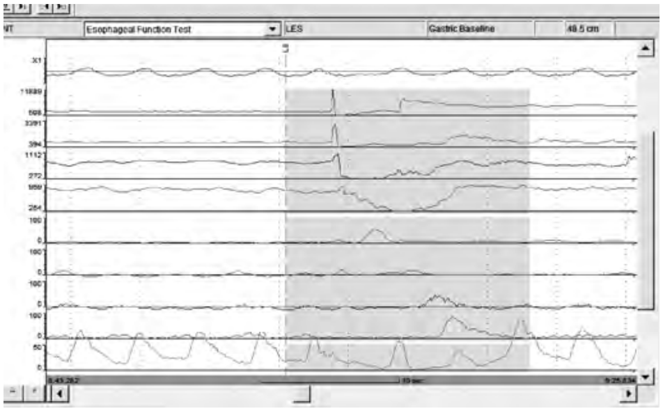

Figure 24.2. Impedance Manometry with Waveform Plot

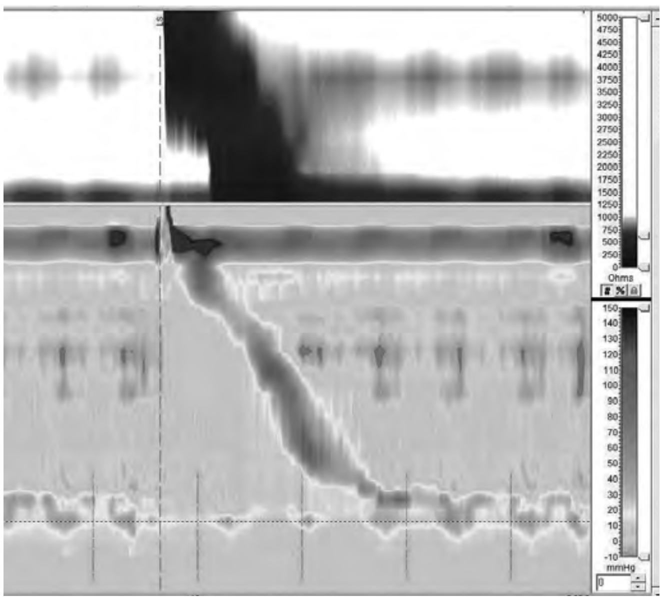

Figure 24.3. High-Resolution Impedance Manometry with Clouse Plot

High resolution manometry with esophageal pressure topography plotting

High resolution manometry (HRM) with the evolving technology of esophageal pressure topography (EPT) plotting (Clouse plot) has introduced new types of pressure measurements and classifications of manometry abnormalities. The Chicago Classification version 3.0 terminology and values listed below have worldwide acceptance (Kahrilas et al., 2015). As this is evolving technology, new parameters and values may change. These measurement parameters can be obtained with systems using HRM.

Lower esophageal sphincter (LES)—esophagogastric junction (EGJ) pressure. This is a measure of LES competency. The high-pressure zone (HPZ) of the LES is measured over three respiration cycles. The measurement is obtained in a quiet data-acquisition area of the esophagus that is free of artifact such as spontaneous esophageal contractions, swallowing, belching, or coughing.

Integrated relaxation pressure (IRP). This is a measure of LES relaxation in response to liquid test swallows. Mean EGJ pressure is measured in all sensors spanning the LES over four contiguous or noncontiguous seconds of lowest pressure over a 10-second window of relaxation. IRP normal values are catheter-technology specific. A medium value among the test swallows is calculated. The upper limit of normal is 15-20 mmHg (Kahrilas et al., 2015), depending on the catheter technology.

Distal contractile integral (DCI). This is a measure of the distal esophageal contractile vigor. This measurement is the segment spanning from the end of the proximal pressure trough to the EGJ, with a calculation of amplitude multiplied by duration, multiplied by length (mmHg x sec x cm). The DCI measurement identifies hypercontractility and hypocontractility conditions. A DCI <100 is a failed contraction, 100-450 is a weak (ineffective) contraction, and >8,000 defines hypercontractility (Jackhammer esophagus). A measurement between 450 and 8,000 is considered a normal result (Kahrilas et al., 2015).

Peristaltic breaks. Peristaltic wave integrity is assessed by measuring gaps in the contour plot of the peristaltic contraction between the UES and EGJ with a 20 mmHg isocontour line. A break >5 cm is considered fragmented and is associated with incomplete bolus transit (Kahrilas et al., 2015).

Distal latency (DL). This is a measure of peristaltic timing from the onset of the swallow (UES relaxation) to an identifiable point in the distal peristalsis sequence, the contractile deceleration point (CDP). This is an important measurement to assess distal esophageal spasm (DES). A value <4.5 seconds is defined as a premature contraction (Kahrilas et al., 2015).

Care and maintenance

Manometry equipment and protocols vary depending on the type of manometry catheter used. Systems consist of a manometric catheter with several **pressure sensors** that interface with a transducer cable and conditioning circuitry to convert the electrical signal to a digital display. A specific computer program is used to display, edit, and analyze the measurements. A report is generated from the measurements obtained.

The system is calibrated per the manufacturer's instructions. It is important to calibrate before each patient to verify that the equipment is in good working order. If using a water-perfused catheter, the catheter and capillaries should be verified as patent and without air bubbles.

After the procedure, non-disposable catheters and catheters not using a sheath require an initial cleaning with an enzymatic detergent, followed by high-level disinfection and rinsing with water.

The manufacturer's recommended cleaning protocol should be followed. Damage to the electrical connections is prevented by protecting them from getting wet or from exposure to the cleaning solutions.

Procedure

A verbal consent should be obtained, and written consent may be required in some organizations.

Prior to the procedure

Before any manometry procedure, the patient should be fully informed of the:

- Purpose of the test.
- Positioning to be used.
- Effective relaxation methods.
- Procedure techniques to be used.
- Approximate length of the procedure.
- Sensations likely to be experienced.
- Risks of the procedure.
- Importance of patient cooperation.
- Need to remain NPO for 6 hours prior to the procedure.

In addition, the gastroenterology nurse should:

- Obtain a patient history as to presenting symptoms.
- Provide and document post-procedure instructions and recommendations for follow-up care.
- Document patient education and comprehension.
- Verify the patient has had nothing by mouth (NPO) for at least 6 hours.
- Provide information and reassurance to decrease the patient's anxiety regarding the procedure.
- Ensure the equipment is functioning properly.
- Adhere to protocol.
- Pass the catheter into the nostril of the patient's choice.
- Gently advance the catheter into the nasopharynx.
- Give the patient swallows of water to ease passing the catheter.

The purpose of the NPO status is to reduce the chance of vomiting and aspiration during intubation and to prevent the physiologic effects of food on the LES. In patients with suspected achalasia, a liquid diet may be advised for 24 hours before the procedure.

Careful explanation and instruction regarding the procedure and nasal intubation will decrease anxiety and improve patient tolerance. The patient should be informed that he or she may experience brief discomfort in the nose and possible mild gagging during the intubation, but these will resolve quickly. The patient will be able to breathe and speak normally.

Patients should continue nitrates, calcium channel blockers, anticholinergics, and respiratory medications, as these are medications that the patient will take long-term, and the study of the esophagus should be done under normal conditions. However, it is important to consult with the attending physician as to his or her preference.

The procedure is generally done without sedation, as sedatives have a direct effect on LES and esophagus body pressure. However, in children, sedation may be required. The patient's nasal passage may be anesthetized with a topical anesthetic such as 2% viscous xylocaine or lidocaine hydrochloride jelly, applied with a cotton-tipped applicator into the nostril of choice.

Intubation

The catheter is usually inserted transnasally but may be inserted orally if not using a solid-state catheter where the sensors can be damaged by biting. The lubricated catheter is introduced in a sitting position. The catheter is gently advanced over the nasal floor with the patient's chin parallel to the floor. Resistance is felt as the tip of the catheter reaches the posterior nasopharynx. When the catheter passes into the hypopharynx (10-15 cm), the patient's head is tilted toward the chest, and the patient is asked to take several sips of water. The catheter is advanced quickly past the gag reflex point and then advanced continuously to a depth per catheter system protocol.

Verification that the distal sensors are in the stomach is done by asking the patient to take a deep breath. Increased pressure will be seen if the sensors are below the diaphragm.

The patient is then positioned on his or her side or back with the head of the bed <30 degrees. The patient should be allowed time to accommodate to the catheter before continuing with the procedure.

Areas of measurement

Areas of measurement in esophageal manometry include LES, esophageal body, and UES studies. Procedure protocol will vary depending on the catheter configuration, high-resolution manometry, or impedance manometry, but the same areas are measured and assessed.

Lower esophageal sphincter study

The LES is a tonically contracted smooth muscle about 3-5 cm in length. The LES relaxes in response to swallowing to allow passage of the bolus into the stomach. LES pressure, length, location relative to the nares and diaphragm, and relaxation in response to liquid swallows are assessed.

The distal sensors are placed in the HPZ of the LES to measure and assess LES resting pressure. LES relaxation is assessed by giving the patient a 5-ml bolus of room temperature water or normal saline if measuring impedance. The sphincter pressure decreases close to gastric pressure in response to the swallows and remains relaxed as the esophageal contractions progress through the esophagus. LES relaxation is assessed with 10 swallows, allowing a 20-30-second interval between swallows.

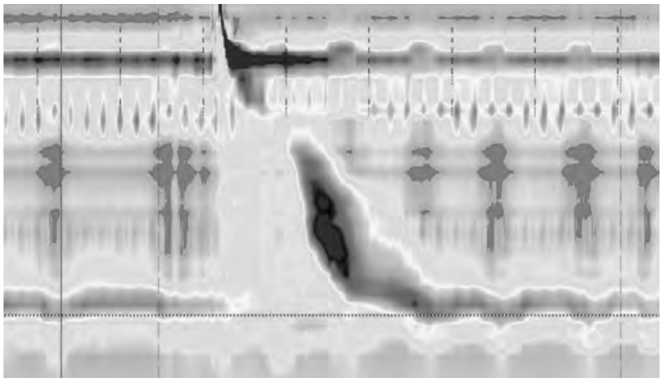

Figure 24.4. HRM Protocol: LES Pressure, Length, and Relaxation in Response to a Liquid Swallow

With conventional catheters, a slow **station pull-through** is performed. The catheter is pulled in 0.5 cm increments to identify the distal border of the LES, the pressure inversion point (PIP), and the proximal border of the LES. With HRM, the LES dynamics are visualized without catheter movement.

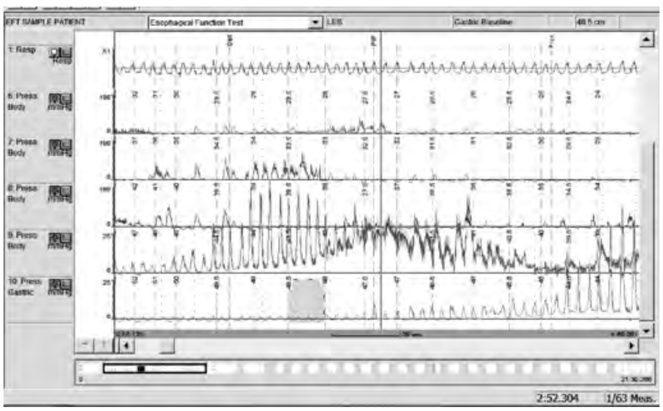

Figure 24.5. LES Pull-Through

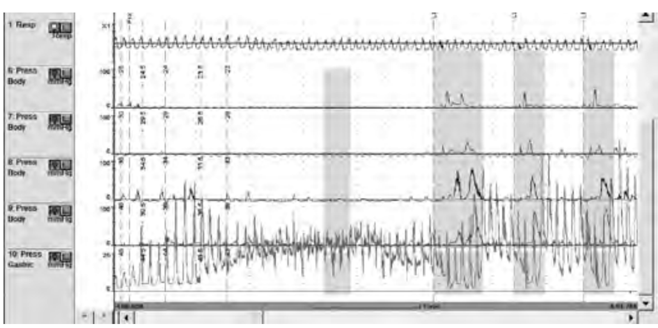

Figure 24.6. LES Pressure and LES Relaxation Measurements

Esophagus body study

The esophagus body study evaluates the esophageal muscle response to controlled swallows. Ten liquid swallows of 5 ml of room temperature water (5 ml of normal saline with impedance studies) are given at 20- to 30-second intervals. At least two sensors are positioned in the distal esophagus at 5 cm and 10 cm above the LES. A normal swallow produces contractions beginning in the proximal esophagus and moving distally with a DCI value between 450 and 8,000.

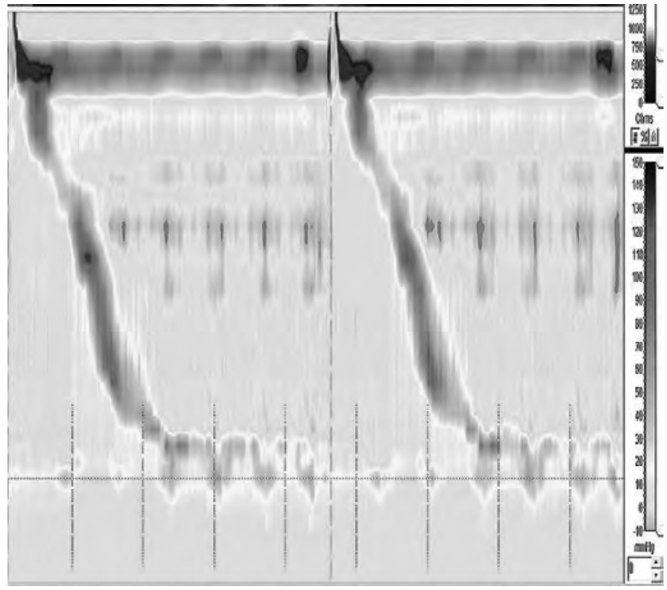

Figure 24.7. HRM Clouse Plot Showing Normal Pressures and Progression with a Liquid Swallow

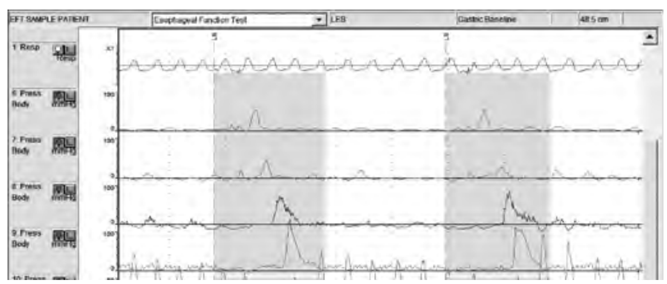

Figure 24.8. Esophagus Body Contractions

Abnormal findings in the esophagus include:

- Simultaneous contractions, where the upslope of the contractions occur at the same time (DL >4.5 seconds).
- Retrograde contractions, which progress from the distal to the proximal esophagus.
- Repetitive contraction with three or more peaks.
- Failed contractions (DCI <100).
- Weak (ineffective) contractions (DCI <450).
- Hypercontractile contractions (DCI >8,000).

It may be necessary to assess the striated muscle in the proximal esophagus in patients with a smooth muscle disorder such as scleroderma or a skeletal muscle disorder such as Parkinson's disease. The proximal sensor is positioned 1 cm below the UES, and five to 10 liquid swallows are performed. It is not necessary to wait 20 seconds between swallows, as the striated muscle recovers quickly.

Upper esophageal sphincter study

The final step of esophageal manometry involves assessing the UES, including UES resting pressure, UES relaxation, and pharyngeal contractions. The UES and pharynx are composed of striated muscle. Pharyngeal contractions and UES relaxations are very rapid, requiring solid-state catheters to accurately record their function.

A circumferential sensor is positioned in the UES, and one to two sensors are positioned in the pharynx. A sensor is positioned in the proximal border of the UES, and five 5-ml liquid swallows are given to assess UES and pharyngeal coordination. The UES should be relaxed while the pharynx is contracting. It is necessary that the measuring sensor is in the proximal border of the UES because the UES moves toward the mouth on swallowing. The UES will move up onto the sensor, and accurate UES relaxation can be recorded. This movement will produce an “M” configuration. Normal UES resting pressure is 30-120 mmHg, and pharyngeal contraction amplitude is >60 mmHg.

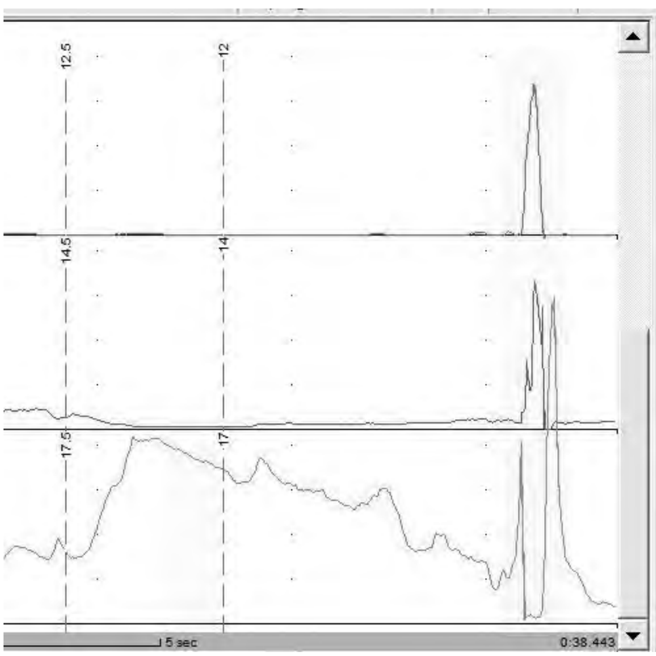

Figure 24.9. Waveform Plot Demonstrating Pharyngeal Contractions and “M” Configuration of UES Relaxation

With HRM, several pressure sensors encompass the UES and pharynx, and a separate UES pull-through is not necessary. A long esophageal length may prevent enough sensors from spanning the UES or pharynx, requiring the catheter to be withdrawn a couple of centimeters. Then, five quick additional swallows are performed. It is not necessary to wait 20 seconds between swallows, as striated muscle recovers quickly.

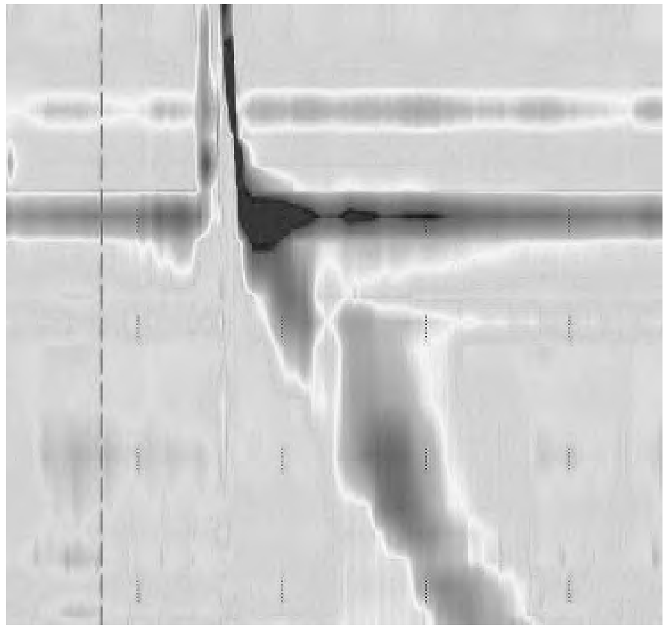

Figure 24.10. Pharynx and UES Relaxation with Clouse Plot

Esophageal Abnormalities

The esophageal abnormalities commonly found on manometry and the manometric plots typical of these abnormalities are discussed below.

Achalasia

Achalasia is a primary esophageal motility disorder. Patients present with a history of slowly progressive, painless dysphagia for solid and liquid food. Nocturnal regurgitation and pulmonary symptoms are common, including aspiration, wheezing, chronic cough, and choking.

The manometric findings in achalasia are:

- Failed peristalsis or spasm in the esophageal body.
- Elevated IRP.

Hypertensive LES, elevated intraesophageal pressure relative to gastric pressure, poor bolus transit, and low distal baseline impedance (DBI) are also common findings.

Achalasia has three subtypes (patterns).

- Type I is identified by 100% failed swallows.
- Type II is identified by >20% of swallows with panesophageal pressurization.
- Type III is identified by >20% of swallows with premature (spastic) contractions.

Occasionally the catheter will not pass through the LES due to a dilated esophagus or hypertensive LES. In this case, the catheter is withdrawn to about 30 cm and reinserted slowly while the patient is swallowing. If the catheter cannot be passed into the stomach, the catheter can be positioned in the esophagus and esophageal swallow data obtained with 10 liquid swallows. This would be considered a failed manometry because stomach measurements cannot be obtained. A direct placement of the manometry catheter under sedation, with visual confirmation or assistance of catheter placement with the endoscope, would be considered for a complete study.

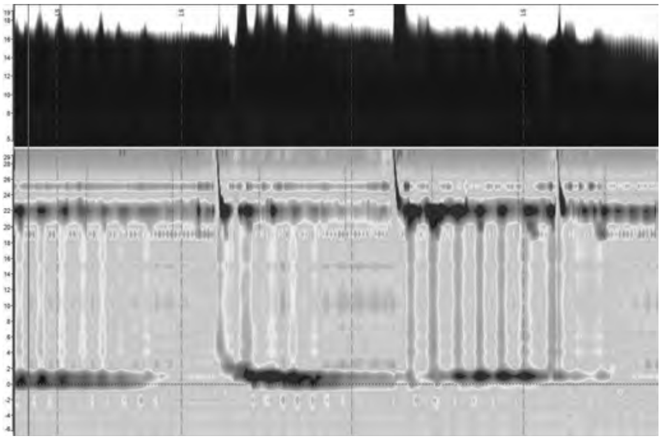

Figure 24.11. Achalasia Displayed with Clouse Plot

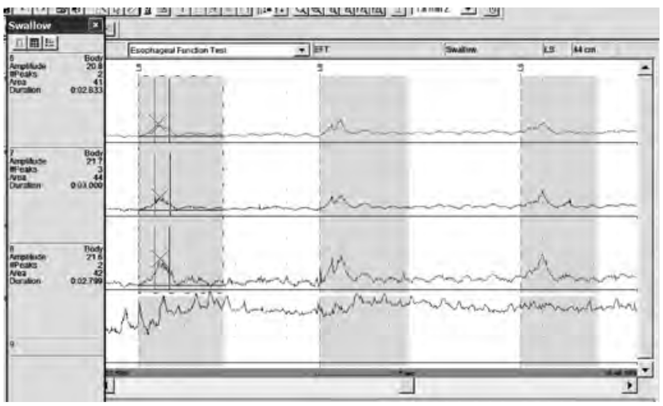

Figure 24.12. Achalasia Waveform

Distal esophageal spasm

Patients with distal esophageal spasm (DES) may present with intermittent symptoms of substernal chest pain and/or dysphagia. The pain may radiate through to the back and shoulder blades. The pain is not associated with swallowing, frequently awakens the patient from sleep, and may be worse during periods of emotional stress. The etiology is not known.

The manometric findings include:

- Normal IRP.
- Reduced distal latency with ≥20% premature swallows (DL <4.5 seconds).

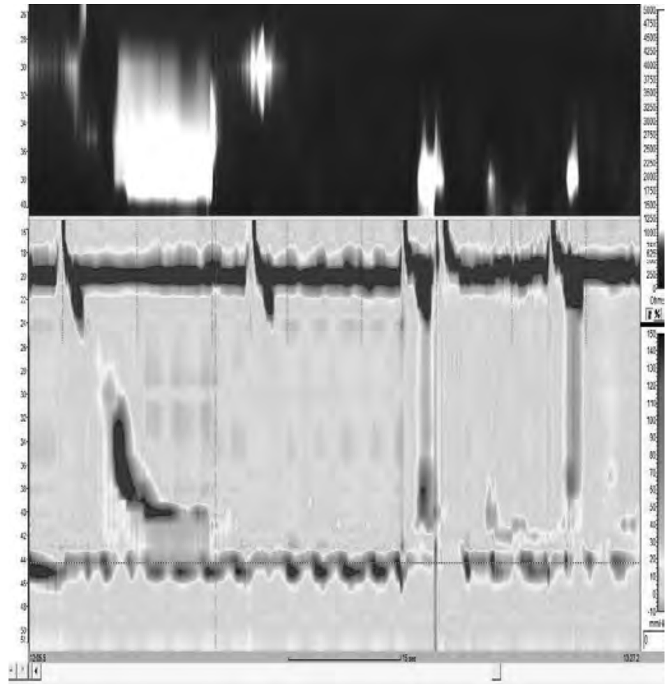

Figure 24.13. Distal Esophageal Spasm Displayed with Clouse Plot

The first swallow is peristaltic, the next two are simultaneous.

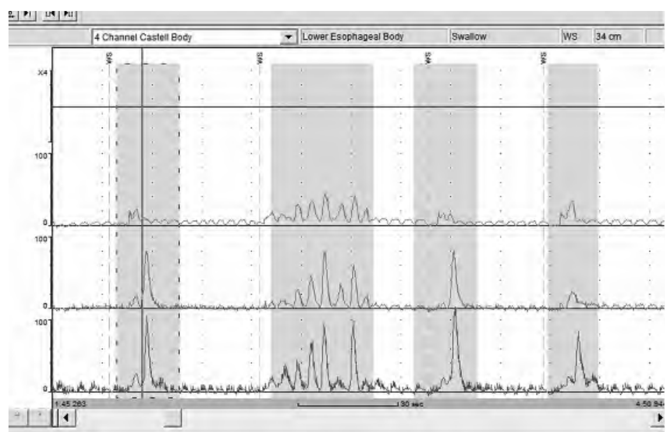

Figure 24.14. Distal Esophageal Spasm

Ineffective esophageal motility

Ineffective esophageal motility (IEM) is defined as ≥50% weak or failed contractions in the distal esophagus. Patients typically present with symptoms of dysphagia. IEM is often associated with GERD.

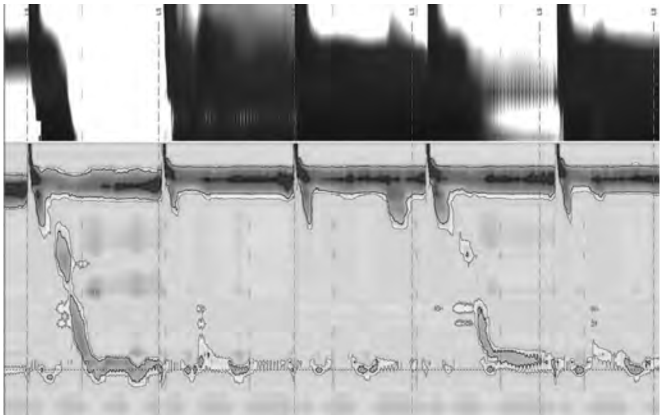

Figure 24.15. IEM Displayed with Clouse Plot

The first swallow has normal amplitude and propagation, while the next four are ineffective or non-transmitted.

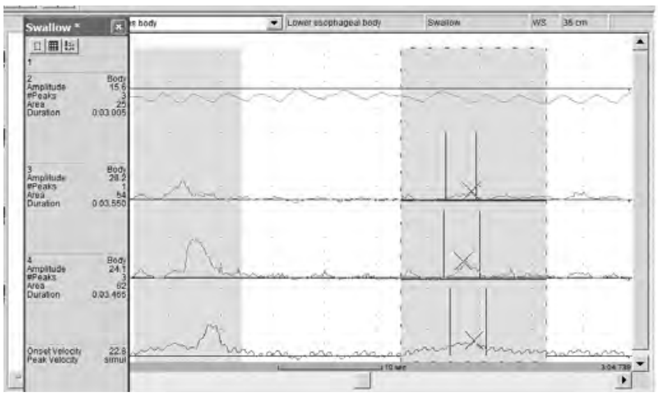

Figure 24.16. Ineffective Esophageal Motility and Incomplete Bolus Transit

Jackhammer (previous termed nutcracker) esophagus

Jackhammer esophagus is an abnormality of contraction wave amplitude identified by ≥20% hypercontractile swallows (DCI >8,000). Jackhammer esophagus may present with elevated IRP. Patients with this disorder complain of chest pain and/or dysphagia. The duration of contractions may be prolonged, but peristaltic progression and bolus transit is normal.

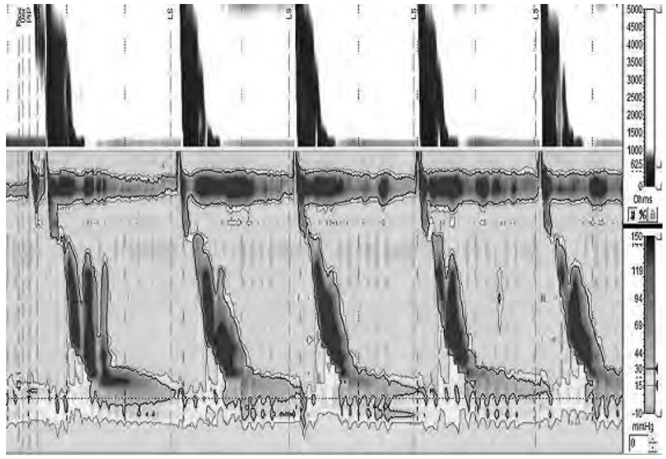

Figure 24.17. Jackhammer Esophagus Displayed with Clouse Plot

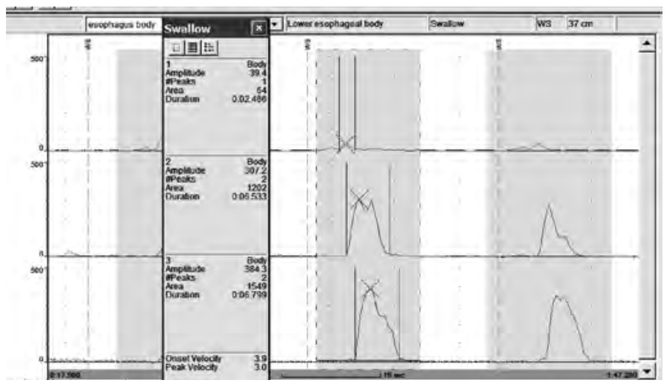

Figure 24.18. Jackhammer Esophagus with Waveform Plot

Hypertensive lower esophageal sphincter

A hypertensive LES is characterized by abnormally high resting LES pressure of >45 mmHg. LES relaxation may be normal or show elevated IRP (EGJ outflow obstruction). Peristaltic progression is normal. Patients may complain of chest pain or dysphagia.

Scleroderma

Scleroderma, as a manifestation of progressive systemic sclerosis, is a connective tissue disorder characterized by a progressive thickening and induration of the dermis. The majority of patients with typical skin manifestations have evidence of esophageal involvement, characterized by muscle atrophy and fibrosis that affect predominantly the smooth muscle region of the esophagus. The end results are failure of muscle contraction in the distal esophagus and hypotensive LES.

Manometry shows reduced esophageal contractions or aperistalsis in the distal esophageal body and low to absent LES pressure. The striated upper body contractions, the UES, and pharyngeal contractions are normal. Poor bolus transit is typically demonstrated in the impedance channels.

Chronic intestinal pseudo-obstruction

The term chronic intestinal pseudo-obstruction describes a syndrome that is characterized by intermittent signs and symptoms of bowel obstruction in the absence of a demonstrable obstructing lesion. It most commonly affects the small bowel. Esophageal manifestations include vomiting, dysphagia, and reflux Manometry in patients with chronic intestinal pseudo-obstructions may demonstrate esophageal aperistalsis.

Typical findings include incomplete LES relaxation, abnormal esophageal contractions (simultaneous and repetitive), and absent peristalsis.

Gastroesophageal reflux disease

Esophageal manometry has limited usefulness in the diagnosis of gastroesophageal reflux disease (GERD). It may be indicated to confirm adequate peristalsis amplitude in patients who are being considered for antireflux surgery. The presence of a significant motility disorder or abnormality may be a contraindication to surgery.

In general, patients with GERD tend to have decreased LES pressure and lower peristaltic amplitude in the esophageal body, but these findings may also be seen in people without GERD. The finding of a hypotensive LES supports the diagnosis of GERD, although few patients with GERD have a hypotensive resting LES pressure. About 50% of patients with moderate or severe esophageal reflux exhibit some degree of impaired esophageal peristalsis. Current understanding is that the primary mechanism of GERD is transient LES relaxations (TLESR). TLESR is stimulated by gastric distension, causing the LES to relax and reflux into the esophagus, which is not cleared by a swallow.

Ambulatory esophageal pH monitoring (reflux testing) is useful to quantify the amount of esophageal reflux in patients with typical GERD symptoms who are unresponsive to medical therapy, have atypical symptoms (e.g., asthma, cough, laryngitis, sore throat) possibly caused by GERD, or are being considered for antireflux surgery. Before placement of the pH probe, esophageal manometry should be used to document the location of the LES. The pH probe is typically placed 5 cm above the manometrically determined proximal border of the LES. In children, the probe tip is typically placed at 87.5% of the distance to the LES. The proper position of the probe in children may be estimated by a calculation based on the child's height. Once the probe is in place, a chest X-ray is usually done to confirm proper positioning.

ANORECTAL MANOMETRY

Anorectal manometry measures pressures throughout the anal canal and rectum to determine abnormalities in the mechanisms of defecation and continence.

Indications

Anorectal manometry is indicated:

- In chronic constipation, in fecal and/or flatus incontinence, for preoperative evaluation prior to ileoanal pouch, for colorectal anastomosis, before and after rectal surgery, in suspected scleroderma and dermatomyositis, and to rule out Hirschsprung's disease.
- As a screening procedure in newborns who have not passed meconium stool within 48 hours.
- As an operant conditioning (biofeedback) technique to improve bowel control in patients with incontinence secondary to previous anorectal surgery, obstetrical trauma, spinal injury, irritable bowel syndrome, diabetic neuropathy, rectal prolapse, multiple sclerosis, scleroderma, and dyssnergic defecation. Dyssynergic defecation refers to a problem coordinating the muscles and nerves of the pelvic floor in order to pass stool.

Anorectal manometry is contraindicated in patients suffering from infectious diarrhea, anal obstruction, anal stricture, or severe anal pain and in patients unwilling or unable to cooperate.

Procedure

Prior to the procedure

Before the procedure, the gastroenterology nurse should:

- Verify informed consent.
- Review the patient's medical and surgical history and results of the physician's physical examination.
- Provide the patient with a thorough explanation and instruction regarding the procedure and catheter insertion in order to decrease anxiety, including the amount of time the procedure will take, that it is generally a well-tolerated procedure, and that there will be the occasional sensation of the need to defecate.
- Encourage the patient to have a bowel movement before the procedure. The use of enema preps is determined by the referring physician.

During the procedure

The procedure is typically performed with the patient in the left lateral position. The catheter is positioned with the pressure sensors in the anal canal and the rectal stimulation balloon in the rectum. Allow the patient a 3-minute wait time to adjust to the sensation of the catheter before obtaining the study measurements.

Anorectal manometry in infants and older children who are able to understand the procedure and cooperate can usually be performed without sedation. Toddlers and children who are highly anxious or fearful will likely need sedation to accomplish the procedure and obtain interpretable data.

The study has several different components, including resting, squeeze, **rectoanal inhibitory reflex (RAIR)** and rectal sensation, push, and rectal compliance studies. Not all components are performed during the procedure, depending on the patient age (an infant cannot voluntarily squeeze), physician preference, and the type of catheter used.

Resting study

During a resting study, pressures throughout the length of the anal sphincter are measured. The resting pressure measurement is obtained over a period of 1 minute, with the patient relaxed and not moving or talking. With a conventional catheter, this is done with a station pull-through in 0.5 cm or 1 cm increments. With the high-resolution anorectal manometry catheter, there are multiple sensors spanning the entire length of the anal canal, requiring only one resting measurement.

The resting study evaluates:

1. Anal sphincter resting pressure:
   a. The internal anal sphincter, external anal sphincter muscle, and puborectalix muscle contribute to this pressure.
   b. Normal resting pressure is 33-101 mmHg in females and 38-114 mmHg in males.
2. Sphincter length:
   a. Normal sphincter length in females is 2.3-5.0 cm.
   b. Normal sphincter length in males is 2.4-5.1 cm.
3. Resting symmetry:
   a. Directional sensors are used to assess symmetry.
   b. Similar resting pressures are demonstrated on all four quadrants: posterior, left, anterior, and right.

Squeeze study

The patient is asked to voluntarily squeeze tightly for 5 seconds, as if trying to prevent the passage of stool. The entire length of the sphincter is assessed in a pull-through with the conventional catheters or at one location with a high-resolution manometry catheter. A prolonged squeeze (squeeze duration) is also performed to assess for sphincter fatigue.

The squeeze study evaluates:

1. External anal sphincter response:
   a. The external anal sphincter and puborectalis muscle contribute to the increased pressure.
   b. Normal maximum squeeze pressure is 90-397 mmHg above rectal pressure in females and 94-590 mmHg above rectal pressure in males.
2. Squeeze duration:
   a. The patient should be able to maintain 50% of the initial squeeze pressure for 30 seconds.

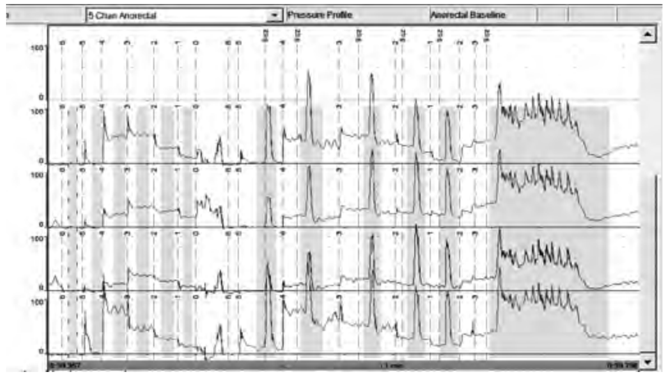

Figure 24.19. Resting and Squeeze Waveform Plot

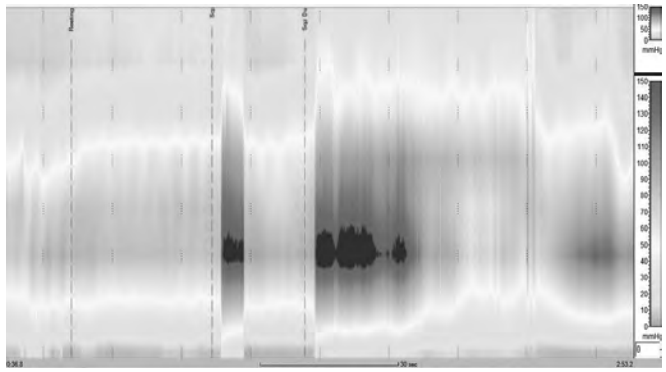

Figure 24.20. Resting, Squeeze, and Squeeze Duration Displayed with Clouse Plot

RAIR and rectal sensation study

Inflation of the rectal balloon results in an autonomic relaxation of the internal anal sphincter in order to measure the rectoanal inhibitory reflex (RAIR). This reflex allows stool or gas to enter the anal canal and provides a signal to make the maneuver to defecate or withhold defecation.

Increasing volumes of air are instilled into the rectal stimulation balloon to determine rectal sensation.

The RAIR and rectal sensation study is performed:

1. As a screening for Hirschsprung's disease. Absence of ganglion cells prevents inhibition of the internal anal sphincter, resulting in an outlet obstruction.
2. To determine rectal sensation. Normal rectal sensation threshold can vary among individuals, but is generally 10-30 ml.

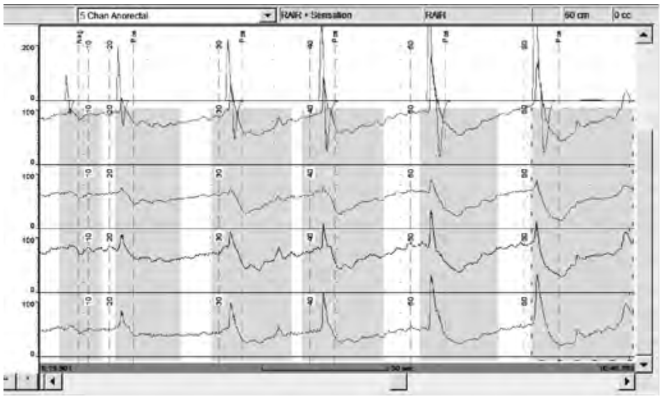

Figure 24.21. RAIR Waveform Plot

Push study

With the sensors in the high-pressure zone (HPZ) of the anal canal, the patient is instructed to bear down for 5 seconds, as if trying to pass stool. The puborectalis muscle and external anal sphincter muscle should relax to allow the stool to pass.

This study is performed:

1. To determine if there is a paradoxical response (increased pressure) in response to pushing. If present, this indicates a fecal outlet obstruction (dyssynergic defecation), resulting in symptoms of incomplete evacuation, excessive pushing, and constipation.
2. With the cough maneuver.
   a. With the sensors in the HPZ of the anal canal, the patient is instructed to forcefully cough once, and then a second time.
   b. Increased pressure is seen in the patient with normal pressure response.

Rectal compliance study

Increasing volumes of air, usually in 50 ml increments in the adult patient, are instilled into the rectal stimulation balloon. Intrarectal pressures are measured after each inflation to determine how well the rectum expands to accommodate the volume.

The test evaluates rectal capacity:

1. The rectum should be able to tolerate 200 ml of inflation before the patient experiences an urgent need to defecate that does not subside.
2. Poor rectal compliance indicates a decreased storage ability, resulting in fecal incontinence and pain.

Neuromuscular retraining

Neuromuscular retraining, or biofeedback, is an instrument-assisted process that increases the awareness of the pelvic muscles' response and improves the muscle functions that control continence and the ability to defecate in a normal manner.

During the procedure:

1. The patient learns to manipulate his or her muscles through trial and error to improve the voluntary control of muscles.
2. The muscle function is demonstrated by either electromyogram (EMG) or pressure sensors positioned in the anal canal.
3. The response is displayed on a computer monitor where the patient can immediately see his or her effort.
4. The health care provider coaches the patient to recognize muscle response and produce the correct response and provides encouragement.

Figure 24.22. Perry EMG Sensor

Figure 24.23. Manometry Catheter

Neuromuscular retraining sessions require four to six visits over 4-6 months and are supplemented by home exercise practiced several times a day. Each session is approximately 45 minutes in length, with the patient actively working his or her muscles for about 20 minutes. The majority of the session time is spent providing patient education regarding the patient's disorder, assessing the patient's progress, and providing emotional support.

The goals of neuromuscular retraining are to:

- Increase the number of muscle fibers innervated by existing nerves.
- Increase the patient's awareness of anal and rectal sensations.
- Improve motor skills.

Creation of new neural pathways is not possible.

Care and maintenance

Anorectal manometry protocol varies considerably with the type of manometry catheter used. The number of channels varies, as well as the type of sensors. Directional sensors measure from one direction, and circumferential sensors provide an average anal canal pressure. Directional sensors are necessary to determine sphincter symmetry. A rectal stimulation balloon on the end of the catheter is inflated with a 60 ml syringe attached to the inflation port during the RAIR and rectal sensation studies and the rectal compliance study. Some of these catheters use a latex rectal stimulation balloon.

The equipment is calibrated before every procedure to ensure the equipment is functioning properly.

Considerations

Nursing considerations for anorectal manometry include at least the following:

1. Obtain patient history pertaining to the defecation complaint.
2. Determine if the patient has a latex allergy.
3. Provide patient education, including the:
   a. Purpose of the test.
   b. Positioning that will be used.
   c. Effective relaxation methods.
   d. Techniques to be used.
   e. Approximate length of the procedure.
   f. Sensations likely to be experienced.
   g. Risks of the procedure.
   h. Importance of patient cooperation.
   i. Post-procedure instructions and recommendations for follow-up care.
4. Document patient education and comprehension.
5. Obtain verbal consent (and written consent if required by the organization).
6. Provide information and reassurance to decrease the patient's anxiety regarding the procedure.
7. Ensure equipment is functioning properly.
8. Adhere to protocol.

ANTRODUODENAL MANOMETRY

Antroduodenal manometry measures gastric and small intestinal contractions. Measuring contraction coordination and contraction amplitudes allows assessment of neural control and the smooth muscle response.

Indications

Antroduodenal manometry is indicated for evaluation of patients with:

- Unexplained nausea and vomiting.
- Delayed gastric emptying.
- Chronic intestinal pseudo-obstruction.

Gastric arrhythmias are associated with:

- Clinical conditions, including idiopathic gastroparesis, gastroparesis secondary to diabetes mellitus or anorexia nervosa, gastric ulcer disease, and gastric adenocarcinoma.
- The effects of certain drugs and hormones, including anticholinergics, methionine-enkephalin, ß-endorphin, epinephrine, glucagon, prostaglandin E2, secretin, and insulin.

Transient abnormal rhythms have been observed postoperatively in otherwise normal individuals. Clinical signs of these disorders include severe gastric retention, nausea, vomiting, and weight loss.

Impaired gastric motor activity in patients with diabetic neuropathy may cause symptoms ranging from vague postprandial abdominal discomfort to constant postprandial nausea and vomiting. Gastric stasis may cause additional, more serious problems such as weight loss, poor diabetic control, and bezoars. The most commonly seen gastric motor abnormality in patients with diabetic neuropathy is an absence of the antral component of the migrating motor complex (MMC), which is normally responsible for clearing indigestible solids. Patients may be helped by prescriptions such as metoclopramide (Reglan) before meals and at bedtime.

Procedure

The patient is NPO for 8 hours prior to the procedure to provide a baseline and to decrease the chance of aspiration during catheter placement. Medications that affect motility, including metoclopramide, erythromycin, narcotics, and anticholinergics, should be discontinued prior to the procedure.

The catheter is passed transnasally and positioned with endoscopic guidance or guide wires. Placement is verified with fluoroscopy.

A fasting period, often 3 hours, is recorded, followed by stimuli from a meal or drugs with a 1-hour recording. Study time varies, up to 24 hours. The patient's mobility is limited to a few feet from the manometry system.

Normal gastric contractions are 3 (contractions per minute) cpm, 11 cpm in the duodenum. Duration of the MMC, propagation velocity, and contraction amplitude are measured in the fasting and fed state.

Care and maintenance

Water-perfused catheters and solid-state catheters are used for antroduodenal manometry. Water-perfused catheters require the patient to remain in a recumbent position during the entire procedure. The solid-state catheters have a faster response time and are not position-dependent.

The spacing of the sensors varies, with closer sensor spacing at the pylorus. A catheter longer than 20 cm is necessary to measure propagation of the MMC.

Considerations

During the procedure, the gastroenterology nurse should ensure the patient is comfortable and adhere to the physician's protocol for testing.

One indication for performing this procedure is gastric paresis, so the gastroenterology nurse should be prepared to address patient discomfort after a meal and potential nausea and vomiting.

SPHINCTER OF ODDI MANOMETRY

The sphincter of Oddi is a circular, smooth muscle that surrounds the ampulla of Vater, the distal common bile duct, and the pancreatic duct (Figure 24.24). During fasting, it exhibits peristaltic-like contractions that propel bile and pancreatic fluids into the duodenum and prevent reflux of duodenal contents. Food causes the release of cholecystokinin from the duodenum, which inhibits contraction of the sphincter of Oddi and lowers the basal pressure.

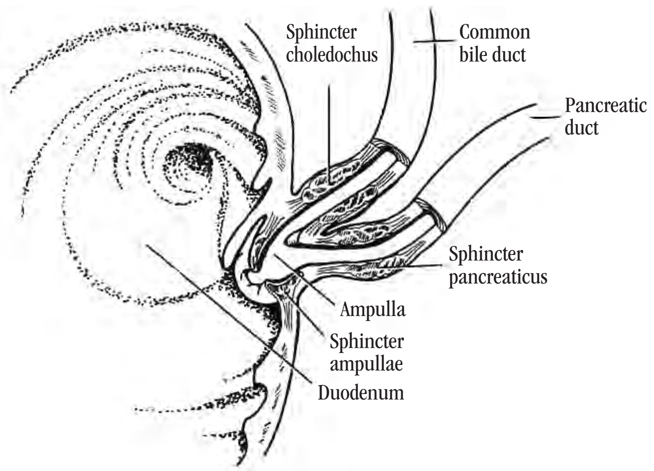

Figure 24.24. Sphincter of Oddi Anatomy

Indications

Sphincter of Oddi manometry is indicated for diagnosis of papillary stenosis and motility disorders of the sphincter of Oddi, which may be associated with choledocholithiasis, biliary pain, abdominal pain, or pancreatitis.

Possible factors causing obstruction of bile flow include spasm of the sphincter of Oddi and retrograde contractions.

Sphincter of Oddi manometry is usually obtained during endoscopic retrograde cholangiopancreatography (ERCP), so like ERCP, sphincter of Oddi manometry is contraindicated in patients who:

- Are unwilling or unable to cooperate.
- Have had a recent myocardial infarction.
- Have acute pancreatitis.
- Have coagulopathy.
- Have barium in the GI tract.
- Are pregnant.
- Have severe pulmonary disease.

Procedure

Sphincter of Oddi manometry is usually obtained during ERCP (Figure 24.25).

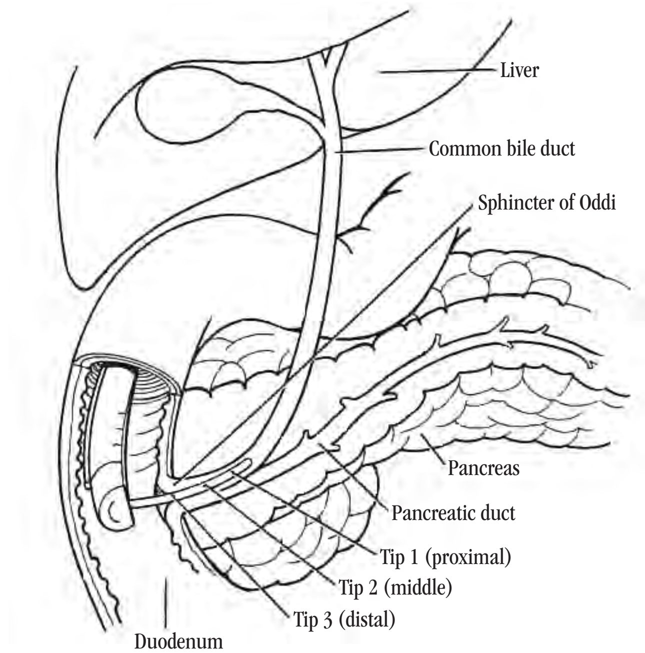

Figure 24.25. Placement of Pressure Catheter in the Sphincter of Oddi.

Prior to the procedure

Before the procedure, the gastroenterology nurse should:

- Verify informed consent and 8 hour NPO status.
- Review the patient's medical history and laboratory results.
- Establish a patent intravenous (IV) line.
- Remove dentures.
- Administer antibiotic prophylaxis, as ordered.
- Record baseline vital signs.

The patient is mildly sedated with anticholinergic drugs. Opiates are not used before manometry because they can alter sphincter pressures.

During the procedure

The catheter is inserted over a guidewire, through the biopsy channel of a side-viewing duodenoscope, into the duodenum. A duodenal baseline recording and measurement is obtained. If possible, this measurement should be obtained in a quiet period without contractions.

The ampulla of Vater is identified and cannulated with the manometry catheter, which is passed through the sphincter of Oddi into the common bile duct. As the catheter is withdrawn slowly at 1 mm increments, a normal tracing will show sphincter basal pressures <40 mmHg, with peristaltic-like phasic waves. If a motility disorder is confirmed (high basal pressures), the physician may choose to perform an endoscopic sphincterotomy.

Procedure tips include:
- Inject sterile water through the biopsy channel prior to inserting the guide wire to ease the passage of the wire and catheter.
- Tape the water-perfused catheter to the patient's shoulder to prevent the catheter from hanging below the water-perfused transducers.
- Attempt to obtain more than a 30-second sphincter pressure for assessment.

After the procedure

Once the procedure is complete, the gastroenterology nurse should:
- Monitor and document the patient's vital signs.
- Observe for abdominal distention and signs of pancreatitis.
- Maintain NPO status until change of status ordered by physician.
- Administer antibiotics, as ordered.
- Remove the IV line.
- Provide written discharge instructions.
- Ensure that outpatients have someone to drive them home.

Care and maintenance

In addition to the ERCP equipment, a manometry recording system and a sphincter of Oddi catheter is necessary. Water-perfused catheters with two to three channels are used.

When using water-perfused catheters, the water chamber is filled with sterile water, and a floating disc is placed on top of the water column. This will decrease air bubbles getting into the tubing. Setting the pressure to 7.5 psi prevents a forceful flow of water into the ducts. Water is perfused through the catheter, ensuring no air bubbles are in the catheter, extension tubing, or perfusion capillaries. The components are kept sterile or as clean as possible.

The system is calibrated before every patient per the manufacturer's instructions to ensure that the system is in good working order.

Considerations

The most common complication of sphincter of Oddi manometry is pancreatitis, which may present as:
- Mild to incapacitating pain, beginning in the midepigastrium and radiating to the patient's back.
- Nausea and vomiting.
- Fever.
- Distended abdomen.
- Decreased bowel sounds.

Less likely complications are:
- Perforation, which may present with abdominal rigidity, rebound tenderness, an increase in pulse rate, and shallow and rapid respirations.
- Bleeding, which may present as a decrease in blood pressure and increased pulse rate.

CASE SITUATION

We met Doris Johnson in Chapter 13. She has suffered significantly with chest discomfort. She has had heartburn; reflux of sour, bitter-tasting liquid up into her mouth; and several chest colds that her physician said were caused by the acid getting into her lungs at night. On upper endoscopy, she was found to have esophagitis. Doris says she has had trouble adhering to her antireflux regimen. She does not have the head of her bed up on blocks yet, but she has been sleeping on several pillows. She thought the pillows would be sufficient, but she admits that when she wakes up in the morning, she has invariably rolled off the pillows. She is in the endoscopy department now for esophageal manometry.

Points to think about

1. The gastroenterology nurse has set Doris up for her motility study. The equipment is all ready, and the gastroenterology nurse is prepared to place the tube into Doris's esophagus. Doris jumps up in alarm and says, "You mean you aren't going to put me to sleep like the last time?" How should the gastroenterology nurse respond?
2. What techniques might help the gastroenterology nurse pass the manometry tube successfully?
3. The gastroenterology nurse would expect that Doris has a hypotensive LES, which allows reflux of gastric material into the esophagus. What else might be seen on the tracing?

Suggested responses

1. When Doris expresses surprise that she will not be sedated, the gastroenterology nurse might respond by:
   a. Explaining that this is not the same as the endoscopy procedure she had before.
   b. Explaining that this test will determine how the muscles in her esophagus contract and that the sedating medication could affect the test by artificially relaxing those muscles.
   c. Reassuring Doris that the tube to be inserted through her nose is much smaller than an ordinary nasogastric (NG) tube and that everything possible will be done to make her comfortable. For example, comfort measures may include providing a topical anesthetic to her nostril, using the nostril that is more open, allowing her to sip on water to help the catheter pass, and giving her a couple of minutes to get used to the catheter before the procedure starts.
2. To help pass the manometry tube successfully, the gastroenterology nurse might:
   a. Pass the catheter through the EGJ with the distal sensors positioned below the diaphragm. Look

at the manometry tracing to identify the high pressure landmarks of the UES and LES to determine proper placement.
   b. If the physician allows, anesthetize the inner nares.
   c. Ask Doris if she breathes better through one side of the nose. If so, that side should be tried first.
   d. Listen to Doris if she says she has had problems in the past with NG tube insertion. Even if the gastroenterology nurse is an expert at passing tubes, listening and being compassionate prior to beginning the insertion process can foster trust and confidence.
   e. Advance the catheter slowly and carefully around the back of the nose and into the back of the throat. The gastroenterology nurse must never push against an obstruction, and if one naris is obstructed, the other side should be tried. After some experience, the gastroenterology nurse will know how to maneuver the catheter (from one side to the other or up or down) and how to curl the tip to get around the back of the nose and throat.
   f. When the tube is ready to enter the esophagus, the gastroenterology nurse should have the patient bend her head forward and swallow. This causes the epiglottis to open the esophagus and close the trachea. Unless there is suspicion of achalasia or other obstruction, it is usually permissible for the patient to take a few sips of water.
   g. Once the patient is able to swallow, the catheter usually passes easily into the stomach. If any obstructions are met, the gastroenterology nurse should stop and check tube placement on the manometry machine. If the tube is pushed against a lower esophageal obstruction, it will curl back on itself.
   h. If the patient gags inordinately, it may be a good idea to have another gastroenterology nurse sit by the patient to calm her and coach her while the tube goes down.
3. In addition to a hypotensive LES, the gastroenterology nurse might expect to see weak or failed contractions.
4. The two types of esophageal motility disorders are:
   a. Primary motility disorders, which involve only the esophagus, such as achalasia, distal esophageal spasm (DES), nutcracker esophagus, hypertensive LES, and nonspecific esophageal motility disorders.
   b. Secondary motility disorders, which occur when the esophagus is affected as part of a more generalized disease process such as diabetes mellitus, scleroderma, or chronic idiopathic pseudo-obstruction.

REVIEW TERMS

bolus transit, disposable catheters, high-resolution manometry (HRM), high resolution impedance manometry (HRIM), IRP (integrated relaxation pressure), DCI (distal contractile integral), DL (distal latency) rectoanal inhibitory reflex (RAIR), maximum squeeze pressure, esophageal peristalsis, LES relaxation

REVIEW QUESTIONS

1. Esophageal manometry is contraindicated in patients with:
   a. Dysphagia.
   b. Recent gastric or esophageal surgery.
   c. Diabetes mellitus.
   d. Noncardiac chest pain.
2. All the following are advantages of high-resolution impedance manometry (HRM) except:
   a. Enables the assessment of bolus transit.
   b. Requires a station pull-through of the LES.
   c. Enables visualization of the entire esophageal body and sphincters.
   d. Technically easier to perform and has better patient tolerance.
3. Esophageal manometry is indicated for each of the following except:
   a. Evaluation of noncardiac chest pain.
   b. Preoperative assessment of peristaltic pressure in the esophagus.
   c. Determination of the location of the LES for pH probe placement.
   d. Evaluation of patients with symptoms of delayed gastric emptying.
4. Motility of the esophageal body is best assessed while the patient:
   a. Performs a series of liquid swallows.
   b. Performs a series of dry swallows.
   c. Takes a series of deep breaths.
   d. Holds his or her breath and does not swallow.
5. Failed peristalsis or spasm of the esophageal body and incomplete relaxation of the LES are manometric symptoms associated with:
   a. Achalasia.
   b. Diffuse esophageal spasm (DES).
   c. Jackhammer esophagus.
   d. Nonspecific esophageal motility disorders.
6. Manometric manifestations of advanced scleroderma include:
   a. Simultaneous and repetitive esophageal contractions.
   b. Aperistalsis in the distal esophageal body and low to absent LES pressure.
   c. Abnormally high resting LES pressure.
   d. A mean contraction amplitude at least two standard deviations above normal.

7. All of the following are true about IEM except:
   a. IEM is defined as low amplitude or non-transmitted contractions in the distal esophagus.
   b. IEM is highly associated with GERD.
   c. Greater than or equal to 50% weak test swallows.
   d. EGJ outlet obstruction always coexists with IEM.
8. All of the following statements are true with regard to sphincter of Oddi manometry except:
   a. Indicated in the diagnosis of papillary stenosis.
   b. Indicated in the diagnosis of motility disorders related to choledocholithiasis and pancreatitis.
   c. Indicated in the diagnosis of scleroderma.
   d. Usually obtained during ERCP.

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Chapter 25

BIOPSY AND CYTOLOGY

This chapter will acquaint the gastroenterology nurse with techniques used to obtain tissue samples during gastrointestinal (GI) endoscopy for the diagnosis and management of GI disease. Techniques include pinch biopsy, endoscopic mucosal resection (EMR), endoscopic submucosal dissection (ESD), endoscopic ultrasound (EUS)-guided fine needle aspiration (FNA), suction biopsy, percutaneous liver biopsy, brush cytology, and combinations of techniques. It is important for the gastrointestinal nurse to understand his or her role in the yield of tissue sampling, including selection and handling of devices, adequacy of the specimens, and specimen preparation.

Learning objectives

After reviewing the content of this chapter, the gastroenterology nurse should be able to:

1. Identify techniques used for endoscopic specimen collection in the GI tract, including biopsy, brush, and aspiration.
2. Discuss indications, complications, and considerations related to endoscopic specimen collection of the GI tract.

BASIC PRINCIPLES

Both biopsy and cytology allow direct sampling of GI tissue for diagnosis and management of diseases of the digestive system.

Biopsy involves excision of pieces of living tissue from a suspected pathological site, with subsequent **histopathological analysis** in the laboratory.

Biopsy specimens may be obtained endoscopically by:

- Passing a biopsy forceps through the accessory channel of an endoscope.
- Using a needle to collect core biopsies endoscopically.
- Using a suction method (e.g., rectal suction biopsy).

Cytology analyzes cells from the body under a microscope to determine what the cells look like, and how they form and function. Cytologic evaluation may also include culture of the cells for infection.

Specimens for cell culture or cytological analysis may be obtained endoscopically by these techniques:

- Brushing.
- **Aspiration**.

ENDOSCOPIC BIOPSY

Biopsies can be obtained through the endoscope during endoscopic procedures of the esophagus, stomach, small bowel, colorectum, and biliary and pancreatic structures.

Indications

Indication for endoscopic biopsy includes confirmation of normal or abnormal mucosal tissue in any portion of the GI tract and/or assessment of tissue response to therapy.

Biopsy may be contraindicated in patients with severe coagulopathy or active bleeding. A bleeding profile should be obtained if clinically indicated. Caution must be exercised in patients who are currently taking:

- Antithrombotic (antiplatelet and/or anticoagulant) medications, including warfarin, heparin, and low molecular weight heparin.
- Antiplatelet agents, including aspirin, nonsteroidal anti-inflammatory agents (NSAIDs), thienopyridine (clopidogrel, ticlopidine), and glycoprotein IIb/IIIa receptor inhibitors.

The American Society for Gastrointestinal Endoscopy (ASGE) (2016a) published a guideline based on the urgency and relative risks of the procedure, the underlying condition, and the indications necessitating the procedure. The guideline is broken down into the following considerations:

- Procedure risks: low risk or high risk.
- Condition risks: low risk, medium risk, or high risk.
- Antithrombotic medications.

The decision to biopsy should be individualized for each patient based on the risks and benefits of the procedure. Elective procedures may need to be delayed pending discontinuation of antithrombotic agents.

Procedure

A variety of **biopsy forceps** are available to retrieve mucosal and submucosal tissue specimens for histology or culture. Biopsy forceps are available in various sizes and lengths, with or without fenestration, with a fixed or moveable head, with a central spike or needle, or with smooth or dentate edges.

The sampling portion of biopsy forceps consists of a pair of small cups set in apposition when closed. The size of the instrument channel determines the maximum size of the forceps, and, therefore, the size of specimen obtained. Large cup or "jumbo" forceps can be used in scopes with a larger-diameter accessory channel to provide larger specimens with less crush artifact compared to standard biopsy forceps.

"Pinch" biopsy, performed with the use of a "cold" biopsy forceps during endoscopy, is the most common form of tissue sampling. Cold biopsy and polypectomy techniques are thought to be safer and are recommended for low-risk small polyps, especially in the right colon.

Forceps with a central spike or needle can help stabilize and direct tissue sampling in order to obtain deeper biopsies than non-spiked versions. Spiked forceps secure the first specimen on the needle while collecting a second or additional specimen during a single passage of the forceps through the accessory channel.

The use of hot biopsy forceps (with cautery), developed for simultaneous tissue biopsy and coagulation of diminutive colon polyps, should not be routinely used. It is typically not recommended due to the risks of post-procedure hemorrhage and delayed perforation.

Obtaining the specimen

To obtain an endoscopic biopsy specimen, an endoscope is inserted into the esophagus, stomach, small bowel, or colon as described in Chapter 23. With the endoscope in place, biopsy forceps are passed through the accessory channel. To avoid potential internal endoscope channel damage, the closed forceps is extended through the distal tip of the accessory channel into position to take a biopsy. The biopsy site is visualized at the time of sampling.

Under the direction of the endoscopist, the assistant opens and closes the forceps to obtain tissue specimens. The opened forceps are pressed against the mucosal surface for tissue sampling. The biopsy forceps are closed against the mucosal surface, and the endoscopist pulls the forceps away from the target mucosa to remove the fragment of tissue. This method often yields samples that include muscularis mucosae, except in regions such as the gastric body where the mucosal folds are quite thick. The submucosa is sampled occasionally with either standard or jumbo forceps.

Specimen retrieval

The accuracy of histopathological interpretation of endoscopic biopsy specimens can usually be improved by obtaining multiple specimens of the suspicious area.

Once collected, the specimen is then retrieved by withdrawing the biopsy forceps through the accessory channel of the endoscope. Failure to completely remove the forceps with the cups in the closed position risks loss of the specimen and internal endoscope accessory channel damage. As the forceps are removed, the shaft should be wiped with a gauze sponge to remove secretions. Care should be taken to control the tip of the forceps as it exits the channel so staff will not be exposed to secretions that might be flicked from the forceps.

Specimen handling

Specimens should be handled, labeled, and submitted according to organization policy and procedure. Specimens for culture should preferably be obtained first. Most GI biopsies do not require specimen orientation for definitive pathologic interpretation. However, properly oriented specimens may be preferred by pathologists according to organization policy and procedure. Care should be used to preserve tissue architecture.

Methods for preparing biopsy samples include:

- **Fixative.** After obtaining, specimens should be prepared and placed in the appropriate fixative. The fixative used for most GI mucosal biopsies is 10% buffered formalin, which provides excellent tissue fixation and allows staining by routine histologic study with optimal results. Complete immersion of biopsies in formalin should always occur immediately on collection. Up to 1 hour is often needed to adequately fix a specimen with a diameter >1.0 mm, and more time may be needed for larger specimens.
- **Non-fixation fluids.** Specimens obtained for specific tests such as tissue culture, molecular tests, and electron microscopy may not require fixative. Many molecular tests may also be performed on formalin-fixed tissue. Specimens that require placement in non-fixation fluids should be handled with a tissue-capture device that has not been previously placed in fixative.
- **Frozen section.** If urgent denial or confirmation of malignancy is required, the endoscopic biopsy specimen in the form of a frozen section may be sent to

the laboratory for immediate microscopic examination by a pathologist. No fixative of any kind is used for a frozen section. Instead, the specimen is placed on mounting material, labeled, and immediately transported to the laboratory.

Many types of tissue artifacts may be introduced into tissues as a result of bowel preparation, endoscopic trauma, or tissue handling. The most common type of artifact is mucosal edema and intramucosal hemorrhage, known as scope trauma. Other artifacts can be the results of crush or cautery trauma during specimen retrieval. Cautery artifact as a result of hot biopsy is a normal and expected component of endoscopic polypectomy with electrocautery.

Care and maintenance

After each patient use, reusable forceps should be cleaned and steam-sterilized in accordance with the manufacturer's recommendations and regulatory and professional society guidelines. Single-use (disposable) biopsy forceps should be discarded in the appropriate sharps disposal container after patient use.

Considerations

Endoscopy-directed diagnostic biopsies are extremely safe. The risk of perforation after diagnostic or therapeutic colonoscopy is extremely low. Excessive bleeding and perforation are the most likely **complications** of endoscopic biopsy.

Post-biopsy patient discharge instructions should include:

- Notifying the physician of signs and symptoms of bleeding or perforation, which include fever or pain.
- Avoiding the use of aspirin or aspirin-containing products for a period of time as recommended by the physician.
- Resuming antithrombotic medications as recommended by the physician.

Tissue samples for assessment of submucosal or deeper lesions may be obtained using techniques of endoscopic mucosal resection (EMR) or endoscopic submucosal dissection (ESD).

ENDOSCOPIC MUCOSAL RESECTION (EMR)

Endoscopic mucosal resection (EMR) is an established technique for the endoscopic removal of precancerous and early-stage malignant lesions in the mucosal and submucosal layers of the esophagus, stomach, duodenum, and colon. EMR can remove larger histologic specimens compared with the standard biopsy forceps.

The endoscopist may remove the lesion in a piecemeal fashion or attempt an en bloc resection with the potential to completely resect an early malignant lesion of no more than 1.5-2 cm in diameter. Endoscopic ultrasonography (EUS) is often used to assess the depth of a particular lesion before performing EMR.

Indications

In the esophagus, EMR is indicated to treat Barrett's esophagus-associated neoplasia and intermucosal carcinoma, as well as superficial esophageal cancer. In the stomach, EMR is used to resect early gastric cancer. It has also been used to effectively resect type 1 gastric carcinoids (associated with chronic atrophic gastritis). In the duodenum, EMR can be used to remove nonampullary duodenal adenomas. However, since the duodenum has a thin wall and increased vascularity, duodenal EMR carries an increased risk of bleeding and perforation. In the colon, EMR is used to resect large or flat lesions.

Procedure

Several endoscopic techniques have been developed for EMR:

- **Injection-assisted EMR**, often called the saline solution lift-assisted polypectomy, allows snaring and resection of lesions after injection of a substance to lift the specimen. This technique minimizes mechanical or electrocautery damage to the deeper layers of the GI wall.
- **Cap-assisted EMR** uses a cap mounted on the tip of a forward-viewing endoscope. To decrease the potential risk of perforation, a submucosal injection is used to separate the mucosa from the muscle layer. A specially designed snare is then pre-looped in the cap. After the endoscope is positioned over the target lesion, suction is used to retract the mucosa into the cap. The snare is closed to capture the lesion. Electrocautery is then applied to resect the lesion, which is then removed by maintaining suction inside the cap.
- **Underwater EMR** is a newer technique that is particularly useful for salvage EMR. Filling the bowel lumen with water, rather than air, allows "floating" the mucosa and submucosa away from the deeper muscularis propria layer to allow EMR without requiring submucosal injection. This technique theoretically eliminates any risk of tracking neoplastic cells into deeper layers of the GI tract wall by the injection needle and making capture of flat lesions easier. This method has also been reported to be effective in managing recurrences after previous EMR, as well avoiding submucosal fibrosis in patients with previous partial resections and biopsies of lesions, which makes lifting the lesion with submucosal injection difficult.
- **Ligation-assisted EMR** uses a band ligation device attached to the endoscope and connected to a banding cap set on the distal tip of the scope. The device is positioned over the target area with or without previous submucosal injection. Suction is applied to retract the lesion into the banding cap, creating a pseudopolyp. A band is then deployed to capture the lesion, after which an electrocautery snare is used to resect the pseudopolyp above or below the band. Available multiband ligator kits allow potential resec-

tion of many mucosal sites without changing the device after resecting each lesion.

Considerations

Major complications of EMR include bleeding (the most common adverse event), perforation, and strictures, depending on the size and location of the lesion and the experience of the endoscopist. Clinically significant bleeding can usually be addressed using standard hemostasis techniques discussed in Chapter 28. Applying endoscopic clips to a defect in the resection site can an effective method to mitigate the risk of post-procedure bleeding.

Bleeding rates after colonic EMR have been reported to range from 2% to 11%. Rare perforation after EMR of colonic lesions (<1%) has been reported (ASGE, 2015a).

It is uncommon to experience bleeding after esophageal EMR. Perforation rates have been reported at <0.5%, depending on the experience of the endoscopist (ASGE, 2015a). Rates of stenosis following esophageal EMR have been reported to range from 6% to 88% of patients (ASGE, 2015a). Esophageal dilation can successfully treat these strictures.

Intraprocedural bleeding during gastric EMR reportedly ranges from 0% to 11%, while 5% of gastric EMR patients experience delayed rebleeding after gastric EMR. Intraprocedural bleeding is the best predictor of delayed bleeding. According to one study, there is a 1% risk of perforation related to gastric EMR (ASGE, 2015a).

There is an increased risk of bleeding and perforation associated with EMR of the duodenum. The reported risk of intraprocedural bleeding during duodenal EMR ranges from 11.5% to 19.3% for lesions <3 cm and as high as 57.8% for lesions >3cm (ASGE, 2015a). Data from one study from a center performing a high volume of duodenal EMRs demonstrated a 2% perforation rate (Shibagaki, Ishimura, & Kinoshita, 2017).

ENDOSCOPIC SUBMUCOSAL DISSECTION (ESD)

Endoscopic submucosal dissection (ESD) is an established effective treatment modality for premalignant and early stage malignant lesions of the stomach, esophagus, and colorectum. ESD allows for en bloc removal of mucosal lesions >2cm; flat lesions; and lesions in deeper submucosal layers in the stomach, esophagus, and colorectum that cannot be removed by other endoscopic methods. Both EMR and ESD involve injection of a substance under the targeted lesion that acts as a cushion. However, after injection, rather than using a snare to remove the lesion, ESD requires a specialized ESD knife (modified needle knife) to remove the lesion by dissecting through the submucosa. This technique enables removal of larger and potentially deeper lesions than EMR, with a curative intent.

Indications

Using EMR to resect tumors >2 cm often results in piecemeal resections, which is associated with higher local recurrence rates. Compared with EMR, ESD is generally associated with higher rates of en bloc and curative resections and a lower rate of local recurrence. Oncologic outcomes with ESD compare favorably with competing surgical interventions. Also, ESD can serve as a T-staging tool to identify noncurative resections requiring further treatment (Mayo Clinic, n.d.; Ning, Abdelfatah, & Othman, 2017).

Procedure

A substance is injected under the submucosa of the targeted lesion to act as a cushion. Then the submucosa is carefully dissected under the lesion with a specialized electrocautery knife. Adding dye such as indigo carmine or methylene blue to the injection solution can help in identifying the submucosal layer to determine margins of the dissection. The specimen is prepared and sent for histologic study.

Considerations

Because of the technical difficulty of ESD, there is a higher rate of adverse events, including bleeding, perforation, and esophageal stricture, compared to other endoscopic procedures such as EMR. Intraprocedure bleeding is common and may be expected during ESD. Coagulation current delivered through the ESD knife can be used to treat minor oozing from small vessels. Hemostatic forceps can be used to treat more significant bleeding. Prophylactic coagulation with hemostatic forceps may be used on larger nonbleeding submucosal vessels identified during the dissection. Delayed bleeding after ESD is more common in gastric areas than in colorectal or esophageal sites.

In a series of meta-analyses, the rate of perforation of gastric ESD was approximately 4.5% and 4.8% for colorectal ESD. Data reviews suggest a 2.3% esophageal ESD perforation rate (ASGE, 2015a). Devices used for successful perforation closure include endoscopic clips, over-the-scope clips, and endoscopic suturing devices to close perforations or defects, if necessary.

Esophageal strictures may develop in 12% to 17% of patients who undergo esophageal ESD. Risk factors include circumference and length of the resection. ESD resections >75% of the circumference of the esophagus are at highest risk for developing esophageal strictures (ASGE, 2015b).

Strictures are often prophylactically managed by serial dilation; intralesional steroid injection or topical steroid gel; and placement of a fully-covered, self-expandable metal stent.

ENDOSCOPIC ESOPHAGEAL BIOPSY

The orientation of the esophagus makes obtaining good quality specimens more difficult than those from the stomach.

Indications

Indications for endoscopic biopsy of mucosal abnormalities of the esophagus include diagnosis and/or evaluation of:

- Radiographically demonstrated stricture.
- Suspected esophageal carcinoma.
- Barrett's esophagus.
- Dyspepsia.
- Chronic or acute esophagitis.
- Gastroesophageal reflux disease (GERD).
- Eosinophilic esophagitis.
- Esophageal ulcer.
- Infectious esophagitis, such as herpes simplex virus (HSV) or cytomegalovirus (CMV).
- Esophageal candidiasis—cytologic brushing may be more sensitive than histology for diagnosis.

Procedure

A specimen for endoscopic esophageal biopsy can be obtained by:

- Biopsy forceps.
- Fine-needle aspiration.
- EMR.
- ESD.

Recommendations for biopsy of specific conditions include the following:

- **GERD.** Esophageal biopsy specimens from patients with chronic esophageal reflux often show thickening of the squamous epithelium. An established biopsy protocol for GERD in the absence of recognized Barrett's metaplasia does not exist (ASGE, 2013b). Biopsies of normal-appearing distal esophageal mucosa in patients with GERD symptoms may reveal nonspecific changes, but biopsy of endoscopically normal mucosa in GERD, when other diagnoses are not suspected, is not recommended.
- **Esophagitis.** Findings of polymorphonuclear leukocytes, eosinophils, and ulceration in esophageal biopsy specimens provide strong support for a diagnosis of active esophagitis. For eosinophilic esophagitis (EoE), two to four biopsy samples should be obtained from both the proximal and the distal esophagus, even if the esophageal mucosa appears normal (ASGE, 2013b). Biopsy samples should also be taken from the gastric antrum and duodenum when there is a suspicion of eosinophilic gastroenteritis.
- **Barrett's esophagus.** In patients with suspected Barrett's esophagus, obtaining at least eight random biopsies is recommended. This provides the best results to identify intestinal metaplasia. If the area of suspected Barrett's esophagus is short and eight biopsies are unattainable, it may be necessary to take fewer specimens at specific intervals and from specific tissues. The American College of Gastroenterology (ACG) provides specific recommendations for biopsy sample collection in endoscopic surveillance for Barrett's esophagus. Mucosal abnormalities should be sampled separately, preferably with EMR (ACG, 2016).
- **Esophageal cancer.** To accurately diagnose esophageal cancer, any strictured lesions should be dilated sufficiently to allow passage of the endoscope to its distal margin. Numerous biopsy specimens should be obtained from tissue that is clearly abnormal but not necrotic. High-definition endoscopes, chromoendoscopy, and EUS may be used to help identify biopsy sites. Brush cytology can also be helpful in some cases. Specimens for cytology may be obtained by using a sheathed brush that is inserted through the endoscope. (See the discussion on brush cytology in the section on Cytology and Culture of the GI Tract at the end of this chapter.)
- **Cytomegalovirus (CMV).** CMV infects mesenchymal and columnar cells, causing as ulcerative lesions in the esophagus. Samples for biopsy should be taken from the ulcer base for diagnostic accuracy.
- **Herpes simplex virus (HSV).** This virus infects squamous epithelial cells found at the outer margins of ulcers and erosions. Biopsy samples taken from the ulcer margin have the highest diagnostic accuracy. Culture and polymerase chain reaction can aid in diagnosing herpes simplex virus esophagitis.
- **Esophageal candidiasis.** The optimal diagnostic technique for esophageal candidiasis has not be established, but samples are generally taken from the creamy patches that appear on the esophageal mucosa. Taking cytologic brushing, rather than biopsy, may be more sensitive for diagnosis.

ENDOSCOPIC GASTRIC BIOPSY

Gastric tissue can be removed for biopsy during endoscopy.

Indications

Endoscopic gastric biopsy is indicated in the diagnosis of:

- Gastric mucosal abnormalities associated with active and chronic gastritis.
- Metaplastic (chronic) atrophic gastritis (e.g., environmental metaplastic atrophic gastritis, autoimmune atrophic metaplastic gastritis [AMAG]).
- Gastric epithelial polyps.
- Malignancy.
- Gastric ulcers.
- Gastric lymphoma.
- Evaluation of dyspepsia.
- *Helicobacter pylori* (*H. pylori*) gastritis.

Procedure

The procedure for biopsy will depend on the suspected disease process.

Gastric polyps

All polyps of the stomach, especially those polyps >5 mm, should be examined by endoscopic biopsy. Choice of biopsy technique depends on the type and size of the protruding lesion and the risks inherent to removal for the individual patient. If possible, polyps should be removed endoscopically. If polyps cannot be removed, biopsies should be performed. Adenomatous polyps, large hyperplastic polyps, and any polyp with a stalk should be removed using an endoscopic snare technique as described in Chapter 30.

Gastric neoplasm

Most neoplasms of the stomach are adenocarcinomas. For patients with suspected gastric adenocarcinoma, the appearance of the lesion is assessed and biopsied. It is possible to detect lesions as small as 2-3 mm in diameter and to obtain a histological diagnosis.

For submucosal tumors, the mucosa overlying the tumor can be lifted off with biopsy forceps, but this will not always provide a positive diagnosis of the tumor itself. ESD, a large-particle biopsy, or a lift-and-cut biopsy employing a snare may also be used to obtain submucosal tissue. Fine needle aspiration or (FNA) or fine needle biopsy (FNB) with ultrasound guidance may be used to obtain tissue from submucosal nodules. (See the in-depth discussion on Fine-Needle Aspiration (FNA) and Fine-Needle Biopsy (FNB) later in this chapter.)

Gastric ulcer

The benign appearance of a gastric ulcer should always be confirmed as benign by multiple biopsies and **exfoliative brush cytology** (Figure 25.1). Biopsies of the ulcer edges are necessary to ascertain whether or not the lesion is malignant. When endoscopy is performed, six to ten biopsy specimens should be obtained from the ulcer margin in a circumferential pattern.

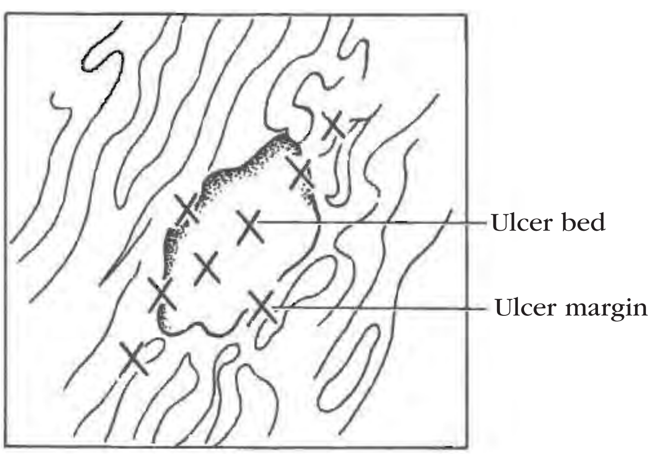

Figure 25.1. Recommended Ulcer Biopsy Sites

H. pylori

Endoscopic tests for patients with suspected *H. pylori* infection include tissue urease activity, histologic examination, and microbial culture.

A standard biopsy forceps may be used to obtain the specimen from the dependent portion of the antrum along the greater curvature. Using a rapid urease test, a sample can then be inserted into an agar gel test kit that contains urea, a pH indicator, buffers, and a bacteriostatic agent. A rapid color change indicates the presence of the urease enzyme typically produced by *H. pylori*. Tissue may also be sent to the pathology lab and stained for *H. pylori*. The gastroenterology nurse should request appropriate laboratory staining per organization protocol.

The sensitivity of these tests may be decreased by proton pump inhibitors, bismuth compounds, antibiotics, and acute GI bleeding. In situations in which test sensitivity is reduced, a negative urease test result should be confirmed with a different test for *H. pylori* infection, such as sending sample tissue to the pathology lab for staining. *H. pylori* culture allows identification of the bacterial strain and antimicrobial resistance patterns but is difficult to obtain and performed at only a few centers.

Considerations

After an endoscopic gastric biopsy, the patient should be observed for signs and symptoms of complications such as bleeding and perforation. Symptoms include abdominal pain, tenderness, distention, nausea, vomiting, chills, hypotension, and temperature elevation.

ENDOSCOPIC SMALL BOWEL BIOPSY

Endoscopic small bowel biopsy provides important diagnostic information. The direct visualization that is permitted by endoscopic biopsy is advantageous in patients with focal disease.

Indications

Small bowel biopsy is indicated for the evaluation of abnormal tissue and differential diagnoses of conditions such as:

- Celiac disease.
- Whipple's disease, which gives rise to highly specific lesions.
- Malabsorption syndromes.
- Unexplained steatorrhea.
- Small intestinal bacterial overgrowth (SIBO), a syndrome of diarrhea and nutritional deficiencies.
- Tropical sprue.
- Eosinophilic gastroenteritis (rare).
- Intestinal lymphangiectasia.
- Amyloidosis.
- Scleroderma and other conditions that cause chronic diarrhea or weight loss.
- Agammaglobulinemia, severe hypogammaglobulinemia, and abetalipoproteinemia.
- *Giardia*.

Indications for enteroscopy evaluation include:
- Abnormal GI bleeding or GI bleeding when the small intestine has been identified as the source of bleeding.
- Suspected small bowel malignancies (e.g., adenocarcinoma, lymphoma, GI stromal tumors, metastatic tumors).
- Suspected small intestinal bleeding with objective evidence of recurrent or obscure GI bleeding (e.g., iron-deficiency anemia, positive fecal occult blood test, or visible bleeding) in patients who have had upper and lower GI endoscopies (EGD and colonoscopy) that failed to identify a bleeding source.
- Elevated white blood cell count.
- Small bowel obstruction.
- Unexplained diarrhea.
- Abnormal X-ray.
- Known or suspected Crohn's disease.

Procedure

Requirements for small bowel biopsy specimens to be of maximum diagnostic value include:
- Precise localization of the biopsy site.
- Proper orientation and prompt fixation of biopsy specimens.
- Careful study of serial sections of the central half or two-thirds of each biopsy specimen.
- Obtaining the specimen from the region of the duodenal bulb and more distal duodenum, including in the area of the ligament of Treitz.

A number of instruments of various designs can be used for small bowel biopsy. Tools used for endoscopic small bowel biopsy include:
- Upper endoscope.
- Small bowel push enteroscope.
- Deep balloon-assisted enteroscope.
- Endoscopic ultrasound (EUS).

Deep balloon-assisted and push enteroscopy procedures are done under direct visualization for both diagnostic and therapeutic intervention to the entire small bowel.
- Deep balloon-assisted enteroscopy uses an antegrade and/or retrograde approach to examine the small bowel, including the jejunum and ileum. This procedure requires a specially designed 200 cm-long balloon enteroscope in conjunction with an overtube to advance the scope.
- Push enteroscopy is commonly performed using a 160-180 cm colonoscope or with a dedicated 220-250 cm enteroscope. Push enteroscopy evaluates the esophagus, stomach, duodenum, and proximal jejunum approximately 50-150 cm beyond the ligament of Treitz.

Considerations

Capsule endoscopy is typically the next step for patients with unexplained overt GI bleeding and negative upper endoscopy and colonoscopy results. Push enteroscopy can be performed if capsule endoscopy reveals a lesion in the proximal jejunum. Deep balloon-assisted enteroscopy can be performed if a lesion is found in the mid small intestine.

ENDOSCOPIC DIAGNOSIS OF BILIARY AND PANCREATIC NEOPLASMS

Biliary and pancreatic masses can be malignant, benign, or of unknown etiology. Diagnosis can be accomplished using various endoscopic tissue sampling options.

Indications

Cytologic brushings, biopsy forceps, and/or bile aspiration may be used to sample suspected bile duct tissue during endoscopic retrograde cholangiopancreatography (ERCP).

Procedure

- **Brush cytology** is the most commonly used method for tissue sampling during ERCP. Recent studies estimate that cytologic brush sensitivity ranges from 35%-70% and that specificity generally exceeds 90% (Korc & Sherman, 2016).

  Tumor type influences the accuracy of brush cytology. Typically, primary biliary tumors demonstrate higher sensitivity than those that do not originate from the biliary tract resulting from mucosal changes that facilitate cell collection. The endoscopist's technique and the cytologist's interpretation of results can also influence the accuracy of brush cytology.
- **Forceps Biopsy** can be used to acquire tissue during ERCP. Reports estimate that the sensitivity of biopsy forceps ranges from 43% to 88% and that specificity generally exceeds 90%, comparable to brush cytology sensitivity (Korc & Sherman, 2016).

  Specially designed forceps are used to obtain samples from biliary strictures. However, the location and characteristics of the tumors and strictures can affect the adequacy of tissue collection. Additionally, it is often difficult to properly position and control the forceps to sample the targeted area.

Considerations

Because of the difficulty obtaining adequate biopsy samples, brush cytology remains the technique of choice to obtain tissue samples during ERCP.

Gallbladder lesions observed on ERCP may be amenable to endoscopic biopsy, although this is rarely performed in clinical practice.

FINE-NEEDLE ASPIRATION (FNA) AND FINE-NEEDLE BIOPSY (FNB)

Conventional fine-needle aspiration (FNA) needles allow for the acquisition of cells for microscopic analysis. However, cytologic interpretation of FNA samples can be affected by:
- the site and size of the lesion.
- difficulties in obtaining an adequate aspirate.

- the presence of blood and benign epithelial cells.
- the availability of a cytopathologist to offer an onsite diagnosis.

Needle sizes range from 25-gauge to 19-gauge.

To overcome these potential FNA limitations, larger-caliber FNB needles can be used to obtain core tissue biopsy for histologic analysis that preserves tissue architecture. FNB needles are designed with either a special geometry of the cutting tip or a side-slot in the distal portion of the needle to obtain core biopsies for histologic analysis.

A specially designed needle, stylet, and handle have been developed for echoendoscopic use. The need to protect the echoendoscopic channel and allow for flexibility during tip deflection of the echoendoscope dictated the design of the needle. The needle encased in an outer sheath is passed through the biopsy channel and exits above the optic lens. The needle can be seen both endoscopically and with ultrasound as it is advanced from the sheath. If used, the balloon is deflated to avoid puncture.

FNA and FNB are used to diagnose and stage solid and cystic lesions in the GI tract and bordering organs, including masses in the pancreas, liver, and bile duct, especially in which prior conventional endoscopic biopsies and other non-invasive imaging modalities have been non-diagnostic or the identified lesions are too well-encased by surrounding vascular structures to allow percutaneous biopsy. Fine-needle sampling is also used for GI subepithelial lesions, lung lesions, mediastinal lymphadenopathy, and adrenal masses.

Indications

FNA and FNB are indicated to:

- Obtain biopsy and cytology specimens of mucosal and submucosal lesions.
- Aspirate fluid from cystic lesions, pseudocysts, and fluid collections for both diagnosis and therapy.

Procedure

For patients with small focal mass lesions, diffuse enlargement of the pancreas with calcification, or equivocal findings on ultrasonogram or CT scan, ERCP is often performed first or in conjunction with EUS.

Several attempts at aspiration may be needed to obtain an adequate number of samples, depending on the skill of the operator and how promptly the specimins are prepared.

The aspirated material is sent for cytological examination according to organization policy and procedure.

Endoscopic ultrasound-guided fine needle aspiration (EUS-FNA) and endoscopic ultrasound-guided fine needle biopsy (EUS-FNB)

The performance of ultrasound-guided FNA and FNB is similar to other aspiration techniques. It is used for the diagnosis and management of patients with pancreatic mass lesions. EUS-FNA is generally used to obtain tissue from masses or lymph nodes. It is also used to aspirate the contents of cystic structures (e.g., pancreatic cysts).

- After both endoscopic and EUS evaluation, the needle may be passed into the targeted tissue.
- In EUS-guided FNA, the assistant is often responsible for the aspiration of the cytological or histological material. This is necessary because of the endoscopist's need to maintain the position of the endoscope with the transducer in contact with the GI wall during needle puncture.
- Under ultrasound guidance, the needle punctures the targeted lesion, the stylet is removed, and suction is applied when directed by the physician.
- While maintaining constant suction, the needle is moved back and forth within the lesion.
- Suction is released while the needle is within the lesion to reduce the risk of aspiration of surrounding tissue.
- The entire needle assembly is removed from the endoscope.
- Specimens, including salvage wash, are prepared according to institution policy and procedure.
- FNB samples may also be fixed before and sent to the lab for further histologic examination.

Considerations

The presence of a cytopathologist in the procedure suite greatly improves the accuracy of diagnosis. When the cytopathologist is present, fewer passes may be needed to obtain adequate specimens for diagnosis. Cazacu et al (2018) reported studies demonstrating that, for pancreatic masses, maximal diagnostic yield can be obtained after two to five passes.

According to a 2017 ASGE report on devices for use with EUS, most patients tolerate these procedures extremely well. Fewer 2% of all patients having EUS-FNA experience complications such as bleeding, bacteremia, and pancreatitis, and the overall mortality rate is 0.98%. They also report studies have demonstrated that FNB devices have shown no significant difference in adverse events compared to FNA devices.

Most EUS-FNA procedures are performed on an outpatient basis. When these are combined with other procedures such as EGD and ERCP, however, patients may need additional recovery time.

ENDOSCOPIC COLORECTAL BIOPSY

Although colonoscopy permits visualization of the entire inner surface of the colon, mucosal biopsy is often necessary to arrive at a specific diagnosis. Biopsies of the colonic mucosa and lesions are obtained routinely for diagnostic purposes. The main advantages of taking biopsy specimens through an endoscope are the ability to sample the entire colon and precisely target the area of interest. In addition, multiple specimens are easily obtained endoscopically.

Indications

Colorectal biopsies are indicated for:

- Suspected neoplastic lesions of the rectum and colon.
- Suspected collagenous or microscopic colitis.
- Inflammatory bowel disease:
  - Suspected Crohn's disease. The typical mucosal biopsy shows a focal ulcerative and inflammatory process, rather than the diffuse abnormality seen with ulcerative colitis. Histological documentation of granulomatous inflammation of bowel mucosa argues strongly for a diagnosis of Crohn's disease, particularly when discrete noncaseating granulomas are found and other causes of granulomatous disease have been excluded.
  - Suspected ulcerative colitis. Biopsy shows characteristic inflammation of the colonic mucosa, with changes ranging from mild degrees of inflammation to microscopic erosions of the epithelium and crypt abscesses. A complete colonoscopy with biopsy is especially useful when prior sigmoidoscopic findings are equivocal.
  - Suspected pouchitis. This is a complication in patients with ulcerative colitis. Symptoms include abdominal cramps, urgency, night incontinence, and fever. Crohn's disease should be considered if macroscopic changes are present in the afferent loop. The diagnosis can be confirmed by histology. Treatment of pouchitis is proctocolectomy with ileoanal anastomosis.
- As an adjunct to cultures and smears of rectal mucosa for detecting an infectious process in men and women who engage in anal intercourse and may be at risk for acquiring human immunodeficiency virus (HIV) infection. Mucosal biopsy may demonstrate focal crypt epithelial cell degeneration, which has been described as a characteristic pathological feature in acquired immunodeficiency syndrome (AIDS). When performing a culture for AIDS-related diseases, the tissue is also examined for CMV, HSV, other sexually transmitted diseases, ***Cryptosporidium***, ***Micrococcus***, and amoeba.
- Diagnosis of a suspected neuronal lipidoses or unexplained signs of a degenerative nervous system disorder. Morphological and histochemical staining characteristics of the biopsy specimen can help diagnose certain neuronal diseases, including Niemann-Pick disease, Tay-Sachs disease, and Hurler syndrome.
- Suspected schistosomiasis, in which the characteristic ova of the parasite can be identified in the biopsy material. However, because the eggs may go undetected, a serologic test may also be required.
- Suspected amebiasis, in which volcano-like lesions are seen. The center of the "volcano" may have organisms.
- Assessment of progress in patients who are undergoing therapy.

Procedure

Obtaining useful diagnostic information from a colorectal biopsy specimen requires a few careful steps:

- Intestinal location of the biopsy specimen(s) must be clearly defined by anatomical area (e.g., rectum, descending colon, splenic flexure). Centimeter designation is not usually accurate, especially if that number will be used to subsequently return to the location of the area in question.
- Separate containers must be used for tissue specimens from different biopsy sites.
- Specimens sent to pathology must stipulate that multiple sections should be made of all specimens for examination.
- All specimens must be clearly labeled according to the organization's policies.

Considerations

It is important to note that significant and potentially diagnostic abnormalities may be present in tissue obtained by colorectal biopsy, despite a normal endoscopic appearance of the mucosa.

RECTAL SUCTION BIOPSY

Most intraluminal biopsies of the GI tract go no deeper than the muscularis mucosa, which is adequate for most diagnoses. Suction biopsy more consistently penetrates the submucosa.

Indications

Rectal suction biopsy is obtained for suspected Hirschsprung's disease, in which evidence of the myenteric ganglia is sought within the submucosa.

Procedure

Rectal biopsy is the definitive diagnostic test for Hirschsprung's disease, which demonstrates the absence of ganglion cells in the submucosal and myenteric plexuses. The biopsy specimen is obtained from above the dentate line and below the peritoneal reflection to reduce the chance of perforation.

Specialized mechanical suction biopsy device kits have been developed that offer the advantage of reproducibly obtaining a standard volume of tissue sample. Suction mucosal biopsies can be done without anesthesia at the patient's bedside.

The capsule of the biopsy tool is placed within the rectum and held against the rectal wall. Using a manometer to measure suction pressure, the tissue is suctioned into the capsule's side hole. While the suction is maintained, the trigger is fired, and the tissue is cut in a guillotine fashion. The specimen is sent to pathology per the organization's policies.

Considerations

Suction biopsy offers a low risk of perforation or bleeding, but complications may include bleeding. Rectal biopsy is not recommended in premature infants due to poor diagnostic specificity and higher risk of complications.

Endoscopic rectal biopsies may not provide sufficient submuosal tissue to diagnose Hirschsprung's disease and may require sedation in a surgical or endoscopy suite. Open rectal biopsy, in which a surgical excision of a segment of the rectal wall is required to obtain a specimen, is performed in an operating room with the patient placed under general anesthesia. Punch biopsies and full-thickness biopsies obtain a deeper tissue specimen.

PERCUTANEOUS LIVER BIOPSY

Percutaneous liver biopsy is a needle aspiration of liver tissue for histological analysis. The procedure is indicated for diagnosis of liver disease or suspected malignancy, assessment of prognosis (disease staging), and/or to assist in making therapeutic management decisions. In most cases, liver biopsy is performed percutaneously, guided by imaging such as ultrasonography. It can also be done under direct vision via laparoscopy or surgery or via EUS guidance.

Indications

Indications for percutaneous liver biopsy related to evaluation include:

- Abnormal liver tests of unknown etiology.
- Fever of unknown origin.
- Focal or diffuse abnormalities on imaging studies.
- Viral hepatitis and its sequelae.
- Alcoholic hepatitis.
- Evaluation of a liver mass that does not exhibit typical imaging features of hepatocellular carcinoma (HCC).
- Quantitative estimation of iron in hemochromatosis.
- Quantitative estimation of copper in Wilson's disease.
- Estimation of the severity of alcoholic liver disease.
- Evaluation of drug toxicity.
- Evaluation of the suitability of a donor liver for transplantation.
- Diagnosis and staging of nonalcoholic fatty liver diseases (NAFLD), including nonalcoholic steatohepatitis (NASH).
- Evaluation of unexplained jaundice.
- Diagnosis of cholestatic liver disease.
- Evaluation of infiltrative or granulomatous disorders.
- Evaluation of liver injury from immunosuppressive agents (e.g., methotrexate).

Indications for percutaneous liver biopsy related to prognosis include:

- Staging of known parenchymal liver disease.

Indications for percutaneous liver biopsy related to management/surveillance during treatment include:

- Developing treatment plans based on histologic analysis.
- Follow-up evaluation while on antiviral treatment for chronic hepatitis C (rare).
- Monitoring of disease activity of autoimmune hepatitis during treatment (may assist in determining if therapy can be discontinued).

Indications for percutaneous liver biopsy relevant to the patient who has undergone a liver transplant include:

- Diagnosis of acute cellular rejection.
- Diagnosis of chronic rejection.
- Diagnosis of recurrent hepatitis C.
- Diagnosis of posttransplant lymphoproliferative disorder.
- Diagnosis of CMV hepatitis.
- Protocol biopsies to monitor for fibrosis or inflammation (particularly in patients who received liver transplants to treat liver failure in chronic hepatitis C).

Percutaneous liver biopsy may be absolutely contraindicated in patients who:

- Are unable or unwilling to cooperate or are confused.
- Have severe coagulopathy or thrombocytopenia (uncorrected prior to the liver biopsy).
- Have infection of the hepatic bed.
- Have extrahepatic biliary obstruction.
- Have suspected vascular lesions (e.g. hemangioma, vascular tumor, echinococcal cyst).
- Have not discontinued NSAID use (including aspirin) within the last 7-10 days.
- Are not able to have the biopsy site identified by percussion and/or ultrasound.
- Refuse blood transfusion or support.
- Have hepatoma.

Relative contraindications to percutaneous liver biopsy may include:

- Ascites.
- Morbid obesity.
- Amyloidosis (because of the possible risk of liver rupture).
- Cystic echinocccosis (Hydatid disease).
- Infection within the right pleural cavity and/or below the right hemidiaphragm.

Procedure

Prior to the procedure

Before to the procedure, the gastroenterology nurse should:

- Establish proper clotting factors, adequate blood volumes, and absence of infection. Laboratory studies commonly done before performing percutaneous liver biopsy include complete blood count (CBC), **platelet count**, **prothrombin time (PT)**, international normalized ratio (INR), and (in some organizations) a partial thromboplastin time and/or cutaneous bleeding time.
- Review medication history. Medications that prolong bleeding time, such as clopidogrel, aspirin, and NSAIDs, should be discontinued at least 1 week prior to the liver biopsy. Alternative and complimentary

therapies, such as ginkgo biloba and fish oil, should also be stopped at least 1 week prior to the liver biopsy. Patients who inadvertently took aspirin or other NSAIDs during the week before the procedure can be rescheduled, or the liver biopsy can be performed if the patient has a normal bleeding time immediately before the procedure.

- Review anticoagulant use. For patients on pharmacologic anticoagulation, antiplatelet medications, and anticoagulant medication, these medications should be discontinued up to 10 days prior to the procedure. There is question about whether new oral anticoagulants such as dabigatran (Pradaxa), rivaroxaban (Xarelto), and (Eliquis) with a much shorter half-life than warfarin could be discontinued closer to the procedure, but clinical guidelines are lacking. Management of specific agents should be individualized and carefully weighed against the need for liver biopsy and the potential risk of bleeding during or after liver biopsy. According to the American Association for the Study of Liver Diseases (AASLD) (2009), antiplatelet therapy may be restarted 48-72 hours after liver biopsy and warfarin may be restarted the day following liver biopsy.
- Complete a nursing admission database that also may include obtaining baseline vital signs, blood typing and crossmatching for blood transfusion, administration of prophylactic vitamin K, and chest and abdominal radiographs and scans. Organization policy and procedure, as well as professional society evidence-based guidelines related to fasting prior to liver biopsy, should be followed.
- Determine NPO status.

Preparing the patient for a percutaneous liver biopsy procedure should include:

- Education about the procedure. Based on organization protocol, the patient should be informed of the recommendation to remain close to the facility after the procedure in case of potential late complications. The patient should be provided information on the purpose of the test, techniques to be used, and sensations the patient is likely to experience. The patient will have to remain in bed for several hours after the biopsy, so he or she should be encouraged to void before the procedure. Breathing techniques needed during the procedure should be explained, practiced, and reinforced. After-care instructions should be discussed with the patient and a responsible adult caregiver.
- Written informed consent. The patient should be educated about his or her liver condition and about investigations other than liver biopsy (if any) that may also provide diagnostic and prognostic information. Patients must be carefully informed about the procedure itself, including alternatives, risks, benefits, and limitations.
- Review of patient history. The patient's medical history should be reviewed, including last use of anticoagulants, aspirin, or NSAIDs.
- Determination on laboratory studies (INR, PTT, bleeding time, platelet count). Ascertain whether those performed during the previous 7 days are within acceptable parameters for performing the biopsy.
- Medication maintenance. The patient should take other prescribed drugs with a few sips of water the morning of the procedure, according to the physician's orders or institution

During the procedure

On the day of the procedure:

An IV line or saline lock may be started in order to provide sedation or analgesia, according to organization policy and procedure, and/or as a precaution should there be significant pain or bleeding after the procedure.

Administer any ordered blood products prior to procedure.

Draw blood for type and cross-match as ordered.

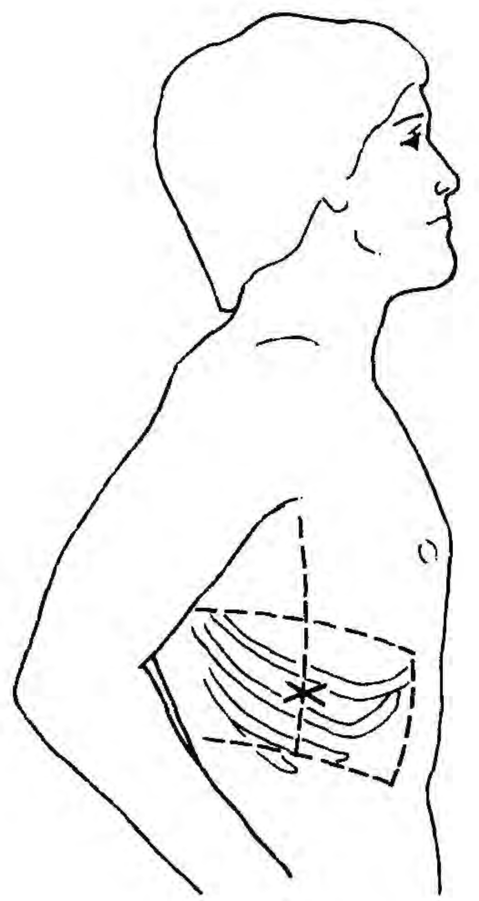

Figure 25.2. Liver Biopsy Site

Percutaneous liver biopsy is performed using a palpation/percussion-guided method or an ultrasound, computed tomography (CT), or magnetic resonance imaging (MRI) image-guided method. If the apparent lesion is of a size or in a location unlikely to be reached with the standard blind-needle technique, image-guided biopsy techniques using ultrasonography, or CT may be more appropriate. Ultrasound guidance is also helpful for patients who are obese, have a history of prior upper abdominal surgery, or are suspected of having advanced cirrhosis with possible atrophy of the right lobe of the liver based on clinical and laboratory findings. Ultrasound guidance is also helpful for less experienced operators.

- The patient is placed in a supine position near the right side of the bed with a pillow under the right side. The right arm is placed comfortably under the head, and the head is turned toward the left. Because pediatric patients cannot be expected to cooperate, they must be adequately sedated. Infants may be restrained on a pediatric immobilizer or held by assistants.
- To identify the biopsy site, the physician percusses over the right hemithorax at the mid-axillary line in order to recognize, then mark, the area of maximum liver dullness, typically found between the sixth and ninth intercostal space. A bedside ultrasound is often used to confirm the location.
- The surrounding skin is prepped. Sterile drapes are arranged around the biopsy site (Figure 25.2).
- The skin, subcutaneous tissues, and deeper intercostal muscles at the insertion site are infiltrated with a local anesthetic, such as lidocaine (Xylocaine), assuming the patient has no known allergy to local anesthetics. A small scalpel incision is made in the skin over the biopsy site.
- Biopsy needles are selected based on physician preference, technique used, and/or patient factors. Three types of needles are available to obtain a percutaneous liver biopsy:
    - Suction needles (Menghini, Klatskin, Jamshidi). For suction biopsies, a suction biopsy needle is attached to a syringe containing a small amount of sterile saline.
    - Spring-loaded cutting needles with triggering mechanisms (various vendors). When using a spring-loaded or automated needle, the device is passed through the skin incision and advanced into the liver to obtain the biopsy specimen.
    - Cutting needles (Vim-Silverman, Trucut). Like suction needle biopsies, cutting needles utilize a similar approach but do not require a saline flush.
- The needle attached to the syringe is pushed through the skin incision and advanced into the intercostal muscles. A trochar may be used to create the tract. Pain with insertion of the biopsy needle indicates inadequate local anesthesia. The saline is injected to expel any tissue fragments from the needle.
- To prevent pleural cavity or diaphragm puncture, the patient is asked to hold his or her breath at full expiration while the biopsy needle is advanced into the liver. For pediatric patients, the biopsy should be timed to the respiratory cycle.
- While constant aspiration suction is maintained, the needle is advanced 4-5 cm, and then quickly withdrawn. During the procedure, the needle is actually in the liver for only a fraction of a second. The patient may resume breathing 5-10 seconds after the specimen is obtained. A second biopsy through the same incision at a slightly different angle may be necessary.

The biopsy sample is expelled from the needle and then sent to the laboratory for analysis per organization protocol. If lymphoma is suspected, accurate diagnosis requires that a portion of the biopsy specimen be placed in saline, not formalin. The minimal amount of tissue necessary for evaluation by the pathologist is usually a core sample longer than 1 cm.

After the procedure

After the procedure is completed, an adhesive bandage is applied to the insertion site, and the patient is turned onto his or her right side against a firm support (e.g., bath blanket, sandbag, pillow). The patient maintains that position for the first 1-2 hours, followed by a supine position for another hour. AASLD (2009) recommends that vital signs should be monitored frequently at least every 15 minutes for the first hour. The recommended observation time after percutaneous liver biopsy is 2-4 hours, depending on local expertise and protocol. The risk of bleeding is greatest immediately after liver biopsy. If bleeding from the site persists, a pressure dressing may be needed. The insertion site must be observed closely for any changes in condition.

Patients may be allowed clear liquids shortly after biopsy. Solid food may be reintroduced after several hours if no serious complications have developed.

For outpatient procedures, when the patient meets discharge criteria according to organization policy and procedure, he or she may be discharged home accompanied by a responsible adult.

Typical liver biopsy discharge instructions may include notifying the patient and responsible adult caregiver that the patient:

- Must be accompanied by a reliable person the first night after the procedure.
- Cannot drive until the day after the biopsy anesthesia was given.
- Cannot shower for 24 hours after the biopsy. If desired, the patient can take a bath with a sponge or washcloth, but cannot get the biopsy dressing wet. When able to shower, the biopsy site should not be scrubbed. The area may be gently washed and patted dry.
- Can remove the bandage covering the biopsy site 48 hours after the procedure.
- Should not engage in strenuous exertion and exercise for 5-7 days following the biopsy to allow the body to heal.
- May not lift anything heavier than 10 pounds for up to 1 week (per organization policy) after the proce-

dure to avoid increasing intra-abdominal pressure, which could cause bleeding from the puncture site.
- Should discuss with the physician when it would be appropriate to return to work. Barring evident complications or significant pain that requires a narcotic, patients typically should be able to return to work the day after the procedure.
- May not start taking an antithrombotic without clear instructions from the physician.

Considerations

The physician should be notified immediately if the patient experiences:
- Increased pulse and decreased systolic blood pressure.
- Prolonged pain that radiates to the back, abdomen, or shoulder.
- Abdominal distention or obvious bleeding or drainage from the biopsy site.
- Increased temperature.
- Dizziness or lightheadedness.
- Shaking chills.
- Yellow eyes or skin.
- Vomiting of blood.
- Sudden chest pain or increased shortness of breath.

Complications

A 1% overall rate of serious complications following liver biopsy has been reported (Boyum et al., 2016; De Peape et al., 2018). Overall mortality risk has been estimated to be 0.2 % (West & Card, 2010).

Pain is the most common complication following percutaneous liver biopsy, typically in the right upper quadrant or right shoulder. Pain can usually be managed with small dosages of narcotics. Puncture of the liver with subsequent bleeding and stretching of its capsule causes the most common and concerning type of pain. The pain is usually dull and mild, worsens with inspiration, and resolves completely within a few hours. Moderate to severe pain should raise the possibility of complications, including gallbladder puncture or bleeding.

Intraperitoneal hemorrhage is the most common serious complication of liver biopsy. Because the liver is highly vascular and must be punctured in the course of the biopsy, all patients have bleeding from the liver after biopsy, but it usually stops within several minutes. Severe bleeding, demonstrated by a change in vital signs with radiographic evidence of intraperitoneal bleeding, requires hospitalization, probable transfusion, potential radiologic intervention, or surgery.

Other reported post-procedure complications include:
- Hematoma.
- Hemobilia.
- Infection—local or systemic (bacteremia, abscess, sepsis).
- Pneumothorax.
- Hemothorax
- Bile peritonitis.
- Malignant tumor track seeding.
- Needle breakage.
- Liver laceration.

Complications of liver biopsy can be minimized by avoiding biopsy in high-risk patients and strictly adhering to protocol for obtaining the biopsy.

Other methods of liver biopsy

Percutaneous liver biopsy is the classic method for performing liver biopsy. However, other methods may be used, including:
- EUS-FNB, performed through the stomach, which can be used to biopsy solid liver lesions.
- FNA, performed by an interventional radiologist using ultrasound or CT scan guidance to sample a focal or targeted liver lesion. This procedure is contraindicated if a safe access route to the lesion cannot be identified or in patients with underlying coagulopathy.
- **Transjugular liver biopsy**, an angiographic approach that may be a safe and efficient alternative to obtain adequate liver tissue specimens in patients with diffuse liver disease when a percutaneous procedure is either suboptimal or presents contraindications. These contraindications may include high-volume ascites, morbid obesity, anticoagulant use, or coagulopathies that place patients at increased risk for intraperitoneal hemorrhage that may result from standard liver biopsy transection of the liver capsule.
- **Surgical/laparoscopic liver biopsy** refers to the surgical or laparoscopic examination and biopsy of lesions or masses on the liver surface, with direct and immediate control of bleeding under direct visualization. This procedure offers an alternative for patients with contraindications to percutaneous liver biopsy or for patients who have other indications for laparoscopy. However, its use has declined in favor of less invasive radiological procedures.

CYTOLOGY AND CULTURE OF THE GI TRACT

Cytology refers to the collection of cells for microscopic examination by brushing or washing a particular site.

Cytologic evaluation analyzes cells from the body under a microscope to determine what the cells look like, how they form, and how they function. The test is usually used to look for cancers and precancerous changes. It may also be used to look for viral infections in cells. The test differs from a biopsy because only cells are examined, not pieces of tissue.

Indications

Indications for obtaining endoscopic cytology specimens include:
- Suspected malignancy.
- Suspected infection.

Indications for obtaining cultures include:
- Symptoms of infection.
- Abnormal X-ray.
- Endoscopic image suspicious for infection.

Procedure

During this procedure, specimens are collected and analyzed for viruses, fungi, bacteria (anaerobic, aerobic, and acid-fast bacilli) in the GI tract. Cells needed for cytology or culture may be obtained using a cytology brush, washing, or aspiration technique.

Brush cytology

Brushing using a cytology brush is a method used for broad sampling of the mucosal surface, which is usually performed for:
- Diagnosis of infectious processes such as candidiasis in the upper GI tract.
- Lesions that are inaccessible for biopsy such as tight strictures in the esophagus.
- Strictures in the pancreaticobiliary tree.

Cytology brushes share a common design, including a metal shaft that runs lengthwise through a protective sheath and a head of bristles at the distal tip. Sheathed brushes are commercially available in various sizes, lengths, and stiffness, as well as single- or multi-lumen and with or without a flexible guide tip. Appropriate selection depends on the location of the lesion and the endoscope length and channel diameter being used.

The cytology brush is passed through the endoscope in much the same way as biopsy forceps. The brush is lined up with the suspicious area, extended beyond its plastic sheath, and rubbed across the mucosa. After harvesting cells from the suspect lesion, the head of the brush is withdrawn into the outer sheath before it is removed from the endoscope. Use of a sheathed brush minimizes the shedding of cells into the suction/biopsy channel.

After the brush is removed from the endoscope, thin smears of cells are made on clean microscope slides. The brush is gently rotated over the slide, brushing in one direction only. Specimens should be fixed with a cellular fixative, labeled, and sent to the laboratory per organization protocol. Alternatively, the tips of brushes may be cut off after the sample has been taken and sent intact to the laboratory. They should be moistened with one or two drops of sterile saline for culture or sent to the laboratory in a container of three to six drops of sterile saline for cytological examination. For the diagnosis of *Candida* infections, a simple smear can be submitted for fungal staining. Organization protocol should be followed.

Fluorescence in situ hybridization (FISH) is an increasingly used technique that can detect chromosomal aneuploidy in specimens from biliary brushing. FISH can be used with any biliary brush cytology tissue specimens obtained from patients with suspected pancreaticobiliary cancer or primary sclerosing cholangitis, who are at increased risk for cholangiocarcinoma. Combining FISH results and biliary cytology offers higher diagnostic yield.

Washing technique

Washing is indicated for cell collection for cytology or culture analysis. A specimen trap is attached to the endoscope suction port. When collecting a washing for cytology, 20-30 ml sterile saline is injected through the endoscope biopsy channel onto the lesion. The specimen is then collected by aspirated the fluid through the endoscope into a specimen trap. The specimen is immediately sent to the lab to prevent lysis of the cells. Alternatively, an amount of fixative is added to the specimen equal to the fluid aspirated or in a volume determined by the organization's policy and procedure.

When collecting a washing for culture, a specimen trap is attached to the endoscope suction port, and the specimen s aspirated through the endoscope into the trap. If difficulty obtaining the specimen is encountered, 10-20 ml of sterile saline is injected through the biopsy channel. The specimen is immediately sent to the lab with or without fixative, according to the organization's policy and procedure and the test that is ordered.

Needle aspiration cytology

The aspiration technique is also used to collect specimens for culture. A catheter is inserted through the endoscope biopsy channel with a syringe attached and closed. The specimen is aspirated, injected into a sterile specimen container, and sent for analysis according to the organization's policy and procedure.

An aspiration needle can be used to retrieve cytological material from the upper GI tract. As previously discussed, using EUS-FNA, aspiration cytology is indicated for pancreatic masses and neuroendocrine tumors, pancreatic cystic lesions, peri-intestinal and mediastinal lymphadenopathy, mucosal and submucosal tumors, or infiltrating malignancy in the gastric wall.

Rectal culture

Rectal culture is used to isolate and identify organisms that can cause symptoms or disease in this area of the GI tract. It may be performed initially or when a stool sample is difficult to obtain.

The main pathogens that are isolated are bacterial or parasitic enterocolitis, gonorrhea infection, and vancomycin-resistant enterococci (VRE).

A rectal culture is a simple to perform and quick and normally painless to the patient. A cotton swab is gently inserted into the rectum and rotated completely, then removed. A smear from the swab is placed in culture media to grow microorganisms for evaluation. As growth occurs, organisms can be identified.

Colonic tissue can be biopsied during colonoscopy and sent to the lab for culture.

CASE SITUATION

This is the second hospital admission for Gary Mason, age 28, who is diagnosed with acquired immune deficiency syndrome (AIDS). Initially, Gary was admitted for severe dysphagia (difficulty swallowing) and odynophagia (pain on swallowing). His upper endoscopy at that time revealed severe esophageal inflammation with whitish patches. Specimens taken at that time revealed candidiasis. Gary now has diarrhea and weight loss, in addition to some continuing odynophagia. He is scheduled for both upper and lower endoscopy, with biopsy and cytology.

Points to think about

1. Patients who are immunocompromised are susceptible to a host of GI problems. What are some of the gut's normal defenses against invasion by pathological organisms, and how are these defenses affected by AIDS?
2. On upper endoscopy, Gary is found to have dense patches of creamy, waxy exudate; esophagitis; and two very large, deep ulcerations. His duodenum is normal, but there are two raised, dark red, 10-mm lesions in his stomach. What types of specimens will help diagnose the cause of these problems?
3. The gastroenterology nurse is involved in specimen collection. Why is it important to develop a good working relationship with laboratory personnel?
4. Diarrhea can have many causes in the patient with AIDS, including amebiasis, shigellosis, gonorrhea, CMV, HSV, syphilis, lymphogranuloma venereum, cryptosporidiosis, and chlamydia. What tests would help determine the cause of Gary's diarrhea?
5. In dealing with infectious specimens, what special practices should be employed?

Suggested responses

1. The normal defenses of the GI tract against pathological organisms include the bactericidal effect of gastric acid, intestinal motility, normal bowel flora, intact mucosa, humoral immunity, lymphocytes, macrophages, and neutrophils. In patients with AIDS, deficiencies in the immune system (neutropenia and alterations in hormonal and cell-mediated immunity) affect the normal defense mechanisms of the GI tract. In addition, infections can disrupt intact mucosa, decrease motility, and overrun normal flora. Because of this, the gut is susceptible to local and systemic infections.
2. In patients with AIDS, the esophagus is often involved with opportunistic infections such as *Candida*, HSV, and CMV. Comprehensive testing is recommended, as the scope is already in place, allowing easy specimen acquisition. Specimens that will help diagnose the cause of Gary's symptoms include:
   a. Brush cytology of the creamy patches to test for *Candida*. Disposable brushes are packaged sterile. It is important to know the organization's routine for preparing specimens. Often, the brush may be delivered to the laboratory in its package with orders for fungal culture and stains, and the laboratory will prepare it. Although symptoms of *Candida* infections may improve or disappear with therapy, this does not always indicate disappearance of mucosal lesions. Follow-up endoscopies are often necessary.
   b. Biopsy of the deep esophageal ulcerations. If the pathologist knows the suspected diagnosis, special stains can be applied to identify fungal, viral, or parasitic invasion.
   c. Viral cultures of esophagitis. Special media are usually obtainable from the laboratory for viral cultures. They are kept refrigerated and have a relatively short shelf life. A forceps biopsy specimen is placed in the viral culture medium and labeled.
   d. Biopsy of the dark red lesions in the stomach. These lesions are most likely Kaposi's sarcoma. They can appear anywhere in the gut and vary in color from red to deep purple. Fifty percent of patients with AIDS who present with Kaposi's skin lesions will also have GI tract lesions, indicating a much poorer prognosis (Lee et al., 2015). In the gut, these lesions lie submucosally. Standard forceps biopsy, which only penetrates mucosal tissue, may be negative. It can be helpful to use the largest forceps that will fit through the channel of the scope. It may be appropriate to use the snare lift-and-cut technique in this situation. Care should be taken in case there is excessive bleeding from these lesions at biopsy.
3. Laboratory personnel can be very helpful. Specimens must often be taken to different departments within the laboratory (e.g., pathology for biopsy and cytology, microbiology for cultures and staining). There are many different ways to order tests, and many specimens need special collection methods and media. Having this knowledge before the procedure can save time and can help avoid mistakes, loss of specimens, and repeat procedures.
4. Tests that can help specify the cause of Gary's diarrhea include:
   a. Colonoscopy to examine the whole large bowel, ascertain the extent of disease, and collect specimens.

b. Stool cultures and microscopic examinations to determine what agent is causing the colitis. In the colon, this may be difficult as the list of possible agents is long. Many of these organisms may be present simultaneously, or they may not appear in every specimen. This means that stool cultures and microscopic examinations may need to be done on a serial basis and must often be repeated.
c. Specimens for gram stain, bacterial culture, ova and parasites, viral cultures, and special staining techniques. Again, each organization uses different methods and media. It is important to be familiar with the organization's procedures and to give the pathologist and laboratory personnel as much information as possible regarding the potential diagnosis.
d. Biopsy to help differentiate colitis from Crohn's disease or ulcerative colitis.
e. Biopsy of the characteristic red lesions of Kaposi's sarcoma.

5. Universal precautions apply to the handling of all specimens. In addition, the contaminated hands or clothing of health care personnel may spread infections to other patients who might have an immune deficiency. It is important to remember that patients with AIDS are immunocompromised and, therefore, are more susceptible to infection than most patients.

   Standard precautions when caring for all patients and handling equipment and specimens should be followed, including:
   a. Implement universal precautions for blood and body fluids for all patients. This eliminates the need for special warning labels on specimens because all specimens should be considered infective.
   b. Wear gloves, protective gowns, masks, and protective eyewear when collecting specimens.
   c. Wash hands thoroughly when gloves are removed and dispose of personal protective equipment appropriately before exiting the procedure room.
   d. Put all specimens in well-constructed containers with secure lids to prevent leaking during transport.
   e. Collect each specimen carefully to avoid contaminating the outside of the container and the laboratory form accompanying the specimen.
   f. Subject endoscopes used on all patients, regardless of their condition or diagnosis, to the same scrupulous cleaning and disinfecting regimen.
   g. Use sterile biopsy forceps or single-use forceps to collect specimens. Reusable forceps are sterilized after each use by ultrasonic cleaning and steam under pressure.
   h. Use single-use cytology brushes.

REVIEW TERMS

aspiration, biopsy, biopsy forceps, brush cytology, culture, cytology, endoscopic mucosal resection (EMR), endoscopic mucosal dissection (ESD), exfoliative brush cytology, fine-needle aspiration (FNA), fine-needle biopsy (FNB), frozen section, histopathological analysis, percutaneous liver biopsy, suction biopsy

REVIEW QUESTIONS

1. Endoscopic biopsy is contraindicated in patients with:
   a. Carcinoma.
   b. Severe coagulopathy.
   c. Inflammatory bowel disease.
   d. GI polyps.
2. Frozen sections are used when:
   a. The biopsy specimen will be preserved for a long period of time.
   b. A fixative is needed to preserve the specimen.
   c. Immediate denial or confirmation of malignancy is required.
   d. A pathologist is not immediately available.
3. The most likely complication of endoscopic biopsy is:
   a. Excessive bleeding.
   b. Infection.
   c. Tumor seeding.
   d. Nausea and vomiting.
4. Suspected esophageal carcinoma is most often sampled using what technique?
   a. Endoscopic mucosal resection (EMR).
   b. Needle aspiration.
   c. Endoscopic biopsy.
   d. Polypectomy.
5. Specimens for biopsy of the upper portion of the small bowel are usually taken from what general area?
   a. The duodenum.
   b. The jejunum.
   c. The ileum.
   d. The ligament of Treitz.
6. During EUS-guided FNA, aspiration of tissue is accomplished using suction applied with:
   a. A syringe.
   b. A manometer.
   c. A special cannula.
   d. A proprietary device.
7. The length of time a patient should remain on his or her right side following a liver biopsy is:
   a. 1-2 hours.
   b. 4-6 hours.
   c. 6-8 hours.
   d. 8-10 hours.

8. To prevent pleural cavity or diaphragm puncture during a percutaneous liver biopsy, insertion of the needle in pediatric patients should be:
   a. Timed to the patient's respiratory cycle.
   b. Done while the patient holds his or her breath on expiration.
   c. Done slowly.
   d. Done under ultrasonography or CT guidance.
9. If single-use cytology brushes are sent intact to the laboratory, they should be moistened with:
   a. Sterile saline.
   b. Glutaraldehyde.
   c. Isopentane.
   d. Cellular fixative.
10. Endoscopic submucosal dissection (ESD) requires:
   a. Use of a specialized knife.
   b. Use of a cytology brush.
   c. Use of a sphincterotome.
   d. Use of a guidewire.

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Chapter 26

OTHER PROCEDURES AND TESTS

This chapter will acquaint the gastroenterology nurse and/or technician with non-endoscopic diagnostic tests and procedures used in the practice of gastroenterology. Radiographic and non-radiographic imaging techniques are discussed, including plain and contrast radiography, ultrasonography, magnetic resonance imaging (MRI), and computerized tomography (CT) scan. Studies involving the analysis of gastrointestinal (GI) secretions are described, and laboratory tests involving the analysis of blood, urine, and fecal samples are reviewed.

Learning objectives

After reviewing the content of this chapter, the gastroenterology nurse will be able to:

1. Describe radiographic and non-radiographic imaging studies performed on gastroenterology patients, including indications, contraindications, and patient care instructions.
2. Discuss a variety of diagnostic tests and procedures used to analyze GI secretions.
3. Briefly explain the diagnostic tests used most frequently to analyze the blood, urine, or stool of gastroenterology patients.

BASIC PRINCIPLES

In addition to the diagnostic techniques discussed in the preceding chapters, gastroenterology patients are often required to undergo a variety of other laboratory tests and procedures.

The importance of discussing these tests with the patient cannot be overemphasized. To educate and prepare patients for such tests, it is essential to understand the role of these tests in the diagnostic process and communicate this information to the patient. Complete instructions and explanations by the gastroenterology nurse may help ensure the patient's compliance and allay any feelings of anxiety. Most patients do better during a test when they have received information about the procedure they will undergo and the sensations they are likely to experience. It may be helpful to provide patient instructions when a member of the patient's family or significant other is present to offer support and reinforce the teaching. After completion of the education, the patient's learning and comprehension should be ascertained and documented.

The gastroenterology nurse should review the patient's health records, including medical and surgical history, medications, and allergies before any diagnostic testing is conducted, and verify compliance with any necessary preparations such as duration of nothing-by-mouth (NPO) status, withholding of certain medications, and or preliminary blood or urine work. Written informed consent should be obtained from the patient when appropriate.

During the procedure, the gastroenterology nurse may be responsible for patient positioning, handling specimens or test samples, gathering instruments and supplies, and calibrating any recording devices.

After the procedure is completed, the gastroenterology nurse should monitor the patient for adverse effects and provide discharge instructions, patient teaching on how to obtain test results, and any follow-up recommendations.

This chapter provides general information regarding the included procedures and tests. Individual organizational policies, patient instructions, and physician preferences regarding these procedures and tests may vary. The gastroenterology nurse should adhere to those guidelines, instructions, and preferences.

RADIOGRAPHIC AND NON-RADIOGRAPHIC IMAGING

Radiographic studies of the GI tract help detect abnormalities and anomalies such as obstructions, strictures, ulcers, or structural changes. Supine and lateral plane X-ray films of the abdomen and chest are often the first tests performed on patients with acute abdominal symptoms. In most cases, other tests are also required.

Undergoing radiological exams may be considered easier for the patient than endoscopy and involve fewer complications. However, endoscopy has greater diagnostic potential. The local availability of radiologic and endoscopic expertise may also influence a physician's choice of diagnostic method.

Contrast media

A variety of X-ray and CT techniques are used for the GI tract that involve administration of contrast agents. Administration of a contrast medium before or during radiographic studies accentuates differences in the densities of abdominal regions and structures, thus facilitating interpretation of the radiographs.

Contrast materials can be administered:

- Orally.
- By enema.
- Through intestinal accesses (e.g., intestinal stomas, catheters, G-tubes).
- By injection per vein or artery.
- Endoscopically during a procedure, such as ERCP.

Following an imaging exam with contrast material, the material is absorbed by the body or eliminated through urine or stool.

The American College of Radiology (ACR) (2018) states that the approach to patients about to undergo a contrast-enhanced examination has four general goals:

- To ensure that the administration of contrast is appropriate for the patient and the indication.
- To balance the likelihood of an adverse event with the benefit of the examination.
- To promote efficient and accurate diagnosis and treatment.
- To be prepared to treat a reaction should one occur.

Barium

The most common contrast material for radiographic tests for GI disorders is **barium sulfate**. Barium is a chalky, radiopaque, inert, nonallergenic substance that enables **fluoroscopic examination** and **X-ray (roentgenographic) examination** of the esophagus, stomach, and small and large intestines. It is available in several forms: powder (which is mixed with water before administration), liquid, paste, and tablet. It may be administered orally, rectally, or via intestinal stoma. Barium substances can be administered through enema, tubes, or catheters.

Orally administered barium contrast media can also be used for GI opacification during routine abdominopelvic CT scan.

Any routine X-ray examinations, lower GI studies, ultrasounds, planned endoscopy, or oral cholecystograms should be done before barium studies because the ingested barium may obscure other films or visualization.

Barium studies are contraindicated in patients with digestive tract obstruction or perforation.

Education for patients undergoing barium studies includes instructions to:

- Report failure to pass the barium within 2-3 days, so cathartics or enemas may be prescribed to avoid constipation or obstruction. An enema or a laxative may be given routinely after barium studies.
- Increase intake of oral fluids.
- Expect stools to be chalky and light-colored for 24-72 hours after the barium study.

Iodinated water-soluble contrast media

The use of iodinated water-soluble contrast media such as diatrizoate meglumine and diatrizoate sodium solution (Gastrografin, MD-Gastroview), is available, but its use is limited to suspected bowel obstruction or bowel perforation such as leaks, abscesses, and fistulas. Water-soluble oral contrast media also may be used prior to an endoscopic procedure or anticipated surgery.

Diatrizoate meglumine and diatrizoate sodium solution is hypertonic and may lead to hypovolemia and hypotension due to fluid loss from the intestine. This solution is usually used undiluted in the upper GI tract in adults. In some children and elderly adults, however, the loss of plasma fluid may be sufficient to cause a shock-like state. For these patients, the contrast material can be diluted with water.

As with conventional fluoroscopic imaging, there are a few specific clinical situations in which water-soluble contrast agents are strongly favored over barium agents for use in CT scan. Such situations include adhesive bowel obstruction or possible bowel perforation.

Intravenous iodine-based (iodinated) contrast media

Iodine-based contrast media are administered intravenously (IV) for radiographs of the gallbladder, pancreas, spleen, and the various ducts. Many types of CT scan also require an injection of iodinated contrast. Iodinated contrast during CT scan enhances the blood vessels and structure of some organs such as the liver, brain, spine, kidney, and pancreas. Iodinated contrast supplied for IV use can also be administered safely by mouth or by rectum.

Radiographic studies with iodine-based contrast media require careful review of the patient's history for allergies to iodine, seafood, or iodine-based contrast agents. Studies also require careful patient preparation before the test and close observation afterward for delayed hypersensitivity reactions.

Gadolinium

Gadolinium contrast media, sometimes called an MRI contrast media or dye, is injected into the body to enhance MRI images. It can be administered IV by hand or automated injector.

MRI with gadolinium contrast injection improves diagnostic accuracy when diagnosing inflammatory or infectious diseases of the brain, spine, soft tissue, and bones. The extent of some cancers and benign tumors is best seen and assessed with gadolinium contrast injection. MRI scans using gadolinium can also show real-time function of the blood vessels.

Clinical and research guidelines urge caution when using gadolinium-based contrast during MRI, as evidence suggests it can accumulate in the brain (Gulani et al., 2017).

Barium swallow

A **barium swallow** permits radiographic examination of the pharynx, esophagus, and lower esophageal sphincter after ingestion of a thick barium solution.

Indications

This test can reveal the presence or absence of foreign bodies, diverticula, ulcerations, varices, polyps, tumors, hiatal hernia, or motility disorders. It may discover, but is not sensitive for, esophagitis or Barrett's esophagus.

A barium swallow is contraindicated in patients with intestinal obstruction. For patients with suspected perforations, a water-soluble agent such as Gastrografin is the preferred contrast media.

Procedure

Prior to the procedure

Before the scheduled barium swallow, the patient may be given instructions to:

- Fast before the exam for a given period of time.
- Consult the ordering physician to determine, which, if any oral medications should be taken prior to the procedure.

During the procedure

During the test, the patient swallows a barium mixture while in a supine position. The Trendelenburg position may be used to detect gastric reflux or a sliding hiatal hernia. For patients with dysphagia, a barium tablet or a food or marshmallow bolus should be used in addition to the liquid barium so that strictures can be better identified.

Considerations

Complications related to barium swallow include:

- Potential allergies to solution. Barium is nonallergenic, but various ingredients are added to improve coating, flow, taste, and color.
- Possible constipation.
- Accidental aspiration into the trachea in children and patients with swallowing disorders.
- Barium leakage into the thoracic cavity or peritoneum.

The presence of bowel obstruction may increase the potential for complications.

Barium swallow testing should be avoided in pregnant women, if possible, due to radiation exposure during X-ray.

Upper gastrointestinal series

In an **upper gastrointestinal (UGI) series** with small bowel follow-through, the esophagus, the stomach, and the entire length of small intestine are examined after ingestion of a barium solution.

Indications

A UGI series can be used to aid in diagnosis of hiatal hernia, diverticula, varices, strictures, ulcers, tumors, Crohn's disease, malabsorption syndromes, or motility disorders. Gastric X-ray films may show masses or ulcers. Films of the small intestine may indicate obstruction, regional ileitis, or diverticula.

Procedure

Prior to the procedure

Prior to the procedure, patients may be instructed to:

- Remain NPO after midnight before the procedure.
- Follow a low-residue diet for 2-3 days.
- Avoid anticholinergic or narcotic medications for 24 hours.
- Refrain from smoking after midnight.
- Avoid antacids for several hours before the test.

During the procedure

After swallowing the barium, the patient assumes various positions on the X-ray examination table so that the barium will outline all parts of the gastric wall. In addition, the abdomen may be palpated or compressed.

The patient may be asked to swallow a gas-producing substance in the form of powder, pills, or a carbonated beverage. This will introduce air into the GI tract to improve visualization of small lesions that can be missed when using barium alone. This is called a double-contrast UGI.

Considerations

Complications related to UGI include:

- Potential allergies to solution. Barium is nonallergenic, but various ingredients are added to improve coating, flow, taste, and color.
- Possible constipation.
- Accidental aspiration into the trachea in children and patients with swallowing disorders.
- Barium leakage into the thoracic cavity or peritoneum.

The presence of a bowel obstruction may increase the potential for complications.

UGI series should be avoided in pregnant women, if possible, due to radiation exposure during X-ray.

Enteroclysis

Enteroclysis is an imaging study that shows how contrast material moves through the small intestine.

Indications

When small bowel follow-through fails to provide a satisfactory study, enteroclysis may be used to detect subtle small intestine disease.

CT scan provides visualization of features of the GI wall, while conventional enteroclysis can distend the small intestine for better visualization of the lumen. CT enteroclysis combines the advantages of both procedures. CT enteroclysis is particularly useful in evaluating symptomatic Crohn's disease, particularly in the detection of a fistula, an abscess, skip lesions, lymphadenopathy, and conglomeration of small intestine loops. MRI enteroclysis is also available.

Procedure

Prior to the procedure

The patient scheduled for enteroclysis should:

- Receive a bowel prep (enemas are not recommended as some fluid may be retained in the small intestine).
- Have someone present to drive home, if sedation is needed.

During the procedure

- Sedation may be necessary to ensure patient comfort during the examination.
- A flexible feeding tube is placed either intranasally or orally. The tip of the tube is positioned in the duodenum.
- Barium is injected through the tube into the small intestine under fluoroscopic guidance.
- Barium is then followed by an injection of water and/or methylcellulose solution to give a **double-contrast effect**.
- Multiple spot films of the various segments of the small intestine are obtained, including the ileocecal valve.

Considerations

Complications related to enteroclysis include:

- Potential refusal to allow tube placement.
- If sedating the patient, medication reaction or complications.
- Potential allergies to solution. Barium is nonallergenic, but various ingredients are added to improve coating, flow, taste, and color.
- Possible constipation.
- Barium leakage into the peritoneum through any leaks or perforations.

The presence of bowel obstruction may increase the potential for complications.

Enteroclysis should be avoided in pregnant women, if possible, due to radiation exposure during X-ray.

Barium enema

Barium enema is a common and valuable diagnostic test for patients with colon disorders. Both single-contrast and double-contrast barium enemas are generally safe and well-tolerated without premedication, although double-contrast enemas are generally the method of choice. In double-contrast studies, a thicker barium solution is used. After rotation by the patient to cover the bowel wall, air is introduced into the bowel so that smaller lesions can be seen.

Indications

A barium enema is indicated for evaluation of the large intestine; evaluation of inflammatory disease; and detection of polyps, masses, diverticula, and structural changes.

The procedure is contraindicated in patients with fulminant ulcerative colitis associated with systemic toxicity, toxic megacolon, or suspected perforation or obstruction.

Procedure

Prior to the procedure

The patient scheduled for a barium enema should:

- Receive complete information about the procedure and expectations, as full cooperation is essential for a successful exam.
- Follow a clear liquid diet for 1-2 days prior to the exam.
- Complete a bowel-cleansing regimen to allow for adequate examination of the bowel.

Unprepped barium enema examination is indicated for children when Hirschsprung's disease is suspected.

During the procedure

The barium is administered as an enema, and the patient is encouraged to retain the barium, despite any cramping or urge to defecate. In some cases, the barium is administered through a Foley-type catheter, and the balloon is inflated slightly to prevent the patient from expelling the solution. The use of a catheter, however, may increase the urge to defecate and exacerbate patient discomfort.

The patient is instructed to move about to allow the barium to spread throughout the colon. In some cases, a tilt table may be used to achieve a semi-erect or Trendelenburg position that will allow the barium to cover all areas of the colon.

After the procedure

A conventional cathartic and enema preparation or a balanced electrolyte lavage solution is given after the procedure to ensure elimination of the barium. Defecation of barium should be monitored, and laxatives given if barium is not passed within 2-3 days.

Considerations

Complications related to barium enema may include:

- Constipation.

- Rectal trauma.
- Barium leakage into the peritoneum through any leaks or perforations.

Barium enema should be avoided in pregnant women, if possible, due to radiation exposure during X-ray.

Capsule endoscopy

Capsule endoscopy is a technique that employs a disposable video capsule swallowed by the patient, allowing visualization of much of the small intestine not within reach of standard upper and lower endoscopy.

Indications

Indications for capsule endoscopy include:

- Aiding in the diagnosis of GI bleeding, especially when upper endoscopy and colonoscopy are negative.
- Evaluation of conditions of the small intestine that cause diarrhea, pain, or weight loss (e.g., Crohn's disease).

Procedure

Prior to the procedure

In order for the physician to get the most accurate information from capsule endoscopy, patients should receive and follow pre-procedure instructions. The patient should be instructed to:

- Avoid iron tablets for 7 days prior to the procedure, as iron will turn the stool dark, making it harder to visualize the bowel during the procedure. Some physicians may also request no vitamins for the same time frame.
- Complete a bowel prep per the physician's order.
- Abstain from smoking 24 hours prior to swallowing the capsule.
- Remain NPO after midnight the night before the procedure or per the physician's order.
- Shave the abdomen 6 inches above and below the navel the night before the procedure, if hairy. This helps to ensure that the sensors applied to the abdomen with adhesive pads will stick. These sensors are connected to the recorder that is worn in a belt around the waist.
- Take simethicone prior to the procedure to decrease gas bubbles and increase visualization, if ordered.
- Do not undergo an MRI before the capsule has been verified as excreted from the body, as undergoing an MRI with the capsule still in the intestine can result in serious damage to the intestinal tract or abdominal cavity.

The gastroenterology nurse should explain that the capsule is disposable and should be evacuated naturally in a bowel movement.

During the procedure

The patient swallows a plastic capsule with a sip of water. The capsule is approximately the size of a large vitamin and contains a disposable miniature camera. Images of the small intestine captured by the camera are transmitted to a number of sensors attached to the patient's torso and recorded digitally on a device worn around the patient's waist.

The patient should be instructed to:

- Remain NPO for at least 2 hours after ingesting the capsule. After 4 hours, have a light meal. After completion of the study, return to a normal diet.
- Contact the facility immediately for any abdominal pain, nausea, or vomiting any time after ingesting the capsule.
- Avoid any strenuous physical activity to avoid sweating, bending, or stooping during the procedure. This helps to prevent the adhesive pads from coming off.
- Treat the data recorder with care. The belt should remain secure throughout the procedure. The recorder is a small computer.
- During capsule endoscopy, verify every 15 minutes that the small light on top of the data recorder is blinking. If, for some reason it stops blinking, record the time and contact the facility for instructions.
- Record the nature and time of any activity or event such as eating, drinking, or feeling unusual sensations. Give these notes to the staff at the time the equipment is returned.

After approximately 8 hours, the capsule is expected to have progressed through the small intestine. At this time, the recorder and the sensors can be removed from the patient. If the patient is allowed to remove data recorder on his or her own, the patient should be given detailed instruction on how to do so and provided with a bag or storage device for all cables and the data recorder.

After the procedure

The data is downloaded to a workstation, and software processes the images to produce a video of the small intestine for interpretation by the physician.

The patient is instructed to:

- Watch for excretion of the capsule in the stool.
- Report development of unexplained abdominal pain, vomiting, or other symptoms of obstruction.
- Report the inability to positively verify excretion of the capsule.

If the patient cannot verify excretion of the capsule, the physician will do an evaluation and may order an abdominal X-ray to determine if the capsule is still present in the GI tract. In the rare case that the capsule is not excreted naturally, it will have to be removed endoscopically or surgically.

Considerations

A history of Crohn's disease, obstructions, narrowing of the bowel, and previous bowel surgery could prevent an accurate test result.

Patency test capsule

Prior to swallowing a video endoscopy capsule, a patency test of the intestinal tract can be done using a **patency capsule**, specifically to aid in the diagnosis of abnormal passage, obstructions, or strictures in the GI tract.

Esophageal capsule endoscopy

The esophageal capsule is quite similar to the small video endoscopy capsule in many ways, although it possesses an imaging unit on both ends of the capsule.

Indications

Esophageal capsule endoscopy is used to diagnose esophageal diseases when EGD is not possible.

Contraindications are similar to those of the small bowel capsule endoscopy procedure.

Procedure

The test may be completed in less than 30 minutes, with the ability to interpret the recording shortly afterward.

Before the procedure

Preparation is not as intense as a small bowel capsule procedure, although the patient is instructed to:

- Abstain from food and fluids after midnight before the procedure.
- Take routine medications with a sip of water the morning of the procedure, up to 2 hours before the exam time.
- Not apply lotions or powders to the chest area so that the recording device can be securely applied.
- Refrain from applying lipstick, lip gloss, or other facial lotions or treatments to prevent accidental smearing on the capsule lens when it is ingested.
- Wear lose-fitting, two-piece clothing to facilitate the application of the recording device.
- Not undergo an MRI before the capsule has been verified as excreted from the body, as undergoing an MRI with the capsule still in the intestine can result in serious damage to the intestinal tract or abdominal cavity.

The gastroenterology nurse should explain that the capsule is disposable and should be evacuated naturally in a bowel movement.

During the procedure

For capsule ingestion, the patient will:

- Stand in an upright position while drinking 100 ml of water to cleanse the mouth of saliva.
- Lie in a lateral right-side position with the head on a pillow.
- Swallow the capsule with a minimal sip of water using a 10 ml syringe.
- Take additional sips of water (15 ml) from the syringe every 30 seconds for 7 minutes.
- After 7 minutes, sit upright and drink another sip of water.
- Remain seated in the examination room or move to the waiting room for 20 minutes, with no eating or drinking during this time.

After the procedure

The recording system may be removed when the indicator lamp stops blinking. The data is then downloaded.

Before discharge, instruct the patient:

- To monitor his or her stool for the elimination of the capsule.
- Not to schedule an MRI until passage of the capsule has been confirmed with certainty.
- Seek medical attention if unexplained nausea, abdominal pain, or vomiting develops.

Considerations

The examination may be incomplete due to technical difficulties, any food retention in esophagus, or any esophageal abnormalities. No test, including esophageal capsule endoscopy, can offer 100% accuracy for diagnosis.

Wireless motility capsule testing

The wireless motility capsule is an ingestible device that uses sensor technology to measure pH, pressure, temperature, and transit time within the entire GI tract. Before administering the wireless motility capsule, a physiological or mechanical GI obstruction must be ruled out as a cause of the patient's symptoms.

Indications

Indications for wireless motility capsule testing include gastroparesis, postprandial fullness, nausea, early satiety, vomiting, delayed chronic transit, irritable bowel syndrome (IBS), chronic abdominal pain, chronic bloating, fecal incontinence, and constipation.

The wireless motility capsule system provides valuable diagnostic information, including:

- Gastric emptying time.
- Small intestine transit time.
- Colonic transit time.
- Whole gut transit time.
- Pressure patterns from the antrum and duodenum.

Procedure

Once the patient ingests the capsule, it transmits information from the capsule within the GI tract to the wireless collecting data receiver. The single-use capsule is capable of transmitting data continuously for more than 5 days and is excreted naturally from the body in the stool.

Prior to the procedure

Before the procedure, the patient may be asked to:

- Discontinue proton-pump inhibitors (PPIs), histamine2 (H_2) blockers, antacids, and drugs that affect GI motility (pain medications, sedatives, tranquilizers, antispasmodics, and promotility medications).
- Refrain from smoking the day before the examination
- Remain NPO after midnight or 8 hours before testing.
- In the morning, take other medications with a sip of water.

During the procedure

The test involves application of a recording device, eating a specific meal or food bar prior to the examination, swallowing the capsule, and returning the appliance in 3-5 days.

- Immediately after the test meal is ingested, the capsule is ingested.
- The patient is advised not to eat for 6 hours after consuming the test meal. Small sips of water, up to 4 ounces, may be taken during the initial 6 hours.
- After the 6 hours, another test meal or drink may be ordered.
- A normal diet may be started 8 hours after the capsule is ingested.

The patient is instructed to:

- Not use laxatives, bowel cathartics, and antidiarrheal medications until the capsule is passed.
- Avoid vigorous exercise and prolonged aerobic activity greater than 15 minutes throughout the testing period.
- Keep the data receiver within 5 feet of him- or herself at all times to ensure the information is recorded.
- Not get the data receiver wet when showering or bathing.
- Maintain a diary for the test interval, as well as to use the event button to mark bowel movements and other occurrences and symptoms that may be useful to the physician when reviewing the test data.

After the procedure

Upon discharge, instruct the patient to:

- Monitor his or her stool for elimination of the capsule.
- Schedule any MRIs only after passage of the capsule has been confirmed with certainty.
- Seek medical attention if unexplained nausea, abdominal pain, or vomiting develops.

Considerations

The examination may be incomplete due to technical difficulties. No technology, including a wireless motility capsule, can offer 100% accuracy for diagnosis. Significant data dropout can occur in patients who are severely obese (>40 BMI).

The risks of the capsule testing include capsule retention and aspiration.

Magnetic resonance imaging

Magnetic resonance imaging (MRI) is a noninvasive diagnostic imaging test that uses an external magnetic field to create a picture of soft tissues in the body. The test does not use radiation. An injection of a gadolinium-based contrast may be given in order to improve imaging.

Magnetic resonance cholangiopancreatography

Magnetic resonance cholangiopancreatography (MRCP) is an application of MRI. This test evaluates the patient's hepatobiliary and pancreatic ductal systems for strictures, stones, and neoplasms using radio waves and a magnetic field with special software.

Intravenous contrast-enhanced MRCP may be performed using gadolinium-based contrast agents excreted into the bile. Agents such as gadobenate dimeglumineor gadoxetate disodium are used for this purpose. The use of gadolinium-based MRI contrast agents requires precautions in patient selection and renal function screening to preclude later development of nephrogenic systemic fibrosis.

Indications

MRCP can produce images similar to those obtained from a more invasive approach using endoscopic retrograde cholangiopancreatography (ERCP) without the added risk of pancreatitis, sedation, or perforation. MRCP also can be used in preoperative anatomic mapping of the biliary tree prior to living donor liver transplantation and in detection of postoperative biliary leak assessment of patients who have received liver transplants.

The procedure does not allow for therapeutic intervention. The patient would require an ERCP if therapeutic intervention can treat the diagnosis that is obtained by MRCP.

Procedure

Prior to the procedure:

The patient's medical and surgical history, medication allergies, medication history, required labs, and previous MRI or other radiology films are obtained if available.

Confirm whether the patient has any of the following:

- Cardiac pacemaker or artificial heart valve.
- Metal plate, pin or other metal implant.
- Insulin pump or other infusion pump.
- Aneurysm clips.
- Stents.
- Intrauterine device (IUD).
- Inner ear implant.
- Tattoo history.
- Piercings in any location.
- Any other internal or external device.

During the procedure

The patient is injected with gadolinium intravenously, dosed at 1 ml for every 10 pounds of body weight. Preparation for imaging includes:

- Having the patient remove all metal.
- Positioning the patient on the exam table, lying supine.
- Providing earplugs or headphones with music for patient comfort.
- Informing the patient to hold his or her breath when instructed.

The exam usually takes between 20 and 30 minutes.

An MRCP may be enhanced by the use of synthetic secretin. Secretin can be administered to stimulate pancreatic fluid and bicarbonate secretion during the test. This improves ductal delineation, as the increased generated ductal fluid volume results in greater ductal distention. Secretin also relaxes the sphincter of Oddi and opens pancreatic duct orifices. Secretin is injected intravenously at the time of the MRCP. Images are then taken every 30 seconds for 10 minutes. Maximum output of pancreatic fluid is optimal between 6 and 8 minutes.

After the procedure

The patient can return to normal activity.

Considerations

In a patient with previous moderate or severe allergic-like reactions to a specific gadolinium, it may be prudent to use a different gadolinium and pre-medicate for subsequent MRI examinations.

Secretin is safely administered to most patients, but its use should be avoided in patients with acute pancreatitis, as symptoms can be exacerbated. Other immediate side effects may be encountered, with the most common symptoms including flushing of the face, neck, and chest. Less commonly, some patients may develop vomiting, diarrhea, fainting, blood clot, fever, and tachycardia. Allergic-like reactions are rare, but symptoms such as hives, redness of the skin, and even anaphylaxis have been reported.

Magnetic resonance enterography

Magnetic resonance enterography (MRE) uses MRI technology to create images of the intestinal tract. MRE can pinpoint inflammation, bleeding, and other problems in the intestinal tract and create detailed images of organs.

Indications

MRE can evaluate various GI disorders and inflammatory bowel diseases such as Crohn's disease. Because it is a noninvasive imaging test, it also can be used in patients with GI bleeding. MRE does not use radiation, making it a good choice for young patients or for patients who may need to have other studies that use radiation.

Percutaneous transhepatic cholangiography

Percutaneous transhepatic cholangiography (PTC) is a procedure used to visualize the biliary ductal system.

Indications

Purely diagnostic, percutaneous transhepatic cholangiography is performed when other less invasive methods of imaging the biliary tree (e.g., MRCP, ERCP, CT cholangiography) have failed. PTC may also be done to:

- Distinguish between obstructive and nonobstructive jaundice.
- Determine the location, extent, and cause of mechanical bile duct obstruction, which may be caused by stones, tumors, inflammation, stenosis, or stricture.
- Place a drainage catheter for the treatment of obstructive jaundice.
- Delineate bile leaks.
- Place a percutaneous biliary stent.
- Dilate postoperative stricture.
- Remove stones.

Procedure

Prior to the procedure

The patient's medical and surgical history, medication allergies, medication history (in particular, blood thinners), required labs (coagulation in particular), and previous procedural and radiological reports related to this procedure are obtained. Antibiotics are administered per the physician's order.

During the procedure

Under fluoroscopic visualization, a "skinny" or fine needle is introduced into the liver through the locally-anesthetized seventh or eighth intercostal space, parallel to the plane of the table. The needle is slowly withdrawn until a green fluid is aspirated, indicating that a bile duct has been entered. A radiopaque iodine dye is then injected, and radiographs are taken to define the site, cause, and extent of bile duct obstruction.

After the procedure

After completion of the procedure:

- Carefully observe the patient. Vital signs should be checked regularly, and the patient should be observed for signs of hemorrhage from the site. Signs and symptoms of hemorrhage include a decrease in blood pressure, an increase in pulse rate and respirations, restlessness, pallor, or diaphoresis.
- Position the patient on the right side for the time ordered.
- Monitor the patient for pain, temperature elevation, chills, abdominal distention, peritonitis, bile leakage, or septicemia.
- Administer antibiotics per the physician's order.

Considerations

Bleeding may result following PTC and can be massive.

Abdominal ultrasonography

Ultrasonography is noninvasive, does not involve patient preparation or radiation, and is simple to perform and interpret. It is indicated to show the size and configuration of organs. It identifies abnormalities of the structures and spaces within the abdomen or pelvis through the transmission of sound waves. Sound waves are used to create images of organs.

Indications

Ultrasonography is indicated as:

- The optimal test for the detection of gallstones.

- Screening for biliary dilation.
- Detecting changes caused by appendicitis, such as a dilated appendix, thickened wall, or periappendiceal fluid accumulation.
- Acute cholecystitis.

Procedure

The skin over the area to be studied is exposed and lubricated to seal out air pockets and keep the transducer from rubbing the patient's skin. While the patient lies still, a microphone transducer is slowly rubbed over the skin surface for approximately 20-30 minutes. Images of the structures within the target area are produced on a screen throughout the test.

Considerations

Ultrasound waves are disrupted by air or gas; therefore, ultrasound is not an ideal imaging technique for air-filled bowels or organs obscured by the bowel. Large patients are more difficult to image by ultrasound because greater amounts of tissue attenuate the sound waves as they pass deeper into the body and return to the transducer for analysis.

For standard diagnostic ultrasound, there are no known harmful effects on humans.

Oral cholecystography

Oral cholecystography is an X-ray procedure used to evaluate the gallbladder. This test has largely been superseded by ultrasonography and MRCP.

Indications

Although ultrasonography is the initial diagnostic procedure of choice for the detection of gallstones, oral cholecystography is an acceptable alternative when ultrasonography is unavailable, not diagnostic, or when the cholecystogram will be performed in conjunction with an UGI series. Oral cholecystography may be superior to ultrasonography for demonstrating the patency of the cystic duct, assessing gallbladder function, and showing the apparent density of stones. It is also necessary before oral dissolution therapy of gallstones to be certain that the cystic duct is open.

Procedure

The patient scheduled for oral cholecystography should be instructed that:

- The test involves oral administration of pills containing iodinated radiocontrast material.
- Films are taken 10-14 hours after ingestion of the pills.
- Fat may be ingested following the initial gallbladder films in order to cause contraction of the gallbladder, filling the common bile duct.
- Radiography takes 30-45 minutes.
- After the test, a normal diet may be resumed and prescribed medications may be continued.
- After the test, the patient should drink plenty of fluids for several days, unless otherwise contraindicated.

Considerations

This test exposes the patient to iodine and X-ray.

Arteriography

Arteriography, also called **angiography**, involves the catheterization of selected arteries, followed by injection of an iodinated dye via the catheter.

GI arteriography tests include:

- Aortic angiography (chest or abdomen).
- Mesenteric angiography (colon or small intestine).
- Pelvic angiography (pelvis).

Indications

X-ray films taken after injection can help to locate the source of GI bleeding, investigate poor blood supply, evaluate cirrhosis and portal hypertension, or determine the extent of vascular damage to the liver and spleen after abdominal trauma. Vascular tumors may be located by selective celiac and superior mesenteric arteriograms.

Selective arteriography can also be therapeutic. Autologous clots or vasoconstricting drugs can be injected into appropriate vessels to control bleeding, or chemotherapeutic agents can be delivered directly to a tumor.

Arteriography is used to detect several conditions and abnormalities, including:

- Aneurysms.
- Blockages.
- Hemorrhages.
- Inflammation.
- Narrowing of blood vessels.
- Thrombosis.
- Tumors.

Procedure

During the procedure

The specific procedure for an arteriogram will depend on the body part or system being studied. Generally, the procedure includes:

- A small incision made in the arm or groin into which a small catheter will be inserted.
- A small catheter inserted into an artery in the patient's arm or groin.
- Contrast dye given through catheter.
- X-rays taken as dye runs through the vessels.
- The catheter removed and a pressure dressing applied over the insertion site.

After the procedure

The patient will remain flat in bed for a designated time.

Instruct the patient how to monitor for bleeding and to report any unusual pain, discoloration, or temperature change of the extremity with the puncture site. Instructions regarding physical activity and wound care will be procedure-specific, but in general, the patient should avoid strenuous physical activity for up to a week. The patient should keep the bandage on the insertion site dry for approximately 2 days.

Considerations

Risks of an arteriography include:

- Pain.
- Bleeding.
- Infection at the insertion site.
- Blood clots.
- Damage to blood vessels.
- Allergic reaction to the contrast dye.

Computerized tomography scan

Computerized tomography (CT) scan is a noninvasive, radiological scanning technique that relies on differences in tissue density to reflect organ configurations. Because it gathers information in all three dimensions, CT scan provides a more detailed and precise view of the area being scanned than does abdominal ultrasonography. However, CT scan can also require more personnel and expensive equipment to operate than ultrasonography, and it uses ionizing radiation.

CT scan may be done with or without contrast, although the diagnostic accuracy of CT scan can be enhanced by IV injection of contrast material to visualize the biliary system or by oral ingestion of a contrast agent to delineate the GI system. Dosages, contrast media, and application of contrast depends on the specific procedure. Orally administered contrast media are used for GI opacification during routine abdominopelvic CT scan. CT scan has several options for opacifying the GI tract of the abdomen and pelvis, with barium-based and iodine-based contrast media being the most commonly used.

Indications

A CT scan is indicated for evaluation of gallbladder carcinoma, common bile duct stones, liver masses, pancreatic adenocarcinoma, intra-abdominal abscess, intestinal obstruction, Crohn's disease, aortic aneurysms, appendicitis, abdominal tumors, and complications of acute pancreatitis.

Procedure

Prep depends on the reason for CT scan. The patient may be given oral contrast of a liquid containing barium or a water-soluble media. The patient may be given intravenous iodine-based contrast if the CT scan is to highlight blood vessels, organs, and other structures.

Considerations

Radiation exposure is much higher than with regular X-ray. Safety precautions related to X-ray, especially with regard to children and pregnant women, should be observed.

Scintigraphy

Scintigraphy is a nuclear medicine diagnostic test. Radioactive isotopes or radiopharmaceuticals are administered to the patient, and the emitted radiation is captured by external detectors or a gamma camera to form two-dimensional images of the target organs. The hepatobiliary (HIDA) scan, also known as cholescintigraphy and hepatobiliary scintigraphy, is an imaging procedure used to diagnose problems of the liver, gallbladder, and bile ducts.

Hepatic scintigraphy

Hepatic scintigraphy with technetium 99m (^{99m}Tc) sulfur colloid is widely used for noninvasive evaluation of the liver. The Kupffer cells and other phagocytic cells in the liver and spleen absorb the radiolabeled chemical. Because most intrahepatic masses do not contain these phagocytes, such masses appear as "cold" spots on the hepatic scintigram.

Indications

The most common use of hepatic scintigraphy is for diagnosing and following the progression of metastatic disease. In addition, its sensitivity for hepatocellular dysfunction or diffuse infiltrating diseases, including alcoholic liver disease, cirrhosis, lymphoma, amyloidosis, or infiltrating metastatic disease, may be higher than other imaging methods.

Biliary scintigraphy or cholescintigraphy

In patients with normal biliary function, a ^{99m}Tc-labeled iminodiacetic acid derivative is taken up by the hepatocytes and excreted into bile. Scans outline the bile ducts, gallbladder, and proximal small bowel.

Indications

This test is performed to detect obstruction of the bile ducts. The test may also be ordered to detect a bile leak.

Procedure

For biliary scintigraphy, the patient must be NPO after midnight before the procedure. A ^{99m}Tc-labeled iminodiacetic acid derivative is administered IV. Scans are obtained continuously with 4- to 6-minute images for filming or digital display, viewed for up to 60 minutes for most procedures. For biliary leaks, hepatocellular dysfunction, common bile duct obstruction, or suspected biliary atresia, delayed images up to 24 hours may be obtained after radionuclide injection.

Radionuclide evaluation of gastric emptying

Indications

For evaluation of gastric emptying, radionuclide testing is preferred over intubation methods. Gastric emptying studies may also be done without radionuclides;

however, the ability of radionuclides to tag a solid test meal offers quantitative results that are not possible with barium radiography.

Procedure

Either egg salad or chicken liver is labeled with ^{99m}Tc, which serves as a marker of the solid of the meal. The liquid component of the meal is labeled with indium 111(^{111}In).

Following ingestion of the radiolabeled solid and liquid meal, the patient reclines under a scintiscanner that measures the initial level of intragastric radioactivity and the rate of passage of radioactivity out of the stomach (i.e., gastric emptying). In this manner, differential rates of solid and liquid emptying can be detected.

Gastrointestinal bleeding scintigraphy (GIBS)

GIBS is a diagnostic radionuclide imaging study performed with ^{99m}Tc-labeled red blood cells using the patient's own blood to detect active bleeding into the GI lumen

Historically, GI bleeding was diagnosed through the use of one of two scintigraphic scanning techniques using technetium:

- Using ^{99m}Tc-labeled sulfur colloid.
- Using ^{99m}Tc-labeled red blood cells using the patient's own blood.

Using ^{99m}Tc-labeled red blood cells is the radiopharmaceutical of choice for performing GIBS, because of an intravascular half-life that allows for continuous imaging over many hours providing superior studies.

Inflammatory foci may be identified using ^{99m}Tc-labeled white blood cells.

Indications

Clinical signs of GI bleeding include hematochezia, melena or hematemesis, obvious bright red bleeding during endoscopy procedure, current and recent hemoglobin, hematocrit, and blood urea nitrogen changes, number of recent transfusions needed, current blood pressure and heart rate changes, and presence of orthostatic vital signs.

Scanning for GI bleeding is helpful in diagnosing active GI bleeding as a result of such conditions as diverticulosis, post polypectomy bleeding, arteriovenous malformations, malignancies, colitis, radiation proctitis, angiodysplasias, and inflammatory foci.

Procedure

After injection with the labeled substance, the patient is placed in a supine position, and rapid images of the abdomen and pelvis are obtained at intermittent intervals. When the labeled substance enters the bleeding site, its location is identified through nuclear scanning. Images are obtained over 20 to 60 minutes. Active hemorrhage is seen within the first 5 to 10 minutes. Bleeding in the splenic flexure or transverse colon regions is difficult to visualize due to intense uptake by the liver and spleen.

Inflammatory foci may be identified using labeled white blood cells and employing a similar technique as that of labeled-red blood cell studies. The patient's white blood cells are isolated, and technetium is added before the substance is administered via IV. Nuclear scanning techniques allow for identification of inflammatory foci (as seen in inflammatory bowel disease [IBD]) in areas where the labeled white blood cells cluster.

Considerations

GIBS is only reliable if the patient is actively bleeding at the time of the procedure. The most common false positive results occur when there is rapid transit of luminal blood that is detected in the colon, although it originated in the upper GI tract. If no bleeding site is identified on the initial images, delayed images can be acquired by rescanning the patient for up to 24 hours, especially if recurrent GI bleeding is evident.

Transjugular intrahepatic portosystemic shunt

Transjugular intrahepatic portosystemic shunt (TIPS) is a percutaneously created low-resistance channel between the portal vein and the hepatic vein. The goal of TIPS is to reduce portal pressure by shunting blood from the portal to the systemic circulation, bypassing the liver.

Indications

TIPS is used to treat:

- Complications of portal hypertension, including variceal bleeding uncontrolled by medical or endoscopic management that normally drains the stomach, esophagus, or intestines into the liver.
- Portal gastropathy in the wall of the stomach.
- Severe ascites and/or hepatic hydrothorax.
- Budd-Chiari syndrome.

TIPS should not be performed in patients who have severe heart failure or pulmonary hypertension, due to an increased risk of life-threatening pulmonary congestion. Other contraindications are multiple hepatic cysts, uncontrolled systemic infections or sepsis, and unrelieved biliary obstruction.

Procedure

Before the procedure

The procedure is explained to the patient, and informed consent is obtained. Moderate sedation is provided to the patient. The gastroenterology nurse or physician preps and drapes the right side of the neck.

During the procedure

The procedure is performed under fluoroscopic guidance.

- A wire is introduced into the right hepatic vein via the right internal jugular vein.

- The catheter is replaced by a sheath, through which contrast is injected to confirm entry into the portal vein.
- The parenchymal tract is dilated with an angioplasty balloon.
- The balloon is deflated, and a wall stent catheter is advanced over a wire across the hepatic vein to the portal vein.
- The metallic stent is left in place.
- Venography is performed and shows hepatoportal flow to the inferior vena cava with collapse of the varices.

After the procedure

The patient must be monitored for possible complications following the procedure. Such complications include inadvertent puncture of the gallbladder, colon, or other abdominal organ; blood loss; and infection.

Doppler ultrasound is used to monitor shunt patency approximately every 3 months.

Considerations

Two main limitations of TIPS are shunt dysfunction and hepatic encephalopathy, which is the main symptom of treatment failure.

Complications related to the puncture site include intraperitoneal hemorrhage, portal vein perforation, and hepatic artery or bile duct injury, which may lead to fistula formation.

SECRETORY STUDIES

Secretory studies evaluate the secretion of bile and various GI enzymes, with or without external stimulation.

Pancreatic stimulation

Exocrine pancreatic insufficiency occurs in any condition that blocks the pancreatic ducts or damages the cells that produce elastase.

Pancreatic stimulation involves the determination of the volume, bicarbonate content, and pancreatic enzyme concentration in the duodenal juice before and after direct IV stimulation of the pancreas with the hormones secretin and/or cholecystokinin (CCK).

Indirect pancreatic (stool elastase) test is used to measure the exocrine pancreas responsible for producing elastase along with other enzymes. This test is only positive in patients with exocrine pancreatic insufficiency, as seen with chronic pancreatitis and pancreatic cancer.

Indications

Pancreatic stimulation is indicated for:

- Evaluation of possible early exocrine dysfunction in patients with abdominal pain.
- The determination of the etiology of steatorrhea.
- Increases in duodenal secretion (volume and/or bicarbonate level) may occur with chronic alcoholism, hepatic or biliary cirrhosis, gastrinomas, or hemochromatosis.

Decreases in secretions are associated with pancreatic carcinoma, collagen disease, diabetes mellitus, and Billroth I and II surgeries. Patients with cystic fibrosis have a substantial reduction in volume and bicarbonate and an 80% to 90% reduction or absence of enzymes. Normal to slightly reduced volume, reduced bicarbonate concentration, and greatly reduced enzyme secretions are found in patients with pancreatic exocrine insufficiency not caused by cystic fibrosis.

Indirect pancreatic (stool elastase) test is indicated for:

- A decrease in elastase, associated with chronic pancreatitis and sometimes pancreatic cancer. In children, a decrease in elastase is mostly associated with cystic fibrosis or Shwachman-Diamond syndrome.

Procedure

Pancreatic stimulation test can be performed by two methods.

- An orally passed radiopaque duodenal tube is guided under fluoroscopic control through the duodenum to the ligament of Treitz. Secretin or CCK is administered intravenously over a one-minute period, and samples of duodenal fluid are collected in ice-cooled flasks for over approximately 50 minutes. Each specimen is examined for volume; bicarbonate concentration; pH; and the enzymes trypsin, chymotrypsin, lipase, and amylase.
- A secretin endoscopic pancreatic function test advances the endoscope to the post-bulbar duodenum. The first specimen is collected prior to secretin or CCK, and then secretin or CCK is administered intravenously over a 1-minute period. Samples of duodenal fluid are collected through upper endoscope every 15 minutes for 1 hour. 3-5 cc of duodenal fluid is collected in a trap, with each specimen collected separately and kept on ice as soon as collected. The tip of scope is kept in the post-bulbar duodenum.

Indirect test, the pancreatic elastase test, collects stool samples.

Considerations

If the patient is taking pancreatic enzymes, they must be discontinued a week prior to either test.

Pancreatic stimulation results are approximate because the duodenal aspirates consist of pancreatic and duodenal juices and bile. In addition, not all of the secreted juice will be aspirated. A finding of less than 90 mEq/L of bicarbonate is 74% to 90% sensitive and 80% to 90% specific for chronic pancreatitis.

For the indirect pancreatic test, the stool must be formed, not watery, and not contaminated with urine or water.

Both of these tests share one major drawback: They will usually not be positive until the pancreas has developed pancreatic insufficiency, a process that takes years. The tests are likely to be inaccurate in the early or less-advanced stages of the disease.

This type of testing is not standardized, and each of the few facilities that perform the test has to develop a protocol and normal ranges.

Gastric analysis

The purpose of gastric analysis is the objective measurement of gastric acid secretions.

Indications

Gastric analysis is indicated in patients with intractable peptic ulcer symptoms; suspected Zollinger-Ellison syndrome (hypersecretory disease); achlorhydria (also known as hypochlorhydria); in the presence of elevated gastrin, which indicates impairment of acid output such as occurs in pernicious anemia, atrophic gastritis, and Ménétrier disease; and after inhibition of gastric acid secretion by potent antisecretory drugs.

Gastric analysis may be indicated to assess the efficacy of treatment regimens in peptic ulcer disease.

For patients with suspected achlorhydria or gastric hypersecretory states, it may also be helpful to obtain a serum gastrin level, as discussed in the section on Blood Tests later in this chapter.

Gastric analysis is contraindicated in patients with gastric outlet obstruction, recent upper GI tract bleeding, or allergy to the stimulating agent; when there is an inability to freely pass a nasogastric tube; or when food particles or fresh blood are present in the gastric aspirate.

Procedure

Before the procedure

Patient preparation includes the following requirements:

- Remain NPO for 12 hours.
- Withhold H_2-receptor antagonists, anticholinergics, tricyclic antidepressants, sucralfate (Carafate), and antacids for 48 hours before testing.
- Withhold PPIs for a minimum of 72 hours before testing.
- Refrain from smoking for 2 hours before the test.

If the test is being done to check the effectiveness of the medications, the medications should be continued until 12 hours prior to the test.

During the procedure

After lubricating the tube, a radiopaque polyethylene nasogastric tube is passed—fluoroscopy may be used if needed—so that the tip lies in the most dependent part of the stomach and along the greater curvature. Proper placement is confirmed by installing 30 ml of water with immediate recovery of at least 80%.

With the patient lying on the left side, suction is used to aspirate the stomach contents as completely as possible, then the aspirate is discarded. Pentagastrin, the usual drug of choice to induce gastric secretions, is given either intravenously or subcutaneously. Four separate 15-minute basal acid output samples are then obtained. Each sample is analyzed for volume, pH, free acid, color, consistency, and occult blood.

Considerations

Complications of gastric analysis are rare but may include nosebleeds, tube misplacement, or an adverse reaction to the gastric stimulant.

The patient's nose and throat may feel irritated, so gargling with warm salt water or sucking on throat lozenges may help.

Multichannel intraluminal impedance

Multichannel intraluminal impedance (MII) is a technique usually done in combination with esophageal manometry to detect intraluminal bolus movement and esophageal contractions during typical eating and drinking. MII calculates the bolus transit parameters, evaluates the bolus clearance, and investigates the relationship between bolus transit and LES relaxation in the esophagus. When done with pH testing, MII allows for the detection of gastroesophageal reflux independent of pH, including both acid and non-acid reflux.

Indications

Indications for MII include evaluation of patients with dysphagia, non-cardiac chest pain, heartburn, and regurgitation; preoperative evaluation before anti-reflux surgery or endoscopic anti-reflux procedures; and location of the LES prior to pH catheter placement. In patients with non-obstructive dysphagia, MII may offer better sensitivity for detection of motility disturbances than standard manometry.

Procedure

An impedance catheter, a thin wire with sensors, is inserted through nose and down the esophagus. The external end is hooked into an external recorder box for approximately 18-24 hours. The catheter measures changes in resistance to alternating electrical current when a bolus passes by a pair of metallic rings mounted on the catheter. The level of electrical current alternates in response to different substances based on their electrical conductivity. For example, liquid-filled boluses passing by the impedance measuring segments show a lower impedance value, and gas will produce a rise in impedance.

The patient continues his or her usual daily routine and pushes a button on the recording device during certain designated activities such as eating or lying down. Upon completion of the test, the patient returns to the facility to have the catheter and recorder removed and the data downloaded.

Considerations

Complications of esophageal manometry are rare but may include nosebleeds or tube misplacement.

The patient's nose and throat may feel irritated, so gargling with warm salt water or sucking on throat lozenges may help post-procedure.

Twenty-four-hour pH monitoring

Measuring the pH using **ambulatory pH monitoring** will show the degree of acidity or alkalinity of gastric secretions. Prolonged esophageal pH monitoring is the most sensitive method of directly detecting esophageal reflux, quantifying it, and correlating symptoms with reflux events.

Indications

Ambulatory pH monitoring is indicated in patients with:

- Suspected pulmonary complications of esophageal reflux.
- Atypical presentations of esophageal reflux such as non-ulcer dyspepsia, wheezing, and hoarseness.
- Typical presentation of esophageal reflux with a negative workup.
- Esophageal reflux after medical or surgical therapy.
- Episodes of apnea or bradycardia to determine if they are associated with reflux—in infants.
- Chest pain to determine a noncardiac source for pain.

Ambulatory pH monitoring may be performed in patients before proposed anti-reflux surgery to prevent unnecessary surgery in patients with atypical symptoms who do not have esophageal reflux.

Ambulatory pH monitoring is contraindicated for patients with any esophageal condition that would prevent correct placement of the probe.

Procedure

Prior to the procedure

Patient preparation includes:

- Remaining NPO after midnight or at least 6 hours before the procedure for adults; in children, fasting for 2-4 hours before the procedure.
- Discontinuing H_2 2-receptor antagonist for 24 hours before the procedure.
- Discontinuing PPIs for at least 72 hours before the procedure.
- Discontinuing antacids 6 hours before the procedure.

Sometimes pH monitoring is done to assess the adequacy of acid suppression. In these cases, the patients should not discontinue medication before the procedure.

During the procedure

The pH probe and equipment must be calibrated for each patient. For the test:

- A pH probe is inserted through the patient's nostril to a position 5 cm above the lower esophageal sphincter (LES). The patient may be asked to take a few sips of water to help advance the probe.
- The LES is located by esophageal manometry, fluoroscopic placement, endoscopic visualization, or by measuring the pH in the stomach and withdrawing the probe until it enters the esophagus where the pH is higher. The proper position of the probe in children may be estimated by a calculation based on the child's height. Proper placement of the probe may need to be confirmed by radiographic examination.
- The tube is taped securely to the nose and side of the face.
- Some types of pH monitoring equipment require the use of an external reference electrode secured to a flat area of skin surface (usually the chest or back). The area may need to be shaved before the electrode is applied to allow optimal conduction.
- The probe is connected to an external recording device.
- The patient proceeds with his or her daily routine while wearing the device for 24 hours.

The patient is instructed to:

- Not take a tub bath or shower until the equipment is removed. The equipment cannot get wet.
- Record symptoms (i.e., chest pain, heartburn, or coughing) and activities in a diary, along with pressing a button to mark the event on the recording device. Children may need to be hospitalized overnight if the parent or guardian is unable to record events accurately.
- Avoid strenuous activities because of the possibility of damaging the monitor, dislodging the probe, or disrupting the recording.

After the procedure

At the completion of the monitoring period, the stored data are downloaded into a computer that analyzes the information and prints a graphical display of esophageal pH over the recording period, along with a summary of these values. Normal pH in the esophagus ranges between 6.5 and 7.0. Acid reflux is defined as a fall in intraesophageal pH below 4.0.

Considerations

Complications of 24-hour pH monitoring are rare but may include nosebleeds or tube misplacement.

The patient's nose and throat may feel irritated, so gargling with warm salt water or sucking on throat lozenges may help post-procedure.

Wireless pH monitoring

A wireless, catheter-free pH monitoring system is designed to minimize the discomfort associated with transnasal catheters, which are uncomfortable, induce social embarrassment, and interrupt daily activity during the pH monitoring and impede the sensitivity of the test.

A wireless pH sensing capsule, which is attached to the mucosa of the distal esophagus, shows improved patient tolerability, enables the patient to perform daily activities, and has the capability to record extended periods of monitoring up to 96 hours. An advantage of wireless pH monitoring is the ability to study pH levels both off and on PPIs during the 96-hour monitoring.

Indications

Wireless pH monitoring is indicated in refractory GERD.

Procedure

The preparation for the procedure is the same as for upper endoscopy.

A small transmitter is placed at the end of the esophagus and attached to the surface of the esophagus. This may be done with endoscopic assistance. The transmitter sends signals to a belt clip recorder for up to 96 hours and then passes out of the body naturally.

Patients are monitored while they go about their daily routine. They are asked to keep a journal and press one of the three buttons on the recorder to indicate if they feel heartburn, when they start and stop eating, and when they lie down.

Considerations

Complications of wireless pH monitoring are rare. Some patients may experience a vague sensation that something is in esophagus, especially when eating or drinking. Some may experience discomfort, and endoscopic removal may be required.

The capsule may cause a tear in lining of esophagus causing bleeding and may require endoscopic intervention. Perforation is possible. The capsule may dislodge early, or it may not completely pass on its own, requiring intervention to remove it.

The capsule contains a small magnet; therefore, the patient should not have an MRI study within 30 days of undergoing placement.

BLOOD TESTS

Blood tests are frequently ordered for gastroenterology patients to test for coagulopathy, to reveal alterations in basic metabolic functions, and/or to indicate the severity of a disorder.

Table 26.1 shows the reference range for some of the more commonly used blood tests. It is important to note that reference ranges are determined by testing a large group of healthy individuals. The values that fall within 2 standard deviations of the mean are then taken as the normal range, thus encompassing 95% of the sample studied. The range established by this method is dependent on two important factors: the method used, including reagents and instrumentation, and the geographic location of the population tested. It may be necessary to establish more than one reference interval for any given test (e.g., for different age groups or for males vs. females). Not all values that fall outside the reference range are clinically significant. A clinically significant change in a test result is a numerical change that also reflects a change in the patient's condition.

Table 26.1. Normal Values for Selected Blood Tests

Normal values for selected blood tests* **Diagnostic tests**	**Normal ranges**
Hematocrit (%)	F: 38–47 M: 40–54
Hemoglobin (g/dl)	F: 12–16 M: 13.5–18
Prothrombin time (seconds)*	11–14
Activated partial thromboplastin time (seconds)*	32–47
INR	<1
Platelets (mm3)	150,000–350,000
Albumin (g/dl)	3.6–5.1
Globulin (g/dl)	2.1–3.8
Albumin/globulin ratio	1:2
Total protein (g/dl)	6.4–8.3
Glucose (mg/dl)	65–115
Cholesterol (mg/dl)	150–240
Total cholesterol (mg/dl)	<200
LDL (mg/dl)	Optimal: <100 Borderline high: 100–159 High: >160
HDL (mg/dl)	M: >37 F: >47
Triglycerides (mg/dl)	<150
VLDL (mg/dl)	2–30
AST (mU/ml)	6–41
ALT (mU/ml)	8–50
Alkaline phosphatase (IU/ml)	0–15 yrs: 50–300 Adult: 30–115
Amylase (u/1)	23–85
Lipase (u/1)	0–160
Total bilirubin (mg/dl)	0.1–1.3
Indirect bilirubin (mg/dl)	0–0.4
CEA (mcg/1)	0–2.5
Ca 19-9 (u/ml)	1–36

**Laboratories established their own reference range for these tests based on its instrumentation and reagents.*

Hemoglobin and hematocrit

Hemoglobin is the iron-containing, oxygen-transport protein in red blood cells. It is formed by the developing red blood cells in bone marrow. **Hematocrit** or **packed cell volume (PCV)** refers to the volume percentage of erythrocytes in whole blood.

Indications

Both hemoglobin and hematocrit levels are decreased when the patient is anemic and increased when the patient is dehydrated, resulting in a concentration of cells proportionate to fluid volume.

Prothrombin level

Prothrombin is a glycoprotein in the plasma that is normally converted to thrombin on activation. A deficiency in this factor leads to hypoprothrombinemia.

Indications

Decreased prothrombin levels may be associated with impaired absorption of vitamin K from the intestine. It also may be related to decreased availability of vitamin K, resulting from the absence of bile salts or a decrease in intestinal flora caused by antibiotic suppression. Low levels of prothrombin are seen in infectious hepatitis, cirrhosis, biliary tract obstructions, and small bowel disorders.

Prothrombin time

Prothrombin time (PT) measures the rapidity of blood clotting through examination of factors I, II, V, VII, and X.

Indications

A prolonged PT is seen in patients with poor nutrition, vitamin K deficiency from decreased absorption, liver disease leading to decreased synthesis of clotting factors, warfarin sodium (Coumadin, Jantoven, etc.) therapy, and inherited blood disorders.

Activated partial thromboplastin time (APTT)

Activated partial thromboplastin time (APTT) also measures the rapidity of blood clotting. It reflects clotting time by examining factors I, II, V, VIII, IX, X, XI, and XII.

Indications

A prolonged APTT is observed in patients who are taking heparin and in those with liver disease or inherited coagulopathies.

International Normalized Ratio

The International Normalized Ratio (INR) is the ratio of the patient's PT compared to the mean PT for a group of individuals.

Indications

Use of the INR allows physicians to determine the level of anticoagulation in a patient independent of the laboratory reagents used.

Bleeding time

If the platelet count is extremely low, clotting mechanisms may be impaired. To assess platelet function, bleeding time is measured by making small cuts in the patient's arm and monitoring the time until bleeding stops.

Indications

Aspirin products and nonsteroidal anti-inflammatory drugs (NSAIDs) such as ibuprofen can prolong bleeding time for up to 14 days after ingestion.

Platelet count

Platelets play an important role in blood clotting. A platelet count measures the number of platelets present in the blood.

Indications

Platelet count is indicated for these possible platelet disorders:

- Thrombocythemia (overproduction of platelets due to myeloproliferative neoplasm) and reactive thrombocytosis (overproduction of platelets due to inflammatory disorder).
- Thrombocytopenia (decrease in platelets due to decreased production, increased splenic sequestration, increased platelet destruction or consumption, and dilution).
- Platelet dysfunction (may be hereditary or acquired).

Serum albumin, globulin, and albumin/globulin ratio

Serum albumin and globulin occur in inverse proportion to one another (i.e., as one increases, the other decreases proportionately).

Indications

These tests are indicated for patients who are experiencing:

- Significant or sudden weight loss.
- Symptoms of kidney or liver disorder
- Edema: ascites, peri-orbital edema, pleural effusion, and /or anasarca.

A low albumin/globulin ratio can indicate conditions causing overproduction of globulins, like an autoimmune disease (lupus), or conditions causing underproduction or loss of albumins (nephrotic syndrome, liver disease, malabsorption, malnutrition, carcinomas, etc.).

A high albumin/globulin ratio indicates conditions causing underproduction of globulins (leukemia, hypothyroidism, some genetic disorders, etc.).

Total protein

The total protein test measures the total amount of albumin and globulin found in the fluid portion of the blood.

Indications

These tests are indicated for patients who are experiencing:
- Significant or sudden weight loss.
- Symptoms of kidney or liver disorder
- Edema: ascites, peri-orbital edema, pleural effusion, and /or anasarca.

Low total protein levels indicate serious malnutrition, conditions producing malabsorption of proteins (celiac disease, inflammatory bowel disease, etc.), and kidney- or liver-related diseases.

High levels of total protein indicate kidney or liver disease, chronic inflammations and/or infections (viral hepatitis, HIV, etc.), and bone marrow disorders (multiple myeloma, etc.).

Galactose tolerance

This liver function test is based on the ability of the liver to convert galactose to glycogen. Following oral ingestion of 40 g of galactose, less than 3 g should appear in the urine within 5 hours after the ingestion. In the IV galactose tolerance test, galactose is administered intravenously, and blood is drawn at half-hour intervals for 2 hours. In a normal test result, all galactose will have been eliminated from the blood 45 minutes after its injection.

Indications

The presence of large amounts of galactose in the serum after the specified periods of time is indicative of liver disease (failure of the liver to convert galactose to glycogen).

Glucose level

Normal levels vary slightly, depending on the resource referenced:

Fasting blood sugar (glucose):

Normal for person without diabetes: 70-99 mg/dl (3.9–5.5 mmol/L)

American Diabetes Association (ADA) recommendation for someone with diabetes: 80-130 mg/dl (4.4–7.2 mmol/L)

Blood sugar 2 hours after meals (postprandial):

Normal for person without diabetes: <140 mg/dl (7.8 mmol/L)

ADA recommendation for someone with diabetes: <180 mg/dl (10.0 mmol/L)

HbA1c (3-month blood sugar average):

Normal for person without diabetes: <5.7%

ADA recommendation for someone with diabetes: <7.0%.

Indications

Glucose level testing can detect high blood glucose (hyperglycemia) and low blood glucose (hypoglycemia).

Glucose tolerance test

In the glucose tolerance test, a set amount of oral glucose is given to patient during a fasting patient, and blood samples and urine specimens are taken at intervals of 30, 60, 120, and 180 minutes.

Indications

Failure of the glucose level to return to normal in 1-2 hours is indicative of impairment of glucose utilization by the body tissues.

In some patients, it may be more appropriate to use an IV glucose test, which measures the ability of the liver to maintain glucose levels in the blood.

Serum d-xylose absorption

This test measures the ability of the intestine to absorb d-xylose, a simple sugar, as an indication of whether nutrients are being properly absorbed.

Indications

This test is used to evaluate the ability to absorb carbohydrates and help determine the cause of malabsorption.

Considerations

Renal failure will give falsely low urine xylose levels. A delay in gastric emptying time also is a major reason for falsely low serum and urine levels of xylose.

Serum cholesterol

Serum cholesterol is an important normal body constituent used in the structure of cell membranes, synthesis of bile acids, and synthesis of steroid hormones.

Indications

Elevated cholesterol levels may be caused by biliary cirrhosis, familial hyperlipidemia, high cholesterol diet, nephritic syndrome, and diabetes. Low cholesterol levels may be caused by liver disease, malabsorption, malnutrition, pernicious anemia, and hyperthyroidism. This blood test is done after a 9- to 12-hour fast.

The serum cholesterol test measures four cholesterol levels:

1. Low density lipoproteins (LDL). This is also known as the "bad cholesterol" because high levels can cause buildup of plaque in blood vessels, increasing the risk of heart disease and stroke.
2. High density lipoproteins (HDL). This is also known as the "good cholesterol" because its function is to take excess cholesterol to the liver for excretion in the bile.

3. Triglycerides. These are compounds used by the body to move fatty acids through the blood.
4. Very low density lipoprotein (VLDL). This is a type of blood fat. It is considered one of the "bad" forms of cholesterol. Sixty percent of a VLDL particle is a triglyceride.

It also tests total cholesterol. This is the sum of LDL, HDL, and VLDL.

Serum aspartate aminotransferase and alanine aminotransferase

The extent of liver disease may be determined by measuring the microsomal enzyme serum aspartate aminotransferase (AST) and the cytosolic enzyme alanine aminotransferase (ALT). These cytoplasmic enzymes are released into the blood following disruption of cell membranes.

Indications

Very high levels of ALT (i.e., as much as 100 times normal) indicate viral or drug-induced hepatitis with extensive cirrhosis. ALT levels in the thousands may indicate hepatic ischemia (shock liver), which is related to prolonged hypotension. Moderate to high ALT levels (i.e., less than 10 times normal) are suggestive of other forms of liver disease, such as biliary obstruction. Slight to moderate levels of ALT indicate hepatocellular injury. ALT may be used to differentiate between alcoholic and viral hepatitis, with alcoholic hepatitis having a lower elevation of ALT.

AST is not as specific to liver damage as ALT and must be used in conjunction with other liver tests. Increases in AST reflect cell permeability and cellular response to injury.

Serum bilirubin

A total serum bilirubin test measures both the secretory and excretory functions of the liver. The normal level in adults is 0.3-1 mg/dL (5.1-17 micromol/L).

Unconjugated (indirect) bilirubin measures the secretory function of the liver. The normal level in adults, the elderly, and children is 0.2-0.8 mg/dL (3.4-12.0 mmol/L).

Bilirubin measures excretory function. The normal level for adults is 0.1-0.3 mg/dL (1.7-5.1 micromol/L). In neonates, the normal level is <1 mg/dL with normal total bilirubin.

Indications

Abnormal elevations of total bilirubin indicate hepatocellular disease, biliary tract disease, or overproduction of bilirubin at a rate beyond the liver's capacity to metabolize it.

- Elevations in unconjugated (indirect) bilirubin usually indicate hemolysis or decreased cell permeability such as hemolytic or pernicious anemia, transfusion reaction, and cirrhosis.
- Elevated direct bilirubin is indicative of obstruction in the common bile duct such as gallstones in the bile ducts, tumors, and scarring of the bile ducts. Sickle cell disease can also cause an elevated direct bilirubin.
- Several inherited chronic conditions such as Gilbert syndrome, Dubin-Johnson syndrome, Rotor syndrome, and Crigler-Najjar syndrome increase bilirubin levels in the blood.
- Physiologic jaundice and hemolytic disease have increased unconjugated (indirect) bilirubin in newborns. It is important that an elevated level of bilirubin in a newborn be identified and quickly treated because excessive unconjugated bilirubin damages developing brain cells.
- In an infant, an elevated conjugated (direct) bilirubin level is seen in the rare conditions of biliary atresia and neonatal hepatitis.

Serum alkaline phosphatase

Serum alkaline phosphatase (ALP) is used to detect and monitor diseases of the liver (specifically the biliary tract) and/or bone. ALP primarily originates from these tissues. It is also present in minor amounts in the intestine, kidney, and placental tissue.

Indications

Increased ALP levels may reflect post-hepatic obstructive jaundice, hepatic metastasis, or fatty infiltration. Marked elevations usually indicate cholestatic processes such as mechanical biliary obstruction or drug-induced secretory failure. ALP may also rise after administration of vitamin D, albumin, or drugs such as barbiturates, oral contraceptives, or phenothiazides. Levels of ALP are also elevated in osteoblastic diseases, Paget's disease, hyperparathyroidism, and normally during pregnancy and growth.

If there is complete obstruction of the common bile duct, both ALP and bilirubin are usually elevated. If a single hepatic duct is obstructed, only ALP is elevated.

Gamma-glutamyl transferase

Gamma-glutamyl transferase (GGT) is an enzyme found in many organs throughout the body, with the highest concentrations in the liver. GGT is elevated in the blood in most diseases that cause damage to the liver or bile ducts.

Indications

If GGT is normal in conjunction with ALP elevation, the source of ALP elevation is probably bone. If GGT is also elevated, it pinpoints the liver as the source of ALP elevation.

Serum amylase

Amylase is an enzyme that helps digest carbohydrates. It is made in the pancreas and the glands that make saliva. When the pancreas is diseased or inflamed, amylase releases into the blood. Serum amylase measures the level of this enzyme in the blood.

Indications

The amount of amylase activity in serum may increase in patients with penetrating or perforated peptic ulcer, peritonitis, bowel obstruction, biliary tract disease, liver disease, pelvic inflammatory disease, diabetic ketoacidosis, acute pancreatitis, parotid or other salivary gland disease, or ischemic bowel. In patients with pancreatic pseudocysts, serum amylase is elevated. Amylase levels may be elevated as early as 6 hours after the onset of acute pancreatitis, remain high for 3-5 days, and drop to normal in about 7 days.

Serum lipase

Lipase is produced in the pancreas and secreted into the duodenum, where it converts triglycerides and other fats into fatty acids and glycerol. Serum lipase measures the level of this enzyme in the blood.

Indications

Serum lipase elevation is most common in acute pancreatitis but also may occur in perforated peptic ulcer, intestinal obstruction, acute cholecystitis, infectious hepatitis, cirrhosis, or jaundice. In patients with pancreatitis, it is not uncommon for serum lipase levels to reach 100 times the normal level. Lipase increases later and remains elevated longer than amylase, usually decreasing 7-10 days after an acute event.

Serum gastrin

Gastrin is a hormone that is secreted in the pyloric antral mucosa of the stomach. It stimulates secretion of hydrochloric acid by the parietal cells of the gastric glands. Levels can be tested using the gastrin stimulation test.

Indications

The gastrin stimulation test is indicated in patients with severe or unusual gastric or duodenal ulcerations, suspected Zollinger-Ellison syndrome, and suspected gastric hypersecretion.

Procedure

The gastrin stimulation test requires patients to fast for 12 hours before the test. H_2-receptor antagonists and anticholinergics should be withheld for 24 hours. PPIs should be withheld for at least 72 hours.

Timed serum gastrin levels are drawn after stimulation with IV secretin. Blood specimens are drawn 10 minutes before; immediately before; and at 1, 2, 5, 10, and 30 minutes after injection of secretin. For each sample, 5 ml of blood is withdrawn and discarded, followed by collection of a 10-ml specimen for testing. Serum gastrin levels are calculated for all seven samples.

Considerations

Potential complications include infiltration of the IV fluid, inability to withdraw the correct amount of blood for the sample, and a local or systemic allergic reaction to administration of secretin.

Hepatitis antibodies and antigens

Hepatitis antibodies are cells produced by the body to help fight the different types of hepatitis. These antibodies are of vital importance: They help the body control the invading disease, and they also serve as markers that can help diagnose the presence of the disease. Antibodies are abbreviated as Ab in lab results.

Antigens are any substance capable of inducing a specific immune response and reacting with the products of that response with specific antibody or specifically sensitized T-lymphocytes, or both. Antigens may be soluble substances such as toxins and foreign proteins or particulates such as bacteria and tissue cells. Only the portion of the protein or polysaccharide molecule known as the antigen determinant combines with an antibody or a specific receptor on a lymphocyte. Antigens are abbreviated as Ag in lab results.

Indications

Hepatitis B

Hepatitis B surface antigen (HBsAg) is a measure for hepatitis B virus (HBV) infection. Testing is positive in carriers and in those with acute and chronic infections.

Hepatitis B surface antibody (HBsAb) is seen in patients who have recovered from hepatitis B and/or have formed antibodies by other means. HBsAb titers may be valuable in confirming immunity after the hepatitis B vaccine series.

Hepatitis B core antibody (HBcAb) is present in both acute and recovered disease states. Immunoglobulins must be assessed to make a differential diagnosis. If immunoglobulin M (IgM) is positive, the patient's disease is active and infectious. If immunoglobulin G (IgG) is positive, the patient has developed the antibody and is no longer infective.

Hepatitis A

If hepatitis A antibody (HAVAb) is negative, there is no history of or currently active hepatitis A virus (HAV). HAVAb is positive in both acute infections and in those who have recovered from infection. As with hepatitis B, immunoglobulins must be assessed to make a differential diagnosis. Patients with acute infectious hepatitis A are IgM-positive. Patients who have recovered from hepatitis A are IgG-positive.

Hepatitis C

The diagnosis of hepatitis C virus (HCV) infection in patients with chronic hepatitis (defined as infection persisting for more than 6 months) entails the detection of anti-HCV in serum, confirmed with a positive recombinant immunoblot assay (RIBA-2).

The diagnosis of acute HCV infection is more problematic because anti-HCV may not appear in serum for 2-6 months. In addition, immunocompromised patients may not mount an anti-HCV response.

Hepatitis D

Because hepatitis D virus (HDV) occurs only in patients infected with HBV, the initial step in diagnosis of HDV is confirmation that the patient is seropositive for HBsAg. Anti-HDV IgG or IgM is found in the acute and chronic phases of HDV infection, respectively.

Carcinoembryonic antigen (CEA)

Carcinoembryonic antigen (CEA) is a glycoprotein that is normally produced in GI tissue during fetal development, but the production stops before birth.

CEA is usually present at very low levels in the blood of healthy adults (about 20 ng/mL). The serum levels are raised in some types of cancer, which means that it can be used as a tumor marker in clinical tests.

Indications

In adults, plasma CEA levels may be raised in colorectal, gastric, pancreatic, lung, breast, and medullary thyroid carcinomas, as well as some non-neoplastic conditions like ulcerative colitis, pancreatitis, cirrhosis, chronic obstructive pulmonary disease (COPD), Crohn's disease, hypothyroidism. Smoking also increases CEA levels.

CEA measurement is mainly used as a tumor marker to monitor colorectal carcinoma treatment effectiveness, to identify recurrences after surgical resection, for staging, or to localize cancer spread. Elevated CEA levels should return to normal after successful surgical removal of the tumor and can be used in follow-up, especially of colorectal cancers.

Cancer antigen 19-9

The cancer antigen 19-9 (CA 19-9) test is a commercially available monoclonal antibody to a tissue glycoprotein that also recognizes a mucin-type glycoprotein in serum.

Indications

This test is sensitive for gastric, pancreatic, and hepatobiliary tumors. This test is not a useful screening method and needs further study.

Serum antibodies

Antibodies sometimes develop to cytoplasmic constituents, including antimitochondrial antibodies and anti-smooth muscle antibodies (both part of autoimmune liver disease panel). Indirect immunofluorescence is used to demonstrate the presence of these antibodies.

Indications

These antibodies seem to have no etiological significance but are useful markers for a limited range of conditions of probable immune etiology, nearly all of which involve the liver.

Anti-smooth muscle antibody (ASMA) detects antibodies that attack smooth muscle. ASMA is primarily used to detect autoimmune hepatitis. It can also be elevated with chronic HCV; cirrhosis; liver, melanoma, breast, and ovarian cancer, and infectious mononucleosis. ASMA is also used to distinguish between autoimmune hepatitis and systemic lupus erythematosus.

Antimitochondrial antibody (AMA) test is primarily used to detect an autoimmune condition, primary biliary cholangitis, but can be positive in chronic active hepatitis.

Neither antibody, however, is specific for the single disease.

Celiac screening panel

A celiac panel refers to serologic tests that look for three antibodies common in celiac disease. The panel includes anti-tissue transglutaminase (tTG) antibodies, anti-endomysial antibodies (EMA), and deamidated gliadin (DGP) antibodies. Most of these are either IgG or IgA antibodies.

Indications

Although detection of these antibodies has a valuable role in initial screening, individuals with celiac disease may have selective deficiencies of total IgA and may not produce these antibodies. Also, a number of these antibodies, such as anti-gliadin and anti-endomysial antibodies, may occur in other enteropathies.

These antibodies disappear with mucosal healing on a gluten-free diet, so they may be used to monitor patient compliance with the dietary regimen.

Small intestine biopsy is necessary to confirm the diagnosis of celiac disease. Intestinal biopsy shows density of intra-epithelial lymphocytes, crypt hyperplasia with decreased villi ration, blunted or atrophic villi, and epithelial changes, including structural abnormalities in the epithelial cells, so the celiac disease can be classified/graded.

URINALYSIS

In gastroenterology patients, urine samples may be analyzed to determine levels of bilirubin or urobilinogen, vitamin B_{12} absorption, or the extent of biliary obstruction. Most urinalysis is semiquantitative in nature and is done on random clean catch samples.

Urine bilirubin

Only direct bilirubin can be measured in the urine. The urine specimen for a bilirubin test must be collected in a dark bottle and immediately taken to the laboratory because light decreases the bilirubin level. Alternatively, the specimen may be shaken at bedside where brown urine with yellow foam production is a positive result.

Urine urobilinogen

Urine specimens that will be analyzed for urobilinogen levels should be put in a dark brown container and immediately taken to the laboratory.

Intrinsic factor

Intrinsic factor (IF) is a glycoprotein that is essential for vitamin B_{12} absorption. IF is produced by the parietal cells in the stomach. Vitamin B_{12} is absorbed in the ileum after binding with intrinsic factor. Vitamin B_{12} is involved in metabolism of every cell of the human body by playing a key role in DNA synthesis and formation, red blood cell creation, and brain functioning.

Indications

IF is indicated if deficiency is related to:

- Pernicious anemia.
- Congenital pernicious anemia.
- Neurological abnormalities.

Congenital pernicious anemia (CPA), or intrinsic factor deficiency, is a rare genetic disorder characterized by lack of gastric intrinsic factor in the normal gastric secretions and mucosal tissue.

CPA signs and symptoms, which usually occur before the age of 5 years, include failure to thrive, symptoms of anemia, and nervous system damage. In adults, other symptoms may be confusion, depression, fatigue, weakness, and numbness and tingling in the hands and/or feet.

This IF blood test is frequently drawn along with vitamin B_{12} and folate to help with diagnosis.

FECAL ANALYSIS

Fecal analysis is an examination of fecal matter for diagnostic purposes by chemical, physical, or microscopic means and includes performing chemical screening tests and screening for microorganisms or parasites.

Indications

Stool specimens should be analyzed for patients with diarrhea, steatorrhea, constipation, bleeding, or persistent abdominal discomfort. Usually, only a small amount of stool needs to be collected on a tongue blade and placed in a disposable container. If an enema is needed to obtain a specimen, only tap water or normal saline should be used.

The length of time it takes for stool passage can be determined by the administration of a carmine marker or charcoal marker when a meal is ingested. The color, form, odor, consistency, content, and pH of the stool may also be important.

The gastroenterology nurse should explain to the patient the method to be used to collect the stool specimen and whether it should be delivered fresh or how it should be stored. The patient should also be instructed to record the times of bowel movements and to note stool color; consistency; and any associated pain, discomfort, or bleeding.

Fecal occult blood test

Fecal occult blood testing (FOBT) is a guaiac-based screening for occult blood in the feces. FOBT is used for colorectal cancer screening.

A test for the detection of **occult blood** in the feces is **Hemoccult**, which uses guaiac-impregnated paper slides or guaiac tape and a developing solution. It is readily available, convenient, easy to use, and inexpensive. Hemoccult has numerous disadvantages, however, as some fruits, vegetables, red meat in the recent diet, and some medications can cause false positives. A pre-cancerous polyp that is not bleeding or recent ingestion of vitamin C can cause false negatives.

Procedure

Before the procedure

Before occult blood testing, the patient must:

- Adhere to a high-fiber diet for 48 hours, with no rare red meat.
- Avoid foods that are high in peroxidase such as turnips, broccoli, or horseradish.
- Avoid iron preparation, bromides, iodides, salicylates, NSAIDs, or high doses of ascorbic acid.
- Ideally, aspirin and NSAIDs should be avoided for 1 week before Hemoccult testing.

During the procedure

Three stool samples must be collected without being contaminated by urine or toilet tissue. The delay between collection and laboratory testing should not exceed 6 days, and the specimens should not be refrigerated.

For analysis, a small piece of stool is placed on the slide or tape and is wetted with the appropriate developing solution. A blue color appears in 30-60 seconds if blood is present.

After the procedure

Positive results indicate GI bleeding and may be important in the early detection of colorectal cancer. False positive results may be caused by patient noncompliance with dietary and/or medication restrictions. Vitamin C (ascorbic acid) can cause false negative results.

All three separate stool specimens should be returned to the facility identified by the physician who distributed the test. Follow instructions for returning the specimens.

Fecal immunochemical test

The fecal immunochemical test (FIT) is a screening test for colorectal cancer by identifying blood in the stool. FIT only identifies human blood in the lower intestines. Medicines and other blood contaminants do not interfere with the test, so it tends to have fewer false positive result than Hemoccult.

Indications

FIT is a screening test for colon cancer. It tests for hidden blood in the stool, which can be an early sign of cancer.

Procedure

Instructions for use:

- Flush the toilet prior to collection.
- If toilet paper is used, toss it in trash can
- After having a stool, use the brush provided to brush the surface of the stool.
- Dip brush in water.
- Touch the brush on the indicator card.
- The test may need to be performed more than once.
- Follow instructions for returning the specimen.

Stool DNA

DNA is continuously shed from cells in the intestinal lining and passed into the stool. Both precancerous and cancer cells shed DNA, so the test can be an early indicator of the presence of colon cancer or precancerous lesions.

Cologuard is an example of a Medicare-approved colon cancer screening tool for average risk patients. Cologuard uses stool DNA technology for qualitative detection of colorectal cancer neoplasia associated with specific DNA markers and for presence of occult blood in stools. The test is not intended for patients with a history of colorectal or other cancers, inflammatory bowel disease (IBD), ulcerative colitis, Crohn's disease, familial adenomatous polyposis (FAP), relevant familial hereditary cancer syndromes, or family history of colorectal cancer. Cologuard can have false positive and false negative results.

Indications

Stool DNA testing is a screening tool for colorectal cancer.

Procedure

No special diet or medication adjustments are needed prior to testing. Patients should be instructed to:

- Store the testing kit in a cool dry place and keep it away from sunlight.
- Not collect the specimen if any bleeding or diarrhea is occurring.
- Collect the specimen in the sample container, following the instruction guide provided in the package.
- Send the package to laboratory.

Considerations

Stool DNA testing should not be the screening tool of choice for patients with blood in the stool, abdominal pain, change in bowel habits, or any histories mentioned above.

Fecal fat

For a qualitative test of stool fecal fat, a random sample of stool is examined for evidence of malabsorption and various fats. Neutral fat, which accounts for 20% to 30% of total stool fat in pancreatic insufficiency, turns red with Sudan stain and appears as globules, whereas fatty acids do not stain with Sudan and appear as crystals.

Indications

Little, if any, dietary fat is unabsorbed when bile and pancreatic secretions are adequate. The presence of abnormal fecal fat content (steatorrhea) is significant in malabsorptive disorders, such as Crohn's disease and blind loop syndrome. Typically, fatty stools are greasy, pale, frothy, floating, and foul-smelling.

Procedure

An accurate method of testing fat malabsorption is 72-hour fecal fat collection. The patient is kept on a high-fat diet, with no alcohol, throughout the stool collection period. A diet record is kept and analyzed for the amount of fat ingested during the collection period.

Some test centers require the administration of a nonabsorbable marker (e.g., charcoal) at the beginning and end of the collection period.

Total amount of fat in the stool collection is measured and compared with the patient's intake. The test result demonstrates the amount of fat that was not absorbed.

Fecal chymotrypsin

Chymotrypsin is an enzyme that digests protein in the small intestine. This test measures the amount of chymotrypsin in stool to help evaluate pancreatic function.

Indications

Fecal chymotrypsin determination is a reliable, semiquantitative test for exocrine pancreatic insufficiency.

Procedure

Stools are collected for 72 hours between two nonabsorbable markers, such as charcoal. Specimens are frozen as soon as possible after they are passed and are pooled in a pre-weighed container. Enzyme activity in the stool samples is measured titrimetrically or by using an ultraviolet spectrophotometer. Fecal chymotrypsin determination can be applied to the same 72-hour stool collection procedure used for quantitation of fecal fat.

Fecal Alph$_1$ antitrypsin

Alpha$_1$ antitrypsin (A$_1$AT) is a natural plasma protein that resists proteolytic digestion.

Indications

The purpose of measuring fecal A$_1$AT is to detect protein loss. Its concentration in the stool can be measured simply and accurately by immunological methods. In addition, measurement of A$_1$AT does not require the use of radioisotopes.

Another way of detecting excessive protein loss is to inject ^{131}I-labeled polyvinylpyrrolidone (povidone) intravenously and collect stool samples over a 72-hour period. A patient with a disease that involves protein loss will excrete large amounts of polyvinylpyrrolidone in the stool.

Alternatively, ^{61}Cr-albumin may be injected intravenously and all stools collected for a 96-hour period, taking care not to contaminate the stool specimens with urine. Normal excretion is less than 1% of the administered dose of tagged albumin. In patients with protein-losing enteropathy, between 4% and 20%, but as much as 40% of the administered dose, is recovered in the stool.

Fecal organisms

Stool may be collected and examined for organisms.

Indications

Testing is indicated for the discovery of bacteria, parasites, and other organisms.

Procedure

Patients should be provided instructions on specimen collection by the laboratory designated to perform the test.

- Bacteria. Testing for bacterial infection (e.g., *Salmonella*, *Shigella*, *Campylobacter*, *Escherichia coli*) usually involves collecting fresh stool and placing it into culture media immediately.
- Parasites. Specimens to detect parasites (e.g., *Giardia*, *Cryptosporidium*) may require placement of the specimen into media designed to preserve both the ova and parasites, such as formalin.
- *Clostridium difficile*. Specimens may need to be collected fresh and then either kept in a container on ice for immediate delivery to the laboratory or frozen.

HYDROGEN/METHANE BREATH TEST

The objective of the **hydrogen/methane breath test** is to measure the volume of hydrogen that is produced in the colon, absorbed from the colon into the blood, and expelled in the breath. Methane is also measured because there are patients with severe malabsorption issues who do not produce hydrogen but methane.

Hydrogen/methane is produced in the body by bacterial fermentation of carbohydrates. Under normal circumstances, lactose is broken down into glucose and galactose and is absorbed in the small intestine, which has virtually no anaerobic bacteria. In patients who are deficient in lactase or who have delayed transit times, mucosal damage, or colonic flora in the upper GI tract, hydrogen gas evolves when lactose comes into contact with the fermenting bacteria in the colon.

Indications

Hydrogen/methane breath tests are used in the diagnosis of small intestinal bacterial overgrowth (SIBO), short bowel syndrome (rapid transit of food through the small bowel), sugar intolerance (lactose, lactulose, fructose, or sucrose), and tests of fat absorption. The breath test also is used for other GI disorders (bloating, cramps, diarrhea) related to IBS.

The test is contraindicated in patients who are unable or unwilling to cooperate.

Procedure

Patients must fast for 12 hours and follow other specific diet instructions prior to the test, and they must refrain from smoking or second-hand smoke exposure after midnight the night before the test.

End-expired tidal air is collected twice before the test and at designated times for 3 hours after ingestion of a carbohydrate drink. Hydrogen gas concentration is determined by gas-liquid chromatography. Test results are normal if the patient expires <20 parts per million (ppm) of hydrogen. Patients with malabsorption syndromes expire >80 ppm.

To avoid false positive results, a positive lactose test may be followed by a glucose breath test to differentiate between true lactose intolerance and bacterial overgrowth. A portion of the population lacks hydrogen-producing bacteria in the colon. To avoid false negative diagnosis, a negative lactose test may be followed by a lactulose test to confirm the presence of hydrogen-producing colonic bacteria.

Considerations

Patents should have at least 28 days between a breath test and any procedure that involves a bowel prep. Antibiotic and probiotics should be discontinued at least 4 weeks before test. Laxative should not be taken for at least 1 week prior to test.

HELICOBACTER PYLORI TESTS

Helicobacter pylori (*H. pylori*) is diagnosed through blood, breath, stool, and/or GI biopsy tests. Blood and breath tests are most common because they are least invasive.

Indications

The urea breath test, *H. pylori* serology, and *H. pylori* stool antigen test are used to detect *H. pylori* infection.

Procedure

Urea breath test. This test is usually used in combination with the blood test for diagnosing *H. pylori*, or after treatment to determine the effectiveness of treatment. The patient drinks a urea solution that contains a special carbon atom. If *H. pylori* is present, it breaks down the urea, releasing the carbon. The blood carries the carbon to the lung, where the patient exhales it. The urea breath test has high diagnostic accuracy for detecting *H. pylori* in patients. Patients must be NPO for at least 4 hours and not take any antibiotics, PPIs, or bismuth compounds for 1 month prior to the test.

***H. pylori* serology.** This is a blood test that detects antibodies to the *H. pylori* bacterium. The *H. pylori* serology test is highly sensitive and specific in diagnosing initial infection. The negative implication with this test is that, once a patient makes antibodies, he or she will always be positive.

***H. pylori* stool antigen (HpSA) test.** This stool test may be used to detect *H. pylori* infection in a patient's fecal matter. The patient must be NPO for 4 hours and off all antibiotics, PPIs, and bismuth compounds for 1 month prior to the test.

CASE SITUATION

Ms. Jennifer Roth, a 28-year-old female, is scheduled for a gastroscopy on Tuesday. She is being treated for alcoholism and has GI bleeding and acute pain.

Points to think about

1. What tests would most commonly be ordered before a procedure for Jennifer?
2. What laboratory tests might be increased or decreased, and why?

Suggested responses

1. The following tests would probably be ordered for Jennifer:
   a. Liver profile, including ALT and AST.
   b. Complete blood count, including hematocrit, hemoglobin, white cell count, the proportions of the different white cells, red blood cell indexes and morphology, and platelet count.
   c. PT, APTT, and INR.
   d. Urinalysis, including bilirubin and urobilinogen.
2. Given Ms. Roth's condition, the gastroenterology nurse might expect to see the following alterations:

Table 26.2. Patient Test Alterations and Causes

Abnormal test results	Possible causes
ALT	Cirrhosis or hepatitis, depending on the degree of alcoholism
AST	
Red blood cell count	GI bleeding resulting in iron deficiency
Hemoglobin	
Hematocrit	
Platelet count	Impaired platelet production because of toxic effects of alcohol on the bone marrow
PT	Liver disease impairs production of protein synthesis
APTT	
Urinary bilirubin	Normal to elevated, depending on the degree of bleeding and the stage of liver disease (the liver converts the hemoglobin from the red cells to bilirubin)
Fecal occult blood test (FOBT)	Positive for occult blood in the stool due to GI bleeding

REVIEW TERMS

activated partial thromboplastin time (APTT), ambulatory pH monitoring, angiography, arteriography, barium enema, barium sulfate, barium swallow, pH monitoring system, capsule endoscopy, cholangiography, computerized tomography (CT) scan, double-contrast effect, enteroclysis, fluoroscopic examination, fecal occult blood test (FOBT), hematocrit, hemoglobin, hydrogen/methane breath test, intrinsic factor, magnetic resonance cholangiopancreatography (MRCP), magnetic resonance enterography (MRE), magnetic resonance imaging (MRI), occult blood, oral cholecystography, packed cell volume (PCV), patency capsule, percutaneous transhepatic cholangiography (PTC), prothrombin time (PT), radiographic studies, scintigraphy, ultrasonography, transjugular intrahepatic portosystemic shunt (TIPS), upper gastrointestinal (UGI) series, X-ray (roentgenographic) examination

REVIEW QUESTIONS

1. The usual position of the patient during a barium swallow is:
   a. Standing.
   b. Sitting.
   c. Prone.
   d. Supine.
2. Which of the following is true about the TIPS procedure?
   a. A percutaneously created low-resistance channel between the portal vein and the hepatic vein.
   b. Performed under fluoroscopic guidance.
   c. Effective in lowering portal pressure.
   d. All of the above.
3. Patient teaching prior to capsule endoscopy should include which of the following?
   a. NPO after midnight.
   b. Iron tablets avoided 1 week prior to the test.
   c. MRIs must be avoided until confirmed excretion of the capsule.
   d. Abdominal pain, nausea, or vomiting require calling the physician.
   e. All of the above.
4. The three-dimensional image produced by CT scan is based on:
   a. Differences in tissue density.
   b. Differential transmission of sound waves by different tissues.
   c. Differential uptake of radioactive isotopes.
   d. Visualization of radiopaque structures.
5. Which of the following is true about magnetic resonance cholangiography?
   a. There is more radiation exposure than with a chest X-ray.
   b. A major shortcoming is the lack of ability to perform therapeutic interventions.
   c. The dye (gadolinium) used causes severe nausea.
   d. The patient must remain NPO after midnight prior to the procedure.
6. Gastric analysis is indicated in patients with all of the following, except:
   a. Intractable peptic ulcer symptoms.
   b. Suspected Zollinger-Ellison syndrome.
   c. Achlorhydria.
   d. Intolerance to lactose, lactulose, fructose, or sucrose.
7. An increase in platelet levels could indicate:
   a. Myeloproliferative neoplasm.
   b. Systemic lupus erythematosus (SLE).
   c. Cirrhosis.
   d. Crohn's disease.
8. Extremely high levels of ALT are indicative of:
   a. Viral or drug-induced hepatitis.
   b. Biliary obstruction.
   c. Hepatocellular injury.
   d. Hepatic metastasis.
9. Intrinsic factor is used to evaluate:
   a. Disaccharide tolerance.
   b. Vitamin B_{12} absorption.
   c. Immunity to hepatitis.
   d. Bilirubin levels.
10. The hydrogen/methane breath test is used for all of the following except:
   a. Small intestinal bacterial overgrowth.
   b. Short bowel syndrome.
   c. Diseases of the liver (specifically the biliary tract).
   d. Sugar intolerance (lactose, lactulose, fructose, or sucrose).

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SECTION VI

Therapeutic Procedures

The final section covers the many therapeutic procedures used for treating GI aberrancies. The procedures reviewed in this section include dilatation, hemostasis and tumor ablation, intubation and drainage, excision and extraction, complications and emergencies, and surgical interventions.

Chapter 27

DILATATION

This chapter will acquaint the gastroenterology nurse with the procedures used for endoscopic dilatation of gastrointestinal (GI) strictures.

Learning objectives

After reviewing the content of this chapter, the gastroenterology nurse will be able to:

1. Describe the indications, contraindications, techniques, and potential complications of using shear force with silicone bougies or tapered polyvinyl chloride dilators to dilate GI strictures.
2. Explain the indications, contraindications, techniques, and potential complications of using hydrostatic or pneumatic balloons for GI dilatation.
3. Discuss nursing considerations pertinent to each of these techniques.

BASIC PRINCIPLES OF GI DILATATION

There are two basic types of GI **dilatation**:

- **Radial-force dilatation** involves inflating a balloon within the lumen, creating a radial force that accomplishes the dilation.
- **Shear-force dilatation** involves forcing a dilator through a narrowed area and using axial force to enlarge the lumen.

Obstructing tumors may also be ablated using a laser or bipolar probe, as discussed in Chapter 28.

Radial-force dilators

Radial-force dilators are hydrostatic or pneumatic balloon dilators. The balloon is inflated within the strictured lumen, providing a radial force that accomplishes the dilation to the stricture. Multidiameter balloon dilators allow for gradual increases in radial pressures against the strictured lumen. Dilator balloon sizes range from 8-10 mm to 18-20 mm.

Endoscopic balloon dilatation is the standard treatment modality for severe benign strictures of the esophagus.

Shear-force dilators

There are two different types of dilators that cause a shearing effect:

- Tungsten-filled silicone Hurst or Maloney dilators, often called "bougies."
- Tapered, polyvinyl chloride Savary-Gilliard or American dilators, which replaced the early Eder Puestow and Celestin dilators.

The sizes of esophageal dilators are expressed in terms of their diameter in millimeters (mm) or in **French units** (Fr), which is based on the circumference of the instrument.

PROCEDURE

Endoscopic dilatation is performed on patients who present with strictures.

Indications for dilatation

Dysphagia resulting from surgical anastomotic strictures, achalasia, peptic strictures, Schatzki's ring, post-radiation strictures, webs, injury, radiation, spasms, scleroderma, or neoplasm are among the indications for dilatation (American Society for Gastrointestinal Endoscopy [ASGE], 2014; Nishikawa et al., 2016). The procedure may be performed on patients based on absolute and relative indications.

The esophagus is the most common target of GI dilatation, with the primary goal of restoring the patient's ability to eat and drink normally. In most cases, dysphagia in response to solid food will not be completely relieved until a minimum luminal diameter equivalent to 38 or 40 Fr (14 mm diameter) is achieved. If the esophageal lumen can be fully dilated to 52 Fr (17 mm diameter), the patient should be able to eat a normal diet. Patients with a lumen less than 39 Fr will require a modified diet.

If the patient takes blood thinners such as warfarin (Coumadin, Jantoven), clopidogrel (Plavix), rivaroxaban (Xarelto), apixaban (Eliquis), and enoxaparin (Lovenox), the patient should contact the physician at least 7 days before the procedure to discuss how to manage the blood thinners.

Day before the procedure

The American Society of Anesthesiologists (ASA) (2017) guidelines provide direction for the gastroenterologist. The guidelines recommend that prior to elective procedures requiring general or procedural sedation, the patient should be nothing by mouth (NPO):

- 6 hours or more after a light meal or nonhuman milk.
- Between 2 and 4 hours after clear liquids for adults and children.
- Between 4 and 6 hours after breast milk and formula.

Patients who have had bariatric surgery or an esophagectomy should have a clear liquid diet for 24 hours prior to the procedure.

Day of the procedure

Prior to GI dilatation, a signed informed consent is verified. A medical history, including allergies, current medications, and any other information pertinent to the current complaint, is obtained. Dentures should be removed, and a topical anesthetic or gargle may be given. A patent intravenous (IV) line is established, and baseline vital signs are obtained. The physician should be notified if the patient is currently on anticoagulant, acetylsalicylic acid (ASA), or nonsteroidal anti-inflammatory drug (NSAID) therapy, or if labs are abnormal.

Endoscopy units may use anesthesia staff to provide monitored anesthesia care (MAC) to patients presenting for endoscopic procedures with medical comorbidities (ASGE, 2018). If MAC will be used, the patient should arrive a few hours prior to the procedure start time to be evaluated by the anesthesiologist. If the patient presenting for dilatation has a debilitating or chronic disease, anesthesia may require laboratory tests or an ECG prior to the procedure.

During the procedure

During the procedure, the gastroenterology nurse should consider the following.

Patient positioning

- The patient is positioned in a sitting, standing, or left lateral position. It is the gastroenterology nurse's responsibility to ensure that the correct position is maintained throughout the procedure. If the dilators deviate from the midline during the procedure, perforation of the pharynx, esophagus, or stomach could occur.
- When dilating patients with esophageal stricture, the bed should be in the reverse Trendelenberg position, slightly tilted down toward the feet. The patient's head should always be elevated at least 30 degrees.

Sedation

Sedation may depend on the procedure performed:

- Bougienage with Hurst or Maloney dilators alone is usually performed without premedication in adults.
- Other forms of dilation are performed using sedation and endoscopy.
- Children typically require sedation.

Nursing considerations

The responsibilities of the gastroenterology nurse assisting the physician include:

- Assembling the dilators and monitor the patient's response to the procedure, providing nursing interventions such as suctioning, as indicated.
- Assisting the endoscopist. When a guidewire is used, the gastroenterology nurse passes the guidewire to the endoscopist, ensuring that there are no kinks or bends.
- Controlling the loose end of the guidewire while the spring tip is inserted.
- Stabilizing the guidewire to maintain the correct position during advancement and withdrawal of the dilator. The distal 5-6 inches of the dilator should be lubricated with a water-soluble lubricant. The lubricated dilator is fed onto the correctly placed guidewire.
- Confirming the correct placement using the guidewire markings.
- Notifying the endoscopist if there is any blood on the used dilator or in the patient's secretions.
- Documenting the sizes of the dilators used.

Note that some dilatation procedures are performed with fluoroscopic guidance, particularly when carcinomas or long strictures are being dilated.

The procedures are generally begun with a dilator that is one or two sizes smaller than the diameter of the stricture. Dilatation can be repeated daily, every other day, or several times per week, depending on the patient's response, the severity of the stricture, and the sense of resistance that is felt. On subsequent sessions, the endoscopist may begin with a dilator that is one size smaller than the largest dilator passed at the previous session. Generally, not more than three increasing French sizes should be used in any one session.

After the procedure

After any upper GI dilatation procedure:

- If local anesthesia is used, keep the patient NPO until the gag reflex returns.
- Monitor and document vital signs and the patient's condition.
- Instruct the patient to notify the endoscopist immediately in the event of chest pain, fever, regurgitation of blood, pain on swallowing, back pain, shoulder pain, abdominal pain, abdominal distention, chills, black stools, or nausea.

Complications

Potential complications of GI dilatation include perforation, bleeding, bacteremia, and aspiration.

Perforation

The most common site of perforation is the area immediately proximal to a stricture. With dilators that use guidewires, the risk of perforation is more likely related to the guidewire than to the dilator. Fluoroscopy is generally advised to be sure that the guidewire does not bend to form a sharp point or loop and to verify correct placement of the dilator at the stricture. Perforation may also occur if the guidewire is misplaced or damaged. In patients who have undergone previous surgery, the use of guidewires is riskier and more complicated because the anatomy may be distorted.

It is critically important that perforation be identified as soon as possible. There are a number of notable signs and symptoms of esophageal perforation.

- The most common symptom of esophageal perforation is persistent pain, even if it is mild, after dilatation.
- Esophageal perforation at the cervical level is usually obvious. It is associated with fever, chest pain, leukocytosis, pain on swallowing, change in the quality of the voice, and air in the subcutaneous tissues of the neck (subcutaneous emphysema).
- Radiographic examination usually reveals air in the retroesophageal space.

When an esophageal perforation is suspected, diagnosis is made by:

- Upright and lateral X-ray films of the chest. These should be obtained to serve as a baseline and to search for pneumothorax, pleural effusion, or free air under the diaphragm.
- An X-ray examination performed using a water-soluble contrast medium.
- If the patient is well-sedated and unable to tolerate a barium swallow, a computerized tomography (CT) scan of the chest is performed to rule out perforation. Caution should be used because aspiration of contrast media can cause pulmonary toxicity.

Treatment for esophageal perforation includes:

- Conservative management, which includes administering antibiotics or using nasogastric suctioning and keeping the patient NPO.
- Surgery, which may be necessary to repair the perforation.

Bleeding

Bleeding is a rare complication of dilatation done for the management of benign lesions. When bleeding occurs, it is usually minor, and transfusions are seldom needed.

If blood and/or mucus are present on the dilator on removal from the affected site, this observation should be documented. If there is a question about malignancy of the stricture, any blood or mucus that is present may be rinsed into a bottle with cytological fixative and submitted for cytological examination.

Bacteremia

Bacteremia, a potential complication after dilatation procedures, is a serious problem in immunocompromised patients or patients with prosthetic valves or a history of endocarditis. The bacteria in the blood may originate from either the dilator or the oropharynx. It is important to clean and disinfect bougienage dilators adequately after each use, according to the manufacturer guidelines.

Aspiration

Aspiration of food and fluid during dilatation constitutes an emergency. The risk of aspiration may be reduced by:

- Maintaining the patient's NPO status before the procedure per ASA guidelines.
- Having airway suction available and using it as needed.
- Using an overtube during the procedure.
- Elevating the head of the bed or placing the patient in a left lateral position.

TYPES OF DILATORS

Indications for the use of dilators, techniques for insertion, and specific risks associated with each technique vary by clinical presentation and type of dilator.

Dilators for bougienage

Bougienage is dilation of the esophagus using tungsten-filled, soft, silicone dilators (**bougies**) of graduated sizes, ranging from 18 to 60 Fr. Dilators less than 32 Fr are so flexible that little effective pressure can be applied from above without bending them. The tungsten provides the proper combination of rigidity and flexibility so the bougies can be swallowed. The force applied to the shaft of the dilator can be transmitted to the stricture.

There are two types of bougies: tapered **Maloney dilators** and blunt or rounded **Hurst dilators**. The

Maloney dilator with its tapered end may be easier for patients to swallow. The tapered tip of the Maloney dilator is more likely to enter a small-diameter stricture but is also more likely to bend, causing the dilator to be misdirected.

Indications

Indications for bougienage include:

- Esophageal strictures, including peptic, postsurgical, post-radiation, or malignant strictures.
- Chemically induced strictures caused by lye ingestion or a sclerosing agent. Esophageal injury associated with a gray-white exudate, ulcers, and hemorrhage can have nonforceful dilation 3-7 days after injury. Esophageal injury that is associated with extensive ulceration, gray-black exudate, mucosal sloughing, and a dilated and atonic lumen without peristalsis should not be dilated until mucosal healing is noted endoscopically.
- Esophageal rings or webs. In bougienage for lower esophageal rings, one pass of a large dilator, such as a 50 Fr, is generally preferred. If this treatment is not successful, it may be preferable to use a pneumatic dilator similar to that used in patients with achalasia.
- Diffuse esophageal spasm with a normal lower esophageal sphincter (LES).
- Scleroderma.

Procedure

A topical anesthetic may be applied to the patient's throat. The patient is usually placed in a 30-degree upright position, but a left lateral position may be used to avoid aspiration and facilitate endoscopic examination preceding dilatation or if fluoroscopy is to be used.

Maloney and Hurst dilators are passed into the hypopharynx and esophagus in much the same way as a flexible endoscope. Before use, the distal end of the bougie is well lubricated. The physician introduces the bougie into the mouth and uses a forefinger to direct it into the pharynx. A swallowing motion allows gentle passage into the upper esophagus. An experienced physician can feel when the dilator engages the stricture. When it does engage a stricture, the dilator is held lightly between the thumb and fingers, and gentle force is applied to dilate the stricture.

Aspiration of oral secretions is usually necessary during the procedure. The physician should be notified of any blood on the used dilator or in the patient's secretions. The sizes of the dilators used are documented.

Care and maintenance

Before each use, the bougies should be thoroughly inspected. Most bougies are stamped with an expiration date from the manufacturer. If the silicone cracks, splits, or has expired, it should be replaced. Most manufacturers have a trade-in policy.

Always refer to the manufacturer guidelines on cleaning. Additional cleaning considerations include:

- Remove silicone dilators from the disinfecting solution promptly. Prolonged soaking will cause deterioration of the silicone.
- After disinfection, thoroughly rinse and dry bougies.
- Store bougies horizontally, as straight as possible, in a closed, clean, and dry area.

Considerations

A notable issue with bougienage techniques is that they are dependent on an externally-applied axial force. If the stricture is too tight, the axial force applied may cause the dilator to bend laterally and rupture the esophagus, rather than push the dilator through the stricture. Bougies are of little benefit when the stricture is unyielding, is angulated, or causes severe luminal narrowing.

Polyvinyl chloride dilators

Strictures may be dilated by inserting a guidewire past the affected area and introducing a semiflexible, tapered, polyvinyl chloride dilator over the wire. Polyvinyl chloride dilators are easily passed through the mouth, and the flexible tip readily traverses the pharynx. There are two types of polyvinyl chloride dilators:

Savary-Gilliard dilators consist of a series of semiflexible, tapered, polyvinyl chloride dilators that have a lumen for the guidewire in the center. The dilators range in size from 5 to 20 mm (15 to 60 Fr). The dilators fit over a stainless-steel guidewire that may range from 200 to 360 cm in length. The wire has a graduated, flexible spring tip that reduces the risk of retroflexion (turning on itself) or causing luminal perforation. A radiopaque marker is incorporated into the dilator to aid in fluoroscopic monitoring.

American dilators are an adaptation of the Savary-Gilliard dilators. The distal tapered end of the dilator shaft is shorter, and the dilators are completely radiopaque. American dilators range in size from 5 to 18 mm (15 to 54 Fr).

Indications

Polyvinyl chloride dilators are for esophageal strictures with the following indications:

- Dilation of peptic, malignant, postsurgical, or postradiation strictures.
- Dilation of obstructing tumors to permit passage of an endoscope, through which laser therapy can begin at the distal tumor margin.
- Dilation of esophageal webs or rings, diffuse esophageal spasm, chemically-induced strictures, or scleroderma.
- Pediatric patients who may have the aforementioned indications.
- Placement of a stent.

Procedure

Guidewires should be introduced via the biopsy channel of the endoscope, with the endoscopist observing the route of the wire. The scope is removed, leaving the wire in place. Dilators are then inserted over the guidewire with or without fluoroscopic guidance.

Considerations

Potential complications of dilatation using polyvinyl chloride dilators include perforation, bleeding, aspiration, and bacteremia.

The polyvinyl chloride dilators should never be used as a bougie or without the assistance of the guidewire.

Hydrostatic balloon dilators

Hydrostatic balloon dilators are made of a specially treated polyethylene, which is altered to greatly increase strength and minimize stretching. A pressure gauge keeps the applied pressure within recommended limits. The balloon is filled with fluid such as water or diluted radiopaque dye. Radiopaque dye is always diluted, as full-strength contrast material crystallizes easily and occludes the lumen of the dilator, rendering it useless.

There are two types of hydrostatic balloon dilators. Both are inflated using the appropriate manufacturer's device.

- **Through-the-scope (TTS) balloons.** TTS balloons have an inflated diameter of 4-25 mm. They are passed through the biopsy channel of the endoscope. Dilation is accomplished under direct visualization.
- **Over-the-guidewire balloons.** These balloon dilators have an inflated diameter of 4-40 mm. A guidewire is passed using the endoscope, then the endoscope is removed, and the dilators are passed over the guidewire. Fluoroscopy should be used to visualize the dilators during the procedure.

Indications

Esophageal strictures that lend themselves to conventional bougienage should be treated in this manner because bougienage is less complicated, less time-consuming, and less expensive than balloon dilatation.

Hydrostatic balloon dilators are used for:

- Dilation of benign or malignant esophageal, pyloric, duodenal, rectal, or left-colon strictures.
- Pyloric stenosis. (This is not used for congenital pyloric stenosis seen in infants.)
- Biliary tract strictures.
- Food impactions above an esophageal stricture.
- Temporarily opening the rectosigmoid luminal pathway when it is obstructed by a tumor before endoscopic laser therapy. (For rectal cancers, it may be easier to use a rigid proctoscope and hand-held laser fiber.)

Procedure

Performing dilatation using a hydrostatic balloon dilator includes the following:

- Before any balloon dilatation procedure, the balloon should be inflated with water to check for leakage. Spraying the balloon with silicone may assist passage through the scope.
- Throughout the procedure, the gastroenterology nurse assists the endoscopist with guidewire control, when used.
- After the balloon is in place, a syringe pressure gauge assembly is used to inflate the balloon to a fixed pressure, according to the manufacturer guidelines or as directed by the endoscopist.
- X-ray contrast material may be used and followed by fluoroscopy to ensure placement and dilation.
- The balloon is then left inflated for a short period of time, as determined by the endoscopist. After the appropriate time has elapsed, the balloon is deflated.
- Patient discomfort may serve as a guide to the number of dilations attempted during a single session.
- With the use of three progressively larger inflatable balloons, esophageal peptic strictures can be dilated dramatically in one session.
- Once the balloon is removed, it should be checked for signs of bleeding.
- The schedule for subsequent dilatation sessions is based on the type of stricture, its response to initial and subsequent dilation, and the patient's tolerance of the procedure. Each stricture requires an individually tailored approach.

In the patient with a food impaction above an esophageal stricture, the dilator is passed and inflated, and the bolus is then pushed into the stomach. This procedure avoids the potentially greater complications of aspiration secondary to removal of the foreign object. Once the bolus is dislodged, the guidewire is passed through the stricture. For all patients who have a meat impaction above the stricture, an overtube is used.

Care and maintenance

After use, balloon dilators should be reprocessed and stored according to the manufacturer guidelines. If contrast material was used to inflate the balloon, it must be rinsed out completely before reprocessing. If contrast material is not removed, the balloon walls may adhere and destroy the balloon. Hydrostatic balloons can withstand temperatures up to 65 degrees C (149 degrees F), which allows for sterilization with ethylene oxide.

Considerations

Potential complications of dilatation using a hydrostatic balloon dilator are the same as those of other dilatation procedures.

Hydrostatic balloon dilatation of pyloric stenosis

In patients with pyloric stenosis, a through-the-scope (TTS) or over-the-guidewire technique may be used to insert the balloon.

- In the TTS technique, a well-lubricated balloon is passed through the biopsy channel of an endoscope and into the stricture. The balloon is inflated to maximum pressure with water or diluted contrast medium using a pressure gauge. Dilation is repeated three or four times, maintaining maximum inflation for at least 1 minute during each dilation. After dilation is complete, the balloon is withdrawn, and the endoscope is passed through the pyloric ring for inspection of the duodenum.
- For over-the-guidewire balloon dilation, a heavy-gauge guidewire with a spring tip is passed through the stricture and dilatation far enough to serve as an anchor. Dilatation may be carried out with fluoroscopic guidance, filling the balloon with dilute radiographic contrast material. Following dilatation, endoscopy is used to evaluate the pyloric ring and duodenal bulb.

Hydrostatic balloon dilatation of biliary tract strictures

Bile duct strictures may be dilated with hydrostatic balloon catheters during endoscopic retrograde cholangiopancreatography (ERCP) or percutaneous transhepatic cholangiography (PTC). After dilatation, a biliary stent may be placed through the stricture either endoscopically during ERCP or percutaneously during PTC.

For dilatation of the biliary tract, a no-profile dilating balloon is selected by the physician. Choice of the balloon is determined by the size and length of the strictured area. These balloons come in a variety of sizes measuring 2-4 cm in length, with a diameter of 4-10 mm. The guidewire is used in conjunction with a stent or nasobiliary tube to achieve correct placement of the balloon.

Indications for balloon dilatation of the biliary tract include:

- Inflammatory strictures.
- Postoperative strictures, especially after liver transplantation.
- Benign or malignant strictures.
- Stenosis at a choledochoenterostomy site.
- Sclerosing cholangitis.
- Sphincter of Oddi dysfunction and biliary dyskinesia.
- Before stent placement.

Before dilatation, a diagnostic ERCP is performed to determine the size, location, and anatomy of the stricture. Once the ERCP is completed:

- The ERCP cannulating device is flushed with saline to clear the contrast media.
- A guidewire that is compatible with the inner diameter of the cannulating device is placed into the cannulating device and advanced into the common bile duct.
- The cannulating device is then removed, leaving the guidewire in place.
- With the guidewire in place, the dilating balloon is sprayed with silicone and then fed over the guidewire, through the biopsy channel of the scope, and into the common bile duct.
- The gastroenterology nurse keeps tension on the guidewire to make passage easier. The physician will position the balloon over the center of the strictured area.
- A pressure gauge is attached to the interlocking connecter end of the balloon. These gauges typically show pressures from 0 to 300 pounds per square inch (psi) or 0 to 20 atmospheres (atm). The maximum amount of pressure for each balloon should be listed on the balloon or the package, and this maximum pressure should not be exceeded during the dilatation.
- Under fluoroscopy using a syringe with dilute contrast medium, the balloon is inflated to the correct pressure and held for 1 minute. When fully inflated, the balloon may show a waist-shape at the site of the stricture.
- Following dilatation, the balloon is deflated, withdrawn from the scope, and discarded.
- Repeated dilatations with increasingly larger balloons may be necessary to achieve the desired patient outcome.
- In patients with common bile duct strictures, a stent may be placed across the stricture for up to 3 months to promote re-epithelialization of the duct lumen and prevent recurrent strictures.
- At the end of the procedure, document the balloon sizes, maximum inflation pressures, and lengths of inflation time according to institutional policy.

Following balloon dilation, a clinical improvement in jaundice and a drop in serum bilirubin should be observed. Use of dilating balloons for temporary dilation, along with the placement of a long-term stent, produces the best patient outcomes.

Complications include:

- Pancreatitis and injury to the bile duct. Most cases of pancreatitis respond to conservative treatment.
- Laceration of the bile duct, which often resolves spontaneously when the lacerated area is bridged by a stent.

Pneumatic balloon dilators

Pneumatic balloon dilators are air-filled, cylinder-shaped balloons used for forceful dilation. The most common use is for achalasia.

Indications

Pneumatic dilatation may be used for:

- Achalasia. Pneumatic balloons are used for forceful stretching of the LES in patients with achalasia.

Forceful dilation to a diameter of approximately 3 cm is necessary to tear the circular muscle, weakening the LES and producing a lasting reduction in LES pressure. The success rate in treatment of achalasia with pneumatic dilators is 70% to 80% (Jayasekeran, Holt, & Bourke, 2013).

- Diffuse esophageal spasm with a hypertensive LES. The object of this procedure is to decrease the resistance at the LES to allow the esophagus to empty.
- Dilation of lower esophageal rings, if bougienage is unsuccessful. One successful dilation should cure this condition permanently. Although the ring may still be visible on radiographic examination, the patient's dysphagia should not recur.

Procedure

The patient may be hospitalized or be an outpatient for the procedure. A liquid diet should be instituted before dilatation per physician and anesthesia guidelines.

Before the procedure:

- Instruct the patient that moderate to severe discomfort may be experienced, with reassurance that additional pain medication will be administered if necessary.
- The procedure is done under MAC or conscious sedation.
- Diazepam (Valium) or midazolam may be administered for conscious sedation.
- If anesthesia is administering care, propofol (Diprivan) is generally used to provide sedation, and the patient may given a narcotic to assist with pain relief.
- The esophagus should be lavaged and emptied of retained material, if present.
- The pressure gauge is set according to the manufacturer guidelines.

Nursing considerations:

- The collapsed pneumatic balloon is introduced over a guidewire, and the procedure is performed under fluoroscopic control.
- When the center of the balloon is positioned at the level of the LES, the balloon bag is inflated rapidly with air to a preset pressure (at least 6 psi).
- After the length of time determined by the endoscopist, usually 15-60 seconds, the balloon is deflated and removed.
- The dilator is checked for evidence of blood.
- During the procedure, the patient is observed for severe chest pain, and additional medication is administered as ordered.
- After the procedure, the head of the patient's bed is elevated.
- Immediately notify the physician if the patient experiences chest pain, fever, regurgitation of blood, pain on swallowing, back pain, shortness of breath, shoulder pain, or chills.

Some physicians immediately follow dilatation with a radiographic contrast study to identify any distal esophageal leaks near the region of the esophagogastric junction. If no leak is seen, the patient is observed over the subsequent 6 hours and may gradually resume a usual diet.

Considerations

Because of the importance of monitoring the patient for the existence, the severity, and the character of chest pain, the physician may prefer inpatient observation for 24 hours after the procedure. In addition, the physician may want the patient to remain on a clear liquid diet for 24 hours after the procedure.

Potential complications of pneumatic balloon dilation include perforation, bleeding, and aspiration. Risk of perforation is somewhat higher than in standard dilation procedures, occurring in 1% to 4% of patients (Elhanafi et al., 2013).

Patients with small perforations and contrast material extending beyond the normal esophageal lumen seen on a follow-up radiographic contrast study may be managed conservatively with antibiotics and close observation for signs of worsening pain and fever. Clinical deterioration or the presence of free-flowing contrast material into the mediastinum mandates immediate thoracotomy and repair. If the tear is small, the repair and a Heller myotomy may be performed in the same operation.

CASE SITUATION

Mrs. Doris Johnson, whose case was presented in Chapter 13 and Chapter 24, was diagnosed with esophageal reflux 2 years ago. Her husband died 18 months ago, and she was unable to comply with her treatment regimen during the grieving period. Emotional recovery has been slow, and she has not returned to the prescribed therapy. Now she is experiencing dysphagia in reaction to solid food but not to liquids. She is scheduled to have an esophageal endoscopy and possible dilatation.

Points to think about

1. Given Doris's history and symptoms, what might the gastroenterology nurse expect to find on endoscopic examination?
2. Based on these expectations, what equipment should the gastroenterology nurse have ready?
3. The gastroenterology nurse discovers that Doris has a knowledge deficit in relation to her current diagnosis and treatment. What should the gastroenterology nurse review with her?
4. Doris might be considered noncompliant in relation to her esophageal reflux medical regimen. What are the ethical issues to be considered in labeling a patient as noncompliant?
5. In the nursing diagnosis "noncompliance in relation to esophageal reflux medical recommenda-

tions," what one factor is inherent in the definition and must be known before an individual can be labeled as noncompliant?

6. What must be assessed before a gastroenterology nurse can assist Doris in overcoming noncompliance?
7. What might the gastroenterology nurse tell Doris regarding post- dilatation follow-up?

Suggested responses

1. The endoscopic findings demonstrate a stricture of the distal esophagus. This type of stricture may or may not permit passage of the endoscope.
2. The equipment the gastroenterology nurse should have available for the dilatation procedure to be performed on Doris includes:
   a. Biopsy forceps and cytology brush. All strictures should be diagnosed histologically to rule out a malignancy or Barrett's esophagus, which is the presence of gastric mucosa above the squamocolumnar junction.
   b. Dilators. Depending on the type of stricture and/or the endoscopist's preference, the gastroenterology nurse may need hydrostatic TTS balloon dilators, polyvinyl chloride dilators (Savary-Gilliard dilators), and/or tungsten-filled bougies (Maloney dilators).
3. Information the gastroenterology nurse should discuss with Doris includes:
   a. The current diagnosis and ways the problem can be alleviated so the gastroenterology nurse can ascertain Doris's recall of what her physician has told her. In addition, the gastroenterology nurse should answer questions or refer her to the physician for more information.
   b. The physiologic process of reflux and ways to manage it, including not eating late at night, not lying down after meals, not wearing constricting clothing, and elevating the head of the bed on 6-inch blocks.
   c. The need to reduce weight, which will help with managing reflux. The gastroenterology nurse should refer Doris for counseling in this area.
   d. Dietary and medication restrictions, such as eliminating caffeine, alcohol, acetylsalicylic acid (ASA), and nonsteroidal anti-inflammatory drugs (NSAIDs).
4. The label "noncompliant" can carry a judgmental connotation and place blame on the patient for failing to comply with a therapeutic recommendation. Health care providers must be able to identify all of the variables or factors that contribute to or interfere with a person's ability to comply with recommendations before labeling a patient as noncompliant.
5. The definition of noncompliance involves the expressed desire and intent not to adhere to therapeutic recommendations. Without this expression, a gastroenterology nurse cannot apply this type of label. It is important to identify not only the variables for nonadherence but also the degree to which compliant behavior has occurred.
6. Before attempting to assist Doris in helping her comply with medical recommendations, the gastroenterology nurse should assess the following:
   a. Defining characteristics of noncompliance, such as observation of noncompliant behavior or results of objective measures that reveal noncompliant behavior (e.g., physiological measures, development of complications, or increase in symptoms).
   b. Personal factors, which include values and beliefs about health, illness, consequences, and the prescribed therapy.
   c. Interpersonal factors, such as support from others and satisfaction gained from it.
   d. Environmental factors, including barriers to compliance in the environment in which the patient must exist (e.g., economic difficulties).
7. Instructions for follow-up after esophageal dilatation might include:
   a. Symptoms of complications such as increased chest pain, vomiting of blood, difficulty breathing, scapular pain, and fever.
   b. Dietary modifications needed (e.g., eat slowly and chew food well).
   c. Prescribed medications.
   d. Required follow-up visits.

REVIEW TERMS

American dilator, bougienage, bougies, dilators, French units, Hurst dilators, hydrostatic balloon dilators, Maloney dilators, pneumatic balloon dilators, Savary-Gilliard dilators, through the scope (TTS) balloon dilators, polyvinyl chloride dilators, radial force, shearing force, over-the-guidewire balloon dilators

REVIEW QUESTIONS

1. The primary goal of most esophageal dilatation procedures is to:
   a. Permit passage of the endoscope.
   b. Allow the patient to eat and drink normally.
   c. Treat achalasia.
   d. Cure esophageal cancer.
2. Potential complications of esophageal dilation include all of the following except:
   a. Perforation.
   b. Aspiration.
   c. Diarrhea.
   d. Bleeding.

3. A tungsten-filled silicone bougie with a tapered tip is called a:
   a. Maloney dilator.
   b. Hurst dilator.
   c. Balloon dilator.
   d. Savary-Gilliard dilator.
4. After bougienage with local anesthesia, the patient should remain NPO:
   a. For 2 hours.
   b. Until the gag reflex returns.
   c. Until the possibility of perforation has been ruled out.
   d. For 8 hours.
5. Indications for bougienage include all of the following except:
   a. Chemically-induced strictures.
   b. Esophageal rings or webs.
   c. Bile duct strictures.
   d. Esophageal strictures.
6. The method used most often for dilatation of pyloric stenosis is:
   a. A polyvinyl chloride dilator.
   b. A pneumatic balloon.
   c. A hydrostatic balloon.
   d. A silicone bougie.
7. A hydrostatic dilating balloon is filled with dilute contrast media because:
   a. Full-strength contrast medium could crystallize, damaging the balloon.
   b. Full-strength contrast medium may cause an allergic reaction.
   c. It helps stimulate peristalsis, thus aiding passage of the balloon through the upper GI tract.
   d. It exerts more pressure than water or saline.
8. Indication for hydrostatic balloons of the biliary tract include all of the following except:
   a. Inflammatory strictures.
   b. Postoperative strictures, especially after liver transplantation.
   c. Sphincter of Oddi dysfunction.
   d. Diffuse esophageal spasm with a hypertensive LES.
9. In pneumatic dilatation, the balloon remains inflated:
   a. For up to 1 minute.
   b. For up to 10 minutes.
   c. Until the patient experiences chest pain.
   d. Until the preset pressure is obtained.

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Chapter 28

HEMOSTASIS AND TUMOR ABLATION

This chapter will acquaint the gastroenterology nurse with endoscopic techniques that are used to stop gastrointestinal (GI) bleeding and treat tumors throughout the GI tract. The indications, contraindications, techniques, and potential complications of electrosurgery, heater probes, laser therapy, argon plasma coagulation, hemostatic grasping forceps, photodynamic therapy, brachytherapy, endoscopic mucosal resection, endoscopic endoluminal radiofrequency ablation, injection therapy, endoscopic variceal ligation, endoscopic clipping, hemostatic powder, esophagogastric tamponade, endoscopic cryotherapy, and confocal laser endomicroscopy are all described in detail. Nursing considerations for managing patients undergoing these procedures are outlined as they relate to the role of the gastroenterology nurse.

Learning objectives

After reviewing the content of this chapter, the gastroenterology nurse will be able to:

1. Discuss the general principles of endoscopic hemostasis.
2. Explain indications, contraindications, and techniques to control bleeding in the GI tract.
3. Discuss therapeutic treatments and endoscopic procedures for GI bleeding, tumors, and lesions.
4. Relate nursing considerations for managing patients undergoing these procedures.

BASIC PRINCIPLES OF ENDOSCOPIC HEMOSTASIS

In planning endoscopic therapy for GI bleeding, the location of the hemorrhage (upper or lower GI tract) must be confirmed, and other non-GI causes such as epistaxis or hemoptysis must be excluded.

The most common causes of bleeding in the GI tract are:

- Esophageal, gastric, duodenal, and colonic ulcers.
- Erosive esophagitis, gastritis, and duodenitis.
- Mallory-Weiss tears.
- Varices.
- Tumors.
- Arteriovenous malformations (AVMs).
- Colitis.

Most of the causes of upper GI bleeding and some of the causes of lower GI bleeding are amenable to endoscopic therapy. Although approximately 70% to 80% of upper GI bleeding episodes are self-limited, the inpatient mortality rate for patients with upper GI bleeding is 2% to 6% (Abougergi, 2015). Factors that increase this risk include the severity of bleeding, significant comorbid medical illness, age above 60 years, the need for surgery, and continued or recurrent bleeding.

At the time of diagnostic endoscopy, the physician and gastroenterology nurse should be prepared to treat the bleeding site with injection therapy, argon plasma coagulation, laser photocoagulation, electrosurgery with monopolar or bipolar accessories, radiofrequency ablation, hemostatic powder, and/or endoscopic clipping.

Several factors affect the timing of the endoscopic examination.

- The likelihood of finding the source of the bleeding is higher when the procedure is done within 24 hours of the bleed.
- Ongoing upper GI bleeding requires urgent endoscopy as soon as the patient is medically stable.
- For upper GI bleeding that is not thought to be active, endoscopy should be performed as soon as personnel and equipment become available.

- For active lower GI bleeding, colonoscopy should not be performed until the colon has been cleansed properly.

Causes of lower GI bleeding include rectal hemorrhoids, diverticulitis, polyps, cancer, AVMs, colitis, colonic ischemia, post-polypectomy bleeding, and hemorrhage from the upper intestinal tract (found in 15% of patients in whom the provisional diagnosis of lower GI bleed has been made) (Strate & Gralnek, 2016). Of these, endoscopic management is most often used for bleeding cancers, AVMs, and polyps.

Premedication for endoscopic hemostasis is similar to that administered for diagnostic endoscopy. The use of meperidine (Demerol) during procedural sedation has been shown to mask the detection of angiodysplasias because of a reduction in mucosal blood flow. The use of Demerol should be avoided if AVMs are suspected, or an opiate antagonist administration may be considered during careful examination of the colon. Because patients with major blood loss may be hypotensive, however, the blood pressure-lowering effects of commonly used narcotics and sedatives must be considered. Antiperistaltic agents such as glucagon may be administered to diminish motility when searching for or treating nonbleeding angiodysplasia.

A number of endoscopic methods are available to control GI hemorrhage, including:

- Monopolar and bipolar electrocautery.
- Heater probes.
- Endoscopic clipping.
- Radiofrequency ablation.
- Argon plasma coagulation.
- Laser therapy.

Variceal bleeding may be treated with:

- Injection sclerotherapy.
- Injection of hypertonic or epinephrine solutions.
- Esophageal variceal band ligation.
- Esophageal-gastric tamponade.

These endoscopic methods, as well as medications such as somatostatin and/or beta-blockers, are commonly used for the treatment of acute variceal bleeding.

Injection therapy is also used to treat bleeding ulcers.

The diagnosis and treatment of tumors within the GI tract has become the most dynamic area of gastroenterology. The development of endoscopic ultrasound (EUS), advances in new endoscopes—such as confocal endomicroscopy and high magnification—and new techniques in the treatment and palliation of tumors have changed patient management. The development of narrow-band imaging (NBI) and Fuji Intelligent Color Enhancement (FICE) chromoendoscopy allows the endoscopist the ability to alter the color band of the white light to examine the surface of the mucosa. The altered light wavelength enhances the surface contours and blood vessels, which allows for better examination of tissue.

ELECTROSURGERY

The use of an **electrosurgical unit (ESU)** or electrosurgical generator in gastroenterology is common in both the acute care setting and the outpatient setting.

- **Electrocautery** refers to a direct current where electrons flow in one direction and the current does not enter the patient's body.
- **Electrosurgery** uses an alternating current. The current enters the patient's body, and the patient is part of the circuit.

The terms electrocautery and electrosurgery are sometimes used interchangeably.

A good understanding of electricity is important in working with electrosurgery. The electrical circuit is the basis for understanding electrosurgery.

- A circuit is the complete path of an electrical current. A completed circuit must be present for the flow of electrons.
- The current is the flow of electrons during a given period of time and is measured in amperes.
- Voltage is the electromotive force that pushes the current through the resistance and is measured in volts.
- The resistance is the impedance or the obstacle to the flow of current.

The basic circuit used in gastroenterology is electricity that flows from the wall to the ESU through the active cord, via an accessory (snare), into the tissue, through the patient, and back to the ESU via a grounding pad. Remember that electricity will always seek the path of least resistance. Anything that can alter the pathway or change the flow of current must be identified (e.g., jewelry; gurney; neurostimulators; gastric stimulators; joint prosthesis; other implantable devices such as pacemakers or defibrillators).

An electrosurgical generator is needed for electrosurgery. Electrosurgery utilizes an alternating current. A current can alternate at different frequencies and is measured in cycles per second or hertz (Hz). The standard household electrical current from the wall cycles at 60 Hz. This electricity from the wall could be used to remove a polyp, but when a person contacts a 60-Hz current (as when sticking a finger in a wall socket), the tonic-clonic type movement of the body (Faradic effect) causes electrocution. The 60-cycle-per-second current causes neuromuscular stimulation and a most unpleasant sensation. The neuromuscular stimulation from an electrical current cease at about 100,000 cycles per second. The electrosurgical generator is designed to take the 60-Hz wall current and increase the frequency to more than 200,000 cycles per second, so that the current can pass through the patient without neuromuscular stimulation or risk of electrocution.

The other important feature of the electrosurgical generator is the ability to produce different types of waveforms. These waveforms provide the different tissue effects when cut or **coagulation** is selected.

- In a cut setting, the electrosurgical generator produces a waveform that is designed to create a spark. The thermal effect leads to rapid vaporization, and the cell membrane ruptures.
- In a coagulation setting, the electrosurgical generator produces a waveform in which the thermal effects lead to slow vaporization of cellular liquids and vascular occlusion. This causes coagulation of proteins, **fulguration**, **desiccation** and shrinkage of tissue, and devitalization of tumors and lesions.

When both **cutting waveform** and **coagulating waveform** are applied, the resulting energy is referred to as a "blended" current.

In each of the different settings, the manufacturer provides power curves or graphs that show how much energy is delivered at each power setting. This energy results in thermal damage to the tissue.

A number of different types of grounding pads are available. Grounding pads used today are disposable single-patient use. Disposable pads with gel and adhesive edges are preferred because they conform to patient contours and do not have to be repositioned if the patient's position is changed.

Grounding pads can be either single or split. A single grounding pad allows current to return to the electrosurgical generator. A split grounding pad is divided into two sections and increases reliability and patient safety by using impedance to alarm out and disable the generator when unsafe patient contact is recognized.

Correct application of the grounding pad includes:

- Verifying the expiration date prior to use.
- Not removing from the original package until immediately prior to use. Grounding pads that are removed from the package and left out risk having the gel dry out and decreasing its effectiveness, which may not allow for good patient contact when applied.
- Visually inspecting prior to placement on the patient.
- Applying according to the manufacturer's guidelines.
- Placing as close to the surgical site as possible.
- Placing on clean and dry skin over a large muscle mass.
- Having generous contact on healthy tissue. Excessive hair should be removed from the site of attachment, if necessary.
- Avoiding placement over bony prominences and large scars that could decrease skin area contact.
- Placing smoothly on the skin surface, avoiding tenting, gaps, or folds.
- Reassessing prior to each delivery of energy. The grounding pad can become dislodged or have poor skin contact as patients are repositioned during a procedure. In some cases, the patient may remove the grounding pad unintentionally.

Incorrect application of a grounding pad may result in "hot spots" where current density remains high because of limited skin contact. Spot burning can also occur. To avoid pacemaker malfunction and burns from internal metal prostheses, the grounding pad should be as far away as possible from any pacemaker, hip pin, or artificial joint and as close as possible to the site of use.

Electrosurgery may be applied in two ways:

1. **Monopolar modality.** Electrical current flows from a small, active electrode that is in contact with the target tissue, through the patient, and toward a grounding pad that is attached to the patient's skin. The site chosen should be muscular; well-vascularized; and not over bony prominences, joint prosthesis, or anywhere that circulation is likely to be impaired. The preferred site is the upper thigh. The high current density at the relatively small electrode site generates a significant amount of heat in the resistor (the target tissue), thereby causing coagulation. The current density at the grounding

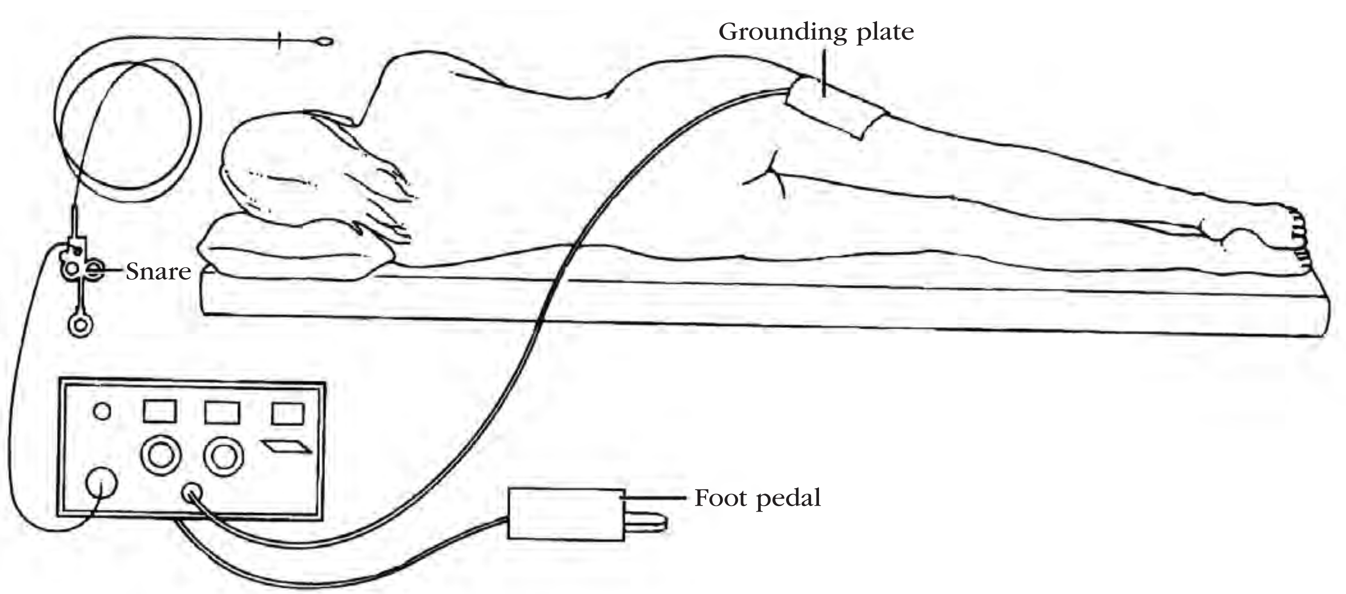

Figure 28.1. Patient Setup for Electrocautery

pad is lower because the amount of tissue in contact with the grounding plate is much larger.

2. **Bipolar modality.** The current flows between two small electrodes that are in contact with the tissue and are separated by a space of only a few millimeters. The localized current pathway between the two electrodes results in tissue heating. The bipolar modality does not require a grounding pad.

Monopolar electrosurgery

There are certain safety factors associated with the use of the grounding pad in monopolar electrocoagulation. Before the procedure, the gastroenterology nurse should explain the purpose of the grounding pad to the patient. The patient should be instructed not to touch side rails, intravenous (IV) poles, or other metal objects during the procedure (Figure 28.1).

With the use of high-frequency currents, current leakage can be a problem, and the following may occur:

- Current may pass through the accessory, leak through the endoscope, and pass back to the endoscopist, causing burns to the operator.
- Current may pass through the accessory, leak through the scope, and pass to the patient at an internal point in which the patient is in contact with the scope, and then continue to the grounding pad, creating the potential for a burn at a scope-patient contact point.
- Current may pass through the accessory, through the tissue at the site of electrosurgery, and through the patient's body to a part that is in contact with a grounded piece of metal (e.g., table edges, Mayo stands, IV poles).

Any personnel in contact with equipment or the patient during use of the generator can become part of the electrical circuit.

To avoid these problems, the gastroenterology nurse should take the following steps:

- Securely attach the grounding pad to the patient.
- Wear gloves to decrease conduction and contact with the scope.
- Ensure that a nonconductive material covers the eyepiece if a fiberoptic instrument is being used. Videoendoscopy has replaced fiberoptics in many departments.
- Avoid touching the patient during activation of the ESU.
- Place metal objects far enough away from the X-ray table or examination table to prevent accidental patient contact.
- Prevent patient contact with metal railings or edges of the gurney or exam table.
- Attach a grounding cable to the endoscope to decrease the 60-cycle interference.

Bipolar electrocoagulation

A bipolar electrode is a specialized, hemostatic probe that is inserted through an endoscope channel to control GI hemorrhage and bleeding. The bipolar probe does not require a grounding pad because the current travels back through the bipolar electrode. Because there are two electrodes, coagulation takes place only between the two points. One point delivers the current and the other closes the circuit returning to the generator. Electrical energy is converted into thermal energy on contact with the tissue, producing a more predictable depth of injury than with monopolar therapy.

Features of a bipolar cautery device include the following:

- The tip of the bipolar probe consists of an array of longitudinal or circumferential microelectrodes that are wrapped around a porcelain tip. The porcelain acts as the ground and needs to be intact to complete the circuit. It is, therefore, important to inspect all bipolar accessories for intact circuits, bends or breaks in the catheter, and correct connectors.
- The probes are 7 French (Fr) and 10 Fr in size. The 10 Fr probe can only be used with a large-channel therapeutic endoscopic retrograde cholangiopancreatography (ERCP) scope, therapeutic gastroscope, or colonoscope.
- Through a hole in the center of the tip, a powerful water jet can be delivered intermittently or constantly to irrigate the bleeding lesion and to increase the precision of targeting. The use of an irrigation pump is better than hand irrigation.
- Coagulation is possible with the top and sides of the probe tip. Thermal energy can be delivered from the sides of the probes.
- Bipolar products with injection needles in the same device are available.
- Bipolar probes can **tamponade** a vessel before delivering thermal energy

Bipolar electrocoagulation has certain advantages over the monopolar modality:

- Depth of tissue penetration is limited, thus reducing the risk of perforation and avoiding full-thickness burns. There is a rapidly decreasing energy and heating effect at the short distance from the electrodes.
- Studies have shown that, although bipolar electrocoagulation is technically more difficult to use, it produces less mucosal injury and is equally effective as monopolar electrocoagulation.

The use of the bipolar probe is indicated:

- For actively bleeding lesions in the upper or lower GI tract.
- To destroy neoplastic cells and relieve symptoms of obstruction.
- To destroy hemorrhoidal tissue.

The use of a bipolar probe is contraindicated:

- In patients who are unwilling or unable to cooperate.
- In cases of massive hemorrhage that require immediate surgery.
- Where visualization of the bleeding site is inadequate.
- When free peritoneal air is observed on X-ray films.

Prior to the procedure using bipolar electrocautery, the gastroenterology nurse should:

- Fill the water bottle with sterile water, and connect the bottle to the port on the machine.
- Connect the probe and foot pedal to the unit.
- Prime the probe with water.
- Test the probe and unit to ensure proper function. Place several drops of normal saline on a glass slide and put the tip of the probe in the saline, then depress the bipolar pedal. If the probe is working properly, the saline will heat and bubbles will form. Following a successful test, the unit should be turned off until needed during the procedure.
- Lubricate the probe with silicone to facilitate smooth passage through the endoscope, as well as to prevent tissue from adhering to the probe.

For application of a bipolar probe, the electrode is placed within 2-3 mm of the vessel, and firm pressure is applied. The probe is activated in 1- to 2-second bursts until circumferential coagulation has occurred.

During the procedure, the nurse:

- Maintains the water level in the water bottle.
- Sets the energy levels and water pressure designated by the physician.
- Wipes secretions from the probe with gauze as it is withdrawn from the endoscope.
- Cleans the probe tip as needed during the procedure.

After the procedure:

- The bipolar cautery unit is cleaned and disassembled, and the water is completely flushed from the unit.
- The water bottle is emptied and sterilized in accordance with organizational policy.
- Probes are maintained and processed according to the manufacturer's instructions.

A disposable water pump tubing that attaches to a sterile water bottle is available.

Potential complications of bipolar electrocoagulation include perforation, delayed hemorrhage, and deep ulcerations.

Detachable loop ligating device

Polypectomy is a common procedure in endoscopy. In certain cases, the use of a **detachable loop ligating device** prior to polypectomy can reduce the risk of bleeding. Many studies have shown that ligation of the polyp stalk, followed by polypectomy, is safer than conventional polypectomy.

During the procedure:

- The ligating loop looks similar to a snare and is placed over the stalk of the polyp.
- The loop is slowly closed until the blood supply to the polyp is decreased or occluded.
- The ligating loop is then released from the device handle.
- The polypectomy snare is place over the polyp but above the ligating loop on the polyp stalk.
- The polyp is removed, and the ligating loop remains in place until the polyp stalk remnant is reabsorbed and the ligating loop is passed in the stool.

If the ligating loop is partially deployed or deployed in the wrong area, a special ligating loop cutter is available. The detachable loop ligating devices are available in both a single-use and reusable version.

Indications

Electrosurgery is used:

- To treat active bleeding from a visible vessel, visible vessels without bleeding, or blood oozing from a clot in the base of an ulcer.
- In conjunction with the excision of polyps, large mucosal biopsies, and endoscopic retrograde sphincterotomy. (Polypectomy and sphincterotomy are discussed in detail in Chapter 30.)
- For treatment of hereditary hemorrhagic telangiectasia (Rendu-Osler-Weber syndrome) of the GI tract.
- For treatment of acquired vascular abnormalities of the stomach, duodenum, or cecal area (angiodysplasia).
- In the treatment of AVMs in the stomach and duodenum. **Electrocoagulation** is achieved by placing an electrode on the peripheral border of the malformation, applying moderate pressure, and moving circumferentially until the AVM has been completely encircled. If the center of an AVM is electrocoagulated before its periphery has been treated, torrential bleeding may occur.

Electrosurgery is contraindicated in patients with excessive bleeding, esophageal varices, or coagulopathy. It may be performed safely with the majority of newer-model pacemakers, but a case-by-case determination must be made before the procedure. The patient's pacemaker and defibrillator may need to be deactivated using a magnet prior to the use of cautery. The patient's cardiovascular status must be managed during the procedure using other methods while the pacemaker and defibrillator are deactivated. The current should be applied intermittently and for the shortest time possible. A bipolar accessory should be considered whenever possible.

Procedure

Prior to the procedure

Before attempting electrosurgery in the upper GI tract, the stomach must be relatively free of blood, and the patient's condition should be stabilized. The risk for aspiration is always increased, and an overtube may be used in certain situations. Oral suction and proper positioning of the patient is essential throughout the procedure. An oral bite block should be placed, and sedation and analgesia should be achieved as appropriate. Diagnostic esophagogastroduodenoscopy (EGD), using a therapeutic scope, if available, should be performed to determine the cause of the bleeding.

Before electrosurgery of the colon or rectum, it is important that the colon be well-prepped to eliminate the risk of igniting any hydrogen and methane gases that are normally present in the stool. Because the colon wall is thinner than the upper GI tract, the risk of perforation is greater, particularly if the lumen is overly distended.

Both before and during electrosurgical procedures, the gastroenterology nurse is responsible to:

- Inspect all equipment and components to make sure they work properly and are intact.
- Assess the patient and surroundings for possible alternative paths for the current to travel. Remove metal jewelry, assess for pacemakers and other implanted electrical devices, and visually assess the surroundings for other points of contact with the patient (e.g., gurney, other members of the team).
- Check the expiration date on the grounding pad. Remove the grounding pad from the original packaging only when ready for placement. These steps will ensure the grounding pad gel is moist and not dried out.
- Select an appropriate grounding pad site:
- Choose a well-vascularized location near the surgical site.
- Avoid scar tissue, adipose tissue, metallic implants, and other implantable devices.
- Place the grounding pad according to the manufacturer's recommendations. Position the grounding pad on the patient—for monopolar electrosurgery and argon plasma coagulation (APC)—and document placement and skin condition.
- Validate physician/manufacturer recommendations for implanted cardiac devices and the use of electrosurgical generators.

During the procedure

- Check the placement of the foot pedal.
- Turn on the power when the physician is ready to use the ESU, and turn it off immediately after use. Another option is to disconnect the active cord from the accessory. Without a completed circuit, the accessory is not hot.
- Verbally confirm the physician's orders for mode of operation (coagulation, cutting, or blended current) and energy settings.
- Verify settings with each use of the ESU.
- Reassure and encourage the patient to stay as still as possible.

After the procedure

After the procedure is completed, the gastroenterology nurse is responsible to:

- Check for skin damage or burns near and under the grounding pad (when using the monopolar modality).
- Monitor the patient's vital signs and document results.
- Monitor the patient for abdominal pain and/or distention.
- Document the power settings, the status of the grounding pad site, and the model number and serial number of the ESU.

Care and maintenance

Electrosurgical units are all different. The gastroenterology nurse must thoroughly understand the tissue effects caused by the use of each unit. Each ESU has diagrams called power curves or charts that show how much energy or thermal damage is delivered at each power setting. If the ESU has the ability to change waveforms, each waveform will have a separate energy curve. These are useful when more energy delivery is required but not necessarily by increasing the power.

All electrosurgical equipment should be inspected regularly for frayed or cracked cords, loose connections, or deterioration in the power cord and plug. Twisting and tightly wrapping cords should be avoided, as this causes increased resistance and results in damaging the insulation and the wires. A working backup for the ESU should be readily available. Unit functioning should be checked and documented before each use.

Considerations

Potential complications of electrosurgery include thermal injury, hemorrhage, perforation, transmural burns, and explosion.

HEATER PROBES

The **heater probe** is very similar in application to the bipolar probe. It consists of a hollow aluminum cylinder with an inner heat coil and an outer Teflon coating. The aluminum has high thermal conductivity, which provides for a precise and uniform distribution of heat to tissue from its end or sides. A coaxial channel is provided to wash away blood and debris. Heater probes are available with diameters of 3.2 or 2.4 mm. The 3.2-mm size must be used with a large-channel endoscope.

Indications

In addition to controlling GI bleeding, heater probes have been used successfully to treat patients with hemorrhoids.

Procedure

Steps in the application of the heater probe unit and bipolar unit are very similar, with the exception of testing. By submerging the tip in water and depressing the pedal labeled "Coag," the heater probe can be tested to determine if the unit is functioning properly. Should a malfunction occur, an alarm will sound and a fault light will be activated.

The heater probe is applied directly to a vessel with firm pressure. Several brief applications of thermal energy may be required to produce hemostasis.

Care and maintenance

The attributes of the heater probe are its portability, coaxial channel for application of a water jet, capability for controlling and presetting the rate of pulses, effectiveness, low cost, and capability for use at angles other than directly vertical. Like the bipolar probe, the heater probe also permits direct tamponade of bleeding sites. Moreover, the tip of the probe is covered with Teflon, which prevents sticking and eliminates the need for frequent removal of the probe to wipe off adherent blood and tissue.

Considerations

A disadvantage of the heater probe is that delivery of the maximum amount of thermal energy requires up to 8 seconds, during which time contact between the probe and the bleeding point must be maintained. Furthermore, the probe must be allowed to cool before it is withdrawn through the scope. If it is still hot, it can melt the channel lining.

LASER THERAPY

The word "**laser**" is an acronym for light amplification by stimulated emission of radiation. Because laser light is coherent and collimated, it can be intensely focused, which allows it to be precisely aimed. Argon, neodymium:yttrium-aluminum-garnet (**Nd:YAG**), and carbon dioxide lasers have been widely used in endoscopy.

In endoscopic applications, laser light energy is transmitted through a flexible quartz waveguide, which is usually protected by a plastic catheter passed through a flexible endoscope. Between the fiber and the catheter is free space through which coaxial air, carbon dioxide, or helium flows, keeping the fiber and the surface of the treatment site clear of blood and other debris. Because this coaxial gas can lead to problems with overdistention, a two-channel endoscope is preferred so that the gas can be exhausted through the suction channel. More than 95% of the lasers used for GI endoscopy procedures are Nd:YAG lasers.

Table 28.1. Comparison of Nd:YAG and Argon Lasers

Nd:YAG Laser	Argon Laser
Not absorbed by hemoglobin and will penetrate clots	Absorbed by hemoglobin so will not penetrate clots
Penetration depth approximately 4 mm	Penetration depth approximately 1 mm
Beam is not visible and requires an additional xenon or helium-neon aiming beam	Beam is visible

The Nd:YAG laser carries the greater risk of potential damage to the eye of the examiner, observer, or patient.

The temperature generated at the treatment site determines the effect of the laser on tissue. Generally speaking, protein coagulates at 60° celcius (C) (photocoagulation), and tissue vaporizes at 100° C (photovaporization).

- **Photocoagulation** creates a white, blanched appearance with edema. The coagulative effect of lasers allows them to be used to achieve hemostasis for acute GI bleeding and to treat GI lesions that are not actively bleeding (e.g., angiodysplasia or ulcers with visible vessels).
- **Photovaporization** may cause a divot, charring of tissue, and smoke. The photovaporization effect of lasers allows them to destroy neoplastic tissue and to cut through normal tissue to achieve therapeutic goals.

Conventional laser therapy differs from heater probe and electrocoagulation treatment in that it is a noncontact method. The laser waveguide does not come into contact with the tissue. Sapphire endoprobes, however, permit the laser to be used as a contact device. These tips, which attach to the tip of the standard laser waveguide, serve as lenses to concentrate the energy at the tip of the waveguide so much lower wattages are required.

Indications

Indications for laser treatment include:

- Hemorrhagic conditions of the GI tract, such as Mallory-Weiss tears, bleeding peptic ulcers, angiodysplasia, and Osler-Weber-Rendu syndrome.
- Stigmata of recent hemorrhage, including active bleeding, a visible vessel, or a fresh clot.
- Laser photovaporization for neoplastic disease, benign esophageal webs and anastomotic strictures, intrahepatic and extrahepatic biliary obstruction, and gallstones.
- In the colorectal area, for benign pedunculated polyps or sessile lesions, tumor vaporization, and photocoagulation of hemorrhoids.

Endoscopic laser therapy is contraindicated in patients who are unwilling or unable to cooperate or in patients with coagulopathy, extremely large vessels in the field, or inaccessible lesions.

Procedure

Prior to the procedure

Before laser therapy, the gastroenterology nurse should:

- Check the power emission from the laser probe.
- Ensure the water-cooling system is in operation.
- Check the power emission from the laser probe.
- Place "laser in use" warning signs at all doors.
- Note that a smoke evacuator may be beneficial.
- Provide the patient with IV sedation and analgesia, if ordered.
- Place the patient in the left lateral decubitus position.

- Provide everyone in the room, including the patient, with safety glasses or goggles. Endoscopists may use goggles or, if using a fiberoptic endoscope, may rely on the protective ocular lens cover once the laser is inserted.
- Ensure all personnel wear laser masks to protect against smoke and possible aerosolization of tissue particles.

During the procedure

During the procedure, the gastroenterology nurse should:

- Set power and duration as ordered by the physician.
- Maintain the laser on standby mode when it is not in the firing position.
- Clean the tip of the fiber frequently with hydrogen peroxide and a soft-bristled brush. Removal of the fiber from the biopsy channel allows an excellent opportunity to remove excess smoke and debris.
- Assist the physician with use of biopsy forceps for possible debridement of the treatment site.
- Monitor the patient during the procedure for abdominal distention and possible vasovagal reaction.
- Observe the patient's pain level so additional IV moderate sedation can be administered as necessary.

After the procedure

Following laser treatment:

- Keep the patient NPO until the physician's orders permit resumption of oral intake.
- Check vital signs as per organizational guidelines for care of patients receiving IV sedation and analgesia.

Care and maintenance

After the procedure is completed, reprocess fibers according to the manufacturer's instructions.

- Wipe the entire exterior of a reusable fiber, including the brass tip, with a gauze pad saturated with hydrogen peroxide. Some laser equipment is disposable, so it is important to refer to the manufacturer's instructions.
- Clean the interior by lavaging with hydrogen peroxide in a syringe without a needle, while a constant flow of carbon dioxide is maintained. The flow of carbon dioxide prevents the peroxide from traveling more than a few centimeters up the fiber.
- Remove the peroxide by depressing the foot pedal with the laser lamps turned off to produce a burst of carbon dioxide gas.

Flexible endoscopes may be damaged if laser energy is reflected from the target surface to the tip of the endoscope. This can be avoided with the following preventive measures:

- Use endoscopes that are designed for laser therapy and are manufactured with stainless-steel or white porcelain reflective tips rather than black tips.
- Make sure that the laser fiber is well outside the endoscope channel and clearly visualized before activating the laser.
- Do not work too closely to the target surface during extensive coagulation.

Considerations

Swallowing is often worse immediately after laser therapy of the esophagus because of edema. If there is odynophagia, antacids may relieve symptoms. If ordered by the physician, a nasogastric tube may be inserted to relieve abdominal distention. Narcotic analgesics may be necessary to relieve chest pain. Chest X-ray films and/or abdominal films may be ordered to exclude the possibility of perforation. It is not uncommon for patients to experience a low-grade fever and mild leukocytosis 12-36 hours after laser therapy.

To obtain adequate patency of a tumor stricture with laser therapy, several treatments may be required. After the final treatment, a water-soluble X-ray contrast swallow may be obtained to rule out perforation and to document the effects of therapy. The first follow-up endoscopy is usually carried out in 3-4 weeks.

Potential complications of endoscopic laser treatments include hemorrhage, tissue slough, perforation, gaseous abdominal distention, ulceration, delayed healing, and fistula or stricture formation. If perforation and/or bleeding are suspected, they should be treated as described in Chapter 31. Complications from endoscopic laser therapy increase with the amount of energy delivered and the length of the procedure. Minor increases in bleeding during laser therapy are relatively common, but laser-induced massive GI bleeding is also possible and may require emergency surgical intervention.

Laser treatment of upper gastrointestinal bleeding

Before laser photocoagulation of upper GI bleeding, a large-bore tube is used for thorough lavage of the stomach. A snare or grasper should be available for transecting and removing large adherent clots. A syringe irrigator may be used to remove smaller clots. A blood pump is necessary to allow rapid blood infusion if needed.

For the treatment of bleeding ulcers, the laser fiber is held 1-2 cm from the target, and pulses of high power and short duration are used for coagulation. If a vessel is visible, the beam is aimed circumferentially around it. When visible edema develops, the vessel itself is treated.

Laser therapy for vascular abnormalities

Endoscopic obliteration and clinical benefit have been reported in patients with angiodysplasia after use of both the argon and the Nd:YAG laser. When the laser beam strikes a large, abnormal vessel, it may initially cause a dormant lesion to bleed, but further laser therapy can bring this hemorrhage under control. Although angiodysplastic lesions may be seen in both the stomach and intestines, gastric and proximal duodenal lesions are responsible for the majority of upper GI bleeding.

Angiodysplasia of the colon, particularly the cecum, has also been treated with argon and Nd:YAG laser therapy. Results are best when it has been established that the colonic lesion has bled and the lesion is not just an incidental finding. Risk of perforation is greater in the right colon because it is thinner than other areas of the GI tract.

In patients with Rendu-Osler-Weber syndrome, laser therapy may reduce treatment time when many lesions are present. Laser treatment has also been shown to significantly reduce the potential for rebleeding and requirement for transfusions.

Laser coagulation of hemorrhoids

Nd:YAG laser photocoagulation and obliteration is a viable alternative to surgical removal of internal and external hemorrhoids. This technique delivers extensive heat therapy directly to the hemorrhoidal structures. It attempts to both fix the mucosa and remove some or all of the internal hemorrhoidal tissue. Compared with a surgical approach, laser therapy requires only local, rather than general, anesthesia; there is less trauma; and laser therapy causes less pain.

The infrared photocoagulator is another method used for the treatment of hemorrhoids. This device is not a laser. It focuses infrared radiation on the tissue via a specially made polymer tip. The source is a low-voltage tungsten halogen lamp. The light source is directed at the base of the hemorrhoid, and heat is used to produce a visible eschar that cuts off the blood supply to the hemorrhoid. The procedure is generally painless for patients receiving treatment on an outpatient basis. The patient may experience mild discomfort, a heat sensation, and a sense of fullness in the rectum after the procedure. In 7-10 days, the patient will notice bleeding from the rectum as the hemorrhoid falls off. Only one hemorrhoid at a time may be treated. If this method fails, then the patient may require surgery.

Laser treatment of GI tumors

Endoscopic laser treatment is also indicated for palliative or curative treatment of benign or malignant tumors of the esophagus, stomach, duodenum, ampulla, colon, or rectum. When palliative treatment is undertaken, the goal is usually to relieve obstruction or to reduce blood loss. Attempts to relieve pain have generally not been successful. Blood loss may be reduced by coagulating tumor bleeding sites or by destroying the tumor vasculature. Palliation of bleeding and/or obstruction with laser photocoagulation appears to be a true alternative to bypass surgery. In some cases, intended curative treatment has been achieved.

The use of laser therapy may be indicated in the following situations and types of malignant and premalignant lesions:

- Benign polyps and large villous adenomas that are sessile, at least 5 mm in transverse diameter, and not amenable to electrosurgical snaring.
- Periodic removal of polyps to prevent development of rectal cancer in patients with Gardner syndrome or familial adenomatous polyposis who have had subtotal colectomies with ileorectal anastomoses.
- Palliative treatment of malignancies in which there is widespread disease and no chance for surgical cure.
- To reestablish luminal patency and relieve dysphagia in patients with incurable esophageal cancer.
- To relieve gastric outlet obstruction, control chronic bleeding, or reduce tumor bulk in patients with gastric cancer.
- To relieve obstruction, control bleeding, or reduce tumor bulk in patients with colorectal cancer who are not surgical candidates or who decline surgery.

To determine whether a patient with GI cancer is a candidate for laser therapy, the following three preliminary examinations are required:

1. A barium swallow to define the location and extent of the tumor in the upper GI tract.
2. Diagnostic endoscopy to view the gross appearance of the tumor and to obtain biopsies and cytology.
3. A computerized tomography (CT) scan or magnetic resonance imaging (MRI) to define the extent of the disease.

If there is any possibility of a cure for GI cancer, and there are no medical contraindications, exploratory surgery may be performed. In patients whose functional status is so poor that even opening the lumen would not greatly improve the quality of life, laser therapy should not be considered.

As in thermal treatment, it is preferable to begin at the distal margin of the tumor. If a two-channel endoscope cannot be advanced to the distal margin of the tumor, the esophagus may be dilated. Dilatation is usually sufficient to allow for treatment to begin at the distal tumor margin. If even the smallest endoscope will not pass beyond the obstruction, the endoscopist may begin treatment at the proximal tumor margin.

To begin laser therapy, the quartz wave-guide that carries the laser beam is passed through the biopsy channel and beyond the distal tip of the scope. The beam is aimed at the portion of the neoplastic tumor closest to the lumen, and treatment progresses in increasingly larger concentric circles toward, but not to, the wall of the target organ. Usually the tumor is destroyed by vaporization. Some endoscopists prefer to accomplish tumor destruction by coagulation because, although this is a slower process, it generates less smoke. At the end of treatment, the tissue is often edematous and charred black secondary to thermal damage. Major blood loss is not usually a problem.

It is uncommon for any patient with an esophageal tumor to require more than three laser sessions. Patients are usually treated every other day until optimal luminal opening is achieved. The 48-hour period between treatments allows for maximal tissue necrosis and is well-tolerated by most patients. In subsequent sessions, surveillance endoscopy is used to observe the effects of the last treatment. If necessary, necrotic tissue can be pushed distally into the stomach, and dilatation may be performed.

Patients with rectal tumors are generally more amenable to outpatient treatment than patients with esophageal cancer, who are often more debilitated. When the colonic lumen is almost completely obliterated by the tumor, a temporary opening may be provided by dilatation with a through-the-scope balloon dilator. For rectal cancers, a rigid proctoscope and hand-held laser fiber may be used. If the lesion has advanced to the anal verge, treatment with the Nd:YAG laser is painful, and a saddle block may be required. If a rectal lesion is circumferential, treating only 270 degrees of the lesion can reduce the risk of rectal stenosis.

ARGON PLASMA COAGULATION

Argon plasma coagulation (APC) was adapted from the surgical arena for use in gastroenterology in 1991 (Malick, 2006). Electrical energy is delivered to the tissue by ionizing argon gas, creating a plasma. Plasma is a gas that has been partially or completely ionized and is a collection of charged particles containing about equal numbers of positive ions and electrons. Plasma is considered a state of matter just like gas, liquid, or solid.

The advantages of APC compared to monopolar or bipolar electrosurgery are the ability to control the depth of burn and the fact that contact with the tissue is not essential. The advantages of argon plasma over the laser are cost and portability.

Indications

The clinical applications of APC are both hemostasis and thermal devitalization.

Procedure

APC uses a special electrosurgical unit to create electrical energy that is passed through a wire via a catheter. A flow of argon gas is delivered via the same catheter. At the tip of the catheter, the electrical energy ionizes the argon gas to form a plasma. Since the argon plasma is electrically conductive, the current can jump from the tip of the catheter to the tissue and create a thermal effect. The coagulation depth is usually 1-3 mm. The depth depends on the power output from the generator and how long the current is activated. The power settings and flow of the gas can be adjusted to vary the gap the plasma can cover. A grounding pad is required during APC use.

Smoke Plume Hazard

Each of the therapeutic modalities—laser, APC, and electrosurgery—can create smoke plumes. During the application of energy, heat and smoke are generated. Many organizations, including the Association of periOperative Registered Nurses (AORN) and the National Institute for Occupational Safety and Health (NIOSH), are looking at the safe evacuation of the smoke plume.

Smoke plumes can contain toxic gases; bioareosols; dead and live cellular material, including blood fragments; and viruses. It is important that staff are aware of this possibility and take precautions in their practice. In gastroenterology departments, the smoke is removed by suctioning through the endoscope, although the suction canisters do not have high-efficiency filters to manage the particulates in the smoke.

As solutions to this issue are researched, many states are working on establishing standards and recommendations to protect employees from the hazards of surgical smoke.

There are no specific recommendations from OSHA to use smoke evacuation devices. While the American National Standards Institute (ANSI) recommends using portable smoke evacuators, there are no smoke evacuation devices for use in the GI setting (American National Standard Institute [ANSI], 2011). To reduce the inhalation of smaller particles in smoke plumes by staff, using a surgical respirator (N95 respirator) is recommended.

PHOTODYNAMIC THERAPY

Photodynamic therapy (PDT) has been used worldwide. This technology is effective as a minimally invasive technique to treat various cancers. When compared to surgical intervention, PDT is a treatment with outcomes that include lower morbidity, mortality, and cost (Wineland & Strack, 2008). The advantage of this treatment is that it does not involve the use of heat or the vaporization of tissue caused by conventional laser therapy.

Indications

PDT is used to treat superficial esophageal cancers, high-grade dysplasia, Barrett's esophagus, and superficial adenocarcinomas of the colon.

Procedure

PDT drugs called photosensitizers are injected into the patient's body where they collect naturally in hyperprolific cells. The dose of photosensitizer is calculated according to the patient's weight and is administered intravenously.

Once the hyperprolific cells have absorbed the photosensitizing agent and converted the light energy to chemical energy, the laser is used as a method to deliver light to the cells. Approximately 24-48 hours after injection, a red light laser is directed endoscopically at the tumor or lesion. Cytotoxicity occurs directly during the light application and indirectly through a series of chemical processes.

The first observable changes are blanching of the tumor tissue and diffuse hemorrhage. Necrosis and eschar formation occur over 2-3 weeks, followed by healing after approximately 6-8 weeks. Patients may require multiple treatments.

Considerations

Due to the absorption by hyperprolific cells, the patient must take precautions to avoid sunlight once they have been injected with the photosensitizing agent. This photosensitivity lasts approximately 30 days. Skin precautions include, but are not limited to, wearing:

- A close-weaved, wide brimmed hat.
- Sunglasses.
- A long-sleeved shirt.
- Long pants.
- Socks.

Patients are also instructed to:

- Avoid direct sunlight by sitting in the middle of the back seat of the car when traveling.
- Not sitting in front of a window when inside a building.
- Avoid exam lights and reading lights.

Patients also should be instructed that sunscreen products will not protect them.

The three major side effects of PDT treatments to the esophagus are:

- Discomfort in the back, chest, or abdomen due to local edema and inflammation.
- Strictures in the esophagus at the site of treatment.
- Photosensitivity.

Side effects can be managed with pain medication, intravenous fluids, and possibly dilation of the esophagus.

On the 31st day after the photosensitizing agent has been injected, the patient should be instructed to check his or her photosensitivity. This can be done by placing a hand in a paper bag that has a 2-inch hole cut in it, placing the hole on top of the patient's hand, and then exposing this area to direct sunlight for 10 minutes.

If there is redness or blistering within 24 hours, light precautions should be continued for an additional 2 weeks, and then the test should be repeated. If there is no reaction, the patient can gradually increase exposure to sunlight while continuing to observe for reaction.

BRACHYTHERAPY

Brachytherapy, also known as sealed source radiotherapy or endocurietherapy, is a form of radiotherapy where a radioactive source is placed inside or next to the area requiring treatment.

Indications

This method of treating GI tumors is growing with the ability to reach and implant small radioactive seeds into the tissues. The placement of catheters can be coordinated with the radiation oncology department to deliver radiation to tumors sites within the GI tract, the most common being the esophagus.

Considerations

Nursing considerations involve exposure to radiation for other patients and staff. The radiation safety department or radiation oncologist will establish the guidelines which must be followed. These guidelines may include:

- Distance from the patient, such as for care or removal of supplies and linen.
- Documentation of the length of time in contact with the patient.
- Location of the bed away from the wall to keep radiation exposure to adjacent rooms at a minimum.

Institutions should have an established policy on how to deal with a radioactive source that has become dislodged, as well as a procedure for documenting the gastroenterology nurse's actions in such a case.

ENDOSCOPIC MUCOSAL RESECTION

Endoscopic mucosal resection (EMR) is a technique to remove small nodules or flat lesions within the GI tract down to the submucosal layer.

Procedure

Once the lesion has been identified, one of two methods can be used:

- The lesion is raised by injecting sterile saline with or without contrast dye and/or epinephrine underneath the site. The lesion can then be snared and removed.
- A small cap is fitted on the endoscope that allows the tissue to be suctioned into the cap. This process is similar to variceal band ligation. Once the lesion is inside the cap, a snare and cautery are used to remove the tissue.

Endoscopic submucosal dissection

Endoscopic mucosal resection, along with various staining methods, have led to variations of removing larger areas of tissue within the GI tract. Endoscopic submucosal dissection (ESD) has the advantage of treating localized lesions to the mucosa and submucosa layers by being less invasive and less costly than conventional surgery. The technique requires a high level of endoscopic skills from the physician and staff to safely perform the procedure.

ESD enables resection of large superficial tumors in the upper gastrointestinal (GI) tract, including the esophagus and stomach.

By contrast, ESD for the colorectal neoplasm has been regarded as a technically difficult therapeutic procedure because of the anatomical and histological differences between the upper GI and the colorectum. This technique uses a needle knife or other specially designed knives to dissect out large areas of tissue. The complications of this procedure are bleeding and perforation.

The ability to provide endoscopic clipping and suturing has improved outcomes from these complications.

ENDOSCOPIC ENDOLUMINAL RADIOFREQUENCY ABLATION

Radiofrequency ablation works by sending an alternating current through the tissue, which creates increased intracellular temperatures and localized interstitial heat.

Barrett's esophagus can lead to low- to high-grade dysplasia and even adenocarcinoma. The options for treating this dysplasia have become varied, from surgery to photodynamic therapy (PDT) to tumor ablation with electrosurgical, laser, or argon plasma coagulation.

A radiofrequency ablation catheter array that uses a special energy generator to ablate the esophageal mucosa has been shown to be highly effective in treating Barrett's esophagus. Current studies have shown that this technique is successful in up to 70% to 95% of patients (Luigiano et al., 2016).

Indications

The goal in Barrett's esophagus treatment is to locate areas of dysplasia and treat those areas. Regular surveillance with EGD and random biopsies is the best method to date, although it has limitations due to biopsy sampling errors, cost, patient compliance, and failure to avert adenocarcinoma.

Chromoendoscopy is a procedure that involves using different staining techniques (methylene blue, indigo carmine, or Lugol's solution) and examining the esophagus for dysplasia. This can be used with a standard endoscope or a high magnification endoscope. Other technologies being used include scanning confocal fluorescence microscopy endoscopes and probes. Narrow-band imaging and multi-band imaging are computerized enhancements that produce images similar to staining tissues. However, staining allows the dye to enter into the cracks and pits of the tissue that computer enhancements cannot reproduce.

Procedure

Prior to the procedure

Prior to the procedure date, the gastroenterology nurse should explain the procedure and answer any questions the patient may have. Post-procedure diet and pain management plan for the patient should also be discussed.

During the procedure

Adequate sedation is important to the success of the procedure. Decreased patient movement and adequate pain control is essential during the radiofrequency ablation procedure. The procedure time is lengthy due to the time to measure the esophagus, spray the acetylcysteine (Mucomyst) solution, pass a guidewire, ablate the tissue, remove the coagulum, and repeat the ablation process.

The esophagus is measured using a sizing catheter to determine the optimal catheter size. Ablation catheters come in 18, 22, 25, 28, and 31 mm diameters. The newest catheter is self-adjusting and eliminates the sizing process. There are radiofrequency catheters with a pad that is 13 mm x 20 mm, 10 mm x 15 mm, or 13 mm x 40 mm that can be used to ablate smaller areas and to treat radiation proctitis and watermelon stomach—gastric antral vascular ectasia (GAVE). These catheters are designed to fit on the distal end of the flexible endoscope with an outside diameter of 8.6 mm to 9.8 mm. The newest catheter can be placed down the working channel of a flexible gastroscope and has a pad size of 7.5 mm x 15.7 mm.

The radiofrequency ablation generator computer controls the temperature created, which limits the depth of burn. The 360-degree ablation catheter is 4 cm long, has a 3 mm ablation section, and takes less than one second to ablate the tissue. This catheter creates a burn that is limited to 1 mm in depth. This process is repeated at different levels of the esophagus.

After the procedure

After ablating the intestinal metaplasia, the normal squamous epithelium is allowed to re-establish itself. The ablation process may need to be repeated at 6- to 8-week intervals. The procedure can be repeated as needed.

Considerations

The nursing considerations for radiofrequency ablation post-procedure are dysphagia and pain. It is important that the patient continue to take proton pump inhibitor (PPI) medication and follow the standard gastric reflux instructions. The physician may prescribe antacids, oral lidocaine, sucralfate (Carafate), and pain medication to assist with post-procedure pain. The dysphagia is managed by consuming only liquids in the first 24-48 hours post-procedure, then the diet can progress to a regular diet as tolerated.

Complications with this procedure are perforation, esophageal strictures, and dysphagia.

INJECTION THERAPY

Endoscopic injection therapy was first used to stop variceal bleeding in 1939. After being supplanted temporarily by portacaval shunt techniques, it was reintroduced in 1974, and has become increasingly popular (Rajoriya & Tripathi, 2014).

Indications

Injection therapy involves the injection of a chemical agent through a needle injector into or around a bleeding site to stop bleeding through variceal thrombosis or local edema.

Injection therapy is contraindicated in patients who are unwilling or unable to cooperate or in patients with severe coagulopathy.

Procedure

The procedure is performed under direct endoscopic visualization, most often by using a flexible upper endoscope. A double-channel endoscope is helpful to permit additional suction when the patient is actively bleeding.

Injection therapy needle assemblies are disposable. The greatest flexibility is obtained with a simple tubular system made of synthetic material, with a needle fixed to the distal tip. A second outer sheath is added, into which the needle may be withdrawn or retracted when not in use.

In addition to the choice of agent, the volume, rate, and placement of injections are all important in determining the overall effect.

Prior to the procedure

Before the procedure, the gastroenterology nurse should:

- Verify the patency of a large-bore IV line, verify blood type and cross match, and obtain the results of laboratory testing.
- Instruct the patient to remain still and refrain from coughing.
- Administer IV sedation and analgesia to maximize patient cooperation and minimize patient movement.

Because some of the agents used in this procedure can cause corneal ulcerations, the patient's eyes should be covered by a small towel during the procedure, and all personnel should wear protective eyewear, gloves, and masks.

During the procedure

- A 5- or 10-ml syringe containing the chemical agent is attached to the proximal end of the needle injector. The injector is flushed with the agent to rid it of air and to check for patency and leaks. The needle injector should be checked to ensure that the needle properly protrudes from and retracts into the sheath.
- A complete endoscopic examination of the esophagus, stomach, and duodenum is performed to locate the bleeding site or varices.
- The injector is passed through the biopsy port of the endoscope. Holding a gauze sponge around the injector-syringe connection will prevent leakage.
- With the patient in the left lateral position, the endoscope is positioned at the bleeding site.
- Air is insufflated to distend the esophageal lumen to enhance visualization.
- The needle is advanced. Once the needle tip is visualized, it should be maneuvered into position, and the needle should be thrust directly into the desired site by rapidly advancing the injector and plunging the needle into the wall up to its junction with the sheath.
- As soon as entry has been achieved, at the physician's direction, the gastroenterology nurse injects 0.5-3 ml of the chemical agent, using slow, steady pressure. The gastroenterology nurse should verbally state the amount of injection in increments of 0.5 ml and should say "stop" as soon as the injection has been completed.
- Throughout the procedure, the patient should be monitored for abdominal distention, and the number, type, and amount of injections should be documented.
- Following the injection, the needle is withdrawn into the sheath. The entire assembly is then removed from the accessory channel and discarded in an appropriate sharps container.
- The stomach is suctioned to decompress the gastric distention caused by continuous insufflation of air.

Nonvariceal injection therapy

Injection of hypertonic saline-epinephrine is a relatively simple maneuver and has satisfactory hemostatic efficacy in the endoscopic treatment of nonvariceal GI bleeding caused by vasoconstriction and platelet aggregation effects.

Indications

Nonvariceal GI bleeding, as seen with conditions such as Mallory-Weiss tears, angiodysplasias, ulcers, and post-polypectomy bleeds, has been treated with a hypertonic saline-epinephrine solution.

A common sclerosing agent used for hemostatic control of bleeding ulcers is dehydrated (98%) ethanol (alcohol). Alcohol works by dehydrating and fixating the exposed blood vessel and surrounding tissue. The resulting local vasoconstriction, vascular wall degeneration, and endothelial destruction lead to thrombosis.

Procedure

Injection therapy is often combined with thermal treatment to control bleeding. Bipolar probes are available that allow for injection therapy and bipolar electrocoagulation in one device.

Variceal sclerotherapy

Injection **sclerotherapy** involves the injection of a sclerosing (hardening) agent into a blood vessel (see Figure 28.2).

Indications

Variceal sclerotherapy is indicated for:

- Temporary control of acute bleeding from esophageal varices, pending shunt placement, or in patients who are poor surgical risks.
- Eradication of esophageal varices to prevent rebleeding.
- Treatment of small hemorrhoids with bleeding.
- Temporary control of acute hemorrhage from gastric varices.

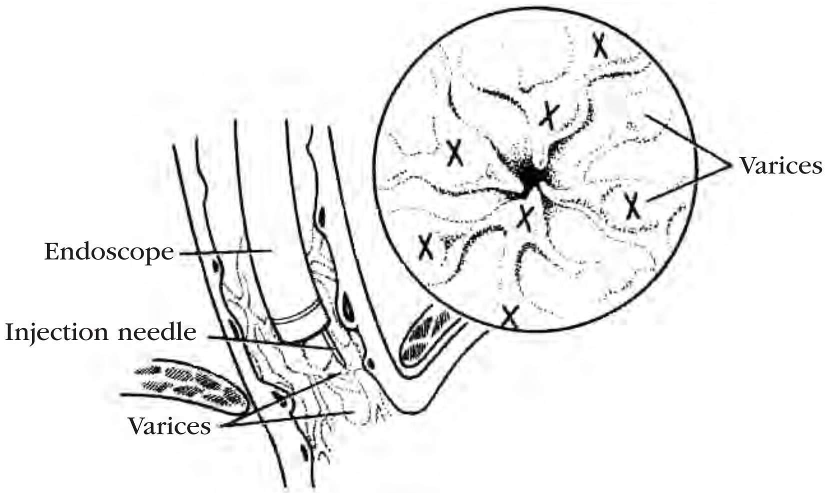

Figure 28.2. Variceal Sclerosing
Insert shows accepted sites for sclerosing.

Gastric varices do not respond as well to sclerotherapy as do those in the esophagus. Needle puncture of a gastric varix may be associated with prolonged bleeding from the injection site that is difficult to tamponade. There are several studies in which n-butyl-cyanoacrylate has been used to obliterate gastric varices (Saraswat & Verma, 2012; Chandra, Holm, El Abiad, & Gerke, 2018). The glue acts as a tissue adhesive that polymerizes on contact with blood in a gastric varix. The most recent studies also demonstrate there have been complications due to multiple pulmonary emboli.

Injection sclerotherapy is rarely used as a prophylactic measure for patients who have never bled from varices. In children who are awaiting liver transplantation, however, grade III or IV varices are injected prophylactically.

Procedure

Prior to the procedure

Sclerotherapy is difficult if there is active hemorrhage because blood must be cleared from the field to make precise injections. The procedure is usually started 4-6 hours after relative hemodynamic stability has been achieved by transfusion and/or balloon tamponade. Pharmacological therapy to control acute bleeding with a splanchnic vasoconstrictor such as octreotide (Sandostatin) leads to a decrease in portal venous inflow and to a decrease in portal pressure. This procedure usually relieves acute bleeding in 75% to 90% of cases and allows the patient to be stabilized (American Society for Gastrointestinal Endoscopy [ASGE], 2014b; Bhutta & Garcia-Tsao, 2015).

If skin or mucus membranes come into contact with the sclerosant, it should be rinsed immediately with water. Eye protection for the patient and the endoscopy staff is paramount to prevent damage from an accidental exposure to the sclerosant.

The physician chooses the sclerosing agent. Ideally, it should induce rapid thrombosis, lead to intimal damage to the vein, and end in obliteration, with minimal damage to the underlying esophageal musculature. It should be innocuous when it reaches the extraesophageal vascular system and other organs. The sclerosing agents used most commonly in the United States for control of variceal bleeding are sodium tetradecyl sulfate (Sotradecol) and ethanolamine oleate (Ethamolin). Ethanolamine oleate may be less ulcerogenic than sodium tetradecyl sulfate.

During the procedure

The endoscopist determines the number and pattern of injections, which may be given intravariceally, paravariceally, or using a combination of both. Paravariceal (submucosal) injection can be recognized if the site blanches after injection. Intravariceal injection results in ballooning of the blood vessel with a characteristic dark blue color within a few minutes after injection.

As soon as the sclerosant is injected, the needle is withdrawn and rapidly reinserted into another site. The needle is fully withdrawn into the sheath after each injection. If undue pressure is needed to inject, it usually means that the needle is against the wall or that the varix has been injected previously. Forcing sclerosant into the wall of the esophagus will probably result in a superficial esophageal ulcer.

In patients with active hemorrhage, when the bleeding point can be identified, the first injection should be made immediately above the bleeding point in the same column, and then to the right and left. As the flow subsides, an injection may be made distally to the bleeding point. Even if injection therapy is unsuccessful in stopping active bleeding, it is important to make an injection below the varix.

After each injection, there is frequently some backbleeding. Oozing is possible and stops in 1-2 minutes. If a stream of blood arises from a puncture site, the injection puncture pattern for active hemorrhage should be followed.

After the procedure

After the procedure, an extended period of observation follows, in which the patient should be monitored for signs of GI blood loss and esophageal perforation. A patent IV line should be maintained until discharge to facilitate the delivery of additional pain medication, as ordered. The physician may order laboratory work and antacids or sucralfate (Carafate) slurry.

Complete eradication of varices may require several endoscopic sessions with an average of five sessions for complete eradication of varices.

Considerations

Transient side effects of injection sclerotherapy include:

- Mild to severe chest pain, dysphagia, and fever. Chest pain usually subsides within 24 hours. Severe pain that lasts more than 24 hours may be indicative of perforation.
- A low-grade fever, which usually subsides in 24 hours.
- Esophageal ulceration, which is considered a side effect of sclerotherapy because up to 78% of patients have some degree of ulceration.
- Esophageal stricture, which is a long-term complication.
- Chronic dysphasia, which may occur in many patients.

More serious potential complications of injection sclerotherapy include:

- Hemorrhage, aspiration, necrosis, mediastinitis, esophageal perforation, pleural effusion, sepsis, or portal vein thrombosis.
- Perforation, manifested by persistent chest pain, dysphagia, fever, and pleural effusion.
- Septicemia, which is rare, but prophylaxis is probably warranted for patients with clinically significant heart disease, prosthetic heart valves, or ascites.
- Pleural effusion, which seems to be related to the amount of sclerosant used and the amount of chest pain the patient experiences, and usually resolves without therapy.

Varices recur in 13% of patients, even once they are varix-free. Annual check-ups are recommended (Floch, 2019).

ENDOSCOPIC VARICEAL LIGATION

In 1988, **endoscopic variceal ligation (EVL)** was developed. EVL is used for the treatment of bleeding esophageal varices (Suzuki & Murashima, 2019). It has also been used successfully for the eradication of rectal hemorrhoids.

Indications

With EVL, latex-free bands or O-rings are applied around the varices. The tensile strength created by the bands is sufficient to eradicate the varices. Over time, the varix shrinks, and the area where the O-ring is applied ulcerates. The band falls off, and the ulcer heals.

Multi-band ligation (MBL) is also used to treat esophageal varices. There are multiple products that range from 4- to 12-band placement per intubation. The setup includes a ligator unit, handle unit with tripwire and scope fastener, and irrigation device.

Advantages of MBL include:

- No overtube-associated complications such as trauma and increased patient discomfort.
- No need for multiple intubations or removal of cassettes, which results in time savings and less exposure of staff to blood and body fluids.
- Various diameters of MBL kits to accommodate the different diameters of endoscopes. The gastroenterology nurse must ensure that the MBL correlates to the endoscope being used.
- The option for MBL kits with latex-free bands.

Although the advantages are numerous, it should be noted that MBL is considerably more expensive than the single-banding technique.

Procedure

Manufacturer information should be followed in setting up all devices. The handle unit controls the deployment of bands.

The physician identifies the varix, makes contact, suctions, and deploys one band by turning the handle unit. Varices should be ligated in a distal-to-proximal sequence to avoid obscuring the endoscopist's field of vision.

With each single-band ligator, the outer cylinder of a friction-fit adaptor is carefully attached to the distal tip of the endoscope. A tripwire is inserted down the biopsy channel of the scope. The inner cylinder of the adaptor, with an O-ring attached, is attached to the tripwire and then to the outer cylinder.

With the overtube properly positioned and the need for EVL confirmed, the preloaded ligator and endoscope are introduced. The varix is aspirated into the inner cylinder, and the tripwire is pulled to release the O-ring. Suction is released, and the endoscope is withdrawn. At the physician's discretion, serial ligations may be performed after removing the spent inner cylinder and reloading a new one.

Considerations

Disadvantages of this technique are poor visibility with profuse bleeding, having to reload bands, and overtube trauma. Compared to injection sclerotherapy, however, EVL seems to result in less substernal discomfort and stricture formation.

Gastroenterology nurses need to be aware that the device decreases the ability to aspirate secretions, so the patient could be at higher risk for aspiration.

ENDOSCOPIC CLIPPING

There are many different versions of endoscopic clips. The concept evolved from laproscopic surgeons using metal staples to ligate vessels or close incisions.

Indications

In gastroenterology, the clips were originally used in active bleeding situations where electrosurgery or injection methods failed. Once the bleeding vessel or site was identified, the clips were deployed to stop the bleeding. Endoscopic clips have been used alone or in conjunction with electrosurgery and injection therapy. Endoscopic clips also have been used to close perforations, provide markers for radiological and endoscopic procedures, and secure tubes (feeding or rectal).

Endoscopic clips are produced using different types of metals and are MRI-conditional, meaning the clip can be scanned safely under certain conditions outlined by the manufacturer. MRI parameters should be checked with the clip manufacturer, including static magnetic field strength, special gradient field, and maximum whole-body average specific absorption rate (SAR). These parameters vary depending on clip manufacturer.

Procedure

The clips are available in different sizes and lengths, and they can be rotated to orient the jaws relative to the site. Clips can be through-the-scope or over-the-scope. The over-the-scope clip is able to provide full-thickness closure compared to through-the-scope clips. This allows for closure of GI defects (fistulas, anastomic leaks, perforations), large ulcers, and polypectomy sites.

The clip is applied to the endoscope and deployed in a similar fashion as a band ligator device. The over-the-scope clip allows passage of a grasping device to help in alignment of tissue prior to clip deployment.

HEMOSTATIC POWDER

In May 2018, the U.S. Food and Drug Administration (FDA) approved the hemostatic powder TC-325 to help control GI bleeding (FDA, 2018). The powder is inert, contains no human or animal proteins or botanicals, and has no known allergens. The powder is not absorbed by the body and passes through the GI tract within 24-72 hours. The powder has been in use outside the United Sates since 2011.

Indications

The use of hemostatic powder is nonthermal, nontraumatic, and noncontact, and it does not require the precise targeting of the bleeding site. The hemostatic powder can be applied to all types of bleeding sites, including large surface tumors, with a high rate of hemostasis. Studies have shown a range of 88% to 100% rate of hemostasis in treating all types of GI bleeding with hemostatic powder (Chen & Barkun, 2015).

Hemostatic powder has been used on acute variceal bleeding and post-banding ulcers in a couple of small sample-size studies (Ibrahim et al., 2018; Chen & Barkun, 2015). Despite this demonstrated efficacy, it is not recommended as the sole modality in the treatment because of its short acting time of 24 hours. It can bridge until a transjugular intrahepatic portosystemic shunt (TIPS) procedure or embolization can be done in some circumstances.

Procedure

The hemostatic powder is sprayed over the site using a pressurized carbon dioxide gas delivery system and applied with a 7 Fr or 10 Fr through-the-scope catheter. The powder is applied using 1- to 2-second bursts until hemostasis is achieved.

Care and maintenance

In order to prevent catheter occlusion and avoid direct contact with body fluids and/or water:

- Flush the accessory channel with air prior to introducing the catheter.
- Eliminate mucosal or vessel contact.
- Avoid applying suction while the catheter is in the endoscope channel.
- Avoid placing the catheter in pooled blood or water.
- Ensure that the trigger is pressed for at least 1-2 seconds to prevent catheter occlusion.

Hemostatic powder achieves hemostasis by cohesively and adhesively forming a mechanical barrier over the bleeding site. This occurs by absorption of water to concentrate clotting factors, activation of the clotting cascade, and the creation of the mechanical barrier.

Considerations

Although the powder has a good safety profile, powder impaction, arterial embolization, and allergic reaction are potential adverse advents.

ESOPHAGOGASTRIC TAMPONADE

Esophagogastric tamponade involves the insertion of specialized tubes to provide pressure on bleeding areas of the esophagus or esophagogastric junction.

Indications

Esophagogastric tamponade is indicated in patients with acute upper GI bleeding from esophageal varices or tears at the esophagogastric junction. Because of the variety of potential complications, it is generally performed in patients with variceal bleeding after the administration of octreotide and beta-blockers, sclerotherapy, or EVL have failed. It is used rarely for Mallory-Weiss tears that are unresponsive to medical therapy.

Esophagogastric tamponade is contraindicated:

- For patients with cardiopulmonary failure or recent surgical trauma to the esophagogastric junction.
- When variceal bleeding has stopped.

- When the source of upper GI bleeding cannot be identified.
- In patients who are surgical candidates and in whom prior tamponade has failed.

Procedure

Prior to the procedure

The gastroenterology nurse should establish and confirm a patent oral airway, monitor vital signs, and establish one or two large-bore IV lines for fluid and blood replacement.

Because patients with acute upper GI bleeding may become extremely agitated, the gastroenterology nurse should explain the benefits of the procedure, assure the patient of constant nursing care and monitoring, and explain that some pressure will be felt when the tube balloons are inflated.

The patient should be placed in a semi-Fowler's or left lateral position.

During the procedure

Before insertion, all balloons should be inflated and examined for defects. They should be checked for air leaks by submerging in water.

The following three different types of tubes may be used for this procedure:

1. **Sengstaken-Blakemore tube** is a three-lumen tube with esophageal and gastric balloons. It provides a gastric aspiration port to allow drainage from below the gastric balloon. Esophageal aspiration is accomplished by a secondary tube, such as a Levin or Salem sump tube that is inserted orally or nasally to rest above the esophageal balloon. A manometer is used to insufflate the balloons and to measure pressure.
2. **Linton tube** is a three-lumen tube that uses a gastric balloon, but no esophageal balloon, and provides ports for both esophageal and gastric aspiration. Elimination of the esophageal balloon reduces the risk of esophageal necrosis. A manometer is used to measure pressures from the gastric inflation port.
3. **Minnesota tube** is a rubber, radiopaque, 18 Fr, four-lumen, double-balloon tube. The four lumens are used for gastric lavage and aspiration, esophageal aspiration, esophageal tamponade, and gastric tamponade. Medications may be instilled through the gastric lavage port. Lavage and gastric and esophageal suction may be performed while the balloon is inflated.

If a Sengstaken-Blakemore tube is used, a nasogastric tube should be inserted above the esophageal balloon to provide esophageal suction. Intermittent suction should be attached to the nasogastric tube.

The lubricated balloon tube is inserted through the patient's mouth or nose, and the position of the tube is confirmed by X-ray examination. Gastric contents are aspirated, and the gastric balloon is inflated in increments of 100 ml of air until the balloon is full (250-500 ml for a Sengstaken-Blakemore tube, 450-500 ml for a Minnesota tube, and 700-800 ml for a Linton tube). If the patient complains of sudden substernal pain or if insufflation of air is not audible over the epigastric area, inflation should be stopped immediately.

Once the gastric balloon is inflated, the tube is pulled back gently until resistance is felt against the gastroesophageal junction. The balloon inlet is clamped, and the tube is secured with 1-2 pounds of traction. Traction may be initiated by the use of a nasal sponge guard, an over-the-bed traction set-up with 1.0 pound of weight, and a football helmet or catcher's mask. External traction may cause ulceration of the nasal mucosa and should not be used for prolonged periods.

Suction is connected to the gastric and esophageal lumens, and drainage from each port should be observed.

If bleeding continues after the gastric balloon is inflated, the esophageal balloon should be attached to a manometer and inflated with air to 25-45 mm Hg. The esophageal balloon inlet is clamped, and the nasogastric tube is changed to constant suction. If bleeding is clearly from esophageal varices above the gastroesophageal junction, the esophageal balloon may be inflated simultaneously with the gastric balloon. If bleeding continues during esophageal tamponade, it may be from a gastric varix.

The procedure for insertion of a Linton tube is the same as for a Sengstaken-Blakemore tube, except that there is no esophageal balloon, and the esophageal suction lumen obviates the need for a secondary nasogastric tube. The Minnesota tube also has an esophageal suction lumen.

Considerations

Regardless of the type of tube used, a pair of scissors should be taped to the head of the bed so it is within easy reach in case transection and emergency removal of the tube become necessary.

While the tube is in place, it is important to check the traction on the tube regularly, monitor vital signs, irrigate the gastric tube as ordered and assess fluid return, recheck and adjust the esophageal balloon pressure every 2 hours, provide good oral care, and keep the patient NPO.

After 24 hours, if bleeding is controlled, the esophageal balloon should be deflated. The gastroenterology nurse should check for rebleeding per the physician's orders. The gastric balloon should be deflated if no bleeding recurs over the next 6-12 hours. Frequent oral care should be provided, and frequent coughing and deep breathing should be encouraged.

Potential complications of esophagogastric tamponade include aspiration, rupture of the esophagus from pressure secondary to a misplaced gastric balloon, tissue necrosis from excessive pressure and/or prolonged inflation time, and airway obstruction secondary to dislodgement of the esophageal tube.

ENDOSCOPIC CRYOTHERAPY

Cryotherapy is a therapeutic method of treating tissue by freezing in situ to achieve a specific response.

Indications

Endoscopic cryotherapy has many clinical applications in gastroenterology and is being used for ablation of GI tumors and lesions, radiation induced proctitis, watermelon stomach, hemostasis, and AVMs.

Procedure

Cryotherapy uses a catheter to apply extreme cold temperatures to the GI mucosa to produce a therapeutic effect. There are several different cryogens, such as liquid nitrogen and liquid carbon dioxide (CO_2). They are released via a catheter, resulting in a depth of injury of 2-3 mm. The mucosa becomes necrotic and is followed by regeneration of normal mucosa in 7-10 days.

Care and maintenance

Damage to the endoscope can result from the extreme cold.

Considerations

The most common complications are perforation and stricture formation due to over-freezing.

CONFOCAL LASER ENDOMICROSCOPY

Confocal laser endomicroscopy involves combining a standard upper endoscope or colonoscope with a confocal scanner.

Indications

The use of confocal laser endomicroscopy is growing with the introduction of confocal probes to enhance the practice in gastroenterology.

Procedure

The confocal scanner delivers a laser light to the tissue, and confocal imaging is returned to a processor. This confocal laser scanning produces in-focus images of thick specimens, a process known as optical sectioning. This technology allows the gastroenterologist to acquire instant in vivo histology imaging. The images have sufficient resolution and contrast to distinguish pit architecture, as well as cellular and some subcellular structures.

Shining a light on some molecules shows a light of a different color emitted from those molecules. This is known as fluorescence. Fluorescence excitation is used in conjunction with the confocal endoscope. A fluorescent is a molecule designed to bind with various intracellular structures when absorbed by the cells. The laser light excites the molecules, and a color change is noted in the image. This acts like tissue staining to see the nucleus and other cellular structures.

NURSING CONSIDERATIONS

Patients with GI bleeding require special attention. GI bleeding is usually the result of another, less obvious problem.

A thorough nursing history is of paramount importance. It is necessary to obtain as much information as possible about the patient's current health problems. In addition, a family health history can help to determine whether there is any family history of alcohol abuse or a history of problems known to cause GI bleeding. The patient's past and current work environments should be assessed to determine if exposure to chemicals or inhalants of any type might be a causative factor.

A physical assessment, including weight, skin quality, and muscle tone, provides data specific to the general appearance of the patient. Any discoloration of the skin should be documented. The Joint Commission requirements include nutritional assessment and information that may be related to other concurrent disease processes (Lim et al., 2014).

Once a total history and physical assessment is completed and documented, the gastroenterology nurse can formulate the nursing diagnosis. Nursing diagnoses that may apply to patients with GI bleeding include the following:

- Anxiety, related to uncertainty of prognosis, multiple diagnostic procedures, and treatment regimen.
- Ineffective breathing pattern, related to decreased lung expansion, decreased energy, and fatigue.
- Potential for injury, with bleeding related to altered clotting mechanisms.
- Knowledge deficit with respect to treatment regimen.

The gastroenterology nurse must work with the patient and family members to identify outcomes that will be realistic for the patient. The overall goals are to prevent further damage and to improve the patient's overall physical and mental condition. Examples of desired outcomes for patients with GI bleeding might include:

- Patient is free of injuries and bleeding.
- Skin integrity is good.
- Patient has normal respiratory function.
- Patient is free of infection.

Patient care must be planned specifically for the individual patient. For the patient with bleeding problems, nursing care is implemented that assists in controlling hemorrhage. The patient's comfort is maintained by assessing the need for pain medication. Fluid and electrolyte status is constantly assessed. The patient and family members and/or significant others are taught appropriate self-care techniques. Potential problems and solutions are reviewed. The patient is made aware of helpful community resources, as appropriate.

CASE SITUATION

Mr. Ben Harrison, age 54, had a physical at his family's urging. As a routine part of his physical, a stool test was done to check for blood. The stool was positive for occult blood, and a small polyp was found at 30 cm from the anus on sigmoidoscopy. Ben's physician knows that findings of stool positive for occult blood and a polyp in the sigmoid may indicate a synchronous lesion higher in the large bowel. For this reason, he referred Ben to the gastroenterology department for a colonoscopy.

Points to think about

1. Ben exhibits the following signs and symptoms:
   a. Increased heart and respiratory rates.
   b. Increased systolic blood pressure.
   c. Profuse palmar sweating.
   d. Increased muscular tension.
   e. Urinary frequency and urgency.
   f. Anger concerning the need for various diagnostic tests.

   To what nursing diagnosis do these data seem most applicable?
2. What interventions might the gastroenterology nurse use before and during the colonoscopy to reduce Ben's anxiety?
3. The procedure has started, and the endoscopist has passed the first small polyp at 30 cm. The patient is tolerating the procedure very well. As the physician advances to the ascending colon, he finds a larger, more vascular appearing polyp. He moves on to the cecum, which is normal, and comes back to the ascending colon to remove the polyp. What are the responsibilities of the gastroenterology nurse operating the ESU?
4. The body uses its own mechanisms to achieve hemostasis. Three mechanisms are local reactions that are a response to cell disruption such as a cut. What are they?
5. By what mechanism does electrosurgery help to provide hemostasis?
6. The heat produced by a monopolar electrode is dependent on what four factors?
7. When using an electrosurgical unit during a procedure, what should the gastroenterology nurse document on the patient record specific to electrocautery?
8. Colonoscopy with polypectomy is a safe, outpatient procedure. Barring any complication, Ben will go home when he has recovered from sedation. What instructions will the gastroenterology nurse give him?

Suggested responses

1. The most applicable nursing diagnosis is anxiety related to a threat to bodily health and lack of control over events.
2. Appropriate anxiety-reducing interventions the gastroenterology nurse may take include:
   a. Provide a relaxing, reassuring, professional atmosphere.
   b. Discuss with Ben the types of sensations he may experience during the examination, including slight cramping, abdominal pain caused by air insufflation, or stretching of the bowel by moving the colonoscope.
   c. Reassure Ben that the gastroenterology nurse will be there throughout the procedure to take care of his needs.
   d. Speak softly to Ben during the procedure and offer comforting touch.
3. The gastroenterology nurse responsible for the ESU should:
   a. Set up and check the unit before the procedure and establish that it is working correctly.
   b. Place the grounding pad on the patient and check that the pad is flat and secure in the proper location.
   c. Be sure that the unit is plugged in, that all connections are secure, and that all personnel are wearing gloves.
   d. Hand the snare to the physician to feed through the scope, and then open and close the snare per the physician's instructions.
   e. Once the polyp is removed, turn the machine to stand-by while removing the snare, or disconnect the active cord from the snare.
   f. Document the site of polypectomy, placement of grounding pad, skin condition at grounding pad site pre- and post-procedure, and ESU model and serial number.
4. The body's own mechanisms for achieving hemostasis include:
   a. Vessel spasm. Vessels contract in an effort to diminish blood loss. Vessel spasm lasts from a few seconds to 30 minutes, so suspected bleeding vessels must be watched.
   b. Platelet aggregation. Circulating platelets have an affinity for the moist, irregular endothelium that is exposed in a broken vessel and will adhere there, forming a platelet plug.
   c. Clot formation. Within seconds or minutes after damage, platelet aggregates and altered blood structure release activator substances. This triggers formation of fibrin threads, forming a network in which plasma and red blood cells can coalesce into a clot.
5. Electrosurgery helps provide hemostasis by the following mechanism: kinetic energy relayed to the cells from the ESU excites the ions in the cells, causing the ions to collide with other cellular particles, creating heat in the cell. The effect on tissue is thermal, not electrical.
6. The heat produced by a monopolar electrode depends on these factors:
   a. The power setting.
   b. Tissue resistance. The heat in the tissue increases in relation to the resistance of the tissue. Tissue

with a lot of water content has low resistance. Dry tissue has higher resistance. When heat dries out the tissue, it may take a higher setting to continue cutting.
   c. Current density. Heat varies according to current density. When the snare loop is closed, it increases the current density and raises the temperature to the adjacent tissue. In the monopolar modality, the current density diminishes as it spreads out to the larger surface area of the dispersive electrode (grounding pad). If the surface area is large enough, the return electrode disperses the current, and the patient does not feel the heat. If the large electrode were to come loose at the edges and have only a small contact with skin, the current density going to that small spot could be great enough to cause a burn.
   d. Time. The longer the physician depresses the foot pedal, the greater the heating depth will be.
7. When using an electrocautery device, the following should be documented:
   a. The equipment and accessories used, including brand name and whether it was monopolar or bipolar.
   b. A description of the lesion and any tissue that was recovered and sent to pathology.
   c. Machine settings and number of applications.
   d. How the lesion looked after heat application and whether any bleeding is present.
   e. How the patient tolerated the procedure and any complications.
8. Instructions the gastroenterology nurse should give Ben when he goes home include:
   a. Do not drink alcohol, drive a car, or operate machinery until the next day.
   b. Observe any dietary or medication restrictions the physician has ordered, for example, restrict the use of aspirin or nonsteroidal anti-inflammatory drugs (NSAIDs) after polypectomy because they interfere with blood clotting.
   c. Some bloating is normal because of the air instilled during the procedure, but it will subside as air is expelled.
   d. Call the physician if any rectal bleeding occurs, or if there is severe abdominal pain or temperature elevation.

REVIEW TERMS

argon laser, argon plasma coagulation (APC), bipolar electrocoagulation, bipolar probe, coagulating current, coagulation, cryotherapy, cutting current, desiccation, electrocautery, electrocoagulation, electrosurgical unit (ESU), endoscopic variceal ligation (EVL), esophagogastric tamponade, detachable loop ligating device, fulguration, heater probe, laser, Linton tube, Minnesota tube, monopolar electrocoagulation, Nd:YAG, photocoagulation, photovaporization, radiofrequency ablation, sclerotherapy, Sengstaken-Blakemore tube, tamponade, hemospray

REVIEW QUESTIONS

1. Thermal coagulation of bleeding vessels is achieved in gastroenterology patients by using:
   a. Monopolar electrocautery.
   b. Bipolar electrocautery.
   c. Lasers.
   d. All of the above.
2. Leakage of current in monopolar electrocoagulation can cause burns to the:
   a. Endoscopist.
   b. Patient.
   c. Gastroenterology nurse or associate.
   d. All of the above.
3. When using bipolar electrocoagulation, all of the following are true except:
   a. There is less mucosal injury than monopolar.
   b. A grounding pad is needed.
   c. Silicone prevents tissue adherence.
   d. Perforation is a potential complication.
4. The type of laser used most often in endoscopic applications is:
   a. Carbon dioxide.
   b. Nd:YAG.
   c. Argon.
   d. KTP/532.
5. Laser fibers used in endoscopic applications are cleaned with:
   a. Alcohol.
   b. Glutaraldehyde.
   c. Hydrogen peroxide.
   d. Sterile water.
6. The specialized three-lumen tube with esophageal and gastric balloons used for esophagogastric tamponade is called a:
   a. Sengstaken-Blakemore tube.
   b. Linton tube.
   c. Minnesota tube.
   d. Salem sump tube.
7. In esophagogastric tamponade, the esophageal balloon should be inflated to what pressure?
   a. 2.5-4.5 mm Hg.
   b. 25-45 mm Hg.
   c. 125-145 mm Hg.
   d. 1.0 psi.
8. The first line of treatment in patients with actively bleeding esophageal varices is usually:
   a. Hemoclip.
   b. Balloon tamponade.
   c. Band ligation.
   d. Electrocoagulation.

9. Severe chest pain that persists for more than 24 hours in patients who have undergone sclerotherapy for esophageal varices is most likely a result of:
   a. Myocardial infarction.
   b. Residual effects of the sclerosing agent.
   c. Aspiration pneumonia.
   d. Perforation.
10. 10. Which of the following statements is false?
    a. Esophageal varices can be treated with sclerotherapy.
    b. Multiple-band ligation is often complicated by overtube trauma.
    c. Gastric varices do not respond well to invasive treatments.
    d. Eradication of varices may require an average of five sessions.

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Chapter 29

INTUBATION AND DRAINAGE

This chapter will acquaint the gastroenterology nurse with indications, contraindications, and techniques for gastrointestinal (GI) intubation and drainage. A variety of intubation and drainage procedures are covered, including nasogastric (NG) intubation, gastric lavage, nasobiliary and nasopancreatic catheters, biliary and pancreatic stents, self-expanding metal stent (SEMS), self-expanding plastic stent (SEPS), temporary expandable esophageal stent, colon decompression, and abdominal paracentesis.

Intubation for esophagogastric tamponade is discussed in Chapter 28. The use of feeding tubes and percutaneous endoscopic gastrostomy (PEG) and percutaneous endoscopic jejunostomy (PEJ) devices for long-term enteral nutrition are discussed in Chapter 22.

Learning objectives

After reviewing the content of this chapter, the gastroenterology nurse will be able to:

1. Discuss techniques for the insertion of a nasogastric tube.
2. Describe the use of nasobiliary and nasopancreatic catheters and biliary and pancreatic stents.
3. Discuss procedures for gastric lavage, colon decompression, and abdominal paracentesis.
4. Identify the indications for endoscopic stent placement and discuss special nursing care before, during, and after SEMS, SEPS, and temporary expanding esophageal stent placement.

NASOGASTRIC TUBE INSERTION

The size of the **nasogastric (NG) tube** generally used for adults is 12 French (Fr) to 18 Fr in diameter and 55 cm to 66 cm in length. The term **French** in reference to circumference is three times the millimeter size—for example, 2 mm is equal to 6 Fr, 3 mm is equal to 9 Fr. Nasogastric suction requires a larger tube, especially if blood or clots are present.

When choosing an NG tube for a pediatric patient, consider the size of the child and the reason for the tube. Feedings of thick formulas require a larger tube than feedings of thin formulas. A wide range of diameters are available, for example, 3.5 Fr for neonates and 5 Fr, 8 Fr, 10 Fr, and 12 Fr for children of all age groups. Smaller diameter tubes are appropriate and more comfortable for children who require long-term nasogastric intubation such as for decompression or feeding.

The length of tubing needed to reach the stomach should be determined by measuring from the patient's nose to the tip of the patient's earlobe, and then down to the xiphoid process.

Types of NG tubes include the following: Levin tube, Salem Sump tube, Cantor tube, and Miller-Abbott tube. Each has benefits that are individually unique for their purpose—for example, extended-use nasogastric feeding tubes or tubes for gastric suctioning of stomach contents.

- **Levin tubes** have only one lumen. If a vacuum forms, causing the tube to adhere to the stomach lining, the gastric mucosa may be damaged. Intermittent low suction is recommended for the Levin tube. As a small-bore tube, it is appropriate for administration of medication and nutrition. The Bard Levin tube is an example.
- **Salem Sump tubes** have a primary suction-drainage lumen and a smaller vent lumen. Continuous airflow through the vent lumen prevents a vacuum from forming. When a Salem Sump tube is used with

suction, the larger lumen is connected to the suction equipment. Intermittent high suction or continuous low suction may be used with this tube. The Salem Sump tube is generally used for removal of gastric contents, decompression, irrigation, and medication delivery. The Argyle Salem Sump tube and the Covidien Salem Sump tube are examples.

- **Extended-use nasogastric feeding tubes** are made of soft, flexible plastic material, with either a weighted or unweighted tip. The size is usually 8 Fr or 10 Fr. These tubes may require the use of a guidewire to facilitate insertion. Some manufacturers state that the tube may be left in place for up to 30 days before requiring replacement.
- **Gastrostomy tubes** are triple-lumen tubes with a gastric retention balloon, a port for distal duodenal feedings, and several esophageal, gastric, and proximal duodenal aspiration ports. This tube is surgically introduced through the abdominal and gastric walls. Gastrostomy tubes are indicated for decompression and simultaneous enteral feedings. The Moss Gastrostomy tube is an example.
- **Combination tubes** are 9 Fr nasojejunal feeding tubes combined with an 18 Fr gastric suction port lumen. The gastric port serves for decompression and drainage, and the other port allows for administration of medications. Use of fluoroscopy is recommended for placement. The manufacturer's instructions should be referenced for placement options. These tubes are commonly known as replacement gastrostomy tubes. One example is the Compat Gastrotube. It is a low profile tube intended for patients requiring intermittent or continuous enteral administration of nutrition, fluid, and medication.

Indications

Nasogastric intubation is indicated for:

- Treatment of gastric distention or gastric outlet obstruction.
- Assessment, including localization and treatment, of upper GI bleeding.
- Certain gastric and esophageal tests.
- Gastric lavage.
- Aspiration of gastric secretions.
- Administration of medications and feedings, including for neurological conditions that may or may not cause dysphagia.
- Emptying the upper GI tract before emergency surgery.
- Gastric decompression, including maintaining decompression after endotracheal intubation or major surgery. Tubes for this use are often placed through the mouth.
- Aspiration of gastric contents from recent ingestion of toxic material.
- Neurological conditions that may or may not cause dysphagia.
- Administration of radiographic contrast to the GI tract.

Insertion of an NG tube must be performed cautiously in pregnant patients and in patients with an aortic aneurysm, recent myocardial infarction, gastric hemorrhage, Zenker's diverticulum, or recent nasal surgery.

NG tube insertion is contraindicated in patients with nasopharyngeal or esophageal obstruction, severe maxillofacial trauma, severe uncontrolled coagulopathy, esophageal stricture or varices, recent banding or cautery of esophageal varices, or alkaline ingestion.

It is extremely important to avoid irritation and nasal erosion of nasal mucous by the NG tube when prolonged treatment is necessary.

Procedure

Prior to the procedure

Before to the procedure, the gastroenterology nurse should:

- Explain the procedure, risks, benefits, alternatives, and complications to the patient or the responsible adult representative and explain that the procedure may cause some nasal discomfort and possibly a gag reflex or watering eyes. Having the patient swallow during insertion will aid in advancement of the NG tube. The gastroenterology nurse may find it helpful to agree on a hand signal if the patient needs to briefly pause the insertion.
- Provide the patient with privacy during the procedure.
- Verify the patient's nothing by mouth (NPO) status as indicated by institutional policy.
- Remove dentures, partials, or removable bridges to prevent them from being dislocated if the patient gags. Verify if the patient has any loose teeth, bridgework, or oral piercing jewelry that is could be dislodged.
- Check the patient's nares. Question the patient about any previous nasal surgery, trauma, or a deviated septum. Inspect the nostrils with a penlight for any obvious obstruction. The optimal naris can be determined by occluding one nostril at a time while the patient breathes through the nose. The tube should be inserted in the nostril with the greatest airflow.
- Prepare the NG tube. In preparing the NG tube for insertion, the tube should be inspected for possible defects such as closed lumens or rough edges. The patency of the tube is checked by flushing it with water. If the tube is too stiff, the tube can be coiled around gloved fingers for a few seconds or dipped into warm water to increase its flexibility and ease the insertion. If the tube is too limp, it may be placed on ice for about 3 minutes. Some NG tubes, particularly feeding tubes, are placed during an esophagogastroduodenoscopy (EGD) through the working channel of the gastroscope. The scope channel must be large enough to accommodate the tube.

- Apply a topical anesthetic per institutional policy. Various topical anesthetic agents for nasogastric intubation have been found to be effective in relieving the pain that can be associated with the insertion of an NG tube and improving the success of the procedure. Topical anesthetics include:
  - Viscous lidocaine (sniff-and-swallow method) has the ability to reduce pain and the gagging sensation.
  - Lidocaine 1% or 4% administered via face mask (4 mg/kg; not to exceed 200 mg/dose in adults) should be preservative-free to minimize the risk of an allergic response.
  - Anesthetic sprays such as benzocaine or tetracaine/benzocaine can be sprayed onto the oropharyngeal or nasal mucosa. Incidences of methemoglobinemia after a single use of benzocaine topical sprays have been reported to the U.S. Food and Drug Administration (FDA).

During the procedure

To ease in the insertion of the NG tube, the gastroenterology nurse should stand at the patient's right side if the gastroenterology nurse is right-handed and left side if left-handed.

Drape the patient with a towel to protect his or her gown and offer the patient tissues to wipe secretions.

- Position the patient. Ideally the patient should be sitting upright with the head tilted slightly back, but the patient may also be in a high Fowler's or left lateral decubitus position. Small children may require bundling or holding for NG tube insertion. Children who are unable to cooperate by sitting up should be held supine.
- Insert the tube into the chosen nostril and rotate gently toward the center. The neck should be slightly flexed.
- Continue passage of the tube. When the tube is in the back of the patient's throat, the chin should be tipped toward the chest to close the trachea and open the esophagus. If the patient's condition or test permits, the patient may sip water through a bendable straw while the tube is guided gently down the esophagus and into the gastric antrum.
- Confirm the correct placement of the tube in the stomach. (See below for methods to confirm correct placement.)
- Tape the tube to the nose. The tube should not be taped until placement has been confirmed. Taping the tubing just below the naris (as opposed to in an upward direction) reduces friction and damage to the nose. Also, a manufactured tube holder may be used to secure the tube. Refer to the manufacturer's instructions for specific application directions.
- Secure the tubing. To secure the tubing, place tape around the distal part of the tube and pin the tape to the patient's clothing or gown. This will support the weight of the tube. Adequate slack in the tubing should be allowed so the patient has free movement and turning.
- Attach tubing to the appropriate device. Once proper placement is confirmed, the tube should be clamped or attached to intermittent suction, gravity drainage, or a feeding, as ordered by the physician.

In some patients such as those with a deviated septum, the tube can be placed orally. In such cases, the distal tip of the tube should be placed on the back of the patient's tongue. The patient should be asked to tip the chin toward the chest and swallow, keeping the upper and lower teeth slightly apart. It may be helpful to stroke the patient's neck to facilitate passage down the esophagus. The tube should be inserted gently with each swallow.

The gastroenterology nurse can pass an NG tube through the mouth in patients who are sedated or paralyzed by placing two to three fingers in the patient's mouth and into the oropharynx. The nurse can use the fingers to guide the NG tube into the hypopharynx. If the patient is unconscious or cannot swallow for any reason, the tube should be advanced between respirations. The gastroenterology nurse can also stroke the unconscious patient's neck to encourage the swallowing reflex and facilitate movement down the esophagus. Another method that assists with placement is to lift the thyroid cartilage anterior and upward, which opens the esophagus and allows passage of the tube into the proximal esophagus. An alternative practice includes tilting the chin toward the chest to close the trachea.

To confirm correct NG tube placement:

- Ask the patient to talk. If the patient cannot talk, the tube may be coiled in the throat or may have passed through the vocal cords.
- Use a tongue depressor and penlight to confirm that the tube is not curled in the mouth or throat, especially in an unconscious patient.
- Inject 10-30 ml of air through the tube and use a stethoscope placed over the epigastrium area to auscultate a "whooshing" sound just below the xiphoid process. If the patient belches, the tube may be in the esophagus.
- Attempt to aspirate the stomach contents. If no contents are obtained, change the patient's position to the left side in order to move the stomach's contents to the greater curvature and attempt to aspirate again. If unsuccessful, advance the NG tube 1-2 inches (2.5-5 cm) and attempt to aspirate again.
- Ensure the gastric pH is between equal to or <5 by testing the aspirate. This provides confirmation of placement of the gastric tube.
- Obtain an X-ray examination or fluoroscopy.

If the patient coughs frequently or experiences dyspnea during insertion, the tube should be removed immediately because it may be in the trachea or coiled in the pharynx. If the tube is malpositioned in the trachea, the patient could aspirate. Water without bubbles does not confirm appropriate placement. The end of the tube should never be placed in water.

After the procedure

Caring for the in-place NG tube includes the following:

- Check patency. Patency of the tube should be confirmed periodically by irrigating the tube with normal saline. The frequency of irrigation and the amount of solution should be specified by the physician. Fluid inserted during irrigation should be removed, measured, and examined. A measurement of the gastric pH should be done, ensuring it is equal to or <5.0.
- Clean the nares and mouth. The nares should be cleansed, old tape should be removed, and the tube retaped as necessary. Mouth care should be provided every 8 hours or as needed. Depending on the patient's condition, lemon-glycerin swabs may be used to clean the teeth, or the patient may brush. The patient's lips can be coated with lip balm to prevent dryness. The patient may be allowed to chew gum, suck on hard candy, or use throat lozenges to promote comfort, if indicated by the patient's condition and the physician's order. Care should be taken to avoid irritation and erosion of nasal mucus when prolonged treatment is necessary.
- Check tube function regularly. When nasogastric suction is used, bowel sounds should be assessed regularly to check GI function. Nausea, vomiting, a feeling of fullness, epigastric discomfort, and distention may be indications that the tube is not patent.
- Observe and document secretions. The color, consistency, and odor of gastric drainage should be observed. Examine the aspirate and measure gastric pH, ensuring it is equal to or <5. Normal gastric secretions are colorless or yellow-green from bile and have a mucoid consistency. Gastric drainage that has the color of coffee grounds may be an indication of bleeding and should be reported immediately.
- Assess patient tolerance. Patients should be observed for fluid and electrolyte imbalances. In critically ill patients, the tape should be periodically removed, and the tube rotated gently, then retaped. This helps prevent serious mucosal breakdown. Repositioning the tip of the tube may lessen the chance of gastric bleeding due to chronic suction in one area.
- Properly instill medications. If medications are to be instilled through the tube, the tube should be irrigated with at least 20-30 ml of water before and after medication instillation. Suction should be omitted for 30-45 minutes after instillation to permit absorption of the medication. The patient should be positioned with the head of the bed elevated during this process.

Removing the NG tube includes these steps:

- Drape the patient's chest with a towel prior to removal.
- Flush the NG tube with 10 ml of air before removing the tube to clear it of stomach contents that could cause irritation during removal.
- Place the patient in a semi-Fowler's position.
- Untape the tube from the patient's nose.
- Instruct the patient not to breathe to ensure closure of the epiglottis.
- Withdraw the tube gently and steadily until the distal end reaches the nasopharynx, at which point it can be pulled quickly.
- If possible, cover and quickly remove the tube from in front of the patient.
- Assist the patient with mouth care and clean tape residue from the nose with adhesive remover.

Considerations

If signs of GI dysfunction recur after removal of the NG tube, reinsertion of the tube may be necessary.

Potential complications of NG tube insertion include respiratory distress caused by incorrect tube placement, pulmonary aspiration, epistaxis (nosebleed), esophageal perforation, necrosis of the nasal mucous membrane caused by incorrect taping of the tube, skin erosion at the nostril, sinusitis, esophagitis, esophagotracheal fistula, gastric ulceration, pulmonary and oral infection, and esophageal or gastric hemorrhage or perforation. Otitis media may be caused by eustachian tube irritation.

The use of suction can cause electrolyte imbalance and dehydration.

GASTRIC LAVAGE

Gastric **lavage** involves insertion of a tube through the nose or mouth into the stomach to irrigate and remove gastric contents.

A smaller-diameter nasogastric tube is used to localize the bleeding. A large-bore tube (often in sizes 30 Fr to 36 Fr) is placed orally for instillation of aliquots of fluid and removal of residual gastric contents. This process forcibly cleans the area, allowing for clearer vision when intubating with an endoscope and the ability to localize the area of concern.

A number of gastric tubes are available, including:

- A **double-lumen orogastric "stomach pump" tube** is indicated in situations where intermittent gastric lavage and evacuation are required. The larger lumen is used for evacuation of gastric contents, and the smaller lumen is used for instillation of an irrigant. This tube is especially suitable for emergency removal of toxic agents, overdose of oral medications, or ingestion of hazardous substances. Activated charcoal slurries may be administered through the large suction lumen. In situations where there is a high risk of aspiration—such as loss of consciousness, seizures, or delirium—a cuffed endotracheal tube should be inserted before the double-lumen tube.
- A **single-lumen tube** has several openings at the distal end and is usually passed orally but can be inserted nasally. Single-lumen tubes allow rapid lavage and evacuation of large volumes of fluid, but

continuous irrigation is not possible because the same lumen must be used for both instillation and evacuation of fluid. During an emergency, the single-lumen tube may be used to aspirate large amounts of gastric contents quickly.

Indications

Gastric lavage is indicated in patients with acute GI bleeding, when preparing the stomach for endoscopy after barium or food ingestion, and for evacuating the stomach after ingestion of toxic substances.

In patients with acute GI bleeding, gastric lavage gives a good indication of the rapidity of bleeding, localizes the source, and cleanses the stomach before endoscopy. In patients with cirrhosis, the procedure also removes blood to lessen the likelihood of hepatic encephalopathy.

Gastric lavage is controversial for the routine management of a GI hemorrhage. It is used more as a diagnostic tool and preparatory phase before treatment. The physician weighs the risks against the benefits when making this decision.

When gastric lavage is performed in conjunction with an EGD, an overtube may be used to isolate the esophagus from the airway. This is particularly advisable if the patient is sedated because it decreases the potential for aspiration of gastric contents or lavage solution.

Gastric lavage is contraindicated in patients with possible esophageal or gastric perforation, known esophageal obstruction, or maxillofacial trauma. It is also contraindicated after ingestion of corrosive substances such as lye and some cleaning compounds because the nasogastric tube may perforate the esophagus.

Procedure

Prior to the procedure

The setup required for gastric lavage includes a container of irrigation solution attached to tubing ending in a Y-connector. One side of the Y-connector is attached to the patient's nasogastric or orogastric tube, and the other side is attached to another piece of tubing that is connected to a calibrated collection container.

Before inserting a gastric tube for lavage:

- Obtain baseline vital signs.
- Ensure an adequate airway.
- Establish a large-bore IV line for volume replacement, as ordered.
- Inform the patient that the tube may cause some gagging, but that he or she will still be able to breathe. Emphasize the importance of remaining on the left side during the procedure.

During the procedure

Inserting the gastric tube includes these steps:

- Provide frequent patient reassurance and support during the procedure.
- Place the patient in the left lateral position with the head of the bed at 15 degrees. This will allow the tube tip to lie in the greater curvature of stomach. If an NG tube is being used, the patient may be in semi-Fowler's position.
- Insert the gastric lavage tube in the same manner as an NG tube. If the tube is inserted orally, the well-lubricated tube should be guided into the back of the mouth while the patient is encouraged to suck on the tube and swallow.
- Check tube placement by injecting 20-40 ml of air (5 ml in children) while auscultating the epigastric area with a stethoscope.
- Aspirate stomach contents before instilling any irrigation solution to ensure correct placement. Test the pH of the aspirate.

If the patient begins to cough or experiences dyspnea, the tube may be in the trachea. The tube should be removed immediately, and another insertion attempted.

Performing gastric lavage includes the following:

- The irrigating solution container is hung from the highest level of an IV pole.
- The inflow tube is unclamped, and 250 ml of irrigation solution is instilled gradually to evaluate the patient's tolerance and prevent vomiting. Iced saline lavage is no longer recommended because it has a tendency to lower the patient's core temperature and stimulate the vagus nerve, triggering hydrochloric acid (HCL) secretion. To prevent excessive stimulation of gastric motility, always flush with room temperature fluids.
- After instillation, continuous negative pressure should be applied with a syringe or by removing the clamp on the outflow tube, allowing the fluid to flow via gravity into a collection container that is lower than the patient's head.
- The procedure should be repeated, increasing the amount of irrigation solution to 500 ml until the return is clear or of acceptable clarity. In patients with upper GI bleeding, lavage should be continued until bleeding stops or until it becomes evident that other measures will be necessary.
- If the amount instilled is significantly greater than the amount recovered, the tube should be repositioned.
- When lavaging the stomach after ingestion of poisons or drugs, all return fluid should be saved for possible laboratory analysis.

When the return is of acceptable clarity, the tube should be withdrawn slowly. Instruct the patient to exhale slowly as the tube is withdrawn.

Considerations

The patient should never be left alone during gastric lavage. The patient should be observed continuously for changes in level of consciousness, and vital signs should be monitored frequently. Throughout lavage, frequent suctioning of the oral cavity may be needed to prevent aspiration.

The most dangerous risk of gastric lavage is aspiration of gastric contents. If the airway is not protected, pneumonia may result. This is likely to happen in patients who are groggy, as their airway is not protected and their gag reflex is lessened. Additional risks are bradyarrhythmias, hypoxia, and laryngospasm. If the gastric tube becomes clogged, unrelieved gastric distention may occur, potentially resulting in fatal shock.

NASOBILIARY AND NASOPANCREATIC CATHETERS

Nasobiliary catheters (NBCs) and **nasopancreatic catheters (NPCs)** are used for short-term decompression or perfusion within the biliary and pancreatic ductal systems. As a result, the transpapillary pressure gradient is reduced, improving the transpapillary flow and reducing bile leakage.

These catheters are long, thin polyethylene tubes that are placed endoscopically over a guidewire into the common bile duct or the pancreatic duct. In this procedure, the use of the terms **proximal** and **distal** are used opposite from other procedures. The distal end of the catheter is placed within the duct of choice. The proximal end of the tube is routed from the duct, through the duodenal bulb, stomach, and esophagus, exiting the patient's mouth. It is then rerouted from the patient's mouth, through the patient's nose, and connected to a drainage bag.

The tips of these catheters vary. NBCs come with either a pigtail or straight tip, whereas NPCs only come in a straight tip. Both catheters have side ports that are placed within the duct to facilitate drainage and to prevent reflux from the duodenum.

Indications

Indications for NBC placement:

- Decompression of an obstructed bile duct in acute suppurative cholangitis.
- Temporary or short-term decompression of the common bile duct, similar to that which follows unsuccessful stone extraction after endoscopic sphincterotomy.
- Prevention of stone impaction after endoscopic sphincterotomy.
- Infusion of contrast medium for repeat cholangiography.
- Instillation of various therapeutic solutions, including medications for gallstone dissolution, antibiotics for acute bacterial cholangitis, corticosteroids for sclerosing cholangitis, or saline to flush sludge or small stones after endoscopic sphincterotomy.
- Preoperative biliary decompression to decrease jaundice, reducing the risk of perioperative complications in patients undergoing elective biliary tract surgery.
- Temporary biliary decompression in patients who are septic or have severe coagulopathy. Once infection has been controlled or coagulopathy corrected, sphincterotomy can be performed, and a large stent can be inserted for long-term therapy.
- Access for intraluminal radiation therapy.
- Aspiration of bile for chemical and bacteriological studies. For example, to identify causative agents in bacterial cholangitis or to determine the lithogenicity of bile in patients with cholestasis.
- Facilitate the healing process in traumatic or surgical biliary fistulas.
- Management of common bile duct stones. The NBC is used to outline the common bile duct for targeting the stones.

Risks associated with NBC catheter placement include tube displacement, vasovagal reaction, epistaxis, and collapse of the tube by twisting. There may also be a loss of electrolytes and fluid or discomfort for the patient. Most patients can tolerate an NBC for up to 14 days.

NBC placement is contraindicated in patients with coagulopathy, sepsis, active pancreatitis or other infection, recent food ingestion, or any other contraindication to EGD. Some endoscopists will insert an NBC in the presence of the first three contraindications if leaving the patient untreated would cause further harm.

Indications for NPC placement:

- Repeated or traumatic cannulation of the pancreatic duct.
- Decompression and drainage of a pancreatic pseudocyst or ductal obstruction.
- Post-endoscopic balloon or catheter dilation or endoscopic sphincterotomy of the pancreatic duct.
- Treatment of a pancreatic duct stone during extracorporeal shock-wave lithotripsy.
- Infusion of contrast medium for repeated pancreatogram without scope insertion.

Procedure

During the procedure

The procedure endoscopic retrograde cholangiopancreatography (ERCP) is performed with the side-viewing scope. A cholangiogram and/or pancreatogram are obtained. Once a diagnosis is confirmed, and a decision to place an NBC or NPC is made, nasobiliary or nasopancreatic catheter placement is conducted during the ERCP.

- Performing endoscopic sphincterotomy is optional.
- The duct of choice is cannulated, and a guidewire is inserted through the existing cannula.
- Once the guidewire is properly positioned, the cannula is withdrawn by the endoscopist using fluoroscopic guidance, while the gastroenterology nurse continues to feed or advance the guidewire

to maintain wire position within the duct. The gastroenterologist and the assistant must be diligent in matching movements to ensure the wire does not perforate the liver capsule, perforate the substance of the pancreas, or loop in the duodenum (which could cause the guidewire to whip out of the duct).

- The portion of the guidewire that protrudes from the duodenoscope is wiped with a moistened gauze pad.
- The nasocatheter of choice is flushed with sterile saline or sterile water before passage so that it slides more easily over the guidewire. The nasocatheters are supplied in 7 Fr and 10 Fr.
- The endoscopist advances the catheter over the guidewire, through the scope, and into position within the duct as the gastroenterology nurse applies gentle back pressure to the guidewire. Fluoroscopy is used to check guidewire and catheter position throughout placement.
- Once the NBC or NPC is in proper position within the desired duct, the guidewire is removed.
- The endoscopist begins to advance the catheter through the biopsy and accessory channel as the gastroenterology nurse withdraws the endoscope in 1-to 2-cm increments. The NBC or NPC should remain stationary throughout the scope removal.
- When the end of the endoscope is visible in the patient's mouth, the endoscope and mouth guard are removed.
- At this point, the NBC or NPC proximal tip is within the duct of choice, and the proximal portion exits the patient's mouth.
- The NBC or NPC is then rerouted through the patient's nose to allow normal ingestion of food and liquid while the nasocatheter is in place.
  - A specially designed 25-30 cm, 14 Fr nasopharyngeal tube is advanced through the nose and brought out through the patient's mouth.
  - The end of the NBC or NPC is threaded inside the oral end of the nasopharyngeal tube and advanced until it exits through the nasal end of the larger tube.
  - The tube in the oropharynx is held firmly by the endoscopist or assistant to maintain its position during withdrawal through the nose, thereby preventing dislodgement of the catheter from the bile or pancreatic duct.
  - The nasopharyngeal tube is then withdrawn slowly through the nostril until the tip of the NBC or NPC emerges from the nose.
  - The nasopharyngeal tube is discarded.
- The excess portion of the NBC or NPC at the nose is transected, and a Luer Lock valve is attached, creating an adapter for the selected biliary tract drainage system.
- The end of the NBC or NPC is taped to the patient's cheek, avoiding sharp angles and kinks.
- The NBC or NPC is taped to both the adapters and a drainage bag on the catheter, and the end of the apparatus is secured to the patient's gown, allowing enough slack in the tubing to prevent traction when the patient turns his or her head.

Following insertion of the NBC or NPC, it is important to monitor and record the patient's vital signs, observe the patient for abdominal pain or distention, and administer antibiotics as ordered.

The patient may experience some minor throat discomfort, and green or yellow fluid may appear in the catheter. The patient may be permitted a diet as ordered 2-4 hours after the procedure. Some movement of the catheter is to be expected during eating and drinking.

To check catheter patency, catheter output is strictly monitored and documented. The physician should be notified if no output is recorded, as this could indicate a catheter position shift or a resolution of a leak if the catheter has been used for bile duct decompression in common bile duct injuries. On occasion, bile collected from this catheter will be fed back to the patient to prevent bile salt depletion.

If irrigations are ordered to maintain patency, they should be performed with slow, gentle pressure. For the NBC, this will avoid driving stone fragments or infected bile into the intrahepatic ducts. Using excessive pressure when irrigating an NPC could force pancreatic secretions into the acini of the pancreas, leading to pancreatitis.

Considerations

To provide continuous access to the biliary tree or to the pancreatic duct only, a specialized catheter with mesh should be used to preventing clogging.

In addition to complications associated with endoscopic sphincterotomy, insertion of an NBC or NPC could result in blockage of the catheter, nasal irritation, or sore throat.

Minor nose or throat complications can be managed by proper care of the nostril and use of throat lozenges and mild analgesics. To prevent mucus from plugging the side holes of the NBC or NPC, it may be helpful to irrigate the NBC or NPC with 10 ml sterile saline every 3-4 hours. To straighten out a kink, a guidewire may be passed through the entire NBC or NPC under fluoroscopic control, followed by irrigation. If this fails, the NBC or NPC must be replaced.

BILIARY STENTS

To provide palliative biliary drainage, a thin, hollow Teflon or polyethylene tube can be placed endoscopically during ERCP (Figure 29.1). The objective of a **biliary stent** or endoprosthesis is to create a bridge between two normal anatomical sites that bypasses the diseased or obstructed portion of the duct.

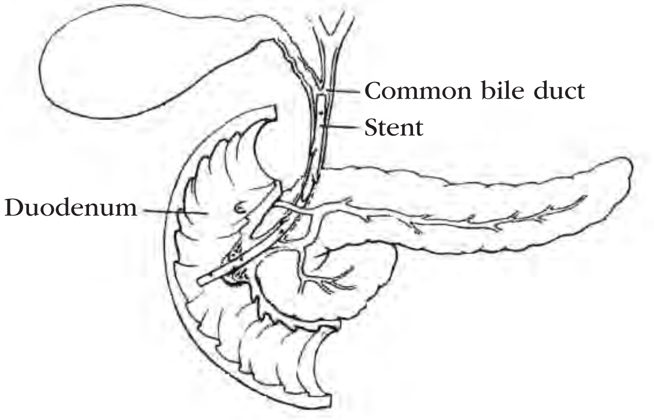

Figure 29.1. Straight Stent Placement in the Common Bile Duct Draining into the Duodenum

Two basic types of biliary stents are currently in use: the barbed stent and the pigtail stent. The scope channel size must accommodate the diameter of the stent. Each stent has a number of side holes to facilitate drainage.

- The **barbed stent** has projections, or "barbs," at each end that result from a diagonal cut of the stent wall. The length of the stent between the barbs may be 5-15 cm (2-6 inches). Barbed stents range in size from 7 Fr to 12 Fr and can be inserted through a lateral-viewing endoscope. Barbed stents are used primarily for strictures of the common and/or pancreatic duct. Barbed stents have been shown to decrease morbidity; maintain their position with infrequent dislodgement; provide maximal flow; and decrease the risk for occlusion by sludge, stones, or tumor.
- The **pigtail stent** has one or both ends of the tube that are coiled. This coiled shape disappears when the stent is pulled taut but quickly returns when allowed to relax. Pigtail stents range in size from 5 Fr to 7 Fr and can be introduced through a lateral-viewing endoscope. Because of their small diameter, these stents provide inadequate drainage in many cases and are used primarily to facilitate drainage when unremovable intraductal stones are present.

Regardless of the type of biliary stent used, one end of the tube is seated above the ductal obstruction, and the other end protrudes into the duodenum. The configuration of the stent itself serves to secure it. The purpose of the stent is to allow free drainage or transhepatic **decompression**. Once the continuity of the duct is reestablished, the patient's condition should improve.

Indications

Indications for a placement of a biliary stent include:

- Relief of obstructive jaundice in patients with benign or malignant strictures of the bile duct.
- Palliative treatment of inoperable or metastatic pancreatic or periampullary neoplasms.
- Preoperative decompression to decrease complications associated with high bilirubin levels.
- Prevention of stone impaction in the ampulla in surgical candidates who have had unsuccessful endoscopic sphincterotomy and stone extraction.
- Maintenance of biliary decompression in cases of sclerosing cholangitis with stricture of the extrahepatic bile ducts.
- Protection of the area into which a biliary fistula drains.
- Post-cholecystectomy biliary leak of the common bile duct (CBD).

Procedure

Prior to the procedure

Before the procedure, it is important to determine whether the patient has had biliary surgery in the past. At least two gastroenterology staff members will assist with the procedure: one gastroenterology nurse monitors the patient, and one gastroenterology nurse assists the endoscopist. Endoscopy staff should review labeling on the stent packaging, which reflects circumference first and length second.

ERCP is performed first to define the level and extent of the obstruction. A sphincterotomy may be performed to allow easier passage of the stent (especially larger stents) and/or to facilitate drainage.

During the procedure

The procedural maneuvers are all performed under fluoroscopic guidance.

When placing a 5 Fr or 7 Fr stent:

- A cannula is passed through a side-viewing duodenoscope with at least a 2.8-mm instrument channel and into the diseased duct.
- A 480-cm guidewire is passed through the cannula.
- The cannula is removed over the guidewire, leaving the guidewire in place.
- The portion of the guidewire that protrudes from the duodenoscope is wiped with a gauze pad dampened with sterile saline or sterile water to remove residual contrast medium.
- The stent is advanced over the guidewire, followed by a pusher tube that is the same diameter as the stent but of a different color.
- Once the stent has bypassed the obstructed area, the pusher tube and the guidewire are slowly removed.

At this point, the distal end of the stent anchors itself in the duct, the midportion of the stent traverses the obstruction, and the proximal end lies free in the duodenum.

When placing a 10 Fr, 11.5 Fr, or 12 Fr stent:

- A guide catheter is used. A guide catheter is a long, polyethylene tube with a tapered tip. Most guide catheters measure 350 cm in length and 6 Fr to 8

Fr. Guide catheters with increasing diameters are available to progressively dilate strictures in order to pass larger stents. The guide catheter has radiopaque markers on one end, which must be passed over the guidewire first. If not, the endoscopist cannot use the markers for proper stent placement. Special attention is warranted if repeated attempts are made to place the stent.
- The guide catheter is passed over the guidewire through the channel of a 4.2-mm-channel duodenoscope.
- The endoscopist guides the catheter through the stricture.
- A stent of predetermined size is passed over the guide catheter, followed by a pusher tube of the same size. The pusher tube should slide freely over the guide catheter but not over the stent.
- The stent is advanced over the guide catheter, into the common bile duct, and through the stricture. The properly positioned stent should protrude 1 cm into the duodenum.
- The gastroenterology nurse must assist the endoscopist with coordinated movements and constant tension on the guidewire and guide catheter.
- After proper placement is determined, the pusher tube, guide catheter, and guidewire are slowly removed.
- Bile mixed with contrast medium should immediately escape through the stent.

Because endoscopic placement of biliary stents can be a lengthy process, as little fluoroscopy as possible should be used during the procedure. To simplify stent placement, decrease fluoroscopy time, and shorten overall procedure time, manufacturers have developed one-step stent kits. These stent-placement devices are preassembled guiding catheters and pusher tubes that feed directly over the guidewire. Some are preloaded with a stent as well. Those without a stent must have a stent backloaded onto the distal tip of the delivery device (the end that will be put through the scope) until it is butted against the pusher tube, before passage over the guidewire and into the duct. Once the guiding catheter is advanced into the duct, the pusher tube is freed from the locked position with a twisting motion so that the endoscopist can "push" the stent up the duct into position. The gastroenterology nurse facilitates the stent delivery by pulling back on the guiding catheter as the stent is advanced into position. Once the stent is in place, the endoscopist holds the stent with the pusher tube as the guiding catheter and guidewire are removed.

There are also systems that use short guidewires and have locking devices that attach to the biopsy channel. These systems hold the guidewire firmly in the biopsy channel, allowing the assistant freedom to assemble other devices without the accidental displacement of the guidewire. This system can also be used with a long guidewire.

Considerations

Considerations include:
- Before the procedure, the gastroenterology nurse should document any existing pain.
- Before and during the procedure, monitoring should include continuous electrocardiogram (ECG).
- During and after the procedure, the gastroenterology nurse should monitor and record the patient's vital signs and observe the patient for abdominal pain or distention.
- Prophylactic antibiotics may be considered prior to placement. According to the American Society for Gastrointestinal Endoscopy (ASGE, 2015), prophylactic antibiotic use may be appropriate for patients who are at high risk for endocarditis and have primary sclerosing cholangitis, bile duct obstruction, and previously manipulated ducts.
- Serial blood studies are usually performed after placement of large stents in order to follow the patient's progress and monitor the effectiveness of decompression. The serum bilirubin level usually declines progressively until nearly normal.
- The patient may be observed in the hospital or on an outpatient basis following the procedure. If a sphincterotomy has been performed, overnight hospital observation is recommended for high-risk patients, patients with complications during the procedure such as bleeding, and patients with recovery issues such as difficult pain management.
- The patient usually remains NPO for 2-4 hours after ERCP. The diet is then increased slowly as tolerated, preferably with a clear liquid diet for the first 24 hours post-ERCP.
- Some endoscopists recommend changing a stent every 3-6 months to avoid obstruction or breakage of the stent material. Because the proximal portion of the stent protrudes into the duodenum, it can be withdrawn endoscopically by using a snare or stent-retrieving device, and it can be replaced fairly easily.
- After biliary stent placement and before discharge, patients should be instructed to notify their physician when any symptoms of stent occlusion occur, such as pruritis, fever, pain, or jaundice.

Failure to place a stent properly is most commonly caused by either a tumor involving the papilla of Vater, in which case the bile duct orifice cannot be identified, or hepatic neoplasms that cause a tight stricture of the intrahepatic ducts.

Potential early complications of stent placement include bleeding from the sphincterotomy or a tumor, cholangitis, pancreatitis, trauma to the biliary tract or duodenum, and obstruction of the pancreatic duct.

Recurrent jaundice is the most common delayed complication of biliary stent placement and is usually caused by a clogged stent. Rarely, it may be a result of tumor growth along the prosthesis or metastatic spread of a malignant tumor to the liver. If occlusion of the prosthesis by cell detritus and viscous bile causes recurrent cholestasis and cholangitis, the endoprosthesis can be extracted with the aid of a polypectomy snare and replaced by a new prosthesis during the same session. Other delayed complications can include migration of the stent, erosion of the duodenal wall with ulcer formation, or duodenal perforation.

PANCREATIC STENTS

In addition to pain relief from strictures, stones, or papillary stenosis, stenting of the pancreatic duct is beneficial in the relief of ductal obstruction. The placement and deployment of stents in the pancreatic duct, the most common treatment for patients with biliary and/or pancreatic obstruction, can be life-changing.

Indications

A **pancreatic stent** placement is indicated for:
- Unresolved pancreatitis.
- Idiopathic acute pancreatitis.
- Pancreas divisum with symptomatology.
- Pancreatic duct disruption: traumatic, carcinoid, and idiopathic.
- Prevention of post-ERCP pancreatitis.
- Chronic pancreatitis.
- Pancreatic strictures and/or stones.
- Sphincter of Oddi dysfunction.

Pancreatic stents are made from radiopaque polyethylene and come in 3 Fr, 5 Fr, and 7 Fr. The length is available from 3 cm to 15 cm. The stent is measured from the distal barb, which is positioned within the duodenum, to the proximal tip of the stent, which is positioned within the duct.

Because of anatomical variations between the pancreatic duct and the biliary system, equipment has been modified for pancreatic therapeutics, as pancreatic and biliary stents are not interchangeable. To facilitate drainage, both pancreatic and biliary stents can be placed during the same procedure to facilitate difficult drainage.

Procedure

Placement of a pancreatic stent is similar in technique to the insertion of a biliary stent.
- The pancreatic duct is cannulated using the endoscopist's preferred catheter.
- A pancreatogram is obtained, and a diagnosis is confirmed.
- The endoscopist approves gentle flushing of the cannulating catheter with sterile saline or sterile water to clear the catheter of contrast medium and allow easier advancement of the guidewire. The endoscopist should clarify the amount of sterile saline or water to be used. It is important to note that, when flushing the catheter, direct visualization with fluoroscopy is performed to decrease the possibility of duct overfill.
- The guidewire of choice is advanced within the pancreatic duct. Choices in guidewires need to be considered to accommodate proper advancement of the catheter. For example, if a long tapered or dilating catheter has been used, a guidewire with a smaller diameter may be necessary.
- Once the guidewire position is confirmed, the endoscopist withdraws the catheter as the gastroenterology nurse advances the guidewire to maintain its duct position. This is a synchronized effort in which communication between the endoscopist and the gastroenterology nurse plays a key role.
- The protruding guidewire is wiped with moist gauze to remove any remaining contrast medium and to provide ease of insertion.
- The preselected 3-Fr, 5-Fr, or 7-Fr stent is passed over the guidewire.
- A pusher tube follows the stent, pushing the stent along the guidewire and up into the pancreatic duct.
- As the endoscopist inserts the stent, the gastroenterology nurse applies gentle back pressure on the guidewire.
- Fluoroscopy is used intermittently throughout.
- With the stent position verified, the guidewire is removed as the pusher tube holds the stent in place. Then the pusher tube is removed.

If a 10-Fr stent is the stent of choice, the addition of a guide catheter is necessary, just as with biliary stents. The procedure order then becomes: 1. guidewire placement; 2. guide catheter over the guidewire; 3. stent over the guide catheter; and 4. pusher tube to push the stent along the guide catheter.

Considerations

Post-procedure care is consistent with that of biliary stenting.

Complications associated with pancreatic stenting include abdominal pain and mild pancreatitis. Sepsis rarely occurs. Clogging of the stent may occur anywhere from 6 weeks to 4 months from placement date. Stent migration, fracture, and dislocation are relatively rare complications associated with pancreatic stenting. Prolonged stent placement causes ductal changes similar to chronic pancreatitis, but these changes appear to be temporary and resolve once the stent is removed.

SELF-EXPANDING STENTS

Self-expanding plastic stent (SEPS)

Self-expanding plastic stents (SEPSs) for use in the esophagus are made from polyflex, a polyester self-expanding braid that is completely covered in a silicone membrane with embedded radiopaque markers to facilitate accurate placement. Its flared ends ensure complete occlusion while preventing ingrowth and overgrowth of granulation tissue, making it easy to reposition and retrieve.

Currently, SEPSs are available in various sizes and diameters to ensure adequate occlusion. The exact size of the stent chosen will depend on the site and the size of the leakage, the operator's preference, and the size of the esophagus.

Self-expanding metal stent (SEMS)

Self-expanding metal stents (SEMSs) are intended to help hold a structure open.

SEMSs are available in a closed-cell or open-cell type and are further classified as crosswire, hookwire, and hook- and crosswire structures. Classification is largely dependent on the weave of metallic fibers. The majority of deployed stents are the closed-cell type, taking advantage of its ability to keep its shape and prevent shortening.

SEMSs are available as uncovered and covered stents.

- Uncovered SEMSs are prone to tumor ingrowth, as they become completely buried inside the tumor with the intent of preventing tumor growth. If tumor ingrowth occurs, an additional stent (metal or polyethylene) may be placed within the existing stent to prevent further tumor growth. The disadvantages of uncovered SEMSs are that it is impossible to change or remove them once placed and that early occlusion can occur due to tumor ingrowth.
- Covered stents have a greater potential for migration. Manufacturers of covered stents compensate by flaring either end of the stent to anchor it in place and/or by only covering the center column of the stent, leaving the ends uncovered so they will bond with the GI tract. These features are important because covered stents are more effective in the treatment of tracheosophageal fistulas (TEFs).

While covered SEMSs are more likely to decrease occlusion, there has been no data to support prolonged patency between the covered and uncovered types.

Indications

As management for malignant esophageal strictures or perforation, SEPSs provide an alternative to surgery alone.

SEMSs are currently available for esophageal, tracheobronchial, duodenal, biliary, and colonic endoscopic placement for palliative care.

Due to medical and technological advancements, SEMSs and SEPSs are now being inserted endoscopically for relief of benign or malignant obstructions of the biliary tract. Deployment of the SEMS is treatment for benign obstruction, but in the majority of patients, deployment of the SEMS is palliative for those diagnosed with an obstructing neoplasm.

Most symptoms of GI cancers come from the obstructive nature of the tumor, which prevents natural passage of oral intake through the GI tract or blocks bile flow. Placed within a malignant stricture, the SEMS, with its wire memory, expends a greater radial force against an encroaching tumor. Maintaining luminal patency decreases symptoms, allows continued oral intake, and thus improves the patient's quality of life.

Procedure

SEMSs are loaded, compressed, and stretched on their own delivery devices. As a result, some shorten 25% to 50% on deployment.

Before stent placement, the stricture must be located, measured, and marked—either internally with contrast injection or instillation, or externally with radiopaque markers. Dilation of the area to be stented may be done just before placement to ensure that the stent delivery device can pass through the stricture, deploy the stent, and be removed successfully. The one absolute exception to this is colonic stent placement. Pre-placement dilation of colonic neoplasm is contraindicated because of the potential for perforation or rupture.

Each manufacturer's stent design and delivery device is slightly different. Until the endoscopist and endoscopy staff are experienced and comfortable with the stent and its delivery device, consider having a manufacturer's representative present during placement.

Stent deployment should be slow, with diligent attention to the fluoroscopic image and endoscopic image where applicable. When placing a stent across the gastroesophageal junction in a distal biliary or ampullary stricture with a markedly dilated biliary tree above or in a colonic stricture with significant colonic dilation proximal to the malignant obstruction, the stent has the tendency to shoot above the stricture into the dilated area. Depending on the stent and the device used, the stent may be recaptured and/or repositioned if 50% or less has been deployed.

Special considerations for placement of esophageal stents include the following:

- Esophageal stent placement should be placed at least 2-3 cm below the cricopharynx to avoid interfering with the patient's airway. Placing the stent too high in the esophagus will cause coughing and gagging. It may also result in poor patient tolerance because of odynophagia or a constant sensation of an esophageal foreign body.
- If the stent bridges the gastroesophageal junction, the patient should be instructed to follow antireflux measures and should be started on antireflux medications.

Considerations

Patient instruction should include reassurance that the metal stent will not set off a metal detector at the airport. Manufacturers of these stents provide endoprosthesis registration forms to track devices after stent implantation.

Instructions following SEMS placement vary by location of the stent.

- **Esophageal SEMS placement.** Patients should be instructed to notify their gastroenterologist about recurrent dysphagia, significant dyspnea, or persistent cough, as these may indicate a fistula, tumor ingrowth, stent migration, or aspiration pneumonia.

- **Biliary SEMS placement.** As with other biliary stents, patients should be instructed to notify their gastroenterologist if symptoms of recurrent obstruction occur, such as pruritis, pain, or jaundice.
- **Colonic SEMS placement.** Bowel control measures and dietary modifications are recommended based on stent location.
- **Sigmoid or left colon SEMS placement.** A daily stool softener, increased oral fluid intake, and a regular diet minimizing fresh vegetables and tough meats are recommended. For each day the patient does not have a bowel movement, milk of magnesia or one tablespoon of mineral oil should be taken.
- **Rectal SEMS placement.** A daily stool softener, a daily tablespoon of mineral oil, and a soft diet without any fresh vegetables are recommended. An enema is indicated after 2 days with no bowel movement or if a pressure sensation is experienced in the rectum.

One distinct advantage to the SEPS is that immediate occlusion allows earlier oral/enteral feeding, which may decrease extended hospital stays. The high migration risk, along with challenges in assembly, may make the SEPS less favorable over the covered SEMS for relieving an esophageal obstruction.

Uncovered metal stents are difficult to remove, as they may become embedded in the tissue and typically require surgical removal. Because stents do not expand to full diameter for hours once placed in body, this presents an opportunity for the endoscopist to remove covered and coiled stents after placement if location or stent patency is questionable.

TEMPORARY EXPANDABLE ESOPHAGEAL STENTS

Temporary expandable stents can be made of metal, plastic, or biodegradable material. The flared ends of the stent can be sutured into place with a blue removal suture on the proximal end to facilitate stent removal. Expandable polyester silicone-covered stents can be removed or repositioned. Temporary stents tend not to be self-expanding, as they are not intended to have tissue embedment.

Indications

Benign esophageal strictures are thought to occur from deep esophageal injury. This injury then stimulates the production of fibrous tissue and deposition of collagen. The damage to the esophagus involves the submucosa and occasionally the muscularis propria. Stricture development necessitates repeated dilation using bougie or balloon dilators. When the esophageal stricture does not respond to conventional therapy, a temporary stent can be used.

Temporary expandable esophageal stents are especially useful in treating refractory benign esophageal strictures, which were historically treated with repeated dilations. With the continuous expansive pressure, the stricture will relax. The length of time varies, but the longest recommended length of placement for the stent is 6 weeks.

Procedure

The insertion is done over a guidewire, which is passed during endoscopy. The stricture is marked with contrast using an injection needle. There are different versions of temporary expandable stents available. One version must be loaded into the introduction sheath by hand, taking care to not deform the stent. The apparatus then resembles a large Savary-Gilliard dilator. Another pre-loaded version is less complex to use.

The stent can be repositioned or removed using a rat tooth forceps. When longitudinal tension is applied to the stent, it narrows easy removal or repositioning.

Considerations

Complications include post-procedure chest pain, stent migration, bleeding, perforation, stent occlusion due to a bolus of food, gastroesophageal reflux, esophagitis, edema, mucus membrane ulceration, fever, fistula development, and sepsis.

COLON DECOMPRESSION

Colon decompression involves placement of a tube into the rectum or colon to relieve colonic distention.

Indication

Colon decompression is indicated in patients with colonic pseudo-obstruction (nontoxic megacolon or Ogilvie's syndrome), post-operative ileus, or colon distention secondary to flexible sigmoidoscopy or colonoscopy. Ogilvie's syndrome, also known as acute colonic pseudo-obstruction, commonly occurs in elderly patients who have a pre-existing disease that necessitates bedrest. While this rare disorder appears as an obstruction, there are no physical barriers affecting the large intestine (colon) and causing the obstruction. Without decompression, cecal perforation may result. Insertion of a rectal tube may be ordered every 2 to 3 hours for this condition.

Massive colonic distention may also be relieved by colonoscopic suction.

Colon decompression is contraindicated in patients with recent rectal surgery, organic colon obstruction, recent surgical anastomosis, recent myocardial infarction, or diseases of the rectal mucosa.

Procedure

Colon decompression can be accomplished using a 22-Fr to 32-Fr rectal tube of soft rubber or plastic, a small-lumen decompression tube, a large-lumen decompression tube, or an over-the-guidewire decompression tube. Tubes are available in kits from various manufacturers. Colon decompression tubes are used temporarily until the patient experiences a relief of symptoms, decompression of the colon occurs, or the physician indicates removal.

- **Rectal tube.** The patient is placed in the Sims position and draped appropriately for privacy. The tube is lubricated with a water-soluble lubricant. The patient breathes slowly and deeply, and then bears down as if for a bowel movement. This relaxes the anal sphincter and facilitates insertion. The tube is inserted into the rectum and then advanced 15 cm. The proximal end of the tube should be taped to the lower buttock, and the end of the tube should be covered with a waterproof absorbent pad or connected to a drainage bag. Rectal tubes should be left in place for a maximum of 30 minutes. If no gas has been expelled, the procedure may be repeated in 2-3 hours. Rectal tubes attached to a drainage system may also be used to minimize excoriation and skin breakdown in patients who are incontinent of stool and have watery diarrhea, or in patients who are incontinent and have significant lower GI bleeding.
- **Small-lumen decompression tube.** This tube can be passed through the biopsy channel of a colonoscope to the cecum. The tube can be preloaded with a wire for insertion, if desired. The colonoscope is removed while the tube is continually advanced through the channel. Tube placement is confirmed fluoroscopically, and the proximal end is secured to the buttock with tape and attached to a drainage bag.
- **Large-lumen decompression tube.** Biopsy forceps are passed through the channel of the colonoscope, and a suture is tied around the distal end of the decompression tube. The suture is grasped with the biopsy forceps, and the colonoscope and tube are passed side-by-side. Once the desired area is reached, the tube is released, and the forceps are withdrawn from the colonoscope. The colonoscope is then withdrawn with care to avoid dislodging the tube. Tube placement is confirmed fluoroscopically. The proximal end of the tube is secured to the buttock with tape and attached to a drainage bag.
- **Over-the-guidewire decompression tube.** The colonoscope is inserted, and a 450-cm guidewire is passed through the channel. The colonoscope is removed while advancement of the guidewire is continued. Once the guidewire is in place and confirmed by fluoroscopy, if requested by endoscopist, the decompression tube is advanced over the guidewire, and placement is confirmed fluoroscopically. The guidewire is removed, and the proximal end of the tube is secured to the buttock with tape and attached to a drainage bag.

Considerations

After placement of a rectal or colon decompression tube, it is important to ensure the patency of the tube and to assess the patient for relief of symptoms. The color, consistency, and amount of drainage, and the character of the patient's abdomen should be assessed and documented. The patient should be told to expect drainage from the tube.

The physician may order a rectal tube to be removed once the abdomen is flat and soft, or if the tube is ineffective after 30 minutes.

Potential complications of decompression include perforation, necrosis of the colon, and clogging of the tube with stool.

ABDOMINAL PARACENTESIS

Abdominal **paracentesis** involves withdrawal of fluid from the peritoneal space for diagnostic and therapeutic purposes using a large-bore needle (i.e., 22 G [gauge] by 1.5 inches) or a trocar inserted in the abdominal wall. It may be performed at bedside or in a treatment room. Ultrasound-guided paracentesis is often used in patients with previous abdominal surgeries and in patients who are obese. A longer needle (i.e., 22 G by 3.5 inches) is necessary for patients who are obese.

Indications

Abdominal paracentesis is indicated for:

- Evaluation of ascites to identify the cause and to assess for infection.
- Determination of a perforated viscus following blunt trauma or symptoms of acute abdomen to verify there is no blood in the abdomen.
- Relief of dyspnea or abdominal pain secondary to **tense ascites**.

Abdominal paracentesis is contraindicated in patients who are unable to cooperate and for patients with:

- Severe coagulopathy.
- Thrombocytopenia.
- Intestinal obstruction.
- Abdominal wall infection.
- Previous multiple abdominal surgeries.
- Portal hypertension with abdominal collateral circulation.

Paracentesis may be performed in patients with coagulopathy if small-gauge needles in the vascular midline are used. It must be performed cautiously in patients who are pregnant and in patients with unstable vital signs.

Procedure

Prior to the procedure

Before paracentesis, the patient should void to reduce the risk of accidental injury to the bladder when the needle or trocar is inserted. Baseline vital signs, weight, and abdominal girth at the umbilical level should be recorded. Patients should be on a blood pressure monitor to alert the gastroenterology nurse to developing hypotension. Pulse and respiratory status should be monitored throughout the procedure.

The gastroenterology nurse should explain the procedure to the patient in order to alleviate anxiety and encourage cooperation. An informed consent should be obtained from the patient for the procedure. The patient should not feel any pain during the procedure other than a slight stinging sensation with the injection of the local anesthetic. The patient may also experience a pressure sensation from the needle or trocar or during the aspiration of the abdominal fluid.

Vacuum collection bottles (500 ml or 1,000 ml) and appropriate tubing often facilitate ease of fluid collection. If greater volumes are to be withdrawn, a larger-gauge needle or catheter and needle assembly should be used. Only a small amount of fluid is needed for diagnostic purposes.

During the procedure

Depending on the physician's preference, the patient may be positioned in Fowler's position, the knee-hand position, or sitting on the side of the bed with the feet supported. The preferred position is usually recumbent with a 30-degree elevation of the head. In this position, gravity causes fluid to accumulate in the lower abdominal cavity, and the pressure created by the abdominal organs facilitates fluid flow.

Specimen containers should be ready to receive fluid.

- The patient is draped with a sheet, exposing the abdomen.
- The gastroenterology nurse disinfects the skin, assists the physician in establishing and maintaining a sterile field, and draws up the local anesthetic.
- The patient is cautioned to remain as still as possible to avoid injury from the needle or trocar. The gastroenterology nurse should assist the patient with remaining still during the procedure to reduce the potential for accidental perforation of abdominal organs.
- The patient receives local anesthesia at the selected insertion site.
- The physician may make a small incision with the scalpel before inserting the needle or trocar, usually 1-2 inches below the umbilicus.
- Using sterile technique, the needle or trocar is introduced in the midline between the umbilicus and pubis (Fig. 29.2). The risk of hemorrhage is reduced by entering through the avascular linea alba, but the catheter may be introduced laterally if necessary. The rectus muscles, upper abdomen, collateral venous channels, and areas of surgical scars should be avoided.
- The first 10 ml of fluid should be collected separately for analysis, followed by 50 ml aliquots. Ascitic fluid should be cultured using blood culture media bottles.

After the procedure

After abdominal paracentesis is completed, an elasticized, adhesive dressing is applied to the site. The patient is helped into a comfortable position, and vital signs should be monitored and documented every 15 minutes. The patient is observed closely for vertigo, faintness, diaphoresis, pallor, height-

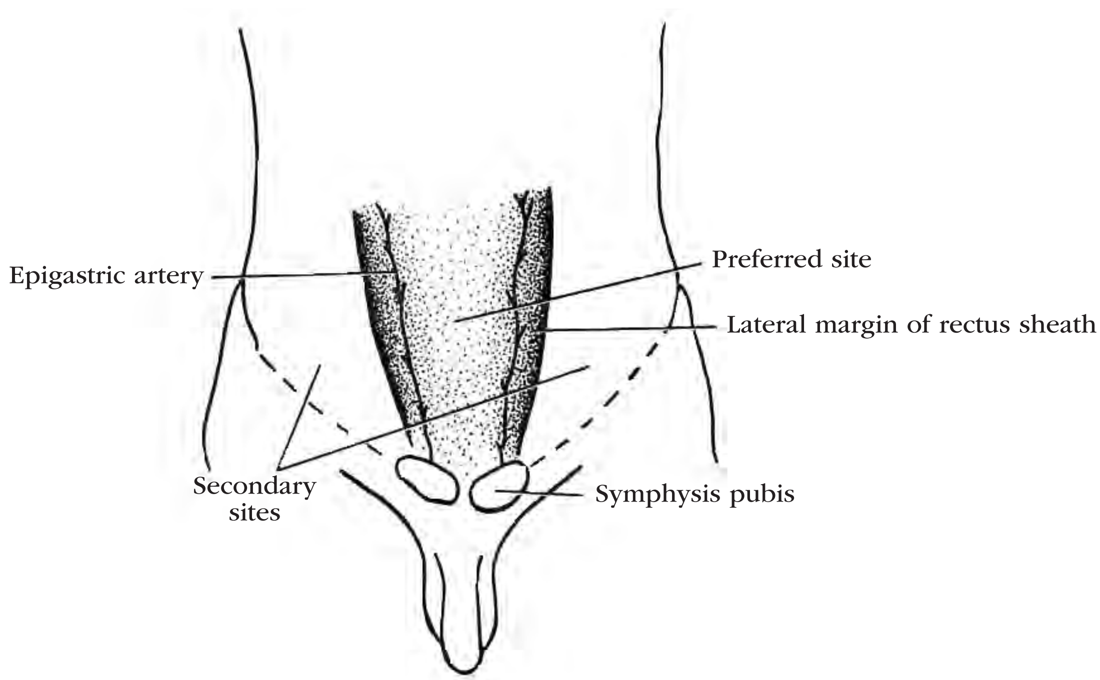

Figure 29.2. Sites for Paracentesis

ened anxiety, tachycardia, dyspnea, and hypotension. The physician is notified if the patient's pulse rate increases and systolic blood pressure decreases, respiratory status changes, temperature is elevated, peritoneal fluid leaks from the site, or scrotal edema develops.

The quality of any drainage is noted, including color, amount, viscosity, and odor. Specimens are labeled and sent to the laboratory with the appropriate paperwork. The patient's weight and abdominal girth are compared with baseline figures.

Considerations

If the patient shows signs of hypovolemic shock during the procedure, the drainage rate should be slowed by reducing the vertical distance between the needle or trocar and the collection container. If necessary, the drainage may be stopped altogether. Limit the amount of fluid withdrawn to 1,500 ml to 2,000 ml to prevent fluid shifts or hypovolemia. If the drainage slows, reposition the patient to facilitate drainage. A large amount of fluid removal may also cause hypotension, hyponatremia, and oliguria. If more than 2 liters of fluid is removed, ascetic fluid may form again by drawing fluid from extracellular tissues throughout the body. Administration of albumin is recommended with large-volume paracentesis (LVP) of >5 liters of fluid drainage.

Complications are rare in abdominal paracentesis but may include hemorrhage, perforation of the abdominal organs by the needle or the trocar, peritonitis, hepatic coma from decreased systemic circulation and reduced tissue perfusion, and incision site infection.

CASE SITUATION

Mr. Allan Ruggles came to his internist complaining of vague abdominal pain that is causing him some distress. He is willing to undergo whatever tests are necessary to find the cause of the problem.

Points to think about

1. With regard to Allan's abdominal pain, what imaging test might his internist order first?
2. If the internist refers Allan to a gastroenterologist, what type of diagnostic test might the gastroenterologist perform?
3. If the gastroenterologist believes that Allan has inflammation caused by overproduction of gastric acid, what tests could he perform to confirm his suspicions?
4. If Allan is not treated properly, what might be the consequences?
5. If Allan must go through a lengthy series of tests and therapeutic procedures, he may become disheartened and discouraged. Periodically, the gastroenterology nurse must reevaluate and, if necessary, intervene in Allan's struggle to cope during this stressful time. What are some behaviors that might indicate intervention is needed?

Suggested responses

1. After a thorough history, physical exam, and appropriate lab test work-up, the internist sends Allan for a computerized tomography (CT) scan of the abdomen. Lab tests are within normal limits. The films reviewed by the radiologist are normal. The internist knows that these normal results do not necessarily mean there is nothing wrong with Allan, so the internist refers Allan to a gastroenterologist for additional investigation.
2. The gastroenterologist conducts a physical exam, reviews Allan's symptoms, and decides that an EGD is in order. The EGD is done, and the diagnosis is moderate duodenitis, gastritis, and esophagitis.
3. To confirm his suspicion that Allan is producing too much gastric acid, the gastroenterologist may order one or more of the following tests:
   a. *Helicobacter pylori* (*H. pylori*) testing is done during endoscopy to identify (*H. pylori*), a submucosal bacteria that can cause inflammation and pain. (See Chapter 26.)
   b. Gastric analysis involves using an NG tube to obtain a small sample of stomach acid after fasting to assess whether the stomach is producing too much or too little acid. Use of gastrin, a gastric acid stimulant, may to be used during testing. (See Chapter 26.)
   c. Twenty-four-hour pH monitoring will show the degree of acidity or alkalinity of gastric secretions. Prolonged esophageal pH monitoring is the most sensitive method of directly detecting esophageal reflux, quantifying it, and correlating symptoms with reflux events. (See Chapter 26.)
   d. Esophageal manometry evaluates motility and will show whether the esophagus contracts properly to clear food. (See Chapter 24.)
   e. A Bernstein test, while not typically used, may be ordered to differentiate between cardiac and non-cardiac chest pain for the diagnosis of GERD. This test requires precise placement of the tip of the NG tube in the esophagus. (See Chapter 13.)
4. If Allan is not treated properly, he could develop:
   a. Scarring and strictures from reflux of acid in the esophagus. This could require dilation of the esophagus, perhaps on an ongoing basis.
   b. Peptic ulcerations from increased acidity in the stomach, particularly in the pre-pyloric and pyloric channel area. This can lead to scarring and deformity or stenosis of the pyloric channel,

causing gastric outlet obstruction, which would need to be dilated. Because dilation of the pylorus is not usually as effective as dilation of the esophagus, gastric surgery may be required to correct the problem.

   c. Erosive esophagitis, which can result from chronic nonerosive reflux disease. This can progress to Barrett's esophagus, which is a precursor to esophageal adenocarcinoma. Erosive esophagitis is associated with a fivefold increased risk for the development of Barrett's esophagus.
   d. Bleeding and its attendant complications, which can result from ulcerations developed from inflammation. This can result if medications do not control Allan's acid production and may require surgical intervention to control acid production (vagotomy and pyloroplasty, vagotomy and antrectomy, highly selective vagotomy, or fundoplication).

5. These difficult tests for the patient require the gastroenterology nurse to show patience and caring while establishing a supportive, trusting relationship with the patient. Nasogastric intubation can be uncomfortable, but a skilled and caring gastroenterology nurse can make the procedure much more tolerable. Some indications that Allan is not effectively coping with the stress of his testing include:
   a. Nonperformance of activities of daily living.
   b. Purposelessness.
   c. Self-absorption.
   d. Inflexibility.
   e. Hopelessness.
   f. Excessive use of denial.

Indifference to social supports and demoralization are two negative manifestations of coping with the stress caused by illness. The nursing diagnosis "ineffective individual coping" is defined as an impairment of adaptive behaviors and a lack of problem-solving abilities in meeting life's demands and roles. Coping is a continual process influenced by numerous factors, so the assessment of a person's coping mechanisms becomes an ongoing nursing responsibility.

REVIEW TERMS

barbed stent, biliary stents, colon decompression, distal orientation, French (Fr) (in reference to circumference), gastric lavage, nasobiliary catheters (NBCs), nasogastric (NG) tube, nasopancreatic catheters (NPCs), pancreatic stents, paracentesis, pigtail stent, proximal orientation, self-expanding metal stents (SEMSs), self-expanding plastic stents (SEPSs)

REVIEW QUESTIONS

1. If the patient belches after air is injected into a nasogastric (NG) tube, what is the most likely cause?
   a. The tube is curled in the patient's mouth or throat.
   b. The tube is in the patient's esophagus.
   c. Stomach contents are blocking the tube.
   d. The tube is blocking the patient's airway.
2. Which of the following statements is false with regard to Ogilvie's syndrome?
   a. The condition commonly occurs in young children.
   b. It is also known as colonic pseudo-obstruction.
   c. Cecal perforation may result if decompression is not initiated.
   d. There are no physical barriers affecting the large intestine.
3. After the patient's tolerance of gastric lavage has been established, what volume of irrigation solution is used for each subsequent instillation?
   a. 50 ml.
   b. 250 ml.
   c. 500 ml.
   d. 1,000 ml.
4. For biliary decompression, a polyethylene tube (inserted via a duodenoscope) exits and drains to the outside through the patient's:
   a. Mouth.
   b. Nose.
   c. Abdominal wall.
   d. Gastrostomy tube.
5. The most common delayed complication of biliary stent placement is:
   a. Recurrent jaundice.
   b. Duodenal perforation.
   c. Hemorrhage.
   d. Pancreatitis.
6. A rectal tube should be left in place:
   a. Until decompression is achieved.
   b. For a maximum of 3 minutes.
   c. For a maximum of 30 minutes.
   d. For a maximum of 3 hours.
7. Before abdominal paracentesis, why is it important for the patient to void?
   a. To keep the patient comfortable during the procedure.
   b. To reduce intra-abdominal pressure.
   c. To give a more accurate measure of abdominal girth.
   d. To avoid injury to the bladder when the needle is inserted.
8. Complications of temporary esophageal expandable stents include all of the following except:
   a. Post-procedure chest pain.
   b. Stent occlusion due to a bolus of food.
   c. Esophagitis.
   d. Confusion.

9. One complication that is unique to pancreatic stenting is:
   a. Stent migration.
   b. Stent occlusion.
   c. Stent-induced pancreatic ductal change.
   d. Gastroesophageal reflux.
10. Indications for pancreatic stent insertion include all of the following except:
   a. Pancreatic duct disruption.
   b. Sphincter of Oddi dysfunction.
   c. Unresolved pancreatitis.
   d. Colonic pseudo-obstruction.

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Chapter 30

EXCISION AND EXTRACTION

This chapter will acquaint the gastroenterology nurse with endoscopic procedures that are used for the removal of foreign bodies, polyps, and retained stones from the common bile duct and pancreatic duct. The use of lithotripsy for the disruption of gallstones is also discussed.

Learning objectives

After reviewing the content of this chapter, the gastroenterology nurse will be able to:

1. Describe techniques used for removal of foreign bodies and bezoars from the gastrointestinal (GI) tract.
2. Explain the indications, contraindications, procedures, and potential complications of endoscopic polypectomy.
3. Describe the indications, contraindications, procedures, and risks of endoscopic sphincterotomy.
4. Discuss the use of biliary lithotripsy for the disruption of gallstones and the use of pulsed-dye laser lithotripsy.

FOREIGN-BODY REMOVAL

Endoscopic techniques may be used for extraction of foreign bodies from the esophagus, stomach, duodenum, or colon. Foreign bodies may be deliberately or accidentally swallowed or may be introduced into the lower GI tract from the rectum.

Most foreign-body obstructions involve the esophagus. Foreign objects usually lodge at areas of anatomic or physiologic narrowing such as the cricopharyngeal sphincter or lower esophageal sphincter (LES), but they may lodge above a benign or malignant stricture, web, or ring. If a foreign body passes the esophagus, it will usually pass through the remainder of the GI tract without incident. Other sites of potential hang-up are the pylorus, the duodenal C-loop, the ligament of Treitz, the ileocecal valve, and the anus.

True foreign-body (i.e. non-food objects) ingestion occurs most frequently in young children between the ages of 6 months and 4 years. Children most often ingest coins, toys, magnets, buttons, and batteries. Adults may present with bones or meat impacted in the esophagus. Adults with dentures or those who are inebriated, have psychiatric conditions, or are cognitively impaired are at a higher risk for foreign-body ingestion. The incidence of intentional ingestion of foreign objects is higher in incarcerated individuals than in the general population (Evans et al., 2015).

In the small intestine, an endoscopy capsule can act as a foreign body if it does not pass through the GI tract due to a small bowel obstruction or adhesions. In the lower GI tract, foreign objects are found predominantly in males who are between the ages of 24 and 65. Of the foreign bodies that enter the GI tract, 80% to 90% pass through without incident, often within 48 hours; 10% to 20% need to be removed endoscopically; and less than 1% requires surgery (Birk et al., 2016).

Foreign bodies in the stomach cause few, if any, symptoms and most often warrant conservative management. Most ingested objects that have passed into the stomach will pass uneventfully through the pylorus and the rest of the GI tract. In children, ingested pennies, nickels, and dimes will pass, but quarters may not.

Objects that have passed beyond the second portion of the duodenum cannot be retrieved endoscopically. Their progress may be followed radiologically, with surgical exploration as a last resort.

Swallowed objects that hang up in the cecum or sigmoid colon may be retrieved with a colonoscope.

Biopsy forceps, pronged polyp-retrieval forceps, special foreign-body forceps, and polyp retrieval nets may be useful in removing foreign bodies from the colon, but the polypectomy snare is the most versatile instrument.

Objects that have been inserted into the rectum and the sigmoid colon may be retrieved using a flexible or rigid sigmoidoscope. In the case of large objects, the patient may require general anesthesia for cooperation and for relaxation of the anal sphincter. No cathartics or enemas should be given. Objects lying below the peritoneal reflection of the rectum can usually be grasped and removed using a rigid anoscope or proctoscope. Early surgical consultation is advised for high colorectal foreign bodies.

Table 30.1. Indications for Endoscopic or Surgical Removal of Foreign Bodies

Endoscopic or surgical removal of foreign bodies is indicated for:
Most foreign bodies lodged in the esophagus (with the possible exception of round objects or food boluses in the distal esophagus, which may pass spontaneously).
Sharp or pointed objects that could result in obstruction or perforation, including pins, toothpicks, and bones, even if they have entered the stomach.
Long, narrow objects, such as wires (more than 6 cm in length for children and more than 13 cm for adults) that may not be able to navigate the fixed duodenal angles.
Gastric foreign bodies >2 cm in diameter, or any gastric foreign bodies that do not pass within an observation period of 3-4 weeks.
Toxic foreign bodies such as alkaline button batteries or small disk batteries.
Duodenal foreign bodies that do not pass within 6 days of ingestion.

Esophageal foreign objects and food bolus impactions in the esophagus should be removed within 24 hours. Delay of greater than 24 hours decreases the likelihood of successful removal and increases the risk of complications.

Endoscopic removal of a foreign body is contraindicated when the risk associated with removing the object is greater than the risk posed by the object itself, in patients who are unable or unwilling to cooperate, or in patients with a known or suspected perforated viscus. Surgical removal is required in these cases.

Prior to the procedure

The physician should confirm the location of the foreign body with X-ray films of the neck, chest, and abdomen. Serial films may be helpful to monitor the progression of the object. It is also important to establish the time of ingestion of any food or fluids. Barium swallows should be avoided if there is any evidence that the object is located at or just below the cricopharynx, but a thin suspension of barium may be given in small sips to help locate radiolucent foreign bodies in the lower esophagus.

A wide assortment of endoscopic tools and retrieval devices is available, including:

- **Laryngoscopes and curved forceps** for removal of objects that are lodged in the hypopharynx or are accidentally dropped into the hypopharynx during extraction.
- **Alligator or grasping forceps** (rat-tooth or shark-tooth) to grasp and secure flat, metallic objects or objects that may be difficult to remove.
- **Tripod-type or retriever grasping forceps** to remove food boluses.
- **W-shaped forceps, polypectomy snares, and endoscopic nets** to remove coins.
- **Polypectomy snares** to remove long, narrow objects such as opened paper clips, stiff pieces of wire, or injector razor blades, or to remove oddly shaped objects such as jacks or jewelry.
- **Pelican-type grasping forceps,** which are available in graduated sizes, for breaking up and removing food obstructions.
- **Stainless-steel wire basket-type forceps** for a variety of round objects such as marbles or stones, or for meat boluses that can be removed in one piece.
- **Snares, grasping forceps, and baskets** for removal of gastric foreign bodies.
- **Rubber tip-type grasping forceps** for grasping needles and nails.
- **Grasping forceps and extraction baskets** for removal of metallic objects (although caution is advised when withdrawing the object through the cricopharynx).
- **Endoscope hood** for capture of sharp objects such as razor blades and paper clips in order to prevent mucosal injury.

It is important to confirm that the retrieval device will fit the channel size of the endoscope. It may be helpful to obtain an object similar to what was ingested and practice grabbing it with various forceps, snares, and baskets to simulate the endoscopic situation and to determine which instrument is best suited in a particular case.

Use of a foreign body hood or **overtube** may be indicated. Endoscopic overtubes are polyvinyl sleeves that are used to facilitate various upper endoscopic procedures, including extraction of foreign bodies. The overtube protects the esophageal and pharyngeal mucosa and the airway as the object is pulled out with the endoscope. An overtube should be used when sharp or pointed foreign bodies must be removed or when the endoscope must be passed several times, as for piecemeal removal of a soft bolus of meat. Overtubes should not be used in the duodenum.

If an overtube is needed, one with an inner diameter approximately 2 mm larger than the outside diameter of the endoscope should be chosen. This size prevents mucosa from catching between the overtube and the endoscope as the scope is advanced or withdrawn. After the overtube has been lubricated both inside and outside, it is slipped over the endoscope insertion tube. The patient is intubated in the usual manner with the overtube covering the endoscope as it is passed into the patient's esophagus. A mouth guard or bite block prevents the overtube from slipping into the patient's mouth.

In situations where a large or dangerous foreign body cannot be pulled far enough into the overtube, it is safer to use an endoscopic hood. The hood is placed on the distal tip of the scope and folded back toward the controls for insertion. When the endoscope is withdrawn into the esophagus, the hood catches on the LES and is forced distally, thereby covering the object.

During the procedure

For removal of foreign bodies from the upper GI tract, the patient is placed in the left lateral position. Moderate sedation is administered and the patient is monitored per organizational policy. Infants, children, or uncooperative patients may require general anesthesia with endotracheal intubation to protect their airway and provide patient safety.

During the procedure, the nurse should monitor the patient's vital signs; the color, warmth, and dryness of the skin; the level of consciousness; and pain tolerance. It is also important to observe the patient for symptoms of perforation (increased restlessness, pain, or tachycardia). Oral secretions should be suctioned. The airway must be protected and maintained, and steps should be taken to prevent aspiration.

Extraction of a foreign body from the esophagus requires good visibility, a firm grasp of the object, and removal without force. Pointed objects should be withdrawn with the point trailing. If the object is pointed on both ends, the proximal sharp end must be completely covered by the grasping forceps. If the object has a single pointed end that is directed toward the head, it can be carried into the stomach and turned so that the pointed end trails before it is removed. Objects with sharp edges should be extracted with the aid of an overtube or hood.

Great care must be taken to avoid dropping the foreign body into the laryngopharynx on withdrawal. A laryngoscope with an appropriate-sized blade and Magill forceps should be readily available in the event that the foreign body is dropped into the patient's airway. The object should be grasped quickly from the patient's mouth to prevent it from falling into the trachea.

Organizational policies should be followed for disposal of the retrieved foreign body.

For patients with food impacted above an esophageal stricture, hydrostatic balloon dilatation of the stricture is an alternative method of treatment, providing the food bolus does not totally occlude the lumen. Once the stricture is dilated, the bolus can be pushed into the stomach. Medications such as glucagon should be available to promote relaxation of the esophagus, thereby facilitating passage of the foreign object into the stomach.

After the procedure

After the object has been removed, the patient's vital signs are monitored and the patient is observed for bleeding, vomiting, abdominal or chest pain, continued abdominal distention, subcutaneous emphysema, or aspiration. When it is determined that the object has been removed without complications, the patient should be reevaluated to rule out any underlying disease that may have caused the obstruction, especially in the esophagus.

Considerations

Although rare, potential complications of endoscopic foreign-body removal include perforation, impaction of the foreign object, hemorrhage, localized inflammation or pressure necrosis, and aspiration of the object.

It is important to act quickly if a patient has swallowed a small, button-type battery. The alkaline substance from the battery acts rapidly on the mucosa, causing direct corrosive action that results in burns and pressure necrosis. Direct corrosive activity frequently leads to perforation. Potentially catastrophic complications such as esophagotracheal or esophagoaortic fistula can occur.

Concealment of illegal drugs in the GI tract (also known as "body packing") is seen in regions of high drug trafficking and has been reported in both children and adults. The illegal drug packets are wrapped in plastic or contained in latex condoms or balloons. Packets are swallowed or inserted into the rectum. Rupture and/or leakage of the contents can be fatal, so endoscopic removal should not be attempted. Surgical intervention is indicated when packets fail to progress or if signs of intestinal obstruction are present. Symptomatic individuals who present with either signs of intoxication or bowel obstruction require surgery.

BEZOAR REMOVAL

Bezoars are concretions of food or foreign matter that have undergone digestive change(s) in the GI tract. They are usually composed of either vegetable and plant material (phytobezoar), ingested medication (pharmacobezoar), milk curd (lactobezoar), or hair that has been chewed (trichobezoar).

Symptoms

Symptoms associated with the presence of a bezoar vary and depend largely on the size, location, and degree of disturbance in gastric physiology. They can include abdominal pain, fullness, and discomfort; difficulty swallowing; and nausea and vomiting. Symptoms such as hematemesis, bloody or tarry stools, anemia, and fainting are signs of the development of ulceration in the gastric mucosa due to pressure necrosis induced by the bezoar. The best way to diagnose and identify the type of bezoar is by gastroscopy.

Procedure

Standard treatment methods for phytobezoars include:
- Endoscopic therapy, which involves fragmenting the bezoar with water, suction, lavage, and endoscopic internal fragmentation using biopsy forceps and polypectomy snares.
- Administration of a carbonated beverage via gastric lavage, alone or in conjunction with an endoscopy.
- Surgical intervention when endoscopic therapy is unsuccessful and for patients with complications of obstruction and significant bleeding.

Trichobezoars cannot be dissolved in vivo. Treatment of these large intragastric masses requires surgical intervention.

Considerations

Gastric outlet and intestinal obstruction are common complications of a bezoar.

POLYPECTOMY

GI polyps are lesions that may project from the mucosal surface into any part of the GI lumen. These can include colonic or gastric polyps.

Indications

Some polyps are **pedunculated**, meaning they are attached to the mucosa by a stem-like pedicle or stalk. **Sessile polyps** are attached to the mucosa by a broad base. Because of their protrusion into the lumen and the stresses of the fecal stream to which colonic polyps are subject, polyps occasionally bleed or cause abdominal pain or obstruction. Symptomatic polyps are uncommon; however, the greatest concern is their potential to become malignant.

Procedure

Most polyps are removed by wire snares or with hot biopsy forceps, which are opened and closed by a gastroenterology nurse or associate. Commercial **polypectomy** snares come in various sizes and shapes.

Polyps are usually transected by use of a high-frequency current produced by a generator attached to the sheathed snare. The current is applied for brief pulses until the polyp is transected. Electrical currents that have a pure cutting effect are never used in colonoscopic polypectomy. Electrocoagulation current alone may be used if the polyp is attached by a thin pedicle, but a blend of cutting and coagulation current is usually applied to thick-based polyps.

Considerations

It is important to remember that different electrosurgical units (ESUs) may not provide the same current output. In addition, the characteristics of the snare used may alter the power setting on a given ESU. For example, a thin wire cuts through tissue more quickly than a thick wire. The operator must be familiar with the units that are available and guidelines for each unit.

To guard against electrical hazards, the electrocautery equipment should be grounded. The ESU, the connections, and the grounding pad should be checked before each use. The snare should be checked to ensure that it is in working order and is conducting current. One way of testing the circuitry is to set the ESU at half power, and then check for sparking between the side of the snare loop and the testing plate. ESU cutting and coagulation controls should be set in accordance with the physician's instructions.

COLONIC POLYPS

The most common types of polyps encountered in the distal portion of the colon or rectum are hyperplastic and metaplastic polyps.
- Hyperplastic polyps are characterized by an abnormal multiplication or increase in the number of normal cells in a tissue.
- Metaplasic polyps are characterized by a change in the adult cells in a tissue to an abnormal form.
- Hamartomatous polyps, including juvenile polyps and Peutz-Jeghers syndrome polyps, are those in which the cells of a circumscribed area grow faster than those of surrounding areas.
- Inflammatory polyps usually result from ulcerative colitis.

Indications

Colonoscopic **polypectomy** is indicated for all polyps with a diameter of 1 cm or larger. Smaller polyps can usually be removed with cold or hot biopsy forceps, particularly if the polyp is sessile.

Contraindications to colonic polypectomy may include:
- Anticoagulant and antiplatelet use.
- Coagulopathy.
- Polyps that appear malignant and are probably invasive.
- Inadequate bowel preparation.
- Uncooperative patients.

Prior to the procedure

Coagulation screening, blood count, and appropriate blood chemistry studies may be needed, in addition to a history and physical examination. The patient should be told how to obtain the results of pathological studies. Compliance with instructions for thorough bowel preparation is critical and must be confirmed.

During the procedure

Before the start of the procedure:

- The patient is placed in the left lateral recumbent position.
- A grounding pad is placed on a muscular site that is closest to the treatment site. The left or right flank is preferred. Never attach the pads over implants metal or bony protuberances, broken skin, tattoos, or scar tissue.
- All leads on the ESU are checked for secure attachment. Cords and attachments should be inspected for fraying and wear. Cut and coagulation controls are set according to the physician's instructions, and verbal orders regarding settings are repeated back to the operator.

A colonoscope is advanced as it would be for diagnostic colonoscopy. If diagnostic colonoscopy to the cecum has not been performed previously, it may be carried out before polypectomy. At a minimum, the scope should be advanced 20-25 cm beyond the polyp to remove fecal fluid.

When the polyp is in view, its shape, size, and length of its stalk is evaluated. Based on these factors, the appropriate technique for polypectomy can be determined:

- **Small sessile polyps less than 8 mm in diameter** may be completely removed and recovered for biopsy examination, or ablated with hot biopsy forceps. The polyp is grasped between the jaws of the insulated forceps and lifted away from the intestinal wall by manipulating the angulation controls. This maneuver pulls the mucosa away from the submucosa and muscularis propria so the application of current causes only slight heating of the deeper tissues. With the grasped polyp pointed toward the lumen, coagulation current is applied, removing the polyp. When the whitish area encircling the polyp base is 1-2 mm wide, the current is discontinued, and the polyp is pulled off its base.
- **Sessile polyps less than 1 cm in diameter** can be removed in one piece by using the snare-cautery technique if their bases are not wide and if a reasonable "pseudo-stalk" can be created at the base.
- **Most pedunculated polyps** require that a polypectomy snare is advanced so that a wire loop is created and the polyp is "lassoed" with the snare wire. The tip of the catheter is advanced to the base of the polyp, and the loop is gently tightened. The polyp is tented slightly into the center of the lumen to be certain that no adjacent normal mucosa is caught in the loop. If the polyp has a long stalk, it is best to leave at least 1 cm of the stalk. Polyps with short stalks are ensnared as close to their necks as possible. Pedunculated polyps are usually transected within 2 to 4 seconds. Mucosal blanching is noted adjacent to the snare wire during transection. Following closure of the snare, the polyp will fall into the lumen and the coagulated stalk will be visible. A suction specimen trap may be placed between the endoscope suction port and the suction tubing to prevent accidental loss of the specimen.
- **Pedunculated polyps with large or lobulated heads** may require segmental resection of the head before complete polypectomy can be done.
- **Broad-based pedunculated polyps** may be bunched to acceptable resection size to permit single transection, or they may be managed by segmental resection.
- **Large sessile polyps more than 2 cm in diameter** are removed in a piecemeal fashion, beginning with two oblique cuts made at right angles to each other across one-quarter to one-third of the polyp. The remaining base of the polyp may be transected during the initial polypectomy or 4-8 weeks later, after the area of mucosal ulceration has healed. Using an injection (sclerotherapy) needle, injecting 1-2 cc of normal saline close to the base of the polyp can help to raise the base prior to the polypectomy.
- **Large, flat, and sessile colorectal lesions** are primarily removed by endoscopic mucosal resection (EMR). EMR using electrocautery or cold snare excision is a safe and effective procedure for removing polyps greater than 3 cm. The procedure can be performed with or without lifting agents such as saline and hydroxypropyl methylcellulose.

If a polyp is awkwardly located for snaring, it may be helpful to bypass it and advance the colonoscope through the rest of the colon. As the instrument is removed, the colon is straightened, providing a better view of the polyp for removal. The patient may also be repositioned to improve the orientation of the polyp.

Snaring and retrieving multiple polyps may require passing the colonoscope several times.

When the physician is ready to use the ESU, the nurse should turn the power on and then turn it off immediately after use. The nurse or associate must also open and close the snare or biopsy forceps at the request of the physician. It is important to close the snare slowly while maintaining continual communication with the physician. Communication between the physician and the nurse or associate is essential during this process. Visualization of this procedure using the video monitor is an absolute necessity.

After polypectomy is complete, the next important step is retrieval of the polyp. The polyp may be retrieved by:

- Removing it in the cup of the forceps, if cold and hot biopsy forceps were used for polypectomy.
- Placing the tip of the colonoscope flush against the head of the polyp and applying suction.

- Resnaring the cut polyp and withdrawing it by use of the colonoscope. If the resnared polyp is kept 3-5 cm from the instrument tip, the colon can be visualized during withdrawal.
- Entrapping the polyp with a basket by a three-pronged polyp retriever or polyp net.
- Aspirating the polyp through the suction line and into a suction specimen trap or a filtered polyp retrieval trap.

To locate a lost polyp, a bolus of water may be squirted through the biopsy channel to identify the path of gravity in the colon to determine the probable location of the resected polyp. To dislodge a small polyp that entered the scope's suction channel, a bolus of water may be squirted through the biopsy channel. A specimen trap must be in place prior to retrieving the small polyp lodged in the suction channel.

It may not be possible to pull large polyps through the anus with the colonoscope, but if the patient bears down, the polyp can often be expelled. Sometimes, the polyp must be removed from the rectum by digital exam, or by passing a rigid sigmoidoscope and grasping it with forceps.

Once the polyp has been retrieved, the colonoscope may be reinserted to inspect and verify that there is no serious complication at the polypectomy site. If there is slight oozing of blood at the polypectomy site, the physician may request an endoscopic clipping device to place one or several clips to control or prevent further bleeding. Additional application of thermal heat via the polypectomy snare or heater probe may be requested.

After the procedure

A thorough histological examination of the entire polyp is essential to determine whether there is the possibility of malignant change. The specimen should be prepared and labeled in accordance with organizational policy. The polyp itself should be fixed in formalin solution. A pathologist will examine the specimen by serial section for the presence of epithelial atypical or frank cancer within the polyp or invading the stalk. A final determination must be made by the pathologist.

After the polypectomy procedure is completed, it is important to observe the patient for abdominal pain or distention. The nurse should monitor vital signs per organizational policy and instruct the patient regarding any dietary and medication restrictions per physician orders.

There is no evidence that a routine follow-up examination is necessary for patients with hyperplastic, inflammatory, or hamartomatous polyps. Carcinoma in situ does not recur or metastasize and requires no treatment other than polyp removal. Patients with sessile polyps that harbor invasive carcinoma or pedunculated polyps with invasive carcinoma and no clear margin of resection should undergo surgery.

Considerations

Polyps in the rectum and lower sigmoid colon can be resected more safely than those in the cecum and right side of the colon where the wall is thinner. There are a number of potential complications of colonoscopic polypectomy, including:

- **Bleeding,** the most common complication, which may occur immediately or as late as 21 days after polypectomy.
- **Adverse reactions to sedation,** which include hypotension, respiratory depression, bradycardia, nausea, vomiting, and sweating.
- **Vasovagal attack,** which usually occurs when colonoscopy causes serious discomfort or pain or excessive abdominal distention or pressure. Clinical manifestations include hypotension; bradycardia; and cold, clammy skin. Depending on the extent of the adverse response, intervention may include reducing the amount of air in the colon, scope withdrawal, IV anticholergic administration, or termination of the procedure.
- **Transmural burns,** which are manifested by abdominal pain, leukocytosis, and fever, without evidence of free air or diffuse peritoneal signs. If any portion of the polyp head is permitted to come in contact with the intestinal wall, heat may be transferred through this point of contact, causing a burn of the wall adjacent to the polyp. To avoid this, the polyp may be jiggled to move a small point of contact to various areas on the wall. Pedunculated polyps may be manipulated with the tip of the catheter or with suction, or the patient's position may be changed to provide better access and to bring the head of the polyp away from the bowel wall. If no perforation is present, the patient should be observed and treated with antibiotics, if ordered, and solid food should be withheld. The syndrome usually resolves in 24-48 hours. Transmural burns from excessive coagulation are a more serious and more common complication with sessile polyps than with pedunculated polyps.
- **Perforation,** which can occur if a portion of the wall of the colon is ensnared, if too much current is applied, or if there is substantial disruption of tissue by mechanical force during colonoscopy. Exploratory laparotomy with closure of the perforation is the procedure of choice for the management of free perforation. Perforation from excessive coagulation is a more serious and more common complication with sessile polyps than with pedunculated polyps.
- **Explosion of flammable gases** such as hydrogen and methane, which can be minimized by good bowel preparation and by avoiding electrocautery in the presence of stool.
- **Current leakage from the ESU,** which can cause thermal injury to the physician, patient, nurse, or associate.

HEMORRHAGE

Hemorrhage is a potential occurrence during any polypectomy, most likely to occur when the polyp is large, in a difficult position, and/or has a thick stalk. Once the snare has been tightened around the polyp or its stalk, it should not be released or loosened. If the tissue has been partially excised with the wire but cannot be removed, bleeding may occur, which may obscure the physician's view of the site. Malfunctioning electrocoagulation equipment may also be a cause of post-polypectomy bleeding.

Hemorrhage is considered an emergency situation. During this situation, at least two nurses should be in attendance. One nurse (or associate) prepares equipment for coagulation, while another attends to the patient and checks vital signs at frequent intervals. The patient should be calmly reassured, warm, and in a flat supine position to encourage blood flow to the brain and other vital organs. Oxygen is administered via nasal cannula.

If blood is actively pumping from the stalk of a pedunculated polyp following polypectomy, the stalk may be regrasped with the snare, which is then tightened until the bleeding is stopped. Coagulation current should be reapplied at successive intervals until hemostasis is achieved. If bleeding occurs following transection of a sessile polyp, an injection needle may be inserted through the colonoscope and a sclerosing agent injected into the tissue. A bipolar probe or heater probe, sclerotherapy, or a clipping device may also be used to stop the bleeding.

If bleeding is suspected, IV fluids should be maintained. IV fluids act to restore volume, transport oxygen, and remove waste products. Blood loss is assessed based on the amount seen in the suction canister and any rectal discharge. Intake and output records are established immediately.

If not done before the procedure, laboratory work is ordered, including hematocrit, hemoglobin, and coagulation studies. Blood is typed and cross-matched.

Most of the time, bleeding can be controlled without surgery, but a surgeon should be notified in case surgery is needed. The patient should be prepared for admission to the hospital for observation.

PEUTZ-JEGHERS SYNDROME

If the GI hamartomatous polyps typical of Peutz-Jeghers syndrome are fairly well localized to a short segment of intestine, segmental resection may be all that is required. Usually, however, polyposis is extensive, and the patient requires multiple enterotomies throughout his or her life. The physician targets the larger polyps, which are most likely to be responsible for symptoms.

GASTRIC POLYPS

Gastric polyps are uncommon. Most are solitary, small, hyperplastic polyps that may be either sessile or pedunculated. Gastric polyps are usually discovered in radiographic or endoscopic examinations in patients experiencing nausea, abdominal pain, and other GI symptoms, although it is unlikely that the polyps are responsible for these symptoms.

Indications

Gastroscopic polypectomy is indicated for most gastric polyps, but surgical excision is indicated for:

- Sessile or broad-based polyps, in which no definitive diagnosis can be made by endoscopic biopsy examination.
- Intramural polypoid lesions such as leiomyomas.
- Any polypoid lesion that is believed to be responsible for symptoms and cannot be removed by endoscopic polypectomy.

Procedure

To enhance the success in retrieving resected polyps after polypectomy in the antrum and duodenum, a peristalsis-inhibiting agent such as glucagon may be given immediately before resection.

Gastric mucosa is more vascular than that of the colon, so slower closure of the snare handle is important during electrocautery to ensure hemostasis. Once located and removed, the polyp should be grasped with a wire snare or net to provide additional traction as the polyp and instrument pass back through the cricopharynx. A foreign-body hood can be helpful during polyp removal to keep the polyp from falling into the trachea.

Considerations

Following removal of the gastric polyp, the denuded mucosa creates an iatrogenic ulcer. An antiulcer regimen with both histamine2 blockers and proton pump inhibitors (PPIs) for 4 weeks following gastric polypectomy may be recommended.

Patients with adenomatous gastric polyps should undergo repeat endoscopic surveillance every 1-2 years. Patients with hyperplastic gastric polyps do not require surveillance.

SPHINCTEROTOMY

Endoscopic retrograde **sphincterotomy**, also known as **papillotomy** is an electrosurgical incision of the ampulla of Vater and the fibers of the sphincter of Oddi during endoscopic retrograde cholangiopancreatography (ERCP). The terms sphincterotomy and papillotomy are often used interchangeably. The term papillotomy may be used to indicate only a mucosal cut rather than a submucosal cut, but this is difficult to determine endoscopically.

Indications

The objective of sphincterotomy is to sever the sphincter fibers and any soft tissue that impede the passage of bile and/or common duct stones. It is the procedure of choice for the management of recurrent common bile duct (CBD) stones after cholecystectomy.

Table 30.2. Indications for Endoscopic Sphincterotomy
Endoscopic sphincterotomy is indicated for:
Post-cholecystectomy choledocholithiasis.
Choledocholithiasis in patients with an intact gallbladder who are at poor surgical risks or who will undergo a laparoscopic cholecystectomy.
Papillary stenosis in patients with prolonged symptoms and severe disability who have been unresponsive to symptomatic treatment. Sphincterotomy in these patients serves to enlarge the papillary opening and decrease ductal pressure.
Obstruction of the CBD by ampullary tumors or distal CBD lesions. In this case, sphincterotomy is done to relieve ductal obstruction caused by tumor growth and to reduce the resultant jaundice. It may be performed in preparation for more extensive surgery or as palliation.
Gallstone pancreatitis.
Acute suppurative (purulent) cholangitis.
Sphincter of Oddi dysfunction.
Choledochocele.
Sump syndrome, a rare clinical entity involving the accumulation of gallstones or food debris in the defunctionalized segment of the distal CBD in patients who have had a side-to-side choledochoduodenostomy.
Preparation for stent placement or nasobiliary catheterization.
HIV-related hepatobiliary disease. A sphincterotomy can relieve pain by reducing ductal pressure caused by AIDS cholangiopathy.
Pressure reduction in the CBD from a bile duct leak after laparoscopic cholecystectomy injury.

Contraindications for sphincterotomy include:

- A patient who is unwilling or unable to cooperate. The patient must be able to lie still and heed directions for posturing during radiographs and to follow other instructions as needed.
- Significant coagulopathy.
- Recent myocardial infarction or severe pulmonary disease.
- Allergy to the contrast medium, although the procedure may sometimes be done using a lower-ionized contrast medium with premedication of an IV steroid.
- The presence of an extremely large stone (greater than 20-25 mm in diameter), unless a lithotripter is available or stent placement is planned instead of surgery.
- Inability to properly position the **sphincterotome** (**papillotome**).

Periampullary diverticula do not necessarily constitute a contraindication to sphincterotomy, but patients with this condition are at additional risk.

Prior to procedure

Patients should be instructed on pre-procedure fasting per organizational policy. Although ERCP can be safely performed on an outpatient basis, patients may be hospitalized post-sphincterotomy because of the risks involved with the procedure.

Patients with underlying coagulation abnormalities warrant precautions such as injections of vitamin K, administration of fresh frozen plasma, or disbursement of specific coagulation factors. Patients with abnormal coagulation studies should also be treated during the healing phase for 7-10 days following the procedure.

Prophylactic antibiotics usually are not necessary, unless the patient has underlying valvular heart disease, sepsis, biliary tract obstruction, or a pancreatic pseudocyst.

Equipment is checked prior to use:

- A side-viewing duodenoscope with a nonmetal head should be checked for movement of the tip; good visibility; and functioning of the forceps elevator, suction, and water channels.
- If the forceps elevator sticks, the tip of the scope should be soaked in tepid water until freed. Once it moves freely, silicone should be applied to maintain its motion.
- The cannulating catheters used for dye injection should be filled with radiopaque contrast material and cleared of air bubbles.
- Sphincterotomes should be flexed to confirm full range of motion and smoothness of function. The sphincterotome should also be filled with the contrast medium.
- Retrieval balloon catheters may be filled with air or contrast material. Catheters should be tested prior to use by using air for balloon inflation.

Having two nurses (or a nurse and an associate) assist during the preparations and the sphincterotomy procedure itself is recommended to help ensure patient safety.

Before the start of the procedure:

- An IV line is started, preferably in the patient's right hand or forearm, and run at a keep-open rate.
- The patient is usually placed in a prone position on the X-ray table with his or her head turned toward the right to aid the passage of the endoscope. The patient may be placed in the left lateral position to the far-right side of the table, making it relatively easy to roll the patient into the prone position once the scope is passed into the duodenum. The patient's left arm may be placed behind the body to further facilitate mov-

ing to the prone position. Pillows may be needed to support the patient in the required position.
- Pressure injury prevention is critical in positioning patients for ERCP. Adjuncts such as foam dressings, pillows, wedges, gel rolls, and foam position devices are used to maintain skin integrity.
- The grounding pad is applied by the nurse in preparation for sphincterotomy.
- The ESU is set up as described in the prior section on polypectomy. The cutting currents used in sphincterotomy are composed of continuous sinusoidal waves or bursts. These cutting currents produce intense heat at the point of contact, thus vaporizing and exploding cells.
- The patient's dentures or dental bridges are removed, and a bite block is used to protect the teeth and the scope.
- Topical anesthetic may be used to numb the throat.
- The patient is sedated before intubation with the endoscope and during the procedure. Pediatric patients may require general anesthesia. Monitored anesthesia care (MAC) or general anesthesia is often used with patients with several comorbidities or an American Society of Anesthesiologists (ASA) Physical Status Classification System classification of ASA III or IV (ASA, 2018).

Medications used for the procedure vary based on the physician's preference. Glucagon or another agent to reduce duodenal motility may be given before cannulation and sphincterotomy. Opiates are generally avoided when sphincter of Oddi manometric studies are anticipated because these agents may cause sphincter spasms, which alter the sphincter pressure and make cannulation difficult.

Narcotic and sedative antagonists should be on hand to reverse the drug effects. Emergency medications and emergency equipment should be available in the event of respiratory or cardiac arrest.

During the procedure

During the procedure, the nurse should help the patient lie as still as possible; adjust patient position as needed; give emotional support; monitor the patient's vital signs, including skin color, warmth, and dryness; and monitor oral secretions.

Cannulation:
- Once the ampulla of Vater is sighted, glucagon may be administered, and a high-quality cholangiogram is obtained.
- The appropriate sphincterotome is selected, preflushed with contrast medium, inserted into the scope, and introduced into the CBD.
- The sphincterotome should be advanced only a short distance before contrast material is injected to confirm placement in the CBD.
- A wire-guided sphinctertome may be used to help direct the sphinctertome into the proper position.
- Fluoroscopy demonstrates proper placement of the sphincterotome within the bile duct.
- The physician then directs the nurse or associate to flex the sphincterotome slowly, and the cannula is withdrawn from the duct far enough so that approximately one-half to two-thirds of the wire is visible in the duodenum outside the papillary orifice.
- The wire is oriented in a 12-o'clock position in relation to the papilla, so the wire is held against the roof of the papilla during cutting.

Sphincterotomy:
Before starting the electrocautery incision, voice checks are made between the physician and the nurse or associate concerning:
- Application of the grounding pad on the patient and its connection (on most units, an alarm warns personnel of problems).
- Attachment of the sphincterotome handle to the ESU.
- Presetting of the ESU with the desired current setting.
- Correct positioning of the foot pedal.
- Control of duodenal motility, using glucagon as indicated.
- Degree of wire flexion required on the sphincterotome.
- Switching on the power just before cutting.
- Removal of guidewire from sphinctertome.

The entire procedure is performed under direct visual guidance. Cutting is usually done with a partially flexed sphincterotome, with very short bursts of current (less than 1 second) to carry the incision through the sphincter 1-2 mm at a time.

The length of the sphincterotomy should be tailored to the size of the CBD and the stones. The apparent length of the intraduodenal segment of the CBD and the length of the narrow portion of the duct until the point where the duct becomes dilated are of great importance. The sphincterotomy must be extended to the dilated portion of the duct, but it must remain within the intraduodenal segment of the duct. The upper extent of the sphincterotomy is preferred no further than the transverse fold, seen approximately 1 cm above the papilla. The usual length of the incision is 10-12 mm. As the current is being applied, the physician may order changes in the flexion of the sphincterotome.

In patients with Billroth II gastrectomies, endoscopic access to the papilla and its subsequent cannulation and sphincterotomy can be difficult. These cases require specially designed sphincterotomes that cut in a downward direction instead of the usual 12-o'clock position.

The completion of a sphincterotomy is usually signaled by a gush of bile that may contain some blood, indicating that the sphincter fibers have been severed and usually that the sphincterotomy is adequate. After the sphincterotomy is completed, the physician determines the size of the opening and patency to the bile duct by using a flexed sphincterotome or an inflated balloon catheter.

The ESU should be turned off and the active cord detached as soon as the sphincterotomy is complete.

If stones are present and do not pass spontaneously, a retrieval device is passed into the duct to retrieve the stones.

Retrieval devices may include the following:

- An **occlusion balloon catheter** is used to extract stones up to 12 mm in diameter. Balloons are passed up beyond the stone, and then inflated to pull the stone down and drag it through the sphincterotomy incision. Balloons also may be used for determining the size of the incision, determining the presence of and removing gravel, and performing cholangiography after sphincterotomy or choledochoduodenostomy. Considerable traction is often required to pull a stone through a sphincterotomy incision.
- A **basket** is used to trap stones up to 15 mm in diameter. A basket is more difficult to introduce into the CBD than a balloon or sphincterotome. Occasionally, the physician has difficulty entrapping stones in the basket. There is a risk of impacting baskets with entrapped stones in the distal portion of the CBD, and patients may require surgical intervention for removal. Most baskets have breakable wires that aid release of the basket if it becomes impacted. A mechanical lithotripter with a pair of wire cutters should be readily available when baskets are used in the duct to remove stones. Only baskets compatible with the mechanical lithotripter should be used.
- A **modified sphincterotome,** similar to a polypectomy snare, is used to hook the stone in a wire loop and extract it.
- A **mechanical lithotripter,** which functions on the same principle as the basket, is used to capture the stone and apply pressure against the trapped stone, causing fragmentation. The device is composed of a basket with high tensile strength wires with a Teflon sheath that can be removed and replaced by a coil spring sheath to allow application of traction on the stone. Mechanical cutting through the stone is accomplished by using wires for mechanical cutting and a coil spring sheath for firm traction on the stone.

Electrohydraulic or laser lithotripsy, which has been useful for removing stones from the urinary tract, can sometimes be effective in the removal of gallstones. This procedure is generally performed under direct visualization.

If stones cannot be removed immediately after sphincterotomy, a nasobiliary catheter or plastic stent may be placed as a prophylactic measure to avoid obstruction. Balloon dilation of the papilla of the sphincterotomy may be considered to facilitate the removal of large stones. (Nasobiliary tubes are discussed in Chapter 29.) The patient is reexamined in 5-14 days to determine whether the stones have passed. During this second examination, the physician may again consider extending the sphincterotomy and attempting extraction of the stones.

Occasionally, if all attempts to remove the stones fail, large stones that cause ductal obstruction and cannot be removed endoscopically may be bypassed with a biliary stent. The stent can be inserted to facilitate biliary drainage and keep the stone from impacting and causing biliary obstruction. (Biliary and pancreatic duct stent placement are discussed in Chapter 29.)

After the procedure

Following endoscopic sphincterotomy, antibiotics may be administered according to the physician's order. The patient's blood pressure, pulse, respirations, and oxygen saturation should be monitored per organization protocol. Any abdominal discomfort, nausea, or vomiting should be noted and treated per the physician's order. Clear fluids are usually given on the same evening. On the following day, if there are no adverse reactions, the diet is advanced to solid food.

Considerations

The mortality rate for endoscopic sphincterotomy is considerably lower than for surgical removal of retained common duct stones. Reported complications include:

- **Bleeding,** which is the most common complication. Bleeding can usually be managed by observation, balloon tamponade, injection of sclerosing agent, or thermal therapy, and/or transfusion.
- **Pancreatitis,** which usually can be adequately treated with antibiotics and conservative follow-up.
- **Free retroduodenal perforation,** which requires either conservative medical treatment for microscopic perforation or immediate surgical intervention.
- **Cholangitis,** which can be prevented by placing a stent in the bile duct or by adding a nasobiliary catheter.
- **Entrapment of a basket,** which can be eliminated by using a balloon to extract stones or by ensuring that the sphincterotomy is adequate to allow the passage of the basket with an entrapped stone.

If the papilla becomes edematous as a result of excessive probing or electrocoagulation, the retroduodenal artery may be displaced. Accidental severing of the artery caused by anatomical aberration may occur.

If methods to remove a stone are unsuccessful, cholesterol stones may be dissolved using oral **ursodeoxycholic acid.** This method dissolves the stones or may soften large calculi so they can be extracted later as "mud." The time required for complete stone dissolution has wide variation. Treatment should be stopped if the patient shows no radiologic evidence of gallstone dissolution after 6 months.

PANCREATIC SPHINCTEROTOMY

Pancreatic sphincterotomy is an electrosurgical incision of the pancreatic duct muscles controlling drainage.

Indications

Pancreatic sphincterotomy may be indicated for the following diagnoses:
- Symptomatic pancreatic obstruction.
- Pancreatic calculi.
- Pancreatic duct strictures, leaks, or pseudocysts.
- Pancreas divisum.
- Pain relief for chronic panceatitis.

Procedure

Endoscopic pancreatic sphincterotomy or stent placement is done with small, specially shaped pancreatic stents and wire-guided sphincterotomes (pull-sphincterotomes) designed to provide accurate positioning into the pancreatic duct. Pre-cut or needle-knife sphincterotomes may also be used when getting into the duct is difficult. Both types require placement of a pancreatic stent to provide guidance for correct positioning.

Nursing care for endoscopic pancreatic sphincterotomy or stent placement is the same as for biliary duct interventions. Voice checks and teamwork between the physician and the nurse or associate are essential. These procedures should be undertaken only by highly skilled physicians with the assistance of experienced biliary gastroenterology nurses.

EXTRACORPOREAL SHOCK WAVE LITHOTRIPSY

Lithotripsy is a procedure that uses shock waves to break up stones. For selected patients, **extracorporeal shock wave lithotripsy (ESWL)** is a noninvasive alternative to cholecystectomy.

Indications

Candidates for ESWL are patients who have fewer than three gallstones. Other favorable predictors for success are low body mass and good gallbladder function.

Procedure

ESWL is usually done on an outpatient basis under local anesthesia and moderate sedation.

Lithotripsy is generally applied in combination with dissolution therapy. The success of stone fragmentation is dependent on the size, composition, and number of stones. Stone clearance depends on the ability of the gallbladder to contract and expel the stone fragments. ESWL has been most successful resolving solitary stones that are less than 2 cm in size.

ESWL involves the use of shock waves generated in a specially designed table to fragment larger stones into smaller particles, most of which can then be passed spontaneously. Lithotriptors use either ultrasound or fluoroscopy to identify the target stones.

Considerations

Serious complications from ESWL or significant damage to surrounding organs have not been reported. Petechiae (tiny brown-purple dots that form from subdermal bleeding) on the abdominal wall is common, and a small percentage of patients develop microscopic hematuria. If large stone fragments remain following ESWL, CBD obstruction is a potential complication. Endoscopic sphincterotomy may be required in these patients. Compared with patients treated with cholecystectomy, patients who undergo ESWL generally appear to have less pain, a shorter recovery period, and less chance of infection.

PULSED-DYE LASER LITHOTRIPSY

Stones in the gallbladder and CBD can be destroyed with a pulsed-dye laser beam. Pulsed laser systems reduce the risk of thermal injury since power peaks are reached within fractions of a second. It permits precise targeting, thereby reducing the risk of bile duct injury.

Procedure

A quartz fiber is pushed up against the stone, and the laser fires. The beam creates a high-energy shock wave (a photoacoustic effect) at the point of contact, and the stone is fragmented. Adjustments can be made to the laser fiber by emitting an acoustic signal that indicates the fiber is in contact with the stone or tissue. It is highly effective and safe for the fragmentation and extraction of difficult-to-treat gallstones.
- During ERCP, **laser lithotripsy** can be used to treat ductal stones, either by threading the laser catheter through a smaller, flexible "baby" scope that is passed through the duodenoscope or by introducing a flexible radiopaque laser catheter through the duodenoscope.
- In a percutaneous approach to fragmentation of ductal stones, the procedural cannula that leads the laser fiber to the stones can be inserted through the tract formed by a T-tube left in place after cholecystectomy or through the skin under fluoroscopic guidance.
- Stones in the gallbladder can be fragmented by catheterizing the gallbladder through its free wall (cholecystolithotomy) and, 2 weeks later, replacing this drainage catheter with a procedural catheter through which an endoscope and a laser fiber are inserted.

Considerations

Compared to ESWL, laser lithotripsy is less time-consuming, eliminates the need for long-term dissolution therapy, and can be used for patients with a greater number of stones.

CASE SITUATION

Dylan Rice is a 69-year-old man with a history of insulin-dependent diabetes. During his last visit to his internist, he complained of abdominal discomfort with early satiety and a general feeling of fullness in his stomach. The internist ordered an upper GI series. The X-ray film showed a gastric mass in the antrum, so the internist referred Dylan to a gastroenterologist, who will perform an esophagogastroduodenoscopy (EGD).

Points to think about

1. The gastroenterology nurse completes an assessment on Dylan before his EGD. Based on the knowledge of his symptoms and history, the most likely causes of the patient's symptoms are either a tumor or a bezoar. What equipment will the nurse set up for the procedure?
2. The nurse knows that this is a potentially time-consuming and resource-intensive procedure. What can be done to plan ahead?
3. The EGD confirms that Dylan has a phytobezoar. The gastroenterologist breaks up the bezoar, allows it to pass, and withdraws the scope. What might happen next?
4. Why would the gastroenterologist schedule Dylan for a repeat endoscopy?
5. What patient education would be helpful for Dylan?

Suggested responses

1. In setting up for Dylan's EGD, the gastroenterology nurse should consider:
   a. The upper GI series shows a mass in the antrum. This could be a malignant lesion, so the nurse will set up to collect specimens for biopsy and cytology.
   b. Because Dylan is diabetic, he may have decreased gastric motility caused by diabetic autonomic neuropathy and gastroparesis. This condition can allow food to remain in the stomach and collect into a phytobezoar. The nurse will set up the equipment to be used to break up this concretion and allow it to pass through the stomach (e.g., rat-tooth forceps, snare, and grasper).
   c. Some bezoars form because pyloric strictures prevent the passage of food into the duodenum. It is possible that the nurse will need to include balloons to dilate the pyloric sphincter in the setup.
2. To plan for a potentially time-consuming and resource-intensive procedure, the nurse should complete the following steps:
   a. Place the procedure at the end of the schedule, rather than during the busiest part of the day, to avoid feeling rushed.
   b. Have all equipment readily available and checked with regard to proper working order.
   c. The procedure is much easier if the patient is comfortable and cooperative. The nurse should assess medication needs throughout the procedure and give additional doses as needed. The nurse should also monitor the patient's vital signs carefully and vigilantly, maintain the airway, and suction the secretions.
3. After the bezoar has been broken up, and the endoscope has been removed, the nurse might expect:
   a. The particles to pass through if the physician has been able to ascertain that the pyloric sphincter is not stenosed. If the particles do not pass, the pyloric sphincter may need to be dilated.
   b. The physician to place Dylan on a full liquid diet for several days to help passage.
   c. Metoclopramide (Reglan) to be ordered to increase peristalsis.
   d. A repeat endoscopy to be ordered.
4. A repeat endoscopy may be ordered for the following reasons:
   a. Properly visualizing the mucosa may be difficult because of the bezoar, and it may be difficult to get past the bezoar to view the antrum and pylorus.
   b. Pyloric stenosis may require further evaluation and dilation.
   c. Gastric carcinomas may be found distal to the bezoar, causing obstruction. These should be examined by biopsy.
   d. Ulcerations may have been missed because of the presence of the bezoar.
   e. It is a good idea to check to be sure that the material has passed successfully through the pyloric sphincter.
5. Patient education considerations for Dylan could include:
   a. Poor dentition is often a problem with phytobezoars. The nurse might question Dylan about teeth and gum problems. He may have ill-fitting dentures or teeth in poor repair that impact chewing. The nurse might suggest a trip to the dentist.
   b. Poor eating habits may contribute to the formation of bezoars. The nurse might question Dylan about his diet and caution him to chew his food very well before swallowing. Consultation with a dietitian may be in order.

REVIEW TERMS

Extracorporeal shock wave lithotripsy, laser lithotripsy, lithotripsy, overtube, papillotomy, pedunculated polyps, polypectomy, polypectomy snares, sessile polyps, sphincterotome (papillotome), sphincterotomy, ursodeoxycholic acid, foreign body hood.

REVIEW QUESTIONS

1. The safest way to remove a large or dangerous foreign body that cannot be pulled through an overtube is by using:
   a. A laryngoscope and curved forceps.
   b. A grasping device.
   c. A foreign-body hood.
   d. Surgical intervention.
2. Alkaline substances from a small, button-type battery act rapidly on the mucosa and could lead to:
   a. Burns and pressure necrosis.
   b. Perforation.
   c. Esophagotracheal or esophagoaortic fistula.
   d. All the above.
3. A polyvinyl overtube is useful for:
   a. Removing foreign bodies from the duodenum.
   b. Removing small, button-type batteries.
   c. Protecting esophageal mucosa and the airway.
   d. Breaking up food obstructions.
4. Most pedunculated polyps are transected by use of:
   a. Pure cutting-current on ESU.
   b. Cold biopsy forceps.
   c. Polypectomy snare.
   d. Surgical intervention only.
5. The next important step immediately after a polypectomy is:
   a. Removal of the grounding pad to check skin integrity.
   b. Observation of the patient for abdominal pain.
   c. Retrieval of the polyp.
   d. Placement of an endoscopic clipping device to the polypectomy site.
6. Surgical excision of a gastric polyp is indicated for:
   a. Peutz-Jeghers syndrome.
   b. Intramural polypoid lesions, such as leiomyomas.
   c. Coagulopathy.
   d. Small sessile polyps.
7. A sphincterotomy is contraindicated in patients with:
   a. Recent myocardial infarction or severe pulmonary disease.
   b. Allergy to glucagon.
   c. Periampullary diverticula.
   d. Acute pancreatitis.
8. Which equipment should always be readily available when using a basket to retrieve large ductal stones?
   a. A nasobiliary drainage kit.
   b. A mechanical lithotriptor with a pair of wire cutters.
   c. An ampullary balloon dilator.
   d. An ESU.
9. Candidates for extracorporeal shock wave lithotripsy are patients who:
   a. Have multiple large common bile duct stones.
   b. Have had a cholecystectomy.
   c. Have fewer than three stones.
   d. Have larger body mass and poor gallbladder function.
10. The difference between ESWL and laser lithotripsy is that laser lithotripsy:
   a. Does not require long-term dissolution therapy.
   b. Is faster.
   c. Can be used for patients with a greater number of stones.
   d. All of the above.

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Chapter 31

COMPLICATIONS AND EMERGENCIES

This chapter will acquaint the gastroenterology nurse with some of the complications and emergencies that can occur in gastroenterology settings and will describe appropriate interventions for each situation. The topics covered include perforations of the upper and lower gastrointestinal (GI) tract, GI hemorrhage, shock, adverse drug reactions, cardiac complications and arrest, respiratory depression, vasovagal syncope, postural hypotension, aspiration, and pediatric complications and emergencies.

Learning objectives

After reviewing the content of this chapter, the gastroenterology nurse will be able to:

1. Discuss the risk factors and causes of the most important emergencies and complications encountered in the gastroenterology laboratory.
2. Identify the signs and symptoms of these complications.
3. Discuss the recommended steps in treating emergencies associated with endoscopic procedures in adult and pediatric populations.

PERFORATION OF THE UPPER GASTROINTESTINAL TRACT

Perforation of the upper GI tract can occur in the esophagus, stomach, or duodenum. Perforation during diagnostic endoscopy is extremely rare.

Esophageal perforation rates vary not only by the nature of the esophageal disorder, but also with the aggressiveness of the management approach. There is an overall mortality rate of approximately 30% associated with this complication (Markar et al., 2015).

Causes and risk factors

Perforation can be the result of blunt trauma, increased intra-esophageal pressure, underlying pathology, ingestion of a foreign body, or mechanical trauma during upper endoscopy. Therapeutic endoscopy carries a much greater risk of perforation.

Esophageal perforation during therapeutic endoscopy is seen most frequently with dilation of any type of stricture, including benign esophageal strictures and strictures resulting from radiation therapy, caustic ingestion, and malignancy. Perforation can also be a result of instrument trauma (e.g., dilators, biopsy forceps, endoscope, overtube).

Gastric perforation is less common than esophageal perforation and may be related to ulcer disease, biopsies of ulcerated lesions, and impaction of the endoscope in the hiatal hernia sac. With the advent of push enteroscopy and the use of an overtube, cases of gastric mucosal stripping and duodenal perforations have been reported.

Duodenal perforations can result from the use of electrocautery in the treatment of a bleeding ulcer in the duodenal bulb. Retroduodenal perforations can occur with sphincterotomy during endoscopic retrograde cholangiopancreatography (ERCP).

Predisposing factors to duodenal perforations include anterior cervical osteophytes, Zenker's diverticulum, stricture, malignancy, and a difficult esophageal intubation. A patient who is unwilling or unable to cooperate with an endoscopy procedure may also be at increased risk. Some research indicates that eosinophilic esophagitis predisposes to a higher incidence of esophageal tears and perforation, even during diagnostic procedures (American Society of Gastrointestinal Endoscopy [ASGE], 2012).

Symptoms

The most consistent clinical sign of upper GI perforation is pain. The type of pain and associated symptoms are determined by the site of the perforation.

- **Perforation in the cervical region of the esophagus** results in dysphagia, crepitus, stiffness of the neck, and throat and neck pain aggravated by swallowing or moving the cervical spine. The patient may also experience fever, tenderness in the affected area, or neck swelling.
- **Perforation of the thoracic region of the esophagus** typically results in substernal or epigastric pain that increases with respirations and movement of the trunk. Upper thoracic esophagus perforation commonly presents with pleural effusion. Shortness of breath, cyanosis, back pain, or pleuritic chest pain may also be present.
- **Perforation at the distal esophagus area** (near the diaphragm) results in shoulder pain, dyspnea, severe back and abdominal pain, tachycardia, cyanosis (due to mediastinal shift or pneumo-mediastinum), diaphoresis, and hypotension.
- **Spontaneous perforation (nontraumatic) of the esophagus** (Boerhaave's syndrome) is commonly associated with dyspnea after a vomiting episode. It may cause severe chest and/or abdominal pain. The patient appears acutely ill and may present with hypotension, fever, subcutaneous emphysema, and unilateral absence of breath sounds or evidence of pleural effusion.
- **Gastric perforation** results in severe back and abdominal pain, tachycardia, cyanosis, diaphoresis, and hypotension, with a drop in temperature followed by a high fever. Patients may also experience prolonged distention of the abdomen or disappearance of liver dullness on percussion following gastroscopy.
- **Duodenal perforation** may initially present with stable vital signs. The patient then experiences sudden local or generalized abdominal pain. Although a brief period of improvement follows, peritonitis develops. The abdomen becomes rigid, and the patient develops a high fever, hypotension, tachycardia, and severe pain that inhibits abdominal movement and the ability to breathe deeply.

Both gastric and duodenal perforation can produce indications of leaking gastric or duodenal contents, such as acute upper abdominal pain and subsequent signs of guarding, rebound tenderness, or absent bowel sounds.

Previously, contrast esophagography was the mainstay for diagnosis of esophageal perforation, with a false negative rate in of 15% to 25% (Han et al., 2018). Today, esophageal perforation is more commonly diagnosed with the use of chest and neck computerized tomography (CT) and the use of Gastrografin contrast.

Interventions

Early identification and management of perforations have been shown to decrease mortality. Delay in diagnosis or inappropriate selection of a conservative management approach is likely to result in a negative patient outcome. Consultation with a thoracic surgeon should be sought immediately if perforation is suspected.

Close observation and symptomatic treatment are critical. Perforations that are recognized early are sometimes treated conservatively with nasogastric or pharyngeal suction, strict intake and output (I &O), parenteral nutrition, and broad-spectrum antibiotics. Conservative management is contraindicated when a major leak is confirmed radiologically or if perforation occurs through an ulcer or tumor.

If necessary, perforations in the cervical region may be sutured and the patient treated with intravenous (IV) antibiotics. Contained perforations in the thoracic region may be treated conservatively. Surgery is often recommended for intrathoracic perforations, which have a poorer prognosis. If an intrathoracic perforation is not contained, urgent thoracotomy with chest and mediastinal drainage and esophageal suturing is indicated.

Small gastric perforations into the lesser sac of the peritoneal cavity may be managed conservatively. Anterior gastric perforations with peritoneal signs and abdominal free air confirmed radiologically require surgical management.

Retroduodenal perforations may be managed conservatively with bowel rest and duodenal suction. Perforations resulting from ERCP with sphincterotomy may require placement of a nasobiliary tube or stent. The endoscopist can use endoclips and seal the perforation if it is recognized at the time of the procedure. The patient will require IV fluids, strict I &O, and IV antibiotics.

PERFORATION OF THE LOWER GASTROINTESTINAL TRACT

Perforation rates for colonoscopy vary widely (ASGE, 2015). Lower GI perforations can be spontaneous or mechanical. They occur most often from mechanical trauma related to manipulation of an instrument during colonoscopy and may be related to the level of skill of the physician performing the procedure and whether the colonoscopy is done for screening or for diagnostic or therapeutic indications (Polter, 2015). Perforation rates in screening patients tend to be lower because the patients are generally healthy and seldom have associated colon conditions that have been associated with perforation.

Causes and risk factors

Pneumatic perforation may result from overdistention with insufflated air. The cecum is most susceptible to pneumatic rupture because the colonic wall tension is highest in the cecum. Initially, a linear tear of the serosa forms in the overdistended segment of bowel. The mucosa may then either herniate through the opening and burst or become progressively thinner until it is permeable to air. This is more likely if transmural inflammation or ulceration weakens the colon wall (Onwubiko et al., 2015).

The risk of perforation with polypectomy and diagnostic colonoscopy is usually related to electrical injury of the bowel wall. Types of electrical injuries include:

- Excessive current applied to the base of a polyp before cutting.
- Incidental injury to the opposite wall.
- Entrapment of normal mucosa in snare polypectomy ("tenting").

The risk of post polypectomy perforation may be increased with large polyps, sessile polyps, and removal of lipomas. Perforations secondary to polypectomy may occasionally be delayed and have been reported to occur up to 2 weeks after the procedure. In addition to polypectomy, thermal injury perforation may also occur with laser therapy and a variety of thermal devices such as gold probes for treatment of arteriovenous malformations (AVMs).

Certain factors are believed to put patients at higher risk of lower GI tract perforations. These include ischemic colitis, adhesions, strictures, radiation, diverticular disease, obstruction, inflammatory bowel disease (IBD), malignancy, and preexisting partial tears or necrosis. Patient factors that increase the risk for perforations include an inability or unwillingness to cooperate or a poorly prepped bowel. Studies have also shown an increase in colon perforations in patients over 75 years of age due to decreased colonic wall strength (de'Angelis et al., 2018).

Symptoms

Increasing abdominal pain following a colon exam may also indicate a perforation. Initially, patients may experience little or no pain due to the effects of sedation administered during the procedure. As the effects of sedation subside, pain may progressively worsen.

Signs of lower GI perforation include sudden and severe abdominal pain (that becomes generalized), possibly accompanied by signs of peritonitis, abdominal distention, pneumoperitoneum, increased tympany, loss of hepatic dullness, malaise, fever, a change in vital signs, and bloody or mucopurulent rectal drainage.

A hole in the colon may be seen by the endoscopist or the omentum may be visualized. On occasion, a colonic perforation is confirmed radiographically using plain abdominal X-rays in the supine position. A microperforation can only be diagnosed by a CT scan of the abdomen.

Interventions

Most free-air colonoscopic perforations need prompt surgical repair. Patients with retroperitoneal perforations, which are usually pneumatic perforations, should receive symptomatic treatment and close observation. Surgical intervention is indicated only if signs progress.

A rare but potentially life-threatening colonoscopic complication is a splenic injury. This may result as the endoscope is advanced around the splenic flexure and the capsule surrounding the spleen is torn secondary to instrument manipulation. The presence of adhesions near the spleen may play a role in this rare complication. Splenic injury may be difficult to diagnose because the bleeding from this type of injury occurs outside of the colon into the peritoneal cavity. Symptoms may range from abdominal tenderness to a rigid acute abdomen, hypotension, and exsanguination. Minor tears may be managed medically, but larger tears will require emergent surgical intervention and possible splenectomy.

GASTROINTESTINAL HEMORRHAGE

GI **hemorrhage** may be associated with an underlying disease state or trauma or may arise as a rare complication of diagnostic endoscopy (Figure 31.1).

Causes and risk factors

Therapeutic endoscopic procedures, including sclerotherapy, polypectomy, laser therapy, tumor ablation, and dilatation, are more likely to cause bleeding. Hemorrhage is the most common complication of polypectomy. Inexperience or lack of expertise on the part of the electrosurgical unit (ESU) operator and/or malfunction of the ESU can contribute to the risk of polypectomy-related hemorrhage.

Upper GI bleeding

Peptic ulcer disease is the leading cause of upper GI hemorrhages, accounting for approximately 50% of cases (Laine, 2016). Esophageal variceal hemorrhage accounts for an estimated 10% of upper GI bleeding (Dehal & Tessier, 2014; Solomon, 2016). Less common causes include Boerhaave's syndrome and Osler-Weber-Rendu disease.

Precipitating factors for upper GI bleeds include traumatization of recently bleeding esophageal varices; esophagitis, gastritis, or duodenitis; mucosal tears in patients with strictures; cardioesophageal (Mallory-Weiss) tears; and biopsy performed on a recently bleeding lesion or at the base of ulcers where larger vessels may be exposed. Hereditary hemorrhagic telangiectasia—a rare genetic disorder that leads to abnormal blood vessel formation in the skin, mucous membranes, and often in organs—may also increase risk.

Lower GI bleeding

Common causes of lower GI bleeding include diverticulosis, IBD, colonic angiodysplasia, and hemorrhoids. Less common causes include intussusception, rectal trauma, and anal disorders.

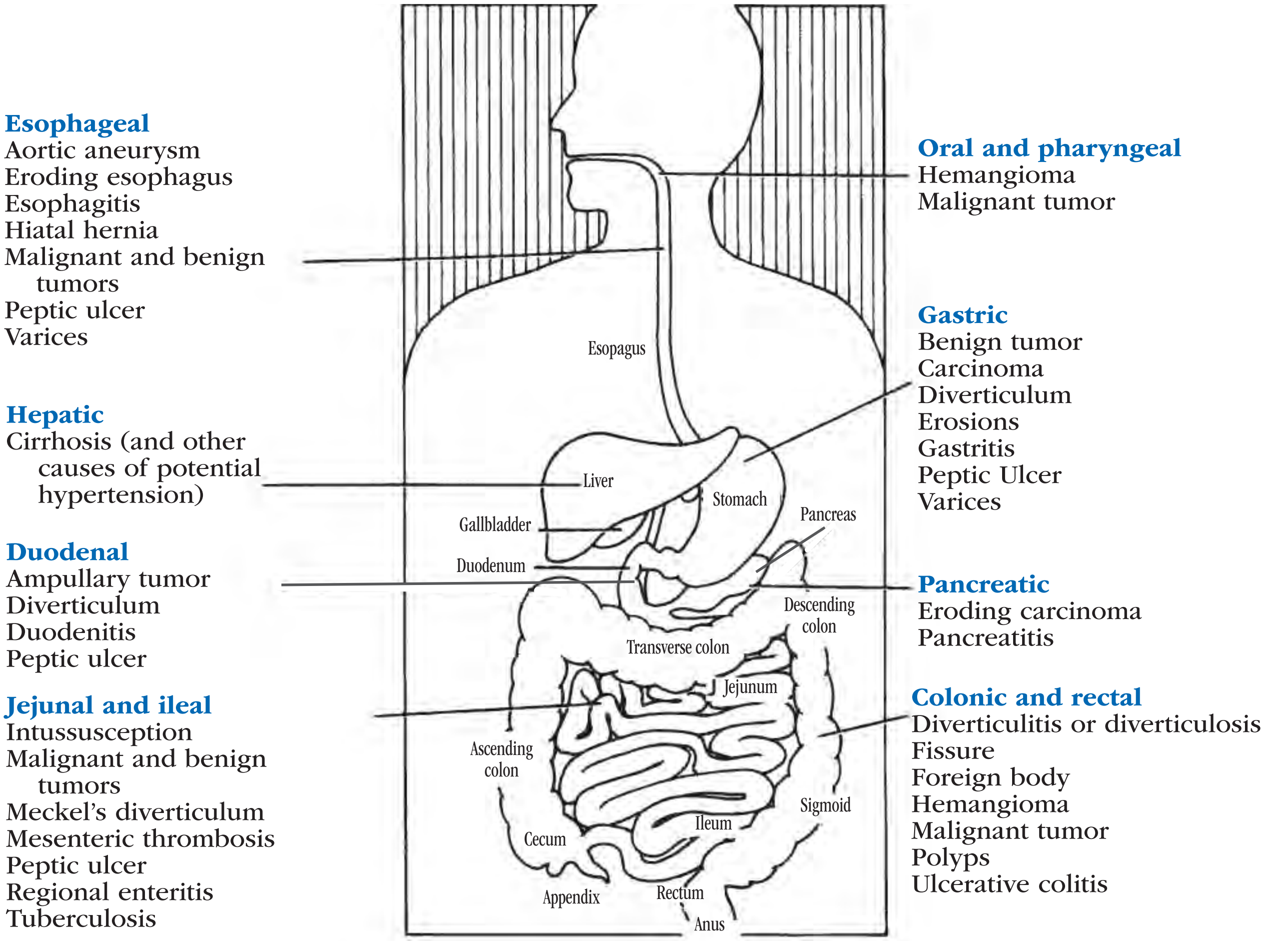

Figure 31.1. Common Sites and Causes of GI Bleeding

Symptoms

Upper GI bleeding is indicated by hematemesis, hypotension, tachycardia, or melena (liquid, tarry, foul-smelling black stools).

Lower GI bleeding is associated with tachycardia, hypotension, weakness, pallor, bright red rectal bleeding (hematochezia), maroon or black stools, and possibly abdominal pain and distention. Bleeding as a complication of polypectomy is most often intracolonic, resulting in obvious hematochezia.

Complications of GI bleeding may include:

- Anemia from blood loss.
- Hypovolemic shock from severe volume depletion.
- Exsanguination from rapid massive intravascular blood loss.
- Myocardial or cerebral infarction from acute hemoglobin depletion.
- Disseminated intravascular coagulation from shock and clotting factor loss.
- Peritonitis and sepsis, if accompanied with bowel rupture.
- Aspiration from massive upper GI bleeding.

Interventions

The American Society for Gastrointestinal Endoscopy (ASGE) has advised that routine screening via laboratory testing is not recommended for all GI endoscopy patients (ASGE, 2014). In patients with known or suspected liver disease or bleeding disorders, or who are taking anticoagulants, some laboratory tests may be useful in minimizing the risk of bleeding during endoscopic procedures.

Evaluating the patient with suspected GI bleeding involves:

- Taking a thorough history, including any history of hematemesis, hematochezia, pain or abdominal tenderness, and severe retching.
- Completing a physical examination, including a rectal examination, and noting the presence of any pallor of the lips, gums, ear lobes, and nail beds.
- Identifying routine medication usage, including over-the-counter drugs.
- Reviewing any concomitant heart, lung, renal, liver, or central nervous system (CNS) disease.
- Obtaining and monitoring vital signs, including postural signs.
- Monitoring labs, including hemoglobin and hematocrit.

Invasive procedures may be indicated to make a prompt diagnosis. It is imperative that patients with bleeding be managed aggressively because of the high mortality rate associated with continued bleeding or rebleeding. This is especially true for patients with esophageal and/or gastric varices.

Determination of upper GI bleeding from such conditions as peptic ulcer disease (PUD) with an extremely rapid transit time and severe pylorospasm may present as lower GI bleeding. Endoscopists generally do not rely solely on clinical presentation and may order a combination of upper and lower endoscopy.

Upper GI bleeding

Initial management of patients with upper GI bleeding should focus on maintaining the airway, providing oxygenation, monitoring serum hemoglobin (Hgb) and hematocrit (Hct) levels every 6 hours during the acute phase, establishing a large-bore IV catheter, maintaining adequate fluid volume, stabilizing the patient's condition, stopping the bleeding, and beginning appropriate therapy to prevent further bleeding.

Continued management of patients with upper GI bleeding may include:

- Frequent assessment for signs and symptoms of electrolyte disturbances and continued or recurrent bleeding.
- Central venous line insertion and serial laboratory studies, as required.
- Administration of crystalloids (such as normal saline or Lactated Ringer's solution) to maintain intravascular volume and colloids (whole blood; packed red blood cells; and blood products such as fresh-frozen plasma, platelets, and albumin) as ordered by the physician.
- Nasogastric aspiration and gastric lavage using a large-lumen tube such as an Ewald tube or BioVac to remove clots from the stomach, if better visualization is needed. If the return fluid from the nasogastric tube is clear initially, or if it clears promptly with lavage, the tube can be removed. If the return is bloody, lavage should be continued until clear.
- Use of supplemental oxygen, pulse oximetry, and electrocardiogram (ECG) monitoring may be needed in patients with hypovolemia to continually assess and combat possible ischemic effects.

Depending on the cause and site of upper GI bleeding, the physician may treat it with gastric lavage, drug therapy, balloon tamponade, injection sclerotherapy, thermal coagulation methods, endoscopic variceal ligation (banding), endoclip or endoloop application, or surgery. Endoscopic techniques for the management of upper GI bleeding are discussed in detail in Chapter 28, including thermal coagulation methods, injection sclerotherapy, banding, and esophagogastric tamponade. Gastric lavage is discussed in Chapter 29.

In patients with acute upper GI bleeding, upper endoscopy may reveal stigmata of recent hemorrhage, including a visible vessel, fresh blood clot, black eschar, or active bleeding. If an obvious bleeding source is identified, the endoscopist may choose to treat the source prophylactically to prevent a rebleed, "tease" the bleeding site to determine the potential for rebleeding and treat if bleeding occurs, or treat medically or conservatively and monitor the patient for signs of recurrent bleeding. If no signs of recent hemorrhage are visible, medical treatment may be sufficient. In patients with ongoing bleeding or stigmata of recent hemorrhage, therapeutic endoscopy or surgery may be indicated, in addition to further diagnostic tests.

Conditions such as PUD can cause an extremely rapid transit time and severe pylorospasm presenting as lower GI bleeding. Endoscopists generally do not rely solely on clinical presentations and, in conjunction, order a combination of upper and lower endoscopy.

Lower GI bleeding

The following steps may be taken to determine the source of lower GI bleeding:

- A thorough history is obtained to determine signs and symptoms that may pinpoint the cause of the bleeding. A painless bleed may indicate a diverticular bleed, arteriovenous malformation (AVM), hemorrhoids, or neoplasm. Pain preceding the bleed may indicate ischemic bowel disease, IBD, or infection. Information collected should include the color of the stool (e.g., black, maroon, bright red) and the consistency characteristics of the stool (e.g., sticky, tarry, mucousy, watery). This description can help identify the possible cause and location of the bleed.
- A physical examination should include a rectal exam, as well as anoscopy and sigmoidoscopy.
- An esophagogastroduodenoscopy (EGD) can be done to exclude an upper GI lesion if no obvious bleeding site is found and the patient continues to bleed. There is a slight possibility that severe hematochezia is related to an upper GI source.
- A technetium-99m-labeled red blood cell (Tc99m RBC) scan can detect bleeding rates as low as 0.1 ml/min. Limitations with Tc99m RBC scans are encountered in patients experiencing intermittent bleeding, as seen in AVMs.
- A colonoscopy using a standard colon prep may be ordered, if the patient is hemodynamically stable.
- A mesenteric arteriography may be performed if bleeding remains brisk.
- If extravasation of contrast is seen, octreotide may be given, or embolization therapy considered. If bleeding persists despite octreotide or embolization, replacement transfusion can be given or continued and surgery performed.

- If no extravasation is noted, but significant bleeding persists, the physician may consider exploratory laparotomy.

In patients with postpolypectomy hemorrhage, management involves stabilizing the patient; monitoring blood pressure, urine output, and central venous pressure in unstable patients; and transfusion in 40% of patients (Wu et al., 2013). Endoscopic techniques for management of a lower GI bleed are discussed in Chapter 28.

Emergency endoscopy

If emergency endoscopy is indicated to control GI bleeding, the gastroenterology nurse should:

- Verify signed informed consent.
- Obtain the patient's medical history, including current medications, allergies, and a history of the present bleeding episode.
- Remove dentures or partials.
- Monitor vital signs at least every 15 minutes, or more frequently according to the organizational policy; monitor cardiac status using a heart monitor and oxygen saturation using a pulse oximeter; and consider capnography if the patient has a history of sleep apnea or chronic obstructive pulmonary disease (COPD).
- Establish a large-bore IV line(s).
- Obtain blood samples for complete blood count (CBC) and type and crossmatch, as ordered.
- Start oxygen via nasal cannula, as ordered. Staff should be prepared to perform gastric lavage using a large-bore orogastric or nasogastric tube.

After an emergency endoscopy, it is important for the gastroenterology nurse to:

- Monitor vital signs.
- Observe the patient for bleeding, vomiting, change in vital signs, pain, and possible abdominal distention.
- Document the type, amount, and times of administrated IV replacement fluids.

SHOCK

Shock is a condition of acute peripheral circulatory failure caused by the derangement of circulatory control or loss of circulating fluid. Shock may be endocrine, neurogenic, bacterial, cardiogenic, or hypovolemic in origin, but it is always characterized by decreased tissue perfusion and increased peripheral vasoconstriction. The two most common forms of shock in gastroenterology patients are hypovolemic shock and septic shock (bacteremia).

Hypovolemic shock

Hypovolemia causes the body to respond to the perceived volume loss by decreasing urinary output and constricting peripheral blood vessels. At the same time, blood flow to the cerebral and cardiovascular systems is increased. Glycogen breakdown increases, insulin is suppressed, and hyperglycemia occurs. If cerebral blood flow is decreased, a subsequent loss of consciousness may occur, and the central regulating centers may shut down as a result.

Causes and risk factors

In hypovolemic shock, the loss of circulating volume may result from loss of blood or plasma, intestinal obstruction, nephrotic syndrome, dehydration, or trauma. In patients undergoing paracentesis, hypovolemic shock may result from the rapid shift of fluid from the circulatory system to the peritoneum as the body attempts to replace the aspirated fluid.

Symptoms

Signs of shock include altered mental status; restlessness; increased anxiety; pale, cool, clammy skin; collapse of peripheral neck veins; and tachycardia. Decreased capillary refill time may be present, which is assessed by depressing the fingernail bed and releasing. In the early stages, hypertension may occur. Hypotension is an advanced sign. The patient may also show diaphoresis, piloerection (goose bumps), muscle weakness, decreased urinary output, hyperventilation, abdominal distention, and electrolyte imbalance.

Interventions

Treatment should focus on restoring tissue perfusion and reducing peripheral vasoconstriction.

- Position the patient in a way that facilitates proper blood supply to the brain and promotes venous return from the rest of the body.
- To promote respiration, the patient should lie flat with legs elevated at the hips to 20 degrees or with the head elevated slightly.
- To decrease oxygen use, activity should be minimized.
- Provide intravascular fluid administration (IV fluids), which maintains patency of the vascular system, restores volume, maintains an oxygen transport system, and removes wastes. Depending on the patient's hematocrit (Hct), the fluid given may be blood, plasma, plasma expanders, or a balanced salt solution (Lactated Ringer's).

Pharmacology

- Administer vasoactive agents or steroids, which decrease peripheral resistance, increase tissue perfusion, increase venous return, and have a positive effect on the heart muscle.
- Correct metabolic acidosis, which is generally corrected by the administration of fluids.

Respiratory

- Endotracheal intubation or a volume respirator may be needed to sufficiently maintain oxygenation.

Hemostasis

- Depending on the bleeding site and the severity of the bleed, a gastroenterologist, interventional radiologist, and a surgeon may be involved in the care of the patient with a GI bleed.
- Methods of controlling external bleeding are direct pressure, ligation, cautery, or application of topical medication.
- Central venous pressure readings may be ordered to monitor the course and effectiveness of treatment.

Septic shock

Septic shock, or bacteremia, is produced by a direct invasion of the bloodstream by microorganisms or their toxins.

Causes and risk factors

Septic shock may be associated with peritonitis, infections such as cholecystitis or necrotizing pancreatitis, bowel surgery or trauma, reduction of volvulus, or hemorrhage. Patients undergoing biliary stent replacements may experience bacteremia (or even sepsis) as a result of the "bacterial shower" into the blood stream that occurs as the obstructed stent is removed.

Symptoms

Initial symptoms of septic shock can be chills; fever; warm, flushed skin; mild hypotension; and an increased cardiac output. This is sometimes referred to as warm shock. Without treatment, warm shock can rapidly progress to the symptoms exhibited in hypovolemic shock.

Interventions

Treatment focuses on removing the cause and providing antibiotic therapy. Preventing system failure during the initial phase, before the antibiotics have had a chance to reach effective serum levels, may include the use of vasopressors such as dopamine or dobutamine.

Although GI endoscopy, dilatation, and sclerotherapy are associated with transient episodes of bacteremia, antibiotic prophylaxis for prevention of endocarditis is controversial. Most experts agree that prophylaxis for high-risk patients (those with prosthetic valves or a surgically constructed systemic pulmonary shunt or a history of endocarditis) undergoing high-risk procedures, including dilation, sclerotherapy, or ERCP with bile duct obstruction, may help avoid septic shock due to complications associated with bacteremia.

ADVERSE DRUG REACTIONS

The gastroenterology nurse must be aware of potential side effects, toxic reactions, and allergic responses whenever a drug is administered.

- A **side effect** is any drug effect that is not intended. Some side effects are transient and subside as the patient develops a tolerance to the drug. In some patients, adjusting the drug dosage may control undesirable side effects, but in others, side effects contraindicate the use of a drug.
- A **toxic reaction** can be acute, resulting from excessive drug doses, or chronic, resulting from progressive accumulation of a drug in the body. Toxic reactions also can result from impaired metabolism or excretion that can cause elevated blood levels.
- A **drug allergy or hypersensitivity** results from an antigen-antibody immune reaction in susceptible patients. Skin lesions (urticaria) and respiratory distress are the most common symptoms of drug allergies, but cardiovascular dysfunction and potentially fatal **anaphylaxis** may also occur. It is vitally important to check for drug allergies before administering the patient any medications.

See Chapter 20 for a discussion of drugs commonly used in the gastroenterology setting.

Causes and risk factors

Most adverse drug reactions in the gastroenterology suite are related to the substances used for moderate sedation and analgesia. Sedatives can cause excessive depression of the central nervous system, particularly in elderly patients and those with hypoalbuminemia. It is important to obtain a complete patient history and physical prior to medication administration.

- Benzodiazepines. Patients who have a known or suspected benzodiazepine dependence should be monitored carefully for seizure activity when receiving flumazenil as a reversal agent for benzodiazepines. All patients receiving flumazenil should also be monitored for resedation because sedation may outlast reversal effects. The patient may need to be monitored longer post procedure.
- Narcotic analgesics. Patients should be monitored for adverse effects, including respiratory depression associated with the administration of all opioids. Narcotic antagonists such as naloxone reverse the side effects of meperidine, but as with flumazenil, patients should be closely observed for resedation. Staff should be alert to the danger of drug withdrawal when narcotic antagonists are given to patients who are being treated with narcotics or have a history of narcotic drug misuse.

All patients receiving a reversal agent should be monitored after each dose for resedation because sedation may outlast reversal effects. Patients may need longer post-procedure monitoring in accordance with organizational policies. Always check organizational policies for guidelines in the use of sedating and reversal agents.

Anticholinergic agents and hormones such as atropine, glycopyrrolate, and glucagon have been associated with reports of ileus, urinary retention, arrhythmias (bradycardia, tachycardia, and atrioventricular block), vomiting, and hyperglycemia. Atropine can increase intraocular pressure in patients with narrow-angle glaucoma.

For patients with a documented or possible latex allergy, every effort must be made to minimize exposure to products that contain latex. As latex allergies and sensitivities have become more prevalent in patients and health care workers, most facilities have chosen latex-free alternatives to minimize latex exposure. Examples of items in the gastroenterology setting that may contain latex include bite blocks straps, syringe plungers, adhesive tape, tourniquets, feeding tubes, and endoscopic balloons (through-the-scope [TTS] balloons).

Symptoms

When administering sedation, the patient should be observed for signs of laryngospasm, bradycardia, excessive diaphoresis, hypotension, respiratory depression, or apnea. Often, EKG monitoring and capnography are utilized when procedural sedation or moderate sedation is administered.

Interventions

Medication administration

To avoid adverse medication effects, all medications should be administered with caution. Titration in incremental dosages should be performed slowly to the level of sedation required.

Anaphylaxis

Early diagnosis and treatment of anaphylactic reactions are necessary to avert death, which can occur within 15 minutes. There must be constant attention to an adequate airway and adequate perfusion. Epinephrine and an antihistamine are the usual treatment.

Allergies to devices containing latex have been identified as a cause of anaphylaxis. Additional emergency treatment for a latex allergy anaphylactic reaction includes removal of the suspected antigens and changing IV catheters and tubing. All other treatments are aimed at correcting respiratory compromise, hypotension, and cardiovascular collapse.

Resuscitation

Resuscitation equipment and reversal agents should always be readily available on every endoscopy unit. A "crash cart" with equipment for cardiopulmonary resuscitation should include airway, bag-valve-mask resuscitation setup, oxygen supply tubing, suction, laryngoscope, IV infusion setup, medications, and a defibrillator. This cart should be checked daily, after each use, or in accordance with organizational policy. The defibrillator should be tested according to organizational policy, and the cart should be restocked after each use and if drugs are outdated. An oxygen supply must be available when resuscitation is attempted.

CARDIAC COMPLICATIONS AND ARREST

Another complication of upper endoscopy (after medication reactions) is cardiopulmonary complications (ASGE, 2012).

Causes and risk factors

Upper GI endoscopy

A mild to moderate amount of cardiovascular stress can be associated with upper GI endoscopy. Minor ECG changes during the procedure are common. These are primarily sinus tachycardia, ST segment changes, and premature atrial or ventricular contractions. These changes are more likely to occur in the elderly and in patients with ischemic heart disease or chronic lung disease. Serious ventricular arrhythmias and myocardial infarction are uncommon in the endoscopy setting. Some ECG changes are related to anxiety and premedication rather than to the procedure itself.

Lower GI endoscopy

According to American Society for Gastrointestinal Endoscopy (ASGE, 2011), ECG changes also occur during lower endoscopy, but these changes are usually transient. During colonoscopy, the gastroenterology nurse must be mindful of grounding pad placement and the use of electric cautery for patients with a pacemaker and/or automatic internal cardiac defibrillator (AICD).

Bipolar or multipolar accessories are preferred to monopolar devices whenever available. Argon plasma coagulation (APC) is one of the monopolar devices and should be used with the lowest possible power available and brief, intermittent bursts.

Cardiac arrest

Some of the major causes of **cardiac arrest** in the gastroenterology unit are:

- Too-rapid administration or overdosage of medications.
- Obstruction of the respiratory tract.
- Preexisting conditions, including acute anxiety states, anemia, cardiac disease, dehydration, pulmonary edema, and shock.

Symptoms

Signs and symptoms of impending cardiac arrest include:

- Symptoms of respiratory obstruction (cyanosis, gasping respiration, and increased rate and shallowness of respiration).
- Pulse irregularities and rate changes.
- Muscle twitching.
- Cold and clammy skin.

Initial signs of collapse of cardiopulmonary function are absent pulse and/or respiration and lifeless appearance. If these conditions are present, cardiopulmonary resuscitation (CPR) must begin immediately. In any situation, the most important principles of CPR include establishment of a patent airway, provision of ventilation, and restoration of cardiac pumping action.

Interventions

Cardiac arrest

When a patient experiences cardiac arrest, the gastroenterology nurse must act quickly. Brain damage or death can result if circulation is not restored within 3 to 6 minutes after cardiac and respiratory arrest.

- First, alert the resuscitation team or another source of assistance.
- Place the patient on a flat, firm surface and initiate CPR.
- Attach the patient to an electrocardiograph or a defibrillator. Defibrillation should be performed by a qualified person.
- Start an IV line using a large-bore catheter.
- Set up the oxygen apparatus and administer oxygen.
- Prepare the tracheal suction apparatus and catheters and suction the patient as needed.
- Assist with intubation, if needed.
- Monitor vital signs.

Events immediately preceding the crisis should be thoroughly documented and the necessary data provided to the resuscitation team. Documentation continues during resuscitation in accordance with institutional policy. The attending physician should be notified if he or she is not in attendance.

Any excess equipment and personnel should be removed from the area. Family members and visitors are escorted to a waiting area and provided with emotional support. Other patients should be screened from activities.

If the resuscitation effort is successful, the patient will be transferred to a monitored area, and the physician should speak with the family. If it is unsuccessful, the physician should notify the family and provide emotional support. The physician may request consent for autopsy. If the family agrees to an autopsy, consent for an autopsy must be obtained. The established organizational protocol for a death in the gastroenterology setting should be followed.

Resuscitation equipment should be checked according to organizational policies and restocked after each use to ensure all necessary equipment is available.

RESPIRATORY DEPRESSION

When treating patients with medications that have been associated with **respiratory depression**, including narcotics and benzodiazepines, the drugs should be titrated to achieve maximum effect with the minimum dose.

Causes and risk factors

In patients receiving anesthesia, sedatives, or narcotics, respiratory depression can occur, potentially leading to cardiovascular collapse.

Symptoms

Symptoms of respiratory depression include a decrease in respiratory rate and/or tidal volume; oxygen desaturation, as measured by pulse oximetry; increase in carbon dioxide (CO_2) when using capnography for patients with compromised respiratory status (sleep apnea, COPD); Cheyne-Stokes respiration (rhythmic waxing and waning of respirations); periods of apnea; and cyanosis. These symptoms may be accompanied by extreme somnolence; skeletal muscle flaccidity; cold, clammy skin; and sometimes bradycardia and hypotension.

Interventions

Diligence should be paid in the event of decreasing oxygen saturation. Inform the endoscopist of the decreasing saturation.

Increasing the oxygen flow rate and performing the head-tilt and chin-lift maneuver may be all that is necessary to increase the oxygen saturation level to within acceptable limits and avoid serious complications. Elevated CO_2 levels noted with capnography can often be managed by stimulating the patient to breathe.

If that does not successfully increase oxygen flow, respiratory depression may be reversed with reversal agents such as naloxone and/or flumazenil. Elderly patients and patients with underlying pulmonary disease may require smaller doses.

Additional caution should be used when sedating the elderly. Due to a slower metabolism, the sedation may require a longer time to take effect. If slower metabolism is not taken into account, the elderly have an increased potential for oversedation.

Patients who have hepatic and/or renal dysfunction may take twice as long to clear the drugs from their system. For this reason, patients with these coexisting conditions should be monitored more closely following their procedures when they are no longer being stimulated by the endoscope. Patients with hepatic disorders do better with sedation if using fentanyl rather than meperidine.

If the patient stops breathing, the physician should be notified immediately. The most important factors are the establishment and maintenance of the patient's airway. The nurse should:

- Place the patient in a supine position and tilt the head back, taking care to avoid hyperextension and subsequent closing of the airway.
- Insert an oropharyngeal airway.
- Administer oxygen by a bag-valve-mask apparatus.
- Assist with intubation of the patient, if necessary.

- Administer an appropriate IV dose of naloxone and/or flumazenil in conjunction with efforts at respiratory resuscitation.
- Check for a pulse. If the carotid pulse cannot be palpated, chest compressions should be initiated.

Usually, once a reversal agent takes effect, the patient begins to awaken and breathe independently. The patient should be monitored for resedation after each dose.

VASOVAGAL SYNCOPE

A vasovagal episode is a transient vascular and neurogenic reaction marked by a rapid drop in blood pressure. When blood pressure falls below a critical level, this results in loss of consciousness known as **vasovagal syncope**, or fainting.

Causes and risk factors

Vasovagal syncope is most often evoked by emotional stress associated with fear or pain. This response can also be seen prior to the procedure with IV insertion. Vasovagal episodes have been reported during colonoscopy. There is no clear agreement as to the cause, but several theories include stretching of the mesentery and the presence of diverticular disease.

Symptoms

A vasovagal episode is marked by pallor, nausea, diaphoresis, bradycardia, a rapid fall in blood pressure, vasovagal syncope, and characteristic electroencephalographic changes.

Interventions

To treat vasovagal syncope, the patient is positioned with the head level with or lower than the rest of the body. Consciousness usually returns quickly once the patient is recumbent. Additional treatment may include IV fluids at an increased rate, supplemental oxygen, and IV atropine.

If the vasovagal syncope is a response to patient discomfort that causes bearing down and holding his or her breath, the patient should be encouraged to breathe. Once the heart rate and blood pressure are adequate, additional sedation or analgesia may be necessary to increase patient tolerance.

POSTURAL HYPOTENSION

Postural hypotension may occur when the patient suddenly assumes an upright position.

Causes and risk factors

Postural hypotension is commonly associated with prolonged bed rest, volume depletion, neurologic disease, and the use of narcotics in ambulatory patients. Dehydration can occur quickly in children and the elderly due to the fragile nature of their fluid and electrolyte balance.

Symptoms

A fall in systolic and diastolic blood pressure occurs, sometimes accompanied by weakness, dizziness, or syncope.

Interventions

When discharging a patient who has undergone a bowel prep for a GI procedure, great care should be taken to prevent postural hypotension.

ASPIRATION

Aspiration occurs when liquids or solids mistakenly enter the pulmonary system. The incidence of aspiration in gastroenterology is rare, but the mortality rate of this complication may reach 10%. Patients can aspirate saliva, gastric secretions, blood, or retained gastric contents or food residual. It is important to have suction equipment readily available to maintain a patent airway.

Causes and risk factors

Topical pharyngeal anesthesia, sedation, and supine positioning may be contributing factors in aspiration and its sequelae. Other factors include a full stomach, active bleeding, and retained gastric secretions. The elderly and other patients who have depressed cough and gag reflexes are at increased risk of aspiration. Aspiration is also a problem in patients who are receiving enteral nutrition.

Interventions

Enteral nutrition

Aspiration of gastric secretions may be minimized by placing a feeding tube well beyond the pylorus into the duodenum, by controlling gastric volumes, and by elevating the patient's head and shoulders.

The question of whether percutaneous endoscopic gastrostomy tubes prevent or cause aspiration is debatable. Most of the literature agrees that the method of enteral formula administration (continuous versus intermittent infusion) does not affect the incidence of aspiration. It is difficult to determine, in cases of aspiration pneumonia, whether the source is oropharyngeal secretions or gastric secretions.

Upper endoscopy

During upper endoscopy, a suction apparatus should be available to suction the patient's airway. The gastroenterology nurse should suction the patient, if necessary, to avoid aspiration. The risk of aspiration is also reduced by performing the procedure with the patient positioned on the left side, with the head tilted to the side to allow secretions to run out.

PEDIATRIC COMPLICATIONS AND EMERGENCIES

Upper GI endoscopy and colonoscopy in pediatric patients are useful tools for diagnosis and management of a number of conditions. Diagnostic endoscopic procedures generally are safe, with a rate of less than 1% for serious complications (ASGE, 2012). As with the adult population, therapeutic procedures carry higher rates of complications but usually are also accomplished without significant problems.

Causes and risk factors

Complications of upper endoscopy in children are the same as in the adult population, with the following additions:

- Oral trauma caused by the presence of loose baby teeth. Pediatric patients should be assessed for orthodontic appliances, which could be dislodged and potentially aspirated.
- Upper airway obstruction caused by selection of incorrect scope size. Observe the pediatric patient for inspiratory stridor, which could be caused by trachea compression by the endoscope.
- Oxygen desaturation caused by crying and struggling.
- Hypoxemia as a result of methemoglobinemia, which can develop after exposure to benzocaine or related medications.
- Overdistension of the abdomen, which can lead to respiratory compromise.
- Arrhythmias, which are rare during endoscopy in the pediatric population.

Pediatric complications during lower endoscopy have been reported infrequently, and data used have been extrapolated from the adult population.

Most problems during endoscopic procedures center around sedation and analgesia. A systematic review indicates that sedation in the pediatric intensive care setting has been found to be less than optimal. The review also suggests that oversedation is more common than undersedation (Vet et al., 2013).

Interventions

In the event of significant decreased oxygen saturation during upper endoscopy, reposition the head, assess for abdominal distension, and consider aborting the procedure until the cause of respiratory compromise is identified and treated.

Sedation

In upper endoscopy, patients age 3 to 9 prove to be the most difficult to sedate, and patients age 6 to 9 require the highest doses of midazolam and meperidine.

During lower endoscopy, doses follow more closely with the recommendations using body weight. Age itself is not a factor.

Procedures are contraindicated when patients cannot be adequately sedated. General anesthesia or monitored anesthesia care (MAC) should be considered for these patients.

Depending on the procedure and condition of the patient, the gastroenterologist may decide that the child requires sedation in the operating room, where the child can be safely monitored by the anesthesiologist.

CASE SITUATION

It's midnight on a Friday. The gastroenterology nurse on call is awakened by her beeper. She checks the number and calls the hospital. The intensive care unit (ICU) informs the gastroenterology nurse that they have just admitted a John Doe who is vomiting bright red blood. The gastroenterologist on call is on his way in and has requested that the gastroenterology nurse meet him as soon as possible in the ICU.

While driving to the hospital, the gastroenterology nurse goes over what she will need to do. Late at night is not an optimal time to do emergency endoscopy in the gastroenterology unit because there are usually not enough personnel present to handle any potential complications. The patient will therefore be cared for at the bedside in the ICU. It is often difficult to get supply items from central service or medications from the pharmacy in the middle of the night. It is essential that the gastroenterology nurse be prepared for any and all possible complications.

Points to think about

1. Because this is a new patient, the gastroenterology nurse has no medical history on which to rely and must think of all possible causes of upper GI bleeding that the gastroenterologist may be able to treat endoscopically. What will she take to the bedside?
2. Because this patient is a John Doe, he is either too disoriented to provide his name or is unconscious and has no identification. If the patient is conscious, some quick questions might help the gastroenterology nurse assess possible problems. What would she ask?
3. The gastroenterologist arrives and, together with the gastroenterology nurse, plans the course of action. The gastroenterology nurse prepares the patient for endoscopy. During the procedure, the patient will be at risk for the serious complication of aspiration. How should the gastroenterology nurse address this issue?
4. Assessing the amount of blood lost is important when judging the amount of fluid needed to reverse volume depletion. Using laboratory data, what can the gastroenterology nurse learn about this?
5. What should the gastroenterology nurse know about how each body system responds to massive GI bleeding?

6. John Doe's inability to provide his name and address leads the gastroenterology nurse to the nursing diagnosis "altered thought processes caused either by loss of consciousness or disorientation to time, place, and person." (Data have not been provided to determine whether this patient is unconscious or disoriented.) What are the seven areas of cognitive assessment in which a gastroenterology nurse ordinarily endeavors to obtain data for this nursing diagnosis?

Suggested responses

1. The gastroenterology nurse should bring the following to the bedside:
   a. Banding equipment, at least two injector needles for sclerosing agents, and alternative medications for therapeutic injection, should the sclerosant fail to control bleeding. Bright red emesis may indicate variceal bleeding.
   b. Gastric lavage equipment (Ewald tube and large irrigating syringes). Large amounts of blood in the stomach make it difficult to visualize bleeding sites.
   c. Electrocautery device the gastroenterology department uses and at least one extra bipolar and/or heater probe. Ensure the properly sized probe (7Fr or 10Fr) is selected for the intended scope, as the channel diameter can vary with scopes. Argon or YAG lasers are used for GI bleeding.
   d. Moderate sedation and emergency medications. The ICU may stock these medications, but it is better to be prepared because the pharmacy may not be open.
   e. Personal protection equipment (PPE), such as gowns, eye shields, masks, and gloves, enough for the gastroenterology nurse, the gastroenterologist, and the ICU nurse who may need to be involved.
   f. The largest-channel gastroscope available or a double-channel gastroscope. The gastroenterology nurse should ensure that equipment is checked and working properly before going to the bedside.
   g. A second scope compatible with banding equipment and/or capable of better flexibility to visualize and treat bleeding in the duodenal bulb.
   h. Tubes for esophagogastric tamponade, should all measures fail to control the bleeding (three-lumen Sengstaken-Blakemore tube or the four-lumen Minnesota Sump tube), along with a manometer, clamps, and other supplies as necessary for use with tubes.
   i. Endoclips.
2. If the gastroenterology nurse were able to ask John Doe some questions, she might ask:
   a. Whether he has ever had a similar bleeding episode or was previously diagnosed or treated for ulcer disease. Recurrent ulcerations are common; however, in some patients, the new bleeding will stem from a different lesion.
   b. If he is allergic to any medication and what prescription and over-the-counter (OTC) medications he has been taking. Nonsteroidal anti-inflammatory drugs (NSAIDS) can prolong bleeding time and predispose the patient to conditions that could be the source of their bleeding.
   c. If he vomited any gastric contents before the onset of vomiting blood. Forceful vomiting can cause a tear in the lower esophagus or gastric cardia, which can result in profuse bleeding (Mallory-Weiss tear).
   d. If he consumes alcohol. It may be helpful to ask about specific quantities, for example, "Do you drink beer? If so, one or two six-packs a day? Do you drink vodka, whiskey, or rum? If so, one or two pints a day?" If a family member is present, he or she may give more accurate information than the patient. Excessive alcohol consumption can cause gastritis or ulcers that bleed excessively. Alcoholic liver damage may be evident on physical examination (ascites, spider angiomas, enlarged liver, gynecomastia). If liver damage is probable, the gastroenterology nurse may suspect esophageal varices as a cause of bleeding. Clotting mechanisms would also be affected. In this patient, an INR and platelet count should be assessed prior to endoscopy, as time and urgency permit. It should also be noted that patients with previously documented varices may be bleeding from a second source, such as a duodenal ulcer, so the gastroenterology nurse must be prepared for all possibilities.
   e. What the state of his general health is, with particular attention to questions regarding cardiac, respiratory, or renal problems. Patients with concomitant health problems are at higher risk for complications.
3. To address the possibility of aspiration during endoscopy, the gastroenterology nurse should take the following precautions:
   a. Recognize that all patients undergoing upper endoscopy are at risk of aspirating saliva, gastric contents, secretions, and blood. Patients most at risk are those who have recently eaten a meal, those who have active upper GI bleeding, those with recent stroke, and those with retention of gastric contents secondary to pyloric obstruction or gastroparesis. Airway protection becomes extremely critical, and many experienced endoscopists will ask anesthesia to pre-

emptively intubate a patient in the ICU before performing an endoscopy in an effort to protect the airway and reduce the risk of aspiration if therapeutic modalities are performed.

b. Recognize that an overtube may be employed to help protect the airway and minimize gagging, especially if the scope and/or gastric lavage tube may be passed more than once.

c. Use a topical throat anesthetic sparingly, if at all. Local pharyngeal anesthesia interferes with swallowing and gag reflex and alters coordination of the pharynx and upper esophageal sphincter.

d. Minimize the use of sedative medications in patients at high risk for aspiration. IV sedation and analgesia cause depression of the respiratory system and central nervous system, as well as suppression of the laryngeal closure reflex for 5-10 minutes.

e. Keep the patient in the left lateral position until sedation wears off. Have functional and adequate oral suction available with Yankauer suction tips. Monitor the patient carefully throughout the procedure and during recovery, with special attention to color, blood pressure, secretions, and sensorium.

4. Loss of 10% to 25% of blood volume is considered a moderate bleed; over 25% is considered a massive bleed. The normal adult has a circulating blood volume of 75 ml/kg body weight. A man who weighs 154 lb (70 kg) will have about 5,250 ml of circulating blood volume (20% of that is equivalent to 1 liter). Therefore, if he loses a single liter, he will experience a moderate bleed, with a corresponding drop in blood pressure, rapid pulse, labored breathing, loss of sensorium, anxiety, or a feeling of impending doom.

a. **Blood pressure.** Orthostatic blood pressures can help determine the extent of depleted circulating blood, shock, and hypotension. If, upon standing, the blood pressure falls 30 mm Hg or more from baseline, or the apical pulse rate increases by 20-30 beats per minute, there is significant postural hypotension. If the patient is "shocky" in the recumbent position, he may already have lost more than 50% of his circulating volume. This will be a more accurate indication of the significance of the bleed.

b. **Hemoglobin and hematocrit.** Hemoglobin and hematocrit levels are always done and may give an indication of the duration of hemorrhage, rather than provide information about the amount of acute blood loss. The patient can exsanguinate with a normal hemoglobin and hematocrit. It can take 12-36 hours for hemoglobin and hematocrit values to drop in acute bleeding.

c. **Blood urea nitrogen (BUN).** A BUN level above 40 mg (in patients without previous renal disease) indicates a significant bleed. Blood in the gut is partially digested and proteins are absorbed, which elevates BUN, and volume depletion produces a prerenal azotemia with consequent elevation of BUN.

d. **Coagulation studies.** Coagulation studies are important because clotting factors will be used up quickly, and patients with underlying liver disease will have difficulty keeping up with the demand.

e. **Arterial blood gases (ABGs).** ABGs should be drawn to check for lactic acidosis, which can occur as a result of severe tissue hypoxia due to less blood carrying less oxygen to cells.

5. Hypoperfusion causes potential complications to each body system as a result of hemorrhage. The different body systems respond to massive GI bleeding in the following ways:

a. **Cardiovascular system.** The heart's workload increases during periods of hypovolemia and reduced tissue oxygenation. A diseased heart may not be able to provide the increased output needed to maintain adequate tissue perfusion. The heart itself needs a lot of blood and oxygen and can suffer from myocardial ischemia during GI bleeding.

b. **Respiratory system.** Hemorrhage can worsen preexisting pulmonary disease. Ventilation and perfusion may be impaired by white blood cells clumping in the lungs' tiny blood vessels, and adult respiratory distress syndrome may result. This may occur 12-28 hours after hemorrhage. ABGs should be monitored.

c. **Hematological system.** Coagulation problems frequently occur following hemorrhage because the clotting factors are used more rapidly than they are produced.

d. **Renal system.** Acute tubular necrosis can occur when large myoglobin molecules in the bloodstream are released from other cells being destroyed and become lodged in tiny tubules. The gastroenterology nurse should monitor for decreased urine output. The serum creatinine is a more reliable indication of kidney function than the BUN.

e. **Metabolic system.** When the compensatory hormone regulation fails to maintain fluid and electrolyte concentrations, metabolic derangement can follow. This may cause a decrease in serum pH (acidosis), decrease in potassium (hypokalemia), and an increase in sodium (hypernatremia), and a state of serum hyperosmolality may occur. Hyperosmolality causes the movement of fluid into the vascular space from the extravascular space as a compensatory measure to maintain circulating volume.

6. The seven areas of cognitive assessment are:
   a. Level of consciousness.
   b. Orientation.
   c. Memory.
   d. Judgment and intellectual functioning.
   e. Thought flow.
   f. Thought content.
   g. Perception.

REVIEW TERMS

anaphylaxis, aspiration, cardiac arrest, hemorrhage, hypovolemia, perforation, respiratory depression, shock, vasovagal syncope

REVIEW QUESTIONS

1. Substernal or epigastric pain that increases with respirations and movement of the trunk is associated with perforation of what portion of the GI tract?
   a. Cervical esophagus.
   b. Thoracic esophagus.
   c. Distal esophagus, near the diaphragm.
   d. Stomach.
2. Conservative management of upper GI perforations may include:
   a. Suction.
   b. IV nutrition.
   c. Administration of antibiotics.
   d. All of the above.
3. Hematochezia is generally a symptom of:
   a. Bleeding esophageal varices.
   b. Bleeding ulcers.
   c. Gastritis.
   d. Lower GI bleeding.
4. In a patient with upper GI bleeding, if the return fluid from gastric lavage promptly becomes clear, the next step is to:
   a. Continue lavage.
   b. Remove the nasogastric tube.
   c. Use an Ewald tube to remove clots from the stomach.
   d. Begin balloon tamponade.
5. In patients suffering from hypovolemic shock, circulation to the heart and the brain initially:
   a. Increases.
   b. Decreases.
   c. Remains the same.
   d. Stops.
6. An important side effect of topical pharyngeal anesthesia is:
   a. Vomiting.
   b. Impaired gag reflex
   c. Allergic reactions.
   d. Superficial phlebitis.
7. When a patient experiences cardiac arrest, what is the first action the gastroenterology nurse should take?
   a. Call for help.
   b. Initiate CPR.
   c. Attach the patient to a monitoring device.
   d. Remove family members and visitors from the room.
8. What drug is used to reverse narcotic-induced respiratory depression?
   a. Atropine.
   b. Naloxone and/or flumazenil.
   c. Midazolam
   d. Epinephrine.
9. To treat a vasovagal attack, it is important to:
   a. Offer fluids.
   b. Begin CPR.
   c. Position the patient with the head level with or below the rest of the body.
   d. Keep the patient warm.
10. To avoid aspiration during upper endoscopy, the gastroenterology nurse usually:
    a. Maintains the patient in the supine position.
    b. Encourages the patient to breathe deeply.
    c. Provides suction as necessary.
    d. Intubates the patient.

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Chapter 32

SURGICAL INTERVENTIONS

This chapter will acquaint the gastroenterology nurse with surgical interventions for the gastrointestinal (GI) tract. An overview of indications, procedures, and possible associated complications will be discussed. Understanding these surgical interventions will help the gastroenterology nurse anticipate patient needs when providing outpatient follow-up and assist with endoscopy and other related procedures.

The gastroenterology nurse is an integral part of the team responsible for management of the GI surgical patient. In the outpatient clinic setting, the gastroenterology nurse is often the first person the surgical patient encounters before any diagnostic workup is done. From that point, the surgical patient continues to interact with gastroenterology nurses throughout the preoperative period and during diagnostic testing, medical intervention, and therapeutic endoscopy procedures. In the postoperative period, the surgical patient encounters gastroenterology nurses in the outpatient clinic, as well as during any endoscopic procedures performed to evaluate the effectiveness of the surgery or to provide treatment for postoperative complications. The GI surgery patient has a tremendous need for education and emotional support, both of which the gastroenterology nurse provides during all phases of care.

Learning objectives

After reviewing the content of this chapter, the gastroenterology nurse will be able to:

1. Identify common GI disorders that require surgical intervention.
2. Describe common surgical procedures performed for GI disorders.
3. Identify complications associated with surgical interventions of the GI tract.

SURGICAL INTERVENTIONS FOR ESOPHAGEAL DISORDERS

Many esophageal disorders are endoscopically diagnosed and initially treated by medical management. When medical management fails or when the disorder is chronic, the patient and physician may choose endoscopic or surgical correction of the disorder. After surgical intervention, the patient may undergo endoscopy for evaluation of the surgical repair or recurrence of disease.

Common problems requiring endoscopic or surgical intervention include gastroesophageal reflux disease (GERD), achalasia, esophageal cancer, perforation of the esophagus, and atresia.

Gastroesophageal reflux disease

Gastroesophageal reflux disease (GERD) is caused when the lower esophageal sphincter (LES) does not close properly, allowing stomach contents to flow backwards into the esophagus. GERD can be treated endoscopically or surgically.

The concept of intraluminal **gastroplication** was first introduced in 2001. A plication is a pleat in tissue that is secured by an intraluminal suturing device. Plications are placed in the stomach at or just below the squamocolumnar junction to enhance the competency of the LES and prevent reflux of gastric contents into the esophagus.

Indications

Endoscopic or surgical intervention for GERD is indicated when medical management fails to adequately treat symptoms, leading to complications such as Barrett's esophagus and bleeding.

Some symptoms of GERD are potentially harmful or life-threatening, including aspiration pneumonia, exacerbation of pulmonary disease, and bradycardia.

Infants are especially prone to GERD due to immaturity of the LES and liquid diets, but GERD typically resolves by age 2 or when solids are introduced into the diet (Jackson & Kane, 2013). Therefore, surgical correction in infants is usually reserved for those with harmful or life-threatening symptoms.

Procedure

Endoscopic procedures

Treating GERD endoscopically has been challenging. Multiple devices have made their way through to the market but have been pulled for ineffectiveness and safety reasons (Nabi & Reddy, 2016).

Current endoscopic modalities to treat GERD include:

- Radiofrequency ablation. The application of energy to the LES muscle and gastric cardia potentially causes hypertrophy of the muscularis propria and reduces LES relaxation.
- Transoral incisionless fundoplication (TIF). The fundus of the stomach is folded up and around the distal esophagus, and the esophagus is pulled down and anchored below the diaphragm.
- Medigus ultrasonic surgical endostapler (MUSE). An endoscopic ultrasound and stapler is used to create a fold similar to fundoplication.

All three of these procedures for endoscopic treatment of GERD continue to be researched for efficacy and complications.

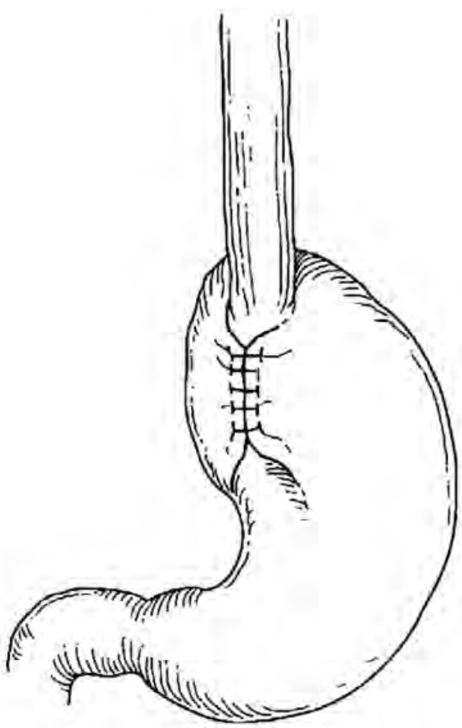

Figure 32.1. Nissen Fundoplication to Treat GERD
Redrawn from Ellett, M. (1988). Pediatric esophageal reflux: A nursing perspective.

Surgical procedures

The most common surgical procedure used to treat GERD in adults and children is the **Nissen fundoplication** or one of its variants. The Nissen procedure can be performed through laparotomy or laparoscopy. In either procedure, the gastric fundus is wrapped 360 degrees around the distal esophagus and sutured into place, creating a tightened LES (Figure 32.1). Current research indicates potential future changes in fundoplication calling for a 180-degree or 270-degree wrap around the distal esophagus (Du et al., 2017).

Considerations

Complications associated with Nissen fundoplication include:

- Inability to belch or vomit.
- Gas bloat syndrome (distention, inability to vomit, abdominal pain, and severe irritability).
- Recurrence of symptoms over time, with a need to repeat the surgical procedure (a failed or slipped fundoplication).

Achalasia

Achalasia occurs when the nerves of the esophagus do not function properly, resulting in problems swallowing and passing food through the LES.

Indications

Surgical intervention to treat **achalasia** is generally reserved for patients who have not responded adequately to the pharmacological or endoscopic measures. (See Chapter 13.)

Procedure

Surgical procedure

The standard surgical management for achalasia is surgical myotomy, in which the circular muscle fibers of the LES are divided.

Endoscopic procedures

While initially performed as an open procedure, the development of laparoscopic techniques for myotomy has led to a decrease in complications, morbidity, and recovery convalescence. The most popular and frequently performed is laparoscopic Heller myotomy.

Peroral endoscopic myotomy (POEM) is an endoscopic method for treating achalasia. This endoscopic procedure accomplishes the myotomy without surgery. Since 2010, POEM has been supported as safe and effective (Misra, Fukami, Nikolic, & Trentman, 2017).

Under general anesthesia, the endoscopist:

- Finds the LES and marks it with an injection of dye.
- Makes a longitudinal incision with an electrocautery knife.
- Tunnels the scope into the submucosal space. The tunnel is created beyond the LES into the proximal stomach, exposing the inner circular muscle layer.

- Performs the myotomy, on average 8-10 cm above the gastroesophageal junction (GEJ) and 2-3 cm into the cardia.

For those patients with severe achalasia and failure of myotomy, an esophagectomy is considered. An esophagectomy is considered a last resort because of the high risk to the patient.

Considerations

The most common complication associated with myotomy is GERD. The recommendation by the Society of American Gastrointestinal and Endoscopic Surgeons (SAGES) is to perform a fundoplication at the same time as the myotomy (Krill, Naik, & Vaezi, 2016).

Esophageal cancer

Successful surgical intervention for esophageal cancer requires accurate diagnosis and careful staging, which are frequently performed in the endoscopy suite using endoscopic ultrasound (EUS). Preoperative staging of esophageal cancer using EUS has been shown to be a well-tolerated and cost-effective tool.

Indications

Early stage esophageal cancers involving the epithelium or mucosa without lymph node involvement may be treated endoscopically with **endoscopic mucosal resection (EMR)**.

Surgical resection is the treatment of choice, in conjunction with radiation and chemotherapy, depending on lymph node involvement and metastasis.

Procedure

Endoscopic procedure

EMR involves using submucosal saline injection with a specialized cautery loop technique to remove cancerous tissue. The procedure is performed through a two-channel therapeutic endoscope. In some instances, EMR is combined with **photodynamic therapy (PDT)** or radiofrequency ablation (RFA) for treatment of Barrett's esophagus with neoplasia. EMR is recommended for lesions less than 2 cm in size.

Surgical procedures

Several techniques may be used to remove the diseased esophagus and restore continuity to the alimentary canal. The type of procedure used depends on the size and location of the tumor.

The two most common techniques are:

- **Transhiatal esophagectomy (THE).** This procedure does not require a thoracotomy and uses a small incision in the cervical neck to remove the esophagus after stabilizing the stomach.
- **Transthoracic esophagectomy (TTE).** This procedure uses a thoracotomy approach, as well as an abdominal approach, to remove the esophagus.

Considerations

Complications associated with THE and TTE procedures include anastomotic leaks, strictures, gastric reflux, delayed gastric emptying, dumping syndrome, and chylothorax (leakage of lymphatic fluid into the thorax). Complications of EMR include narrowing of esophagus, perforation, and bleeding.

Despite the multiple treatment options for patients with esophageal cancer, the 5-year survival rate varies from 15% to 20%. Based on data from the National Cancer Institute, the anticipated incidence of esophageal cancer in the United States is expected to grow by 140% by the year 2025 (Napier et al., 2014).

Perforation of the esophagus

Esophageal perforation is a potentially life-threatening condition that requires prompt recognition and may require immediate surgical intervention. In fact, if the perforation is not repaired, mortality rates are approximately 56% after 24 hours of injury and 75% to 89% if longer than 48 hours after injury (Reardon & Martin, 2015).

Symptoms of esophageal perforation include fever, abdominal or chest pain, dyspnea, abdominal distention, abdominal rigidity, increased heart rate, and increased respiratory rate. These signals all warrant receive immediate attention. Subcutaneous emphysema may also be noted. Hypotension is a late and ominous sign of impending shock and circulatory collapse.

Perforation is confirmed by X-ray examination, demonstrating free air within the chest or abdominal cavities.

Indications

Perforation of the esophagus may be associated with esophageal instrumentation (including endoscopes, esophageal dilators, guidewires, and biopsy forceps), surgery, swallowed foreign bodies, penetrating trauma, peptic ulceration, ingestion of caustic substances, infections (herpes simplex virus (HSV) and tuberculosis), malignancy, vascular abnormalities (aortic aneurysm and aberrant right subclavian artery), and Boerhaave's syndrome.

Boerhaave's syndrome is an uncommon but catastrophic event in which the lower thoracic esophagus is completely torn away from the gastric cardia. While some patients present with vomiting, difficulty breathing, and increased heart rate, symptoms will depend on the location of the perforation:

- Cervical—neck pain and muscle spasm, cervical motion pain, dysphonia, hoarseness, and cervical dysphagia.
- Thoracic—dysphagia, odynophagia dyspnea, cyanosis, and chest discomfort.

Procedure

Early treatment of esophageal perforation includes stabilization of respiratory status, antibiotic therapy, volume replacement, total parenteral nutrition (TPN), and chest tube drainage.

When surgery becomes necessary, the site of perforation dictates the surgical approach. Generally, the esophageal tear is closed with sutures. The sutured area is reinforced with pleural or intercostal flaps in a thoracic esophageal injury and with a diaphragmatic flap in a lower esophageal tear.

Considerations

Although the physician and gastroenterology nurse may see the esophageal tear during the course of a procedure, many perforations are not detected until after the procedure is complete. As a result, patients who undergo endoscopy on an outpatient basis must receive detailed instructions about reporting symptoms of possible perforation.

Complications associated with repair of esophageal perforation depend on the site of repair. Dysphagia and airway difficulties may be associated with repair of the upper esophagus, whereas gastroesophageal reflux may be associated with repair of the lower esophagus. Stricture due to scar tissue formation is a potential complication of repair to any area of the esophagus.

Esophageal atresia

Esophageal atresia (EA) is a congenital malformation affecting one in 2,500 live births and is slightly more common in boys (Cartabuke, Lopez, & Thota, 2016). In EA, the esophagus ends in a blind pouch instead of attaching to the stomach.

Indications

In most cases, EA is associated with **tracheoesophageal fistula (TEF)**. TEF is an open communication channel between the trachea and esophagus. EA and TEF are frequently associated with other congenital anomalies.

Procedure

Endoscopic procedure

Endoscopic management using self-expanding stents is the preferred method of treatment. (See Chapter 29.)

Surgical procedure

The surgical management of EA involves surgical ligation of the TEF and an end-to-end anastomosis of the esophagus through a right thoracotomy. Due to the increased complication rate associated with the surgical procedure, the endoscopic approach is more accepted (Ke, Wu, & Zeng, 2015). Steps of surgical ligation of the TEF include the following:

- The site of the TEF is identified and carefully cut free from surrounding structures.
- The fistula is divided and sutured or suture-ligated.
- The upper pouch of the esophagus is then identified and connected to the lower portion of the esophagus (primary anastomosis).

In cases where the gap between the ends of the esophagus is too long for primary anastomosis, primary repair may be delayed until after the infant has gained sufficient weight.

Considerations

Complications associated with repair of EA include dysphagia, anastomotic leak, recurrent TEF, gastroesophageal reflux, and esophageal strictures. When TEF repair is also performed, complications may include respiratory compromise and the inability to manage respiratory secretions.

SURGICAL INTERVENTIONS FOR DISORDERS OF THE STOMACH

Many disorders of the stomach are endoscopically diagnosed and initially treated by medical management. Depending on the nature of the surgical correction, many patients require a follow-up endoscopy after surgery.

Common disorders of the stomach that may require surgical intervention include hiatal hernia, morbid obesity, peptic ulcer disease (PUD), perforated peptic ulcer (PPU), pyloric stenosis, and gastric cancer.

Hiatal hernia

Hiatal hernia is a common disorder, frequently associated with gastroesophageal reflux symptoms.

Indications

There are two types of hiatal hernias:

- **Sliding hiatal hernia.** The most common type of hiatal hernia is a sliding hiatal hernia. A sliding hiatal hernia occurs when the GEJ moves above the diaphragm. The stomach remains in its normal position with the fundus below the GEJ. Sliding hiatal hernias account for more than 95% of hiatal hernias (SAGES, 2013).
- **Paraesophageal hiatal hernia.** This type accounts for the remaining 5% of hiatal hernias (SAGES, 2013). With this type of hiatal hernia, the fundus herniates through the diaphragmatic hiatus next to the esophagus, but the GEJ remains in its normal position. These hernias can become quite large, and other organs may be herniated with the stomach, including the small intestine, spleen, or colon.

Procedure

Initially, pharmacological treatment may be used for the treatment of sliding hernia symptoms. When associated GERD becomes refractory to medication, surgery is indicated.

A Nissen fundoplication is the surgery of choice and most often is performed laparoscopically. An open approach may be needed in the presence of paraesophageal hernias. Mesh may be used during either surgery to help reinforce the placement of the GEJ below the diaphragm. In the pediatric patient, the laparoscopic approach is indicated.

Considerations

Laparoscopic repair of hiatal hernias has the advantages of same-day surgery or decreased hospital stays, as well as decreased complications and morbidity when compared to an open surgical approach.

Morbid obesity

Morbid obesity is prevalent in nearly one-third of Americans and is associated with heart disease, stroke, diabetes, and cancer (Hales et al., 2017). While multiple methods of weight reduction have been introduced, the surgical approach to morbid obesity is a radical yet valid option for the majority of individuals who are obese.

Indications

Bariatric surgery is recommended for patients with a body mass index (BMI) greater than 40 kg/min^2 or a BMI greater than 30 kg/min^2 with multiple co-morbidities. Bariatric surgery has demonstrated a reduction of cardiovascular risk factors and remission of type 2 diabetes (McGrice & Don Paul, 2015; Pham et al., 2014). Several procedural options exist, with each requiring careful consideration. The patient and surgeon usually include the family and significant others in the decision-making process.

Procedure

Bariatric surgery may be restrictive, malabsorptive, or a combination of both.

- Restrictive surgery limits the amount of food that may be ingested.
- Malabsorptive surgeries impact the absorptive qualities of the GI system so weight loss can occur rapidly.

Current trends in bariatric surgery reflect a predominance for sleeve gastrectomy and Roux-en-Y gastric bypass {American Society for Metabolic and Bariatric Surgery (ASBMS, 2019)}. When bariatric surgery was first introduced, the gastric band approach was the popular method; however, according to data supplied by the ASBMS (2018), this procedure is rarely performed today due to concerns regarding band erosion and slippage.

Sleeve gastrectomy

The sleeve gastrectomy procedure is a nonreversible, restrictive, and malabsorptive laparoscopic surgical procedure in which approximately 80% of the stomach fundus is removed, leaving a tube resembling a banana (ASMBS, 2019). It is believed to work by decreasing the area of the stomach responsible for producing hormones that control the feeling of hunger.

Roux-en-Y gastric bypass

Roux-en-Y gastric bypass is regarded as both a restrictive and a malabsorptive procedure and can be performed as a laparoscopic or open surgical procedure. With laparoscopic surgery, the patient is typically ready for discharge within 1-2 days after surgery. This surgery is not reversible.

In Roux-en-Y surgery, the stomach is first sectioned off into a 30 ml pouch. Then the duodenum is divided off of the jejunum, and the jejunal limb (Roux) is brought up and attached to the stomach pouch. The duodenal limb is then anastomosed to the remaining jejunum.

Because the Roux-en-Y surgery causes immediate and dramatic weight loss, with the patients losing approximately 60% to 80% of excess body weight, careful nutritional education and lifestyle changes are required (ASMBS, 2019). A high-protein, low-carbohydrate diet and small portion sizes are required. Lifelong vitamin supplementation—Vitamin B_{12}, iron, calcium, and folate—are required, with special attention to Vitamin B_{12} due to the decreased production of intrinsic factor by the dissected stomach.

Laparoscopic adjustable gastric band

A bariatric surgery procedure that is performed in the endoscopy setting is the laparoscopic adjustable gastric band. A silastic band is placed around the upper portion of the stomach, adjusted via a subcutaneously implanted port. This is a purely restrictive procedure. This procedure has fallen out of favor due to the high incidences of band erosion and band slippage.

Considerations

Complications associated with sleeve gastrectomy include leakage along the suture line and potential for long-term nutritional deficiencies.

Complications associated with Roux-en-Y surgery may include leakage along suture lines, anastomotic ulcers and stenosis, nutritional deficiencies, excessive weight loss, or dumping syndrome.

In the gastroenterology suite, the gastroenterology nurse may see patients with laparoscopic adjustable bands, which can cause an upper GI bleed. Treatment for the failed gastric band is removal of the band. The patient may undergo a subsequent Roux-en-Y surgery.

Peptic ulcer disease

Although antral and duodenal ulcers may have different clinical presentations, they are usually discussed together because they share common pathophysiology and surgical intervention. Surgery was once the definitive treatment for peptic ulcer disease (PUD). The introduction of effective medical therapy such as proton pump inhibitors (PPIs), the recognition of the adverse role of *Helicobacter pylori (H. pylori)*, and the avoidance of aspirin and nonsteroidal anti-inflammatory drugs (NSAIDs) in the presence of PUD has made the need for surgical intervention less common.

Indications

Surgery is required in instances of perforations, bleeding, or obstruction. The key signs of gastric ulcer perforation include abdominal pain, tachycardia, and abdominal rigidity. Perforation requires emergency surgery.

Procedure

Surgical approaches to PUD and perforation have varied over the years.

Vagotomy

This procedure was frequently performed to decrease the amount of gastric acid secretion and gastrin release by transecting the vagus nerve. The vagus nerve is responsible for the stimulation of gastric acid production. The advent of proton pump inhibitors (PPIs), histamine2 (H_2) blockers, and *H. pylori* treatments, along with an estimated 42% post-procedure reoccurrence rate, have made this procedure no longer a viable option for treatment (Chung & Shelat, 2017).

Laparoscopic omentum patching

Omentum patching involves the placement of a gelatin sponge plug (patch) or fibrin glue over the site of the ulcerated lesion by a laparoscopic approach.

Partial gastrectomy (gastric resection)

This procedure is currently performed for the treatment of gastric cancers or large perforated ulcers but not for control of PUD, for many of the same reasons why vagotomy is no longer used for treatment. Approximately 10% of patients with perforated ulcers will require gastric resection rather than omentum patching (Chung & Shelat, 2017).

Billroth I and II

These procedures are still used for treatment of PPU but are less frequently performed, as laparoscopic approaches with omentum patching have demonstrated fewer complications and a shorter length of hospital stay.

- In the classic Billroth I procedure, the antrum is resected and the duodenum is reanastomosed to the gastric remnant.
- In the Billroth II, the distal portion of the stomach and proximal duodenum are resected, and the jejunum is reanastomosed to the gastric remnant.

Considerations

Complications of laparoscopic omentum patching include gastric stasis, anastamotic leakage, or peritonitis. Complications of partial gastrectomy include reflux, diarrhea, leakage at the site, and nausea and vomiting. Complications associated with Billroth I and II include anastomotic and suture leaks, resulting in peritonitis and or sepsis. Side effects include dumping syndrome, diarrhea, and bile reflux gastritis.

Pyloric stenosis

Hypertrophic **pyloric stenosis** is a common disorder of infancy. It occurs in approximately 3 in 1,000 live births in the United States and more commonly in white male infants (Humphries & Steele, 2012).

Indications

Symptoms of hypertrophic pyloric stenosis are usually present by 3-6 weeks of age and include projectile, nonbilious emesis. This can lead to dehydration, metabolic alkalosis, and malnutrition. Pyloric stenosis is diagnosed with upper GI series or ultrasound.

In adults, chronic ulceration with subsequent scarring in the pyloric channel and duodenum can result in pyloric stenosis and gastric outlet obstruction. Pyloric stenosis can also be caused by inflammatory edema surrounding an acute channel ulcer.

Procedure

The treatment of choice for infantile hypertrophic pyloric stenosis is pyloromyotomy, which involves incision of the muscle surrounding the pylorus. The procedure is done via an abdominal incision.

Pyloric stenosis in adults may be treated by dilation. Surgical intervention is indicated when pyloric stenosis is secondary to PUD. A gastric outlet drainage procedure (Billroth I) may be indicated to treat the underlying PUD.

Considerations

The incidence of complications from pyloromyotomy is low but may include prolonged vomiting, leakage from site, and site infection.

Gastric cancer

Gastric cancer is the fifth most common cancer in the world, yet it is the third most common cause of cancer death (Mihmanli, Ilhan, Idiz, Alemdar, & Demir, 2016). Early detection and treatment before metastasis to the lymph node is an indicator for survival rate.

Indications

EUS has been shown to be an accurate and cost-effective tool in preoperative staging of gastric cancer. Accurate staging prior to surgical intervention enables the surgeon to plan the most appropriate treatment.

Procedure

Endoscopic procedures

Early-stage cancers confined to the mucosal layer and without lymph node involvement may be removed by EMR as described in the section on esophageal cancer or by endoscopic submucosal resection (ESR). Successfully performed, this eliminates the need for surgical intervention.

Surgical procedures

Gastric resection: This procedure is dictated by the location and extent of the lesion. Many surgeries can now be performed laparoscopically. Local resection includes laparoscopic wedge resection of the stomach and intragastric mucosal resection.

Total gastrectomy: This is indicated when the length of the neoplasm is more than that required to obtain good margins, when the cancer involves two or all three sections of the stomach, and when the carcinoma is diffuse. A Billroth I procedure is usually performed for cancers in the distal portion of the stomach. If total gastrectomy is performed, a Roux-en-Y procedure may be indicated.

Considerations

Total gastrectomy is usually combined with resection of adjacent lymph nodes and appears to be sufficient for lesions in the distal stomach. Advanced disease in patients with nodal metastases has poor results with surgery alone, and adjuvant and neoadjuvant therapy (i.e., chemotherapy, radiation therapy, or a combination of these two) is needed.

Inoperable cancers of the stomach or of the pancreatobiliary system may cause duodenal obstruction and may be palliated by the formation of an antecolic gastroenterostomy at the greater curvature of the stomach.

Complications associated with gastrectomy for gastric cancer include weight loss, iron-deficiency anemia, macrocytic anemia, reflux gastritis, and diarrhea. Gastric outlet syndrome may occur as a result of surgical adhesions and may be treated endoscopically with an expanding metal stent.

SURGICAL INTERVENTIONS FOR PANCREATIC DISORDERS

Two general conditions require pancreatic surgery: chronic pancreatitis and pancreatic cancer.

Chronic pancreatitis

In all cases of pancreatitis, there are three elements to treatment:

1. Determining the definitive diagnosis.
2. Treating any underlying jaundice or duodenal obstruction with an endoscopic approach.
3. Removing the inflamed area or tumor of the pancreas.

Indications

Resection is based on the belief that removal of diseased tissue diminishes or eliminates pain associated with pancreatitis, as well as the risk of further complications.

Procedure

Surgical options include:

- Pylorus-preserving pancreaticoduodenectomy.
- Duodenal-preserving pancreatic head resection.

Both surgeries allow drainage from the biliary tree and the pancreatic duct to occur.

Considerations

The severe pain associated with chronic pancreatitis can be effectively treated with surgical management with minimal long-term negative effects. Surgical management early in chronic pancreatitis has been demonstrated to decrease the amount of opioid medication required to manage pain (Bliss et al., 2015).

Pancreatic cancer

Pancreatic cancer is the fourth leading cause of cancer death and has a 5-year survival rate of <5% (Ansari, Gustafsson, & Andersson, 2015). At the time of diagnosis, only 50% of patients with pancreatic cancer are free of distant metastases, and 10% to 15% of these patients have localized disease amenable to curative resection (Ansari et al., 2015).

Indications

The majority of patients with pancreatic cancer have disease in the head, neck, or uncinate process of the gland. There are three stages associated with pancreatic cancer disease; resectable disease, locally advanced disease, and metastatic disease.

Procedure

Potentially curative surgical treatments for patients with pancreatic cancer are **pancreaticoduodenectomy**, also known as the **Whipple procedure** (Figure 32.2); distal pancreatectomy; and total pancreatectomy. The decision as to which surgery to perform is based on tumor size and location.

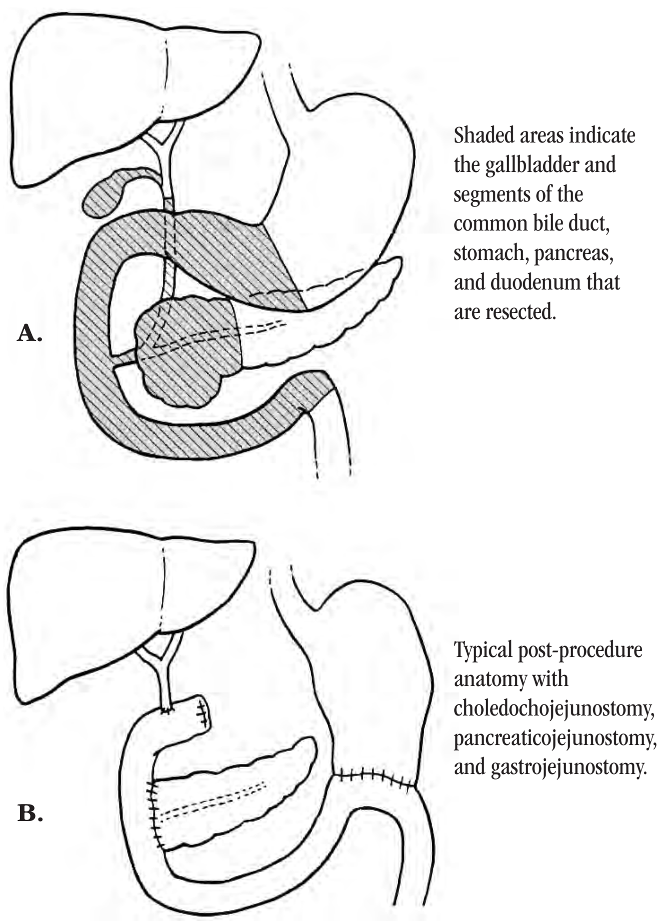

Figure 32.2. Whipple Procedure (Pancreaticoduodenectomy)

Whipple procedure (pancreaticoduodenectomy)

The most common procedure performed is the Whipple procedure. It involves removal of approximately 50% of the stomach, all of the duodenum, and the proximal jejunum. The pancreatic head, neck, and uncinate process are resected, along with the gallbladder and distal biliary tree.

In the pylorus-preserving pancreaticoduodenectomy, the second, third, and fourth portions of the duodenum; the neck, head, and uncinate process of the pancreas; and the gallbladder and distal biliary tree are removed. Three anastomoses are necessary for reconstruction: an end-to-end pancreaticojejunostomy, an end-to-end hepaticojejunostomy, and an end-to-end duodenojejunostomy.

Modified Whipple procedure

The **modified Whipple procedure** leaves the entire stomach and 2-4 cm of the proximal duodenum, preserving peptic acid-inhibiting hormones and preventing postoperative peptic ulceration.

Pancreatectomy

Distal pancreatectomy involves the removal of the tail and portion of the body of the pancreas, whereas a total pancreatectomy is the removal of the entire pancreas.

Considerations

Outcomes in managing pancreatic cancer have improved over the past decade. If performed before nodal spread, the Whipple procedure (pancreaticoduodenectomy) is an effective treatment. The survival rate is increased if adjunctive therapy consisting of radiation therapy and chemotherapy is given. Many patients fare better if chemotherapy and/or radiation therapy are used to shrink the tumor mass prior to surgery.

Along with the advantage of diagnosing and staging pancreatic malignancies with EUS, diagnosis can be made much earlier using fine needle aspiration (FNA). Advances made in genetic and biomarker testing also offer early diagnosis and accurate staging, allowing planning for optimal care and a greater chance for survival.

Complications from the Whipple procedure include leaking from the various connections between organs, infection, bleeding, difficulty with stomach emptying after eating, difficulty digesting some foods, weight loss, changes in bowel habits, and diabetes.

SURGICAL INTERVENTIONS FOR DISORDERS OF THE BILIARY TRACT

Indications

Cholecystitis, usually resulting from cholelithiasis, is the most common indication for surgery of the biliary tract.

Cholecystitis

The gold standard of treatment for cholelithiasis is cholecystectomy.

Procedure

Open cholecystectomy

The traditional approach has been open cholecystectomy performed through a right, subcostal incision, dividing the rectus muscle to expose the gallbladder. Calot's triangle (the cystic duct, cystic artery, and common hepatic duct) is identified, and both the cystic duct and cystic artery are ligated before removing the gallbladder. Other procedures associated with open cholecystectomy include the following:

- **Intraoperative cholangiography** is often performed by inserting a catheter through the cystic duct into the common bile duct. Radiopaque contrast is injected into the common bile duct and X-rays are taken to detect gallstones or other obstructions in the common bile duct.
- In the presence of stones or obstruction, a **choledochostomy** is performed to explore the common bile duct.
- **Sphincteroplasty** may also be performed if intractable obstruction or dense strictures are found in the ampulla of Vater.
- When the gallbladder has been previously removed and recurrent obstruction of the common bile duct occurs, **choledochoenterostomy** is indicated. This procedure involves a side-to-side anastomosis of the common bile duct to the first part of the duodenum.

Laparoscopic cholecystectomy

Laparoscopic cholecystectomy is a popular approach and is frequently performed. Laparoscopic cholecystectomy allows for surgical resection of the gallbladder without a large abdominal incision.

The gallbladder is approached by laparoscope through a small incision just below the umbilicus. Additional small puncture wounds are made in the upper abdomen to allow for insufflation of carbon dioxide and insertion of surgical instruments. Dissection of the gallbladder is performed with either laser or diathermy, and the gallbladder is removed through the small incision just below the umbilicus.

Gallstones remaining in the common bile duct can be removed endoscopically at a later time. Endoscopic retrograde cholangiopancreatography (ERCP) may be requested by the surgeon prior to surgery to remove stones from the ducts.

Considerations

Laparoscopic cholecystectomy has demonstrated decreased hospitalization stays (in some instances, same-day surgery), decreased pain, and smaller incisions, allowing for more rapid recovery and return to daily activities of normal living (Karim & Kadyal, 2015).

Laparoscopic cholecystectomy is the first surgery of choice in acute cholelithiasis. Some patients with acute calculous cholecystitis may require open emergent surgery due to clinical deterioration or complications of the disease.

Complications of cholecystectomy include injury to the common bile duct and anastomotic strictures of the common bile duct.

SURGICAL INTERVENTIONS FOR DISORDERS OF THE SMALL INTESTINE

Common surgical interventions involving the small intestine include resection due to a variety of diseases and disorders, including Crohn's disease.

Disorders necessitating small intestine resection

Removal of portions of the small intestine may be necessary for a variety of reasons.

Indications

At birth, congenital anomalies such as duodenal atresia, jejunal atresia, ileal atresia, gastroschisis, and omphalocele may dictate the removal of affected parts of the small intestine. In infancy, necrotizing enterocolitis may lead to resection. In children and adults, problems that may require small intestine resection include trauma, obstruction, infection, ischemia, Crohn's disease, and cancer.

Procedure

In general, resection of the small intestine involves the removal of the affected portion of bowel and end-to-end anastomosis of the remaining healthy segments. The ability to create a primary anastomosis (performed at the time of initial surgery) depends on the length of resection and the integrity of the remaining tissue. Creation of a temporary or permanent ostomy may be necessary with resection or poor tissue integrity.

Considerations

Complications associated with small intestine resection include stricture, adhesions and scarring, diarrhea, and malnutrition. The degree of malnutrition associated with small intestine resection depends on the location and the length of the resection. Massive small intestine resection may result in the need to maintain nutrition by TPN on a long-term or permanent basis.

Crohn's disease

Surgical resection of the small intestine in Crohn's disease is reserved for patients who have not responded to aggressive pharmacological treatment.

Indications

Indications for small intestine surgery in Crohn's disease include intestinal obstruction, fistula, abscess, uncontrolled hemorrhage, perforation, toxic megacolon, and growth failure in children who have localized disease.

Procedure

Some surgical resections can now be performed laparoscopically or at least with the assistance of laparoscopic devices.

Diversion surgery (ileostomy) may be indicated when a patient has severe unremitting sepsis related to anorectal Crohn's disease, with or without abscess.

Resection or strictureplasty may be performed when the stricture occurs in the small intestine. Strictureplasty involves a longitudinal incision made down the area of stricture. The incision is then sutured in a horizontal direction, creating a wider lumen (Figure 32.3).

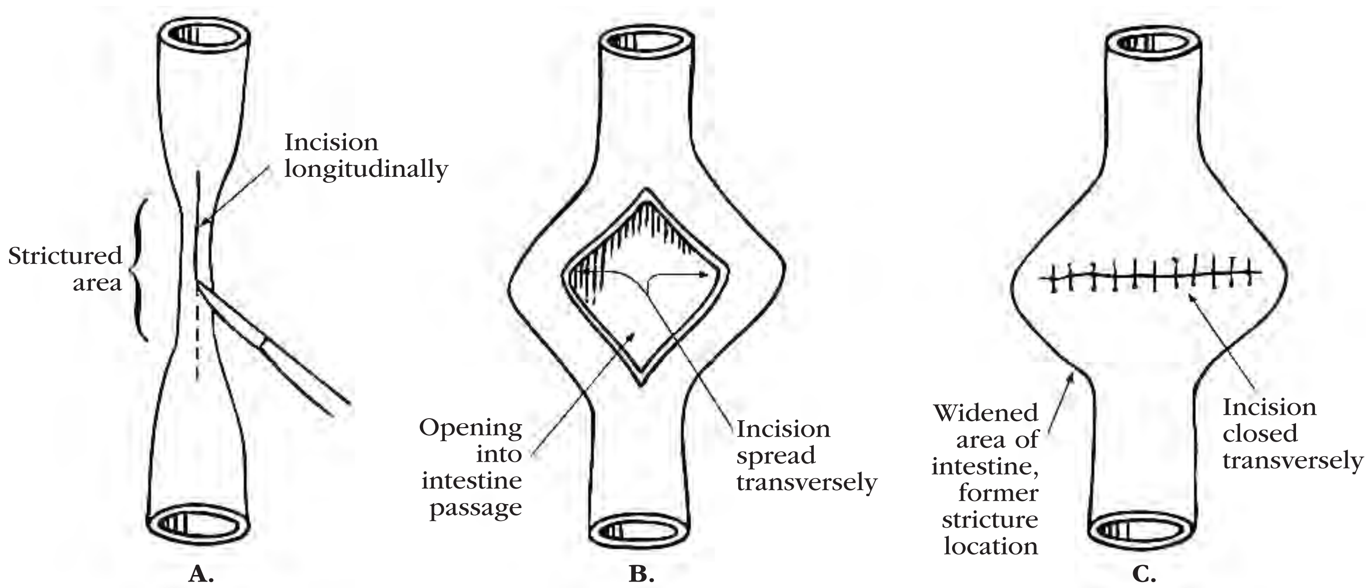

Figure 32.3. Surgical Treatment for Small Intestine Strictures

A. The strictured area is identified, and the bowel is opened longitudinally. **B.** The incision is spread transversely. **C.** The incision is closed transversely.

Considerations

Complications associated with surgery for Crohn's disease include diarrhea, weight loss, and recurrence of disease.

Although prolonged remission may occur after resection, approximately one-third of patients will require additional surgery within a few years. Strictureplasty leads to a higher reoccurrence for surgery than resection (Toh et al., 2016).

SURGICAL INTERVENTIONS FOR DISORDERS OF THE COLON AND RECTUM

Indications for surgical intervention in the colon and rectum are numerous and fall into several broad categories. These categories include, but are not limited to, congenital anomalies, trauma, inflammatory diseases, and neoplastic disease. Common disorders include Hirschsprung's disease, inflammatory bowel disease (Crohn's disease and ulcerative colitis), colorectal cancer, and perforation.

Hirschsprung's disease

Hirschsprung's disease is the congenital absence of intramural ganglia in the intestinal tract and hypertrophic nerve trunks, most frequently of the anorectum and variable lengths of the distal colon. The incidence of Hirschsprung's disease is 1 in 5,000 live births, with males being four times more likely to be affected than females (Graneli, Dahlin, Borjesson, Ambjormsson, & Stenstrom, 2017).

Indications

Treatment of Hirschsprung's disease involves surgical removal of the affected portion of the colon.

Procedure

The most commonly recommended surgical procedures to treat Hirschsprung's disease are:

- Retrorectal transanal pull-thorough (Duhamel's procedure). The aganglionic rectum is left in place, and the normal ganglionic bowel is pulled down behind the rectum and through an incision in the posterior rectal wall at the level of the internal sphincter.
- Transanal endorectal pull-through (TERPT). Resection of the aganglionic rectum occurs just above the dentate line, and a muscular rectal cuff is formed.

Considerations

In either procedure, the recommendations for treatment include biopsy of the rectal and intestinal wall to determine the extent of disease.

Complications associated with surgical intervention for Hirschsprung's disease include anal stenosis, obstructive symptoms, fecal incontinence, constipation, and enterocolitis.

Inflammatory bowel disease

Although Crohn's disease may affect any part of the GI tract, ulcerative colitis affects only the large intestine. As a result, the colon may be the site of disease activity for both of these chronic illnesses. In ulcerative colitis (UC), surgery is a definitive cure for the disease.

Indications

Surgery is indicated when pharmaceutical therapy for UC fails to prevent significant physical and emotional morbidity. Approximately 25% to 45% of patients with UC will need surgery at some point during their disease progression (Devaraj & Kaiser, 2015).

Devaraj and Kaiser (2015) have identified four categories of indicators for surgery in UC:

- Category 1: Life-threatening complications—including toxic megacolon, colonic perforation, and hemorrhage.
- Category 2: Cancer-related conditions—including localized cancer, high-grade dysplasia, low-grade dysplasia with flat polyps, colonic stricture, and treatment-induced secondary tumors.
- Category 3: Insufficient treatment response—including treatment refractoriness, lack of mucosal healing, persistent symptoms, frequent recurrence and flares, hospitalizations despite pharmacological therapy, chronic corticosteroid dependence, unacceptable lifestyle, and growth retardation in children.
- Category 4: Treatment side effects—including intolerance and complications related to medications, including cataracts, Cushing's syndrome, osteoporosis, hypertension, hyperglycemia, pancreatitis, neurological issues, and skin or extra-intestinal malignancy.

Three goals associated with surgery for the patient with UC are:

1. Cure from the disease and/or toxic effects of the medications.
2. Reconstruction with low morbidity and improved quality of life.
3. Minimization of morbidity or mortality of any procedure.

Procedure

Proctocolectomy is the gold standard of surgical management for UC. By removing the rectum and the colon, a cure is achieved in a majority of patients (Davaraj & Kaiser, 2015). The proctocolectomy is performed by creating a cuff in the remaining rectum and allowing for an anastomosis. This procedure is performed in two stages: First, the loop ileostomy is created as a temporary diversion, then later the ileostomy is closed. If the ileostomy is to be permanent, the anal canal must have the epithelial tissue removed in order to avoid rectal seepage.

Total abdominal colectomy is typically performed on an emergent basis, and the final surgery with pouch creation and anastomosis is not performed until the patient is healed and stable.

Considerations

Complications associated with either surgery include wound infection, anastomotic leak, and pouchitis. However, long-term studies of patients with UC have demonstrated an increased quality of life after surgery (Davaraj & Kaiser, 2015).

Colorectal cancer

Surgical treatment of colorectal cancer typically involves the removal of the affected segment of colon, as well as the adjacent mesentery and 12 regional lymph nodes in an attempt to cure or palliate.

Indications

A right hemicolectomy is indicated for cancers found in the cecum and ascending colon. A left hemicolectomy is performed when indicated by a lesion in the area of the splenic flexure. Lesions in the sigmoid or rectosigmoid colon are usually treated with an anterior resection.

Procedure

In a right hemicolectomy:

- The mesocolic vessels are isolated and divided, and the ileum is divided above the ileocecal valve.
- The transverse colon is divided at a point beyond the hepatic flexure.
- An end-to-end ileocolonic anastomosis is usually made.
- An ileosigmoidal anastomosis is then formed.

An extended right hemicolectomy is similar, but it sacrifices the terminal ileum and the portion of the colon that lies distal to the sigmoid.

In a left hemicolectomy, the procedure is done in a similar fashion as a right hemicolectomy. End-to-end anastomosis of the transverse colon and rectosigmoid is usually made to retain bowel continence.

In an anterior resection for lesions in the sigmoid or rectosigmoid colon, the sigmoid and a segment of the rectum are resected, with the descending colon anastomosed to the rectal remnant. Total mesorectal excision with a distal surgical margin of at least 2 cm is recommended for rectal cancers.

The current trend is toward a laparoscopic approach to colon resection, rather than an open approach. Multiple studies have shown the benefits of laparoscopic surgery over open surgery, which include earlier bowel function return, earlier intake of oral fluids postoperatively, decreased blood loss, decreased length of hospital stay, and decreased morbidity (Lee et al., 2017; Lorenzon et al., 2014).

Considerations

Treatment depends on the stage of cancer and may include surgery, chemotherapy, and/or radiation therapy. Complications from hemicolectomy include infection, bleeding, blood clots, constipation, and bowel obstruction. Post-colectomy outcomes are comparable in right-sided and left-sided resections, though surgical site infections are less common in right-sided resections.

Perforation of the colon

Perforation of the colon may occur secondary to acute inflammatory bowel disease, inserted foreign bodies, and penetrating trauma, as well as during surgical and endoscopic procedures. Perforation during an endoscopic procedure can occur secondary to instrumentation, such as use of the endoscope or biopsy forceps, or secondary to overinflation of the colon. The risk of perforation during endoscopy is increased when a polypectomy is performed in patients with diminished mucosal integrity, such as those with inflammatory bowel disease.

Indications

Perforation of the colon is a potentially life-threatening condition that requires prompt recognition and may require immediate surgical intervention. Endoscopic visualization of an open defect in the bowel wall, visible fatty tissue (such as the omentum), or abdominal organs is clear indication of a perforation. Perforation is confirmed by X-ray, demonstrating free air under the diaphragm.

Unfortunately, not all perforations of the colon are easily detected. As a result, patients should be closely monitored for signs and symptoms of perforation after the procedure is completed.

Symptoms include fever, abdominal or rectal pain, abdominal distention, abdominal rigidity, increased heart rate, and increased respiratory rate. Any and all of these should receive immediate attention. Hypotension is a late and ominous sign of impending shock and circulatory collapse.

Procedure

Treatment for colon perforation depends on the patient's status, amount of free air, and size of the perforation. If the patient is stable but presents with minimal pain, treatment consists of bowel rest and antibiotics, with repeat X-rays to ensure healing. If the patient becomes unstable, surgical therapy is needed.

When surgery becomes necessary, the tear is generally closed with sutures, and the abdominal cavity is thoroughly rinsed with an antibiotic solution. In some cases of perforation, the tear may be closed with endoscopic mucosal clips. In cases of severe injury to the colon and surrounding structures, partial or total colectomy with creation of an ostomy may be indicated. Ability to create an anastomosis to the rectum after the patient has stabilized will depend on the degree of injury to the rectum.

Considerations

Patients who undergo colonoscopy on an outpatient basis must receive detailed instructions on reporting symptoms of possible perforation, as well as other possible complications related to the procedure.

Complications associated with repair of a perforated colon depend on the site and the degree of repair. Anastomotic strictures may form at the repair site. Repair in the rectal area may result in bowel incontinence and rectal strictures.

TRANSGASTRIC SURGERY

Refinements in laparoscopic surgery have allowed some previously open surgical procedures to be performed in a minimally invasive setting. Gastroenterologists team with surgical colleagues who are experts in laparoscopic surgical techniques to use natural orifices for the entry point.

Natural orifice translumenal endoscopic surgery (NOTES) uses natural orifices such as the mouth, vagina, and rectum as the entry point for access to the peritoneum, making **transgastric surgery** possible. It was noted in a study that 86% of reported NOTES cases took place abroad and 14% took place in the United States (Clark et al., 2012).

Indications

Advances made in NOTES procedures allow laparoscopic procedures such as cholecystectomy, appendectomy, and bariatric surgery to be performed via a natural orifice.

Procedure

NOTES is performed in a sterile environment, using a sterile double-channel endoscope with a sterile overtube inserted into the stomach. An incision is dilated with an 18 mm balloon dilator. The surgical procedure is then performed, using specialized instruments, some of which are still being developed. The incision is closed with clips, sutures, or possibly surgical adhesive.

REVIEW TERMS

achalasia, Billroth I procedure, Billroth II procedure, Boerhaave's syndrome, endoscopic mucosal resection (EMR), endoscopic ultrasound (EUS), esophageal atresia (EA), gastroplication, gastroesophageal reflux disease (GERD), Heller's myotomy, Whipple procedure, Nissen fundoplication, pancreaticoduodenectomy, pyloric stenosis, Roux-en-Y gastric bypass, sliding hiatal hernia, tracheoesophageal fistula (TEF), transgastric surgery

REVIEW QUESTIONS

1. Complications from laparoscopic fundoplication include all except:
   a. Gas bloat syndrome.
   b. Inability to belch.
   c. Lower success rates than laparotomy fundoplication.
   d. Recurrence of symptoms.
2. Esophageal perforation may result from all the following except:
   a. *Helicobacter pylori (H. pylori)*.
   b. Peptic ulcer disease.
   c. Boerhaave's syndrome.
   d. Use of esophageal instruments.
3. A surgical procedure in which the distal portion of the stomach and proximal duodenum have been resected, with the jejunum reanastomosed to the gastric remnant, is referred to as:
   a. Billroth I.
   b. Billroth II.
   c. Roux-en-Y.
   d. Vagotomy.
4. A common infant gastric disorder requiring surgical intervention is:
   a. Foreign body removal.
   b. Esophageal atresia.
   c. Hiatal hernia.
   d. Hypertrophic pyloric stenosis.
5. A Whipple procedure involves removal of:
   a. The pylorus.
   b. The vagus nerve.
   c. The lower one-third of the esophagus.
   d. Fifty percent of the stomach and all of the duodenum.
6. Reasons for small bowel resection include all of the following except:
   a. Congenital anomalies.
   b. Irritable bowel syndrome (IBS).
   c. Crohn's disease.
   d. Trauma.
7. Hirschsprung's disease:
   a. Is the congenital absence of intramural ganglia in the intestinal tract.
   b. Can be cured with medications.
   c. Cannot be treated surgically.
   d. Occurs in 1 in 1,000 live births.
8. Crohn's disease:
   a. Is primarily prevalent in people with type A personality.
   b. Causes severe constipation.
   c. Is limited to the colon.
   d. May affect any part of the digestive tract.
9. A cancerous lesion within the sigmoid colon or retrosigmoid are commonly treated with:
   a. Colon transplant.
   b. Total colectomy.
   c. Whipple procedure.
   d. Rectal resection and descending colon anastomoses of the rectal remnant.

10. In transgastric surgery, entry points for access to the peritoneum may include which of the following:
    a. Mouth.
    b. Vagina.
    c. Rectum.
    d. All of the above.

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Answers to Review Questions

Chapter 1

1. c
2. a
3. b
4. a
5. d
6. b
7. b
8. c
9. a

Chapter 2

1. a
2. a
1. a
2. b
3. c
4. b
5. c
6. b
7. c
8. d

Chapter 3

1. d
2. c
3. b
4. d
5. b
6. a
7. b
8. c
9. c
10. b

Chapter 4

1. a
2. b
3. d
4. d
5. d
6. a
7. d
8. a
9. d

Chapter 5

1. d
2. a
3. d
4. c
5. c
6. b
7. a
8. a
9. a
10. d

Chapter 6

1. b
2. d
3. d
4. d
5. d
6. False
7. a
8. b
9. c
10. a
11. a

Chapter 7

1. a
2. b
3. b
4. b
5. d
6. a
7. c
8. c
9. b
10. a
11. a
12. a
13. c
14. b
15. c
16. d
17. c
18. a
19. c
20. b
21. a
22. a

Chapter 8

1. a
2. d
3. d
4. b
5. d
6. c
7. a
8. c
9. d
10. c

Chapter 9

1. c
2. b
3. c
4. b
5. a
6. b
7. c
8. b
9. c

Chapter 10

1. a
2. d
3. d
4. a
5. d
6. c
7. c
8. c
9. a
10. b

Chapter 11

1. a
2. c
3. d
4. d
5. c
6. b
7. d
8. a
9. d
10. b

Chapter 12

1. a
2. d
3. b
4. d
5. a
6. b
7. c
8. a
9. c

Chapter 13

1. b
2. c
3. d
4. a
5. c
6. a
7. c
8. d
9. d
10. c

Chapter 14

1. a
2. c
3. b
4. c
5. b
6. a
7. a
8. d
9. c
10. c

Chapter 15

1. c
2. b
3. c
4. c
5. d
6. b
7. a
8. b
9. d
10. c

Chapter 16

1. b
2. a
3. d
4. b
5. d
6. b
7. c
8. b
9. d
10. b

Chapter 17

1. b
2. b
3. c
4. d
5. c
6. b
7. a
8. d
9. c
10. a
11. d

Chapter 18

1. a
2. b
3. d
4. b
5. b
6. d
7. d
8. a
9. d
10. a
11. d
12. a
13. c
14. c

Chapter 19

1. b
2. c
3. a
4. d
5. d
6. a
7. c
8. c
9. d
10. b
11. c
12. c
13. c

Chapter 20

1. c
2. a
3. a
4. c
5. c
6. c
7. a
8. a
9. d
10. a

Chapter 21

1. c
2. b
3. a
4. c
5. b
6. d
7. b
8. a
9. b
10. b

Chapter 22

1. b
2. d
3. a
4. a
5. a
6. a
7. d
8. c
9. d
10. c
11. d
12. d
13. d
14. b
15. a

Chapter 23

1. c
2. b
3. d
4. c
5. a
6. b
7. d
8. b
9. a
10. c

Chapter 24

1. b
2. b
3. d
4. a
5. a
6. b
7. d
8. c

Chapter 25

1. b
2. c
3. a
4. c
5. d
6. a
7. a
8. a
9. a
10. a

Chapter 26

1. d
2. d
3. e
4. a
5. b
6. d
7. a
8. a
9. b
10. c

Chapter 27

1. b
2. c
3. a
4. b
5. c
6. c
7. a
8. d
9. a

Chapter 28

1. d
2. d
3. b
4. b
5. c
6. a
7. b
8. c
9. d
10. b

Chapter 29

1. b
2. a
3. c
4. a
5. a
6. c
7. d
8. d
9. c
10. d

Chapter 30

1. a
2. d
3. c
4. c
5. c
6. b
7. a
8. b
9. c
10. d

Chapter 31

1. b
2. d
3. d
4. b
5. a
6. b
7. a
8. b
9. c
10. c

Chapter 32

1. c
2. a
3. b
4. d
5. d
6. b
7. a
8. d
9. d
10. d

Glossary

abetalipoproteinemia. A hereditary syndrome characterized by a lack of beta-lipoproteins in the blood, acanthocytosis, hypocholesterolemia, progressive ataxic neuropathy, atypical retinitis pigmentosa and malabsorption.

achalasia. A combined defect of absent peristalsis of the esophageal body and elevated lower esophageal sphincter pressure.

achlorhydria. Absence of free hydrochloric acid in the stomach. May be caused by gastric cancer, ulcer, pernicious anemia, adrenal insufficiency or chronic gastritis.

acinus (hepatic). A small saclike dilatation, especially a functional unit of the liver, which is supplied by terminal branches of the portal vein and the hepatic artery, and drained by a terminal branch of the bile duct.

activated partial thromboplastin time. A measure of the rapidity of blood clotting, which examines factors I, II, V, VIII, IX, X, XI and XII.

actual health problem. A health condition that is identified as presently causing some difficulty for the patient.

adenomatous polyp. Benign neoplastic tissue originating in the glandular epithelium.

advocacy. The act of speaking or writing in support of another or in protection of another's rights.

Alagille's syndrome. A rare congenital syndrome in which arteriohepatic dysplasia is associated with developmental anomalies of the face, heart, kidneys, muscle and nervous system. *See also* intrahepatic biliary dysplasia.

α_1-antitrypsin deficiency. Lack of a plasma protein that is produced in the liver.

ambulatory pH monitoring. A 24-hour test that records fluctuations in esophageal pH and correlates them with symptoms of esophageal reflux.

amebiasis. The state of being infected with Entamoeba histolytica.

American Board of Certification for Gastroenterology Nurses (ABCGN). A volunteer non-profit organization whose purpose is to maintain and improve the knowledge, understanding and skill of nurses in the fields of gastroenterology and endoscopy by developing and administering a certification program.

American dilator. One of a series of radiopaque, tapered, polyvinyl dilators that are passed over a guidewire for the purpose of widening a gastrointestinal lumen.

American Nurses Association (ANA). A professional society for nursing in the United States.

amino acid. A class of organic compounds containing an amino group and a carboxyl group. Amino acids form the chief structural components of proteins, and several are essential in human nutrition.

ampulla of Vater. The dilatation formed by the junction of the common bile duct and the pancreatic duct proximal to their opening into the duodenum.

anabolism. Any constructive process by which simple substances are converted by living cells into more complex compounds, especially into living matter.

anaphylaxis. An unusual or exaggerated allergic reaction to a foreign protein or other substance.

anemia. A reduction below normal in the number of erythrocytes. *See also* pernicious anemia.

anesthetic. A drug or agent used to abolish the sensation of pain, particularly before surgery or other painful procedures.

angiography. The roentgenographic visualization of blood vessels following introduction of contrast material.

annular pancreas. A developmental anomaly in which the pancreas forms a ring entirely surrounding the duodenum.

anorexia. Lack or loss of appetite for food.

anoscopy. Examination of the anus and lower rectum using a specially designed speculum.

antacid. A substance that counteracts or neutralizes acidity, usually gastric acidity.

antibiotic. An agent that inhibits the growth of or kills microorganisms, used in the treatment of infectious diseases.

anticholinergic. An agent that blocks the parasympathetic nerves.

antidiarrheal. An agent that combats abnormally frequent and liquid fecal discharges.

antiemetic. An agent that prevents or alleviates nausea and vomiting.

antiflatulent. An agent that disperses or prevents the formation of air or gas pockets in the gastrointestinal tract.

antifungal. An agent that is destructive to fungi, suppresses their growth or reproduction, or is effective against fungal infections.

antrum. The constricted, elongated, lower portion of the stomach.

anus. The terminal orifice of the gastrointestinal tract.

argon. An inert gas that is used in lasers.

arteriography. Roentgenography of an artery after injection of a contrast medium.

ascending colon. The portion of the large intestine between the cecum and the hepatic flexure.

ascites. The effusion and accumulation of serous fluid in the abdominal cavity.

aspiration. (1) The act of inhaling, including the accidental inhalation of solids or liquids; (2) the removal of fluids or gases from a cavity by the application of suction. *See also* fine-needle aspiration.

aspiration biopsy. A biopsy in which the tissue is obtained by the application of suction through a needle attached to a syringe.

assessment. Continuous, systematic collection, validation, and communication of patient data for the purpose of planning, implementing, and evaluating nursing care directed toward the attainment of specific patient outcomes.

atresia. Congenital absence or closure of a normal body orifice or tubular organ.

audit. A review of documentation for the purpose of determining whether or not specific objectives were met (e.g., patient goals were achieved, nursing standards of care were met, or structural or environmental criteria were attained) during the period of time outlined in a goal or standard. *See also* concurrent audit, nursing audit, *and* retrospective audit.

Auerbach's plexus. The part of the enteric plexus that is within the muscularis. Also called the myenteric plexus.

authority. The legal or rightful power to command or act.

balloon tamponade. Esophageal-gastric tamponade, involving exertion of pressure against bleeding esophageal varices by inflation of esophageal and usually gastric balloons.

barbed stent. A stent with projections or "barbs" at each end that result from a diagonal cut in the stent wall and serve to hold the stent in place.

barium enema. A suspension of barium that is injected into the rectum and retained in the intestines during roentgenological examination. Also called a contrast enema.

barium sulfate. A bulky, fine, white powder without odor or taste, and free from grittiness, that is used as a contrast medium in roentgenography of the digestive tract.

barium swallow. Ingestion of a thick barium solution for the purpose of radiographic examination of the esophagus.

Barrett's esophagus. Replacement of the normal squamous epithelium of the esophagus by columnar epithelium.

bird's beak. Tapering of the esophagus at the gastric cardia seen on barium esophagram, characteristic of patients with achalasia. *See also* achalasia.

Bernstein test. Attempted simulation of noncardiac chest pain by instillation of hydrochloric acid through one of the ports of a manometry catheter or a nasogastric tube that is positioned in the esophagus.

bezoar. A concretion of foreign material that builds up in the stomach.

bile. An alkaline golden brown to greenish-yellow fluid that is secreted by the liver and poured into the small intestine via the bile ducts. Important constituents include conjugated bile salts, cholesterol, phospholipid, bilirubin diglucuronide and electrolytes.

biliary colic. Paroxysms of pain and other severe symptoms resulting from the passage of gallstones along the bile duct.

biliary stent. A stent inserted into the common bile duct or pancreatic duct.

Billroth I procedure. Surgical procedure sacrificing the distal portion of the stomach, pylorus and duodenal bulb. The duodenum is then reattached by anastomosis with the gastric remnant.

Billroth II procedure. Surgical procedure sacrificing the distal portion of the stomach and a portion of the proximal duodenum. The proximal duodenum is closed, and a segment of proximal jejunum is attached to the gastric remnant with end-to-end or side-to-end anastomosis.

biopsy. The removal and examination, usually microscopic, of tissue from the living body, performed to establish a precise diagnosis. *See also* aspiration biopsy, percutaneous liver biopsy *and* suction biopsy.

biopsy forceps. An instrument that can be passed through the biopsy channel of an endoscope for the purpose of excising pieces of living tissue from a suspected pathological site. *See also* hot biopsy forceps.

bipolar electrocoagulation. An electrocoagulation method in which the electrical current flows between two small electrodes on the tip of the probe, both of which are in contact with the target tissue.

bipolar probe. A specialized bipolar hemostatic probe that is inserted through the instrument channel of an endoscope.

body. The largest and most important part of the stomach, lying between the fundus and the antrum.

Boerhaave's syndrome. Catastrophic event in which the lower thoracic esophagus is completely torn away from the gastric cardia.

borborygmi. Rumbling noises caused by the propulsion of gas through the intestines.

bougie. A slender, flexible, cylindrical instrument for introduction into a tubular organ, usually for the purpose of dilating a constricted area. *See also* Hurst bougie *and* Maloney bougie.

bougienage. The passage of a slender, flexible cylindrical instrument into a tubular organ to dilate a stricture.

Brunner's gland. A tubulo-alveolar gland in the submucosa of the duodenum, which opens into a crypt of Lieberkühn.

candidiasis. Infection with a fungus of the genus Candida.

cannula. A tube for insertion into a duct or cavity; sometimes passed over a guidewire.

capital budget. Planning document used to anticipate costs for durable equipment with purchase price (usually) greater than $500 and expected life of greater than 5 years.

carbohydrate. An aldehyde or ketone derivative of a polyhydric alcohol; the hydrogen and oxygen are usually in the proportion to form water. The most important carbohydrates are the starches, sugars, celluloses and gums.

cardia. The portion of the stomach surrounding the esophagogastric junction, which contains cardiac glands but lacks parietal and chief cells.

cardiac arrest. Sudden cessation of cardiac function, with disappearance of arterial blood pressure, connoting either ventricular fibrillation or ventricular standstill.

cardiac gland. A gland located distal to the esophagogastric junction that secretes mucus and pepsinogens.

cardiac sphincter. *See* lower esophageal sphincter.

care conference. A collaborative meeting of nurses and possibly other health and allied health professionals for the purposes of planning and evaluating nursing management of a patient's health problem or set of problems. It represents a brainstorming effort to generate creative, comprehensive, or more aggressive approaches to care, usually for long-term patients with complicated problems whose previous management has failed to bring about desired outcomes.

catabolism. Any destructive process by which complex substances are converted by living cells into more simple compounds.

cathartic. An agent that causes evacuation of the bowels by increasing bulk (bulk cathartic), stimulating peristaltic action (stimulant cathartic), softening the feces and reducing friction between them and the intestinal wall (lubricant cathartic), or increasing fluidity of the intestinal contents by retention of water by osmotic forces and indirectly increasing motor activity (saline cathartic).

catheter. A tubular, flexible surgical instrument for withdrawing fluids from, or introducing fluids into, a cavity of the body. *See also* nasobiliary catheter (NBC), solid-state catheter *and* water perfusion catheter.

cecum. The first part of the large intestine, forming a dilated pouch into which open the ileum, the colon and the vermiform appendix.

celiac sprue. A malabsorption syndrome affecting both children and adults, precipitated by the ingestion of gluten-containing foods. Pathologically, the proximal intestinal mucosa loses its villous structure, surface epithelial cells exhibit degenerative changes, and their absorptive function is severely impaired.

cell. Any one of the minute protoplasmic masses that make up organized tissue, consisting of a nucleus surrounded by cytoplasm that contains the various organelles and is enclosed in the cell or plasma membrane. A cell is the fundamental structural and functional unit of living organisms. *See also* chief cell, enterochromaffin cell, G cell, goblet cell, Kupffer cell, oxyntic cell, Paneth's cell, parietal cell, red blood cell, white blood cell *and* zymogen cell.

Centers for Disease Control and Prevention (CDC). The federal public health agency within the Department of Health and Human Services that investigates specific disease outbreaks and formulates general guidelines for disease control.

central tendency. The grouping or score that occurs with the greatest frequency, used in describing a mass of data.

certification. The process by which a nongovernmental agency or association grants recognition to an individual who has met certain qualifications that have been predetermined by that agency or association.

chief cell. A cell located in the parietal glands of the stomach; chief cells secrete pepsinogens.

cholangiogram. A roentgenogram of the gallbladder and bile ducts, following intravenous injection of contrast medium. *See also* percutaneous transhepatic cholangiogram.

cholangitis. An inflammation of a bile duct. *See also* primary sclerosing cholangitis.

cholecystitis. Inflammation of the gallbladder.

cholecystokinin. A polypeptide hormone secreted by the mucosa of the upper small bowel, which stimulates contraction of the gallbladder (with release of bile) and secretion of pancreatic enzymes.

choledocholithiasis. The presence of gallstones in the common bile duct.

cholelithiasis. The presence or formation of gallstones.

cholestasis. Stoppage or suppression of the flow of bile, having either intrahepatic or extrahepatic causes.

cholinergic. Stimulated, activated or transmitted by choline (acetylcholine); a term applied to nerve fibers that liberate acetylcholine at a synapse when a nerve impulse passes; an agent that produces such effects.

chyme. A relatively homogeneous semiliquid combination of food and digestive juices found in the stomach and small bowel.

cirrhosis. A liver disease characterized pathologically by loss of the normal microscopic lobular architecture, with fibrosis and nodular regeneration.

coagulating current. An electric current that is applied for the purpose of coagulating tissue.

coagulation. The process of clot formation; in surgery, the disruption of tissue by physical means to form an amorphous residuum, as in electrocoagulation and photocoagulation.

colitis. Inflammation of the colon. *See also* Crohn's colitis, ischemic colitis, pseudomembranous colitis *and* ulcerative colitis.

collaborative diagnosis. Statements of actual or potential health problems that occur from complications of disease, diagnostic studies, or therapeutic procedures, for which the nurse identifies a need to work with other members of the health care team toward resolution. *See also* medical diagnosis *and* nursing diagnosis.

colloid. A state of matter made up of very small, insoluble, nondiffusible particles that remain in suspension in a dispersion medium. The particles in a colloid are larger than ordinary crystalloid molecules, but they are not large enough to settle out under the influence of gravity. *See also* crystalloid.

colon. The part of the large intestine that extends from the cecum to the rectum. *See also* ascending colon, descending colon, sigmoid colon, *and* transverse colon.

colonoscopy. Endoscopic examination of the colon.

common bile duct. The duct formed by the union of the cystic duct and the hepatic duct.

comparison group. A group of subjects whose scores on a dependent variable are used as the basis for evaluating the scores of an experimental group or the group of primary interest. "Comparison group" is used rather than "control group" when the investigation does not use a true experimental design. *See also* control group.

computed tomography. Also based on the variable absorption of x-rays by different tissues, computed tomography (CT) imaging, also known as "CAT scanning" (Computerized Axial Tomography), provides a different form of imaging known as cross-sectional imaging. A CT imaging system produces cross-sectional images or "slices" of anatomy.Also called a CT scan.

concurrent audit. An evaluation of nursing care and patient outcomes performed while the patient is receiving care. It is performed by using direct observation of nursing care, patient interview and/or chart review.

constipation. Infrequent or difficult evacuation of feces; passage of unduly hard or dry fecal material.

consultation. A meeting of two or more professionals to exchange ideas concerning patient care or to seek advice, instruction or information.

contrast roentgenography. Roentgenography performed after the administration of a contrast medium, often barium sulfate, which facilitates interpretation of the film by accentuating differences in the densities of different regions and structures.

control group. The subjects not receiving an experimental treatment or intervention, whose performance provides a baseline against which the effects of the treatment can be measured. *See also* comparison group.

corticosteroid. Any of the steroids elaborated by the adrenal cortex (excluding sex hormones of adrenal origin) in response to the release of corticotropin by the pituitary gland, or any of the synthetic equivalents of these steroids.

counseling. The act of rendering short-term, long-term, or motivational guidance to a patient/significant other, an act that may involve the patient in problem solving.

criterion. A measurable quality, attribute, behavior or characteristic that specifies a skill, knowledge or health state that is met at the point a health goal is achieved. Plural: criteria.

critical item. An instrument or object that is introduced directly into the bloodstream or into other normally sterile areas of the body.

Crohn's colitis. Crohn's disease, confined to the colon.

Crohn's disease. A chronic granulomatous inflammatory disease involving any part of the GI tract, but commonly involving the terminal ileum, with scarring and thickening of the bowel wall. It frequently leads to intestinal obstruction and fistula and abscess formation and has a high rate of recurrence after treatment. Also known as regional enteritis.

cryoprecipitate. Any one of a group of serum proteins, including factors VIII, XIII, and fibrinogen, that settle out of solution at temperatures below 20° C.

crypt of Lieberkühn. A simple tubular gland in the mucous membrane of the intestine, opening between the bases of the villi and containing argentaffin cells.

crystalloid. A substance that, in solution, passes readily through animal membranes, lowers the freezing point of the solvent containing it and is generally capable of being crystallized. *See also* colloid.

culture. The propagation of microorganisms or of living tissue cells in special media conducive to their growth.

Curling's ulcer. A stress ulcer that appears in patients with serious burn injuries.

Cushing's ulcer. A stress ulcer that appears in patients with intracranial trauma.

cutting current. An electrical current applied for the purpose of dissection or fulguration.

cystic duct. The passage connecting the neck of the gallbladder and the common bile duct.

cystic fibrosis. A hereditary disorder of infants, children and young adults, in which there is widespread dysfunction of the exocrine glands. It is characterized by signs of chronic pulmonary disease caused by excess mucus production in the respiratory tract, pancreatic deficiency, abnormally high levels of electrolytes in the sweat and occasionally by biliary cirrhosis.

cytology. The study of cells, their origins, structure, function and pathology. *See also* exfoliative cytology.

cytology brush. A sheathed, disposable brush that can be passed through the biopsy channel of an endoscope for the purpose of obtaining specimens for microscopic examination.

data. The material or collection of facts upon which a discussion or an inference is based. *See also* objective data *and* subjective data.

database. A foundation of subjective and objective patient information that enables the design and implementation of a comprehensive and effective plan of care.

decompression. The removal of pressure, as in the removal of excess gas from the intestinal tract.

decontamination. The removal of gross soils and the reduction of the number of microorganisms to the point where an item may be considered safe for handling.

dependent intervention. Nursing action performed under the supervision or direction of a physician.

dependent variable. A concept capable of taking on different values whose value is affected by, or determined by, other variables.

descending colon. The portion of the colon between the splenic flexure and the sigmoid colon at the pelvic brim.

desiccation. The act of drying up, especially the treatment of a tumor or other disease by drying up the part by the application of laser or electrical energy.

dextrose. D-Glucose monohydrate. A monosaccharide that occurs as colorless crystals or as a white, crystalline or granular powder; used chiefly as a fluid and nutrient replenisher, usually administered by IV infusion. Also used as a diuretic and alone or in combination with other agents for other clinical purposes.

diaphragmatic hiatus. An opening in the diaphragm where the esophagus enters the abdominal cavity.

diarrhea. Abnormally frequent and liquid fecal discharges.

diffuse esophageal spasm. Repetitive, prolonged simultaneous contractions along the length of the esophagus, with intermittent normal peristalsis.

dilator. An instrument that is used to enlarge an orifice or canal by stretching. *See also* American dilator, bougie, Hurst bougie, Maloney bougie *and* Savary-Gilliard dilator.

disaccharide. Any of a class of sugars that yield two monosaccharides on hydrolysis and have the general formula $C_n(H_2O)_n\_1$

disinfection. A physical or chemical process that kills or destroys most pathogenic microorganisms, but rarely kills all spores.

dispersive electrode. Grounding pad.

diverticulitis. Inflammation of a diverticulum, especially inflammation related to colonic diverticula, which may undergo perforation with abscess formation.

diverticulosis. The presence of diverticula, particularly colonic diverticula, in the absence of inflammation.

diverticulum. An outpouching of one or more layers of the wall of a tubular organ. *See also* Meckel's diverticulum.

documentation. The act of collecting, abstracting and coding of patient data and therapeutic processes for the purposes of communicating patient care, supplying a supporting reference concerning the status or progress of a patient and archiving evidence of care rendered.

double-blind. An experiment in which neither subjects nor investigators are aware of which subjects are in the experimental group and which subjects are in the control group.

double-contrast roentgenography. Mucosal relief roentgenography; involves injection and evacuation of a barium enema, followed by inflation of the intestine with air under light pressure. The light coating of barium on the walls of the inflated intestine in the roentgenogram clearly reveals even small abnormalities.

dry swallow. Performing the action of swallowing during esophageal manometry without ingesting liquid. *See also* wet swallow.

duct. A passage with well-defined walls, especially a tube for the passage of excretions or secretions. *See also* common bile duct, cystic duct, duct of Santorini, duct of Wirsung *and* hepatic duct.

duct of Santorini. The minor pancreatic duct, draining a part of the head of the pancreas into the minor duodenal papilla.

duct of Wirsung. Pancreatic duct; the main excretory duct of the pancreas, which usually unites with the common bile duct before entering the duodenum at the major duodenal papilla (papilla of Vater).

dumping syndrome. A group of disabling symptoms associated with rapid gastric emptying that mimic the symptoms of hypoglycemia.

duodenum. The first, or proximal, portion of the small bowel, extending from the pylorus to the jejunum.

dyspepsia. Impairment of the power or function of digestion, usually applied to epigastric discomfort following meals.

dysphagia. A sensation of difficulty in swallowing.

edrophonium chloride. A cholinesterase inhibitor that is administered by IV bolus in a provocative test designed to reproduce noncardiac chest pain caused by esophageal dysmotility.

electrocautery. An instrument used to destroy tissue, using an electrical current.

electrocoagulation. Coagulation of tissue, using either a monopolar or a bipolar electrical current. *See also* bipolar electrocoagulation *and* monopolar electrocoagulation.

electrolyte. A substance that dissociates into ions when fused or in solution and thus becomes capable of conducting electricity.

electrosurgical unit (ESU). An apparatus for cutting or coagulating tissue, using a high-frequency electrical current.

endogenous. Produced or arising from within a cell or organism.

endoscopic retrograde cholangiopancreatography (ERCP). An endoscopic technique for radiological visualization of the biliary and/or pancreatic ducts.

endoscopic variceal ligation (EVL). The endoscopic introduction of rubber bands or O-rings for the treatment of bleeding varices.

endoscopy. Visual inspection of any cavity of the body by means of an endoscope.

enema. A liquid injected into the rectum. *See also* barium enema.

enteral nutrition. Administration of a prescribed diet by means of a flexible tube inserted into the stomach or small bowel transnasally, surgically or endoscopically.

enteric plexus. A plexus of autonomic nerve fibers within the wall of the digestive tube, and made up of the submucosal, myenteric and subserosal plexuses.

enteritis. Inflammation of the intestine, especially of the small bowel. *See also* radiation enteritis *and* regional enteritis.

enterochromaffin cell. A basal granular cell whose granules stain readily with silver and chromium salts and which is a site of synthesis and storage of serotonin; includes argentaffin cells and agyrophilic cells.

enteroclysis. The injection of a nutrient or a medicinal liquid into the bowel.

enterocolitis. Inflammation involving both the small bowel and the colon.

erythrocyte. A red blood cell; one of the elements found in peripheral blood; normally, in humans, the mature form is a nonnucleated, yellowish, biconcave disk, adapted, by virtue of its configuration and its hemoglobin content, to transport oxygen.

esophageal atresia. Birth defect characterized by a markedly dilated blind upper esophageal pouch, a variable esophageal defect, and a lower pouch terminating as a fistula communicating with the posterior trachea.

esophageal reflux. Reflux of gastric or duodenal contents back into the esophagus.

esophageal rings and webs. Thin, circumferential mucosal shelves appearing in the esophagus. *See also* Schatzki's ring.

esophagitis. An inflammation of the esophageal mucosa.

esophagogastroduodenoscopy (EGD). Endoscopic examination of the esophagus, stomach and duodenum.

esophagus. The musculomembranous tubular portion of the GI tract that extends from the pharynx to the stomach.

ethylene oxide. A colorless, flammable gas used to sterilize instruments.

evaluative statement. A statement defining an actual outcome; for example, skills developed, knowledge obtained or change in health status.

exfoliative cytology. Microscopic examination of cells desquamated from the body surface or a lesion as a means of detecting malignancy and microbiological changes, to measure hormonal levels, etc. Cells may be obtained by such procedures as aspiration, washing, smears, and scraping, and the technique may be applied to vaginal secretions, sputum, urine, abdominal fluid, prostatic secretions, etc.

exogenous. Originating outside an organ or part.

experimental group. The subjects receiving an experimental treatment or intervention.

familial polyposis. Multiple adenomatous polyps with high malignant potential lining the mucous membrane of the intestine, particularly the colon, beginning about puberty.

fatty acid. Any monobasic aliphatic acid containing only carbon, hydrogen, and oxygen and made up of an alkyl radical attached to the carboxyl group. Saturated fatty acids have the general formula $C_nH_2nO_2$. There are also several series of unsaturated fatty acids having one or more double bonds, and a few cyclic acids.

fiber optics. The transmission of an image along flexible bundles of coated parallel fibers that propagate light by internal reflections.

fine-needle aspiration. Sampling of pancreatic tissue for the purpose of cytological examination. Used in the diagnosis of pancreatic cancer.

fistula. An abnormal passage between two internal organs. *See also* pancreatic fistula.

fluoroscopy. Examination of deep structures by means of roentgen rays; uses a screen covered with crystals of calcium tungstate, on which are projected the shadows of x-ray beams passing through the body from the source of irradiation.

Food and Drug Administration (FDA). The federal regulatory agency responsible for controlling the safety and effectiveness of drugs, devices, and instrumentation and approving products for disinfectant registration by review of labeling and supporting data submitted by the registrants.

French unit. A unit for denoting the size of catheters or other tubular instruments, each unit being roughly equivalent to 0.3 mm in diameter. (e.g., 18 French [Fr] indicates a diameter of 6 mm).

frozen section. A tissue biopsy obtained during endoscopy that is sent for immediate microscopic examination by a pathologist to determine the type of abnormal tissue present.

fulguration. Destruction of living tissue by electric sparks generated by a high-frequency current.

fulminant hepatic failure. Massive liver cell death that occurs within 2 months of the development of acute hepatitis.

functional organization. A form of organizational structure that is designed to allow specialists in given areas to give and enforce recommendations within a clearly defined scope.

fundus. The proximal portion of the stomach, which lies above and to the left of the lower esophageal sphincter.

G cell. A cell type located in the pyloric glands of the stomach; G cells secrete gastrin.

gallbladder. The pear-shaped reservoir for bile on the posteroinferior surface of the liver, between the right and the quadrate lobe; from its neck, the cystic duct projects to join the common bile duct.

Gardner syndrome. Familial polyposis of the colon (with malignant potential), supernumerary teeth, fibrous dysplasia of the skull, osteomas, fibromas and epithelial cysts.

gastric baseline. Manometric tracing showing a relatively flat, smooth tracing with a small pressure increase on inspiration or abdominal pressure. Indicates all catheter recording ports are in the patient's stomach.

gastric ulcer. Ulcer of the gastric mucosa.

gastritis. An inflammation of the gastric mucosa.

gastroenterology associate. A non-RN health care professional with varied educational background who is engaged in the field of gastroenterology.

gastroenterology nurse. A registered nurse who specializes in the field of gastroenterology.

gastroesophageal reflux disease (GERD). Backward flow of gastric contents into the esophagus when the pressure in the stomach is greater than in the esophagus. Associated with pregnancy, obesity or incompetence of the lower esophageal sphincter.

gastroesophageal sphincter. *See* lower esophageal sphincter.

giardiasis. Infection with the flagellate protozoan Giardia lamblia, characterized by protracted, intermittent diarrhea with symptoms suggesting malabsorption, and by abdominal pain, distention, and flatulence; light infections are usually asymptomatic.

gland. An aggregation of cells, specialized to secrete or excrete materials not related to their ordinary metabolic needs. *See also* Brunner's gland, cardiac gland *and* pyloric gland.

glucose. A monosaccharide, $C_6H_{12}O_6$, found in certain foodstuffs, especially fruits, and in the normal blood of all animals. It is the chief energy source for living organisms, its utilization being controlled by insulin.

glutaraldehyde. A high-level disinfectant that is effective against vegetative gram-positive, gram-negative, and acid-fast bacteria, some bacterial spores, some fungi and viruses.

glycerol. A trihydric sugar alcohol that is the alcoholic component of the fats; it is soluble in water and alcohol and is an intermediate in the metabolism of fatty acids.

glycogen. A polysaccharide that is the chief carbohydrate storage material in animals. It is a long-chain polymer of glucose, formed in and largely stored in the liver and to a lesser extent in muscles, being depolymerized to glucose and liberated as needed.

goal. (1) A desired outcome that should reflect the mission statement of an organization. (2) A desired patient outcome, which must be realistic, usable, observable and specific.

goblet cell. A unicellular mucous gland found in the epithelium of various mucous membranes, especially in the respiratory passages and the intestines. Droplets of mucigen collect in the upper part of the cell and distend it, while the basal end remains slender and the cell assumes the shape of a goblet.

greater curvature. The lower lateral border of the stomach.

greater omentum. A layer of visceral peritoneum that hangs from the greater curvature of the stomach over the anterior side of the abdominal viscera.

grounding pad. A dispersive electrode that is securely attached to the patient's skin and serves to complete the current flow from a monopolar electrosurgery probe, through the patient's body, and back to the generator. Also known as a grounding plate.

halon. Bromotrafluoromethane. A commercial product used in fire extinguishers that are safe for use in areas containing sensitive electrical equipment.

haustrum. Sacculation in the wall of the colon produced by adaptation of its length to that of the tenia coli, or by the arrangement of the circular muscle fibers. Plural: haustra.

health problem. A condition related to health that requires intervention if disease or illness is to be prevented or resolved and if coping and wellness are to be promoted. *See also* actual health problem, possible health problem *and* potential health problem.

heartburn. A retrosternal sensation of warmth or burning that occurs in waves and tends to rise toward the neck. Also known as pyrosis.

heater probe. A hollow aluminum cylinder with an inner heat coil and an outer coating of Teflon that is applied directly to a bleeding vessel to produce hemostasis using bipolar electrocoagulation.

Helicobacter pylori. A gram-negative curved or special rod that is microaerophilic. Formerly Campylobacter pylori.

Heller's myotomy. Surgical procedure performed to treat achalasia.

hematochezia. The passage of bloody stools.

hematocrit. The volume percentage of red blood cells in whole blood.

Hemoccult. The trademark for a modification of the guaiac test for occult blood, in which guaiac-impregnated filter paper is used; the test is positive if the specimen turns blue.

hemochromatosis. A disorder of iron metabolism characterized by excess deposition of iron in the tissues, especially in the liver and pancreas, and by bronze pigmentation of the skin, cirrhosis, diabetes mellitus, and associated bone and joint changes.

hemoglobin. The oxygen-carrying pigment of the red blood cells.

hemolysis. The liberation of hemoglobin from the red blood cells and its appearance in the plasma.

hemorrhage. Bleeding; the escape of blood from the blood vessels.

hepatic duct. The duct that is formed by the union of the right and left hepatic ducts and in turn joins the cystic duct to form the common bile duct.

hepatic encephalopathy. A condition usually occurring secondary to advanced liver disease but also seen in the course of any severe disease or in patients with portacaval shunts. Marked by disturbances of consciousness that may progress to deep coma (hepatic coma), psychiatric changes, flapping tremor and fetor hepaticus. Also called portal-systemic encephalopathy.

hepatic flexure. The right flexure of the colon; the bend in the large intestine at which the ascending colon becomes the transverse colon.

hepatitis. Inflammation of the liver.

hepatocyte. A parenchymal liver cell.

hepatorenal syndrome. A syndrome characterized by functional renal failure, oliguria, and low urinary sodium concentration, without pathological renal changes, associated with cirrhosis and ascites or with obstructive jaundice.

hiatal hernia. Occurs when a portion of the stomach protrudes through the diaphragmatic hiatus into the thoracic cavity.

high-level disinfection. Process of cleaning instruments that destroys all microorganisms with the exception of low levels of bacterial spores.

highly selective vagotomy. Surgical procedure interrupting the nerve fibers to the antrum but preserving the innervation of the pyloric region.

Hirschsprung's disease. Megacolon caused by congenital absence of myenteric ganglion cells in a distal segment of the colon. The resultant loss of motor function causes massive hypertrophic dilatation of the normal proximal colon; the aganglionic segment usually remains narrowed but may dilate passively. Also known as congenital megacolon or aganglionic megacolon.

histamine. A decarboxylation product of histidine found in all body tissues. Cellular receptors of histamine include H_1 receptors, which mediate the effects of histamine on smooth muscle and capillaries; and H_2 receptors, which mediate the acceleration of heart rate and the promotion of gastric acid secretion.

histamine$_2$ (H_2) blocker. An agent that blocks the cellular receptor site for histamine that is responsible for stimulating the heart rate and gastric secretion.

histology. The study of the minute structure, composition and function of the tissues; also called microscopical anatomy.

hot biopsy forceps. A type of biopsy forceps that is insulated by a nonconducting sheath and attached to an electrocoagulation snare handle.

Hurst bougie. One of a series of blunt-tipped, mercury-filled tubes of graded diameter used for dilating esophageal strictures.

hydrogen breath test. A measure of the amount of hydrogen expelled in the breath after ingestion of a carbohydrate drink; used to detect carbohydrate malabsorption, abnormal gastrointestinal transit time or bacterial overgrowth in the small bowel.

hydrostatic balloon. A polyethylene balloon that can be inserted into the GI tract and inflated with fluid to a specified pressure; used primarily for the dilatation of strictures.

hypertonic. A term denoting a solution that, when bathing body cells, causes a net flow of water across the semipermeable cell membrane out of the cell. Also denotes a solution having a greater tonicity than another solution (e.g., the blood), to which it is being compared.

hypoalbuminemia. An abnormally low albumin content in the blood.

hypoglycemia. An abnormally low glucose content in the blood, which may lead to tremulousness, cold sweat, piloerection, hypothermia, and headache, accompanied by confusion, hallucinations, bizarre behavior, and ultimately, convulsions and coma.

hypopharyngeal sphincter. *See* upper esophageal sphincter.

hypothesis. A declarative conjectured statement posing a relationship between two or more variables; hypotheses lead to empirical studies that seek to confirm or disconfirm the relationships. *See also* null hypothesis.

hypotonic. A term denoting a solution that, when bathing body cells, causes a net flow of water across the semipermeable cell membrane into the cell; also denotes a solution having less tonicity than another solution (e.g., the blood), to which it is being compared.

hypovolemia. Abnormally decreased volume of circulating fluid in the body.

ileocecal valve. A functional valve at the junction of the ileum and cecum, consisting of circular muscle of the terminal ileum.

ileum. The distal portion of the small intestine, extending from the jejunum to the cecum.

independent intervention. Nursing action initiated without direction or supervision of other health care professionals. Independent nursing intervention is instituted as the result of a nursing assessment.

independent variable. A concept capable of taking on different values whose value is unaffected by other variables in a given study.

indicator. A measurable variable that is used to measure the degree to which standards are met.

infantile hypertrophic pyloric stenosis. Congenital obstruction of the pyloric lumen caused by pyloric muscular hypertrophy.

infectious waste. Waste capable of producing an infectious disease; includes human, animal, or biological wastes and any items that may be contaminated with pathogens.

infiltration. The diffusion or accumulation in a tissue or cells of substances not normal to it or in amounts in excess of the normal.

inflammatory bowel disease. A general term for inflammatory diseases of the bowel of unknown etiology, including Crohn's disease and ulcerative colitis.

informed consent. An interaction between physician and patient in which a meaningful exchange of information concerning an impending health care administration occurs. Consent is not valid without fulfillment of these four requirements: full disclosure, competent judgment and decision-making ability, comprehension of the procedure and its associated risks and after effects, and free will.

insufflation. The act of blowing a vapor, gas or air into a body cavity.

interdependent intervention. Nursing action performed in concert with the efforts of other health care professionals.

intervention. All those activities the nurse identifies that directly relate to the nursing diagnosis and are directed toward improving the patient's problem. *See also* dependent intervention, independent intervention *and* interdependent intervention.

interview. A meeting of patient and nurse to gather data concerning a patient's health status, health problems, risks, weaknesses, strengths and need for nursing care.

intestinal pseudoobstruction. A condition characterized by constipation, colicky pain, and vomiting, but without evidence of organic obstruction.

intrahepatic biliary dysplasia. A rare autosomal-dominant liver disease that incorporates a combination of anomalies in conjunction with chronic cholestasis.

intussusception. The prolapse of one part of the intestine into the lumen of an immediately adjoining part.

ionizing radiation. High-energy radiation that interacts with matter to produce ion pairs.

irritable bowel syndrome. A chronic noninflammatory disease characterized by excessive secretion of mucus and disordered colonic motility with consequent colic, constipation and/or diarrhea with the passage of mucus. It is a common disorder with a psychophysiological basis.

ischemic colitis. Acute vascular insufficiency of the colon usually involving the portion supplied by the inferior mesenteric artery. The classic radiological sign is thumbprinting caused by localized elevation of the mucosa by submucosal hemorrhage or edema. Ulceration may follow.

islet of Langerhans. One of the irregular microscopic structures scattered throughout the pancreas and comprising the endocrine portion of the pancreas.

isotonic. A term denoting a solution in which body cells can be bathed without a net flow of water across the semipermeable cell membrane. Also denotes a solution having the same tonicity as another solution (e.g., the blood), to which it is being compared.

jaundice. A syndrome characterized by hyperbilirubinemia and deposition of bile pigment in the skin, mucous membranes and sclera with resulting yellow appearance of the patient.

jejunum. The portion of the small bowel that extends from the duodenum to the ileum.

jackhammer esophagus. Esophageal peristalsis with a contractile amplitude two to three times the normal volume. Also known as nutcracker esophagus.

Joint Commission (JC). An independent credentialing agency that grants approval to health care facilities that voluntarily comply with the agency's standards for public and patient health and safety.

Kupffer cell. A large, star-shaped or pyramidal cell with a large oval nucleus and a small prominent nucleolus. These intensely phagocytic cells line the walls of the sinusoids of the liver and form part of the reticuloendothelial system.

lactase deficiency. A deficiency in the brush-border enzyme lactase, which causes malabsorption of the disaccharide lactose; patients typically experience distention, flatulence, cramping and diarrhea within minutes of ingesting milk or milk products.

lamina propria. The connective tissue coat of a mucous membrane.

laparoscope. A fiberoptic instrument that permits inspection of the peritoneal cavity.

laparoscopy. Examination of the interior of the abdomen using a laparoscope.

laparotomy. A surgical incision made through the abdomen.

laser. Light Amplification by Stimulated Emission of Radiation. A device that transforms light of various frequencies into an extremely intense, small and nearly nondivergent beam of monochromatic radiation in the visible region with all the waves in phase. Capable of mobilizing immense heat and power when focused at close range, it is used as a tool in surgical procedures, in diagnosis and in physiological studies.

lavage. The irrigation or washing out of an organ, such as the stomach or bowel.

laxative. An agent that acts to promote evacuation of the bowel.

lesser curvature. The upper lateral border of the stomach.

lesser omentum. A layer of visceral peritoneum that attaches the lesser curvature of the stomach to the underside of the liver.

leukocyte. A white blood cell.

liability. Legal responsibility for one's acts (or failure to act), including the responsibility for financial restitution in the event of demonstrable damages resulting from negligent acts.

ligament of Treitz. Suspensory muscle of the duodenum; a flat band of smooth muscle originating from the diaphragm and continuous with the muscular coat of the duodenum at its junction with the jejunum.

line organization. A traditional form of organizational structure, in which each position has authority over a lower one in the organization.

Linton tube. A three-lumen tube used for esophageal-gastric tamponade; it has a gastric balloon but no esophageal balloon, and ports for both gastric and esophageal aspiration.

lipid. A fat or fatlike substance that is easily stored in the body and serves as a source of fuel. They include the fatty acids, neutral fats, waxes and steroids; compound lipids include glycolipids, lipoproteins and phospholipids.

lithotripsy. The crushing of gallstones or bladder calculi, either by using a mechanical lithotripter or by focusing shock waves on the stone.

lower esophageal sphincter (LES). A group of thickened circular muscles at the distal end of the esophagus, which regulate the entry of food into the stomach. Also known as the cardiac sphincter or gastroesophageal sphincter.

malabsorption. Impaired intestinal absorption of nutrients.

maldigestion. Impaired digestion.

Mallory-Weiss tear. A mucosal tear at the gastroesophageal junction that is associated with prolonged forceful vomiting.

malnutrition. Any disorder of nutrition, whether caused by unbalanced or insufficient diet or by defective assimilation or utilization of foods.

Maloney bougie. One of a series of mercury-filled bougies similar to the Hurst bougie but with a conical tip.

malrotation. Failure of normal rotation of an organ, as of the gut, during embryonic development.

manometry. Measurement of pressure or contraction, especially within the GI tract.

matrix organization. A type of organizational structure that looks at individual subsystems within a complex structure. These subsystems can be viewed anywhere on the continuum from totally dependent to totally autonomous.

mean. The index of central tendency usually referred to as the average; it is the sum of the values in a set, divided by the total number of elements in the set.

Meckel's diverticulum. An occasional sacculation or appendage of the ileum, derived from an unobliterated yolk stalk.

median. That point in a set of values above which and below which 50% of the values lie.

medical diagnosis. Classification of a patient's medical condition, based on interpretation of data related to pathology and etiology; usually implies a course of treatment.

megacolon. Abnormally large or dilated colon; the condition may be congenital or acquired, acute or chronic. *See also* toxic megacolon.

Meissner's plexus. The part of the enteric plexus that is situated in the submucosa. Also called the submucosal plexus.

microorganism. A minute living organism, usually microscopic, including bacteria, viruses, fungi and protozoa.

microvillus. A minute cylindrical process on the free surface of a cell, especially in the intestinal epithelium.

mineral. A nonorganic, homogeneous solid substance, usually a constituent of the earth's crust.

minimum effective concentration. Threshold strength below which a disinfecting solution should not be used.

Minnesota tube. A four-lumen tube used for esophageal-gastric tamponade; it has both gastric and esophageal balloons and ports for gastric and esophageal aspiration.

mission statement. A statement that describes the intent of a specific organization. The statement should include the unit's overall goals, objectives, services and the intent of the quality to be delivered.

mode. The numerical value in a set of values that occurs most frequently.

modified Whipple's procedure. Pylorus-preserving pancreaticoduodenectomy.

monitoring. The measurement of physiological parameters, including the use of mechanical devices and clinical observations.

monoglyceride. A compound consisting of one molecule of fatty acid esterified to glycerol.

monopolar electrocoagulation. An electrocoagulation method in which the electrical current flows between a small, active electrode that is in contact with the target tissue and a larger grounding pad that is attached to the patient's skin.

monosaccharide. A simple sugar; a carbohydrate that cannot be decomposed by hydrolysis. The monosaccharides are colorless crystalline substances, with a sweet taste, and which have the general formula CH_2O.

narcotic. An agent that depresses the central nervous system, reduces pain and sometimes produces sleep.

narcotic antagonist. An agent that opposes the action of narcotics on the nervous system.

nasobiliary catheter (NBC). A catheter that is inserted endoscopically into the common bile duct during ERCP, with the opposite end brought out through the patient's nostril. Its purpose is to provide drainage or to allow the instillation of therapeutic solutions.

nasoenteric intubation. Insertion of a tube that is passed through the naris, into the stomach, and then into the intestinal tract; used primarily to remove intestinal contents or to provide for tube feeding (enteral nutrition).

nasogastric tube. A soft rubber or plastic tube that is inserted through a nostril and into the stomach, for instilling liquid foods or other substances, or for withdrawing gastric contents.

nasopancreatic catheter (NPC). Long, thin, polyethylene tube placed endoscopically into the pancreatic duct and routed through the nose for short-term decompression or perfusion.

Nd:YAG (neodymium:yttrium-aluminum-garnet). A mineral crystal that is used as a laser medium to produce 1060-nm light.

needle. A sharp instrument used for suturing or puncturing.

Nissen fundoplication. Open abdominal surgical anti-esophageal reflux procedure.

nitrogen balance. A state of the body in regard to ingestion and excretion of nitrogen. In negative nitrogen balance the amount of nitrogen excreted is greater than the quantity ingested; in positive nitrogen balance the amount excreted is smaller than the amount ingested.

noncritical item. An item that either does not ordinarily touch the patient or touches only intact skin. Washing with a detergent is often sufficient cleaning for these items.

normal saline. An isotonic solution of sodium chloride for temporarily maintaining living cells. Also known as physiological saline.

null hypothesis. The hypothesis that states that there is no relationship between the variables under study. It is used primarily in connection with tests of statistical significance as the hypothesis to be rejected.

nursing audit. The method of evaluating care, the outcomes of care or the process by which these outcomes are achieved by using a review of patient records.

nursing diagnosis. A statement of an actual or potential health problem that can be alleviated or prevented by independent nursing intervention.

nursing examination. A physical assessment focused on functional abilities, usually performed in head-to-toe format, during which objective data about a patient's health status is gathered; steps in the examination include inspection, auscultation, percussion and palpation.

nursing history. An interview-style assessment that is performed to evaluate a patient's health status, health problems, risks, weaknesses, strengths and need for nursing care.

nursing order. A prescription for the nursing care that is to be given to achieve patient health goals.

nursing process. A systematic approach to nursing care using problem-solving techniques. It encompasses assessment, diagnosis, outcome identification, planning, implementation and evaluation.

nutcracker esophagus. *See* jackhammer esophagus.

nutrition. The processes involved in taking nutriments and assimilating and using them. *See also* enteral nutrition, peripheral parenteral nutrition *and* total parenteral nutrition.

objective. An observable activity that is developed to help achieve the established goals of an organization.

objective data. Facts perceptible by the senses of one observer that can be verified by another person observing the same data.

observation. Systematic, deliberate use of the five senses to gather data.

obstipation. Intractable constipation.

occult blood. Blood present in such small quantities that it can be detected only by chemical tests of suspected material, or by microscopic or spectroscopic examination.

Occupational Safety and Health Administration (OSHA). The federal regulatory agency responsible for enforcing safety and health regulations in the workplace.

odynophagia. Painful swallowing.

oral. Pertaining to the mouth; taken through or applied in the mouth.

organizational structure. A structure that determines the process by which a group of people distribute responsibilities, establish lines of communication, identify relationships and establish accountability. *See also* functional organization, line organization, matrix organization, project organization *and* staff organization.

osmosis. The passage of a solvent from a solution of lesser to one of greater solute concentration when the two solutions are separated by a membrane that selectively prevents the passage of solute molecules but is permeable to the solvent.

outcome. The end product of nursing care; measurable changes in a patient's health or behavior.

outcome identification. An actual or potential health problem exhibited by an individual through the process of clinical reasoning and judgement functions that nurses by virtue of their education and experience are capable and licensed to treat independently.

outcome standard. A patient-focused standard that addresses changes in the patient's health status or the results of nursing care.

overtube. A polyvinyl sleeve that fits over an endoscope and serves to protect the esophageal mucosa and the airway during various upper GI procedures, including extraction of foreign bodies.

oxyntic cell. *See* parietal cell.

pancreas. A large, elongated gland situated transversely behind the stomach, between the spleen and the duodenum.

pancreas divisum. A developmental anomaly in which the pancreas is present as two separate structures, each with its own duct.

pancreatic enzyme insufficiency. A deficiency in pancreatic exocrine function, leading to malabsorption of fats and other nutrients.

pancreatic fistula. An abnormal passage between the pancreas and another organ or, more often, between the pancreas and the exterior, often following pancreatic trauma, external drainage of a pseudocyst or pancreatic surgery.

pancreaticoduodenectomy. Surgical procedure indicated as therapy for chronic pancreatitis and its inherent complications and as a therapy for pancreatic cancer.

pancreatitis. Acute or chronic inflammation of the pancreas.

Paneth's cell. A narrow, pyramidal or columnar epithelial cell with a round or oval nucleus close to the base of the cell, occurring in the fundus of the crypts of Lieberkühn; Paneth's cells contain large secretory granules that may contain peptidase.

papillotome. A cutting instrument for incising the papilla of Vater.

papillotomy. Incision of a papilla.

paracentesis. Surgical puncture of a cavity for the aspiration of fluid, especially the abdominal cavity.

paralytic ileus. Obstruction of the intestines resulting from inhibition of bowel motility, which may be produced by numerous causes, most frequently by peritonitis.

parenteral. Administration of medications or nutrition by an injection route, such as subcutaneous, intramuscular or intravenous.

parietal cell. A cell type located in the parietal glands of the stomach and that secretes hydrochloric acid and intrinsic factor. Also known as oxyntic cells.

pathology. The structural or functional manifestations of disease.

pedunculated polyp. A polyp that is attached to the mucosa by a stemlike pedicle or stalk.

peptic ulcer. An ulceration of the mucous membrane of the esophagus, stomach or duodenum, caused by the action of the acid gastric juice.

peracetic acid. A chemical solution used for high-level disinfection.

percutaneous endoscopic gastrostomy (PEG). A technique for the endoscopic insertion of a gastrostomy feeding tube, for the purpose of providing enteral feeding.

percutaneous endoscopic jejunostomy (PEJ). A technique for the endoscopic insertion of a feeding tube into the jejunum, for the purpose of providing enteral feeding.

percutaneous liver biopsy. Aspiration biopsy of the liver by using a needle that has been inserted through a small incision in the skin.

percutaneous transhepatic cholangiogram. A roentgenogram of the hepatic and biliary ductal systems following injection of contrast directly into an intrahepatic bile duct, using a needle that is introduced percutaneously into the liver, through the eighth or ninth intercostal space.

perforation. A hole made through a body part.

performance improvement. *See* process improvement.

peripheral parenteral nutrition (PPN). Intravenous administration of a prescribed diet by means of a catheter inserted into a peripheral vein.

peristalsis. A distally progressive band of circular muscle contraction that causes the gradual progression of digestive contents through the GI tract.

peritoneoscopy. Examination of the peritoneal cavity by an instrument (laparoscope) that is inserted through the abdominal wall.

peritoneum. The serous membrane that lines the abdominopelvic walls and holds the viscera in place.

pernicious anemia. A megaloblastic anemia occurring in children or more commonly in later life, characterized by histamine-fast achlorhydria; laboratory and clinical manifestations are based on malabsorption of vitamin B_{12} because of a failure of the gastric mucosa to secrete adequate and potent intrinsic factor.

Peutz-Jeghers syndrome. A hereditary syndrome characterized by gastrointestinal polyposis associated with excessive melanin pigmentation of the skin and mucous membranes; gastrointestinal bleeding and intussusception are common complications.

Peyer's patch. An oval elongated area of lymphoid tissue on the mucosa of the small intestine, composed of many lymphoid nodules closely packed together.

philosophy. Major beliefs held by an individual or the members of a group.

phlebitis. Inflammation of a vein.

photocoagulation. Condensation of protein material by the controlled use of an intense beam of light.

physiograph. Device that produces a graphic display of test results.

pigtail stent. A stent that is coiled at one or both ends. The coiled shape straightens out when the stent is pulled taut but returns when it is allowed to relax.

placebo effect. An assumed psychological response to administration of a treatment suggested by the process of taking a medicine; usually encountered in drug testing, these effects are discounted from the real effects of the drug under study.

planning. The development of patient goals based on nursing diagnoses for the purpose of preventing, reducing or resolving health problems through nursing intervention.

plasma. The fluid portion of the blood, in which the particulate components are suspended.

plasmolysis. Contraction or shrinking of a cell caused by the loss of water by osmotic action.

platelet. A disk-shaped structure found in the blood of all mammals and chiefly known for its role in blood coagulation.

plexus. A general term for a network of lymphatic vessels, nerves or veins. *See also* Auerbach's plexus, enteric plexus *and* Meissner's plexus.

pneumatic balloon. A balloon that is inserted over a guidewire into the lower esophageal sphincter and then inflated to a preset pressure and left in place for a period of time; used in the treatment of patients with achalasia.

pneumoperitoneum. The presence of gas or air in the peritoneal cavity; it may occur spontaneously, as in a subphrenic abscess, or be deliberately introduced as an aid to radiological examination and diagnosis.

polyp. A protruding growth from any mucous membrane; includes gastric polyps. *See also* adenomatous polyp, pedunculated polyp *and* sessile polyp.

polypectomy. Surgical or endoscopic removal of a polyp.

polypectomy snare. A sheathed wire loop that can be passed through the instrument channel of an endoscope; it may be attached to an electrosurgical unit and used to apply coagulation current for removal of gastrointestinal polyps, or it may be used to remove foreign bodies.

polyposis. The development of multiple polyps on a part. *See also* familial polyposis.

population. All of the members of a group in which a survey researcher is interested; the entire set of people, objects, etc., with characteristics in common.

porphyria. Any of a group of disturbances of porphyrin metabolism, characterized by marked increase in formation and excretion of porphyrins or their precursors.

portal hypertension. Abnormally increased blood pressure in the portal venous system, a frequent complication of cirrhosis of the liver.

portal triad. The grouping of the tributaries of the hepatic artery, hepatic vein and bile duct at the angles of the lobules of the liver.

position description. A delineation of the responsibilities of an individual in an organization, including the title of the position, the department, the person to whom the individual is responsible, a job summary, job qualifications and specific duties and functions.

possible health problem. A health condition that has a high probability of developing because of an existing condition or disease.

potential health problem. A health condition that does not presently exist, but because of the presence of identified risk factors, requires that the nurse take preventive measures.

pressure transducer. A device that translates one form of energy to another; in the case of manometric pressure transducers, changes in pressure are translated into electrical signals.

primary sclerosing cholangitis. A rare and serious condition in which inflammation involves the entire biliary tract; often related to GI or biliary tract infection.

priority setting. The activity concerned with ranking nursing diagnoses in order of actual or potential threat to the patient's well-being.

process improvement. A systematic approach to the way work is designed and performance is measured, assessed and improved.

process standard. A standard that focuses on the nature and sequence of activities carried out by nurses implementing the nursing process; describes an acceptable level of performance of nursing actions.

proctoscopy. Inspection of the rectum with a speculum or tubular instrument with appropriate illumination.

proctosigmoidoscopy. Examination of the rectum and sigmoid colon with an instrument designed for illuminating and viewing those areas.

project organization. An organizational structure that is designed to complete a specific task.

prostaglandin. A group of naturally occurring, chemically related, long-chain hydroxy fatty acids that stimulate contractility of smooth muscle and have the ability to lower blood pressure, regulate acid secretion of the stomach, regulate body temperature and platelet aggregation, and control inflammation and vascular permeability; they also affect the action of certain hormones. There are six types: A, B, C, D, E and F, with the degree of saturation of the side chain of each being designated by subscripts 1, 2 and 3.

protein. Any of a group of complex organic compounds, which contain carbon, hydrogen, oxygen, nitrogen, and usually sulfur, the characteristic element being nitrogen. Proteins are of high molecular weight and consist essentially of combinations of α-amino acids in peptide linkages.

prothrombin. Coagulation factor II, a protein present in the plasma that is converted to thrombin. *See also* prothrombin time.

prothrombin time. A measure of the rapidity of blood clotting that examines coagulation factors I, II, V, VII and X.

pseudocyst. An abnormal or dilated space resembling a cyst but not lined by epithelium as is a true cyst. A pancreatic pseudocyst is an encapsulated collection of pancreatic juice and cellular debris that has escaped from the pancreas, the wall being formed by inflammatory fibrosis of serosal surfaces of adjacent organs; pseudocysts most commonly occur in the lesser sac of the peritoneum.

pseudomembranous colitis. An acute inflammation of the bowel mucosa with the formation of pseudomembranous plaques overlying an area of superficial ulceration and the passage of the pseudomembranous material in the feces; may result from shock and ischemia or be associated with antibiotic therapy. Also called necrotizing enterocolitis.

pyloric gland. A gland located in the antrum or pylorus of the stomach; contains mucous cells and G cells.

pyloric sphincter. The thickened muscular sphincter that controls the passage of food from the stomach into the duodenum.

pyloric stenosis. Obstruction of the pyloric sphincter at the outlet of the stomach.

pylorus. The most distal portion of the stomach, lying between the antrum and the duodenum.

pyrosis. *See* heartburn.

qualitative. Pertains to describing or analyzing qualities, attributes or characteristics.

quality. In health care, the degree to which actions taken or not taken maximize the probability of beneficial outcomes.

quantitative. Pertaining to or measuring quantity.

radiation enteritis. Radiation injury to the intestines, usually occurring as a result of radiotherapy for pelvic, intraabdominal or retroperitoneal malignancies.

radiography. The making of film records (radiographs) of internal structures of the body by passage of x-rays or gamma rays through the body to act on specially sensitized film. *See also* roentgenography.

random sample. Selection from the population (or a subpopulation) at large performed in such a way that each member of the population has an equal probability of being included in the sample.

range. The highest score (or value) minus the lowest score in a given set of values.

rapid pull-through. A technique whereby a manometry catheter is withdrawn steadily through the esophagus while the patient is not breathing or swallowing.

rectosigmoidoscopy. Endoscopic visualization of the lower portion of the sigmoid colon and the upper portion of the rectum.

rectum. The distal portion of the colon, beginning anterior to the third sacral vertebra as a continuation of the sigmoid and ending at the anal canal.

red blood cell. *See* erythrocyte.

referral. The process of sending a patient to another source for aid or to another professional for appropriate action; also, a patient received from another source.

regional enteritis. *See* Crohn's disease.

regurgitation. A backward flowing of undigested food.

respiratory depression. A decrease in the rapidity and depth of respirations.

respiratory inversion point. During station pull-through, manometry of the lower esophageal sphincter, the point at which the tracing goes down (rather than up) with an inspiration. Indicates the point at which the recording port moves from the abdominal cavity into the thoracic cavity.

retrospective audit. An evaluation of nursing care and patient outcomes performed after discharge of the patient. Postdischarge questionnaires, interviews (over the telephone or face-to-face) or chart review are retrospective auditing techniques.

Ringer's solution. A sterile solution containing sodium chloride, potassium chloride and calcium chloride in water for injection; used as a topical physiological salt solution.

roentgenography. The making of a record (roentgenogram) of internal body structures by passing x-rays through the body to act on specially sensitized film. *See also* contrast roentgenography *and* double-contrast roentgenography.

role delineation. A statement of the behaviors that are expected of an individual in a certain position, as of a gastroenterology nurse or nursing assistive personnel.

Roux-en-Y gastric bypass. Surgical procedure performed to treat morbid obesity.

ruga. A wrinkled ridge in the interior wall of the stomach. Plural: rugae.

sample. The portion of a group or population that is targeted for a survey. *See also* random sample.

sanitation. A process capable of destroying or reducing the number of microbial contaminants to a relatively safe level, as judged by public health requirements.

Savary-Gilliard dilator. One of a series of semiflexible, tapered polyvinyl chloride bougies that are passed over a guidewire for the purpose of widening a gastrointestinal lumen.

Schatzki's ring. One of a series of thin, concentric membranes located at the esophagogastric junction.

Schilling test. A test for gastrointestinal absorption of vitamin B_{12}, in which a measured amount of radioactive vitamin B_{12} is given orally and the percentage of radioactivity in the urine excreted over a 24-hour period is determined.

Schindler, Gabriele. The wife of pioneer gastroscopist Dr. Rudolph Schindler. She assisted her husband with gastroscopic procedures and is considered a role model for today's professional gastroenterology nurses and associates.

Shwachman-Diamond syndrome. Pancreatic insufficiency, cyclic neutropenia, metaphyseal dysostosis and growth retardation. Second most common cause of pancreatic insufficiency in children.

scintigraphy. The production of two-dimensional images of the distribution of radioactivity in tissues after the internal administration of radionuclide, with the images obtained by a scintillation camera.

sclerotherapy. The injection of sclerosing solutions in the treatment of hemorrhoids, varicose veins or esophageal varices.

scope of practice. A statement of the dimensions of a professional practice that outlines the functions of individuals in that profession.

secretin. A strongly basic polypeptide hormone secreted by the mucosa of the duodenum and jejunum when acid chyme enters the intestine. Carried by the blood, it stimulates the secretion of a watery pancreatic juice high in salt content but low in enzymes. It has a lesser stimulatory effect on bile and intestinal secretion.

sedation and analgesia. Describes a state that allows patients to tolerate unpleasant procedures while maintaining protective reflexes. Refers to a continuum of states ranging from minimal sedation (anxiolysis) through general anesthesia.

sedative. An agent that allays excitement.

self-expanding metal stents. Compressed and stretched metal devices that increase in diameter and decrease in length automatically on deployment to relieve strictures in lumens of various body structures.

semicritical item. An item or instrument (including endoscopes) that may come in contact with intact mucous membranes but does not ordinarily penetrate body surfaces. Meticulous physical cleaning followed by high-level disinfection is required for these items.

Sengstaken-Blakemore tube. A three-lumen tube used for esophageal-gastric tamponade; it has both gastric and esophageal balloons and a port for gastric aspiration.

serum. The cell-free portion of the blood from which the fibrinogen has been separated in the process of clotting.

sessile polyp. A polyp that is attached to the mucosa by a broad base.

shock. (1) A condition of acute peripheral circulatory failure caused by derangement of circulatory control or loss of circulating fluid; marked by hypotension, coldness of the skin, usually tachycardia and often anxiety. (2) An extreme stimulation of the nerves, muscles, etc., accompanying the passage of electrical current through the body.

short bowel syndrome. Any of the malabsorption syndromes resulting from massive resection of the small bowel, the degree and kind of malabsorption depending on the site and extent of the resection; characterized by diarrhea, steatorrhea and malnutrition.

sigmoid colon. The S -shaped part of the colon, lying in the pelvis, extending from the pelvic brim to the third segment of the sacrum, and continuous above with the descending (iliac) colon and below with the rectum.

sigmoidoscopy. Inspection of the sigmoid colon through the use of an endoscope.

sinusoid. A form of terminal blood channel consisting of a large, irregular anastomosing vessel; found in the liver, suprarenals, heart, parathyroid, carotid gland, spleen, hemolymph glands and pancreas.

sliding hiatal hernia. Common type of hiatal hernia in which the gastroesophageal junction and a portion of the stomach slide upward into the mediastinum.

small bowel. The proximal portion of the intestine.

small bowel enteroscopy. Visualization of the small bowel with a long, thin, extremely flexible endoscope.

Society of Gastroenterology Nurses and Associates, Inc. (SGNA). The professional organization of nurses and associates dedicated to the safe and effective practice of gastroenterology and endoscopy nursing.

solid-state catheter. A long, flexible manometry catheter that contains a series of miniature pressure transducers that directly record gastrointestinal contractions.

sphincter. A ring-like band of muscle fibers that constricts a passage or closes a natural orifice. *See also* cardiac sphincter, gastroesophageal sphincter, hypopharyngeal sphincter, lower esophageal sphincter, pyloric sphincter, sphincter of Oddi *and* upper esophageal sphincter.

Sphincter of Oddi. The sheath of muscle fibers surrounding bile and pancreatic ducts as they pass through the wall of the duodenum.

sphincterotome. An electrosurgical instrument for cutting through a sphincter, specifically the sphincter of Oddi.

sphincterotomy. Division of a sphincter, especially division of the sphincter of Oddi during ERCP.

splenic flexure. The left flexure of the colon; the bend at which the transverse colon becomes the descending colon.

sprue. A chronic form of malabsorption syndrome that occurs in both tropical and celiac forms. *See also* celiac sprue *and* tropical sprue.

staff organization. A type of organizational structure that requires staff to assist management but allows them no authority.

standard. An acceptable, expected level of performance established by authority, custom or consent. Standards in nursing define optimum levels of actual and expected performance. *See also* outcome standard, process standard *and* structural standard.

standard deviation. An average-size spread among values in a set around the average value in the set; how far away the numbers in a list are from their average.

standard for practice. An authoritative statement of the expected outcomes of professional practice, established through research and/or professional consensus.

standard of care. A measurable statement that defines the means to accomplish a practice outcome.

standard precautions. The minimum infection prevention practices that apply to all patient care, regardless of suspected or confirmed infection status of the patient, in any setting where health care is delivered. Standard Precautions are developed by the Centers for Disease Control and Prevention.

station pull-through. A technique whereby a manometry catheter is withdrawn through the esophagus in a step-wise fashion while the patient breathes slowly and evenly.

steatorrhea. Excessive amounts of fat in the feces, as in malabsorption syndromes.

stent. A hollow tube or endoprosthesis that is inserted for the purpose of bypassing diseased or obstructed parts of a duct or tubular organ. *See also* barbed stent.

sterilization. A process resulting in the complete elimination or destruction of all forms of microbial life.

stoma. An opening established in the abdominal wall by colostomy, ileostomy, etc.

stress ulcer. A form of acute gastritis that is related to a severe trauma, illness or chronic ingestion of certain drugs.

stricture. A narrowing of a canal, duct or other passage as a result of scarring or deposition of abnormal tissue.

structural standard. A standard concerned with the environment in which care is provided.

subjective data. Perceptions by an affected person that cannot be perceived or verified experimentally.

suction biopsy. A method of obtaining tissue specimens from the rectum or small bowel, by creating a vacuum in a specially designed capsule or tube.

syncope. A temporary suspension of consciousness caused by generalized cerebral ischemia; faint.

syndrome. A set of symptoms that occur together; the sum of signs of any morbid state; a symptom complex.

tamponade. Compression of a part. *See also* balloon tamponade.

tenesmus. Straining, especially ineffectual and painful straining at stool or in urination.

tenia coli. Three thickened flat bands, about one-sixth shorter than the colon, formed by the longitudinal fibers in the muscular tunic of the colon and extending from the root of the vermiform appendix to the rectum, where they spread out and form a continuous layer encircling the tube.

threshold. Acceptable rates of activity that determine when to evaluate care.

tonicity. The effective osmotic pressure equivalent.

topical. Pertaining to a particular surface.

total parenteral nutrition (TPN). The intravenous administration of the total nutrient requirements of a patient with gastrointestinal dysfunction, accomplished via a central venous catheter, usually inserted in the superior vena cava.

toxic megacolon. Acute dilatation of the colon associated with amebic or ulcerative colitis; it may precede perforation of the colon.

tracheoesophageal fistula. Abnormal passage between the trachea and esophagus.

transverse colon. The portion of the colon that runs transversely across the upper part of the abdomen, from the right to the left colic flexure.

triglyceride. A compound consisting of three molecules of fatty acid esterified to glycerol; it is a neutral fat synthesized from carbohydrates for storage in animal adipose cells. On enzymatic hydrolysis, it releases free fatty acids in the blood.

trocar. A sharp-pointed instrument contained in a cannula, used to puncture the wall of a body cavity; usually used for insertion of the cannula.

tropical sprue. A malabsorption syndrome occurring in the tropics and subtropics. Protein malnutrition is usually precipitated by malabsorption, and anemia caused by folic acid insufficiency is particularly common.

ulcer. A local defect, or excavation, of the surface of an organ or tissue, which is produced by the sloughing of inflammatory necrotic tissue. *See also* Curling's ulcer, Cushing's ulcer, gastric ulcer, peptic ulcer *and* stress ulcer.

ulcerative colitis. Chronic, recurrent ulceration in the colon, chiefly of the mucosa and submucosa, of unknown cause; manifested clinically by cramping, abdominal pain, rectal bleeding, and loose discharges of blood, pus, and mucus with scanty fecal particles.

ultrasonography. Mechanical radiant energy with a frequency greater than 20,000 hertz (cycles per second); ultrasonography is the visualization of deep structures of the body by recording the reflections (echoes) of ultrasonic waves directed into the tissues. Diagnostic ultrasonography uses a frequency range of 1 million to 10 million Hz, or 1 to 10 MHz.

upper gastrointestinal (UGI) series. A series of radiographs taken to visualize the esophagus, stomach, and sometimes the small bowel, following the ingestion of a barium solution.

upper esophageal sphincter (UES). The sphincter located at the upper end of the esophagus. Also known as the hypopharyngeal sphincter.

vagotomy. Surgical deenervation of vagus nerve.

validation. Verification; confirmation.

Valsalva maneuver. Forcible exhalation against a closed glottis, resulting in an increase in intrathoracic pressure.

variability. A concept concerned with how spread out or dispersed the data values are about the mean; the degree to which subjects in a sample vary from one another with respect to some critical attribute.

variable. A measured concept or construct; a characteristic or attribute of a person or object that takes on different values within the population under study. *See also* dependent variable *and* independent variable.

varix. An enlarged and tortuous vein or artery. Plural, varices.

vasovagal reaction. A transient vascular and neurogenic reaction marked by pallor, nausea, sweating, bradycardia and rapid fall in arterial blood pressure which, when below a critical level, results in loss of consciousness and characteristic EEG changes. It is most often evoked by emotional stress associated with fear or pain.

vermiform appendix. A worm-like diverticulum of the cecum, ranging from 3 to 6 inches in length.

veress needle. A disposable or reusable needle that is used in laparoscopic procedures for the creation of pneumoperitoneum.

vertical banded gastroplasty. Surgical procedure performed to treat morbid obesity.

villus. A small vascular process or protrusion, especially such a protrusion from the free surface of a membrane; the intestinal villi are the numerous threadlike projections that cover the surface of the mucosa of the small bowel and serve as the sites of absorption of fluids and nutrients.

vision. A statement that sets direction for a unit, department or institution.

vitamin. An organic substance that occurs in foods in small amounts and that is necessary in trace amounts for the normal metabolic functioning of the body.

volvulus. Intestinal obstruction caused by a knotting and twisting of the bowel.

washing. Collection of a specimen for culture or cytology by injecting and then aspirating 20 to 30 ml of nonbacteriostatic saline.

water perfusion catheter. A long, multilumen manometry catheter that is continuously perfused with water. Each lumen has a separate recording port that is attached to a separate external pressure transducer; when a port is occluded by gastrointestinal contractions, the resulting pressure change is recorded on a physiograph.

webs. *See* esophageal rings and webs.

wet swallow. Swallowing 3 to 5 ml of water during esophageal manometry. *See also* dry swallow.

Whipple's disease. A malabsorption syndrome characterized by diarrhea, steatorrhea, skin pigmentation, arthralgia and arthritis, lymphadenopathy and central nervous system lesions.

Whipple's procedure. Pancreaticoduodenectomy.

white blood cell. *See* leukocyte.

whole blood. Blood from which none of the elements have been removed.

Wilson's disease. Hepatolenticular degeneration. A rare progressive disease, inherited as an autosomal-recessive trait, and caused by a defect in the metabolism of copper; a pigmented ring at the outer margin of the cornea is pathognomonic.

x-ray. Electromagnetic vibrations of short wavelengths that are produced when high-velocity electrons impinge on various substances. X-rays are able to penetrate some substances much more readily than others and to affect a photographic plate, thus making them useful for taking roentgenograms of various parts of the body. They also cause certain substances to fluoresce, allowing fluoroscopic observation of the size, shape and movements of various organs.

Zollinger-Ellison syndrome. A triad comprising intractable, sometimes fulminating and in many ways atypical peptic ulcers; extreme gastric hyperacidity; and gastrin-secreting, nonbeta islet cell tumors of the pancreas.

zymogen cell. *See* chief cell.

For specific references, refer to the Bibliography at the end of each chapter.

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